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Fuzhen Zhang

Matrix Theory

Basic Results and Techniques

Third Edition

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Fuzhen Zhang

Matrix Theory

Basic Results and Techniques

Third Edition

Springer

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To my beloved wife Cheng and our children Sunny, Andrew, and Alan

Preface to the Third Edition

This is the third edition of *Matrix Theory: Basic Results and Techniques*.

Over the years, as I revised and expanded this book, my goal has remained the same: to provide a comprehensive text for a second course in linear algebra and matrix theory where many topics are covered briefly. This text is intended for advanced undergraduate and early graduate students. Through this course, students learn, practice, and master fundamental matrix results and techniques—what I like to call *matrix kung fu*—which are essential for applications in various fields, including mathematics, statistics, physics, computer science, data science, engineering, etc. I also hope that the book serves as a valuable reference for researchers in the field as well as for those who use matrices.

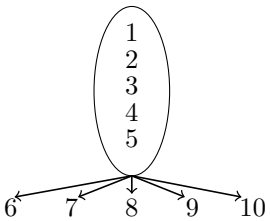
New Features in This Edition:

This edition introduces more than 200 examples. These examples help make theoretical concepts—such as theorems and proofs—more accessible to students. Additionally, this edition includes approximately 1,500 exercises of varying difficulty levels. With a focus on basic matrix theory and an emphasis on partitioned matrices, this edition significantly revises and expands upon its predecessors.

Notes to the Instructor:

Matrix theory (or a second course in linear algebra) can be approached in various ways. The material in this text reflects my own interests and preferences: The results are basic, important, and elegant. While the content is condensed, I strive to make it clear, concise, and accessible.

Covering all chapters in detail in a semester class is impractical. The first five chapters establish fundamental concepts, while Chapters 6–9 focus on special matrices, including unitary, positive semidefinite, Hermitian, and normal matrices. Chapter 10 explores majorization inequalities (of eigenvalues, singular values, diagonal entries, and norms), which can be used as a supplementary material or as a graduate-level topic in pure or applied mathematics.



My emphasis is on basic results and techniques rather than standard materials in a traditional way. Some key results are presented with multiple proofs, while certain topics are tailored for classroom instruction. For instance, in Sect. 3.4, the Jordan canonical form is described without a rigorous proof. A formal proof (e.g., via λ-matrices) requires substantial class time and overlaps with similar material typically covered in elementary number theory and abstract algebra courses. Here, the theorem itself is prioritized over its proof, and the concepts are conveyed through examples.

Some results are too important to omit. For example, the second part of Geršgorin's theorem is an elegant and intuitive result involving discs, but its rigorous proof requires complex analysis (specifically the argument principle). Another example concerns the continuity of eigenvalues. If the entries of a matrix depend continuously on a parameter over a domain, one might naturally expect the eigenvalues to also vary continuously with the parameter, forming smooth curves that evolve on the complex plane. However, this is not necessarily the case, or at least, not as easy as it may seem (Sect. 2.6). The material is included with explanation rather than formal proofs.

Notes to the Reader:

Some knowledge in precalculus, elementary linear algebra, and calculus is needed as prerequisite. In particular, one is assumed to know

(a) Polar representation of complex numbers:
$z = re^{i\theta}$, $r \geq 0$, $\theta \in (-\pi, \pi] \subset \mathbb{R}$;

(b) Fundamental theorem of algebra:
A polynomial of degree n (≥ 1) has n roots (including multiplicities) in the complex number field $\mathbb{C}$;

(c) Extreme value theorem:
A real-valued continuous function on a closed interval (a compact set) attains its maximum and minimum values; and

(d) Mathematical induction:
Understanding induction and its variations is essential for mathematical proofs, particularly for the proofs involving positive integers (such as the dimension of a vector space or the order of a matrix).

If an unfamiliar symbol or term arises, consult the Notation section (p. 495) or the Index (p. 499). Below are some frequently used notations in this text:

- $|\cdot|$: Absolute value, modulus, and determinant. If z is a complex number, then $|z|$ is the absolute value (i.e., modulus) of z. If M is a large or partitioned matrix, say $M = \left(\begin{smallmatrix} A & B \\ C & D \end{smallmatrix}\right)$, where A, B, C, and D are block matrices, we denote the determinant of M by $\det M = \left|\begin{smallmatrix} A & B \\ C & D \end{smallmatrix}\right|$ for simplicity. For a matrix $A = (a_{ij})$, $|A|$ may have two other very different meanings: (1) $|A| = (|a_{ij}|)$, the entrywise modulus matrix (mostly in nonnegative matrix theory; Sect. 5.7), and (2) $|A| = (A^*A)^{1/2}$, the operator modulus of A (in operator theory and matrix analysis), where A^* is the conjugate transpose of A. We adopt (2) in this book except in Sect. 5.7. The intended meaning of $|\cdot|$ should be clear from context.

- $\Leftrightarrow$: Logical equivalence. If statement A is equivalent to statement B, we say that A (is true) *if and only if* B (is true), and write $\mathrm{A} \Leftrightarrow \mathrm{B}$. That is, A implies B, i.e., $\mathrm{A} \Rightarrow \mathrm{B}$, and A is implied from B, i.e., $\mathrm{A} \Leftarrow \mathrm{B}$.
- $\mathbb{M}_n$: Matrix notation. This is the set of all $n \times n$ complex matrices. We write $A \in \mathbb{M}_n$ to mean that A is an $n \times n$ matrix with complex (including real) entries. We also call an $n \times n$ matrix an n-square matrix, or a matrix of order n or of size $n \times n$.
- i: Imaginary unit and index. This letter is widely used in math for the *imaginary* unit $i = \sqrt{-1}$ of complex numbers as well as for the *index* of a sequence (or similar kind). In the complex number $z = a+bi$, where $a = \Re(z)$ and $b = \Im(z)$ are real, $i = \sqrt{-1}$, while in the summation $\sum_{i=1}^{n} a_i$, i is the (subscript) index of the sum. Likewise, if $A = (a_{ij})$ is a matrix, a_{ij} denotes the entry of A in row i and column j.
- Plural forms. We often see statements like "Let $u_1, u_2, \ldots, u_n$ be vectors in V (a vector space)." We may simply state this as "Let u_i's (or u's) be vectors in V." To avoid typographical ambiguity, we write u_i's or u's instead of us. Similarly, we use v's (not vs) for the plural form of v. A matrix with zero and one entries can be written as a matrix of 0's and 1's, or of 0s and 1s (both usages are common in math writing).
- **Proof.** The proofs of theorems begin with **Proof**, whereas some examples containing mathematical statements that require "proofs" begin with an indented *Proof* (within the Example environment).
- ∎: This symbol marks the end of a proof for a theorem or an example.

Matrix theory is one of the most widely used mathematical disciplines across scientific fields. While this book does not cover all advanced topics, interested readers can find additional references at the end of the book.

I wish to express my gratitude to all those who offered feedback on the first and second editions. I am indebted to the following individuals for their comments and suggestions on this edition's manuscript: Yunshyong Chow, Yongjian Hu, Huajun Huang, Yongtao Li, Mohammad Sal Moslehian, Meiyue Shao, and Qingxiang Xu. My special thanks go to Roger Horn for his invaluable insights and fruitful discussions on a wide range of topics.

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February 18, 2026

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Frequently Used Notation and Terminology

$\mathbb{R}^n$, 3	vectors (n-tuples) of n real components
$\mathbb{C}^n$, 3	vectors (n-tuples) of n complex components
$\mathbb{R}_n[x]$, 3	real polynomials in x of degree less than or equal to n
$\dim V$, 5	dimension of vector space V
$\mathbb{M}_n$, 12	$n \times n$ (i.e., n-square) matrices with real or complex entries
$\mathbb{M}_{m\times n}$, 12	$m \times n$ matrices with real or complex entries
$A = (a_{ij})$, 12	matrix A with (i,j)-entry a_{ij}
I, I_n, 13	identity matrix, identity matrix of order n
A^{-1}, 13	inverse of matrix A
A^T, 14	transpose of matrix A
$\overline{A}$, 14	conjugate of matrix A
A^*, 14	conjugate transpose of matrix A, i.e., $A^* = \overline{A}^T$
$\mathrm{Ker}(A)$, 16, 32	kernel or null space of A, i.e., $\mathrm{Ker}(A) = \{x : Ax = 0\}$
$\mathrm{Im}(A)$, 16, 32	image space of A, i.e., $\mathrm{Im}(A) = \{Ax : \text{for all } x\}$
$\mathrm{rank}(A)$, 18	rank of matrix A
$\det A$, 20	determinant of matrix A
$\lvert A\rvert$, 20, 124, 241	$\det A$ if A is large in size, or $(A^*A)^{\frac{1}{2}}$, or $(\lvert a_{ij}\rvert)$ in Sect. 5.7
$\mathrm{tr}\, A$, 38	trace of matrix A
$\langle u, v\rangle$, 46	inner product of vectors u and v
$\lVert\cdot\rVert$, 47	norm of a vector or a matrix
$A \geq 0$, 120	A is positive semidefinite (or all $a_{ij} \geq 0$ in Sect. 5.7)
$A \geq B$, 120	$A - B$ is positive semidefinite (or $a_{ij} \geq b_{ij}$ in Sect. 5.7)
$\rho(A)$, 166	spectral radius of matrix A
$\sigma_{\max}(A)$, 166	largest singular value (spectral norm) of matrix A
$\lambda_{\max}(A)$, 170	largest eigenvalue of matrix A
$A \circ B$, 182	Hadamard (entrywise) product of matrices A and B
$A \otimes B$, 182	Kronecker (tensor) product of matrices A and B
$x \prec_w y$, 422	weak majorization, i.e., $\sum_{i=1}^k x_i^{\downarrow} \leq \sum_{i=1}^k y_i^{\downarrow}$, $k = 1, \ldots, n$
$x \prec_{\mathrm{wlog}} y$, 446	weak log-majorization, i.e., $\prod_{i=1}^k x_i^{\downarrow} \leq \prod_{i=1}^k y_i^{\downarrow}$, $k = 1, \ldots, n$

An $n \times n$ matrix A is said to be

Hermitian	if $A = A^*$, i.e., $a_{ji} = \overline{a_{ij}}$ for all i, j;
normal	if $A^*A = AA^*$, i.e., A and A^* commute;
positive semidefinite	if $x^*Ax \geq 0$ for all column vectors $x \in \mathbb{C}^n$;
unitary	if $AA^* = A^*A = I$, i.e., $A^{-1} = A^*$;
upper-triangular	if all entries below the main diagonal equal zero;
diagonalizable	if $P^{-1}AP$ is diagonal for some invertible matrix P;
similar to B	if $P^{-1}AP = B$ for some invertible matrix P; and
unitarily similar to B	if $U^*AU = B$ for some unitary matrix U.

Frequently Used Theorems

- **Cauchy-Schwarz inequality** (p. 46): Let V be an inner product space over a scalar field ($\mathbb{R}$ or $\mathbb{C}$) with inner product $\langle \cdot\,, \cdot \rangle$. Then for all $x, y \in V$

$$|\langle x, y\rangle|^2 \leq \langle x, x\rangle \langle y, y\rangle.$$

  Equality holds if and only if x and y are linearly dependent.

- **Theorem on the eigenvalues of AB and BA** (p. 87): Let A and B be $m \times n$ and $n \times m$ matrices, respectively. Then AB and BA have the same nonzero eigenvalues, counted according to multiplicities. As a consequence,

$$\operatorname{tr}(AB) = \operatorname{tr}(BA).$$

- **Schur triangularization theorem** (p. 119): Let A be an $n \times n$ matrix with eigenvalues $\lambda_1, \ldots, \lambda_n$. Then there exists an $n \times n$ unitary matrix U such that

$$A = U^* T U,$$

  where T is an upper-triangular matrix with $\lambda_1, \ldots, \lambda_n$ on the diagonal.

- **Jordan decomposition theorem** (p. 145): For any n-square matrix A, there exists an n-square invertible matrix P (of complex entries) such that

$$A = P^{-1}(J_1 \oplus \cdots \oplus J_k)P,$$

  where $J_1, \ldots, J_k$ are the Jordan blocks of A (with eigenvalues on the diagonals).

- **Spectral decomposition theorem** (p. 121): Let $A \in \mathbb{M}_n$ be a normal matrix with eigenvalues $\lambda_1, \ldots, \lambda_n$. Then there is a unitary matrix $U \in \mathbb{M}_n$ such that

$$A = U^* \operatorname{diag}(\lambda_1, \ldots, \lambda_n) U.$$

  In particular, if A is positive semidefinite, then all $\lambda_i \geq 0$; if A is Hermitian, then all λ_i are real; and if A is unitary, then all $|\lambda_i| = 1$, $i = 1, \ldots, n$.

- **Singular value decomposition theorem (SVD)** (p. 124): Let A be an $m \times n$ matrix with rank $r \geq 1$. Then there exist an m-square unitary matrix U and an n-square unitary matrix V such that

$$A = U D V,$$

  where D is an $m \times n$ matrix whose diagonal (i, i)-entries, $i = 1, \ldots, r$, are the nonzero singular values of A and whose other entries are all equal to zero.

CHAPTER 1

Elementary Linear Algebra Review

Introduction: We briefly review, mostly without proofs, the fundamental concepts and results typically covered in an introductory linear algebra course. The topics include vector spaces, bases and dimensions, matrices and determinants, linear transformations, eigenvalues, and inner product spaces.

1.1 Vector Spaces

A vector space comprises two sets and two operations. Let V be a nonempty set of elements (referred to as *vectors*) and let $\mathbb{F}$ be a field (of numbers, called *scalars*). On the elements of V, there is an operation $+$, called *vector addition*, that takes any pair of elements, say $u, v \in V$, to a unique element in V, called the *sum* of u and v,

$$u + v \in V, \quad \text{where } u, v \in V.$$

On the scalars in $\mathbb{F}$ and the elements in V, there is an operation $\cdot$, called *scalar multiplication*, that assigns any $c \in \mathbb{F}$ and $v \in V$ a unique element in V, denoted by $c \cdot v$, or simply cv (in which the dot $\cdot$ is usually omitted),

$$cv \in V, \quad \text{where } c \in \mathbb{F},\ v \in V.$$

We say that V is a *vector space* over $\mathbb{F}$ if the following axioms hold:

(1) $u + v = v + u$ for all $u, v \in V$.
(2) $(u + v) + w = u + (v + w)$ for all $u, v, w \in V$.
(3) There is an element $0 \in V$ such that $v + 0 = v$ for all $v \in V$.
(4) For each $v \in V$, there is an element $-v \in V$ such that $v + (-v) = 0$.
(5) $c(u + v) = cu + cv$ for all $c \in \mathbb{F}$ and $u, v \in V$.
(6) $(a + b)v = av + bv$ for all $a, b \in \mathbb{F}$ and $v \in V$.
(7) $(ab)v = a(bv)$ for all $a, b \in \mathbb{F}$ and $v \in V$.
(8) $1v = v$ for all $v \in V$.

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In other words, V is a vector space over $\mathbb{F}$ if (a) the vector addition is commutative and associative, and has an (unique) additive identity 0 and additive inverse $-v$ in V for each $v \in V$, and (b) the scalar multiplication is associative and distributive, and the number 1 satisfies $1v = v$ for all $v \in V$.

We call the elements of a vector space *vectors* and call 0 the *zero vector*. A vector space is also known as *linear space*. In addition, the field $\mathbb{F}$ of scalars is usually the real numbers $\mathbb{R}$ or the complex numbers $\mathbb{C}$ in this book. So, in a vector space, vectors can be added and can be multiplied by numbers.

Note that for convenience and simplicity of exposition, we have used the symbol + for the addition of vectors and the addition of scalars. For instance, in (6), $a + b$ is the sum of scalars a and b, while $av + bv$ is the sum of vectors av and bv. Likewise, 0 may represent the zero vector or the zero scalar. One can tell which is which in the context; no confusion is caused.

Example 1.1.1 $\mathbb{R}^2$, the set of all ordered real pairs (x, y), is a vector space over $\mathbb{R}$ with respect to the addition and scalar multiplication defined by

$$(x_1, y_1) + (x_2, y_2) = (x_1 + x_2, y_1 + y_2), \quad c(x, y) = (cx, cy).$$

One checks all the conditions (1)–(8). For instance, for condition (6),

$$\begin{aligned} (a + b)(x, y) &= \big((a + b)x, (a + b)y\big) = (ax + bx, ay + by) \\ &= (ax, ay) + (bx, by) = a(x, y) + b(x, y). \end{aligned}$$

The zero vector is $(0, 0)$, and the negative element of (x, y) is $(-x, -y)$. ∎

Intuitively, we may view $\mathbb{R}^2$ as the Cartesian xy-plane, the complex plane $\mathbb{C}$, or the set of all arrows (directed line segments) in a plane. Two arrows are added by the parallelogram rule and multiplied by scalars, say cu (where $c \in \mathbb{R}$, $u \in \mathbb{R}^2$), cu is the arrow whose length is $|c|$ times the length of u, pointing in the same direction as u if $c \geq 0$, and otherwise pointing in the opposite direction. [Note: two arrows are considered to be equal if they have the same length and point in the same direction.] Similarly, $\mathbb{R}^3 = \{(x, y, z) : x, y, z \in \mathbb{R}\}$ is a vector space over $\mathbb{R}$. $\mathbb{R}^2$ and $\mathbb{R}^3$ are premier examples that help us understand and visualize many concepts of vector spaces.

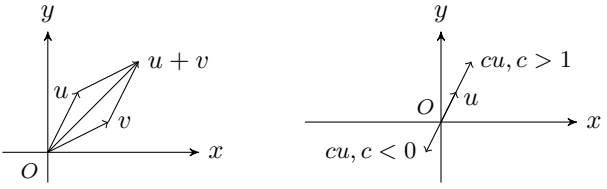


Fig. 1.1. Vector addition and scalar multiplication

$\mathbb{F}^n = \{(x_1, x_2, \ldots, x_n) : x_1, x_2, \ldots, x_n \in \mathbb{F}\}$, the set of n-tuples with components in $\mathbb{F}$, is a vector space over $\mathbb{F}$ with respect to the vector addition

$$(x_1, x_2, \ldots, x_n) + (y_1, y_2, \ldots, y_n) = (x_1 + y_1, x_2 + y_2, \ldots, x_n + y_n)$$

and the scalar multiplication

$$c\,(x_1, x_2, \ldots, x_n) = (cx_1, cx_2, \ldots, cx_n), \quad c \in \mathbb{F}.$$

$\mathbb{R}^n$ and $\mathbb{C}^n$ are frequently used. Note that in the future when dealing with matrix-vector product, we often write the n-tuples as column vectors

$$x = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix} = (x_1, x_2, \ldots, x_n)^T \quad (^T \text{ for transpose}).$$

Example 1.1.2 Let V be the set of real-valued functions defined on a domain $\mathbb{D}$ (usually a real interval or $\mathbb{R}$). The functions in V are added and multiplied by scalars in the usual way, that is, for $f, g \in V$ and $c \in \mathbb{R}$,

$$(f + g)(t) = f(t) + g(t), \quad (cf)(t) = cf(t), \quad t \in \mathbb{D}.$$

V is a vector space over $\mathbb{R}$ (or we say that V is a *real vector space*). ∎

One checks that the set of even functions on $\mathbb{R}$ (i.e., $f(-t) = f(t)$, $t \in \mathbb{R}$) forms a real vector space. What about the set of odd functions?

Example 1.1.3 Let $\mathbb{R}_n[x]$ be the set of real polynomials in x of degree $\leq n$. For technical reasons, it is convenient to define the degree of the zero polynomial 0 to be $-\infty$ and adopt usual conventions such as $-\infty < n$, etc. So, $0 \in \mathbb{R}_n[x]$. $\mathbb{R}_n[x]$ is a real vector space with respect to the usual addition and scalar multiplication for polynomials. $\mathbb{R}[x]$, the set of real polynomials of any finite degrees and the zero polynomial, is also a real vector space. ∎

Note that the set of the polynomials of degree equal to (a fixed) n is not a vector space because the set is not closed with respect to the addition.

Example 1.1.4 Let V be a vector space over $\mathbb{F}$. Let 0 be the zero element of $\mathbb{F}$ and let O be the zero vector of V. We show that $0v = O$ for all $v \in V$.

First observe that if $u + w = u$ for some vector $u \in V$, then $w = O$. This is because $O = u + (-u) = (u + w) + (-u) = (w + u) + (-u) = w + (u + (-u)) = w + O = w$. Since $0v = (0 + 0)v = 0v + 0v$, we get $0v = O$.

In a similar way, one shows that $cO = O$ for all $c \in \mathbb{F}$. ∎

Exercise: If $cv = O$, where $c \in \mathbb{F}$, $v \in V$, show that $c = 0$ or $v = O$.

A vector space is assumed to have an additive binary operation. For three vectors u, v, w in a vector space V, we can add u and v first, then add $u + v$ and w. Thus, $(u+v)+w$ is well-defined as an element in V; so is $u+(v+w)$. Because of associativity (2), $(u+v)+w = u+(v+w)$. We write it as $u+v+w$. For four vectors, say v_1, v_2, v_3, v_4, one proves that

$$(v_1 + v_2) + (v_3 + v_4) = v_1 + \big(v_2 + (v_3 + v_4)\big) = v_1 + (v_2 + v_3 + v_4).$$

Likewise,

$$v_1 + (v_2 + v_3) + v_4 = v_1 + ((v_2 + v_3) + v_4) = v_1 + (v_2 + v_3 + v_4).$$

Inductively, for more vectors, the associativity of addition[①]ensures that repeated applications of the operation produce the same result regardless how we group the vectors by parentheses. We denote by $v_1 + v_2 + \cdots + v_k$ the unique sum of vectors $v_1, v_2, \ldots, v_k$. Moreover, the sum is independent of the order of the vectors because vector addition is commutative.

Let S be a subset of a vector space V over a field $\mathbb{F}$. Denote by $\operatorname{Span} S$, called the *span* of S, the set of all *linear combinations* of finitely many vectors in S, that is, $\operatorname{Span} S$ consists of all vectors of the form

$$c_1 v_1 + c_2 v_2 + \cdots + c_p v_p,$$

where p is a positive integer, $c_1, c_2, \ldots, c_p \in \mathbb{F}$, and $v_1, v_2, \ldots, v_p \in S$.

For the sake of convenience, we define $\operatorname{Span} S = \{0\}$ if S is the empty set.

Apparently, $S \subseteq \operatorname{Span} S \subseteq V$. If $\operatorname{Span} S = V$, then every vector in V can be expressed as a (finite) linear combination of the vectors in S. In this case, we say that the set S *spans* or generates the vector space V.

Vectors $v_1, v_2, \ldots, v_k$ are said to be *linearly independent* if

$$c_1 v_1 + c_2 v_2 + \cdots + c_k v_k = 0 \tag{1.1}$$

only when $c_1 = c_2 = \cdots = c_k = 0$. Equivalently, for linearly independent vectors $v_1, v_2, \ldots, v_k$, (1.1) holds if and only if all coefficients c's are zero. Vectors that are not linearly independent are said to be *linearly dependent*.

Note: We also say that the set $\{v_1, v_2, \ldots, v_k\}$ is linearly (in)dependent if the vectors $v_1, v_2, \ldots, v_k$ in the set are linearly (in)dependent. The vectors in an indexed set can be equal. For instance, $\{v_1, v_2, v_3\}$ is a linearly dependent set in $\mathbb{R}^2$, where $v_1 = (1, 0)$, $v_2 = (0, 1)$, and $v_3 = (0, 1) = v_2$.

An infinite set S of vectors is said to be *linearly independent* if any nonempty finite subset of S is linearly independent.

If a set contains the zero vector, then the set is linearly dependent.

[①]This is the so-called *generalized associative law*. See J.J. Rotman, *A First Course in Abstract Algebra with Applications*, 3rd edition, Prentice Hall, 2005, p. 130.

Example 1.1.5 Span and linear independence.

(a) $(1,0)$ and $(0,1)$ span $\mathbb{R}^2$. $(1,0)$, $(0,1)$, and $(1,1)$ also span $\mathbb{R}^2$. Thus,

$$\mathbb{R}^2 = \text{Span}\{(1,0),(0,1)\} = \text{Span}\{(1,0),(0,1),(1,1)\}.$$

$(1,0)$ and $(0,1)$ are linearly independent, while $(1,0)$, $(0,1)$, and $(1,1)$ are linearly dependent because $(1,1) = (1,0) + (0,1)$.

(b) $\alpha = (1,-1,0)$, $\beta = (-1,0,1)$, and $\gamma = (0,t,-1) \in \mathbb{R}^3$ are linearly independent if and only if $t \neq 1$. For any $t \neq 1$, α, β, and γ span $\mathbb{R}^3$.

[Note: Determining the linear dependence of α, β, γ invokes a discussion of solutions of the system of linear equations $x_1\alpha + x_2\beta + x_3\gamma = 0$.]

(c) $1, x, x^2, \ldots, x^n$ span $\mathbb{R}_n[x]$, that is, $\mathbb{R}_n[x] = \text{Span}\{1, x, x^2, \ldots, x^n\}$. $1, x, x^2, \ldots, x^n$ are linearly independent. $1, x, 1 + x^2$ are linearly independent, while $1, x, 1 + 2x$ are linearly dependent. ∎

A *basis* of a vector space V is a set of vectors that satisfy two criteria:

(1) The vectors are linearly independent;

(2) The vectors span the vector space V.

If V possesses a basis of n vectors, where n is a positive integer, we say that V has *dimension* n, written as $\dim V = n$. If $\{u_1, u_2, \ldots, u_n\}$ and $\{v_1, v_2, \ldots, v_m\}$ are linearly independent sets that each span V, then $m = n$ (Problem 21). If V is not spanned by any finite subset of V, then we say that V is *infinite-dimensional* and write $\dim V = \infty$ (positive infinity). Unless otherwise stated, we assume that the vector spaces are finite-dimensional, even though some results hold for infinite-dimensional vector spaces.

Conventionally, the zero vector space has dimension zero, and the zero vector space has no basis (or we say that its basis is the empty set).

Let V be a vector space of dimension n (> 0), and let $\{v_1, v_2, \ldots, v_k\}$ be a linearly independent subset of V. Then $k \leq n$. If $k < n$, then there exists a vector $v_{k+1} \in V$ such that the set $\{v_1, v_2, \ldots, v_k, v_{k+1}\}$ is linearly independent (otherwise, V would be spanned by $v_1, v_2, \ldots, v_k$). It follows that a linearly independent set of vectors can be extended to a basis.

Any $n+1$ vectors in a vector space of dimension n are linearly dependent.

Let $\{u_1, u_2, \ldots, u_n\}$ be a basis of a vector space V. Then every v in V can be uniquely expressed as a linear combination of the basis vectors:

$$v = x_1u_1 + x_2u_2 + \cdots + x_nu_n,$$

where $x_1, x_2, \ldots, x_n$ are scalars. The n-tuple $(x_1, x_2, \ldots, x_n)$ is the *coordinate* of vector v with respect to the ordered basis $\{u_1, u_2, \ldots, u_n\}$. We also say that $x_1, x_2, \ldots, x_n$ are the coordinates of v (relative to the basis).

Note: The coordinates are relative to the order of the vectors in the basis. So, when coordinates are under consideration, the basis vectors are assumed to be already arranged in certain order.

Example 1.1.6 Dimension, basis, and coordinate.

(a) $\mathbb{C}$ is a vector space of dimension 1 over $\mathbb{C}$ with basis $\{1\}$.

$\mathbb{C}$ is a vector space of dimension 2 over $\mathbb{R}$ with basis $\{1, i\}$, where $i = \sqrt{-1}$. Note that $\{2, i\}$ also serves as a basis because $2 \in \mathbb{C}$ and $i \in \mathbb{C}$ are linearly independent over $\mathbb{R}$. Let $a + bi \in \mathbb{C}$, $a, b \in \mathbb{R}$. Since

$$a + bi = a \cdot 1 + b \cdot i = \tfrac{a}{2} \cdot 2 + b \cdot i,$$

the coordinate of $a + bi \in \mathbb{C}$ with respect to the basis $\{1, i\}$ is (a, b); the coordinate of $a + bi \in \mathbb{C}$ with respect to the basis $\{2, i\}$ is $(\frac{a}{2}, b)$.

$\mathbb{C}$ is a vector space of infinite dimension over $\mathbb{Q}$, the rational numbers.

Note: For $z = a + bi$, $a, b \in \mathbb{R}$, $|z| = \sqrt{a^2 + b^2}$, $a = \Re(z)$, $b = \Im(z)$.

(b) Let V be the vector space spanned (generated) by $1, \sqrt{2}, 1 - i$, and $1 + i$ over $\mathbb{F}$. If $\mathbb{F} = \mathbb{Q}$, then $\dim V = 3$. $\{1, \sqrt{2}, 1 - i\}$ is a basis (so is $\{1, \sqrt{2}, 1 + i\}$). If $\mathbb{F} = \mathbb{R}$, then $\dim V = 2$. $\{1, 1 - i\}$ is a basis.

(c) $\mathbb{R}^n$ is a vector space of dimension n over $\mathbb{R}$, $\mathbb{C}^n$ is a vector space of dimension n over $\mathbb{C}$, both have the *standard basis*

$$e_1 = (1, 0, \ldots, 0, 0),\ e_2 = (0, 1, \ldots, 0, 0),\ \ldots,\ e_n = (0, 0, \ldots, 0, 1).$$

(d) $\mathbb{R}_n[x]$ has dimension $n + 1$, and $1, x, \ldots, x^n$ comprise a basis.

(e) $\dim \mathbb{R}[x] = \infty$ because $\mathbb{R}[x]$ is not spanned by any set of finitely many vectors. Note that $\{1, x, \ldots, x^n, \ldots\}$ is a linearly independent set and every element in $\mathbb{R}[x]$ can be expressed as a (finite) linear combination of $1, x, \ldots, x^n, \ldots$. Thus, $\{1, x, \ldots, x^n, \ldots\}$ is a basis of $\mathbb{R}[x]$. ∎

Let W be a nonempty subset of a vector space V. If W is also a vector space under the addition and scalar multiplication of V, then W is called a *subspace* of V. $\{0\}$ and V are trivial subspaces. One checks (Problem 12) that $W \subseteq V$ is a subspace of V if and only if W is closed under the addition and scalar multiplication for V. For a subset S of V, $\operatorname{Span} S$ is a subspace.

Given subspaces V_1 and V_2 (of V), the *sum* of V_1 and V_2 defined by

$$V_1 + V_2 = \{v_1 + v_2 : v_1 \in V_1,\ v_2 \in V_2\}$$

is a subspace; so is the intersection $V_1 \cap V_2 = \{v : v \in V_1, v \in V_2\}$. Note that

$$V_1 \cap V_2 \subseteq V_1 \cup V_2 \subseteq V_1 + V_2 = \operatorname{Span}(V_1 \cup V_2).$$

The sum $V_1 + V_2$ is called a *direct sum*, symbolized by $V_1 \oplus V_2$, if

$$v_1 + v_2 = 0,\ \ v_1 \in V_1,\ v_2 \in V_2 \quad \Rightarrow \quad v_1 = v_2 = 0.$$

One checks that in the case of a direct sum, every vector v in $V_1 \oplus V_2$ is uniquely written as a sum of a vector v_1 in V_1 and a vector v_2 in V_2.

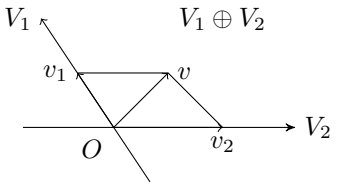


Fig. 1.2. Direct sum of subspaces

Example 1.1.7 Sum of subspaces.

(a) Any straight line passing through the origin in the xy-plane is viewed as a subspace of $\mathbb{R}^2$, and $\mathbb{R}^2$ is the direct sum of any two such (distinct) lines. In particular, $\mathbb{R}^2$ is the direct sum of the x- and y-axes.

(b) In $\mathbb{R}^3$, any straight line through the origin or any plane that contains the origin is (viewed as) a subspace. $\mathbb{R}^3$ is the direct sum of the xy-plane and any straight line that passes the origin and is not contained in the xy-plane. Can $\mathbb{R}^3$ be the direct sum of two lines or two planes?

(c) Let $W = \{(x_1, x_2, x_3, x_4) \in \mathbb{R}^4 : x_2 = 2x_1, x_3 + x_4 = 0\}$. Then W is a subspace of $\mathbb{R}^4$ with $\dim W = 2$, because W is closed under the addition and scalar multiplication, and for $w = (x_1, x_2, x_3, x_4) \in W$,

$$w = (x_1, 2x_1, x_3, -x_3) = x_1(1, 2, 0, 0) + x_3(0, 0, 1, -1).$$

$\{(1, 2, 0, 0), (0, 0, 1, -1)\}$ is a basis of W. Therefore, W can be written (decomposed) as a direct sum of two "lines" (one-dimensional spaces).

(d) $\mathbb{R} \subset \mathbb{R}_1[x] \subset \cdots \subset \mathbb{R}_n[x] \subset \cdots \subset \mathbb{R}[x]$. $\dim \mathbb{R}_n[x] = n + 1, \dim \mathbb{R}[x] = \infty$. ∎

Theorem 1.1.1 (Dimension Identity) *Let V be a finite-dimensional vector space, and let V_1 and V_2 be subspaces of V. Then*

$$\dim V_1 + \dim V_2 = \dim(V_1 + V_2) + \dim(V_1 \cap V_2).$$

A proof goes as follows: First choose a basis $\{u_1, \ldots, u_k\}$ for $V_1 \cap V_2$, extend it to a basis $\{u_1, \ldots, u_k, v_{k+1}, \ldots, v_s\}$ for V_1 and a basis $\{u_1, \ldots, u_k, w_{k+1}, \ldots, w_t\}$ for V_2, then show that $\{u_1, \ldots, u_k, v_{k+1}, \ldots, v_s, w_{k+1}, \ldots, w_t\}$ is a basis for $V_1 + V_2$. This reveals the desired dimension identity.

If $\dim V_1 + \dim V_2 > \dim V$, then $\dim(V_1 \cap V_2) > 0$. Thus, the subspaces V_1 and V_2 contain nonzero common vectors. By this, if $\{u_1, \ldots, u_p\}$ and $\{v_1, \ldots, v_q\}$ are two linearly independent sets in a vector space V of dimension n, and if $p + q > n$, then the subspace spanned by $\{u_1, \ldots, u_p\}$ and the subspace spanned by $\{v_1, \ldots, v_q\}$ contain a nonzero common vector.

Example 1.1.8 Consider $\mathbb{R}^3$. Let P_1 and P_2 be any two planes in $\mathbb{R}^3$ that contain the origin. Since $\dim P_1 + \dim P_2 = 4$ and because $\dim \mathbb{R}^3 = 3$, there exists a nonzero vector v (a line) that is contained in both P_1 and P_2. ∎

Problems

1. Referring to Example 1.1.1, show that $(-1)(x, y) = -(x, y) = (-x, -y)$ and that $(ab)(x, y) = a(b(x, y)) = (abx, aby)$ for $(x, y) \in \mathbb{R}^2$, $a, b \in \mathbb{R}$.

2. Let $\mathbb{R}^2 = \{(x, y) : x, y \in \mathbb{R}\}$ be the set of all ordered real pairs.

   (a) Show that $\mathbb{R}^2$ is a vector space over $\mathbb{Q}$, the rational numbers.

   (b) Is $\mathbb{R}^2$ a vector space over $\mathbb{C}$ with respect to the usual addition and the scalar multiplication $c(x, y) = (cx, cy)$, $c \in \mathbb{C}$? What if it is instead defined as $c(x, y) = (ax, by)$, where $c = a + bi$, $a, b \in \mathbb{R}$?

   (c) Show that $\mathbb{R}^2$ is a vector space over $\mathbb{C}$ with respect to the usual addition and the scalar multiplication (newly) defined by

   $$c(x, y) = (ax + by, ay - bx), \quad \text{where } c = a + bi, \ a, b \in \mathbb{R}.$$

   (d) Is $\mathbb{R}^2$ a vector space over $\mathbb{C}$ with respect to the usual addition and the scalar multiplication defined by

   $$c(x, y) = (ax + by, ax - by), \quad \text{where } c = a + bi, \ a, b \in \mathbb{R}?$$

3. Can a vector space have two different additive identities? Why (not)?

4. Let V be a vector space over a field $\mathbb{F}$. Let $u, v \in V$, $\lambda \in \mathbb{F}$. Show that

   (a) $-v$ is unique for each v.

   (b) $-v = (-1)v$.

   (c) $-(\lambda v) = (-\lambda)v = \lambda(-v)$ (denoted by $-\lambda v$).

   (d) $\lambda(u - v) = \lambda u - \lambda v$, where $u - v$ denotes $u + (-v)$.

5. Show that $\mathbb{F}_n[x]$, the set of polynomials over a field $\mathbb{F}$ with degree at most n (including $-\infty$), is a vector space over $\mathbb{F}$ under the ordinary addition and scalar multiplication. Is $\mathbb{F}[x]$, the set of zero and polynomials with any finite degrees, a vector space over $\mathbb{F}$? What is the dimension of $\mathbb{F}_n[x]$ or $\mathbb{F}[x]$? Is $1 + x + x^2 + \cdots$ an element in $\mathbb{F}[x]$?

6. Determine whether the vectors $v_1 = 1 + x - 2x^2$, $v_2 = 2 + 5x - x^2$, and $v_3 = x + x^2$ in $\mathbb{R}_2[x]$ are linearly independent.

7. Show that $\{(1, i), (i, -1)\}$ is a linearly independent subset of $\mathbb{C}^2$ over the real numbers $\mathbb{R}$ but not over the complex numbers $\mathbb{C}$.

8. Let $u_1, \ldots, u_m \in \mathbb{R}^n$. Show that $u_1, \ldots, u_m$ are linearly (in)dependent over $\mathbb{C}$ if and only if they are linearly (in)dependent over $\mathbb{R}$.

9. Let $\mathcal{C}(\mathbb{R})$ be the vector space of all real-valued continuous functions over $\mathbb{R}$. Show that $\{1, \cos x, \cos 2x\}$ is a linearly independent set, while $\{1, \cos^2 x, \cos 2x\}$ is a linearly dependent set.

10. Find the dimension and a basis for each of the following vector spaces.
    (a) U is generated by 1, $\sqrt{2}$, and $\sqrt{5}$ over $\mathbb{Q}$ (the rational numbers).
    (b) V is generated by 1, $\sqrt{2}$, $1+i$, and $1-i$ over $\mathbb{Q}$ (here $i=\sqrt{-1}$).
    (c) W is generated by 1, $\sqrt{2}$, $1+i$, and $1-i$ over $\mathbb{R}$.

11. Let V be a vector space. If $u, v, w \in V$ satisfy $au + bv + cw = 0$ for some scalars a, b, c, $ac \neq 0$, show that $\text{Span}\{u, v\} = \text{Span}\{v, w\}$.

12. Let V be a vector space over a field $\mathbb{F}$ and let W be a subset of V. Show that W is a subspace of V if and only if for all $u, v \in W$ and $c \in \mathbb{F}$

$$u + v \in W \quad \text{and} \quad cu \in W.$$

13. Is the set $\{(x, y) \in \mathbb{R}^2 : 2x - 3y = 0\}$ a subspace of $\mathbb{R}^2$? How about $\{(x, y) \in \mathbb{R}^2 : 2x - 3y = 1\}$? Give a geometric explanation.

14. Show that the set $\{(x, y-x, y) : x, y \in \mathbb{R}\}$ is a subspace (with the usual operations) of $\mathbb{R}^3$. Find the dimension and a basis of the subspace.

15. Find a basis for $\text{Span}\{u, v, w\}$ in $\mathbb{R}^3$, where $u = (1, 1, 0)$, $v = (1, 3, -1)$, and $w = (1, -1, 1)$. Find the coordinate of $(8, 2, 3)$ under the basis.

16. Let $W = \{(x_1, x_2, x_3, x_4) \in \mathbb{R}^4 : x_3 = x_1 + x_2 \text{ and } x_4 = x_1 - x_2\}$.
    (a) Prove that W is a subspace of $\mathbb{R}^4$.
    (b) Find a basis for W. What is the dimension of W?
    (c) Prove that $\{c(1, 0, 1, 1) : c \in \mathbb{R}\}$ is a subspace of W.
    (d) Is $\{c(1, 0, 0, 0) : c \in \mathbb{R}\}$ a subspace of W?

17. In $\mathbb{R}^3$, let $W_1 = \text{Span}\{u_1, u_2\}$ and $W_2 = \text{Span}\{v_1, v_2\}$, where

$$u_1 = \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix}, \; u_2 = \begin{pmatrix} 2 \\ 1 \\ 1 \end{pmatrix}, \; v_1 = \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix}, \; v_2 = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}.$$

    Find the dimension of $W_1 + W_2$ and find a basis of $W_1 \cap W_2$.

18. In $\mathbb{R}^4$, let $S_1 = \text{Span}\{u_1, u_2\}$ and $S_2 = \text{Span}\{v_1, v_2\}$, where

$$u_1 = \begin{pmatrix} 1 \\ 1 \\ 2 \\ 1 \end{pmatrix}, \; u_2 = \begin{pmatrix} -1 \\ 1 \\ 1 \\ 1 \end{pmatrix}, \; v_1 = \begin{pmatrix} 2 \\ 0 \\ -1 \\ 0 \end{pmatrix}, \; v_2 = \begin{pmatrix} 1 \\ 3 \\ -1 \\ 3 \end{pmatrix}.$$

    Find the dimension of $S_1 + S_2$ and find a basis of $S_1 \cap S_2$.

19. Show that each of the following sets is a vector space over $\mathbb{R}$.

    (a) $\mathcal{C}[a, b]$, the set of all continuous functions on the interval $[a, b]$.

    (b) $\mathcal{C}'(\mathbb{R})$, the set of all functions of continuous derivatives on $\mathbb{R}$.

    (c) The set of all even functions.

    (d) The set of all odd functions.

    (e) The set of all functions f that satisfy $f(0) = 0$.

20. Show that if W is a subspace of vector space V of dimension n, then $\dim W \leq n$. Is it possible that $\dim W = n$ for a proper subspace W? [Note: By a *proper subspace* W, we mean W is neither $\{0\}$ nor V.]

21. Let $\{u_1, \ldots, u_s\}$ and $\{v_1, \ldots, v_t\}$ be two sets of vectors in a vector space. If $s > t$ and every u_i, $i = 1, \ldots, s$, can be expressed as a linear combination of $v_1, \ldots, v_t$, show that $u_1, \ldots, u_s$ are linearly dependent.

22. Let V be a vector space over a field $\mathbb{F}$. Show that $cv = 0$, where $c \in \mathbb{F}$ and $v \in V$, if and only if $c = 0$ or $v = 0$. [Note: The scalar 0 and the vector 0 are usually different. For simplicity, we use 0 for both. The meaning of the zero is easily referred from the context.]

23. Let V_1 and V_2 be subspaces of a finite-dimensional vector space V and let $W = V_1 + V_2$. Show that the following statements are equivalent.

    (a) $V_1 + V_2$ is a direct sum, i.e., $0 \in W$ has a unique decomposition.

    (b) Every vector $w \in W$ has a unique decomposition, i.e., if $w = x_1 + y_1 = x_2 + y_2$, $x_i \in V_1$, $y_i \in V_2$, $i = 1, 2$, then $x_1 = x_2$, $y_1 = y_2$.

    (c) If $\{u_1, \ldots, u_s\}$ is a basis of V_1 and $\{v_1, \ldots, v_t\}$ is a basis of V_2, then $\{u_1, \ldots, u_s, v_1, \ldots, v_t\}$ is a basis for $V_1 + V_2$.

    (d) $\dim(V_1 + V_2) = \dim V_1 + \dim V_2$.

    (e) $V_1 \cap V_2 = \{0\}$.

    [Note: (a)–(d) can be easily extended for multiple subspaces of V.]

24. Let V_1 and V_2 be subspaces of a finite-dimensional vector space V.

    (a) If $V_1 \oplus W = V_2 \oplus W$ for some subspace W of V, must $V_1 = V_2$?

    (b) Suppose that $\dim V_1 = \dim V_2$. Show that there exists a subspace W such that $V = V_1 \oplus W = V_2 \oplus W$.

25. Let $u_1, \ldots, u_s$ and $v_1, \ldots, v_t$ be vectors in a vector space. Show that

$$\text{Span}\{u_1, \ldots, u_s\} + \text{Span}\{v_1, \ldots, v_t\} = \text{Span}\{u_1, \ldots, u_s, v_1, \ldots, v_t\}.$$

    Does this imply that $\dim(\text{Span}\{u_1, \ldots, u_s, v_1, \ldots, v_t\}) = s + t$?

26. Let V_1 and V_2 be subspaces of a vector space of finite dimension such that $\dim(V_1 + V_2) = \dim(V_1 \cap V_2) + 1$. Show that $V_1 \subseteq V_2$ or $V_2 \subseteq V_1$.

27. Show by an example that it is not true in general that $(S_1 + S_2) \cap S_3 = S_1 \cap S_3 + S_2 \cap S_3$, where S_1, S_2, and S_3 are subspaces of a vector space.

28. Let S_1, S_2, S_3 be subspaces of a vector space of dimension n. Show that

$$(S_1 + S_2) \cap (S_1 + S_3) = S_1 + (S_1 + S_2) \cap S_3.$$

29. Let S_1, S_2, S_3 be subspaces of a vector space of dimension n. Show that

$$\dim(S_1 \cap S_2 \cap S_3) \geq \dim S_1 + \dim S_2 + \dim S_3 - 2n.$$

30. Let $S_1 = \{(s_1, s_2, \dots) : s_i \in \mathbb{R}, i = 1, 2, \dots\}$ (real infinite sequences) and let $S_2 = \{(s_1, s_2, \dots) : s_i \in \mathbb{R}$ and $s_i \neq 0$ for finitely many $i\}$.

    (a) Show that S_1 and S_2 are vector spaces over $\mathbb{R}$ (with respect to the ordinary addition and scalar multiplication as for $\mathbb{R}^n$).

    (b) Show that $\{\varepsilon_1, \varepsilon_2, \dots\}$ is a basis of S_2, where

$$\varepsilon_k = (0, \dots, 0, \overset{k\text{th}}{1}, 0, \dots), \quad k = 1, 2, \dots.$$

    (c) Find a vector that is in S_1 but not in S_2.

    [Note: An infinite sum of vectors is not permitted in a vector space.]

31. (*Cartesian product of vector spaces*) Let V and W be vector spaces over a field $\mathbb{F}$ with bases $\{v_1, \dots, v_m\}$ and $\{w_1, \dots, w_n\}$, respectively. Let $V \times W = \{(v, w) : v \in V, w \in W\}$ (called *Cartesian product* of V and W). Define the vector addition and the scalar multiplication by

$$(v_1, w_1) + (v_2, w_2) = (v_1 + v_2, w_1 + w_2), \quad c(v, w) = (cv, cw), \ c \in \mathbb{F}.$$

    (a) Show that $V \times W$ is a vector space with respect to the operations.

    (b) If $m, n \geq 2$, show that $(v_1, w_1) - (v_1, w_2) - (v_2, w_1) + (v_2, w_2) = 0$.

    (c) Is $\{(v_i, w_j) : i = 1, \dots, m, \ j = 1, \dots, n\}$ a basis for $V \times W$?

    (d) Is $\{(v_1, 0), \dots, (v_m, 0), (0, w_1), \dots, (0, w_n)\}$ a basis for $V \times W$?

    (e) Is $(u + v, w) = (u, w) + (v, w)$ for all $u, v \in V$ and $w \in W$?

    (f) Show that $\dim(V \times W) = \dim V + \dim W$.

    (g) Find $\mathbb{R} \times \mathbb{R}^2$.

. ——————— ⊙ ——————— .

1.2 Matrices and Determinants

A *matrix* is a rectangular array of scalars (from a number field $\mathbb{F}$), arranged in rows and columns. A matrix A with m rows and n columns is denoted by

$$A = \begin{pmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{pmatrix}.$$

We also write $A = (a_{ij})$ in which a_{ij} represents the entry of A in the ith row and the jth column, i.e., the (i, j)-entry of A. We say that A has *size* $m \times n$. If $m = n$, A is called an *n-square* matrix, and n is the *order* of A.

Two matrices are equal if and only if they have the same size and the same corresponding entries in all positions.

The set of all $m \times n$ matrices with entries from a field $\mathbb{F}$, denoted by $\mathbb{M}_{m\times n}(\mathbb{F})$, or $\mathbb{M}_n(\mathbb{F})$ if $m = n$, is a vector space over $\mathbb{F}$ with respect to the *matrix addition* by adding the corresponding entries and the *scalar multiplication* by multiplying each entry of the matrix by the scalar:

$$A + B = (a_{ij} + b_{ij}), \quad \lambda A = (\lambda a_{ij}),$$

where $A = (a_{ij}), B = (b_{ij}) \in \mathbb{M}_{m\times n}(\mathbb{F})$, and $\lambda \in \mathbb{F}$.

A *zero matrix* is a matrix whose entries are all zero. We use $0_{m\times n}$ or 0_n if $m = n$ to denote the $m \times n$ zero matrix; most of time, we simply use 0 when the size is unimportant or it can be referred from the context. Note that two zero matrices are not the same if they have different sizes.

The dimension of the space $\mathbb{M}_{m\times n}(\mathbb{F})$ (over $\mathbb{F}$) is mn, and the matrices with (i, j)-entry 1 and other entries 0 ($1 \le i \le m, 1 \le j \le n$) form a basis.

Note that $\mathbb{M}_{n\times 1}(\mathbb{F})$ and $\mathbb{M}_{1\times n}(\mathbb{F})$ are the sets of column vectors and row vectors, respectively. As vector spaces, they can be identified with $\mathbb{F}^n$.

Example 1.2.1 $\mathbb{M}_n(\mathbb{C})$ is a vector space over $\mathbb{C}$, $\mathbb{M}_n(\mathbb{R})$ is a vector space over $\mathbb{R}$, each has dimension n^2, and the $n \times n$ matrices E_{ij} whose (i, j)-entry is 1 and 0 elsewhere, $1 \le i, j \le n$, comprise a basis. In particular, for $\mathbb{M}_2(\mathbb{C})$, $\{E_{11} = \left(\begin{smallmatrix} 1 & 0 \\ 0 & 0 \end{smallmatrix}\right), E_{12} = \left(\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right), E_{21} = \left(\begin{smallmatrix} 0 & 0 \\ 1 & 0 \end{smallmatrix}\right), E_{22} = \left(\begin{smallmatrix} 0 & 0 \\ 0 & 1 \end{smallmatrix}\right)\}$ is a basis. Note that

$$A = (a_{ij}) = \sum_{i,\,j} a_{ij} E_{ij}. \ \blacksquare$$

We use $\mathbb{M}_n$ for the space of complex (including real) $n \times n$ matrices.

Besides addition and scalar multiplication, matrices (of appropriate sizes) can be multiplied. The *product* AB of matrix $A = (a_{ij})$, of size $m \times n$, and

matrix $B = (b_{ij})$, of size $n \times p$, is defined to be the matrix $C = (c_{ij})$, of size $m \times p$, whose (i, j)-entry is given by

$$c_{ij} = a_{i1}b_{1j} + a_{i2}b_{2j} + \cdots + a_{in}b_{nj} = \sum_{t=1}^{n} a_{it}b_{tj}.$$

(I.e., c_{ij} is row i of A "times" column j of B, $i = 1, 2, \ldots, m$, $j = 1, 2, \ldots, p$.) Thus, to multiply A by B (from the right-hand side), i.e., in AB, the number of columns of A must equal the number of rows of B. (We say that A and B are *compatible for product* or A and B have *appropriate sizes.*) For example,

$$A = \begin{pmatrix} 1 & -1 \\ 0 & 2 \end{pmatrix}, \quad B = \begin{pmatrix} 3 & 4 & 5 \\ 6 & 0 & 8 \end{pmatrix},$$

$$AA = \begin{pmatrix} 1 & -3 \\ 0 & 4 \end{pmatrix}, \quad AB = \begin{pmatrix} -3 & 4 & -3 \\ 12 & 0 & 16 \end{pmatrix}.$$

However, $A + B$, BA, and BB are undefined. [Note: $AB \neq BA$ in general.]

The $n \times n$ *identity* matrix I_n, or simply I (when the size of the matrix is clear from the context), is the n-square matrix with all diagonal entries equal to 1 and off-diagonal entries 0. A *scalar* matrix is a multiple of I, i.e., cI, where c is a scalar. For any $m \times n$ matrix A, $I_m A = A = AI_n$.

Example 1.2.2 Let A, B, and C be matrices of appropriate sizes. Then

$$(AB)C = A(BC).$$

Proof. Write $A = (a_{is})$, $B = (b_{st})$, $C = (c_{tj})$. Computations show that the (i, j)-entry of $(AB)C$ is exactly the (i, j)-entry of $A(BC)$ for any (i, j):

$$\sum_t \Big(\sum_s (a_{is}b_{st}) \Big) c_{tj} = \sum_{s,\,t} a_{is}b_{st}c_{tj} = \sum_s a_{is} \Big(\sum_t b_{st}c_{tj} \Big).$$

In particular, for $x = (x_1, \ldots, x_m)$, $A = (a_{ij}) \in \mathbb{M}_{m \times n}$, and $y = (y_1, \ldots, y_n)^T$,

$$xAy = \sum_{i,\,j} x_i \, a_{ij} \, y_j. \ \blacksquare \tag{1.2}$$

The above example says that the matrix product is associative. Thus, for k matrices $A_1, A_2, \ldots, A_k$ of appropriate sizes, the product $A_1A_2 \cdots A_k$ is well-defined by $((A_1A_2)A_3) \cdots A_k$. (In fact, the pairs of parentheses can be inserted in anyways.) For an $n \times n$ matrix A, we denote

$$A^0 = I_n, \quad A^k = A \cdots A \ \ (k \text{ copies of } A).$$

Exercise: Show that for matrices A, B, and C of appropriate sizes,

$$A(B + C) = AB + AC, \quad (A + B)C = AC + BC.$$

Let $A \in \mathbb{M}_n$. If there exists a matrix $B \in \mathbb{M}_n$ such that $AB = BA = I_n$, then we call B the *inverse* of A and denote it by $B = A^{-1}$. Such a B is unique. If A has an inverse, we say that A is *invertible* or *nonsingular.*

The *transpose* of an $m \times n$ matrix $A = (a_{ij})$ is an $n \times m$ matrix, denoted by A^T, whose (i,j)-entry is a_{ji}, and the *conjugate* of A is a matrix of the same size as A, symbolized by $\overline{A}$, whose (i,j)-entry is $\overline{a_{ij}}$. Apparently, $(\overline{A})^T = \overline{A^T}$. We denote the conjugate transpose $(\overline{A})^T$ of A by A^*. Of many properties (Problem 11) of conjugate and transpose of matrices, we will repeatedly use

$$(A+B)^* = A^* + B^*, \quad (AB)^* = B^*A^*.$$

The following terms are frequently used. A matrix $A = (a_{ij}) \in \mathbb{M}_n$ is

diagonal	if $a_{ij} = 0,\ i \neq j$,
upper-triangular	if $a_{ij} = 0,\ i > j$,
lower-triangular	if A^T is upper-triangular,
symmetric	if $A^T = A$,
Hermitian	if $A^* = A$,
positive semidefinite	if $x^*Ax \geq 0$ for all $x \in \mathbb{C}^n$,
normal	if $A^*A = AA^*$,
unitary	if $A^*A = AA^* = I$, and
real orthogonal	if A is real and $A^TA = AA^T = I$.

Example 1.2.3 Let a, b, c, d, e, and f be complex numbers.

(a) $\begin{pmatrix} a & 0 & 0 \\ 0 & b & 0 \\ 0 & 0 & c \end{pmatrix}$ is a diagonal matrix, written as $\operatorname{diag}(a,b,c)$.

(b) $\begin{pmatrix} a & b & c \\ 0 & d & e \\ 0 & 0 & f \end{pmatrix}$ is upper-triangular.

(c) $\begin{pmatrix} a & b & c \\ b & d & e \\ c & e & f \end{pmatrix}$ is symmetric (*real symmetric* if the entries are all real).

(d) $\begin{pmatrix} a & b & \bar{c} \\ \bar{b} & d & e \\ c & \bar{e} & f \end{pmatrix}$ is Hermitian (and normal) if a, d, and f are real.

(e) $\begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix}$ is normal (but not Hermitian).

(f) $\frac{1}{\sqrt{2}}\begin{pmatrix} 1 & i \\ i & 1 \end{pmatrix}$ is unitary (where $i = \sqrt{-1}$).

(g) $\begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$ is real orthogonal, where $\theta \in \mathbb{R}$. ∎

A *submatrix* of a given matrix is a matrix whose entries lie in the intersections of specified rows and columns of the given matrix. Let

$$A = \begin{pmatrix} 0 & 1 & 2 \\ i & 3 & \frac{1}{4} \\ \pi & \sqrt{3} & -1 \end{pmatrix}.$$

Then $\begin{pmatrix} 1 & 2 \\ 3 & \frac{1}{4} \end{pmatrix}$ is the submatrix of A lying in rows one and two and columns two and three; (-1) is the submatrix in row three and column three.

A matrix can be divided (partitioned) into submatrices in different ways. For instance, let $B = \binom{0}{i}$, $C = \begin{pmatrix} 1 & 2 \\ 3 & \frac{1}{4} \end{pmatrix}$, $D = (\pi)$, and $E = (\sqrt{3}, -1)$. Then

$$A = \left(\begin{array}{c|cc} 0 & 1 & 2 \\ i & 3 & \frac{1}{4} \\ \hline \pi & \sqrt{3} & -1 \end{array}\right) = \begin{pmatrix} B & C \\ D & E \end{pmatrix}.$$

The matrix on the right is called a *partitioned* or *block* form of A, or we say that A is a *partitioned* (or *block*) matrix. Note that B and C have the same number of rows, and B and D have the same number of columns.

A *principal submatrix* of a matrix A is a submatrix lying in the intersections of the same rows and columns of A. A *leading principal submatrix* of order k is the principal submatrix in rows $1, \ldots, k$ and columns $1, \ldots, k$.

In the previous example, $(0), (3)$, and (-1) are the principal submatrices of A of order 1; $\begin{pmatrix} 0 & 1 \\ i & 3 \end{pmatrix}$, $\begin{pmatrix} 0 & 2 \\ \pi & -1 \end{pmatrix}$, and $\begin{pmatrix} 3 & \frac{1}{4} \\ \sqrt{3} & -1 \end{pmatrix}$ are the principal submatrices of A of order 2; A is the principal submatrix of A of order 3. (0), $\begin{pmatrix} 0 & 1 \\ i & 3 \end{pmatrix}$, and A are the leading principal submatrices of A, while $\begin{pmatrix} 0 & 2 \\ \pi & -1 \end{pmatrix}$ is not leading.

Working skillfully with partitioned (block) matrices is important in matrix theory. We shall be focused on the partitioned matrices in Chapter 2.

Example 1.2.4 Matrices can be written (partitioned) in terms of their rows and columns. Let A and B be $m \times n$ and $n \times p$ matrices, respectively. Let

$$A = (a_1, a_2, \ldots, a_n) = \begin{pmatrix} r_1 \\ r_2 \\ \vdots \\ r_m \end{pmatrix}, \; B = (b_1, b_2, \ldots, b_p) = \begin{pmatrix} s_1 \\ s_2 \\ \vdots \\ s_n \end{pmatrix},$$

where a_i's and b_i's, and r_i's and s_i's are the columns and rows of A and B, respectively. Sometimes it is convenient to write the product AB as

$$AB = (Ab_1, Ab_2, \ldots, Ab_p) = \big(r_i b_j\big) = \sum_{t=1}^{n} a_t s_t = \begin{pmatrix} r_1 B \\ r_2 B \\ \vdots \\ r_m B \end{pmatrix}. \; \blacksquare$$

Example 1.2.5 Let $A \in \mathbb{M}_{m\times n}(\mathbb{F})$. The solutions to $Ax = 0$ (the matrix-vector product form of the homogeneous system of linear equations) form a subspace of $\mathbb{F}^n$, called the *solution space* or *null space of* A and denoted by

$$\mathrm{Nul}(A) = \{x \in \mathbb{F}^n : \; Ax = 0\}. \; \blacksquare$$

The dimension of the null space of A is referred to as the *nullity* of A.

Another important space associated with $A \in \mathbb{M}_{m\times n}(\mathbb{F})$ is the *column space* of A, denoted by $\mathrm{Col}(A)$. It is the subspace of $\mathbb{R}^m$ spanned by the columns of the matrix A. If $c_1, \ldots, c_n$ are the columns of A, then

$$\mathrm{Col}(A)=\{Ax : x \in \mathbb{F}^n\} = \{x_1c_1 + \cdots + x_nc_n : x_1, \ldots, x_n \in \mathbb{F}\}.$$

A matrix can be viewed as a linear transformation (see the next section). In this regard, the null space of matrix A is also called the *kernel* of A, denoted by $\mathrm{Ker}(A)$, and the column space is also known as the *range* or *image* of A, denoted by $\mathrm{Im}(A)$. So $\mathrm{Ker}(A) = \mathrm{Nul}(A)$ and $\mathrm{Im}(A) = \mathrm{Col}(A)$.

It is a useful and important fact that for $A \in \mathbb{M}_{m\times n}$ and $B \in \mathbb{M}_{n\times p}$,

$$\mathrm{Im}(AB) \subseteq \mathrm{Im}(A). \tag{1.3}$$

A general system of linear equations is written in a matrix-vector product form, say, $Ax = b$, where A is the matrix of the coefficients of the linear equations, x is the (column) vector of unknowns, and b is a given (column) vector. Solving the system (whether it has a solution or not) is equivalent to row-reducing the augmented matrix (A, b) to a row echelon form.

The following are *elementary row operations* performed on a matrix:

(i) Interchange two rows.
Say, interchange row i and row j.
For example, $\left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) \to \left(\begin{smallmatrix} c & d \\ a & b \end{smallmatrix}\right)$.

(ii) Multiply a row by a nonzero constant.
Say, multiply row i by a nonzero number k.
For example, $\left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) \to \left(\begin{smallmatrix} ka & kb \\ c & d \end{smallmatrix}\right)$.

(iii) Add a multiple of a row to another row.
Say, add row i times a number k to row j.
For example, $\left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) \to \left(\begin{smallmatrix} a & b \\ c+ka & d+kb \end{smallmatrix}\right)$.

Elementary column operations are similarly defined. Row and column operations for partitioned matrices will be discussed in Sect. 2.1, Chap. 2.

An n-square matrix is called an *elementary matrix* if it is obtained from I_n by a single elementary row or column operation. Elementary operations can be represented by elementary matrices. Let E be the elementary matrix by performing an elementary row (or column) operation on I_m (or I_n). If the same elementary row (column) operation is performed on an $m \times n$ matrix A, then the resulting matrix from A via the elementary row (column) operation is given by EA (or AE). We write $A \to EA$ (or AE, respectively).

An elementary matrix is referred to as a *permutation matrix* if it is obtained from the identity matrix by applying several operations (i). A permutation matrix has exactly one 1 in each row and each column (see Sect. 5.6).

Each of the elementary operations (i)–(iii) can be reversed, and the reversed operation is the same type, i.e., (i), (ii), or (iii), respectively. For example, if E is a type (iii) operation, say, adding the ith row times k to the jth row, then its reversed operation, represented by E^{-1} (called the *inverse* of E), adds the ith row times $-k$ to the jth row. In other words, elementary matrices are *invertible*, meaning $E^{-1}E = EE^{-1}$ is an identity matrix.

Elementary row and column operations do not change the nullity of a matrix. For rows, if E is an elementary matrix, then EA and A have the same null space because $(EA)x = 0 \Leftrightarrow Ax = E^{-1}(EA)x = 0$. For columns, let F be an elementary matrix. If $u_1, \ldots, u_p \in \mathrm{Nul}(AF)$ are linearly independent, then $v_1 = Fu_1, \ldots, v_p = Fu_p \in \mathrm{Nul}(A)$ are linearly independent. Suppose otherwise $l_1v_1 + \cdots + l_pv_p = 0$, where l's are not all equal to zero, then $F^{-1}(l_1v_1 + \cdots + l_pv_p) = l_1u_1 + \cdots + l_pu_p = 0$, a contradiction. Therefore, EAF has the same nullity as A. In a similar argument, elementary row and column operations do not change the dimension of the column space of a matrix. (In fact, as we proceed, we will see later that PAQ and A have the same nullity and the same rank for nonsingular matrices P and Q.)

Example 1.2.6 Perform elementary row and column operations on the 2×3 matrix $A = \left(\begin{smallmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{smallmatrix}\right)$ to get a matrix that has as many zeros as possible.

$$
\begin{aligned}
A &= \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix} \left[\begin{array}{l} \text{multiply row one by } (-4) \\ \text{then add it to row two} \end{array}\right] \\
&\to \begin{pmatrix} 1 & 2 & 3 \\ 0 & -3 & -6 \end{pmatrix} \left[\text{ divide row two by } -3 \right] \\
&\to \begin{pmatrix} 1 & 2 & 3 \\ 0 & 1 & 2 \end{pmatrix} \left[\begin{array}{l} \text{multiply row two by } (-2) \\ \text{then add it to row one} \end{array}\right] \\
&\to \begin{pmatrix} 1 & 0 & -1 \\ 0 & 1 & 2 \end{pmatrix} \left[\text{ add the 1st column to the last column}\right] \\
&\to \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 2 \end{pmatrix} \left[\begin{array}{l} \text{multiply column two by } (-2) \\ \text{then add it to the last column} \end{array}\right] \\
&\to \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix} \left[\text{ echolon form}\right].
\end{aligned}
$$

The corresponding elementary matrices in the above five steps are

$$
R_1 = \begin{pmatrix} 1 & 0 \\ -4 & 1 \end{pmatrix}, \quad R_2 = \begin{pmatrix} 1 & 0 \\ 0 & -\frac{1}{3} \end{pmatrix}, \quad R_3 = \begin{pmatrix} 1 & -2 \\ 0 & 1 \end{pmatrix},
$$

$$
C_1 = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad C_2 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & -2 \\ 0 & 0 & 1 \end{pmatrix}.
$$

A is brought to $\left(\begin{smallmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{smallmatrix}\right)$, which cannot be reduced further. Put in equation:

$$
R_3R_2R_1 \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix} C_1C_2 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}. \blacksquare
$$

This leads to the following *rank factorization theorem.*

Theorem 1.2.1 *Let A be an $m \times n$ matrix. Through elementary row and column operations, A can be brought to a unique matrix in the form $\left(\begin{smallmatrix} I_r & 0 \\ 0 & 0 \end{smallmatrix}\right)$. ($I_r$ is absent if $A = 0$). Equivalently, there exist an $m \times m$ matrix P and an $n \times n$ matrix Q, both are products of some elementary matrices, such that*

$$PAQ = \begin{pmatrix} I_r & 0 \\ 0 & 0 \end{pmatrix}. \tag{1.4}$$

The block matrix $\left(\begin{smallmatrix} I_r & 0 \\ 0 & 0 \end{smallmatrix}\right)$ in (1.4) is unique, that is, it is independent of P and Q. Suppose $P_1AQ_1 = \left(\begin{smallmatrix} I_{r_1} & 0 \\ 0 & 0 \end{smallmatrix}\right)$ and $P_2AQ_2 = \left(\begin{smallmatrix} I_{r_2} & 0 \\ 0 & 0 \end{smallmatrix}\right)$. Then $n - r_1$ is the nullity of P_1AQ_1, and $n - r_2$ is the nullity of P_2AQ_2. Since elementary operations do not change the nullity of a matrix, we conclude $n-r_1 = n-r_2$. So $r_1 = r_2$ (see Problem 39 for a different proof). Moreover, if A is a real (or rational) matrix, then P and Q can be chosen to be real (resp., rational).

The size, or order r, of the identity matrix I_r in (1.4) is called the *rank* of A, denoted by $\text{rank}(A)$. Obviously, $\text{rank}(A) = 0$ if and only if $A = 0$.

For $A \in \mathbb{M}_n$, $\text{rank}(A) = n$, i.e., $PAQ = I_n$ in (1.4), if and only if A is the product of some elementary matrices (by applying the inverses of the elementary matrices in P and Q. Note: the inverse of an elementary matrix is an elementary matrix). Throughout the book, by the rank of a matrix, we mean the rank of the matrix over the complex number field $\mathbb{C}$.

If P is an elementary matrix, so are P^T, $\overline{P}$, and P^*. By (1.4), we have

$$\text{rank}(A^T) = \text{rank}(\overline{A}) = \text{rank}(A^*) = \text{rank}(A). \tag{1.5}$$

The rank of a matrix can be defined in different but equivalent ways (such as using minors, row rank, and column rank; see Sect. 2.3, Chap. 2).

Elementary column operations do not change the column space of a matrix. Suppose that F is a column operation on matrix $A \in \mathbb{M}_{m\times n}(\mathbb{F})$. Then we have $\text{Col}(AF) = \{(AF)x : x \in \mathbb{F}^n\} = \{A(Fx) : x \in \mathbb{F}^n\} \subseteq \text{Col}(A)$ and $\text{Col}(A) = \text{Col}((AF)F^{-1}) \subseteq \text{Col}(AF)$. It follows that $\text{Col}(AF)$=$\text{Col}(A)$.

Elementary row operations do not change the dimension of the column space of a matrix. Let E be an elementary row operation on $A \in \mathbb{M}_{m\times n}$. Then $E^{-1}E = I_m$. Let $A = (c_1, \ldots, c_n)$. Then $EA = (Ec_1, Ec_2, \ldots, Ec_n)$. The columns of A, say, $c_{i_1}, \ldots, c_{i_k}$, are linearly independent if and only if $Ec_{i_1}, \ldots, Ec_{i_k}$ are linearly independent. Thus, $\dim \text{Col}(EA) = \dim \text{Col}(A)$.

By (1.4), we have $\dim \text{Col}(A) = \dim \text{Col}(PAQ) = \dim \text{Col}\left(\begin{smallmatrix} I_r & 0 \\ 0 & 0 \end{smallmatrix}\right) = r$. So, the rank and nullity of a matrix remain the same under elementary operations. Theorem 1.2.1 gives rise to the following result immediately.

Theorem 1.2.2 *Let $A \in \mathbb{M}_{m\times n}$ have rank r (as given in (1.4)). Then*

$$r = \text{rank}(A) = \dim \text{Col}(A), \quad \dim \text{Nul}(A) = n - r, \quad \textit{and}$$
$$\dim \text{Col}(A) + \dim \text{Nul}(A) = n.$$

Example 1.2.7 Let $A \in \mathbb{M}_{m\times n}$ and $B \in \mathbb{M}_{n\times p}$. If $AB = 0$, then

$$\operatorname{rank}(A) + \operatorname{rank}(B) \le n.$$

Proof. Because $AB = 0$, the columns of B are all contained in the solution space of $Ax = 0$, that is, $\operatorname{Col}(B) \subseteq \operatorname{Nul}(A)$. It follows that $\dim \operatorname{Col}(B) \le \dim \operatorname{Nul}(A)$, implying $\operatorname{rank}(B) \le n - \operatorname{rank}(A)$. ∎

To find the rank of a matrix A, one can apply both elementary row and column operations on A, while to find (a basis of) the null space of A, only row (reduction) operations can be applied to A. Moreover, given a column vector b (of m components), $Ax = b$ has a solution(s), equivalently, b is contained in the column space of A, if and only if $\operatorname{rank}(A) = \operatorname{rank}(A, b)$:

$$Ax = b \text{ for some } x \quad \Leftrightarrow \quad \operatorname{rank}(A) = \operatorname{rank}(A, b).$$

Example 1.2.8 Consider the 3×4 matrix

$$A = \begin{pmatrix} 1 & 2 & -1 & 1 \\ 0 & -4 & 5 & -2 \\ 2 & 0 & 3 & 0 \end{pmatrix}.$$

(a) To find the rank of A, perform both row and column operations on A:

$$A \to B = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} = \begin{pmatrix} I_2 & 0 \\ 0 & 0 \end{pmatrix}. \text{ So } r = \operatorname{rank}(A) = 2.$$

By Theorem 1.2.2, the nullity of A is $n - r = 4 - 2 = 2$.

(b) Part (a) reveals that the dimension of the column space $\operatorname{Col}(A)$ (a subspace of $\mathbb{R}^3$) is 2 (the rank of A). We want to find a basis of $\operatorname{Col}(A)$.

The column space $\operatorname{Col}(A)$, i.e., $\operatorname{Im}(A)$, is spanned by the columns $c_1, \ldots, c_n$ of A, and some columns of A, say $c_{i_1}, \ldots c_{i_k}$ are linearly independent if and only if $Rc_{i_1}, \ldots Rc_{i_k}$ are linearly independent, where R is an elementary matrix. Note that $RA = (Rc_1, \ldots, Rc_n)$. Thus, to find a basis of the column space, we perform row operations on A:

$$A \xrightarrow{\text{row}} C = \begin{pmatrix} 1 & 0 & \frac{3}{2} & 0 \\ 0 & -4 & 5 & -2 \\ 0 & 0 & 0 & 0 \end{pmatrix}.$$

The first two columns of C are linearly independent, and any three columns of C are linearly dependent. Thus, the first two columns $(1, 0, 2)^T$ and $(2, -4, 0)^T$ of A form a basis of $\operatorname{Col}(A)$. Because the first column $(1, 0, 0)^T$ of C is not a linear combination of the columns of A, $\operatorname{Col}(A) \neq \operatorname{Col}(C)$. So, row operations change column spaces.

Bases of the column space can also be obtained through column operations because $\mathrm{Col}(AQ)=\mathrm{Col}(A)$ for any elementary matrix Q.

$$A \xrightarrow{\text{column}} D = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 2 & 1 & 0 & 0 \end{pmatrix}, \quad \mathrm{Col}(A)=\mathrm{Col}(D).$$

Thus, $(1,0,2)^T$ and $(0,1,1)^T$ constitute a basis for $\mathrm{Col}(A)$.

In summary, to find a basis of the column space of a matrix, row operations show which columns of the original matrix are linearly independent, while column operations yield bases that span the column space. Note: In general, one cannot use both row and column operations to find the bases of the column space.

(c) The *row space* of A, denoted by $\mathrm{Row}(A)$, is spanned by the rows of A. Any two rows of the matrix A are linearly independent, so any two rows of A form a basis of $\mathrm{Row}(A)$. For a basis of the row space, one may also work on A^T as in part (b).

(d) To find (a basis of) the null space $\mathrm{Nul}(A)$, perform row operations:

$$A \xrightarrow{\text{row}} G = \begin{pmatrix} 1 & 0 & \frac{3}{2} & 0 \\ 0 & 1 & -\frac{5}{4} & \frac{1}{2} \\ 0 & 0 & 0 & 0 \end{pmatrix}.$$

Thus, $Ax = 0$ if and only if $Gx = 0$, where $x = (x_1, x_2, x_3, x_4)^T$ with $x_1 = -\frac{3}{2}x_3$ and $x_2 = \frac{5}{4}x_3 - \frac{1}{2}x_4$. Setting $x_3 = 1, x_4 = 0$ and $x_3 = 0, x_4 = 1$, respectively, reveals a basis of the null space $\mathrm{Nul}(A)$: $(-\frac{3}{2}, \frac{5}{4}, 1, 0)^T$ and $(0, -\frac{1}{2}, 0, 1)^T$. The nullity of A is 2.

(e) Determine if $b = (1,-1,1)^T \in \mathbb{R}^3$ is s linear combination of the columns of A. Yes, because $\mathrm{rank}(A) = \mathrm{rank}(A, b) = 2$, $Ax = b$ has a solution. So $b \in \mathrm{Col}(A)$. In fact, $b = (1,-1,1)^T = \frac{1}{2}(1,0,2)^T + \frac{1}{2}(1,-2,0)^T$. ∎

Note: In the above example, A is a 3×4 matrix, $\mathrm{Col}(A)$ is a subspace of $\mathbb{R}^3$, whereas $\mathrm{Row}(A)$ (written in columns) and $\mathrm{Nul}(A)$ are subspaces of $\mathbb{R}^4$.

Let A be a square matrix. The *determinant* of A, denoted by $\det A$, or $|A|$ (preferred if A is large or in a partitioned form), is a number associated with A which characterizes some properties of A. It can be defined in different but equivalent ways such as by recursive formula or via permutations.

For 2×2 matrix $A = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)$, define $\det A = ad - bc$. For $n \times n$ matrix A, $n = 3, 4, \ldots$, the determinant of A is defined inductively by the formula

$$\det A = \sum_{j=1}^{n} (-1)^{1+j} a_{1j} \det A(1|j), \tag{1.6}$$

where $A(1|j)$ is the submatrix of A obtained by deleting row 1 and column j of A. This formula is referred to as the *Laplace expansion formula* along the first row. Determinant can also be expanded along other rows or columns.

The determinant of a submatrix of a given matrix is called a *minor*.

The determinant defined via permutations is sometimes convenient.

A *permutation* of $\{1, 2, \ldots, n\}$ is an arrangement of $1, 2, \ldots, n$. A permutation p of $\{1, 2, \ldots, n\}$ is *even* if p can be restored to natural order by an even number of interchanges. Otherwise, p is *odd*. For example, consider the permutations of $\{1, 2, 3, 4\}$. The permutation p: $p(1) = 2$, $p(2) = 1$, $p(3) = 4$, $p(4) = 3$, written as $p = \left(\begin{smallmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{smallmatrix}\right)$, is even because it becomes $\left(\begin{smallmatrix} 1 & 2 & 3 & 4 \\ 1 & 2 & 3 & 4 \end{smallmatrix}\right)$ after interchanging 2 and 1 in column 2; 4 and 3 in column 4 (two interchanges):

$$\begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \to \begin{pmatrix} 1 & 2 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix} \to \begin{pmatrix} 1 & 2 & 3 & 4 \\ 1 & 2 & 3 & 4 \end{pmatrix}.$$

Note that $\left(\begin{smallmatrix} 1 & 2 & 3 & 4 \\ 1 & 4 & 3 & 2 \end{smallmatrix}\right)$ is odd as one interchange of 4 and 2 gives $\left(\begin{smallmatrix} 1 & 2 & 3 & 4 \\ 1 & 2 & 3 & 4 \end{smallmatrix}\right)$.

Let S_n be the set of all $(n!)$ permutations of $\{1, 2, \ldots, n\}$. For $p \in S_n$, define $\text{sign}(p) = 1$ if p is even and $\text{sign}(p) = -1$ if p is odd. Then the determinant of an n-square matrix $A = (a_{ij})$ is given by

$$\det A = \sum_{p \in S_n} \text{sign}(p) \prod_{t=1}^{n} a_{tp(t)}. \tag{1.7}$$

Simply put, the determinant is the sum of all $n!$ possible "signed" products in which each product involves n entries of A chosen from different rows and different columns (i.e., exactly one entry from each row and each column). It is immediate that the determinant of a (lower- or upper-) triangular matrix is the product of the diagonal entries. In particular, the determinant of a diagonal matrix is the product of the diagonal entries.

Elementary operations do not change the rank of a matrix, but they may change the determinant of a (square) matrix. To be precise, the determinant of a square matrix has the following properties.

(a) The determinant changes sign if two rows (columns) are interchanged.

(b) The determinant is unchanged if a scalar multiple of a row (column) is added to another row (column).

(c) If a row (column) is multiplied by a scalar, say c, then the determinant is c times the original determinant.

Example 1.2.9 Let $a, b, c, d \in \mathbb{C}$. Compute the determinants of matrices

$$A = \begin{pmatrix} 1 & a & 0 \\ 0 & 1 & b \\ c & d & 1 \end{pmatrix}, \; B = \begin{pmatrix} a & 1 & 1 & 1 \\ 0 & b & 0 & 0 \\ 0 & 1 & c & 1 \\ 0 & 1 & 1 & d \end{pmatrix}, \; C = \begin{pmatrix} a & 1 & 1 & 1 \\ 1 & b & 0 & 0 \\ 1 & 0 & c & 0 \\ 1 & 0 & 0 & d \end{pmatrix}.$$

$\det A = 1 + abc - bd$, $\det B = ab(cd - 1)$, and $\det C = abcd - bc - bd - cd$. A has 9 2×2 minors, 3 of which are $\left|\begin{smallmatrix} 1 & a \\ 0 & 1 \end{smallmatrix}\right| = 1$, $\left|\begin{smallmatrix} 0 & 1 \\ c & d \end{smallmatrix}\right| = -c$, $\left|\begin{smallmatrix} a & 0 \\ d & 1 \end{smallmatrix}\right| = a$. ∎

The following two important facts will be repeatedly used:

$$\det(AB) = \det A \det B, \quad A, B \in \mathbb{M}_n; \tag{1.8}$$

$$\det \begin{pmatrix} A & B \\ 0 & C \end{pmatrix} = \det A \det C, \quad A \in \mathbb{M}_n,\ C \in \mathbb{M}_m. \tag{1.9}$$

Note about the notation: For simplicity and for the reason of typography, when a matrix is in a block form or the matrix is large, we usually use $|\cdot|$ for $\det(\cdot)$. For instance, we use $\left|\begin{smallmatrix} A & B \\ 0 & C \end{smallmatrix}\right|$ for $\det\left(\begin{smallmatrix} A & B \\ 0 & C \end{smallmatrix}\right)$; $\left|\begin{smallmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{smallmatrix}\right|$ for $\det\left(\begin{smallmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{smallmatrix}\right)$.

(1.8) can be shown in various ways (usually by elementary matrices). In Sect. 2.2, Chap. 2, we present a proof of (1.8) by using (1.9).

(1.9) follows from (1.7). Here is an explanation. Note that the determinant is the sum of all (signed) products each of which contains one and only one entry from every row and every column. In the expanded form of $\left|\begin{smallmatrix} A & B \\ 0 & C \end{smallmatrix}\right|$, we sort the terms (products) into the ones containing entries of B and the ones that contain only the entries of A and C. Every product (of $m+n$ entries) with an entry of B contains a zero entry from the zero submatrix in the lower-left corner, so it vanishes. For the remaining terms, a permutation that produces a sign in the expanded form for $\left|\begin{smallmatrix} A & B \\ 0 & C \end{smallmatrix}\right|$ consists of "disjoint"permutations for $\det A$ and $\det C$ (as A and C have no rows nor columns in common). This leads to (1.9) (see Sect. 2.2 for a proof).

Recall that a square matrix A is said to be *invertible* or *nonsingular* if there exists a matrix B of the same order such that $AB = BA = I$. Such a B is unique and it is denoted by A^{-1}. If A is invertible, (1.8) shows immediately that $\det A \neq 0$ because $\det A \det B = 1$. The converse is also true.

Let $A = (a_{ij}) \in \mathbb{M}_n$. For each entry a_{ij}, we call $(-1)^{i+j} \det A(i|j)$ the *cofactor* of a_{ij}, $1 \leq i, j \leq n$. We construct an $n \times n$ matrix whose (j,i)-entry is the cofactor of a_{ij} (with reverse subscripts), and call it the *cofactor matrix* or the *adjugate* of A, written as $\operatorname{adj}(A)$. Upon computation, we have that $A \operatorname{adj}(A) = \operatorname{adj}(A) A = (\det A)\, I_n$. So, if $\det A \neq 0$, then A is invertible and

$$A^{-1} = \frac{1}{\det A} \operatorname{adj}(A). \tag{1.10}$$

Fact: Let A and B be $n \times n$ matrices. We have $AB = I_n \Leftrightarrow BA = I_n$.

For the 2×2 case, if $A = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)$ is invertible ($\Leftrightarrow \det A = ad - bc \neq 0$), then

$$A^{-1} = \frac{1}{ad - bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}. \tag{1.11}$$

For a nonsingular upper-triangular matrix A, A and A^{-1} take the forms

$$A = \begin{pmatrix} \lambda_1 & & \star \\ & \ddots & \\ 0 & & \lambda_n \end{pmatrix} \quad \text{and} \quad A^{-1} = \begin{pmatrix} \lambda_1^{-1} & & \star \\ & \ddots & \\ 0 & & \lambda_n^{-1} \end{pmatrix}.$$

[Note: We use $\star$'s to represent unspecified entries that are unimportant.]

If A and B are invertible matrices, both $n \times n$, then AB is invertible, and

$$(AB)^{-1} = B^{-1}A^{-1}. \tag{1.12}$$

A square matrix A is nonsingular if and only if A can be brought to the identity matrix through elementary operations. Equivalently, A is invertible if and only if it is a product of some elementary matrices. An effective way to find the inverse of A is to apply elementary row operations to the matrix (A, I) to get a matrix in the form (I, B). Then $B = A^{-1}$ (Problem 37):

$$(A, I) \xrightarrow{\text{row operations}} (I, A^{-1}).$$

Example 1.2.10 Let a, b, c, d be nonzero numbers. Find the inverse of

$$A = \begin{pmatrix} 0 & a & 0 & 0 \\ 0 & 0 & b & 0 \\ 0 & 0 & 0 & c \\ d & 0 & 0 & 0 \end{pmatrix}.$$

Since $\det A = -abcd \neq 0$, A is invertible. We present three ways to find A^{-1}. Way I works for this particular matrix; Way II uses the cofactor matrix which requires labor of calculation; Way III is via row operations.

Way I. Let $X = (x_{ij}) \in \mathbb{M}_4$ be the inverse of A, that is, $AX = I_4$. Then

$$ax_{21} = 1, \;\; bx_{32} = 1, \;\; cx_{43} = 1, \;\; dx_{14} = 1, \;\; \text{and all other } x_{ij} = 0.$$

Thus

$$A^{-1} = X = \begin{pmatrix} 0 & 0 & 0 & \frac{1}{d} \\ \frac{1}{a} & 0 & 0 & 0 \\ 0 & \frac{1}{b} & 0 & 0 \\ 0 & 0 & \frac{1}{c} & 0 \end{pmatrix}.$$

Way II. $A^{-1} = \frac{1}{\det A} \operatorname{adj}(A)$. First compute the cofactors (there are 16 of them!). For example, the cofactors of a_{12} and a_{22} are respectively

$$(-1)^{1+2}A(1|2) = \begin{vmatrix} 0 & b & 0 \\ 0 & 0 & c \\ d & 0 & 0 \end{vmatrix} = -bcd, \;\; (-1)^{2+2}A(2|2) = \begin{vmatrix} 0 & 0 & 0 \\ 0 & 0 & c \\ d & 0 & 0 \end{vmatrix} = 0.$$

So, the $(2,1)$-entry of A^{-1} is $\frac{-bcd}{\det A} = \frac{1}{a}$, and the $(2,2)$-entry of A^{-1} is 0. Similarly, one finds the cofactors of other entries of A, then constructs A^{-1}.

Way III. We perform elementary row operations on (A, I_4) to produce a matrix in the form (I_4, B), that is, $(A, I_4) \to (I_4, B)$. Then $B = A^{-1}$.

$$
\begin{aligned}
(A, I_4) &= \begin{pmatrix} 0 & a & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & b & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & c & 0 & 0 & 1 & 0 \\ d & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix} \\
&\to \begin{pmatrix} d & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & a & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & b & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & c & 0 & 0 & 1 & 0 \end{pmatrix} \\
&\to \begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{d} \\ 0 & 1 & 0 & 0 & \frac{1}{a} & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & \frac{1}{b} & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & \frac{1}{c} & 0 \end{pmatrix} \\
&= (I_4, A^{-1}). \ \blacksquare
\end{aligned}
$$

By Theorem 1.2.1, $\text{rank}(A) = \text{rank}(PAQ)$ for invertible matrices P and Q (of appropriate sizes). It can also be shown that the rank of A is the largest number of linearly independent columns (rows) of A. See Sect. 2.3 of Chapter 2 for more discussions on rank, row rank, and column rank. In addition, the rank of A is r if and only if A has a nonsingular r-square submatrix, and all $(r+1)$-square submatrices of A are singular (Problem 38).

Theorem 1.2.3 *Let $A \in \mathbb{M}_n$. The following statements are equivalent.*

(1) *A is invertible, i.e., $AB = BA = I$ for some $B \in \mathbb{M}_n$.*
(2) *$AB = I$ (or $BA = I$) for some $B \in \mathbb{M}_n$.*
(3) *A has rank n (full rank).*
(4) *A is a product of some elementary matrices.*
(5) *$Ax = 0$ has only the trivial solution $x = 0$.*
(6) *$Ax = b$ has a unique solution for each $b \in \mathbb{C}^n$.*
(7) $\det A \neq 0$.
(8) *The columns of A are linearly independent.*
(9) *The rows of A are linearly independent.*

Consider $Ax = b$, where $A \in \mathbb{M}_{m\times n}$ has rank r. Row reductions on the augmented matrix (A, b) yield an equivalent system of equations in the form

$$
\left\{ \begin{array}{lll}
x_{i_1} & & +c_{1\,r+1}x_{i_{r+1}} + \cdots + c_{1n}x_{i_n} = d_1 \\
\quad \ddots & & \vdots \\
& x_{i_r} & +c_{r\,r+1}x_{i_{r+1}} + \cdots + c_{rn}x_{i_n} = d_r \\
& & 0 = d_{r+1},
\end{array} \right. \tag{1.13}
$$

in which $x_{i_1}, \ldots, x_{i_r}, x_{i_{r+1}}, \ldots, x_{i_n}$ is a relist (or reordering) of $x_1, \ldots, x_n$.

If d_{r+1} is not equal to 0, then $Ax = b$ has no solution. If d_{r+1} is 0, then $Ax = b$ has a solution(s): it has a unique solution if $r = n$; it has infinitely many solutions if $r < n$, and $x_{i_{r+1}}, \ldots, x_{i_n}$ are free variables.

Example 1.2.11 (**Cramer's Rule**) Let $A \in \mathbb{M}_n$ be an invertible matrix and $b \in \mathbb{C}^n$. Then $Ax = b$ has the unique solution $x = A^{-1}b$. The components of $x = (x_1, \ldots, x_n)^T$ can be calculated through determinants as follows.

Write $A = (c_1, \ldots, c_n)$ in columns. Let $A_k(b)$ be the matrix obtained by replacing the kth column of A with b. Likewise, $I_k(x)$ is the matrix obtained from the identity matrix $I = (e_1, \ldots, e_n)$ with the kth column replaced by x, $k = 1, \ldots, n$. Note that $Ae_t = c_t$ for $t = 1, \ldots, n$. Then

$$A\, I_k(x) = (Ae_1, \ldots, Ax, \ldots, Ae_n) = (c_1, \ldots, b, \ldots, c_n) = A_k(b).$$

Take determinants for both sides. Since $\det I_k(x) = x_k$, we have for each k

$$x_k \det A = \det A_k(b), \quad \text{that is,} \quad x_k = \frac{\det A_k(b)}{\det A}. \ \blacksquare$$

Cramer's rule reveals the inverse formula (1.10) via adjugate as follows. Let $X = (x_{ij})$ be the inverse of A. Partition $X = (y_1, \ldots, y_n)$ in columns. Then $AX = (Ay_1, \ldots, Ay_n) = I$ implies $Ay_j = e_j$, $j = 1, \ldots, n$. By Cramer's rule, the ith component of y_j is $x_{ij} = \frac{\det A_i(e_j)}{\det A}$. Since $\det A_i(e_j) = (-1)^{i+j} \det A(j|i)$, the cofactor of a_{ji} in A, we have $A^{-1} = X = \frac{1}{\det A} \operatorname{adj}(A)$.

Problems

1. Let $A = \left(\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right)$, $B = \frac{i}{\sqrt{2}} \left(\begin{smallmatrix} 1 & 1 \\ 1 & -1 \end{smallmatrix}\right)$, and $C = \left(\begin{smallmatrix} 1 & 2 \\ 3 & 4 \end{smallmatrix}\right)$. Compute

$$A^2, \quad B^*B, \quad C^{-1}, \quad A + C, \quad AC, \quad CA.$$

2. Find the rank of $\left(\begin{smallmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 2 & 1 & 0 \end{smallmatrix}\right)$ by performing elementary operations.

3. Let $A = \left(\begin{smallmatrix} 1 & 2 & -1 & 1 \\ 0 & -4 & 5 & -2 \\ 2 & 0 & 3 & 0 \end{smallmatrix}\right)$. Determine whether or not

   (a) $x = (1, -1, 1)^T$ is in the column space $\operatorname{Col}(A)$.

   (b) $x = (1, -1, 1, 0)$ is in the row space $\operatorname{Row}(A)$.

   (c) $y = (1, -1, 1, 0)^T$ is in the null space $\operatorname{Nul}(A)$.

4. Referring Theorem 1.2.1, if A has entries from $\mathbb{F}$ (= $\mathbb{C}$, $\mathbb{R}$, or $\mathbb{Q}$), show that P and Q have entries from $\mathbb{F}$. In particular, if A is real, then P and Q can be chosen to be real. [Note: $\mathbb{F} = \mathbb{C}$ unless otherwise stated.]

5. Let A be a real matrix. Show that a set of columns of A is linearly independent over $\mathbb{C}$ if and only if the set is linearly independent over $\mathbb{R}$.

6. Let $A = (a_{ij}) \in \mathbb{M}_n$. Let $e_1, \ldots, e_n \in \mathbb{R}^n$ (in columns) be the standard basis of $\mathbb{R}^n$, and let $E_{ij} \in \mathbb{M}_n$, $1 \le i, j \le n$, be the standard basis of $\mathbb{M}_n$. Denote by $e = (1, \ldots, 1)^T \in \mathbb{R}^n$ the all-ones column vector.
   (a) Compute $e^T A e$.
   (b) Compute $e_i^T A e_j$.
   (c) Find $E_{ip}E_{qj}$, $E_{ip}A$, and AE_{qj}.
   (d) Show that the (i,j)-entry of $E_{ip}AE_{qj}$ is a_{pq}.

7. Evaluate the determinants (where $x, y, z, \lambda \in \mathbb{C}$)
$$\begin{vmatrix} 2 & -3 & 10 \\ 1 & 2 & -2 \\ 0 & 1 & -3 \end{vmatrix}, \quad \begin{vmatrix} 1+x & 1 & 1 \\ 1 & 1+y & 1 \\ 1 & 1 & 1+z \end{vmatrix}, \quad \begin{vmatrix} \lambda & 1 & 0 \\ 1 & \lambda & 1 \\ 0 & 1 & \lambda \end{vmatrix}.$$

8. Let $a_1, \ldots, a_n \in \mathbb{C}$.
   (a) Show the order 3 Vandermonde determinant identity
$$\begin{vmatrix} 1 & 1 & 1 \\ a_1 & a_2 & a_3 \\ a_1^2 & a_2^2 & a_3^2 \end{vmatrix} = (a_3 - a_1)(a_3 - a_2)(a_2 - a_1),$$
   and use it to evaluate the determinant
$$\begin{vmatrix} 1 & a_1 & a_1^2 - a_2a_3 \\ 1 & a_2 & a_2^2 - a_3a_1 \\ 1 & a_3 & a_3^2 - a_1a_2 \end{vmatrix}.$$
   (b) Show by induction the order n *Vandermonde determinant* identity
$$\begin{vmatrix} 1 & 1 & 1 & \cdots & 1 \\ a_1 & a_2 & a_3 & \cdots & a_n \\ a_1^2 & a_2^2 & a_3^2 & \cdots & a_n^2 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ a_1^{n-1} & a_2^{n-1} & a_3^{n-1} & \cdots & a_n^{n-1} \end{vmatrix} = \prod_{j>i} (a_j - a_i).$$
   (See Sect. 5.4 for more results on the Vandermonde determinant.)

9. Show that the determinant (with letters for arbitrary scalars) is zero:
$$\begin{vmatrix} a & b & 0 & 0 & 0 \\ c & d & 0 & 0 & 0 \\ e & f & 0 & 0 & 0 \\ g & h & i & j & k \\ l & m & n & o & p \end{vmatrix} = 0.$$

10. Let A be an n-square matrix. Show that

    (a) $\det A = \det A^T$.

    (b) $\det A^* = \overline{\det A}$.

    (c) $\det(cA) = c^n \det A$, where c is a scalar.

    (d) $\det(A^k) = (\det A)^k$, where k is a positive integer.

    (e) $\det(A^{-1}) = (\det A)^{-1}$ if A^{-1} exists.

11. Let A and B be complex matrices of appropriate sizes. Show that

    (a) $\overline{AB} = \overline{A}\,\overline{B}$,

    (b) $(AB)^T = B^T A^T$,

    (c) $(AB)^* = B^* A^*$, in particular, $(xy)^* = y^* x^*$ (for $x, y^T \in \mathbb{C}^n$), and

    (d) $(AB)^{-1} = B^{-1}A^{-1}$ if A and B are invertible.

12. If A is a Hermitian matrix, show that the determinant of A is real.

13. If A an $n \times n$ real matrix, where n is odd, show that $A^2 \neq -I_n$.

14. Let $A \in \mathbb{M}_n$. Show that $A^* + A$ is Hermitian and $A^* - A$ is normal.

15. (*Cartesian Decomposition*) Let $A \in \mathbb{M}_n$. Show that

$$A = \mathcal{H}(A) + \mathfrak{I}(A)i,$$

    where $\mathcal{H}A) = \frac{A+A^*}{2}$ and $\mathfrak{I}(A) = \frac{A-A^*}{2i}$ are both Hermitian.

16. Show that $M = \left(\begin{smallmatrix} 1 & i \\ i & 1 \end{smallmatrix}\right)$ and $N = \left(\begin{smallmatrix} i & i \\ i & -1 \end{smallmatrix}\right)$ are both (complex) symmetric and that M is normal and N is not normal. Find M^{-1} and N^{-1}.

17. Show that $\left(\begin{smallmatrix} r & \sqrt{1-r^2} \\ \sqrt{1-r^2} & -r \end{smallmatrix}\right)$ is real orthogonal, where $r \in [0, 1]$.

18. Let A be a matrix. Show that $\operatorname{rank}(A) = 1$ if and only if $A = CR$, where C is a nonzero column vector and R is a nonzero row vector.

19. Find the inverse for each of the following matrices.

$$\begin{pmatrix} 1 & a & 0 \\ 0 & 1 & 0 \\ 0 & b & 1 \end{pmatrix}, \quad \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix}, \quad \begin{pmatrix} 1 & 1 & 0 \\ 1 & 1 & 1 \\ 0 & 1 & 1 \end{pmatrix}.$$

20. Let k be a positive integer. Compute

$$\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}^k, \quad \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}^k, \quad \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}^k, \quad \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}^k, \quad \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}^k.$$

21. Let A and B be square matrices of the same size. Show that
$$A^*A - B^*B = \tfrac{1}{2}\big((A+B)^*(A-B) + (A-B)^*(A+B)\big).$$

22. Let A and B be $n \times n$ matrices. If $A+B$ is nonsingular, show that
$$A(A+B)^{-1}B = B(A+B)^{-1}A.$$

23. If $AB = A+B$ for matrices A and B, show that A and B commute, i.e.,
$$AB = A+B \quad \Rightarrow \quad AB = BA.$$
Find such a pair of (i) diagonal, (ii) nondiagonal matrices of A and B.

24. Let $A, B \in \mathbb{M}_n$. If A and B commute, i.e., $AB = BA$, show that
$$(A+B)^k = A^k + kA^{k-1}B + \tfrac{k(k-1)}{2}A^{k-2}B^2 + \cdots + B^k, \ k = 1, 2, \ldots.$$

25. Let $A, B \in \mathbb{M}_n$ and let m be a positive integer. Show that
$$(A+B)^m = A^m + \sum_{k=0}^{m-1} A^k B(A+B)^{m-k-1}.$$

26. Let $A, B \in \mathbb{M}_n$, $z \in \mathbb{C}$, and let m be a positive integer. Show that
$$(A+zB)^m = A^m + \sum_{k=1}^{m-1} z^k S_k^m(A,B) + z^m B^m,$$
where $S_k^m(A,B)$ denotes the sum of all products that each contain k copies of B's and $m-k$ copies of A's. For example, if $m=4$ and $k=2$, then $S_2^4(A,B) = A^2B^2 + BA^2B + B^2A^2 + ABAB + AB^2A + BABA$.

27. Let $A, B \in \mathbb{M}_n$, let m be a positive integer, and let ω be the mth *primitive root of unity*, i.e., $\omega^m = 1$, $\omega^k \neq 1$ if $1 \leq k < m$. Show that
$$A^m + B^m = \tfrac{1}{m}\big((A+B)^m + (A+\omega B)^m + \cdots + (A+\omega^{m-1}B)^m\big).$$

28. Let A be a square matrix and let k be a positive integer. Show that
$$I - A^{k+1} = (I-A)(I + A + A^2 + \cdots + A^k).$$

29. Determine whether each of the following statements is true.
    (a) The sum of Hermitian matrices is Hermitian.
    (b) The product of Hermitian matrices is Hermitian.
    (c) The sum of unitary matrices is unitary.
    (d) The product of unitary matrices is unitary.
    (e) The sum of normal matrices is normal.
    (f) The product of normal matrices is normal.

30. For $P = \left(\begin{smallmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{smallmatrix}\right)$ and $A = \left(\begin{smallmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{smallmatrix}\right)$, compute AP, PA, P^TAP, and PAP.

31. Let A, B, C, and D be $n \times n$ matrices. Multiply (in terms of A, B, C, D)

$$A(B,C), \quad (A,B)\begin{pmatrix} C \\ D \end{pmatrix}, \quad \begin{pmatrix} A \\ B \end{pmatrix} C, \quad \begin{pmatrix} A \\ B \end{pmatrix} (C,D).$$

32. Let $p(x) = a_n x^n + \cdots + a_1 x + a_0$ be a polynomial in x, where a_i's are scalars from a field $\mathbb{F}$. Let $A \in \mathbb{M}_m(\mathbb{F})$. The *matrix polynomial* $p(A)$ is defined to be $p(A) = a_n A^n + \cdots + a_1 A + a_0 I$. (a) Find $p(A)$ for $p(x) = 2x^3 - 3x^2 - ix + 5$ and $A = \left(\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right)$. (b) Show that $p(A)$ is upper-triangular (normal) if A is upper-triangular (resp., normal).

33. Find the rank and a basis for the column space of matrix A, where

$$A = \begin{pmatrix} 1 & 1 & -1 & 2 \\ 0 & 2 & 1 & 1 \\ 1 & 3 & 0 & 3 \\ 0 & 2 & 1 & 1 \end{pmatrix}.$$

34. Let A and B be matrices having the same number of columns, say n. If $\mathrm{Nul}(A) \subseteq \mathrm{Nul}(B)$, that is, $Ax = 0 \Rightarrow Bx = 0$ $(x \in \mathbb{C}^n)$, show that

$$\mathrm{rank}(B) \le \mathrm{rank}(A), \quad \mathrm{rank}\begin{pmatrix} A \\ B \end{pmatrix} = \mathrm{rank}(A),$$

and that $B = CA$ for some matrix C. When is C invertible?

35. Let $A \in \mathbb{M}_n$ and $B = I_n - A$. Show that $\mathrm{Nul}(A) \subseteq \mathrm{Col}(B)$.

36. Show that any two of the following three properties imply the third:

$$\text{(a)}\ \ A = A^*, \qquad \text{(b)}\ \ A^* = A^{-1}, \qquad \text{(c)}\ \ A^2 = I.$$

37. Let $A, B \in \mathbb{M}_n$. If $B(A, I) = (I, B)$, show that $B = A^{-1}$. Explain why A^{-1}, if it exists, can be obtained by row operations, that is, if

$$(A, I) \ \text{ row-reduces to } \ (I, B),$$

then matrix $B = A^{-1}$ is the inverse of A. Use this approach to find

$$\begin{pmatrix} 2 & 7 & 3 \\ 3 & 9 & 4 \\ 1 & 5 & 3 \end{pmatrix}^{-1}.$$

38. Show that the following statements are equivalent for $A \in \mathbb{M}_{m \times n}$.

    (a) $PAQ = \left(\begin{smallmatrix} I_r & 0 \\ 0 & 0 \end{smallmatrix}\right)$ for some invertible matrices P and Q.

    (b) The largest number of column (row) vectors of A that constitute a linearly independent set is r.

    (c) A contains an $r \times r$ nonsingular submatrix, and every $(r+1)$-square submatrix of A has determinant zero.

39. Let $A \in \mathbb{M}_{m\times n}$. Let P_1, P_2 and Q_1, Q_2 be invertible matrices such that $P_1AQ_1 = \begin{pmatrix} I_{r_1} & 0 \\ 0 & 0 \end{pmatrix}$ and $P_2AQ_2 = \begin{pmatrix} I_{r_2} & 0 \\ 0 & 0 \end{pmatrix}$. Show that $r_1 = r_2$.

40. Let $A, B \in \mathbb{M}_{m\times n}$. Show that $\operatorname{rank}(A) = \operatorname{rank}(B)$ if and only if there are nonsingular matrices $P \in \mathbb{M}_m$ and $Q \in \mathbb{M}_n$ such that $A = PBQ$.

41. (*Rank Factorization*) Let A be an $m \times n$ matrix of rank $r \geq 1$. Show that $A = B_{m\times r}C_{r\times n}$, where $B_{m\times r}$ is an $m \times r$ matrix of rank r (full column rank) and $C_{r\times n}$ is an $r \times n$ matrix of rank r (full row rank).

42. (*LU Factorization*) Let $A \in \mathbb{M}_n$. If A and all leading principal submatrices of A are nonsingular, then $A = LU$, where L and U are some n-square lower- and upper-triangular matrices, respectively.

43. Let A and B be $m \times r$ matrices with rank r (i.e., full column rank). Show that $PA = B$ for some nonsingular matrix P.

44. If A is a submatrix of a matrix M, show that $\operatorname{rank}(A) \leq \operatorname{rank}(M)$.

45. Let A and $B \in \mathbb{M}_{m\times n}$ have ranks p and q, respectively. Show that there exist $R \in \mathbb{M}_m$ and $S \in \mathbb{M}_n$, both are nonsingular, such that $RAS = \begin{pmatrix} A_1 \\ 0 \end{pmatrix}$ and $RBS = (B_1, 0)$, where $A_1 \in \mathbb{M}_{p\times n}$, $B_1 \in \mathbb{M}_{m\times q}$.

46. Let $A = \begin{pmatrix} S_{r\times s} & B \\ 0 & C \end{pmatrix}$ be a partitioned square matrix, in which $S_{r\times s}$ is a submatrix of size $r \times s$. If $r < s$, show that A is singular.

47. Prove Theorem 1.2.3.

48. Let $x_1, \ldots, x_n$ be complex numbers (not all zero) and let V be the vector space generated by $x_1, \ldots, x_n$ over the field of rational numbers. If $\dim V = m$, show that $(x_1, \ldots, x_n) = (y_1, \ldots, y_m)Q$ for some $m \times n$ rational matrix Q, where $\{y_1, \ldots, y_m\}$ is a basis of V. Use this to show that if the homogeneous equation system $Ax = 0$ has a complex solution, where A is a rational matrix, then it has a rational solution.

49. (a) Let A be an $m \times n$ real matrix and let $b \in \mathbb{R}^m$. If there exists a complex $x \in \mathbb{C}^n$ such that $Ax = b$, show that there exists a real $y \in \mathbb{R}^n$ such that $Ay = b$. (b) Let A and b have rational entries (components). If $Ax = b$ has a complex solution, show that it has a rational solution.

50. Let $a, b, c \in \mathbb{C}$ and let $A = \begin{pmatrix} a & b \\ \bar{b} & c \end{pmatrix}$. Show that $(\bar{z}, 1)A \begin{pmatrix} z \\ 1 \end{pmatrix} = a|z|^2 + b\bar{z} + \bar{b}z + c \geq 0$ for all $z \in \mathbb{C}$ if and only if $a \geq 0$, $c \geq 0$, and $ac \geq |b|^2$.

51. Let $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$, where $a, b, c, d \in \mathbb{C}$. Suppose that $\det A = ad - bc = 1$. Write A as a product of some 2×2 elementary matrices (with a, b, c, d).

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1.3 Linear Transformations, Matrices, and Their Eigenvalues

Let V and W be vector spaces over (the same) field $\mathbb{F}$. A function (or mapping or map) $\mathcal{A}$ from V to W is called a *linear transformation* or a *linear map* if

$$\mathcal{A}(u+v) = \mathcal{A}(u) + \mathcal{A}(v), \quad \mathcal{A}(cv) = c\mathcal{A}(v), \quad \text{for all } u, v \in V,\ c \in \mathbb{F}.$$

A linear transformation from V to V is also called a *linear operator*.

Example 1.3.1 Linear transformations.

(a) The *zero map* that maps every vector to zero: $\mathcal{O}(v) = 0$, $v \in V$.

(b) The *identity map* that maps each vector v to itself: $\mathcal{I}(v) = v$, $v \in V$.

(c) The *differential operator* $\mathcal{D}_x$ defined by

$$\mathcal{D}_x(f) = \frac{df(x)}{dx}, \quad f \in \mathcal{C}^\infty[a,b]$$

is a linear operator on $\mathcal{C}^\infty[a,b]$, the space of functions on $[a,b]$ whose derivatives of all orders exist (i.e., infinitely differentiable functions).

(d) $\mathcal{L} : (x,\, y,\, z) \mapsto (x+y,\, x-y,\, x+y+z)$ is a linear operator on $\mathbb{R}^3$.

(e) Define $\mathcal{T} : \mathbb{M}_n \to \mathbb{C}$, where $A = (a_{ij}) \in \mathbb{M}_n$, by $\mathcal{T}(A) = a_{11} + \cdots + a_{nn}$. Then $\mathcal{T}$ is a linear transformation (functional) from $\mathbb{M}_n$ to $\mathbb{C}$. ∎

Let $\mathcal{A}$ be a linear transformation from V to W. The *image* of $\mathcal{A}$

$$\mathrm{Im}(\mathcal{A}) = \{\mathcal{A}(v) : v \in V\}$$

is a subspace of W, also called the *range* of $\mathcal{A}$, and the *kernel* of $\mathcal{A}$

$$\mathrm{Ker}(\mathcal{A}) = \{v \in V : \mathcal{A}(v) = 0 \in W\}$$

is a subspace of V, also known as the *null space* of $\mathcal{A}$.

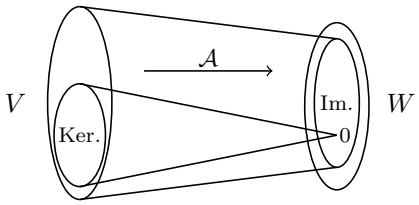


Fig. 1.3. Image and Kernel

Theorem 1.3.1 (Im-Ker Dimension Identity) *Let $\mathcal{A}$ be a linear transformation from a vector space V of finite dimension to a vector space W. Then*

$$\dim \mathrm{Im}(\mathcal{A}) + \dim \mathrm{Ker}(\mathcal{A}) = \dim V.$$

A standard proof goes as follows: Take a basis $\{u_1, \ldots, u_s\}$ for $\mathrm{Ker}(\mathcal{A})$ and extend it to a basis $\{u_1, \ldots, u_s, v_1, \ldots, v_t\}$ for V, where $s + t = \dim V$. It is routine to show that $\{\mathcal{A}(v_1), \ldots, \mathcal{A}(v_t)\}$ is a basis of $\mathrm{Im}(\mathcal{A})$ (Problem 40).

Given an $m \times n$ matrix A with entries in a field $\mathbb{F}$, one can define a linear transformation $\mathcal{A}$ from $\mathbb{F}^n$ to $\mathbb{F}^m$ by

$$\mathcal{A}(x) = Ax, \quad x \in \mathbb{F}^n. \tag{1.14}$$

Conversely, linear transformations on finite-dimensional spaces can be represented by matrices. Consider, for example, $\mathcal{A} : \mathbb{R}^2 \to \mathbb{R}^3$ defined by

$$\mathcal{A}(x_1,\, x_2)^T = (3x_1,\; 2x_1 + x_2,\; -x_1 - 2x_2)^T.$$

One checks that $\mathcal{A}$ is a linear transformation. We may write $\mathcal{A}$ in the form

$$\mathcal{A}(x) = Ax,$$

where

$$A = \begin{pmatrix} 3 & 0 \\ 2 & 1 \\ -1 & -2 \end{pmatrix}, \quad x = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}.$$

In (1.14), $\mathrm{Im}(\mathcal{A})$ is the space spanned by the columns of the matrix A, that is, the *column space* of A (also known as the image or range of A). $\mathrm{Ker}(\mathcal{A})$ is the solution space of $Ax = 0$, namely, the *kernel* or the *null space* of A. Note that the column space and the null space of A are also commonly denoted by $\mathrm{Col}(A) = \mathrm{Im}(A)$ and $\mathrm{Nul}(A) = \mathrm{Ker}(A)$, respectively (see the previous section). Theorem 1.3.1 reveals (Theorem 1.2.2) that for $m \times n$ matrix A,

$$\dim \mathrm{Im}(A) + \dim \mathrm{Ker}(A) = \dim \mathrm{Col}(A) + \dim \mathrm{Nul}(A) = n. \tag{1.15}$$

Example 1.3.2 Let A be an $m \times n$ matrix. Then for any $l \times m$ matrix X,

$$\mathrm{rank}(A) = \mathrm{rank}(XA) + \dim \big(\mathrm{Col}(A) \cap \mathrm{Nul}(X)\big). \tag{1.16}$$

Proof. Let $V = \mathrm{Im}(A) = \mathrm{Col}(A)$ and $W = \mathrm{Im}(XA)$. Define a linear map $\mathcal{X} : V \to W$ that sends $Ax \in V$ to $\mathcal{X}(Ax) = X(Ax) = (XA)x \in W$. Then

$$\dim V = \dim \mathrm{Im}(\mathcal{X}) + \dim \mathrm{Ker}(\mathcal{X}).$$

Since $\mathrm{Im}(\mathcal{X}) = \mathrm{Im}(XA)$ and $\mathrm{Ker}(\mathcal{X}) = \mathrm{Im}(A) \cap \mathrm{Ker}(X)$, (1.16) follows. ∎

Let $\mathcal{A}$ be a linear transformation from V to W, both of finite dimension. Once the bases for V and W have been chosen, $\mathcal{A}$ has a unique matrix representation A as in (1.14) determined by the images of the basis vectors of V under $\mathcal{A}$, and there is a one-to-one correspondence between the linear transformations and their matrices. We consider the case of linear operators,

i.e., $V = W$. A linear operator may have different matrices (representations) under different bases. We show that these matrices are similar.

Let $A, B \in \mathbb{M}_n(\mathbb{F})$. We say that A and B are *similar* over $\mathbb{F}$ if $PAP^{-1} = B$ for some invertible $P \in \mathbb{M}_n(\mathbb{F})$. Similar matrices have many properties in common, such as having the same determinant: $\det(PAP^{-1}) = \det A$.

Let $\mathcal{A}$ be a linear operator on a vector space V of dimension n and let $\{u_1, \ldots, u_n\}$ be a basis of V. Since each $\mathcal{A}(u_i)$ is a vector in V, we can write

$$\mathcal{A}(u_i) = \sum_{j=1}^{n} a_{ji} u_j, \quad i = 1, \ldots, n. \tag{1.17}$$

We call $A = (a_{ij})$ the *matrix representation*, or simply the *matrix of* $\mathcal{A}$, with respect to, or under, the (ordered) basis $\{u_1, \ldots, u_n\}$. [Note: The indexes j, i of a_{ji} in (1.17) are swapped from the indexes of the entry a_{ij}.]

Because $\{u_1, \ldots, u_n\}$ is a basis, a linear combination of $\mathcal{A}(u_i)$ in (1.17) is unique. Thus, the matrix A is unique. Write (1.17) conventionally as

$$\mathcal{A}(u_1, \ldots, u_n) = \big(\mathcal{A}(u_1), \ldots, \mathcal{A}(u_n)\big) = (u_1, \ldots, u_n)A. \tag{1.18}$$

Let $v \in V$. If $v = x_1 u_1 + \cdots + x_n u_n$, where x_i's are scalars, then

$$\mathcal{A}(v) = \sum_{i=1}^{n} x_i \mathcal{A}(u_i) = \big(\mathcal{A}(u_1), \ldots, \mathcal{A}(u_n)\big)x = (u_1, \ldots, u_n)Ax,$$

where x is the column vector $(x_1, \ldots, x_n)^T$. In the event of $\mathbb{R}^n$ or $\mathbb{C}^n$ with the standard basis $u_1 = e_1, \ldots, u_n = e_n$, we have $\mathcal{A}(v) = Ax$. (See (1.14).)

Let $\{v_1, \ldots, v_n\}$ be also a basis of V. Expressing each u_i as a linear combination of $v_1, \ldots, v_n$, $i = 1, \ldots, n$, gives an $n \times n$ matrix T such that

$$(u_1, \ldots, u_n) = (v_1, \ldots, v_n)T. \tag{1.19}$$

The matrix T is called the *change-of-basis matrix* between the bases (or from u's to v's). It can be shown (Problem 15) that T is invertible because $\{u_1, \ldots, u_n\}$ is a linearly independent set. It follows by using (1.17) that

$$\begin{aligned} \mathcal{A}(v_1, \ldots, v_n) &= \mathcal{A}\big((u_1, \ldots, u_n)T^{-1}\big) \\ &= (u_1, \ldots, u_n)AT^{-1} \\ &= (v_1, \ldots, v_n)(TAT^{-1}). \end{aligned}$$

Therefore, the matrix of $\mathcal{A}$ with respect to the basis $\{v_1, \ldots, v_n\}$ is TAT^{-1}. So, the matrices of a linear map under different bases are similar.

Note: In some other texts, the transpose of the matrix A in (1.17) is called the matrix representation of $\mathcal{A}$; the transpose of the matrix T in (1.19) is called the change-of-basis matrix of the bases. T is also known as the *transition matrix* from basis $\{v_1, \ldots, v_n\}$ to basis $\{u_1, \ldots, u_n\}$.

Example 1.3.3 Consider $\mathbb{R}^2$ with the bases $\{u_1, u_2\}$ and $\{v_1, v_2\}$, where

$$u_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \ u_2 = \begin{pmatrix} 1 \\ 1 \end{pmatrix} \quad \text{and} \quad v_1 = \begin{pmatrix} 2 \\ 3 \end{pmatrix}, \ v_2 = \begin{pmatrix} 3 \\ 4 \end{pmatrix}.$$

(1) In (1.19), let $P = (u_1, \ldots, u_n)$, $Q = (v_1, \ldots, v_n) \in \mathbb{M}_n$. Then $P = QT$. If R is the inverse of Q, then $RP = RQT = T$. Thus, $R(Q, P) = (RQ, RP) = (I, T)$. Since R is invertible and is the product of some elementary matrices, we see that the change-of-basis matrix T can be obtained from row-reducing the augmented matrix (Q, P) to (I, T).

$$(Q, P) = (v_1, v_2, u_1, u_2) = \begin{pmatrix} 2 & 3 & 1 & 1 \\ 3 & 4 & 0 & 1 \end{pmatrix} \to \begin{pmatrix} 1 & 0 & -4 & -1 \\ 0 & 1 & 3 & 1 \end{pmatrix}.$$

Thus, $(u_1, u_2) = (v_1, v_2)T$, where $T = \left(\begin{smallmatrix} -4 & -1 \\ 3 & 1 \end{smallmatrix}\right)$ is the change-of-basis matrix from $\{u_1, u_2\}$ to $\{v_1, v_2\}$. (One may also find T via $T = Q^{-1}P$.)

(2) If v is a vector having coordinate $(5, -6)$ with respect to $\{u_1, u_2\}$, then

$$\begin{aligned} v &= 5u_1 - 6u_2 = (u_1, u_2) \begin{pmatrix} 5 \\ -6 \end{pmatrix} \\ &= (v_1, v_2)B \begin{pmatrix} 5 \\ -6 \end{pmatrix} = (v_1, v_2) \begin{pmatrix} -4 & -1 \\ 3 & 1 \end{pmatrix} \begin{pmatrix} 5 \\ -6 \end{pmatrix} \\ &= (v_1, v_2) \begin{pmatrix} -14 \\ 9 \end{pmatrix} = -14v_1 + 9v_2. \end{aligned}$$

It follows that $(-14, 9)$ is the coordinate of v with respect to $\{v_1, v_2\}$.

(3) Define $\mathcal{A}(u_1) = v_1$ and $\mathcal{A}(u_2) = v_1 - v_2$. $\mathcal{A}$ is a linear transformation on $\mathbb{R}^2$. We find the matrix A of $\mathcal{A}$ with respect to the basis $\{u_1, u_2\}$:

$$\begin{aligned} \big(\mathcal{A}(u_1), \mathcal{A}(u_2)\big) &= (v_1, v_1 - v_2) = (v_1, v_2) \begin{pmatrix} 1 & 1 \\ 0 & -1 \end{pmatrix} \\ &= (u_1, u_2)T^{-1} \begin{pmatrix} 1 & 1 \\ 0 & -1 \end{pmatrix} \\ &= (u_1, u_2) \begin{pmatrix} -1 & -1 \\ 3 & 4 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 0 & -1 \end{pmatrix} \\ &= (u_1, u_2) \begin{pmatrix} -1 & 0 \\ 3 & -1 \end{pmatrix}. \end{aligned}$$

So $A = \left(\begin{smallmatrix} -1 & 0 \\ 3 & -1 \end{smallmatrix}\right)$. Likewise, we find the matrix of $\mathcal{A}$ under $\{v_1, v_2\}$:

$$\begin{aligned} \big(\mathcal{A}(v_1), \mathcal{A}(v_2)\big) &= \big(\mathcal{A}(u_1), \mathcal{A}(u_2)\big)T^{-1} = (u_1, u_2)AT^{-1} \\ &= (v_1, v_2)TAT^{-1} = (v_1, v_2) \begin{pmatrix} 2 & 3 \\ -3 & -4 \end{pmatrix}. \end{aligned}$$

Or directly, by the change-of-basis matrix T from u's to v's, one computes $TAT^{-1} = \left(\begin{smallmatrix} 2 & 3 \\ -3 & -4 \end{smallmatrix}\right)$, which is the matrix of $\mathcal{A}$ relative to $\{v_1, v_2\}$.

(4) What is the matrix of $\mathcal{A}$ in (3) relative to the standard basis $\{e_1, e_2\}$? Since $(e_1, e_2) = I_2 = (u_1, u_2)(u_1, u_2)^{-1} = (u_1, u_2)\left(\begin{smallmatrix} 1 & -1 \\ 0 & 1 \end{smallmatrix}\right)$, we have

$$\begin{aligned} \big(\mathcal{A}(e_1), \mathcal{A}(e_2)\big) &= \big(\mathcal{A}(u_1), \mathcal{A}(u_2)\big) \begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix} = (u_1, u_2) A \begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix} \\ &= \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} -1 & 0 \\ 3 & -1 \end{pmatrix} \begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix} \\ &= \begin{pmatrix} 2 & -3 \\ 3 & -4 \end{pmatrix} = (e_1, e_2) \begin{pmatrix} 2 & -3 \\ 3 & -4 \end{pmatrix}. \end{aligned}$$

Thus, the matrix of $\mathcal{A}$ relative to the standard basis $\{e_1, e_2\}$ is $\left(\begin{smallmatrix} 2 & -3 \\ 3 & -4 \end{smallmatrix}\right)$.

Alternatively, by computation we have $\mathcal{A}(e_1) = \mathcal{A}(u_1) = v_1 = 2e_1 + 3e_2$ and $\mathcal{A}(e_2) = \mathcal{A}(u_2 - u_1) = \mathcal{A}(u_2) - \mathcal{A}(u_1) = -v_2 = -3e_1 - 4e_2$. It follows that $(\mathcal{A}(e_1), \mathcal{A}(e_2)) = (e_1, e_2)\left(\begin{smallmatrix} 2 & -3 \\ 3 & -4 \end{smallmatrix}\right)$, revealing the matrix. ∎

Given a linear transformation on a vector space, it is a central theme of linear algebra to find a basis of the vector space so that the matrix representation of the transformation with respect to the basis has a special or desired structure, say, containing as many zeros as possible. In other words, we want to reduce a matrix to its canonical form via similarity. (Canonical forms will be studied in Chapter 3.) This is concerned with some vectors especially associated with the linear transformation.

Let $\mathcal{A}$ be a linear operator on a vector space V over field $\mathbb{F}$. A nonzero vector $v \in V$ is called an *eigenvector* of $\mathcal{A}$ associated to *eigenvalue* $\lambda \in \mathbb{F}$ if

$$\mathcal{A}(v) = \lambda v, \quad v \neq 0.$$

The pair (λ, v) is referred to as an *eigenpair*. The set of all eigenvectors v in V belonging to λ plus the zero vector of V forms a subspace, i.e., the set

$$V_\lambda = \{v \in V : \mathcal{A}(v) = \lambda v\}$$

(including 0) is a subspace of V. It is called the *eigenspace* of eigenvalue λ.

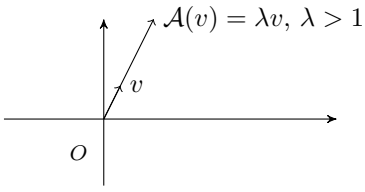


Fig. 1.4. Eigenvalue and eigenvector

Example 1.3.4 Linear operators, matrix representations, and eigenvalues.

(1) $\mathcal{A} : (x, y) \mapsto (y, x)$ is a linear operator on $\mathbb{R}^2$.

(i) $\mathrm{Im}(\mathcal{A}) = \mathbb{R}^2$, $\mathrm{Ker}(\mathcal{A}) = \{\mathcal{O} = (0, 0)\}$.

(ii) $\mathcal{A}$ has two eigenvalues 1 and -1. (x, x) is an eigenvector belonging to the eigenvalue 1 for any $x \neq 0$. The eigenvectors corresponding to -1 are $(x, -x)$ for any $x \neq 0$.

(iii) The matrix of $\mathcal{A}$ with respect to the standard basis is $\left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$.

(iv) The matrix of $\mathcal{A}$ with respect to the basis $\{(1, 1), (1, -1)\}$ is $\left(\begin{smallmatrix} 1 & 0 \\ 0 & -1 \end{smallmatrix}\right)$.

(2) $\mathcal{D} : p(x) \mapsto \frac{dp(x)}{dx}$ (the derivative of $p(x)$) is a linear map on $\mathbb{R}_n[x]$.

(i) $\mathrm{Im}(\mathcal{D}) = \mathbb{R}_{n-1}[x]$, $\mathrm{Ker}(\mathcal{D}) = \mathbb{R}$.

(ii) $0 \in \mathbb{R}$ is the only eigenvalue of $\mathcal{D}$. Any nonzero real number is an eigenvector of $\mathcal{D}$ associated with the eigenvalue 0. Thus, $V_0 = \mathbb{R}$.

(iii) The matrix of $\mathcal{D}$ with respect to the basis $\{1, x, x^2, \ldots, x^n\}$ is

$$\begin{pmatrix} 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \ddots & \vdots \\ 0 & 0 & 0 & \ddots & n \\ 0 & 0 & 0 & \cdots & 0 \end{pmatrix}.$$

(3) $\mathcal{L} : \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) \mapsto \left(\begin{smallmatrix} \frac{a+d}{2} & 0 \\ 0 & \frac{a+d}{2} \end{smallmatrix}\right)$ is a linear map on $\mathbb{M}_2(\mathbb{F})$, where $\mathbb{F} = \mathbb{R}$ or $\mathbb{C}$.

(i) $\mathrm{Im}(\mathcal{L}) = \{kI_2 : k \in \mathbb{F}\}$, $\mathrm{Ker}(\mathcal{L}) = \{\left(\begin{smallmatrix} a & b \\ c & -a \end{smallmatrix}\right) : a, b, c \in \mathbb{F}\}$.

(ii) $0 \in \mathbb{F}$ is an eigenvalue of $\mathcal{L}$ with eigenvectors $\left(\begin{smallmatrix} a & b \\ c & -a \end{smallmatrix}\right) \neq 0$, and $1 \in \mathbb{F}$ is an eigenvalue with eigenvectors kI_2, $0 \neq k \in \mathbb{F}$.
Does $\mathcal{L}$ have other eigenvalues than 0 and 1? No. Why?

(iii) To find a matrix representation of $\mathcal{L}$ with respect to the (ordered) basis $\{E_{11}, E_{12}, E_{21}, E_{22}\}$ (see page 12), we write, for example,

$$\mathcal{L}(E_{11}) = \begin{pmatrix} \frac{1}{2} & 0 \\ 0 & \frac{1}{2} \end{pmatrix} = \frac{1}{2}E_{11} + 0E_{12} + 0E_{21} + \frac{1}{2}E_{22}.$$

The matrix of $\mathcal{L}$ with respect to the basis $\{E_{11}, E_{12}, E_{21}, E_{22}\}$ is

$$\begin{pmatrix} \frac{1}{2} & 0 & 0 & \frac{1}{2} \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ \frac{1}{2} & 0 & 0 & \frac{1}{2} \end{pmatrix}. \blacksquare$$

If λ_1 and λ_2 are different eigenvalues of a linear map $\mathcal{A}$ with respective eigenvectors u and v, then u and v are linearly independent, because if

$$l_1 u + l_2 v = 0 \tag{1.20}$$

for some scalars l_1 and l_2, then applying $\mathcal{A}$ to both sides yields

$$l_1\lambda_1 u + l_2\lambda_2 v = 0. \tag{1.21}$$

Multiplying both sides of (1.20) by λ_1, we have

$$l_1\lambda_1 u + l_2\lambda_1 v = 0. \tag{1.22}$$

Subtracting (1.21) from (1.22) results in

$$l_2(\lambda_1 - \lambda_2)v = 0.$$

So $l_2 = 0$, thus $l_1 = 0$ by (1.20), that is, u and v are linearly independent.

Furthermore, if $u_1, \ldots, u_s$ are linearly independent eigenvectors associated with (eigenvalue) λ_1, and if $v_1, \ldots, v_t$ are linearly independent eigenvectors associated with λ_2, $\lambda_1 \neq \lambda_2$, then $u_1, \ldots, u_s, v_1, \ldots, v_t$ are linearly independent. This can be generalized to the case of more eigenvalues by induction. Simply put: The eigenvectors of distinct eigenvalues are linearly independent.

Let $\mathcal{A}$ be a linear transformation on a vector space V of dimension n. If $\mathcal{A}$ happens to have n linearly independent eigenvectors, say, $u_1, u_2, \ldots, u_n$, belonging to (not necessarily distinct) eigenvalues $\lambda_1, \lambda_2, \ldots, \lambda_n$, then these linearly independent eigenvectors comprise a basis of V, and with respect to the basis, $\mathcal{A}$ has a diagonal matrix representation $\operatorname{diag}(\lambda_1, \lambda_2, \ldots, \lambda_n)$:

$$\begin{aligned} \mathcal{A}(u_1, u_2, \ldots, u_n) &= \big(\mathcal{A}(u_1), \mathcal{A}(u_2), \ldots, \mathcal{A}(u_n)\big) \\ &= (\lambda_1 u_1, \lambda_2 u_2, \ldots, \lambda_n u_n) \\ &= (u_1, u_2, \ldots, u_n)\operatorname{diag}(\lambda_1, \lambda_2, \ldots, \lambda_n). \end{aligned}$$

It is immediate that if the eigenvalues of $\mathcal{A}$ are all distinct, then $\mathcal{A}$ has a diagonal matrix representation (under the basis of eigenvectors). Since the matrices of $\mathcal{A}$ under different bases are similar to each other, if A is a matrix of $\mathcal{A}$ (with respect to some basis), then there exists a nonsingular matrix P such that $P^{-1}AP$ is diagonal. We say that $\mathcal{A}$ (and A) is *diagonalizable.*

We summarize the above discussions in a theorem.

Theorem 1.3.2 *Linear independence and diagonalization.*

(1) *The eigenvectors of distinct eigenvalues are linearly independent.*

(2) *If all eigenvalues are distinct, then the linear map is diagonalizable.*

Let $\alpha = \{\alpha_1, \alpha_2, \ldots, \alpha_n\}$ be a basis of the vector space V. Let A be the matrix representation of $\mathcal{A}$ with respect to the basis. For a vector $v \in V$, suppose the coordinate of v under α is x (in column). We write $v = \alpha x$. Then $\mathcal{A}(v) = \mathcal{A}(\alpha x) = \alpha(Ax)$. Thus, for an eigenpair (λ, v), we see that

$$\mathcal{A}(v) = \lambda v \quad \Leftrightarrow \quad \alpha Ax = \lambda\alpha x \quad \Leftrightarrow \quad Ax = \lambda x.$$

It follows that the eigenvalues of $\mathcal{A}$ are the scalars $\lambda \in \mathbb{F}$ that satisfy

$$\det(\lambda I - A) = 0, \tag{1.23}$$

and that the eigenvectors of $\mathcal{A}$ associated with λ are the vectors whose coordinates under the basis α are the solutions to the linear equation system $Ax = \lambda x$. Do the eigenvalues of $\mathcal{A}$ depend on the choices of the bases? No, because the matrices of a linear map under different bases are similar. In view of (1.23), if B is similar to A, then $\det(\lambda I - B) = 0 \Leftrightarrow \det(\lambda I - A) = 0$.

In Example 1.3.4 (3), the eigenvalues of $\mathcal{L}$ are 1, 0, 0, 0 (we say that 0 is a repeated eigenvalue, or an eigenvalue of *algebraic multiplicity* 3).

Suppose that A is an $n \times n$ complex matrix. The polynomial in λ,

$$p_A(\lambda) = \det(\lambda I_n - A), \tag{1.24}$$

is called the *characteristic polynomial* of A, and the zeros of the polynomial are called the *eigenvalues of* A. The fundamental theorem of algebra ensures that every n-square real or complex matrix has n eigenvalues over $\mathbb{C}$ (including the repeated ones). [Note: This need not be true over $\mathbb{R}$.]

Example 1.3.5 Eigenvalue and eigenvector.

(a) Let $A = \begin{pmatrix} a & u \\ 0 & B \end{pmatrix}$, where $a \in \mathbb{C}$, $B \in \mathbb{M}_{n-1}$, and u is a row vector. Then a is an eigenvalue of A because $\det(aI_n - A) = \det \begin{pmatrix} 0 & -u \\ 0 & aI_{n-1} - B \end{pmatrix} = 0$.

(b) Let $u, v \in \mathbb{C}^n$ (in columns), $u \neq 0$. Let $A = uv^T$ and $a = v^T u$. Then a is an eigenvalue of A and u is an eigenvector associated to a because

$$Au = (uv^T)u = u(v^T u) = (v^T u)u = au. \ \blacksquare$$

Let $A \in \mathbb{M}_n$ and let $\lambda_1, \ldots, \lambda_n$ be the eigenvalues of A. Then

$$p_A(\lambda) = \det(\lambda I_n - A) = (\lambda - \lambda_1) \cdots (\lambda - \lambda_n). \tag{1.25}$$

The sum of the eigenvalues of A is called the *trace* of A, denoted by $\operatorname{tr} A$. Let A have diagonal entries $a_{11}, \ldots, a_{nn}$. From (1.25) by expansions, we see

$$\operatorname{tr} A = \lambda_1 + \cdots + \lambda_n = a_{11} + \cdots + a_{nn} \tag{1.26}$$

and

$$\det A = \prod_{i=1}^{n} \lambda_i. \tag{1.27}$$

Theorem 1.3.3 (Properties of Similar Matrices) *Similar matrices have the same rank; the same characteristic polynomial, consequently, the same (set of) eigenvalues, the same trace, and the same determinant.*

Proof. Let $A, B \in \mathbb{M}_n$ and let $A = P^{-1}BP$ for some invertible matrix P. By Theorem 1.2.3, P is a product of some elementary matrices; so is P^{-1}. Thus, A is obtained from B through elementary operations, and A has the same rank as B. For the characteristic polynomial, by (1.24),

$$\begin{aligned} p_A(\lambda) &= \det(\lambda I_n - A) = \det(\lambda I_n - P^{-1}BP) = \det(P^{-1}(\lambda I_n - B)P) \\ &= \det(P^{-1})\det(\lambda I_n - B)\det P = \det(\lambda I_n - B) \;=\; p_B(\lambda). \end{aligned}$$

It follows that similar matrices have the same characteristic polynomial, consequently, the same eigenvalues, trace, and determinant. In symbols:

$$\operatorname{rank}(A) = \operatorname{rank}(B),\;\; \det A = \det B,\;\; \operatorname{tr} A = \operatorname{tr} B,\;\; \lambda(A) = \lambda(B). \;\blacksquare$$

Theorem 1.3.4 *If A is an $m \times n$ matrix and B is an $n \times m$ matrix, then*

$$\operatorname{tr}(AB) = \operatorname{tr}(BA). \tag{1.28}$$

This is known as the *trace cyclic property.* Here is a proof in one line:

$$\operatorname{tr}(AB) = \sum_{i=1}^{m}\sum_{j=1}^{n} a_{ij}b_{ji} = \sum_{j=1}^{n}\sum_{i=1}^{m} b_{ji}a_{ij} = \operatorname{tr}(BA).$$

Example 1.3.6 Find the eigenvalues and eigenvectors of matrices

$$A = \begin{pmatrix} 2 & 1 & 0 \\ 1 & 3 & 1 \\ 0 & 1 & 2 \end{pmatrix} \quad \text{and} \quad B = \begin{pmatrix} -4 & 0 & -10 \\ 3 & 1 & 6 \\ 1 & 0 & 3 \end{pmatrix}.$$

Solve the equation $\det(\lambda I - A) = 0$ for λ to obtain the eigenvalues of A: $\lambda = 1, 2, 4$. Then solve the linear system $(\lambda I - A)x = 0$ with each λ to get the corresponding eigenvectors $x = (1, -1, 1)$, $(1, 0, -1)$, $(1, 2, 1)$ (not unique).

Similarly, B has eigenvalues $\lambda = 1,\ 1,\ -2$. Solve $(\lambda I - B)x = 0$ to get corresponding eigenvectors $x = (0, 1, 0)$, $(-2, 0, 1)$, $(5, -3, -1)$ (not unique).

$\operatorname{tr} A = 1 + 2 + 4 = 2 + 3 + 2 = 7$, $\det A = 8$; $\operatorname{tr} B = 0$, $\det B = -2$. $\blacksquare$

Theorem 1.3.5 *An $n \times n$ matrix A is diagonalizable, i.e., A is similar to a diagonal matrix, if and only if A has n linearly independent eigenvectors.*

Proof. If A is diagonalizable, i.e., $P^{-1}AP = D$ is a diagonal matrix for some invertible matrix P, then $AP = PD$. Partition P in columns as $P = (u_1, \ldots, u_n)$. Since P is invertible, by Theorem 1.2.3, $u_1, \ldots, u_n$ are linearly independent. On the other hand, let $D = \operatorname{diag}(\lambda_1, \ldots, \lambda_n)$. Then $AP = PD$ reveals $Au_i = \lambda_i u_i$ for each i, that is, the u_i's are eigenvectors.

Conversely, if $Au_i = \lambda_i u_i$ for $i = 1, \ldots, n$, and if $u_1, \ldots, u_n$ are linearly independent, let $P = (u_1, \ldots, u_n)$. Then $AP = P\operatorname{diag}(\lambda_1, \ldots, \lambda_n)$, i.e., $P^{-1}AP = \operatorname{diag}(\lambda_1, \ldots, \lambda_n)$ is a diagonal matrix, so A is diagonalizable. $\blacksquare$

Note that the λ_i's in the above proof need not be distinct.

Exercise: Both A and B in Example 1.3.6 are diagonalizable.

Let $\mathcal{A}$ be a linear transformation (operator) on a vector space V. Let W be a subspace of V. If for every $w \in W$, $\mathcal{A}(w) \in W$ (i.e., $\mathcal{A}(W) \subseteq W$), we say that W is *invariant* under $\mathcal{A}$. Obviously $\{0\}$ and V are invariant under $\mathcal{A}$. It is easy to check that $\mathrm{Ker}(\mathcal{A})$ and $\mathrm{Im}(\mathcal{A})$ are also invariant under $\mathcal{A}$.

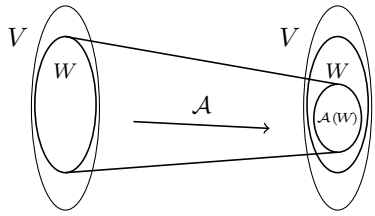


Fig. 1.5. Invariant subspace

Let V be a vector space over a field. Consider all linear operators on V and denote the set by $L(V)$. Then $L(V)$ is a vector space with respect to the following addition and scalar multiplication:

$$(\mathcal{A}+\mathcal{B})(v) = \mathcal{A}(v) + \mathcal{B}(v), \quad (k\mathcal{A})(v) = k\mathcal{A}(v).$$

The zero vector in $L(V)$ is the zero transformation $\mathcal{O} : v \mapsto 0 \in V$, and for every $\mathcal{A} \in L(V)$, $-\mathcal{A}$ is the operator $(-\mathcal{A})(v) = -(\mathcal{A}(v))$. For two operators $\mathcal{A}$ and $\mathcal{B}$ on V, define the *product* of $\mathcal{A}$ and $\mathcal{B}$ by

$$(\mathcal{A}\mathcal{B})(v) = \mathcal{A}\big(\mathcal{B}(v)\big), \quad v \in V.$$

Then $\mathcal{A}\mathcal{B}$ is again a linear transformation on V. The *identity* $\mathcal{I}$ for the multiplication is the transformation such that $\mathcal{I}(v) = v$ for all $v \in V$.

Example 1.3.7 For linear operators $\mathcal{A} : \left(\begin{smallmatrix} x \\ y \end{smallmatrix}\right) \mapsto \left(\begin{smallmatrix} x+y \\ x-y \end{smallmatrix}\right) = \left(\begin{smallmatrix} 1 & 1 \\ 1 & -1 \end{smallmatrix}\right)\left(\begin{smallmatrix} x \\ y \end{smallmatrix}\right)$ and $\mathcal{B} : \left(\begin{smallmatrix} x \\ y \end{smallmatrix}\right) \mapsto \left(\begin{smallmatrix} y \\ x \end{smallmatrix}\right) = \left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)\left(\begin{smallmatrix} x \\ y \end{smallmatrix}\right)$ on $\mathbb{R}^2$, $\mathcal{A}\mathcal{B} : \left(\begin{smallmatrix} x \\ y \end{smallmatrix}\right) \mapsto \left(\begin{smallmatrix} y+x \\ y-x \end{smallmatrix}\right) = \left(\begin{smallmatrix} 1 & 1 \\ -1 & 1 \end{smallmatrix}\right)\left(\begin{smallmatrix} x \\ y \end{smallmatrix}\right)$. ∎

Let V be a vector space of finite dimension. Then every $\mathcal{A} \in L(V)$ has a matrix representation A with respect to a fixed basis. The linear operators on V may be viewed as matrices, i.e., $L(V)$ is essentially the same as $\mathbb{M}_n$.

Problems

1. Show that the transformation $\mathcal{A}$ from $\mathbb{R}^3$ to $\mathbb{R}^3$ defined by

$$\mathcal{A}(x, y, z) = (x + y, x - y, z)$$

is linear. Find a matrix of $\mathcal{A}$ with respect to the standard basis of $\mathbb{R}^3$.

2. Find the dimension and a basis for $\mathrm{Im}(\mathcal{A})$ and for $\mathrm{Ker}(\mathcal{A})$, where $\mathcal{A}$ is the linear transformation on $\mathbb{R}^3$ defined by

$$\mathcal{A}(x, y, z) = (x - 2z, y + z, 0).$$

3. Define a linear transformation $\mathcal{A} : \mathbb{R}^2 \to \mathbb{R}^2$ by

$$\mathcal{A}(x, y) = (y, 0).$$

   (a) Find $\mathrm{Im}(\mathcal{A})$ and $\mathrm{Ker}(\mathcal{A})$.
   (b) Find a matrix representation of $\mathcal{A}$.
   (c) Verify that $\dim \mathbb{R}^2 = \dim \mathrm{Im}(\mathcal{A}) + \dim \mathrm{Ker}(\mathcal{A})$.
   (d) Is $\mathrm{Im}(\mathcal{A}) + \mathrm{Ker}(\mathcal{A})$ a direct sum?
   (e) Does $\mathbb{R}^2 = \mathrm{Im}(\mathcal{A}) + \mathrm{Ker}(\mathcal{A})$?

4. Show that the matrix similarity is an *equivalence relation* $\sim$ between square matrices of the same order, namely, (1) $A \sim A$ for every matrix A, (2) if $A \sim B$, then $B \sim A$, (3) if $A \sim B$ and $B \sim C$, then $A \sim C$.

5. Find the eigenvalues and eigenvectors of the differential operator $\mathcal{D}_x$:

$$\mathcal{D}_x(f(x)) = f'(x), \text{ where } f \text{ is infinitely differentiable on } \mathbb{R}.$$

6. Find the eigenvalues and corresponding eigenvectors for matrices

$$A = \begin{pmatrix} 1 & 4 \\ 2 & 3 \end{pmatrix}, \quad B = \begin{pmatrix} 1 & 4 \\ 3 & 2 \end{pmatrix}, \quad C = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}.$$

7. Find the traces, determinants, and eigenvalues of the matrices

$$A = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, B = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}, C = \begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & 0 \\ 0 & 1 & 1 \end{pmatrix}, D = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}.$$

8. Let $A = \begin{pmatrix} 1 & \alpha \\ \beta & -1 \end{pmatrix}$ and $B = \begin{pmatrix} \alpha & 1 \\ -1 & \beta \end{pmatrix}$. For what α and β does A have only zero eigenvalues? For what α and β does B have only zero eigenvalues?

9. Let λ be an eigenvalue of linear map $\mathcal{A}$ on a vector space V, and let

$$V_\lambda = \{v \in V : \mathcal{A}(v) = \lambda v\},$$

   called the *eigenspace* of λ. Show that V_λ is an *invariant subspace* of V under $\mathcal{A}$, that is, it is a subspace and $\mathcal{A}(v) \in V_\lambda$ for every $v \in V_\lambda$.

10. Define linear transformations $\mathcal{A}$ and $\mathcal{B}$ on $\mathbb{R}^2$ respectively by

$$\mathcal{A}(x, y) = (x + y, y), \quad \mathcal{B}(x, y) = (x + y, x - y).$$

    Find all eigenvalues of $\mathcal{A}$ and $\mathcal{B}$ and determine their eigenspaces.

11. Let A be an $n \times n$ matrix and let λ be an eigenvalue of A. Show that

$$\dim V_\lambda = \dim \operatorname{Ker}(\lambda I - A) = n - \operatorname{rank}(\lambda I - A).$$

If the characteristic polynomial $p(x) = \det(xI - A)$ of A is factored as $p(x) = (x-\lambda)^k q(x)$ for some polynomial $q(x)$ with $q(\lambda) \neq 0$, show that

$$k \geq \dim V_\lambda.$$

[Note: Such a k is called the *algebraic multiplicity* of λ; $\dim V_\lambda$ is referred to as the *geometric multiplicity* of λ. When we say multiplicity of λ, we usually mean the algebraic multiplicity unless otherwise stated.]

12. Find the geometric and algebraic multiplicities of the eigenvalues of

$$A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \quad \text{and} \quad B = \begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix}.$$

13. If λ is an eigenvalue of $A \in \mathbb{M}_n$, show that $\operatorname{rank}(\lambda I - A) + \dim V_\lambda = n$.

14. Let $u \neq 0, v, w$ be three vectors in a vector space V over the rational number field $\mathbb{Q}$. If there is a linear transformation $\mathcal{L}$ on V such that

$$\mathcal{L}(u) = v, \;\; \mathcal{L}(v) = w, \;\; \mathcal{L}(w) = u + v,$$

show that u, v, w are linearly independent over $\mathbb{Q}$. What if over $\mathbb{R}$?

15. Let $\{u_1, \ldots, u_n\}$ be a basis of a vector space V and let $\{v_1, \ldots, v_k\}$ be a set of k vectors in V. If $v_i = \sum_{j=1}^n a_{ij} u_j$, $i = 1, \ldots, k$, show that
    (a) $\dim \operatorname{Span}\{v_1, \ldots, v_k\} = \operatorname{rank}(A)$, where $A = (a_{ij}) \in \mathbb{M}_{k \times n}$.
    (b) Matrix $A = (a_{ij}) \in \mathbb{M}_{k \times n}$ is unique (for the given $v_1, \ldots, v_k$).
    (c) If $k = n$ and if $\{v_1, \ldots, v_n\}$ is also a basis, then A is invertible. [Note: A (or A^T in some texts) is called *change-of-basis matrix*.]

16. If $\lambda_1, \ldots, \lambda_n$ are the eigenvalues of an n-square matrix A, what are the eigenvalues of A^k, pA, and $qI_n - A$ (where $p, q \in \mathbb{C}$, $k = 1, 2, \ldots$)?

17. Let $A \in \mathbb{M}_n$ be an invertible matrix. If $\lambda_1, \ldots, \lambda_n$ are the eigenvalues of A, show that $\lambda_1^{-1}, \ldots, \lambda_n^{-1}$ are the eigenvalues of A^{-1}.

18. Let $A \in \mathbb{M}_n$. Show that A is similar to $-A$ if and only if $AX + XA = 0$ for some nonsingular $X \in \mathbb{M}_n$.

19. Show that the eigenvalues of a diagonal matrix, an upper-triangular matrix, and a lower-triangular matrix are the scalars on the diagonal.

20. Show that every linear map on a vector space of dimension n over $\mathbb{C}$ has n eigenvalues (with multiplicities). Is this true over $\mathbb{R}$?

21. Let $A \in \mathbb{M}_n$ and let $S \in \mathbb{M}_n$ be nonsingular. If the first column of $S^{-1}AS$ is $(\lambda, 0, \ldots, 0)^T$, show that λ is an eigenvalue of A and that the first column of S is an eigenvector of A associated with λ.

22. Let v_1 and v_2 be eigenvectors of matrix A associated with distinct eigenvalues λ_1 and λ_2, respectively, and let $w = v_1 + v_2$. Show that
    (a) w is not an eigenvector of A (of any eigenvalue).
    (b) w and v_1 are linearly independent.
    (c) w and v_2 are linearly independent.
    (d) w, v_1, and v_2 are linearly dependent.
    (e) Find all eigenvectors of A that lie in $\text{Span}\{v_1, v_2\}$.

23. Let $x \in \mathbb{C}^n$. Find the eigenvalues and eigenvectors of the matrices
$$A_1 = xx^* \quad \text{and} \quad A_2 = \begin{pmatrix} 0 & x^* \\ x & 0 \end{pmatrix}.$$

24. Let $x \in \mathbb{C}^n$. Show that $I_n + xx^*$ is invertible. Find the inverse.

25. If $x \in \mathbb{C}^n$ is an eigenvector of $A \in \mathbb{M}_n$ belonging to the eigenvalue λ, show that for any $y \in \mathbb{C}^n$, $\lambda + y^*x$ is an eigenvalue of $A + xy^*$.

26. Let R be an $n \times n$ real matrix. Prove or disprove (by counterexample):
    (a) $\det(R^2) \geq 0$.
    (b) $\text{tr}(R^2) \geq 0$.
    (c) If λ is an eigenvalue of R^2, then $\lambda \geq 0$.

27. Let $\mathcal{A}$ be a linear transformation on a vector space of dimension n with matrix representation A. Show that
$$\dim \text{Im}(\mathcal{A}) = \text{rank}(A) \quad \text{and} \quad \dim \text{Ker}(\mathcal{A}) = n - \text{rank}(A).$$

28. Referring to Example 1.3.2, show that $\text{Ker}(\mathcal{X}) = \text{Im}(A) \cap \text{Ker}(X)$ and $\text{Im}(\mathcal{X}) = \text{Im}(XA)$. Find an example for $\text{Ker}(XA) \neq \text{Im}(A) \cap \text{Ker}(X)$.

29. Let A be a square matrix. Show that
$$\text{rank}(A) = \text{rank}(A^2) + \dim\big(\text{Col}(A) \cap \text{Nul}(A)\big).$$

30. Let $A \in \mathbb{M}_n$. A *principal submatrix* of A is a submatrix lying in the intersection of the same rows and columns of A, and the determinant of a principal submatrix is called a *principal minor* of A. Show that
$$\det(\lambda I - A) = \lambda^n - \delta_1\lambda^{n-1} + \delta_2\lambda^{n-2} - \cdots + (-1)^n \det A,$$
where δ_i is the sum of all principal minors of order i, $i = 1, 2, \ldots, n-1$.

31. Let $\left(\begin{smallmatrix} a & c \\ \bar{c} & b \end{smallmatrix}\right)$ be a 2×2 Hermitian matrix with eigenvalues λ, μ. Show that
$$\max\{2|c|, |a-b|\} \leq |\lambda - \mu| \leq 2|c| + |a-b|.$$

32. If each row sum (i.e., the sum of all entries in a row) of matrix A is equal to 1, show that 1 is an eigenvalue of A.

33. A linear transformation $\mathcal{A}$ on a vector space V is said to be *invertible* if there exists a linear transformation $\mathcal{B}$ such that $\mathcal{AB} = \mathcal{BA} = \mathcal{I}$, the identity map. If $\dim V < \infty$, show that the following are equivalent.
    (a) $\mathcal{A}$ is invertible.
    (b) If $\mathcal{A}(x) = 0$, then $x = 0$, that is, $\mathrm{Ker}(\mathcal{A}) = \{0\}$.
    (c) If $\{u_1, \ldots, u_n\}$ is a basis for V, then so is $\{\mathcal{A}u_1, \ldots, \mathcal{A}u_n\}$.
    (d) $\mathcal{A}$ is one-to-one.
    (e) $\mathcal{A}$ is onto, that is, $\mathrm{Im}(\mathcal{A}) = V$.
    (f) $\mathcal{A}$ has a nonsingular matrix representation under some basis.

34. Show that $\mathcal{L} : X \mapsto \mathrm{tr}(X)$ is a linear map from $\mathbb{M}_n$ to $\mathbb{C}$. What are the dimensions of $\mathrm{Im}(\mathcal{L})$ and $\mathrm{Ker}(\mathcal{L})$? Find a basis for $\mathrm{Ker}(\mathcal{L})$.

35. Show that $\mathcal{M} : \left(\begin{smallmatrix} x \\ y \end{smallmatrix}\right) \mapsto \left(\begin{smallmatrix} 1 & 2 \\ 4 & 3 \end{smallmatrix}\right)\left(\begin{smallmatrix} x \\ y \end{smallmatrix}\right)$ is a linear operator on $\mathbb{R}^2$. Find a matrix representation of $\mathcal{M}$ with respect to the basis $\{\left(\begin{smallmatrix} 1 \\ 0 \end{smallmatrix}\right), \left(\begin{smallmatrix} 1 \\ 1 \end{smallmatrix}\right)\}$.

36. Define a transformation from $\mathbb{M}_2$ to $\mathbb{M}_2$ by $\mathcal{N} : \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) \mapsto \left(\begin{smallmatrix} 1 & 2 \\ 4 & 3 \end{smallmatrix}\right)\left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)$.
    (a) Show that $\mathcal{N}$ is a linear map.
    (b) How many eigenvalues (with multiplicities) does $\mathcal{N}$ have?
    (c) Find all eigenvalues (with multiplicities) of $\mathcal{N}$.
    (d) Find a matrix representation of $\mathcal{N}$ with respect to the basis $\{E_{11} = \left(\begin{smallmatrix} 1 & 0 \\ 0 & 0 \end{smallmatrix}\right), E_{21} = \left(\begin{smallmatrix} 0 & 0 \\ 1 & 0 \end{smallmatrix}\right), E_{12} = \left(\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right), E_{22} = \left(\begin{smallmatrix} 0 & 0 \\ 0 & 1 \end{smallmatrix}\right)\}$.
    (e) Find a basis of $\mathbb{M}_2$ such that the matrix of $\mathcal{N}$ with respect to the basis is diagonal.

37. Consider $\mathbb{R}^3$ with the standard basis $e = \{e_1, e_2, e_3\}$. Let
$$\alpha_1 = (1, 0, 0), \quad \alpha_2 = (1, 1, 0), \quad \alpha_3 = (1, 1, 1).$$
    (a) Show that $\alpha = \{\alpha_1, \alpha_2, \alpha_3\}$ is a basis of $\mathbb{R}^3$.
    (b) Find the change-of-basis matrix from basis e to basis α.
    (c) If $v \in \mathbb{R}^3$ has the coordinate $(1, 2, 3)$ with respect to the standard basis e, what is the coordinate of v with respect to the basis α?
    (d) If $\mathcal{A}$ is a line transformation on $\mathbb{R}^3$ defined by
$$\mathcal{A}(e_1) = \alpha_1, \quad \mathcal{A}(e_2) = \alpha_2, \quad \mathcal{A}(e_3) = \alpha_3,$$
    find the matrix representation of $\mathcal{A}$ with respect to the basis α.

38. For $f(x) = a_n x^n + a_{n-1}x^{n-1} + \cdots + a_1 x + a_0 \in \mathbb{R}[x]$, define
$$\mathcal{S}(f(x)) = \frac{a_n}{n+1}x^{n+1} + \frac{a_{n-1}}{n}x^n + \cdots + \frac{a_1}{2}x^2 + a_0 x$$
and
$$\mathcal{T}(f(x)) = n a_n x^{n-1} + (n-1)a_{n-1}x^{n-2} + \cdots + a_1.$$
Show that (a) $\mathbb{R} \subseteq \mathrm{Ker}(\mathcal{T})$. (b) $\mathcal{TS}$ is the identity map. (c) $\mathcal{ST} \neq \mathcal{TS}$.

39. Define $\mathcal{P} : (x_1, \ldots, x_n) \mapsto (\varepsilon_1 x_1, \ldots, \varepsilon_n x_n)$, where each $\varepsilon_i = 0$ or 1. Show that $\mathcal{P}$ is a linear map on $\mathbb{C}^n$ and that $\mathcal{P}^2 = \mathcal{P}$. What is $\mathrm{Ker}(\mathcal{P})$?

40. Let $\mathcal{A}$ be a linear transformation from a vector space V of dimension n to a vector space W (of any dimension).

    (a) If $\{u_1, \ldots, u_k\}$ is a basis of $\mathrm{Ker}(\mathcal{A})$, show that there exist $u_{k+1}, \ldots, u_n \in V$ such that $\{u_1, \ldots, u_k, u_{k+1}, \ldots, u_n\}$ is a basis of V.
    (b) Show that $\{\mathcal{A}(u_{k+1}), \ldots, \mathcal{A}(u_n)\}$ is a basis of $\mathrm{Im}(\mathcal{A})$.
    (c) Show that $\dim V = \dim \mathrm{Im}(\mathcal{A}) + \dim \mathrm{Ker}(\mathcal{A})$.
    (d) Find an example for $V \neq \mathrm{Im}(\mathcal{A}) + \mathrm{Ker}(\mathcal{A})$.

41. Let $\mathcal{A}$ be a linear transformation on a finite-dimensional vector space V, and let W be a subspace of V. Denote

$$\mathcal{A}(W) = \{\mathcal{A}(w) : w \in W\}.$$

    Show that $\mathcal{A}(W)$ is a subspace of V. Furthermore, show that

$$\dim\big(\mathcal{A}(W)\big) + \dim\big(\mathrm{Ker}(\mathcal{A}) \cap W\big) = \dim W.$$

42. Let V be a vector space of dimension n over $\mathbb{C}$ and let $\{u_1, \ldots, u_n\}$ be a basis of V. For $x = x_1 u_1 + \cdots + x_n u_n \in V$, $x_1, \ldots x_n \in \mathbb{C}$, define

$$\mathcal{T}(x) = (x_1, \ldots, x_n) \in \mathbb{C}^n.$$

    Show that $\mathcal{T}$ is an *isomorphism*, that is, $\mathcal{T}$ is one-to-one, onto, and

$$\mathcal{T}(ax + by) = a\mathcal{T}(x) + b\mathcal{T}(y), \quad x, y \in V, \ a, b \in \mathbb{C}.$$

43. Let $\mathcal{A}$ be a linear operator on a vector space V of dimension n. Let

$$V_0 = \cup_{i=1}^{\infty} \mathrm{Ker}(\mathcal{A}^i), \quad V_1 = \cap_{i=1}^{\infty} \mathrm{Im}(\mathcal{A}^i).$$

    Show that V_0 and V_1 are invariant under $\mathcal{A}$ and that $V = V_0 \oplus V_1$.

44. Let V be a vector space of dimension n. A linear operator $\mathcal{L}$ on V is said to be *diagonalizable* if its matrix under some basis is diagonal. Show that $\mathcal{L}$ is diagonalizable if and only if there exists a basis of eigenvectors of $\mathcal{L}$ and if and only if V is the direct sum of the eigenspaces.

45. Prove the following statements (about non-existence of eigenvalues).

    (a) $\mathcal{S} : \left(\begin{smallmatrix} x \\ y \end{smallmatrix}\right) \mapsto \left(\begin{smallmatrix} y \\ -x \end{smallmatrix}\right)$ is a linear map on $\mathbb{R}^2$. $\mathcal{S}$ has no eigenvalues in $\mathbb{R}$.
    (b) $\mathcal{T} : (a_1, a_2, \ldots) \mapsto (0, a_1, a_2, \ldots)$ is a linear map on the vector space of infinite complex sequences. $\mathcal{T}$ has no eigenvalues in $\mathbb{C}$.

    Do (a) and (b) contradict the statement that every linear transformation on a finite-dimensional vector space has an eigenvalue in $\mathbb{C}$?

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1.4 Inner Product Spaces

An *inner product space* is a vector space with an inner product. The inner product of two vectors in the space, say u and v, is a scalar, often denoted with angle brackets as $\langle u, v\rangle$. Let V be a vector space over $\mathbb{F}$ ($= \mathbb{R}$ or $\mathbb{C}$).

The inner product on V satisfies, for all $u, v, w \in V$ and all $c \in \mathbb{F}$,

(1) $\langle u, u\rangle \geq 0$, and $\langle u, u\rangle = 0$ if and only if $u = 0$,
(2) $\langle u + v, w\rangle = \langle u, w\rangle + \langle v, w\rangle$,
(3) $\langle cu, v\rangle = c\langle u, v\rangle$, and
(4) $\overline{\langle u, v\rangle} = \langle v, u\rangle$.

Note: An inner product $\langle\ ,\ \rangle$ is in fact a function from the Cartesian product space (Problem 31, Sect. 1.1) $V \times V$ to $\mathbb{F}$ which assigns every ordered pair $(u, v) \in V \times V$ a scalar $\langle u, v\rangle \in \mathbb{F}$ satisfying the conditions (1)–(4).

An inner product space over $\mathbb{R}$ is usually called a *Euclidean space*, and an inner product space over $\mathbb{C}$ is also referred to as a *unitary space*.

Example 1.4.1 The following are a few exemplary inner product spaces.

(1) $\mathbb{R}^n$ is an inner product space (over $\mathbb{R}$) with the inner (dot $\cdot$) product

$$\langle x, y\rangle = x \cdot y = x_1\, y_1 + \cdots + x_n\, y_n.$$

[Note: It is understood from the context that $z = (z_1, \ldots, z_n)$.]

(2) $\mathbb{C}^n$ is an inner product space (over $\mathbb{C}$) with the inner product

$$\langle x, y\rangle = y^* x = \overline{y_1}\, x_1 + \cdots + \overline{y_n}\, x_n.$$

In particular, $x^* x \geq 0$, and equality holds if and only if $x = 0$.

(3) $\mathbb{M}_{m\times n}$ is an inner product space with the *Frobenius inner product*

$$\langle A, B\rangle = \operatorname{tr}(B^* A) = \sum_{i,j} a_{ij}\, \overline{b_{ij}}, \quad \text{where } A = (a_{ij}),\ B = (b_{ij}).$$

In particular, $\operatorname{tr}(A^* A) \geq 0$, and equality holds if and only if $A = 0$.

Is $\langle A, B\rangle' = \operatorname{tr}(AB)$ or $\operatorname{tr}(A^* B)$ an inner product of $\mathbb{M}_n$ over $\mathbb{C}$?

(4) $C[a, b]$, the real-valued continuous functions on the interval $[a, b]$, is an inner product space (over $\mathbb{R}$) with respect to the inner product

$$\langle f,\, g\rangle = \int_a^b f(t) g(t) dt. \ \blacksquare$$

Theorem 1.4.1 (Cauchy-Schwarz Inequality) *Let V be an inner product space with the inner product $\langle\ ,\ \rangle$. Then for all $x, y \in V$,*

$$|\langle x, y\rangle|^2 \leq \langle x, x\rangle\, \langle y, y\rangle. \tag{1.29}$$

Equality holds if and only if x and y are linearly dependent.

The Cauchy-Schwarz inequality for an inner product space is one of the most useful and important results in mathematics; it can be shown in a number of ways. Here is a simple proof. We may assume $x \neq 0$. Let

$$z = y - \frac{\langle y, x\rangle}{\langle x, x\rangle} x.$$

One checks that $\langle z, x\rangle = 0$. The desired inequality (1.29) follows from

$$0 \leq \langle z, z\rangle = \langle z, y\rangle = \langle y, y\rangle - \frac{|\langle x, y\rangle|^2}{\langle x, x\rangle}.$$

The inequality is an equality if and only if $z = 0$, i.e., y is a multiple of x.

Another elementary proof is to consider the quadratic form in real t

$$f(t) = \langle tx + cy, tx + cy\rangle = \langle x, x\rangle t^2 + 2|c|^2 t + \langle y, y\rangle |c|^2,\ x \neq 0,\ c = \langle x, y\rangle.$$

Since $f(t) \geq 0$ for all real t, the nonpositive discriminant Δ yields (1.29) via

$$\Delta = 4|c|^4 - 4\langle x, x\rangle\langle y, y\rangle |c|^2 \leq 0.$$

Inequality (1.29) is equivalent to $\begin{vmatrix} \langle x,x\rangle & \langle y,x\rangle \\ \langle x,y\rangle & \langle y,y\rangle \end{vmatrix} \geq 0$ (nonnegative determinant). A matrix proof is left as an exercise for the reader (Problem 21).

For a vector x in an inner product space, the positive square root of $\langle x, x\rangle$ is called the *length* or (*induced*) *norm* of the vector x and is denoted by

$$\|x\| = \sqrt{\langle x, x\rangle}. \tag{1.30}$$

Unit vectors have length (norm) 1. If x is a nonzero vector, then x divided by its length (via scaling or *normalization*), i.e., $\frac{1}{\|x\|}x$, is a unit vector.

The Cauchy-Schwarz inequality is rewritten in terms of norm as

$$|\langle x, y\rangle| \leq \|x\|\,\|y\|. \tag{1.31}$$

Equality holds if and only if x and y are linearly dependent.

Example 1.4.2 Let $c_k \in \mathbb{C}$ and write $c_k = e^{i\theta_k}|c_k|$, $\theta_k \in (-\pi, \pi]$, $k = 1, \ldots, n$. Set $x = (\sqrt{|c_1|}, \ldots, \sqrt{|c_n|})$ and $y = (\sqrt{|c_1|}e^{i\theta_1}, \ldots, \sqrt{|c_n|}e^{i\theta_n})$. Then (1.31) shows that $|c_1 + \cdots + c_n| = |\langle x, y\rangle| \leq \|x\|\|y\| = |c_1| + \cdots + |c_n|$, where equality holds if and only if there exists $\theta \in \mathbb{R}$ such that $e^{i\theta}c_k = |c_k| \geq 0$ for all k. ∎

Example 1.4.3 $\mathbb{M}_{m\times n}$ is an inner product space with respect to the Frobenius inner product $\langle A, B\rangle = \operatorname{tr}(B^*A)$. For $A = (a_{ij}) \in \mathbb{M}_{m\times n}$, the norm of A derived from the inner product (called the *Frobenius norm* or *2-norm*) is

$$\|A\|_F = \sqrt{\operatorname{tr}(A^*A)} = \Big(\sum_{i,\,j} |a_{ij}|^2\Big)^{\frac{1}{2}}. \tag{1.32}$$

The Cauchy-Schwarz inequality (for $\mathbb{M}_{m\times n}$ as an inner product space) implies

$$|\operatorname{tr}(B^*A)|^2 \leq \operatorname{tr}(A^*A)\operatorname{tr}(B^*B).$$ ∎

Theorem 1.4.2 *Let V be an inner product space over $\mathbb{C}$ (or $\mathbb{R}$). The norm $\|\cdot\|$ on V derived from* (1.30) *is a function from V to $[0,\infty)$ that satisfies, for all $x, y \in V$ and $c \in \mathbb{C}$ (or $\mathbb{R}$),*

(i) $\|x\| > 0$ *if* $x \neq 0$, (ii) $\|cx\| = |c|\,\|x\|$, and (iii) $\|x+y\| \leq \|x\| + \|y\|$.

The last inequality is known as the *triangle inequality*, which is derived via the Cauchy-Schwarz inequality by showing $\|x+y\|^2 \leq (\|x\| + \|y\|)^2$:

$$\begin{aligned} \|x+y\|^2 &= \langle (x+y), (x+y) \rangle \\ &= \langle x, x \rangle + \langle x, y \rangle + \langle y, x \rangle + \langle y, y \rangle \\ &= \|x\|^2 + 2\,\Re\langle x, y \rangle + \|y\|^2 \\ &\leq \|x\|^2 + 2|\langle x, y \rangle| + \|y\|^2 \\ &\leq \|x\|^2 + 2\|x\|\,\|y\| + \|y\|^2 \\ &= (\|x\| + \|y\|)^2. \end{aligned}$$

Note: The norm (1.30) is derived from an inner product $\langle \cdot, \cdot \rangle$. In general, a vector space is said to be a *normed space* if it is equipped with a function $\|\cdot\|$ (called norm) satisfying (i), (ii), and (iii) in Theorem 1.4.2. A norm of a normed space need not be derived from an inner product (Problem 42).

Example 1.4.4 Consider $\mathbb{R}^2$. For $x = (x_1, x_2) \in \mathbb{R}^2$, define $\|x\|_1 = |x_1| + |x_2|$, $\|x\|_2 = \sqrt{x_1^2 + x_2^2}$, and $\|x\|_\infty = \max\{|x_1|, |x_2|\}$. They are all norms on $\mathbb{R}^2$. $\|\cdot\|_\infty$ (called ∞- or *sup-norm*) is not induced by an inner product on $\mathbb{R}^2$. ∎

Two important properties about norm of a normed space follow: norm is a continuous function and norms are equivalent (for finite-dimensional spaces).

Theorem 1.4.3 *Let V be a normed space over $\mathbb{C}$ (or $\mathbb{R}$) with norm $\|\cdot\|$. Then*

(1) $|\,\|x\| - \|y\|\,| \leq \|x-y\|$ *for all $x, y \in V$.* (Norm continuity)

(2) *If x_k is convergent to x in V (with respect to the norm), that is, $\|x_k - x\| \to 0$ (as $k \to \infty$), then $\|x_k\|$ is convergent to $\|x\|$ (as $k \to \infty$).*

(3) *If V is finite-dimensional, then any two norms on V are equivalent. Equivalently, if $\|\cdot\|_a$ is also a norm on V, then constants c_1 and c_2 exist such that $c_1\|x\| \leq \|x\|_a \leq c_2\|x\|$ for all $x \in V$.* (Norm equivalence)

Proof. (1) $x = (x-y)+y$ implies $\|x\| \leq \|x-y\| + \|y\|$, or $\|x\| - \|y\| \leq \|x-y\|$. Similarly, $\|y\| - \|x\| \leq \|x-y\|$. Thus, $|\,\|x\| - \|y\|\,| \leq \|x-y\|$ for all $x, y \in V$.

This says that norm is a continuous function at each $x \in V$, that is, for any given $\varepsilon > 0$, there exists $\delta(= \varepsilon) > 0$ such that $\|x-y\| < \delta \;\Rightarrow\; |\,\|x\| - \|y\|\,| < \varepsilon$.

(2) By (1), $|\,\|x_k\| - \|x\|\,| \leq \|x_k - x\|$. Thus, $\|x_k\|$ is convergent to $\|x\|$.

(3) We present a proof for $V = \mathbb{C}^n$ (as our V is usually $\mathbb{C}^n$ or $\mathbb{M}_{m\times n}$).

[Note: The proof for a general V is essentially the same, but requires more background on topology, which is beyond our scope; it's omitted here.]

Since $\|\cdot\|$ is a continuous function on $\mathbb{C}^n$, the set $S = \{x \in \mathbb{C}^n : \|x\| = 1\}$ (called a *sphere*) is a compact set (a closed and bounded set). Because $\|\cdot\|_a$ is a continuous function on $\mathbb{C}^n$, it attains its minimum and maximum values on S by the extreme value theorem. Let $c_1 = \min_{u\in S} \|u\|_a$ and $c_2 = \max_{u\in S} \|u\|_a$.

If $x = 0$, then $c_1\|x\| \le \|x\|_a \le c_2\|x\|$. Let $x \ne 0$, then $\frac{1}{\|x\|}x \in S$. So

$$c_1 \le \|\tfrac{1}{\|x\|}x\|_a \le c_2 \quad \text{or} \quad c_1\|x\| \le \|x\|_a \le c_2\|x\|. \blacksquare$$

Note: Exchanging the two norms in the above theorem, we see that there exist constants d_1 and d_2 such that $d_1\|x\|_a \le \|x\| \le d_2\|x\|_a$ for all $x \in V$.

Let V be an inner product space of dimension n, and let $\{u_1, \ldots, u_n\}$ be a basis of V. For $u = x_1u_1 + \cdots + x_nu_n$ and $v = y_1u_1 + \cdots + y_nu_n$ in V,

$$\langle u, v\rangle = \Big\langle \sum_i x_iu_i, \sum_j y_ju_j \Big\rangle = \sum_{i,j} x_i\overline{y_j}\langle u_i, u_j\rangle = x^T\big(\langle u_i, u_j\rangle\big)\bar{y}, \qquad (1.33)$$

where $x = (x_1, \ldots, x_n)^T$, $y = (y_1, \ldots, y_n)^T \in \mathbb{C}^n$, which equals y^*Gx, where

$$G = \begin{pmatrix} \langle u_1, u_1\rangle & \cdots & \langle u_n, u_1\rangle \\ \vdots & \ddots & \vdots \\ \langle u_1, u_n\rangle & \cdots & \langle u_n, u_n\rangle \end{pmatrix} \qquad (1.34)$$

is the *Gram matrix* of $u_1, \ldots, u_n$. Note that the (i, j)-entry of G is $\langle u_j, u_i\rangle$ (or $u_i^*u_j = \langle u_j, u_i\rangle$ in the case of $\mathbb{C}^n$ with the standard inner product).

A basis whose Gram matrix is the identity matrix is of great interest.

In an inner product space, a unit vector is a vector having length (norm) 1. If u is a nonzero vector, then $\frac{1}{\|u\|}u$ is a unit vector. Vectors u and v are said to be *orthogonal* or *perpendicular* if $\langle u, v\rangle = 0$. An *orthogonal set* is a set in which any two vectors are orthogonal. An orthogonal set of unit vectors is called *orthonormal set*. An *orthonormal basis* of an inner product vector space is an orthonormal set that spans the whole space.

Example 1.4.5 Unit vectors, orthogonal, and orthonormal sets.

(a) Any vector $(a, b) \in \mathbb{R}^2$, where $a^2 + b^2 = 1$, is a unit vector of $\mathbb{R}^2$.

(b) $\{u_1, u_2\}$ is an orthonormal basis of $\mathbb{R}^2$, where $u_1 = (1, 0), u_2 = (0, -1)$; so is $\{v_1, v_2\}$, where $v_1 = \big(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\big)$, $v_2 = \big(\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}}\big)$.

(c) The standard basis $\{e_1, \ldots, e_n\}$ is an orthonormal basis of $\mathbb{R}^n$ (and $\mathbb{C}^n$).

(d) Vectors $u_1, \ldots, u_m \in \mathbb{C}^n$ are orthonormal if and only if $u_i^*u_j = \delta_{ij}$, where δ_{ij} is the *Kronecker delta*, i.e., $\delta_{ij} = 0$ if $i \ne j$; $\delta_{ij} = 1$ if $i = j$.

(e) Orthogonal vectors are linearly independent.

(f) The column (row) vectors of a unitary matrix are orthonormal.

(g) A set of n vectors in an inner product space is orthonormal if and only if the Gram matrix of the vectors is the identity matrix I_n. $\blacksquare$

Through an orthonormal basis we may "identify" an inner product space of dimension n with $\mathbb{C}^n$ (or $\mathbb{R}^n$). For instance, if $\{u_1, \ldots, u_n\}$ is an orthonormal basis of the inner product space V, then the Gram matrix in (1.34) is I_n, and $\langle u, v\rangle$ for V is (equal to) the same as $\langle x, y\rangle$ for $\mathbb{C}^n$. That is, (1.33) reveals

$$\langle u, v\rangle = y^* G x = y^* x = \overline{y_1}\, x_1 + \cdots + \overline{y_n}\, x_n = \langle x, y\rangle. \tag{1.35}$$

Note: $\langle x, y\rangle$ on the right-hand side is the usual inner product of $\mathbb{C}^n$ or $\mathbb{R}^n$. The coordinates of x and y are relative to the orthonormal basis $e_1, \ldots, e_n$. Moreover, $x_i = \langle u, u_i\rangle = \langle x, e_i\rangle$, $y_i = \langle v, u_i\rangle = \langle y, e_i\rangle$ for $i = 1, \ldots, n$, and

$$\|u\| = \|x\| = \sqrt{\langle x, x\rangle} = \left(|x_1|^2 + \cdots + |x_n|^2\right)^{\frac{1}{2}}. \tag{1.36}$$

The norm in (1.36) is called the *Euclidean norm* or ℓ_2*-norm* of $x \in \mathbb{C}^n$.

Let $\mathcal{A}$ be a linear operator on the inner product space V. An important linear operator associated to $\mathcal{A}$ is its *adjoint* $\mathcal{A}^*$ (relative to the inner product): $\langle \mathcal{A}(u), v\rangle = \langle u, \mathcal{A}^*(v)\rangle$ for all $u, v \in V$. Adjoint can be obtained in different ways (such as by the Riesz representation theorem for more general setting). We introduce it by matrix representations.

Let A be the matrix of $\mathcal{A}$ with respect to orthonormal basis $\{u_1, \ldots, u_n\}$, i.e., $\mathcal{A}(u_1, \ldots, u_n) = (u_1, \ldots, u_n)A$. The coordinates of u, $\mathcal{A}(u)$, and v with respect to $\{u_1, \ldots, u_n\}$ are x, Ax, and y, respectively. By (1.35), we have

$$\langle \mathcal{A}(u), v\rangle = \langle Ax, y\rangle. \tag{1.37}$$

Since $y^*(Ax) = (A^*y)^*x$, we have (for the usual inner product of $\mathbb{C}^n$)

$$\langle Ax, y\rangle = \langle x, A^*y\rangle, \quad x,\, y \in \mathbb{C}^n. \tag{1.38}$$

The matrix A^* (the conjugate transpose of matrix A) defines (via the orthonormal basis) a unique linear map $\mathcal{A}^*$ by $\mathcal{A}^*(u_1, \ldots, u_n) = (u_1, \ldots, u_n)A^*$. $\mathcal{A}^*$ is called the *Hermitian adjoint*, or simply *adjoint*, of $\mathcal{A}$, and it satisfies

$$\langle \mathcal{A}(u), v\rangle = \langle u, \mathcal{A}^*(v)\rangle, \quad u,\, v \in V. \tag{1.39}$$

Example 1.4.6 For $\mathbb{C}^2$ (with the usual inner product), define $\mathcal{A} : \left(\begin{smallmatrix} x \\ y \end{smallmatrix}\right) \mapsto \left(\begin{smallmatrix} x+iy \\ 2y-x \end{smallmatrix}\right) = \left(\begin{smallmatrix} 1 & i \\ -1 & 2 \end{smallmatrix}\right)\left(\begin{smallmatrix} x \\ y \end{smallmatrix}\right)$. Then $\mathcal{A}^* : \left(\begin{smallmatrix} x \\ y \end{smallmatrix}\right) \mapsto \left(\begin{smallmatrix} x-y \\ -ix+2y \end{smallmatrix}\right) = \left(\begin{smallmatrix} 1 & -1 \\ -i & 2 \end{smallmatrix}\right)\left(\begin{smallmatrix} x \\ y \end{smallmatrix}\right)$. ∎

A linear transformation preserves linearity of a vector space. If a vector space V is equipped with an inner product, what are the linear transformations that preserve the inner product? In other words, for what $\mathcal{A}$ does

$$\langle \mathcal{A}(u), \mathcal{A}(v)\rangle = \langle u, v\rangle, \quad u,\, v \in V? \tag{1.40}$$

First of all, such a map preserves the length of a vector, namely,

$$\|\mathcal{A}(v)\| = \sqrt{\langle \mathcal{A}(v), \mathcal{A}(v)\rangle} = \sqrt{\langle v, v\rangle} = \|v\|, \quad v \in V. \tag{1.41}$$

Second, because $\langle \mathcal{A}(u), \mathcal{A}(v)\rangle = \langle Ax, Ay\rangle$ and $\langle u, v\rangle = \langle x, y\rangle$, we see that

$$\langle Ax, Ay\rangle = \langle A^*Ax, y\rangle = \langle x, y\rangle, \quad x,\, y \in \mathbb{C}^n, \tag{1.42}$$

implying $A^*A = I_n$, that is, A is a unitary matrix.

The transformation defined by (1.40) is known as a *unitary transformation* (or *orthogonal transformation* if V is a real vector space).

With an orthonormal basis, we reduce an inner product space of finite dimension to $\mathbb{C}^n$ (or $\mathbb{R}^n$). Does every inner product space (of finite dimension) have an orthonormal basis? The answer is affirmative.

Let S be a nonempty subset of an inner product space V. Denote by $S^\perp$ the set of the vectors in V that are orthogonal to all vectors in S, i.e.,

$$S^\perp = \{v \in V : \langle v, s\rangle = 0 \ \text{ for all } s \in S\}.$$

Obviously, $\{0\}^\perp = V$ and $V^\perp = \{0\}$. For any (nonempty) subset S of V, $S^\perp$ is always a subspace of V. $S^\perp$ is called the *orthogonal complement* of S. If S contains only one element, say x, we simply use $x^\perp$ for $S^\perp$. For two subsets S_1 and S_2, if $\langle x, y\rangle = 0$ for all $x \in S_1$ and $y \in S_2$, we write $S_1 \perp S_2$.

Let V be an inner product space of finite dimension and let S be a subspace of V. If $\dim V = n$ and $\dim S = k$, then $\dim S^\perp = n - k$, and

$$V = S \oplus S^\perp. \tag{1.43}$$

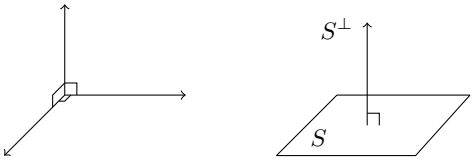


Fig. 1.6. Orthogonality

Example 1.4.7 We view an $n \times n$ matrix A as a linear map on $\mathbb{C}^n$ defined by $x \mapsto Ax$ with $\mathrm{Im}(A) = \{Ax : x \in \mathbb{C}^n\}$ and $\mathrm{Ker}(A) = \{x \in \mathbb{C}^n : Ax = 0\}$. By (1.38), we have $\mathrm{Ker}(A^*) = (\mathrm{Im}(A))^\perp$ (and $\mathrm{Im}(A^*) = (\mathrm{Ker}(A))^\perp$). Thus,

$$\mathbb{C}^n = \mathrm{Im}(A) \oplus \mathrm{Ker}(A^*). \ \blacksquare \tag{1.44}$$

Example 1.4.8 Orthogonal projection.

(1) Let $u = (-1, 2, 0), v = (-2, 1, 1) \in \mathbb{R}^3$. We can find all vectors that are orthogonal (perpendicular) to u and v. Let $w = (x, y, z)$. Then $w \perp u$ implies $-x + 2y = 0$; $w \perp v$ yields $-2x + y + z = 0$. Thus, $x = 2y$ and $z = 3y$. So, $w = y(2, 1, 3)$ is orthogonal to both u and v, that is,

$$\{u, v\}^\perp = (\mathrm{Span}\{u, v\})^\perp = \{k(2, 1, 3) : k \in \mathbb{R}\}.$$

Let $w = (2, 1, 3)$. Then w is a vector orthogonal to u and v.

(2) The vectors u, v, w in (1) are linearly independent; they constitute a basis for $\mathbb{R}^3$. However, the basis $\{u, v, w\}$ is not orthonormal (yet).

Let $\gamma = \frac{1}{\sqrt{14}}(2, 1, 3)$. Then γ is a unit vector orthogonal to u and v.

To convert the basis to an orthonormal basis, we can find a unit vector in the plane spanned by u and v that is orthogonal to v (or u). This can be done by an operation called *orthogonal projection* of u onto v:

$$\mathrm{Proj}_v u = \frac{\langle u, v\rangle}{\langle v, v\rangle} v \ \text{ (which is on the same line as } v).$$

Let $\tilde{u} = u - \mathrm{Proj}_v u = (1, 4, -2)$. Then $\tilde{u} \perp v$. Normalize $\tilde{u}$ and v and let

$$\alpha = \frac{1}{\|\tilde{u}\|}\tilde{u} = \frac{1}{\sqrt{21}}(1, 4, -2), \quad \beta = \frac{1}{\|v\|}v = \frac{1}{\sqrt{6}}(-2, 1, 1).$$

It follows that $\{\alpha, \beta, \gamma\}$ is an orthonormal basis for $\mathbb{R}^3$. ∎

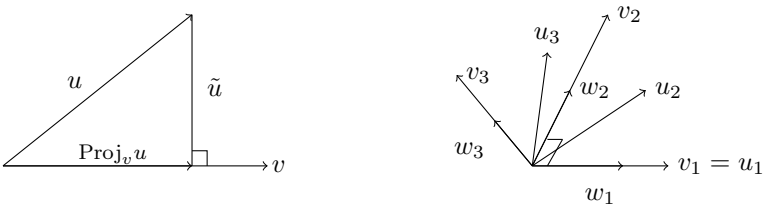


Fig. 1.7. Orthogonal projection and Gram–Schmidt process

As illustrated above, given two linearly independent vectors, we can construct two orthonormal vectors via orthogonal projection and normalization. For more linearly independent vectors, say $u_1, u_2, \ldots, u_n$, we can apply the *Gram-Schmidt process* to produce mutually orthogonal vectors $v_1, v_2, \ldots, v_n$ through projections, then by normalization to obtain orthonormal vectors $w_1, w_2, \ldots, w_n$. The Gram-Schmidt process goes as follows.

$$\begin{array}{ll}
v_1 = u_1 & w_1 = \frac{1}{\|v_1\|}v_1 \\
v_2 = u_2 - \frac{\langle u_2, v_1\rangle}{\langle v_1, v_1\rangle}v_1 & w_2 = \frac{1}{\|v_2\|}v_2 \\
v_3 = u_3 - \frac{\langle u_3, v_1\rangle}{\langle v_1, v_1\rangle}v_1 - \frac{\langle u_3, v_2\rangle}{\langle v_2, v_2\rangle}v_2 & w_3 = \frac{1}{\|v_3\|}v_3 \\
\cdots\cdots\cdots\cdots\cdots\cdots\cdots\cdots & \cdots\cdots\cdots\cdots \\
v_n = u_n - \frac{\langle u_n, v_1\rangle}{\langle v_1, v_1\rangle}v_1 - \frac{\langle u_n, v_2\rangle}{\langle v_2, v_2\rangle}v_2 - \cdots - \frac{\langle u_n, v_{n-1}\rangle}{\langle v_{n-1}, v_{n-1}\rangle}v_{n-1} & w_n = \frac{1}{\|v_n\|}v_n.
\end{array}$$

Note that (the given) $u_1, u_2, \ldots, u_n$ are linearly independent, $v_1, v_2, \ldots, v_n$ are orthogonal, and $w_1, w_2, \ldots, w_n$ are orthonormal. Moreover, for each k,

$$\mathrm{Span}\{u_1, u_2, \ldots, u_k\} = \mathrm{Span}\{v_1, v_2, \ldots, v_k\} = \mathrm{Span}\{w_1, w_2, \ldots, w_k\}.$$

Theorem 1.4.4 (Gram-Schmidt QR Form) *Let A be an $m \times n$ matrix. If the columns of A are linearly independent, i.e.,* $\operatorname{rank}(A) = n$ *(full column rank), then A can be factored as $A = QR$, where Q is an $m \times n$ matrix whose columns form an orthonormal basis for the column space of A, and R is an $n \times n$ upper-triangular matrix with positive entries on the diagonal. Moreover, If A is real, then both Q and R can be chosen to be real; If $m = n$, then Q is unitary (or real orthogonal if A is real).*

Proof. Write $A = (u_1, u_2, \ldots, u_n)$ with the linearly independent columns $u_1, u_2, \ldots, u_n$. The Gram-Schmidt process in the above discussion gives

$$\begin{aligned}
(u_1, u_2, \ldots, u_n) &= (v_1, v_2, \ldots, v_n) \begin{pmatrix} 1 & s_{21} & s_{31} & \ldots & s_{n1} \\ 0 & 1 & s_{32} & \ldots & s_{n2} \\ 0 & 0 & 1 & \ldots & s_{n3} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \ldots & 1 \end{pmatrix} \\
&\qquad \text{(in which } s_{ij} = \tfrac{\langle u_i, v_j \rangle}{\langle v_j, v_j \rangle},\ i > j) \\
&= (w_1, w_2, \ldots, w_n) \begin{pmatrix} \|v_1\| & r_{21} & r_{31} & \ldots & r_{n1} \\ 0 & \|v_2\| & r_{32} & \ldots & r_{n2} \\ 0 & 0 & \|v_3\| & \ldots & r_{n3} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \ldots & \|v_n\| \end{pmatrix} \\
&\qquad \text{(in which } r_{ij} = \tfrac{\langle u_i, v_j \rangle}{\langle v_j, v_j \rangle} \|v_j\|,\ i > j) \\
&= QR,
\end{aligned}$$

where Q is the $m \times n$ matrix with orthonormal columns $w_1, w_2, \ldots, w_n$, and $\operatorname{Col}(A) = \operatorname{Span}\{u_1, u_2, \ldots, u_n\} = \operatorname{Span}\{w_1, w_2, \ldots, w_n\}$. R is an $n \times n$ upper-triangular matrix with positive numbers on its diagonal. ∎

Every finite-dimensional vector space has a basis. The Gram-Schmidt process ensures that every inner product space has an orthonormal basis.

As we saw in the first section, a set of linearly independent vectors of a vector space of a finite dimension can be extended to a basis for the vector space. Likewise, a set of orthogonal vectors of an inner product space of a finite dimension can be extended to an orthogonal basis of the space. The same is true for a set of orthonormal vectors. Consider $\mathbb{R}^3$. Let u_1 be a unit vector in $\mathbb{R}^3$. Take a unit vector u_2 in $u_1^{\perp}$. Then u_1 and u_2 are orthonormal. $\operatorname{Span}\{u_1, u_2\}$ has dimension 2; its orthogonal complement in $\mathbb{R}^3$ has dimension 1. Let u_3 be a unit vector in $(\operatorname{Span}\{u_1, u_2\})^{\perp}$ (equivalently, $\langle u_1, u_3 \rangle = 0$ and $\langle u_2, u_3 \rangle = 0$). Then u_1, u_2, u_3 are orthonormal and form an orthonormal basis for $\mathbb{R}^3$. This idea works for $\mathbb{R}^n$ as well as $\mathbb{C}^n$. We summarize this as a theorem for $\mathbb{C}^n$, which will be freely and frequently used in the book.

Theorem 1.4.5 (Nonsingular Matrix Extension Theorem) *If $u_1, \ldots, u_k$ are k linearly independent column vectors in $\mathbb{C}^n$, $1 \le k < n$, then there exist $n-k$ column vectors $u_{k+1}, \ldots, u_n$ in $\mathbb{C}^n$ such that the matrix*

$$P = (u_1, \ldots, u_k, u_{k+1}, \ldots, u_n)$$

is nonsingular. Moreover, if $u_1, \ldots, u_k$ are orthonormal vectors in $\mathbb{C}^n$, then there exist $n-k$ orthonormal vectors $u_{k+1}, \ldots, u_n$ in $\mathbb{C}^n$ such that the matrix

$$U = (u_1, \ldots, u_k, u_{k+1}, \ldots, u_n)$$

is unitary. In particular, for any unit vector u in $\mathbb{C}^n$, there exist orthonormal vectors $u_2, \ldots, u_n$ in $\mathbb{C}^n$ such that $U = (u_1, u_2, \ldots, u_n)$ is unitary.

The above statements hold true for the real case, that is, if $u_1, \ldots, u_k$ are real (in $\mathbb{R}^n$), then $u_{k+1}, \ldots, u_n$ can be selected to be real (in $\mathbb{R}^n$). The statements can be similarly stated for row vectors.

We end this section by presenting an inequality of the angles between nonzero vectors in a Euclidean space. This inequality is intuitive and obvious in $\mathbb{R}^2$ (the xy-plane) and in $\mathbb{R}^3$ (the 3D space). The good part of this theorem is the idea in its proof of reducing the problem to $\mathbb{R}^2$ or $\mathbb{R}^3$.

Let V be an inner product space over $\mathbb{R}$. For any two nonzero vectors u and v, define the (measure of) *angle* between u and v by

$$\Theta_{u,\,v} = \cos^{-1} \frac{\langle u,\, v\rangle}{\|u\|\,\|v\|} \in [0, \pi].$$

[Note: For unitary spaces, define $\Theta_{u,\,v}$ with $\langle u,\, v\rangle$ replaced by $\Re(\langle u,\, v\rangle)$.]

Theorem 1.4.6 *For any nonzero vectors u, v, and w in a Euclidean space,*

$$\Theta_{u,\,v} \le \Theta_{u,\,w} + \Theta_{w,\,v}. \tag{1.45}$$

Equality occurs if and only if $w = au + bv$ for some $a, b \ge 0$.

Proof. Because the inequality only involves the vectors u, v, and w, we may focus on the subspace $\mathrm{Span}\{u,\, v,\, w\}$, which has dimension at most 3. We can further choose an orthonormal basis (a unit vector in the case of dimension one) for this subspace. Let u, v, and w have coordinate vectors α, β, and γ with respect to the basis, respectively. Then the inner product, say $\langle u, v\rangle$, is equal to $\langle \alpha, \beta\rangle$; see (1.35) and (1.36). Thus, the inequality (1.45) holds if and only if it holds for real vectors α, β, and γ in $\mathbb{R}$, $\mathbb{R}^2$, or $\mathbb{R}^3$ depending on the dimension of $\mathrm{Span}\{u,\, v,\, w\}$ being 1, 2, or 3, respectively. For $\mathbb{R}$, the assertion is nearly trivial. For $\mathbb{R}^2$, draw the vectors α, β, and γ in the xy-plane and verify the inequality (1.45) by considering the angles between them. The proof is completed by a similar argument for $\mathbb{R}^3$. ∎

Note: We may assume that the vectors u, v, and w are unit. Related inequalities about unit vectors and angles are shown in Sect. 6.5, Chap. 6.

Problems

1. Let V be an inner product space (with $\langle \cdot\,, \cdot \rangle$) over $\mathbb{C}$. Show that

$$\langle x, cy \rangle = \bar{c}\langle x, y \rangle \text{ and } \langle x, y \rangle\langle y, x \rangle = |\langle x, y \rangle|^2 \text{ for all } x, y \in V,\ c \in \mathbb{C}.$$

2. Let u, v, w be vectors in an inner product space. Prove or disprove that

   (a) $|\langle u, v \rangle| \leq \langle u, u \rangle\langle v, v \rangle$.
   (b) $|\langle u, w \rangle| \leq |\langle u, v \rangle| + |\langle v, w \rangle|$.

3. Find all vectors in $\mathbb{R}^2$ (with the usual inner product) that are orthogonal to $(1, 1) \in \mathbb{R}^2$. Is $(1, 1)$ a unit vector? Is $(\frac{1}{2}, \frac{1}{2})$ a unit vector?

4. For $x = (x_1, x_2)$, $y = (y_1, y_2) \in \mathbb{R}^2$, define $\langle x, y \rangle_2 = x_1 y_1 + 2x_2 y_2$. Show that $\langle\ ,\ \rangle_2$ is an inner product on $\mathbb{R}^2$. Are $(1, 0)$ and $(0, 1)$ unit vectors?

5. Show that $\mathbb{R}_3[x]$ is an inner product space with respect to

$$\langle f, g \rangle = \int_0^1 f(x)g(x)dx, \quad f, g \in \mathbb{R}_3[x].$$

   Find $1^\perp$, $x^\perp$, $(1 + x)^\perp$, and $\{1, x\}^\perp$.

6. What are the unit vectors in the space $\mathbb{M}_{m\times n}(\mathbb{C})$ with the Frobenius inner product? Find $\|E\|_F$, where E is the all-ones $m \times n$ matrix.

7. Let V be an inner product space over $\mathbb{R}$ or $\mathbb{C}$ and $x, y \in V$. Show that

$$\langle x, y \rangle = 0 \quad \Rightarrow \quad \|x + y\|^2 = \|x\|^2 + \|y\|^2.$$

   Show that the converse is true if V is over $\mathbb{R}$ but not true over $\mathbb{C}$.

8. Let V be an inner product space over $\mathbb{R}$ and $x, y \in V$. Show that

$$\|x\| = \|y\| \quad \Leftrightarrow \quad \langle x + y,\, x - y \rangle = 0.$$

   Does this remain true if the inner product space V is over $\mathbb{C}$?

9. Let V be an inner product space over $\mathbb{C}$ and $x, y \in V$. Show that

$$\langle x, y \rangle = \|x\|\|y\| \quad \Leftrightarrow \quad \|x\|\, y = \|y\|\, x.$$

10. (*Parallelogram law*) Let V be an inner product space. Show that

$$\|x - y\|^2 + \|x + y\|^2 = 2(\|x\|^2 + \|y\|^2) \text{ for all } x, y \in V.$$

11. Let $c_k \in \mathbb{C}$ and $p_k > 0$, $k = 1, \ldots, n$. Use the Cauchy-Schwarz inequality to show that $|p_1 c_1 + \cdots + p_n c_n| \leq p_1|c_1| + \cdots + p_n|c_n|$, where equality holds if and only if all c_k's have the same argument, in other words, there exists $\theta \in (-\pi, \pi]$ such that $e^{i\theta} c_k \geq 0$ for all k.

12. Is $[\ ,\ ]$ defined as $[x,\, y] = y^T x = \sum_i x_i y_i$ an inner product for $\mathbb{C}^n$?

13. For what diagonal $D \in \mathbb{M}_n$ is $[x,\, y] = y^* D x$ an inner product for $\mathbb{C}^n$?

14. Let $\{u_1, \ldots, u_n\}$ be a basis of an inner product space V and $x, y \in V$. Show that $\langle u_i, x\rangle = 0$ for $i = 1, \ldots, n$ if and only if $x = 0$, and that $\langle u_i, x\rangle = \langle u_i, y\rangle$ for $i = 1, \ldots, n$ if and only if $x = y$.

15. Let $G = (g_{ij})$ be the *Gram matrix* of vectors $v_1, \ldots, v_n$ in an inner product space, that is, $g_{ij} = \langle v_j, v_i\rangle$ for all i, j. Show that G is singular, i.e., $\det G = 0$, if and only if $v_1, \ldots, v_n$ are linearly dependent.

16. Let $A \in \mathbb{M}_n$. If $\operatorname{tr}(AX) = 0$ for all $X \in \mathbb{M}_n$, show that $A = 0$.

17. Let $A = (a_{ij}) \in \mathbb{M}_n$ and let $x = (x_1, \ldots, x_n)^T \in \mathbb{C}^n$. Show that

$$\langle Ax, x\rangle = x^* A x = \sum_{i,\,j} a_{ij} \overline{x_i} x_j$$

and that (with $e_1, \ldots, e_n$ for the standard basis vectors of $\mathbb{C}^n$)

$$\operatorname{tr} A = \sum_{i=1}^{n} a_{ii} = \sum_{i=1}^{n} \langle Ae_i, e_i\rangle.$$

18. Let $A \in \mathbb{M}_n$. Show that for any unit vector $x \in \mathbb{C}^n$ (in columns),

$$|x^* A x|^2 \le x^* A^* A x.$$

19. Let $A = (a_1, \ldots, a_n)$ and $B = (b_1, \ldots, b_n)$ be $m \times n$ matrices partitioned according to their columns. Show that $\langle A, B\rangle = \sum_{i=1}^{n} \langle a_i, b_i\rangle$.

20. Let $A = (a_{ij}) \in \mathbb{M}_{m\times n}$. Show that $\operatorname{tr}(A^*A) = \operatorname{tr}(AA^*) = \sum_{i,j} |a_{ij}|^2$, concluding that $\operatorname{tr}(A^*A) = 0$ if and only if $A = 0$.

21. Let A be a matrix. Use the fact that $\operatorname{tr}(A^*A) \ge 0$ with equality if and only if $A = 0$ to show the Cauchy-Schwarz inequality for $\mathbb{C}^n$ by taking $A = xy^* - yx^*$ (where x and y are column vectors in $\mathbb{C}^n$).

22. Let A be a square matrix. Show by examples that neither $\operatorname{tr}(A^*A) \le \operatorname{tr}(A^*)\operatorname{tr}(A)$ nor $\operatorname{tr}(A^*)\operatorname{tr}(A) \le \operatorname{tr}(A^*A)$ holds in general.

23. Let $A \in \mathbb{M}_{m\times n}$ and $B \in \mathbb{M}_{n\times m}$. Show that

$$|\operatorname{tr}(AB)| \le \big(\operatorname{tr}(A^*A)\operatorname{tr}(B^*B)\big)^{1/2} \le \tfrac{1}{2}\big(\operatorname{tr}(A^*A) + \operatorname{tr}(B^*B)\big).$$

24. Let A and B be n-square matrices. Show that

$$\langle Ax, x\rangle = \langle Bx, x\rangle \text{ for all } x \in \mathbb{C}^n \text{ implies } A = B.$$

Does it follow that $A = B$ if $\langle Ax, x\rangle = \langle Bx, x\rangle$ for all $x \in \mathbb{R}^n$?

25. Let A be an $m \times n$ matrix. Show that $(\operatorname{Row}(A))^\perp = \operatorname{Nul}(\overline{A})$ and that

$$(\operatorname{Col}(A))^\perp = \operatorname{Nul}(A^*), \quad \text{i.e., } (\operatorname{Im}(A))^\perp = \operatorname{Ker}(A^*).$$

26. Let A and B be $m \times n$ matrices. Show that

$$\mathrm{Im}(A) \perp \mathrm{Im}(B) \quad \Leftrightarrow \quad A^*B = 0.$$

27. Let A be an n-square matrix and denote $B = A^*$. Show that

$$\mathbb{C}^n = \mathrm{Im}(A^k) \oplus \mathrm{Ker}(B^k), \quad k = 1, 2, \ldots$$

and that $\mathbb{C}^n = \mathrm{Im}(A) \oplus \mathrm{Ker}(A) \Leftrightarrow \mathrm{rank}(A^2) = \mathrm{rank}(A)$.

28. Let θ_i's and λ_i's be $2n$ positive numbers, and $\sum_{i=1}^n \theta_i = 1$. Prove

(a) $1 \le \big(\sum_{i=1}^n \theta_i \lambda_i\big)\big(\sum_{i=1}^n \theta_i \lambda_i^{-1}\big)$.

(b) $\big(\sum_{i=1}^n \theta_i \lambda_i\big)^2 \le \sum_{i=1}^n \theta_i \lambda_i^2$.

(c) $\big(\sum_{i=1}^n \theta_i \lambda_i\big)^2 \le \big(\sum_{i=1}^n \theta_i^a \lambda_i^b\big)\big(\sum_{i=1}^n \theta_i^{2-a} \lambda_i^{2-b}\big)$, where $a, b \in \mathbb{R}$.

(d) $\big(\sum_{i=1}^n \lambda_i\big)\big(\sum_{i=1}^n \lambda_i^{-1}\big) \ge n^2$. When does equality hold?

29. Let P and Q be any two permutation matrices of order n. Show that

$$\langle I - P, I - Q \rangle \ge 0.$$

30. Let u be a nonzero vector in an inner product space V. If $v_1, v_2, \ldots, v_k$ are vectors in V such that (a) $\langle v_i, u \rangle > 0$ for all i and (b) $\langle v_i, v_j \rangle \le 0$ whenever $i \ne j$, show that $v_1, v_2, \ldots, v_k$ are linearly independent.

31. Let $\{u_1, \ldots, u_n\}$ be an orthonormal basis of an inner product space V. Show that $x_1, \ldots, x_m \in V$ are pairwise orthogonal if and only if

$$\sum_{k=1}^n \langle x_i, u_k \rangle \overline{\langle x_j, u_k \rangle} = 0, \quad i \ne j.$$

32. Let V be an inner product space of dimension n and let $\{u_1, \ldots, u_k\}$ be a set of orthonormal vectors in V. Show that $k \le n$ and that

$$\|x\|^2 \ge \sum_{i=1}^k |\langle x, u_i \rangle|^2, \quad x \in V.$$

33. Let $\{u_1, \ldots, u_n\}$ and $\{v_1, \ldots, v_n\}$ be orthonormal bases of an inner product space. Show that there is a unique unitary $U \in \mathbb{M}_n$ such that

$$(u_1, \ldots, u_n) = (v_1, \ldots, v_n)U.$$

Discuss and state explicitly for the cases of $\mathbb{C}^n$ and $\mathbb{R}^n$.

34. Let $\alpha = \{u_1, \ldots, u_n\}$ and $\beta = \{v_1, \ldots, v_n\}$ be two orthonormal bases of an inner product space V. Let $u, v \in V$. If the coordinates of u with respect to α and β are respectively x_α and x_β, and the coordinates of v with respect to α and β are respectively y_α and y_β, show that

$$\langle x_\alpha, y_\alpha \rangle = \langle x_\beta, y_\beta \rangle.$$

35. Let $\{u_1, \ldots, u_n\}$ and $\{v_1, \ldots, v_n\}$ be two bases of an inner product space. If A is the Gram matrix of $\{u_1, \ldots, u_n\}$ and B is the Gram matrix of $\{v_1, \ldots, v_n\}$, show that there exists a nonsingular matrix T such that $B = T^*AT$. (B is said to be *-*congruent* to A.)

36. Define the *angle* between the one-dimensional subspaces ℓ_x and ℓ_y spanned respectively by nonzero vectors x and y in an inner product space by $\theta_{x,y} = \cos^{-1} \frac{|\langle x, y\rangle|}{\|x\| \, \|y\|} \in [0, \frac{\pi}{2}]$. By $\ell_x \perp \ell_y$ we mean $x \perp y$. Show that $\ell_x \perp \ell_y \Leftrightarrow \theta_{x,y} = \frac{\pi}{2}$. Explain why the law of cosines fails.

37. Find the angle between $x = (1, 2, 3)$ and $y = (1, 0, -1)$ in $\mathbb{R}^3$, and the angle between the lines ℓ_x and ℓ_y (containing x and y, respectively).

38. Let u and v be unit vectors in a Euclidean space. Show that $\theta_{u,v} = \Theta_{u,v}$ if $\langle u, v\rangle \geq 0$ and that $\theta_{u,v} = \pi - \Theta_{u,v}$ if $\langle u, v\rangle < 0$.

39. Let u and v be unit vectors in a Euclidean space. Show that

$$\theta_{u,v} = \min_{|p|=1} \Theta_{pu,v} = \min_{|q|=1} \Theta_{u,\,qv} = \min_{|p|=|q|=1} \Theta_{pu,\,qv}.$$

40. Let u, v, and w be unit vectors in a Euclidean space. Show that

$$\theta_{u,v} \leq \theta_{u,w} + \theta_{w,v}.$$

41. Let V be an inner product space and let $\|\cdot\|$ be the induced norm, that is, $\|u\| = \sqrt{\langle u, u\rangle}$, $u \in V$. Show that for vectors x and y in V,

$$\|x + y\| = \|x\| + \|y\|$$

if and only if one of the vectors is a nonnegative multiple of the other.

42. Show by example that it is possible that $\|x + y\| = \|x\| + \|y\|$ for some linearly independent vectors x and y in a normed vector space, concluding that this norm is not induced (derived) from an inner product.

43. Consider $\mathbb{R}^2$. For $x = (x_1, x_2) \in \mathbb{R}^2$, define $\|x\|_a = \sqrt{x_1^2 + x_2^2}$, $\|x\|_b = |x_1| + |x_2|$, and $\|x\|_c = \max\{|x_1|, |x_2|\}$. Show that they are all norms on $\mathbb{R}^2$. Sketch the unit balls $S_t = \{x \in \mathbb{R}^2 : \|x\|_t \leq 1\}$ for $t = a, b, c$.

44. Let V be a vector space of dimension n and let $\{u_1, \ldots, u_n\}$ be a basis. For $x = x_1u_1 + \cdots + x_nu_n \in V$, define $\|x\|_1 = |x_1| + \cdots + |x_n|$, where $x_1, \ldots, x_n$ are scalars. (a) Show that $\|\cdot\|_1$ is a norm on V. (b) If $\|\cdot\|$ is also a norm on V, show that $\|x\| \leq C\|x\|_1$, where $C = \max_i\{\|u_i\|\}$.

45. Let $\|\cdot\|$ be a norm on $\mathbb{C}^n$ and let $x \in \mathbb{C}^n$. Show that $x \to 0$ (componentwise) if and only if $\|x\| \to 0$.

46. Let V_1 and V_2 be subspaces of a finite-dimensional inner product space. Show that

    (a) $(V_1^{\perp})^{\perp} = V_1$.
    (b) $V_1 \subseteq V_2 \Leftrightarrow V_2^{\perp} \subseteq V_1^{\perp}$.
    (c) $V_1 = V_2 \Leftrightarrow V_2^{\perp} = V_1^{\perp}$.
    (d) $(V_1 + V_2)^{\perp} = V_1^{\perp} \cap V_2^{\perp}$ and $(V_1 \cap V_2)^{\perp} = V_1^{\perp} + V_2^{\perp}$.

47. Let V_1 and V_2 be subspaces of a finite-dimensional inner product space. If $\dim V_1 > \dim V_2 > 0$, show that there is a subspace V_3 such that

$$V_3 \subsetneq V_1, \quad V_3 \perp V_2, \quad \dim V_3 \geq \dim V_1 - \dim V_2.$$

    Give a geometric explanation of this statement in $\mathbb{R}^3$.

48. Let $\binom{a}{b} \in \mathbb{C}^2$ and $r = \sqrt{|a|^2 + |b|^2}$. For $r > 0$, define $Q = \frac{1}{r}\begin{pmatrix} \bar{a} & \bar{b} \\ -b & a \end{pmatrix}$. Show that $Q \in \mathbb{M}_2$ is unitary and $Q\binom{a}{b} = \binom{r}{0}$. Generalize this for $\mathbb{C}^n$: Let $v = \begin{pmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{pmatrix} \in \mathbb{C}^n$. Show that there exists a unitary matrix $Q \in \mathbb{M}_n$ such that $Qv = Q\begin{pmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{pmatrix} = \begin{pmatrix} r \\ 0 \\ \vdots \\ 0 \end{pmatrix}$ with $r = \sqrt{|a_1|^2 + \cdots + |a_n|^2} = \|v\|$.

49. Define $\mathcal{A} : \begin{pmatrix} x \\ y \\ z \end{pmatrix} \mapsto \begin{pmatrix} x-z \\ iy \\ z-y \end{pmatrix} = \begin{pmatrix} 1 & 0 & -1 \\ 0 & i & 0 \\ 0 & -1 & 1 \end{pmatrix}\begin{pmatrix} x \\ y \\ z \end{pmatrix}$ on $\mathbb{C}^3$, where $i = \sqrt{-1}$. Find the adjoint of $\mathcal{A}$ (with respect to the usual inner product of $\mathbb{C}^3$).

50. Let V be a vector space of dimension n with a basis $\alpha = \{\alpha_1, \ldots, \alpha_n\}$. Let $\mathcal{A}$ be a linear transformation on V with the matrix representation A under the basis α. Show that $\mathcal{A}^2 = \mathcal{A}$ if and only if $A^2 = A$. In addition, if V is an inner product space and $\mathcal{A}$ is *self-adjoint*, i.e., $\mathcal{A} = \mathcal{A}^*$, meaning $\langle \mathcal{A}(x), y\rangle = \langle x, \mathcal{A}(y)\rangle$ for all $x, y \in V$, does it follow that $A = A^*$? What condition on the basis α is needed for $A = A^*$?

51. Let A be an n-square matrix. Show that for any $x, y \in \mathbb{C}^n$ (in columns),

$$4\langle Ax, y\rangle = \langle As, s\rangle - \langle At, t\rangle + i\langle Au, u\rangle - i\langle Av, v\rangle,$$

    where $s = x + y$, $t = x - y$, $u = x + iy$, $v = x - iy$, and $i = \sqrt{-1}$. Derive

$$4\langle x, y\rangle = \|x + y\|^2 - \|x - y\|^2 + i\|x + iy\|^2 - i\|x - iy\|^2.$$

52. For any nonzero vectors x and y in $\mathbb{C}^n$, show that the inequality holds:

$$\|x - y\| \geq \frac{1}{2}(\|x\| + \|y\|)\left\|\frac{1}{\|x\|}x - \frac{1}{\|y\|}y\right\|.$$

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CHAPTER 2

Partitioned Matrices, Ranks, and Eigenvalues

Introduction: We begin with partitioned (i.e., block) matrices and the elementary operations on the partitioned matrices. We discuss the determinants and present explicit forms of the inverses of 2×2 block matrices. We study the ranks of sum $A+B$ and product AB of matrices A and B (of appropriate sizes). We then give four different proofs for the (very important) theorem that the products AB and BA have the same nonzero eigenvalues. At the end we show the Geršgorin disc theorem for localizing eigenvalues and discuss the continuity technique that is often used in studying matrices.

2.1 Partitioned Matrices and Their Elementary Operations

The operations on partitioned (block) matrices resemble those for ordinary matrices of scalars. Referring to Sect. 1.2 of Chap. 1, if $A = (a_{ij})$ and $B = (b_{ij})$ are matrices of the same size, we add A and B by adding the corresponding entries a_{ij} and b_{ij}, that is, $A+B = (a_{ij}+b_{ij})$. If A and B are partitioned in blocks (i.e., submatrices): $A = (A_{ij})$ and $B = (B_{ij})$, where A_{ij} and B_{ij} have the same size for each pair of i, j, then $A+B = (A_{ij}+B_{ij})$.

For product AB (where A and B^T are required to have the same number of columns), the (i, j)-entry of AB is $\sum_t a_{it}b_{tj}$. For block matrices, the (i, j)-block of AB is $\sum_t A_{it}B_{tj}$, provided that A_{it} and B_{tj} can be multiplied.

We demonstrate the operations with 2×2 block partitions:

$$A = \begin{pmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{pmatrix}, \quad B = \begin{pmatrix} B_{11} & B_{12} \\ B_{21} & B_{22} \end{pmatrix}.$$

If A_{ij} and the corresponding block B_{ij} have the same size, $1 \le i, j \le 2$, then

$$A + B = \begin{pmatrix} A_{11} + B_{11} & A_{12} + B_{12} \\ A_{21} + B_{21} & A_{22} + B_{22} \end{pmatrix}.$$

If A_{it} can be multiplied by B_{tj} from the right, that is to say, the number of

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columns of A_{it} is the same as the number of rows of B_{tj}, $1 \le i, t, j \le 2$, then

$$AB = \begin{pmatrix} A_{11}B_{11} + A_{12}B_{21} & A_{11}B_{12} + A_{12}B_{22} \\ A_{21}B_{11} + A_{22}B_{21} & A_{21}B_{12} + A_{22}B_{22} \end{pmatrix}.$$

Note: When we add matrices, say A and B, a precondition for $A + B$ is that A and B have the same size; when we multiply matrices, a precondition for AB is that the number of columns of A is the same as the number of rows of B. (That is, A and B are compatible for the operations.) Likewise, when we add or multiply block matrices, we assume that the matrices are already *conformally* (preferred, or *conformably*, or *compatibly*, or *appropriately*) partitioned, and we say that the (partitioned) matrices have appropriate sizes. For simplicity, we do not always state the sizes of some (sub)matrices explicitly, or the sizes of the (sub)matrices can be easily referred from the context.

Example 2.1.1 Suppose that matrices A, B, C, D, X, and Y have appropriate sizes in each of the following identities. Direct verifications show that (conventionally, k, m, and n are positive integers unless otherwise stated)

(1) $A(B, C) = (AB, AC)$, $\begin{pmatrix} A \\ B \end{pmatrix} C = \begin{pmatrix} AC \\ BC \end{pmatrix}$, and $(A, B)^* = \begin{pmatrix} A^* \\ B^* \end{pmatrix}$.

(2) $(A, B)\begin{pmatrix} C \\ D \end{pmatrix} = AC + BD$ and $\begin{pmatrix} A \\ B \end{pmatrix}(C, D) = \begin{pmatrix} AC & AD \\ BC & BD \end{pmatrix}$.

(3) $\begin{pmatrix} A & B \\ C & D \end{pmatrix}^T = \begin{pmatrix} A^T & C^T \\ B^T & D^T \end{pmatrix}$ and $\begin{pmatrix} A & B \\ C & D \end{pmatrix}^* = \begin{pmatrix} A^* & C^* \\ B^* & D^* \end{pmatrix}$.

(4) $\begin{pmatrix} A & C \\ 0 & B \end{pmatrix}^k = \begin{pmatrix} A^k & \star \\ 0 & B^k \end{pmatrix}$ ($\star$ represents unspecified entries).

(5) $\begin{pmatrix} A & C \\ 0 & B \end{pmatrix}^{-1} = \begin{pmatrix} A^{-1} & \star \\ 0 & B^{-1} \end{pmatrix}$ if A and B are invertible.

(6) $\begin{pmatrix} I_m & X \\ 0 & I_n \end{pmatrix}^{-1} = \begin{pmatrix} I_m & -X \\ 0 & I_n \end{pmatrix}$ and $\begin{pmatrix} I_m & 0 \\ Y & I_n \end{pmatrix}^{-1} = \begin{pmatrix} I_m & 0 \\ -Y & I_n \end{pmatrix}$. ∎

Recall the elementary row and column operations (i)–(iii) for scalar matrices (Sect. 1.2, Chap. 1). For example, consider the 2×2 matrix

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix}, \quad a, b, c, d \in \mathbb{C}.$$

By an elementary row operation, say type (iii), adding the first row multiplied by a scalar x to the second row results in $\begin{pmatrix} a & b \\ c+xa & d+xb \end{pmatrix}$, that is,

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \to \begin{pmatrix} a & b \\ c + xa & d + xb \end{pmatrix}.$$

Writing this in a matrix identity, we have

$$\begin{pmatrix} 1 & 0 \\ x & 1 \end{pmatrix}\begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} a & b \\ c + xa & d + xb \end{pmatrix}.$$

Generalized elementary row (*column*) *operations* for block matrices are:

(I) Interchange two block rows (columns).

(II) Multiply a block row (column) from the left (right) by a nonsingular matrix of appropriate size.

(III) Multiply a block row (column) by a matrix from the left (right), then add it to another row (column, resp.).

For the case of 2×2 block matrices, say, a type (III) elementary row operation by adding the first row (in blocks) pre-multiplied by a matrix X (of size $n \times m$ if $A \in \mathbb{M}_{m\times p}, D \in \mathbb{M}_{n\times q}$) to the second row (in blocks) shows

$$\begin{pmatrix} A & B \\ C & D \end{pmatrix} \to \begin{pmatrix} A & B \\ C + XA & D + XB \end{pmatrix}.$$

Note that A is multiplied by X from the left (when row operations are performed, and X is on the right for column operations). One can verify that

$$\begin{pmatrix} I_m & 0 \\ X & I_n \end{pmatrix} \begin{pmatrix} A & B \\ C & D \end{pmatrix} = \begin{pmatrix} A & B \\ C + XA & D + XB \end{pmatrix}.$$

If $A \in \mathbb{M}_m$ is invertible, we set $X = -CA^{-1}$. Then $XA + C = 0$ and

$$\begin{pmatrix} I_m & 0 \\ X & I_n \end{pmatrix} \begin{pmatrix} A & B \\ C & D \end{pmatrix} = \begin{pmatrix} A & B \\ 0 & D - CA^{-1}B \end{pmatrix}.$$

Generalized elementary matrices are the block matrices that are obtained by applying a single elementary operation (I), (II), or (III) to the identity matrix. For example, the following are generalized elementary matrices of types (I) and (III) (with $X \in \mathbb{M}_{n\times m}$ and $Y \in \mathbb{M}_{m\times n}$), respectively,

$$\text{(I)}: \begin{pmatrix} 0 & I_m \\ I_n & 0 \end{pmatrix} \quad \text{and} \quad \text{(III)}: \begin{pmatrix} I_m & 0 \\ X & I_n \end{pmatrix}, \begin{pmatrix} I_m & Y \\ 0 & I_n \end{pmatrix}.$$

The above discussion can be extended for the partitioned matrices with more block rows and block columns, and it leads to the following result.

Theorem 2.1.1 *Let G be the generalized elementary matrix obtained by performing an elementary row* (*column*) *operation on I. If that same elementary row* (*column*) *operation is performed on a* (*suitably*) *partitioned matrix A, then the resulting matrix is given by the product GA* (*or AG, resp.*).

A generalized elementary matrix (type (I), (II), or (III)) is the product of elementary matrices (types (i), (ii), and (iii); see Sect. 1.2).

For example, let $X = (x_{ij})$ be an $m\times n$ matrix. We write $X = \sum_{i,j} x_{ij}E_{ij}$, where E_{ij} is the $m \times n$ matrix with (i, j)-entry equal to 1 and elsewhere 0,

$i = 1, \ldots, m$, $j = 1, \ldots n$. Then $\left(\begin{smallmatrix} I_m & x_{ij}E_{ij} \\ 0 & I_n \end{smallmatrix}\right)$ (having at most one nonzero off-diagonal entry) is an elementary matrix of type (iii), and there are mn such matrices. One checks that (Problem 14)

$$\begin{pmatrix} I_m & X \\ 0 & I_n \end{pmatrix} = \begin{pmatrix} I_m & \sum_{i,j} x_{ij}E_{ij} \\ 0 & I_n \end{pmatrix} = \prod_{i,j} \begin{pmatrix} I_m & x_{ij}E_{ij} \\ 0 & I_n \end{pmatrix}. \tag{2.1}$$

Generalized elementary operations (I), (II), and (III) do not change the rank of a (partitioned) matrix, and a type (III) operation does not change the determinant of a square block matrix.

The manipulation of partitioned matrices is a basic skill in matrix theory and is efficient for solving many problems. The "rules" for manipulating these block matrices are analogous to those used for ordinary matrices of scalars.

Example 2.1.2 Let $A \in \mathbb{M}_m$ and $B \in \mathbb{M}_n$ be invertible matrices, and let $C \in \mathbb{M}_{m\times n}$. We find (show) the inverse of $\left(\begin{smallmatrix} A & C \\ 0 & B \end{smallmatrix}\right)$ in two ways as follows.

Way I. Recall from Sect. 1.2 that, to find the inverse of a matrix M, we can apply elementary row operations on the augmented matrix of the form (M, I) to get a matrix in the form (I, N). Then $N = M^{-1}$.

Now we apply generalized elementary row operations on the matrix of the form (M, I) in blocks to get (I, N). Then $N = M^{-1}$. We proceed

$$\begin{pmatrix} A & C & I_m & 0 \\ 0 & B & 0 & I_n \end{pmatrix} \to \begin{pmatrix} I_m & A^{-1}C & A^{-1} & 0 \\ 0 & I_n & 0 & B^{-1} \end{pmatrix} \to \begin{pmatrix} I_m & 0 & A^{-1} & -A^{-1}CB^{-1} \\ 0 & I_n & 0 & B^{-1} \end{pmatrix}.$$

[Note: The last step was adding row two pre-multiplied by $-A^{-1}C$ to the first row.] It follows that $\left(\begin{smallmatrix} A & C \\ 0 & B \end{smallmatrix}\right)^{-1} = \left(\begin{smallmatrix} A^{-1} & -A^{-1}CB^{-1} \\ 0 & B^{-1} \end{smallmatrix}\right)$.

Way II. Applying generalized elementary (row and column) operations on the block matrix, we deduce the matrix to the identity matrix, then take inverses. To be explicit, post-multiply the first column of $\left(\begin{smallmatrix} A & C \\ 0 & B \end{smallmatrix}\right)$ by $-A^{-1}C$, then add it to the second column to get $\left(\begin{smallmatrix} A & 0 \\ 0 & B \end{smallmatrix}\right)$. Now multiply row one by A^{-1} and row two by B^{-1} to get I_{m+n}. Display these steps as

$$\begin{pmatrix} A & C \\ 0 & B \end{pmatrix} \to \begin{pmatrix} A & 0 \\ 0 & B \end{pmatrix} \to \begin{pmatrix} I_m & 0 \\ 0 & I_n \end{pmatrix},$$

or in an identity,

$$\begin{pmatrix} A^{-1} & 0 \\ 0 & B^{-1} \end{pmatrix} \begin{pmatrix} A & C \\ 0 & B \end{pmatrix} \begin{pmatrix} I & -A^{-1}C \\ 0 & I \end{pmatrix} = I_{m+n}.$$

Taking the inverses of both sides and keeping $\left(\begin{smallmatrix} A & C \\ 0 & B \end{smallmatrix}\right)^{-1}$ on the left, we get

$$\begin{pmatrix} A & C \\ 0 & B \end{pmatrix}^{-1} = \begin{pmatrix} I & -A^{-1}C \\ 0 & I \end{pmatrix} \begin{pmatrix} A^{-1} & 0 \\ 0 & B^{-1} \end{pmatrix} = \begin{pmatrix} A^{-1} & -A^{-1}CB^{-1} \\ 0 & B^{-1} \end{pmatrix}. \blacksquare$$

Example 2.1.3 Let $A, B \in \mathbb{M}_n$. If A is nonsingular, then the block matrix $\left(\begin{smallmatrix} A & B \\ 0 & A^{-1} \end{smallmatrix}\right)$ equals a product of triangular matrices with 1's on the diagonals.

To prove the statement, it is sufficient to show that $\left(\begin{smallmatrix} A & B \\ 0 & A^{-1} \end{smallmatrix}\right)$ is a product of block matrices of the forms $\left(\begin{smallmatrix} I & X \\ 0 & I \end{smallmatrix}\right)$ and $\left(\begin{smallmatrix} I & 0 \\ Y & I \end{smallmatrix}\right)$. This can be done by performing type (III) operations on the block matrix.

Add the first row of $\left(\begin{smallmatrix} A & B \\ 0 & A^{-1} \end{smallmatrix}\right)$ pre-multiplied by A^{-1} to the second row:

$$\begin{pmatrix} A & B \\ I & A^{-1} + A^{-1}B \end{pmatrix}.$$

Add the second row pre-multiplied by $I - A$ to the first row:

$$\begin{pmatrix} I & A^{-1} + A^{-1}B - I \\ I & A^{-1} + A^{-1}B \end{pmatrix}.$$

Subtract row one from row two to get the upper-triangular matrix

$$\begin{pmatrix} I & A^{-1} + A^{-1}B - I \\ 0 & I \end{pmatrix},$$

whose diagonal entries are all 1's. Putting these steps in an identity shows

$$\begin{pmatrix} I & 0 \\ -I & I \end{pmatrix} \begin{pmatrix} I & I - A \\ 0 & I \end{pmatrix} \begin{pmatrix} I & 0 \\ A^{-1} & I \end{pmatrix} \begin{pmatrix} A & B \\ 0 & A^{-1} \end{pmatrix} = \begin{pmatrix} I & A^{-1} + A^{-1}B - I \\ 0 & I \end{pmatrix}.$$

Pre-multiplying the inverses of the three matrices on the left, we obtain

$$\begin{pmatrix} A & B \\ 0 & A^{-1} \end{pmatrix} = \begin{pmatrix} I & 0 \\ -A^{-1} & I \end{pmatrix} \begin{pmatrix} I & A - I \\ 0 & I \end{pmatrix} \begin{pmatrix} I & 0 \\ I & I \end{pmatrix} \begin{pmatrix} I & A^{-1} + A^{-1}B - I \\ 0 & I \end{pmatrix}$$

as a product of type III generalized elementary matrices, implying that $\left(\begin{smallmatrix} A & B \\ 0 & A^{-1} \end{smallmatrix}\right)$ is a product of triangular matrices with 1's on the diagonals. ∎

Problems

1. Let $E[i(c) \to j]$ denote the n-square elementary matrix obtained from the identity matrix I_n by adding row i times c to row j.
   (a) Show that $\big(E[i(c) \to j]\big)^* = E[j(\bar{c}) \to i]$.
   (b) Show that $\big(E[i(c) \to j]\big)^{-1} = E[i(-c) \to j]$.
   (c) How is $E[i(c) \to j]$ obtained from I_n via a column operation?
   (d) Explain how to obtain EAE^* and EAE^{-1} from A through elementary operations, where $A = (a_{ij}) \in \mathbb{M}_n$, $E = E[i(c) \to j]$.

2. Let $A, P \in \mathbb{M}_n$, where $P = (p_1, \ldots, p_n)$ is partitioned according to columns. Let $D = \operatorname{diag}(\lambda_1, \ldots, \lambda_n)$, where $\lambda_1, \ldots, \lambda_n \in \mathbb{C}$. Show that

$$AP = (Ap_1, \ldots, Ap_n) \text{ and } PD = (\lambda_1 p_1, \ldots, \lambda_n p_n).$$

3. Let X and Y be matrices. Show that
   (a) $\left(\begin{smallmatrix} 0 & X \\ 0 & 0 \end{smallmatrix}\right)^2 = 0$ and $\left(\begin{smallmatrix} 0 & 0 \\ Y & 0 \end{smallmatrix}\right)^2 = 0$ for any X and Y.
   (b) $\left(\begin{smallmatrix} 0 & X \\ Y & 0 \end{smallmatrix}\right)^2 = \left(\begin{smallmatrix} XY & 0 \\ 0 & YX \end{smallmatrix}\right)$, where X is $m \times n$ and Y is $n \times m$.

4. Let $A = (A_{ij})$ and $B = (B_{ij})$ be block matrices given in the forms

$$A = \begin{pmatrix} A_{11} & A_{12} & \ldots & A_{1n} \\ A_{21} & A_{22} & \ldots & A_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ A_{m1} & A_{m2} & \ldots & A_{mn} \end{pmatrix}, \quad B = \begin{pmatrix} B_{11} & B_{12} & \ldots & B_{1q} \\ B_{21} & B_{22} & \ldots & B_{2q} \\ \vdots & \vdots & \ddots & \vdots \\ B_{p1} & B_{p2} & \ldots & B_{pq} \end{pmatrix}.$$

   (a) Compute $A + B$ when $m = p$, $n = q$, and A_{ij} and B_{ij} have the same size for every pair i, j.
   (b) Compute AB when $n = p$, and the number of columns of A_{it} is equal to the number of rows of B_{tj} for every triple i, t, j.
   (c) Compute λA, where $\lambda \in \mathbb{C}$.

5. Let A and B be matrices of approximate sizes.
   (a) What generalized (block) elementary row or column operation is applied to $\left(\begin{smallmatrix} I_m & A \\ 0 & I_n \end{smallmatrix}\right)$ to obtain the identity matrix $\left(\begin{smallmatrix} I_m & 0 \\ 0 & I_n \end{smallmatrix}\right)$?
   (b) If A is an $m \times m$ invertible matrix, what generalized (block) elementary row or column operation is applied to $\left(\begin{smallmatrix} A & B \\ 0 & I_n \end{smallmatrix}\right)$ to obtain a matrix in the form $\left(\begin{smallmatrix} I_m & C \\ 0 & I_n \end{smallmatrix}\right)$ (where C is some matrix)?
   (c) What generalized (block) elementary row or column operations are applied to $\left(\begin{smallmatrix} A & 0 \\ 0 & B \end{smallmatrix}\right)$ to obtain the matrix $\left(\begin{smallmatrix} A & A+B \\ 0 & B \end{smallmatrix}\right)$?

6. Let X be an $n \times m$ matrix. Show that

$$\begin{pmatrix} 0 & I_m \\ I_n & X \end{pmatrix}^{-1} = \begin{pmatrix} -X & I_n \\ I_m & 0 \end{pmatrix}, \quad \begin{pmatrix} X & I_n \\ I_m & 0 \end{pmatrix}^{-1} = \begin{pmatrix} 0 & I_m \\ I_n & -X \end{pmatrix}.$$

   In particular, $\left(\begin{smallmatrix} 0 & I_n \\ I_m & 0 \end{smallmatrix}\right)^{-1} = \left(\begin{smallmatrix} 0 & I_m \\ I_n & 0 \end{smallmatrix}\right)$. When is $\left(\begin{smallmatrix} 0 & I_n \\ I_m & 0 \end{smallmatrix}\right)^{-1} = \left(\begin{smallmatrix} 0 & I_n \\ I_m & 0 \end{smallmatrix}\right)$?

7. Let A and B be respectively m- and n-square invertible matrices. Show that $\left(\begin{smallmatrix} 0 & A \\ B & 0 \end{smallmatrix}\right)$ is invertible and $\left(\begin{smallmatrix} 0 & A \\ B & 0 \end{smallmatrix}\right)^{-1} = \left(\begin{smallmatrix} 0 & B^{-1} \\ A^{-1} & 0 \end{smallmatrix}\right)$.

8. Let X and Y be matrices with the same number of rows. Multiply

$$\begin{pmatrix} X & Y \\ 0 & 0 \end{pmatrix} \begin{pmatrix} X^* & 0 \\ Y^* & 0 \end{pmatrix} \quad \text{and} \quad \begin{pmatrix} X^* & 0 \\ Y^* & 0 \end{pmatrix} \begin{pmatrix} X & Y \\ 0 & 0 \end{pmatrix}.$$

9. Let X and Y be matrices of the same size. Verify that

$$\begin{pmatrix} I + XX^* & X + Y \\ X^* + Y^* & I + Y^*Y \end{pmatrix} = \begin{pmatrix} I & X \\ Y^* & I \end{pmatrix} \begin{pmatrix} I & Y \\ X^* & I \end{pmatrix} =$$
$$\begin{pmatrix} X & I \\ I & Y^* \end{pmatrix} \begin{pmatrix} X^* & I \\ I & Y \end{pmatrix} = \begin{pmatrix} X \\ I \end{pmatrix} (X^*, I) + \begin{pmatrix} I \\ Y^* \end{pmatrix} (I, Y).$$

10. Let M, A, B, C, D, X, and Y be n-square matrices. Compute

(a) $\begin{pmatrix} X^* \\ Y^* \end{pmatrix} M(X, Y)$.

(b) $(X^*, Y^*) \begin{pmatrix} A & B \\ C & D \end{pmatrix} \begin{pmatrix} X \\ Y \end{pmatrix}$.

(c) $\begin{pmatrix} X^* & 0 \\ 0 & Y^* \end{pmatrix} \begin{pmatrix} A & B \\ C & D \end{pmatrix} \begin{pmatrix} X & 0 \\ 0 & Y \end{pmatrix}$.

11. Let A, B, and C be $n \times n$ matrices, $M = \begin{pmatrix} I & B \\ A & C \end{pmatrix}$, and $P = \begin{pmatrix} I & 0 \\ -A & I \end{pmatrix}$.

(a) Compute PMP^*, PMP^{-1}, and $PM\tilde{P}$, where $\tilde{P} = \begin{pmatrix} I & -A \\ 0 & I \end{pmatrix}$.

(b) Explain (a) in terms of elementary operations applied to M.

(c) Show that the matrices in (a) have the same determinant as M.

(d) Do the matrices in (a) have the same eigenvalues as M?

12. Let $a_1, a_2, \ldots, a_n \in \mathbb{C}$. Find $\det A$ and A^{-1} if $a_1 \neq 0$, where

$$A = \begin{pmatrix} 0 & 1 & 0 & \ldots & 0 \\ 0 & 0 & 1 & \ldots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \ldots & 1 \\ -a_1 & -a_2 & -a_3 & \ldots & -a_n \end{pmatrix}.$$

13. Let $a_1, a_2, \ldots, a_n \in \mathbb{C}$. Find $\det A$ and A^{-1} if all $a_i \neq 0$, where

$$A = \begin{pmatrix} 0 & a_1 & 0 & \ldots & 0 \\ 0 & 0 & a_2 & \ldots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \ldots & a_{n-1} \\ a_n & 0 & 0 & \ldots & 0 \end{pmatrix}.$$

14. Let A and B be $m \times n$ matrices. Show that

$$\begin{pmatrix} I_m & A \\ 0 & I_n \end{pmatrix} \begin{pmatrix} I_m & B \\ 0 & I_n \end{pmatrix} = \begin{pmatrix} I_m & B \\ 0 & I_n \end{pmatrix} \begin{pmatrix} I_m & A \\ 0 & I_n \end{pmatrix} = \begin{pmatrix} I_m & A + B \\ 0 & I_n \end{pmatrix}.$$

15. Let $f(x)$ and $g(x)$ be polynomials that have no common roots over $\mathbb{C}$ (that is, f and g are coprime). Let A be an $n \times n$ matrix. Show that

$$\operatorname{rank}\big(f(A)\big) + \operatorname{rank}\big(g(A)\big) = n + \operatorname{rank}\big(f(A)g(A)\big).$$

[Hint: There exist $u(x)$ and $v(x)$ such that $f(x)u(x) + g(x)v(x) = 1$.]

16. Find $\begin{pmatrix} A & 0 \\ C & B \end{pmatrix}^{-1}$ and $\begin{pmatrix} 0 & A \\ B & C \end{pmatrix}^{-1}$ for invertible matrices A, B, and any C.

17. Find the ranks of the block matrices by elementary operations:

$$\begin{pmatrix} I_n & I_n \\ I_n & I_n \end{pmatrix}, \quad \begin{pmatrix} I_n & I_n & 0 \\ I_n & I_n & I_n \\ 0 & I_n & I_n \end{pmatrix}, \quad \begin{pmatrix} I_n & I_n & 0 \\ I_n & 2I_n & I_n \\ 0 & I_n & I_n \end{pmatrix}.$$

18. Let A and B be matrices. Show that $\operatorname{rank}\begin{pmatrix} A & 0 \\ 0 & B \end{pmatrix} = \operatorname{rank}(A) + \operatorname{rank}(B)$.

19. Let $A, B \in \mathbb{M}_n$. Apply elementary operations to $\begin{pmatrix} A & 0 \\ 0 & B \end{pmatrix}$ (written as $A \oplus B$, called the *direct sum* of A and B) to get $\begin{pmatrix} A+B & B \\ B & B \end{pmatrix}$ and show that

$$\operatorname{rank}(A+B) \le \operatorname{rank}(A) + \operatorname{rank}(B).$$

20. Let A and B be $n \times n$ matrices. Apply elementary operations to $\begin{pmatrix} I_n & A \\ B & I_n \end{pmatrix}$ to get $\begin{pmatrix} I_n - AB & 0 \\ 0 & I_n \end{pmatrix}$ and $\begin{pmatrix} I_n & 0 \\ 0 & I_n - BA \end{pmatrix}$. Draw the conclusion that $I_n - AB$ and $I_n - BA$ have the same rank for any $n \times n$ matrices A and B.

21. Let A, B, C, and D be matrices and let A be invertible. Show that

$$\operatorname{rank}\begin{pmatrix} A & B \\ C & D \end{pmatrix} = \operatorname{rank}(A) + \operatorname{rank}(D - CA^{-1}B).$$

In particular, for any nonsingular matrix A,

$$\operatorname{rank}\begin{pmatrix} A & B \\ 0 & D \end{pmatrix} = \operatorname{rank}(A) + \operatorname{rank}(D)$$

and for any n-square matrices X and Y,

$$\operatorname{rank}\begin{pmatrix} I_n & I_n \\ X & Y \end{pmatrix} = n + \operatorname{rank}(X - Y).$$

22. Let $A, B \in \mathbb{M}_n$. Find elementary operations for $\begin{pmatrix} A & 0 \\ I_n & B \end{pmatrix} \to \begin{pmatrix} AB & 0 \\ 0 & I_n \end{pmatrix}$.

23. Let $A, B \in \mathbb{M}_n$. Find elementary operations for $\begin{pmatrix} A & B \\ B & A \end{pmatrix} \to \begin{pmatrix} A+B & 0 \\ B & A-B \end{pmatrix}$.

24. Show the determinant identities by interchanging rows (or columns):

$$\begin{vmatrix} 0 & I_m \\ I_n & 0 \end{vmatrix} = \begin{vmatrix} 0 & I_n \\ I_m & 0 \end{vmatrix} = (-1)^{mn}.$$

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2.2 Determinants and Inverses of Partitioned Matrices

We revisit two important facts about determinants that we stated in Sect. 1.2: $\det(AB) = \det A \det B$ and $\det(AB) = \left|\begin{smallmatrix} A & X \\ 0 & B \end{smallmatrix}\right| = \det A \det B$. [Note: For block matrices, we use vertical bars to denote their determinants; see Sect. 1.2.]

Theorem 2.2.1 *Let $A \in \mathbb{M}_m$, $B \in \mathbb{M}_n$. Then $\left|\begin{smallmatrix} A & X \\ 0 & B \end{smallmatrix}\right| = \left|\begin{smallmatrix} A & 0 \\ Y & B \end{smallmatrix}\right| = \det A \det B$, where X is any $m \times n$ matrix and Y is any $n \times m$ matrix.*

Proof. We use the Laplace expansion formula (Sect. 1.2) and induction on the order of A. If $m = 1$, say $A = (a)$, then $\left|\begin{smallmatrix} A & X \\ 0 & B \end{smallmatrix}\right| = a \det B$ by the Laplace formula (expansion along the first column of the block matrix).

Suppose that the determinant identity holds for any matrix A of order $m > 1$. We show that the identity holds if A has order $m + 1$.

Let $A = (a_{ij}) \in \mathbb{M}_{m+1}$ and let $A(j|1) \in \mathbb{M}_m$ be obtained from A by deleting the jth row and the first column. Expanding the determinant along the first column of the block matrix and by induction on m, we have

$$\begin{aligned}
\begin{vmatrix} A & X \\ 0 & B \end{vmatrix} &= \sum_{j=1}^{m} (-1)^{1+j} a_{j1} \begin{vmatrix} A(j|1) & \star \\ 0 & B \end{vmatrix} \ (\star \text{ for unspecified entries}) \\
&= \sum_{j=1}^{m} (-1)^{1+j} a_{j1} \det A(j|1) \det B \ \ (\text{by induction on } m) \\
&= \Big(\sum_{j=1}^{m} (-1)^{1+j} a_{j1} \det A(j|1) \Big) \det B \\
&= \det A \det B.
\end{aligned}$$

Similarly, $\left|\begin{smallmatrix} A & 0 \\ Y & B \end{smallmatrix}\right| = \det A \det B$ by expansion along rows. ∎

Example 2.2.1 Let $A, B \in \mathbb{M}_n$. Then $\det(AB) = \det A \det B$.

Consider $\left(\begin{smallmatrix} A & 0 \\ I & B \end{smallmatrix}\right)$. Adding the second row pre-multiplied by $-A$ to the first row, we obtain $\left(\begin{smallmatrix} 0 & -AB \\ I & B \end{smallmatrix}\right)$. Since the determinant remains unchanged under a type (III) operation (as it does under a type (iii) operation; see Sect. 1.2), we have $\left|\begin{smallmatrix} A & 0 \\ I & B \end{smallmatrix}\right| = \left|\begin{smallmatrix} 0 & -AB \\ I & B \end{smallmatrix}\right|$. On the other hand, move the last n rows up to the top (there are n moves for each row, a total of n^2 moves; each move changes the sign of the determinant once). Thus,

$$\begin{aligned}
\det A \det B &= \begin{vmatrix} A & 0 \\ I & B \end{vmatrix} = \begin{vmatrix} 0 & -AB \\ I & B \end{vmatrix} = (-1)^{n^2} \begin{vmatrix} I & B \\ 0 & -AB \end{vmatrix} \\
&= (-1)^{n^2} \det(-AB) = (-1)^{n^2+n} \det(AB) = \det(AB). \ ∎
\end{aligned}$$

Let M be a square matrix partitioned as $M = \left(\begin{smallmatrix} A & B \\ C & D \end{smallmatrix}\right)$, where A and D are m- and n-square matrices, respectively. We discuss the determinants and

inverses of the block matrices in this general form. The results are fundamental and used almost everywhere in matrix theory, such as in the areas of matrix computations and matrix inequalities. The method of continuity argument and the way to find inverses deserve special attention.

Theorem 2.2.2 *Let* $M = \left(\begin{smallmatrix} A & B \\ C & D \end{smallmatrix}\right)$, *where* $A \in \mathbb{M}_m$ *and* $D \in \mathbb{M}_n$. *Then*

$$\det M = \det A \det(D - CA^{-1}B), \quad \text{if } A \text{ is invertible,} \tag{2.2}$$

and

$$\det M = \det(AD - CB), \quad \text{if } AC = CA \ \ (m = n). \tag{2.3}$$

Proof. If A is nonsingular, we can easily verify that

$$\begin{pmatrix} I_m & 0 \\ -CA^{-1} & I_n \end{pmatrix} \begin{pmatrix} A & B \\ C & D \end{pmatrix} = \begin{pmatrix} A & B \\ 0 & D - CA^{-1}B \end{pmatrix}. \tag{2.4}$$

By taking determinants for both sides, we have

$$\det M = \begin{vmatrix} A & B \\ 0 & D - CA^{-1}B \end{vmatrix} = \det A \det(D - CA^{-1}B).$$

For the second part, if A and C commute, then A, B, C, and D have the same size. We show the identity by the so-called *continuity argument*.

Note that for $A \in \mathbb{M}_m$, $p(t) = \det(tI - A)$ is a polynomial of degree m. Thus, $p(t)$ has m roots in $\mathbb{C}$, and $p(t)$ is a continuous function in t.

First consider the case where A is invertible. By (2.2) and the fact that $\det(XY) = \det X \det Y$ for square matrices X and Y of the same size, we have

$$\begin{aligned} \det M &= \det A \det(D - CA^{-1}B) \\ &= \det(AD - ACA^{-1}B) \\ &= \det(AD - CAA^{-1}B) \\ &= \det(AD - CB). \end{aligned}$$

Suppose that A is singular and consider $A + \epsilon I$. Since $\det(A + \epsilon I)$ as a polynomial in ϵ has a finite number of zeros, we can choose $\delta > 0$ such that

$$\det(A + \epsilon I) \neq 0, \quad 0 < \epsilon < \delta,$$

that is, $A + \epsilon I$ is an invertible matrix for all $\epsilon \in (0, \delta)$. Let

$$M_\epsilon = \begin{pmatrix} A + \epsilon I & B \\ C & D \end{pmatrix}, \quad 0 < \epsilon < \delta.$$

Since A and C commute, $A + \epsilon I$ and C commute. Thus,

$$\det M_\epsilon = \det\big((A + \epsilon I)D - CB\big), \quad 0 < \epsilon < \delta.$$

Observe that both sides of the above identity are continuous functions of ϵ. Letting $\epsilon \to 0^+$ results in $\det M = \det(AD - CB)$. ∎

Note that (2.3) need not be true if $AC \neq CA$ (Problem 7).

It follows from Theorem 2.2.2 that for $X \in \mathbb{M}_{n\times m}$ and $Y \in \mathbb{M}_{m\times n}$,

$$\begin{vmatrix} I_m & Y \\ X & I_n \end{vmatrix} = |I_n - XY| = |I_m - YX| = \begin{vmatrix} I_n & X \\ Y & I_m \end{vmatrix} \tag{2.5}$$

and that for nonzero $a \in \mathbb{C}$, x, $y^T \in \mathbb{C}^n$ (in columns), and $Z \in \mathbb{M}_n(\mathbb{C})$,

$$\begin{vmatrix} a & y \\ x & Z \end{vmatrix} = \begin{vmatrix} a & y \\ 0 & Z - \frac{1}{a}xy \end{vmatrix} = a\Big|Z - \tfrac{1}{a}xy\Big| = \frac{1}{a^{n-1}}|aZ - xy|. \tag{2.6}$$

Example 2.2.2 Let A be an $n \times n$ matrix with diagonal entries $x_1, \ldots, x_n$ and off-diagonal entries all equal to 1. Find the determinant of A.

If two (or more) x's are equal to 1, then A has two rows of 1's, and $\det A = 0$. If one x, say x_1 is 1, by subtracting the first row from other rows, we get $\det A = (x_2 - 1)\cdots(x_n - 1)$. Suppose that none of x's is 1.

Let $e = (1, \ldots, 1)^T \in \mathbb{R}^{n-1}$ and $\Lambda = \operatorname{diag}(x_2 - 1, \ldots, x_n - 1)$. Subtracting the first row from other rows of A, then adding the second block column post-multiplied by $-(1 - x_1)\Lambda^{-1}e$ to the first column, we obtain

$$\begin{aligned} \det A &= \begin{vmatrix} x_1 & e^T \\ e & \Lambda + ee^T \end{vmatrix} = \begin{vmatrix} x_1 & e^T \\ (1-x_1)e & \Lambda \end{vmatrix} \\ &= \begin{vmatrix} x_1 - (1-x_1)e^T\Lambda^{-1}e & e^T \\ 0 & \Lambda \end{vmatrix} \\ &= \big(x_1 - (1-x_1)e^T\Lambda^{-1}e\big)\det\Lambda \\ &= \Big(1 + \sum_{i=1}^{n}\frac{1}{x_i - 1}\Big)\prod_{j=1}^{n}(x_j - 1) \\ &= \prod_{j=1}^{n}(x_j - 1) + \sum_{i=1}^{n}\prod\widehat{x_i - 1}, \end{aligned}$$

where $\prod\widehat{x_i - 1}$ represents $\prod_{j=1}^{n}(x_j - 1)$ with $x_i - 1$ absent. ∎

We now turn our attention to the inverses of partitioned matrices.

Theorem 2.2.3 *Suppose that the partitioned matrix*

$$M = \begin{pmatrix} A & B \\ C & D \end{pmatrix}, \quad A \in \mathbb{M}_m,\ D \in \mathbb{M}_n,$$

is invertible and let the inverse be conformally partitioned as

$$M^{-1} = \begin{pmatrix} X & Y \\ U & V \end{pmatrix}, \quad X \in \mathbb{M}_m,\ V \in \mathbb{M}_n.$$

Then

$$\det A = \det V \det M. \tag{2.7}$$

Proof. This is immediate by taking the determinants of both sides of

$$\begin{pmatrix} A & B \\ C & D \end{pmatrix} \begin{pmatrix} I_m & Y \\ 0 & V \end{pmatrix} = \begin{pmatrix} A & 0 \\ C & I_n \end{pmatrix}. \blacksquare$$

Given that $\det M \neq 0$, by (2.7), A is singular if and only if V is singular. Moreover, if A, B, C, and D are all $n \times n$ matrices, then (Problem 32)

$$\det B = (-1)^{n^2} \det Y \det M. \tag{2.8}$$

Example 2.2.3 Let $S = (s_{ij}) \in \mathbb{M}_n$ be a real orthogonal matrix. We have $S^T = S^{-1}$ and $\det S = \pm 1$. If S contains a zero entry in the first row, say $s_{11} = 0$, i.e., $S = \left(\begin{smallmatrix} 0 & x \\ y & Z \end{smallmatrix}\right)$, x^T, $y \in \mathbb{R}^{n-1}$, $Z \in \mathbb{M}_{n-1}$, then Z is singular.

In general, if $s_{ij} = 0$, then the submatrix of S by deleting the ith row and jth column is singular. Note: If T is real orthogonal, so is PTQ for type (i) elementary matrices P and Q (i.e., permutation matrices). $\blacksquare$

Theorem 2.2.4 *Under the conditions in Theorem 2.2.3, if A^{-1} exists, then*

$$\begin{aligned} X &= A^{-1} + A^{-1}B(D - CA^{-1}B)^{-1}CA^{-1}, \\ Y &= -A^{-1}B(D - CA^{-1}B)^{-1}, \\ U &= -(D - CA^{-1}B)^{-1}CA^{-1}, \\ V &= (D - CA^{-1}B)^{-1}. \end{aligned}$$

Proof. Every invertible matrix is a product of elementary matrices (Theorem 1.2.1). Since $M^{-1}(M, I) = (I, M^{-1})$, we can view M^{-1} as a sequence of elementary operations and perform row reductions on (M, I) to get (I, M^{-1}) which reveals the inverse M^{-1} (Sect. 1.2).

We now apply row operations to the augmented block matrix

$$\begin{pmatrix} A & B & I & 0 \\ C & D & 0 & I \end{pmatrix}.$$

Multiply row one by A^{-1} (from the left for row operations) to get

$$\begin{pmatrix} I & A^{-1}B & A^{-1} & 0 \\ C & D & 0 & I \end{pmatrix}.$$

Subtract row one multiplied by C from row two to get

$$\begin{pmatrix} I & A^{-1}B & A^{-1} & 0 \\ 0 & D - CA^{-1}B & -CA^{-1} & I \end{pmatrix}.$$

Multiply row two by $(D - CA^{-1}B)^{-1}$ (which exists; why?) to get

$$\begin{pmatrix} I & A^{-1}B & A^{-1} & 0 \\ 0 & I & -(D - CA^{-1}B)^{-1}CA^{-1} & (D - CA^{-1}B)^{-1} \end{pmatrix}.$$

Subtracting row two times $A^{-1}B$ from row one reveals a block matrix in the form $\left(\begin{smallmatrix} I & 0 \\ 0 & I \end{smallmatrix}\right)$, in which $\left(\begin{smallmatrix} \star & \star \\ \star & \star \end{smallmatrix}\right)$ on the right-hand side is the inverse of M:

$$\begin{pmatrix} A^{-1} + A^{-1}B(D - CA^{-1}B)^{-1}CA^{-1} & -A^{-1}B(D - CA^{-1}B)^{-1} \\ -(D - CA^{-1}B)^{-1}CA^{-1} & (D - CA^{-1}B)^{-1} \end{pmatrix}.$$

Comparing to the inverse M^{-1} in the block form, we have X, Y, U, and V with the desired expressions in terms of A, B, C, and D. ∎

As an alternative proof, we apply type (III) row and column operations on $\left(\begin{smallmatrix} A & B \\ C & D \end{smallmatrix}\right)$ to obtain $\left(\begin{smallmatrix} A & 0 \\ 0 & D - CA^{-1}B \end{smallmatrix}\right)$. To display this explicitly, we have

$$\begin{pmatrix} I & 0 \\ -CA^{-1} & I \end{pmatrix} \begin{pmatrix} A & B \\ C & D \end{pmatrix} \begin{pmatrix} I & -A^{-1}B \\ 0 & I \end{pmatrix} = \begin{pmatrix} A & 0 \\ 0 & D - CA^{-1}B \end{pmatrix}.$$

Taking the inverses and keeping only M^{-1} on the left-hand side, we arrive at

$$M^{-1} = \begin{pmatrix} I & -A^{-1}B \\ 0 & I \end{pmatrix} \begin{pmatrix} A^{-1} & 0 \\ 0 & (D - CA^{-1}B)^{-1} \end{pmatrix} \begin{pmatrix} I & 0 \\ -CA^{-1} & I \end{pmatrix}.$$

Multiply out the right-hand side and compare with the blocks in M^{-1}.

If D is nonsingular in the theorem, a similar argument implies

$$X = (A - BD^{-1}C)^{-1}.$$

It follows that if A, D, and $M = \left(\begin{smallmatrix} A & B \\ C & D \end{smallmatrix}\right)$ are nonsingular, then

$$(A - BD^{-1}C)^{-1} = A^{-1} + A^{-1}B(D - CA^{-1}B)^{-1}CA^{-1}. \tag{2.9}$$

Below is a direct proof of identity (2.9). More similar identities are derived by means of partitioned matrices in Sect. 7.4 of Chap. 7.

Theorem 2.2.5 *Let $A \in \mathbb{M}_m$, $B \in \mathbb{M}_n$, $C \in \mathbb{M}_{m\times n}$, and $D \in \mathbb{M}_{n\times m}$. Suppose that A, B, and $A + CBD$ are nonsingular. Then*

$$(A + CBD)^{-1} = A^{-1} - A^{-1}C(B^{-1} + DA^{-1}C)^{-1}DA^{-1}. \tag{2.10}$$

Proof. First observe that $B^{-1} + DA^{-1}C$ is nonsingular, because (by (2.5))

$$\begin{aligned} \det(B^{-1} + DA^{-1}C) &= \det B^{-1} \det(I_n + BDA^{-1}C) \\ &= \det B^{-1} \det(I_m + A^{-1}CBD) \\ &= \det B^{-1} \det A^{-1} \det(A + CBD) \neq 0. \end{aligned}$$

We now prove (2.10) by a direct verification:

$$\begin{aligned}
&(A+CBD)\big(A^{-1}-A^{-1}C(B^{-1}+DA^{-1}C)^{-1}DA^{-1}\big)\\
&\quad= I_m - C(B^{-1}+DA^{-1}C)^{-1}DA^{-1} + CBDA^{-1}\\
&\qquad - CBDA^{-1}C(B^{-1}+DA^{-1}C)^{-1}DA^{-1}\\
&\quad= I_m - C\big((B^{-1}+DA^{-1}C)^{-1} - B\\
&\qquad + BDA^{-1}C(B^{-1}+DA^{-1}C)^{-1}\big)DA^{-1}\\
&\quad= I_m - C\big((I_n+BDA^{-1}C)(B^{-1}+DA^{-1}C)^{-1} - B\big)DA^{-1}\\
&\quad= I_m - C\big(B(B^{-1}+DA^{-1}C)(B^{-1}+DA^{-1}C)^{-1} - B\big)DA^{-1}\\
&\quad= I_m - C(B-B)DA^{-1} = I_m. \ \blacksquare
\end{aligned}$$

Many matrix identities involving inverses can be derived from (2.10).

Example 2.2.4 Let $X \in \mathbb{M}_{m\times n}$ and $Y \in \mathbb{M}_{n\times m}$. Then $I_n - YX$ is invertible if and only if $I_m - XY$ is invertible. By (2.10) or by a direct verification,

$$(I_m - XY)^{-1} = I_m + X(I_n - YX)^{-1}Y. \ \blacksquare$$

Example 2.2.5 Let $A, B \in \mathbb{M}_n(\mathbb{R})$ be nonsingular matrices such that $A+iB$ is nonsingular. Find $C, D \in \mathbb{M}_n(\mathbb{R})$ such that $C+iD = (A+iB)^{-1}$.

Way I. Compute $(A+iB)(C+iD) = (AC-BD)+i(AD+BC)$. If $C+iD$ is the inverse of $A+iB$, then $AC-BD = I$ and $AD+BC = 0$. Because A and B are nonsingular, we have $C = A^{-1}+A^{-1}BD$ and $D = -A^{-1}BC$. Thus, $C = A^{-1} - A^{-1}BA^{-1}BC$, so $(A^{-1}BA^{-1}B + I)C = A^{-1}$, implying

$$C = (I + A^{-1}BA^{-1}B)^{-1}A^{-1} = (A + BA^{-1}B)^{-1}.$$

Since $C+iD = (A+iB)^{-1}$ gives $D - iC = (-B+iA)^{-1}$, we obtain

$$D = -(B + AB^{-1}A)^{-1}.$$

Way II. Represent a complex matrix by a block real matrix. To be exact, $C+iD$ is the inverse of $A+iB$ if and only if $\left(\begin{smallmatrix} C & D \\ -D & C \end{smallmatrix}\right)$ is the inverse of $\left(\begin{smallmatrix} A & B \\ -B & A \end{smallmatrix}\right)$ (Problem 12). Theorem 2.2.4 (in which $X = V$ and $U = -Y$) reveals

$$\begin{aligned}
C &= (A+BA^{-1}B)^{-1} = A^{-1} - A^{-1}B(A+BA^{-1}B)^{-1}BA^{-1},\\
D &= -CBA^{-1} = -A^{-1}BC = -(B+AB^{-1}A)^{-1}.
\end{aligned}$$

Way III. One can obtain C and D in terms of A and B by verifying that

$$(A+iB)\big((B^{-1}A - iI)(B+AB^{-1}A)^{-1}\big) = I.$$

Note that $B^{-1}A(B+AB^{-1}A)^{-1} = (A+BA^{-1}B)^{-1}$. $\blacksquare$

Example 2.2.6 Let A and B be $n \times n$ (real or complex) matrices. If $A + iB$ is nonsingular ($i = \sqrt{-1}$), then $A + rB$ is nonsingular for some real $r \in \mathbb{R}$.

If $\det A \neq 0$, then take $r = 0$. Otherwise, let $p(z) = \det(A + zB)$, $z \in \mathbb{C}$. Then $p(0) = 0$ and $p(i) \neq 0$. Thus, $p(z)$ is a non-constant polynomial in z, and it has a finite number of zeros. Take an $r \in \mathbb{R}$ such that $p(r) \neq 0$. ∎

Problems

1. Let A and B be n-square nonsingular matrices. Show that
$$A^{-1} - B^{-1} = B^{-1}(B - A)A^{-1}.$$

2. Let A be an $n \times n$ nonsingular matrix, $a \in \mathbb{C}$, α^T, $\beta \in \mathbb{C}^n$. Show that
$$(\det A)^{-1} \begin{vmatrix} a & \alpha \\ \beta & A \end{vmatrix} = a - \alpha A^{-1}\beta.$$

3. Let $A = \begin{pmatrix} a & b & c \\ d & e & f \\ g & h & i \end{pmatrix}$ and $P = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix}$. Compute P^TAP and PAP.

4. (a) Show that simultaneous permutations of rows and columns of a square matrix do not change the eigenvalues of the matrix. That is, if $A \in \mathbb{M}_n$ and $P \in \mathbb{M}_n$ is a permutation matrix, then P^TAP have the same eigenvalues as A. (b) Does PAP have the same eigenvalues as A? (c) If A is upper-triangular, is P^TAP necessarily upper-triangular?

5. Refer to Theorem 2.2.2 and assume that $AC = CA$. Does it follow that
$$\begin{vmatrix} A & B \\ C & D \end{vmatrix} = \det(AD - BC)?$$

6. For matrices A, B, C of appropriate sizes, evaluate the determinants
$$\begin{vmatrix} A & I_n \\ I_m & 0 \end{vmatrix}, \quad \begin{vmatrix} 0 & A \\ A^{-1} & 0 \end{vmatrix} \text{ (if } A^{-1} \text{ exists)}, \quad \begin{vmatrix} 0 & A \\ B & C \end{vmatrix}, \quad \begin{vmatrix} I_n & A \\ B & C \end{vmatrix}.$$

7. Show by examples that any two of the following determinants cannot be equal for arbitrary square matrices A, B, C, and D of the same size:
$$|AD - CB|, \ |AD - BC|, \ |DA - CB|, \ |DA - BC|, \ \begin{vmatrix} A & B \\ C & D \end{vmatrix}.$$

8. Find the determinant and the inverse of the $2m \times 2m$ block matrix
$$A = \begin{pmatrix} aI_m & bI_m \\ cI_m & dI_m \end{pmatrix}, \quad ad - bc \neq 0.$$

9. Let X, Y, Z be matrices such that the block matrix $\left(\begin{smallmatrix} I & X \\ Y & Z \end{smallmatrix}\right)$ is a square matrix. Show that $\left(\begin{smallmatrix} I & X \\ Y & Z \end{smallmatrix}\right)$ is invertible if and only if $Z - YX$ is invertible. If the inverses exist, find $\left(\begin{smallmatrix} I & X \\ Y & Z \end{smallmatrix}\right)^{-1}$ in terms of X, Y, Z (and I).

10. Let A and B be m- and n-square matrices, respectively. Show by (generalized) elementary operations (or permutations of blocks) that

$$\begin{vmatrix} X & A \\ B & Y \end{vmatrix} = \begin{vmatrix} Y & B \\ A & X \end{vmatrix} = (-1)^{mn} \begin{vmatrix} A & X \\ Y & B \end{vmatrix}, \quad X \in \mathbb{M}_{m\times n},\ Y \in \mathbb{M}_{n\times m}.$$

11. Let U be a unitary matrix. If u_{ij} is the (i,j)-entry of U and $U(i|j)$ is the submatrix obtained from U by deleting row i and column j, show that $|u_{ij}| = |\det U(i|j)|$, that is, every entry of a unitary matrix and the co-factor of the entry are equal in absolute value.

12. Let M be an n-square nonsingular (real or complex) matrix. Write

$$M = A + iB, \quad M^{-1} = C + iD, \quad (i = \sqrt{-1}\,)$$

where A, B, C, and D are real matrices, and let

$$N = \begin{pmatrix} A & B \\ -B & A \end{pmatrix}.$$

Show that N is nonsingular, the inverse of N is

$$N^{-1} = \begin{pmatrix} C & D \\ -D & C \end{pmatrix},$$

and N is normal (real orthogonal) if and only if M is normal (unitary).

13. Let $M = \left(\begin{smallmatrix} N & P \\ 0 & 0 \end{smallmatrix}\right)$ be a partitioned matrix in which each block is a square. If M is normal, show that N is normal and $P = 0$. If M^k is normal for some positive integer $k > 1$ and if N is nonsingular, show that $P = 0$.

14. Let A and C be m- and n-square invertible matrices, respectively. Show that for any $m \times n$ matrix B and any $n \times m$ matrix D,

$$\det(A + BCD) = \det A \det C \det(C^{-1} + DA^{-1}B).$$

What can be deduced from this identity if $A = I_m$ and $C = I_n$?

15. Let A and B be real square matrices of the same size. Show that

$$\begin{vmatrix} A & -B \\ B & A \end{vmatrix} = |\det(A + iB)|^2.$$

16. Let $M = \left(\begin{smallmatrix} A & B \\ B & A \end{smallmatrix}\right)$, where A and B are n-square matrices. Show that

    (a) M is similar to $\left(\begin{smallmatrix} A+B & 0 \\ B & A-B \end{smallmatrix}\right)$ via invertible matrix $P = \left(\begin{smallmatrix} I & I \\ 0 & I \end{smallmatrix}\right)$.

    (b) M is similar to $\left(\begin{smallmatrix} A+B & 0 \\ 0 & A-B \end{smallmatrix}\right)$ via unitary matrix $U = \frac{1}{\sqrt{2}} \left(\begin{smallmatrix} I & I \\ -I & I \end{smallmatrix}\right)$.

    (c) The eigenvalues of M are the eigenvalues of $A + B$ and $A - B$.

    (d) $\det M = \det(A + B) \det(A - B)$.

17. Let A and B be n-square matrices. Find a formula for $\left(\begin{smallmatrix} A & B \\ 0 & I_n \end{smallmatrix}\right)^k$, where k is any integer (positive or negative if A is invertible).

18. Let $X \in \mathbb{M}_{m\times n}$, $Y \in \mathbb{M}_{n\times p}$, and $Z \in \mathbb{M}_{p\times n}$. Find the inverses

$$\begin{pmatrix} I_m & X & 0 \\ 0 & I_n & Y \\ 0 & 0 & I_p \end{pmatrix}^{-1} \quad \text{and} \quad \begin{pmatrix} I_m & X & 0 \\ 0 & I_n & 0 \\ 0 & Z & I_p \end{pmatrix}^{-1}.$$

19. Let $A = (B, C)$, where $B \in \mathbb{M}_{m\times n}$ and $C \in \mathbb{M}_{m\times p}$. Show that

$$\begin{vmatrix} I_n & B^* \\ B & AA^* \end{vmatrix} = \begin{vmatrix} I_n & B^* \\ B & BB^* + CC^* \end{vmatrix} = \det(CC^*)$$

and that if $C^*B = 0$, then $\det(A^*A) = \det(B^*B)\det(C^*C)$.

20. Let $A_{11} \in \mathbb{M}_m$ and $A_{22} \in \mathbb{M}_n$. Show that for any $m \times m$ matrix B

$$\begin{vmatrix} BA_{11} & BA_{12} \\ A_{21} & A_{22} \end{vmatrix} = \begin{vmatrix} A_{11}B & A_{12} \\ A_{21}B & A_{22} \end{vmatrix} = (\det B) \begin{vmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{vmatrix}$$

and that for any $n \times m$ matrix C

$$\begin{vmatrix} A_{11} & A_{12} \\ A_{21} + CA_{11} & A_{22} + CA_{12} \end{vmatrix} = \begin{vmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{vmatrix}.$$

21. Let $A \in \mathbb{M}_{m\times n}$. If $\rho_1, \ldots, \rho_n$ are the eigenvalues of A^*A, show that

$$\begin{vmatrix} I_m & A \\ A^* & I_n \end{vmatrix} = \prod_{j=1}^{n} (1 - \rho_j).$$

22. Let $A \in \mathbb{M}_n$. Show that there exists a diagonal matrix D with diagonal entries ± 1 such that $\det(A + D) \neq 0$. [Hint: Show by induction.]

23. Let A be a square matrix. If $I + A$ is invertible, show that $(I + A)^{-1}$ and $I - A$ commute. If A is also invertible, show that

$$(I + A)^{-1} + (I + A^{-1})^{-1} = I.$$

24. Let A be an $m \times n$ matrix. Show that $I_n + A^*A$ is invertible and that

$$\begin{aligned} (I_n + A^*A)^{-1} &= I_n - (I_n + A^*A)^{-1}A^*A \\ &= I_n - A^*A(I_n + A^*A)^{-1} \\ &= I_n - A^*(I_m + AA^*)^{-1}A. \end{aligned}$$

25. Let A and B be $m \times n$ and $n \times m$ matrices, respectively. Show that

    (a) $I_m + AB$ is nonsingular if and only if $I_n + BA$ is nonsingular.
    (b) $(I_n + BA)^{-1}B = B(I_m + AB)^{-1}$ if the inverses exist.
    (c) $(I_m + AB)^{-1} = I_m - A(I_n + BA)^{-1}B$ if the inverses exist.
    (d) $(I_m + AA^*)^{-1} = I_m - A(I_n + A^*A)^{-1}A^*$ for any $A \in \mathbb{M}_{m\times n}$.
    (e) $(I_m - AA^*)^{-1} = I_m + A(I_n - A^*A)^{-1}A^*$ if $I - A^*A$ is nonsingular.

26. Let $A \in \mathbb{M}_n$ and $\alpha, \beta \in \mathbb{C}$. If the involved inverses exist, show that

$$(\alpha I - A)^{-1} - (\beta I - A)^{-1} = (\beta - \alpha)(\alpha I - A)^{-1}(\beta I - A)^{-1}.$$

27. Let $u, v \in \mathbb{C}^n$. Show that the adjugate matrix of $I - uv^*$ is equal to

$$\operatorname{adj}(I - uv^*) = uv^* + (1 - v^*u)I.$$

28. Let $u, v \in \mathbb{C}^n$ with $v^*u \neq 0$. Write $v^*u = p^{-1} + q^{-1}$. Show that

$$(I - puv^*)^{-1} = I - quv^*.$$

    Derive that for any unit vector u in $\mathbb{C}^n$,

$$(I - 2uu^*)^{-1} = I - 2uu^*, \quad (I + uu^*)^{-1} = I - \tfrac{1}{2}uu^*.$$

29. Let A be an $n \times n$ nonsingular matrix. Let $u, v \in \mathbb{C}^n$ be such that $v^*A^{-1}u$ is not equal to -1. Show that $A + uv^*$ is invertible and that

$$(A + uv^*)^{-1} = A^{-1} - \delta^{-1}A^{-1}uv^*A^{-1}, \text{ where } \delta = 1 + v^*A^{-1}u.$$

30. Let $A \in \mathbb{M}_m$, $U, V \in \mathbb{M}_{m\times n}$. If the involved inverses exist, show that

$$(A + UV^*)^{-1} = A^{-1} - A^{-1}U(I + V^*A^{-1}U)^{-1}V^*A^{-1}.$$

31. Recall from Sect. 1.3 that $\lambda \in \mathbb{C}$ is an eigenvalue of a matrix M if $Mv = \lambda v$ for some vector $v \neq 0$. Let $A \in \mathbb{M}_m$, $B \in \mathbb{M}_n$, and $C \in \mathbb{M}_{m\times n}$.

    (a) Show (without using determinants) that λ is an eigenvalue of $\left(\begin{smallmatrix} A & C \\ 0 & B \end{smallmatrix}\right)$ if and only if λ is an eigenvalue of A or (and) B.
    (b) Let $\lambda_1, \ldots, \lambda_m$ be the eigenvalues of A and let $\mu_1, \ldots, \mu_n$ be the eigenvalues of B. Show that $\lambda_1, \ldots, \lambda_m$, $\mu_1, \ldots, \mu_n$ are the eigenvalues of $\left(\begin{smallmatrix} A & C \\ 0 & B \end{smallmatrix}\right)$. (Note that the eigenvalues need not be distinct.)

32. Let $M = \left(\begin{smallmatrix} A & B \\ C & D \end{smallmatrix}\right)$ be nonsingular and let $M^{-1} = \left(\begin{smallmatrix} X & Y \\ U & V \end{smallmatrix}\right)$, where the submatrices in the partitioned M and M^{-1} are all $n \times n$. Show that

$$\det B = (-1)^{n^2} \det Y \det M.$$

33. Let $M = \left(\begin{smallmatrix} A & B \\ C & D \end{smallmatrix}\right)$ in which A and D are square and A is nonsingular. Denote $M/A = D - CA^{-1}B$ (called the *Schur complement* of A in M).

    (a) Show that M is nonsingular if and only if M/A is nonsingular.

    (b) Let $M = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix}$ and $A = \left(\begin{smallmatrix} 1 & 2 \\ 4 & 5 \end{smallmatrix}\right)$. Find M/A. Is M invertible?

    (c) Let $E = (1)$. Find A/E and show that $(M/E)/(A/E) = M/A$.

34. Let A and B be real square matrices. If $AP = PB$ for some invertible complex matrix P, show that $AR = RB$ for some invertible real matrix R. In other words, if two real square matrices are similar over $\mathbb{C}$, then they are similar over $\mathbb{R}$.

35. Let $M = \left(\begin{smallmatrix} A & B \\ C & D \end{smallmatrix}\right)$. Assume that all involved inverses exist, and denote

$$S = D - CA^{-1}B, \quad T = A - BD^{-1}C.$$

    Show that each of the following expressions with blocks equals M^{-1}.

    (a) $\begin{pmatrix} A^{-1} + A^{-1}BS^{-1}CA^{-1} & -A^{-1}BS^{-1} \\ -S^{-1}CA^{-1} & S^{-1} \end{pmatrix}$.

    (b) $\begin{pmatrix} T^{-1} & -T^{-1}BD^{-1} \\ -D^{-1}CT^{-1} & D^{-1} + D^{-1}CT^{-1}BD^{-1} \end{pmatrix}$.

    (c) $\begin{pmatrix} T^{-1} & -A^{-1}BS^{-1} \\ -D^{-1}CT^{-1} & S^{-1} \end{pmatrix}$.

    (d) $\begin{pmatrix} T^{-1} & (C - DB^{-1}A)^{-1} \\ (B - AC^{-1}D)^{-1} & S^{-1} \end{pmatrix}$.

    (e) $\begin{pmatrix} I & -A^{-1}B \\ 0 & I \end{pmatrix} \begin{pmatrix} A^{-1} & 0 \\ 0 & S^{-1} \end{pmatrix} \begin{pmatrix} I & 0 \\ -CA^{-1} & I \end{pmatrix}$.

    (f) $\begin{pmatrix} A^{-1} & 0 \\ 0 & 0 \end{pmatrix} + \begin{pmatrix} A^{-1}B \\ -I \end{pmatrix} S^{-1}(CA^{-1}, -I)$.

36. Deduce the following inverse identities from the previous problem.

$$\begin{aligned} (A - BD^{-1}C)^{-1} &= A^{-1} + A^{-1}B(D - CA^{-1}B)^{-1}CA^{-1} \\ &= -C^{-1}D(B - AC^{-1}D)^{-1} \\ &= -(C - DB^{-1}A)^{-1}DB^{-1} \\ &= C^{-1}D(D - CA^{-1}B)^{-1}CA^{-1} \\ &= A^{-1}B(D - CA^{-1}B)^{-1}DB^{-1}. \end{aligned}$$

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2.3 The Ranks of Product AB and Sum $A + B$

This section is concerned with the ranks of the product AB and the sum $A + B$ in terms of the ranks of the matrices A and B.

Let $A \in \mathbb{M}_{m\times n}$. Through elementary operations, A can be brought to a matrix in the form $\left(\begin{smallmatrix} I_r & 0 \\ 0 & 0 \end{smallmatrix}\right)$. Equivalently, there are nonsingular matrices $P \in \mathbb{M}_m$ and $Q \in \mathbb{M}_n$ such that $PAQ = \left(\begin{smallmatrix} I_r & 0 \\ 0 & 0 \end{smallmatrix}\right)$ (Theorem 1.2.1). The nonnegative number r, i.e., the order of the identity matrix I_r, is uniquely determined by A (if $r = 0$, that is, $A = 0$, I_r is absent). Such an r associated to A is called the *rank* of A, and it is denoted by $\text{rank}(A)$. Note that for any nonsingular $S \in \mathbb{M}_m$ and $T \in \mathbb{M}_n$ (as products of elementary matrices)

$$\text{rank}(A) = \text{rank}(SAT). \tag{2.11}$$

The rank of a matrix can be defined in different but equivalent ways (Problem 38 of Sect. 1.2). For instance, it can be defined by *row rank* or *column rank*. The row (column) rank of a matrix A is the dimension of the vector space spanned by the rows (columns) of A, denoted by $\text{Row}(A)$ (resp. $\text{Col}(A)$). The row rank and column rank of a matrix are equal. This is obvious for $\left(\begin{smallmatrix} I_r & 0 \\ 0 & 0 \end{smallmatrix}\right)$. For the general case, let the row rank of an $m \times n$ matrix A be p and let the column rank be q. We show $p = q$. Suppose that columns $a_{i_1}, a_{i_2}, \ldots, a_{i_q}$ of A are linearly independent and span the column space of $A = (a_1, a_2, \ldots, a_n)$. For the jth column a_j of A, $j = 1, 2, \ldots, n$, we write

$$a_j = d_{1j}a_{i_1} + d_{2j}a_{i_2} + \cdots + d_{qj}a_{i_q} = Cd_j,$$

where $C = (a_{i_1}, a_{i_2}, \ldots, a_{i_q})$ and $d_j = (d_{1j}, d_{2j}, \ldots, d_{qj})^T$, $j = 1, 2, \ldots, n$.

Let $D = (d_{ij}) = (d_1, d_2, \ldots, d_n) \in \mathbb{M}_{q\times n}$. Then $A = CD$. It follows that every row of A is a linear combination of the rows of D. Thus, $p \le q$. A similar argument on A^T shows $q \le p$. Therefore, $p = q$.

We can also see that the row rank equals the column rank as follows. Denote the row and column ranks of a matrix X by $rr(X)$ and $cr(X)$, respectively, that is, $cr(X) = \dim \text{Im}\, X$ and $rr(X) = \dim \text{Im}\, X^T$.

Since $Ax = 0 \Leftrightarrow (Ax)^*(Ax) = 0 \Leftrightarrow A^*Ax = 0$ for $x \in \mathbb{C}^n$, some columns (say, columns $i_1, \ldots, i_k$) of A are linearly independent if and only if the corresponding columns $i_1, \ldots, i_k$ of A^*A are linearly independent. We obtain

$$cr(A) = cr(A^*A) = \dim \text{Im}(A^*A) \le \dim \text{Im}\, A^* = cr(A^*).$$

Hence, $cr(A^*) \le cr((A^*)^*) = cr(A)$. So $cr(A) = cr(A^*)$.

Since $cr(A) = rr(A^T)$, we have $q = cr(A) = cr(A^*) = rr(\overline{A}) = rr(A) = p$.

An elementary row operation does not change row space; an elementary column operation does not change column space. So, for the nonsingular P and Q, $cr(A) = rr(A) = rr(PA) = cr(PA) = cr(PAQ) = cr\left(\begin{smallmatrix} I_r & 0 \\ 0 & 0 \end{smallmatrix}\right) = r$.

Theorem 2.3.1 *The following rank identities hold for any matrix A.*

$$\text{rank}(A) = \text{rank}(A^T) = \text{rank}(\overline{A}) = \text{rank}(A^*) = \text{rank}(A^*A) = \text{rank}(AA^*).$$

We point out that $\text{rank}(A) \neq \text{rank}(AA^T)$ in general (unless A is real). For example, if $A = \begin{pmatrix} 1 & i \\ i & -1 \end{pmatrix}$, then $\text{rank}(A) = 1$ (over $\mathbb{C}$ as our underlying field) while $AA^T = 0$. Note that the rows of A are linearly independent over $\mathbb{R}$, but linearly dependent over $\mathbb{C}$. See also (1.5) of Sect. 1.2 for the rank identities.

We summarize some basic facts about the rank of a matrix as follows.

(i) The rank of matrix A is r, that is, $PAQ = \begin{pmatrix} I_r & 0 \\ 0 & 0 \end{pmatrix}$ for some invertible (products of elementary) matrices P and Q. (Theorem 1.2.1)

(ii) $r = \dim \text{Col}(A) = \dim \text{Im}\, A$. (Theorem 1.2.2; Theorem 1.3.1)

(iii) $r = \dim \text{Row}(A) = \dim \text{Col}(A^T)$. ((1.5) of Sect. 1.2; Theorem 2.3.1)

(iv) Some r columns (rows) of A are linearly independent, and any $r+1$ columns (rows) of A are linearly dependent. (Problem 38 of Sect. 1.2)

(v) Some $r \times r$ submatrix of A is nonsingular, and any submatrix of order $r+1$ is singular. Equivalently, A has an $r \times r$ nonzero minor, and all minors of order $r+1$ or higher are zero. (Problem 38 of Sect. 1.2)

Example 2.3.1 Assume that the matrices below have appropriate sizes.

(1) $\text{rank}(A) \leq \text{rank}(M)$ if A is a submatrix of a matrix M.

Reason: If $a_1, \ldots, a_k$ are linearly independent columns of A, then the k columns of M that contain $a_1, \ldots, a_k$ are linearly independent.

(2) $\text{rank}(A+B) \leq \text{rank}(A) + \text{rank}(B)$.

This is because of the fact that $\text{Im}(A+B) \subseteq \text{Im}\, A + \text{Im}\, B$ and

$$\begin{aligned} \text{rank}(A+B) &= \dim \text{Im}(A+B) \leq \dim(\text{Im}\, A + \text{Im}\, B) \\ &\leq \dim \text{Im}\, A + \dim \text{Im}\, B = \text{rank}(A) + \text{rank}(B). \end{aligned}$$

Alternatively, see Problem 19 of Sect. 2.1 for another proof.

(3) $\text{rank}(A) + \text{rank}(B) \leq \text{rank} \begin{pmatrix} A & X \\ 0 & B \end{pmatrix}$ for any X. Equality holds if $X = 0$.

Let A have n columns. Observe that if columns $i_1, \ldots, i_p$ of A are linearly independent and columns $j_1, \ldots, j_q$ of B are linearly independent, then columns $i_1, \ldots, i_p, n+j_1, \ldots, n+j_q$ of $\begin{pmatrix} A & X \\ 0 & B \end{pmatrix}$ are linearly independent. Likewise, we have $\text{rank}(A) + \text{rank}(B) \leq \text{rank} \begin{pmatrix} 0 & A \\ B & X \end{pmatrix}$.

(4) $\text{rank}(AB) \leq \text{rank}(A)$; $\text{rank}(AB) \leq \text{rank}(B)$.

Note that $\text{Im}(AB)$ is a subspace of $\text{Im}\, A$ (Sect. 1.2). Thus,

$$\text{rank}(AB) = \dim \text{Im}(AB) \leq \dim \text{Im}\, A = \text{rank}(A);$$

$$\text{rank}(AB) = \text{rank}(AB)^T = \text{rank}(B^T A^T) \leq \text{rank}(B^T) = \text{rank}(B).$$

Another proof: Because $\operatorname{rank}(PAQ) = \operatorname{rank}(A)$ for nonsingular matrices P and Q, without loss of generality, we assume that $A = \left(\begin{smallmatrix} I_a & 0 \\ 0 & 0 \end{smallmatrix}\right)$, where $a = \operatorname{rank}(A)$. Let B_1 consist of the first a rows of B and let $B = \left(\begin{smallmatrix} B_1 \\ B_2 \end{smallmatrix}\right)$. Then $AB = \left(\begin{smallmatrix} B_1 \\ 0 \end{smallmatrix}\right)$. It follows that $\operatorname{rank}(AB) = \operatorname{rank}(B_1) \leq \operatorname{rank}(B)$. Similarly, $\operatorname{rank}(AB) \leq \operatorname{rank}(A)$.

(5) $\operatorname{rank}(A) + \operatorname{rank}(B) \leq \operatorname{rank}(AB) + n$, where A has n columns.

This follows from part (3) and by the elementary operations:

$$\begin{pmatrix} I_n & 0 \\ 0 & AB \end{pmatrix} \to \begin{pmatrix} I_n & 0 \\ A & AB \end{pmatrix} \to \begin{pmatrix} I_n & -B \\ A & 0 \end{pmatrix}.$$

Another proof: We assume again that $A = \left(\begin{smallmatrix} I_a & 0 \\ 0 & 0 \end{smallmatrix}\right)$ and use the inequality in part (2). It is sufficient to observe that $\operatorname{rank}\left(\begin{smallmatrix} 0 \\ B_2 \end{smallmatrix}\right) \leq n - a$ and that $\operatorname{rank}(B) = \operatorname{rank}\left(\begin{smallmatrix} B_1 \\ B_2 \end{smallmatrix}\right) = \operatorname{rank}\left(\left(\begin{smallmatrix} B_1 \\ 0 \end{smallmatrix}\right) + \left(\begin{smallmatrix} 0 \\ B_2 \end{smallmatrix}\right)\right) \leq \operatorname{rank}\left(\begin{smallmatrix} B_1 \\ 0 \end{smallmatrix}\right) + \operatorname{rank}\left(\begin{smallmatrix} 0 \\ B_2 \end{smallmatrix}\right) = \operatorname{rank}(AB) + \operatorname{rank}\left(\begin{smallmatrix} 0 \\ B_2 \end{smallmatrix}\right) \leq \operatorname{rank}(AB) + n - \operatorname{rank}(A)$.

(6) $\operatorname{rank}(ABC) \geq \operatorname{rank}(AB) + \operatorname{rank}(BC) - \operatorname{rank}(B)$.

Proof. Use part (3) and observe that

$$\begin{pmatrix} I & -A \\ 0 & I \end{pmatrix} \begin{pmatrix} 0 & AB \\ BC & B \end{pmatrix} \begin{pmatrix} I & 0 \\ -C & I \end{pmatrix} = \begin{pmatrix} -ABC & 0 \\ 0 & B \end{pmatrix}. \blacksquare$$

Inequality (5) is known as the *Sylvester rank inequality*, and (6) is called the *Frobenius rank inequality*. The two inequalities are in fact equivalent.

Now we present $\operatorname{rank}(AB)$ and $\operatorname{rank}(A + B)$ in identity forms.

Theorem 2.3.2 *Let $A \in \mathbb{M}_{m\times n}$ and $B \in \mathbb{M}_{n\times p}$. Then*

$$\operatorname{rank}(AB) = \operatorname{rank}(B) - \dim(\operatorname{Im} B \cap \operatorname{Ker} A). \tag{2.12}$$

Proof. This is in essence Example 1.3.2 of Sect. 1.3, Chap. 1. ∎

Note that Sylvester's rank inequality is derived from (2.12) because

$$\dim \operatorname{Im} A + \dim(\operatorname{Im} B \cap \operatorname{Ker} A) \leq \dim \operatorname{Im} A + \dim \operatorname{Ker} A = n.$$

We turn our attention to the rank of sum $A + B$.

Theorem 2.3.3 *Let $A, B \in \mathbb{M}_{m\times n}$, $C = (I_m, I_m)$, and $D = \left(\begin{smallmatrix} A \\ B \end{smallmatrix}\right)$. Then*

$$\begin{aligned} \operatorname{rank}(A + B) \;=\; & \operatorname{rank}(A) + \operatorname{rank}(B) - \dim(\operatorname{Ker} C \cap \operatorname{Im} D) \\ & - \dim(\operatorname{Im} A^* \cap \operatorname{Im} B^*). \end{aligned} \tag{2.13}$$

Proof. Write

$$A + B = (I_m, I_m) \begin{pmatrix} A \\ B \end{pmatrix} = CD.$$

By the previous theorem, we have

$$\operatorname{rank}(A+B) = \operatorname{rank}(CD) = \operatorname{rank}(D) - \dim(\operatorname{Im} D \cap \operatorname{Ker} C). \tag{2.14}$$

We compute

$$\begin{aligned}
\operatorname{rank}(D) &= \operatorname{rank}(D^*) = \operatorname{rank}(A^*, B^*) \\
&= \dim \operatorname{Im}(A^*, B^*) \quad \text{(by Problem 9)} \\
&= \dim(\operatorname{Im} A^* + \operatorname{Im} B^*) \\
&= \dim \operatorname{Im} A^* + \dim \operatorname{Im} B^* \\
&\quad - \dim(\operatorname{Im} A^* \cap \operatorname{Im} B^*) \\
&= \operatorname{rank}(A^*) + \operatorname{rank}(B^*) - \dim(\operatorname{Im} A^* \cap \operatorname{Im} B^*) \\
&= \operatorname{rank}(A) + \operatorname{rank}(B) - \dim(\operatorname{Im} A^* \cap \operatorname{Im} B^*).
\end{aligned}$$

Substituting $\operatorname{rank}(D)$ in (2.14) with the above expression reveals (2.13). ∎

Problems

1. Let A and B be n-square matrices. If $\operatorname{rank}(A) < \frac{n}{2}$ and $\operatorname{rank}(B) < \frac{n}{2}$, show that $\det(A + \lambda B) = 0$ for any complex number λ.
2. Find $\operatorname{rank}(2I - A)$ and $\operatorname{rank}(2I - B)$ for $A = \begin{pmatrix} 2 & 1 & 0 \\ 0 & 2 & 1 \\ 0 & 0 & 2 \end{pmatrix}$ and $B = \begin{pmatrix} 2 & 0 & 0 \\ 0 & 2 & 1 \\ 0 & 0 & 2 \end{pmatrix}$.
3. Prove the statement: If B is an $n \times n$ invertible matrix, then $\operatorname{rank}(AB) = \operatorname{rank}(A)$ for any $m \times n$ matrix A. Show that the converse is also true.
4. Is the sum of two singular matrices singular? How about the product?
5. Let A be an $m \times n$ matrix. If $\operatorname{rank}(A) = m$, show that there exists an $n \times m$ matrix B such that $AB = I_m$; if $\operatorname{rank}(A) = n$, show that there exists an $m \times n$ matrix B such that $BA = I_n$. If $m < n$, does there exist an $n \times m$ matrix B such that $BA = I_n$?
6. Let A be an $m \times n$ matrix. Show that for any $s \times m$ matrix X with rank m and any $n \times t$ matrix Y with rank n,

$$\operatorname{rank}(A) = \operatorname{rank}(XA) = \operatorname{rank}(AY) = \operatorname{rank}(XAY).$$

   In particular, $\operatorname{rank}(PAQ) = \operatorname{rank}(A)$ for invertible matrices P and Q.
7. Let $A \in \mathbb{M}_{m\times n}$ and $B \in \mathbb{M}_{n\times p}$. If $\operatorname{rank}(AB) = \operatorname{rank}(B)$, show that

$$ABX = ABY \;\Leftrightarrow\; BX = BY \;\; (\text{where } X, Y \in \mathbb{M}_{p\times q})$$

   and if $\operatorname{rank}(AB) = \operatorname{rank}(A)$, show that

$$XAB = YAB \quad \Leftrightarrow \quad XA = YA \;\; (\text{where } X, Y \in \mathbb{M}_{l\times m}).$$

8. Find examples of matrices A and B.
   (a) Matrix A contains no zero columns, the columns of A are mutually orthogonal, but the rows of A are not mutually orthogonal.
   (b) Can the rows of the matrix A in (a) be linearly dependent?
   (c) Why does such a matrix A in (a) have at least two different singular values? That is, the singular values of A cannot be all the same.
   (d) The columns of B are mutually orthogonal and the rows of B are also mutually orthogonal, but B is not a normal matrix.

9. For any matrices A and B of the same size, show that
$$\operatorname{Im}(A, B) = \operatorname{Im}(A) + \operatorname{Im}(B).$$

10. Let A be an $m \times n$ matrix. Show that for any $n \times m$ matrix B,
$$\dim \operatorname{Im}(A) + \dim \operatorname{Ker}(A) = \dim \operatorname{Im}(BA) + \dim \operatorname{Ker}(BA).$$
Is it true that
$$\dim \operatorname{Im}(BA) + \dim \operatorname{Ker}(BA) = \dim \operatorname{Im}(B) + \dim \operatorname{Ker}(B)?$$

11. Let A and B be $m \times n$ and $n \times m$ matrices, respectively. Show that
$$\det(AB) = 0 \quad \text{if } m > n.$$

12. For matrices A and B of the same size, show that
$$|\operatorname{rank}(A) - \operatorname{rank}(B)| \leq \operatorname{rank}(A \pm B).$$

13. Let A and B be square matrices of the same size. Show that
$$\operatorname{rank}(AB - I) \leq \operatorname{rank}(A - I) + \operatorname{rank}(B - I).$$

14. Let A be an n-square matrix. Show that $\operatorname{rank}(A^2) \geq 2\operatorname{rank}(A) - n$.

15. Let A be an n-square matrix. Prove each of the following statements.
   (a) If $A^2 = 0$, then $\operatorname{rank}(A + I_n) = n$.
   (b) If $A^2 = I_n$, then $\operatorname{rank}(A + I_n) + \operatorname{rank}(A - I_n) = n$.
   (c) If $A^2 = A$, then $\operatorname{rank}(A) + \operatorname{rank}(I_n - A) = n$.

16. Let A and B be n-square matrices. Show that
$$\operatorname{rank}(A - ABA) = \operatorname{rank}(A) + \operatorname{rank}(I_n - BA) - n.$$

17. Let A be a square matrix. If $(A^*A)^k = (A^*)^k A^k$ for some positive integer k, show that $\operatorname{rank}(A^p) = \operatorname{rank}(A)$ for all positive integers p.

18. Let A and B be n-square matrices. If $A + B = I_n$ and $\text{rank}(A) + \text{rank}(B) = n$, show that $A^2 = A$, $B^2 = B$, and $AB = BA = 0$.

19. Let A and B be $n \times p$ and $q \times n$ matrices, respectively. Show that

$$\text{rank}\begin{pmatrix} A & I_n \\ 0 & B \end{pmatrix} = \text{rank}\begin{pmatrix} 0 & B \\ A & I_n \end{pmatrix} = n + \text{rank}(BA).$$

Can $\text{rank}(BA)$ on the right-hand side be changed to $\text{rank}(AB)$?

20. Let A, C be $m \times n$ matrices and let B, D be $m \times p$ matrices. Show that

$$\text{rank}\begin{pmatrix} AA^* + BB^* & AC^* + BD^* \\ CA^* + DB^* & CC^* + DD^* \end{pmatrix} = \text{rank}\begin{pmatrix} A & B \\ C & D \end{pmatrix}.$$

Show also that the block matrix on the left is positive semidefinite.

21. Let $\left(\begin{smallmatrix} I_n & X \\ Y & Z \end{smallmatrix}\right)$ be a block matrix. Find nonsingular block matrices P and Q such that $P\left(\begin{smallmatrix} I_n & X \\ Y & Z \end{smallmatrix}\right)Q$ is in the form $\left(\begin{smallmatrix} I_n & 0 \\ 0 & W \end{smallmatrix}\right)$, where W is some matrix.

22. Let A be an $m \times n$ matrix having $\text{rank}(A) = r$. Show that for every positive integer k, $r \le k \le n$, there exists an $n \times n$ matrix B such that $AB = 0$ and $\text{rank}(A) + \text{rank}(B) = k$. In particular, there exists an $n \times n$ matrix B of rank $n - r$ such that $AB = 0$.

23. Let A be an $m \times n$ matrix with rank n. If $m > n$, show that there is a matrix B of size $(m - n) \times m$ and a matrix C of size $m \times (m - n)$, both of rank $m - n$, such that $BA = 0$ and (A, C) is nonsingular.

24. Let $A, B \in \mathbb{M}_n$. If $AB = 0$, show that $\dim \text{Ker}(A) + \dim \text{Ker}(B) \ge n$.

25. Let $A, B \in \mathbb{M}_n$. Show that $\text{Ker}(A) = \text{Ker}(B)$ if and only if $A = PB$ and $B = QA$ for some $n \times n$ matrices P and Q.

26. Let A and B be matrices of appropriate sizes. Show that

    (a) $\text{Col}(A) \subseteq \text{Col}(B) \Leftrightarrow A = BC$ for some matrix C.

    (b) $\text{Row}(A) \subseteq \text{Row}(B) \Leftrightarrow A = RB$ for some matrix R.

27. Let A and B be $m \times n$ matrices. Show that

$$\text{Col}(A + B) \subseteq \text{Col}(A) + \text{Col}(B)$$

and that the following statements are equivalent:

    (a) $\text{Col}(A) \subseteq \text{Col}(A + B)$.

    (b) $\text{Col}(B) \subseteq \text{Col}(A + B)$.

    (c) $\text{Col}(A + B) = \text{Col}(A) + \text{Col}(B)$.

28. Let A be an $m \times n$ matrix factored as $A = UDV$, where $U = (u_1, \ldots, u_r)$ (partitioned in columns), $D = \operatorname{diag}(d_1, \ldots, d_r)$, and $V = \begin{pmatrix} v_1 \\ \vdots \\ v_r \end{pmatrix}$ (partitioned in rows). Show that $A = d_1 u_1 v_1 + \cdots + d_r u_r v_r$.

29. Let A be an m-square complex matrix. Show that there exists a polynomial p over $\mathbb{C}$ of degree at most m^2 such that $\operatorname{rank}(p(A)) = 0$.

30. Prove or disprove that for any n-square matrices A and B,

$$\text{(a) } \operatorname{rank}\begin{pmatrix} A \\ B \end{pmatrix} = \operatorname{rank}(A, B). \quad \text{(b) } \operatorname{rank}\begin{pmatrix} A \\ B \end{pmatrix} = \operatorname{rank}(A^T, B^T).$$

31. Let A be an $m \times n$ matrix and let B be a $p \times n$ matrix. Show that

$$\operatorname{Ker}(A) \cap \operatorname{Ker}(B) = \operatorname{Ker}\begin{pmatrix} A \\ B \end{pmatrix}.$$

32. Let A, B, and C be matrices with the same number of rows. Show that

$$\begin{aligned} \operatorname{rank}(A, B, C) &\leq \operatorname{rank}(A, B) + \operatorname{rank}(B, C) - \operatorname{rank}(B) \\ &\leq \operatorname{rank}(A) + \operatorname{rank}(B) + \operatorname{rank}(C). \end{aligned}$$

33. Let A and B be matrices of the same size. Prove the rank inequalities

$$\operatorname{rank}(A + B) \leq \operatorname{rank}\begin{pmatrix} A \\ B \end{pmatrix} \leq \operatorname{rank}(A) + \operatorname{rank}(B)$$

and

$$\operatorname{rank}(A + B) \leq \operatorname{rank}(A, B) \leq \operatorname{rank}(A) + \operatorname{rank}(B)$$

by writing

$$A + B = (A, B)\begin{pmatrix} I \\ I \end{pmatrix} = (I, I)\begin{pmatrix} A \\ B \end{pmatrix}.$$

Additionally, show that

$$\operatorname{rank}\begin{pmatrix} A & B \\ C & D \end{pmatrix} \leq \operatorname{rank}(A) + \operatorname{rank}(B) + \operatorname{rank}(C) + \operatorname{rank}(D).$$

34. Let $\mathcal{A}$ be a linear map on a finite-dimensional vector space. Show that

$$\operatorname{Ker}(\mathcal{A}) \subseteq \operatorname{Ker}(\mathcal{A}^2) \subseteq \operatorname{Ker}(\mathcal{A}^3) \subseteq \cdots$$

and that

$$\operatorname{Im}(\mathcal{A}) \supseteq \operatorname{Im}(\mathcal{A}^2) \supseteq \operatorname{Im}(\mathcal{A}^3) \supseteq \cdots.$$

Further show that there are only finite proper inclusions in each chain.

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2.4 The Eigenvalues of AB, BA, and $\left(\begin{smallmatrix} 0 & A \\ A^* & 0 \end{smallmatrix}\right)$

Let A and B be $n \times n$ matrices. The products AB and BA need not be equal; AB and BA may have different ranks. But what do they have in common?

Example 2.4.1 The eigenvalues of AB and BA.

(a) Let $A = \left(\begin{smallmatrix} 1 & 0 \\ 1 & 0 \end{smallmatrix}\right)$, $B = \left(\begin{smallmatrix} 0 & 0 \\ 1 & 1 \end{smallmatrix}\right)$. Then $AB = \left(\begin{smallmatrix} 0 & 0 \\ 0 & 0 \end{smallmatrix}\right)$, $BA = \left(\begin{smallmatrix} 0 & 0 \\ 2 & 0 \end{smallmatrix}\right)$. AB and BA have the eigenvalues $0, 0$. $\operatorname{rank}(AB) = 0$, while $\operatorname{rank}(BA) = 1$.

(b) Let $A = \left(\begin{smallmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \end{smallmatrix}\right)$, $B = \left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \\ 0 & 0 \end{smallmatrix}\right)$. Then $AB = \left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right)$, $BA = \left(\begin{smallmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{smallmatrix}\right)$. The eigenvalues of AB are $i, -i$; the eigenvalues of BA are $i, -i, 0$.

(c) Let $A = \left(\begin{smallmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{smallmatrix}\right)$, $B = \left(\begin{smallmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{smallmatrix}\right)$. Then $AB = \left(\begin{smallmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{smallmatrix}\right)$, $BA = \left(\begin{smallmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{smallmatrix}\right)$. The eigenvalues of AB are $1, 1, 0$, so are the eigenvalues of BA. ∎

The above examples suggest that if A and B are square, AB and BA have the same (multi)set of eigenvalues (we simply say that AB and BA have the same eigenvalues); if A and B are rectangular, AB and BA have the same nonzero eigenvalues. Are these coincidences? Or can one find an example such that AB has only zero eigenvalues but BA has some nonzero eigenvalues? The following theorem gives a negative answer to the question.

Theorem 2.4.1 *Let A and B be $m \times n$ and $n \times m$ matrices, respectively. Then AB and BA have the same nonzero eigenvalues, including (algebraic) multiplicities. If $m = n$, then AB and BA have the same eigenvalues.*

Proof 1. Use determinants. Notice that

$$\begin{pmatrix} I_m & -A \\ 0 & \lambda I_n \end{pmatrix} \begin{pmatrix} \lambda I_m & A \\ B & I_n \end{pmatrix} = \begin{pmatrix} \lambda I_m - AB & 0 \\ \lambda B & \lambda I_n \end{pmatrix}$$

and that

$$\begin{pmatrix} I_m & 0 \\ -B & \lambda I_n \end{pmatrix} \begin{pmatrix} \lambda I_m & A \\ B & I_n \end{pmatrix} = \begin{pmatrix} \lambda I_m & A \\ 0 & \lambda I_n - BA \end{pmatrix}.$$

Taking determinants and equating the right-hand sides, we obtain that

$$\lambda^n \det(\lambda I_m - AB) = \lambda^m \det(\lambda I_n - BA). \tag{2.15}$$

Thus, $\det(\lambda I_m - AB) = 0$ if and only if $\det(\lambda I_n - BA) = 0$ when $\lambda \neq 0$. It is immediate that AB and BA have the same nonzero eigenvalues, including multiplicities (by factorizations of the characteristic polynomials).

If $m < n$ and if $\lambda_1, \ldots, \lambda_m$ are the eigenvalues of AB, then the eigenvalues of BA are $\lambda_1, \ldots, \lambda_m$ plus $n - m$ zeros. (The case $m > n$ is similar.)

Proof 2. Use matrix similarity. Start with the block matrix $\left(\begin{smallmatrix} 0 & 0 \\ B & 0 \end{smallmatrix}\right)$. Add the second row multiplied by A from the left to the first row (to get AB), and perform the similar operations for columns (to get BA):

$$\begin{pmatrix} 0 & 0 \\ B & 0 \end{pmatrix} \to \begin{pmatrix} AB & 0 \\ B & 0 \end{pmatrix}, \quad \begin{pmatrix} 0 & 0 \\ B & 0 \end{pmatrix} \to \begin{pmatrix} 0 & 0 \\ B & BA \end{pmatrix}.$$

Write these in identity forms:

$$\begin{pmatrix} I_m & A \\ 0 & I_n \end{pmatrix} \begin{pmatrix} 0 & 0 \\ B & 0 \end{pmatrix} = \begin{pmatrix} AB & 0 \\ B & 0 \end{pmatrix},$$

$$\begin{pmatrix} 0 & 0 \\ B & 0 \end{pmatrix} \begin{pmatrix} I_m & A \\ 0 & I_n \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ B & BA \end{pmatrix}.$$

It follows that

$$\begin{pmatrix} I_m & A \\ 0 & I_n \end{pmatrix}^{-1} \begin{pmatrix} AB & 0 \\ B & 0 \end{pmatrix} \begin{pmatrix} I_m & A \\ 0 & I_n \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ B & BA \end{pmatrix}.$$

Thus, $\left(\begin{smallmatrix} AB & 0 \\ B & 0 \end{smallmatrix}\right)$ and $\left(\begin{smallmatrix} 0 & 0 \\ B & BA \end{smallmatrix}\right)$ are similar. So $\left(\begin{smallmatrix} AB & 0 \\ 0 & 0 \end{smallmatrix}\right)$ and $\left(\begin{smallmatrix} 0 & 0 \\ 0 & BA \end{smallmatrix}\right)$ have the same eigenvalues. Consequently, AB and BA have the same nonzero eigenvalues counted according to multiplicities. (Are AB and BA similar?)

Proof 3. Use the continuity argument. We first deal with the case $m = n$. If A is nonsingular, then $BA = A^{-1}(AB)A$. Thus, AB and BA are similar and have the same eigenvalues (including zeros and multiplicities).

Let A be singular. Note that $\det(\epsilon I + A)$ as a polynomial in ϵ has n zeros. Let $\delta > 0$ be such that $\epsilon I + A$ is nonsingular for ϵ, $0 < \epsilon < \delta$. Then

$$(\epsilon I + A)B \quad \text{and} \quad B(\epsilon I + A)$$

are similar and have the same characteristic polynomials. Therefore,

$$\det\big(\lambda I_n - (\epsilon I_n + A)B\big) = \det\big(\lambda I_n - B(\epsilon I_n + A)\big), \quad 0 < \epsilon < \delta.$$

Since both sides are continuous functions of ϵ, letting $\epsilon \to 0$ yields

$$\det(\lambda I_n - AB) = \det(\lambda I_n - BA).$$

Thus, AB and BA have the same eigenvalues.

If $m \neq n$, assuming $m < n$, we add zero rows to A and zero columns to B such that $A_1 = \left(\begin{smallmatrix} A \\ 0 \end{smallmatrix}\right)$ and $B_1 = (B, 0)$ are n-square matrices. Then

$$A_1 B_1 = \begin{pmatrix} AB & 0 \\ 0 & 0 \end{pmatrix} \quad \text{and} \quad B_1 A_1 = BA.$$

It follows that $A_1 B_1$ and $B_1 A_1$ have the same eigenvalues. As a result, AB and BA have the same nonzero eigenvalues (including multiplicities).

Proof 4. Treat matrices as operators. We show that if $\lambda I_m - AB$ is singular, so is $\lambda I_n - BA$. It may be assumed that $\lambda = 1$ (by multiplying $\frac{1}{\lambda}$ otherwise).

If $I_m - AB$ is invertible, let $X = (I_m - AB)^{-1}$. We compute

$$\begin{aligned}(I_n - BA)(I_n + BXA) &= I_n + BXA - BA - BABXA \\ &= I_n + (BXA - BABXA) - BA \\ &= I_n + B(I_m - AB)XA - BA \\ &= I_n + BA - BA = I_n.\end{aligned}$$

Thus, $I_n - BA$ is invertible. Note that this approach gives no information about the multiplicities of the nonzero eigenvalues. ∎

As a side product, observing that the eigenvalues of $I + X$ are 1 plus the eigenvalues of X, or simply put, $\lambda(I + X) = 1 + \lambda(X)$, we have that (see also (2.5) of Sect. 2.2), for $A \in \mathbb{M}_{m\times n}$ and $B \in \mathbb{M}_{n\times m}$,

$$\det(I_m + AB) = \det(I_n + BA).$$

Thus, $I_m + AB$ is invertible if and only if $I_n + BA$ is invertible. Moreover,

$$\operatorname{tr}(AB) = \operatorname{tr}(BA). \text{ (Trace cyclic property; see Theorem 1.3.4.)}$$

Example 2.4.2 AB and BA have the same nonzero eigenvalues.

(1) Let $e = (1, \ldots, 1) \in \mathbb{R}^n$ be the all-ones row vector and let E be the all-ones $n \times n$ matrix. Then $E = e^T e$. The nonzero eigenvalue of E is $ee^T = n$. The eigenvalues of E are n and $n - 1$ zeros. $\operatorname{tr}(E) = n$.

(2) For matrix $A \neq 0$, the nonzero eigenvalues of A^*A and AA^* are the same, and $\operatorname{tr}(A^*A) = \operatorname{tr}(AA^*)$. In addition, if (λ, x) is an eigenpair of A^*A, i.e., $A^*Ax = \lambda x$, $x \neq 0$, then $0 \leq x^*A^*Ax = \lambda x^*x$, and $\lambda \geq 0$.

(3) If A is a matrix of rank one, the only nonzero eigenvalue of A is $\operatorname{tr}(A)$. Because $\operatorname{rank}(A) = 1$, all rows of A are multiples of each other. We can write $A = xy^T$ for column vectors x and y. Then the eigenvalues of A are y^Tx and 0's, and $\operatorname{tr}(A) = \operatorname{tr}(xy^T) = \operatorname{tr}(y^Tx) = y^Tx$.

(4) Let $A, B, C \in \mathbb{M}_n$. Then $\operatorname{tr}(ABC) = \operatorname{tr}(CAB) = \operatorname{tr}(BCA)$.

(5) Let A and B be matrices having the same number of rows. Let $M = AA^* + BB^*$ and $N = \begin{pmatrix} A^*A & A^*B \\ B^*A & B^*B \end{pmatrix}$. Then M and N have the same nonzero eigenvalues (including multiplicities). The nonzero eigenvalues are positive, because $M = (A, B)\begin{pmatrix} A^* \\ B^* \end{pmatrix}$ and $N = \begin{pmatrix} A^* \\ B^* \end{pmatrix}(A, B)$. By part (2), the eigenvalues of M and N are nonnegative. ∎

We end the section with an important theorem about the eigenvalues of a block Hermitian matrix in terms of so-called singular values. Let A be an $m \times n$ matrix. The nonnegative square roots of the eigenvalues of A^*A are called *singular values* of A, denoted by $\sigma_i(A) = \sqrt{\lambda_i(A^*A)}$, $i = 1, \ldots, n$.

Example 2.4.3 Let $A = \left(\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right)$, $B = \left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right)$, and $C = \left(\begin{smallmatrix} 0 & 1 & i \\ 0 & 0 & 0 \end{smallmatrix}\right)$. Then the singular values of A, B, and C are $1, 0$; $1, 1$; and $\sqrt{2}, 0, 0$, respectively. ∎

If the rank of A is $r \neq 0$, then A has r nonzero singular values. This is seen as follows. Let $A = P\begin{pmatrix} B \\ 0 \end{pmatrix}$, where $P \in \mathbb{M}_m$ is nonsingular (via row operations), and $B \in \mathbb{M}_{r\times n}$ has rank r. By the QR decomposition (Theorem 1.4.4), let $P = QR$, where $Q \in \mathbb{M}_m$ is unitary, R is nonsingular and upper-triangular. Partition $R = \left(\begin{smallmatrix} R_1 & R_2 \\ 0 & R_3 \end{smallmatrix}\right)$, where R_1 is $r \times r$. Then

$$AA^* = P\begin{pmatrix} B \\ 0 \end{pmatrix}(B^*, 0)P^* = QR\begin{pmatrix} BB^* & 0 \\ 0 & 0 \end{pmatrix}R^*Q^* = Q\begin{pmatrix} R_1BB^*R_1^* & 0 \\ 0 & 0 \end{pmatrix}Q^*.$$

We see that AA^* has the same eigenvalues as $\begin{pmatrix} R_1^*B^*BR_1 & 0 \\ 0 & 0 \end{pmatrix}$, which has r nonzero eigenvalues because $R_1^*B^*BR_1$ (of rank r) is nonsingular.

We arrange the singular values of $A \in \mathbb{M}_{m\times n}$ of rank r in the order:

$$\sigma_{\max}(A) = \sigma_1(A) \geq \cdots \geq \sigma_r(A) \geq \cdots \geq \sigma_n(A) = \sigma_{\min}(A).$$

We will study more about A^*A (and its square root) in later chapters.

Theorem 2.4.2 *Let $M = \begin{pmatrix} 0 & A \\ A^* & 0 \end{pmatrix}$, where $A \in \mathbb{M}_{m\times n}$. If $\sigma_1, \ldots, \sigma_r$ are the nonzero singular values of A, $r = \operatorname{rank}(A)$, then the eigenvalues of M are*

$$\sigma_1, \ldots, \sigma_r, \overbrace{0, \ldots, 0}^{m+n-2r}, -\sigma_r, \ldots, -\sigma_1.$$

Proof. We compute the characteristic polynomial of the $(m+n)$-square matrix M. Note that $\lambda I_{m+n} - M = \begin{pmatrix} \lambda I_m & -A \\ -A^* & \lambda I_n \end{pmatrix}$ and for any nonzero $\lambda \in \mathbb{C}$,

$$\begin{pmatrix} I_m & 0 \\ \frac{1}{\lambda}A^* & I_n \end{pmatrix}\begin{pmatrix} \lambda I_m & -A \\ -A^* & \lambda I_n \end{pmatrix} = \begin{pmatrix} \lambda I_m & -A \\ 0 & \lambda I_n - \frac{1}{\lambda}A^*A \end{pmatrix}.$$

Taking the determinants of both sides, we obtain, for $\lambda \neq 0$,

$$\det(\lambda I_{m+n} - M) = \lambda^{m-n}\det(\lambda^2 I_n - A^*A).$$

Thus, the nonzero eigenvalues λ of M are the nonzero solutions λ to

$$\det(\lambda^2 I_n - A^*A) = 0,$$

that is, they are the r nonzero singular values $\sigma_i(A)$ of A and their negatives $-\sigma_i(A)$. The remaining $m + n - 2r$ eigenvalues of M are zeros. ∎

Example 2.4.4 Let $A = \left(\begin{smallmatrix} 0 & 1 & i \\ -1 & 0 & 0 \end{smallmatrix}\right)$. The singular values of A are $\sqrt{2}$, 1, 0. $\begin{pmatrix} 0 & A \\ A^* & 0 \end{pmatrix}$ is a 6×6 Hermitian matrix with eigenvalues $\sqrt{2}$, 1, 0, 0, -1, $-\sqrt{2}$. ∎

Problems

1. Show that $\operatorname{tr}(XY) = \operatorname{tr}(YX)$ for any $X \in \mathbb{M}_{m\times n}$ and any $Y \in \mathbb{M}_{n\times m}$. In particular, $\operatorname{tr}(X^*X) = \operatorname{tr}(XX^*)$, and $\operatorname{tr}(xy^*) = y^*x$ for $x, y \in \mathbb{C}^n$.

2. What are the singular values of a unitary matrix?

3. Show that row and column permutations do not change the singular values of a matrix (or $\sigma(PAQ) = \sigma(A)$ for permutation matrices P, Q).

4. Let $x \in \mathbb{C}^n$ be a column vector. Show that

   (a) $xx^* \in \mathbb{M}_n$ has eigenvalues x^*x, $0, \ldots, 0$ ($n-1$ 0's).

   (b) $\left(\begin{smallmatrix} 0 & x^*x \\ x^*x & 0 \end{smallmatrix}\right) \in \mathbb{M}_2$ has eigenvalues x^*x, $-x^*x$.

   (c) $\left(\begin{smallmatrix} 0 & x \\ x^* & 0 \end{smallmatrix}\right) \in \mathbb{M}_{n+1}$ has eigenvalues $(x^*x)^{1/2}$, $-(x^*x)^{1/2}$, $0, \ldots, 0$.

   (d) $\left(\begin{smallmatrix} 0 & xx^* \\ xx^* & 0 \end{smallmatrix}\right) \in \mathbb{M}_{n+1}$ has eigenvalues x^*x, $-x^*x$, $0, \ldots, 0$.

   (e) $\left(\begin{smallmatrix} x^*x & x^* \\ x & I_n \end{smallmatrix}\right) \in \mathbb{M}_{n+1}$ has eigenvalues $1 + x^*x$, $1, \ldots, 1$.

5. For any square matrices A and B of the same size, show that

$$\operatorname{tr}(A+B)^2 = \operatorname{tr} A^2 + 2\operatorname{tr}(AB) + \operatorname{tr} B^2.$$

   Does it follow that $(A+B)^2 = A^2 + 2AB + B^2$?

6. Answer true or false for $A \in \mathbb{M}_{m\times n}, B \in \mathbb{M}_{n\times m}$ with $m = n$ or $m \neq n$.

   (a) $\det(AB) = \det(BA)$.

   (b) $\operatorname{tr}(AB) = \operatorname{tr}(BA)$.

   (c) $\operatorname{rank}(AB) = \operatorname{rank}(BA)$.

   (d) $\lambda(AB) = \lambda(BA)$, i.e., AB and BA have the same eigenvalues.

   (e) $\sigma(AB) = \sigma(BA)$, i.e., AB and BA have the same singular values.

7. Let A and B be $m \times n$ and $n \times m$ matrices, respectively, $m < n$. If the eigenvalues of AB are $\lambda_1, \ldots, \lambda_m$, what are the eigenvalues of BA?

8. Let A and B be 3×5 and 5×3 complex matrices, respectively. If the eigenvalues of AB are 1, $\sqrt{2}$, and $1+i$, what are the eigenvalues (with multiplicities) of BA? Find $\operatorname{tr}(AB)$, $\det(AB)$, and $\det(BA)$.

9. Let A and B be 4×5 and 5×4 real matrices, respectively. It is given that 1, $\sqrt{2}$, and $1+i$ are (some) eigenvalues of AB, what are the eigenvalues of BA? Why is AB invertible? Is BA also invertible?

10. Let $A \in \mathbb{M}_{5\times9}$. If AA^* is invertible, show that A^*A has a 5×5 nonsingular submatrix and that any 6×6 submatrix of A^*A is singular.

11. Let A and B be $n \times n$ matrices. Show that for every integer $k \geq 1$,

$$\operatorname{tr}(AB)^k = \operatorname{tr}(BA)^k.$$

Does

$$\operatorname{tr}(AB)^k = \operatorname{tr}(A^k B^k)?$$

12. Let $x, y \in \mathbb{C}^n$ (in columns). Find $\operatorname{rank}(I_n + xy^*)$ and show that

$$\operatorname{tr}(I_n + xy^*) = n + y^*x, \quad \det(I_n + xy^*) = 1 + y^*x.$$

13. Let $x, y \in \mathbb{C}^n$ (in columns). Let $A \in \mathbb{M}_n$ be nonsingular. Show that

$$\det(A + xy^*) = (1 + y^*A^{-1}x) \det A.$$

14. Let x_j be nonzero numbers and let $y_j = \frac{1}{x_j}$, $j = 1, 2, \ldots, n$. Compute

$$\begin{vmatrix} 1 + x_1y_1 & x_1y_2 & \cdots & x_1y_n \\ x_2y_1 & 1 + x_2y_2 & \cdots & x_2y_n \\ \vdots & \vdots & \vdots & \vdots \\ x_ny_1 & x_ny_2 & \cdots & 1 + x_ny_n \end{vmatrix}.$$

15. Let $A, B, C \in \mathbb{M}_n$. Show that $(AB)^2$ and $(BA)^2$ have the same eigenvalues (including zeros and multiplicities) and that ABC, CAB, and BCA have the same eigenvalues. Is it true that ABC and CBA have the same eigenvalues? Is it true that $\operatorname{tr}(ABC) = \operatorname{tr}(ACB)$?

16. Let $A \in \mathbb{M}_n$. Do A^* and A have the same (a) rank? (b) determinant? (c) eigenvalues? (d) singular values? Discuss (a)–(d) for A^*A and AA^*.

17. Let $A, B \in \mathbb{M}_n$. Show that $\begin{pmatrix} 0 & A \\ B & 0 \end{pmatrix}$ is similar to $\begin{pmatrix} 0 & B \\ A & 0 \end{pmatrix}$ via $\begin{pmatrix} 0 & I \\ I & 0 \end{pmatrix}$.

18. Are $\begin{pmatrix} AB & 0 \\ 0 & 0 \end{pmatrix}$ and $\begin{pmatrix} 0 & 0 \\ 0 & BA \end{pmatrix}$ similar for all $A, B \in \mathbb{M}_n$?

19. Let $A \in \mathbb{M}_n$. Find the eigenvalues of $\begin{pmatrix} A & A \\ A & A \end{pmatrix}$ in terms of those of A.

20. For any square matrices A and B of the same size, show that

$$A^2 + B^2 \quad \text{and} \quad \begin{pmatrix} A^2 & AB \\ BA & B^2 \end{pmatrix}$$

have the same nonzero eigenvalues. Further show that the latter block matrix always have zero eigenvalues. How many zeros in general?

21. Let $A, B \in \mathbb{M}_n$. It is known (by Theorem 2.4.1) that A^*A and AA^* always have the same eigenvalues; so do B^*B and BB^*. Is it true that $A^*A + B^*B$ and $AA^* + BB^*$ have the same eigenvalues?

22. Let $A = \begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix}$ and $P = \begin{pmatrix} 1 & \varepsilon \\ 0 & 1 \end{pmatrix}$, $\varepsilon \in \mathbb{R}$. Find the eigenvalues and singular values of A, PAP^{-1}, and $P^{-1}AP$. Are the singular values the same?

23. Let $A = \left(\begin{smallmatrix} -1 & 0 & 1 \\ 0 & 1 & 1 \end{smallmatrix}\right)$ and $B = \left(\begin{smallmatrix} 0 & 1 \\ 0 & 0 \\ 1 & 0 \end{smallmatrix}\right)$. Find the eigenvalues of AB. What are the eigenvalues of BA? Find the singular values of A and B.

24. Let A be a 3×2 matrix and let B be a 2×3 matrix such that

$$AB = \begin{pmatrix} 8 & 2 & -2 \\ 2 & 5 & 4 \\ -2 & 4 & 5 \end{pmatrix}.$$

Find the ranks of AB and $(AB)^2$ and show that $BA = \left(\begin{smallmatrix} 9 & 0 \\ 0 & 9 \end{smallmatrix}\right)$.

25. Let $A \in \mathbb{M}_{m\times n}$ have rank r and let $M = \left(\begin{smallmatrix} 0 & A \\ A^* & 0 \end{smallmatrix}\right)$. Show that

(a) M is a Hermitian matrix.
(b) $\operatorname{rank}(M) = 2\operatorname{rank}(A)$.
(c) $\det M = \det(-A^*A) = (-1)^n |\det A|^2$ if A is n-square.
(d) The nonzero singular values of M are $\sigma_1(A)$, $\sigma_1(A)$, $\ldots$, $\sigma_r(A)$, $\sigma_r(A)$ (i.e., the nonzero singular values of A each appearing twice).
(e) $2\left(\begin{smallmatrix} 0 & A \\ A^* & 0 \end{smallmatrix}\right) = \left(\begin{smallmatrix} B & A \\ A^* & C \end{smallmatrix}\right) + \left(\begin{smallmatrix} -B & A \\ A^* & -C \end{smallmatrix}\right)$ for any $B \in \mathbb{M}_m$ and $C \in \mathbb{M}_n$.

26. Find the eigenvalues and singular values of the following matrices:

$$\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \quad \begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix}, \quad \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}, \quad \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}.$$

27. Find the eigenvalues of $\left(\begin{smallmatrix} 0 & A \\ A^* & 0 \end{smallmatrix}\right)$ and $\left(\begin{smallmatrix} A & 0 \\ 0 & A^* \end{smallmatrix}\right)$, where $A = \left(\begin{smallmatrix} 1 & 1 \\ 1 & 1 \end{smallmatrix}\right)$.

28. Find the eigenvalues and singular values of the matrix $\left(\begin{smallmatrix} 1 & 1 & 0 \\ 1 & 1 & 0 \\ 1 & -1 & -1 \end{smallmatrix}\right)$.

29. If D is a diagonal matrix, say $D = \operatorname{diag}(\lambda_1, \ldots, \lambda_n)$, where $\lambda_1, \ldots, \lambda_n$ are complex numbers, what are the singular values of D?

30. Let $D = \left(\begin{smallmatrix} D_r & 0 \\ 0 & 0 \end{smallmatrix}\right) \in \mathbb{M}_{m\times n}$, where $D_r = \operatorname{diag}(d_1, \ldots, d_r)$ with nonzero $d_1, \ldots, d_r$ (r is a positive integer). Let $D^\sim = \left(\begin{smallmatrix} D_r^{-1} & 0 \\ 0 & 0 \end{smallmatrix}\right) \in \mathbb{M}_{m\times n}$. Find the singular values of D and show that $D = (DD^T)D^\sim = D^\sim(D^TD)$.

31. Let $A \in \mathbb{M}_n$. Let $\lambda_1, \ldots, \lambda_n$ be the eigenvalues of A and let $\sigma_1, \ldots, \sigma_n$ be the singular values of A. Show that

$$|\det A| = \left|\prod_{i=1}^n \lambda_i\right| = \prod_{i=1}^n \sigma_i.$$

32. Let A be a square matrix. Show that A has a zero singular value if and only if A has a zero eigenvalue. Does it follow that the number of zero singular values is equal to the number of zero eigenvalues?

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2.5 Localization of Eigenvalues: The Geršgorin Disc Theorem

One of the central topics in linear algebra and matrix theory is to study eigenvalues. Is there a way to locate or estimate the eigenvalues of a matrix by using its entries? The celebrated Geršgorin theorem is a useful result on this; it ensures that the eigenvalues of a matrix lie in certain discs (circles) in the complex plane centered at the diagonal entries of the matrix.

We begin with a simple result which shows that the eigenvalues of a square matrix are all located in one (closed) disc centered at the origin with the largest sum of row entries in absolute values as the radius.

Theorem 2.5.1 *Let $A = (a_{ij}) \in \mathbb{M}_n$. Then for any eigenvalue λ of A,*

$$|\lambda| \leq R = \max_i \sum_{j=1}^{n} |a_{ij}|.$$

Proof. Let (λ, x) be an eigenpair: $Ax = \lambda x$, where $x = (x_1, \ldots, x_n)^T \neq 0$. Let $|x_p| \geq |x_j|$ for all j. Then $|x_p| > 0$ and $\lambda x_p = \sum_{j=1}^{n} a_{pj} x_j$. Thus,

$$|\lambda|\,|x_p| = \Big|\sum_{j=1}^{n} a_{pj} x_j\Big| \leq \sum_{j=1}^{n} |a_{pj}|\,|x_j| \leq |x_p| \sum_{j=1}^{n} |a_{pj}|.$$

Dividing both sides by $|x_p|$ reveals the desired inequality. ∎

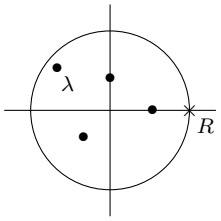


Fig. 2.1. Eigenvalues lie in (or on) the circle

Example 2.5.1 $A = \left(\begin{smallmatrix} 0 & 1 \\ t & 0 \end{smallmatrix}\right)$, $t \in \mathbb{R}$, has eigenvalues $\pm\sqrt{t}$ and $|\pm\sqrt{t}| \leq \max\{1, |t|\}$. If $t = \frac{1}{4}$, then the two eigenvalues $\pm\frac{1}{2}$ lie in the circle $|z| = 1$. If $t = -4$, then the eigenvalues $\pm 2i$ $(i = \sqrt{-1})$ lie in the circle $|z| = 4$. ∎

A slight modification of the proof of Theorem 2.5.1 yields the *Geršgorin disc* (or *circle*) *theorem*. Let $A = (a_{ij}) \in \mathbb{M}_n$ and let r_i be the sum of the entries in absolute values in row i excluding the diagonal entry a_{ii}, that is,

$$r_i = \sum_{j=1,\, j\neq i}^{n} |a_{ij}|, \quad i = 1, \ldots, n.$$

We call the following set in the complex plane a (*row*) *Geršgorin disc*:

$$\{z \in \mathbb{C} : |z - a_{ii}| \le r_i\} \text{ (or simply } |z - a_{ii}| \le r_i).$$

So, for an $n \times n$ matrix, there are at most n different Geršgorin discs.

Theorem 2.5.2 (Geršgorin Disc Theorem, Part 1) *Let $A = (a_{ij}) \in \mathbb{M}_n$. Then every eigenvalue of A lies in a Geršgorin disc, i.e., the eigenvalues of A are located in the union of all Geršgorin discs (obtained from rows)*

$$G(A) = \cup_{i=1}^{n} \{z \in \mathbb{C} : |z - a_{ii}| \le r_i\}.$$

Proof. Let λ be an eigenvalue of A, and let x be an eigenvector of A associated with λ. Suppose that x_p is the largest component of x in absolute value, that is, $|x_p| \ge |x_j|$, $j = 1, \ldots, n$. Then $x_p \neq 0$.

The equation $Ax = \lambda x$ gives $\sum_{j=1}^{n} a_{pj} x_j = \lambda x_p$, rewritten as

$$x_p(\lambda - a_{pp}) = \sum_{j=1,\, j \neq p}^{n} a_{pj} x_j.$$

By taking absolute values of both sides, we have

$$\begin{aligned} |x_p||\lambda - a_{pp}| &= \left| \sum_{j=1,\, j\neq p}^{n} a_{pj} x_j \right| \le \sum_{j=1,\, j\neq p}^{n} |a_{pj} x_j| \\ &= \sum_{j=1,\, j\neq p}^{n} |a_{pj}||x_j| \le |x_p| \sum_{j=1,\, j\neq p}^{n} |a_{pj}| = |x_p|\, r_p. \end{aligned}$$

So $|\lambda - a_{pp}| \le r_p$, i.e., λ lies in the closed disc $\{z \in \mathbb{C} : |z - a_{pp}| \le r_p\}$. ∎

An application of the theorem to A^T gives a version of the theorem for the columns of A. As a result, all eigenvalues of A lie in $G(A) \cap G(A^T)$.

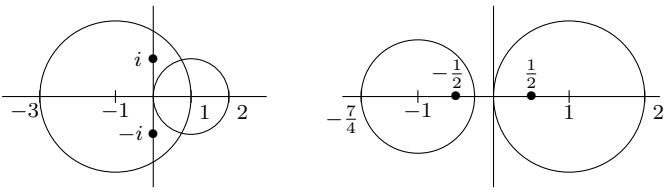


Fig. 2.2. Geršgorin circles for $t = 2$ and $t = \frac{3}{4}$ in Example 2.5.2

Example 2.5.2 Let $A = \begin{pmatrix} 1 & -1 \\ t & -1 \end{pmatrix}$, $t \in \mathbb{R}$. Then $\det(\lambda I - A) = \lambda^2 + t - 1$. The eigenvalues of A are $\pm\sqrt{1-t}$. One may parameterize the eigenvalues by $\lambda_1(t)$ and $\lambda_2(t)$, $\lambda_1(t) \ge \lambda_2(t)$. Then $\lambda_1(t) = \sqrt{1-t}$ and $\lambda_2(t) = -\sqrt{1-t}$.

The Geršgorin disc obtained from the first row of A is $G_1 : |z-1| \leq 1$, and from the second row of A is $G_2 : |z+1| \leq |t|$. The cases $t = 2$ (with eigenvalues $\pm i$) and $t = \frac{3}{4}$ (with eigenvalues $\pm\frac{1}{2}$) are illustrated in Fig. 2.2. Note that for $t = 2$, the smaller disc contains no eigenvalues of A. It is a good exercise to draw the diagrams for the cases of $t = 0$, 1, -1, -2, 3, -3, and 5. Note that it is possible that the eigenvalues lie on the circles. ∎

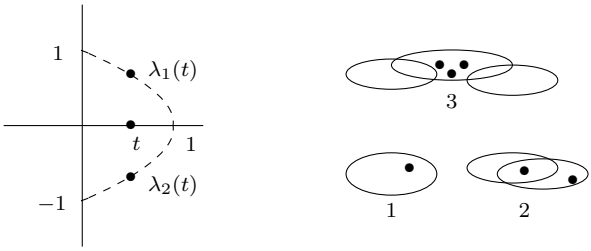


Fig. 2.3. (a) Eigenvalues with parameter t (b) Part 2

Theorem 2.5.3 (Geršgorin Disc Theorem, Part 2) *Let $A = (a_{ij}) \in \mathbb{M}_n$. If the union, R_k, of k of the n Geršgorin discs is disjoint from the other $n-k$ discs, then R_k contains exactly k eigenvalues of A (with multiplicities).*

We include the second part of the Geršgorin disc theorem because it is useful and easy to understand. However, a rigorous proof[①] of it requires advanced machinery (argument principle) of complex analysis that is not assumed as our prerequisite. Thus, we skip the proof by the argument principle, and offer an explanation of a proof using eigenvalue continuity.

Let $D = \text{diag}(a_{11}, \ldots, a_{nn})$ be the diagonal matrix of the diagonal entries of A, and let $T(t) = D + t(A - D)$, $t \in [0, 1]$. Then $T(0) = D$ and $T(1) = A$. The diagonal entries of $T(t)$ are exactly those of A for all $t \in [0, 1]$. So the discs of $T(t)$ and those of A have the same centers $a_{11}, \ldots, a_{nn}$. Moreover, every disc of $T(t)$ is contained in the corresponding disc of A.

Assuming that the eigenvalues of $T(t)$ are each continuous functions in t, we see that as t increases from $t = 0$ to $t = 1$, the curves of the eigenvalues of $T(t)$ as functions in t start from the centers $a_{11}, \ldots, a_{nn}$ to the eigenvalues of A. By the first part of the Geršgorin disc theorem, all these curves are contained in the discs of $T(t)$, consequently, they lie inside the discs of A. If the union, R_k, of k of the n discs of A is disjoint from the other $n - k$ discs, then R_k contains exactly k eigenvalues (curves) of A.

Here are two critical points[②] about the method of eigenvalue continuity: (1) eigenvalues are continuous functions of the entries of matrices (why? see

[①] R. Horn and C. Johnson, *Matrix Analysis*, Cambridge Univ. Press, 2013, pp. 388–390.
[②] C.-K. Li and F. Zhang, *Eigenvalue continuity and Geršgorin's theorem,* Electronic Journal of Linear Algebra, Volume 35, 2019, pp. 619–625.

the next section for more discussions about eigenvalue continuity) and (2) the sum of the algebraic multiplicities of the eigenvalues remains constant on each connected Geršgorin region as the parameter varies (why?).

If a disc is separate from others, then the disc contains an eigenvalue.

Example 2.5.3 Let

$$A = \begin{pmatrix} 1 & 1 & 0 \\ 0.25 & 2 & 0.25 \\ 0.25 & 0 & 3 \end{pmatrix}.$$

A has three Geršgorin discs obtained by using rows:

$$\begin{aligned} R_1 &= \{z \in \mathbb{C} : \ |z-1| \le 1\}, \\ R_2 &= \{z \in \mathbb{C} : \ |z-2| \le 0.5\}, \\ R_3 &= \{z \in \mathbb{C} : \ |z-3| \le 0.25\}. \end{aligned}$$

The first two discs are connected and disjoint with R_3. It follows that A has two eigenvalues in $R_1 \cup R_2$, and one eigenvalue in R_3.

There are three Geršgorin discs obtained by using columns:

$$\begin{aligned} C_1 &= \{z \in \mathbb{C} : \ |z-1| \le 0.5\}, \\ C_2 &= \{z \in \mathbb{C} : \ |z-2| \le 1\}, \\ C_3 &= \{z \in \mathbb{C} : \ |z-3| \le 0.25\} = R_3. \end{aligned}$$

Since C_1, C_2, C_3 are connected, we cannot tell if A has an eigenvalue in R_3. ∎

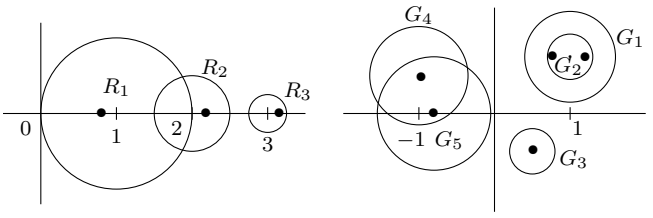


Fig. 2.4. Examples 2.5.3 and 2.5.4

Example 2.5.4 Let

$$A = \begin{pmatrix} 1+0.75i & 0.3 & 0 & -0.3 & 0 \\ 0.15 & 1+0.75i & 0.15 & 0 & 0 \\ 0.1 & 0.1 & 0.5-0.5i & 0 & 0.1 \\ 0.25 & 0 & 0.2 & -1+0.5i & 0.2 \\ 0.25 & 0.25 & 0.25 & 0 & -0.8 \end{pmatrix}.$$

A has five Geršgorin circles (or discs with interior) obtained from the rows:

$$\begin{aligned} G_1&: \ |z-1-0.75i| = 0.6,\\ G_2&: \ |z-1-0.75i| = 0.3,\\ G_3&: \ |z-0.5+0.5i| = 0.3,\\ G_4&: \ |z+1-0.5i| = 0.65,\\ G_5&: \ |z+0.8| = 0.75. \end{aligned}$$

The eigenvalues are represented by the $\bullet$ dots in the circles in Fig. 2.4. ∎

Let $A = (a_{ij}) \in \mathbb{M}_n$. A is said to be (*row*) *diagonally dominant* if $|a_{ii}| \geq r_i = \sum_{j=1,\, j\neq i}^{n} |a_{ij}|$ for $i = 1, \ldots, n$; A is said to be *doubly* (*row*) *diagonally dominant* if $|a_{ii}|\,|a_{jj}| \geq r_i r_j$ for all $i \neq j$. If all strict inequalities hold, we say that A is *strictly* diagonally dominant and *strictly* doubly diagonally dominant. The terms are similarly defined for columns.

Example 2.5.5 (1) *Levy–Desplanques theorem* and (2) *Brauer theorem.*

(1) Strictly diagonally dominant matrices are nonsingular.

This is immediate from the Geršgorin disc theorem (Part 1) because none of the Geršgorin discs contains the origin.

Alternatively, it can be shown directly as follows.

If $A = (a_{ij}) \in \mathbb{M}_n$ is singular, then zero is an eigenvalue of A, i.e., $Ax = 0$ for some nonzero $x = (x_1, \ldots, x_n)^T$. Let $x_p \neq 0$ be such that $|x_p| \geq |x_j|$ for all j. Then $Ax = 0$ gives $\sum_{j=1}^{n} a_{pj}x_j = 0$. Thus,

$$|a_{pp}|\,|x_p| = |-a_{pp}x_p| = \Big|\sum_{j\neq p} a_{pj}x_j\Big| \leq |x_p| \sum_{j\neq p} |a_{pj}|,$$

implying $|a_{pp}| \leq \sum_{j\neq p} |a_{pj}|$, a contradiction to diagonal dominance.

(2) Strictly doubly diagonally dominant matrices are nonsingular.

In the proof of (1), the vector x has at least two nonzero components. Otherwise, say $x_1 \neq 0$ and $x_i = 0, i > 1$. Then $a_{11}x_1 = 0$, so $a_{11} = 0$, contradicting $|a_{ii}|\,|a_{jj}| > r_i r_j$ for all $i \neq j$. (In fact, all $a_{kk} \neq 0$.)

Now let x_p, x_q be nonzero and $|x_p| \geq |x_q| \geq |x_j|$ for all $j \neq p, q$. Then

$$|a_{pp}|\,|x_p| = |-a_{pp}x_p| = \Big|\sum_{j\neq p} a_{pj}x_j\Big| \leq |x_q| \sum_{j\neq p} |a_{pj}| = |x_q| r_p,$$

$$|a_{qq}|\,|x_q| = |-a_{qq}x_q| = \Big|\sum_{j\neq q} a_{qj}x_j\Big| \leq |x_p| \sum_{j\neq q} |a_{qj}| = |x_p| r_q.$$

So

$$|a_{pp}|\,|x_p|\,|a_{qq}|\,|x_q| \leq |x_q| r_p\,|x_p| r_q.$$

Dividing by $|x_p|\,|x_q|$ reveals $|a_{pp}|\,|a_{qq}| \leq r_p r_q$, a contradiction.

(3) All eigenvalues of $A = (a_{ij})$ are contained in the union of the regions

$$O_{ij} = \{z \in \mathbb{C} : |z - a_{ii}|\,|z - a_{jj}| \le r_i r_j\}, \quad i \ne j. \tag{2.16}$$

Reason: If c is such a number that $cI - A$ is singular, then $cI - A$ cannot be strictly doubly diagonally dominant. Thus, every eigenvalue of A lies in a region (2.16), referred to as an *oval of Cassini.* ∎

Problems

1. Draw the row and column Geršgorin discs and find eigenvalues for

$$\begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix}, \quad \begin{pmatrix} 1 & 1 \\ -1 & -1 \end{pmatrix}, \quad \begin{pmatrix} 1 & 1 \\ 3 & 3 \end{pmatrix}.$$

2. Apply the Geršgorin disc theorem (for rows and columns) to the following matrix A, and draw the discs (there are six), where

$$A = \begin{pmatrix} 1 & 1 & 0 \\ -1 & -1 & 0 \\ 1 & 2 & 4 \end{pmatrix}.$$

3. Show by the Geršgorin theorem that A has three different eigenvalues:

$$A = \begin{pmatrix} 1 & \frac{1}{2} & \frac{i}{2} \\ \frac{1}{2} & 3 & 0 \\ 0 & 1 & 5 \end{pmatrix} \quad \text{or} \quad \begin{pmatrix} 1 & i & 0 \\ \frac{1}{2} & 4 & \frac{i}{2} \\ 1 & 0 & 7 \end{pmatrix}.$$

Conclude that A is diagonalizable (see Sect. 1.3, Chap. 1).

4. Draw a graph to illustrate the Geršgorin disc theorem by the matrix

$$A = \begin{pmatrix} 2 & -\frac{1}{2} & \frac{1}{4} & -\frac{1}{4} \\ \frac{1}{4} & 1+2i & 0 & \frac{1}{4} \\ -\frac{1}{2} & \frac{1}{4} & -1 & \frac{1}{2} \\ \frac{1}{4} & -\frac{1}{2} & \frac{1}{2} & -2-2i \end{pmatrix}.$$

5. Let $A(t) = \begin{pmatrix} 1 & -1 \\ t & -1 \end{pmatrix}$, $t \in \mathbb{R}$. Find the eigenvalues and draw the Geršgorin discs using rows for the cases of $t = 0$, 1, -1, -3, and 5, respectively.

6. Find the eigenvalues of $T(t) = \begin{pmatrix} 0 & 1-t \\ t-1 & 0 \end{pmatrix}$ in terms of $t \in [0, 1]$.

7. Let $A_1 = \left(\begin{smallmatrix} 1 & 0 \\ 0 & 1 \end{smallmatrix}\right)$, $A_2 = \left(\begin{smallmatrix} 1 & 0 \\ 0 & 2 \end{smallmatrix}\right)$, and $B = \left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$. Let $t \in [-1, 1]$.
   (a) Explain why the eigenvalues of $A_1 + tB$ and $A_2 + tB$ are all real.
   (b) Find the eigenvalues $\lambda_1(t)$ and $\lambda_2(t)$ of $A_1 + tB$ in terms of t.
   (c) Find the eigenvalues $\mu_1(t)$ and $\mu_2(t)$ of $A_2 + tB$ in terms of t.
   (d) Parameterize the eigenvalues as $\lambda_1(t) \geq \lambda_2(t)$, $\mu_1(t) \geq \mu_2(t)$.
   (e) Show that $\mu_1(t)$ and $\mu_2(t)$ are differential functions of t.
   (f) Show that $\lambda_1(t)$ and $\lambda_2(t)$ are not differentiable at $t = 0$.
   (g) Draw the graphs of the eigenvalues as functions of t.
   (h) How do the Geršgorin discs change as t increases from 0 to 1?

8. Construct a 4×4 complex matrix such that it contains no zero entries and that the four different eigenvalues of the matrix lie in the circles centered at 1, -1, i, and $-i$, all with diameter 1.

9. State and prove the first part of Geršgorin theorem for columns.

10. Let $A \in \mathbb{M}_n$. Show that A and A^T have the same eigenvalues.

11. Let $A = (a_{ij}) \in \mathbb{M}_n$. For each diagonal entry a_{ii} of A, there are two Geršgorin discs centered at a_{ii}, one obtained by row and the other one by column. Select the smaller disc and denote it by S_i, $i = 1, 2, \ldots, n$. Is it true that all eigenvalues of A lie in the union $\cup_{i=1}^n S_i$?

12. Find the eigenvalues and draw the Geršgorin discs for the matrices

$$\begin{pmatrix} 1 & 1 & 0 & 0 \\ -1 & -1 & 0 & 0 \\ 0 & 0 & i & 1 \\ 0 & 0 & 1 & -i \end{pmatrix}, \quad \begin{pmatrix} 1 & 1 & 0 & 0 \\ 1 & -1 & 0 & 0 \\ 0 & 0 & i & 1 \\ 0 & 0 & 1 & -i \end{pmatrix}, \quad \begin{pmatrix} 1 & 1 & 0 & 0 \\ 1 & -1 & 0 & 0 \\ 0 & 0 & i & -1 \\ 0 & 0 & 1 & -i \end{pmatrix}.$$

13. Let $A \in \mathbb{M}_n$ and $D = \operatorname{diag}(A)$, that is, the diagonal matrix of A. Denote $B = A - D$. If λ is an eigenvalue of A and it is not a diagonal entry of A, show that 1 is an eigenvalue of $(\lambda I - D)^{-1}B$.

14. Let $A = (a_{ij}) \in \mathbb{M}_n$. Show that for any eigenvalue λ of A

$$|\lambda| \geq \min_i \{|a_{ii}| - r_i\}, \quad \text{where } r_i = \sum_{j=1,\, j \neq i}^{n} |a_{ij}|.$$

Derive that if A is diagonally dominant, then

$$|\det A| \geq (\min_i \{|a_{ii}| - r_i\})^n.$$

15. Let $A = (a_{ij}) \in \mathbb{M}_n$. Show that for any eigenvalue λ of A

$$|\lambda| \leq \min \left\{ \max_i \sum_{s=1}^{n} |a_{is}|, \ \max_j \sum_{t=1}^{n} |a_{tj}| \right\}.$$

16. Let $A = (a_{ij}) \in \mathbb{M}_n$. Show that for any eigenvalue λ of A

$$|\lambda| \leq n \max_{i,j} |a_{ij}|.$$

17. Let $A = (a_{ij}) \in \mathbb{M}_n$. Show that for any eigenvalue λ of A and for any positive numbers $d_1, \ldots, d_n$,

$$|\lambda| \leq \min \left\{ \max_i \frac{1}{d_i} \sum_{s=1}^{n} d_s |a_{is}|, \ \max_j \frac{1}{d_j} \sum_{t=1}^{n} d_t |a_{tj}| \right\}.$$

18. Let H be a Hermitian matrix with nonnegative diagonal entries. If H is diagonally dominant, show that all eigenvalues of H are nonnegative, and if H is strictly diagonally dominant, show that all eigenvalues of H are positive. Is this true if H is not Hermitian?

19. Let $A = (a_{ij}) \in \mathbb{M}_n$ and denote $\rho_i = \sum_{j=1}^{n} |a_{ij}|$ for $i = 1, 2, \ldots, n$. Suppose that A does not have a zero row. Show that

$$\operatorname{rank}(A) \geq \sum_{i=1}^{n} \frac{|a_{ii}|}{\rho_i}.$$

20. Let $A = (a_{ij}) \in \mathbb{M}_n$. Denote $\delta = \min_{i \neq j} |a_{ii} - a_{jj}|$ and $\epsilon = \max_{i \neq j} |a_{ij}|$. If $\delta > 0$ and $\epsilon \leq \delta/(2n)$, show that each Geršgorin disc contains exactly one eigenvalue of A.

. ——————— ⊙ ——————— .

2.6 The Continuity Argument and Matrix Functions

We often see matrices whose entries are functions of a variable (or parameter) and encounter certain functions such as determinant (or trace, etc) of the matrices with the parameter. For instance, the characteristic polynomial $p(\lambda) = \det(\lambda I - A)$ of $A \in \mathbb{M}_n(\mathbb{C})$ is a complex-valued function in λ. Some of these functions are continuous functions of the entries of matrices.

Theorem 2.6.1 *Let A be an $n \times n$ matrix. Then there exists $\delta > 0$ such that $A + \epsilon I$ is nonsingular for all $\epsilon \in (0, \delta)$, and $\lim_{\epsilon \to 0^+} \det(A + \epsilon I) = \det A$.*

Proof. The polynomial $\det(\lambda I + A)$ in λ has at most n different roots. If they are all 0, we can take δ to be any positive number. Otherwise, let δ be the smallest nonzero λ in modulus. This δ serves the purpose.

Note that the determinant is a continuous function of its entries. ∎

We used the so-called *continuity argument* in the proof of the determinant identity (2.3) of Theorem 2.2.2. We used the continuity again in Proof 3 of Theorem 2.4.1 which states that AB and BA have the same eigenvalues when A and B are square matrices of the same size. The proof goes as follows. First consider the case where A is invertible (the easy case) and show that AB and BA are similar due to the fact that $AB = A(BA)A^{-1}$.

If A is singular, consider $A + \epsilon I$ in place of A. Choose $\delta > 0$ such that $A + \epsilon I$ is nonsingular, $0 < \epsilon < \delta$. Thus, $(A + \epsilon I)B$ and $B(A + \epsilon I)$ have the same eigenvalues for each $\epsilon \in (0, \delta)$. Equate the characteristic polynomials:

$$\det(\lambda I - (A + \epsilon I)B) = \det(\lambda I - B(A + \epsilon I)), \quad 0 < \epsilon < \delta.$$

Since both sides are continuous functions of ϵ, letting $\epsilon \to 0^+$ results in

$$\det(\lambda I - AB) = \det(\lambda I - BA).$$

Thus, AB and BA have exactly the same (multiset of) eigenvalues.

The proof was done in three steps:

1. Show the assertion for nonsingular A.
2. Replace singular A by nonsingular $A + \epsilon I$.
3. Use continuity of the det function in ϵ to get the desired conclusion.

Continuity argument is a useful and effective technique for many matrix problems when a singular matrix is involved and the nonsingular case is easy to be dealt with. The setting in which the technique is used is rather important, and the *continuity* (of what?) should be carefully used. Sometimes the result for nonsingular matrices need not be valid for the singular case. Here is an example for which the continuity argument fails.

Example 2.6.1 Let C and D be n-square matrices such that

$$CD^T + DC^T = 0.$$

If D is nonsingular, then for any n-square matrices A and B

$$\begin{vmatrix} A & B \\ C & D \end{vmatrix} = \det(AD^T + BC^T). \tag{2.17}$$

Proof. Since $\left(\begin{smallmatrix} A & B \\ C & D \end{smallmatrix}\right)\left(\begin{smallmatrix} D^T & 0 \\ C^T & I \end{smallmatrix}\right) = \left(\begin{smallmatrix} AD^T+BC^T & B \\ 0 & D \end{smallmatrix}\right)$ and $\det D = \det D^T \neq 0$, taking the determinants of both sides gives the desired determinant identity.

The identity does not hold in general if D is singular. For example, take

$$A = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \quad B = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}, \quad C = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \quad D = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix},$$

where D is singular. Computations show that $CD^T + DC^T = 0$, $\left|\begin{smallmatrix} A & B \\ C & D \end{smallmatrix}\right| = 1$, while $\det(AD^T + BC^T) = -1$; the determinant identity (2.17) is false. ∎

The trace and determinant of a matrix depend continuously on the entries of the matrix because the trace is the sum of the diagonal entries and the determinant is the sum of all signed products of the entries from different rows and columns. So, we may simply say that trace and determinant are continuous functions of matrices. On the other hand, the trace is the sum of eigenvalues and the determinant is the product of eigenvalues. Are eigenvalues continuous functions of a matrix? In other words, do the eigenvalues of a matrix depend continuously on the entries of the matrix?

What does it really mean to say that eigenvalues are continuous functions? One may imagine continuous curves of the eigenvalues (as functions) evolving on the complex plane, or one may trace the curves continuously. But that is not as easy as it sounds. First, ordering eigenvalues with a parameter can be difficult; second, the eigenvalue curves may merge and the multiplicities of eigenvalues can change as the parameter varies.

Let $A(t)$ be a square matrix with entries depending on a variable t and let $f(t)$ be a complex-valued function of the variable t. We say that $f(t)$ is an *eigenvalue function* of $A(t)$ if $f(t)$ is an eigenvalue of $A(t)$ for every t. [Note: Different from our eigenvalue function, *eigenfunction* is another mathematical term that has had its own meaning as an eigenvector function.]

Example 2.6.2 Let $A(t) = \left(\begin{smallmatrix} 0 & t \\ t & 0 \end{smallmatrix}\right)$, $t \in [-1, 1]$. For each fixed t, $A(t)$ has eigenvalues t and $-t$. It is natural to order the eigenvalues of $A(t)$ as $\lambda_1(t) = t$, $\lambda_2(t) = -t$. Notice that $A(t)$ is real symmetric. For real symmetric (or complex Hermitian) matrices, we usually want the eigenvalues to be in a non-increasing (or non-decreasing) order. So we would order the eigenvalues

as $\mu_1(t) = |t| \geq \mu_2(t) = -|t|$. $\lambda_1(t)$, $\lambda_2(t)$, $\mu_1(t)$, and $\mu_2(t)$ are all continuous functions of t. Unlike $\lambda_1(t)$ and $\lambda_2(t)$, $\mu_1(t)$ and $\mu_2(t)$ are not differentiable.

We may order or parameterize the eigenvalues in other ways such as

$$\nu_1(t) = \begin{cases} t, & \text{if } t \text{ is irrational,} \\ -t, & \text{if } t \text{ is rational;} \end{cases} \quad \nu_2(t) = \begin{cases} -t, & \text{if } t \text{ is irrational,} \\ t, & \text{if } t \text{ is rational.} \end{cases}$$

For each $t \in [-1, 1]$, $\nu_1(t)$ and $\nu_2(t)$ are eigenvalues of $A(t)$. So, $\nu_1(t)$ and $\nu_2(t)$ are also considered as eigenvalue functions of $A(t)$. Obviously, they are not continuous (and of course they are not our interest). We are interested in the eigenvalue functions with continuity (or differentiability). ∎

The eigenvalues $\lambda_1(t)$ and $\lambda_2(t)$ (or $\mu_1(t)$ and $\mu_2(t)$) in Example 2.6.2 are parameterized as continuous functions of t. This idea was used in the preceding section to explain (the second part of) the Geršgorin theorem. Is this always possible? Precisely, let $A(t)$ be a family of $n \times n$ matrices whose entries depend continuously on a parameter t over a domain in the complex plane or on a real interval. Then each t produces a discrete set of eigenvalues of $A(t)$. Do there exist n continuous complex functions of t (i.e., a set of continuous functions on the domain) that represent the eigenvalues of $A(t)$? The answer is negative in general.

Example 2.6.3 Let $A(t) = \left(\begin{smallmatrix} 0 & 1 \\ t & 0 \end{smallmatrix}\right)$, where $t \in \mathbb{C}$. Then $\det(\lambda I - A(t)) = \lambda^2 - t$. Borrowing a fact[3] from complex analysis that no continuous function f satisfies $(f(z))^2 = z$ for all $z \in \mathbb{C}$, we see immediately that it is impossible to have continuous functions representing the eigenvalues of $A(t)$. ∎

It is known[4], however, that the continuous eigenvalue functions exist if (i) the parameter t runs over a real interval or (ii) the eigenvalues are always real. Below we provide a weaker result than this statement with both (i) and (ii) being met. In particular, it applies to Hermitian matrices[5] (including real symmetric and positive semidefinite matrices). Interested reader may further study the perturbation theory (of eigenvalues) for operators.

Theorem 2.6.2 *Let $A(t) = (a_{ij}(t)) \in \mathbb{M}_n$, in which all entries $a_{ij}(t)$ are continuous functions of t over an real interval $\mathbb{I}$. Suppose that $A(t)$ has only real eigenvalues for each $t \in \mathbb{I}$. Then there exist n continuous functions $\lambda_1(t), \ldots, \lambda_n(t)$ over $\mathbb{I}$ that serve as the eigenvalues of $A(t)$. Furthermore, they can be assumed to be in the order $\lambda_1(t) \geq \cdots \geq \lambda_n(t)$ for all $t \in \mathbb{I}$.*

Proof. The characteristic polynomial of $A(t)$ is a monic polynomial whose coefficients are continuous functions and whose roots are all real. The continuity of eigenvalues as functions is immediate from the following statement:

[3] E. Freitag and R. Busam, *Complex Analysis*, Springer, 2005, p. 41.
[4] T. Kato, *Perturbation Theory for Linear Operators*. Springer, 1995, pp. 108–110.
[5] F. Rellich, *Perturbation Theory of Eigenvalue Problems*, CRC, 1969, pp. 39–41.

Let $a_1(t), \ldots, a_n(t)$ be continuous functions over an interval $\mathbb{I} \subseteq \mathbb{R}$. Let $p_t(x) = x^n + a_1(t)x^{n-1} + \cdots + a_{n-1}(t)x + a_n(t)$ (a monic polynomial in x). Suppose that for each $t \in \mathbb{I}$, the n roots of $p_t(x)$ are all real. Denote and order the roots as $\lambda_1(t) \geq \cdots \geq \lambda_n(t)$ (each of which is well-defined and uniquely determined). We show that each $\lambda_i(t)$ is a continuous function of t on $\mathbb{I}$.

First, each $\lambda_i(t)$ in ordered $\lambda_1(t) \geq \cdots \geq \lambda_n(t)$ is a well-defined function (without suitable ordering, the $\lambda_i(t)$'s need not be continuous), and

$$p_t(x) = x^n + a_1(t)x^{n-1} + \cdots + a_{n-1}(t)x + a_n(t) = \big(x - \lambda_1(t)\big) \cdots \big(x - \lambda_n(t)\big).$$

Let $c_0 \in \mathbb{I}$. We show that each $\lambda_i(t)$ is continuous at c_0, that is,

$$\lim_{t \to c_0} \lambda_i(t) = \lambda_i(c_0). \tag{2.18}$$

(If c_0 is an endpoint of $\mathbb{I}$, then it is a one-sided limit; and the proof is similar.)

We prove the statement by contradiction. Suppose some $\lambda_j(t)$ is discontinuous at c_0, i.e., either $\lambda_j(t)$ has no limit at c_0 or $\lim_{t \to c_0} \lambda_j(t) \neq \lambda_j(c_0)$. There exist a number $\delta > 0$ and a sequence $\{c_k\}$ convergent to c_0 such that

$$|\lambda_j(c_k) - \lambda_j(c_0)| > \delta \quad \text{for all } k. \tag{2.19}$$

Choose $a, b \in \mathbb{I}$ such that $c_0 \in (a, b) \subset \mathbb{I}$. (If c_0 is, say the left endpoint of $\mathbb{I}$, then consider $[c_0, b]$.) We may assume that $c_k \in (a, b)$ for all k. Since $a_1(t), \ldots, a_n(t)$ are continuous on $[a, b]$, they are uniformly bounded, that is, there exists a constant L such that $|a_i(t)| \leq L$ for all i and all $t \in [a, b]$. By Problem 20, the roots $\lambda_1(t), \ldots, \lambda_n(t)$ are uniformly bounded by $1 + L$. Thus, the n sequences $\{\lambda_1(c_k)\}, \ldots, \{\lambda_n(c_k)\}$ are uniformly bounded. It follows that there exists a subsequence $\{c'_k\}$ of $\{c_k\}$ tending to c_0 such that every sequence $\{\lambda_i(c'_k)\}$ is convergent (as $k \to \infty$), say, to μ_i, $i = 1, \ldots, n$. That is,

$$\lim_{k \to \infty} \lambda_i(c'_k) = \mu_i, \quad i = 1, \ldots, n.$$

Because $\lambda_1(c'_k) \geq \cdots \geq \lambda_n(c'_k)$, we have $\mu_1 \geq \cdots \geq \mu_n$. Moreover,

$$p_{c'_k}(x) = x^n + a_1(c'_k)x^{n-1} + \cdots + a_{n-1}(c'_k)x + a_n(c'_k) = \big(x - \lambda_1(c'_k)\big) \cdots \big(x - \lambda_n(c'_k)\big).$$

Taking limits of both sides gives (as all $a_i(t)$ are continuous functions of t)

$$p_{c_0}(x) = x^n + a_1(c_0)x^{n-1} + \cdots + a_{n-1}(c_0)x + a_n(c_0) = (x - \mu_1) \cdots (x - \mu_n).$$

On the other hand, $p_{c_0}(x) = (x - \lambda_1(c_0)) \cdots (x - \lambda_n(c_0))$. Because $\lambda_1(c_0) \geq \cdots \geq \lambda_n(c_0)$, we have $\mu_1 = \lambda_1(c_0) \geq \cdots \geq \mu_n = \lambda_n(c_0)$. Thus,

$$\lim_{k \to \infty} \lambda_i(c'_k) = \lambda_i(c_0), \quad i = 1, \ldots, n.$$

This contradicts (2.19) for λ_j. ∎

See Corollary 10.5.1, Chap. 10, for another proof of Theorem 2.6.2.

Example 2.6.4 Eigenvalue continuity and singular value continuity.

(a) $A(t) = \begin{pmatrix} t^2+1 & -1 \\ -1 & t+1 \end{pmatrix}$ is real symmetric for all $t \in \mathbb{R}$. So, the eigenvalues of $A(t)$ can be parameterized as continuous functions of t on $\mathbb{R}$.

(b) $B(t) = \begin{pmatrix} -1 & t+i & t \\ t-i & 0 & ti \\ t & -ti & 1 \end{pmatrix}$ is Hermitian for all $t \in [0,1]$. Thus, there exist continuous functions $\lambda_1(t)$, $\lambda_2(t)$, and $\lambda_3(t)$ on $[0,1]$ such that $\lambda_1(t) \geq \lambda_2(t) \geq \lambda_3(t)$ and they serve as the eigenvalues of $B(t)$ for each $t \in [0,1]$.

(c) If a parameterized matrix $A(t)$ is entrywise continuous on a real interval, then there exists a representation of the eigenvalues for the Hermitian matrix $A(t)^*A(t)$ as continuous functions of t. [Note: The eigenvalues of $A(t)^*A(t)$ are nonnegative and the square root is continuous.] It follows that there exists a representation of the singular values by continuous functions. In particular, the ordered singular values $\sigma_1(t) \geq \cdots \geq \sigma_n(t)$ of $A(t)$ are continuous functions of t. We may simply say that the singular values are continuous functions of the parameter t. ∎

We may refer to the eigenvalue continuity discussed above as *functional continuity*. There is another eigenvalue continuity known as *matching* (or *topological*) continuity. Roughly speaking, if t_1 and t_2 are close, then the eigenvalues of $A(t_1)$ and $A(t_2)$ are close after being suitably matched (or as a whole via sets). To be exact, the map $\mathcal{T} : A(t) \mapsto \mathrm{sp}(A(t))$ is continuous (from the matrix space to the unordered tuples with respect to their metrics), where $\mathrm{sp}(A(t))$ is the set of the eigenvalues of $A(t)$, i.e., the *spectrum* of $A(t)$. Simply out: eigenvalues are always topologically continuous. The difference between topological continuity and functional continuity is that the eigenvalues (as a whole) are always topologically continuous but need not be continuous as individual functions. The two continuities for $A(t)$ are equivalent (i.e., one implies the other) when the parameter t belongs to a real interval.

Example 2.6.5 Let $A(t) = \left(\begin{smallmatrix} 0 & 1 \\ t & 0 \end{smallmatrix}\right)$. For $t \in \mathbb{C}$, as stated in Example 2.6.3, the eigenvalues $\lambda_1(t)$ and $\lambda_2(t)$ of $A(t)$ cannot be expressed (or parameterized) as continuous functions of t. For $t \in (0, \infty)$, we can take $\lambda_1(t) = \sqrt{t}$ and $\lambda_2(t) = -\sqrt{t}$, both are continuous functions of t. Note that $A(t) \to A(0) = \left(\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right)$ which has eigenvalues $0, 0$ (at the origin). The eigenvalue topological continuity says that the eigenvalues $\lambda_1(t)$ and $\lambda_2(t)$ of $A(t)$ are close to the origin when t is close to the origin (i.e., $|t|$ is small enough). (Problem 15). ∎

Let $A_0 = (a_{ij}) \in \mathbb{M}_n$ have eigenvalues $\alpha_1, \ldots, \alpha_n$. Suppose that $A(t) = \big(a_{ij}(t)\big)$ is entrywise continuous on a real interval and $\lim_{t \to t_0} A(t) = A_0$. Then there exist eigenvalues $\lambda_1(t), \ldots, \lambda_n(t)$ of $A(t)$, each of which is a continuous function of t, such that $\lim_{t \to t_0} \lambda_1(t) = \alpha_1, \ldots, \lim_{t \to t_0} \lambda_n(t) = \alpha_n$. Moreover, $\rho(t) = \rho\big(A(t)\big) = \max_i |\lambda_i\big(A(t)\big)|$ (*spectral radius*) and $\sigma_{\max}\big(A(t)\big)$ (*spectral norm*, i.e., the largest singular value) are continuous functions of t. (See Sect. 4.1, Chap. 4, for more on spectral radius and spectral norm.)

A *sequence* $\{A_k\}$ of matrices converges (entrywise) to a matrix A if for every pair i and j, the (i,j)-entry of A_k converges to the (i,j)-entry of A (as $k \to \infty$), written as $\lim_{k\to\infty} A_k = A$, or $A_k \to A$ (as $k \to \infty$). For example,

$$A_k = \begin{pmatrix} \frac{1}{k} & \frac{k-1}{k} \\ 0 & \frac{1}{k} \end{pmatrix} \to \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \quad \text{as } k \to \infty.$$

We state a fact from matrix analysis (and omit its proof[®], which is a bit technical) about the eigenvalues of a convergent matrix sequence. Let $\lambda_1, \ldots, \lambda_n$ be the eigenvalues of $A \in \mathbb{M}_n$. If $A_k \to A$, then the eigenvalues $\lambda_{k,i}$ of A_k, $i = 1, 2, \ldots, n$, can be suitably numbered so that $\lambda_{k,i} \to \lambda_i$ for each i (as $k \to \infty$). One may explore more on the spectral theory of matrices.

Let $A \in \mathbb{M}_n$. The matrix power sequence $I, A, A^2, \ldots, A^k, \ldots$, or the matrix power series $\sum_{k=0}^{\infty} A^k$, is convergent if it is convergent entrywise. For example, if $A = \begin{pmatrix} 1 & 1 \\ 0 & -\frac{1}{2} \end{pmatrix}$, then $A^k = \begin{pmatrix} 1 & \frac{2}{3}(1-(-\frac{1}{2})^k \\ 0 & (-\frac{1}{2})^k \end{pmatrix} \to \begin{pmatrix} 1 & \frac{2}{3} \\ 0 & 0 \end{pmatrix}$ (as $k \to \infty$), and

$$\sum_{k=0}^{\infty} \begin{pmatrix} \frac{1}{2} & 1 \\ 0 & \frac{1}{3} \end{pmatrix}^k = \begin{pmatrix} 2 & 3 \\ 0 & \frac{3}{2} \end{pmatrix}. \quad \text{(See Problem 24 or Sect. 4.1.)}$$

We see *matrix functions* often in the literature. What is a matrix function after all? A *matrix function* or *function of a matrix* can have several different meanings. It can be an operation on a matrix producing a scalar, such as $\mathrm{tr}A$ and $\det A$; it can be a mapping from a matrix space to a matrix space, like $f(A) = A^2$, or $g(A) = e^A$; it can also be an entrywise operation on the matrix, for instance, $h(A) = (a_{ij}^2)$. In this book we use the term matrix function in a general (loose) sense, that is, a matrix function is a mapping $f : A \mapsto f(A)$ as long as $f(A)$ is well-defined for every matrix A (in a matrix space or a set), where $f(A)$ can be a scalar or a vector or a matrix.

Given a matrix $A \in \mathbb{M}_n$, the square of A, A^2, is well-defined. How about a *square root* of A? By a square root of $A \in \mathbb{M}_n$ we mean a matrix $X \in \mathbb{M}_n$ such that $X^2 = A$. Take $A = \left(\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right)$, for example. We can check that there does not exist a matrix X satisfying $X^2 = A$. So, this is not as easy as expected. In fact, generalizing a function $f(z)$ of a scalar variable $z \in \mathbb{C}$ to a matrix function $f(A)$ is a serious business and it takes great effort. One may refer to Lagrange-Sylvester interpolation theory for further studies.

Many terminologies in elementary math and calculus can be defined or extended for matrices. For instance, if $p(z)$ is a polynomial of $z \in \mathbb{C}$, then $p(A)$ is a polynomial of matrix $A \in \mathbb{M}_n(\mathbb{C})$ by replacing z with A; one may also define a matrix polynomial entrywise by $q(A) = (q(a_{ij}))$, where $q(z)$ is a polynomial in z and $A = (a_{ij})$. Interested readers may continue to study matrix calculus and matrix analysis.

[®]M. Artin, *Algebra*, 2nd edition, Pearson, 2017, p. 138

Problems

1. Let $A = \left(\begin{smallmatrix} 2 & 1 \\ 2 & 3 \end{smallmatrix}\right)$. Find a 2×2 matrix X such that $X^2 = A$.

2. Show that there is no 3×3 matrix X of rank 2 such that $X^2 = 0$.

3. Show that matrix $A = \left(\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right)$ has no square root, that is, there does not exist a matrix X satisfying $X^2 = A$. Show (by finding examples) that B and C have square roots, where

$$B = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \quad \text{and} \quad C = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix}.$$

   [Hint: Consider $(0, 0, 1)$ and $(0, 0, 1, 0)$ as first rows, respectively.]

4. Use a continuity argument to show the adjugate identity

$$\operatorname{adj}(AB) = \operatorname{adj}(B) \operatorname{adj}(A), \quad \text{where } A, B \in \mathbb{M}_n.$$

5. Explain why the continuity argument fails Example 2.6.1.

6. Let C and D be real matrices such that $CD^T + DC^T = 0$. If C is *skew-symmetric* (i.e., $C^T = -C$), show that DC is skew-symmetric.

7. Refer to the proof of Theorem 2.6.2. (i) Show that the roots can be ordered as $\lambda_1(t) \leq \cdots \leq \lambda_n(t)$. (ii) If the roots are ordered in moduli as $|\lambda_1(t)| \geq \cdots \geq |\lambda_n(t)|$, must each $\lambda_i(t)$ be continuous?

8. Show that the identity $A_\epsilon = P_\epsilon J_\epsilon P_\epsilon^{-1}$ holds for any $\epsilon \neq 0$, where

$$A_\epsilon = \begin{pmatrix} \epsilon & 0 \\ 1 & 0 \end{pmatrix}, \quad P_\epsilon = \begin{pmatrix} 0 & \epsilon \\ 1 & 1 \end{pmatrix}, \quad J_\epsilon = \begin{pmatrix} 0 & 0 \\ 0 & \epsilon \end{pmatrix}.$$

   What happens to the identity if $\epsilon \to 0$? Is A_0 similar to J_0?

9. If A is a nonsingular matrix, then $\operatorname{rank}(A^2) = \operatorname{rank}(A)$. Does it follow that $\operatorname{rank}(A^2) = \operatorname{rank}(A)$ for all square matrices A by a continuity argument (as $\operatorname{rank}(A_t^2) = \operatorname{rank}(A_t)$, where $A_t = A + tI$ is nonsingular)?

10. Show that the eigenvalues of A are independent of ϵ, where

$$A = \begin{pmatrix} \epsilon - 1 & -1 \\ \epsilon^2 - \epsilon + 1 & -\epsilon \end{pmatrix}.$$

11. Denote by $\sigma_{\max}$ and $\sigma_{\min}$ respectively the singular values of matrix $A = \left(\begin{smallmatrix} 1 & \epsilon \\ \epsilon & 1 \end{smallmatrix}\right)$, $\epsilon > 0$, $\sigma_{\max} \geq \sigma_{\min}$. Show that $\lim_{\epsilon \to 1^-} \sigma_{\max}/\sigma_{\min} = +\infty$.

12. Let $A \in \mathbb{M}_n$. What are the eigenvalues of $A + tI$? Explain why the (suitably parameterized) eigenvalues are continuous functions of t.

13. Let $A(t) = \left(\begin{smallmatrix} 0 & 1 \\ 1 & t \end{smallmatrix}\right)$, $t \in \mathbb{R}$. Find the eigenvalues of $A(t)$ for each $t \in \mathbb{R}$.

14. Let $A(t) = \left(\begin{smallmatrix} 1+t & 0 \\ 0 & 1-t \end{smallmatrix}\right)$, $t \in \mathbb{R}$. Find the eigenvalues of $A(t)$ in terms of $t \in \mathbb{R}$, and order the eigenvalues as $\lambda_1(t) \geq \lambda_2(t)$.

15. Let $A(t) = \left(\begin{smallmatrix} 0 & 1 \\ t & 0 \end{smallmatrix}\right)$, $t \in \mathbb{C}$. Show that for every given $\varepsilon > 0$, there exists $\delta > 0$ such that $|t| < \delta \Rightarrow |\lambda| < \varepsilon$, where λ is any eigenvalue of $A(t)$.

16. Let $p(x)$ and $q(x)$ be polynomials. If $q(A)$ is invertible for $A \in \mathbb{M}_n$, show that $p(A)(q(A))^{-1} = (q(A))^{-1}p(A)$. Conclude that if A has no eigenvalue equal to 1, then $(I - A)^{-1}(I + A^2) = (I + A^2)(I - A)^{-1}$.

17. Let $A = (a_{ij})$ be a nonsingular matrix with $A^{-1} = B = (b_{ij})$. Let $A(t)$ be a matrix with parameter t and $\lim_{t\to 0} A(t) = A$. Show that b_{ij} are continuous functions of the entries of A, and show that

$$\lim_{t\to 0} \big(A(t)\big)^{-1} = A^{-1}.$$

Conclude that

$$\lim_{\lambda\to 0} (A - \lambda I)^{-1} = A^{-1}$$

and that for $X \in \mathbb{M}_{m\times n}$ and $Y \in \mathbb{M}_{n\times m}$ (both are independent of ϵ),

$$\lim_{\epsilon\to 0} \begin{pmatrix} I_m & \epsilon X \\ \epsilon Y & I_n \end{pmatrix}^{-1} = I_{m+n}.$$

18. Let $A = \left(\begin{smallmatrix} -1 & 1 \\ 0 & -1 \end{smallmatrix}\right)$. Show that $\sum_{k=0}^{\infty} \frac{1}{k^2} A^k$ is convergent.

19. Let n be a positive number and let x be a real number. Define

$$A_n(x) = \begin{pmatrix} 1 & -\frac{x}{n} \\ \frac{x}{n} & 1 \end{pmatrix}.$$

Show that

$$\lim_{x\to 0} \Big(\frac{1}{x} \lim_{n\to\infty} \big(I - \big(A_n(x)\big)^n\big)\Big) = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}.$$

[Hint: The columns (and rows) of $A_n(x)$ are orthogonal.]

20. Let $f(x) = x^n + a_1 x^{n-1} + \cdots + a_n$ be a polynomial, where $a_1, \ldots, a_n \in \mathbb{C}$ are constants. Show that the roots $r_1, \ldots, r_n$ of f are bounded by

$$\tau = 1 + \max_{1\leq k\leq n} |a_k|, \quad \text{i.e.,} \quad |r_i| \leq \tau, \quad i = 1, \ldots, n.$$

[Hint: Consider $|f(r)/r^n|$ and show that $|f(r)/r^n| > 0$ if $|r| > \tau$.]

21. Let $f(x) = x^n + a_1 x^{n-1} + \cdots + a_n$ be a polynomial, where $a_1, \ldots, a_n \in \mathbb{C}$ are constants. Show that the roots $r_1, \ldots, r_n$ of f are bounded by

$$\gamma = 2 \max_{1 \le k \le n} |a_k|^{1/k}, \quad \text{i.e.,} \quad |r_i| \le \gamma, \quad i = 1, \ldots, n.$$

22. Let $A(t)$ be a square matrix whose entries are continuous functions of t in a real interval and assume that the eigenvalues of $A(t)$ are all real for each t. Show that $\rho(t) = \rho\big(A(t)\big) = \max_i |\lambda_i\big(A(t)\big)|$ (*spectral radius*) and $\sigma_{\max}\big(A(t)\big)$ (*spectral norm*) are continuous functions of t.

23. Let $X \in \mathbb{M}_n$. Show that the series $I + X + \frac{X^2}{2!} + \frac{X^3}{3!} + \cdots$ is convergent. (We denote $e^X = \sum_{k=0}^{\infty} \frac{1}{k!} X^k$.) Find e^A, e^B, $e^A e^B$, and $e^B e^A$, where

$$A = \begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix}, \quad B = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}.$$

Verify that $e^A e^B \neq e^B e^A$. Is $e^{X+Y} = e^X e^Y$ for all X and $Y \in \mathbb{M}_n$?

24. Show that $\sum_{k=0}^{\infty} \begin{pmatrix} \frac{1}{2} & 1 \\ 0 & \frac{1}{3} \end{pmatrix}^k = \sum_{k=0}^{\infty} \begin{pmatrix} (\frac{1}{2})^k & 6\big((\frac{1}{2})^k - (\frac{1}{3})^k\big) \\ 0 & (\frac{1}{3})^k \end{pmatrix} = \begin{pmatrix} 2 & 3 \\ 0 & \frac{3}{2} \end{pmatrix}$.

25. Let $A = \begin{pmatrix} 1 & 1 \\ 0 & -\frac{1}{2} \end{pmatrix}$.

    (a) Show that $A^k = \begin{pmatrix} 1 & \frac{2}{3}\big(1 - (-\frac{1}{2})^k\big) \\ 0 & (-\frac{1}{2})^k \end{pmatrix}$, where $k = 1, 2, \ldots$.

    (b) Show that A^k is convergent to $\begin{pmatrix} 1 & \frac{2}{3} \\ 0 & 0 \end{pmatrix}$ as $k \to \infty$.

    (c) Is $I + A + A^2 + \cdots$ convergent?

    (d) Find e^A.

26. Let $A, B \in \mathbb{M}_n$. Prove each of the following statements.

    (a) If $A = 0$, then $e^A = I$.

    (b) If $A = I$, then $e^A = eI$.

    (c) If $AB = BA$, then $e^{A+B} = e^A e^B = e^B e^A$.

    (d) The converse of (c) is false. [Hint: Take $A = B^* = \begin{pmatrix} 2\pi i & 1 \\ 0 & 0 \end{pmatrix}$.]

    (e) $e^{-A} = (e^A)^{-1}$.

    (f) $e^{ABA^{-1}} = A\, e^B A^{-1}$ (if A is nonsingular).

    (g) If λ is an eigenvalue of A, then e^λ is an eigenvalue of e^A.

    (h) $\det e^A = e^{\operatorname{tr} A}$.

    (i) $(e^A)^* = e^{A^*}$.

    (j) If A is Hermitian, then e^{iA} is unitary.

    (k) If A is real skew-symmetric, then e^A is (real) orthogonal.

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CHAPTER 3

Matrix Polynomials and Canonical Forms

Introduction: This chapter is devoted to matrix decompositions. We study the Schur decomposition, spectral decomposition, singular value decomposition (SVD), polar decomposition, and the Jordan decomposition. We begin with commuting matrices, study the polynomials that annihilate matrices, especially the minimal and characteristic polynomials, and discuss the similarity of the matrices associated with a given matrix.

3.1 Commuting Matrices

Matrices do not commute in general. One can easily find square matrices A and B of the same size, say $A = \left(\begin{smallmatrix} 1 & 0 \\ 0 & 0 \end{smallmatrix}\right)$ and $B = \left(\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right)$, such that $AB \neq BA$. Any square matrix A, however, commutes with the polynomials in A.

A question arises: If a matrix B commutes with A, is it true that B can be expressed as a polynomial in A? The answer is negative by taking A to be the $n \times n$ identity matrix I_n and B to be an $n \times n$ nondiagonal matrix. I_n commutes with all $n \times n$ matrices and it has n repeated eigenvalues 1. The following result says if A has no repeated eigenvalues and if B commutes with A, then B indeed can be written as a polynomial of A.

Theorem 3.1.1 *Let A and B be $n \times n$ matrices such that $AB = BA$. If all the eigenvalues of A are distinct (also said to be simple), then B can be expressed uniquely as a polynomial in A with degree no more than $n - 1$.*

Proof. Recall from Sect. 1.3, Chap. 1, that the eigenvectors belonging to different eigenvalues are linearly independent. If $\lambda_1, \lambda_2, \ldots, \lambda_n$ are the (distinct) eigenvalues of A with corresponding eigenvectors $u_1, u_2, \ldots, u_n \in \mathbb{C}^n$ (in columns), then $u_1, u_2, \ldots, u_n$ are linearly independent. Set

$$P = (u_1, u_2, \ldots, u_n).$$

Then P is an invertible matrix and $AP = P\,\mathrm{diag}(\lambda_1, \lambda_2, \ldots, \lambda_n)$, that is,

$$P^{-1}AP = \mathrm{diag}(\lambda_1, \lambda_2, \ldots, \lambda_n).$$

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Let

$$P^{-1}AP = C \quad \text{and} \quad P^{-1}BP = D.$$

Then

$$AB = BA \quad \text{if and only if} \quad CD = DC.$$

The diagonal entries of C are distinct, therefore D must be a diagonal matrix too (Problem 10). Let $D = \operatorname{diag}(\mu_1, \mu_2, \ldots, \mu_n)$, $\mu_1, \mu_2, \ldots, \mu_n \in \mathbb{C}$.

Consider the linear equation system of unknowns $x_0, x_1, \ldots, x_{n-1}$:

$$\begin{aligned} x_0 + \lambda_1 x_1 + \cdots + \lambda_1^{n-1} x_{n-1} &= \mu_1, \\ x_0 + \lambda_2 x_1 + \cdots + \lambda_2^{n-1} x_{n-1} &= \mu_2, \\ &\vdots \\ x_0 + \lambda_n x_1 + \cdots + \lambda_n^{n-1} x_{n-1} &= \mu_n. \end{aligned}$$

Since the coefficient matrix is a Vandermonde matrix that is nonsingular if $\lambda_1, \lambda_2, \ldots, \lambda_n$ are distinct (Problem 8 of Sect. 1.2 or Theorem 5.4.1 of Chap. 5), the system has a unique solution, say, $(a_0, a_1, \ldots, a_{n-1})$.

Define a polynomial with degree less than or equal to $n-1$ by

$$p(x) = a_0 + a_1 x + a_2 x^2 + \cdots + a_{n-1} x^{n-1}.$$

Then

$$p(\lambda_i) = \mu_i, \quad i = 1, 2, \ldots, n,$$

implying $p(C) = D$, so $P(p(C))P^{-1} = PDP^{-1}$. Therefore, $p(A) = B$. The polynomial $p(x)$ is unique because the solution to the system is unique. ∎

Such a method of finding a polynomial $p(x)$ of the given pairs (λ_i, μ_i) is known as *interpolation*. The proof also shows that there exists an invertible matrix P such that $P^{-1}AP$ and $P^{-1}BP$ are both diagonal.

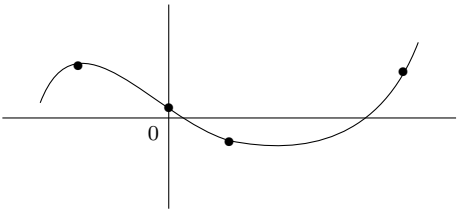


Fig. 3.1. Interpolation

Example 3.1.1 Commuting matrices

(a) $A = \begin{pmatrix} 1 & -1 \\ 2 & 0 \end{pmatrix}$ and $B = \begin{pmatrix} 1 & -3 \\ 6 & -2 \end{pmatrix}$ commute. A has two simple eigenvalues. B can be written as a polynomial $p(x)$ of A. ($p(x) = 3x - 2$.)

(b) $A = \left(\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right)$ and $B = \left(\begin{smallmatrix} 1 & 0 \\ 0 & 1 \end{smallmatrix}\right)$ commute; A is not diagonalizable.

(c) $R = \left(\begin{smallmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{smallmatrix}\right)$ is real orthogonal, i.e., $R^T R = RR^T = I_2$. Thus, $R^T = \left(\begin{smallmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{smallmatrix}\right) = p(R)$ for a polynomial p. ($p(x) = -x + 2\cos\theta$.)

(d) If A is an invertible matrix with distinct eigenvalues, then A^{-1} can be written as a polynomial of A, i.e., $A^{-1} = p(A)$ for some polynomial p. (In fact, the eigenvalues need not be distinct; see Sect. 3.3.) ∎

The previous theorem requires that the eigenvalues of A be distinct (i.e., simple). What can be said about A and B without this condition?

Theorem 3.1.2 *Let $A, B \in \mathbb{M}_n$. If $AB = BA$, then there exists a unitary matrix $U \in \mathbb{M}_n$ such that U^*AU and U^*BU are both upper-triangular.*

Proof. We use induction on n. If $n = 1$, we have nothing to show. Suppose that the assertion is true for any two commuting matrices of order $n - 1$.

For the case of n, we consider matrices as linear transformations on $\mathbb{C}^n$ (or on a subspace V of $\mathbb{C}^n$). Recall that every linear transformation on a finite-dimensional vector space over $\mathbb{C}$ has an eigenvalue (Sect. 1.3).

For each (given) eigenvalue μ of B, consider the eigenspace of B

$$V_\mu = \{v \in \mathbb{C}^n : Bv = \mu v\},$$

which is a subspace of $\mathbb{C}^n$. If A and B commute, then for every $v \in V_\mu$,

$$B(Av) = (BA)v = (AB)v = A(Bv) = A(\mu v) = \mu(Av).$$

Thus, $Av \in V_\mu$, that is, V_μ is an invariant subspace of A. Considered as a linear transformation on V_μ defined by $A(v) = Av$, A has an eigenvalue λ and a corresponding eigenvector v_1 in V_μ. Put these in equations:

$$Av_1 = \lambda v_1, \quad Bv_1 = \mu v_1, \quad v_1 \in V_\mu.$$

We may assume that v_1 is a unit vector. Extend v_1 to a unitary matrix $U_1 \in \mathbb{M}_n$ whose first column is v_1 (Sect. 1.4). By computation, we have

$$U_1^* A U_1 = \begin{pmatrix} \lambda & \alpha \\ 0 & C \end{pmatrix} \quad \text{and} \quad U_1^* B U_1 = \begin{pmatrix} \mu & \beta \\ 0 & D \end{pmatrix},$$

where $C, D \in \mathbb{M}_{n-1}$, and α, β are some row vectors of $n - 1$ components.

$AB = BA$ implies $CD = DC$. By induction hypothesis there is a unitary $U_2 \in \mathbb{M}_{n-1}$ such that $U_2^* C U_2$ and $U_2^* D U_2$ are both upper-triangular. Let

$$U = U_1 \begin{pmatrix} 1 & 0 \\ 0 & U_2 \end{pmatrix}.$$

Note that the product of unitary matrices is a unitary matrix. Thus, U is unitary. It follows that both U^*AU and U^*BU are upper-triangular. ∎

Example 3.1.2 Let $A = \begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix}$ and $B = \begin{pmatrix} 1 & -2 & -2 \\ -2 & 1 & -2 \\ -2 & -2 & 1 \end{pmatrix}$. One checks that $AB = BA$ by a simple computation. (In fact, $A = -\frac{1}{2}(B - I)$.) The eigenvalues of B are $-3, 3, 3$. For instance, take $\mu = -3$. Then the solutions to $(\mu I - B)v = 0$ are the vectors $(x, x, x)^T = xe$, where $x \in \mathbb{C}$ and $e^T = (1, 1, 1)$, that is, $V_{-3} = \{xe : x \in \mathbb{C}\}$. Clearly, e^T is an eigenvector associated to -3.

As a linear map on V_{-3}, A maps xe to $A(xe) = xAe = x(2e) = 2(xe)$. Thus, 2 is an eigenvalue of A and $e^T \in V_{-3}$ is an eigenvector associated to 2. Let $v_1 = \frac{1}{\sqrt{3}}e^T$. Then v_1 is a unit vector, $Av_1 = 2v_1$, and $Bv_1 = -3v_1$.

Let $\{v_1, v_2, v_3\}$ be an orthonormal basis for $\mathbb{C}^3$ (Sect. 1.4) and let $U_1 = (v_1, v_2, v_2)$. Then U_1 is unitary. Thus, $U_1^{-1} = U_1^*$. Write Av_2, Av_3, Bv_2, and Bv_3 as linear combinations of v_1, v_2, v_3. We can express

$$(Av_1, Av_2, Av_3) = (2v_1, Av_2, Av_3) = (v_1, v_2, v_2) \begin{pmatrix} 2 & \star \\ 0 & C \end{pmatrix}, \quad C \in \mathbb{M}_2$$

($\star$ for unspecified entries) and

$$(Bv_1, Bv_2, Bv_3) = (-3v_1, Bv_2, Bv_3) = (v_1, v_2, v_2) \begin{pmatrix} -3 & \star \\ 0 & D \end{pmatrix}, \quad D \in \mathbb{M}_2.$$

It follows that

$$U_1^* A U_1 = \begin{pmatrix} 2 & \star \\ 0 & C \end{pmatrix} \quad \text{and} \quad U_1^* B U_1 = \begin{pmatrix} -3 & \star \\ 0 & D \end{pmatrix}.$$

Since A and B commute, C and D commute. Repeating the above argument for C and D brings C and D to upper-triangular matrices. ∎

Theorem 3.1.2 asserts that commuting matrices can be transformed to upper-triangular matrices through simultaneous unitary similarity.

Example 3.1.3 If A is a Hermitian matrix, then there exists a unitary matrix U such that U^*AU is diagonal. This is because A and A^* commute. Hence, U^*AU and U^*A^*U are upper-triangular for some unitary U. On the other hand, $(U^*AU)^* = U^*A^*U$ is lower-triangular. It follows that U^*AU is diagonal. It is immediate that the diagonal entries of $U^*AU = U^{-1}AU$ are the eigenvalues of A; and the eigenvalues of A are all real. [Note: See the next section for more discussions on matrix decompositions.] ∎

Example 3.1.4 Let $A, B \in \mathbb{M}_{m\times n}$, $C \in \mathbb{M}_{n\times m}$, $a, b \in \mathbb{C}$, $ab \neq 0$. We have

$$ACB = aA + bB \;\; \Rightarrow \;\; ACB = BCA.$$

In particular, if A and B are n-square and $AB = A + B$, then $AB = BA$.

Proof. The special case where $C = I_n$ and $a = b = 1$ (Problem 23, Sect. 1.2) states that $AB = A + B \Rightarrow AB = BA$. This is shown directly:

$$\begin{aligned} AB = A + B &\Rightarrow (A - I_n)(B - I_n) = AB - A - B + I_n = I_n \\ &\Rightarrow (B - I_n)(A - I_n) = I_n \Rightarrow BA - A - B + I_n = I_n \\ &\Rightarrow BA = A + B = AB. \end{aligned}$$

With a slight modification, we can prove the general case. On the one hand,

$$\begin{aligned} (CA - bI_n)(CB - aI_n) &= CACB - aCA - bCB + abI_n \\ &= C(aA + bB) - aCA - bCB + abI_n = abI_n. \end{aligned}$$

Since $ab \neq 0$, we have $CA - bI_n = ab(CB - aI_n)^{-1}$. On the other hand,

$$\begin{aligned} ACB = aA + bB &\Rightarrow ACB - aA = bB \\ &\Rightarrow A(CB - aI_n) = bB. \end{aligned}$$

We obtain $A = bB(CB - aI_n)^{-1} = bB(ab)^{-1}(CA - bI_n) = a^{-1}(BCA - bB)$. It follows that $aA = BCA - bB$, that is, $BCA = aA + bB = ACB$. ∎

Problems

1. Let $A = \left(\begin{smallmatrix} 1 & 0 \\ 0 & 0 \end{smallmatrix}\right)$ and $B = \left(\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right)$. Find the matrices that commute with (a) A, (b) B, (c) A, B, and $A + B$, (d) B^2, or (e) $B + B^T$.

2. Let $A, B \in \mathbb{M}_n$ and let m be a positive integer. How many terms are there in the expansion of $(A+B)^m$ if (a) $AB = BA$ and (b) $AB \neq BA$?

3. Let $A = \left(\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right)$. Show that a 2×2 matrix B commutes with A (i.e., $AB = BA$) if and only if $B = xI_2 + yA$ for some scalars x and y.

4. Let $A = \left(\begin{smallmatrix} 0 & 0 \\ 0 & 2 \end{smallmatrix}\right)$, $B = \left(\begin{smallmatrix} 1 & 0 \\ 0 & 0 \end{smallmatrix}\right)$. Find a polynomial f such that $B = f(A)$.

5. Let $A_\varepsilon = \operatorname{diag}(0, \varepsilon, 1)$, $\varepsilon \in \mathbb{R}$, and $B = \operatorname{diag}(1, 2, 0)$. Show that

   (a) $A_\varepsilon B = BA_\varepsilon$.

   (b) A_0 can be expressed as a polynomial of B.

   (c) B cannot be expressed as a polynomial of A_0.

   (d) $B = f(A_\varepsilon)$ for some polynomial f (depending on ε) if $\varepsilon \neq 0, 1$.

6. If A and B are matrices such that both $AB = I_m$ and $BA = I_n$ hold, show that $m = n$, $AB = BA = I$, and B is the inverse of A.

7. Let $A = \left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right)$ and $B = \left(\begin{smallmatrix} 1 & 1 \\ -1 & 1 \end{smallmatrix}\right)$. (i) Show that $AB = BA$, (ii) find a unitary matrix U such that both U^*AU and U^*BU are diagonal, and (iii) show that such a matrix U cannot be a real matrix.

8. Let $A = \begin{pmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}$ and $B = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 2 \end{pmatrix}$.
   (a) Show that A and B commute (i.e., $AB = BA$).
   (b) For the eigenvalue $\mu = 1$ of B, find the eigenspace V_μ.
   (c) Show that V_μ is an invariant subspace of $\mathbb{C}^3$ under A.
   (d) Define a linear map $A|_{V_\mu}$ on V_μ by $(A|_{V_\mu})(v) = Av$. Find a matrix representation $A|_{V_\mu}$ relative to a basis of V_μ.
   (e) Show that every eigenpair of $A|_{V_\mu}$ is an eigenpair of A.
   (f) Find all eigenvalues of $A|_{V_\mu}$ over $\mathbb{C}$.
   (g) Find corresponding eigenvectors of $A|_{V_\mu}$ in $\mathbb{C}^3$ associated with the eigenvalues found in (f).
   (h) Of all the parts above, which ones hold over $\mathbb{R}$ (in place of $\mathbb{C}$)?

9. Let $A = \begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix}$ and $B = \begin{pmatrix} 1 & -2 & -2 \\ -2 & 1 & -2 \\ -2 & -2 & 1 \end{pmatrix}$.
   (a) Show that 3 is an eigenvalue of B. Find the eigenspace V_3.
   (b) Show that $AB = BA$ but A and B have no common eigenvalues.
   (c) Find a matrix of A considered as a linear map $x \mapsto Ax$ on V_3.
   (d) Find the eigenvalue(s) of the matrix in (c).
   (e) Find a common vector for A and B (of different eigenvalues).

10. Let A be a diagonal matrix with distinct diagonal entries. If B is a matrix such that $AB = BA$, show that B is also diagonal.

11. Let $A, B \in \mathbb{M}_n$ and let A have n distinct eigenvalues. Show that $AB = BA$ if and only if there exist n linearly independent vectors as the eigenvectors of A and B (of possibly different eigenvalues).

12. Let A and B be commuting matrices, i.e., $AB = BA$. If A has k distinct eigenvalues, show that B has at least k linearly independent eigenvectors. Does it follow that B has k distinct eigenvalues?

13. Let $A \in \mathbb{M}_n(\mathbb{R})$. If all eigenvalues of A are real, show that there exists an orthogonal matrix $Q \in \mathbb{M}_n(\mathbb{R})$ such that $Q^T AQ$ is upper-triangular.

14. Let $A \in \mathbb{M}_n$. Show that $(P^{-1}AP)^k = P^{-1}A^kP$ for $k = 2, 3, \ldots$, where $P \in \mathbb{M}_n$ is invertible. If $f(x)$ is a polynomial, show that
$$f(P^{-1}AP) = P^{-1}f(A)P.$$
Conclude that if A and B are similar, then $f(A)$ and $f(B)$ are similar.

15. Let $A \in \mathbb{M}_n$. Show that there exists a unitary $U \in \mathbb{M}_n$ such that U^*AU and $U^*f(A)U$ are upper-triangular for all polynomials f.

16. Show that the adjugates, inverses (if they exist), sums, products, and polynomials of upper-triangular matrices are upper-triangular.

17. Let $A, B, C \in \mathbb{M}_n$. If $AB = CA$, show that for any polynomial $f(x)$,

$$Af(B) = f(C)A.$$

18. Let A and B be Hermitian matrices of the same size. Show that if $AB = BA = 0$, then $\operatorname{rank}(A + B) = \operatorname{rank}(A) + \operatorname{rank}(B)$. Show by example that this need not be true for non-Hermitian matrices.

19. Let $A_1, A_2, \ldots, A_m \in \mathbb{M}_n$ be mutually commuting matrices, that is, $A_iA_j = A_jA_i$ for all i, j. Show that there exists a unitary matrix $U \in \mathbb{M}_n$ such that $U^*A_1U, U^*A_2U, \ldots, U^*A_mU$ are all upper-triangular.

20. Let $A, B \in \mathbb{M}_n$. If $AB = BA$, what are the eigenvalues of $A + B$ and AB in terms of those of A and B? (Write an explicit statement.)

21. Find an example of matrices A and B for which $AB = BA$, λ is an eigenvalue of A, μ is an eigenvalue of B, but $\lambda + \mu$ is not an eigenvalue of $A + B$, and $\lambda\mu$ is not an eigenvalue of AB.

22. Let $A, B \in \mathbb{M}_n$. If $AB = BA$, find the eigenvalues of $A^2 + B^2$ and the eigenvalues of $\left(\begin{smallmatrix} A & B \\ B & -A \end{smallmatrix}\right)$ in terms of the eigenvalues of A and B.

23. Consider the $n \times n$ matrices. What matrices commute with all diagonal matrices? With all Hermitian matrices? With all matrices?

24. Let A and B be $n \times n$ matrices. If A commutes with B and B^*, show that A commutes with B^*B and that $A + A^*$ commutes with $B + B^*$.

25. Apply Theorem 3.1.2 to each of the following cases.

    (a) $A \in \mathbb{M}_n, B = I_n$.

    (b) A is normal (i.e., $AA^* = A^*A$).

    (c) A is unitary (i.e., $AA^* = A^*A = I_n$).

26. Let A and B be n-square and m-square matrices, respectively, $m \leq n$. If $AP = PB$ for some $n \times m$ matrix P of full column rank, that is, $\operatorname{rank}(P) = m$, show that every eigenvalue of B is an eigenvalue of A.

27. If A and B are nonsingular matrices such that $AB - BA$ is singular, show that 1 is an eigenvalue of $A^{-1}B^{-1}AB$.

28. Let A and B be $n \times n$ matrices such that $\operatorname{rank}(AB - BA) \leq 1$. Show that A and B have a common eigenvector (associated with possibly different eigenvalues of A and B). Find a common eigenvector for

$$A = \begin{pmatrix} 1 & -1 \\ 1 & -1 \end{pmatrix} \text{ and } B = \begin{pmatrix} 1 & 0 \\ 1 & 0 \end{pmatrix}.$$

29. Let $S = \{AB - BA : A, B \in \mathbb{M}_n\}$. Show that $\dim(\operatorname{Span} S) = n^2 - 1$.

30. Let A and B be 2×2 block matrices in which each block is $n \times n$:

$$A = \begin{pmatrix} A_{11} & 0 \\ 0 & A_{22} \end{pmatrix}, \quad B = \begin{pmatrix} B_{11} & B_{12} \\ B_{21} & B_{22} \end{pmatrix}.$$

If A commutes with B (i.e., $AB = BA$), show that (a) A_{11} commutes with B_{11}, (b) A_{22} commutes with B_{22}, and (c) for any polynomial $f(x)$

$$f(A_{11})B_{12} = B_{12}f(A_{22}), \quad f(A_{22})B_{21} = B_{21}f(A_{11}).$$

In particular, if $A_{11} = a_1 I$, $A_{22} = a_2 I$, $a_1 \neq a_2$, then $B_{12} = B_{21} = 0$, thus $B = \left(\begin{smallmatrix} B_{11} & 0 \\ 0 & B_{22} \end{smallmatrix}\right)$. This also holds true if $f(A_{11}) = 0$ (or nonsingular) and $f(A_{22})$ is nonsingular (or respectively $f(A_{22}) = 0$).

31. The $n \times n$ *backward identity* matrix is the $n \times n$ matrix with 1 in the positions $(1, n)$, $(2, n-1)$, $\ldots$, $(n, 1)$, and 0 elsewhere, that is,

$$S = \begin{pmatrix} 0 & 0 & \ldots & 0 & 1 \\ 0 & 0 & \ldots & 1 & 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ 0 & 1 & \ldots & 0 & 0 \\ 1 & 0 & \ldots & 0 & 0 \end{pmatrix}.$$

Show that $S^{-1} = S^T = S$, equivalently, $S^T = S$, $S^2 = I$, $S^T S = I$. What is $\det S$? When $n = 3$, compute SAS for $A = (a_{ij}) \in \mathbb{M}_3$.

32. Let $A = (a_{ij})$ be an $n \times n$ matrix and let S be the $n \times n$ backward identity. Denote $\hat{A} = (\hat{a}_{ij}) = SAS$. Show that $\hat{a}_{ij} = a_{n-i+1,n-j+1}$.

33. Let $A, B \in \mathbb{M}_n$. Show that A and B are simultaneously triangularizable if and only if there exists a basis $\{v_1, \ldots, v_n\}$ of $\mathbb{C}^n$ such that $V_i = \operatorname{Span}\{v_1, \ldots, v_i\}$ is invariant under both A and B (see Sect. 1.3).

34. Let $f(x) = ax^3 + bx^2 + cx + d$ be a real-coefficient polynomial. Determine the values of a, b, c, d such that the graph of $f(x)$ contains the points $(0, 1)$, $(1, 1)$, $(2, 11)$, and $(3, 37)$ in the first quadrant of the xy-plane.

35. Let $a_1, \ldots, a_n \in \mathbb{C}$. Prove the existence of the polynomials p.

    (a) $p(a_i) = \overline{a_i}$, $i = 1, \ldots, n$.
    (b) $p(a_i) = |a_i|$, $i = 1, \ldots, n$.
    (c) $p(a_i) = \sqrt{a_i}$, $i = 1, \ldots, n$, if all a_i are nonnegative.
    (d) $p(a_i) = \frac{1}{a_i}$, $i = 1, \ldots, n$ if all a_i are nonzero.

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3.2 Matrix Decompositions

Decomposing matrices into some special forms of matrices via similarity are of fundamental importance in matrix theory. We study the following decompositions of matrices in this section: the Schur decomposition, spectral decomposition, singular value decomposition, and polar decomposition. We shall continue our study of Jordan decomposition in later sections.

Theorem 3.2.1 (Schur Decomposition; Schur Triangular Form) *Let* $\lambda_1, \lambda_2, \ldots, \lambda_n \in \mathbb{C}$ *be the eigenvalues of* $A \in \mathbb{M}_n(\mathbb{C})$ *(in any prescribed order). Then there exists a unitary matrix* $U \in \mathbb{M}_n(\mathbb{C})$ *such that* U^*AU *is upper-triangular, i.e.,*

$$U^*AU = \begin{pmatrix} \lambda_1 & & & \star \\ & \lambda_2 & & \\ & & \ddots & \\ 0 & & & \lambda_n \end{pmatrix} \quad (\star \textit{ for unspecified entries}).$$

Proof. This follows from Theorem 3.1.2 with $B = I$. Note that in the proof of Theorem 3.1.2, A is considered as a linear map on an eigenspace of B. We present below a matrix proof without using linear transformations.

If $n = 1$, there is nothing to show. Let $n > 1$ and suppose that the statement is true for the matrices with orders less than n. We show by induction that it is true for the matrices of order n.

Let x_1 be a unit eigenvector of A associated with eigenvalue λ_1:

$$Ax_1 = \lambda_1 x_1, \quad \|x_1\| = (x_1^* x_1)^{\frac{1}{2}} = 1.$$

Let $S = (x_1, y_2, \ldots, y_n)$ be a unitary matrix (or a real orthogonal matrix if x_1 is real) with x_1 as its first column (see Theorem 1.4.5). Then

$$\begin{aligned} AS &= (Ax_1, Ay_2, \ldots, Ay_n) \\ &= (\lambda_1 x_1, Ay_2, \ldots, Ay_n) \\ &= S(u, S^{-1}Ay_2, \ldots, S^{-1}Ay_n), \end{aligned}$$

where $u = (\lambda_1, 0, \ldots, 0)^T$. Thus, S^*AS has u as the first column. We write

$$S^*AS = S^{-1}AS = \begin{pmatrix} \lambda_1 & v \\ 0 & B \end{pmatrix},$$

where $v \in \mathbb{C}^{n-1}$ is a row vector and $B \in \mathbb{M}_{n-1}(\mathbb{C})$ has eigenvalues $\lambda_2, \ldots, \lambda_n$.

Applying the induction hypothesis on B, we have a unitary matrix T of order $n-1$ such that T^*BT is an upper-triangular matrix. Let $U = S \left(\begin{smallmatrix} 1 & 0 \\ 0 & T \end{smallmatrix}\right)$. Then U is unitary, U^*AU is upper-triangular and the diagonal entries $\lambda_1, \lambda_2, \ldots, \lambda_n$ of the upper-triangular matrix are the eigenvalues of A. ∎

Slightly modifying the proof, we see that if A is real and if all eigenvalues of A are real, then the unitary matrix U can be chosen to be real orthogonal.

The Schur decomposition theorem, also known as the *Schur triangularization*, says that every square matrix can be brought to an upper-triangular form via a unitary similarity. Two $n \times n$ matrices A and B are said to be *unitarily similar* if $A = U^*BU$ for some $n \times n$ unitary matrix U.

Example 3.2.1 Unitary similarity to an upper-triangular matrix.

(1) $\left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right)$ with eigenvalues $\pm i$ is unitarily similar to $\left(\begin{smallmatrix} i & \star \\ 0 & -i \end{smallmatrix}\right)$.

(2) $\left(\begin{smallmatrix} 1 & 1 \\ -1 & -1 \end{smallmatrix}\right)$ with two 0 eigenvalues is unitarily similar to $\left(\begin{smallmatrix} 0 & \star \\ 0 & 0 \end{smallmatrix}\right)$.

(3) $\left(\begin{smallmatrix} 1 & 0 \\ 2 & 3 \end{smallmatrix}\right)$ is unitarily similar to $\left(\begin{smallmatrix} 1 & 2 \\ 0 & 3 \end{smallmatrix}\right)$ (but not through permutations).

(4) $(1,1,1)^T(1,2,3)$ with rank one is unitarily similar to $\left(\begin{smallmatrix} 6 & \star & \star \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{smallmatrix}\right)$. ∎

Schur triangularization is one of the most important theorems in linear algebra and matrix theory. It is used repeatedly in this book. As an application, we see (by taking the conjugate transpose) that any Hermitian matrix A (i.e., $A^* = A$) is unitarily diagonalizable. The same is true for normal matrices A (i.e., $A^*A = AA^*$), because the normality, together with the Schur form of A, implies the diagonal form of U^*AU (Problem 7).

Let $A \in \mathbb{M}_n$. A is said to be *positive semidefinite* if $x^*Ax \geq 0$ for all $x \in \mathbb{C}^n$. A is said to be *positive definite* if $x^*Ax > 0$ for all nonzero $x \in \mathbb{C}^n$.

A positive semidefinite matrix is necessarily Hermitian. Here is why:

Since $x^*Ax \geq 0$ for all $x \in \mathbb{C}^n$, setting x to be the column vector with the sth component 1, the tth component $c \in \mathbb{C}$, and 0 elsewhere, reveals

$$x^*Ax = a_{ss} + a_{tt}|c|^2 + a_{ts}\bar{c} + a_{st}c \geq 0.$$

It follows that each diagonal entry a_{ss} is nonnegative by putting $c = 0$, and that $a_{st} = \overline{a_{ts}}$ or $A^* = A$ by putting $c = 1$ and i, respectively.

Alternatively, using the fact (Problem 11) that if $y^*By = 0$ for all $y \in \mathbb{C}^n$, then $B = 0$, we obtain $A^* = A$ because $x^*(A^* - A)x = 0$ for all $x \in \mathbb{C}^n$.

It is immediate that the eigenvalues λ of a positive semidefinite A are nonnegative since $x^*Ax = \lambda x^*x \geq 0$ for eigenvectors x associated with λ.

Let $A, B \in \mathbb{M}_n$ be Hermitian matrices. We write

$A \geq 0$ if A is positive semidefinite, i.e., $x^*Ax \geq 0$ for all $x \in \mathbb{C}^n$,
$A > 0$ if A is positive definite, i.e., $A \geq 0$ and A is nonsingular,
$A \geq B$ if $A - B \geq 0$, and
$A > B$ if $A - B > 0$.

Note: In some texts and in Sect. 5.7 (on nonnegative matrices), $A \geq 0$ and $A > 0$ mean entrywise positivity. This is easy to tell from the contexts. For convenience, we sometimes write $A \geq 0$ as $0 \leq A$, and $A > 0$ as $0 < A$.

Example 3.2.2 Positive semidefinite matrices.

(1) $I_n \geq 0$ and $E_n \geq 0$, where E_n is the all-ones n-square matrix.

(2) $\begin{pmatrix} 2 & -1 \\ -1 & 3 \end{pmatrix} \geq \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$.

(3) $A^*A \geq 0$ for any $A \in \mathbb{M}_{m\times n}$.

(4) $I_n + A^*A \geq A^* + A$ for any $A \in \mathbb{M}_n$.

(5) $AA^* + BB^* \geq 0$ and $\begin{pmatrix} A^*A & A^*B \\ B^*A & B^*B \end{pmatrix} \geq 0$ for $A \in \mathbb{M}_{m\times p}$, $B \in \mathbb{M}_{m\times q}$. ∎

A positive semidefinite matrix is Hermitian, and a Hermitian matrix is normal. We summarize the above discussions in the following theorem.

Theorem 3.2.2 (Spectral Decomposition) *Let $A \in \mathbb{M}_n$ have eigenvalues $\lambda_1, \lambda_2, \ldots, \lambda_n \in \mathbb{C}$. A is normal (i.e., $A^*A = AA^*$) if and only if A is unitarily diagonalizable, i.e., there exists a unitary matrix $U \in \mathbb{M}_n$ such that*

$$U^*AU = \operatorname{diag}(\lambda_1, \lambda_2, \ldots, \lambda_n). \tag{3.1}$$

Furthermore, A is Hermitian if and only if all λ_i's in (3.1) *are real, and A is positive semidefinite if and only if all λ_i's in* (3.1) *are nonnegative. Additionally, the columns $u_1, u_2, \ldots, u_n$ of U in* (3.1) *are orthonormal eigenvectors associated with the eigenvalues $\lambda_1, \lambda_2, \ldots, \lambda_n$, respectively.*

The above spectral diagonalization decomposition (of a normal matrix), the Schur decomposition, and the Jordan decomposition (Sect. 3.4) are about factoring a matrix into a canonical form in terms of its eigenvalues (and eigenvectors). These decompositions are also known as *eigendecomposition.*

Example 3.2.3 Let A be an n-square matrix. Let $\lambda_1, \lambda_2, \ldots, \lambda_n$ and $\sigma_1, \sigma_2, \ldots, \sigma_n$ be the eigenvalues and singular values of A, respectively. Show that

$$|\lambda_1|^2 + |\lambda_2|^2 + \cdots + |\lambda_n|^2 \leq \sigma_1^2 + \sigma_2^2 + \cdots + \sigma_n^2 \tag{3.2}$$

and that the inequality is an equality if and only if A is normal.

Proof. Let $A = U^*TU$ be a Schur decomposition of A, where U is unitary, $T = (t_{ij})$ is upper-triangular, and $\lambda_k = t_{kk}$, $k = 1, 2, \ldots, n$. We show that T is diagonal. Note that $\sigma_1^2, \sigma_2^2, \ldots, \sigma_n^2$ are the eigenvalues of A^*A (Sect. 2.4). Therefore, $\operatorname{tr}(A^*A) = \sigma_1^2 + \sigma_2^2 + \cdots + \sigma_n^2$. On the other hand, we compute

$$\begin{aligned} \operatorname{tr}(A^*A) &= \operatorname{tr}(U^*T^*TU) = \operatorname{tr}(T^*T) = \sum_{i,j} |t_{ij}|^2 \\ &= |\lambda_1|^2 + |\lambda_2|^2 + \cdots + |\lambda_n|^2 + \sum_{i<j} |t_{ij}|^2. \end{aligned}$$

If equality (3.2) holds, then $t_{ij} = 0$ for all $i \neq j$, that is, T is diagonal. It follows that $T^*T = TT^*$, implying $A^*A = AA^*$, i.e., A is normal.

Conversely, if A is normal, by the spectral decomposition of A, the singular values of A are $|\lambda_1|, |\lambda_2|, \ldots, |\lambda_n|$. Equality in (3.2) holds. ∎

(3.2) is one of the many conditions on matrix normality (Sect. 9.1).

It is immediate from (3.1) that if $A \geq 0$ then $\det A \geq 0$ (with equality if and only if A is singular) and $\operatorname{tr} A \geq 0$ (with equality if and only if $A = 0$). The rank of a normal matrix is equal to the number of nonzero eigenvalues.

The spectral decomposition theorem says that $A \in \mathbb{M}_n$ is a normal matrix if and only if there exist n orthonormal eigenvectors $u_1, u_2, \ldots, u_n$ corresponding to the eigenvalues $\lambda_1, \lambda_2, \ldots, \lambda_n$ of A, respectively, that is,

$$AU = U \operatorname{diag}(\lambda_1, \lambda_2, \ldots, \lambda_n), \text{ or } Au_i = \lambda_i u_i, \quad i = 1, 2, \ldots, n.$$

We can rewrite (3.1) as $A = U \operatorname{diag}(\lambda_1, \lambda_2, \ldots, \lambda_n)U^*$. With this form, one easily finds the *square roots* of positive semidefinite matrices.

Theorem 3.2.3 *Let A be an $n \times n$ positive semidefinite matrix. Then there exists a unique $n \times n$ positive semidefinite matrix B such that $B^2 = A$.*

Proof. Let $A = U \operatorname{diag}(\lambda_1, \lambda_2, \ldots, \lambda_n)U^*$ be a spectral decomposition of A. Take $B = U \operatorname{diag}(\sqrt{\lambda_1}, \sqrt{\lambda_2}, \ldots, \sqrt{\lambda_n})U^*$. Then $B \geq 0$ and $B^2 = A$.

For uniqueness, let $C \geq 0$ and $C^2 = A$. By the spectral decomposition, let $C = V \operatorname{diag}(\mu_1, \mu_2, \ldots, \mu_n)V^*$, where V is unitary and μ_i's are nonnegative scalars (actually $\mu_i = \sqrt{\lambda_i}$). Then $A = C^2$ implies that

$$U \operatorname{diag}(\lambda_1, \lambda_2, \ldots, \lambda_n)U^* = V \operatorname{diag}(\mu_1^2, \mu_2^2, \ldots, \mu_n^2)V^*,$$

or $W \operatorname{diag}(\lambda_1, \lambda_2, \ldots, \lambda_n) = \operatorname{diag}(\mu_1^2, \mu_2^2, \ldots, \mu_n^2)W$, where $W = (w_{ij}) = V^*U$. This results in $w_{ij}\lambda_j = \mu_i^2 w_{ij}$ for all i, j. Therefore, $w_{ij}\sqrt{\lambda_j} = \mu_i w_{ij}$. So, $W \operatorname{diag}(\sqrt{\lambda_1}, \sqrt{\lambda_2}, \ldots, \sqrt{\lambda_n}) = \operatorname{diag}(\mu_1, \mu_2, \ldots, \mu_n)W$, implying $U \operatorname{diag}(\sqrt{\lambda_1}, \sqrt{\lambda_2}, \ldots, \sqrt{\lambda_n})U^* = V \operatorname{diag}(\mu_1, \mu_2, \ldots, \mu_n)V^*$, i.e., $B = C$. ∎

The matrix B in Theorem 3.2.3 is called the *square root* of A and is denoted by $A^{1/2}$. If $A = U \operatorname{diag}(\lambda_1, \lambda_2, \ldots, \lambda_n)U^*$, where all $\lambda_i \geq 0$, then $A^{1/2} = U \operatorname{diag}(\sqrt{\lambda_1}, \sqrt{\lambda_2}, \ldots, \sqrt{\lambda_n})U^*$. Similarly, we define (Problem 50)

$$A^p = U \operatorname{diag}(\lambda_1^p, \lambda_2^p, \ldots, \lambda_n^p)U^*, \quad p \geq 0. \tag{3.3}$$

The Gram-Schmidt process (Sect. 1.4) ensures that a matrix, say A, of full column rank can be factored in a QR form, that is, $A = QR$, where Q is a matrix whose columns are orthonormal and R is an upper-triangular matrix whose diagonal entries are nonnegative (Theorem 1.4.4).

We now present the QR decomposition for a general matrix.

Theorem 3.2.4 (QR Decomposition) *Let A be an $m \times n$ matrix. Let R represent an $n \times n$ upper-triangular matrix with nonnegative diagonal entries. If $m = n$, there exist a unitary $Q \in \mathbb{M}_n$ and an $R \in \mathbb{M}_n$ such that $A = QR$. If $m > n$, there exist a unitary $Q \in \mathbb{M}_m$ and an $R \in \mathbb{M}_n$ such that $A = Q \begin{pmatrix} R \\ 0 \end{pmatrix}$. If $m < n$, then then there exist a partially unitary $Q \in \mathbb{M}_{m \times n}$, i.e., $QQ^* = I_m$ (or the rows of Q are orthonormal) and an $R \in \mathbb{M}_n$ such that $A = QR$.*

Besides, if A is real, then factors Q and R can be chosen to be real.

Proof. We first show the statement for square matrices, i.e., $m = n$, then generalize it to the cases $m > n$ and $m < n$. We proceed by induction on n.

If $n = 1$, let $A = (c)$ and $c = e^{i\theta}|c|$. Then $A = (e^{i\theta})(|c|)$ is a QR form.

Let $n > 1$. Suppose that a square matrix of order less than n has a QR form. Let $A \in \mathbb{M}_n$. We partition A according to its columns as $A = (a_1, a_2, \ldots, a_n)$. If $a_1 = 0$, then $A = \left(\begin{smallmatrix} 0 & \star \\ 0 & A_1 \end{smallmatrix}\right)$, where A_1 has order $(n-1)$ (and $\star$ represents unspecified entries). The induction hypothesis applies to A_1.

If $a_1 \neq 0$, then $\frac{a_1}{\|a_1\|} \in \mathbb{C}^n$ is a unit (column) vector. By Theorem 1.4.5 (or by Problem 48, Sect. 1.4.), there exist orthonormal vectors $v_2, \ldots, v_n$ such that $Q_1 = (\frac{a_1}{\|a_1\|}, v_2, \ldots, v_n) \in \mathbb{M}_n$ is unitary (real orthogonal if a_1 is real). Observe that $a_1 = Q_1 \begin{pmatrix} \|a_1\| \\ 0 \end{pmatrix}$ (in which the 0 means $(0, \ldots, 0)^T \in \mathbb{C}^{n-1}$). We have $Q_1^{-1}A = (Q_1^{-1}a_1, Q_1^{-1}a_2, \ldots, Q_1^{-1}a_n) = \begin{pmatrix} \|a_1\| & \star \\ 0 & A_1 \end{pmatrix}$, where $A_1 \in \mathbb{M}_{n-1}$.

By induction hypothesis, there exist a unitary matrix $Q_2 \in \mathbb{M}_{n-1}$ and an upper-triangular matrix $R_2 \in \mathbb{M}_{n-1}$ with nonnegative diagonal entries such that $A_1 = Q_2 R_2$. We arrive at $A = Q_1 \begin{pmatrix} \|a_1\| & \star \\ 0 & A_1 \end{pmatrix} = Q_1 \begin{pmatrix} 1 & 0 \\ 0 & Q_2 \end{pmatrix} \begin{pmatrix} \|a_1\| & \star \\ 0 & R_2 \end{pmatrix}$. Put $Q = Q_1 \begin{pmatrix} 1 & 0 \\ 0 & Q_2 \end{pmatrix}$ and $R = \begin{pmatrix} \|a_1\| & \star \\ 0 & R_2 \end{pmatrix}$. Then $A = QR$, as desired.

For the case $m > n$, let $B = (A, 0)$ be an m-square matrix (by adding zero columns to A). By the above argument for square matrices, B has a QR decomposition $B = QR$. Partition $R = \begin{pmatrix} R_1 & \star \\ 0 & R_2 \end{pmatrix}$, where $R_1 \in \mathbb{M}_n$ is upper-triangular and has nonnegative diagonal entries. Note that the first n columns of QR are $Q \begin{pmatrix} R_1 \\ 0 \end{pmatrix}$ and first n columns of B are A. So, $A = Q \begin{pmatrix} R_1 \\ 0 \end{pmatrix}$.

For the case $m < n$, let $C = \begin{pmatrix} A \\ 0 \end{pmatrix} \in \mathbb{M}_n$. Suppose $C = QR$ is a QR decomposition of C. Let $Q = \begin{pmatrix} Q_1 \\ Q_2 \end{pmatrix}$, where Q_1 consists of the first m rows of the unitary matrix Q. The rows of Q_1 are orthonormal, i.e., $Q_1 Q_1^* = I_m$. Since $QR = \begin{pmatrix} Q_1 \\ Q_2 \end{pmatrix} R = \begin{pmatrix} Q_1 R \\ Q_2 R \end{pmatrix}$ and as $C = \begin{pmatrix} A \\ 0 \end{pmatrix} = QR$, we get $A = Q_1 R$. ∎

Example 3.2.4 $A = \begin{pmatrix} 1 & \sqrt{2} \\ -1 & 0 \end{pmatrix} = \begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{pmatrix} \cdot \begin{pmatrix} \sqrt{2} & 1 \\ 0 & 1 \end{pmatrix}$ is a QR form of A. ∎

Now we turn our attention to one of the most important theorems in matrix theory – the singular value decomposition (SVD) theorem.

Recall from Sect. 2.4 that the *singular values* of an $m \times n$ matrix A are the nonnegative square roots of the eigenvalues of A^*A which is always positive semidefinite because $x^*(A^*A)x = (Ax)^*(Ax) \geq 0$. Let σ_i be the ith largest singular value of A and let λ_i be the ith largest eigenvalue of A^*A. Then

$$\sigma_i(A) = \sqrt{\lambda_i(A^*A)}, \quad i = 1, 2, \ldots, n.$$

It follows that $\sigma_j(A^*) = \sqrt{\lambda_j(AA^*)}$, $j = 1, 2, \ldots, m$. Since A^*A and AA^* have the same nonzero eigenvalues, $\sigma_k(A^*) = \sigma_k(A)$ for $k \leq \min\{m, n\}$.

Additionally, because $\operatorname{rank}(A^*A) = \operatorname{rank}(A) = r$ (Theorem 2.3.1), we see that A has r positive singular values and $n - r$ zero singular values.

Theorem 3.2.5 (Singular Value Decomposition (SVD)) *Let $A \in \mathbb{M}_{m\times n}$ be a nonzero matrix having rank r and positive singular values $\sigma_1, \ldots, \sigma_r$. Then there exist a unitary $U \in \mathbb{M}_m$ and a unitary $V \in \mathbb{M}_n$ such that*

$$A = UDV = U_r D_r V_r = \sigma_1 u_1 v_1 + \cdots + \sigma_r u_r v_r, \tag{3.4}$$

where $D = \begin{pmatrix} D_r & 0 \\ 0 & 0 \end{pmatrix} \in \mathbb{M}_{m\times n}$ with $D_r = \operatorname{diag}(\sigma_1, \ldots, \sigma_r)$, U_r comprises the first r columns $u_1, \ldots, u_r$ of U, and V_r comprises the first r rows $v_1, \ldots, v_r$ of V.

Proof. If $m = n = 1$ and $A = (c) \in \mathbb{C}$, then $|c|$ is the singular value of A, and $A = |c|e^{i\theta}$ for some $\theta \in \mathbb{R}$. If A is a nonzero row or column vector, say, $A = (a_1, \ldots, a_n)$, then σ_1 is the length of the vector. Let V be an $n \times n$ unitary matrix with $\frac{1}{\sigma_1}(a_1, \ldots, a_n)$ as its first row. Then $A = (\sigma_1, 0, \ldots, 0)V$.

Let $m, n > 1$. Let (σ_1^2, w_1) be an eigenpair of A^*A with $\|w_1\| = 1$, i.e.,

$$(A^*A)w_1 = \sigma_1^2 w_1, \quad w_1^* w_1 = 1, \ \ w_1 \in \mathbb{C}^n. \tag{3.5}$$

Let

$$u_1 = \tfrac{1}{\sigma_1} A w_1 \in \mathbb{C}^m. \tag{3.6}$$

By computations, we have $\|u_1\| = 1$, $A^*u_1 = \sigma_1 w_1$, and $w_1^* A^* u_1 = \sigma_1$.

Let $W_1 = (w_1, x_2, \ldots, x_n) \in \mathbb{M}_n$ and $U_1 = (u_1, y_2, \ldots, y_m) \in \mathbb{M}_m$ be unitary matrices. Then $A^*u_1 = \sigma_1 w_1$ gives $x_j^* A^* u_1 = \sigma_1 x_j^* w_1 = 0$ for $j \geq 2$. Similarly, $w_1^* A^* = (Aw_1)^* = \sigma_1 u_1^*$ implies $w_1^* A^* y_j = \sigma_1 u_1^* y_j = 0$ for $j \geq 2$. Thus, the first column of $W_1^* A^* U_1$ is $W_1^* A^* u_1 = \sigma_1 W_1^* w_1 = (\sigma_1, 0, \ldots, 0)^T$, and the first row of $W_1^* A^* U_1$ is $w_1^* A^* U_1 = \sigma_1 u_1^* U_1 = (\sigma_1, 0, \ldots, 0)$. Write

$$W_1^* A^* U_1 = \begin{pmatrix} \sigma_1 & 0 \\ 0 & B^* \end{pmatrix} \quad \text{or} \quad A = U_1 \begin{pmatrix} \sigma_1 & 0 \\ 0 & B \end{pmatrix} W_1^*, \tag{3.7}$$

where $B \in \mathbb{M}_{(m-1)\times(n-1)}$. The singular values of B are also singular values of A. If $B = 0$, take $U = U_1$ and $V = W_1^*$; otherwise, repeat the above process for B. The last two identities in (3.4) are left as Problem 39. ∎

Note that the unitary matrices in SVD are not unique in general.

If A is a real matrix, U and V in Theorem 3.2.5 can be chosen to be real. Besides, if A is an $n \times n$ matrix, then U and V are $n \times n$ unitary matrices, and $A = UVV^*DV = WP$, where $W = UV$ is unitary and $P = V^*DV$ is positive semidefinite. Since $A^*A = PW^*WP = P^2$, we see $P = (A^*A)^{1/2}$, which is uniquely determined by A. For a general $A \in \mathbb{M}_{m\times n}$, we denote

$$|A| = (A^*A)^{1/2}$$

and call $|A|$ the *modulus* of A. Note that $|A|$ is always positive semidefinite. the eigenvalues of $|A|$ are the singular values of A (or simply $\lambda(|A|) = \sigma(A)$).

It is evident that a square matrix A is positive semidefinite if and only if $A = |A|$ and that if H is a Hermitian matrix, then $|H| = (H^2)^{1/2}$.

Note about the notation: We have also used the vertical bars for the determinant of a block matrix such as $\left|\begin{smallmatrix} A & B \\ C & D \end{smallmatrix}\right|$ (Sect. 1.2) for the typographical reason. One can easily tell what is intended from the context.

Example 3.2.5 Compute $|A| = (A^*A)^{1/2}$ and $|A^*| = (AA^*)^{1/2}$.

(a) For $A = \left(\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right)$, $|A| = \left(\begin{smallmatrix} 0 & 0 \\ 0 & 1 \end{smallmatrix}\right)$ and $|A^*| = \left(\begin{smallmatrix} 1 & 0 \\ 0 & 0 \end{smallmatrix}\right)$.
Note: $|A|^2 \neq A^2 = 0$, $|A|^2 \neq |A^2|$, $A|A| \neq |A|A$.

(b) For $A = \left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right)$, $|A| = |A^*| = \left(\begin{smallmatrix} 1 & 0 \\ 0 & 1 \end{smallmatrix}\right)$.

(c) For $A = \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}$, $A = |A| = |A^*| = \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}$.

(d) For $A = \left(\begin{smallmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \end{smallmatrix}\right)$, $|A| = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix}$ and $|A^*| = \left(\begin{smallmatrix} 1 & 0 \\ 0 & 0 \end{smallmatrix}\right)$.

(e) For $A = (0, 1, 0)$, $|A| = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix}$ and $|A^*| = (1)$. ∎

Note that in some other texts a singular value decomposition (SVD) of A in Theorem 3.2.5 is written as $A = U\Sigma V^*$, where Σ is nonnegative diagonal, U and V are unitary. Moreover, steps (3.5) and (3.6) in the proof show how to get started with constructing an SVD of a matrix (see also Problem 52).

Theorem 3.2.6 (Polar Decomposition) *Let A be an $m \times n$ matrix. Then there exists an $m \times n$ (unitary or partially unitary) matrix W such that*

$$A = W|A| = |A^*|W \quad \textit{and} \quad |A| = W^*A,\ |A^*| = AW^*,$$

where $W^*W = I_n$ if $m > n$, $WW^* = I_m$ if $m < n$ (such a W is referred to as partially unitary or a partial isometry), and W is unitary if $m = n$.

Proof. For $m = n$, let $A = UDV$ be an SVD of A, where U and V are unitary, and D is nonnegative diagonal. Then $|A| = V^*DV$ and $|A^*| = UDU^*$. We have $A = UDV = (UV)(V^*DV) = (UDU^*)(UV) = W|A| = |A^*|W$, where $W = UV$. Note that W is unitary (for the case $m = n$, i.e., A is square).

If $m > n$, let $B = (A, 0) \in \mathbb{M}_m$. Then $BB^* = AA^*$ and $B^*B = \left(\begin{smallmatrix} A^*A & 0 \\ 0 & 0 \end{smallmatrix}\right)$, which imply $|B^*| = |A^*|$ and $|B| = \begin{pmatrix} |A| & 0 \\ 0 & 0 \end{pmatrix}$. By the above argument for square matrices, there is an $m \times m$ unitary matrix R such that $B = R|B| = |B^*|R$. Let W consist of the first n columns of R and let $R = (W, P)$. Then $B = R|B| = |B^*|R$ is the same as $(A, 0) = (W, P)\begin{pmatrix} |A| & 0 \\ 0 & 0 \end{pmatrix} = |A^*|(W, P)$. By comparing the first n columns, we arrive at $A = W|A| = |A^*|W$. The columns of W (taken from the unitary R) are orthonormal, so $W^*W = I_n$.

The case of $m \leq n$ is obtained by applying the above argument to A^*.

We show $|A| = W^*A$. Since $A = W|A| \Rightarrow A^* = |A|W^* \Rightarrow |A|^2 = A^*A = |A|W^*W|A|$, we have $|A|(I_n - W^*W)|A| = 0$, implying $(I_n - W^*W)|A| = 0$ as $I_n - W^*W \geq 0$ (Problem 49). Hence, $|A| = W^*W|A| = W^*A$. ∎

Example 3.2.6 Let $A \in \mathbb{M}_{m\times n}$. If $m < n$, then there exists a unitary matrix $U \in \mathbb{M}_n$ such that $A^*A = U^* \left(\begin{smallmatrix} AA^* & 0 \\ 0 & 0 \end{smallmatrix}\right) U$; if $m = n$, then $A^*A = U^*(AA^*)U$.

Proof. If $m = n$, A^*A and AA^* are both unitarily similar to the diagonal matrix $\operatorname{diag}(\sigma_1^2(A), \ldots, \sigma_n^2(A))$. For $m < n$, make $\left(\begin{smallmatrix} A \\ 0 \end{smallmatrix}\right)$ an n-square matrix. Then $\left(\begin{smallmatrix} A \\ 0 \end{smallmatrix}\right)(A^*, 0) = \left(\begin{smallmatrix} AA^* & 0 \\ 0 & 0 \end{smallmatrix}\right)$ is unitarily similar to $(A^*, 0)\left(\begin{smallmatrix} A \\ 0 \end{smallmatrix}\right) = A^*A$. Alternatively, let $A = |A^*|W$ be a polar decomposition of A, where $W \in \mathbb{M}_{m\times n}$ and $WW^* = I_m$. Let $U = \left(\begin{smallmatrix} W \\ V \end{smallmatrix}\right) \in \mathbb{M}_n$ be unitary. This U is as desired. ∎

Example 3.2.7 Let $A = W|A|$ be a polar decomposition of $A \in \mathbb{M}_n$. Write $\operatorname{tr}(W|A|) = \operatorname{tr}(W|A|^{1/2} \cdot |A|^{1/2}) = \langle |A|^{1/2}, |A|^{1/2}W^* \rangle$ as a Frobenius inner product. Using the Cauchy-Schwarz inequality, we obtain $|\operatorname{tr} A|^2 = |\operatorname{tr}(W|A|)|^2 \le (\operatorname{tr}|A|)(\operatorname{tr}(W|A|W^*)) = (\operatorname{tr}|A|)^2$. Thus, $|\operatorname{tr} A| \le \operatorname{tr}|A|$. ∎

We showed the polar decomposition by the singular value decomposition. One may also prove the singular value decomposition (SVD) by using the polar decomposition and the spectral decomposition. For the nonsingular case of matrix A, by Theorem 3.2.2, let $A^*A = V^*D^2V$, where V is unitary and $D = \operatorname{diag}(\sigma_1, \sigma_2, \ldots, \sigma_n)$. A is invertible, so is D. Let $U = AV^{-1}D^{-1}$. Then U is unitary because $U^*U = D^{-1}VA^*AV^*D^{-1} = I_n$. Thus, $A = UDV$.

Example 3.2.8 Spectral decomposition, SVD, and polar decomposition.

(a) Let $A = \left(\begin{smallmatrix} 1 & 1 \\ 1 & 1 \end{smallmatrix}\right)$. A is positive semidefinite, $\operatorname{rank}(A) = 1$, and $\operatorname{tr} A = 2$. It is easy to find the eigenvalues and orthonormal eigenvectors.

$$A = |A| = |A^*| = U^* \begin{pmatrix} 2 & 0 \\ 0 & 0 \end{pmatrix} U, \text{ where } U = \tfrac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix}.$$

(b) Let $A = \left(\begin{smallmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \end{smallmatrix}\right)$. Compute $A^*A = \left(\begin{smallmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{smallmatrix}\right)$ and find the eigenvalues of A^*A: 2, 1, 0. Thus, the singular values of A are $\sqrt{2}, 1, 0$.

In view of (3.5) and (3.6) with $\sigma_1^2 = 2$, $A^*Av_1 = 2v_1$ and $u_1 = \frac{1}{\sigma_1}Av_1$, we obtain $v_1 = \frac{1}{\sqrt{2}}(1, 0, 1)^T$ and $u_1 = \frac{1}{\sqrt{2}}Av_1 = (1, 0)^T$. For $\sigma_2 = 1$, we have $v_2 = (0, 1, 0)^T$ and $u_2 = Av_2 = (0, 1)^T$. It follows that $U = (u_1, u_2) = \left(\begin{smallmatrix} 1 & 0 \\ 0 & 1 \end{smallmatrix}\right)$. Observe that $\frac{1}{\sqrt{2}}(-1, 0, 1)$ is orthogonal to v_1 and v_2. Setting $V = (v_1, v_2, v_3)$ gives the singular value decomposition

$$A = U \begin{pmatrix} \sqrt{2} & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix} V^T, \text{ where } U = I_2,\ V = \tfrac{1}{\sqrt{2}} \begin{pmatrix} 1 & 0 & -1 \\ 0 & \sqrt{2} & 0 \\ 1 & 0 & 1 \end{pmatrix}.$$

(c) Let $A = \frac{1}{5}\begin{pmatrix} 6 & 0 & 2 \\ 0 & 5 & 0 \\ -7 & 0 & 6 \end{pmatrix}$. A has a singular value decomposition:

$$A = \frac{1}{\sqrt{5}} \begin{pmatrix} 1 & 0 & 2 \\ 0 & \sqrt{5} & 0 \\ -2 & 0 & 1 \end{pmatrix} \cdot \begin{pmatrix} 2 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \cdot \frac{1}{\sqrt{5}} \begin{pmatrix} 2 & 0 & -1 \\ 0 & \sqrt{5} & 0 \\ 1 & 0 & 2 \end{pmatrix}. \quad ∎$$

Problems

1. Let $c \in \mathbb{C}$. What are the eigenvalues and singular values of $\left(\begin{smallmatrix} 1 & c \\ 0 & 1 \end{smallmatrix}\right)$?
2. Let $A = \left(\begin{smallmatrix} 1 & 2 \\ 4 & 3 \end{smallmatrix}\right)$.
   (a) Find the eigenvalues of A.
   (b) Find the singular values of A.
   (c) Find a matrix P such that $P^{-1}AP$ is diagonal.
   (d) Compute A^5.
   (e) Does there exist a unitary matrix U such that U^*AU is diagonal?
3. A matrix $A \in \mathbb{M}_n(\mathbb{F})$ is said to be *diagonalizable* over the field $\mathbb{F}$ if $P^{-1}AP$ is a diagonal matrix for some invertible matrix $P \in \mathbb{M}_n(\mathbb{F})$. If A is not diagonalizable, we say that A is *non-diagonalizable*. Show that
   (a) The real matrix $\left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right)$ is diagonalizable over $\mathbb{C}$.
   (b) The real matrix $\left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right)$ is non-diagonalizable over $\mathbb{R}$.
   (c) The complex symmetric matrix $\left(\begin{smallmatrix} 1 & i \\ i & -1 \end{smallmatrix}\right)$ is non-diagonalizable over $\mathbb{C}$.
4. Find the singular values and moduli of $\left(\begin{smallmatrix} 1 & i \\ -i & 1 \end{smallmatrix}\right)$, $\left(\begin{smallmatrix} 1 & i \\ i & -1 \end{smallmatrix}\right)$, and $\left(\begin{smallmatrix} 1 & i \\ i & 1 \end{smallmatrix}\right)$.
5. Let $A = \left(\begin{smallmatrix} 1 & 2 \\ 2 & 1 \end{smallmatrix}\right)$. Compute the eigenvalues and singular values of A, and find a spectral decomposition and the modulus $|A|$ of A.
6. Find a QR decomposition of $A = \left(\begin{smallmatrix} 3 & 1 \\ 4 & 2 \end{smallmatrix}\right)$.
7. If A is an upper-triangular normal matrix, show that A is diagonal.
8. Let A, B, and P be n-square matrices and let P be invertible.
   Do AB and BA have the same singular values (as for eigenvalues)?
   Do PAP^{-1} and A have the same singular values (as for eigenvalues)?
9. Let A be an invertible matrix. Show that $|A^{-1}| = |A^*|^{-1}$.
10. Let $A = \left(\begin{smallmatrix} 1 & 0 \\ -1 & 2 \end{smallmatrix}\right)$. Show that $x^TAx \geq 0$ for all $x \in \mathbb{R}^2$ but $A \not\geq 0$.
11. Let A be an n-square (real or complex) matrix. Show that
    (a) $x^*Ax = 0$ for all $x \in \mathbb{C}^n \Leftrightarrow A = 0$.
    (b) $x^TAx = 0$ for all $x \in \mathbb{R}^n \Leftrightarrow A^T = -A$ (i.e., A is skew-symmetric).
    (c) $x^*Ax \in \mathbb{R}$ for all $x \in \mathbb{C}^n \Leftrightarrow A^* = A$ (i.e., A is Hermitian).
    (d) If A is real symmetric, then $x^TAx \geq 0$ for all $x \in \mathbb{R}^n \Leftrightarrow z^*Az \geq 0$ for all $z \in \mathbb{C}^n$, that is, A is positive semidefinite (over $\mathbb{C}$).
12. Let $a, b, c \in \mathbb{C}$. Show that $\left(\begin{smallmatrix} a & b \\ \bar{b} & c \end{smallmatrix}\right) \geq 0$ if and only if $a, c \geq 0$, $ac \geq |b|^2$.
13. Let A be an $m \times n$ matrix, $m > n$. Show that $A = Q_1R$, where Q_1 is some $m \times n$ partially unitary matrix, i.e., $Q_1^*Q_1 = I_n$, and R is an $n \times n$ upper-triangular matrix with nonnegative diagonal entries.

14. Let A be an n-square matrix and let f be a polynomial.
    (a) If λ is an eigenvalue of A, show that $f(\lambda)$ is an eigenvalue $f(A)$. In particular, λ^k is an eigenvalue of A^k for $k = 2, 3, \ldots$.
    (b) If $\alpha \in \mathbb{C}$ is an eigenvalue of $f(A)$, show that $\alpha = f(\lambda)$ for some eigenvalue λ of A.

15. Let A be an n-square positive semidefinite matrix. Show that
    (a) $\operatorname{tr} A \geq 0$; and $\operatorname{tr} A = 0$ if and only if $A = 0$.
    (b) $\det A \geq 0$; and $\det A = 0$ if and only if A is singular.
    (c) $X^*AX \geq 0$ for any $X \in \mathbb{M}_{n\times m}$.
    (d) Every principal submatrix of A is positive semidefinite.

16. Let $A \in \mathbb{M}_n$. Denote by A_{ii} the diagonal (i,i)-entry of A for each i. Show by example that $|A_{ii}| \leq |A|_{ii}$ need not be true.

17. Let $A = \left(\begin{smallmatrix} a & \alpha \\ \alpha^* & B \end{smallmatrix}\right) \geq 0$. Find the $(1,1)$-entries of A^2 and A^{-1} (if it exists).

18. If v is a row (or column) of a matrix A, show that $\|v\| \leq \sigma_{\max}(A)$.

19. Let $A = \left(\begin{smallmatrix} 0 & 1 & 0 \\ 0 & 0 & -1 \\ 0 & 0 & 0 \end{smallmatrix}\right)$ and $B = \left(\begin{smallmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{smallmatrix}\right)$.
    (a) Show that $A^3 = B^3 = C^3 = 0$, where $C = \lambda A + \mu B$, $\lambda, \mu \in \mathbb{C}$.
    (b) Does there exist an integer k such that $(AB)^k = 0$?
    (c) Does there exist a nonsingular matrix P such that $P^{-1}AP$ and $P^{-1}BP$ are both upper-triangular?

20. Prove each of the following statements.
    (a) The eigenvalues of a matrix are exactly the singular values of the matrix if and only if the matrix is positive semidefinite.
    (b) The diagonal entries are exactly the singular values if and only if the matrix is diagonal and the diagonal entries are nonnegative.
    (c) The eigenvalues of $\left(\begin{smallmatrix} 2 & 2 & -2 \\ 1 & 4 & 1 \\ 2 & 2 & 6 \end{smallmatrix}\right)$ are 2, 4, and 6 (the diagonal entries).

21. If A is an $m \times n$ matrix with a singular value decomposition $A = UDV$, what is a singular value decomposition of $(A, 0)$ (obtained by adding some zero columns to A)? Discuss the similar question for $\left(\begin{smallmatrix} A \\ 0 \end{smallmatrix}\right)$.

22. Let A be a normal matrix of k nonzero eigenvalues $\lambda_1, \ldots, \lambda_k$. Show that $\operatorname{rank}(A) = k$. What are the singular values of A?

23. Let $A \in \mathbb{M}_{m\times n}$, $m \geq n$. If $A = WP$, where P is positive semidefinite and W satisfies $W^*W = I_n$, show that $P = |A|$. Is this true if $m < n$?

24. What are the square matrices whose singular values are all the same?

25. How do permutations (matrices) P and Q applied to a matrix A affect the Schur decomposition, singular value decomposition, and polar decomposition of the matrix A? In other words, if these decompositions of PAQ are known, how does one obtain the decompositions of A?

26. If $T \in \mathbb{M}_n$ is upper-triangular, does it necessarily follow that $P^T TP$ is upper-triangular for every permutation matrix $P \in \mathbb{M}_n$?

27. Find a Schur triangular form and a singular value decomposition for
$$A = \begin{pmatrix} 1 & 0 & 0 & 1 \\ 0 & 2 & 3 & 0 \\ 0 & 0 & 2 & 0 \\ 1 & 0 & 0 & 1 \end{pmatrix}.$$

28. Let A and B be $n \times n$ positive semidefinite matrices. Show that the eigenvalues of AB are all nonnegative. Is AB positive semidefinite?

29. Let A be an n-square matrix and let P be an n-square nonsingular matrix such that $P^{-1}AP$ is upper-triangular. Write $P^{-1}AP = D - U$, where D is a diagonal matrix and U is an upper-triangular matrix with zeros on the diagonal. Show that if A is invertible, then
$$A^{-1} = PD^{-1}\big(I + UD^{-1} + (UD^{-1})^2 + \cdots + (UD^{-1})^{n-1}\big)P^{-1}.$$

30. Show that two Hermitian matrices are similar if and only if they are unitarily similar, and if and only if they have the same eigenvalues.

31. Let $A \in \mathbb{M}_n$ have rank r. Show that A is normal if and only if
$$A = \sum_{i=1}^{r} \lambda_i u_i u_i^*,$$
where λ_i are complex numbers and u_i are column vectors of a unitary matrix. Show further that A is Hermitian if and only if all λ_i are real, and A is positive semidefinite if and only if all λ_i are nonnegative.

32. Let $u_1, \ldots, u_n \in \mathbb{C}^n$ be orthonormal vectors and let $\lambda_1, \ldots, \lambda_n$ be nonzero scalars. Let $A = \sum_{i=1}^{n} \lambda_i u_i u_i^*$. Show that A is invertible and
$$A^{-1} = \sum_{i=1}^{n} \lambda_i^{-1} u_i u_i^*.$$

33. Let $A \in \mathbb{M}_n$. If A has k nonzero eigenvalues and r nonzero singular values, show that $k \leq r$, furthermore, if $k = n - 1$ or n, then $k = r$.

34. Let A and B be upper-triangular matrices with positive diagonal entries. If $A = UB$ for some unitary U, show that $U = I$ and $A = B$.

35. Let $A, B \in \mathbb{M}_n$. Show that $|A| = |B| \Leftrightarrow A = UB$ for some unitary U.

36. Let $A = P_1U_1 = P_2U_2$ be two polar decompositions of $A \in \mathbb{M}_n$, where P_1, P_2 are positive semidefinite, and U_1, U_2 are unitary. Show that $P_1 = P_2$, and $U_1 = U_2$ if A is nonsingular. What if A is singular?

37. Let P be an n-square positive definite matrix and let U and V be n-square unitary matrices such that $UP = PV$. Show that $U = V$.

38. Show that if A is an n-square matrix, then there exist nonsingular matrices P and Q such that $(PA)^2 = PA$ and $(AQ)^2 = AQ$.

39. Let A be an $m \times n$ matrix with rank r ($r > 0$). Show that

    (a) A has r nonzero singular values.

    (b) A has at most r nonzero eigenvalues (for the case $m = n$).

    (c) $A = U_rD_rV_r$, where U_r is an $m \times r$ matrix, $D_r = \operatorname{diag}(\sigma_1, \ldots, \sigma_r)$, V_r is an $r \times n$ matrix, all of rank r, and $U_r^*U_r = V_rV_r^* = I_r$.

    (d) $A = \sum_{i=1}^{r} \sigma_i u_i v_i^*$, where the σ_i are the singular values of A, and u_i and v_i are column vectors of some unitary matrices.

40. Let D be an $m \times n$ matrix whose entries in the positions $(1,1), \ldots, (r,r)$ are nonzero numbers $d_1, \ldots, d_r$, $r \leq \min\{m, n\}$, and whose other entries are all zero. Let $D^{\sim}$ be the $m \times n$ matrix whose entries in the positions $(1,1), \ldots, (r,r)$ are $d_1^{-1}, \ldots, d_r^{-1}$, respectively, and whose other entries are all zero. Show that $D = (DD^T)D^{\sim} = D^{\sim}(D^TD)$. Find the singular values and a singular value decomposition of D.

41. Let A be an $m \times n$ matrix of rank r. Show that there exist unitary matrices $U \in \mathbb{M}_m$ and $V \in \mathbb{M}_n$ such that, with $W = U \left(\begin{smallmatrix} I_r & 0 \\ 0 & 0 \end{smallmatrix}\right)_{m \times n} V$,

$$A = W|A| = |A^*|W, \quad |A| = W^*A, \quad |A^*| = AW^*.$$

42. Let A be an $m \times n$ nonzero matrix. Show that $A = (AA^*)Z = Z(A^*A)$ for some $m \times n$ matrix Z whose nonzero singular values are the reciprocals of the nonzero singular values of A.

43. State the singular value decomposition theorem for the vectors in $\mathbb{C}^n$.

44. Let $A = (a_{ij}) \in \mathbb{M}_n$ have singular values $\sigma_1, \ldots, \sigma_n$. Show that

$$\sum\nolimits_i |\Re(a_{ii})| \leq \sum\nolimits_i \sigma_i = \operatorname{tr}|A|.$$

45. Let A be an $n \times n$ matrix. If there exist n entries of A in different rows and different columns (for example, the diagonal entries of A) that are equal to the singular values of A, what can be said about the matrix?

46. (**The Hadamard determinant inequality**) Let $A = (a_{ij})$ be an $n \times n$ positive semidefinite matrix. Show that $\det A \leq a_{11} \cdots a_{nn}$. [Hint: First consider the case $a_{11} = \cdots = a_{nn} = 1$, and use the AM-GM inequality.]

47. Let $A, P \in \mathbb{M}_n$. If $AP = PA$ and $P \geq 0$, show that $AP^{1/2} = P^{1/2}A$.

48. Let $A \in \mathbb{M}_n$. Show that A is positive semidefinite if and only if there exists a matrix $B \in \mathbb{M}_n$ such that $A = B^*B$ and that A is positive definite if and only if such a matrix B is nonsingular.

49. Let $A, B \in \mathbb{M}_n$ and $A, B \geq 0$. If $ABA = 0$, show that $AB = BA = 0$.

50. Let $A \in \mathbb{M}_n$ be a normal matrix with a spectral decomposition $A = U^* \operatorname{diag}(\lambda_1, \lambda_2, \ldots, \lambda_n)U$, where U is unitary and λ_i's are the eigenvalues of A. Let $f(x)$ be a (single-valued) function whose domain contains all λ_i's. Define $f(A) = U^* \operatorname{diag}(f(\lambda_1), f(\lambda_2), \ldots, f(\lambda_n))U$. Show that $f(A)$ is well-defined, that is, $f(A)$ is independent of the choices of U and the orders of λ_i's. In other words, if $A = V^* \operatorname{diag}(\mu_1, \mu_2, \ldots, \mu_n)V$ is another spectral decomposition of A, where V is unitary and the μ_i's are a rearrangement of the λ_i's, then $U^* \operatorname{diag}(f(\lambda_1), f(\lambda_2), \ldots, f(\lambda_n))U = V^* \operatorname{diag}(f(\mu_1), f(\mu_2), \ldots, f(\mu_n))V$. In particular, for $A \geq 0$, A^r is well-defined for $r \geq 0$, which generalizes A^k for positive integers k.

51. Let $A, B \geq 0$ and let U be unitary, all $n \times n$. Show that for any $r \geq 0$,

    (a) $(U^*AU)^r = U^*A^rU$.

    (b) If $AB = BA$, then $AB \geq 0$ and $(AB)^r = A^rB^r$.

52. Let A be an $m \times n$ matrix with rank $r > 0$ and let $\Sigma = \operatorname{diag}(\sigma_1, \ldots, \sigma_r)$, where $\sigma_1, \ldots, \sigma_r$ are the positive singular values of A. Verify the following steps for construction of a singular value decomposition of A.

    (a) A^*A is an $n \times n$ (Hermitian) positive semidefinite matrix. Let $v_1, \ldots, v_r, v_{r+1}, \ldots, v_n \in \mathbb{C}^n$ be orthonormal eigenvectors associated respectively with the eigenvalues $\sigma_1^2, \ldots, \sigma_r^2, 0, \ldots, 0$ of A^*A.

    (b) Define $u_i = \frac{1}{\sigma_i} Av_i \in \mathbb{C}^m$ for $i = 1, \ldots, r$. Then $\{u_1, \ldots, u_r\}$ is an orthonormal set. Extend it to an orthonormal basis of $\mathbb{C}^m$: $\{u_1, \ldots, u_r, u_{r+1}, \ldots, u_m\}$ (where $u_{r+1}, \ldots, u_m$ are free to pick).

    (c) u_i and v_i satisfy the following relations for $i = 1, \ldots, r$,

    $$Av_i = \sigma_i u_i, \quad A^*u_i = \sigma_i v_i, \quad \|Av_i\| = \sigma_i = \|A^*u_i\|.$$

    (d) $Av_i = 0$ for $i = r+1, \ldots, n$. [Hint: $(A^*A)v_i = 0$ for $i \geq r+1$.]

    (e) Let $V = (v_1, \ldots, v_r) \in \mathbb{M}_{n \times r}$, $U = (u_1, \ldots, u_r) \in \mathbb{M}_{m \times r}$. Then

    $$U^*AV = \Sigma.$$

    (f) Let $L = (u_1, \ldots, u_r, u_{r+1} \ldots, u_m)$, $R = (v_1, \ldots, v_r, v_{r+1}, \ldots, v_n)$. L and R are m- and n-square unitary matrices, respectively.

    (g) $A = L \left(\begin{smallmatrix} \Sigma & 0 \\ 0 & 0 \end{smallmatrix} \right) R^*$ (a *singular value decomposition* (SVD) of A).

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3.3 Annihilating Polynomials of Matrices

Given a polynomial in λ with scalar coefficients a_m, a_{m-1}, $\ldots$, a_0,

$$f(\lambda) = a_m\lambda^m + a_{m-1}\lambda^{m-1} + \cdots + a_1\lambda + a_0,$$

one can always define a *matrix polynomial* for $A \in \mathbb{M}_n$ by

$$f(A) = a_mA^m + a_{m-1}A^{m-1} + \cdots + a_1A + a_0I.$$

We consider in this section the annihilating polynomials of a matrix, that is, the polynomials $f(\lambda)$ for which $f(A) = 0$. The characteristic and minimal polynomials are such polynomials and are of particular interest.

To begin, let $A \in \mathbb{M}_n(\mathbb{F})$, where $\mathbb{F} = \mathbb{R}$ or $\mathbb{C}$. Since $\mathbb{M}_n(\mathbb{F})$ is a vector space over $\mathbb{F}$ of dimension n^2, any $n^2 + 1$ vectors (matrices) in $\mathbb{M}_n(\mathbb{F})$ are linearly dependent. In particular, the matrices $I, A, A^2, \ldots, A^{n^2}$ are linearly dependent. So, there exist $a_0, a_1, a_2, \ldots, a_{n^2} \in \mathbb{F}$, not all zero, such that

$$a_0I + a_1A + a_2A^2 + \cdots + a_{n^2}A^{n^2} = 0.$$

Set

$$f(\lambda) = a_0 + a_1\lambda + a_2\lambda^2 + \cdots + a_{n^2}\lambda^{n^2}. \tag{3.8}$$

Then $f(A) = 0$. [Note: Conventionally, $A^0 = I$ for a square matrix A.]

Theorem 3.3.1 (Cayley-Hamilton Theorem) *Let $A \in \mathbb{M}_n$ and let $p_A(\lambda) = \det(\lambda I_n - A)$ be the characteristic polynomial of A. Then $p_A(A) = 0$.*

Proof. Let the eigenvalues of A be $\lambda_1, \lambda_2, \ldots, \lambda_n \in \mathbb{C}$. By the Schur triangularization theorem, there exists an invertible (or unitary) matrix P such that

$$A = P^{-1}TP,$$

where T is an upper-triangular matrix with $\lambda_1, \lambda_2, \ldots, \lambda_n$ on the diagonal.

Let $p(\lambda) = p_A(\lambda)$. Then $p(A) = p(P^{-1}TP) = P^{-1}p(T)P$. We show $p(T) = 0$. Factor and expand the polynomial $p(\lambda)$ in terms of $\lambda_1, \lambda_2, \ldots, \lambda_n$:

$$\begin{aligned} p(\lambda) &= (\lambda - \lambda_1)(\lambda - \lambda_2)\cdots(\lambda - \lambda_n) \\ &= \lambda^n - (\lambda_1 + \cdots + \lambda_n)\lambda^{n-1} + \cdots + (-1)^n\lambda_1\cdots\lambda_n. \end{aligned} \tag{3.9}$$

We have

$$\begin{aligned} p(T) &= T^n - (\lambda_1 + \cdots + \lambda_n)T^{n-1} + \cdots + (-1)^n(\lambda_1\cdots\lambda_n)I_n \\ &= (T - \lambda_1I_n)(T - \lambda_2I_n)\cdots(T - \lambda_nI_n). \end{aligned}$$

It can be shown inductively that $(T - \lambda_1I_n)\cdots(T - \lambda_kI_n)$ has the first k columns equal to zero, $1 \le k \le n$. Thus, $p(T) = 0$, implying $p(A) = 0$. ∎

Note: It is false to show the Cayley-Hamilton theorem by directly plugging A in $p_A(\lambda) = \det(\lambda I_n - A)$ to get $p_A(A) = \det(AI_n - A) = \det 0 = 0$.

Example 3.3.1 Let $A = \left(\begin{smallmatrix} 1 & 0 \\ 0 & 0 \end{smallmatrix}\right)$ and let $f(\lambda) = \det(\lambda I - A)$. In general, $\det f(B) \neq \det(B - A)$ for a matrix B. For example, take $B = 2I$. Then $f(\lambda) = \det(\lambda I - A) = \lambda^2 - \lambda$, $f(B) = 2I$, $\det f(B) = 4$, $\det(B - A) = 2$. ∎

The degree of the polynomial in (3.8) is possibly n^2; the degree of the characteristic polynomial $p_A(\lambda) = \det(\lambda I_n - A)$ is n. It is possible that an annihilating polynomial of a square matrix has a smaller degree than n. A monic polynomial $m_A(\lambda)$, or simply $m(\lambda)$, is called the *minimal polynomial* of a square matrix A if $m(A) = 0$ and it has the smallest degree in the set

$$\{f(\lambda) : f(A) = 0\}.$$

It is immediate that if $f(A) = 0$, then $m(\lambda)$ divides $f(\lambda)$, that is, $m(\lambda)|f(\lambda)$, because otherwise we can write (by long division)

$$f(\lambda) = q(\lambda)m(\lambda) + r(\lambda), \quad r(\lambda) \neq 0.$$

$r(\lambda)$ has a smaller degree than $m(\lambda)$ and $r(A) = 0$, a contradiction. In other words, the minimal polynomial is the "smallest" polynomial among the annihilating polynomials (of the matrix). In particular, the minimal polynomial divides the characteristic polynomial. Note that both minimal and characteristic polynomials are uniquely determined by the matrix.

Let A be a square matrix and let f be a polynomial. It is easy to show that $f(A^T) = (f(A))^T$. It follows that $f(A) = 0$ if and only if $f(A^T) = 0$. Thus, A and A^T have the same characteristic (and minimal) polynomial.

Example 3.3.2 Characteristic and minimal polynomials.

(1) Let $A = \left(\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right)$. Then $p_A(\lambda) = m_A(\lambda) = \lambda^2$.

(2) Let $A = \left(\begin{smallmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{smallmatrix}\right)$. Then $p_A(\lambda) = \lambda^3$, $m_A(\lambda) = \lambda^2$.

(3) Let $A = \left(\begin{smallmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{smallmatrix}\right)$. Then $p_A(\lambda) = (\lambda + 1)(\lambda - 1)^2$, $m_A(\lambda) = \lambda^2 - 1$.

(4) Let $A = \left(\begin{smallmatrix} 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{smallmatrix}\right)$ and $B = \left(\begin{smallmatrix} 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \end{smallmatrix}\right)$. Then A and B have the same characteristic polynomial $(\lambda - 1)^4$, and the same minimal polynomial $(\lambda - 1)^2$. A and B are not similar as $\operatorname{rank}(I - A) \neq \operatorname{rank}(I - B)$. ∎

Example 3.3.3 Find the minimal polynomial of $A = \left(\begin{smallmatrix} 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \end{smallmatrix}\right)$.

Observe that $A^3 = 4A$. It follows that $f(A) = 0$, where $f(\lambda) = \lambda^3 - 4\lambda$. $f(\lambda)$ has factors λ, $\lambda+2$, $\lambda-2$. Since $m(\lambda)$ divides $f(\lambda)$ and A is not a scalar matrix, the possible minimal polynomial is $\lambda(\lambda+2)$, $\lambda(\lambda-2)$, or $(\lambda+2)(\lambda-2)$. It is easy to check that $m(\lambda) = \lambda(\lambda - 2) = \lambda^2 - 2\lambda$. ∎

Theorem 3.3.2 *Similar matrices have the same characteristic polynomial and the same minimal polynomial.* (*The converse is not true.*)

Proof. The first part is Theorem 1.3.3. Here is a reproof. Let A and B be matrices such that $A = P^{-1}BP$ for some nonsingular matrix P. Then

$$\begin{aligned} p_A(\lambda) &= \det(\lambda I - A) = \det(\lambda I - P^{-1}BP) = \det P^{-1} \det(\lambda I - B) \det P \\ &= (\det P)^{-1} \det(\lambda I - B) \det P = \det(\lambda I - B) = p_B(\lambda). \end{aligned}$$

Let $m_A(\lambda)$ and $m_B(\lambda)$ be the minimal polynomials of A and B. Then

$$m_B(A) = m_B(P^{-1}BP) = P^{-1}m_B(B)P = 0.$$

Thus, $m_A(\lambda)$ divides $m_B(\lambda)$. Similarly, $m_B(\lambda)$ divides $m_A(\lambda)$. Since they both have leading coefficient 1 (i.e., monic), $m_A(\lambda) = m_B(\lambda)$. ∎

The converse of the theorem is not true in general as Example 3.3.2 (4) shows. So, matrices having the same characteristic and minimal polynomials need not be similar. In order to completely characterize the similarity of matrices, we will introduce the Jordan canonical forms (see next section).

Example 3.3.4 Given a polynomial of degree n with scalar coefficients

$$c(x) = x^n + c_{n-1}x^{n-1} + \cdots + c_1 x + c_0,$$

we can construct an n-square matrix with the coefficients c_0, c_1, …, c_{n-1}:

$$C = \begin{pmatrix} 0 & \dots & 0 & -c_0 \\ 1 & \dots & 0 & -c_1 \\ \vdots & \ddots & \vdots & \vdots \\ 0 & \dots & 1 & -c_{n-1} \end{pmatrix}.$$

Such a matrix C is referred to as the *companion matrix* of $c(x)$. Expanding $\det(xI - C)$ shows that $c(x)$ is the characteristic polynomial of C. It is also the minimal polynomial of C. To see this, let $e_1, e_2, \dots, e_n$ be the column vectors with the 1st, 2nd, …, nth component 1, respectively, and all other components equal to 0. Then $C = C(e_1, e_2, \dots, e_n) = (Ce_1, Ce_2, \dots, Ce_n)$. On the other hand, $e_2, e_3, \dots, e_n$ are the first $n-1$ columns of C. We have

$$Ce_1 = e_2,\ Ce_2 = e_3 = C^2e_1,\ \dots,\ Ce_{n-1} = e_n = C^{n-1}e_1$$

and

$$Ce_n = -c_0e_1 - c_1e_2 - \cdots - c_{n-1}e_n.$$

If $d(x) = x^m + d_{m-1}x^{m-1} + d_{m-2}x^{m-2} + \cdots + d_1x + d_0$ is a polynomial such that $d(C) = 0$, $m < n$, we compute $d(C)e_1 = 0$ to get

$$\begin{aligned} 0 &= C^m e_1 + d_{m-1}C^{m-1}e_1 + d_{m-2}C^{m-2}e_1 + \cdots + d_1Ce_1 + d_0e_1 \\ &= e_{m+1} + d_{m-1}e_m + d_{m-2}e_{m-1} + \cdots + d_1e_2 + d_0e_1. \end{aligned}$$

This says that $e_1, e_2, \dots, e_n$ are linearly dependent, a contradiction. ∎

Example 3.3.5 $\left(\begin{smallmatrix} 0 & 0 & c \\ 1 & 0 & b \\ 0 & 1 & a \end{smallmatrix}\right)$ and $\left(\begin{smallmatrix} x & 0 & 0 \\ 0 & x & 0 \\ 0 & 0 & y \end{smallmatrix}\right)$ are not similar for any $a, b, c, x, y \in \mathbb{C}$, because their minimal polynomials are respectively $m_1(\lambda) = \lambda^3 - a\lambda^2 - b\lambda - c$ and $m_2(\lambda) = (\lambda - x)(\lambda - y) = \lambda^2 - (x + y)\lambda + xy$, and $m_1(\lambda) \neq m_2(\lambda)$ for any $a, b, c, x, y \in \mathbb{C}$. It follows that if $\left(\begin{smallmatrix} 0 & 0 & c \\ 1 & 0 & b \\ 0 & 1 & a \end{smallmatrix}\right)$ is similar to a diagonal matrix $\operatorname{diag}(\alpha, \beta, \gamma)$, then α, β, and γ must be distinct. ∎

An effective method of computing minimal polynomials is given in the next section in conjunction with the discussion of Jordan canonical forms.

Problems

1. Let $A = \left(\begin{smallmatrix} 1 & 2 \\ 3 & 1 \end{smallmatrix}\right)$ and $f(x) = 3x^2 - 5x - 2$. Compute $f(A)$.
2. Let $A = \left(\begin{smallmatrix} 1 & -1 \\ 2 & 0 \end{smallmatrix}\right)$. Find a nonzero polynomial $p(x)$ such that $p(A) = 0$.
3. Let $R = \left(\begin{smallmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{smallmatrix}\right)$. Find a polynomial $p(x)$ such that $R^{-1} = p(R)$.
4. Find the characteristic and minimal polynomials of the 2×2 matrices
$$\begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}, \quad \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \quad \begin{pmatrix} 0 & 1 \\ 0 & 1 \end{pmatrix}, \quad \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \quad \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}.$$
5. Find the characteristic and minimal polynomials of the matrices
$$\begin{pmatrix} 0 & 0 & 4 \\ 1 & 0 & 3 \\ 0 & 1 & 2 \end{pmatrix}, \quad \begin{pmatrix} 2 & 1 & 0 \\ 0 & 2 & 1 \\ 0 & 0 & 2 \end{pmatrix}, \quad \begin{pmatrix} 2 & 1 & 0 \\ 1 & 2 & 0 \\ 0 & 0 & 2 \end{pmatrix}, \quad \begin{pmatrix} 2 & 1 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{pmatrix}.$$
6. Find the characteristic and minimal polynomials of the matrices
$$\begin{pmatrix} 1 & 1 & 0 & 1 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \quad \begin{pmatrix} 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \quad \begin{pmatrix} 1 & 1 & 0 & 1 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \\ 1 & 0 & 0 & 1 \end{pmatrix}.$$
7. Let A be an $n \times n$ matrix. Prove each of the following statements.
   (a) If the entries of A are real, then there is a polynomial f of degree at most n with real coefficients such that $f(A) = 0$.
   (b) If the eigenvalues of A are real, then there is a polynomial f of degree at most n with real coefficients such that $f(A) = 0$.
   (c) If the entries of A are rational, then there is a polynomial f of degree at most n with integer coefficients such that $f(A) = 0$.
   (d) If the entries of A are complex, then there is a polynomial f of degree at most $2n^2$ with real coefficients such that $f(A) = 0$.

8. Find a nonzero polynomial $f(x)$ such that $f(A) = 0$, where

$$A = \begin{pmatrix} 2 & -2 & 0 \\ -2 & 1 & -2 \\ 0 & 2 & 0 \end{pmatrix}.$$

9. Compute $\det(AB - A)$ and $f(A)$, where $f(\lambda) = \det(\lambda B - A)$ and

$$A = \begin{pmatrix} 1 & 0 \\ 1 & 0 \end{pmatrix}, \quad B = \begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix}.$$

10. Let A and B be $n \times n$ matrices and let $f(\lambda) = \det(\lambda I - B)$. Show that $f(A)$ is invertible if and only if A and B have no common eigenvalues.

11. Let A and B be $n \times n$ matrices. If A and B are similar, show that $f(A) = 0$ if and only if $f(B) = 0$. Is the converse true?

12. If A and B are $n \times n$ upper-triangular matrices of rank $n - 1$ with all diagonal entries equal to zero, show that $\operatorname{rank}(AB) = n - 2$.

13. Let $A_1, \ldots, A_m$ be $n \times n$ upper-triangular matrices. If their diagonal entries are all equal to zero, show that $A_1 \cdots A_m = 0$ when $m \geq n$.

14. Let $A \in \mathbb{M}_n$. Explain what is wrong with the following proof of the Cayley-Hamilton theorem: Because $p(\lambda) = \det(\lambda I - A)$, plugging A for λ directly on both sides results in $p(A) = \det(A - A) = \det 0_n = 0$.

15. As is known from Sect. 2.4, Chap. 2, for any square matrices A and B of the same order, AB and BA have the same characteristic polynomial. Do AB and BA have the same minimal polynomial?

16. Let A be a nonsingular n-square matrix. Show that there exists a polynomial $f(x)$ of degree at most n such that $A^{-1} = f(A)$.

17. Let A be a square matrix. Show that A and A^T have the same characteristic polynomial and the same minimal polynomial. Does A^* have the same characteristic or minimal polynomial as A?

18. Let A be a square matrix and let $f(x)$ be a polynomial. (a) If $f(A) = 0$, show that $f(\lambda) = 0$ for any eigenvalue λ of A. (b) If $\det(f(A)) = 0$, show that $f(\mu) = 0$ for some eigenvalue μ of A. Are the converses true?

19. Let $A \in \mathbb{M}_n$ and $v \in \mathbb{C}^n$. If $f(\lambda)$ is the monic polynomial with the smallest degree such that $f(A)v = 0$, show that $f(\lambda)$ divides $m_{A}(\lambda)$.

20. Let A and B be n-square matrices. Show that

$$AX - XB = 0 \quad \Rightarrow \quad f(A)X - Xf(B) = 0$$

for every polynomial f. In addition, if A and B have no common eigenvalues, then $AX - XB = 0$ has only the trivial solution $X = 0$.

21. Let A and B be $2n \times 2n$ matrices partitioned conformally as

$$A = \begin{pmatrix} A_{11} & 0 \\ 0 & A_{22} \end{pmatrix}, \; B = \begin{pmatrix} B_{11} & B_{12} \\ B_{21} & B_{22} \end{pmatrix}, \; A_{11}, A_{22}, B_{11}, B_{22} \in \mathbb{M}_n.$$

If $AB = BA$ and A_{11} and A_{22} have no common eigenvalue, show that

$$B_{12} = B_{21} = 0.$$

22. Let A and B be square matrices of the same order. If $AB = BA$, show that $g(A)h(B) = h(B)g(A)$ for any polynomials g and h.

23. Let $A, B, C \in \mathbb{M}_n$. If $X \in \mathbb{M}_n$ satisfies $AX^2 + BX + C = 0$ and if λ is an eigenvalue of X, show that $\lambda^2 A + \lambda B + C$ is singular.

24. Let $c_0, c_1, \ldots, c_{n-1} \in \mathbb{C}$ and let

$$C = \begin{pmatrix} 0 & \ldots & 0 & -c_0 \\ 1 & \ldots & 0 & -c_1 \\ \vdots & \ddots & \vdots & \vdots \\ 0 & \ldots & 1 & -c_{n-1} \end{pmatrix}, \quad D = \begin{pmatrix} -c_{n-1} & \ldots & -c_1 & -c_0 \\ 1 & \ldots & 0 & 0 \\ \vdots & \ddots & \vdots & \vdots \\ 0 & \ldots & 1 & 0 \end{pmatrix}.$$

Show that $SCS = D^T$, where S is the backward identity, that is, the entries of S are 1's in the positions $(1, n)$, $(2, n-1)$, $\ldots$, $(n, 1)$, and 0's elsewhere. Show that the matrices C, C^T, D, and D^T all have

$$p(x) = x^n + c_{n-1}x^{n-1} + \cdots + c_1 x + c_0$$

as their characteristic and minimal polynomials.

25. Let C be the companion matrix of the polynomial $p(x) = x^n + c_{n-1}x^{n-1} + \cdots + c_1 x + c_0$. Show that C is nonsingular if and only if $c_0 \neq 0$. In the case where C is nonsingular, find the inverse of C.

26. Let $A \in \mathbb{M}_n$, $p(x) = \det(xI - A)$, and $P(x) = \operatorname{adj}(xI - A)$. Define $Q_k(x) = x^k P(x) - P(x)A^k$, $k = 1, 2, \ldots$. Show that every entry of $Q_k(x)$ is divisible by $p(x)$ for $k = 1, 2, \ldots$.

27. Let A be an n-square matrix with eigenvalues all equal to λ. Show that

$$e^{tA} = e^{t\lambda} \sum_{k=0}^{n-1} \frac{t^k}{k!} (A - \lambda I)^k, \quad t \in \mathbb{C}.$$

In particular, if A is a 3×3 matrix of the repeated eigenvalue λ, then

$$e^{tA} = e^{t\lambda}\big(I + t(A - \lambda I) + \tfrac{1}{2}t^2(A - \lambda I)^2\big).$$

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3.4 λ-matrices and the Jordan Canonical Form

This section is devoted to the similarity of matrices. Given two matrices A and B of order n, we describe necessary and sufficient conditions for A and B to be similar (over $\mathbb{C}$), i.e, $A = P^{-1}BP$ for some invertible matrix P of order n. A basic result states that A and B are similar if and only if they have the same *Jordan canonical form* (also referred to as *Jordan decomposition*).

For this purpose, we decompose a square matrix into its Jordan canonical form (or simply called *Jordan form*). Our primary questions are

(i) What is a Jordan form?
(ii) How is a Jordan form constructed?
(iii) Why is a Jordan form unique?
(iv) Why is a matrix similar to its Jordan form?

What is a Jordan form? A Jordan form is a block-diagonal matrix

$$J = \begin{pmatrix} J_1 & & & \\ & J_2 & & \\ & & \ddots & \\ & & & J_k \end{pmatrix}, \tag{3.10}$$

where $J_1, J_2, \ldots, J_k$ are the so-called *Jordan blocks*. A Jordan block is an upper-triangular matrix in which the diagonal entries are the same, the entries directly above and to the right of the main diagonal (i.e., *superdiagonal*) are 1's, and all other entries are 0's. In (3.10), if the Jordan block J_i is $n_i \times n_i$ and has the diagonal entries $\lambda_i \in \mathbb{C}$, we write J_i in the upper-triangular form

$$J_i = J_{n_i}(\lambda_i) = \begin{pmatrix} \lambda_i & 1 & & \\ & \lambda_i & \ddots & \\ & & \ddots & 1 \\ & & & \lambda_i \end{pmatrix}, \quad i = 1, 2, \ldots, k. \tag{3.11}$$

(Note that in some texts, 1's are placed below the diagonal.) If J is $n \times n$, then $n_1 + \cdots + n_k = n$. The block-diagonal matrix in (3.10) is also written as

$$J = \operatorname{diag}(J_1, J_2, \ldots, J_k) = J_1 \oplus J_2 \oplus \cdots \oplus J_k.$$

That is, a Jordan form (or a *Jordan matrix*) is a direct sum of Jordan blocks.

Example 3.4.1 Some (non-)Jordan blocks and Jordan forms. Let $\lambda \in \mathbb{C}$.

(a) Jordan blocks: (0), (λ), $\left(\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right)$, $\left(\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}\right)$, $\left(\begin{smallmatrix} \lambda & 1 \\ 0 & \lambda \end{smallmatrix}\right)$, $\left(\begin{smallmatrix} 2 & 1 & 0 \\ 0 & 2 & 1 \\ 0 & 0 & 2 \end{smallmatrix}\right)$, $\left(\begin{smallmatrix} \lambda & 1 & 0 & 0 \\ 0 & \lambda & 1 & 0 \\ 0 & 0 & \lambda & 1 \\ 0 & 0 & 0 & \lambda \end{smallmatrix}\right)$.

(b) Non-Jordan blocks: $\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$, $\begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix}$, $\begin{pmatrix} 1 & 1 \\ 0 & 2 \end{pmatrix}$, $\begin{pmatrix} 2 & 1 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{pmatrix}$, $\begin{pmatrix} 2 & 1 & 1 \\ 0 & 2 & 1 \\ 0 & 0 & 2 \end{pmatrix}$, $\begin{pmatrix} \lambda & 1 & 0 & 0 \\ 0 & \lambda & 0 & 0 \\ 0 & 0 & \lambda & 1 \\ 0 & 0 & 0 & \lambda \end{pmatrix}$.

The 1st, 4th, and 6th matrices are Jordan matrices (forms).

(c) Let $J_1 = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$, $J_2 = (2)$, $J_3 = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$, $J_4 = \begin{pmatrix} 2 & 1 & 0 \\ 0 & 2 & 1 \\ 0 & 0 & 2 \end{pmatrix}$, $J_5 = (-1)$, and let

$$J = \operatorname{diag}(J_1, J_2, J_3, J_4, J_5) = \begin{pmatrix} 0 & 1 & & & & & & & \\ 0 & 0 & & & & & & & \\ & & 2 & & & & & & \\ & & & 0 & 1 & & & & \\ & & & 0 & 0 & & & & \\ & & & & & 2 & 1 & 0 & \\ & & & & & 0 & 2 & 1 & \\ & & & & & 0 & 0 & 2 & \\ & & & & & & & & -1 \end{pmatrix}.$$

Then J is a Jordan matrix as each diagonal block is a Jordan block. ∎

Why does every (square) matrix have a Jordan form? Why is it unique? How does one find it? And why is a matrix similar to its Jordan form?

The Jordan forms of square matrices can be achieved by taking different paths. The existence and uniqueness of the Jordan forms are usually shown

(1) by vector space decompositions (into certain invariant subspaces)[①],

(2) by λ-matrix theory (matrices over a polynomial ring)[②], or

(3) by pure matrix theory (through the Weyr characteristic)[③].

Each of the above approaches is technical and requires diligent effort. We outline (without detailed proofs) the approach of using λ-matrices as a tool of the study (here λ is thought of as an indeterminate) and work through examples. The idea is to use elementary operations to transform a λ-matrix to a standard form (called the *Smith canonical form* or simply *Smith form*). The Smith form is uniquely determined by the λ-matrix. Two matrices A and B are similar if and only if their λ-matrices $\lambda I - A$ and $\lambda I - B$ can be transformed to the same Smith form. A Smith form produces a Jordan form (through elementary divisors obtained from the Smith form). A is similar to the Jordan form of A; B is similar to the Jordan form of B. A and B are similar if and only if they have the same Jordan form (Jordan blocks).

A λ-*matrix* is a matrix whose entries are polynomials in λ. For example,

$$\begin{pmatrix} 1 & \lambda^2 - \sqrt{2} & (\lambda+1)^2 \\ \frac{1}{2}\lambda - 1 & 0 & 1 - 2\lambda - \lambda^2 \\ -1 & 1 & \lambda - i \end{pmatrix}$$

is a λ-matrix (over $\mathbb{C}$). Matrices of scalars (Sect. 1.2) are special λ-matrices.

[①] P. Halmos, *Finite-Dimensional Vector Spaces*, Springer, 1958, pp. 112–115.

[②] P. Lancaster and M. Tismenetsky, *The Theory of Matrices*, 2nd Ed., Academic Press, 1985, Chap. 7.

[③] S. Garcia and R. Horn, *Matrix Mathematics*, 2nd Ed., Cambridge Univ. Press, 2023, Chap. 12.

We perform operations (i.e., addition and multiplication) on λ-matrices in the same way as for matrices of scalars. Of course, the polynomials obey their usual rules of operations; but division by a (non-scalar) polynomial is not permitted as the set of λ-matrices is not closed under division.

Elementary operations (for rows and columns) on λ-matrices are similar to those on the matrices of scalars: (I) interchange two rows (columns), (II) multiply a row (column) by a nonzero constant, and (III) add a row (column) multiplied by a polynomial $p(\lambda)$ to another row (column). *Elementary λ-matrices* are similarly defined as those of the matrices of scalars. For example, $\left(\begin{smallmatrix} 1 & 0 \\ p(\lambda) & 1 \end{smallmatrix}\right)$ is an elementary λ-matrix, where $p(\lambda)$ is any polynomial in λ.

A square λ-matrix $A(\lambda)$ is *invertible* if there exists a λ-matrix $B(\lambda)$ such that $A(\lambda)B(\lambda) = B(\lambda)A(\lambda) = I$. For example, $\left(\begin{smallmatrix} 1 & 0 \\ p(\lambda) & 1 \end{smallmatrix}\right)$ is invertible and its inverse is $\left(\begin{smallmatrix} 1 & 0 \\ -p(\lambda) & 1 \end{smallmatrix}\right)$. If $A(\lambda)$ is invertible, by taking determinants, we see that $\det A(\lambda)$ is a nonzero constant. Conversely, if $\det A(\lambda)$ is a nonzero constant, then $(\det A(\lambda))^{-1} \operatorname{adj}(A(\lambda))$ is a λ-matrix and it is the inverse of $A(\lambda)$.

Recall that every matrix A of scalars can be brought to a (diagonal) matrix with 1's and 0's on the diagonal by elementary operations (Sect. 1.2). In the same spirit (with some more detailed work), by elementary operations on λ-matrices, every λ-matrix $A(\lambda)$ can be brought to a *Smith canonical form*

$$S(\lambda) = \begin{pmatrix} d_1(\lambda) & & & & & \\ & \ddots & & & & \\ & & d_r(\lambda) & & & \\ & & & 0 & & \\ & & & & \ddots & \\ & & & & & 0 \end{pmatrix}, \tag{3.12}$$

where $d_i(\lambda)$ divides $d_{i+1}(\lambda)$, $i = 1, \ldots, r-1$, and each $d_i(\lambda)$ is 1 or a monic polynomial. (r is the *rank* of $A(\lambda)$.) Therefore, for any λ-matrix $A(\lambda)$ there exist elementary λ-matrices $P_s(\lambda), \ldots, P_1(\lambda)$ and $Q_1(\lambda), \ldots, Q_t(\lambda)$ such that

$$P_s(\lambda) \cdots P_1(\lambda) A(\lambda) Q_1(\lambda) \cdots Q_t(\lambda) = S(\lambda).$$

A square λ-matrix is invertible if and only if its Smith form is the identity matrix and if and only if it is a product of some elementary λ-matrices.

Example 3.4.2 Find the Smith canonical form for the λ-matrix

$$A(\lambda) = \begin{pmatrix} 1-\lambda & \lambda^2 & \lambda \\ \lambda & \lambda & -\lambda \\ 1+\lambda^2 & \lambda^2 & -\lambda^2 \end{pmatrix}.$$

We perform elementary operations on $A(\lambda)$ (with c for column and r for row):

$$A(\lambda) \xrightarrow{c_3+c_1} \begin{pmatrix} 1 & \lambda^2 & \lambda \\ 0 & \lambda & -\lambda \\ 1 & \lambda^2 & -\lambda^2 \end{pmatrix} \xrightarrow{r_3-r_1} \begin{pmatrix} 1 & \lambda^2 & \lambda \\ 0 & \lambda & -\lambda \\ 0 & 0 & -\lambda^2-\lambda \end{pmatrix}$$

$$\overset{-r_3}{\longrightarrow} \begin{pmatrix} 1 & \lambda^2 & \lambda \\ 0 & \lambda & -\lambda \\ 0 & 0 & \lambda^2+\lambda \end{pmatrix} \overset{c}{\longrightarrow} \begin{pmatrix} 1 & 0 & 0 \\ 0 & \lambda & 0 \\ 0 & 0 & \lambda(\lambda+1) \end{pmatrix} = S(\lambda).$$

Let $Q_1(\lambda)$, $P_1(\lambda)$, $Q_2(\lambda)$, $Q_3(\lambda)$, and $Q_4(\lambda)$ be respectively the λ-matrices

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{pmatrix}, \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -1 & 0 & 1 \end{pmatrix}, \begin{pmatrix} 1 & -\lambda^2 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \begin{pmatrix} 1 & -\lambda & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix}.$$

Then

$$P_1(\lambda)A(\lambda)Q_1(\lambda)Q_2(\lambda)Q_3(\lambda)Q_4(\lambda) = S(\lambda). \blacksquare$$

Because our interest is the Jordan canonical decomposition of a square matrix, we now focus on the λ-matrices associated with matrices of scalars.

For an n-square matrix A of scalars, the Smith form of the λ-matrix $\lambda I - A$ is $S(\lambda) = \operatorname{diag}(d_1(\lambda), \ldots, d_n(\lambda))$ in (3.12). The $d_i(\lambda)$ are called the *invariant factors* of A, and the divisors of $d_i(\lambda)$ factored into the forms $(\lambda - x)^t$, where x is a scalar and t is a positive integer, are the *elementary divisors* of A.

For each elementary divisor $(\lambda - x)^t$, we construct a Jordan block

$$J_t(x) = \begin{pmatrix} x & 1 & & \\ & x & \ddots & \\ & & \ddots & 1 \\ & & & x \end{pmatrix} \in \mathbb{M}_t. \tag{3.13}$$

Let J be the direct sum of all Jordan blocks obtained from the elementary divisors of A. We call J a *Jordan canonical form* (or just *Jordan form*) of A.

The Smith form of $\lambda I_t - J_t(x)$ is $S(\lambda) = \operatorname{diag}(1, \ldots, 1, (\lambda - x)^t)$.

Here are the steps for finding a Jordan form of a square matrix A:

$$\lambda I - A \to \text{Smith form} \to \text{invariant factors } d_i(\lambda) \text{ factored completely}$$
$$\to \text{elementary divisors} \to \text{Jordan blocks} \to \text{Jordan form}.$$

Example 3.4.3 We demonstrate how to find the Smith canonical form, invariant factors, elementary divisors, and Jordan canonical form for matrix

$$A = \begin{pmatrix} -1 & 0 & 0 \\ 1 & 1 & 2 \\ 3 & 0 & 1 \end{pmatrix}.$$

We perform elementary row and column operations on the λ-matrix

$$\lambda I - A = \begin{pmatrix} \lambda+1 & 0 & 0 \\ -1 & \lambda-1 & -2 \\ -3 & 0 & \lambda-1 \end{pmatrix}.$$

Interchange row 1 and row 2 times -1 to get a 1 for the $(1,1)$ position:

$$\begin{pmatrix} 1 & -\lambda+1 & 2 \\ \lambda+1 & 0 & 0 \\ -3 & 0 & \lambda-1 \end{pmatrix}.$$

Subtract row 1 times $(\lambda+1)$ from row 2, and add row 1 times 3 to row 3:

$$\begin{pmatrix} 1 & -\lambda+1 & 2 \\ 0 & (\lambda-1)(\lambda+1) & -2(\lambda+1) \\ 0 & -3(\lambda-1) & \lambda+5 \end{pmatrix}.$$

Add row 3 times 2 to row 2 (to get a nonzero number, i.e., 8):

$$\begin{pmatrix} 1 & -\lambda+1 & 2 \\ 0 & \lambda^2-6\lambda+5 & 8 \\ 0 & -3(\lambda-1) & \lambda+5 \end{pmatrix}.$$

Interchange column 2 and column 3:

$$\begin{pmatrix} 1 & 2 & -\lambda+1 \\ 0 & 8 & \lambda^2-6\lambda+5 \\ 0 & \lambda+5 & -3(\lambda-1) \end{pmatrix}.$$

Subtract the second row times $\frac{1}{8}(\lambda+5)$ from row 3 to get

$$\begin{pmatrix} 1 & 2 & -\lambda+1 \\ 0 & 8 & \lambda^2-6\lambda+5 \\ 0 & 0 & -\frac{1}{8}(\lambda-1)^2(\lambda+1) \end{pmatrix},$$

which gives the Smith canonical form at once (by column operations)

$$S(\lambda) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & (\lambda+1)(\lambda-1)^2 \end{pmatrix}. \tag{3.14}$$

Thus, the invariant factors of A are

$$d_1(\lambda) = 1, \quad d_2(\lambda) = 1, \quad d_3(\lambda) = (\lambda+1)(\lambda-1)^2,$$

and the elementary divisors of A are

$$\lambda+1, \quad (\lambda-1)^2.$$

The Jordan blocks corresponding to the elementary divisors $\lambda+1$ and $(\lambda-1)^2$ are respectively $J_1(-1) = (-1)$ and $J_2(1) = \left(\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}\right)$. The Jordan form of A is

$$J = \begin{pmatrix} -1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix} = (-1) \oplus \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}.$$

One checks that A is similar to J as $AP = PJ$, where $P = \left(\begin{smallmatrix} 2 & 0 & 0 \\ 2 & 2 & 0 \\ -3 & 0 & 1 \end{smallmatrix}\right)$. ∎

Exercise: Find the Smith forms of $\left(\begin{smallmatrix} \lambda-1 & \lambda \\ 0 & \lambda+1 \end{smallmatrix}\right)$ and $\left(\begin{smallmatrix} \lambda-1 & \lambda \\ \lambda & \lambda+1 \end{smallmatrix}\right)$.

Example 3.4.4 Find examples of matrices A, B, and C such that their elementary divisors are λ, λ, λ; λ, λ^2; and λ^3, respectively.

A, B, and C are 3×3 matrices. The zero matrix $A = 0_3$ is the only matrix that has elementary divisors λ, λ, λ. Take $B = \left(\begin{smallmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{smallmatrix}\right)$ and $C = \left(\begin{smallmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{smallmatrix}\right)$. ∎

Example 3.4.5 Let A be a 4×4 matrix. If $\lambda I_4 - A$ has the Smith form

$$S(\lambda) = \begin{pmatrix} 1 & & & \\ & 1 & & \\ & & \lambda & \\ & & & \lambda^3 - \lambda^2 \end{pmatrix}, \tag{3.15}$$

then the invariant factors are 1, 1, λ, $\lambda^2(\lambda - 1)$, and the elementary divisors are λ, λ^2, $\lambda - 1$. Thus, the corresponding Jordan blocks are (0), $\left(\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right)$, and (1), respectively. It follows that A has the Jordan canonical form

$$J = \begin{pmatrix} 0 & & & \\ & 0 & 1 & \\ & 0 & 0 & \\ & & & 1 \end{pmatrix} = (0) \oplus \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \oplus (1). \ ∎$$

If J is a Jordan form of matrix A, then $\lambda I - A$ and $\lambda I - J$ have the same Smith form. We show this by the following example. (A rigorous proof is usually accomplished through so-called *determinant divisors* for λ-matrices.)

Example 3.4.6 Continuing with the above example, we show that the Smith form of $\lambda I - J$ is also $S(\lambda)$ in (3.15). We perform elementary operations:

$$\begin{aligned}
\lambda I - J &= \begin{pmatrix} \lambda & 0 & 0 & 0 \\ 0 & \lambda & 1 & 0 \\ 0 & 0 & \lambda & 0 \\ 0 & 0 & 0 & \lambda - 1 \end{pmatrix} \xrightarrow{(-\lambda)\mathrm{r}_2 + \mathrm{r}_3} \begin{pmatrix} \lambda & 0 & 0 & 0 \\ 0 & \lambda & 1 & 0 \\ 0 & -\lambda^2 & 0 & 0 \\ 0 & 0 & 0 & \lambda - 1 \end{pmatrix} \\
&\overset{\mathrm{c}}{\to} \begin{pmatrix} \lambda & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & \lambda^2 & 0 \\ 0 & 0 & 0 & \lambda - 1 \end{pmatrix} \begin{pmatrix} \text{Note that } \lambda^2 - (\lambda - 1)(\lambda + 1) = 1. \\ \text{Add column 3 to column 4, then} \\ \text{add row 4 times } -(\lambda + 1) \text{ to row 3.} \end{pmatrix} \\
&\to \begin{pmatrix} \lambda & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & \lambda^2 & 1 \\ 0 & 0 & 0 & \lambda - 1 \end{pmatrix} \xrightarrow{-(\lambda-1)\mathrm{r}_3 + \mathrm{r}_4} \begin{pmatrix} \lambda & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & \lambda^2 & 1 \\ 0 & 0 & -\lambda^2(\lambda - 1) & 0 \end{pmatrix} \\
&\overset{\mathrm{c}}{\to} \begin{pmatrix} \lambda & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & \lambda^2(\lambda - 1) & 0 \end{pmatrix} \overset{\mathrm{c,r}}{\longrightarrow} \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & \lambda & 0 \\ 0 & 0 & 0 & \lambda^2(\lambda - 1) \end{pmatrix} = S(\lambda). \ ∎
\end{aligned}$$

One may check that for the Jordan form $K = (0) \oplus (1) \oplus \left(\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right)$ obtained from J in Example 3.4.5 by reordering the diagonal blocks, $\lambda I_4 - K$ has the same Smith form $S(\lambda)$ as in (3.15). In general, the Smith canonical form of a λ-matrix (in the form of $\lambda I - A$) corresponds to a family of Jordan blocks by which the constructed Jordan canonical forms are similar to each other.

It follows from the above discussions that for matrix A and its Jordan form J, there exist invertible λ-matrices $P_i(\lambda)$ and $Q_i(\lambda)$, $i = 1, 2$, such that

$$P_1(\lambda)(\lambda I - A)Q_1(\lambda) = P_2(\lambda)(\lambda I - J)Q_2(\lambda) = S(\lambda). \tag{3.16}$$

A profound result states that every square matrix A is similar to its Jordan canonical form J. This is because of (3.16) and the following theorem.

Theorem 3.4.1 *Let A and B be $n \times n$ matrices. Then A and B are similar if and only if $\lambda I - A$ and $\lambda I - B$ have the same Smith canonical form. (We say that $\lambda I - A$ and $\lambda I - B$ are equivalent.) In other words, A and B are similar if and only if there exist invertible λ-matrices $P(\lambda)$ and $Q(\lambda)$ such that*

$$P(\lambda)(\lambda I - A)Q(\lambda) = \lambda I - B. \tag{3.17}$$

We sketch the proof of Theorem 3.4.1 as follows.

Proof outline. If A and B are similar, then there exists an invertible scalar matrix P (as a special λ-matrix) such that $PAP^{-1} = B$. It follows that

$$P(\lambda I - A)P^{-1} = \lambda I - B.$$

To show the reverse direction, let $P(\lambda)$ and $Q(\lambda)$ be invertible λ-matrices such that (we use $Q(\lambda)$ for $Q(\lambda)^{-1}$ on the right-hand side for convenience)

$$P(\lambda)(\lambda I - A) = (\lambda I - B)Q(\lambda). \tag{3.18}$$

Using the remainder theorem of polynomials, we write (Problem 16)

$$P(\lambda) = (\lambda I - B)P_1(\lambda) + P, \quad Q(\lambda) = Q_1(\lambda)(\lambda I - A) + Q,$$

where $P_1(\lambda)$ and $Q_1(\lambda)$ are λ-matrices, and P and Q are matrices of scalars.

Identity (3.18) implies $P_1(\lambda) - Q_1(\lambda) = 0$, $Q = P$, and $PA = BP$.

It remains to show that P is invertible. We assume that $R(\lambda)$ is the inverse of $P(\lambda)$, i.e., $P(\lambda)R(\lambda) = I$. Write $R(\lambda) = (\lambda I - A)R_1(\lambda) + R$, where R is a scalar matrix. With $PA = BP$, we compute $P(\lambda)R(\lambda) = I$ to get

$$(\lambda I - B)T(\lambda) + PR = I, \tag{3.19}$$

where

$$T(\lambda) = P_1(\lambda)(\lambda I - A)R_1(\lambda) + P_1(\lambda)R + PR_1(\lambda).$$

Since the right-hand side of (3.19) contains no λ, $T(\lambda)$ on the left-hand side must be equal to zero. Therefore, $I = PR$, and P is nonsingular. ∎

Theorem 3.4.1 implies our main result about the Jordan decomposition of a square matrix, which is one of the most useful results in matrix theory.

Theorem 3.4.2 (Jordan Decomposition) *Let A be an $n \times n$ matrix. Then there exists an $n \times n$ invertible matrix P (with complex entries) such that*

$$P^{-1}AP = J_1 \oplus J_2 \oplus \cdots \oplus J_k = J, \tag{3.20}$$

where J_i's are the Jordan blocks of A and J is the Jordan form of A.

The Jordan blocks in (3.20) are uniquely determined by A, and the Jordan form of A is unique up to ordering (permutations) of the diagonal Jordan blocks. The uniqueness follows from the uniqueness of the Smith canonical form (3.12) (which holds true in a more general algebraic setting – Ring theory). The uniqueness is also shown by considering $\text{rank}(\lambda_i I - J)^t$, $t = 1, 2, \ldots$, for each eigenvalue λ_i of A in Theorem 3.4.2 (Problem 32).

Note that the decomposition (3.20) is over $\mathbb{C}$. Even though A is real, P need not be real unless the eigenvalues of A are all real (Problem 17). In addition, P is not unitary in general. Recall that the Schur theorem (Sect. 3.2) ensures that a square matrix is unitarily similar to an upper-triangular matrix (Schur form). Both Schur form and Jordan form are upper-triangular matrices with the eigenvalues of the matrix on the diagonal. The Jordan form of a matrix appears to have more zero entries than its Schur form.

We summarize our discussions in the following theorem.

Theorem 3.4.3 (Matrix Similarity Theorem) *Let A and B be $n \times n$ matrices (of real or complex entries). The following statements are equivalent.*

(1) *A and B are similar.*
(2) *A and B have the same Jordan form (Jordan blocks).*
(3) *$\lambda I - A$ and $\lambda I - B$ are equivalent.*
(4) *$\lambda I - A$ and $\lambda I - B$ have the same Smith form.*
(5) *$\lambda I - A$ and $\lambda I - B$ have the same invariant factors.*
(6) *$\lambda I - A$ and $\lambda I - B$ have the same elementary divisors.*

Example 3.4.7 Determine similarity of matrices via Jordan blocks.

(a) $\left(\begin{smallmatrix} 1 & x \\ 0 & 1 \end{smallmatrix}\right)$ and $\left(\begin{smallmatrix} 1 & 0 \\ y & 1 \end{smallmatrix}\right)$ are similar for all $x, y \in \mathbb{C}$, $xy \neq 0$.

(b) $\left(\begin{smallmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{smallmatrix}\right)$ and $\left(\begin{smallmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{smallmatrix}\right)$ are not similar.

(c) $\left(\begin{smallmatrix} 2 & 1 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 3 & 1 \\ 0 & 0 & 0 & 3 \end{smallmatrix}\right)$ and $\left(\begin{smallmatrix} 2 & 1 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 3 & 0 \\ 0 & 0 & 0 & 3 \end{smallmatrix}\right)$ are not similar.

(d) $\left(\begin{smallmatrix} 2 & 1 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 3 & 1 \\ 0 & 0 & 0 & 4 \end{smallmatrix}\right)$ and $\left(\begin{smallmatrix} 2 & 1 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 3 & 0 \\ 0 & 0 & 0 & 4 \end{smallmatrix}\right)$ are similar as $\left(\begin{smallmatrix} 3 & 1 \\ 0 & 4 \end{smallmatrix}\right)$ is similar to $\left(\begin{smallmatrix} 3 & 0 \\ 0 & 4 \end{smallmatrix}\right)$.

(e) $\left(\begin{smallmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{smallmatrix}\right)$ and $\left(\begin{smallmatrix} 1 & -1 & 0 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{smallmatrix}\right)$ are similar to the Jordan form $\left(\begin{smallmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{smallmatrix}\right)$. ∎

The minimal and characteristic polynomials of a matrix, say A of order n, can be easily obtained from the Smith form of $\lambda I - A$. Reduce $\lambda I - A$ by elementary operations to its Smith form $S(\lambda) = \text{diag}(d_1(\lambda), \ldots, d_n(\lambda))$, where $d_i(\lambda)$ divides $d_{i+1}(\lambda)$, $i = 1, \ldots, n-1$. Thus, $\det(\lambda I - A) = \det S(\lambda) = d_1(\lambda) \cdots d_n(\lambda)$ (monic). One shows that $d_n(\lambda)$ is the minimal polynomial of the Jordan form of A (Problem 27). As similar matrices have the same minimal polynomial (Theorem 3.3.2), $d_n(\lambda)$ is the minimal polynomial of A.

Theorem 3.4.4 *Let $p(\lambda)$ and $m(\lambda)$ be respectively the characteristic and minimal polynomials of matrix A of order n. Let $d_1(\lambda), \ldots, d_n(\lambda)$ be the invariant factors of A, where $d_i(\lambda)$ divides d_{i+1}, $i = 1, \ldots, n-1$. Then*

$$p(\lambda) = d_1(\lambda) \cdots d_n(\lambda) \quad \textit{and} \quad m(\lambda) = d_n(\lambda).$$

In Example 3.4.3, the minimal and characteristic polynomials of A are the same: $p(\lambda) = m(\lambda) = (\lambda + 1)(\lambda - 1)^2 = \lambda^3 - \lambda^2 - \lambda + 1$.

In Example 3.4.5, the minimal and characteristic polynomials of A are $m(\lambda) = \lambda^3 - \lambda^2$ and $p(\lambda) = \lambda(\lambda^3 - \lambda^2) = \lambda^4 - \lambda^3$, respectively.

Example 3.4.8 Let $f(x) = x^3 - ax^2 - bx - c$, where $a, b, c \in \mathbb{C}$, and let

$$A = \begin{pmatrix} 0 & 0 & c \\ 1 & 0 & b \\ 0 & 1 & a \end{pmatrix}.$$

One performs operations on $\lambda I - A$ (by adding row 2 times λ to row 1, then adding row 3 times λ^2 to row 1) to find the Smith canonical form $S(\lambda) = \text{diag}(1, 1, d_3(\lambda))$, where $d_3(\lambda) = f(\lambda) = \lambda^3 - a\lambda^2 - b\lambda - c$. We have

$$p(\lambda) = m(\lambda) = d_3(\lambda) = \lambda^3 - a\lambda^2 - b\lambda - c.$$

The elementary divisors depend on the values of a, b, and c. For example, if $a = b = 1$, $c = -1$, then the elementary divisors are $\lambda + 1$ and $(\lambda - 1)^2$ with corresponding Jordan blocks $J_1 = (-1)$ and $J_2 = \left(\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}\right)$, respectively. If $a = c = 6$, $b = -11$, then the elementary divisors are $\lambda - 1$, $\lambda - 2$, and $\lambda - 3$ with corresponding Jordan blocks $J_1 = (1), J_2 = (2)$, and $J_3 = (3)$, respectively. In this case, A is diagonalizable. ∎

Note: In Example 3.4.8, A is the companion matrix of $f(x)$. The characteristic and minimal polynomials of A are the same, i.e., $f(x)$. Thus, alternatively, one easily obtains the Smith canonical form of $\lambda I - A$: $S(\lambda) = \text{diag}(1, 1, d_3(\lambda))$, where $d_3(\lambda) = f(\lambda) = \lambda^3 - a\lambda^2 - b\lambda - c$. Moreover, A is diagonalizable if and only if A has three distinct eigenvalues (Example 3.3.5).

A square matrix is unitarily diagonalizable (over $\mathbb{C}$) if and only if it is a normal matrix (the spectral decomposition theorem, i.e., Theorem 3.2.2). Using Jordan decomposition, one can obtain conditions for a matrix to be diagonalizable (not necessarily unitarily diagonalizable); see Problem 48.

Problems

1. Let $A = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}$, $B = \begin{pmatrix} 2 & 1 & 0 \\ 0 & 2 & 1 \\ 0 & 0 & 2 \end{pmatrix}$, $C = \begin{pmatrix} 2 & 0 & 0 \\ 0 & 2 & 1 \\ 0 & 0 & 2 \end{pmatrix}$, and $D = \begin{pmatrix} -2 & 0 & 0 \\ 0 & 2 & 1 \\ 0 & 0 & 2 \end{pmatrix}$.
   (a) Find A^2, A^3, $(2I_3 - B)^4$, $(2I_3 - C)^2$, and $(2I_3 - D)^2$.
   (b) Compute the ranks of $(2I_3 - B)^k$ and $(2I_3 - D)^k$ for $k = 1, 2, 3$.
2. Find the Smith form of $\lambda I - A$ for $A = \begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix}$, $\begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}$, $\begin{pmatrix} 2 & 1 \\ 0 & 2 \end{pmatrix}$, $\begin{pmatrix} 2 & 1 \\ 0 & -2 \end{pmatrix}$.
3. Find the Smith forms of $\begin{pmatrix} -1 & \lambda \\ 0 & 1 \end{pmatrix}$, $\begin{pmatrix} -1 & \lambda \\ 0 & \lambda \end{pmatrix}$, $\begin{pmatrix} \lambda - 1 & \lambda \\ 0 & \lambda + 1 \end{pmatrix}$, and $\begin{pmatrix} \lambda - 1 & \lambda \\ \lambda & \lambda + 1 \end{pmatrix}$.
4. Let $A(\lambda) = \begin{pmatrix} \lambda & 1 \\ -1 & \lambda \end{pmatrix}$. Discuss the invertibility of $A(\lambda)$ as a λ-matrix.
5. Show that a square λ-matrix is invertible if and only if it is a product of some elementary λ-matrices, and if and only if its Smith form is I.
6. Find the Smith form for the λ-matrix $A(\lambda) = \begin{pmatrix} \lambda^2 & 1 - \lambda & \lambda \\ \lambda & \lambda & -\lambda \\ \lambda^2 & 1 + \lambda^2 & -\lambda^2 \end{pmatrix}$.
7. Consider the Jordan form $J = \begin{pmatrix} -1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix}$ of matrix A in Example 3.4.3. Perform elementary operations on $\lambda I - J$ to find its Smith form.
8. Find the invariant factors, elementary divisors, characteristic and minimal polynomials, and Jordan canonical form associated with the matrix
$$A = \begin{pmatrix} 3 & 1 & -3 \\ -7 & -2 & 9 \\ -2 & -1 & 4 \end{pmatrix}.$$
9. Compute A^k, where k is a positive integer, $a, b, c \in \mathbb{C}$, and
$$A = \begin{pmatrix} a & 0 & 0 & 0 & 0 & 0 \\ 0 & b & 1 & 0 & 0 & 0 \\ 0 & 0 & b & 0 & 0 & 0 \\ 0 & 0 & 0 & c & 1 & 0 \\ 0 & 0 & 0 & 0 & c & 1 \\ 0 & 0 & 0 & 0 & 0 & c \end{pmatrix}.$$
10. Show that A and B are similar but not unitarily similar, where
$$A = \begin{pmatrix} 0 & 2 \\ 0 & 0 \end{pmatrix}, \quad B = \begin{pmatrix} 0 & 3 \\ 0 & 0 \end{pmatrix}.$$
What is the Jordan canonical form J of A (and B)? Does there exist an invertible matrix P such that $P^{-1}AP = J$ and $P^{-1}BP = J$?

11. Find the Smith canonical forms of $\lambda I - A$, $\lambda I - B$, and $\lambda I - C$, then find the Jordan canonical forms of A, B, and C, where

$$A = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix}, \quad B = \begin{pmatrix} 1 & 1 & -1 \\ 1 & 1 & 1 \\ -1 & 1 & 1 \end{pmatrix}, \quad C = \begin{pmatrix} 3 & 1 & -1 \\ -1 & 1 & 1 \\ 0 & 0 & 2 \end{pmatrix}.$$

12. Determine which of the following matrices are similar to each other:

$$\begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix}, \quad \begin{pmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 0 & 2 & 1 \end{pmatrix}, \quad \begin{pmatrix} 1 & 2 & 0 \\ 2 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad \begin{pmatrix} 1 & 2 & 0 \\ 2 & 1 & 0 \\ 0 & 2 & 1 \end{pmatrix}.$$

13. Show that $\left(\begin{smallmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ a & b & c \end{smallmatrix}\right)$ and $\left(\begin{smallmatrix} x & 0 & 0 \\ 0 & y & 0 \\ 0 & 0 & y \end{smallmatrix}\right)$ are not similar for any $a, b, c, x, y \in \mathbb{C}$.

14. If $A = \left(\begin{smallmatrix} 2 & 1 & 0 \\ 0 & 2 & x \\ 0 & 0 & 2 \end{smallmatrix}\right)$ is similar to $B = \left(\begin{smallmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 3 & 2 \end{smallmatrix}\right)$, what is the value(s) of $x \in \mathbb{C}$?

15. Let

$$Q(\lambda) = \begin{pmatrix} \lambda^3 - 1 & \lambda^2 + 3 \\ 2\lambda^2 - \lambda + 1 & \lambda - 5 \end{pmatrix}, \quad A = \begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix}.$$

Find matrices B, C, D, E, T of scalars, and λ-matrix $S(\lambda)$ such that

$$Q(\lambda) = B\lambda^3 + C\lambda^2 + D\lambda + E, \quad Q(\lambda) = (\lambda I - A)S(\lambda) + T.$$

16. Let $A \in \mathbb{M}_n$. If $P(\lambda)$ is an n-square λ-matrix, show that there exist an n-square λ-matrix $S(\lambda)$ and an n-square scalar matrix T such that

$$P(\lambda) = (\lambda I - A)S(\lambda) + T.$$

[Hint: Let $P(\lambda) = \lambda^m P_m + \cdots + \lambda P_1 + P_0$ and use induction on m.]

17. Let A be a real square matrix. If the eigenvalues of A are all real, show that there exists a real invertible matrix P such that $P^{-1}AP = J$ is the Jordan form of A. Can P be always chosen to be real orthogonal?

18. Find the invariant factors, elementary divisors, and characteristic and minimal polynomials associated with each of the following matrices.

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix}, \quad \begin{pmatrix} -1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix}, \quad \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix}, \quad \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 1 & 1 \end{pmatrix},$$

$$\begin{pmatrix} 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \quad \begin{pmatrix} 1 & -1 & 0 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \quad \begin{pmatrix} \lambda & 0 & 1 & 0 \\ 0 & \lambda & 0 & 1 \\ 0 & 0 & \lambda & 0 \\ 0 & 0 & 0 & \lambda \end{pmatrix}.$$

19. Find a Jordan canonical form of the matrix

$$P = \begin{pmatrix} 0 & 1 & 0 & \dots & 0 \\ 0 & 0 & 1 & \dots & 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ 0 & 0 & 0 & \dots & 1 \\ 1 & 0 & 0 & \dots & 0 \end{pmatrix}.$$

20. Let A be a square matrix with the invariant factors

$$1, \quad \lambda(\lambda - 2), \quad \lambda^3(\lambda - 2).$$

(a) What is the characteristic polynomial of A?

(b) What is the minimal polynomial of A?

(c) What are the elementary divisors of A?

(d) What is the size of A?

(e) What is the rank of A?

(f) What is the trace of A?

(g) What is the Jordan form of A?

21. By using the Smith canonical forms, find an example of matrices A and B of the same size such that A and B have the same minimal polynomial and the same characteristic polynomial, but different Smith canonical forms, concluding that A and B are not similar.

22. Let $A = \left(\begin{smallmatrix} 1 & 1+i \\ 0 & 1 \end{smallmatrix}\right)$. Which of the following matrices are similar to A?

$$\begin{pmatrix} 1 & 0 \\ 1-i & 1 \end{pmatrix}, \begin{pmatrix} 1 & 0 \\ 1+i & 1 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ 1 & 1+i \end{pmatrix}, \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \begin{pmatrix} 1 & 0 \\ 2 & 1 \end{pmatrix}, \begin{pmatrix} 1+i & 2 \\ 2 & 1-i \end{pmatrix}.$$

23. Let x, $y \in \mathbb{C}^n$ (in columns) and let $A = xy^T \in \mathbb{M}_n$. Show that there exists an invertible matrix P such that $P^{-1}AP = \left(\begin{smallmatrix} y^Tx & 0 \\ 0 & 0 \end{smallmatrix}\right)$ if $y^Tx \neq 0$ and $P^{-1}AP = \left(\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right)$ if $y^Tx = 0$, and that there exists a unitary matrix U such that $U^{-1}AU = \left(\begin{smallmatrix} y^Tx & z \\ 0 & 0 \end{smallmatrix}\right)$, where $z \in \mathbb{C}^{n-1}$. Must $z = 0$?

24. Let A be a square matrix. Show that the rank of A is one if and only if A is similar to a square matrix that has exactly one nonzero entry.

25. If $A \in \mathbb{M}_n$ satisfies $A^2 = 0$, what are possible Jordan blocks of A?

26. If $A \in \mathbb{M}_n$ satisfies $A^2 = A$, what are possible Jordan blocks of A?

27. Let $A \in \mathbb{M}_n$. Show that the last invariant factor $d_n(\lambda)$ in the Smith canonical form of $\lambda I - A$ is the minimal polynomial of A.

28. Let $p(x) = x^n + c_{n-1}x^{n-1} + c_{n-2}x^{n-2} + \cdots + c_1x + c_0$. What are the characteristic polynomial, minimal polynomial, invariant factors, and the Smith canonical form of the companion matrix of $p(x)$?

29. Let A be an n-square complex matrix. Show that there exists a real-coefficient polynomial p of degree at most $2n$ such that $p(A) = 0$.

30. Let $A = \begin{pmatrix} 1 & -1 \\ 1 & -1 \end{pmatrix}$, $J = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$, and $J_x = \begin{pmatrix} 0 & x \\ 0 & 0 \end{pmatrix}$, where $x \in \mathbb{C}$. Show that

    (a) There exists a real invertible $P \in \mathbb{M}_2$ such that $P^{-1}AP = J$.
    (b) There does not exist a unitary $U \in \mathbb{M}_2$ such that $U^{-1}AU = J$.
    (c) There is a unitary V such that $V^*AV = J_x$ if and only if $|x| = 2$.
    (d) Find a matrix P in (a) and a unitary matrix V in (c) for $x = 2i$.

31. Let $J = \begin{pmatrix} J_p(x) & 0 \\ 0 & J_q(y) \end{pmatrix}$, where $J_k(z)$ denotes the $k \times k$ Jordan block with z on the diagonal. Find the rank of $(\lambda I_n - J)^m$ in terms of p, q, and m, where p, q, and m are positive integers, $\lambda \in \mathbb{C}$, and $n = p + q$.

32. Let J and K be two $n \times n$ matrices in Jordan forms. Show that

$$\operatorname{rank}(\lambda I_n - J)^k = \operatorname{rank}(\lambda I_n - K)^k \text{ for all } \lambda \in \mathbb{C}, \text{ k=1, 2, } \ldots, \text{ n,}$$

    if and only if J and K are permutation similar (i.e., diagonal-block matrices J and K have the same Jordan blocks up to ordering. This implies the uniqueness of the Jordan form up to similarity).

33. Show that every Jordan block $J \in \mathbb{M}_n$ is similar to its transpose J^T:

$$S^{-1}JS = J^T,$$

    where S is the backward identity matrix, that is, the entries of S are 1's in the positions $(1, n), (2, n-1), \ldots, (n, 1)$, and 0's elsewhere. Conclude that every square matrix is similar to its transpose.

34. Let $A \in \mathbb{M}_n$. If $\operatorname{rank}(A^2) = \operatorname{rank}(A)$, show that $\operatorname{rank}(A^k) = \operatorname{rank}(A)$ for all positive integers k and that $\mathbb{C}^n = \operatorname{Im} A \oplus \operatorname{Ker} A$.

35. Let $A \in \mathbb{M}_n$. Show that $\operatorname{rank}(A^m) = \operatorname{rank}(A^{m+1})$ for some positive integer $m \leq n$ and that $\operatorname{rank}(A^m) = \operatorname{rank}(A^k)$ for all positive integers $k > m$. In particular, $\operatorname{rank}(A^n) = \operatorname{rank}(A^{n+1})$ for any $A \in \mathbb{M}_n$.

36. Show that every matrix $A \in \mathbb{M}_n$ can be written as $A = B + C$, where $C^m = 0$ for some integer m, B is diagonalizable, and $BC = CB$.

37. Let $A \in \mathbb{M}_n$. If $Ax = 0$ whenever $A^2x = 0$, show that A has no Jordan block of order more than 1 associated with the eigenvalue 0.

38. Show that if a square matrix A has all eigenvalues equal to 1, then A^k is similar to A for every positive integer k. Discuss the converse.

39. For $a, b \in \mathbb{C}$, let A be the $n \times n$ Jordan block with diagonal entries a's and let B be the $n \times n$ Jordan block with diagonal entries b's. If $b^m = a$ for some positive integer m, show that B^m is similar to A.

40. Let $A \in \mathbb{M}_n$. Show that the dimension of the vector space of all polynomials in A equals the degree of the minimal polynomial of A.

41. Let A be an $n \times n$ matrix such that $A^k v = 0$ and $A^{k-1} v \neq 0$ for some vector v and positive integer k. Show that $v, Av, \ldots, A^{k-1}v$ are linearly independent. What is a Jordan canonical form of A?

42. Let J be an $n \times n$ Jordan block. Show that there exists a vector $v \in \mathbb{C}^n$ such that $v, Jv, \ldots, J^{n-1}v$ constitute a basis for $\mathbb{C}^n$.

43. Show that the characteristic polynomial of $A \in \mathbb{M}_n$ is equal to the minimal polynomial of A if and only if $v, Av, \ldots, A^{n-1}v$ are linearly independent for some vector $v \in \mathbb{C}^n$. What can be said about the Jordan canonical form (or Jordan blocks) of such A?

44. Let A be an n-square matrix. Show that for every nonzero vector $v \in \mathbb{C}^n$, there exists an eigenvector u of A that is contained in the span of $v, Av, A^2v, \ldots$. [Hint: $v, Av, A^2v, \ldots, A^k v$ are linearly dependent for some k. Find a related polynomial, then factor it out.]

45. Let $J_m(\lambda)$ be an $m \times m$ upper-triangular matrix whose diagonal entries are all equal to λ. Show that the geometric multiplicity of λ is equal to its algebraic multiplicity if and only if $J_m(\lambda)$ is the scalar matrix λI_m, that is, $\dim V_\lambda = m$ if and only if $J_m(\lambda) = \lambda I_m$.

46. If A is a normal matrix, show that (a) the geometric and algebraic multiplicities are equal for every eigenvalue of A and (b) the minimal polynomial of A has no repeated roots. Are the converses true?

47. Let A be an n-square matrix having a Jordan canonical form

$$J = \begin{pmatrix} J_{n_1}(\lambda_1) & & & 0 \\ & J_{n_2}(\lambda_2) & & \\ & & \ddots & \\ 0 & & & J_{n_k}(\lambda_k) \end{pmatrix}, \; J_{n_i}(\lambda_i) = \begin{pmatrix} \lambda_i & 1 & & 0 \\ & \lambda_i & \ddots & \\ & & \ddots & 1 \\ 0 & & & \lambda_i \end{pmatrix},$$

where $J_{n_i}(\lambda_i)$ of order n_i are the Jordan blocks associated with eigenvalues λ_i (which need not be distinct), and $n_1 + n_2 + \cdots + n_k = n$. Let λ be any eigenvalue, i.e., $\lambda_1, \lambda_2 \ldots, \lambda_k$, of A. Denote the algebraic multiplicity of λ by p and the geometric multiplicity by q. Show that

(a) p is the total number of λ on the diagonal of J.

(b) q is the total number of Jordan blocks corresponding to λ.

(c) $p = q$ if and only if A is similar to a matrix in the form $\begin{pmatrix} \lambda I_p & 0 \\ 0 & \star \end{pmatrix}$.

48. (*Diagonalization conditions*) Let A be an n-square matrix having distinct eigenvalues $\lambda_1, \lambda_2, \ldots, \lambda_s \in \mathbb{C}$, and let the characteristic polynomial of A be factored completely on the complex number field $\mathbb{C}$ as

$$\det(\lambda I - A) = (\lambda - \lambda_1)^{r_1}(\lambda - \lambda_2)^{r_2} \cdots (\lambda - \lambda_s)^{r_s},$$

where $r_1, r_2, \ldots, r_s$ are the algebraic multiplicities of $\lambda_1, \lambda_2, \ldots, \lambda_s$, respectively. Show that the following statements are equivalent.

(a) A is diagonalizable, namely, A is similar to a diagonal matrix.
(b) A has n linearly independent eigenvectors.
(c) All the elementary divisors of $\lambda I - A$ are linear.
(d) $m(\lambda) = (\lambda - \lambda_1)(\lambda - \lambda_2) \cdots (\lambda - \lambda_s)$ is the minimal polynomial of A. Equivalently, $m(A) = 0$ and $m(\lambda)$ has no repeated roots.
(e) $\operatorname{rank}(\lambda I - A) = \operatorname{rank}(\lambda I - A)^2$ for every eigenvalue λ.
(f) $\operatorname{rank}(cI - A) = \operatorname{rank}(cI - A)^2$ for any complex number c.
(g) $\operatorname{Ker}(\lambda I - A) = \operatorname{Ker}(\lambda I - A)^2$ for every eigenvalue λ.
(h) $\operatorname{Ker}(cI - A) = \operatorname{Ker}(cI - A)^2$ for any complex number c.
(i) $\dim V_{\lambda_i} = r_i$ for each eigenvalue λ_i with eigenspace V_{λ_i}, i.e., the geometric and algebraic multiplicities of each eigenvalue are equal.
(j) $\mathbb{C}^n = V_{\lambda_1} \oplus V_{\lambda_2} \oplus \cdots \oplus V_{\lambda_s}$.
(k) $\operatorname{rank}(\lambda_i I - A) = n - r_i$ for every eigenvalue λ_i.
(l) $\operatorname{Im}(\lambda I - A) \cap \operatorname{Ker}(\lambda I - A) = \{0\}$ for every eigenvalue λ.
(m) $\operatorname{Im}(cI - A) \cap \operatorname{Ker}(cI - A) = \{0\}$ for any complex number c.

49. Let $\mathcal{A}$ be a linear transformation on a vector space of dimension n. If λ is an eigenvalue of $\mathcal{A}$, show that for some positive integer $m \leq n$,

$$\operatorname{Ker}(\lambda\mathcal{I} - \mathcal{A})^m = \operatorname{Ker}(\lambda\mathcal{I} - \mathcal{A})^{m+1} = \operatorname{Ker}(\lambda\mathcal{I} - \mathcal{A})^{m+2} = \cdots.$$

Conclude that for every eigenvalue λ of $\mathcal{A}$,

$$\operatorname{Ker}(\lambda\mathcal{I} - \mathcal{A})^n = \operatorname{Ker}(\lambda\mathcal{I} - \mathcal{A})^{n+1} = \operatorname{Ker}(\lambda\mathcal{I} - \mathcal{A})^{n+2} = \cdots.$$

50. (a) Compute J^k, $\operatorname{rank}(J^k)$, and J^{-1} (if $x \neq 0$) for the Jordan block

$$J = J_n(x) = \begin{pmatrix} x & 1 & & 0 \\ & x & \ddots & \\ & & \ddots & 1 \\ 0 & & & x \end{pmatrix} \in \mathbb{M}_n,\ x \in \mathbb{C},\ k = 1, 2, \ldots.$$

(b) Find explicit expressions of J^k and J^{-1} for the case where $x = 1$.
(c) Find conditions on x, n, k so that the $(1, n)$-entry of J^k is nonzero.

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3.5 The Matrices A^T, $\overline{A}$, A^*, A^TA, A^*A, and $\overline{A}A$

We have seen the numerical values (scalars) associated with a (square) matrix, such as determinant, trace, rank, eigenvalues, and singular values, etc; they characterize and carry certain properties of the matrix. Given a matrix A, we now consider (the similarities of) the matrices associated with A,

$$A^T,\ \ \overline{A},\ \ A^*,\ \ A^TA,\ \ A^*A,\ \ \overline{A}A,$$

where ^{T}, $-$, and $*$ stand for the transpose, conjugate, and conjugate transpose, respectively. By Theorem 2.3.1, the following rank identities hold:

$$\operatorname{rank}(A) = \operatorname{rank}(A^T) = \operatorname{rank}(\overline{A}) = \operatorname{rank}(A^*) = \operatorname{rank}(A^*A) = \operatorname{rank}(AA^*).$$

Note that $\operatorname{rank}(A)$ and $\operatorname{rank}(AA^T)$ need not be equal unless A is real.

Example 3.5.1 Conjugate transpose.

(a) Let $u = (1, i)$. Then $uu^* = 2$, $u^*u = \begin{pmatrix} 1 & i \\ -i & 1 \end{pmatrix}$. [Note: $\bar{u}u$ is undefined.]

(b) Let $A = \begin{pmatrix} 1 & 1 \\ i & i \end{pmatrix}$. Then $\overline{A}A = \begin{pmatrix} 1+i & 1+i \\ 1-i & 1-i \end{pmatrix}$, $A^TA = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$, $A^*A = \begin{pmatrix} 2 & 2 \\ 2 & 2 \end{pmatrix}$.

(c) Let $A = \begin{pmatrix} 1 & i & 0 \\ 0 & 2 & i \end{pmatrix}$. Then $AA^* = \begin{pmatrix} 2 & 2i \\ -2i & 5 \end{pmatrix}$, $A^*A = \begin{pmatrix} 1 & i & 0 \\ -i & 5 & 2i \\ 0 & -2i & 1 \end{pmatrix}$. ∎

Theorem 3.5.1 *Let A be an n-square matrix. Then*

(1) *A is similar to its transpose A^T.*

(2) *A^*A is unitarily similar to AA^*.*

(3) *$\overline{A}A$ is similar to $A\overline{A}$.*

Proof. For (1), recall from Theorem 3.4.1 that two square matrices X and Y of the same size are similar if and only if $\lambda I - X$ and $\lambda I - Y$ have the same Smith canonical form. It is obvious that the matrices $\lambda I - A$ and $\lambda I - A^T$ have the same Smith canonical form. Thus, A and A^T are similar.

An alternative approach to the proof of (1) is to use the Jordan canonical forms of A and A^T with the fact that $SJS^{-1} = J^T$ for every Jordan block J, where S is the backward identity matrix (Problem 33, Sect. 3.4).

(2) is proven by using a singular value decomposition of A.

To show (3) that $\overline{A}A$ is similar to $A\overline{A}$, we show that $A\overline{A}$ and $\overline{A}A$ have the same Jordan blocks (or decomposition up to permutations).

First $A\overline{A}$ and $\overline{A}A$ have the same real Jordan blocks because $\overline{A}A = \overline{A\overline{A}}$. Thus, $A\overline{A}$ and $\overline{A}A$ have the same Jordan blocks corresponding to zero eigenvalues. For nonzero eigenvalues, the Jordan blocks are nonsingular. By using Problem 24, $A\overline{A}$ and $\overline{A}A$ have the same nonsingular Jordan blocks.

Alternatively, we can show that $A\overline{A}$ and $\overline{A}A$ have the same nonsingular Jordan blocks as follows (or by using elementary divisors). The identity

$$\begin{pmatrix} I & -A \\ 0 & I \end{pmatrix} \begin{pmatrix} A\overline{A} & 0 \\ \overline{A} & 0 \end{pmatrix} \begin{pmatrix} I & A \\ 0 & I \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ \overline{A} & \overline{A}A \end{pmatrix}$$

implies the similarity of the block matrices

$$M = \begin{pmatrix} A\overline{A} & 0 \\ \overline{A} & 0 \end{pmatrix} \quad \text{and} \quad N = \begin{pmatrix} 0 & 0 \\ \overline{A} & \overline{A}A \end{pmatrix}.$$

Therefore, $\lambda I_{2n} - M$ and $\lambda I_{2n} - N$ are similar; so are $(\lambda I_{2n} - M)^k$ and $(\lambda I_{2n} - N)^k$, $k = 1, 2, \ldots$. Note that for any $\lambda \neq 0$ and for each $k = 1, 2, \ldots$

$$\operatorname{rank}(\lambda I_{2n} - M)^k = n + \operatorname{rank}(\lambda I_n - A\overline{A})^k$$

and

$$\operatorname{rank}(\lambda I_{2n} - N)^k = n + \operatorname{rank}(\lambda I_n - \overline{A}A)^k.$$

It follows that for any nonzero eigenvalue λ of A and for $k = 1, 2, \ldots,$

$$\operatorname{rank}(\lambda I_n - A\overline{A})^k = \operatorname{rank}(\lambda I_n - \overline{A}A)^k. \tag{3.21}$$

Replace $A\overline{A}$ and $\overline{A}A$ by their respective Jordan forms in (3.21). Note that $A\overline{A}$ and $\overline{A}A$ have the same eigenvalues. For each nonzero eigenvalue λ, computing both sides of (3.21) with $k = 1, 2, \ldots, n$, we see that $A\overline{A}$ and $\overline{A}A$ have the same nonsingular Jordan blocks. (See also Problem 25.) ∎

By Theorem 3.5.1 (1) and (3), we see that $\overline{A}A$, $A\overline{A}$, and $(A\overline{A})^T = (\overline{A}A)^*$ are similar. Note that for general square matrices X and Y of the same order, XY and YX need not be similar; X and $\overline{X}$ need not be similar.

Example 3.5.2 Every square matrix A is similar to its transpose A^T (Theorem 3.5.1) because A is similar to its Jordan form and every Jordan block J is similar to its transpose J^T via a backward identity matrix (Problem 33, Sect. 3.4). Note that a backward identity matrix is a permutation matrix.

Question: Is every square matrix A similar to its transpose A^T via a permutation matrix? That is, are A and A^T permutation-similar?

The answer is negative. Let $A = \left(\begin{smallmatrix} 0 & 1 & 0 \\ 0 & 0 & 2 \\ 0 & 0 & 0 \end{smallmatrix}\right)$. One can easily verify through computation that $AP \neq PA^T$ for any 3×3 unitary (including permutation) matrix P. Thus, A and A^T are similar but need not be unitarily similar. ∎

Theorem 3.5.2 *Let A be an n-square matrix. Then A is similar to A^* (equivalently, A is similar to $\overline{A}$) if and only if the Jordan blocks (counted with repetition) of the non-real eigenvalues of A occur in conjugate pairs. That is, A is similar to A^* if and only if the Jordan blocks of non-real eigenvalues can be paired in the form $J_i(\lambda) \oplus J_i(\bar{\lambda})$ in the Jordan canonical form of A.*

Proof. Let $J_1, \ldots, J_k$ be the Jordan blocks of A and let

$$P^{-1}AP = J_1 \oplus \cdots \oplus J_k$$

for some invertible matrix P. Taking the conjugate transpose gives

$$P^*A^*(P^*)^{-1} = J_1^* \oplus \cdots \oplus J_k^*.$$

The right-hand side $J_1^* \oplus \cdots \oplus J_k^*$ is similar to $\overline{J_1} \oplus \cdots \oplus \overline{J_k}$.

If A and A^* are similar, then $J_1 \oplus \cdots \oplus J_k$ and $\overline{J_1} \oplus \cdots \oplus \overline{J_k}$ are similar. By the uniqueness (up to permutations of blocks) of Jordan decomposition, we have $\{\overline{J_1}, \ldots, \overline{J_k}\} = \{J_1, \ldots, J_k\}$ (as multisets). If J_i is associated with a non-real eigenvalue and it appears j times, then so does $\overline{J_i}$. Because $J_i \neq \overline{J_i}$ for non-real eigenvalues, we see that the Jordan blocks of non-real eigenvalues of A occur in conjugate pairs (Problem 10).

Conversely, to show sufficiency, we may consider the special case

$$A = J \oplus \overline{J} \oplus R, \tag{3.22}$$

where J and R are Jordan blocks, J is complex, and R is real. Then

$$A^* = \overline{J}^T \oplus J^T \oplus R^T,$$

which is, by permutation, similar to

$$J^T \oplus \overline{J}^T \oplus R^T. \tag{3.23}$$

Theorem 3.5.1 ensures the similarity of A^* and A via (3.22) and (3.23). ∎

If $J_1, \ldots, J_k$ are the Jordan blocks of a matrix A, then $J_1, \ldots, J_k$ are also the Jordan blocks of A^T, and $\overline{J_1}, \ldots, \overline{J_k}$ are the Jordan blocks of $\overline{A}$ and A^*. Following the above proof for the similarity of A and A^*, we consider the matrix with 2×2 Jordan blocks of conjugate pairs

$$\begin{pmatrix} \lambda & 1 & 0 & 0 \\ 0 & \lambda & 0 & 0 \\ 0 & 0 & \overline{\lambda} & 1 \\ 0 & 0 & 0 & \overline{\lambda} \end{pmatrix},$$

which is similar via permutation to

$$\begin{pmatrix} \lambda & 0 & 1 & 0 \\ 0 & \overline{\lambda} & 0 & 1 \\ 0 & 0 & \lambda & 0 \\ 0 & 0 & 0 & \overline{\lambda} \end{pmatrix} = \begin{pmatrix} C(\lambda) & I \\ 0 & C(\lambda) \end{pmatrix}, \text{ where } C(\lambda) = \begin{pmatrix} \lambda & 0 \\ 0 & \overline{\lambda} \end{pmatrix}.$$

If $\lambda = a + bi$ with $a, b \in \mathbb{R}$, then we have by computation

$$\begin{pmatrix} -i & -i \\ 1 & -1 \end{pmatrix} \begin{pmatrix} \lambda & 0 \\ 0 & \overline{\lambda} \end{pmatrix} \begin{pmatrix} -i & -i \\ 1 & -1 \end{pmatrix}^{-1} = \begin{pmatrix} a & b \\ -b & a \end{pmatrix}.$$

Thus, the following two matrices (one is complex, one is real) are similar:

$$\begin{pmatrix} \lambda & 0 & 1 & 0 \\ 0 & \overline{\lambda} & 0 & 1 \\ 0 & 0 & \lambda & 0 \\ 0 & 0 & 0 & \overline{\lambda} \end{pmatrix} \in \mathbb{M}_4(\mathbb{C}), \quad \begin{pmatrix} a & b & 1 & 0 \\ -b & a & 0 & 1 \\ 0 & 0 & a & b \\ 0 & 0 & -b & a \end{pmatrix} \in \mathbb{M}_4(\mathbb{R})$$

These observations (with generalizations) lead to the next theorem.

Theorem 3.5.3 *A square matrix A is similar to A^* if and only if A is similar to a real matrix. Consequently, $\overline{A}A$ is similar to a real matrix.*

Note: A theorem in Chapter 8 (Theorem 8.2.2, Sect. 8.2) states that A is similar to A^* if and only if A is a product of two Hermitian matrices.

Example 3.5.3 Similarity and eigenvalues.

(a) If all eigenvalues of $A \in \mathbb{M}_n$ are real, then A is similar to A^* because A and A^* have the same Jordan blocks J's (and J is similar to J^T).

(b) Consider

$$A = \begin{pmatrix} i & 1 & 0 & 0 \\ 0 & i & 0 & 0 \\ 0 & 0 & -i & 0 \\ 0 & 0 & 0 & -i \end{pmatrix}, \quad A^* = \begin{pmatrix} -i & 0 & 0 & 0 \\ 1 & -i & 0 & 0 \\ 0 & 0 & i & 0 \\ 0 & 0 & 0 & i \end{pmatrix}.$$

A and A^* have the same eigenvalues in conjugate pairs (thus the same characteristic and minimal polynomials), but they have different Jordan blocks. A and A^* are not similar. In fact, it is easy to compute that $\operatorname{rank}(iI + A) = 2$, while $\operatorname{rank}(iI + A^*) = 3$. ∎

Problems

1. Let $A = \left(\begin{smallmatrix} 1 & i \\ i & -1 \end{smallmatrix}\right)$. Compute $A^2 = A^T A = AA^T$, $\overline{A}A = A^*A$, and AA^*.
2. Let $A \in \mathbb{M}_{m\times n}$. Prove or disprove (by counterexample) that
   (a) $\operatorname{rank}(A) = \operatorname{rank}(A^T A)$.
   (b) $\operatorname{rank}(A^*A) = \operatorname{rank}(AA^*)$.
   (c) $A^T A$ is similar to AA^T.
   (d) $\operatorname{rank}(A) = \operatorname{rank}(\overline{A}A)$ (when $m = n$).
3. Let $A \in \mathbb{M}_n$. If A is skew-Hermitian (i.e., $A^T = -A$), show that A is similar to $-A$ and that $AX + XA = 0$ for some nonsingular $X \in \mathbb{M}_n$.
4. Is it possible that $\overline{A}A = 0$ or $A^*A = 0$ for some nonzero $A \in \mathbb{M}_n$?
5. Let $A \in \mathbb{M}_n$. Show that $\operatorname{Row}(A) = \operatorname{Row}(|A|) = \operatorname{Row}(A^*A)$ for the row space. Is it true in general that $\operatorname{Col}(A) = \operatorname{Col}(|A|)$ for the column space?

6. Let $A = \left(\begin{smallmatrix} 1 & 1 \\ 1 & 1 \end{smallmatrix}\right)$ and $B = \left(\begin{smallmatrix} 1 & 1 \\ -1 & -1 \end{smallmatrix}\right)$. (a) Find the eigenvalues of A and B. What are the Jordan forms of A and B? Are A and B similar? (b) Compute AB and BA. Find rank(AB) and rank(BA). What are the Jordan forms of AB and BA? Are AB and BA similar?

7. Find the Jordan canonical forms for J_1^2, J_2^2, J_3^2, and J_3^{-1}, where

$$J_1 = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}, \; J_2 = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix}, \; J_3 = \begin{pmatrix} 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \end{pmatrix}.$$

8. Let J be a Jordan block with diagonal entries equal to $a \in \mathbb{C}$. Find the Jordan canonical forms of $2J$, J^2, J^T, J^*, $\overline{J}$, and J^{-1} (if $a \neq 0$).

9. Let J be the $n \times n$ Jordan block with 1's on the diagonal. Let $f(x)$ be a polynomial such that $\frac{df}{dx}(1) \neq 0$. Show that the Jordan canonical form of $f(J)$ is the $n \times n$ Jordan block with $f(1)$'s on the diagonal.

10. Let $J_1, \ldots, J_k$ be non-real Jordan blocks. Let $S = \{J_1, \ldots, J_k\}$ and $T = \{\overline{J_1}, \ldots, \overline{J_k}\}$ (as multisets). If $S = T$, i.e., $\overline{J_1}, \ldots, \overline{J_k}$ are rearrangements (or re-list) of $J_1, \ldots, J_k$, show that if a J_i appears j times in the list, then its conjugate $\overline{J_i}$ appears exactly j times. Interpret this for the case when the Jordan blocks are all 1×1 (namely, scalars).

11. If the non-real eigenvalues of a square matrix A occur in conjugate pairs, must A be similar to A^*? Equivalently, if the characteristic polynomial of A has only real coefficients, must A and A^* be similar?

12. Show that the characteristic polynomial of $A\overline{A}$ has only real coefficients and the non-real eigenvalues of $A\overline{A}$ occur in conjugate pairs.

13. Let $A = B + iC \in \mathbb{M}_n$ be nonsingular, where B and C are real square matrices. Show that $A^{-1} = \overline{A}$ if and only if $BC = CB$ and $B^2 + C^2 = I$. Find the conditions on B and C if $A^T = \overline{A} = A^{-1}$.

14. If $A \in \mathbb{M}_n$ is similar to A^*, must the eigenvalues of A be real?

15. Let $A \in \mathbb{M}_n$. Show that the following statements are equivalent.
    (a) A is similar to A^*.
    (b) A is similar to $\overline{A}$.
    (c) The elementary divisors of A occur in conjugate pairs.
    (d) The invariant factors of A all have real coefficients.

16. Show that $\left(\begin{smallmatrix} 1 & 0 \\ -i & 1 \end{smallmatrix}\right)$ and $\begin{pmatrix} 1 & i & 0 \\ 0 & 1 & 0 \\ 0 & 1+i & 1 \end{pmatrix}$ are similar (over $\mathbb{C}$) to real matrices.

17. If $p(\lambda) = \lambda(\lambda-2)^2(\lambda^2-2\lambda+2)$ is the characteristic polynomial of matrix $A \in \mathbb{M}_5$, what are the possible Jordan canonical forms of A?

18. Let $A \in \mathbb{M}_n$. When (in terms of Jordan blocks) is A similar to $-A$?

19. Let A be a nonsingular square matrix. When is A similar to A^{-1}?

20. Show that if $A \in \mathbb{M}_n$ is orthogonal ($A^TA = I$), then A is similar to A^{-1}.

21. Show that $A = \left(\begin{smallmatrix} 0 & 1 & 0 \\ 0 & 0 & 2 \\ 0 & 0 & 0 \end{smallmatrix}\right)$ is not unitarily similar to its transpose A^T.

22. Let $A \in \mathbb{M}_n$, $B \in \mathbb{M}_{m\times n}$, and let $M = \left(\begin{smallmatrix} A & 0 \\ B & 0 \end{smallmatrix}\right) \in \mathbb{M}_{m+n}$. Show that M and A have the same nonsingular Jordan blocks, i.e., the nonsingular Jordan blocks of M (if any) are exactly those of A.

23. Let A, B, C, and D be n-square matrices. If the block matrices

$$\begin{pmatrix} A & B \\ 0 & 0 \end{pmatrix} \quad \text{and} \quad \begin{pmatrix} C & D \\ 0 & 0 \end{pmatrix}$$

are similar, show that A and C have the nonsingular Jordan blocks. Find an example showing that A and C are not similar in general.

24. Let $A \in \mathbb{M}_{m\times n}$ and $B \in \mathbb{M}_{n\times m}$. Show that AB and BA have the same nonsingular Jordan blocks. As a result, AB and BA have the same nonzero eigenvalues, counted according to multiplicities.

25. Let A and B be n-square matrices. Show that A and B are similar (over $\mathbb{C}$) if and only if for all $\lambda \in \mathbb{C}$, including $\lambda = 0$, and for $k = 1, 2, \ldots,$

$$\operatorname{rank}(\lambda I_n - A)^k = \operatorname{rank}(\lambda I_n - B)^k.$$

Show by example that the condition "$\lambda = 0$" is needed for the similarity of A and B, that is, A and B need not be similar if the rank identity in the display holds only for all nonzero $\lambda \in \mathbb{C}$ and for $k = 1, 2, \ldots$.

26. If A and $B \in \mathbb{M}_n$ have no common eigenvalues, show that the following three block matrices are similar for any $X, Y \in \mathbb{M}_n$:

$$\begin{pmatrix} A & X \\ 0 & B \end{pmatrix}, \quad \begin{pmatrix} A & 0 \\ 0 & B \end{pmatrix}, \quad \begin{pmatrix} A & 0 \\ Y & B \end{pmatrix}.$$

27. Let A, B, and C be matrices of appropriate sizes. Show that

$$AX - YB = C$$

for some matrices X and Y if and only if the rank identity holds:

$$\operatorname{rank}\begin{pmatrix} A & C \\ 0 & B \end{pmatrix} = \operatorname{rank}\begin{pmatrix} A & 0 \\ 0 & B \end{pmatrix}.$$

28. Let A and B be $n \times n$ matrices and let $M = \begin{pmatrix} A & B \\ -\overline{B} & \overline{A} \end{pmatrix}$. Show that

    (a) The characteristic polynomial of M has only real coefficients.

    (b) The eigenvalues of M occur in conjugate pairs with eigenvectors in the forms $\begin{pmatrix} x \\ y \end{pmatrix} \in \mathbb{C}^{2n}$ and $\begin{pmatrix} -\bar{y} \\ \bar{x} \end{pmatrix} \in \mathbb{C}^{2n}$, where $x, y \in \mathbb{C}^n$.

    (c) The eigenvectors in (b) are linearly independent.

    (d) $\det M \geq 0$. In particular, $\det(I + A\overline{A}) \geq 0$ for any $A \in \mathbb{M}_n$.

29. Let $c_0, c_1, \ldots, c_{n-1} \in \mathbb{C}$ and let C and D be (the companion matrices)

$$C = \begin{pmatrix} 0 & \ldots & 0 & -c_0 \\ 1 & \ldots & 0 & -c_1 \\ \vdots & \ddots & \vdots & \vdots \\ 0 & \ldots & 1 & -c_{n-1} \end{pmatrix}, \quad D = \begin{pmatrix} -c_{n-1} & \ldots & -c_1 & -c_0 \\ 1 & \ldots & 0 & 0 \\ \vdots & \ddots & \vdots & \vdots \\ 0 & \ldots & 1 & 0 \end{pmatrix}.$$

    Show that C, C^T, D, and D^T are all similar to each other. (Recall from Problem 24 of Sect. 3.3 that $SCS = D^T$, where S is the backward identity.) Show in particular by verification that $CT = TC^T$, where

$$T = \begin{pmatrix} c_1 & c_2 & \ldots & c_{n-1} & 1 \\ c_2 & c_3 & \ldots & 1 & 0 \\ \vdots & \vdots & & \vdots & \vdots \\ c_{n-1} & 1 & \ldots & 0 & 0 \\ 1 & 0 & \ldots & 0 & 0 \end{pmatrix}.$$

30. Let A, C, and D be the 4×4 matrices given below and let $B = \overline{A}$.

$$A = \begin{pmatrix} i & 1 & 0 & 0 \\ 0 & i & 0 & 0 \\ 0 & 0 & -i & 0 \\ 0 & 0 & 0 & -i \end{pmatrix}, \; C = \begin{pmatrix} i & 0 & 0 & 0 \\ 0 & i & 1 & 0 \\ 0 & 0 & -i & 0 \\ 0 & 0 & 0 & -i \end{pmatrix}, \; D = \begin{pmatrix} i & 1 & 0 & 0 \\ 0 & i & 0 & 0 \\ 0 & 0 & -i & 1 \\ 0 & 0 & 0 & -i \end{pmatrix}.$$

    (a) Find the Smith forms of $\lambda I - A$, $\lambda I - B$, $\lambda I - C$, and $\lambda I - D$.

    (b) Do A, B, C, and D have the same characteristic polynomial?

    (c) Do A, B, C, and D have the same minimal polynomial?

    (d) What are the invariant factors of A, B, C, and D?

    (e) What are the elementary divisors of A, B, C, and D?

    (f) What are the Jordan blocks of A, B, C, and D?

    (g) Are A, B, C, and D similar to each other?

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CHAPTER 4

Numerical Ranges, Norms, and Special Products of Matrices

Introduction: This chapter introduces a few basic concepts in matrix studies: numerical range, numerical radius, and matrix norm. We also present two important special matrix operations: the Kronecker product and the Hadamard product. We end the chapter with compound matrices.

4.1 Numerical Range and Numerical Radius

Let $x = (x_1, \ldots, x_n)^T \in \mathbb{C}^n$. The Euclidean (or ℓ_2-) norm of x (Sect. 1.4, Chap. 1) is $\|x\| = \|x\|_2 = (\sum_{i=1}^n |x_i|^2)^{1/2}$. Let A be an $n \times n$ matrix. The *numerical range*, also known as the *field of values*, of A is defined to be

$$W(A) = \{x^*Ax : x \in \mathbb{C}^n, \|x\| = 1\} = \left\{ \frac{x^*Ax}{x^*x} : x \in \mathbb{C}^n, x \neq 0 \right\} \subset \mathbb{C}.$$

We show that $W(A)$ is a bounded convex set in the complex plane.

Example 4.1.1 Numerical range.

(a) For the zero matrix, $W(0) = \{0\}$; for the identity matrix, $W(I) = \{1\}$.

(b) For $A = \left(\begin{smallmatrix} 1 & 0 \\ 0 & 0 \end{smallmatrix}\right)$, $W(A) = \{|x_1|^2 : |x_1| \leq 1\}$ is the closed interval $[0, 1]$.

(c) For $B = \left(\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right)$, $W(B) = \{\overline{x_1}x_2 : |x_1|^2 + |x_2|^2 = 1\}$ is the disc $|z| \leq \frac{1}{2}$. In fact, if $x = \left(\begin{smallmatrix} x_1 \\ x_2 \end{smallmatrix}\right) \in \mathbb{C}^2$ is a unit vector, then we have $|x^*Bx| = |\overline{x_1}x_2| \leq \frac{1}{2}(|x_1|^2 + |x_2|^2) = \frac{1}{2}$. Thus, x^*Bx lies in the disc $|z| \leq \frac{1}{2}$. Conversely, for any point $z \in \mathbb{C}$ in the disc $|z| \leq \frac{1}{2}$, write $z = re^{i\theta}$, $0 \leq r \leq \frac{1}{2}$ and $\theta \in \mathbb{R}$. We show that $z = v^*Bv$ for some unit $v \in \mathbb{C}^2$. Take $x_1 = ae^{i\theta}$ and $x_2 = be^{2i\theta}$, where $a \geq 0$ and $b > 0$ are such that $ab = r$ and $a^2 + b^2 = 1$. (It is easy to find $a = r/b$ and $b = \sqrt{(1 + \sqrt{1 - 4r^2})/2}$.) Setting $v = \left(\begin{smallmatrix} x_1 \\ x_2 \end{smallmatrix}\right) \in \mathbb{C}^2$, we obtain $z = v^*Bv$.

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(d) For $C = \text{diag}(-1 - 0.5i, 1 + i, 2 - i)$, $W(C)$ is the triangle (including the interior) with vertices at $(-1, -0.5)$, $(1, 1)$, and $(2, -1)$ because

$$x^*Cx = (-1 - 0.5i)\,|x_1|^2 + (1 + i)\,|x_2|^2 + (2 - i)\,|x_3|^2. \blacksquare$$

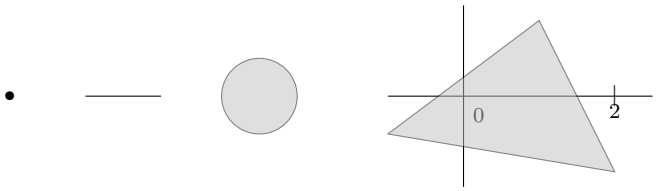


Fig. 4.1. Example 4.1.1

One of the fundamental results about numerical range is the classical and celebrated Toeplitz-Hausdorff convexity theorem. A convex set is a set in which the line between any two points in the set lies entirely in the set.

Theorem 4.1.1 (Toeplitz-Hausdorff Convexity Theorem) *The numerical range of a square matrix is a convex compact subset of the complex plane.*

Proof. If $W(A)$ is a singleton, there is nothing to show. Suppose that $W(A)$ has more than one point. We prove that the line segment joining any two distinct points in $W(A)$ lies completely in $W(A)$, that is, we show that if $u, v \in W(A)$, then $tu + (1 - t)v \in W(A)$ for all $t \in [0, 1]$.

For any complex numbers α and β, it is easy to verify that

$$W(\alpha I + \beta A) = \{\alpha + \beta z : z \in W(A)\} = \alpha + \beta W(A).$$

Intuitively, the convexity of $W(A)$ does not change under shifting, scaling, and rotation. Thus, we may assume that the two points to be considered are 0 and 1 on the x-axis, and show that $[0, 1] \subseteq W(A)$. Write

$$A = H + iK, \text{ where } H = \tfrac{1}{2}(A + A^*),\ K = \tfrac{1}{2i}(A - A^*).$$

Note that H and K are Hermitian. Let $x, y \in \mathbb{C}^n$ be unit vectors such that

$$x^*Ax = 0, \quad y^*Ay = 1.$$

One can easily check that x and y are linearly independent and that

$$x^*Hx = x^*Kx = y^*Ky = 0, \quad y^*Hy = 1.$$

We may further assume that the real part of x^*Ky is 0, otherwise, we replace x by cx, $c \in \mathbb{C}$, and $|c| = 1$, such that the real part of cx^*Ky is 0.

Note that $tx + (1-t)y \neq 0$, $t \in [0,1]$. Define for $t \in [0,1]$

$$z(t) = \frac{1}{\|tx + (1-t)y\|}\big(tx + (1-t)y\big).$$

Then $z(t) \in \mathbb{C}^n$ is a unit vector. Computations show that for all $t \in [0,1]$

$$z(t)^* K z(t) = 0, \quad z(t)^* A z(t) = z(t)^* H z(t) \in \mathbb{R}.$$

Let $f(t) = z(t)^* H z(t)$, $t \in [0,1]$. Because H is Hermitian, $f(t)$ is a real-valued continuous function in t with $f(0) = 1$ and $f(1) = 0$. Thus,

$$[0,1] \subseteq \{f(t) : 0 \leq t \leq 1\} = \{z(t)^* A z(t) : 0 \leq t \leq 1\} \subseteq W(A).$$

It is not difficult to see that the numerical range $W(A)$ of a square matrix A is bounded. More can be said about $W(A)$: it contains all boundary points (thus, $W(A)$ is compact), because $x \mapsto x^* A x$ is a continuous function (from $\mathbb{C}^n$ to $\mathbb{C}$) acting on the compact set $\{x \in \mathbb{C}^n : \|x\| = 1\}$ and because a continuous function maps a compact set to a compact set. ∎

We point out that for the operators on an infinite-dimensional (Hilbert) space, the numerical range is convex but need not be compact.

If λ is an eigenvalue of matrix A, then $\lambda \in W(A)$. Thus all eigenvalues (spectrum) of a matrix lie in the numerical range of the matrix (Problem 8).

Example 4.1.2 The numerical range of a normal matrix is the convex hull (the smallest convex set) of the eigenvalues of the matrix.

Let $A \in \mathbb{M}_n$ be a normal matrix with eigenvalues $\lambda_1, \ldots, \lambda_n$. The spectral decomposition theorem ensures that $A = U^* D U$ for some unitary matrix U, where $D = \operatorname{diag}(\lambda_1, \ldots, \lambda_n)$. Note that if $x \in \mathbb{C}^n$ is a unit vector, so is $y = Ux$. Let $y = (y_1, \ldots, y_n) \in \mathbb{C}^n$. Then $|y_1|^2 + \cdots + |y_n|^2 = 1$. Therefore,

$$W(A) = \{x^* A x : \|x\| = 1\} = \{y^* D y : \|y\| = 1\} = \{|y_1|^2 \lambda_1 + \cdots + |y_n|^2 \lambda_n\}$$

is the convex hull of the eigenvalues $\lambda_1, \ldots, \lambda_n$ of A. ∎

Note that the converse statement in the example is false (Problem 39).

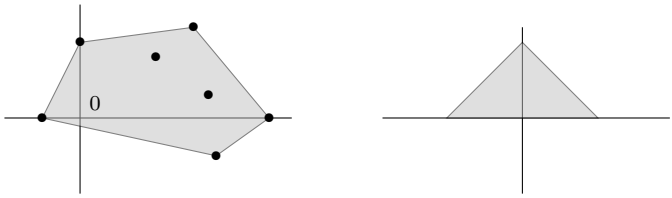


Fig. 4.2. Example 4.1.2 and Example 4.1.3

Example 4.1.3 $A = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ -1 & 0 & 0 \end{pmatrix}$ is a normal matrix. The eigenvalues of A are $1, \pm i$. Thus, the numerical range of A is the closed triangle with vertices at $(1,0)$, $(0,1)$, and $(0,-1)$ on the complex plane. ∎

The convexity of a numeral range or some related problems is often reduced to that of a 2×2 matrix. Below is an explicit description about the numerical range of such a matrix. The ideal of the proof is due to Li.

Theorem 4.1.2 (Elliptical Range Theorem) *Let A be a 2×2 matrix with eigenvalues λ_1 and λ_2. Let $\delta = \sqrt{\mathrm{tr}(A^*A) - |\lambda_1|^2 - |\lambda_2|^2}$. The numerical range $W(A)$ is the elliptical disc with λ_1 and λ_2 as foci and δ as minor axis.*

Proof. Since $W(U^*AU) = W(A)$ for any unitary matrix U and by Schur form, we may assume that A is in an upper-triangular form $A = \begin{pmatrix} \lambda_1 & a \\ 0 & \lambda_2 \end{pmatrix}$.

Case (i): $a = 0$. A is a normal matrix with eigenvalues λ_1 and λ_2. $W(A)$ is the line segment in the complex plane joining λ_1 and λ_2. Since A is normal, $\mathrm{tr}(A^*A) - |\lambda_1|^2 - |\lambda_2|^2 = 0$. The line segment can be viewed as a degenerate ellipse with λ_1 and λ_2 as foci and $\delta = 0$ as (the length of) the minor axis. This includes the case where $W(A)$ is a singleton when A is a scalar matrix.

Case (ii): $a \neq 0$, $\lambda_1 = \lambda_2 = 0$, and $A = \begin{pmatrix} 0 & a \\ 0 & 0 \end{pmatrix}$. By computation, we have

$$W(A) = \{ax_1x_2 : |x_1|^2 + |x_2|^2 = 1,\ x_1, x_2 \in \mathbb{C}\}.$$

Note that $|ax_1x_2| \leq \frac{1}{2}|a|(|x_1|^2 + |x_2|^2) = \frac{1}{2}|a|$. $W(A)$ is a circular disc centered at the origin with radius $\frac{1}{2}|a|$, which is a special elliptical disc.

Case (iii): $a \neq 0$, $A = \begin{pmatrix} \lambda_1 & a \\ 0 & \lambda_2 \end{pmatrix}$ with $\lambda_1 \neq 0$ or $\lambda_2 \neq 0$.

Because $W(\alpha I + A) = \alpha + W(A)$ for any $\alpha \in \mathbb{C}$ (shifting property; Problem 23), we can assume that $\mathrm{tr}A = 0$ (by taking $\alpha = -\frac{1}{2}\,\mathrm{tr}A$) and write $A = \begin{pmatrix} \lambda_1 & a \\ 0 & \lambda_2 \end{pmatrix}$, where $\lambda_1 = -\lambda_2 \neq 0$ and $a \neq 0$. By the scaling property $W(\beta A) = \beta W(A)$, we see that $W(A)$ is a rotation and a dilation (enlarging or shrinking) of $W(B)$, where $B = \beta A$. We further assume that $A = \begin{pmatrix} 1 & a \\ 0 & -1 \end{pmatrix}$ with $a \neq 0$. If $e^{i\theta}a > 0$, pre-multiplying the first row of A by $e^{i\theta}$ and post-multiplying the first column of A by $e^{-i\theta}$, we can assume that $a > 0$.

The proof reduces to the case of $A = \begin{pmatrix} 1 & 2c \\ 0 & -1 \end{pmatrix}$, where $c > 0$. We show that $W(A)$ is an elliptical disc with foci $(\pm 1, 0)$ and minor axis $\delta = 2c$.

Let $C = \frac{1}{2}\big((A + A^*) + \gamma(A - A^*)\big) = \begin{pmatrix} 1 & c(1+\gamma) \\ c(1-\gamma) & -1 \end{pmatrix}$, where $\gamma \in \mathbb{R}$. For $\gamma = \frac{1}{c}\sqrt{1 + c^2}$, C has only zero eigenvalues. Thus, C is unitarily similar to a matrix in the form $\begin{pmatrix} 0 & b \\ 0 & 0 \end{pmatrix}$, $b > 0$. By computing $\mathrm{tr}(C^*C)$, we obtain $b = 2\sqrt{1 + c^2}$. Thus, without loss of generality, we assume that $C = \begin{pmatrix} 0 & 2\sqrt{1+c^2} \\ 0 & 0 \end{pmatrix}$.

For any $v \in \mathbb{C}^2$, we express v^*Av and v^*Cv in $x + yi$ form with $x, y \in \mathbb{R}$:

$$v^*Av = \tfrac{1}{2}v^*(A + A^*)v + \tfrac{1}{2}v^*(A - A^*)v;$$
$$v^*Cv = \tfrac{1}{2}v^*(A + A^*)v + \tfrac{1}{2}\gamma v^*(A - A^*)v.$$

This means that $x + yi \in W(A)$ if and only if $x + \gamma yi \in W(C)$. (Geometrically, $W(A)$ is a dilation of $W(C)$ vertically by $\frac{1}{\gamma}$ on the complex plane.)

As discussed in Case (ii), $W(C)$ is the circular disc with boundary (curve) $(\sqrt{1+c^2}\cos\theta, \sqrt{1+c^2}\sin\theta)$. So, $W(A)$ is the elliptical disc with boundary $(\sqrt{1+c^2}\cos\theta, \sqrt{1+c^2}\frac{1}{\gamma}\sin\theta) = (\sqrt{1+c^2}\cos\theta, c\sin\theta)$, which is an ellipse with foci $(\pm 1, 0)$, vertices $(\pm\sqrt{1+c^2}, 0)$, $(0, \pm c)$, and minor axis $\delta = 2c$. ∎

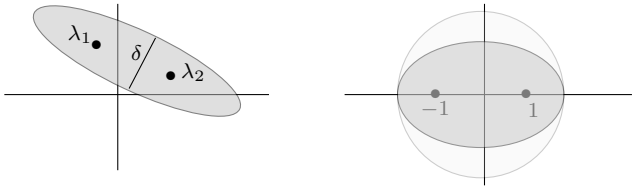


Fig. 4.3. Elliptical disc; Example 4.1.4

Example 4.1.4 For $D = \left(\begin{smallmatrix} 1 & 2 \\ 0 & -1 \end{smallmatrix}\right)$, by the elliptical range theorem, $W(D)$ is the closed ellipse (with interior) centered at the origin, having foci $(-1, 0)$, $(1, 0)$, vertices $(-\sqrt{2}, 0)$, $(\sqrt{2}, 0)$, $(0, 1)$, $(0, -1)$, minor axis $\delta = 2$, and major axis $2\sqrt{2}$. Below we demonstrate a proof through a direct computation.

Let $u = (a, b)^T \in \mathbb{C}^2$, $a = e^{i\alpha}\cos\theta$, $b = e^{i\beta}\sin\theta$, $\alpha, \beta, \theta \in \mathbb{R}$. Then

$$\begin{aligned} u^*Du &= |a|^2 - |b|^2 + 2\bar{a}b \\ &= \cos^2\theta - \sin^2\theta + 2e^{i(\beta-\alpha)}\sin\theta\cos\theta \\ &= \cos\phi + e^{i\gamma}\sin\phi, \quad \phi = 2\theta,\ \gamma = \beta - \alpha \\ &= x + yi, \quad x = \cos\phi + \sin\phi\cos\gamma,\ y = \sin\phi\sin\gamma. \end{aligned} \tag{4.1}$$

It follows that

$$(x - \cos\phi)^2 + y^2 = \sin^2\phi = 1 - \cos^2\phi.$$

This implies

$$2\cos^2\phi - 2x\cos\phi + x^2 + y^2 - 1 = 0, \tag{4.2}$$

revealing (via discriminant)

$$4x^2 - 8(x^2 + y^2 - 1) \geq 0.$$

Thus,

$$\frac{x^2}{2} + y^2 \leq 1. \tag{4.3}$$

This says that u^*Du lies in the elliptic disc $\frac{x^2}{2} + y^2 \leq 1$ in the plane.

It remains to be shown that every point in the region can be expressed as u^*Du for some unit vector u. Let (x, y), or $x + yi \in \mathbb{C}$, be such a point, i.e., $x^2 + 2y^2 \le 2$. As an example, we show the simple case where (x, y) is a point on the x-axis, i.e., $y = 0$ and $0 \le x \le \sqrt{2}$.

The quadratic equation $2t^2 - 2xt + x^2 - 1 = 0$ has two real solutions

$$t = \frac{1}{2}\big(x \pm \sqrt{2 - x^2}\,\big). \tag{4.4}$$

Note that $t \in [-1, 1]$. Set $\cos\phi = \frac{1}{2}\big(x + \sqrt{2 - x^2}\big)$. Then $\sin\phi = \frac{1}{2}\big(x - \sqrt{2 - x^2}\big)$, and $\cos\phi + \sin\phi = x$. Let $\theta = \frac{1}{2}\phi$ and $u = (\cos\theta, \sin\theta)^T$. Then $u^*Du = \cos^2\theta + 2\sin\theta\cos\theta - \sin^2\theta = \cos 2\theta + \sin 2\theta = x$. ∎

When considering the smallest disc centered at the origin that covers the numerical range on the complex plane, we associate $W(A)$ to a number

$$w(A) = \max\{|z| : z \in W(A)\} = \max_{\|x\|=1} |x^*Ax| = \max_{x \ne 0} \frac{|x^*Ax|}{x^*x}$$

and call it the *numerical radius* of $A \in \mathbb{M}_n$. (Note that the maximum "max" is attained at some $z \in W(A)$.) It is immediate that for any $x \in \mathbb{C}^n$

$$|x^*Ax| \le w(A)\|x\|^2. \tag{4.5}$$

For $A \in \mathbb{M}_n$, A has n eigenvalues (counted with multiplicities) and n singular values, r of which are positive, $r = \operatorname{rank}(A)$. We now make comparisons of the numerical radius $w(A)$ to the largest eigenvalue $\rho(A)$ in absolute value, which is referred to as the *spectral radius*, that is,

$$\rho(A) = \max\{|\lambda(A)| : \lambda(A) \text{ is an eigenvalue of } A\},$$

and to the largest singular value $\sigma_{\max}(A)$, i.e., the nonnegative square root of the largest eigenvalue of A^*A, or $\sigma_{\max}(A) = \sqrt{\lambda_{\max}(A^*A)}$. It is equal to the *spectral norm* $\|A\|_{\mathrm{sp}}$ of A (see the next section for matrix norms).

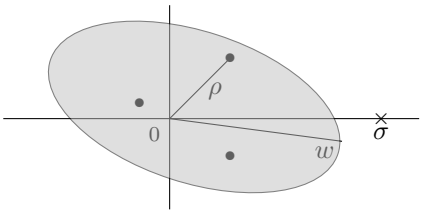


Fig. 4.4. Convexity, numerical radius, spectral radius, and spectral norm

Theorem 4.1.3 (Spectral Norm Max-expression) *Let $A \in \mathbb{M}_{m\times n}$. Then*

$$\sigma_{\max}(A) = \max_{\|x\|=1,\, x\in\mathbb{C}^n} \|Ax\| = \max_{x\ne 0,\, x\in\mathbb{C}^n} \frac{\|Ax\|}{\|x\|} = \max_{\substack{\|x\|=1,\, x\in\mathbb{C}^n \\ \|y\|=1,\, y\in\mathbb{C}^m}} |\langle Ax, y\rangle|. \tag{4.6}$$

Proof. Let the singular values of A be $\sigma_1 \geq \cdots \geq \sigma_n$, $\sigma_1 = \sigma_{\max}(A)$, and let $A^*A = U^* \operatorname{diag}(\sigma_1^2, \ldots, \sigma_n^2)U$ be a spectral decomposition of A^*A, where U is unitary. Set $y = (y_1, \ldots, y_n)^T = Ux$. Since $\|x\| = 1$, we see $\|y\| = 1$. Then

$$\|Ax\|^2 = x^*A^*Ax = x^*U^* \operatorname{diag}(\sigma_1^2, \ldots, \sigma_n^2)Ux = |y_1|^2\sigma_1^2 + \cdots + |y_n|^2\sigma_n^2.$$

It follows that $\|Ax\|^2 \leq \sigma_1^2$, and $\|Ax\| \leq \sigma_1$ for any unit vector $x \in \mathbb{C}^n$.

Conversely, let z be a unit eigenvector of A^*A associated with σ_1^2, i.e., $A^*Az = \sigma_1^2 z$. Then $\|Az\|^2 = z^*A^*Az = \sigma_1^2$. Thus, $\sigma_1 \leq \max\limits_{\|x\|=1} \|Ax\|$. Hence,

$$\sigma_{\max}(A) = \max_{\|x\|=1} \|Ax\|. \tag{4.7}$$

We now show that $\sigma_{\max}(A) = \max_{\|x\|=1,\, \|y\|=1} |\langle Ax, y\rangle|$. On the one hand, by the Cauchy-Schwarz inequality, $|\langle Ax, y\rangle| \leq \|Ax\|\,\|y\| = \|Ax\|$. We have $|\langle Ax, y\rangle| \leq \sigma_1$. On the other hand, by singular value decomposition, let VAW be the $m \times n$ matrix with $\sigma_1, \ldots, \sigma_r$ in the positions $(1,1), \ldots, (r,r)$, where r is the rank of A, and with 0's in other positions. Let v be the first row of (unitary) V and let w be the first column of (unitary) W. Then $vAw = \sigma_1$. This implies $\sigma_1(A) \leq \max_{\|x\|=1,\, \|y\|=1} |\langle Ax, y\rangle|$. Alternatively, take $y = \frac{Ax}{||Ax||}$, where x is a unit eigenvector of A^*A corresponding to the eigenvalue σ_1^2 of A^*A. Then y is unit, and $\langle Ax, y\rangle = \frac{1}{||Ax||}\langle Ax, Ax\rangle = ||Ax|| = \sigma_1$, implying $\max_{\|x\|=1,\, \|y\|=1} |\langle Ax, y\rangle| \geq \sigma_1$. Therefore, the last equality in (4.6) holds. ∎

Note: For the smallest singular value, $\sigma_{\min}(A) \neq \min\limits_{\|x\|=\|y\|=1} |\langle Ax, y\rangle|$.

It follows from the last identity in the theorem that if $A = (a_{ij})$, then

$$|a_{ij}| = |e_i^T A e_j| = |\langle Ae_j, e_i\rangle| \leq \sigma_{\max}(A) \text{ for all } i, j, \tag{4.8}$$

where e_k (in $\mathbb{R}^m$ or $\mathbb{R}^n$) has 1 in the kth position and 0's elsewhere.

Note: One may notice that $\|x\|$ is the ℓ_2-norm of x in $\mathbb{C}^n$, while $\|Ax\|$ is the ℓ_2-norm of $\|Ax\|$ in $\mathbb{C}^m$. It is immediate that for $A \in \mathbb{M}_{m\times n}$ and $x \in \mathbb{C}^n$,

$$\|Ax\| \leq \sigma_{\max}(A)\|x\|, \tag{4.9}$$

which gives rise to an important fact (Problem 32) that for $A, B \in \mathbb{M}_n$,

$$\sigma_{\max}(AB) \leq \sigma_{\max}(A)\sigma_{\max}(B). \tag{4.10}$$

Example 4.1.5 Let $A = \left(\begin{smallmatrix} 0 & 0 \\ 1 & 0 \end{smallmatrix}\right)$. Then $\rho(A) = 0 < w(A) = \frac{1}{2} < \sigma_{\max}(A) = 1$.

Let $B = \left(\begin{smallmatrix} 0 & 0 \\ 1 & 1 \end{smallmatrix}\right)$. Then B has eigenvalues 0 and 1. So $\rho(B) = 1$. It is easy to find $\sigma_{\max}(B) = \sqrt{2}$. $W(B)$ is the closed ellipse with foci at $(0,0)$, $(1,0)$ and major axis $\sqrt{2}$ (Problem 26). We have $w(B) = \frac{\sqrt{2}+1}{2} < \sigma_{\max}(B)$. ∎

Theorem 4.1.4 (Numerical Radius Theorem) *If A is a square matrix, then*

$$\rho(A) \leq w(A) \leq \sigma_{\max}(A) \leq 2w(A). \tag{4.11}$$

Proof. Let $\lambda = \lambda(A)$ be such an eigenvalue of A that $\rho(A) = |\lambda|$, and let u be a unit eigenvector corresponding to λ, i.e., $Au = \lambda u$, $\|u\| = 1$. Then

$$\rho(A) = |\lambda u^* u| = |u^* A u| \le w(A).$$

The second inequality in (4.11) follows from the Cauchy-Schwarz inequality:

$$|x^* A x| = |\langle Ax, x\rangle| \le \|Ax\|\|x\| = \|Ax\|, \quad \|x\| = 1.$$

We next show that $\sigma_{\max}(A) \le 2w(A)$. Recall Problem 51 of Sect. 1.4

$$\begin{aligned} 4\langle Ax, y\rangle &= \langle A(x+y), x+y\rangle - \langle A(x-y), x-y\rangle \\ &\quad + i\langle A(x+iy), x+iy\rangle - i\langle A(x-iy), x-iy\rangle. \end{aligned}$$

Using (4.5) and observing that $\|x+y\|^2 + \|x-y\|^2 = \|x\|^2 + \|y\|^2$, we derive

$$\begin{aligned} 4|\langle Ax, y\rangle| &\le w(A)\left(\|x+y\|^2 + \|x-y\|^2 + \|x+iy\|^2 + \|x-iy\|^2\right) \\ &= 4w(A)\left(\|x\|^2 + \|y\|^2\right) \le 8w(A) \text{ for unit } x, y. \end{aligned}$$

It follows (by the last expression (4.6) of $\sigma_{\max}$ in Theorem 4.1.3) that

$$\sigma_{\max}(A) = \max_{\|x\|=\|y\|=1} |\langle Ax, y\rangle| \le 2w(A). \blacksquare$$

Example 4.1.6 (a) The quantities $\rho(A), w(A)$, and $\sigma_{\max}(A)$ are nonnegative values attained in the complex setting. If restricted to the reals, then (4.11) need not be true. For example, if $A = \left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right)$, then $x^* A x = 0$ for all $x \in \mathbb{R}^2$.

(b) Let $A = \left(\begin{smallmatrix} 0 & \varepsilon \\ 0 & 0 \end{smallmatrix}\right)$, $\varepsilon > 0$. Obviously, $\rho(A) = 0$ and $\sigma_{\max}(A) = \varepsilon$.

To find $w(A)$, we compute $|x^* A x| = |\varepsilon \bar{a} b| \le \varepsilon \frac{|a|^2 + |b|^2}{2} \le \frac{\varepsilon}{2}$ for any unit $x = \left(\begin{smallmatrix} a \\ b \end{smallmatrix}\right) \in \mathbb{C}^2$. Since $|x^* A x| = \frac{\varepsilon}{2}$ when $x = \frac{1}{\sqrt{2}}\left(\begin{smallmatrix} 1 \\ 1 \end{smallmatrix}\right)$, we have $w(A) = \frac{\varepsilon}{2}$. $\blacksquare$

The next result (which is left to the reader as Problem 30) is a useful fact.

Theorem 4.1.5 *Let $A \in \mathbb{M}_n$ and let $\lambda(A)$ be any eigenvalue of A. Then*

$$\sigma_{\min}(A) \le |\lambda(A)| \le \sigma_{\max}(A). \tag{4.12}$$

Theorem 4.1.6 *Let A be a square matrix. Then $A^k \to 0$ (as $k \to \infty$) if and only if $\rho(A) < 1$ (that is, every eigenvalue of A has modulus less than 1).*

Proof. Let $A \in \mathbb{M}_n$ and let $A = P^{-1} T P$ be a Jordan decomposition of A, where P is an invertible matrix and T is a direct sum of Jordan blocks J's with the eigenvalues $\lambda_1, \ldots, \lambda_n$ of A on the diagonal. Then $A^k = P^{-1} T^k P$ and $\rho(A^k) = \rho(T^k) = (\rho(A))^k$. Thus, if A^k tends to zero (entrywise), so does T^k. It follows that $\lambda^k \to 0$ as $k \to 0$ for every eigenvalue λ of A. Therefore, $\rho(A) < 1$. Conversely, suppose $\rho(A) < 1$. We show that $A^k \to 0$ as $k \to \infty$. It suffices to show that $J^k \to 0$ as $k \to \infty$ for each Jordan block J.

Suppose that J is an $m \times m$ Jordan block:

$$J = \begin{pmatrix} \lambda & 1 & & 0 \\ & \lambda & \ddots & \\ & & \ddots & 1 \\ 0 & & & \lambda \end{pmatrix}.$$

Upon direct computations (or by Problem 50, Sect. 3.4), we have

$$J^k = \begin{pmatrix} \lambda^k & \binom{k}{1}\lambda^{k-1} & \cdots & \binom{k}{m-1}\lambda^{k-m+1} \\ & \lambda^k & \ddots & \vdots \\ & & \ddots & \binom{k}{1}\lambda^{k-1} \\ 0 & & & \lambda^k \end{pmatrix}.$$

Observe that every entry of J^k is either a zero or in the form $\binom{k}{l}\lambda^{k-l}$ for some l (with $\binom{k}{l} = 0$ if $k < l$). A fact from calculus ensures that for $|\lambda| < 1$

$$\lim_{k\to\infty} \binom{k}{l}\lambda^{k-l} = 0.$$

It follows that each J^k, thus A^k, is convergent to 0 as $k \to \infty$. ∎

Problems

1. Let A be a square matrix. If A is singular, show that $W(A)$ contains the origin 0. Is the converse true? If $0 \notin W(A)$, show that $0 \notin W(A^{-1})$.
2. What can be said about a matrix A if the spectral radius $\rho(A) = 0$? The numerical radius $w(A) = 0$? Or the spectral norm $\sigma_{\max}(A) = 0$?
3. Find the eigenvalues, singular values, numerical radius, spectral radius, spectral norm, and numerical range for each of the following matrices:
$$A = \begin{pmatrix} 0 & 2 \\ 0 & 0 \end{pmatrix}, \quad B = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \quad C = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \quad D = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}.$$
4. Let $A \in \mathbb{M}_n$. Show that $\rho(A) \geq \frac{|\operatorname{tr} A|}{n}$ and $\rho(A) \geq |\det A|^{\frac{1}{n}}$.
5. Let $A = (a_{ij}) \in \mathbb{M}_n$ and $\epsilon > 0$. If $\max_{i,j} |a_{ij}| \leq \epsilon$, show that $\rho(A) \leq \epsilon$.
6. Let $A \in \mathbb{M}_n$. Show that $W(U^*AU) = W(A)$ for any unitary matrix $U \in \mathbb{M}_n$ and that $W(V^*AV) \subseteq W(A)$ for any partially unitary matrix $V \in \mathbb{M}_{n\times m}$, that is, V is a matrix such that $V^*V = I_m$, $1 \leq m \leq n$.
7. Let $A \in \mathbb{M}_n$. Show that the numerical radius, spectral radius, spectral norm, and numerical range are unitarily invariant. That is, for instance, $w(U^*AU) = w(A)$ for any n-square unitary matrix U.

8. Show that the diagonal entries and the eigenvalues of a square matrix are all contained in the numerical range of the matrix.

9. If the convex hull (polygon) of the eigenvalues of a matrix $A \in \mathbb{M}_n$ contains zero (the origin), prove that $\sigma_{\max}(\lambda A + I_n) \geq 1$ for all $\lambda \in \mathbb{C}$.

10. Let $A \in \mathbb{M}_n$. If the numerical radius $\rho(A) < 1$, show that
$$(I_n - A)^{-1} = I + A + A^2 + \cdots .$$

11. Let $A \in \mathbb{M}_n$. Show that $\sum_{k=0}^{\infty} A^k$ is convergent if and only if $A^k \to 0$ (as $k \to \infty$) which is equivalent to $\rho(A) < 1$ (Theorem 4.1.6).

12. Let $A \in \mathbb{M}_n$. Show that $\frac{1}{n} \operatorname{tr} A$ is contained in $W(A)$. Conclude that for any nonsingular $P \in \mathbb{M}_n$, $W(P^{-1}AP - PAP^{-1})$ contains 0.

13. Let $A \in \mathbb{M}_n$. Show that $W(A^T) = W(A)$ and $W(A^*) = W(\overline{A})$. Denote $\overline{W(A)} = \{\bar{z} : z \in W(A)\}$. Then $W(A^*) = W(\overline{A}) = \overline{W(A)}$.

14. Let A be a real square matrix. Show that the numerical range $W(A)$ is a symmetric region in the xy-plane with respect to the real axis.

15. Let $A \in \mathbb{M}_n$. Show that $W(e^{i\theta}A)$ is a rotation of $W(A)$ by $\theta \in \mathbb{R}$.

16. If A is a Hermitian matrix having the largest eigenvalue $\lambda_{\max}$ and the smallest eigenvalue $\lambda_{\min}$, show that $W(A) = [\lambda_{\min}, \lambda_{\max}]$.

17. Let $A \in \mathbb{M}_n$. Show that $W(A)$ is a subset of the closed right half-plane $\{x + yi : x \geq 0, y \in \mathbb{R}\}$ if and only if $A^* + A$ is positive semidefinite.

18. Let A be an n-square matrix. Prove each of the following statements.
    (a) $W(A)$ is a singleton if and only if $A = cI_n$ for some $c \in \mathbb{C}$.
    (b) $W(A) \subset \mathbb{R}$ if and only if A is Hermitian, i.e., $A^* = A$.
    (c) $W(A) \subset [0, \infty)$ if and only if A is positive semidefinite.
    (d) If A is unitary, then $W(A)$ is a polygon (possibly degenerated) inscribed in the unit circle. Is the converse true?
    (e) If A is real and $x^T Ax \geq 0$ for all $x \in \mathbb{R}^n$, then $W(A)$ is contained in the closed right half-plane $\{x + yi : x \geq 0, y \in \mathbb{R}\}$.
    (f) If A is real, then $x^T(A^T - A)x = 0$ for all $x \in \mathbb{R}^n$.
    (g) If A is complex and $x^*Ax = 0$ for all $x \in \mathbb{C}^n$, then $A = 0$.
    (h) If A is complex and $x^T Ay = 0$ for all $x, y \in \mathbb{R}^n$, then $A = 0$.
    (i) If $A = \left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right)$, then $x^T Ax = 0$ and $x^T(I_2 + A)x \geq 0$ for all $x \in \mathbb{R}^2$.

19. Let $A = \left(\begin{smallmatrix} 1 & \alpha \\ \beta & -1 \end{smallmatrix}\right)$. For what α and β does A have only zero eigenvalues?

20. If B is a principal submatrix of $A \in \mathbb{M}_n$, show that $W(B) \subseteq W(A)$.

21. Let $A \in \mathbb{M}_m, B \in \mathbb{M}_n$, and let $C = \left(\begin{smallmatrix} A & 0 \\ 0 & B \end{smallmatrix}\right) = A \oplus B$. Show that $W(C)$ is the smallest closed convex set that contains both $W(A)$ and $W(B)$.

22. Let $A \in \mathbb{M}_n$. Denote $H = \frac{1}{2}(A + A^*)$ and $S = \frac{1}{2}(A - A^*)$. Show that
$$W(H) = \{\Re(z) : z \in W(A)\}, \quad W(S) = \{\Im(z)i : z \in W(A)\}.$$

23. Let $A \in \mathbb{M}_n$, $a, b \in \mathbb{C}$. Show that $W(aI_n + bA) = \{a + bz : z \in W(A)\}$, denoted by $a + bW(A)$. That is, $W(aI_n + bA) = a + bW(A)$.

24. Let $A, B \in \mathbb{M}_n$. Show that $W(A+B) \subseteq \{a+b : a \in W(A), b \in W(B)\}$, denoted by $W(A) + W(B)$. That is, $W(A + B) \subseteq W(A) + W(B)$.

25. Refer to Example 4.1.4 with $A = \left(\begin{smallmatrix} 1 & 2 \\ 0 & -1 \end{smallmatrix}\right)$. Show that A is not unitarily similar to a matrix in the form $\left(\begin{smallmatrix} 0 & z \\ 0 & 0 \end{smallmatrix}\right)$, $z \in \mathbb{C}$. Find $C = \frac{1}{2}\big((A + A^*) + \gamma(A - A^*)\big)$ and $W(C)$. Explain how to obtain $W(A)$ from $W(C)$.

26. Find and sketch the numerical ranges of $\left(\begin{smallmatrix} -\frac{1}{2} & 0 \\ 1 & \frac{1}{2} \end{smallmatrix}\right)$ and $\left(\begin{smallmatrix} 0 & 0 \\ 1 & 1 \end{smallmatrix}\right)$.

27. Let A be an n-square matrix. Show that $\frac{\|Ax\|}{\|x\|}$ is a (fixed) constant for all nonzero $x \in \mathbb{C}^n$ if and only if the singular values of A are all equal.

28. Let $\sigma_{\min}(A)$ denote the smallest singular value of matrix A. Show by an example of a 2×2 matrix A such that $\sigma_{\min}(A) \neq \min\limits_{\|x\|=\|y\|=1} |\langle Ax, y\rangle|$.

29. Let $\sigma_{\min}(A)$ denote the smallest singular value of matrix A. Show that
$$\sigma_{\min}(A) = \min_{\|x\|=1} \|Ax\|.$$

30. Let $A \in \mathbb{M}_n$ and let $\lambda(A)$ be any eigenvalue of A. Show that
$$\sigma_{\min}(A) \leq |\lambda(A)| \leq \sigma_{\max}(A).$$

31. Show that for any square matrices A and B of the same size,
$$w(A + B) \leq w(A) + w(B).$$
Show by example that $\rho(A + B) \leq \rho(A) + \rho(B)$ need not be true.

32. Show that for any square matrices A and B of the same size,
$$\sigma_{\max}(AB) \leq \sigma_{\max}(A)\sigma_{\max}(B);$$
$$\sigma_{\max}(A + B) \leq \sigma_{\max}(A) + \sigma_{\max}(B).$$

33. If all eigenvalues of a matrix X are real, we denote the largest eigenvalue of X by $\lambda_{\max}(X)$. Let A and B be $n \times n$ positive semidefinite matrices.
    (a) Show that the eigenvalues of AB are all nonnegative.
    (b) Show that $\lambda_{\max}(AB) \leq \lambda_{\max}(A)\lambda_{\max}(B)$.
    (c) Show that $\lambda_{\max}(A + B) \leq \lambda_{\max}(A) + \lambda_{\max}(B)$.
    (d) Show by example that $\lambda_{\max}(AB) = \sigma_{\max}(AB)$ may not be true.
    (e) If $\lambda_{\max}(A + B) = \lambda_{\max}(A)$, is it necessarily true that $B = 0$?

34. Let $A, B \in \mathbb{M}_n$. Show by example that $\|ABx\| \le \|Ax\|\|Bx\|$ is false for some x with $\|x\| = 1$. Explain why the following inequality holds:
$$\max_{\|x\|=1} \|ABx\| \le \big(\max_{\|x\|=1} \|Ax\|\big)\big(\max_{\|x\|=1} \|Bx\|\big).$$

35. Let $A \in \mathbb{M}_n$ and $|A| = (A^*A)^{1/2}$. Show that for unit vectors $x \in \mathbb{C}^n$,
    (a) $|x^*Ax|^2 \le x^*|A|^2x$.
    (b) $(x^*|A|x)^2 \le x^*|A|^2x$.
    (c) $(x^*A^*Ax)^2 \le x^*(A^*A)^2x$.
    (d) $|x^*A^2x| \le x|A|^2x$ need not be true (by an example).
    (e) $|x^*Ax| \le x^*|A|x$ need not be true (by an example).
    (f) $|x^*A^kx| \le \max_{\|x\|=1}(x^*|A|^kx),\ k \ge 0$.

    [Note: See similar inequalities of Jensen-type, Problem 44 in Sect. 10.3.]

36. Let A and B be Hermitian matrices of the same size. Show that
$$\rho\big((AB)^2\big) \le \rho(A^2B^2) \le \rho(A^2)\rho(B^2).$$

37. Let $A = \left(\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}\right)$. Show that $\sigma_1(A^2) < (\sigma_1(A))^2$ and $\sigma_2(A^2) > (\sigma_2(A))^2$.

38. Let $A = \left(\begin{smallmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{smallmatrix}\right)$ and $B = A^2$. Prove $w(A) < 1$. Find $w(B)$ and $w(AB)$.

39. Show that $w(A) = \rho(A) = \sigma_{\max}(A)$ if A is normal. Show that the converse is not true in general by considering the non-normal matrix
$$A = \operatorname{diag}(1, i, -1, -i) \oplus \left(\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right).$$

40. Let B be a submatrix of matrix A. Discuss if each of the following holds:
$$\text{(a) } \rho(B) \le \rho(A), \quad \text{(b) } w(B) \le w(A), \quad \text{(c) } \sigma_{\max}(B) \le \sigma_{\max}(A).$$

41. Show by example that the inequalities below do not hold for $A, B \in \mathbb{M}_n$:
    (a) $\rho(AB) \le \rho(A)\rho(B)$.
    (b) $w(AB) \le w(A)w(B)$.
    (c) $\sigma_{\max}\big((AB)^k\big) \le \sigma_{\max}(A^kB^k)$, where k is a positive integer.
    (d) $\sigma_{\max}(A^kB^k) \le \sigma_{\max}\big((AB)^k\big)$, where k is a positive integer.

42. Find an example of matrix A and positive integers k and m such that
$$w(A^{k+m}) \not\le w(A^k)\, w(A^m).$$

43. Let A be a square matrix. Show that for every positive integer k,
$$\rho(A^k) = \big(\rho(A)\big)^k, \quad \sigma_{\max}(A^k) \le \big(\sigma_{\max}(A)\big)^k, \quad w(A^k) \le \big(w(A)\big)^k.$$

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4.2 Matrix Norms

A matrix can be associated with numerical items in various ways. In addition to determinant, trace, eigenvalues, singular values, numerical radius, and spectral radius, matrix norm is another important one.

Recall from Sect. 1.4 that vectors in a normed space V can be measured by their lengths, i.e., norms. A norm $\|\cdot\| : V \to \mathbb{R}$ is a function that satisfies

(i) $\|v\| \geq 0$ for all $v \in V$ with equality if and only if $v = 0$,
(ii) $\|cv\| = |c|\|v\|$ for all scalars c and vectors v, and
(iii) $\|u + v\| \leq \|u\| + \|v\|$ for all vectors u, v.

If V is an inner product space, then $\|v\| = \sqrt{\langle v, v\rangle}$ is an induced norm.

Like the norms for vectors being introduced to measure the magnitudes of vectors, norms for matrices are used to characterize some properties of matrices. We call a matrix function $\|\cdot\| : \mathbb{M}_{m\times n} \to \mathbb{R}$ a *matrix norm* if for all $A, B \in \mathbb{M}_{m\times n}$ and $c \in \mathbb{C}$, the following conditions are satisfied:

(1) $\|A\| \geq 0$ with equality if and only if $A = 0$,
(2) $\|cA\| = |c|\|A\|$,
(3) $\|A + B\| \leq \|A\| + \|B\|$ (triangle inequality), and
(4) $\|AB\| \leq \|A\|\|B\|$ (submultiplicativity for $m = n$).

The norm $\|\cdot\|$ on matrices satisfying (1)–(3) is usually called a *matrix-vector norm*, or simply a *norm*, which is well-defined for rectangular matrices. (4) requires that A and B be square matrices of the same order. Such a norm (satisfying (1)–(4)) is referred to as a *submultiplicative matrix norm*. We call it a *matrix norm*. So, by a matrix norm, we mean it is submultiplicative.

Example 4.2.1 Consider the norms on $\mathbb{M}_n$. Let $A = (a_{ij}), B = (b_{ij}) \in \mathbb{M}_n$.

(a) ℓ_1*-norm* defined by $\|A\|_{\ell_1} = \sum_{i,j} |a_{ij}|$ is submultiplicative because

$$\|AB\|_{\ell_1} = \sum_{i,j} \Big| \sum_k a_{ik}b_{kj} \Big| \leq \sum_{i,j,k} |a_{ik}b_{kj}| \leq \sum_{i,j,k,l} |a_{ik}b_{lj}| = \|A\|_{\ell_1}\|B\|_{\ell_1}.$$

(b) ℓ_∞*-norm* defined by $\|A\|_{\ell_\infty} = \max_{i,j} |a_{ij}|$ is a matrix-vector norm; it is not submultiplicative. Take $A = B = \left(\begin{smallmatrix} 1 & 1 \\ 1 & 1 \end{smallmatrix}\right)$, for example. Then

$$2 = \|AB\|_{\ell_\infty} > \|A\|_{\ell_\infty}\|B\|_{\ell_\infty} = 1.$$

(c) $\|A\|_{\S} = n\|A\|_{\ell_\infty} = n \max_{i,j} |a_{ij}|$ is a submultiplicative norm on $\mathbb{M}_n$:

$$\begin{aligned} \|AB\|_{\S} &= n \max_{i,\,j} \Big| \sum_k a_{ik}b_{kj} \Big| \leq n\|A\|_{\ell_\infty} \max_j \sum_k |b_{kj}| \\ &\leq (n\|A\|_{\ell_\infty})\,(n\|B\|_{\ell_\infty}) = \|A\|_{\S}\|B\|_{\S}. \end{aligned}$$

∎

Example 4.2.2 From Problems 31, 33, 41 of Sect. 4.1, we have that

(a) ρ, the spectral radius, is not a vector norm on $\mathbb{M}_n$.
(b) w, the numerical radius, is a non-submultiplicative vector norm on $\mathbb{M}_n$.
(c) $\sigma_{\max}$, the spectral norm, is a submultiplicative matrix norm on $\mathbb{M}_n$. ∎

Theorem 4.2.1 (Norm Continuity) *Let $\|\cdot\|$ be a norm on $\mathbb{M}_{m\times n}$. Then*

$$|\,\|A\| - \|B\|\,| \le \|A - B\|, \quad A, B \in \mathbb{M}_{m\times n}.$$

That is, norm is a continuous function (from $\mathbb{M}_{m\times n}$ to $\mathbb{R}$ with usual metrics).

Proof. Observe that $A = (A - B) + B$. By the triangle inequality, we have $\|A\| \le \|A - B\| + \|B\| \Rightarrow \|A\| - \|B\| \le \|A - B\|$. Likewise, $\|B\| - \|A\| \le \|B - A\| = \|A - B\|$. It follows that $|\,\|A\| - \|B\|\,| \le \|A - B\|$.

For any given $\varepsilon > 0$, there exists a $\delta = \varepsilon > 0$ such that $\|A - B\| < \delta \Rightarrow |\,\|A\| - \|B\|\,| < \varepsilon$. That is, the norm $\|\cdot\|$ is a continuous function. ∎

See also Theorem 1.4.3 of Section 1.4 for vector norm continuity. Note that the numerical radius $w(A(t))$ and the spectral norm $\sigma_{\max}(A(t))$ are continuous functions of t if the entries of $A(t)$ are continuous functions of t.

Of many norms for matrices, two stand out: the *Frobenius norm* $\|A\|_F = \sqrt{\mathrm{tr}(A^*A)}$; it is the norm of A as a vector in the matrix space equipped with the Frobenius inner product (see Sect. 1.4, Chap. 1), and the *spectral norm* $\|A\|_{\mathrm{sp}} = \max_{\|x\|=1} \|Ax\|$ which equals the largest singular value of matrix A (Theorem 4.1.3); it is derived from the usual standard inner products of $\mathbb{C}^n$ and $\mathbb{C}^m$ when $A \in \mathbb{M}_{m\times n}$ is viewed as a linear map $(x \mapsto Ax)$ from $\mathbb{C}^n$ to $\mathbb{C}^m$. It is also referred to as the *operator norm* of the matrix. The Frobenius norm is sometimes called *Schur norm* or *Hilbert-Schmidt norm* or *ℓ_2-norm*.

To summarize, let $\sigma_k(A)$'s be the singular values of $A = (a_{ij})$. Then

$$\|A\|_F = \sqrt{\mathrm{tr}(A^*A)} = \sqrt{\sum_{i,j} |a_{ij}|} = \sqrt{\sum_k \sigma_k^2(A)} \tag{4.13}$$

and

$$\|A\|_{\mathrm{sp}} = \max_{x\neq 0} \frac{\|Ax\|}{\|x\|} = \max_{\|x\|=1} \|Ax\| = \sigma_{\max}(A) = \sigma_1(A). \tag{4.14}$$

Note: In the above (4.14), $\|x\|$ and $\|Ax\|$ are two different sorts of norms: the first one is for $\mathbb{C}^n$ while the second one is for $\mathbb{C}^m$. This is clear from the context. Unless stated otherwise, we assume that they are the standard *Euclidean norm* (i.e., ℓ_2-norm), that is, for $x = (x_1, \ldots, x_n)$ (or x^T) in $\mathbb{C}^n$,

$$\|x\| = \sqrt{\langle x, x\rangle} = \left(|x_1|^2 + \cdots + |x_n|^2\right)^{\frac{1}{2}}.$$

It is obvious that $\|A\|_F \geq \|A\|_{\text{sp}}$, where equality holds if and only if $A = 0$ or $\text{rank}(A) = 1$. $\|\cdot\|_F$ and $\|\cdot\|_{\text{sp}}$ depend only on singular values, and both are submultiplicative (see Problem 32 of Sect. 4.1 or see the next theorem).

We state and prove the subadditive and submultiplicative properties of these norms explicitly because they are basic and we will use them often.

Theorem 4.2.2 *Let A and B be matrices of appropriate sizes. Then*

$$\|A+B\|_{\text{sp}} \leq \|A\|_{\text{sp}} + \|B\|_{\text{sp}}; \quad \|A+B\|_F \leq \|A\|_F + \|B\|_F \tag{4.15}$$

and

$$\|AB\|_{\text{sp}} \leq \|A\|_{\text{sp}}\|B\|_{\text{sp}}; \quad \|AB\|_F \leq \|A\|_F\|B\|_F. \tag{4.16}$$

Proof. For $A+B$, let A and B be $m \times n$ matrices. For the spectral norm,

$$\begin{aligned} \|A+B\|_{\text{sp}} &= \max_{\|x\|=1,\, x\in\mathbb{C}^n} \|(A+B)x\| = \max_{\|x\|=1} \|Ax+Bx\| \\ &\leq \max_{\|x\|=1} \|Ax\| + \max_{\|x\|=1} \|Bx\| = \|A\|_{\text{sp}} + \|B\|_{\text{sp}}. \end{aligned}$$

For the Frobenius norm, $\|A+B\|_F \leq \|A\|_F + \|B\|_F$ is nothing but the triangle inequality for the induced norm by the Frobenius inner product (Sect. 1.4).

For AB, let A and B be $m \times n$ and $n \times p$ matrices, respectively. Then

$$\begin{aligned} \|AB\|_{\text{sp}} &= \max_{\|x\|=1,\, x\in\mathbb{C}^p} \|(AB)x\| = \max_{\|x\|=1} \|A(Bx)\| \\ &\leq \|A\|_{\text{sp}} \max_{\|x\|=1} \|Bx\| = \|A\|_{\text{sp}}\|B\|_{\text{sp}}. \end{aligned}$$

For the Frobenius norm, note that $\|X^*X\|_F \leq \|X\|_F^2$ for any matrix X as

$$\|X^*X\|_F^2 = \text{tr}\left((X^*X)(X^*X)\right) \leq \text{tr}(X^*X)\,\text{tr}(X^*X) = \|X\|_F^4.$$

It follows that (via the Cauchy-Schwarz inequality)

$$\begin{aligned} \|AB\|_F^2 &= \text{tr}(B^*A^*AB) = \text{tr}(A^*ABB^*) = \langle BB^*, A^*A\rangle \\ &\leq \|BB^*\|_F\|A^*A\|_F \leq \|B^*\|_F^2\|A\|_F^2 = \|A\|_F^2\|B\|_F^2. \end{aligned}$$

Alternatively, compute directly (and use the Cauchy-Schwarz inequality)

$$\begin{aligned} \|AB\|_F^2 &= \sum_{i,\,j} \Big|\sum_k a_{ik}b_{kj}\Big|^2 \leq \sum_{i,\,j} \Big(\sum_k |a_{ik}|^2\Big)\Big(\sum_k |b_{kj}|^2\Big) \\ &= \Big(\sum_{i,\,k} |a_{ik}|^2\Big)\Big(\sum_{j,\,k} |b_{kj}|^2\Big) = \|A\|_F^2\|B\|_F^2. \end{aligned}$$

Taking the square roots of both sides results in $\|AB\|_F \leq \|A\|_F\|B\|_F$. ∎

In terms of singular values, we have (Problem 32 of Sect. 4.1)

$$\sigma_{\max}(A+B) \leq \sigma_{\max}(A) + \sigma_{\max}(B) \tag{4.17}$$

and

$$\sigma_{\min}(A)\sigma_{\min}(B) \leq \sigma_{\min}(AB) \leq \sigma_{\max}(AB) \leq \sigma_{\max}(A)\sigma_{\max}(B). \tag{4.18}$$

If A and B are positive semidefinite matrices of the same order, then

$$\lambda_1(AB) \le \sigma_1(AB) \le \sigma_1(A)\sigma_1(B) = \lambda_1(A)\lambda_1(B). \tag{4.19}$$

Here $\lambda_1(X)$ denotes the largest eigenvalue of X if all eigenvalues of X are real.

Example 4.2.3 Frobenius norms and spectral norms.

(a) Let $A = \left(\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right)$. Then $\|A\|_{_F} = 1$, $\|A\|_{\rm sp} = 1$.

(b) Let $A = \left(\begin{smallmatrix} 2 & 0 & 0 \\ 3 & 2 & 0 \end{smallmatrix}\right)$. Then $\|A\|_{_F} = \sqrt{17}$, $\|A\|_{\rm sp} = 4$.

(c) $\|I_n\|_{_F} = \sqrt{n}$, $\|I_n\|_{\rm sp} = 1$.

(d) $\|E_n\|_{_F} = \|E_n\|_{\rm sp} = n$, where $E_n \in \mathbb{M}_n$ is the all-ones matrix.

(e) $\|x\|_{_F} = \|x\|_{\rm sp} = \sqrt{|x_1|^2 + \cdots + |x_n|^2}$, where $x = (x_1, \ldots, x_n) \in \mathbb{C}^n$.

(f) Let $A \in \mathbb{M}_n$. Then $\sigma_1(A^k) \le \sigma_1^k(A)$ for positive integers k.

(g) If $A, B \in \mathbb{M}_n$ are positive semidefinite, then $\lambda_1\big((AB)^2\big) \le \lambda_1(A^2B^2)$:

$$\lambda_1\big((AB)^2\big) = \lambda_1^2(AB) \le \sigma_1^2(AB) = \lambda_1(B^*A^*AB) = \lambda_1(A^2B^2). \ \blacksquare$$

Exercise: $\lambda_1(A^2B^2) \le \lambda_1^2(A)\lambda_1^2(B)$ for positive semidefinite $A, B \in \mathbb{M}_n$.

Two observations follow: first, a matrix norm $\|\cdot\| : A \mapsto \|A\|$ is a continuous function on the matrix space $\mathbb{M}_n$ (in the sense stated in Theorem 4.2.1), and second, $\rho(\cdot) \le \|\cdot\|$ for any (submultiplicative) matrix norm $\|\cdot\|$. This is easy to see: If $Ax = \lambda x$, where $x \ne 0$ and $\rho(A) = |\lambda|$, by letting X be the $n \times n$ matrix with all columns equal to the eigenvector x, we have

$$\rho(A)\|X\| = \|\lambda X\| = \|AX\| \le \|A\|\|X\|,$$

implying

$$\rho(A) \le \|A\|. \tag{4.20}$$

Note that the numerical radius $\rho(\cdot)$ is not a submultiplicative matrix norm (why?). Moreover, (4.20) need not be true for non-submultiplicative norms.

The following result reveals a close relation between $\rho(A)$ and $\|A\|$.

Theorem 4.2.3 (Gelfand) *Let $\|\cdot\|$ be a matrix norm and let $A \in \mathbb{M}_n$. Then*

$$\rho(A) = \lim_{k\to\infty} \|A^k\|^{1/k}.$$

Proof. The eigenvalues of A^k are the kth powers of the eigenvalues of A. Since spectral radius is dominated by norm (by (4.20)), for each $k = 1, 2, \ldots,$

$$(\rho(A))^k = \rho(A^k) \le \|A^k\| \quad \text{or} \quad \rho(A) \le \|A^k\|^{1/k}.$$

On the other hand, for any $\epsilon > 0$, let

$$A_\epsilon = \frac{1}{\rho(A) + \epsilon} A.$$

Then $\rho(A_\epsilon) < 1$. By Theorem 4.1.6, A_ϵ^k tends to 0 as $k \to \infty$. Because the norm is a continuous function, there exists a positive number N_ϵ such that

$$\|A_\epsilon^k\| < 1 \quad \text{or} \quad \|A^k\| \le (\rho(A) + \epsilon)^k \ \text{ for } k > N_\epsilon.$$

Thus,

$$\|A^k\|^{1/k} \le \rho(A) + \epsilon,$$

yielding

$$\rho(A) \le \|A^k\|^{1/k} \le \rho(A) + \epsilon \ \ (k > N_\epsilon).$$

By the definition of limit, we arrive at $\lim_{k\to\infty} \|A^k\|^{1/k} = \rho(A)$. ∎

We now introduce an important class of norms for matrices: *unitarily invariant norms*. We call a matrix-vector norm $\|\cdot\|$ on $\mathbb{M}_{m\times n}$ a *unitarily invariant norm* if for all $A \in \mathbb{M}_{m\times n}$, unitary $U \in \mathbb{M}_m$ and unitary $V \in \mathbb{M}_n$,

$$\|UAV\| = \|A\|.$$

Example 4.2.4 The spectral norm and Frobenius norm are unitarily invariant norms. This is because the singular values of a matrix A remain unchanged when A is multiplied by a unitary matrix. To be precise, let $B = UAV$, where U, V are unitary. Then $B^*B = V^*A^*AV$. It follows that B and A have the same (nonzero) singular values. (4.13) and (4.14) show that both norms solely depend on the singular values. ∎

In Example 4.2.1, the ℓ_1- and ℓ_∞-norms are not unitarily invariant.

Actually, the spectral norm and the Frobenius norm belong to two larger families of unitarily invariant norms: the *Fan k-norms* and the *Schatten p-norms*, which we shall study more in Chapter 10.

Fan k-norm on $\mathbb{M}_{m\times n}$: Let $1 \le k \le \min\{m, n\}$ be an integer. Define

$$\|A\|_{(k)} = \sum_{i=1}^{k} \sigma_i(A).$$

Schatten p-norm on $\mathbb{M}_{m\times n}$: Let $p \ge 1$ be a real number. Define

$$\|A\|_p = \left(\sum_{i=1}^{n} \sigma_i^p(A) \right)^{1/p}.$$

Apparently, the Frobenius norm is the Schatten 2-norm, i.e., $\|A\|_F = \|A\|_2$. The Schatten 1-norm is the sum of all singular values; it is also known as the *trace norm* or *nuclear norm*: $\|A\|_1 = \|A\|_{\text{tr}} = \text{tr}\left((A^*A)^{1/2}\right)$. The spectral norm is the Fan norm when $k = 1$; the spectral norm is also equal to the limit of the Schatten norm $\|A\|_p$ as $p \to \infty$, that is, $\|A\|_{\text{sp}} = \|A\|_\infty = \lim_{p\to\infty} \|A\|_p$. Note that in some other texts, $\|A\|_\infty$ (or $\|A\|_{\max}$) is used for the ℓ_∞-norm and $\|A\|_2$ is used for $\sigma_{\max}(A)$. We point out that there is no universal set of notations for the norms of matrices and matrix norms. One should check and be clear about the meaning of a notation in the context.

Problems

1. Let $A = \left(\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right)$. Show that $\|A^2\| \neq \|A\|^2$ for any norm of matrices.

2. Consider the norms on $\mathbb{M}_n$ for $A = (a_{ij}) \in \mathbb{M}_n$ defined by
   $\|A\|_{\ell_1} = \sum_{i,j} |a_{ij}|$,
   $\|A\|_{\ell_\infty} = \max_{i,j} |a_{ij}|$,
   $\|A\|_F = \left(\sum_{i,j} |a_{ij}|^2\right)^{1/2}$,
   $\|A\|_{\mathrm{sp}} = \sigma_{\max}(A)$ (the largest singular value of A), and
   $\|A\|_{\mathrm{tr}} = \sigma_1(A) + \cdots + \sigma_n(A)$ (the sum of all singular values of A).
   Find the norms $\|A\|$, where $\|\cdot\|$ is each of the above norms, and A is I_n, $\left(\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right)$, $\left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$, $\left(\begin{smallmatrix} 1 & 1 \\ 1 & 1 \end{smallmatrix}\right)$, and E_n (the all-ones matrix), respectively.

3. (a) Show that all unitary matrices have the same spectral norm (regardless of order), (b) show that all unitary matrices of the same order, say n, have the same Frobenius norm and the same trace norm (in terms of n), and (c) do they all have the same ℓ_1-norm or ℓ_∞-norm?

4. Let $\|\cdot\|$ be a vector norm on $\mathbb{F}^n$, where $\mathbb{F} = \mathbb{C}$ or $\mathbb{R}$. Define
   $$\|x\|^D = \max\{|\langle x, y\rangle| : y \in \mathbb{F}^n,\ \|y\| = 1\}.$$
   Show that $\|\cdot\|^D$ is a vector norm on $\mathbb{F}^n$ (known as the *dual norm*).

5. Let $A \in \mathbb{M}_{m\times n}$. Show that A, A^*, A^T, and $\overline{A}$ have the same spectral norm and the same Frobenius norm. Show by example that $\|A\| = \|A^*\|$ need not be true for some other matrix norms $\|\cdot\|$.

6. Let $A \in \mathbb{M}_{m\times n}$. Show that $\|A^*A\|_{\mathrm{sp}} = \|AA^*\|_{\mathrm{sp}} = \|A\|_{\mathrm{sp}}^2$ and that $\|A^*A\|_F = \|AA^*\|_F$. However, $\|A^*A\|_F = \|A\|_F^2$ need not be true.

7. Let A be a square matrix and let r be the rank of A. Show that
   $$\|A\|_{\mathrm{sp}} \leq \|A\|_F \leq \sqrt{r}\,\|A\|_{\mathrm{sp}}.$$

8. Let $A \in \mathbb{M}_{m\times n}$. Show that $\|A\|_F = \|A\|_{\mathrm{sp}}$ if and only if $\mathrm{rank}(A) = 1$.

9. Find a matrix $A \in \mathbb{M}_2$ and two norms $\|\cdot\|_a$ and $\|\cdot\|_b$ on $\mathbb{M}_2$ such that $\|A\|_a < 1 < \|A\|_b$. Can such an A have an eigenvalue λ with $|\lambda| > 1$?

10. Let $\|\cdot\|$ be a matrix-vector norm and $k > 0$. Show that $\|\cdot\|' = k\|\cdot\|$ is also a matrix-vector norm. In particular, $\|\cdot\|' = \frac{1}{2}\sigma_1(\cdot)$ is a matrix-vector norm. Is it a submultiplicative norm (for square matrices)?

11. Show that $\|\cdot\|_{\mathrm{rowmax}}$ is a submultiplicative (matrix) norm on $\mathbb{M}_n$:
    $$\|A\|_{\mathrm{rowmax}} = \max_{1\leq i\leq n} \sum_{j=1}^{n} |a_{ij}|, \quad \text{where } A = (a_{ij}) \in \mathbb{M}_n.$$

12. Let $A = (a_{ij}) \in \mathbb{M}_n$ and let $\|\cdot\|$ be a norm on $\mathbb{M}_n$.
    (a) (4.20) shows $\rho(A) \le \|A\|$ for submultiplicative norms of A. Is it true that $w(A) \le \|A\|$, where $w(A)$ is the numerical radius of A?
    (b) Show that $\rho(A) > \|A\|$ for $A = \left(\begin{smallmatrix} 1 & 0 \\ 0 & 0 \end{smallmatrix}\right)$ with $\|A\| = \frac{1}{2}\sigma_1(A)$.

13. Show by examples that neither the ℓ_∞-norm $\|\cdot\|_{\ell_\infty}$ nor the spectral norm $\|\cdot\|_{\mathrm{sp}}$ on $\mathbb{M}_n$ is derived from an inner product on $\mathbb{M}_n$. [Hint: An induced norm obeys the parallelogram law. See Sect. 1.4, Chap. 1.]

14. Let ϵ be a positive constant and let $A = (a_{ij})$ and $B = (b_{ij})$ be n-square matrices such that $|a_{ij} - b_{ij}| \le \epsilon$ for all $1 \le i, j \le n$. Show that
$$|\,\|A\|_F - \|B\|_F\,| \le \|A - B\|_F \le n\epsilon.$$

15. Let A and B be matrices of the same size. Show that
$$\|A + B\|_F^2 = \|A\|_F^2 + \|B\|_F^2 + 2\Re\big(\mathrm{tr}(B^*A)\big).$$

16. Let A be an $m \times n$ matrix and let $x \in \mathbb{C}^n$. Show that
$$\sigma_n(A)\|x\| \le \|Ax\| \le \sigma_1(A)\|x\|.$$

17. Let A and B be matrices of appropriate sizes. Show that
$$\sigma_{\min}(AB) \ge \sigma_{\min}(A)\sigma_{\min}(B).$$
    Is it true that $\sigma_{\min}(A + B) \ge \sigma_{\min}(A) + \sigma_{\min}(B)$?

18. Let $\|\cdot\|$ be a norm on $\mathbb{M}_{m\times n}$. For $A = (a_{ij}) \in \mathbb{M}_{m\times n}$, show that
$$\|A\| \le \sum_{i,\,j=1}^{n} |a_{ij}|\,\|E_{ij}\|,$$
    where $E_{ij} \in \mathbb{M}_{m\times n}$ is the matrix with (i, j)-entry 1 and 0 elsewhere.
    [Note: This is *norm continuity* with respect to the matrix entries.]

19. (*Norm equivalence*) Let $\|\cdot\|_a$ and $\|\cdot\|_b$ be two matrix-vector norms. Show that there exist some positive constants C_1 and C_2 such that
$$C_1\|X\|_a \le \|X\|_b \le C_2\|X\|_a \quad \text{for all (matrices) } X.$$
    [Note: See Theorem 1.4.3, Chapter 1, for vector norm equivalence.]

20. Let A be a square matrix with singular values $\sigma_i(A)$'s and the moduli of the eigenvalues $|\lambda_i(A)|$'s ordered decreasingly. Show that
$$\lim_{k\to\infty} \big(\sigma_1(A^k)\big)^{1/k} = |\lambda_1(A)|.$$

21. Let $A \in \mathbb{M}_n$. Show that for any matrix norm $\|\cdot\|$ and integer $k \ge 1$,
$$\|A^k\| \le \|A\|^k \quad \text{and} \quad \|A^k\|^{-1}\|I\| \le \|A^{-1}\|^k \text{ if } A \text{ is invertible.}$$

22. Let $A \in \mathbb{M}_n$ be such that $\|A\| < 1$ for some matrix norm $\|\cdot\|$. Show that $\lim_{k\to\infty} A^k = 0$ and that $\lim_{k\to\infty} \|A^k\|' = 0$ for any matrix norm $\|\cdot\|'$.

23. Let $A \in \mathbb{M}_n$ and let $\|\cdot\|$ be a matrix norm on $\mathbb{M}_n$. (a) Show that $\lim_{k\to\infty} A^k = 0$ if and only if $\lim_{k\to\infty} \|A^k\| = 0$. (b) Show that $\lim_{k\to\infty} \|A\|^k = 0$ implies $\lim_{k\to\infty} \|A^k\| = 0$. (c) Is the converse of (b) true?

24. Let $A \in \mathbb{M}_n$ and let $\|\cdot\|$ be a matrix norm on $\mathbb{M}_n$. If $\|A\| < 1$, show that $\sum_{k=0}^{\infty} A^k$ is convergent (to $(I - A)^{-1}$). Is the converse true?

25. Let $A \in \mathbb{M}_n$ and let $\|\cdot\|$ be a matrix norm on $\mathbb{M}_n$. Show that the power series $\sum_{k=0}^{\infty} A^k$ is convergent if and only if $\|A^k\| \to 0$ (as $k \to \infty$).

26. Let $\|\cdot\|$ be a matrix norm on $\mathbb{M}_n$ and let $P \in \mathbb{M}_n$ be an invertible matrix. Show that $\|\cdot\|_P : \mathbb{M}_n \to \mathbb{R}$ defined by $\|X\|_P = \|P^{-1}XP\|$ for $X \in \mathbb{M}_n$ is also a matrix norm on $\mathbb{M}_n$.

27. Which of the norms $\|\cdot\|_{\ell_1}$, $\|\cdot\|_{\ell_\infty}$, $\|\cdot\|_F$, $\|\cdot\|_{\mathrm{sp}}$, and $\|\cdot\|_{\mathrm{rowmax}}$ on $\mathbb{M}_n$ are submultiplicative norms? Unitarily invariant norms?

28. Let $\|\cdot\|$ be a submultiplicative (matrix) norm on $\mathbb{M}_n$. Show that $\|I_n\| \le \|A\|\,\|A^{-1}\|$ for any nonsingular $A \in \mathbb{M}_n$.

29. Give an example of matrix-vector norm $\|\cdot\|_v$ on $\mathbb{M}_n$ such that $\|I_n\|_v = \|I_n^2\|_v > \|I_n\|_v\|I_n\|_v$, i.e., $\|I_n^2\|_v > \|I_n\|_v^2$. [Hint: $\|X\|_v = \frac{1}{2}\sigma_{\max}(X)$.]

30. Let $\sigma_{\max}(A)$ be the largest singular value of $A \in \mathbb{M}_n$. Show that
$$\sigma_{\max}(I_n - A) \ge 1 - \sigma_{\max}(A).$$

31. Let $\|\cdot\|_{(k)}$ be the Fan k-norm on $\mathbb{M}_n$, where $1 \le k \le n$. Show that
$$\|I_n - A\|_{(k)} + \|A\|_{(k)} \ge k.$$

32. Find the unit spheres of the matrix space $\mathbb{M}_2$ with various norms.
    (a) Describe the set $S_\infty = \{X \in \mathbb{M}_2 : \|X\|_{\ell_\infty} = 1\}$.
    (b) Describe the set $S_1 = \{X \in \mathbb{M}_2 : \|X\|_{\ell_1} = 1\}$.
    (c) Describe the set $S_F = \{X \in \mathbb{M}_2 : \|X\|_F = 1\}$.

    What are the elements that lie in all three sets S_∞, S_1, and S_F?

33. Let A be a matrix. Prove each of the following statements.
    (a) If a is an entry of A, then $|a| \le \sigma_{\max}(A)$.
    (b) If v is a row or column of A, then $\|v\|_2 \le \sigma_{\max}(A)$.
    (c) If $\left(\begin{smallmatrix} p & q \\ r & s \end{smallmatrix}\right)$ is a submatrix of A, then
    $$\sqrt{|p|^2 + |q|^2 + |r|^2 + |s|^2} \le \|A\|_F.$$
    Is it true that $\sqrt{|p|^2 + |q|^2 + |r|^2 + |s|^2} \le \sigma_{\max}(A)$ (i.e. $\|A\|_{\mathrm{sp}}$)?

34. Let M be a matrix and let A be a submatrix of M. Show that

$$\|A\|_{\text{sp}} \leq \|M\|_{\text{sp}}, \quad \|A\|_F \leq \|M\|_F.$$

[Note: It is tempting to compare $\|A\|$ and $\|M\|$ for a general norm $\|\cdot\|$. However, the comparison makes no sense unless the norm is defined for both A and M. Note that A and M have different sizes in general.]

35. Let $M = \left(\begin{smallmatrix} A & B \\ C & D \end{smallmatrix}\right)$ be a partitioned matrix. Show that

$$\|M\|_F^2 = \|A\|_F^2 + \|B\|_F^2 + \|C\|_F^2 + \|D\|_F^2$$

and

$$\|M\|_{\text{sp}} \leq \|A\|_{\text{sp}} + \|B\|_{\text{sp}} + \|C\|_{\text{sp}} + \|D\|_{\text{sp}}.$$

36. Let A be an $m \times m$ nonsingular normal matrix and let $|A| = (A^*A)^{1/2}$.

    (a) If $\|A^{-n}XA^n\|_{\text{sp}}$ is bounded (finite) for all $X \in \mathbb{M}_m$ and all integers n, show that $|A| = kI_m$ for some $k \geq 0$. Thus, $|A|X = X|A|$.

    (b) Let $Y \in \mathbb{M}_m$ be given. Show that $\|A^{-n}YA^n\|_{\text{sp}}$ is bounded for all integers n if and only if $|A|Y = Y|A|$.

    (c) Let $A = \left(\begin{smallmatrix} 1 & 0 \\ 0 & 2 \end{smallmatrix}\right)$ and $Z = \left(\begin{smallmatrix} 0 & 0 \\ 1 & 0 \end{smallmatrix}\right)$. Compute $A^{-n}YA^n$. Show that $\|A^{-n}ZA^n\|_{\text{sp}}$ is bounded for all positive integers n, but $AZ \neq ZA$.

    (d) Show by example that (b) need not be true if A is not normal.

37. Let $M = \left(\begin{smallmatrix} A & B \\ C & D \end{smallmatrix}\right)$ and $N = \left(\begin{smallmatrix} A & B \\ 0 & D \end{smallmatrix}\right)$, where $A, B, C, D \in \mathbb{M}_n$. Show that

    (a) $\|M\|_F \geq \|N\|_F$.

    (b) $\|M\|_{\text{sp}} < \|N\|_{\text{sp}}$ for $M = \left(\begin{smallmatrix} 1 & 1 \\ -\frac{1}{8} & 1 \end{smallmatrix}\right)$.

    (c) $M^*M \geq N^*N$ if and only if $C^*D = 0$.

    (d) $\det(M^*M - N^*N) = (-1)^n |\det C \det D|^2$.

    (e) $M^*N - N^*N$ has at least k positive eigenvalues if $\text{rank}(C) = k$.

    (f) If C and D are both nonsingular, then $M^*N - N^*N$ has exactly n positive eigenvalues and n negative eigenvalues.

38. Let A, B, and C be $n \times n$ matrices. If all the eigenvalues of B are contained in an open disc $\{z \in \mathbb{C} : |z| < r\}$, and all the eigenvalues of A lie outside the closed disc, i.e., in $\{z \in \mathbb{C} : |z| \geq r + d\}$ for some $d > 0$, show that the matrix equation $AX - XB = C$ has a solution

$$X = \sum_{i=0}^{\infty} A^{-i-1} C B^i.$$

Show further that $\|X\|_{\text{sp}} \leq \frac{1}{d}\|C\|_{\text{sp}}$ if A and B are normal matrices.

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4.3 The Kronecker and Hadamard Products

Matrices can be multiplied in different ways. In addition to the ordinary product, the Kronecker product and Hadamard product are important ones and have applications in other fields. Another basic matrix operation is vectorization of ("vecing") matrices; it is a useful tool in solving matrix equations. We introduce the three concepts and present their properties.

The *Kronecker product*, denoted by $A \otimes B$ and also called the *tensor product* or *direct product*, of $A = (a_{ij}) \in \mathbb{M}_{m\times n}$ and $B \in \mathbb{M}_{p\times q}$ is defined by

$$A \otimes B = \begin{pmatrix} a_{11}B & a_{12}B & \dots & a_{1n}B \\ a_{21}B & a_{22}B & \dots & a_{2n}B \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1}B & a_{m2}B & \dots & a_{mn}B \end{pmatrix}.$$

That is, the Kronecker product $A \otimes B$ is an $(mp) \times (nq)$ matrix, composed of mn blocks with the $p \times q$ matrix $a_{ij}B$ as the (i, j) block, $1 \le i \le m$, $1 \le j \le n$. The element $a_{ij}b_{st}$ appears in $A \otimes B$ on row $(i-1)p + s$ and column $(j-1)q + t$. Note that A and B can have any different sizes.

The *Hadamard product* $A \circ B$, also called the *Schur product*, of matrices $A = (a_{ij})$ and $B = (b_{ij})$ of the same size is defined by the entrywise product

$$A \circ B = (a_{ij}b_{ij}).$$

In particular, for $u = (u_1, u_2, \dots, u_n)$ and $v = (v_1, v_2, \dots, v_n)$ in $\mathbb{C}^n$,

$$u \otimes v = (u_1v_1, \dots, u_1v_n,\ u_2v_1, u_2v_2, \dots, u_2v_n,\ \dots,\ u_nv_1, \dots, u_nv_n) \in \mathbb{C}^{n^2}$$

and

$$u \circ v = (u_1v_1, u_2v_2, \dots, u_nv_n) \in \mathbb{C}^n.$$

Note that $A \circ B = B \circ A$ but $A \otimes B \ne B \otimes A$ (as $AB \ne BA$) in general.

Example 4.3.1 We take, for example, $A = \left(\begin{smallmatrix} 1 & 2 \\ 3 & 4 \end{smallmatrix}\right)$ and $B = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)$. Then

$$A \otimes B = \begin{pmatrix} a & b & 2a & 2b \\ c & d & 2c & 2d \\ 3a & 3b & 4a & 4b \\ 3c & 3d & 4c & 4d \end{pmatrix}, \quad A \circ B = \begin{pmatrix} a & 2b \\ 3c & 4d \end{pmatrix}. \blacksquare$$

The Kronecker product has the following basic properties, each of which can be verified by definition and direct computations.

Theorem 4.3.1 *Let A, B, and C be matrices of appropriate sizes. Then*

(1) $(kA) \otimes B = A \otimes (kB) = k(A \otimes B)$, *where k is a scalar.*

(2) $(A + B) \otimes C = A \otimes C + B \otimes C$.

(3) $A \otimes (B + C) = A \otimes B + A \otimes C$.

(4) $(A \otimes B) \otimes C = A \otimes (B \otimes C)$. (Problem 7)

(5) $A \otimes B = 0$ *if and only if* $A = 0$ *or* $B = 0$.

(6) $(A \otimes B)^T = A^T \otimes B^T$. *If A and B are symmetric, so is $A \otimes B$.*

(7) $(A \otimes B)^* = A^* \otimes B^*$. *If A and B are Hermitian, so is $A \otimes B$.*

Theorem 4.3.2 *Let A, B, and C be matrices of appropriate sizes. Then*

(a) $(A \otimes B)(C \otimes D) = (AC) \otimes (BD)$ (*the mixed product formula*).

(b) $(A \otimes B)^{-1} = A^{-1} \otimes B^{-1}$ *if A and B are invertible.*

(c) $A \otimes B$ *is unitary if A and B are unitary.*

(d) $A \otimes B$ *is normal if A and B are normal.*

(e) $A \otimes B$ *is positive semidefinite if A and B are positive semidefinite.*

Proof. For (a), assume that A has n columns. Then C has n rows as indicated in the product AC on the right-hand side of (a). We write $A \otimes B = (a_{ij}B)$, $C \otimes D = (c_{ij}D)$. Then the (i, j) block of $(A \otimes B)(C \otimes D)$ is

$$\sum_{t=1}^{n} a_{it}Bc_{tj}D = \sum_{t=1}^{n} a_{it}c_{tj}\,BD.$$

This is the (i, j)-entry of AC times BD; it is also exactly the (i, j) block of $(AC) \otimes (BD)$. (a) follows. (b), (c), and (d) are immediate from (a).

To show (e), let $A = U^*\Lambda U$ and $B = V^*\Gamma V$ be spectral decompositions of A and B (Theorem 3.2.2), respectively, where U and V are unitary. Then

$$A \otimes B = (U^*\Lambda U) \otimes (V^*\Gamma V) = (U \otimes V)^*(\Lambda \otimes \Gamma)(U \otimes V).$$

Note that $U \otimes V$ is unitary, and $\Lambda \otimes \Gamma$ is nonnegative and diagonal. ∎

A matrix may be identified with a vector. We express a matrix as a column vector via vectorization by stacking the columns (from left to right) of the matrix vertically. Vectorization of matrices is a linear transformation from matrices to vectors (say, $\mathbb{M}_{m\times n}(\mathbb{C}) \to \mathbb{C}^{mn}$); it is connected to the inner product, Kronecker product, as well as the Hadamard product.

Let $X = (x_{ij})$ be an $m \times n$ matrix. Partition X according to its columns as $X = (x_1, x_2, \ldots, x_n)$, and define a column vector of mn components by

$$\mathrm{vec}(X) = (x_{11}, x_{21}, \ldots, x_{m1}, \ldots, x_{1n}, x_{2n}, \ldots, x_{mn})^T = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix}.$$

Example 4.3.2 Examples of vectorization.

(a) If $A = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)$, then $\mathrm{vec}(A) = \begin{pmatrix} a \\ c \\ b \\ d \end{pmatrix}$.

(b) If $A, B \in \mathbb{M}_{m\times n}(\mathbb{R})$, then $\mathrm{tr}(A^T B) = (\mathrm{vec}(A))^T \mathrm{vec}(B)$.
The left-hand side is equal to the Frobenius product of A and B, while the right-hand side is equal to the usual inner product of two vectors in $\mathbb{R}^{mn}$. This also holds for complex matrices with ^{T} replaced by *.

(c) If $A, B \in \mathbb{M}_{m\times n}$, $c \in \mathbb{C}$, then $\mathrm{vec}(A + cB) = \mathrm{vec}(A) + c\,\mathrm{vec}(B)$.

(d) If $A, B \in \mathbb{M}_{m\times n}$, then $\mathrm{vec}(A \circ B) = \mathrm{vec}(A) \circ \mathrm{vec}(B)$.

(e) $\mathrm{vec}(ABC) = (C^T \otimes A)\,\mathrm{vec}(B)$ (whenever ABC is well-defined.)
To show this, let $A \in \mathbb{M}_{m\times n}$, $B \in \mathbb{M}_{n\times p}$, and $C = (c_{ij}) \in \mathbb{M}_{p\times q}$. Partition $B = (b_1, \ldots, b_p)$ and $C = (c_1, \ldots, c_q)$ according to their columns. For each $j = 1, 2, \ldots, q$, we compute the jth column of ABC

$$\begin{aligned} (AB)c_j &= (Ab_1, \ldots, Ab_p)c_j = \sum_{i=1}^{p} (c_{ij}A)b_i \\ &= (c_{1j}A, c_{2j}A, \ldots, c_{pj}A)\mathrm{vec}(B) = (c_j^T \otimes A)\mathrm{vec}(B). \end{aligned}$$

It follows that (with $v = \mathrm{vec}(B)$ for brevity)

$$\mathrm{vec}(ABC) = \begin{pmatrix} (c_1^T \otimes A)v \\ (c_2^T \otimes A)v \\ \vdots \\ (c_q^T \otimes A)v \end{pmatrix} = \begin{pmatrix} c_1^T \otimes A \\ c_2^T \otimes A \\ \vdots \\ c_q^T \otimes A \end{pmatrix} v = (C^T \otimes A)\mathrm{vec}(B).$$

(f) Let $A \in \mathbb{M}_{m\times n}$, $X \in \mathbb{M}_{n\times p}$, and $Y \in \mathbb{M}_{q\times m}$. Then part (e) yields

$$\mathrm{vec}(AX) = (I_p \otimes A)\mathrm{vec}(X), \quad \mathrm{vec}(YA) = (A^T \otimes I_q)\mathrm{vec}(Y). \blacksquare$$

To perform the Hadamard product, matrices need to have the same size. In the case of square matrices, an interesting and important observation is that the Hadamard product $A \circ B$ is contained in the Kronecker product $A \otimes B$ as a principal submatrix. (Check this with Example 4.3.1.)

Theorem 4.3.3 *Let A and B be $n \times n$ matrices. The Hadamard (or Schur or entrywise) product $A \circ B$ is the principal submatrix of the Kronecker product $A \otimes B$ lying on the intersections of rows and columns $1, n+2, 2n+3, \ldots, n^2$.*

Proof. As usual, let e_i be the column vector of n components with the ith component 1 and 0 elsewhere, $i = 1, 2, \ldots, n$. Then $e_1 \otimes e_1$ has 1 as the first component, $e_2 \otimes e_2$ has 1 as the $(n+2)$-th component, $\ldots$, $e_n \otimes e_n$ has 1 as the n^2-th component, all other components are 0's in these vectors. Let

$$E = (e_1 \otimes e_1, \ldots, e_n \otimes e_n) \in \mathbb{M}_{n^2 \times n}.$$

Then for every pair of i and j, we have by computation

$$a_{ij} b_{ij} = (e_i^T A e_j) \otimes (e_i^T B e_j) = (e_i \otimes e_i)^T (A \otimes B)(e_j \otimes e_j),$$

which is equal to the (i, j)-entry of the matrix $E^T (A \otimes B) E$. Thus,

$$E^T (A \otimes B) E = A \circ B.$$

This says that $A \circ B$ is the principal submatrix of $A \otimes B$ lying on the intersections of rows and columns $1,\ n+2,\ 2n+3, \ldots, n^2$. ∎

The following theorem reveals relations between the eigenvalues of the Kronecker product and the eigenvalues of each of the individual matrices.

Theorem 4.3.4 *Let $A \in \mathbb{M}_m$ and $B \in \mathbb{M}_n$ have eigenvalues λ_i and μ_j, $i = 1, \ldots, m$, $j = 1, \ldots, n$, respectively. Then the eigenvalues of $A \otimes B$ are*

$$\lambda_i \mu_j, \quad i = 1, \ldots, m, \ \ j = 1, \ldots, n,$$

and the eigenvalues of $A \otimes I_n + I_m \otimes B$ are

$$\lambda_i + \mu_j, \quad i = 1, \ldots, m, \ \ j = 1, \ldots, n.$$

Proof. By the Schur decomposition (Theorem 3.2.1), let U and V be unitary matrices of orders m and n, respectively, such that

$$U^* A U = T_1 \quad \text{and} \quad V^* B V = T_2,$$

where T_1 and T_2 are upper-triangular matrices with diagonal entries λ_i and μ_j, $i = 1, \ldots, m$, $j = 1, \ldots, n$, respectively. Then

$$T_1 \otimes T_2 = (U^* A U) \otimes (V^* B V) = (U^* \otimes V^*)(A \otimes B)(U \otimes V).$$

Note that $U \otimes V$ is unitary. Thus $A \otimes B$ is unitarily similar to the upper-triangular matrix $T_1 \otimes T_2$ with eigenvalues $\lambda_i \mu_j$, $i = 1, \ldots, m$, $j = 1, \ldots, n$.

For the eigenvalues of $A \otimes I_n + I_m \otimes B$, let $W = U \otimes V$. Then

$$W^* (A \otimes I_n) W = T_1 \otimes I_n = \begin{pmatrix} \lambda_1 I_n & & \star \\ & \ddots & \\ 0 & & \lambda_m I_n \end{pmatrix}$$

and

$$W^*(I_m \otimes B)W = I_m \otimes T_2 = \begin{pmatrix} T_2 & & 0 \\ & \ddots & \\ 0 & & T_2 \end{pmatrix}.$$

Thus,

$$W^*(A \otimes I_n + I_m \otimes B)W = T_1 \otimes I_n + I_m \otimes T_2$$

is an upper-triangular matrix with eigenvalues $\lambda_i + \mu_j$ for all i, j. ∎

Problems

1. Compute $A \otimes B$ and $B \otimes A$ for

$$A = \begin{pmatrix} 0 & 1 \\ -2 & 3 \end{pmatrix} \quad \text{and} \quad B = \begin{pmatrix} 0 & \sqrt{2} \\ \pi & 2 \\ -1 & 7 \end{pmatrix}.$$

2. Compute $(e_1 - e_2) \otimes (e_1 + 2e_2)$, where $\{e_1, e_2\}$ is a basis of $\mathbb{R}^2$.

3. Let $x, y, z \in \mathbb{C}^n$ (in columns). Show that $(x^*y)z = z(x^*y) = (zx^*)y$ and that $(x^* \otimes y) \otimes z = x^* \otimes (y \otimes z)$. Explain why $(x^*y)z \neq x^*(yz)$.

4. Let $J_2 = \left(\begin{smallmatrix} 1 & 1 \\ 1 & 1 \end{smallmatrix}\right)$. Compute $J_2 \otimes J_2$, $J_2 \circ J_2$, $I_n \otimes J_2$, and $J_2 \otimes I_n$.

5. Let $A \in \mathbb{M}_n$. Show that $A \circ A^*$ is Hermitian. Is $A \otimes A^*$ Hermitian?

6. Let A, B, C, and D be matrices of appropriate sizes. Show that

   (a) $(A \otimes B)^k = A^k \otimes B^k$, where k is a positive integer.

   (b) $\operatorname{tr}(A \otimes B) = \operatorname{tr}A \operatorname{tr}B$.

   (c) $\operatorname{rank}(A \otimes B) = \operatorname{rank}(A) \operatorname{rank}(B)$.

   (d) $\det(A \otimes B) = (\det A)^n (\det B)^m$, where $A \in \mathbb{M}_m$ and $B \in \mathbb{M}_n$.

   (e) If $A \otimes B = C \otimes D \neq 0$, where A and C have the same size (so do B and D), then $A = aC$ and $B = bD$ with $ab = 1$ (and vice versa).

7. Let $X \in \mathbb{M}_{m \times n}$, $Y \in \mathbb{M}_{p \times q}$, and let $M = X \otimes Y \in \mathbb{M}_{mp \times nq}$. Show that the position of $x_{ij}y_{rs}$ in M is row $(i-1)p + r$ and column $(j-1)q + s$. Use this to prove the associative property: $(A \otimes B) \otimes C = A \otimes (B \otimes C)$.

8. Let A and B be m- and n-square matrices, respectively. Show that

$$(A \otimes I_n)(I_m \otimes B) = A \otimes B = (I_m \otimes B)(A \otimes I_n).$$

9. Let $A \in \mathbb{M}_n$ have characteristic polynomial $p(x)$. Show that

$$\det(A \otimes I + I \otimes A) = (-1)^n \det p(-A).$$

10. Let $A, B \in \mathbb{M}_n$. Show that for some permutation matrix $P \in \mathbb{M}_{n^2}$,

$$P^{-1}(A \otimes B)P = B \otimes A.$$

11. Let $x, y, u, v \in \mathbb{C}^n$. With $\langle x, y\rangle = y^*x$ and $\|x\|^2 = x^*x$, show that

$$\langle x, y\rangle\langle u, v\rangle = \langle x \otimes u, y \otimes v\rangle.$$

Derive

$$\|x \otimes y\| = \|x\|\,\|y\|.$$

12. Let $A, B \in \mathbb{M}_n$. Show that $A \circ I_n = \operatorname{diag}(a_{11}, \ldots, a_{nn})$ and that

$$D_1(A \circ B)D_2 = (D_1AD_2) \circ B = A \circ (D_1BD_2)$$

for any n-square diagonal matrices D_1 and D_2.

13. Let A, B, and C be matrices (of any sizes). Show that

$$(A \oplus B) \otimes C = (A \otimes C) \oplus (B \otimes C).$$

(Here $\oplus$ for the matrix direct sum.) But it need not be true that

$$(A \otimes B) \oplus C = (A \oplus C) \otimes (B \oplus C).$$

14. Consider the vector space $\mathbb{M}_2$ of all 2×2 matrices over $\mathbb{C}$.
    (a) What is the dimension of $\mathbb{M}_2$?
    (b) Find a basis for $\mathbb{M}_2$.
    (c) For $A, B \in \mathbb{M}_2$, define

$$\mathcal{L}(X) = AXB, \quad X \in \mathbb{M}_2.$$

    Show that $\mathcal{L}$ is a linear transformation on $\mathbb{M}_2$.
    (d) Show that if λ and μ are eigenvalues of A and B, respectively, then $\lambda\mu$ is an eigenvalue of $\mathcal{L}$.

15. Compute $\operatorname{vec}(A)$, where $A = \left(\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right)$, $\left(\begin{smallmatrix} 1 & 1 \\ 0 & 0 \end{smallmatrix}\right)$, $\left(\begin{smallmatrix} 1 & 2 \\ 2 & 4 \end{smallmatrix}\right)$, or $\left(\begin{smallmatrix} 1 & 2 \\ 3 & 4 \end{smallmatrix}\right)$.

16. Let $A, B \in \mathbb{M}_{m\times n}$. Show that $A = B$ if and only if $\operatorname{vec}(A) = \operatorname{vec}(B)$.

17. Show that $\operatorname{vec}\colon X \mapsto \operatorname{vec}(X)$ is a linear map from $\mathbb{M}_{m\times n}(\mathbb{C})$ to $\mathbb{C}^{mn}$.

18. Find a matrix representation of the "vec" transformation $X \mapsto \operatorname{vec}(X)$ from $\mathbb{M}_2(\mathbb{R})$ to $\mathbb{R}^4$ with respect to the standard bases.

19. Let $A, C \in \mathbb{M}_{m\times n}$ and $B, D \in \mathbb{M}_{p\times q}$. Show that

$$(A \otimes B) \circ (C \otimes D) = (A \circ C) \otimes (B \circ D).$$

20. Let $A, B \in \mathbb{M}_n$, $x, y \in \mathbb{C}^n$. Prove $x^*(A \circ B)y = \operatorname{tr}(D_x^* A D_y B^T)$ and $(yx^*) \circ A = D_y A D_x^*$, where $D_z = \operatorname{diag}(z_1, \ldots, z_n)$ if $z = (z_1, \ldots, z_n)^T$.

21. Let $x \in \mathbb{C}^m$ and $y \in \mathbb{C}^n$, both in columns. Show that

$$x^T \otimes y = yx^T = y \otimes x^T.$$

22. Let $A \in \mathbb{M}_m$, $B \in \mathbb{M}_n$, $x \in \mathbb{C}^m$, and $y \in \mathbb{C}^n$ (in columns). Show that

$$(x^* \otimes y^*)(A \otimes B)(x \otimes y) = (x^*Ax)(y^*By).$$

23. Let $v \in \mathbb{C}^2$ be given. Define a map $\mathcal{L}$ from $\mathbb{C}^2$ to $\mathbb{M}_2$ by $\mathcal{L}(x) = x^T \otimes v$. Show that $\mathcal{L}$ is a linear map from $\mathbb{C}^2$ to $\mathbb{M}_2$. Find the image of $\mathcal{L}$.

24. Let $V = \mathbb{C}^2 \times \mathbb{C}^2$ (the Cartesian product; see Problem 31, Sect. 1.1). Define a map $\mathcal{T}$ from V to $\mathbb{M}_2$ by $\mathcal{T}(x, y) = x^T \otimes y$. Is $\mathcal{T}$ a linear map from V to $\mathbb{M}_2$? What are the elements in the image of $\mathcal{T}$?

25. Let x and y be linearly independent vectors in $\mathbb{C}^n$ (or matrices in $\mathbb{M}_n$). Show that $x \otimes x$, $x \otimes y$, $y \otimes x$, and $y \otimes y$ are linearly independent.

26. Let $\{e_1, \ldots, e_n\}$ be the standard basis of $\mathbb{R}^n$, i.e., e_i has 1 as the ith component, and all other components are 0, $i = 1, \ldots, n$. Show that $e_i \otimes e_j$, $1 \le i, j \le n$, comprise a basis of $\mathbb{R}^{n^2}$.

27. If $\lambda_1, \ldots, \lambda_n$ are the eigenvalues of $A \in \mathbb{M}_n$, what are the eigenvalues of the $(2n) \times (2n)$ matrices $\left(\begin{smallmatrix} A & 0 \\ 0 & 0 \end{smallmatrix}\right)$, $\left(\begin{smallmatrix} 0 & A \\ 0 & 0 \end{smallmatrix}\right)$, $\left(\begin{smallmatrix} 0 & A \\ A & 0 \end{smallmatrix}\right)$, and $\left(\begin{smallmatrix} A & A \\ A & A \end{smallmatrix}\right)$?

28. Let A be a square matrix whose eigenvalues are all real and denote by $\lambda_{\max}(A)$ the largest eigenvalue of A. If A is positive semidefinite, show that $\lambda_{\max}(A^2) = \lambda_{\max}^2(A)$ and that $\lambda_{\max}(A \otimes A) = \lambda_{\max}^2(A)$. Show by example that this need not be true for Hermitian matrices.

29. Let A and B be square matrices (of possibly different sizes). Show that

$$e^{A \otimes I} = e^A \otimes I,\ e^{I \otimes B} = I \otimes e^B,\ e^{A \oplus B} = e^A \oplus e^B,\ e^A \otimes e^B = e^{A \otimes I + I \otimes B}.$$

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4.4 Compound Matrices

We now turn our attention to *compound matrices*. The Kronecker product and the Hadamard product are operations on two (or more) matrices. Unlike the Kronecker and Hadamard products, "compounding" matrix is a matrix operation on a single matrix that arranges in certain order all minors (the determinants of submatrices) of a given size from the given matrix. The compound matrix has some important properties about the original matrix, and it serves as a useful tool in deriving matrix inequalities (in later chapters).

Let $A \in \mathbb{M}_{m\times n}$. Let $\alpha = \{i_1, \ldots, i_s\}$ and $\beta = \{j_1, \ldots, j_t\}$ be index sets, $1 \le i_1 < \cdots < i_s \le m$, $1 \le j_1 < \cdots < j_t \le n$. We denote by $A[i_1, \ldots, i_s|j_1, \ldots, j_t]$, or simply $A[\alpha|\beta]$, for the submatrix of A consisting of the entries in the intersections of rows $i_1, \ldots, i_s$ and columns $j_1, \ldots, j_t$.

Let k be a positive integer and $k \le \min\{m, n\}$. The $m \times n$ matrix A has $\binom{m}{k} \times \binom{n}{k}$ minors (scalars). We construct a matrix, denoted by $C_k(A)$ and called the *kth compound matrix* of A, of size $\binom{m}{k} \times \binom{n}{k}$ by ordering these minors lexicographically (for instance, $\{1, 2, 4\}$ is prior to $\{1, 2, 5\}$, which is prior to $\{1, 3, 4\}$), that is, we arrange the entries of $C_k(A)$ as follows:

$(1,1)$-entry: $\det A[1, \ldots, k|1, \ldots, k]$,

$(1,2)$-entry: $\det A[1, \ldots, k|1, \ldots, k-1, k+1]$ (if $k+1 \le \min\{m, n\}$),

$\vdots$ $\qquad$ $\vdots$

$(2,1)$-entry: $\det A[1, \ldots, k-1, k+1|1, \ldots, k]$,

$\vdots$ $\qquad$ $\vdots$

$\binom{m}{k} \times \binom{n}{k}$-entry: $\det A[m-k+1, \ldots, m-1, m|n-k+1, \ldots, n-1, n]$.

If $A \in \mathbb{M}_n$, it is clear that $C_1(A) = A$ and $C_n(A) = \det A$.

Example 4.4.1 Let $m = n = 3$, $k = 2$, and

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix}.$$

Then

$$\begin{aligned} C_2(A) &= \begin{pmatrix} \det A[1,2|1,2] & \det A[1,2|1,3] & \det A[1,2|2,3] \\ \det A[1,3|1,2] & \det A[1,3|1,3] & \det A[1,3|2,3] \\ \det A[2,3|1,2] & \det A[2,3|1,3] & \det A[2,3|2,3] \end{pmatrix} \\ &= \begin{pmatrix} \left|\begin{smallmatrix} 1 & 2 \\ 4 & 5 \end{smallmatrix}\right| & \left|\begin{smallmatrix} 1 & 3 \\ 4 & 6 \end{smallmatrix}\right| & \left|\begin{smallmatrix} 2 & 3 \\ 5 & 6 \end{smallmatrix}\right| \\ \left|\begin{smallmatrix} 1 & 2 \\ 7 & 8 \end{smallmatrix}\right| & \left|\begin{smallmatrix} 1 & 3 \\ 7 & 9 \end{smallmatrix}\right| & \left|\begin{smallmatrix} 2 & 3 \\ 8 & 9 \end{smallmatrix}\right| \\ \left|\begin{smallmatrix} 4 & 5 \\ 7 & 8 \end{smallmatrix}\right| & \left|\begin{smallmatrix} 4 & 6 \\ 7 & 9 \end{smallmatrix}\right| & \left|\begin{smallmatrix} 5 & 6 \\ 8 & 9 \end{smallmatrix}\right| \end{pmatrix} = \begin{pmatrix} -3 & -6 & -3 \\ -6 & -12 & -6 \\ -3 & -6 & -3 \end{pmatrix}. \end{aligned}$$

∎

Example 4.4.2 Let $m = 3$, $n = 4$, $k = 2$, and

$$A = \begin{pmatrix} 1 & 2 & 3 & 0 \\ 4 & 5 & 6 & 0 \\ 7 & 8 & 9 & 1 \end{pmatrix}.$$

Then

$$\begin{aligned} C_2(A) &= \begin{pmatrix} \left|\begin{smallmatrix}1&2\\4&5\end{smallmatrix}\right| & \left|\begin{smallmatrix}1&3\\4&6\end{smallmatrix}\right| & \left|\begin{smallmatrix}1&0\\4&0\end{smallmatrix}\right| & \left|\begin{smallmatrix}2&3\\5&6\end{smallmatrix}\right| & \left|\begin{smallmatrix}2&0\\5&0\end{smallmatrix}\right| & \left|\begin{smallmatrix}3&0\\6&0\end{smallmatrix}\right| \\ \left|\begin{smallmatrix}1&2\\7&8\end{smallmatrix}\right| & \left|\begin{smallmatrix}1&3\\7&9\end{smallmatrix}\right| & \left|\begin{smallmatrix}1&0\\7&1\end{smallmatrix}\right| & \left|\begin{smallmatrix}2&3\\8&9\end{smallmatrix}\right| & \left|\begin{smallmatrix}2&0\\8&1\end{smallmatrix}\right| & \left|\begin{smallmatrix}3&0\\9&1\end{smallmatrix}\right| \\ \left|\begin{smallmatrix}4&5\\7&8\end{smallmatrix}\right| & \left|\begin{smallmatrix}4&6\\7&9\end{smallmatrix}\right| & \left|\begin{smallmatrix}4&0\\7&1\end{smallmatrix}\right| & \left|\begin{smallmatrix}5&6\\8&9\end{smallmatrix}\right| & \left|\begin{smallmatrix}5&0\\8&1\end{smallmatrix}\right| & \left|\begin{smallmatrix}6&0\\9&1\end{smallmatrix}\right| \end{pmatrix} \\ &= \begin{pmatrix} -3 & -6 & 0 & -3 & 0 & 0 \\ -6 & -12 & 1 & -6 & 2 & 3 \\ -3 & -6 & 4 & -3 & 5 & 6 \end{pmatrix}. \blacksquare \end{aligned}$$

For convenience, we say that the minor $\det A[\alpha, \beta]$ is in the (α, β) position of the compound matrix $C_k(A)$ and denote it by $C_k(A)_{\alpha,\beta}$.

Let $A \in \mathbb{M}_n$. The diagonal entries of $C_k(A)$ are $\det A[\alpha|\alpha]$, i.e., the principal minors of A. For an n-square upper-triangular matrix A, $\det A[\alpha|\beta] = 0$ if α is after β in lexicographic order. This leads to the result that if A is upper (lower)-triangular, then so is $C_k(A)$. Thus, if A is diagonal, then so is $C_k(A)$. In addition, $C_k(I_n) = I_{\binom{n}{k}}$, $C_k(A^T) = \big(C_k(A)\big)^T$, and $C_k(A^*) = \big(C_k(A)\big)^*$.

Example 4.4.3 Let

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 0 & 5 & 6 \\ 0 & 0 & 9 \end{pmatrix}.$$

Then $C_1(A) = A$, $C_3(A) = 45$, and

$$C_2(A) = \begin{pmatrix} 5 & 6 & -3 \\ 0 & 9 & 18 \\ 0 & 0 & 45 \end{pmatrix}. \blacksquare$$

We show that the compound matrix of the product of matrices is the product of their compound matrices. For this purpose, we introduce, without proof, the *Binet-Cauchy formula*[1] for determinant expansion. The Binet-Cauchy formula is a generalization of the Laplace expansion formula (Sect. 1.2) by taking $\alpha = \{i\}$ and $\beta = \{j\}$ and it is used only once in this section.

Theorem 4.4.1 (Binet-Cauchy Formula) *Let $C = AB$, where $A \in \mathbb{M}_{m\times n}$ and $B \in \mathbb{M}_{n\times m}$, $m \le n$, and let $\alpha = \{1, 2, \ldots, m\}$. Then*

$$\det C = \sum\nolimits_{\beta} \det A[\alpha|\beta] \det B[\beta|\alpha],$$

where β runs over all sequences $\{j_1, \ldots, j_m\}$, $1 \le j_1 < \cdots < j_m \le n$.

[1] P. Lancaster and M. Tismenetsky, *The Theory of Matrices*, Academic Press, 2nd Ed., 1985, pp. 36–42.

The following multiplicative theorem is of fundamental importance for compound matrices; it plays an important role in deriving some matrix inequalities involving eigenvalues and singular values in product forms.

Theorem 4.4.2 *Let A be an $m \times p$ matrix and let B be a $p \times n$ matrix. Then $C_k(AB) = C_k(A)C_k(B)$, where k is a positive integer and $k \leq \min\{m, p, n\}$.*

Proof. For $\alpha = \{i_1, \ldots, i_k\}$ and $\beta = \{j_1, \ldots, j_k\}$, $1 \leq i_1 < \cdots < i_k \leq m$, $1 \leq j_1 < \cdots < j_k \leq m$, we compute the entry in the (α, β) position (in lexicographic order) of $C_k(AB)$ by the Binet-Cauchy formula and get

$$\begin{aligned} C_k(AB)_{\alpha,\beta} &= \det\big((AB)[\alpha|\beta]\big) \\ &= \sum\nolimits_{\gamma} \det A[\alpha|\gamma] \det B[\gamma|\beta] \\ &= \big(C_k(A)C_k(B)\big)_{\alpha,\,\beta} \end{aligned}$$

where γ runs over all possible sequences $1 \leq \gamma_1 < \cdots < \gamma_k \leq p$. ∎

A few important facts follow:

(a) If A is invertible, then so is $C_k(A)$, and $C_k(A^{-1}) = (C_k(A))^{-1}$.
(b) If U is unitary, then so is $C_k(U)$, and $C_k(U^{-1}) = (C_k(U))^{-1}$.
(c) If $A = U^*TU$ is in Schur form, then $C_k(A) = (C_k(U))^* C_k(T) C_k(U)$.
(d) If $A = U\Sigma V$ is in SVD form, then $C_k(A) = (C_k(U))^* C_k(\Sigma) C_k(V)$.
(e) If A is normal, Hermitian, or positive semidefinite, then so is $C_k(A)$.

By the above (c), we obtain the following result.

Theorem 4.4.3 *Let $\lambda_1, \ldots, \lambda_n$ be the eigenvalues of $A \in \mathbb{M}_n$. Then the $\binom{n}{k}$ eigenvalues of $C_k(A)$ are the products $\lambda_{i_1} \cdots \lambda_{i_k}$, $1 \leq i_1 < \cdots < i_k \leq n$.*

Proof. Let $A = U^*TU$ be a Schur decomposition of $A \in \mathbb{M}_n$ (Theorem 3.2.1), where $T \in \mathbb{M}_n$ is an upper-triangular matrix with $\lambda_1, \ldots, \lambda_n$ on the diagonal and $U \in \mathbb{M}_n$ is a unitary matrix. Then

$$C_k(A) = C_k(U^*TU) = (C_k(U))^* C_k(T) C_k(U).$$

Since $C_k(U)$ is unitary, $(C_k(U))^* = (C_k(U))^{-1}$. The conclusion follows by noticing that $C_k(T)$ is upper-triangular with $\lambda_{i_1} \cdots \lambda_{i_k}$ on the diagonal. ∎

Example 4.4.4 Let

$$A = \begin{pmatrix} \lambda_1 & 2 & 3 \\ 0 & \lambda_2 & 0 \\ 0 & 4 & \lambda_3 \end{pmatrix}.$$

The eigenvalues of A are λ_1, λ_2, λ_3; the eigenvalues of $C_2(A)$ are $\lambda_1\lambda_2$, $\lambda_1\lambda_3$, $\lambda_2\lambda_3$. Moreover, $\det A = \lambda_1\lambda_2\lambda_3$ and $\det C_2(A) = (\lambda_1\lambda_2\lambda_3)^2 = (\det A)^2$. ∎

Theorem 4.4.4 *Let A be an n-square positive semidefinite matrix with eigenvalues $\lambda_1(A) \geq \cdots \geq \lambda_n(A)$. Then the largest eigenvalue $\lambda_{\max}(C_k(A))$ of $C_k(A)$ is the product of the first k largest eigenvalues of A, that is,*

$$\lambda_{\max}\big(C_k(A)\big) = \prod_{i=1}^{k} \lambda_i(A).$$

Proof. This follows from Theorem 4.4.3. If $A \geq 0$, then so is $C_k(A)$. Note that if $A = U^*TU$ is a spectral decomposition of A, then $C_k(T)$ is nonnegative diagonal with the largest diagonal entry $\lambda_1(A) \cdots \lambda_k(A)$. ∎

Problems

1. Let $a, b, c, d \in \mathbb{C}$, and let

$$A = \begin{pmatrix} 1 & a & 0 \\ 0 & 1 & b \\ c & d & 1 \end{pmatrix}, \quad B = \begin{pmatrix} a & 1 & 1 & 1 \\ 0 & b & 0 & 0 \\ 0 & 1 & c & 1 \\ 0 & 1 & 1 & d \end{pmatrix}.$$

   Compute all six 2×2 minors of A, $\det B[2,3|2,3]$, and $\det B[1,2,3|2,3,4]$.

2. Find $C_2(A)$, where

$$A = \begin{pmatrix} 0 & 1 & 2 & 3 \\ 4 & 5 & 6 & 7 \\ 8 & 9 & 10 & -1 \end{pmatrix}.$$

3. Let $A \in \mathbb{M}_{m\times n}$ and $c \in \mathbb{C}$. Show that $C_k(cA) = c^k C_k(A)$.

4. Show that $(A^*)[\alpha|\beta] = (A[\beta|\alpha])^*$.

5. Let A be an $m \times n$ matrix. Show that $A[\alpha|\beta] = E_\alpha^T A E_\beta$, where $\alpha = \{i_1, \ldots, i_s\}$ and $\beta = \{j_1, \ldots, j_t\}$ are index sets, $s \leq m$, $t \leq n$, E_α is the $m \times s$ matrix whose rows $i_1, \ldots, i_s$ form the identity matrix I_s and whose other entries are 0's, and E_β is the $n \times t$ matrix whose rows $j_1, \ldots, j_t$ form the identity matrix I_t and whose other entries are 0's.

6. Show that $C_k(0_n) = 0_m$ and $C_k(I_n) = I_m$, where $m = \binom{n}{k}$, $1 \leq k \leq n$.

7. Show that $C_k(A^*) = (C_k(A))^*$ and $C_k(A^T) = (C_k(A))^T$.

8. Show that $C_k(A^{-1}) = (C_k(A))^{-1}$ if A is nonsingular.

9. Compute M^{-1}, $C_2(M)$, and $(C_2(M))^{-1}$, where

$$M = \begin{pmatrix} 3 & -2 & -1 \\ -1 & 4 & -2 \\ -1 & -2 & 4 \end{pmatrix}.$$

10. Show that $\det C_k(A) = (\det A)^{\binom{n-1}{k-1}}$ if A is n-square.

11. Let $A \in \mathbb{M}_{m\times n}$ and $\operatorname{rank}(A) = r$. Show that $\operatorname{rank}(C_k(A)) = \binom{r}{k}$ if $k < r$, $\operatorname{rank}(C_k(A)) = 0$ if $k > r$, and $\operatorname{rank}(C_k(A)) = 1$ if $k = r$.

12. Show that if A is unitary, symmetric, positive (semi-)definite, Hermitian, or normal, then so is $C_k(A)$, respectively.

13. If $A = \operatorname{diag}(a_1, \ldots, a_n)$, show that $C_k(A)$ is an $\binom{n}{k} \times \binom{n}{k}$ diagonal matrix with diagonal entries $a_{i_1} \cdots a_{i_k}$, $1 \le i_1 < \cdots < i_k \le n$.

14. If $A \in \mathbb{M}_n$ has singular values $\sigma_1, \ldots, \sigma_n$, show that $C_k(A)$ has singular values $\sigma_{i_1} \cdots \sigma_{i_k}$, $1 \le i_1 < \cdots < i_k \le n$.

15. If $1, 2, 3$ are the eigenvalues of a 3×3 matrix A, what are the eigenvalues of $C_2(A)$? Find $\operatorname{tr} A$, $\det A$, $\operatorname{tr}(C_2(A))$, and $\det(C_2(A))$.

16. If $A \in \mathbb{M}_n$ has eigenvalues $\lambda_1, \ldots, \lambda_n$, show that $\operatorname{tr}(C_k(A))$ equals $\sum_\gamma \lambda_{i_1} \cdots \lambda_{i_k}$, denoted by $s_k(\lambda_1, \ldots, \lambda_n)$ and called kth *elementary symmetric function*, where γ is any sequence $1 \le i_1 < \cdots < i_k \le n$.

17. If A is an n-square positive semidefinite matrix with eigenvalues $\lambda_1 \ge \cdots \ge \lambda_n$, show that $C_k(A)$ is also positive semidefinite, and the smallest eigenvalue of $C_k(A)$ is $\lambda_{\min}(C_k(A)) = \prod_{i=1}^k \lambda_{n-i+1}$.

18. Let A be an $m \times n$ matrix. Show that for $k \le \min\{m, n\}$,

$$\sigma_{\max}\big(C_k(A)\big) = \prod_{i=1}^{k} \sigma_i(A).$$

19. Let $A \in \mathbb{M}_n$ and let $\lambda_1, \ldots, \lambda_k$ be any k eigenvalues of A. Show that

$$\Big|\prod_{i=1}^{k} \lambda_i\Big| \le \prod_{i=1}^{k} \sigma_i(A)$$

for $k = 1, 2, \ldots, n$ and that the inequality becomes an equality if $k = n$.

20. Let $\left(\begin{smallmatrix} A & B \\ B^* & C \end{smallmatrix}\right) \ge 0$ (namely, positive semidefinite), where $A \in \mathbb{M}_m$, $B \in \mathbb{M}_{m\times n}$, and $C \in \mathbb{M}_n$. Show that $\left(\begin{smallmatrix} C_k(A) & C_k(B) \\ C_k(B^*) & C_k(C) \end{smallmatrix}\right) \ge 0$ for $k \le \min\{m, n\}$.

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CHAPTER 5

Special Types of Matrices

Introduction: This chapter studies a few special types of matrices. They are: idempotent matrices (projections), nilpotent matrices, involutory matrices, orthogonal projection matrices, tridiagonal matrices, circulant matrices, Vandermonde matrices, Hadamard matrices, permutation matrices, doubly stochastic matrices, and nonnegative matrices. These matrices are often used in many subjects of mathematics as well as in other fields.

5.1 Idempotence, Nilpotence, Involution, and Projections

We first present three types of matrices that have simple structures under similarity: idempotent (projection) matrices, nilpotent matrices, and involutory matrices. We then turn attention to orthogonal projection matrices.

A square matrix A is said to be *idempotent*, or called a *projection*, if

$$A^2 = A \quad \text{(idempotence)},$$

nilpotent if for some positive integer k

$$A^k = 0 \quad \text{(nilpotence)},$$

and *involutory* if

$$A^2 = I \quad \text{(involution)}.$$

Similar terms are defined for linear transformations.

Example 5.1.1 Idempotent, nilpotent, and involutory matrices.

(a) 0_n and I_n are idempotent, so are $\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$, $\frac{1}{2}\begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}$, and $\begin{pmatrix} 1 & i \\ 0 & 0 \end{pmatrix}$.

(b) 0_n, $\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$, $\begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$, $\begin{pmatrix} 1 & -1 \\ 1 & -1 \end{pmatrix}$, $\begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}$, and $\begin{pmatrix} 0 & 1 & 2 \\ 0 & 0 & 0 \\ 0 & 3 & 0 \end{pmatrix}$ are nilpotent.

(c) I_n, $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$, $\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$, $\frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$, and $\begin{pmatrix} 1 & 0 \\ x & -1 \end{pmatrix}$ $(x \in \mathbb{C})$ are involutory.

(d) A Jordan block with zero diagonal entries is nilpotent.

(e) If all eigenvalues of $A \in \mathbb{M}_n$ are equal to zero, then A is nilpotent. ∎

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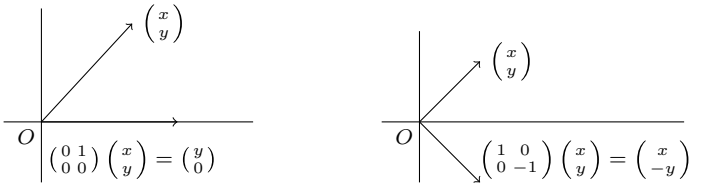


Fig. 5.1. Nilpotence and involution

Theorem 5.1.1 *Let $A \in \mathbb{M}_n$. The following statements hold.*

(1) *A is idempotent if and only if A is similar to* $\operatorname{diag}(1, \ldots, 1, 0, \ldots, 0)$.
(2) *A is nilpotent if and only if the eigenvalues of A are all equal to zero.*
(3) *A is involutory if and only if A is similar to* $\operatorname{diag}(1, \ldots, 1, -1, \ldots, -1)$.

[Note: In (1) and (3), 1, 0, and -1 are allowed to be absent.]

Proof. The sufficiency in (1) is obvious. To see the necessity, let

$$A = P^{-1}(J_1 \oplus \cdots \oplus J_k)P$$

be a Jordan decomposition of A. Then for each i, $i = 1, \ldots, k$,

$$A^2 = A \quad \Rightarrow \quad J_i^2 = J_i.$$

Observe that if J is a Jordan block and if $J^2 = J$, then J has order 1, that is, J is a scalar, 1 or 0. Thus, each J_i is either 1 or 0. It follows that A is similar to a diagonal matrix with 1's and (or) 0's on the diagonal.

For (2), consider the Schur (or Jordan) decomposition of A,

$$A = U^{-1} \begin{pmatrix} \lambda_1 & & \star \\ & \ddots & \\ 0 & & \lambda_n \end{pmatrix} U, \quad (\star \text{ for unspecified entries})$$

where $U \in \mathbb{M}_n$ is unitary, and λ_i's are the eigenvalues of A.

If $A^k = 0$, then each $\lambda_i^k = 0$, and A has only zero eigenvalues. Conversely, it is easy to verify by computation that $A^n = 0$ if all the eigenvalues λ_i's of A are equal to zero (see also Problem 13, Sect. 3.3).

The proof of (3) is similar to that of (1). ∎

Theorem 5.1.2 *Let A and B be nilpotent matrices of the same size. If A and B commute, i.e., $AB = BA$, then AB and $A + B$ are nilpotent.*

Proof. Let $A^m = 0$ and $B^n = 0$, where m and n are some positive integers. Then $A^p = 0$, $B^q = 0$ for all $p \geq m$, $q \geq n$. Thus, $(AB)^k = A^k B^k = 0$, where $k = m + n$, and AB is nilpotent. For the sum, computations show

$$(A + B)^{m+n} = 0,$$

because a term in the expansion of $(A+B)^{m+n}$ is A^{m+n}, B^{m+n}, or A^pB^q, $p+q=m+n$ (so, $p \geq m$ or $q \geq n$). In any case, every term vanishes. ∎

Recall from Sect. 1.3, Chap. 1, that the matrices of a linear transformation with respect to different bases of a (finite-dimensional) vector space are similar. Consider an n-square matrix as a linear map on $\mathbb{C}^n$. We can interpret Theorem 5.1.1 (1) as follows. A matrix $A \in \mathbb{M}_n$ is a projection if and only if there exists a basis of $\mathbb{C}^n$ such that the matrix representation of A with respect to the basis is $\mathrm{diag}(1,\ldots,1,0,\ldots,0)$. This is the same as saying that $\mathbb{C}^n$ can be decomposed (via a suitable basis of $\mathbb{C}^n$) as

$$\mathbb{C}^n = W_1 \oplus W_2, \tag{5.1}$$

where W_1 and W_2 are subspaces such that for all $w_1 \in W_1$ and $w_2 \in W_2$,

$$Aw_1 = w_1, \quad Aw_2 = 0.$$

Thus, if $v = v_1 + v_2 \in \mathbb{C}^n$, where $v_1 \in W_1$ and $v_2 \in W_2$, then

$$Av = Av_1 + Av_2 = v_1.$$

Such a vector v_1 in W_1 is called the *projection* of v onto W_1. Note that

$$W_1 = \mathrm{Im}\, A, \quad W_2 = \mathrm{Ker}\, A = \mathrm{Im}(I-A).$$

Theorem 5.1.3 *Let $A \in \mathbb{M}_n$. The following statements are equivalent.*

(1) *A is a projection (idempotent) matrix, that is, $A^2 = A$.*
(2) *$A = P^{-1}\left(\begin{smallmatrix} I_r & 0 \\ 0 & 0 \end{smallmatrix}\right) P$ for some invertible $P \in \mathbb{M}_n$, where $r = \mathrm{rank}(A)$.*
(3) *$\mathbb{C}^n = \mathrm{Im}\, A + \mathrm{Ker}\, A$ and $Ax = x$ for every $x \in \mathrm{Im}\, A$.*
(4) *$\mathrm{Ker}\, A = \mathrm{Im}(I-A)$.*
(5) *$\mathrm{rank}(A) + \mathrm{rank}(I-A) = n$.*
(6) *$\mathrm{Im}\, A \cap \mathrm{Im}(I-A) = \{0\}$.*

Proof. (1)⇔(2): This is a restatement of Theorem 5.1.1 (1).

We show below that (1) is equivalent to all other statements.

(1) ⇒ (3): For any $z \in \mathbb{C}^n$, $z = Az + (z - Az)$, where $Az \in \mathrm{Im}\, A$ and $z - Az \in \mathrm{Ker}\, A$. So, $\mathbb{C}^n = \mathrm{Im}\, A + \mathrm{Ker}\, A$. Moreover, if $x \in \mathrm{Im}\, A$, by letting $x = Ay$, then $Ax = A^2y = Ay = x$, i.e., A is the identity map on $\mathrm{Im}\, A$.

(3) ⇒ (4): $\mathrm{Im}(I-A) = \{(I-A)z : z \in \mathbb{C}^n\} = \{z - Az : z \in \mathbb{C}^n\}$. Because A is the identity map on $\mathrm{Im}\, A$, $A(z - Az) = 0$. So, $\mathrm{Im}(I-A) \subseteq \mathrm{Ker}\, A$.

For the reverse inclusion, if $v \in \mathrm{Ker}\, A$, then $Av = 0$, and $v = v - Av = (I-A)v \in \mathrm{Im}(I-A)$. It follows that $\mathrm{Ker}\, A = \mathrm{Im}(I-A)$.

(4) ⇒ (5): This is because the rank of A is equal to the dimension of $\mathrm{Im}\, A$, and $\dim \mathrm{Im}\, A + \dim \mathrm{Ker}\, A = n$ (see Theorems 1.2.2 and 1.3.1, Chap. 1).

(5) $\Rightarrow$ (6): (5) says that $\dim \operatorname{Im} A + \dim \operatorname{Im}(I-A) = n$. By the dimension identity (Theorem 1.1.1), the intersection of $\operatorname{Im} A$ and $\operatorname{Im}(I-A)$ is $\{0\}$.

(6) $\Rightarrow$ (1): If $A^2 \neq A$, then $A(I-A) = (I-A)A \neq 0$. It follows that $A(I-A)w = (I-A)Aw \neq 0$ for some $w \in \mathbb{C}^n$. However, $A(I-A)w \in \operatorname{Im} A$ and $(I-A)Aw \in \operatorname{Im}(I-A)$, contradicting $\operatorname{Im} A \cap \operatorname{Im}(I-A) = \{0\}$. ∎

A projection matrix $A \in \mathbb{M}_n$ is further called an *orthogonal projection* if

$$A^2 = A = A^*.$$

For orthogonal projection matrices, the subspaces (of $\mathbb{C}^n$)

$$W_1 = \operatorname{Im} A \quad \text{and} \quad W_2 = \operatorname{Im}(I-A) = \operatorname{Ker} A$$

in (5.1) are orthogonal to each other, that is, for all $w_1 \in W_1$ and $w_2 \in W_2$,

$$\langle w_1, w_2 \rangle = 0. \tag{5.2}$$

In other words, $\langle Ax, (I-A)x \rangle = 0$ for all $x \in \mathbb{C}^n$, this is because

$$\langle Ax, (I-A)x \rangle = \langle x, A^*(I-A)x \rangle = \langle x, A(I-A)x \rangle = \langle x, (A-A^2)x \rangle = 0.$$

Example 5.1.2 Projection and orthogonal projection.

(a) $\left(\begin{smallmatrix} 1 & 0 \\ 1 & 0 \end{smallmatrix}\right)$ is a projection, but is not an orthogonal projection.
(b) $\left(\begin{smallmatrix} 1 & 0 \\ 0 & 0 \end{smallmatrix}\right)$ is an orthogonal projection (it is of course a projection).
(c) $\frac{1}{2}\left(\begin{smallmatrix} 1 & i \\ -i & 1 \end{smallmatrix}\right)$ is an orthogonal projection. ∎

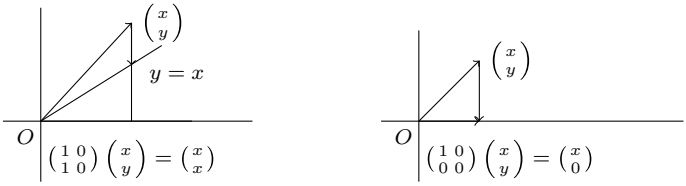


Fig. 5.2. Projection and orthogonal projection

Theorem 5.1.4 *Let $A \in \mathbb{M}_n$. The following statements are equivalent.*

(1) *A is an orthogonal projection matrix, that is, $A^2 = A = A^*$.*
(2) *$A = U^* \left(\begin{smallmatrix} I_r & 0 \\ 0 & 0 \end{smallmatrix}\right) U$ for some unitary $U \in \mathbb{M}_n$, where $r = \operatorname{rank}(A)$.*
(3) *A is a projection, and $\operatorname{Im} A$ and $\operatorname{Ker} A$ are orthogonal.*
(4) *$\|x - Ax\| \leq \|x - Ay\|$ for all $x, y \in \mathbb{C}^n$.*
(5) *$A^2 = A$ and $\|Ax\| \leq \|x\|$ for all $x \in \mathbb{C}^n$.*
(6) *$A = A^*A$.*
(7) *A is positive semidefinite and its eigenvalues are 1's and 0's.*

Proof. We first show (1) ⇒ (2). The direction (2) ⇒ (1) is obvious.

Since A is Hermitian, the spectral decomposition theorem (Theorem 3.2.2) ensures $A = V^* \operatorname{diag}(\lambda_1, \ldots, \lambda_n)V$ for some unitary matrix V, where the λ_i's are the eigenvalues of A. Since A is idempotent, it has only eigenvalues ones and zeros according to Theorem 5.1.3 (1). It follows that

$$A = U^* \operatorname{diag}(\overbrace{1, \ldots, 1}^{r}, 0, \ldots, 0)U = U^* \begin{pmatrix} I_r & 0 \\ 0 & 0 \end{pmatrix} U,$$

where r is the rank of A and U is some n-square unitary matrix.

(1) ⇒ (3): This is because (5.2). For (3) ⇒ (1), we show $A - A^* = 0$. Since A is a projection, $\mathbb{C}^2 = \operatorname{Im} A + \operatorname{Ker} A$. For any $z \in \mathbb{C}^n$, write $z = z_1 + z_2$, where $z_1 \in \operatorname{Im} A$ and $z_2 \in \operatorname{Ker} A$. Since $A^2 = A$, $Az = Az_1 = z_1$. If $\operatorname{Im} A$ and $\operatorname{Ker} A$ are orthogonal, we derive for all $x, y \in \mathbb{C}^n$,

$$\begin{aligned} \langle (A - A^*)x, y\rangle &= \langle Ax, y\rangle - \langle A^*x, y\rangle = \langle Ax, y\rangle - \langle x, Ay\rangle \\ &= \langle x_1, y_1 + y_2\rangle - \langle x_1 + x_2, y_1\rangle = \langle x_1, y_1\rangle - \langle x_1, y_1\rangle = 0. \end{aligned}$$

It follows that $A - A^* = 0$, that is, A is Hermitian.

(3) ⇒ (4): Let A be a projection with $\operatorname{Im} A$ and $\operatorname{Ker} A$ being orthogonal. Let $x = x_1 + x_2$, where $x_1 \in \operatorname{Im} A$, $x_2 \in \operatorname{Ker} A$. Then $\langle x_1, x_2\rangle = 0$, $Ax = Ax_1 = x_1$. Similarly, for $y = y_1 + y_2$, $Ay = y_1$. Note that $x_1 - y_1 \in \operatorname{Im} A$.

Since $\langle u, v\rangle = 0$ implies $\|u\|^2 + \|v\|^2 = \|u + v\|^2$, we have

$$\|x - Ax\|^2 = \|x_2\|^2 \le \|x_2\|^2 + \|x_1 - y_1\|^2 = \|x_2 + (x_1 - y_1)\|^2 = \|x - Ay\|^2.$$

[Note: One may draw a graph on the complex plane to illustrate (4).]

(4) ⇒ (3): Because $v = Av + (I - A)v$ for every $v \in \mathbb{C}^n$, we have

$$\mathbb{C}^n = \operatorname{Im} A + \operatorname{Im}(I - A).$$

We show that $\langle u, v\rangle = 0$ for all $u \in \operatorname{Im} A$ and $v \in \operatorname{Im}(I - A)$, resulting in $A(I - A) = 0$. Suppose otherwise that $\langle (I - A)x, Ay\rangle \neq 0$ for some x and $y \in \mathbb{C}^n$. Let $\langle (I - A)x, Ay\rangle = a < 0$ (or replace x with $e^{i\theta}x$, where θ is real such that $e^{i\theta}a < 0$). Set $z_\epsilon = x - \epsilon\, y$, where $\epsilon > 0$. Then

$$\begin{aligned} \|x - Az_\epsilon\|^2 &= \|(x - Ax) + (Ax - Az_\epsilon)\|^2 \\ &= \|x - Ax\|^2 + \|Ax - Az_\epsilon\|^2 + 2\,\Re\langle (I - A)x, A(x - z_\epsilon)\rangle \\ &= \|x - Ax\|^2 + \|\epsilon Ay\|^2 + 2\epsilon\langle (I - A)x, Ay\rangle \\ &= \|x - Ax\|^2 + \epsilon^2\|Ay\|^2 + 2\epsilon a. \end{aligned}$$

Because $a < 0$, we have $\epsilon^2\|Ay\|^2 + 2\epsilon a < 0$ for ϵ small enough, i.e., $\|x - Az_\epsilon\| < \|x - Ax\|$, a contradiction to the assumption (4).

Now that $\operatorname{Im} A$ and $\operatorname{Im}(I - A)$ are orthogonal. For any $w \in \mathbb{C}^n$, $A(I - A)w = (I - A)Aw \in \operatorname{Im} A \cap \operatorname{Im}(I - A)$. It follows that $A(I - A) = 0$, i.e.,

A is a projection. Moreover, $\mathrm{Im}(I-A) = \mathrm{Ker}\, A$, because $A(I-A)w = 0 \Rightarrow \mathrm{Im}(I-A) \subseteq \mathrm{Ker}\, A$, and if $Az = 0$, then $z = z - Az = (I-A)z \in \mathrm{Im}(I-A)$.

(1) $\Rightarrow$ (5): Since $A^2 = A = A^*$, $\langle Ax, (I-A)x\rangle = 0$ for all $x \in \mathbb{C}^n$. Thus,

$$\|Ax\|^2 \leq \|Ax\|^2 + \|(I-A)x\|^2 = \|Ax + (I-A)x\|^2 = \|x\|^2.$$

(5) $\Rightarrow$ (6): First note that $\|Ax\| \leq \|x\|$ for all $x \in \mathbb{C}^n$ implies $\|A^*y\| \leq \|y\|$ for all $y \in \mathbb{C}^n$ (because A and A^* have the same spectral norm).

If $A \neq A^*A$, then $Az \neq A^*Az$ for some $z \in \mathbb{C}^n$. We compute

$$\begin{aligned} \|Az - A^*Az\|^2 &= \|Az\|^2 + \|A^*Az\|^2 - \langle Az, A^*Az\rangle - \langle A^*Az, Az\rangle \\ &= \|Az\|^2 + \|A^*Az\|^2 - \langle A^2z, Az\rangle - \langle Az, A^2z\rangle \\ &= \|Az\|^2 + \|A^*Az\|^2 - \langle Az, Az\rangle - \langle Az, Az\rangle \\ &= \|A^*Az\|^2 - \|Az\|^2 > 0 \ \ (\text{as } Az - A^*Az \neq 0), \end{aligned}$$

implying $\|Az\| < \|A^*Az\|$, and $\|y\| < \|A^*y\|$ for $y = Az$, a contradiction.

(6) $\Rightarrow$ (1): If $A = A^*A$, then $A^* = (A^*A)^* = A^*A = A$, and

$$A = A^*A = AA = A^2.$$

(7) is obviously equivalent to (2) (via spectral decomposition). ∎

Problems

1. Characterize all 2×2 idempotent, nilpotent, involutory, and orthogonal projection matrices up to similarity (over the complex number field $\mathbb{C}$).
2. Can a nonzero matrix be both idempotent and nilpotent?
3. What is the characteristic polynomial of a nilpotent matrix?
4. What idempotent matrices are nonsingular?
5. Find an example: A and B are nilpotent matrices but $A + B$ is not.
6. If $A \in \mathbb{M}_n$ is nilpotent, show that $\mathrm{tr}A = 0$. Is the converse true?
7. Let $x, y \in \mathbb{R}$. If $A = \left(\begin{smallmatrix} x & 0 \\ y & 0 \end{smallmatrix}\right)$ is a projection matrix, what are the possible values of x and y? And when is A an orthogonal projection matrix?
8. Prove: If $A \in \mathbb{M}_n$ is idempotent, so is $P^{-1}AP$ for invertible $P \in \mathbb{M}_n$.
9. Show that the rank of an idempotent matrix is equal to the number of nonzero eigenvalues of the matrix.
10. Let A and B be $n \times n$ nonsingular idempotent matrices such that ABA is idempotent. Show that B is the inverse of A.

11. Let $A \in \mathbb{M}_n$. Show that (a) if A is idempotent, so is $I - A$, (b) $I + A$ is not impotent unless $A = 0$, (c) $\frac{1}{2}(I + A)$ is idempotent if and only if A is involutory, and (d) $I - A$ is nilpotent if all eigenvalue of A are 1's.

12. Let $A \in \mathbb{M}_n$. Show that $A^2 = A$ if and only if $\text{rank}(A) = \text{tr}A$ and $\text{rank}(I - A) = \text{tr}(I - A)$. If $A^2 + A = 0$, show that $\text{rank}(A) + \text{tr}A = 0$.

13. Let A be an idempotent matrix (i.e., a projection, $A^2 = A$). Show that

$$A = A^* \quad \Leftrightarrow \quad \text{Im}\, A = \text{Im}\, A^*.$$

Is this true if the condition $A^2 = A$ is dropped?

14. Show that a Hermitian idempotent matrix is positive semidefinite and that $I_n - \frac{1}{y^*y} yy^*$ is positive semidefinite for any nonzero $y \in \mathbb{C}^n$.

15. Show that $\mathcal{T}$ is a nilpotent linear map, where $\mathcal{T}$ is defined on $\mathbb{M}_2$ by

$$\mathcal{T}(X) = TX - XT, \quad \text{where } T = \left(\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right), \ X \in \mathbb{M}_2.$$

Find the Jordan form of a matrix representation of $\mathcal{T}$ (under a basis).

16. Let $A \in \mathbb{M}_n$. Define a linear transformation on $\mathbb{M}_n$ by

$$\mathcal{T}(X) = AX - XA, \quad X \in \mathbb{M}_n.$$

Show that if A is nilpotent then $\mathcal{T}$ is nilpotent and that if A is diagonalizable then the matrix representation of $\mathcal{T}$ is diagonalizable.

17. Let A and B be square matrices of the same size. If A is nilpotent and $AB = BA$, show that AB is nilpotent. Is the converse true?

18. Let A be an n-square nonsingular matrix. If $X \in \mathbb{M}_n$ and $\lambda \in \mathbb{C}$ are such that $AXA^{-1} = \lambda X$, show that $|\lambda| = 1$ or X is nilpotent.

19. Find a 2×2 matrix such that $A^2 = I$ but $A^*A \neq I$.

20. Let $A^2 = A$. Show that $(A + I)^k = I + (2^k - 1)A$ for $k = 1, 2, \ldots$.

21. Find a matrix that is a projection but not an orthogonal projection.

22. Let A and B be square matrices of the same size. If $AB = A$ and $BA = B$, show that A and B are projection matrices.

23. Let A be a projection matrix. Show that A is Hermitian if and only if $\text{Im}\, A$ and $\text{Ker}\, A$ are orthogonal, that is, $\text{Im}\, A \perp \text{Ker}\, A$.

24. Prove Theorem 5.1.3 along the line: $(1) \Leftrightarrow (3) \Rightarrow (4) \Rightarrow (5) \Rightarrow (6) \Rightarrow (3)$.

25. Let $A \in \mathbb{M}_n$. Show that A is an orthogonal projection if and only if $A = B^*B$ for some $B \in \mathbb{M}_{r \times n}$ satisfying $BB^* = I_r$, where $r = \text{rank}(A)$.

26. Let $A_1, \ldots, A_m$ be $n \times n$ idempotent matrices. If

$$A_1 + \cdots + A_m = I_n,$$

show that

$$A_i A_j = 0, \quad i \neq j.$$

[Hint: Show that $\mathbb{C}^n = \operatorname{Im} A_1 \oplus \cdots \oplus \operatorname{Im} A_m$ by using trace.]

27. Let $\mathcal{L}$ be a linear transformation on a finite-dimensional vector space V. If $\mathcal{L}^2 = \mathcal{L}$, then $\mathcal{L}$ is said to be a *projection* (onto its range $\operatorname{Im} \mathcal{L}$).

(a) Show that $V = \operatorname{Im}(\mathcal{L}) \oplus \operatorname{Ker}(\mathcal{L})$.

(b) Show that for every $v \in V$, there exist unique $w \in \operatorname{Im}(\mathcal{L})$ (the projection of v onto $\operatorname{Im}(\mathcal{L})$) and $u \in \operatorname{Ker}(\mathcal{L})$ such that $v = w + u$.

(c) Show that there exists a basis of V such that the matrix representation of $\mathcal{L}$ with respect to the basis is $\operatorname{diag}(1, \ldots, 1, 0, \ldots, 0)$.

(d) Show that if A is a matrix representation of $\mathcal{L}$, then $A^2 = A$.

(e) Show that the converse of (d) is true, i.e., $A^2 = A \Rightarrow \mathcal{L}^2 = \mathcal{L}$.

(f) If $V = \operatorname{Im}(\mathcal{L}) \oplus K$, where K is a subspace of V, is $K = \operatorname{Ker}(\mathcal{L})$?

(g) Is $V = \operatorname{Im}(\mathcal{T}) \oplus \operatorname{Ker}(\mathcal{T})$ for all linear transformations $\mathcal{T}$ on V?

28. Let W be a subspace of a finite-dimensional vector space V and let $V = W \oplus U$, where U is some subspace of V. Define $\mathcal{T}(v) = w$ if $v = w + u$, $w \in W, u \in U$. Show that $\mathcal{T}$ is a linear transformation on V and that $\mathcal{T}$ is a projection onto W, that is, $\mathcal{T}^2 = \mathcal{T}$ and $\operatorname{Im}(\mathcal{T}) = W$.

29. Let V be a finite-dimensional inner product space with an inner product $\langle \cdot, \cdot \rangle$ and let $\mathcal{A}$ be a linear transformation on V. Show that $\mathcal{A}$ is idempotent and self-adjoint (i.e., $\mathcal{A}^2 = \mathcal{A}$ and $\langle \mathcal{A}(x), y \rangle = \langle x, \mathcal{A}(y) \rangle$ for all $x, y \in V$) if and only if a matrix representation A of $\mathcal{A}$ with respect to an orthonormal basis satisfies $A^2 = A = A^*$. (Such a linear map $\mathcal{A}$ is referred to as an *orthogonal projection* of V onto its image $\operatorname{Im}(\mathcal{A})$.) Simply put, a linear map is an orthogonal projection if and only if its matrix (under an orthonormal basis) is an orthogonal projection.

30. Let V be a finite-dimensional inner product space with an inner product $\langle \cdot, \cdot \rangle$ and let $\mathcal{A}$ be an orthogonal projection (onto $\operatorname{Im}(\mathcal{A})$). Show that

(i) $V = \operatorname{Im}(\mathcal{A}) \oplus \operatorname{Ker}(\mathcal{A})$.

(ii) $\operatorname{Im}(\mathcal{A})$ and $\operatorname{Ker}(\mathcal{A})$ are orthogonal.

(iii) $\|v - \mathcal{A}(v)\| \leq \|v - \mathcal{A}(u)\|$ for any $v, u \in V$.

(iv) $\|v\|^2 = \|\mathcal{A}(v)\|^2 + \|\mathcal{A}(v) - v\|^2$ for any $v \in V$.

31. When does equality in Theorem 5.1.3 (4) hold?

32. Let V be an inner product space of finite dimension and $V = W \oplus W^{\perp}$, where W is a subspace of V and $W^{\perp}$ is the orthogonal complement of W in V. Define a transformation $\mathcal{A}$ on V by

$$\mathcal{A}(v) = w, \quad \text{if } v = w + w^{\perp},\ w \in W,\ w^{\perp} \in W^{\perp}.$$

Show that $\mathcal{A}$ is an orthogonal projection, $\mathrm{Im}(\mathcal{A}) = W$, and $\mathrm{Ker}(\mathcal{A}) = W^{\perp}$. Is $\mathcal{B} = \mathcal{A}$ if $\mathcal{B}$ is a map such that $\mathrm{Im}(\mathcal{B}) = W$ and $\mathrm{Ker}(\mathcal{B}) = W^{\perp}$?

33. Find an example of an $n \times n$ matrix A such that $\mathbb{C}^n = \mathrm{Im}\, A \oplus \mathrm{Ker}\, A$, and $\mathrm{Im}\, A$ and $\mathrm{Ker}\, A$ are orthogonal, but A is not a projection.

34. Let $A \in \mathbb{M}_n$. If $\mathrm{Im}\, A \subseteq \mathrm{Im}\, A^*$, show that $\mathrm{Im}\, A = \mathrm{Im}\, A^* = (\mathrm{Ker}\, A)^{\perp}$.

35. Let $\mathcal{R} : (x_1, \ldots, x_n) \in \mathbb{C}^n \mapsto (0, x_1, \ldots, x_{n-1}) \in \mathbb{C}^n$ and $\mathcal{L} : (x_1, \ldots, x_n) \in \mathbb{C}^n \mapsto (x_2, \ldots, x_n, 0) \in \mathbb{C}^n$. Show that $\mathcal{R}$ and $\mathcal{L}$ are nilpotent. Find their matrix representations (with respect to the standard basis of $\mathbb{C}^n$).

36. Let A and B be $m \times n$ matrices of full column rank, i.e., $\mathrm{rank}(A) = n = \mathrm{rank}(B)$. Show that the matrix $A(A^*A)^{-1}A^*$ is idempotent and that

$$A(A^*A)^{-1}A^* = B(B^*B)^{-1}B^* \quad \Leftrightarrow \quad A = BX$$

for some nonsingular matrix X. [Hint: For "$\Rightarrow$", post-multiply by A.]

37. Let $A, B \in \mathbb{M}_n$ be orthogonal projections. Show that

(a) $A + B$ is an orthogonal projection if and only if $AB = BA = 0$.

(b) $A - B$ is an orthogonal projection if and only if $AB = BA = B$.

(c) AB is an orthogonal projection if and only if $AB = BA$.

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5.2 Tridiagonal Matrices

Recursion is an important technique in matrix computation. We show this method by computing the determinant of a tridiagonal (banded) matrix, we then go on studying the eigenvalues of the matrices of this kind.

A square matrix $T = (t_{ij})$ is *tridiagonal* if $t_{ij} = 0$ whenever $|i - j| > 1$.

Example 5.2.1 The following are three examples of tridiagonal matrices.

$$\begin{pmatrix} 1 & 2 & 0 \\ 3 & 4 & 5 \\ 0 & 6 & 7 \end{pmatrix}, \quad \begin{pmatrix} a & 1 & 0 & 0 \\ x & b & 2 & 0 \\ 0 & y & c & 3 \\ 0 & 0 & z & d \end{pmatrix}, \quad \begin{pmatrix} a_1 & b_1 & & & \\ c_1 & a_2 & b_2 & & \\ & c_2 & a_3 & b_3 & \\ & & c_3 & a_4 & b_4 \\ & & & c_4 & a_5 \end{pmatrix}. \blacksquare$$

[Note: In the last matrix, the unspecified entries in the blank are 0's.]

We compute the determinant of the $n \times n$ special tridiagonal matrix

$$T_n = \begin{pmatrix} a & b & & & & \\ c & a & b & & & \\ & c & a & b & & \\ & & \ddots & \ddots & \ddots & \\ & & & c & a & b \\ & & & & c & a \end{pmatrix}, \quad a, b, c \in \mathbb{C}, \tag{5.3}$$

with $T_1 = (a)$ and $T_2 = \left(\begin{smallmatrix} a & b \\ c & a \end{smallmatrix}\right)$. Obviously, $\det T_1 = a$ and $\det T_2 = a^2 - bc$.

Theorem 5.2.1 *Let T_n be defined as in* (5.3). *Then*

$$\det T_n = \begin{cases} a^n & \text{if } bc = 0, \\ (n+1)(a/2)^n & \text{if } a^2 = 4bc, \\ (\alpha^{n+1} - \beta^{n+1})/(\alpha - \beta) & \text{if } a^2 \neq 4bc, \end{cases}$$

where

$$\alpha = \frac{a + \sqrt{a^2 - 4bc}}{2}, \quad \beta = \frac{a - \sqrt{a^2 - 4bc}}{2}. \tag{5.4}$$

Proof. For simplicity, we denote $t_n = \det T_n$. The cases of $n = 1, 2$ are done by direct computations. Let $n \geq 3$. Expanding the determinant along the first row of the matrix in (5.3) reveals the recursive formula

$$t_n = a\, t_{n-1} - bc\, t_{n-2}. \tag{5.5}$$

If $bc = 0$, then $b = 0$ or $c = 0$, and clearly $t_n = a^n$ from (5.3).
If $bc \neq 0$, let α and β be the solutions to $x^2 - ax + bc = 0$. Then

$$\alpha + \beta = a, \quad \alpha\beta = bc, \quad a^2 - 4bc = (\alpha - \beta)^2.$$

Using the recursive formula (5.5), we obtain

$$t_n - \alpha\, t_{n-1} = \beta(t_{n-1} - \alpha t_{n-2})$$

and

$$t_n - \beta t_{n-1} = \alpha(t_{n-1} - \beta t_{n-2}).$$

Denote

$$f_n = t_n - \alpha t_{n-1}, \quad g_n = t_n - \beta t_{n-1}.$$

Then

$$f_n = \beta f_{n-1}, \quad g_n = \alpha g_{n-1},$$

with

$$f_2 = \beta^2, \quad g_2 = \alpha^2.$$

Thus,

$$f_n = \beta^n, \quad g_n = \alpha^n,$$

that is,

$$t_n - \alpha t_{n-1} = \beta^n, \quad t_n - \beta t_{n-1} = \alpha^n. \tag{5.6}$$

Using t_{n+1} in (5.6) and subtracting the two identities, we obtain

$$t_n = \frac{\alpha^{n+1} - \beta^{n+1}}{\alpha - \beta}, \quad \text{if } \alpha \neq \beta \ \text{ (i.e., } a^2 \neq 4bc).$$

If $\alpha = \beta$, namely, $a^2 = 4bc$, then $\alpha = \frac{a}{2}$. Using (5.6), we have

$$t_n = \alpha^n + \alpha t_{n-1} = \cdots = (n-2)\alpha^n + \alpha^{n-2} t_2 = (n+1)\alpha^n. \ \blacksquare$$

Observe that $\det T_n$ in the theorem depends solely on a and the product bc. Thus, if $a \in \mathbb{R}$ and $bc > 0$, we can replace b and c by d and $\bar{d}$, respectively, where $d\bar{d} = bc$, to construct a tridiagonal Hermitian matrix H_n, for which

$$\det(\lambda I - T_n) = \det(\lambda I - H_n). \tag{5.7}$$

It follows that T_n has only real eigenvalues because H_n has only real eigenvalues. In fact, for $a, b, c \in \mathbb{R}$, $bc > 0$, the matrix $R_n = DT_nD^{-1}$ is real symmetric, where D is the diagonal matrix $\operatorname{diag}(1, \delta, \ldots, \delta^{n-1})$, $\delta = \sqrt{b/c}$.

Example 5.2.2 Let $a, b, c \in \mathbb{R}$, and $bc > 0$. Denote $d = \sqrt{bc}$ and let

$$T_4 = \begin{pmatrix} a & b & 0 & 0 \\ c & a & b & 0 \\ 0 & c & a & b \\ 0 & 0 & c & a \end{pmatrix}, \quad R_4 = \begin{pmatrix} a & d & 0 & 0 \\ d & a & d & 0 \\ 0 & d & a & d \\ 0 & 0 & d & a \end{pmatrix}.$$

Then T_4 is similar to R_4. Thus, T_4 has only real eigenvalues (as R_4 does).

Example 5.2.3 Setting $a = 0$ and $b = c = 1$ for the T_n in (5.3), we get

$$S_n = \begin{pmatrix} 0 & 1 & & & & \\ 1 & 0 & 1 & & & \\ & 1 & 0 & 1 & & \\ & & \ddots & \ddots & \ddots & \\ & & & 1 & 0 & 1 \\ & & & & 1 & 0 \end{pmatrix} \in \mathbb{M}_n. \tag{5.8}$$

By Theorem 5.2.1 (with $\alpha = i, \beta = -i$), we obtain $\det S_n = 0$ if n is odd, $\det S_n = 1$ if $n = 4k$, and $\det S_n = -1$ if $n = 4k + 2$ (k is a positive integer).

The n-square matrix S_n is real symmetric, and has rank at least $n - 1$ (because the submatrix of S_n by deleting the first row and last column is nonsingular). The eigenvalues of S_n are all real. Then, what are they?

For each $k = 1, 2, \ldots, n$, let

$$v_k = \Big(\sin \frac{k\pi}{n+1},\ \sin \frac{2k\pi}{n+1},\ \ldots,\ \sin \frac{nk\pi}{n+1} \Big)^T.$$

Using trigonometric identities, one can easily verify (Problem 22) that

$$S_n v_k = 2 \cos \tfrac{k\pi}{n+1}\, v_k. \tag{5.9}$$

It follows that the eigenvalues of S_n are $2 \cos \frac{k\pi}{n+1}$, $k = 1, 2, \ldots, n$. ∎[①]

The eigenvalues of S_n in (5.8) are all real and distinct (i.e., simple).

Theorem 5.2.2 *If T_n is a tridiagonal matrix defined as in* (5.3) *with $a, b, c \in \mathbb{R}$ and $bc > 0$, then the eigenvalues of T_n are all real and distinct* (*simple*)*:*

$$a + 2\sqrt{bc} \cos \tfrac{k\pi}{n+1}, \quad k = 1, 2, \ldots, n.$$

Proof. T_n is similar to $R_n = DT_nD^{-1}$, where $D = \operatorname{diag}(1, \delta, \ldots, \delta^{n-1})$, $\delta = \sqrt{b/c}$. Note that R_n is real symmetric, and $R_n = aI_n + \sqrt{bc}\, S_n$, where S_n is given in (5.8). The conclusion is immediate from Example 5.2.3. ∎

Below is another way to look at the problem, showing that all the eigenvalues of T_n are simple. Since T_n is diagonalizable (i.e., algebraic and geometric multiplicities are equal), it is sufficient to prove that each eigenvalue has only one eigenvector up to a factor (i.e., geometric multiplicity is 1).

Let λ be an eigenvalue of T_n and let $x = (x_1, \ldots, x_n)^T$ be an associated eigenvector. Then $(\lambda I - T_n)x = 0$, $x \neq 0$. Write the equations out explicitly:

$$\begin{aligned} (\lambda - a)x_1 - bx_2 &= 0, \\ -cx_1 + (\lambda - a)x_2 - bx_3 &= 0, \\ &\vdots \\ -cx_{n-2} + (\lambda - a)x_{n-1} - bx_n &= 0, \\ -cx_{n-1} + (\lambda - a)x_n &= 0. \end{aligned}$$

[①] Alternative approaches are using Chebyshev polynomial or graph theory.

Because $b \neq 0$, x_2 is uniquely determined by x_1 (as a multiple of x_1) in the first equation. The second equation shows that x_3 is uniquely determined (linearly) by x_1 and x_2. Successively, x_n is uniquely determined by x_{n-2} and x_{n-1}. If x_1 is replaced by kx_1, then $x_2, x_3, \ldots, x_n$ become $kx_2, kx_3, \ldots, kx_n$. Thus, the eigenvector is unique up to a factor.

One may pursue more for general tridiagonal matrices with different real entries a_k, b_k, c_k, and $b_k c_k > 0$ for all k. Note that our T_n belongs to two larger families of matrices – Toeplitz matrices and Hessenberg matrices.

Problems

1. Let $a, b, c \in \mathbb{C}$. Compute the determinant $\begin{vmatrix} a & b & 0 \\ c & a & b \\ 0 & c & a \end{vmatrix}$.

2. Find the eigenvalues of the matrices $\begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}$ and $\begin{pmatrix} 0 & 1 & 0 \\ -1 & 0 & 1 \\ 0 & -1 & 0 \end{pmatrix}$.

3. Show that $\begin{pmatrix} x & a \\ b & y \end{pmatrix}$ is similar to a symmetric matrix if $x, y \in \mathbb{R}$, $ab > 0$.

4. Let A be the tridiagonal matrix $\begin{pmatrix} 1 & 2 & 0 \\ \frac{1}{2} & 1 & 2 \\ 0 & \frac{1}{2} & 1 \end{pmatrix}$. Find a real symmetric matrix R such that A is similar to R and find the eigenvalues of R and A.

5. Show that the rank of T_n defined in (5.3) with $bc \neq 0$ is at least $n - 1$.

6. Work out in detail the proof that T_n defined in (5.3) is similar to a real symmetric matrix if a is real and $bc > 0$.

7. Let $a \in \mathbb{R}$. Show that the following two matrices T_4 and H_4 are similar:

$$T_4 = \begin{pmatrix} a & 2 & 0 & 0 \\ 8 & a & 2 & 0 \\ 0 & 8 & a & 2 \\ 0 & 0 & 8 & a \end{pmatrix}, \quad H_4 = \begin{pmatrix} a & 4 & 0 & 0 \\ 4 & a & 4 & 0 \\ 0 & 4 & a & 4 \\ 0 & 0 & 4 & a \end{pmatrix}.$$

8. For $n = 5, 6, 7, 8$, compute the $n \times n$ tridiagonal determinants

$$\begin{vmatrix} 0 & 1 & & & & & \\ 1 & 0 & 1 & & & & \\ & 1 & 0 & 1 & & & \\ & & \ddots & \ddots & \ddots & & \\ & & & 1 & 0 & 1 & \\ & & & & 1 & 0 & \end{vmatrix}.$$

9. Compute the $n \times n$ tridiagonal determinant

$$\begin{vmatrix} -3 & 2 & & & & \\ 1 & -3 & 2 & & & \\ & 1 & -3 & 2 & & \\ & & \ddots & \ddots & \ddots & \\ & & & 1 & -3 & 2 \\ & & & & 1 & -3 \end{vmatrix}.$$

10. Compute the $n \times n$ tridiagonal determinants for $\lambda = 0$ and $\lambda = 1$:

$$\begin{vmatrix} \lambda & 1 & & & & \\ -1 & \lambda & 1 & & & \\ & -1 & \lambda & 1 & & \\ & & \ddots & \ddots & \ddots & \\ & & & -1 & \lambda & 1 \\ & & & & -1 & \lambda \end{vmatrix}.$$

11. Compute the $n \times n$ tridiagonal determinant in terms of $\lambda \in \mathbb{C}$

$$\begin{vmatrix} \lambda & 1 & & & & \\ 1 & \lambda & 1 & & & \\ & 1 & \lambda & 1 & & \\ & & \ddots & \ddots & \ddots & \\ & & & 1 & \lambda & 1 \\ & & & & 1 & \lambda \end{vmatrix}$$

and evaluate the determinants for $\lambda = 2$, $\lambda = 0$, $\lambda = -1$, and $n = 100$.

12. Find the inverse of the $n \times n$ tridiagonal matrix

$$\begin{pmatrix} 2 & -1 & & & & & \\ -1 & 2 & -1 & & & & \\ & -1 & 2 & -1 & & & \\ & & -1 & 2 & -1 & & \\ & & & \ddots & \ddots & \ddots & \\ & & & & -1 & 2 & -1 \\ & & & & & -1 & 2 \end{pmatrix}.$$

13. Show that the value of the following determinant is independent of x:

$$\begin{vmatrix} a & x & & & & \\ \frac{1}{x} & a & x & & & \\ & \frac{1}{x} & a & x & & \\ & & \ddots & \ddots & \ddots & \\ & & & \frac{1}{x} & a & x \\ & & & & \frac{1}{x} & a \end{vmatrix}.$$

14. (**Cauchy matrix**) Let $\lambda_1, \ldots, \lambda_n$ be positive numbers and let

$$\Lambda = \left(\frac{1}{\lambda_i + \lambda_j}\right) = \begin{pmatrix} \lambda_1+\lambda_1 & \lambda_1+\lambda_2 & \cdots & \lambda_1+\lambda_n \\ \lambda_2+\lambda_1 & \lambda_2+\lambda_2 & \cdots & \lambda_2+\lambda_n \\ \vdots & \vdots & \ddots & \vdots \\ \lambda_n+\lambda_1 & \lambda_n+\lambda_2 & \cdots & \lambda_n+\lambda_n \end{pmatrix}.$$

Show that

$$\det \Lambda = \frac{\prod_{i>j}(\lambda_i - \lambda_j)^2}{\prod_{i,j}(\lambda_i + \lambda_j)}.$$

[Hint: Subtract the last column from each of the other columns, then factor; do the same operations for rows; apply induction on n.]

15. Show that

$$\begin{vmatrix} \frac{1}{2} & \frac{1}{3} & \cdots & \frac{1}{n+1} \\ \frac{1}{3} & \frac{1}{4} & \cdots & \frac{1}{n+2} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{1}{n+1} & \frac{1}{n+2} & \cdots & \frac{1}{2n} \end{vmatrix} \geq 0.$$

16. Let $A = \begin{pmatrix} a_1 & b_1 & 0 \\ c_1 & a_2 & b_2 \\ 0 & c_2 & a_3 \end{pmatrix} \in \mathbb{M}_3(\mathbb{C})$ with $b_1\overline{c_1} > 0$ and $b_2\overline{c_2} > 0$. Find nonzero scalars x and y such that PAP^{-1} is symmetric, where $P = \operatorname{diag}(1, x, y)$.

17. Show that $\begin{pmatrix} 0 & 4i & 0 \\ i & z & 1 \\ 0 & 2 & i \end{pmatrix} \in \mathbb{M}_3(\mathbb{C})$ is similar to a complex symmetric matrix, but it cannot be similar to a real symmetric matrix for any $z \in \mathbb{C}$.

18. Let $a_t, b_t, c_t \in \mathbb{R}$ for $1 \leq t \leq n$, and let T_n be the tridiagonal matrix

$$T_n = \begin{pmatrix} a_1 & b_1 & & & & \\ c_1 & a_2 & b_2 & & & \\ & c_2 & a_3 & b_3 & & \\ & & \ddots & \ddots & \ddots & \\ & & & c_{n-2} & a_{n-1} & b_{n-1} \\ & & & & c_{n-1} & a_n \end{pmatrix}.$$

If $b_t c_t > 0$ for all t, show that the eigenvalues of T_n are real and simple (i.e., of multiplicity 1). [Hint: DT_nD^{-1} is real and symmetric, where $D = \operatorname{diag}(d_1, d_2, \ldots, d_n)$ with $d_1 = 1$, $d_k = \sqrt{\frac{b_1 \cdots b_{k-1}}{c_1 \cdots c_{k-1}}}$, $k = 2, 3, \ldots, n$.]

19. Find a symmetric matrix R such that $\begin{pmatrix} 1 & -2 & 0 \\ -3 & 4 & -2 \\ 0 & -3 & 5 \end{pmatrix}$ is similar to R.

20. Find a symmetric matrix S such that $\begin{pmatrix} 1 & 2 & 0 \\ 3 & 4 & 5 \\ 0 & 6 & 7 \end{pmatrix}$ is similar to S.

21. Compute the $n \times n$ tridiagonal matrix determinant

$$\begin{vmatrix} 3 & 2 & & & & & \\ 1 & 3 & 1 & & & & \\ & 2 & 3 & 2 & & & \\ & & 1 & 3 & 1 & & \\ & & & \ddots & \ddots & \ddots & \\ & & & & x & 3 & x \\ & & & & & y & 3 \end{vmatrix},$$

in which $x = 1, y = 2$ if n is odd, and $x = 2, y = 1$ if n is even.

22. Use the trigonometric identity $2\cos x \sin y = \sin(x + y) + \sin(y - x)$, $x, y \in \mathbb{R}$, to verify the matrix-eigenvector-eigenvalue identities

$$\begin{pmatrix} 0 & 1 & & & & & \\ 1 & 0 & 1 & & & & \\ & 1 & 0 & 1 & & & \\ & & \ddots & \ddots & \ddots & & \\ & & & 1 & 0 & 1 & \\ & & & & 1 & 0 & \end{pmatrix} \begin{pmatrix} \sin \frac{k\pi}{n+1} \\ \sin \frac{2k\pi}{n+1} \\ \sin \frac{3k\pi}{n+1} \\ \vdots \\ \vdots \\ \sin \frac{nk\pi}{n+1} \end{pmatrix} = 2\cos \frac{k\pi}{n+1} \begin{pmatrix} \sin \frac{k\pi}{n+1} \\ \sin \frac{2k\pi}{n+1} \\ \sin \frac{3k\pi}{n+1} \\ \vdots \\ \vdots \\ \sin \frac{nk\pi}{n+1} \end{pmatrix}$$

for $k = 1, 2, \ldots, n$. (a) Show that $\cos \frac{i\pi}{n+1}$ and $\cos \frac{j\pi}{n+1}$ are not equal for $i \neq j$, where $1 \leq i, j \leq n$. (b) Show that $\cos \frac{\pi}{7} \cos \frac{2\pi}{7} \cdots \cos \frac{6\pi}{7} = -\frac{1}{2^6}$.

23. An $n \times n$ matrix $A = (a_{ij})$ is called an *upper-Hessenberg* matrix if $a_{ij} = 0$ for all i, j with $i > j + 1$. A is said to be *lower-Hessenberg* if A^T is upper-Hessenberg, that is, $a_{ij} = 0$ for all i, j with $j > i + 1$. For example, H_1 is upper-Hessenberg and H_2 is lower-Hessenberg, where

$$H_1 = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 5 & 6 & 7 & 8 \\ 0 & 9 & 0 & 1 \\ 0 & 0 & 1 & 2 \end{pmatrix}, \quad H_2 = \begin{pmatrix} 1 & 4 & 0 & 0 \\ 3 & 4 & 1 & 0 \\ 0 & 2 & 3 & 4 \\ 4 & 5 & 1 & 3 \end{pmatrix}.$$

(a) If $A \in \mathbb{M}_n$ is upper-Hessenberg and $T \in \mathbb{M}_n$ is upper-triangular, show that AT and TA are upper-Hessenberg. (b) Show that a tridiagonal matrix is both upper-Hessenberg and lower-Hessenberg; and the converse is also true. (c) Show that the Hadamard product of an upper-Hessenberg matrix and a lower-Hessenberg matrix is tridiagonal.

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5.3 Circulant Matrices

Let $c_0, c_1, \ldots, c_{n-1} \in \mathbb{C}$. An $n \times n$ *circulant matrix* is a matrix of the form

$$\begin{pmatrix} c_0 & c_1 & c_2 & \cdots & c_{n-1} \\ c_{n-1} & c_0 & c_1 & \cdots & c_{n-2} \\ c_{n-2} & c_{n-1} & c_0 & \cdots & c_{n-3} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ c_1 & c_2 & c_3 & \cdots & c_0 \end{pmatrix}. \tag{5.10}$$

Example 5.3.1 The following are examples of circulant matrices.

$$\begin{pmatrix} a & b & c \\ c & a & b \\ b & c & a \end{pmatrix}, \quad \begin{pmatrix} 1 & i & 2 & -i \\ -i & 1 & i & 2 \\ 2 & -i & 1 & i \\ i & 2 & -i & 1 \end{pmatrix}, \quad \begin{pmatrix} 1 & 2 & 3 & \cdots & n \\ n & 1 & 2 & \cdots & n-1 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 3 & 4 & 5 & \cdots & 2 \\ 2 & 3 & 4 & \cdots & 1 \end{pmatrix}$$

(where $a, b, c \in \mathbb{C}$ in the first matrix; the second matrix is Hermitian), and

$$P_n = \begin{pmatrix} 0 & 1 & 0 & \ldots & 0 \\ 0 & 0 & 1 & \ldots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \ldots & 1 \\ 1 & 0 & 0 & \ldots & 0 \end{pmatrix}. \blacksquare \tag{5.11}$$

The above $n \times n$ circulant matrix P_n (or simply P) is also a permutation matrix (having one 1 in each row and each column). We refer to this circulant permutation matrix P in (5.11) as the $n \times n$ *primary permutation matrix*.

This section presents some basic properties of circulant matrices. The following theorem can be shown by a direct verification.

Theorem 5.3.1 *An n-square matrix C is circulant if and only if*

$$C = PCP^T,$$

where $P = P_n$ is the $n \times n$ primary permutation matrix in (5.11).

A complex number ω is an nth *primitive root* of unity if $\omega^n - 1 = 0$ and $\omega^k - 1 \neq 0$ for every positive integer $k < n$. Since $(\omega^k)^n - 1 = (\omega^n)^k - 1 = 0$, $1, \omega, \omega^2, \ldots, \omega^{n-1}$ are the n distinct roots of $x^n - 1 = 0$. We may express

$$\omega = e^{\frac{2\pi i}{n}} = \cos\tfrac{2\pi}{n} + i \sin\tfrac{2\pi}{n},\ \omega^k = e^{\frac{2k\pi i}{n}} = \cos\tfrac{2k\pi}{n} + i \sin\tfrac{2k\pi}{n},\ 1 \leq k \leq n-1.$$

Note that

$$x^n - 1 = (x-1)(x-\omega)(x-\omega^2)\cdots(x-\omega^{n-1})$$

and

$$\sum_{j=0}^{n-1} \omega^{jk} = \frac{(\omega^k)^n - 1}{\omega^k - 1} = 0.$$

Theorem 5.3.2 *Let C be a circulant matrix in the form* (5.10)*, let $f(\lambda) = c_0 + c_1\lambda + \cdots + c_{n-1}\lambda^{n-1}$, and let ω be an nth primitive root of unity. Then*

(1) *$C = f(P)$, where P is the $n \times n$ primary permutation matrix.*

(2) *C is a normal matrix, that is, $C^*C = CC^*$.*

(3) *The eigenvalues of C are $f(\omega^k)$, $k = 0, 1, \ldots, n-1$.*

(4) *$\det C = f(\omega^0)f(\omega^1)\cdots f(\omega^{n-1})$.*

(5) *$F^*CF = \operatorname{diag}\big(f(\omega^0), f(\omega^1), \ldots, f(\omega^{n-1})\big)$, where F is the unitary matrix with the (p,q)-entry equal to $\frac{1}{\sqrt{n}}\omega^{(p-1)(q-1)}$, $p, q = 1, 2, \ldots, n$.*

Proof. (1) is easy to see by a direct computation. (2) is due to the fact that if matrices A and B commute, so do $g(A)$ and $h(B)$, where g and h are any polynomials (Problem 22, Sect. 3.3). Note that $PP^* = P^*P$.

For (3) and (4), the characteristic polynomial of the matrix P is

$$\det(\lambda I - P) = \lambda^n - 1 = \prod_{k=0}^{n-1}(\lambda - \omega^k).$$

Thus, the eigenvalues of P are ω^k, $k = 0, 1, \ldots, n-1$, and the eigenvalues of $C = f(P)$ are $f(\omega^k)$, $k = 0, 1, \ldots, n-1$ (Problem 14, Sect. 3.2). Moreover,

$$\det C = \prod_{k=0}^{n-1} f(\omega^k).$$

To show (5), for each $k = 0, 1, \ldots, n-1$, let

$$x_k = \big(1,\ \omega^k,\ \omega^{2k},\ \ldots,\ \omega^{(n-1)k}\big)^T.$$

Then

$$Px_k = \big(\omega^k,\ \omega^{2k},\ \ldots,\ \omega^{(n-1)k},\ 1\big)^T = \omega^k x_k,$$

implying

$$Cx_k = f(P)x_k = f(\omega^k)x_k.$$

In other words, x_k are the eigenvectors of P and C corresponding to the eigenvalues ω^k and $f(\omega^k)$, respectively, $k = 0, 1, \ldots, n-1$.

Because $\overline{\omega^t} = \omega^{-t}$ for every integer t (Problem 1), we have

$$\langle x_p, x_q \rangle = x_q^* x_p = \sum_{k=0}^{n-1} \overline{\omega^{qk}}\, \omega^{pk} = \sum_{k=0}^{n-1} \omega^{(p-q)k} = \begin{cases} 0, & p \neq q, \\ n, & p = q. \end{cases}$$

Hence, $x_p^* C x_q = f(\omega^q) x_p^* x_q = n f(\omega^q)$ if $p = q$, or 0 otherwise. Note that

$$\left\{ \frac{1}{\sqrt{n}} x_0,\ \frac{1}{\sqrt{n}} x_1,\ \ldots,\ \frac{1}{\sqrt{n}} x_{n-1} \right\}$$

is an orthonormal basis for $\mathbb{C}^n$. Thus, we obtain a unitary matrix

$$F = \frac{1}{\sqrt{n}} \begin{pmatrix} 1 & 1 & 1 & \ldots & 1 \\ 1 & \omega & \omega^2 & \ldots & \omega^{n-1} \\ 1 & \omega^2 & \omega^4 & \ldots & \omega^{2(n-1)} \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ 1 & \omega^{n-1} & \omega^{2(n-1)} & \ldots & \omega^{(n-1)(n-1)} \end{pmatrix} = \frac{1}{\sqrt{n}} \left(\omega^{(p-1)(q-1)} \right)$$

such that

$$F^* C F = \operatorname{diag}\left(f(\omega^0), f(\omega^1), \ldots, f(\omega^{n-1}) \right).$$

A direct computation shows that F is a unitary matrix. ∎

Note that the matrix F in the proof, called a *Fourier matrix*, is independent of the circulant matrix C. A Fourier matrix is also a Vandermonde matrix. Theorem 5.3.2 says that the circulant matrices are just the matrices of the polynomials of the primary permutation matrices, and they are similar to diagonal matrices (universally) via a Fourier matrix.

Example 5.3.2 The 4×4 Fourier matrix. $n = 4$, $\omega = \cos \frac{2\pi}{4} + i \sin \frac{2\pi}{4} = i$,

$$F = \frac{1}{\sqrt{4}} \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & \omega & \omega^2 & \omega^3 \\ 1 & \omega^2 & \omega^4 & \omega^6 \\ 1 & \omega^3 & \omega^6 & \omega^9 \end{pmatrix} = \frac{1}{\sqrt{4}} \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & i & -1 & -i \\ 1 & -1 & 1 & -1 \\ 1 & -i & -1 & i \end{pmatrix}.$$

By Theorem 5.3.2, a 4×4 matrix C is circulant if and only if $C = F \operatorname{diag}(f(1), f(\omega), f(\omega^2), f(\omega^3)) F^*$ for some polynomial f of degree ≤ 3. ∎

Example 5.3.3 Let $f(\lambda) = 1 + i\lambda + 2\lambda^2 - i\lambda^3$ and let

$$H = \begin{pmatrix} 1 & i & 2 & -i \\ -i & 1 & i & 2 \\ 2 & -i & 1 & i \\ i & 2 & -i & 1 \end{pmatrix}.$$

H is a 4×4 circulant (Hermitian) matrix. Take $\omega = i$. The eigenvalues of H are: $f(1) = 3$, $f(\omega) = f(i) = -3$, $f(\omega^2) = f(-1) = 3$, $f(\omega^3) = f(-i) = 1$. ∎

Problems

1. Let ω be an nth primitive root of unity. Show that

   (a) $\omega\overline{\omega} = 1$.

   (b) $\overline{\omega}^k = \omega^{-k} = \omega^{n-k}$ (k is a positive integer).

   (c) $1 + \omega + \cdots + \omega^{n-1} = 0$.

2. Let ω be an nth primitive root of unity. Show that ω^k (k is a positive integer) is also an nth primitive root of unity if and only if $(n, k) = 1$, that is, n and k have no common positive divisors other than 1.

3. If A is a circulant matrix, show that A^*, A^k (where k is a positive integer), and A^{-1} (if the inverse exists) are circulant matrices.

4. Let A and B be circulant matrices of the same order. Show that A and B commute and that AB is a circulant matrix.

5. Let A be a circulant matrix. Show that for every positive integer k,

$$\operatorname{rank}(A^k) = \operatorname{rank}(A).$$

6. Find a 3×3 Fourier matrix.

7. Find the eigenvalues of the circulant matrices

$$\begin{pmatrix} 1 & 2 & 3 \\ 3 & 1 & 2 \\ 2 & 3 & 1 \end{pmatrix} \quad \text{and} \quad \begin{pmatrix} 1 & \omega & \omega^2 \\ \omega^2 & 1 & \omega \\ \omega & \omega^2 & 1 \end{pmatrix}, \ \omega^3 = 1.$$

8. Find the eigenvalues and the eigenvectors of the circulant matrix

$$\begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \end{pmatrix}$$

   and find the Fourier matrix F that diagonalizes the above matrix.

9. Let P be the $n \times n$ primary permutation matrix. Show that

$$P^n = I, \quad P^T = P^{-1} = P^{n-1}.$$

10. If C is a circulant matrix with $(c_0, c_1, \ldots, c_{n-1})$ as its first row, show that $s = c_0 + c_1 + \cdots + c_{n-1}$ is an eigenvalue of C and $e = (1, 1, \ldots, 1)^T$ (with n components) is an eigenvector corresponding to s.

11. Find all matrices $C \in \mathbb{M}_3$ such that $C = Q^T C Q$, where $Q = \left(\begin{smallmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{smallmatrix}\right)$.

12. Let $c_0, c_1, c_2 \in \mathbb{C}$. Find a 3×3 matrix X such that

$$\begin{pmatrix} c_0 & c_1 & c_2 \\ c_1 & c_2 & c_0 \\ c_2 & c_0 & c_1 \end{pmatrix} = X \begin{pmatrix} c_0 & c_1 & c_2 \\ c_2 & c_0 & c_1 \\ c_1 & c_2 & c_0 \end{pmatrix}.$$

13. Find a 4×4 matrix Y such that

$$Y^T \begin{pmatrix} 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{pmatrix} Y = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \end{pmatrix}.$$

14. Let F be the $n \times n$ Fourier matrix. Show that

    (a) $F^T = F$, that is, F is symmetric.

    (b) $(F^*)^2 = F^2$ and F^2 is a permutation matrix.

    (c) $(F^*)^3 = F$ and $F^4 = I$.

    (d) The eigenvalues of F are ± 1 and $\pm i$ (with some multiplicities).

    (e) $F^* = n^{-1/2} V(1, \overline{\omega}, \overline{\omega}^2, \ldots, \overline{\omega}^{n-1})$, where V stands for the *Vandermonde matrix* (see Problem 8 of Sect. 1.2 or the next section).

    (f) If $F = R + iS$, where R and S are real, then $R^2 + S^2 = I$, $RS = SR$, and R and S are symmetric.

15. Let e_i be the column vectors of n components with the ith component 1 and 0 elsewhere, $i = 1, 2, \ldots, n$, let $c_1, c_2, \ldots, c_n \in \mathbb{C}$, and let

$$A = \operatorname{diag}(c_1, c_2, \ldots, c_n) P,$$

    where P is the $n \times n$ primary permutation matrix. Show that

    (a) $Ae_i = c_{i-1} e_{i-1}$ for each $i = 1, 2, \ldots, n$, where $c_0 = c_n,\ e_0 = e_n$.

    (b) $A^n = cI$, where $c = c_1 c_2 \cdots c_n$.

    (c) $\det(I + A + \cdots + A^{n-1}) = (1 - c)^{n-1}$, where $c = c_1 c_2 \cdots c_n$.

16. It is given that the following matrix H is (complex) Hermitian circulant

$$H = \begin{pmatrix} 1 & i & \times & \times \\ \times & \times & \times & \times \\ 2 & \times & \times & \times \\ \times & \times & \times & \times \end{pmatrix}.$$

    Complete H (i.e. find $\times$'s) and find its eigenvalues and eigenvectors.

17. Find all eigenvalues of the real symmetric circulant matrix

$$R = \begin{pmatrix} 1 & 2 & 3 & 2 \\ 2 & 1 & 2 & 3 \\ 3 & 2 & 1 & 2 \\ 2 & 3 & 2 & 1 \end{pmatrix}.$$

18. Find the eigenvalues of the $n \times n$ symmetric circulant matrix

$$\begin{pmatrix} 0 & 1 & 0 & \dots & 0 & 1 \\ 1 & 0 & 1 & \dots & 0 & 0 \\ 0 & 1 & 0 & \ddots & 0 & 0 \\ \vdots & \vdots & \ddots & \ddots & \ddots & \vdots \\ 0 & 0 & 0 & \ddots & 0 & 1 \\ 1 & 0 & 0 & \dots & 1 & 0 \end{pmatrix}$$

(in which the entries in the positions $(1,n),(n,1),(i,i+1),(i+1,i)$, $i = 1,2,\dots,n-1$, are 1's, and the entries in other positions are 0's).

19. An $n \times n$ matrix $A = (a_{ij})$ is called a *Toeplitz matrix* if the entries of A are constant down the diagonals parallel to the main diagonal, i.e., $a_{ij} = a_{j-i}$, where $a_{1-n},\dots,a_{n-1}$ are $2n-1$ constants, or explicitly,

$$A = \begin{pmatrix} a_0 & a_1 & a_2 & \cdots & a_{n-1} \\ a_{-1} & a_0 & a_1 & \ddots & \vdots \\ a_{-2} & a_{-1} & a_0 & \ddots & a_2 \\ \vdots & \ddots & \ddots & \ddots & a_1 \\ a_{1-n} & \cdots & a_{-2} & a_{-1} & a_0 \end{pmatrix}.$$

For example, matrix $G = (g_{ij})$ with $g_{i,i+1} = 1$, $i = 1,2,\dots,n-1$, and 0 elsewhere, is a Toeplitz matrix. Show that (i) a matrix $A \in \mathbb{M}_n$ is a Toeplitz matrix if and only if A can be written in the form

$$A = \sum_{k=1}^{n-1} a_{-k}(G^T)^k + \sum_{k=0}^{n-1} a_k G^k,$$

(ii) if A is Toeplitz, then $A = SA^TS$, where S is the backward identity, (iii) SA is a symmetric matrix, known as a *Hankel matrix*, (iv) the sum of two Toeplitz matrices is a Toeplitz matrix, (v) a circulant matrix is a Toeplitz matrix, and (vi) AB and BA are similar if A and B are Toeplitz matrices of the same order.

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5.4 Vandermonde Matrices

An $n \times n$ *Vandermonde matrix* is a matrix of the form (with $a_1, a_2, \ldots, a_n \in \mathbb{C}$)

$$\begin{pmatrix} 1 & 1 & \cdots & 1 \\ a_1 & a_2 & \cdots & a_n \\ \vdots & \vdots & \vdots & \vdots \\ a_1^{n-1} & a_2^{n-1} & \cdots & a_n^{n-1} \end{pmatrix} \quad \text{or} \quad \begin{pmatrix} 1 & a_1 & \cdots & a_1^{n-1} \\ 1 & a_2 & \cdots & a_2^{n-1} \\ \vdots & \vdots & \vdots & \vdots \\ 1 & a_n & \cdots & a_n^{n-1} \end{pmatrix}.$$

We denote the Vandermonde matrix by $V_n(a_1, a_2, \ldots, a_n)$ or simply V_n.

Example 5.4.1 The following are examples of Vandermonde matrices.

$$\begin{pmatrix} 1 & 1 & 1 \\ 0 & 2 & 2 \\ 0 & 4 & 4 \end{pmatrix}, \quad \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 2 & 3 & \frac{1}{2} \\ 1 & 4 & 9 & \frac{1}{4} \\ 1 & 8 & 27 & \frac{1}{8} \end{pmatrix}, \quad \begin{pmatrix} 1 & a & a^2 & a^3 \\ 1 & b & b^2 & b^3 \\ 1 & c & c^2 & c^3 \\ 1 & d & d^2 & d^3 \end{pmatrix}. \blacksquare$$

Vandermonde matrices play a role in various areas such as interpolation problems and solving the systems of linear equations, etc. We consider the determinant and the inverse of a Vandermonde matrix in this section.

Example 5.4.2 Compute the 3×3 Vandermonde determinant:

$$\begin{vmatrix} 1 & 1 & 1 \\ a_1 & a_2 & a_3 \\ a_1^2 & a_2^2 & a_3^2 \end{vmatrix} = (a_3 - a_2)(a_3 - a_1)(a_2 - a_1). \blacksquare$$

Theorem 5.4.1 *Let $V_n(a_1, a_2, \ldots, a_n)$ be a Vandermonde matrix. Then*

$$\det V_n(a_1, a_2, \ldots, a_n) = \prod_{1 \le i < j \le n} (a_j - a_i), \tag{5.12}$$

and $V_n(a_1, a_2, \ldots, a_n)$ is invertible if and only if all a_i's are distinct.

Note: The identity (5.12) has appeared as Problem 8 of Sect. 1.2, Chap. 1. It can be shown in different ways. We present two proofs in the following.

Proof 1. We consider $a_1, a_2, \ldots, a_n$ as variables (indeterminates). If we set $a_i = a_j$ in $\det V_n(a_1, a_2, \ldots, a_n)$, then the determinant is zero because it contains two identical columns (or rows). This implies that $\det V_n(a_1, a_2, \ldots, a_n)$ has factors $a_i - a_j$ for all $i \neq j$. Since the total degree of $a_1, a_2, \ldots, a_n$ in each product in the expansion of $\det V_n(a_1, a_2, \ldots, a_n)$ is $n(n-1)/2$, every factor $a_i - a_j$ appears only once. Thus, we can write, for some constant c,

$$\det V_n(a_1, a_2, \ldots, a_n) = c \prod_{1 \le i < j \le n} (a_j - a_i).$$

In the expanded forms of both sides, the term $1a_2 \cdots a_n^{n-1}$ (the product of diagonal entries) is on the left, while $ca_2 \cdots a_n^{n-1}$ is on the right. So $c = 1$.

Proof 2. We present an explicit proof by induction on the order n of the matrix. If $n = 1$ or 2, there is nothing to show. Let $n \geq 3$.

Suppose that the assertion is true when the order of the matrix is $n-1$. For the case of n, we start working from the bottom (last rows) by adding row $n-1$ times $-a_1$ to row n, then adding row $n-2$ times $-a_1$ to row $n-1$ (of the resulting matrix), ..., adding row 1 times $-a_1$ to row 2. We derive

$$\begin{aligned}
\det V_n &= \begin{vmatrix} 1 & 1 & 1 & \cdots & 1 \\ 0 & a_2 - a_1 & a_3 - a_1 & \cdots & a_n - a_1 \\ 0 & a_2(a_2 - a_1) & a_3(a_3 - a_1) & \cdots & a_n(a_n - a_1) \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ 0 & a_2^{n-2}(a_2 - a_1) & a_3^{n-2}(a_3 - a_1) & \cdots & a_n^{n-2}(a_n - a_1) \end{vmatrix} \\
&= \begin{vmatrix} a_2 - a_1 & a_3 - a_1 & \cdots & a_n - a_1 \\ a_2(a_2 - a_1) & a_3(a_3 - a_1) & \cdots & a_n(a_n - a_1) \\ \vdots & \vdots & \vdots & \vdots \\ a_2^{n-2}(a_2 - a_1) & a_3^{n-2}(a_3 - a_1) & \cdots & a_n^{n-2}(a_n - a_1) \end{vmatrix} \\
&= \prod_{j=2}^{n} (a_j - a_1) \det V_{n-1}(a_2, a_3, \ldots, a_n) \\
&= \prod_{j=2}^{n} (a_j - a_1) \prod_{2 \leq i < j \leq n} (a_j - a_i) \ \text{(by the hypothesis)} \\
&= \prod_{1 \leq i < j \leq n} (a_j - a_i). \ \blacksquare
\end{aligned}$$

It is readily seen the Vandermonde matrix is nonsingular, i.e., the determinant $\det V_n(a_1, a_2, \ldots, a_n) \neq 0$, if and only if $a_1, a_2, \ldots, a_n$ are distinct.

Example 5.4.3 Compute the Vandermonde determinant $\det V_5(1, 2, 3, 4, 5)$.

$$\begin{vmatrix} 1 & 1 & 1 & 1 & 1 \\ 1 & 2 & 3 & 4 & 5 \\ 1 & 2^2 & 3^2 & 4^2 & 5^2 \\ 1 & 2^3 & 3^3 & 4^3 & 5^3 \\ 1 & 2^4 & 3^4 & 4^4 & 5^4 \end{vmatrix} = \begin{aligned} &(5-4)(5-3)(5-2)(5-1)(4-3)(4-2)(4-1) \\ &\times(3-2)(3-1)(2-1) = 288. \ \blacksquare \end{aligned}$$

Example 5.4.4 For any integers $k_1 < k_2 < \cdots < k_n$, $\det V_n(k_1, k_2, \ldots, k_n)$ is divisible by $\det V_n(1, 2, \ldots, n)$. We show this statement as follows.

Let f_i be any monic polynomial of degree i, $i = 1, 2, \ldots, n-1$. For example, $f_1(x) = x + c$. Then $f_1(k_j) = k_j + c$, $j = 1, 2, \ldots, n$. The additive

property of determinants (see Sect. 1.2, Chapter 1) reveals that

$$\begin{vmatrix} 1 & 1 & 1 & \cdots & 1 \\ f_1(k_1) & f_1(k_2) & f_1(k_3) & \cdots & f_1(k_n) \\ f_2(k_1) & f_2(k_2) & f_2(k_3) & \cdots & f_2(k_n) \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ f_{n-1}(k_1) & f_{n-1}(k_2) & f_{n-1}(k_3) & \cdots & f_{n-1}(k_n) \end{vmatrix} \tag{5.13}$$

is equal to

$$\begin{vmatrix} 1 & 1 & 1 & \cdots & 1 \\ k_1 & k_2 & k_3 & \cdots & k_n \\ f_2(k_1) & f_2(k_2) & f_2(k_3) & \cdots & f_2(k_n) \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ f_{n-1}(k_1) & f_{n-1}(k_2) & f_{n-1}(k_3) & \cdots & f_{n-1}(k_n) \end{vmatrix}.$$

Inductively, we see that the determinant (5.13) equals $\det V_n(k_1, k_2, \ldots, k_n)$.

Now let $f_i(x) = x(x-1)(x-2)\cdots(x-i+1)$. Then for any integer k,

$$f_i(k) = k(k-1)(k-2)\cdots(k-i+1) = \begin{cases} i!\binom{k}{i}, & k > 0, \\ 0, & k = 0, \\ i!(-1)^i\binom{-k+i-1}{i}, & k < 0. \end{cases}$$

It follows that $f_i(k)$ is divisible by $i!$ for $i = 1, 2, \ldots, n-1$.

In (5.13), factoring out $(i-1)!$ from row i, $i = 2, 3, \ldots, n$, we see that the determinant in (5.13), consequently $\det V_n(k_1, k_2, \ldots, k_n)$, is divisible by the product $\prod_{i=1}^{n}(i-1)!$ which is equal to $\det V_n(1, 2, \ldots, n)$. ∎

Example 5.4.5 Let $A \in \mathbb{M}_n$. By the Vandermonde matrix, we show that

$$A^n = 0 \quad \Leftrightarrow \quad \operatorname{tr} A^k = 0, \;\; k = 1, 2, \ldots, n.$$

Proof. If $A^n = 0$, i.e., A is nilpotent, then A has only zero eigenvalues, so does A^k, revealing $\operatorname{tr} A^k = 0$ for each k. For the reverse direction, let the eigenvalues of A be $\lambda_1, \lambda_2, \ldots, \lambda_n$. Then the trace identities imply

$$\begin{cases} \lambda_1 + \lambda_2 + \cdots + \lambda_n = 0, \\ \lambda_1^2 + \lambda_2^2 + \cdots + \lambda_n^2 = 0, \\ \qquad\qquad \vdots \\ \lambda_1^n + \lambda_2^n + \cdots + \lambda_n^n = 0. \end{cases} \tag{5.14}$$

If all λ_i's are equal, i.e., $\lambda_1 = \lambda_2 = \cdots = \lambda_n$, then $n\lambda_1 = 0$, so all $\lambda_i = 0$.

We show otherwise (if not all λ_i's are the same) there would be a contradiction. Suppose that $\lambda_{i_1}, \lambda_{i_2}, \ldots, \lambda_{i_p}$ $(p > 1)$ are distinct and they occur respectively $l_1, l_2, \ldots, l_p$ times in $\{\lambda_1, \lambda_2, \ldots, \lambda_n\}$. Then we rewrite (5.14) as

$$\begin{cases} l_1\lambda_{i_1} + l_2\lambda_{i_2} + \cdots + l_p\lambda_{i_p} = 0, \\ l_1\lambda_{i_1}^2 + l_2\lambda_{i_2}^2 + \cdots + l_p\lambda_{i_p}^2 = 0, \\ \qquad\qquad \vdots \\ l_1\lambda_{i_1}^n + l_2\lambda_{i_2}^n + \cdots + l_p\lambda_{i_p}^n = 0. \end{cases} \tag{5.15}$$

Use the first p equations in (5.15) and express them as (matrix times vector)

$$V_p(\lambda_{i_1}, \lambda_{i_2}, \ldots, \lambda_{i_p})(l_1\lambda_{i_1}, l_2\lambda_{i_2}, \ldots, l_p\lambda_{i_p})^T = 0.$$

Because $\lambda_{i_1}, \lambda_{i_2}, \ldots, \lambda_{i_p}$ are distinct, the Vandermonde matrix V_p is nonsingular. So $(l_1\lambda_{i_1}, l_2\lambda_{i_2}, \ldots, l_p\lambda_{i_p})^T = 0$, implying $\lambda_{i_1} = \lambda_{i_2} = \cdots = \lambda_{i_p} = 0$, a contradiction to $\lambda_{i_1}, \lambda_{i_2}, \ldots, \lambda_{i_p}$ being distinct. Therefore, all $\lambda_i = 0$. ∎

The idea in the above example applies to the *interpolation problem* (see Sect. 3.1) of finding a polynomial $f(x)$ of degree at most $n-1$ satisfying

$$f(x_i) = y_i, \quad i = 1, 2, \ldots, n, \tag{5.16}$$

where x_i and y_i are given constants and x_i's are distinct (Problem 4). In fact, if $f(x) = a_0 + a_1x + \cdots + a_{n-1}x^{n-1}$, then (5.16) is equivalent to

$$\begin{pmatrix} 1 & x_1 & \cdots & x_1^{n-1} \\ 1 & x_2 & \cdots & x_2^{n-1} \\ \vdots & \vdots & \vdots & \vdots \\ 1 & x_n & \cdots & x_n^{n-1} \end{pmatrix} \begin{pmatrix} a_0 \\ a_1 \\ \vdots \\ a_{n-1} \end{pmatrix} = \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{pmatrix}.$$

Since x_i's are distinct, the coefficient matrix on the left-hand side is an invertible Vandermonde matrix. It follows that $a_0, a_1, \ldots, a_{n-1}$ are uniquely expressed in terms of x_i's and y_i's. Thus, $f(x)$ exists and is unique.

We now discuss the inverse of a Vandermonde matrix.

Let $a_1, a_2, \ldots, a_n$ be given constants. Define a polynomial in x by

$$p(x) = (x + a_1)(x + a_2)\cdots(x + a_n)$$

and expand it as

$$p(x) = s_0x^n + s_1x^{n-1} + \cdots + s_{n-1}x + s_n,$$

where $s_0 = 1$ and for each $k = 1, 2, \ldots, n$,

$$s_k = s_k(a_1, a_2, \ldots, a_n) = \sum_{1 \le p_1 < \cdots < p_k \le n} \prod_{q=1}^{k} a_{p_q}.$$

We refer to s_k (which depend on $a_1, a_2, \ldots, a_n$ symmetrically) as the kth *elementary symmetric function* of $a_1, a_2, \ldots, a_n$ (see Problem 16 of Sect. 4.4).

Example 5.4.6 Elementary symmetric functions. By expansion, we have

$$p(x) = (x + a_1)(x + a_2)(x + a_3) = x^3 + s_1x^2 + s_2x + s_3,$$

where

$$s_1 = a_1 + a_2 + a_3, \quad s_2 = a_1a_2 + a_1a_3 + a_2a_3, \quad s_3 = a_1a_2a_3. \ \blacksquare$$

Theorem 5.4.2 *Suppose that $a_1, a_2, \ldots, a_n \in \mathbb{C}$ are distinct. Then*

$$\big(V_n(a_1, a_2, \ldots, a_n)\big)^{-1} = (\alpha_{ij}),$$

where

$$\alpha_{ij} = \frac{(-1)^{1+j} \sum\limits_{\substack{1 \le p_1 < \cdots < p_{n-j} \le n \\ p_q \ne i,\ q=1,\ldots,n-j}} \prod\limits_{q=1}^{n-j} a_{p_q}}{\prod\limits_{k=1,\,k\ne i}^{n} (a_k - a_i)}.$$

Proof. Denote $V = V_n(a_1, a_2, \ldots, a_n)$. Recall from Sect. 1.2 that

$$V^{-1} = \frac{1}{\det V} \operatorname{adj}(V) = \left(\frac{1}{\det V} c_{ij}\right)^T,$$

where c_{ij} is the cofactor of the (i, j)-entry of V.

We now compute the cofactor c_{ij} of V. Let V_k be the matrix obtained from V by deleting row $k+1$ (the kth powers) and adjoining $(a_1^n, a_2^n, \ldots, a_n^n)$ (the nth powers) as the new nth (last) row, $k = 0, 1, \ldots, n-1$. We show

$$\det V_k = s_{n-k} \det V, \quad k = 0, 1, \ldots, n-1. \tag{5.17}$$

Let W be the augmented matrix by adding $(a_1^n, a_2^n, \ldots, a_n^n)$ to V as the last row and then adding $(1,\ -x,\ (-x)^2,\ \ldots,\ (-x)^n)^T$ as the first column. Then W is a Vandermonde matrix of order $n+1$. It follows that

$$\begin{aligned} \det W &= (x + a_1) \cdots (x + a_n) \det V \\ &= (x^n + s_1x^{n-1} + \cdots + s_{n-1}x + s_n) \det V. \end{aligned} \tag{5.18}$$

Expanding $\det W$ along the first column, we have

$$\det W = \det V_0 + x \det V_1 + \cdots + x^{n-1} \det V_{n-1} + x^n \det V. \tag{5.19}$$

Identity (5.17) is obtained by comparing (5.18) and (5.19).

Now notice that each cofactor c_{ij} of V is a determinant of order $n-1$ in the same form as $\det V_k$. Let $V(\widehat{a_j})$ and $s_k(\widehat{a_j})$ denote, respectively, the $(n-1)$-square Vandermonde matrix and the kth elementary symmetric function of

$a_1, a_2, \ldots, a_n$ with a_j absent. Using (5.17) we have

$$\begin{aligned} c_{ij} &= (-1)^{i+j} \det V(i|j) \\ &= (-1)^{i+j} s_{(n-1)-(i-1)}(\widehat{a_j}) \det V(\widehat{a_j}) \\ &= (-1)^{i+j} s_{n-i}(\widehat{a_j}) \det V(\widehat{a_j}). \end{aligned}$$

Thus,

$$\begin{aligned} \frac{1}{\det V} c_{ij} &= \frac{(-1)^{i+j} s_{n-i}(\widehat{a_j}) \det V(\widehat{a_j})}{\prod_{t>s}(a_t - a_s)} \\ &= \frac{(-1)^{i+j} s_{n-i}(\widehat{a_j})}{\prod_{s<j}(a_j - a_s) \prod_{j<t}(a_t - a_j)} \\ &= \frac{(-1)^{i+1} s_{n-i}(\widehat{a_j})}{\prod_{k=1,\, k\neq j}^{n}(a_k - a_j)} \\ &= \frac{(-1)^{i+1} \sum_{p_1<\cdots<p_{n-i}} \prod_{q=1,\, p_q\neq j}^{n-i} a_{p_q}}{\prod_{k=1,\, k\neq j}^{n}(a_k - a_j)}. \end{aligned}$$

Since $\alpha_{ij} = \frac{1}{\det V} c_{ji}$, we have

$$\alpha_{ij} = \frac{(-1)^{1+j} \sum_{p_1<\cdots<p_{n-j},\, p_q\neq i} \prod_{q=1}^{n-j} a_{p_q}}{\prod_{k=1,\, k\neq i}^{n}(a_k - a_i)}. \quad \blacksquare$$

Problems

1. Find the solution(s) to the determinant equation in $x \in \mathbb{C}$:

$$\begin{vmatrix} x^2 & 4 & 9 \\ x & 2 & 3 \\ 1 & 1 & 1 \end{vmatrix} = 0.$$

2. Evaluate the determinant (in terms of $a, x, y, z \in \mathbb{C}$):

$$\begin{vmatrix} 1 & ax & a^2 + x^2 \\ 1 & ay & a^2 + y^2 \\ 1 & az & a^2 + z^2 \end{vmatrix}.$$

3. Find all solutions to the system of equations in $x_1, \ldots, x_n$:

$$\begin{cases} x_1 + \cdots + x_n = a \\ x_1^2 + \cdots + x_n^2 = a^2 \\ \qquad\vdots \\ x_1^n + \cdots + x_n^n = a^n, \end{cases}$$

where (i) $a = 0$, (ii) $a, x_1, \ldots, x_n \in \mathbb{R}$, (iii) $a, x_1, \ldots, x_n \in \mathbb{C}$.

4. Let $x_1, x_2, \ldots, x_n \in \mathbb{C}$ be distinct numbers. Show that for any set of n numbers $y_1, y_2, \ldots, y_n \in \mathbb{C}$, there exists a unique polynomial $f(x)$ of degree at most $n-1$ such that $f(x_i) = y_i,\ i = 1, 2, \ldots, n$. In particular, for any numbers $\lambda_1, \lambda_2, \ldots, \lambda_n$, there exist polynomials $g(x)$ and $h(x)$ (for $h(x)$, assume all $\lambda_i \geq 0$), both of degree at most $n-1$, such that

$$g(\lambda_i) = \overline{\lambda_i}, \quad h(\lambda_i) = \sqrt{\lambda_i}, \quad i = 1, 2, \ldots, n.$$

5. Let $A = \operatorname{diag}(\lambda_1, \lambda_2, \ldots, \lambda_n)$, where $\lambda_i \neq \lambda_j$ for all $i \neq j$. Show that for every normal matrix $B \in \mathbb{M}_n$, there exist a unitary matrix U and a polynomial f of degree at most $n-1$ such that $B = U^* f(A) U$.

6. Let p be a prime integer and let ω be a pth primitive root of unity. Define a $p \times p$ matrix $U = (u_{ij})$ with (i,j)-entry

$$u_{ij} = \omega^{(i-1)(j-1)}, \quad i, j = 1, 2, \ldots, p,$$

Show that all 2×2 submatrices of U are invertible.

7. Let $S_1 = \{\alpha_i\}_{i=1}^r$ and $S_2 = \{\beta_i\}_{i=1}^s$ be multisets of nonzero complex numbers (i.e., repetition of elements is allowed, say, $\{1, 2, 2, 3\}$). If

$$\sum_{i=1}^{r} \alpha_i^k = \sum_{i=1}^{s} \beta_i^k, \quad \text{for every positive integer } k \leq r + s,$$

show that $r = s$ and $S_1 = S_2$, that is, the two sets are the same.

8. Show that two n-square matrices A and B have exactly the same (multiset of) eigenvalues if and only if $\operatorname{tr}(A^k) = \operatorname{tr}(B^k)$ for $k = 1, 2, \ldots, n$.

9. Find the inverse, if it exists, of the Vandermonde matrix

$$\begin{pmatrix} 1 & 1 & 1 \\ 1 & x & y \\ 1 & x^2 & y^2 \end{pmatrix}, \quad x, y \in \mathbb{C}.$$

10. Expand and find the elementary symmetric functions for

$$p(x) = (x + a_1)(x + a_2)(x + a_3)(x + a_4).$$

11. Let W be the matrix obtained from $V = V_n(a_1, \ldots, a_n)$ by replacing the last row $(a_1^{n-1}, \ldots, a_n^{n-1})$ with $(a_1^n, \ldots, a_n^n)$. Show directly without using identity (5.17) that $\det W = (a_1 + \cdots + a_n) \det V$.

12. Let $f_1(x)$, $f_2(x)$, $\ldots$, $f_n(x)$ be polynomials with degree at most $n-2$. Show that for any complex numbers a_1, a_2, $\ldots$, a_n,

$$\begin{vmatrix} f_1(a_1) & f_1(a_2) & \cdots & f_1(a_n) \\ f_2(a_1) & f_2(a_2) & \cdots & f_2(a_n) \\ \vdots & \vdots & \vdots & \vdots \\ f_n(a_1) & f_n(a_2) & \cdots & f_n(a_n) \end{vmatrix} = 0.$$

13. Let $A = (a_{ij}) \in \mathbb{M}_n(\mathbb{C})$. Define $f_i(x) = a_{1i} + a_{2i}x + \cdots + a_{ni}x^{n-1}$ for each $i = 1, 2, \ldots, n$. Show that

$$\begin{vmatrix} f_1(x_1) & f_1(x_2) & \cdots & f_1(x_n) \\ f_2(x_1) & f_2(x_2) & \cdots & f_2(x_n) \\ \vdots & \vdots & \vdots & \vdots \\ f_n(x_1) & f_n(x_2) & \cdots & f_n(x_n) \end{vmatrix} = \det A \prod_{1 \le i < j \le n} (x_j - x_i).$$

14. Let $a_1, a_2, \ldots, a_n$ be complex numbers. Show that

$$\begin{vmatrix} 1 & \cdots & 1 \\ a_1 & \cdots & a_n \\ \vdots & \vdots & \vdots \\ a_1^{n-2} & \cdots & a_n^{n-2} \\ a_2 a_3 \cdots a_n & \cdots & a_1 a_2 \cdots a_{n-1} \end{vmatrix} = (-1)^{n-1} \det V_n(a_1, \ldots, a_n).$$

15. Let $V = V_n(x_1, \ldots, x_n)$ be the $n \times n$ Vandermonde matrix of $x_1, \ldots, x_n$, where $x_i \neq x_j$ if $i \neq j$. Define $F(x) = \prod_{i=1}^{n}(x - x_i)$ and $f_k(x) = F(x)/(x_k - x)$. Show that $f_k(x_j) = 0$ if $j \neq k$ and $f_k(x_k) = -F'(x_k)$, where F' is the first derivative of F. Let $f(x) = -f_k(x)/F'(x_k)$. Construct a matrix M by the coefficients of expanded $f(x)$ as the kth row of M, $k = 1, \ldots, n$. Show that M is the inverse of V.

16. Let A be an $n \times n$ matrix with eigenvalues $\lambda_1, \lambda_2, \ldots, \lambda_n$. Show that

$$\det(\lambda I - A) = \prod_{i=1}^{n}(\lambda - \lambda_i) = \lambda^n - s_1\lambda^{n-1} + s_2\lambda^{n-2} - \cdots + (-1)^n s_n,$$

where $s_k = s_k(\lambda_1, \lambda_2, \ldots, \lambda_n)$, $k = 1, 2, \ldots, n$. Describe s_k in terms of principal minors (see Problem 30 of Sect. 1.3, Chapter 1).

17. Let $\mathcal{A}$ be a diagonalizable linear map on a vector space V of finite dimension with distinct eigenvalues $\lambda_1, \lambda_2, \ldots, \lambda_m$. For $0 \neq x \in V$, let

$$\begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_m \end{pmatrix} = \begin{pmatrix} 1 & 1 & \ldots & 1 \\ \lambda_1 & \lambda_2 & \ldots & \lambda_m \\ \vdots & \vdots & \ddots & \vdots \\ \lambda_1^{m-1} & \lambda_2^{m-1} & \ldots & \lambda_m^{m-1} \end{pmatrix}^{-1} \begin{pmatrix} x \\ \mathcal{A}(x) \\ \vdots \\ \mathcal{A}^{m-1}(x) \end{pmatrix}.$$

Show that $\mathcal{A}(x_k) = \lambda_k x_k$, $1 \le k \le m$. [Note: x_k need not be nonzero.]

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5.5 Hadamard Matrices

An n-square matrix A is called a *Hadamard matrix* if each entry of A is 1 or -1 and if the rows or columns of A are mutually orthogonal, that is,

$$AA^T = nI \quad \text{or} \quad A^TA = nI.$$

Note that $AA^T = nI$ and $A^TA = nI$ are equivalent (Problem 5).

Example 5.5.1 Below are examples of Hadamard matrices of orders 2 and 4:

$$\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}, \quad \begin{pmatrix} -1 & 1 \\ 1 & 1 \end{pmatrix}, \quad \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & -1 & 1 \\ 1 & -1 & 1 & -1 \end{pmatrix}. \blacksquare$$

Theorem 5.5.1 *Let A be an n-square Hadamard matrix. Then*

$$|\det A| = n^{n/2}.$$

Proof. Take the determinants of both sides of $A^TA = nI$. $\blacksquare$

(1) is considered as a Hadamard matrix of order 1. Does there exist a Hadamard matrix of order 3? How about order 6? The answer is negative.

Let A be an $n \times n$ Hadamard matrix. For any $n \times n$ matrices P and Q having entries $0, \pm 1$, and satisfying $P^TP = Q^TQ = I_n$, we have $(PAQ)^T(PAQ) = nI_n$. It follows that PAQ is a Hadamard matrix. Thus, one may change the -1's (if any) in the first row of A to $+1$'s by multiplying a diagonal matrix with diagonal entries ± 1's. For example, $\left(\begin{smallmatrix} -1 & 1 \\ 1 & 1 \end{smallmatrix}\right)\left(\begin{smallmatrix} -1 & 0 \\ 0 & 1 \end{smallmatrix}\right) = \left(\begin{smallmatrix} 1 & 1 \\ -1 & 1 \end{smallmatrix}\right)$.

Theorem 5.5.2 *Let $n > 2$. A necessary condition for an n-square matrix A (of entries ± 1) to be a Hadamard matrix is that n is a multiple of 4.*

Proof 1. Let $A = (a_{ij})$ be an n-square Hadamard matrix. Note that the entries of A are ± 1. The matrix equation $AA^T = nI$ reveals

$$\sum_{k=1}^{n} a_{ik}a_{jk} = \begin{cases} 0, & \text{if } i \neq j, \\ n, & \text{if } i = j. \end{cases}$$

Upon computation, we have

$$\begin{aligned} \sum_{k=1}^{n}(a_{1k} + a_{2k})(a_{1k} + a_{3k}) &= \sum_{k=1}^{n} a_{1k}^2 + \sum_{k=1}^{n} a_{1k}a_{2k} \\ &\quad + \sum_{k=1}^{n} a_{1k}a_{3k} + \sum_{k=1}^{n} a_{2k}a_{3k} \\ &= \sum_{k=1}^{n} a_{1k}^2 = n. \end{aligned}$$

Observe that the possible values of $a_{1k}+a_{2k}$ and $a_{1k}+a_{3k}$ are $+2$, 0, and -2. Thus, each term in the summation $\sum_{k=1}^{n}(a_{1k}+a_{2k})(a_{1k}+a_{3k})$ is $+4$, 0, or -4. It follows that n is divisible by 4.

Proof 2. Let P be an $n \times n$ diagonal matrix with diagonal entries 1 or -1 such that the first row of AP consists entirely of 1. Note that AP is also a Hadamard matrix. Since the second row of AP is orthogonal to the first row, if this row has, say r, 1's, then it has r (-1)'s. (Thus, $n = 2r$ is an even number.) For the same reason, the third row has r 1's and r (-1)'s.

Let n_{-}^{+} be the number of columns of AP that contain a 1 of row two and a (-1) of row three. In a similar way, define n_{+}^{-}, n_{+}^{+}, and n_{-}^{-}. Then

$$n_{+}^{+} + n_{-}^{+} = n_{+}^{+} + n_{+}^{-} = n_{-}^{-} + n_{-}^{+} = r.$$

Thus,

$$n_{+}^{+} = n_{-}^{-}, \quad n_{-}^{+} = n_{+}^{-}.$$

The orthogonality of rows two and three implies that

$$n_{+}^{+} + n_{-}^{-} = n_{+}^{-} + n_{-}^{+}.$$

It follows that $n_{+}^{+} = n_{-}^{+}$. Therefore, $n = 2r = 4n_{+}^{+}$ is a multiple of 4. ∎

It is an open (hard) problem, known as the *Hadamard matrix conjecture*, that a Hadamard matrix of size $4k \times 4k$ exists for every positive integer k.

The following theorem, which can be verified directly by a simple computation, gives a way to construct Hadamard matrices of larger dimensions.

Theorem 5.5.3 *If A is a Hadamard matrix of order n, then the block matrix $\begin{pmatrix} A & A \\ A & -A \end{pmatrix}$ is a Hadamard matrix of order $2n$.*

Proof. Verify

$$\begin{pmatrix} A & A \\ A & -A \end{pmatrix}^{T} \begin{pmatrix} A & A \\ A & -A \end{pmatrix} = \begin{pmatrix} 2A^{T}A & A^{T}A - A^{T}A \\ A^{T}A - A^{T}A & 2A^{T}A \end{pmatrix} = 2nI_{2n}. \quad ∎$$

Example 5.5.2 Construction of a Hadamard matrix of higher order.

$$A = \begin{pmatrix} -1 & 1 & 1 & 1 \\ 1 & -1 & 1 & 1 \\ 1 & 1 & -1 & 1 \\ 1 & 1 & 1 & -1 \end{pmatrix} \text{ is a Hadamard matrix, and so is}$$

$$B = \begin{pmatrix} A & A \\ A & -A \end{pmatrix} = \begin{pmatrix} -1 & 1 & 1 & 1 & -1 & 1 & 1 & 1 \\ 1 & -1 & 1 & 1 & 1 & -1 & 1 & 1 \\ 1 & 1 & -1 & 1 & 1 & 1 & -1 & 1 \\ 1 & 1 & 1 & -1 & 1 & 1 & 1 & -1 \\ -1 & 1 & 1 & 1 & 1 & -1 & -1 & -1 \\ 1 & -1 & 1 & 1 & -1 & 1 & -1 & -1 \\ 1 & 1 & -1 & 1 & -1 & -1 & 1 & -1 \\ 1 & 1 & 1 & -1 & -1 & -1 & -1 & 1 \end{pmatrix}. \quad ∎$$

We can recursively construct Hadamard matrices H_n of order 2^n with

$$H_1 = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}, \quad H_n = \begin{pmatrix} H_{n-1} & H_{n-1} \\ H_{n-1} & -H_{n-1} \end{pmatrix}, \quad n \geq 2. \tag{5.20}$$

Let

$$x_1 = \begin{pmatrix} 1 \\ c \end{pmatrix} \text{ and } x_n = \begin{pmatrix} x_{n-1} \\ cx_{n-1} \end{pmatrix}, \text{ where } c = -1 + \sqrt{2}. \tag{5.21}$$

A simple computation shows that H_1 has two eigenvalues $\pm\sqrt{2}$ and that x_1 is an eigenvector corresponding to $\sqrt{2}$. This generalizes as follows.

Theorem 5.5.4 *Let H_n be defined as in* (5.20). *Then $2^{n/2}$ and $-2^{n/2}$ are the eigenvalues of H_n each of multiplicity 2^{n-1}, furthermore, x_n defined in* (5.21) *is an eigenvector corresponding to the positive eigenvalue $2^{n/2}$.*

Proof. We use induction on n. The case of $n = 1$ is by a direct computation.

Let $n \geq 2$ and suppose that the assertion is true for $n - 1$. We derive

$$\begin{aligned} \det(\lambda I - H_n) &= \begin{vmatrix} \lambda I - H_{n-1} & -H_{n-1} \\ -H_{n-1} & \lambda I + H_{n-1} \end{vmatrix} \\ &= \det\left((\lambda I - H_{n-1})(\lambda I + H_{n-1}) - H_{n-1}^2\right) \\ &= \det(\lambda^2 I - 2H_{n-1}^2) \\ &= \det(\lambda I - \sqrt{2}H_{n-1}) \det(\lambda I + \sqrt{2}H_{n-1}). \end{aligned}$$

If μ is an eigenvalue of H_{n-1}, then $\det(\sqrt{2}\mu I - \sqrt{2}H_{n-1}) = 0$ and $\det(-\sqrt{2}\mu I + \sqrt{2}H_{n-1}) = 0$. Thus, $\det(\pm\sqrt{2}\mu I - H_n) = 0$. This is to say that each eigenvalue μ of H_{n-1} generates two eigenvalues $\pm\sqrt{2}\mu$ of H_n. The assertion (of case n) then follows by the induction hypothesis that H_{n-1} has eigenvalues $2^{(n-1)/2}$ and $-2^{(n-1)/2}$ each of multiplicity 2^{n-2}.

For the eigenvector part, we observe that (using induction again)

$$\begin{aligned} H_n x_n &= \begin{pmatrix} H_{n-1} & H_{n-1} \\ H_{n-1} & -H_{n-1} \end{pmatrix} \begin{pmatrix} x_{n-1} \\ (-1+\sqrt{2}\,)x_{n-1} \end{pmatrix} \\ &= \begin{pmatrix} \sqrt{2}H_{n-1}x_{n-1} \\ (2-\sqrt{2}\,)H_{n-1}x_{n-1} \end{pmatrix} \\ &= 2^{n/2} \begin{pmatrix} x_{n-1} \\ (-1+\sqrt{2}\,)x_{n-1} \end{pmatrix} \\ &= 2^{n/2} x_n. \end{aligned}$$

∎

Let A be a Hadamard matrix. By negating rows and columns of A, or by interchanging rows and columns of A, we obtain a matrix B which is

also a Hadamard matrix. (A and B are *equivalent* Hadamard matrices.) By equivalence operations we may put "many" 1's together as a block.

Let E_n be the $n \times n$ matrix of all-ones. We give a lower bound for the order of a Hadamard matrix that contains E_n as a submatrix.

Theorem 5.5.5 *If A is an m-square Hadamard matrix that contains E_n (the all-ones matrix of order n) as a submatrix, then $m \geq n^2$.*

Proof. We may assume by permutation that A is partitioned as

$$A = \begin{pmatrix} E_n & X \\ Y & Z_s \end{pmatrix}, \tag{5.22}$$

where Z_s is an s-square matrix of entries ± 1's, and $s = m - n$.

Since A is a Hadamard matrix of order $m = n + s$, we have

$$AA^T = (n+s)I_m,$$

which implies, by using the block form (5.22) of A, that

$$E_n^2 + XX^T = (n+s)I_n.$$

Thus,

$$XX^T = (n+s)I_n - nE_n. \tag{5.23}$$

Note that the eigenvalues of the matrix on the right-hand side of (5.23) are

$$n + s - n^2, \quad n + s, \quad \ldots, \quad n + s.$$

Because XX^T is a positive semidefinite matrix, the eigenvalues of XX^T are all nonnegative. It follows that $n + s - n^2 \geq 0$, that is, $m \geq n^2$. ∎

For a Hadamard matrix to have the 4×4 all-ones matrix E_4 as a submatrix, the size of such a Hadamard matrix has to be at least 16×16.

Problems

1. Show that A is a Hadamard matrix and find A^4, where

$$A = \begin{pmatrix} 1 & 1 & 1 & -1 \\ 1 & 1 & -1 & 1 \\ 1 & -1 & 1 & 1 \\ -1 & 1 & 1 & 1 \end{pmatrix}.$$

2. Find a 2×2 real matrix A of non-integer entries such that $A^TA = 2I_2$.

3. Does there exist a 3×3 Hadamard matrix? Does there exist a 6×6 Hadamard matrix? Construct an 8×8 Hadamard matrix.

4. Let $A = (a_{ij})$ be a 3×3 matrix. Show that if each $a_{ij} = 1$ or -1, then $\det A$ is an even number. What is the maximum value for $\det A$?

5. Let $A \in \mathbb{M}_n$. Show that $AA^T = nI$ if and only if $A^T A = nI$. Conclude that if A is a Hadamard matrix, then $\frac{1}{\sqrt{n}}A$ is an orthogonal matrix.

6. Find the eigenvalues and eigenvectors of the Hadamard matrix

$$\begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & -1 & 1 & -1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & -1 & 1 \end{pmatrix}.$$

7. Show that the Kronecker product of two Hadamard matrices is also a Hadamard matrix. How about the Hadamard product?

8. Let A, B, C, D be $n \times n$ matrices of entries ± 1 such that $AA^T + BB^T + CC^T + DD^T = 4nI_n$ and $XY^T = YX^T$, where X and Y are any two of A, B, C, D. Show that the block matrix H is a Hadamard matrix:

$$H = \begin{pmatrix} A & B & C & D \\ -B & A & -D & C \\ -C & D & A & -B \\ -D & -C & B & A \end{pmatrix}.$$

9. Define recursively for $n = 1, 2, 3, \ldots$

$$H_1 = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}, \quad H_n = \begin{pmatrix} H_{n-1} & H_{n-1} \\ H_{n-1} & -H_{n-1} \end{pmatrix},$$

and let

$$F_1 = \frac{1}{2}(I + 2^{-n/2}H_n), \quad F_2 = -\frac{1}{2}(I - 2^{-n/2}H_n).$$

Show that F_1 and F_2 are idempotent and that $F_1 + F_2 = 2^{-n/2}H_n$.

10. Define recursively for $n = 1, 2, 3, \ldots$

$$G_1 = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \quad G_{n+1} = \begin{pmatrix} 0 & -G_n \\ G_n & 0 \end{pmatrix}.$$

Show that

(a) $G_n^2 = (-1)^n I_{2^n}$.

(b) G_n is symmetric if n is even, and skew-symmetric if n is odd.

(c) $G_n H_n = (-1)^n H_n G_n$, where H_n is defined as in Problem 9.

11. Let A be a square matrix with entries 1, -1, or 0. If each row and column of A contains only one nonzero entry 1 or -1, show that $A^k = I$ for some positive integer k.

12. How many $n \times n$ matrices of 0 and 1 entries are there in which the number of 1's in each row and each column is even?

13. Let $A = (a_{ij}) \in \mathbb{M}_n(\mathbb{C})$.

    (a) If $|a_{ij}| \leq 1$ for all i, j, show that $|\det A| \leq n^{n/2}$.

    (b) If the entries of A are ± 1's, show that $|\det A| = n^{n/2}$ if and only if A is a Hadamard matrix.

    (c) Find a 2×2 matrix A of non-integer entries such that $|\det A| = 2$.

14. Evaluate the determinant of each of the following matrices.

$$\begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix}, \quad \begin{pmatrix} 0 & 1 & -1 \\ -1 & 0 & 1 \\ 1 & -1 & 0 \end{pmatrix}, \quad \begin{pmatrix} 0 & 1 & 1 & 1 \\ -1 & 0 & 1 & 1 \\ -1 & -1 & 0 & -1 \\ -1 & -1 & 1 & 0 \end{pmatrix}.$$

15. In a tournament of n players, each player enters a game where ties are impossible. A player will get 1 point if she/he wins the game, otherwise she/he loses 1 point. The results of the tournament can be represented by an n-square *tournament matrix* whose (i, j)-entry is 1 if player i beats player j, -1 if j beats i, and 0 if $i = j$. Let A be an n-square tournament matrix. Show that

    (a) A is skew-symmetric, i.e., $A = -A^T$.

    (b) $\det A = 0$ if and only if n is odd.

    (c) $\text{rank}(A) = n$ if n is even.

    (d) $\text{rank}(A) = n - 1$ if n is odd.

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5.6 Permutation Matrices and Doubly Stochastic Matrices

A square matrix with 0 and 1 entries is called a *permutation matrix* if it has exactly one entry 1 in each row and each column and all other entries are 0. A square matrix P of entries 0's and 1's is a permutation matrix if and only if $P^T P = I$, equivalently, P is invertible and $P^{-1} = P^T$ (Problem 1). It is easy to check that the product of two permutation matrices of the same order is a permutation matrix. There are $n!$ permutation matrices of order n.

A square matrix is said to be *doubly stochastic* if all entries of the matrix are nonnegative and the sum of the entries in each row and each column is equal to 1. So, permutation matrices are doubly stochastic matrices.

Example 5.6.1 Some examples of permutation matrices:

$$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix}, \quad \begin{pmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}, \quad \begin{pmatrix} 0 & I_n \\ I_m & 0 \end{pmatrix}.$$

Some examples of doubly stochastic matrices:

$$\begin{pmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} \end{pmatrix}, \quad \begin{pmatrix} \frac{1}{4} & \frac{1}{2} & \frac{1}{4} \\ 0 & \frac{1}{4} & \frac{3}{4} \\ \frac{3}{4} & \frac{1}{4} & 0 \end{pmatrix}, \quad \frac{1}{6}\begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \\ 3 & 1 & 2 \end{pmatrix}, \quad \begin{pmatrix} 0.5 & 0.2 & 0.3 \\ 0.1 & 0.4 & 0.5 \\ 0.4 & 0.4 & 0.2 \end{pmatrix}. \blacksquare$$

We show that every permutation matrix is a direct sum of primary permutation matrices under permutation similarity and that every doubly stochastic matrix is a convex combination of permutation matrices.

We say that a square matrix A is *reducible* if it can be brought to a block upper-triangular form by simultaneous row and column permutations, equivalently, there exists a permutation matrix P such that

$$P^T AP = \begin{pmatrix} B & C \\ 0 & D \end{pmatrix}, \tag{5.24}$$

where B and D are *square* matrices of order at least 1.

A matrix is said to be *irreducible* if it is not reducible. Conventionally, any matrix of order 1 is considered to be irreducible.

Let $M = P^T AP = P^{-1}AP$ in (5.24). M is similar to A via permutation matrix P. We say that M and A are *permutation similar*. A is reducible if it is permutation similar to a block upper-triangular matrix; A is irreducible if and only if $Q^T AQ$ is irreducible for any permutation matrix Q.

It is obvious that the diagonal entries of irreducible permutation matrices are all equal to zero. The converse need not be true in general.

Example 5.6.2 Irreducible and reducible matrices.

$$\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix}, \text{ and } \begin{pmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix} \text{ are reducible.}$$

$$\begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}, \begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 0 & 1 & 1 \end{pmatrix}, \text{ and } \begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 0 & 1 \end{pmatrix} \text{ are irreducible. } \blacksquare$$

Theorem 5.6.1 *Every reducible permutation matrix is permutation similar to a direct sum (i.e., block diagonal) of irreducible permutation matrices.*

Proof. Let A be an n-square reducible permutation matrix as in (5.24). We show that matrix C contains only zeros. Suppose otherwise. Let B be $r \times r$ and let D be $s \times s$, $r + s = n$. Then B contains r 1's (in columns) and D contains s 1's (in rows). If C contained a 1, then A would have at least $r + s + 1 = n + 1$ 1's, a contradiction. It follows that $A = B \oplus D$, where B and D are permutation matrices of orders less than n. The assertion follows by repeating the process (or by induction) on B and D. $\blacksquare$

We show (Theorem 5.6.3) that every n-square irreducible permutation matrix is permutation similar to the $n \times n$ *primary permutation matrix*

$$P = \begin{pmatrix} 0 & 1 & 0 & \dots & 0 \\ 0 & 0 & 1 & \dots & 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ 0 & 0 & 0 & \dots & 1 \\ 1 & 0 & 0 & \dots & 0 \end{pmatrix}. \tag{5.25}$$

Example 5.6.3 Eigenvalues of (non-)primary permutation matrices.

(a) The characteristic polynomial of P in (5.25) is $\det(\lambda I - P) = \lambda^n - 1$. Thus, the eigenvalues of the $n \times n$ primary permutation matrix P are exactly the n roots of the equation $\lambda^n = 1$, that is, $1, \omega, \dots, \omega^{n-1}$, where ω is an nth primitive root of unity. To be explicit, they are

$$e^{\frac{2k\pi}{n}i} = \cos\frac{2k\pi}{n} + i\sin\frac{2k\pi}{n}, \quad k = 0, 1, 2, \dots, n-1.$$

(b) The eigenvalues of $\begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix}$ are 1, $\frac{-1+\sqrt{3}\,i}{2}$, $\frac{-1-\sqrt{3}\,i}{2}$.

(c) The eigenvalues of $\begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \end{pmatrix}$ are 1, -1, i, $-i$.

(d) Let P be the primary permutation matrix of order n given in (5.25). Then $PP^T = I_n$, $P^T = P^{-1} = P^{n-1}$, $P^n = I_n$, $P^k = \begin{pmatrix} 0 & I_{n-k} \\ I_k & 0 \end{pmatrix}$ $(k < n)$.

(e) Let $Q = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{pmatrix}$. Q is a non-primary permutation matrix of order 3 with eigenvalues $1, 1, -1$. Computations show $Q^2 = I_3$ and $Q^3 \neq I_3$. ∎

Theorem 5.6.2 *A primary permutation matrix is irreducible.*

Proof. Suppose that the $n \times n$ primary permutation matrix P in (5.25) is reducible. Let $S^T PS = J_1 \oplus \cdots \oplus J_k$, $k \geq 2$, where S is some permutation matrix and J_t is an irreducible matrix with order $n_t < n$, $t = 1, \ldots, k$.

The rank of $P - I$ is $n - 1$, because $\det(P - I) = 0$ and the submatrix of order $n - 1$ in the upper-right corner of $P - I$ is nonsingular. Thus,

$$\operatorname{rank}(S^T PS - I) = \operatorname{rank}\big(S^T (P - I)S\big) = n - 1.$$

On the other hand, since $S^T PS = J_1 \oplus \cdots \oplus J_k$, we have

$$\operatorname{rank}(S^T PS - I) = \sum_{s=1}^{k} \operatorname{rank}(J_t - I_{n_t}) \leq n - k < n - 1.$$

This is a contradiction. The proof is complete. ∎

Theorem 5.6.3 *A permutation matrix is irreducible if and only if it is permutation similar to a primary permutation matrix.*

Proof. Let Q be an $n \times n$ permutation matrix. If Q is permutation similar to P as defined in (5.25), then Q is irreducible by the previous theorem.

For the converse, suppose that Q is irreducible. We show that Q can be brought to P through simultaneous row and column permutations.

Q contains a 1 in the first row. Let the 1 be in the position $(1, i_1)$. Then $i_1 \neq 1$ (since Q is irreducible). If $i_1 = 2$, we proceed to the next step (or $n = 2$), considering the 1 in the second row. Otherwise, $i_1 > 2$. Interchange columns 2 and i_1 so that the 1 is placed in the $(1, 2)$ position.

Interchange rows 2 and i_1 to get a matrix Q_1. This matrix is permutation similar to Q and is also irreducible. If the $(2, 3)$-entry of Q_1 is 1, we go on to the next step. Otherwise, let the $(2, i_2)$-entry be 1, $i_2 \neq 3$. If $i_2 = 1$, then Q_1 would be reducible, because all entries in the first two columns but not in the first two rows are equal to 0. Thus, $i_2 \geq 3$. Interchange columns 3 and i_2 so that the 1 is in the $(2, 3)$ position.

Note that the 1 in the $(1, 2)$ position was not affected by the permutations in the second step. Continuing in this way, one obtains the permutation matrix having 1's in the $(1, 2)$, $(2, 3)$, $\ldots$, $(n - 1, n)$ positions. Since it is a permutation matrix, the $(n, 1)$-entry has to be 1, and this is exactly the primary permutation matrix P in the form of (5.25). The product of a sequence of permutation matrices is also a permutation matrix. As a result, we have a permutation matrix S such that $S^T QS = S^{-1} QS = P$. ∎

Theorem 5.6.4 *An n-square permutation matrix is irreducible if and only if its eigenvalues are the n roots of unity.*

Proof. Let Q be an n-square permutation matrix. If Q is irreducible, then Q is similar to the n-square primary permutation matrix. The latter has eigenvalues $1, \omega, \ldots, \omega^{n-1}$ (the n roots of unity), so does matrix Q.

Conversely, suppose that $1, \omega, \ldots, \omega^{n-1}$ are the eigenvalues of Q. Note that $\omega^k \neq 1$ for any $1 \leq k < n$ (since ω is an nth primitive root of unity). If Q is reducible, then we may write

$$S^T Q S = J_1 \oplus \cdots \oplus J_k,$$

where S is a permutation matrix, and $J_1, \ldots, J_k$ (referred to as J_t's) are primary permutation matrices with orders less than n.

The eigenvalues of J_t's are the eigenvalues of Q, none of which is an nth primitive root of unity, because the order of every J_t is less than n. This is a contradiction to ω being an eigenvalue of Q. Thus, Q is irreducible. ∎

We summarize the above discussions in a theorem with two parts.

Theorem 5.6.5 *Part I: Every permutation matrix is permutation similar to $P_1 \oplus P_2 \oplus \cdots \oplus P_k$, where $k \geq 1$ and $P_1, \ldots, P_k$ are primary permutation matrices (of possibly different orders). Part II: Two permutation matrices of the same order are similar if and only if they are permutation similar, if and only if they have the same eigenvalues (including multiplicities), and if and only if they have the same irreducible blocks.*

Next we study stochastic matrices, a family of matrices that have applications in statistics and in other subjects. We present an important relation between permutation matrices and doubly stochastic matrices.

A *nonnegative matrix* (i.e., of nonnegative entries) is said to be *doubly stochastic* if the entries of each row and each column sum to 1. Equivalently, a nonnegative matrix $A \in \mathbb{M}_n$ is doubly stochastic if for $e = (1, \ldots, 1)^T \in \mathbb{R}^n$,

$$e^T A = e^T \quad \text{and} \quad Ae = e. \tag{5.26}$$

It is readily seen that 1 is an eigenvalue of any doubly stochastic matrix. Moreover, permutation matrices are doubly stochastic and the product of two doubly stochastic matrices is doubly stochastic.

We show that a square matrix is a doubly stochastic matrix if and only if it is a convex combination of (finitely many) permutation matrices. To prove this, we need a result which is of interest in its own right.

Theorem 5.6.6 (Frobenius-König) *Let $A \in \mathbb{M}_n$. Then every product of n entries of A taken from distinct rows and columns equals 0, in symbols,*

$$a_{1i_1} a_{2i_2} \cdots a_{ni_n} = 0 \ \text{ if } \{i_1, i_2, \ldots, i_n\} = \{1, 2, \ldots, n\}, \tag{5.27}$$

if and only if A contains an $r \times s$ zero submatrix, where $r + s = n + 1$.

Proof. First notice that property (5.27) of $A \in \mathbb{M}_n$ remains true when row or column interchanges are applied to A. In other words, an n-square matrix A has property (5.27) if and only if PAQ has the property, where P and Q are any n-square permutation matrices.

Sufficiency: We may assume by permutations of rows and columns that the $r \times s$ zero submatrix is in the lower-left corner, and we write

$$A = \begin{pmatrix} B & C \\ 0_{r\times s} & D \end{pmatrix}.$$

Because $n - r = s - 1$, B has size $(s-1) \times s$. In any product $a_{1i_1} a_{2i_2} \cdots a_{ni_n}$, there are s entries from the first s columns of A. Because the s entries are from different rows and since B has only $s-1$ rows, at least one of the s entries has to be a zero from the zero matrix $0_{r\times s}$. Therefore, every product $a_{1i_1} a_{2i_2} \cdots a_{ni_n}$ contains a zero factor, hence is equal to zero.

Necessity: If $n = 1$ or all the entries of A are zero, there is nothing to prove. Let A have a nonzero entry and $n > 1$. Suppose that the assertion holds for the matrices of order $n-1$. Let A_1 be the submatrix obtained from A by deleting the row and the column that contain the nonzero entry. Then A_1 is a matrix of order $n-1$ having the property (5.27). Applying induction hypothesis to A_1 results in a zero submatrix of size $p \times q$, where $p + q = (n-1) + 1 = n$. By interchanging rows and columns, we write A as

$$A = \begin{pmatrix} B & C \\ 0_{p\times q} & D \end{pmatrix},$$

where B is $q \times q$ and D is $p \times p$. Since every product of the entries of A from different rows and columns is 0, this property is inherited by B or D, say B. By induction, B has an $l \times s$ zero submatrix such that $l + s = q + 1$. Putting (by permutations) this zero submatrix in the lower-left corner of B, we see that A has an $r \times s$ zero submatrix, where $r = p + l$ and $r + s = n + 1$. ∎

Theorem 5.6.7 (Birkhoff Theorem) *A nonnegative square matrix is doubly stochastic if and only if it is a convex combination of permutation matrices.*

Proof. For sufficiency, let A be a convex combination of permutation matrices $P_1, P_2, \ldots, P_m$, i.e., for some positive $t_1, t_2, \ldots, t_m \in \mathbb{R}$ of sum 1,

$$A = t_1 P_1 + t_2 P_2 + \cdots + t_m P_m.$$

Then $P_j e = e$, $j = 1, 2, \ldots, m$, where $e = (1, \ldots, 1)^T$. Thus, $e^T A = e^T$ and $Ae = e$. Note that A is nonnegative. By (5.26), A is doubly stochastic.

For necessity, we apply induction on the number of zero entries of the doubly stochastic matrices (of fixed order n). Let $A \in \mathbb{M}_n$ be doubly stochastic. If A has $n^2 - n$ zero entries (the largest number of zeros as the base case for

induction), then A is a permutation matrix, and we have nothing to show. Suppose that the doubly stochastic matrices with at least k zero entries are convex combinations of permutation matrices. We show that the assertion holds for the doubly stochastic matrices with $k-1$ zero entries.

Let A be an n-square doubly stochastic matrix of $k-1$ zero entries. If every product of the entries of A from distinct rows and columns is zero, by the Frobenius–König theorem, A can be written as (up to permutations)

$$A = \begin{pmatrix} B & C \\ 0_{r\times s} & D \end{pmatrix}, \quad r+s = n+1.$$

Since the entries in each column of A sum to 1, the sum of all entries of B is equal to s. Similarly, by considering rows, the sum of all entries of D is equal to r. Thus, the sum of all entries of A would be at least $r+s = n+1$. This is impossible, because the sum of all entries of A is n. Therefore, some product, say, $a_{1i_1 1}a_{2i_2}\cdots a_{ni_n} \neq 0$. Let $\delta = \min\{a_{1i_1}, a_{2i_2}, \ldots, a_{ni_n}\}$. Then $0 < \delta \leq 1$. If $\delta = 1$, then A is a permutation matrix. So, let $0 < \delta < 1$.

Let P_1 be the permutation matrix with 1 in the positions $(1, i_1)$, $(2, i_2)$, $\ldots$, (n, i_n), and 0 elsewhere. Consider the matrix

$$E = (1-\delta)^{-1}(A - \delta P_1), \quad \delta = \min\{a_{1i_1}, a_{2i_2}, \ldots, a_{ni_n}\}.$$

By (5.26), one can check that E is doubly stochastic. Note that E has at least one more zero entry than A, namely, E has at leat k zero entries. By the induction hypothesis, there are positive numbers $t_2, \ldots, t_m$ with sum equal to 1 and permutation matrices $P_2, \ldots, P_m$ such that

$$E = t_2P_2 + \cdots + t_mP_m.$$

It follows that

$$A = \delta P_1 + (1-\delta)t_2P_2 + \cdots + (1-\delta)t_mP_m,$$

which is a convex combination of permutation matrices $P_1, P_2, \ldots, P_m$. ∎

The set of $n \times n$ doubly stochastic matrices is called the *Birkhoff polytope*. The $n \times n$ permutation matrices serve as the vertices (or extreme points) of the polytope. The Birkhoff polytope has $n!$ vertices.

Example 5.6.4 Convex combinations of permutation matrices.

$$\frac{1}{6}\begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \\ 3 & 1 & 2 \end{pmatrix} = \frac{1}{6}\begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix} + \frac{1}{3}\begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix} + \frac{1}{2}\begin{pmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{pmatrix}.$$

$$\begin{pmatrix} \frac{1}{3} & \frac{7}{12} & 0 & \frac{1}{12} \\ \frac{5}{12} & 0 & \frac{7}{12} & 0 \\ 0 & \frac{1}{12} & 0 & \frac{11}{12} \\ \frac{1}{4} & \frac{1}{3} & \frac{5}{12} & 0 \end{pmatrix} = \frac{1}{4}\begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \end{pmatrix} + \frac{1}{3}\begin{pmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}$$
$$+\frac{1}{3}\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \end{pmatrix} + \frac{1}{12}\begin{pmatrix} 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{pmatrix}. \blacksquare$$

Problems

1. Show that the determinant of a permutation matrix is ± 1 and that permutation matrices are unitary and hence normal.

2. Let $A = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) \in \mathbb{M}_2$. Show that A is reducible if and only if $bc = 0$.

3. Let $A = \left(\begin{smallmatrix} t & 1-t \\ 1-t & t \end{smallmatrix}\right)$ for $t \in [0,1]$. (i) Show that A is doubly stochastic. (ii) What are the eigenvalues of A? (iii) Show that A is diagonalizable.

4. Determine if each of the following matrices is reducible or irreducible:

$$\begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 \\ 1 & 1 \end{pmatrix}, \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}, \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix},$$
$$\begin{pmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \end{pmatrix}, \begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \end{pmatrix}, \begin{pmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \end{pmatrix}, \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}.$$

5. Find permutations that reduce the (reducible) permutation matrix

$$P = \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix}$$

to a direct sum of irreducible matrices. (Note that $P^2 = I_4$.)

6. Show that the determinant below is 0 for any unspecified values $\times$:

$$\begin{vmatrix} 0 & \times & \times & 0 & 0 \\ \times & \times & \times & \times & \times \\ 0 & \times & \times & 0 & 0 \\ \times & \times & \times & \times & \times \\ 0 & \times & \times & 0 & 0 \end{vmatrix}.$$

7. Let $A \in \mathbb{M}_n$. If A contains a $p \times q$ zero submatrix with $p+q > n$, show that A contains an $r \times s$ zero submatrix with $r + s = n + 1$.

8. Let $A = B = \left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$. Show that A and B are irreducible but AB, A^2, and $A \otimes B$ are reducible. Find two irreducible matrices such that the Hadamard (entrywise) product of the matrices is reducible.

9. Let P be an $n \times n$ irreducible permutation matrix. Show that

$$\operatorname{rank}(P - I) = n - 1.$$

10. Let P be an $n \times n$ irreducible permutation matrix. Show that P^k is irreducible if and only if $(n, k) = 1$ (i.e., n and k have no common positive divisors other than 1). Thus, P^k is reducible if $(n, k) > 1$.

11. Let P be an $n \times n$ permutation matrix. Show that P is irreducible if and only if P has the property: $P^m = I_n \Leftrightarrow n|m$ (i.e., n divides m).

12. Show that for any $n \times n$ permutation matrix P, $P^{n!} = I$, and further show that if P is irreducible, then $P^n = I$. Is the converse true?

13. Show that every eigenvalue of an $n \times n$ permutation matrix is a root of $x^{n!} - 1 = 0$. Does it have to be a root of $x^n - 1 = 0$?

14. Let P be an $n \times n$ irreducible permutation matrix. Show that P is diagonalizable over $\mathbb{C}$ for all n, but not over $\mathbb{R}$ if $n \geq 3$.

15. Let Q be an $n \times n$ permutation matrix. Show that 1 is a simple (i.e., non-repeated) eigenvalue of Q if and only if Q is irreducible (equivalently, Q is permutation similar to the primary permutation matrix) and if and only if all eigenvalues of Q are simple.

16. Show that if two permutation matrices A and B are similar, i.e., $S^{-1}AS = B$ for some nonsingular S, then they are permutation similar, namely, S can be chosen to be a permutation matrix.

17. Show that any $n \times n$ symmetric permutation matrix is reducible if $n \geq 3$. Show by an example that this need not be true if $n = 2$.

18. If P and Q are primary permutation matrices of respective orders m and n, what is the smallest positive integer k such that $\left(\begin{smallmatrix} P & 0 \\ 0 & Q \end{smallmatrix}\right)^k = I_{m+n}$?

19. Find the ranks of the matrices Q and $I_{5n} - Q$, where

$$Q = \begin{pmatrix} 0 & 0 & I_n & 0 & 0 \\ 0 & 0 & 0 & I_n & 0 \\ 0 & 0 & 0 & 0 & I_n \\ I_n & 0 & 0 & 0 & 0 \\ 0 & I_n & 0 & 0 & 0 \end{pmatrix} \in \mathbb{M}_{5n}.$$

What is the smallest positive integer k such that $Q^k = I_{5n}$?

20. Show that $\begin{pmatrix} 0&1&1&1\\ 0&0&1&1\\ 0&0&0&1\\ 0&0&0&0 \end{pmatrix}$ is similar but not permutation similar to $\begin{pmatrix} 0&1&0&0\\ 0&0&1&0\\ 0&0&0&1\\ 0&0&0&0 \end{pmatrix}$.

21. Show that an $n \times n$ matrix A with nonnegative entries is reducible if and only if there is a proper subset $\{i_1, \ldots, i_k\}$ of $\{1, \ldots, n\}$ such that

$$\text{Span}\{Ae_{i_1}, \ldots, Ae_{i_k}\} \subseteq \text{Span}\{e_{i_1}, \ldots, e_{i_k}\},$$

where the e_i are the standard basis vectors for $\mathbb{C}^n$.

22. Let A and B be n-square matrices. If c is an entry of A and c is not an entry of B, show that A and B cannot be permutation similar.

23. Show that the product of two doubly stochastic matrices is a doubly stochastic matrix. What about the sum?

24. Show that if $U = (u_{ij})$ is a unitary matrix, then $A = (|u_{ij}|^2)$ is a doubly stochastic matrix. What about $B = (|u_{ij}||v_{ij}|)$, where $V = (v_{ij})$ is a unitary matrix of the same size as U?

25. Find a 3×3 doubly stochastic matrix S such that S is not equal to $(|u_{ij}|^2)$ for any 3×3 unitary matrix $U = (u_{ij})$.

26. If $A \in \mathbb{M}_n$ is nonsingular, how many zero entries can A have at most?

27. Show that 1 is an eigenvalue of any doubly stochastic matrix.

28. Show that $|\lambda| \leq 1$ for any eigenvalue λ of a doubly stochastic matrix.

29. Let S be a 2×2 doubly stochastic matrix. Show that $\sigma_{\max}(S) \leq 1$.

30. Let S be an $n \times n$ doubly stochastic matrix. Show that $\sigma_{\max}(S) \leq \sqrt{n}$.

31. Show that a tridiagonal doubly stochastic matrix is real symmetric.

32. Explain why the product of two permutation matrices (of the same order) is always a permutation matrix, while the sum of two permutation matrices is never a permutation matrix.

33. Let P and Q be permutation matrices of the same order. Show that $\alpha P + (1 - \alpha)Q$ is a doubly stochastic matrix for any $\alpha \in [0, 1]$.

34. Let P and Q be $n \times n$ permutation matrices with p 1's and q 1's on the main diagonals, respectively. If $p + q \geq n + 1$, show that PQ has at least one 1 on the main diagonal (thus PQ is reducible).

35. Show that every nilpotent matrix with nonnegative entries is permutation similar to a strictly upper-triangular matrix. [Hint: Show that a nilpotent (0, 1)-matrix contains a zero column by induction.]

36. A *permutation of an n element set* $\{1, 2, \ldots, n\}$ is a reordering of the elements in the set, say $p_1, p_2, \ldots, p_n$. Let p be the mapping (function)

$$p : 1 \to p_1,\ 2 \to p_2,\ \ldots,\ n \to p_n,$$

written as

$$p = \begin{pmatrix} 1 & 2 & \cdots & n \\ p_1 & p_2 & \cdots & p_n \end{pmatrix}.$$

Assign p a permutation matrix $P = f(p)$, which has 1 in the (k, p_k) position and 0 elsewhere, $k = 1, 2, \ldots, n$. Consider two permutations

$$p = \begin{pmatrix} 1 & 2 & \cdots & n \\ p_1 & p_2 & \cdots & p_n \end{pmatrix} \quad \text{and} \quad q = \begin{pmatrix} 1 & 2 & \cdots & n \\ q_1 & q_2 & \cdots & q_n \end{pmatrix}.$$

Reorder the columns of q so that the 1st line of q matches the 2nd line of p, i.e., $q = \begin{pmatrix} p_1 & p_2 & \cdots & p_n \\ r_1 & r_2 & \cdots & r_n \end{pmatrix}$. Define the product of p and q by

$$pq = \begin{pmatrix} 1 & 2 & \cdots & n \\ r_1 & r_2 & \cdots & r_n \end{pmatrix}.$$

Show that

$$f(pq) = f(p)f(q).$$

For example, $p = \left(\begin{smallmatrix} 1&2&3&4&5 \\ 2&4&1&3&5 \end{smallmatrix}\right)$ and $q = \left(\begin{smallmatrix} 1&2&3&4&5 \\ 3&4&5&2&1 \end{smallmatrix}\right)$. By reordering the columns of q so that the first line of q matches the second line of p, we rewrite $q = \left(\begin{smallmatrix} 2&4&1&3&5 \\ 4&2&3&5&1 \end{smallmatrix}\right)$. It follows that $pq = \left(\begin{smallmatrix} 1&2&3&4&5 \\ 4&2&3&5&1 \end{smallmatrix}\right)$. Thus, we have

$$f(p) = \begin{pmatrix} 0&1&0&0&0 \\ 0&0&0&1&0 \\ 1&0&0&0&0 \\ 0&0&1&0&0 \\ 0&0&0&0&1 \end{pmatrix},\ f(q) = \begin{pmatrix} 0&0&1&0&0 \\ 0&0&0&1&0 \\ 0&0&0&0&1 \\ 0&1&0&0&0 \\ 1&0&0&0&0 \end{pmatrix},\ f(pq) = \begin{pmatrix} 0&0&0&1&0 \\ 0&1&0&0&0 \\ 0&0&1&0&0 \\ 0&0&0&0&1 \\ 1&0&0&0&0 \end{pmatrix} = f(p)f(q).$$

[Note: The pq defined above is qp and $f(p) = P^T$ in some other texts.]

37. A permutation is called an *interchange* if only two elements are permuted. For instance, the following p is an interchange, where

$$p = \begin{pmatrix} 1 & 2 & 3 & 4 & \cdots & n \\ 3 & 2 & 1 & 4 & \cdots & n \end{pmatrix}.$$

Show that every permutation on $\{1, 2, \ldots, n\}$ can be obtained by a sequence of interchanges (permutation of two numbers each time).

38. Show that every $n \times n$ permutation matrix can be expressed as the product of at most $n - 1$ symmetric permutation matrices.

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5.7 Nonnegative Matrices

A *nonnegative matrix* is a matrix whose entries are all nonnegative; A *positive matrix* is a matrix whose entries are all positive. For example, a permutation matrix is a nonnegative matrix; an all-ones matrix is a positive matrix.

If $A = (a_{ij})$ is a nonnegative matrix, that is, all $a_{ij} \geq 0$, we write $A \geq 0$ (or $0 \leq A$). If all entries a_{ij} are positive, we write $A > 0$ (or $0 < A$).

Every matrix $A = (a_{ij})$ (including row and column vectors) can be associated with a nonnegative matrix (vector), written as $|A|$, whose entries are the absolute values of the entries of A, that is, $|A| = (|a_{ij}|)$.

Important note: In this section, and only in this section of the book, $A \geq 0$ and $|A|$ are entrywise. That is, $A \geq 0$ means that A is entrywise nonnegative and $|A|$ stands for the entrywise absolute value matrix. In all other sections, $A \geq 0$ means that A is positive semidefinite and $|A|$ is the modulus of A, i.e., $|A| = (A^*A)^{1/2}$ (Sect. 3.2). Both notations are widely and evenly used in the literature for the two meanings. This is why we prefer to stick with the traditional notations rather than introducing new ones.

For two real matrices A and B of the same size, we write

$$A \geq B \text{ or } B \leq A \quad \text{if } A - B \geq 0;$$

$$A > B \text{ or } B < A \quad \text{if } A - B > 0.$$

Obviously, for any matrices A and B of the same size,

$$|A + B| \leq |A| + |B|$$

and for matrices A and B of sizes $m \times n$ and $n \times p$, respectively,

$$|AB| \leq |A||B|.$$

In particular, for any n-square matrix A and column vector x in $\mathbb{C}^n$,

$$|Ax| \leq |A||x|$$

and for any square matrix A and positive integer k,

$$|A^k| \leq |A|^k.$$

Example 5.7.1 Let $A \in \mathbb{M}_n$ and $x \in \mathbb{C}^n$.

(a) If $A > 0$ and $x \neq 0$, then $A|x| > 0$.
This is nearly trivial.

(b) If $A > 0$ and $|Ax| = A|x|$, then $x = e^{i\theta}|x|$ for some $\theta \in \mathbb{R}$.
Use triangle inequality.

(c) If $x > 0$ and $Ax = |A|x$, then $A = |A|$.
Because $(|A| - \Re(A))x = 0$, where $\Re(A) = \big(\Re(a_{ij})\big) = \frac{1}{2}(\overline{A} + A)$. ∎

We are now ready to present a theorem on comparison of the spectral radii of nonnegative matrices. Recall from Sect. 4.1 that the *spectral radius* $\rho(A)$ of an n-square complex matrix A is defined to be

$$\rho(A) = \max\{|\lambda| : \lambda \text{ is an eigenvalue of } A\}.$$

Note that $\rho(A)$ need not be an eigenvalue of A. Our goal of this section is to show that if A is a positive matrix, then $\rho(A)$ is an eigenvalue of A and there exists a positive eigenvector associated with this eigenvalue.

Theorem 5.7.1 *Let A and B be n-square nonnegative matrices. Then*

$$A \geq B \quad \Rightarrow \quad \rho(A) \geq \rho(B).$$

Proof. Since $A \geq B \geq 0$, i.e., $a_{ij} \geq b_{ij} \geq 0$ for all i and j, we have

$$\sum_{i,\,j} a_{ij}^2 \geq \sum_{i,\,j} b_{ij}^2.$$

It follows that

$$\left(\sum_{i,\,j} a_{ij}^2\right)^{1/2} \geq \left(\sum_{i,\,j} b_{ij}^2\right)^{1/2}$$

or in the Frobenius norm

$$\|A\|_F \geq \|B\|_F.$$

On the other hand (Problem 11), for all positive integers k, we have

$$A \geq B \geq 0 \quad \Rightarrow \quad A^k \geq B^k.$$

This yields

$$\|A^k\|_F \geq \|B^k\|_F$$

implying

$$\|A^k\|_F^{1/k} \geq \|B^k\|_F^{1/k}.$$

Take limits (as $k \to \infty$). By Theorem 4.2.3, we have $\rho(A) \geq \rho(B)$. ∎

Consider n-square nonnegative matrices A and B. Since the Hadamard (entrywise) product $A \circ B$ is a principal submatrix of the Kronecker product $A \otimes B$ (Theorem 4.3.3), by Problem 19, we have

$$\rho(A \circ B) \leq \rho(A \otimes B) = \rho(A)\rho(B). \tag{5.28}$$

Example 5.7.2 $\rho(AB) \leq \rho(A)\rho(B)$ need not be true. Take

$$A = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \quad \text{and} \quad B = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}. \text{ ∎}$$

The following result states that $\rho(AB)$ dominates $\rho(A \circ B)$.

Theorem 5.7.2 *Let A and B be $n \times n$ nonnegative matrices. Then*

$$\rho(A \circ B) \le \rho(AB).$$

Proof. The proof is based on two facts: If $0 \le X \le Y$, then $\rho(X) \le \rho(Y)$ (Theorem 5.7.1), and if P is a principal submatrix of a nonnegative square matrix Q, then $\rho(P) \le \rho(Q)$ (Problem 19). We first show that

$$(A \circ B)(B \circ A) \le (AB) \circ (BA). \tag{5.29}$$

Computing the (i, j)-entry of the right-hand side of (5.29), we have

$$\Big(\sum_s a_{is}b_{sj}\Big)\Big(\sum_t b_{it}a_{tj}\Big) = \sum_{s,\,t} a_{is}b_{it}a_{tj}b_{sj}.$$

Setting $s = t = p$ yields $\sum_p a_{ip}b_{ip}a_{pj}b_{pj}$, which is precisely the (i, j)-entry of the left-hand side of (5.29). Thus, (5.29) holds as $A, B \ge 0$.

Because the Hadamard product is a principal submatrix of the Kronecker product (Theorem 4.3.3), using (5.29), we have

$$\big(\rho(A \circ B)\big)^2 \le \rho(AB \circ BA) \le \rho(AB \otimes BA) = \big(\rho(AB)\big)^2.$$

By taking the square roots, we obtain the desired inequality. ∎

The next result gives lower and upper bounds for the spectral radius of a nonnegative matrix in terms of the entries of the matrix.

Theorem 5.7.3 *Let A be an n-square nonnegative matrix. Then*

$$\min_{1\le i\le n} \sum_{j=1}^{n} a_{ij} \le \rho(A) \le \max_{1\le i\le n} \sum_{j=1}^{n} a_{ij}.$$

In other words, the spectral radius of a nonnegative square matrix is between the smallest row sum and the largest row sum.

Proof. We may assume $A \ne 0$. Denote the ith row sum by r_i, namely, $r_i = a_{i1} + a_{i2} + \cdots + a_{in}$. Let r be the smallest row sum of A, that is, $r = \min\{r_1, \ldots, r_n\}$. We construct a new matrix B such that

$$0 \le B \le A \quad \text{and} \quad r = \rho(B).$$

If $r = 0$, set $B = 0$. Otherwise, all $r_i > 0$. Let $b_{ij} = r(r_i^{-1}a_{ij})$. It is immediate that $0 \le B \le A$, implying $\rho(B) \le \rho(A)$. To see $\rho(B) = r$, first observe that $Be = re$, where $e = (1, \ldots, 1)^T$. This says that r is an eigenvalue of B. So $\rho(B) \ge r$. However, considering the maximum row sum matrix norm $\|\cdot\|_{\text{rowmax}}$ (Problem 11, Sect. 4.2), we have $\rho(B) \le \|B\|_{\text{rowmax}} = r$. Thus, $\rho(B) = r$. The upper bound is similarly shown. ∎

Analogous results hold for columns with A^T in place of A in the theorem.

Example 5.7.3 Lower and upper bounds for spectral radii.

(a) Let $A = \begin{pmatrix} 1&0&1&0 \\ 0&0&0&1 \\ 0&1&0&0 \\ 1&0&0&1 \end{pmatrix}$. Theorem 5.7.3 implies $1 \le \rho(A) \le 2$.

(b) Let $B = \begin{pmatrix} 1&1&0&0 \\ 0&1&1&0 \\ 0&0&1&1 \\ 1&0&0&1 \end{pmatrix}$. Theorem 5.7.3 implies $\rho(B) = 2$.

(c) Let $C = \begin{pmatrix} 1&\frac{1}{2}&\frac{1}{4}&\frac{1}{8} \\ 1&\frac{1}{3}&\frac{1}{9}&\frac{1}{27} \\ 1&\frac{1}{4}&\frac{1}{16}&\frac{1}{64} \\ 1&\frac{1}{5}&\frac{1}{25}&\frac{1}{125} \end{pmatrix}$ (a Vandermonde matrix). The largest row sum of C is $1+\frac{1}{2}+\frac{1}{4}+\frac{1}{8} = \frac{15}{8}$, and the smallest row sum is $1+\frac{1}{5}+\frac{1}{25}+\frac{1}{125} = \frac{156}{125}$. By Theorem 5.7.3, we have that $\frac{156}{125} \le \rho(C) \le \frac{15}{8}$. ∎

We now show a fundamental theorem on nonnegative matrices. We present the theorem for positive matrices. The results can be generalized and amplified to nonnegative matrices (a result of Frobenius).

Theorem 5.7.4 (Perron Theorem) *Let $A \in \mathbb{M}_n$ be a positive matrix. Then*

(1) $\rho(A) > 0$.

(2) $\rho(A)$ *is a simple eigenvalue of* A.

(3) $Ax_0 = \rho(A)x_0$ *for some vector* $x_0 > 0$, $x_0 \in \mathbb{C}^n$.

(4) *If* $Av = \rho(A)v$, $v \in \mathbb{C}^n$, *then* $v = \alpha x_0$ *for some* $\alpha \in \mathbb{C}$.

(5) *If* λ *is an eigenvalue of* A *and* $\lambda \neq \rho(A)$, *then* $|\lambda| < \rho(A)$.

Note: (2) says that the algebraic multiplicity of $\rho(A)$ as an eigenvalue of A is 1; (3) and (4) reveal that the geometric multiplicity of $\rho(A)$ is 1, and the positive eigenvector x_0 is unique up to a scalar.

Proof. (1) Because $A > 0$, Theorem 5.7.3 ensures $\rho(A) > 0$.

To show (2) and (3), let (λ, x) be an eigenpair of A and $|\lambda| = \rho(A)$. We first show that $A|x| = \rho(A)|x|$. Denote $\rho = \rho(A)$ for simplicity. Then

$$\rho|x| = |\lambda||x| = |\lambda x| = |Ax| \le |A||x| = A|x|.$$

Note that $A|u| > 0$ for any $u \neq 0$ since the entries of A are all positive. Thus, $A|x| > 0$ and $|x| > 0$. Let $y = A|x| - \rho|x|$. Then $y \ge 0$. If $y \neq 0$, then

$$0 < Ay = A(A|x|) - \rho A|x|$$

or

$$Az > \rho z, \quad \text{where } z = A|x| > 0.$$

If we write $z = A|x| = (z_1, \ldots, z_n)$, then all z_t are positive and $Az > \rho z$ implies that $\sum_j a_{tj} z_j > \rho z_t$ for every t. Let Z be the diagonal matrix $\operatorname{diag}(z_1, \ldots, z_n)$. Using Theorem 5.7.3, we have

$$\rho(A) = \rho(Z^{-1}AZ) \geq \min_t \frac{1}{z_t} \sum_j a_{tj} z_j > \rho = \rho(A),$$

a contradiction. So $y = 0$, i.e., $A|x| = \rho|x|$, and $|x| > 0$. Take $x_0 = |x|$.

We have shown that for any eigenvector x associated with $\rho(A)$,

$$Ax = \lambda x,\ x \neq 0,\ |\lambda| = \rho(A) \ \Rightarrow\ Ax_0 = \rho(A)x_0,\ x_0 = |x| > 0. \tag{5.30}$$

To show that $\rho = \rho(A)$ as an eigenvalue of A is simple, i.e., ρ has algebraic multiplicity 1, we consider the Jordan block associated with ρ.

We may assume that $\rho = 1$ (or use $\frac{1}{\rho}A$ in the following discussion) and take x_0 to be a positive vector with the sum of components equal to 1. Thus, $Ax_0 = x_0$. It follows that for any positive integer m, $A^m x_0 = x_0$. Since $A > 0$ and $x_0 > 0$, the entries of A^m are bounded as $m \to \infty$.

On the other hand, let $J = \begin{pmatrix} J_\rho & 0 \\ 0 & J_0 \end{pmatrix}$ be a Jordan decomposition of A, where J_ρ is the Jordan block associated with $\rho = 1$, and J_0 is the direct sum of all other Jordan blocks. We write $P^{-1}AP = J$. Then $P^{-1}A^mP = \begin{pmatrix} J_\rho^m & 0 \\ 0 & J_0^m \end{pmatrix}$. If the order of J_ρ is more than 1, then the $(1,2)$-entry of J_ρ^m is m which is unbounded (as $m \to \infty$). However, the entries of $P^{-1}A^mP$ are bounded, resulting in a contradiction. So, J_ρ is 1×1, and ρ is simple.

(4) follows from (2) and (3) immediately because the algebraic multiplicity of an eigenvalue is always greater than or equal to its geometric multiplicity. Since ρ is simple, the dimension of the eigenspace of ρ is 1, and x_0 serves as a basis of the eigenspace. Below is an alternative proof without using the relation of algebraic and geometric multiplicities.

Following the argument prior to (5.30), we have $A|x| = \rho|x|$ and $|x| > 0$ whenever $Ax = \rho x$ and $x \neq 0$. Let $x = (x_1, x_2, \ldots, x_n)^T$. Then

$$\rho|x_t| = \sum_{j=1}^n a_{tj}|x_j|, \quad t = 1, 2, \ldots, n.$$

However, $\rho|x| = |\rho x| = |Ax|$. Thus, we have

$$\rho|x_t| = \Big|\sum_{j=1}^n a_{tj}x_j\Big| \quad \text{and} \quad \Big|\sum_{j=1}^n a_{tj}x_j\Big| = \sum_{j=1}^n a_{tj}|x_j|$$

which holds if and only if all $a_{t1}x_1, a_{t2}x_2, \ldots, a_{tn}x_n$ have the same argument. It follows that $x_1, x_2, \ldots, x_n$ (as $a_{t1}, a_{t2}, \ldots, a_{tn}$ are positive) have the same argument (Problem 11, Sect. 1.4), namely, there exists $\theta \in \mathbb{R}$ such that

$$e^{i\theta}x_j > 0 \text{ for } j = 1, 2, \ldots, n, \quad \text{that is, } |x| = e^{i\theta}x > 0. \tag{5.31}$$

Now suppose v is also an eigenvector associated with the eigenvalue ρ. According to the above discussion, there exists $\theta \in \mathbb{R}$ such that $|v| = e^{i\theta}v > 0$. Let $v = (v_1, v_2, \ldots, v_n)^T$ and $x_0 = (x_1, x_2, \ldots, x_n)^T$. Let $\delta = \min_t |v_t|/x_t$ and define $w = |v| - \delta x_0$. Then w is a nonnegative vector having at least one component 0. On the other hand,

$$Aw = A(|v| - \delta x_0) = A|v| - \delta Ax_0 = \rho|v| - \delta\rho x_0 = \rho w.$$

If $w \neq 0$, then $w > 0$ by (5.30), a contradiction to w containing a zero. Thus $w = 0$ and $|v| = \delta x_0$, revealing $v = \alpha v$, where $\alpha = e^{-i\theta}\delta$.

For (5), if $|\lambda| = \rho$ and $Ax = \lambda x$, $x \neq 0$, then $e^{i\theta}x > 0$ for some $\theta \in \mathbb{R}$ by (5.31). Set $y = e^{i\theta}x$. Then $Ay = \lambda y > 0$. Thus, $\lambda > 0$ and $\lambda = \rho$. In other words, if $\lambda \neq \rho$, then $|\lambda| < \rho$. ∎

Example 5.7.4 Positive matrices have a dominant eigenvalue $\lambda_{\max} = \rho$.

(a) Let $A = \begin{pmatrix} 0.4 & 0.5 & 0.1 \\ 0.3 & 0.3 & 0.4 \\ 0.3 & 0.2 & 0.5 \end{pmatrix}$. Then $\lambda_{\max}(A) = \rho(A) \leq 1$ (by Theorem 5.7.3). A is a positive doubly stochastic matrix. So $\lambda_{\max}(A) = \rho(A) = 1$ is simple and the all-ones vector is a corresponding eigenvector.

(b) Let $B = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 2 & 2 \\ 1 & 2 & 2 \end{pmatrix}$. B is a positive matrix of rank 2. So, B has three (simple) eigenvalues $0 = \lambda_3$, λ_2, and $\lambda_{\max}(B) = \rho(B)$. Theorem 5.7.3 ensures $3 \leq \rho(B) \leq 5$. Moreover, $\operatorname{tr} B = 5$, $0 < \lambda_2 < \rho(B) < 5$.

(c) Let $C = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 2 & 2 \\ 1 & 3 & 2 \end{pmatrix}$. C is a positive matrix of rank 3. So, C has three nonzero eigenvalues λ_3, λ_2, and $\lambda_1 = \rho(C)$. By Theorem 5.7.3, we have $3 \leq \rho(C) \leq 6$. $\rho(C)$ is simple, and λ_3 and λ_2 are a conjugate pair. ∎

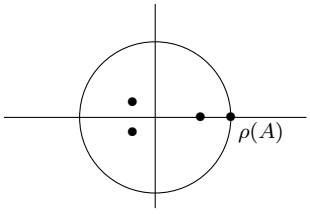


Fig. 5.3. Perron theorem

The Perron theorem states that for a square positive matrix A, the spectral radius $\rho(A)$ is an eigenvalue of A, called the *Perron root*. The Perron root of a positive matrix is the only eigenvalue that attains the spectral radius. Moreover, the positive eigenvectors, referred to as the *Perron vectors*, of the Perron eigenvalue are unique up to magnitude. These statements hold for irreducible nonnegative matrices. However, for a general nonnegative matrix A (with zero entries), it is possible that A has several eigenvalues that have the maximum modulus $\rho(A)$. [Note: This is a large research area.]

Problems

1. Does there exist a positive matrix that is similar to $\left(\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right)$?
2. Apply Theorem 5.7.3 to $A = \left(\begin{smallmatrix} 1 & 2 \\ 5 & 4 \end{smallmatrix}\right)$ and A^T. Find $\rho(A)$.
3. Let $A = \left(\begin{smallmatrix} 2 & 1 & 0 & 0 \\ 0 & 1 & 2 & 0 \\ 0 & 0 & 2 & 1 \\ 2 & 0 & 0 & 1 \end{smallmatrix}\right)$. Find $\rho(A)$.
4. Let $A = \left(\begin{smallmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 1 & 0 \end{smallmatrix}\right)$. Find the smallest positive integer k such that $A^k > 0$.
5. Let $B = \left(\begin{smallmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 1 & 1 & 1 \end{smallmatrix}\right)$. Does there exist a positive integer k such that $B^k > 0$?
6. Find the eigenvalues of $A = \left(\begin{smallmatrix} 1-\alpha & \alpha \\ 1-\alpha & \alpha \end{smallmatrix}\right)$, $0 < \alpha < 1$. Show that the spectral radius $\rho(A)$ can be strictly less than the numerical radius $w(A) = \max_{\|x\|=1} x^*Ax$ for some α.
7. What is the spectral radius of a doubly stochastic matrix?
8. Let $A \geq 0$, $B \geq 0$, both $n \times n$. Is it true that $\rho(A+B) \leq \rho(A) + \rho(B)$?
9. Find a 4×4 nonnegative matrix with eigenvalues $\sqrt{2}$, $\sqrt{2}$, 1, -1.
10. Show that there does not exist a 4×4 nonnegative matrix having eigenvalues 0.9, 0.9, i, $-i$, or having eigenvalues 1.1, 0, i, $-i$.
11. Let A, B, C, and D be square matrices of the same size. Show that

$$0 \leq B \leq A, \quad 0 \leq D \leq C \quad \Rightarrow \quad 0 \leq BD \leq AC.$$

    In particular, $0 \leq B \leq A \Rightarrow B^k \leq A^k$ for any positive integer k.
12. Let A be an $n \times n$ nonnegative matrix. Show that e^A is nonnegative.
13. Let A be an $n \times n$ positive matrix. Show that there exists $x \in \mathbb{C}^n$ such that $Ax = \rho(A)x$ and $x_1 + x_2 + \cdots + x_n = 1$. Show that such an x is unique and all components of x are positive (i.e., $x > 0$).
14. Let P be an $n \times n$ positive matrix and let u and v be two different nonnegative vectors in $\mathbb{R}^n$. If $u \geq v$, show that $Pu > Pv$.
15. Let $A \geq 0$ be a square matrix. If $\alpha x \leq Ax \leq \beta x$ for some scalars $\alpha, \beta \in \mathbb{R}$ and some nonnegative vector $x \neq 0$, show that $\alpha \leq \rho(A) \leq \beta$.
16. Let $A \geq 0$ be an n-square matrix. If $Au = \lambda u$ for some positive vector $u \in \mathbb{R}^n$ and a scalar $\lambda \in \mathbb{C}$, show that $\lambda = \rho(A)$. Is it true that $Av \leq \rho(A)v$ for all positive vectors $v \in \mathbb{R}^n$?

17. Show that $\rho(A) \leq \rho(|A|)$ for any complex square matrix A.

18. Compute $\rho(AB)$, $\rho(A)\rho(B)$, and $\rho(A \circ B)$ for each pair of the following matrices A and B, and conclude that any two of these quantities can be the same whereas the third one may be different:

$$\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}; \quad \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}; \quad \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}.$$

19. Let B be a principal submatrix of a nonnegative square matrix A.

    (a) Show that $\rho(B) \leq \rho(A)$.

    (b) Is (a) true for general real matrices?

    (c) If $A > 0$ and $B \neq A$, show that $\rho(B) < \rho(A)$.

    (d) Show that every diagonal entry of A is less than or equal to $\rho(A)$.

    (e) Find an example: $A > 0$ and B is a non-principal submatrix of A such that $\rho(A) < \rho(B)$.

20. Let A and B be $n \times n$ nonnegative matrices. Show that $B < A \Rightarrow \rho(B) < \rho(A)$. Is this true if the nonnegativity of A and B is dropped?

21. Let A and B be $n \times n$ nonnegative matrices. Show that $(AB)[\alpha] \geq A[\alpha]B[\alpha]$, where $X[\alpha]$ is the principal submatrix of X indexed by α.

22. Let A and B be $n \times n$ positive matrices. Show that $\rho(A \circ B) < \rho(AB)$.

23. Let A and B be $n \times n$ nonnegative matrices. Show that $\|A \circ B\|_{\text{sp}} \leq \rho(A^T B)$, where $\|A \circ B\|_{\text{sp}}$ is the spectral norm of $A \circ B$. Show by example that $\|A \circ B\|_{\text{sp}} \leq \rho(AB)$ does not hold in general.

24. Show that every square positive matrix is similar to a positive matrix whose row sums are all equal to each other.

25. Let $A = (a_{ij})$ be an $n \times n$ nonsingular nonnegative matrix, where $n > 1$.

    (a) If $A > 0$ (i.e., all $a_{ij} > 0$), show that $A^{-1} \ngeq 0$.

    (b) Find an example: $A \geq 0$ and $A^{-1} \geq 0$.

    (c) If all $a_{ii} > 0$ and $A^{-1} \geq 0$, show that A is a diagonal matrix.

    (d) If $A \geq 0$ and $A^{-1} \geq 0$, show that A is a generalized diagonal matrix, that is, AP is diagonal for some permutation matrix P.

26. Let T, V (Vandermonde), and P (Pascal) be the matrices given below. Show that the minors (i.e., the determinants of submatrices) of T, V, and P are all positive, then find the inverses of T, V, and P.

$$T = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 2 & 2 \\ 1 & 2 & 3 \end{pmatrix}, \quad V = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 2 & 4 \\ 1 & 3 & 9 \end{pmatrix}, \quad P = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \\ 1 & 3 & 6 \end{pmatrix}.$$

27. Let $A = \left(\begin{smallmatrix} 1 & 1 \\ 1 & 0 \end{smallmatrix}\right)$ and $B = \left(\begin{smallmatrix} 0 & 1 \\ 1 & 1 \end{smallmatrix}\right)$. Show that $Au \neq Bv$ for any positive vectors u and v. Use this to show that two multiplicative elements (called *words*) of A and B that are written in different ways cannot be equal. For examples, $A^2B \neq ABA$, $A^2B^3ABA^5B \neq B^2A^3BA^3B^2$.

28. Let $A = (a_{ij})$ be an $n \times n$ nonnegative matrix, i.e., $a_{ij} \geq 0$ for all $1 \leq i, j \leq n$. Denote $\max(A) = \max_{i,j} a_{ij}$. If there exist entries a_{1i_1}, $a_{2i_2}, \ldots, a_{ni_n}$ all in different rows and different columns such that the product $a_{1i_1}a_{2i_2} \ldots a_{ni_n} > 1$, show that $\lim_{k\to\infty} \max(A^k) = \infty$.

29. An *M-matrix*, say $A \in \mathbb{M}_n$, is a real matrix that can be written as

$$A = \alpha I_n - P, \quad \text{for some } P \geq 0,\ \alpha \geq \rho(P).$$

Let $A \in \mathbb{M}_n$ be an M-matrix.

(a) Show that all principal submatrices of A are M-matrices.

(b) Show that all real eigenvalues of A are nonnegative.

(c) Show that the real part of each eigenvalue of A is nonnegative.

(d) Show that the determinant of A is nonnegative.

(e) Show that A is nonsingular if and only if $\alpha > \rho(P)$.

(f) Show that $A + tI$ is nonsingular for any $t > 0$.

(g) If A is nonsingular, show that A^{-1} is nonnegative, i.e., $A^{-1} \geq 0$.

(h) Is the inverse of every nonnegative invertible matrix an M-matrix?

(i) If A is nonsingular, show that $Ax > 0$ for some $x > 0$.

(j) If A is nonsingular and $Ax > 0$ for some $x \in \mathbb{R}^n$, show that $x > 0$.

(k) If A is nonsingular, show that there exists a positive diagonal matrix D such that AD has all positive row sums.

(l) If A is nonsingular, show that there exists a positive diagonal matrix D such that AD is strictly diagonally dominant.

(m) If A is nonsingular, show that A can be written as $A = M - N$, where M is positive diagonal, N is nonnegative, and $\rho(M^{-1}N) < 1$.

(n) Show that $A = \left(\begin{smallmatrix} 1 & -1 & -1 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{smallmatrix}\right)$ is an M-matrix and find its inverse.

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CHAPTER 6

Unitary Matrices and Contractive Matrices

Introduction: This chapter studies unitary matrices and contractive matrices (contractions). Sect. 6.1 gives basic properties of unitary matrices, Sect. 6.2 studies the structures of real orthogonal matrices under similarity, Sect. 6.3 discusses the connections of contractions with unitary matrices, Sect. 6.4 concerns the unitary similarity of real matrices, and Sect. 6.5 presents a trace inequality for unitary matrices.

6.1 Properties of Unitary Matrices

A square matrix U (of real or complex entries) is called a *unitary matrix* if

$$U^*U = UU^* = I.$$

A real unitary matrix is usually referred to as a *real orthogonal matrix*, i.e., a real orthogonal matrix V is a real matrix such that $V^TV = VV^T = I$. A complex matrix W is said to be *complex orthogonal* if $W^TW = WW^T = I$.

Example 6.1.1 Real orthogonal and unitary matrices.

(a) $\frac{1}{\sqrt{2}}\begin{pmatrix} 1 & i \\ 1 & -i \end{pmatrix}$ is unitary.

(b) $\frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$ is real orthogonal.

(c) Let $S = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & i \\ i & 1 \end{pmatrix}$. Then $S^T = S$, $S^*S = I_2$, but $S^TS \neq I_2$.

(d) Let $W = \begin{pmatrix} \sqrt{2} & i \\ i & -\sqrt{2} \end{pmatrix}$. Then $W^TW = I_2$, but $W^*W \neq I_2$.

(e) $\operatorname{diag}(e^{i\theta_1}, \ldots, e^{i\theta_n})$ is unitary, where $\theta_1, \ldots, \theta_n \in \mathbb{R}$. ∎

One checks that if U is unitary, then U^{-1}, U^*, U^T, and $\overline{U}$ are unitary too. Moreover, $|\det U| = 1$, and the product of unitary matrices is unitary.

It is an important fact that if λ is an eigenvalue of a unitary matrix, then $|\lambda| = 1$. This is because if $Ux = \lambda x$, $x \neq 0$, then $x^*U^* = \bar{\lambda}x^*$, and $x^*x = x^*U^*Ux = |\lambda|^2 x^*x$. Dividing by x^*x yields $|\lambda|^2 = 1$, i.e., $|\lambda| = 1$.

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Theorem 6.1.1 *Let $U \in \mathbb{M}_n$. The following statements are equivalent.*

(1) *U is unitary.*
(2) *$UU^* = I$.*
(3) *$U^*U = I$.*
(4) *$U^* = U^{-1}$.*
(5) *The column vectors of U form an orthonormal basis for $\mathbb{C}^n$.*
(6) *The row vectors of U form an orthonormal basis for $\mathbb{C}^n$.*
(7) *$\|Ux\| = \|x\|$ for every $x \in \mathbb{C}^n$.*
(8) *$U = V \operatorname{diag}(\lambda_1, \ldots, \lambda_n) V^*$ for a unitary V and $|\lambda_1| = \cdots = |\lambda_n| = 1$.*
(9) *U is a normal matrix with eigenvalues all equal to 1 in modulus.*

Proof. (1) $\Rightarrow$ (2) $\Rightarrow$ (3) $\Rightarrow$ (4) $\Rightarrow$ (1) because $AB = I \Rightarrow BA = I$ for $A, B \in \mathbb{M}_n$.

(5) $\Leftrightarrow$ (3): Let $U = (u_1, \ldots, u_n)$, where u_i is the ith column of U, $i = 1, \ldots, n$. Then the matrix identity (3) $U^*U = I$ is equivalent to

$$\begin{pmatrix} u_1^* \\ \vdots \\ u_n^* \end{pmatrix} (u_1, \ldots, u_n) = \begin{pmatrix} 1 & & 0 \\ & \ddots & \\ 0 & & 1 \end{pmatrix},$$

that is, $\langle u_i, u_j \rangle = u_j^* u_i = 1$ if $i = j$, and 0 otherwise, namely, $\{u_1, \ldots, u_n\}$ is an orthonormal basis for $\mathbb{C}^n$ (i.e., condition (5)).

(6) $\Leftrightarrow$ (2): This is similar to (3) $\Leftrightarrow$ (5), or use U^T for U in the discussion.

(7) $\Leftrightarrow$ (2): $\|Ux\| = \|x\| \Leftrightarrow \|Ux\|^2 = \|x\|^2$ which is equivalent to

$$\langle Ux, Ux \rangle = \langle x, U^*Ux \rangle = \langle x, x \rangle.$$

The above identity holds for all $x \in \mathbb{C}^n$ if and only if $U^*U = I$.

(1) $\Leftrightarrow$ (8): Unitary matrices are normal. Use the spectral decomposition.

(8) $\Leftrightarrow$ (9): This is by the spectral decomposition. So unitary matrices are the normal matrices whose eigenvalues all lie on the unit circle. ∎

Example 6.1.2 If $U \in \mathbb{M}_n$ is a unitary matrix, the adjugate formula gives

$$\operatorname{adj}(U) = (\det U) U^{-1} = (\det U) U^*.$$

If we partition U as

$$U = \begin{pmatrix} a & \alpha \\ \beta & V \end{pmatrix}, \quad a \in \mathbb{C},\ \alpha^T,\ \beta \in \mathbb{C}^{n-1},\ V \in \mathbb{M}_{n-1},$$

then, by comparing the $(1,1)$-entries in $\operatorname{adj}(U)$ and $(\det U)U^*$, we have

$$\det V = \bar{a} \det U \quad \text{and} \quad |\det V| = |a|.$$

(See also Theorem 2.2.3). ∎

The eigenvalues of a unitary matrix are necessarily equal to 1 in modulus. The converse is not true in general by taking $\left(\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}\right)$ as a counterexample.

Theorem 6.1.2 *Let $A \in \mathbb{M}_n$. If the eigenvalues of A are all equal to 1 in modulus and if $|x^*Ax| \leq 1$ for all unit vectors $x \in \mathbb{C}^n$, then A is unitary.*

Proof. The statement is unitarily invariant, that is, the statement is true if and only if it is true when A is replaced by U^*AU for any unitary $U \in \mathbb{M}_n$. Thus, by the Schur triangularization (Theorem 3.2.1), we may assume that A is an upper-triangular matrix with the eigenvalues $\lambda_1, \ldots, \lambda_n$ of A on the diagonal. To show that A is unitary, it is sufficient to show that A is diagonal. We show that every entry above the diagonal is equal to zero.

Suppose that the (p, q)-entry is nonzero c for some $p < q$. Consider the 2×2 submatrix $\left(\begin{smallmatrix} \lambda_p & c \\ 0 & \lambda_q \end{smallmatrix}\right)$ in the rows p, q and columns p, q of A. We can further assume that $\lambda_p = 1$, or we replace A with $\overline{\lambda_p}A$ in the following discussion.

Thus, the statement reduces to the case $A = \left(\begin{smallmatrix} 1 & c \\ 0 & \lambda_q \end{smallmatrix}\right)$ with $|\lambda_q| = 1$ and $|x^*Ax| \leq 1$ for all unit vectors $x = (x_1, x_2)^T \in \mathbb{C}^2$. We show that $c = 0$.

With $|x_1|^2 + |x_2|^2 = 1$ and $\lambda_q = a + bi$, $a, b \in \mathbb{R}$, we compute

$$x^*Ax = |x_1|^2 + \lambda_q|x_2|^2 + c\overline{x_1}x_2 = 1 + (a - 1 + bi)|x_2|^2 + c\overline{x_1}x_2.$$

If $c \neq 0$, we can choose the arguments of x_1 and x_2 such that $c\overline{x_1}x_2 > 0$. Note that $(a-1)|x_2|^2 + c\overline{x_1}x_2 > 0$ when $|x_2|$ is small enough. It follows that

$$|x^*Ax| \geq |\,\Re(x^*Ax)| = |1 + (a - 1)|x_2|^2 + c\overline{x_1}x_2| > 1.$$

This is a contradiction to $|x^*Ax| \leq 1$ for all unit x. Therefore, $c = 0$. ∎

A geometric explanation of Theorem 6.1.2 is that if all eigenvalues (spectrum) of a matrix lie on the unit circle and if the numerical range of the matrix is contained in the (closed) unit disc, then the matrix is unitary.

A slightly weaker result than Theorem 6.1.2 is that if all eigenvalues (spectrum) of a matrix lie on the unit circle and if the matrix as a linear operator maps all unit vectors into the unit sphere of $\mathbb{C}^n$ (that is, $\|Ax\| \leq 1$ for all unit $x \in \mathbb{C}^n$), then the matrix is unitary (Problem 13).

The numerical range (Sect.4.1) of a unitary matrix is precisely the polygon with the eigenvalues of the matrix as vertices inscribed in the unit circle.

We next discuss the singular values of submatrices of a unitary matrix. Recall that the singular values of an $m \times n$ matrix X are the positive square roots of the eigenvalues of X^*X (Sect. 2.4). In symbols, $\sigma_i(X) = \sqrt{\lambda_i(X^*X)}$, $i = 1, 2, \ldots, n$. The singular values of a unitary matrix are all equal to 1. If A is a submatrix of a unitary matrix U, then $\sigma_{\max}(A) \leq \sigma_{\max}(U) = 1$. Conversely, if A is a matrix with $\sigma_{\max}(A) \leq 1$, A can be embedded in a unitary matrix (Theorem 6.3.3 or Problem 26 of Sect. 6.3). We now show a result concerning the singular values of submatrices of a unitary matrix.

Theorem 6.1.3 *Let U be a unitary matrix. If A is a submatrix of U and B is the submatrix by deleting the rows and columns of A in U, then A and B have the same singular values except some 0's and 1's. Precisely, if A is $m \times n$ and B is $p \times q$, where $m + p = n + q$, then the number of the 0's is $|m - n|$ and the number of the 1's is $|m - q|$. In particular, if A is a square submatrix (so is B), then the singular values of A and B are the same except $|m - q|$ 1's; if $m = n = p = q$, then A and B have the same singular values.*

Proof. Since PUQ is unitary for any permutation matrices P and Q of the same size as U, we assume (via permutations of rows and columns) that

$$U = \begin{pmatrix} A & C \\ D & B \end{pmatrix}, \quad A \in \mathbb{M}_{m\times n},\ B \in \mathbb{M}_{p\times q},\ C \in \mathbb{M}_{m\times q},\ D \in \mathbb{M}_{p\times n}.$$

Because U is unitary, the identities $UU^* = I$ and $U^*U = I$ imply

$$AA^* + CC^* = I_m, \quad C^*C + B^*B = I_q.$$

It follows that

$$AA^* = I_m - CC^*, \quad B^*B = I_q - C^*C.$$

Since CC^* and C^*C have the same eigenvalues except $|m-q|$ 0's (Sect. 2.4), $I_m - CC^*$ and $I_q - C^*C$ have the same eigenvalues except $|m - q|$ 1's. Thus, AA^* and B^*B have the same eigenvalues except $|m - q|$ 1's. On the other hand, AA^* and A^*A have the same eigenvalues except $|m-n|$ 0's. Therefore, A^*A and B^*B have the same eigenvalues except $|m - q|$ 1's and $|m - n|$ 0's.

If A is square, i.e., $m = n$, then A and B have the same singular values except $|m - q|$ 1's. In the case where A and B are both square of the same size, they have the same singular values. ∎

As a result, if $U = \left(\begin{smallmatrix} A & C \\ D & B \end{smallmatrix}\right)$ is unitary, where $A \in \mathbb{M}_m$ and $B = \mathbb{M}_p$, then

$$|\det A| = |\det B|. \tag{6.1}$$

Moreover, $\operatorname{rank}(B) = \operatorname{rank}(A) + p - m$ if $m \le p$ and $\|A\|_{\mathrm{sp}} = \|B\|_{\mathrm{sp}}$ if $m = p$.

Example 6.1.3 Consider the identity matrix $U = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$ and apply Theorem 6.1.3 to U with various partitions as follows.

(a) $\left(\begin{array}{c|cc} 1 & 0 & 0 \\ \hline 0 & 1 & 0 \\ 0 & 0 & 1 \end{array}\right) = \left(\begin{smallmatrix} X & A \\ B & Y \end{smallmatrix}\right)$. $m = 1, n = 2, p = 2, q = 1$, $m - q = 0$, $n - m = 1$. The singular values of A and B are $0, 0$; 0, respectively.

(b) $\left(\begin{array}{cc|c} 1 & 0 & 0 \\ 0 & 1 & 0 \\ \hline 0 & 0 & 1 \end{array}\right) = \left(\begin{smallmatrix} A & X \\ Y & B \end{smallmatrix}\right)$. $m = 1, n = 2, p = 2, q = 1$, $m - q = 0$, $n - m = 1$. The singular values of A and B are $1, 0$; 1, respectively.

(c) $\left(\begin{array}{c|cc} 1 & 0 & 0 \\ 0 & 1 & 0 \\ \hline 0 & 0 & 1 \end{array}\right) = \left(\begin{smallmatrix} A & X \\ Y & B \end{smallmatrix}\right)$. $m = 2, n = 1, p = 1, q = 2, m - q = 0$, $m - n = 1$. The singular values of A and B are 1; $1, 0$, respectively.

(d) $\left(\begin{array}{c|cc} 1 & 0 & 0 \\ 0 & 1 & 0 \\ \hline 0 & 0 & 1 \end{array}\right) = \left(\begin{smallmatrix} X & A \\ B & Y \end{smallmatrix}\right)$. $m = n = 2, p = q = 1$, $m - q = 1, m - n = 0$. The singular values of A and B are $1, 0$; 0, respectively. ∎

Problems

1. Show that the product of two unitary matrices (of the same order) is a unitary matrix. Is the sum of two unitary matrices always unitary?

2. Show that $\operatorname{diag}(e^{i\theta_1}, \ldots, e^{i\theta_n})$ is unitary, where $\theta_1, \ldots, \theta_n \in \mathbb{R}$.

3. Show that permutation matrices are unitary and unitary matrices are normal. (Therefore, unitary matrices are unitarily diagonalizable.)

4. What are the eigenvalues and singular values of a unitary matrix?

5. Let $W = \begin{pmatrix} \sqrt{2} & i \\ i & -\sqrt{2} \end{pmatrix}$. Show that $\operatorname{tr} W = 0$, $\operatorname{tr}(W^T W) = 2$, $\operatorname{tr}(W^* W) = 6$.

6. Let $A = \left(\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right)$. Show that $\operatorname{tr}(S^T A S) = 0$ for any 2×2 complex orthogonal matrix S (i.e., $S^T S = I$). Is it true that $\operatorname{tr}(S^* A S) = 0$?

7. If A is a complex orthogonal matrix, show that A and A^{-1} are similar. Is the converse true? Are complex orthogonal matrices normal?

8. Find an $m \times n$ matrix V, $m \neq n$, such that $V^* V = I_n$.

9. Let $A = \left(\begin{smallmatrix} 0 & x \\ 0 & 0 \end{smallmatrix}\right)$, $B = \left(\begin{smallmatrix} 1 & x \\ 0 & 0 \end{smallmatrix}\right)$, and $C = \left(\begin{smallmatrix} 1 & x \\ 0 & 1 \end{smallmatrix}\right)$, where $x \in \mathbb{C}$. What are the eigenvalues and singular values of A, B, and C in terms of x?

10. Show that for any $\theta \in \mathbb{R}$, the following two matrices are similar:

$$\begin{pmatrix} e^{i\theta} & 0 \\ 0 & e^{-i\theta} \end{pmatrix} \quad \text{and} \quad \begin{pmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{pmatrix}.$$

    [Note: They are both unitary; the first is complex; the second is real.]

11. Show that any 2×2 unitary matrix with determinant equal to 1 is similar to a real orthogonal matrix.

12. Find a matrix A such that $|x^* A x| < 1 < \sigma_{\max}(A)$ for unit vectors x.

13. Let $A \in \mathbb{M}_n$. If the eigenvalues of A are all equal to 1 in absolute value and if $\|Ax\| \leq \|x\|$ for all $x \in \mathbb{C}^n$, show that A is unitary.

14. Let $A = (a_{ij})$ be an $n \times n$ unitary matrix. If the entry in the (p, q) position is zero, i.e., $a_{pq} = 0$, show that the submatrix $A(p|q)$ obtained from A by deleting row p and column q is singular.

15. Let A and C be m- and n-square matrices, respectively, and let

$$M = \begin{pmatrix} A & B \\ 0 & C \end{pmatrix}.$$

Show that M is unitary if and only if $B = 0$ and A and C are unitary.

16. Show that the 2×2 block matrix below is real orthogonal:

$$\begin{pmatrix} \sqrt{\lambda}\, I_n & -\sqrt{1-\lambda}\, I_n \\ \sqrt{1-\lambda}\, I_n & \sqrt{\lambda}\, I_n \end{pmatrix}, \quad \lambda \in [0, 1].$$

17. Show that $\frac{1}{\sqrt{2}} \begin{pmatrix} I_n & I_n \\ cI_n & -cI_n \end{pmatrix} \in \mathbb{M}_{2n}$ is unitary for any $c \in \mathbb{C}$ with $|c| = 1$.

18. If $U \in \mathbb{M}_n$ is unitary, show that $V = \frac{1}{2} \begin{pmatrix} I_n+U & I_n-U \\ I_n-U & I_n+U \end{pmatrix} \in \mathbb{M}_{2n}$ is unitary.

19. Let $D = \operatorname{diag}(d_1, d_2, \ldots, d_n)$, where $0 < d_k \leq 1$, $k = 1, 2, \ldots, n$, and let $E = (I - D^2)^{1/2}$. Show that $U = \begin{pmatrix} D & E \\ E & -D \end{pmatrix}$ is a unitary matrix.

20. If A is a unitary matrix with all eigenvalues being real, show that

$$A^2 = I \quad \text{and} \quad A^* = A.$$

21. If A is similar to a unitary matrix, show that A^* is similar to A^{-1}.

22. Let $A \in \mathbb{M}_n$ be Hermitian. Show that $(A - iI)^{-1}(A + iI)$ is unitary.

23. Let A be an $n \times n$ unitary matrix. If $A - I$ is nonsingular, show that $i(A - I)^{-1}(A + I)$ is Hermitian.

24. Let a and b be real numbers such that $a^2 - b^2 = 1$, $ab \neq 0$, and let

$$K = \begin{pmatrix} a & bi \\ -bi & a \end{pmatrix}.$$

(a) Show that K is complex orthogonal but not unitary, that is,

$$K^T K = I \quad \text{but} \quad K^* K \neq I.$$

(b) Let $a = \sqrt{2}$ and $b = 1$. Find the eigenvalues of K.

(c) Let $a = \sqrt{2}$ and $b = 1$. Find the singular values of K.

(d) Let $a = \frac{1}{2}(e + e^{-1})$ and $b = \frac{1}{2}(e - e^{-1})$, where $e = 2.718\ldots$. Show that the eigenvalues of K are e and e^{-1}.

(e) Let $a = \frac{1}{2}(e^t + e^{-t})$ and $b = \frac{1}{2}(e^t - e^{-t})$, $t \in \mathbb{R}$. What are the eigenvalues of K? Is the trace of K bounded?

25. Let U be an $n \times n$ unitary matrix and let V be a $p \times q$ submatrix of U. If $p + q > n$, show that the spectral norm of V is equal to 1.

26. Let $U = \begin{pmatrix} 0 & x \\ y & V \end{pmatrix}$ be an n-square unitary matrix, where $V \in \mathbb{M}_{n-1}$. Show that the singular values of V are $1, \ldots, 1, 0$ ($n-2$ ones and one zero).

27. Let $U = \begin{pmatrix} A & C \\ D & B \end{pmatrix}$ be a unitary matrix, where $A \in \mathbb{M}_m$, $B \in \mathbb{M}_n$.
    (a) Show that $|\operatorname{rank}(A) - \operatorname{rank}(B)| = |m - n|$.
    (b) If $\sigma_{\max}(A) = 1$, show that $\sigma_{\min}(C) = 0$. Is $C = 0$?
    (c) If $\sigma_{\max}(A) < 1$ and $\sigma_{\max}(B) < 1$, show that $m = n$.
    (d) If $m = n$, show that A and B have the same singular values.
    (e) If A, B, C, and D are all (square) nonsingular, show that $\sigma_{\max}(X) < 1$ for $X = A, B, C, D$ (i.e., strict contractions; see next section).

28. Let A be an n-square matrix. Show that $A = 0$ if A satisfies either
$$|\langle Ay, y\rangle| \leq \|y\| \text{ for all } y \in \mathbb{C}^n$$
or
$$|\langle Ay, y\rangle| \leq \|Ay\| \text{ for all } y \in \mathbb{C}^n.$$
[Hint: If $A \neq 0$, then $\langle Ay_0, y_0\rangle \neq 0$ for some $y_0 \in \mathbb{C}^n$.]

29. Let A be an n-square matrix and let $\alpha \in (0, 1)$. Show that $A = 0$ if
$$|\langle Ay, y\rangle| \leq \langle y, y\rangle^\alpha \text{ for all } y \in \mathbb{C}^n.$$

30. Let A be an n-square matrix having the largest and the smallest singular values $\sigma_{\max}(A)$ and $\sigma_{\min}(A)$, respectively. Show that
$$\langle Ay, Ay\rangle \leq \langle y, y\rangle \text{ for all } y \in \mathbb{C}^n \;\Rightarrow\; \sigma_{\max}(A) \leq 1$$
and
$$\langle y, y\rangle \leq \langle Ay, Ay\rangle \text{ for all } y \in \mathbb{C}^n \;\Rightarrow\; \sigma_{\min}(A) \geq 1.$$

31. Let A be an n-square matrix whose eigenvalues are all equal to 1 in modulus. Show that A is a unitary matrix if either
$$|\langle Ay, y\rangle| \leq \|y\|^2 \text{ for all } y \in \mathbb{C}^n$$
or
$$|\langle Ay, y\rangle| \leq \|Ay\|^2 \text{ for all } y \in \mathbb{C}^n.$$

32. Let A be an n-square matrix whose eigenvalues are all equal to 1 in modulus. Show that A is a unitary matrix if for some real $\alpha \neq \frac{1}{2}$,
$$|\langle Ay, y\rangle| \leq \langle Ay, Ay\rangle^\alpha \text{ for all unit } y \in \mathbb{C}^n.$$
[Hint: Assume that A is upper-triangular with $a_{11} = 1$, $a_{1i} > 0$ for some $i > 1$. Take $y = (\cos t, 0, \ldots, 0, \sin t, 0, \ldots, 0)^T$ and consider the behavior of the function $f(t) = \langle Ay, Ay\rangle^\alpha - |\langle Ay, y\rangle|$ near the origin.]

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6.2 Real Orthogonal Matrices

A square (real or complex) matrix A is called an *orthogonal matrix* if $AA^T = A^TA = I$. This section is devoted to real orthogonal matrices, i.e., the unitary matrices of real entries. We discuss the structures of real orthogonal matrices under real orthogonal similarity and show that real orthogonal matrices with a commutativity condition are necessarily involutions.

Note that complex orthogonal matrices are different from real orthogonal matrices. The latter are necessarily unitary. Note that $\left(\begin{smallmatrix} 0 & i \\ -i & 0 \end{smallmatrix}\right)$ is unitary but not orthogonal, while $\begin{pmatrix} \sqrt{2} & i \\ i & -\sqrt{2} \end{pmatrix}$ is orthogonal but not unitary.

Example 6.2.1 We begin discussions with 2×2 real orthogonal matrices. Let

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}, \quad a,\ b,\ c,\ d \in \mathbb{R}.$$

The identity $A^TA = I_2$ implies $a^2 + c^2 = 1$, $b^2 + d^2 = 1$, $ab + cd = 0$, and $\det A = ad - bc = \pm 1$. We set $a = \cos\theta$ and $c = \sin\theta$ for some $\theta \in [0, \frac{\pi}{2})$. Using trigonometric identities, we obtain $b = -\sin\theta$ and $d = \cos\theta$ if $\det A = 1$, or $b = \sin\theta$ and $d = -\cos\theta$ if $\det A = -1$ (Problem 9). As a result, there are only two types of 2×2 real orthogonal matrices:

$$\begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix} \quad \text{and} \quad \begin{pmatrix} \cos\theta & \sin\theta \\ \sin\theta & -\cos\theta \end{pmatrix}, \quad \theta \in \mathbb{R}. \tag{6.2}$$

The first type is called a *rotation* and the second one is a *reflection*.

Note that if $F \in \mathbb{M}_2$ is a reflection, then $F^2 = I_2$ and the eigenvalues of F are ± 1, and that if $R \in \mathbb{M}_2$ is a rotation, then the eigenvalues of R are $e^{\pm i\theta} = \cos\theta \pm i\sin\theta$ (including 1, 1, and -1, -1). Moreover, if R is a rotation such that $R^2 = I_2$, then $R = \pm I_2$ (Problem 7). ∎

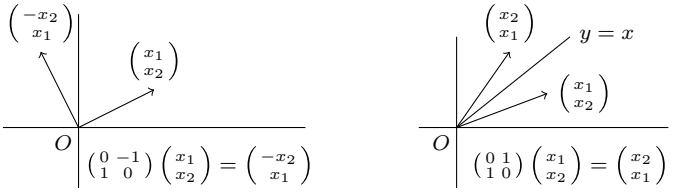


Fig. 6.1. Rotation with $\theta = \frac{\pi}{2}$ and reflection about $y = x$

A real orthogonal matrix is normal, thus it is (unitarily) diagonalizable over $\mathbb{C}$. However, it need not be diagonalizable over $\mathbb{R}$. For example, $\left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right)$ is real orthogonal; it is not diagonalizable over $\mathbb{R}$ because it has non-real eigenvalues $\pm i$. We show that a real orthogonal matrix is similar to a direct sum of real orthogonal matrices of orders 1 and (or) 2.

Theorem 6.2.1 *Every real orthogonal matrix is similar (via a real orthogonal matrix) to a direct sum of real orthogonal matrices of orders 1 and (or) 2.*

Proof. Let A be an $n \times n$ real orthogonal matrix. We apply induction on n. If $n = 1$ or 2, the statement is obviously true. Suppose that $n > 2$.

If A has a real eigenvalue λ with a real unit eigenvector x, then

$$Ax = \lambda x, \;\; x \neq 0 \;\; \Rightarrow \;\; x^T A^T A x = \lambda^2 x^T x.$$

Since $A^T A = I$, we see that $\lambda = \pm 1$, say, 1 (the case for -1 is similar). Extend the real unit eigenvector x to a real orthogonal matrix P. Then $P^T AP$ has $(1,1)$-entry 1 and 0 elsewhere in the first column, or explicitly,

$$P^T AP = \begin{pmatrix} 1 & u \\ 0 & A_1 \end{pmatrix}, \;\; \text{where } u \text{ is a real row vector.}$$

The orthogonality of A implies $u = 0$. Notice that A_1 is also real orthogonal. The conclusion then follows from an induction on A_1.

Assume that A has no real eigenvalues. Then for any nonzero $x \in \mathbb{R}^n$, vectors x and Ax cannot be linearly dependent. Recall from Sect. 1.4 the angle $\Theta_{x,y}$ between two vectors x and y and note that $\Theta_{x,y} = \Theta_{Ax,Ay}$ due to the orthogonality of A. Define $f(x)$ to be the angle function

$$f(x) = \Theta_{x,Ax} = \cos^{-1} \frac{\langle x, Ax \rangle}{\|x\| \, \|Ax\|}.$$

Then $f(x)$ is continuous on the compact set $S = \{x \in \mathbb{R}^n : \|x\| = 1\}$.

Let $x_0 \in S$ be such that $\theta_0 = \Theta_{x_0,Ax_0}$ is the minimum of $f(x)$ on S. Since x_0 and Ax_0 are linearly independent, $0 < \theta_0 < \pi$. Let y_0 be a unit vector in $\mathrm{Span}\{x_0, Ax_0\}$ bisecting the angle between x_0 and Ax_0, that is, $\Theta_{x_0,y_0} = \Theta_{y_0,Ax_0} = \frac{\theta_0}{2}$. Note that x_0 and y_0 are linearly independent. We show that $\mathrm{Span}\{x_0, y_0\}$ is invariant under A, i.e., $Ax_0, Ay_0 \in \mathrm{Span}\{x_0, y_0\}$.

$Ax_0 \in \mathrm{Span}\{x_0, y_0\}$ because $\mathrm{Span}\{x_0, y_0\} = \mathrm{Span}\{x_0, Ax_0\}$ as x_0 and $y_0 \in \mathrm{Span}\{x_0, Ax_0\}$ are linearly independent.

We now show that $Ay_0 \in \mathrm{Span}\{x_0, y_0\}$. By Theorem 1.4.6 (Chap. 1),

$$\theta_0 \leq \Theta_{y_0,Ay_0} \leq \Theta_{y_0,Ax_0} + \Theta_{Ax_0,Ay_0} = \frac{\theta_0}{2} + \frac{\theta_0}{2} = \theta_0,$$

implying that $Ax_0 \in \mathrm{Span}\{y_0, Ay_0\}$. Thus, Ay_0 lies in $\mathrm{Span}\{y_0, Ax_0\} = \mathrm{Span}\{x_0, y_0\}$. Therefore, $\mathrm{Span}\{x_0, y_0\}$ is an invariant subspace under A.

Choose an orthonormal basis of $\mathrm{Span}\{x_0, y_0\}$ and extend it to an orthonormal basis of $\mathbb{R}^n$. Up to real orthogonal similarity (equivalently, via a real orthogonal matrix), we can write $A = \left(\begin{smallmatrix} T_0 & 0 \\ 0 & B \end{smallmatrix}\right)$, where $T_0 \in \mathbb{M}_2(\mathbb{R})$ and $B \in \mathbb{M}_{n-2}(\mathbb{R})$. Since A is orthogonal, T_0 and B are orthogonal. An application of the induction hypothesis to $B \in \mathbb{M}_{n-2}$ completes the proof. ∎

Theorem 6.2.1 can be shown differently. One may consider the eigenvectors of the non-real eigenvalues. Since A is real, the characteristic polynomial of A has real coefficients, and the non-real eigenvalues of A occur in conjugate pairs. Furthermore, their eigenvectors are in the forms $\alpha + \beta i$ and $\alpha - \beta i$, where $\alpha, \beta \in \mathbb{R}^n$, $\langle \alpha, \beta \rangle = 0$, and $A\alpha = a\alpha - b\beta$, $A\beta = b\alpha + a\beta$ for some real scalars a and b with $a^2 + b^2 = 1$. Matrix A will have the desired form (via real orthogonal similarity) by choosing a suitable real orthonormal basis. Another approach is to use the Jordan forms of real orthogonal matrices whose non-real complex Jordan blocks appear in conjugate pairs.

Note that a complex matrix may be similar to a real matrix via complex similarity. For example, $T^* \begin{pmatrix} a+bi & 0 \\ 0 & a-bi \end{pmatrix} T = \begin{pmatrix} a & -b \\ b & a \end{pmatrix}$, where $T = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & i \\ 1 & -i \end{pmatrix}$.

Example 6.2.2 Let $A = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix}$. A is a real orthogonal matrix. The eigenvalues of A are $1, \frac{-1+\sqrt{3}i}{2}, \frac{-1-\sqrt{3}i}{2}$. So, over the complex $\mathbb{C}$, A is unitarily similar to $\operatorname{diag}\left(1, \frac{-1+\sqrt{3}i}{2}, \frac{-1-\sqrt{3}i}{2}\right)$ (via a Fourier matrix; see Sect. 5.3). Over the real $\mathbb{R}$, A is similar to $\begin{pmatrix} 1 & 0 & 0 \\ 0 & -\frac{1}{2} & \frac{\sqrt{3}}{2} \\ 0 & -\frac{\sqrt{3}}{2} & -\frac{1}{2} \end{pmatrix}$ via a real orthogonal matrix which consists of the orthonormal eigenvectors of A.

In fact, $(1,1,1)^T$ is obviously an eigenvector associated to the eigenvalue 1. By computation, we get $\left(\frac{-1+\sqrt{3}i}{2}, \frac{-1-\sqrt{3}i}{2}, 1\right)^T$ as an eigenvector corresponding to the eigenvalue $\frac{-1+\sqrt{3}i}{2}$. We obtain orthogonal real vectors $(1,1,1)^T$, $(-\frac{1}{2}, -\frac{1}{2}, 1)^T$, and $\left(\frac{\sqrt{3}}{2}, -\frac{\sqrt{3}}{2}, 0\right)^T$ which are normalized as $\left(\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}\right)^T$, $\left(-\frac{1}{\sqrt{6}}, -\frac{1}{\sqrt{6}}, \frac{2}{\sqrt{6}}\right)^T$, and $\left(\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}}, 0\right)^T$, respectively. Thus,

$$Q^T A Q = \begin{pmatrix} 1 & 0 & 0 \\ 0 & -\frac{1}{2} & \frac{\sqrt{3}}{2} \\ 0 & -\frac{\sqrt{3}}{2} & -\frac{1}{2} \end{pmatrix}, \quad \text{where } Q = \begin{pmatrix} \frac{1}{\sqrt{3}} & -\frac{1}{\sqrt{6}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{3}} & -\frac{1}{\sqrt{6}} & -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{3}} & \frac{2}{\sqrt{6}} & 0 \end{pmatrix}. \blacksquare$$

If two real orthogonal matrices (or normal matrices) commute, then they are simultaneously unitarily diagonalizable (see Sect. 3.1, Chap. 3). We next show that a real orthogonal matrix with a certain commuting property (with another real orthogonal matrix) is necessarily an involution. For this, we need a result which is of interest in its own right.

Lemma 6.2.1 *If $F \in \mathbb{M}_m$ and $G \in \mathbb{M}_n$ have no common eigenvalues, then the matrix equation $FX - XG = 0$ has the unique solution $X = 0 \in \mathbb{M}_{m \times n}$.*

Proof. $FX = XG$ yields $F^k X = XG^k$ for every positive integer k. Thus

$$f(F)X = Xf(G)$$

for any polynomial f. In particular, if we take f to be the characteristic polynomial of F, then $f(F) = 0$, thus $Xf(G) = 0$. Because F and G have no common eigenvalues, $f(G)$ is a nonsingular matrix, hence $X = 0$. $\blacksquare$

Theorem 6.2.2 *Let $A, U \in \mathbb{M}_n$ be real orthogonal matrices. If the eigenvalues of U are distinct and if $UA = AU^T$, then A is involutory, i.e., $A^2 = I_n$.*

Proof. By the previous theorem, let P be a real orthogonal matrix such that $P^{-1}UP$ is a direct sum of real orthogonal matrices V_i's of order(s) 1 or 2. Note that V_i and V_j, $i \neq j$, have no common eigenvalues because the eigenvalues of U are distinct. The identity $UA = AU^T$ results in

$$(P^{-1}UP)(P^{-1}AP) = (P^{-1}AP)(P^{-1}UP)^T.$$

Partition $P^{-1}AP$ conformally with $P^{-1}UP$ as $P^{-1}AP = (B_{ij})$, where B_{ii} has the same order as V_i. So, the number of rows (and columns) of B_{ij} is 1 or 2. Computing and comparing both sides of $UA = AU^T$, we obtain

$$V_i B_{ij} = B_{ij} V_j^T, \quad i, j = 1, \ldots, k. \tag{6.3}$$

By Lemma 6.2.1, we have

$$B_{ij} = 0, \quad i \neq j.$$

Hence $P^{-1}AP$ is a direct sum of matrices of orders no more than 2:

$$P^{-1}AP = B_{11} \oplus \cdots \oplus B_{kk}.$$

The orthogonality of A, thus of $P^{-1}AP$, implies that each B_{ii} is either a real orthogonal matrix of order 1 (i.e., ± 1) or a real orthogonal matrix of order 2 in the form (6.2). For the latter, $B_{ii}^2 = I_2$ if B_{ii} is a reflection. Suppose B_{ii} is a rotation. Then V_i is not a rotation, otherwise, V_i and B_{ii} are both rotations and hence commute (Problem 4), implying $V_i^2 B_{ii} = B_{ii}$ and $V_i^2 = I_2$. Then $V_i = \pm I_2$ (by Problem 7), contradicting the assumption that U (so V_i) has distinct eigenvalues. Thus, V_i is a reflection, and is real orthogonally similar to $\mathrm{diag}(1, -1)$. By (6.3), B_{ii} is similar to $\pm I_2$. In either case $B_{ii}^2 = I_2$. It follows that $(P^{-1}AP)^2 = I_n$ and $A^2 = I_n$. ∎

Can the condition that the eigenvalues of U be distinct be dropped?

Problems

1. Find a 2×2 real matrix such that $A^2 = I_2$ but $A^*A = A^TA \neq I_2$.
2. When is an upper-triangular matrix (real or complex) orthogonal?
3. If A is a 2×2 real matrix with a complex eigenvalue $\lambda = a + bi$, where $a, b \in \mathbb{R}$, $b \neq 0$, show that A is similar to the real matrix $\left(\begin{smallmatrix} a & b \\ -b & a \end{smallmatrix}\right)$.
4. Verify that the rotations A and B commute, that is, $AB = BA$, where

$$A = \begin{pmatrix} \cos\alpha & \sin\alpha \\ -\sin\alpha & \cos\alpha \end{pmatrix}, \quad B = \begin{pmatrix} \cos\beta & \sin\beta \\ -\sin\beta & \cos\beta \end{pmatrix}, \quad \alpha, \beta \in \mathbb{R}.$$

5. Show that a real matrix P is an orthogonal projection if and only if P is orthogonally similar to a matrix in the form $\mathrm{diag}(1,\dots,1,0,\dots,0)$.

6. Let $R_\theta = \left(\begin{smallmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{smallmatrix}\right)$ (a rotation), where $\theta \in \mathbb{R}$. Show that

   (a) $(R_\theta)^T = R_{-\theta}$,

   (b) $R_\theta R_\phi = R_\phi R_\theta = R_{\theta+\phi}$,

   (c) $R_\theta^n = R_{n\theta}$ for positive integers n, and

   (d) a reflection is expressed as $\left(\begin{smallmatrix} 1 & 0 \\ 0 & -1 \end{smallmatrix}\right) R_\theta$.

   What are the eigenvalues of R_θ in terms of θ?

7. Show that if $R \in \mathbb{M}_2$ is a rotation such that $R^2 = I_2$, then $R = \pm I_2$.

8. Show that the eigenvalues of a 2×2 reflection matrix are ± 1.

9. Show that a 2×2 real orthogonal matrix is either a rotation or a reflection. Can a matrix be both rotation and reflection?

10. (a) Explain the geometric meaning of a 2×2 rotation matrix as a (linear) map applied to a vector on the xy-plane. (b) For the vector v with endpoint $(1,1) \in \mathbb{R}^2$, find a rotation that places v on the positive side of the x-axis. What is the corresponding rotation matrix?

11. (a) Explain the geometric meaning of a 2×2 reflection matrix as a (linear) map applied to a vector on the xy-plane. (b) For the vector v with endpoint $(1,1) \in \mathbb{R}^2$, find a reflection about the line $y = -x$. What is the corresponding reflection matrix?

12. If A is a real orthogonal matrix and $\det A = -1$, show that A has an eigenvalue -1. [Note: This holds for complex orthogonal matrices.]

13. Let $K = \left(\begin{smallmatrix} s & it \\ -it & s \end{smallmatrix}\right)$, where $s = \sec\theta$, $t = \tan\theta$, $\theta \in (0, \frac{\pi}{2})$. Show that

   (a) K is Hermitian.

   (b) K is complex orthogonal, i.e., $K^T K = I$.

   (c) K is not unitary, i.e., $K^* K \neq I$.

   (d) Both singular values of K can be arbitrarily large as $\theta \to \frac{\pi}{2}^-$.

14. If $A = (a_{ij})$ is a 3×3 real orthogonal matrix with $\det A = 1$, show that

$$(\mathrm{tr}A - 1)^2 + \sum_{1 \le i < j \le 3} (a_{ij} - a_{ji})^2 = 4.$$

15. Let $A = (a_{ij})$ be an $n \times n$ real matrix. Let $\Sigma_2(A) = \sum_{i,j=1}^n a_{ij}^2$ be the sum of the squares of all entries of A. Show that A is real orthogonal if and only if $\Sigma_2(A^T X A) = \Sigma_2(X)$ for all $n \times n$ real matrices X.

16. If A and B are two nonsingular real matrices such that $A+B$ is nonsingular and that $(A+B)^{-1} = A^{-1} + B^{-1}$, show that $C^2 + C + I = 0$, where $C = AB^{-1}$. Find a pair of such matrices A and B.

17. Let A be a 3×3 real orthogonal matrix. Show that there exists a real orthogonal matrix B such that $B^{-1}AB$ is in the form $\left(\begin{smallmatrix} \pm 1 & 0 \\ 0 & X \end{smallmatrix}\right)$, where X is a rotation or a reflection.

18. Let $A = (a_{ij})$ be a real symmetric and idempotent matrix. Show that $0 \le a_{ii} \le 1$ for all i and $|a_{ij}| \le \frac{1}{2}$ for all i and j with $i \ne j$. Moreover, if $a_{ii} = 0$ or 1, then $a_{ij} = a_{ji} = 0$ for all j with $j \ne i$.

19. Let A be an $n \times n$ real orthogonal matrix and $\operatorname{rank}(A - I_n) = 1$. Show that A is similar to $\operatorname{diag}(-1, 1, \ldots, 1)$ via a real orthogonal matrix.

20. Let A be an $n \times n$ real orthogonal matrix. If $a_1, \ldots, a_k$ are k columns of A, show that $a_1 a_1^* + \cdots + a_k a_k^*$ has exactly k nonzero eigenvalue 1's.

21. Let $A = (a_{ij}) \in \mathbb{M}_n(\mathbb{C})$ and let C_{ij} be the cofactor of a_{ij}, $i, j = 1, 2, \ldots, n$. Show that A is orthogonal if and only if (i) $\det A = 1$ and $a_{ij} = C_{ij}$ for all i, j, or (ii) $\det A = -1$ and $a_{ij} = -C_{ij}$ for all i, j.

22. Let $A = (a_{ij})$ be an $n \times n$ real matrix and let C_{ij} be the cofactor of a_{ij}, $i, j = 1, 2, \ldots, n$. Suppose that $A \ne 0$ and $n > 2$. Show that A is (real) orthogonal if $a_{ij} = C_{ij}$ for all i, j, or if $a_{ij} = -C_{ij}$ for all i, j.

23. Let

$$A = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 & 0 & -1 & 1 \\ 0 & 0 & 1 & 1 \\ 1 & 1 & 0 & 0 \\ 1 & -1 & 0 & 0 \end{pmatrix}, \quad U = \begin{pmatrix} 0 & 1 & 0 & 0 \\ -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & -1 & 0 \end{pmatrix}.$$

    (a) Show that A is real orthogonal.

    (b) Show that $UA = AU^T$.

    (c) Find the eigenvalues of U.

    (d) Show that $A^2 \ne I$.

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6.3 Contractive Matrices

Let A be an $m \times n$ matrix and consider A as a linear map from $\mathbb{C}^n$ to $\mathbb{C}^m$ defined by the ordinary matrix-vector product, namely, $x \mapsto Ax$, where $x \in \mathbb{C}^n$. We say that A is a *contractive matrix* or a *contraction* if A brings (contracts) the vectors "closer" (in terms of the usual norms of $\mathbb{C}^n$ and $\mathbb{C}^m$). To be exact, A is a contraction if there is a constant c, $0 < c \leq 1$, such that

$$\|Ax - Ay\| \leq c\|x - y\| \quad \text{for all } x, y \in \mathbb{C}^n.$$

We show that a matrix A is contractive if and only if the spectral norm $\|A\|_{\text{sp}} \leq 1$, equivalently, the largest singular value $\sigma_{\max}(A) = \sigma_1(A) \leq 1$.

Theorem 6.3.1 *A matrix A is a contraction if and only if $\sigma_{\max}(A) \leq 1$.*

Proof. Let A be an $m \times n$ matrix. For any $x, y \in \mathbb{C}^n$, we have (Sect. 4.1)

$$\|Ax - Ay\| = \|A(x - y)\| \leq \sigma_{\max}(A)\|x - y\|.$$

It follows that A is a contraction if $\sigma_{\max}(A) \leq 1$. Conversely, suppose that A is a contraction. Then for some c, $0 < c \leq 1$, and for all $x, y \in \mathbb{C}^n$,

$$\|Ax - Ay\| \leq c\|x - y\|.$$

In particular, $\|Ax\| \leq c\|x\|$ for all $x \in \mathbb{C}^n$, implying $\sigma_{\max}(A) \leq c \leq 1$. ∎

Let $A \in \mathbb{M}_{m \times n}$. If $\sigma_{\max}(A) < 1$, we say that A a *strict contraction.* (or a *strictly contractive matrix*). For any nonzero matrix A, $\frac{1}{\sigma_{\max}(A)}A$ is a contraction, and $\frac{1}{\sigma_{\max}(A)+\varepsilon}A$ for any $\varepsilon > 0$ is a strict contraction.

Example 6.3.1 Contractive matrices.

(a) $\left(\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right)$ is contractive, but not strict.

(b) $\left(\begin{smallmatrix} 1 & 0 \\ 0 & 0 \end{smallmatrix}\right)$ is contractive, but not strict.

(c) $\left(\begin{smallmatrix} 0 & 0.7 \\ 0 & 0 \end{smallmatrix}\right)$ is strictly contractive.

(d) $\left(\begin{smallmatrix} 0 & 0.7 \\ 0 & 0.8 \end{smallmatrix}\right)$ is not contractive.

(e) $\left(\begin{smallmatrix} 0 & 0.7 \\ 0.8 & 0 \end{smallmatrix}\right)$ is strictly contractive. ∎

Example 6.3.2 Let $A = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)$ be a real matrix. If $a^2 + b^2 + c^2 + d^2 \leq 1$, then A is a contraction, because $\|A\|_{\text{sp}} \leq \|A\|_F = (a^2 + b^2 + c^2 + d^2)^{1/2} \leq 1$. ∎

We next present a result about partitioned positive semidefinite matrices, from which a variety of matrix inequalities can be derived. The result invokes a contractive matrix. Let $A \geq 0$, i.e., A is a positive semidefinite matrix. Recall two facts from Sect. 3.2: $A = B^*B$ for some matrix B (Problem 48, Sect. 3.2) and A has a positive semidefinite square root $A^{1/2}$ (Theorem 3.2.3).

Theorem 6.3.2 *Let $A \in \mathbb{M}_m$ and $B \in \mathbb{M}_n$ be positive semidefinite. Then*

$$\begin{pmatrix} A & X \\ X^* & B \end{pmatrix} \geq 0 \quad \Leftrightarrow \quad X = A^{1/2}CB^{1/2} \text{ for some contraction } C \in \mathbb{M}_{m\times n}.$$

Proof. Sufficiency: Let $X = A^{1/2}CB^{1/2}$ with $\sigma_{\max}(C) \leq 1$. We write

$$\begin{pmatrix} A & X \\ X^* & B \end{pmatrix} = \begin{pmatrix} A^{1/2} & 0 \\ 0 & B^{1/2} \end{pmatrix} \begin{pmatrix} I & C \\ C^* & I \end{pmatrix} \begin{pmatrix} A^{1/2} & 0 \\ 0 & B^{1/2} \end{pmatrix}.$$

For the positive semidefiniteness, it suffices to notice that (Problem 16)

$$\sigma_{\max}(C) \leq 1 \quad \Leftrightarrow \quad \begin{pmatrix} I & C \\ C^* & I \end{pmatrix} \geq 0.$$

Necessity: Since $\left(\begin{smallmatrix} A & X \\ X^* & B \end{smallmatrix}\right) \in \mathbb{M}_{m+n}$ is positive semidefinite, there exists a matrix D (of order $m+n$) such that $\left(\begin{smallmatrix} A & X \\ X^* & B \end{smallmatrix}\right) = D^*D$ (Problem 48, Sect. 3.2). We partition $D = (S, T)$, where $S \in \mathbb{M}_{(m+n)\times m}$ consists of the first m columns of D, and $T \in \mathbb{M}_{(m+n)\times n}$ has the last n columns of D. We obtain

$$\begin{pmatrix} A & X \\ X^* & B \end{pmatrix} = D^*D = \begin{pmatrix} S^* \\ T^* \end{pmatrix} (S, T) = \begin{pmatrix} S^*S & S^*T \\ T^*S & T^*T \end{pmatrix}.$$

So,

$$A = S^*S = |S|^2, \quad X = S^*T, \quad B = T^*T = |T|^2.$$

By polar decomposition (Theorem 3.2.6), let $S^* = |S|U$ and $T = V|T|$, where $U \in \mathbb{M}_{m\times(m+n)}$ and $V \in \mathbb{M}_{(m+n)\times n}$ are partially unitary. We have

$$X = |S|UV|T| = A^{1/2}CB^{1/2},$$

where $C = UV$ is contractive as $\sigma_{\max}(UV) \leq \sigma_{\max}(U)\sigma_{\max}(V) \leq 1$. ∎

Note that targeting (making zeros in) the submatrices X and X^* in the upper-right and lower-left corners in the given partitioned matrix by row and column elementary operations for block matrices is a basic technique in matrix theory. It is used repeatedly in later chapters of this book.

Unitary matrices are contractions, but they are not strict contractions. Every submatrix of a unitary matrix is a contraction (by considering the singular values). We now present two results connecting contractive matrices and unitary matrices. We first show how to construct a unitary matrix to contain a given contractive matrix as a submatrix; we then show that a square matrix is contractive if and only if it is a convex combination of unitary matrices (of the same order as the square matrix).

Theorem 6.3.3 *A matrix A is a contraction, i.e., $\|A\|_{\text{sp}} \leq 1$, if and only if*

$$U = \begin{pmatrix} A & X \\ Y & Z \end{pmatrix}$$

is a unitary matrix for some matrices X, Y, and Z of appropriate sizes.

Proof. Sufficiency:

$$U^*U = I \;\Rightarrow\; A^*A + Y^*Y = I \;\Rightarrow\; A^*A \le I \;\Rightarrow\; \|A\|_{\mathrm{sp}} \le 1.$$

Necessity: Let

$$U = \begin{pmatrix} A & (I - AA^*)^{1/2} \\ (I - A^*A)^{1/2} & -A^* \end{pmatrix}.$$

We show that U is a unitary matrix (see also Problems 26 and 27).

We assume that A is a square matrix, otherwise we add zero rows or columns to make it square and continue the discussion with the square matrix. Let $A = VDW$ be a singular value decomposition of A, where V and W are unitary matrices, and D is a nonnegative diagonal matrix with diagonal entries (the singular values of A) not exceeding 1. Then

$$(I - AA^*)^{1/2} = V(I - D^2)^{1/2}V^*, \;\; (I - A^*A)^{1/2} = W^*(I - D^2)^{1/2}W.$$

Because D is diagonal, it is easy to check that

$$D(I - D^2)^{1/2} = (I - D^2)^{1/2}D.$$

Multiplying by V and W from the left and right, we have

$$VD(I - D^2)^{1/2}W = V(I - D^2)^{1/2}DW,$$

equivalently,

$$A(I - A^*A)^{1/2} = (I - AA^*)^{1/2}A.$$

With this, a simple computation results in $U^*U = I$. ∎

The preceding theorem can be restated as: A matrix is a contraction if and only if it is a submatrix of some unitary matrix (of larger size).

Example 6.3.3 Let A be a 2×2 matrix with singular values $\frac{1}{2}$ and $\frac{1}{3}$ and let $A = VDW$ be a singular value decomposition of A, where $D = \mathrm{diag}(\frac{1}{2}, \frac{1}{3})$, V and W are some 2×2 unitary matrices. Then A is a contraction, and

$$\begin{pmatrix} D & (I - D^2)^{\frac{1}{2}} \\ (I - D^2)^{\frac{1}{2}} & -D \end{pmatrix} = \begin{pmatrix} \frac{1}{2} & 0 & \frac{\sqrt{3}}{2} & 0 \\ 0 & \frac{1}{3} & 0 & \frac{\sqrt{8}}{3} \\ \frac{\sqrt{3}}{2} & 0 & -\frac{1}{2} & 0 \\ 0 & \frac{\sqrt{8}}{3} & 0 & -\frac{1}{3} \end{pmatrix}$$

is a unitary matrix. Thus, $\begin{pmatrix} V & 0 \\ 0 & I_2 \end{pmatrix} \begin{pmatrix} D & (I-D^2)^{1/2} \\ (I-D^2)^{1/2} & -D \end{pmatrix} \begin{pmatrix} W & 0 \\ 0 & I_2 \end{pmatrix}$ is a unitary matrix containing A (in the upper-left corner) as a submatrix. ∎

Theorem 6.3.4 *A matrix A is contractive if and only if $A = \frac{1}{2}(U + V)$ for some partially unitary matrices U and V. For square A, U and V are unitary.*

Proof. Sufficiency: By Theorem 4.2.2 (or Problem 32, Sect. 4.1), we have $\sigma_{\max}(A) = \frac{1}{2}(\sigma_{\max}(U+V)) \le \frac{1}{2}(\sigma_{\max}(U) + \frac{1}{2}\sigma_{\max}(V)) \le 1$. A is contractive.

Necessity: Since A is contractive, $\sigma_{\max}(A) \le 1$ and $A^*A = |A|^2 \le I$. Let

$$S = |A| + i(I - A^*A)^{\frac{1}{2}}.$$

Note that $|A|(I - A^*A)^{\frac{1}{2}} = (I - A^*A)^{\frac{1}{2}}|A|$. One can easily verify that $S^*S = SS^* = I$, that is, S is a unitary matrix. We have $|A| = \frac{1}{2}(S + S^*)$.

By polar decomposition (Theorem 3.2.6), let $A = W|A|$, where W is a (partially) unitary matrix. We obtain $A = \frac{1}{2}W(S + S^*) = \frac{1}{2}(WS + WS^*) = \frac{1}{2}(U + V)$, where $U = WS$ and $V = WS^*$ are (partially) unitary matrices. ∎

Problems

1. Let λ and μ be positive numbers. Show that for any $t \in [0, 1]$,
$$\lambda\mu \le \big(t\lambda + (1-t)\mu\big)\big(t\mu + (1-t)\lambda\big).$$
2. If $A = (a_{ij})$ is a contractive matrix, show that $|a_{ij}| \le 1$ for all i, j.
3. Show that the product of two $n \times n$ contractions is a contraction.
4. If $A \in \mathbb{M}_n$ is a contraction and $\lambda(A)$ is an eigenvalue of A, show that
$$|\det A| \le 1, \quad |\lambda(A)| \le 1, \quad \text{and} \quad |x^*Ax| \le 1 \ \text{ for any unit } x \in \mathbb{C}^n.$$
5. Let $A \in \mathbb{M}_n$ and let $C \in \mathbb{M}_n$ be a contraction. Show that $|\det(AC)| \le |\det A|$ and $\sigma_{\max}(AC) \le \sigma_{\max}(A)$. Is it true that $|\operatorname{tr}(AC)| \le |\operatorname{tr} A|$?
6. Let $A \in \mathbb{M}_n$ and $C \in \mathbb{M}_{n\times m}$. If A is positive semidefinite and C is contractive, show that $\det(C^*AC) \le \det A$ and that $\operatorname{tr}(C^*AC) \le \operatorname{tr} A$.
7. Can a matrix and its inverse (if it exists) be both strict contractions?
8. If $A \in \mathbb{M}_n$ is a strict contraction, show that $I_n - A^*A$ is a strict contraction and that $I_n - A$ is invertible. Must $I_n - A$ be a contraction?
9. If a contractive matrix contains a unitary submatrix, what can be said about the contractive matrix? [Note: Theorem 6.3.3 states that a contractive matrix can be embedded in a unitary matrix (of larger size).]
10. Let $A, B \in \mathbb{M}_{m\times n}$ satisfy $\sigma_{\max}(A) \le \frac{1}{2}$ and $\sigma_{\max}(B) \le \frac{1}{2}$. Show that $A + B$ is a contraction and that AB is a strict contraction.
11. Let A be a nonsingular matrix. If A is strictly contractive, show that A^{-1} is not strictly contractive, but γA^{-1} is for any $\gamma \in (0, \sigma_{\min}(A))$.
12. Write real number $t \in [0, 1]$ as a convex combination of 1 and -1, and write matrix $\begin{pmatrix} \frac{1}{2} & 0 \\ 0 & \frac{1}{3} \end{pmatrix}$ as a convex combination of unitary matrices.
13. Show that a submatrix of a contractive matrix is contractive.
14. Show that a convex combination of unitary matrices is contractive.
15. If A is a contraction satisfying $A + A^* = 2I$, show that $A = I$.

16. Let $A \in \mathbb{M}_{m\times n}$. Show that the following statements are equivalent.

    (a) A is a contraction, i.e., $\sigma_{\max}(A) \le 1$.
    (b) $A^*A \le I_n$.
    (c) $AA^* \le I_m$.
    (d) $\left(\begin{smallmatrix} I & A \\ A^* & I \end{smallmatrix}\right) \ge 0$.
    (e) $x^*(A^*A)x \le 1$ for all unit $x \in \mathbb{C}^n$.
    (f) $\|Ax\| \le \|x\|$ for all $x \in \mathbb{C}^n$.
    (g) $|x^*Ay| \le 1$ for all unit $x, y \in \mathbb{C}^n$.

    Are they equivalent to $|x^*Ax| \le 1$ for all unit $x \in \mathbb{C}^n$?

17. If A is contractive, show that $S = |A| + i(I - |A|^2)^{1/2}$ is unitary.

18. Let $C = \left(\begin{smallmatrix} 0 & \delta \\ 0 & 0 \end{smallmatrix}\right)$ and $P = \left(\begin{smallmatrix} 1 & 0 \\ 0 & \varepsilon \end{smallmatrix}\right)$, where $0 < \varepsilon < \delta < 1$. Show that

    (a) C and P are contractions.
    (b) PCP^{-1} is not a contraction.
    (c) $P^{-1}CP$ is a contraction.

19. Let $A \in \mathbb{M}_n$ and let $B \in \mathbb{M}_{m\times n}$. Show that

$$\begin{pmatrix} A & B^* \\ B & I_m \end{pmatrix} \ge 0 \;\Leftrightarrow\; \begin{pmatrix} I_m & B \\ B^* & A \end{pmatrix} \ge 0 \;\Leftrightarrow\; A \ge B^*B.$$

    Can $A \ge B^*B$ be changed to $A \ge BB^*$ (when $m = n$)?

20. (a) Let A and B be n-square matrices. Show that $A^*A \le B^*B$ if and only if $A = CB$ for some n-square contractive matrix C.
    (b) Find an example: $A^*A \le B^*B$ but $A \ne BC$ for any matrix C.

21. Let A and B be n-square positive definite matrices. Find the conditions on the invertible n-square matrices X such that both

$$\begin{pmatrix} A & X^* \\ X & B \end{pmatrix} \ge 0 \quad \text{and} \quad \begin{pmatrix} A^{-1} & X^{-1} \\ (X^*)^{-1} & B^{-1} \end{pmatrix} \ge 0.$$

22. Let $A \in \mathbb{M}_n$. If there exists a Hermitian matrix $X \in \mathbb{M}_n$ such that

$$\begin{pmatrix} I + X & A \\ A^* & I - X \end{pmatrix} \ge 0,$$

    show that

$$|\langle Ay, y\rangle| \le \langle y, y\rangle \quad \text{for all } y \in \mathbb{C}^n.$$

23. Let $\left(\begin{smallmatrix} A & B \\ B^* & C \end{smallmatrix}\right) \ge 0$, where A is m-square and C is n-square. Show that there exist $m \times n$ matrices X and Y such that $B = AX = YC$.

24. Let $X, Y \in \mathbb{M}_{m\times n}$. Show that there exists a contraction C such that

$$X + Y = (I_m + XX^*)^{1/2} C (I_n + Y^*Y)^{1/2}.$$

When $m = n$, derive the following determinant inequality:

$$|\det(X + Y)|^2 \le \det(I + XX^*)\det(I + Y^*Y).$$

25. For any two matrices U and V of the same size, show that

$$U^*V + V^*U \le U^*U + V^*V.$$

In particular, if U and V are unitary, then

$$U^*V + V^*U \le 2I.$$

Show also that for any $t \in [0, 1]$ and unitary U and V,

$$\big(tU + (1-t)V\big)^*\big(tU + (1-t)V\big) \le I.$$

26. Let $A \in \mathbb{M}_{m\times n}$ and $\|A\|_{\rm sp} \le 1$. If $A = U\Sigma V^*$ is a singular value decomposition of A (Theorem 3.2.5), show that the block matrix $\left(\begin{smallmatrix} B & A \\ A^* & -C \end{smallmatrix}\right)$ is unitary, where $B = U(I - \Sigma\Sigma^*)^{1/2}U^*$ and $C = V(I - \Sigma^*\Sigma)^{1/2}V^*$.

27. If $A \in \mathbb{M}_{m\times n}$ is contractive, show that the matrices below are unitary:

$$\begin{pmatrix} A & (I - AA^*)^{1/2} \\ -(I - A^*A)^{1/2} & A^* \end{pmatrix}, \quad \begin{pmatrix} (I - AA^*)^{1/2} & A \\ -A^* & (I - A^*A)^{1/2} \end{pmatrix}.$$

28. Let B_m be the backward identity matrix of order m. Show that

$$\frac{1}{\sqrt{2}}\begin{pmatrix} I_m & I_m \\ B_m & -B_m \end{pmatrix} \quad \text{and} \quad \frac{1}{\sqrt{2}}\begin{pmatrix} I_m & 0 & I_m \\ 0 & \sqrt{2} & 0 \\ B_m & 0 & -B_m \end{pmatrix}$$

are unitary matrices of orders $2m$ and $(2m + 1)$, respectively.

29. Let $\sigma_1 \ge \cdots \ge \sigma_r \ge 0$. Show that $\operatorname{diag}(\sigma_1, \ldots, \sigma_r, 0, \ldots, 0) \in \mathbb{M}_n$ is a convex combination of $D_k = \operatorname{diag}(\sigma_1, \ldots, \sigma_1, 0, \ldots, 0) \in \mathbb{M}_n$ in which there are k copies of σ_1 on the diagonal, $k = 1, \ldots, r$.

30. Let $0 \le \sigma \le 1$. Show that $\operatorname{diag}(\sigma, \ldots, \sigma, 0, \ldots, 0) \in \mathbb{M}_n$ (r copies of σ) is a convex combination of the following diagonal (unitary) matrices:

$$I_n,\ -I_n,\ \ G_r = \operatorname{diag}(\overbrace{1, \ldots, 1}^{r}, -1, \ldots, -1) \in \mathbb{M}_n.$$

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6.4 The Unitary Similarity of Real Matrices

We show in this section that if two real matrices are similar over the complex number field $\mathbb{C}$, then they are similar over the real $\mathbb{R}$. The statement also holds for unitary similarity. More precisely, if two real matrices are unitarily similar (over $\mathbb{C}$), then they are real orthogonally similar (over $\mathbb{R}$).

Theorem 6.4.1 *Let A and B be $n \times n$ real matrices. If there exists a complex invertible matrix P such that $A = PBP^{-1}$, then there exists a real invertible matrix Q such that $A = QBQ^{-1}$ (that is, P can be chosen to be real).*

Proof. Let $P = P_1 + P_2 i$, where P_1 and P_2 are real matrices. If $P_2 = 0$, we have nothing to show. Otherwise, rewriting $A = PBP^{-1}$ as $AP = PB$, we have $AP_1 = P_1 B$ and $AP_2 = P_2 B$. It follows that for any real number t,

$$A(P_1 + tP_2) = (P_1 + tP_2)B.$$

Since $\det(P_1 + tP_2) = 0$ for finitely many values of t, there exists a real t such that $Q = P_1 + tP_2$ is invertible. Thus, A and B are similar via Q. ∎

[Note: The above theorem has appeared as Problem 34 of Sect. 2.2.]

Example 6.4.1 Real similarity.

(a) $A = \begin{pmatrix} 1 & \sqrt{2} & 0 \\ 0 & 3 & 0 \\ \pi & \sqrt{3} & 2 \end{pmatrix}$ is similar to $B = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{pmatrix}$ via an invertible $Q \in \mathbb{M}_3(\mathbb{R})$.

(b) If $A \in \mathbb{M}_n$ is a real matrix and has only real eigenvalues, then A is similar to its Jordan form via an invertible real matrix. For instance, $\begin{pmatrix} 1 & \sqrt{2} & 0 & \pi \\ 0 & 1 & 3 & 1 \\ 0 & 0 & -2 & 0 \\ 0 & 0 & \sqrt{3} & 2 \end{pmatrix}$ is similar to $\begin{pmatrix} 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & -2 & 0 \\ 0 & 0 & 0 & 2 \end{pmatrix}$ via an invertible $Q \in \mathbb{M}_4(\mathbb{R})$. ∎

For unitary similarity, we begin with a result about symmetric unitary matrices. For example, $\frac{1}{\sqrt{2}} \begin{pmatrix} i & 1 \\ 1 & i \end{pmatrix}$ and $\begin{pmatrix} \cos\theta & \sin\theta \\ \sin\theta & -\cos\theta \end{pmatrix}$ are symmetric and unitary.

Theorem 6.4.2 *Let U be a symmetric unitary matrix, that is, $U^T = U$ and $U^* = U^{-1}$. Then there exists a complex matrix S that satisfies*

(1) $S^2 = U$,
(2) *S is unitary,*
(3) *S is symmetric, and*
(4) *S commutes with every matrix that commutes with U.*

In other words, a symmetric unitary U has a symmetric unitary square root (not necessarily unique) that commutes with any matrix commuting with U.

Proof. Since U is unitary, it is unitarily diagonalizable (Theorem 6.1.1). Let $U = VDV^*$, where V is unitary, D is diagonal, and all diagonal entries of D have modulus 1. If D is a scalar matrix, say $D = dI$, $|d| = 1$, then $U = dI$, and we take $S = cI$, where $c^2 = d$. In general, let $D = d_1 I_{n_1} \oplus \cdots \oplus d_k I_{n_k}$ with distinct d_j's, $|d_j| = 1$. Let $d_j = e^{i\theta_j}$, $\theta_j \in (-\pi, \pi]$, $j = 1, \ldots, k$.

Let $S = VCV^*$, where $C = c_1 I_{n_1} \oplus \cdots \oplus c_k I_{n_k}$ with $c_j = e^{i\theta_j/2}$. Obviously, S is a unitary matrix and $S^2 = U$. We show that this S is symmetric.

Note that $U = U^T$ is $VDV^* = \overline{V}DV^T$, yielding $(V^TV)D = D(V^TV)$. If ν_{st} denotes the (s,t)-entry of V^TV, then $\nu_{st}(d_s - d_t) = 0$ for all s, t. This reveals $\nu_{st}(c_s - c_t) = 0$ for all s, t, consequently, $(V^TV)C = C(V^TV)$, which gives $VCV^* = \overline{V}CV^T = (VCV^*)^T$, that is, S is symmetric.

If W is a matrix commuting with U, i.e., $WU = UW$, then V^*WV commutes with D. This results in $V^*WV = W_1 \oplus \cdots \oplus W_k$ with each W_j having the same size as I_{n_j} (Problem 4). Then S commutes with W. ∎

Example 6.4.2 Square roots of a symmetric unitary matrix.

(a) Every normal matrix has a square root, so does a unitary matrix.

(b) Let $U = I_2$ and $S = -I_2$. S satisfies all (1)–(4) in the theorem.

(c) Let $U = I_2$ and $T = \left(\begin{smallmatrix} 1 & 0 \\ 0 & -1 \end{smallmatrix}\right)$. T satisfies (1)–(3), but not (4).

(d) Let $U = I_2$ and $Z = \left(\begin{smallmatrix} 0 & i \\ -i & 0 \end{smallmatrix}\right)$. Z satisfies (1) and (2), not (3) and (4).

(e) Let $U = I_2$. Take $S = I_2$ or $-I_2$. S is not unique in the theorem. ∎

Theorem 6.4.3 *Let $A, B \in \mathbb{M}_n(\mathbb{R})$. If $A = UBU^*$ for some unitary matrix U, then there exists a real orthogonal matrix Q such that $A = QBQ^T$.*

Proof. Since A and B are real, we have $UBU^* = A = \overline{A} = \overline{U}BU^T$. This gives $U^TUB = BU^TU$. Because U^TU is symmetric unitary, the preceding theorem ensures that U^TU has a symmetric unitary square root, say S, that is, $U^TU = S^2$, $S^T = S$, $S^* = S^{-1}$, and that S commutes with B.

Let $Q = US^{-1}$ or $U = QS$. Then Q is also unitary. Notice that

$$Q^TQ = (US^{-1})^T(US^{-1}) = S^{-1}U^TUS^{-1} = I.$$

Q is real because $Q^T = Q^{-1} = SU^{-1} = SU^* = (US^*)^* = Q^*$, yielding $Q = \overline{Q}$. Thus, Q is real orthogonal. Putting these together: S and B commute, S is unitary, and Q is real orthogonal. It follows that

$$\begin{aligned} A &= UBU^* = (US^{-1})(SB)U^* = Q(BS)U^* \\ &= QB(S^{-1})^*U^* = QBQ^* = QBQ^T. \end{aligned}$$ ∎

The theorem simply says that if two real matrices are unitarily similar (over $\mathbb{C}$), then they are real orthogonally similar (over $\mathbb{R}$).

Problems

1. If A^2 is a unitary matrix, must A be a unitary matrix?

2. If A is an invertible matrix with complex, real, rational, or integer entries, is the inverse A^{-1} of A also a matrix with complex, real, rational, or integer entries, respectively?

3. Let A be a normal matrix. Show that there exist normal matrices B and C such that $B^2 = A$ and $C^3 = A$. Are such B and C unique?

4. If a matrix A commutes with matrix $B = b_1 I_{n_1} \oplus \cdots \oplus b_k I_{n_k}$, where $b_1, \ldots, b_k$ are distinct scalars and I_{n_j} is the identity matrix of order n_j, $j = 1, \ldots, k$, show that A has the direct sum form $A = A_1 \oplus \cdots \oplus A_k$, where each A_j has the same size as the corresponding I_{n_j}.

5. Show that A and B are similar via a real invertible matrix Q, where

$$A = \begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix}, \quad B = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}.$$

   Find such a Q. Are A and B real orthogonally (or unitarily) similar?

6. If two matrices A and B with rational entries are similar over the complex $\mathbb{C}$, are they similar over the real $\mathbb{R}$? Over the rational $\mathbb{Q}$?

7. Let Q be a real orthogonal matrix. If $\lambda \in \mathbb{C}$ is a non-real eigenvalue of Q and $u = x + yi$ is a corresponding eigenvector, where x and y are real, show that x and y are orthogonal and have the same length.

8. Let $A = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)$, $a, b, c, d \in \mathbb{C}$. If $|b| \neq |c|$, show that A is not normal.

9. Let $A, B \in \mathbb{M}_n$ and let $B = U^* A U$ for some unitary matrix $U \in \mathbb{M}_n$. If A is Hermitian, i.e., $A^* = A$, show that B is Hermitian. If A is symmetric, i.e., $A^T = A$, is it true that B is also symmetric?

10. Determine if each of the following statements is true or false.

    (a) Similar matrices have the same eigenvalues.
    (b) Similar matrices have the same singular values.
    (c) Unitarily similar matrices have the same singular values.
    (d) Similar matrices have the same Frobenius norm.
    (e) Unitarily similar matrices have the same Frobenius norm.
    (f) Unitarily similar matrices have the same ℓ_∞-norm.

11. Show that if A is a real symmetric matrix, then there exists a real orthogonal Q such that $Q^T A Q$ is real diagonal. Give an example of a real normal matrix that is unitarily similar to a (complex) diagonal matrix but is not real orthogonally similar to any diagonal matrix.

12. Find an example of block matrix $M = \left(\begin{smallmatrix} A & B \\ C & D \end{smallmatrix}\right)$, where A, B, C, and D are square, such that $\sigma_1(M) = \sigma_1(A) = \sigma_1(B) = \sigma_1(C) = \sigma_1(D) = 1$.

13. Let
$$A = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 2 & 1 \\ 0 & 0 & 2 \end{pmatrix}, \quad B = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 2 & \pm\sqrt{2} \\ 0 & 0 & 2 \end{pmatrix}.$$
    (a) What are the eigenvalues and eigenvectors of A and B?
    (b) Why are A and B similar?
    (c) Show that 1 is a singular value of B but not of A.
    (d) Show that A cannot be unitarily similar to a direct sum of upper-triangular matrices of order(s) 1 or (and) 2.
    (e) Can A and B be unitarily similar?

14. Let A be a nonsingular matrix of odd order. Show that there does not exist a real nonsingular matrix R such that $AR + R^{-1}A = 0$.

15. Let P_1 and P_2 be n-square matrices. If $P_1 + zP_2$ is nonsingular for some $z \in \mathbb{C}$, show that $P_1 + rP_2$ is nonsingular for some $r \in \mathbb{R}$.

16. Find a nonsingular upper-triangular matrix A such that $A^2 = (A^*)^2$ and find a matrix B with non-real eigenvalues such that $B^3 = (B^*)^3$.

17. Let U and V be $n \times n$ unitary matrices. Show that the following statements are equivalent:
    (a) U and V are similar.
    (b) U and V are unitarily similar.
    (c) $U = Z^*VZ$ for some nonsingular Z.

18. Let A be an n-square matrix.
    (a) Show that $A\overline{A} = I_n$ if and only if $A = B(\overline{B})^{-1}$ for some nonsingular matrix B. [Hint: Let $B = cI + \bar{c}A$ for some unit c.]
    (b) For $A = \left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$, find a matrix B such that $A = B(\overline{B})^{-1}$.
    (c) If A is a real matrix such that $A^2 = I_n$, show that there exists a nonsingular matrix B such that $A = B(\overline{B})^{-1}$.
    (d) If $A \neq I_n$, show that the matrix B in (c) cannot be a real matrix.

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6.5 An Inequality of Unit Vectors and Its Applications

We first present an inequality of unit vectors in an inner product space. We then show a trace inequality for unitary matrices, relating the average of the eigenvalues of each of two unitary matrices to that of their product. Recall that the set of complex matrices of the same size forms an inner product space over $\mathbb{C}$ with respect to the Frobenius inner product $\langle A, B\rangle = \mathrm{tr}(B^*A)$.

Theorem 6.5.1 *Let u, v, w be unit vectors in an inner product space. Then*

$$\sqrt{1-|\langle u,v\rangle|^2} \le \sqrt{1-|\langle u,w\rangle|^2} + \sqrt{1-|\langle w,v\rangle|^2}\,. \tag{6.4}$$

Equality holds if and only if w is a multiple of u or v.

Proof 1. Let V be the inner product space of dimension n over $\mathbb{C}$ (or over $\mathbb{R}$; the discussions are similar). Let $e_1 = w$, $e_2, \ldots, e_n$ be an orthonormal basis for V and let x, y, and z be respectively the coordinates of u, v, and w with respect to the basis. Then $z = (1, 0, \cdots, 0)^T$. We write

$$x = \begin{pmatrix} e^{it_1}\cos\alpha \\ x_1 \end{pmatrix} \quad \text{and} \quad y = \begin{pmatrix} e^{it_2}\cos\beta \\ y_1 \end{pmatrix}$$

for some $x_1, y_1 \in \mathbb{C}^{n-1}$, $t_1, t_2 \in \mathbb{R}$, and $\alpha, \beta \in [0, \frac{\pi}{2}]$. It follows that

$$\langle u,v\rangle = y^*x, \;\; \langle u,w\rangle = z^Tx = e^{it_1}\cos\alpha, \;\; \langle w,v\rangle = y^*z = e^{-it_2}\cos\beta.$$

Let $0 \le \gamma \le \frac{\pi}{2}$ be such that $|y_1^*x_1| = \cos\gamma\,\sin\alpha\,\sin\beta$. We derive

$$\begin{aligned} |\langle u,v\rangle|^2 &= |y^*x|^2 \\ &= |e^{i(t_1-t_2)}\cos\alpha\,\cos\beta + y_1^*x_1|^2 \\ &\ge \big||e^{i(t_1-t_2)}\cos\alpha\,\cos\beta| - |y_1^*x_1|\big|^2 \\ &= |\cos\alpha\cos\beta - \cos\gamma\,\sin\alpha\,\sin\beta|^2 \\ &= |\cos(\alpha+\beta) + (1-\cos\gamma)\,\sin\alpha\,\sin\beta|^2. \end{aligned}$$

If $0 \le \alpha+\beta \le \frac{\pi}{2}$, then

$$|\langle u,v\rangle|^2 \ge \cos^2(\alpha+\beta) = 1-\sin^2(\alpha+\beta) \ge 1-(\sin\alpha+\sin\beta)^2.$$

Equality occurs if and only if $\alpha = 0$ ($x_1 = 0$) or $\beta = 0$ ($y_1 = 0$). Thus,

$$\sqrt{1-|\langle u,v\rangle|^2} \le \sin\alpha+\sin\beta = \sqrt{1-|\langle u,w\rangle|^2} + \sqrt{1-|\langle w,v\rangle|^2}.$$

Equality occurs if and only if w is a multiple of u (if $\alpha = 0$) or v (if $\beta = 0$).

If $\alpha + \beta > \frac{\pi}{2}$, namely, $\beta > \frac{\pi}{2} - \alpha \geq 0$, then

$$\begin{aligned}
&\sqrt{1 - |\langle u, w\rangle|^2} + \sqrt{1 - |\langle w, v\rangle|^2} \\
&\quad = \sin\alpha + \sin\beta > \sin\alpha + \sin(\tfrac{\pi}{2} - \alpha) \\
&\quad = \sin\alpha + \cos\alpha \geq \sin^2\alpha + \cos^2\alpha = 1 \\
&\quad \geq \sqrt{1 - |\langle u, v\rangle|^2}.
\end{aligned}$$

Proof 2. Let $w = w_1 + w_2$, where $w_1 \in \mathrm{Span}\{u, v\}$ and $w_2 \in \mathrm{Span}\{u, v\}^{\perp}$. If $w_1 = 0$, the inequality is obvious. Otherwise, let $w' = \frac{w_1}{\|w_1\|}$, $0 < \|w_1\| \leq 1$. Then w' is a unit vector in $\mathrm{Span}\{u, v\}$, we have

$$|\langle u, w\rangle| = |\langle u, w_1\rangle| \leq |\langle u, w'\rangle|$$

and

$$|\langle w, v\rangle| = |\langle w_1, v\rangle| \leq |\langle w', v\rangle|.$$

Thus, $1 - |\langle u, w'\rangle|^2 \leq 1 - |\langle u, w\rangle|^2$ and $1 - |\langle w', v\rangle|^2 \leq 1 - |\langle w, v\rangle|^2$. Hence, it suffices to show (6.4) for the case where w is a unit vector in $\mathrm{Span}\{u, v\}$.

If u and v are linearly dependent, it is trivial. Suppose that u and v are linearly independent and let $\{u, z\}$ be an orthonormal basis of $\mathrm{Span}\{u, v\}$. Let $v = \mu u + \lambda z$ and $w = \alpha u + \beta z$ for some $\mu, \lambda, \alpha, \beta \in \mathbb{C}$. Then we have

$$|\lambda|^2 + |\mu|^2 = 1 \quad \text{and} \quad |\alpha|^2 + |\beta|^2 = 1.$$

By using the above relations and the arithmetic mean-geometric mean inequality, together with $|c| \geq \Re(c)$ for any complex number c, we compute

$$\begin{aligned}
|\lambda\beta| &= \frac{1}{2}|\lambda\beta|(|\mu|^2 + |\lambda|^2 + |\alpha|^2 + |\beta|^2) \\
&\geq |\lambda\beta|(|\lambda\beta| + |\alpha\mu|) \\
&= |\lambda\beta|^2 + |\lambda\beta\alpha\mu| \\
&= |\lambda\beta|^2 + |\lambda\bar{\beta}\alpha\bar{\mu}| \\
&\geq |\lambda\beta|^2 + \Re(\lambda\bar{\beta}\alpha\bar{\mu}).
\end{aligned}$$

It follows that $-2|\lambda\beta| \leq -2|\lambda\beta|^2 - 2\,\Re(\lambda\bar{\beta}\alpha\bar{\mu})$. Thus, we have

$$\begin{aligned}
(|\lambda| - |\beta|)^2 &= |\lambda|^2 - 2|\lambda\beta| + |\beta|^2 \\
&\leq |\lambda|^2 + |\beta|^2 - 2|\lambda\beta|^2 - 2\,\Re(\lambda\bar{\beta}\alpha\bar{\mu}) \\
&= |\lambda|^2 + |\beta|^2(1 - |\lambda|^2) - |\lambda\beta|^2 - 2\,\Re(\lambda\bar{\beta}\alpha\bar{\mu}) \\
&= (1 - |\mu|^2) + |\beta|^2|\mu|^2 - |\lambda\beta|^2 - 2\,\Re(\lambda\bar{\beta}\alpha\bar{\mu}) \\
&= 1 - |\mu|^2(1 - |\beta|^2) - |\lambda\beta|^2 - 2\,\Re(\lambda\bar{\beta}\alpha\bar{\mu}) \\
&= 1 - |\mu\alpha|^2 - |\lambda\beta|^2 - 2\,\Re(\lambda\bar{\beta}\alpha\bar{\mu}) \\
&= 1 - |\alpha\bar{\mu} + \beta\bar{\lambda}|^2.
\end{aligned}$$

This gives

$$|\lambda| - |\beta| \le \sqrt{1 - |\alpha\bar{\mu} + \beta\bar{\lambda}|^2}\ ,$$

or

$$|\lambda| \le |\beta| + \sqrt{1 - |\alpha\bar{\mu} + \beta\bar{\lambda}|^2}\ ,$$

which is the same as

$$\sqrt{1 - |\mu|^2} \le \sqrt{1 - |\alpha|^2} + \sqrt{1 - |\alpha\bar{\mu} + \beta\bar{\lambda}|^2}\ .$$

Because $|\mu|^2 = |\langle u, v\rangle|^2$, $|\alpha|^2 = |\langle u, w\rangle|^2$, and

$$|\alpha\bar{\mu} + \beta\bar{\lambda}|^2 = |\langle \alpha u + \beta z, \mu u + \lambda z\rangle|^2 = |\langle w, v\rangle|^2,$$

the inequality (6.4) is proved.

Equality holds for the overall inequality if and only if equality holds at the two points in our derivation where we invoked the arithmetic mean-geometric mean inequality and $|c| \ge \Re(c)$. Thus, equality holds if and only if $|\lambda| = |\beta|$ and $|\alpha| = |\mu|$, as well as $\Re(\lambda\bar{\beta}\alpha\bar{\mu}) = |\lambda\bar{\beta}\alpha\bar{\mu}|$. The former is equivalent to having $\lambda = e^{i\theta}\beta$ and $\mu = e^{i\phi}\alpha$ for some real numbers θ and ϕ, while the latter is then equivalent to $\Re\left(|\alpha\beta|^2(e^{i(\theta-\phi)} - 1)\right) = 0$. Thus, $\alpha = 0$, $\beta = 0$, or $e^{i\theta} = e^{i\phi}$, so equality in (6.4) holds if and only if either w is a multiple of u ($\beta = 0$) or w is a multiple of v ($\alpha = 0$ or $e^{i\theta} = e^{i\phi}$). ∎

As an application, consider the space $\mathbb{M}_n$ with the inner product $\langle A, B\rangle = \operatorname{tr}(B^*A)$ for $A, B \in \mathbb{M}_n$. Let U and V be $n \times n$ unitary matrices. By putting

$$u = \frac{1}{\sqrt{n}}V, \quad v = \frac{1}{\sqrt{n}}U^*, \quad w = \frac{1}{\sqrt{n}}I$$

in (6.4), and writing $m(X) = \frac{1}{n}\operatorname{tr} X$ for the average of the eigenvalues of the matrix $X \in \mathbb{M}_n$, we have the following result.

Theorem 6.5.2 *For any $n \times n$ unitary matrices U and V,*

$$\sqrt{1 - |m(UV)|^2} \le \sqrt{1 - |m(U)|^2} + \sqrt{1 - |m(V)|^2}$$

with equality if and only if U or V is a unitary scalar matrix.

Example 6.5.1 Let P and Q be $n \times n$ permutation matrices. If the numbers of 1's on the diagonals of P and Q are p and q, respectively, then the number of 1's on the diagonal of PQ, denoted by x (i.e., $x = \operatorname{tr}(PQ)$), satisfies

$$\sqrt{n^2 - x^2} \le \sqrt{n^2 - p^2} + \sqrt{n^2 - q^2}.$$ ∎

[Note: One may use majorization (Sect. 10.3) to show the inequality.]

Problems

1. Let P and Q be $n \times n$ permutation matrices. Show that
$$\operatorname{tr}(P) + \operatorname{tr}(Q) - \operatorname{tr}(PQ) \leq n.$$

2. Is it true that $|\langle u, v\rangle| \leq |\langle u, w\rangle| + |\langle w, v\rangle|$ or $|\langle u, w\rangle| + |\langle w, v\rangle| \leq 1 + |\langle u, v\rangle|$ for all unit vectors u, v, and w in an inner product space?

3. Let U be an $m \times n$ matrix such that $U^*U = I_n$. Show that
$$\operatorname{tr}(UAU^*) = \operatorname{tr} A, \quad \text{for any } A \in \mathbb{M}_n.$$

4. Show that for any square matrix A and positive integers p and q,
$$|\operatorname{tr} A^{p+q}|^2 \leq \operatorname{tr}\left((A^*)^p A^p\right) \operatorname{tr}\left((A^*)^q A^q\right),$$
$$\operatorname{tr}(A^*A)^p \leq (\operatorname{tr}(A^*A))^p, \quad \text{and} \quad \operatorname{tr}((A^*)^p A^p) \leq (\operatorname{tr}(A^*A))^p.$$

5. If U is an $n \times n$ nonscalar unitary matrix (i.e., $U \neq kI$, $k \in \mathbb{C}$) with eigenvalues $\lambda_1, \ldots, \lambda_n$, show that the following strict inequality holds:
$$\left| \frac{\lambda_1 + \cdots + \lambda_n}{n} \right| < 1.$$

6. Let U be a unitary matrix and let V be any square submatrix of U. Show that $|\lambda(V)| \leq 1$ for every eigenvalue $\lambda(V)$ of V.

7. Let U and V be $n \times n$ unitary matrices such that $U + V$ is unitary.
   (a) If a, b, and c are complex numbers on the unit circle (i.e., having moduli 1) such that $a + b = c$, what are a and b in terms of c?
   (b) Show that the eigenvalues of U^*V are $-\frac{1}{2} + \frac{\sqrt{3}}{2} i$ and (or) $-\frac{1}{2} - \frac{\sqrt{3}}{2} i$. [Hint: Let $W = U^*(U + V) = I + U^*V$. Observe that if λ is a unit complex number such that $|1 + \lambda| = 1$, then $\lambda = -\frac{1}{2} \pm \frac{\sqrt{3}}{2} i$.]
   (c) Find a pair of U, V such that UV has no eigenvalues $-\frac{1}{2} \pm \frac{\sqrt{3}}{2} i$.
   (d) Find a pair of U, V such that UV has only eigenvalues $-\frac{1}{2} \pm \frac{\sqrt{3}}{2} i$.

8. For unit vectors u and v in an inner product space V over $\mathbb{C}$, define $\theta_{u,v} = \cos^{-1} |\langle u, v\rangle| \in [0, \frac{\pi}{2}]$. Show that for any unit vector $w \in V$,
$$\sin \theta_{u,v} \leq \sin \theta_{u,w} + \sin \theta_{w,v}.$$

9. For unit vectors u and v in an inner product space V over $\mathbb{R}$, let $\Theta_{u,v} = \cos^{-1} \langle u, v\rangle \in [0, \pi]$ (Sect. 1.4). Show that for any unit vector $w \in V$,
$$\sin \Theta_{u,v} \leq \sin \Theta_{u,w} + \sin \Theta_{w,v}.$$

10. Let $D = \operatorname{diag}(d_1, \ldots, d_n)$, where $d_1, \ldots, d_n \in \mathbb{C}$. Show that
$$\min_i |d_i| \leq |\lambda(DU)| \leq \max_i |d_i|$$
for any unitary $U \in \mathbb{M}_n$, where $\lambda(DU)$ is any eigenvalue of DU.

11. Let A and B be n-square matrices.

    (a) Show that $|\operatorname{tr}(AB)| \le \|A\|_{F}\|B\|_{F}$.

    (b) Is it true that $|\operatorname{tr}(AB)| \le \|A\|_{\mathrm{sp}}\|B\|_{\mathrm{sp}}$?

    (c) If A is positive semidefinite, show that $|\operatorname{tr}(AB)| \le (\operatorname{tr}A)\,\|B\|_{\mathrm{sp}}$.

    (d) If A is positive semidefinite, show that $\Re(\operatorname{tr}(A - UAV)) \ge 0$ for all n-square contractive matrices U and V.

    (e) If $\operatorname{tr}A$ is real and $B \ge 0$, show that $\operatorname{tr}(A-B) \le \Re(\operatorname{tr}(A-UBV)) \le \operatorname{tr}(A+B)$ for all n-square contractive matrices U and V.

12. Let $\|X\|_{F} = \sqrt{\operatorname{tr}(X^*X)}$ be the Frobenius norm of $X \in \mathbb{M}_n$. Show that

$$\|A\|_{F}^2 = \left\|\frac{A+A^*}{2}\right\|_{F}^2 + \left\|\frac{A-A^*}{2}\right\|_{F}^2 \quad \text{for any } A \in \mathbb{M}_n.$$

13. Let U be a unitary matrix. If λ and μ are two different eigenvalues of U, show that their eigenvectors u and v are orthogonal. Show that $au + bv$ cannot be an eigenvector of U (for any eigenvalue) if $ab \ne 0$.

14. Let A be an n-square matrix. Show that for all $x, y \in \mathbb{C}^n$ (in columns),

$$|x^*Ax - y^*Ay| \le \|x-y\|(\|x\| + \|y\|)\|A\|_{\mathrm{sp}}.$$

15. Let S be the set of unit vectors in $\mathbb{C}^n$ with the usual inner product. Let

$$d(x,y) = \sqrt{1 - |\langle x, y\rangle|^2}, \quad x,\, y \in S.$$

    Show that $d(x,y) = d(y,x)$ for all x and y in S and that $d(x,y) = 0$ if and only if $x = cy$ for some complex number c with $|c| = 1$.

16. For nonzero vectors $u, v \in \mathbb{C}^2$ (in columns) with $\langle u, v\rangle = v^*u$, define

$$d(u,v) = \sqrt{1 - \frac{|\langle u, v\rangle|^2}{\|u\|^2\|v\|^2}}\ .$$

    Show that for any nonzero u, v, $w \in \mathbb{C}^2$, and nonzero λ, $\mu \in \mathbb{C}$,

    (a) $d(\lambda u, \mu v) = d(u,v)$,

    (b) $d(u,v) \le d(u,w) + d(w,v)$, and

    (c) $d(u,v) = d(z_u, z_v)$, where $z_x = \frac{x_2}{x_1}$ if $x = \binom{x_1}{x_2} \in \mathbb{C}^2$, $x_1 \ne 0$, and

$$d(z_u, z_v) = \frac{|z_u - z_v|}{\sqrt{(1+|z_u|^2)(1+|z_v|^2)}}.$$

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CHAPTER 7

Positive Semidefinite Matrices

Introduction: This chapter studies the positive semidefinite matrices, focusing on the inequalities of the matrices of this type. Our goal is to present the fundamental results and to show some often-used techniques. Sect. 7.1 gives the basic properties, Sect. 7.2 treats the Loewner partial order of positive semidefinite matrices, and Sect. 7.3 presents some inequalities for principal submatrices. Sect. 7.4 derives determinant inequalities of Fischer, Hadamard, and Hua, and Sect. 7.5 investigates the Hadamard product of the positive semidefinite matrices. Finally, Sect. 7.6 shows the Cauchy-Schwarz type inequalities and the inequalities of Choi, Kadison, Kantorovich, and Wielandt.

7.1 Positive Semidefinite Matrices

In Chap. 3 we introduced positive semidefinite matrices and showed some basic results. In this chapter we study more for the matrices of this type.

Let A be an $n \times n$ matrix. We say that A is *positive semidefinite* (PSD) or *nonnegative definite* (Sect. 3.2), if for all complex column vectors $x \in \mathbb{C}^n$,

$$x^*Ax \geq 0, \quad \text{denoted by } A \geq 0. \tag{7.1}$$

A is further called *positive definite* (PD), written as $A > 0$, if the strict inequality in (7.1) holds for all nonzero $x \in \mathbb{C}^n$, i.e., $x^*Ax > 0$ if $0 \neq x \in \mathbb{C}^n$.

Note that $A^*A \geq 0$ (and $AA^* \geq 0$) for any matrix A because

$$x^*(A^*A)x = (Ax)^*(Ax) \geq 0.$$

Example 7.1.1 Positive semidefinite and positive definite matrices.

(a) 0_n and I_n are positive semidefinite. In fact, I_n is positive definite. Every positive definite matrix is also positive semidefinite.

(b) $\left(\begin{smallmatrix} 1 & 0 \\ 0 & 0 \end{smallmatrix}\right)$ and $\left(\begin{smallmatrix} 1 & 1 \\ 1 & 1 \end{smallmatrix}\right)$ are positive semidefinite, not positive definite.

(c) $\left(\begin{smallmatrix} 2 & -1 \\ -1 & 1 \end{smallmatrix}\right) \geq 0$ (> 0 in fact) because if we set $y = \left(\begin{smallmatrix} 1 & -1 \\ -1 & 0 \end{smallmatrix}\right)\left(\begin{smallmatrix} x_1 \\ x_2 \end{smallmatrix}\right)$, then $(\overline{x_1}, \overline{x_2})\left(\begin{smallmatrix} 2 & -1 \\ -1 & 1 \end{smallmatrix}\right)\left(\begin{smallmatrix} x_1 \\ x_2 \end{smallmatrix}\right) = (\overline{x_1}, \overline{x_2})\left(\begin{smallmatrix} 1 & -1 \\ -1 & 0 \end{smallmatrix}\right)\left(\begin{smallmatrix} 1 & -1 \\ -1 & 0 \end{smallmatrix}\right)\left(\begin{smallmatrix} x_1 \\ x_2 \end{smallmatrix}\right) = y^*y \geq 0.$

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(d) None of $\left(\begin{smallmatrix} 3 & -1 \\ 0 & 1 \end{smallmatrix}\right)$, $\left(\begin{smallmatrix} 3 & 1 \\ 1 & -1 \end{smallmatrix}\right)$, $\left(\begin{smallmatrix} 2 & 2 \\ 2 & 1 \end{smallmatrix}\right)$, $\left(\begin{smallmatrix} 0 & 1 \\ 1 & 1 \end{smallmatrix}\right)$, and $\left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right)$ is positive semidefinite. For instance, we compute to get $(1,-2)\left(\begin{smallmatrix} 2 & 2 \\ 2 & 1 \end{smallmatrix}\right)\left(\begin{smallmatrix} 1 \\ -2 \end{smallmatrix}\right) = -2 < 0$, while $(1,-i)\left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right)\left(\begin{smallmatrix} 1 \\ i \end{smallmatrix}\right) = 2i$ is not a nonnegative number. ∎

If $A \in \mathbb{M}_n$ is positive semidefinite, then for any $n \times m$ matrix X and for any $y \in \mathbb{C}^m$, $y^*(X^*AX)y = (Xy)^*A(Xy) \geq 0$. Thus, $X^*AX \geq 0$. Conversely, let $X^*AX \geq 0$ for all $X \in \mathbb{M}_{n\times m}(\mathbb{C})$ (for any or some fixed m). For $x \in \mathbb{C}^n$, let $X = (x, 0) \in \mathbb{M}_{n\times m}$ and take $e_1 = (1, 0 \ldots, 0)^T \in \mathbb{R}^m$. Then $Xe_1 = x$ and $x^*Ax = (Xe_1)^*A(Xe_1) = e_1^T(X^*AX)e_1 \geq 0$, i.e., $A \geq 0$. In summary,

$$A \geq 0 \quad \Leftrightarrow \quad X^*AX \geq 0 \text{ for any } X \in \mathbb{M}_{n\times m}(\mathbb{C}). \tag{7.2}$$

Let $A, B \in \mathbb{M}_n$. It is readily seen that

$$A, B \geq 0 \quad \Rightarrow \quad A + B \geq A \geq 0 \text{ and } A^{1/2}BA^{1/2} \geq 0. \tag{7.3}$$

The following decomposition theorem (by the spectral decomposition theorem, Theorem 3.2.2, Sect. 3.2, Chap. 3) of positive semidefinite (PSD) matrices best characterizes positive semidefiniteness under unitary similarity.

Theorem 7.1.1 (Spectral Decomposition of a PSD Matrix) *Let $A \in \mathbb{M}_n$ and let $\lambda_1, \ldots, \lambda_n$ be the eigenvalues of A. A is positive semidefinite if and only if all λ_i are nonnegative and there exists a unitary $U \in \mathbb{M}_n$ such that*

$$A = U^* \operatorname{diag}(\lambda_1, \ldots, \lambda_n)U, \tag{7.4}$$

and A is positive definite if and only if (7.4) holds and all λ_i are positive.

It is immediate that $A \geq 0 \Rightarrow \det A \geq 0$, and $A > 0 \Rightarrow \det A > 0$.

Positive semidefinite matrices are Hermitian matrices with nonnegative eigenvalues (Sect. 3.2, Chap. 3). Positive definite matrices are positive semidefinite and nonsingular. As a central topic in matrix theory, positive semidefinite matrices have many important properties and applications in various areas. The following theorem is useful in practice to determine if a matrix is positive (semi)definite when the size of the matrix is small.

Theorem 7.1.2 *Let A be an $n \times n$ Hermitian matrix (i.e., $A^* = A$).*

(1) *A is positive semidefinite if and only if the determinant of every (not just the leading) principal submatrix of A is nonnegative.*

(2) *A is positive definite if and only if the determinant of every leading principal submatrix (including A) of A is positive.*

Proof. To show (1), let A_k be a $k \times k$ principal submatrix of $A \in \mathbb{M}_n$. By permuting rows and columns simultaneously we can place A_k in the upper-left corner. In other words, there exists a permutation matrix P such that A_k is the $(1,1)$-block of P^TAP. If $A \geq 0$, then (7.1) holds, and for $x \in \mathbb{C}^k$,

$$x^*A_kx = y^*Ay \geq 0, \quad \text{where } y = P\left(\begin{smallmatrix} x \\ 0 \end{smallmatrix}\right) \in \mathbb{C}^n.$$

This says that A_k is positive semidefinite. Therefore, $\det A_k \geq 0$. The strict inequalities hold for $x \neq 0$ and positive definite A (as it is nonsingular).

Conversely, if every principal submatrix of A has a nonnegative determinant, then the polynomial in λ (Problem 30 of Sect. 1.3)

$$\det(\lambda I - A) = \lambda^n - \delta_1 \lambda^{n-1} + \delta_2 \lambda^{n-2} - \cdots + (-1)^n \delta_n$$

has no negative zeros, because all δ_i are nonnegative, where δ_i is the sum of the determinants of all principal matrices of order i, $i = 1, \ldots, n$.

[Note: The determinant of a square submatrix is called a *minor*.]

For (2), we only need to show the sufficiency, and we show it by induction on n. If $n = 1$, it is obvious. For $n > 1$, partition $A = \begin{pmatrix} A_{n-1} & u \\ u^* & a \end{pmatrix}$ and suppose the assertion holds for A_{n-1} of order $n-1$. The assumption that $\det A > 0$ implies $a - u^* A_{n-1}^{-1} u > 0$. Let $b = a - u^* A_{n-1}^{-1} u > 0$. Then $A = \begin{pmatrix} A_{n-1} & u \\ u^* & u^* A_{n-1}^{-1} u \end{pmatrix} + \begin{pmatrix} 0 & 0 \\ 0 & b \end{pmatrix}$ is the sum of positive semidefinite matrices, and A is positive definite because A is nonsingular (as $\det A > 0$). ∎

For the 2×2 case, here is a useful fact: Let $a, c \in \mathbb{R}$, $b \in \mathbb{C}$. Then

$$\begin{pmatrix} a & b \\ \bar{b} & c \end{pmatrix} \geq 0 \quad \Leftrightarrow \quad a \geq 0,\ c \geq 0, \text{ and } ac \geq |b|^2. \tag{7.5}$$

Example 7.1.2 Positive semidefiniteness and principal submatrices.

(a) $A = \left(\begin{smallmatrix} 1 & 0 \\ 1 & 1 \end{smallmatrix}\right)$ is not Hermitian, so it cannot be positive semidefinite. Note that all principal minors of A are positive.

(b) $\left(\begin{smallmatrix} 0 & 0 \\ 0 & 1 \end{smallmatrix}\right)$ and $\left(\begin{smallmatrix} 1 & -1 \\ -1 & 1 \end{smallmatrix}\right)$ are positive semidefinite, while $\left(\begin{smallmatrix} 0 & 1 \\ 1 & 1 \end{smallmatrix}\right)$ is not.

(c) $\begin{pmatrix} 3 & -1 \\ -1 & 1 \end{pmatrix}$ and $\left(\begin{smallmatrix} 3 & i \\ -i & 1 \end{smallmatrix}\right)$ are positive definite, while $\left(\begin{smallmatrix} 3 & i \\ i & 1 \end{smallmatrix}\right)$ is not.

(d) $B = \left(\begin{smallmatrix} 1 & 1 & 0 \\ 1 & 2 & 1 \\ 0 & 1 & 0 \end{smallmatrix}\right)$ is not positive semidefinite. Note that all leading principal minors of order less than $n-1$ (i.e., 2 in this case) are positive.

(e) $C = \left(\begin{smallmatrix} 0 & 0 & 1 \\ 0 & 0 & 1 \\ 1 & 1 & 0 \end{smallmatrix}\right)$ is not positive semidefinite. Note that all leading principal minors (including $\det C$) of C are nonnegative.

(f) If A is positive (semi)definite, so is every principal submatrix of A.

(g) If $A = (a_{ij}) \geq 0$, then $a_{ii} \geq 0$ for every i and $a_{ii} a_{jj} \geq |a_{ij}|^2$ for all i, j, because the 2×2 principal submatrix $\left(\begin{smallmatrix} a_{ii} & a_{ij} \\ a_{ji} & a_{jj} \end{smallmatrix}\right) \geq 0$, $i \neq j$. ∎

In a positive semidefinite matrix, all diagonal entries are necessarily nonnegative. If a diagonal entry is zero, then the row and the column containing the zero diagonal entry consist entirely of zeros. So, for a positive definite matrix, the diagonal entries are all positive (the converse is untrue).

It is immediate that $A \geq 0 \Rightarrow \operatorname{tr} A \geq 0$ with equality if and only if $A = 0$.

The following theorem is important and useful; it best characterizes the positive semidefiniteness of a square matrix in terms of factorizations.

Theorem 7.1.3 *Let $A \in \mathbb{M}_n$. The following statements are equivalent.*

(1) *A is positive semidefinite.*
(2) *$A = B^*B$ for some matrix $B \in \mathbb{M}_n$.*
(3) *$A = C^*C$ for some upper-triangular matrix $C \in \mathbb{M}_n$.*
(4) *$A = D^*D$ for some upper-triangular matrix $D \in \mathbb{M}_n$ with nonnegative diagonal entries (Cholesky factorization).*
(5) *$A = E^* \left(\begin{smallmatrix} I_r & 0 \\ 0 & 0 \end{smallmatrix}\right) E$, $r = \operatorname{rank}(A)$, for some invertible matrix $E \in \mathbb{M}_n$.*
(6) *$A = F^*F$ for some $r \times n$ matrix F, $r = \operatorname{rank}(A)$ (Rank factorization).*

Proof. (1) $\Rightarrow$ (2): By the spectral decomposition (Theorem 7.1.1), let $A = U^*TU$, where U is unitary and $T = \operatorname{diag}(\lambda_1, \ldots, \lambda_n)$ with all $\lambda_i \geq 0$. Let $T^{1/2} = \operatorname{diag}(\lambda_1^{1/2}, \ldots, \lambda_n^{1/2})$ and let $B = U^*T^{1/2}U$ (which is also PSD). Then $A = U^*TU = U^*T^{1/2}T^{1/2}U = (U^*T^{1/2}U)(U^*T^{1/2}U) = B^*B$, as desired.

(2) $\Rightarrow$ (1): $x^*Ax = x^*B^*Bx = (Bx)^*(Bx) \geq 0$ for any $x \in \mathbb{C}^n$.

(2) $\Rightarrow$ (4) $\Rightarrow$ (3) $\Rightarrow$ (2). It suffices to show (2) $\Rightarrow$ (4). Let $A = B^*B$, where $B \in \mathbb{M}_n$. By the QR decomposition (Problem 3.2.4, Sect. 3.2), let $B = QR$, where Q is unitary and R is upper-triangular and has nonnegative diagonal entries. Take $D = R$. Then $A = B^*B = (R^*Q^*)(QR) = R^*R = D^*D$.

(1) $\Rightarrow$ (5): Let $A = U^*TU$ as in (1) $\Rightarrow$ (2). We may assume that $\lambda_i > 0$ for $i \leq r$ and $\lambda_i = 0$ for $i > r$ (see Theorem 3.2.1, Schur form with any prescribed order of the λ_i's). Let $G = \operatorname{diag}(\lambda_1^{1/2}, \ldots, \lambda_r^{1/2}, 1, \ldots, 1) = G^* > 0$. Then $T = G \left(\begin{smallmatrix} I_r & 0 \\ 0 & 0 \end{smallmatrix}\right) G$. Let $E = GU$. Then E is invertible and $A = E^* \left(\begin{smallmatrix} I_r & 0 \\ 0 & 0 \end{smallmatrix}\right) E$.

(5) $\Rightarrow$ (1): Note that $A = E^* \left(\begin{smallmatrix} I_r & 0 \\ 0 & 0 \end{smallmatrix}\right) E = E^* \left(\begin{smallmatrix} I_r & 0 \\ 0 & 0 \end{smallmatrix}\right) \left(\begin{smallmatrix} I_r & 0 \\ 0 & 0 \end{smallmatrix}\right) E = B^*B$, where $B = \left(\begin{smallmatrix} I_r & 0 \\ 0 & 0 \end{smallmatrix}\right) E$. By (2), A is positive semidefinite.

(5) $\Rightarrow$ (6): Let F be the $r \times n$ matrix consisting of the first r rows of E, i.e., $E = \left(\begin{smallmatrix} F \\ H \end{smallmatrix}\right)$, where $H \in \mathbb{M}_{(n-r)\times n}$. It follows that $A = E^* \left(\begin{smallmatrix} I_r & 0 \\ 0 & 0 \end{smallmatrix}\right) E = E^* \left(\begin{smallmatrix} I_r & 0 \\ 0 & 0 \end{smallmatrix}\right) \left(\begin{smallmatrix} I_r & 0 \\ 0 & 0 \end{smallmatrix}\right) E = (F^*, H^*) \left(\begin{smallmatrix} I_r & 0 \\ 0 & 0 \end{smallmatrix}\right) \left(\begin{smallmatrix} I_r & 0 \\ 0 & 0 \end{smallmatrix}\right) \left(\begin{smallmatrix} F \\ H \end{smallmatrix}\right) = (F^*, 0) \left(\begin{smallmatrix} F \\ 0 \end{smallmatrix}\right) = F^*F$.

(6) $\Rightarrow$ (1): $x^*Ax = x^*(F^*F)x = (Fx)^*(Fx) \geq 0$ for any $x \in \mathbb{C}^n$. ∎

Example 7.1.3 Notes about ^{T} (transpose) and * (conjugate transpose).

(a) Let $A = \left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right)$. Then $x^TAx \geq 0$ (= 0 in fact) for all $x \in \mathbb{C}^2$, but A is not PSD; A has eigenvalues $\pm i$; A cannot be written as B^*B (PSD) nor as B^TB (symmetric) for any matrix B. Note that $\langle Ax, x\rangle \geq 0$ for all x in the Euclidean space $\mathbb{R}^2$ with the usual inner product.

(b) Let $A = \left(\begin{smallmatrix} 1 & 0 \\ 0 & 0 \end{smallmatrix}\right)$. Then $x^*Ax \geq 0$ for all $x \in \mathbb{C}^2$; A is a positive semidefinite matrix. However, $x^TAx = -1 < 0$ for $x = \left(\begin{smallmatrix} i \\ 1 \end{smallmatrix}\right) \in \mathbb{C}^2$. ∎

Every nonnegative number has a unique nonnegative square root. The analogous result for positive semidefinite matrices also holds as we saw in Sect. 3.2 (Theorem 3.2.3). We restate the theorem and present a proof that works for linear operators on finite-dimensional inner product spaces.

Theorem 7.1.4 *For every $A \geq 0$, there exists a unique $B \geq 0$ such that*

$$B^2 = A \quad (B \text{ is denoted by } A^{1/2}).$$

Moreover, $A^{1/2}$ can be expressed as a matrix polynomial in A.

Proof. We view n-square matrices as linear operators on $\mathbb{C}^n$. The spectral theorem ensures the existence of orthonormal eigenvectors $u_1, \ldots, u_n$ associated with the eigenvalues $\lambda_1, \ldots, \lambda_n$ of A, respectively. Then $u_1, \ldots, u_n$ form an orthonormal basis for $\mathbb{C}^n$ and $A(u_i) = \lambda_i u_i$, $\lambda_i \geq 0$. Define a linear operator B by $B(u_i) = \sqrt{\lambda_i}\, u_i$ for $i = 1, \ldots, n$. It is routine to check that $B^2(x) = A(x)$ and $\langle B(x), x\rangle \geq 0$ for all $x \in \mathbb{C}^n$. Hence, $B^2 = A$ and $B \geq 0$.

To show the uniqueness of B, suppose that C is also a linear operator such that $C^2(x) = A(x)$ and $\langle C(x), x\rangle = \langle x, C(x)\rangle \geq 0$ for all $x \in \mathbb{C}^n$. If (μ, v) is an eigenpair of C, then $Cv = \mu v$, $\mu \geq 0$, and $C^2 v = \mu^2 v$. Thus, μ^2 is an eigenvalue of A. It follows that the eigenvalues of C are the nonnegative square roots of the eigenvalues of A, that is, $\sqrt{\lambda_1}, \ldots, \sqrt{\lambda_n}$.

Choose orthonormal eigenvectors $v_1, \ldots, v_n$ corresponding to the eigenvalues $\sqrt{\lambda_1}, \ldots, \sqrt{\lambda_n}$ of C, respectively. Then $v_1, \ldots, v_n$ form an orthonormal basis for $\mathbb{C}^n$. Let $u_i = w_{1i}v_1 + \cdots + w_{ni}v_n$, $i = 1, \ldots, n$. On the one hand,

$$C^2(u_i) = A(u_i) = \lambda_i u_i = w_{1i}\lambda_i v_1 + \cdots + w_{ni}\lambda_i v_n.$$

On the other hand,

$$C^2(u_i) = A(u_i) = w_{1i}\lambda_1 v_1 + \cdots + w_{ni}\lambda_n v_n.$$

The linear independence of $v_1, \ldots, v_n$ implies $w_{ti}\lambda_i = w_{ti}\lambda_t$ for every t. If $w_{ti} \neq 0$, then $\lambda_i = \lambda_t$. Hence, $w_{ti}\sqrt{\lambda_i} = w_{ti}\sqrt{\lambda_t}$ for all w_{ti} and t. Thus,

$$\begin{aligned} C(u_i) &= C(w_{1i}v_1 + \cdots + w_{ni}v_n) \\ &= w_{1i}\sqrt{\lambda_1}v_1 + \cdots + w_{ni}\sqrt{\lambda_n}\, v_n \\ &= w_{1i}\sqrt{\lambda_i}v_1 + \cdots + w_{ni}\sqrt{\lambda_i}\, v_n \\ &= \sqrt{\lambda_i}u_i = B(u_i). \end{aligned}$$

Since $u_1, u_2, \ldots, u_n$ constitute a basis for $\mathbb{C}^n$, we conclude $B = C$.

To see that B is a polynomial of A, let $p(x)$ be a polynomial (by interpolation) such that $p(\lambda_i) = \sqrt{\lambda_i}$, $i = 1, \ldots, n$ (Problem 4, Sect. 5.4). Then $p(A) \geq 0$ and $(p(A))^2 = A$. By the uniqueness of B, we have $p(A) = B$. ∎

The matrix B in Theorem 7.1.4 is called the *square root* of A, denoted by $A^{1/2}$. The theorem states that every positive (semi)definite matrix has a unique square root that is also positive (semi)definite.

Example 7.1.4 Square roots of positive semidefinite matrices.

(a) Let E_n be the $n \times n$ all-ones matrix. Then $E_n^{1/2} = \frac{1}{\sqrt{n}} E_n$.

(b) Let $A = \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}$. A is positive definite. The eigenvalues of A are $3, 1$, and the associated orthonormal eigenvectors are $u_1 = \begin{pmatrix} \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} \end{pmatrix}$ and $u_2 = \begin{pmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{pmatrix}$, respectively. It follows that $A(u_1, u_2) = (Au_1, Au_2) = (3u_1, 1u_2) = (u_1, u_2) \begin{pmatrix} 3 & 0 \\ 0 & 1 \end{pmatrix}$. Let $U = (u_1, u_2)$. Then U is unitary and $A = U \begin{pmatrix} 3 & 0 \\ 0 & 1 \end{pmatrix} U^*$. Taking the square roots of both sides gives $A^{1/2} = U \begin{pmatrix} \sqrt{3} & 0 \\ 0 & 1 \end{pmatrix} U^*$. A computation reveals $A^{1/2} = \frac{1}{2} \begin{pmatrix} 1+\sqrt{3} & 1-\sqrt{3} \\ 1-\sqrt{3} & 1+\sqrt{3} \end{pmatrix}$.

(c) Let $A = \begin{pmatrix} 5 & -4 \\ -4 & 5 \end{pmatrix}$. A is positive definite. $A = P^2 = Q^2 = S^2 = C^T C$, where $P = \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix} \geq 0$, $Q = -P$, $S = \begin{pmatrix} 1 & -2 \\ -2 & 1 \end{pmatrix}$, $C = \frac{1}{\sqrt{5}} \begin{pmatrix} 5 & -4 \\ 0 & 3 \end{pmatrix}$. ∎

Note that A^*A is positive semidefinite for any matrix A and that the eigenvalues of $(A^*A)^{1/2}$ are the singular values of A. We shall further discuss the modulus matrix $|A| = (A^*A)^{1/2}$ in the remaining chapters.

Problems

1. Determine if each of the following matrices is positive semidefinite:
$$\begin{pmatrix} 2 & 2 \\ 2 & 2 \end{pmatrix}, \begin{pmatrix} 2 & 2 \\ 2 & 0 \end{pmatrix}, \begin{pmatrix} 1 & -1 \\ -1 & 2 \end{pmatrix}, \begin{pmatrix} 2 & i \\ -i & 1 \end{pmatrix}, \begin{pmatrix} 2 & i \\ i & 1 \end{pmatrix}.$$

2. If A is a positive semidefinite matrix, show that the matrices $\overline{A}$, A^T, $\mathrm{adj}(A)$, and A^{-1} (if the inverse exists) are also positive semidefinite.

3. If $A \geq 0$, show that $\mathrm{tr}A \geq 0$ and equality holds if and only if $A = 0$.

4. Let $A = (a_{ij}) \in \mathbb{M}_n$ and $\Sigma = \sum_{i,j=1}^n a_{ij}$. If $A \geq 0$, show that $\Sigma \geq 0$.

5. Let $A \in \mathbb{M}_n$ be positive semidefinite. Show that $(\det A)^{1/n} \leq \frac{1}{n} \mathrm{tr}A$.

6. Let $A = \begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix}$. Show that $x^*Ax > 0$ for all nonzero $x \in \mathbb{R}^2$. Is A positive semidefinite? Is it possible that $y^*Ay < 0$ for some $y \in \mathbb{C}^2$?

7. Let $A \in \mathbb{M}_n$. If $S^*AS \geq 0$ for a nonsingular $S \in \mathbb{M}_n$, show that $A \geq 0$.

8. Let $A = \begin{pmatrix} 4 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$ and $B = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix}$.
   (a) Is B similar to A (i.e., $P^{-1}AP = B$ for some nonsingular P)?
   (b) Is B *-congruent to A (i.e., $Q^*AQ = B$ for some nonsingular Q)?
   (c) Is B obtainable from A by elementary row and column operations?

9. For what $t \in \mathbb{R}$ is the $n \times n$ matrix $\begin{pmatrix} 1 & & t \\ & \ddots & \\ t & & 1 \end{pmatrix}$ (in which the diagonal entries are 1 and the off-diagonal entries are t) positive semidefinite?

10. For what x, y, $z \in \mathbb{C}$ are the following matrices positive semidefinite?

$$\begin{pmatrix} 1 & 1 & x \\ 1 & 1 & 1 \\ \bar{x} & 1 & 1 \end{pmatrix}, \quad \begin{pmatrix} 1 & 1 & -1 \\ 1 & 1 & y \\ -1 & \bar{y} & 1 \end{pmatrix}, \quad \begin{pmatrix} 1 & z & 0 \\ \bar{z} & 1 & 1 \\ 0 & 1 & 1 \end{pmatrix}.$$

11. Let $x \in \mathbb{C}$, $u, v \in \mathbb{C}^n$, $\alpha \in [0,1]$. Show that (with assumption $0^0 = 1$)

$$\begin{pmatrix} |x|^{2\alpha} & x \\ \bar{x} & |x|^{2(1-\alpha)} \end{pmatrix} \geq 0, \quad \begin{pmatrix} u^*u & u^*v \\ v^*u & v^*v \end{pmatrix} \geq 0.$$

12. If λ, $\mu \in \mathbb{C}$, show that the following matrices are positive semidefinite:

$$\begin{pmatrix} |\lambda|^2+1 & \lambda+\mu \\ \bar{\lambda}+\bar{\mu} & |\mu|^2+1 \end{pmatrix}, \quad \begin{pmatrix} |\lambda|^2 & \lambda\mu \\ \bar{\lambda}\bar{\mu} & |\mu|^2 \end{pmatrix}, \quad \begin{pmatrix} |\lambda|^2+|\mu|^2 & \lambda+\mu \\ \bar{\lambda}+\bar{\mu} & 2 \end{pmatrix}.$$

13. Show that if A is positive definite, so is every principal submatrix of A. Conclude that the diagonal entries of A are all positive.

14. Let $A = (a_{ij}) > 0$ be $n \times n$. Show that the matrix with (i,j)-entry

$$\frac{a_{ij}}{\sqrt{a_{ii}a_{jj}}}, \quad i,j = 1,2,\ldots,n,$$

is positive definite. What are the diagonal entries of this matrix?

15. Let $A \geq 0$. Show that A can be written as a sum of rank 1 matrices

$$A = \sum_{i=1}^{r} u_i u_i^*,$$

where each u_i is a column vector and $r = \operatorname{rank}(A)$.

16. Let A be an n-square positive semidefinite matrix. Show that

$$x^*Ax = 0 \Leftrightarrow Ax = 0 \Leftrightarrow A^{\frac{1}{2}}x = 0 \Leftrightarrow x^*A^{\frac{1}{2}}x = 0, \text{ where } x \in \mathbb{C}^n.$$

If H is Hermitian and $x^*Hx = 0$ for some $x \neq 0$, must $Hx = 0$?

17. Let A be an n-square positive semidefinite matrix. Show that

$$\lambda_{\min}(A) \leq x^*Ax \leq \lambda_{\max}(A), \text{ for all unit } x \in \mathbb{C}^n.$$

18. If A is an $n \times n$ matrix such that $x^*Ax \geq x^*x$ for every $x \in \mathbb{C}^n$, show that A is nonsingular and that $A \geq I \geq A^{-1} > 0$.

19. Let A be a nonsingular Hermitian matrix. Show that $A \geq A^{-1}$ if and only if all eigenvalues of A lie in $[-1,0) \cup [1,\infty)$.

20. Let A be a positive definite matrix. Show that $\operatorname{tr}(A^{-1}) \geq (\operatorname{tr}A)^{-1}$.

21. Let $A \in \mathbb{M}_n$ be such that $A + A^* \geq 0$. If $\lambda = a + bi$ is an eigenvalue of A, where $a, b \in \mathbb{R}$, show that $a \geq 0$. Is the converse true?

22. Let $A = A^*$. Show that $0 \leq A \leq I \Leftrightarrow$ all eigenvalues of A lie in $[0, 1]$.

23. Let $A = A^*$ have eigenvalues $\lambda_{\max}(A) \geq \cdots \geq \lambda_{\min}(A)$. Show that

    (a) $\lambda_{\max}(A)I \geq A \geq \lambda_{\min}(A)I$.

    (b) $\big(\lambda_{\max}(A)I - A\big)\big(A - \lambda_{\min}(A)I\big) \geq 0$.

24. Show that $A \geq 0$ if and only if $A = Q^*Q$ for some matrix Q with linearly independent rows. How many rows does Q have?

25. Prove Theorem 7.1.3. Show further that if A is a positive definite matrix, then the Cholesky factorization of A is unique.

26. Show that every positive definite matrix is $*$-congruent to its inverse, that is, if $A > 0$, then $A^{-1} = P^*AP$ for some invertible matrix P.

27. Let A be a Hermitian matrix and $a, b \in \mathbb{R}$. If no eigenvalue of A lies in the interval $[a, b]$, show that $A^2 - (a + b)A + abI$ is positive definite.

28. Let A be a Hermitian matrix. Show that all eigenvalues of A lie in the interval $[a, b]$ if and only if $A - aI \geq 0$ and $bI - A \geq 0$. Show also that there exist $\alpha > 0$ and $\beta > 0$ such that $\alpha I + A > 0$ and $I + \beta A > 0$.

29. Let $A \in \mathbb{M}_n$. Write $A = R_1 + iR_2 = H_1 + iH_2$, where R_1, R_2 are real, and H_1, H_2 are Hermitian. Find R_1, R_1, H_1, and H_2 in terms of A. Show that if $A \geq 0$, then $R_1 \geq 0$ and that if $A > 0$, then $R_1 > 0$.

30. Find a matrix $A \in \mathbb{M}_n$ such that all of its principal submatrices of order less than or equal to $n - 1$ are positive semidefinite, but A is not.

31. Find a Hermitian matrix A such that the leading minors are all nonnegative, but A is not positive semidefinite.

32. Let $A \in \mathbb{M}_n$ be a singular Hermitian matrix. If A contains a positive definite principal submatrix of order $n - 1$, show that $A \geq 0$.

33. Let $A \in \mathbb{M}_n$ be a positive definite matrix. Prove $(\mathrm{tr}A)(\mathrm{tr}A^{-1}) \geq n^2$.

34. Show that every normal matrix has a square root that is also normal. Is it unique? Does every complex matrix have a square root?

35. Find the square root of each of the positive semidefinite matrices

$$\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}, \begin{pmatrix} 1 & \frac{1}{2} \\ \frac{1}{2} & 1 \end{pmatrix}, \begin{pmatrix} a & 1 \\ 1 & a^{-1} \end{pmatrix} (a > 0), \begin{pmatrix} 1 & b \\ b & b^2 \end{pmatrix} (0 \leq b \leq 1).$$

36. Let A be a Hermitian matrix and let $k > 0$ be an odd number. Show that there exists a unique Hermitian matrix B such that $A = B^k$. Show further that if $AP = PA$ for some matrix P, then $BP = PB$.

37. Let $A \in \mathbb{M}_n$ be a nonzero Hermitian matrix. Show that
    (a) If $\frac{\mathrm{tr}A}{(\mathrm{tr}A^2)^{1/2}} \geq \sqrt{n-1}$, then $A \geq 0$.
    (b) If $A \geq 0$, then $\frac{\mathrm{tr}A}{(\mathrm{tr}A^2)^{1/2}} \geq 1$.

38. Let $A, B \in \mathbb{M}_n$ and let $A \geq 0$.
    (a) If $AB = BA$, show that $A^{1/2}B = BA^{1/2}$.
    (b) Show that A commutes with its square root $A^{1/2}$.
    (c) Show that $|B|$ commutes with $I - (B^*B)^r$ for all $r \geq 0$.
    (d) Does $|B|$ commute with $I - (BB^*)^r$ for all $r \geq 0$?
    (e) Does B commute with $|B|$? I.e., $B|B| = |B|B$?

39. Let $A \geq 0$. Find a matrix X such that $A^{1/2} = AX = XA$.

40. Let $A > 0$. Show that $(A^{-1})^{1/2} = (A^{1/2})^{-1}$ (denoted by $A^{-1/2}$).

41. Does a contractive matrix really make matrices "smaller"? To be precise, if A is a positive definite matrix and C is a contraction, i.e., $\sigma_{\max}(C) \leq 1$, both of size $n \times n$, is it true that $A \geq C^*AC$?

42. Let $A \geq 0$ and let B be a principal submatrix of A. Show that B is singular (i.e., $\det B = 0$) if and only if the rows (and columns) of A that contain B are linearly dependent.

43. Let A be an $n \times n$ positive definite matrix. Show that
$$(\det A)^{1/n} = \min_{X>0,\, \det X=1} \frac{\mathrm{tr}(AX)}{n},$$
where the minimum is taken over all n-square $X > 0$ with $\det X = 1$.

44. Let A be a Hermitian matrix. Show that neither A nor $-A$ is positive semidefinite if and only if at least one of the following statements holds.
    (a) A has a principal minor of even order with negative sign.
    (b) A has two principal minors of odd order with opposite signs.

45. Let $A, B \in \mathbb{M}_n$. Show that $A^*A = B^*B$ if and only if $B = UA$ for some unitary matrix $U \in \mathbb{M}_n$. Can "$B = UA$" be changed to "$B = AU$"?

46. Denote by X_{ii} the diagonal entry in the (i, i) position of square matrix X. Let A be an n-square positive semidefinite matrix. Show that
$$(A_{ii})^2 \leq (A^2)_{ii}, \;\; (A^{1/2})_{ii} \leq (A_{ii})^{1/2}, \;\; (A_{ii})^{-1} \leq (A^{-1})_{ii} \;\; (\text{if } A > 0)$$
and show that each of the above inequalities becomes an equality if and only if the ith row and column consist entirely of zeros except for A_{ii}.

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7.2 A Pair of Positive Semidefinite Matrices

Inequality is one of the central topics in matrix theory. In this section we present some inequalities involving two positive semidefinite matrices.

Let A and B be Hermitian matrices of the same order. Referring to Sect. 3.2, if $A - B \geq 0$, i.e., $A - B$ is a positive semidefinite matrix, we write

$$A \geq B \quad \text{or} \quad B \leq A.$$

The relation $\geq$ on Hermitian matrices is a partial order, referred to as *Loewner* (*partial*) *order*, namely, the following statements hold true.

(1) $A \geq A$ for every Hermitian matrix A.
(2) If $A \geq B$ and $B \geq A$, then $A = B$.
(3) If $A \geq B$ and $B \geq C$, then $A \geq C$.

Let $A = (a_{ij})$ and $B = (b_{ij})$ be $n \times n$ Hermitian matrices. We have

$$A \geq B \;\Rightarrow\; a_{ii} \geq b_{ii},\ i = 1, 2, \ldots, n.$$

Example 7.2.1 The partial order $\geq$ (or $\leq$) for Hermitian matrices.

(a) $\begin{pmatrix} 1 & -1 \\ -1 & 2 \end{pmatrix} \geq 0$.

(b) $\begin{pmatrix} 1 & -1 \\ -1 & 2 \end{pmatrix} \geq \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$. However, $\begin{pmatrix} 1 & -1 \\ -1 & 2 \end{pmatrix} \not\geq \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$.

(c) Neither $\begin{pmatrix} 1 & -1 \\ -1 & 2 \end{pmatrix} \geq \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$ nor $\begin{pmatrix} 1 & -1 \\ -1 & 2 \end{pmatrix} \leq \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$. ∎

It is obvious that $A + B \geq B$ if $A \geq 0$ and $B^* = B$. The assertion that $A \geq 0 \Leftrightarrow X^*AX \geq 0$ in (7.2) (in Sect. 7.1) immediately generalizes to

$$A \geq B \quad \Leftrightarrow \quad X^*AX \geq X^*BX \tag{7.6}$$

for every complex matrix X of appropriate size. Note that if A and B are both positive semidefinite, then $(A^{1/2})^* = A^{1/2}$ and $A^{1/2}BA^{1/2} \geq 0$.

Theorem 7.2.1 *Let A and B be $n \times n$ positive semidefinite matrices. Then*

(1) *The trace of the product AB is less than or equal to the product of the traces* $\operatorname{tr} A$ *and* $\operatorname{tr} B$*, that is,* $\operatorname{tr}(AB) \leq \lambda_{\max}(A) \operatorname{tr}(B) \leq \operatorname{tr}(A) \operatorname{tr}(B)$.
(2) *The eigenvalues of AB are all nonnegative. Furthermore, AB is positive semidefinite if and only if A and B commute, i.e., $AB = BA$.*
(3) *If α and β are the largest eigenvalues of A and B, respectively, then*

$$-0.25\,\alpha\beta I_n \leq AB + BA \leq 2\alpha\beta I_n.$$

Proof. To show (1), note that for any n-square unitary matrix U,

$$\operatorname{tr}(AB) = \operatorname{tr}(U^*ABU) = \operatorname{tr}(U^*AU\,U^*BU).$$

Since A is positive semidefinite, by the spectral decomposition, there exists a unitary $U \in \mathbb{M}_n$ such that $U^*AU = \operatorname{diag}(\lambda_1, \ldots, \lambda_n)$, where λ_i are the eigenvalues of A. Let $C = (c_{ij}) = U^*BU$. Then $\operatorname{tr}(C) = \operatorname{tr}(B)$. We compute

$$\begin{aligned}
\operatorname{tr}(AB) &= \operatorname{tr}(U^*AU \cdot U^*BU) \\
&= \lambda_1 c_{11} + \cdots + \lambda_n c_{nn} \\
&\le \lambda_{\max}(A)(c_{11} + \cdots + c_{nn}) \ \text{(as } A, C \ge 0) \\
&= \lambda_{\max}(A)\operatorname{tr}(C) \le (\lambda_1 + \cdots + \lambda_n)\operatorname{tr}(C) \\
&= \operatorname{tr}(A)\ \operatorname{tr}(C) = \operatorname{tr}(A)\ \operatorname{tr}(B).
\end{aligned}$$

For (2), since XY and YX have the same eigenvalues if X and Y are square matrices of the same size (Theorem 2.4.1, Sect. 2.4), $AB = A^{1/2}(A^{1/2}B)$ has the same (nonnegative) eigenvalues as the positive semidefinite $A^{1/2}BA^{1/2}$.

AB is not positive semidefinite in general, since it need not be Hermitian. However, if A and B commute, then AB is Hermitian, because

$$(AB)^* = B^*A^* = BA = AB$$

is Hermitian, so $AB \ge 0$. Conversely, if $AB \ge 0$, then it is Hermitian, and

$$AB = (AB)^* = B^*A^* = BA.$$

To show (3), we assume that $A \ne 0$ and $B \ne 0$. Dividing both sides of the inequalities by $\alpha\beta$, we see that the statement is equivalent to the case of $\alpha = 1$ and $\beta = 1$, that is, we need to show $-\frac{1}{4}I_n \le AB + BA \le 2I_n$ for $0 \le A, B \le I_n$. Note that $0 \le A \le I_n$ implies $0 \le A^2 \le A \le I_n$. We derive

$$\begin{aligned}
0 &\le \left(A + B - \tfrac{1}{2}I_n\right)^2 \\
&= (A+B)^2 - (A+B) + \tfrac{1}{4}I_n \\
&= A^2 + B^2 + AB + BA - A - B + \tfrac{1}{4}I_n \\
&\le AB + BA + \tfrac{1}{4}I_n,
\end{aligned}$$

therefore, $AB + BA \ge -\frac{1}{4}I_n$. To show that $AB + BA \le 2I_n$, we compute

$$0 \le (A-B)^2 = A^2 + B^2 - AB - BA \le 2I_n - AB - BA. \ \blacksquare$$

Example 7.2.2 (a) Let $A \in \mathbb{M}_n$ be a positive semidefinite matrix. Then for any positive definite matrix $X \in \mathbb{M}_n$, $\operatorname{tr}(A) \le \operatorname{tr}(AX)\operatorname{tr}(X^{-1})$, because

$$\operatorname{tr}(A) = \operatorname{tr}(X^{\frac{1}{2}}AX^{-\frac{1}{2}}) = \operatorname{tr}(X^{\frac{1}{2}}AX^{\frac{1}{2}} \cdot X^{-1}) \le \operatorname{tr}(AX)\operatorname{tr}(X^{-1}).$$

(b) Let $A = \frac{1}{4}\begin{pmatrix} 1 & \sqrt{3} \\ \sqrt{3} & 3 \end{pmatrix}$ and $B = \frac{1}{4}\begin{pmatrix} 1 & -\sqrt{3} \\ -\sqrt{3} & 3 \end{pmatrix}$. A and B are real symmetric. The eigenvalues of A are 0 and 1, so are the eigenvalues of B. Thus, $0 \le A, B \le I_2$. A computation gives $AB + BA = \frac{1}{4}\begin{pmatrix} -1 & 0 \\ 0 & 3 \end{pmatrix}$. $AB + BA$ has a negative eigenvalue $-\frac{1}{4}$ (which is best possible for the lower bound). $\blacksquare$

Example 7.2.3 The product of three positive semidefinite matrices may have non-real eigenvalues. For example, the eigenvalues of ABC are 0, $8+i$ for

$$A = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}, \quad B = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}, \quad C = \begin{pmatrix} 2 & i \\ -i & 1 \end{pmatrix}. \blacksquare$$

The following theorem about simultaneous *-congruence of two positive semidefinite matrices is the main result of this section. We use this result to reduce many inequality problems involving a pair of positive definite matrices to a problem for two diagonal matrices.

Theorem 7.2.2 *Let A and B be n-square positive semidefinite matrices. Then there exists an invertible n-square matrix P such that*

$$P^*AP \quad \text{and} \quad P^*BP$$

are both nonnegative diagonal matrices. In addition, if A is nonsingular, then P can be chosen so that $P^*AP = I_n$ and P^*BP is nonnegative diagonal.

Proof. Let $\operatorname{rank}(A+B) = r$ and let S be a nonsingular matrix such that

$$S^*(A+B)S = \begin{pmatrix} I_r & 0 \\ 0 & 0 \end{pmatrix}.$$

Let $r > 0$, or there is nothing to show. Conformally partition S^*BS as

$$S^*BS = \begin{pmatrix} B_{11} & B_{12} \\ B_{21} & B_{22} \end{pmatrix}, \quad B_{11} \in \mathbb{M}_r.$$

By (7.6), we have $S^*(A+B)S \geq S^*BS$. This implies

$$B_{22} = 0, \quad B_{12} = 0, \quad B_{21} = 0.$$

Because $B_{11} \geq 0$, $T^*B_{11}T$ is diagonal for some unitary $T \in \mathbb{M}_r$. Put

$$P = S \begin{pmatrix} T & 0 \\ 0 & I_{n-r} \end{pmatrix}.$$

Then P^*BP and $P^*AP = P^*(A+B)P - P^*BP$ are both diagonal.

If A is invertible, we write $A = C^*C$ for some invertible C. Because $(C^{-1})^*BC^{-1}$ is positive semidefinite, we have a unitary $U \in \mathbb{M}_n$ such that

$$(C^{-1})^*BC^{-1} = UDU^*,$$

where D is a diagonal matrix with nonnegative diagonal entries.

Let $P = C^{-1}U$. Then $P^*AP = I_n$ and $P^*BP = D$. $\blacksquare$

Some results can be derived by simultaneous *-congruence reduction of positive semidefinite matrices A and B to diagonal matrices, and further to nonnegative numbers, to which some elementary scalar inequalities may apply. The following two theorems are immediate from the preceding theorem by writing $A = P^*D_1P$ and $B = P^*D_2P$, where P is an invertible matrix, and D_1 and D_2 are diagonal matrices with nonnegative entries.

Theorem 7.2.3 *If $A \geq B \geq 0$, then*

(1) $\text{rank}(A) \geq \text{rank}(B)$,

(2) $\det A \geq \det B$, *and*

(3) $B^{-1} \geq A^{-1}$ *if A and B are nonsingular.*

Theorem 7.2.4 *The following determinant and inverse inequalities hold.*

(1) *Let $A, B \in \mathbb{M}_n$ be positive semidefinite, $n \geq 2$. Then*

$$\det(A+B) \geq \det A + \det B. \tag{7.7}$$

Equality occurs if and only if $A+B$ is singular or $A=0$ or $B=0$.

(2) *Let $A, B \in \mathbb{M}_n$ be positive definite, $a+b=1$, $a>0$, $b>0$. Then*

$$(aA+bB)^{-1} \leq aA^{-1} + bB^{-1}. \tag{7.8}$$

Equality occurs if and only if $A=B$. (See Example 7.5.5, Sect. 7.5.)

Proof. (7.7) is equivalent to the inequality $\prod_i (a_i+b_i) \geq \prod_i a_i + \prod_i b_i$, where all a_i's and b_i's are nonnegative real numbers. Equality holds if and only if $a_i = b_i = 0$ for some i, or $a_i = 0$ for all i, or $b_i = 0$ for all i.

(7.8) is by the fact that $(ax+by)^{-1} \leq ax^{-1} + by^{-1}$, $x, y > 0$, because

$$(ax+by)(ax^{-1}+by^{-1}) = a^2 + ab(xy^{-1}+x^{-1}y) + b^2 \geq a^2 + 2ab + b^2 = 1. \blacksquare$$

Example 7.2.4 Let $A \in \mathbb{M}_n$. If $A \geq 0$, then $A + \varepsilon I > 0$ for any $\varepsilon > 0$, and $\det(A+\varepsilon I) \geq \det A + \varepsilon^n$. Moreover, if $\alpha = \frac{\beta+\gamma}{2}$, where $\beta > 0$, $\gamma > 0$, then

$$2(A+\alpha I)^{-1} \leq (A+\beta I)^{-1} + (A+\gamma I)^{-1}. \blacksquare$$

Example 7.2.5 Let A, B, and C be $n \times n$ positive definite matrices. Then

$$\left(\tfrac{A+B}{2}\right)^{-1} \leq \tfrac{A^{-1}+B^{-1}}{2} \quad \text{and} \quad \left(\tfrac{A+B+C}{3}\right)^{-1} \leq \tfrac{A^{-1}+B^{-1}+C^{-1}}{3}.$$

The second one is obtained by (7.8) with $\left(\frac{A+B+C}{3}\right)^{-1} = \left(\frac{2}{3}\frac{A+B}{2} + \frac{1}{3}C\right)^{-1}$. $\blacksquare$

Every positive semidefinite matrix has a unique positive semidefinite square root. The square root is a monotone matrix function for positive semidefinite matrices in the sense that the Loewner partial order is preserved when taking the square root. Precisely, it is stated as follows.

Theorem 7.2.5 *Let A and B be $n \times n$ positive semidefinite matrices. Then*

$$A \geq B \quad \Rightarrow \quad A^{1/2} \geq B^{1/2}.$$

Proof 1. We assume that A is positive definite by continuity (or use $A+\varepsilon I$, $\varepsilon > 0$. Problem 9). Let $C = A^{1/2}$, $D = B^{1/2}$, and $E = C - D$. We have to establish $E \geq 0$. For this purpose, since E is Hermitian, it is sufficient to show that the eigenvalues of E are all nonnegative. Notice that

$$0 \leq A - B = C^2 - D^2 = C^2 - (C-E)^2 = CE + EC - E^2.$$

It follows that $CE + EC \geq 0$ because E is Hermitian and $E^2 \geq 0$.

Let (λ, u) be an eigenpair of E. Then λ is real (as E is Hermitian), and

$$0 \leq u^*(CE + EC)u = 2\lambda(u^*Cu).$$

Since $C > 0$, we have $u^*Cu > 0$, thus $\lambda \geq 0$. Hence, $E \geq 0$, namely, $C \geq D$.

Proof 2. First, $A^{1/2} - B^{1/2}$ is Hermitian, thus has only real eigenvalues. We show that the eigenvalues of $A^{1/2} - B^{1/2}$ are nonnegative. Let (λ, x) be an eigenpair: $(A^{1/2} - B^{1/2})x = \lambda x$. So $B^{1/2}x = A^{1/2}x - \lambda x$, $\lambda \in \mathbb{R}$, and

$$x^*A^{1/2}B^{1/2}x = x^*A^{1/2}(A^{1/2}x - \lambda x) = x^*Ax - \lambda x^*A^{1/2}x \in \mathbb{R}.$$

By the Cauchy-Schwarz inequality (Sect. 1.4, Chap. 1), we derive

$$\begin{aligned} x^*Ax &= (x^*Ax)^{1/2}(x^*Ax)^{1/2} \geq (x^*Ax)^{1/2}(x^*Bx)^{1/2} \\ &\geq |x^*A^{1/2}B^{1/2}x| \geq x^*A^{1/2}B^{1/2}x \\ &= x^*A^{1/2}(A^{1/2}x - \lambda x) = x^*Ax - \lambda x^*A^{1/2}x. \end{aligned}$$

Therefore, $\lambda x^*A^{1/2}x \geq 0$. If $\lambda < 0$, we deduce $x^*A^{1/2}x = 0 \Rightarrow A^{1/2}x = 0$ (as $A^{1/2} \geq 0$) $\Rightarrow x^*Ax = 0 \Rightarrow x^*Bx = 0$ (due to $A \geq B \geq 0$) $\Rightarrow B^{1/2}x = 0$. Hence, $0 = B^{1/2}x = A^{1/2}x - \lambda x = \lambda x$, contradicting $\lambda < 0$ and $x \neq 0$.

Proof 3. If A is positive definite, then by using (7.6) and by multiplying both sides of $B \leq A$ by $A^{-1/2} = (A^{-1/2})^*$, we obtain

$$A^{-1/2}BA^{-1/2} \leq I,$$

rewritten as

$$(B^{1/2}A^{-1/2})^*(B^{1/2}A^{-1/2}) \leq I,$$

which implies

$$\sigma_{\max}(B^{1/2}A^{-1/2}) \leq 1.$$

(Here $\sigma_{\max}(\cdot)$ means the largest singular value.) Thus (by Theorem 4.1.4),

$$\lambda_{\max}(A^{-1/4}B^{1/2}A^{-1/4}) = \lambda_{\max}(B^{1/2}A^{-1/2}) \leq \sigma_{\max}(B^{1/2}A^{-1/2}) \leq 1,$$

where $A^{-1/4}$ is the square root of $A^{-1/2}$. We have (by Problem 22, Sect. 7.1)

$$0 \leq A^{-1/4}B^{1/2}A^{-1/4} \leq I.$$

Multiplying both sides by $A^{1/4}$ (i.e., the square root of $A^{1/2}$), we see that

$$B^{1/2} \leq A^{1/2}.$$

The case for singular A follows from a continuity argument. ∎

Theorem 7.2.6 *Let A and B be positive semidefinite matrices. Then*

$$A \geq B \quad \Rightarrow \quad A^p \geq B^p, \quad 0 \leq p \leq 1.$$

This result, known as the Loewner-Heinz inequality, has various different proofs. It can be shown in a similar way as the above proof 3, in other words, one can prove that if the inequality holds for $s, t \in [0,1]$, then it holds for $(s+t)/2$, concluding that the set of numbers in $[0,1]$ for which the inequality holds is a convex set, so it is $[0,1]$. It can also be shown by integral representation for operator-monotone functions (which is beyond our scope).

Problems

1. Let $A = \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}$, $B = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$, $C = \begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix}$, $D = \begin{pmatrix} 0 & 1 \\ 1 & -2 \end{pmatrix}$, $E = \begin{pmatrix} 0 & 1 \\ 0 & -2 \end{pmatrix}$. Show that $A \geq B \geq D$, but neither $A \geq C$ nor $A \leq C$. Is $D \geq E$?
2. Find an example: $A, B \in \mathbb{M}_2$, $A \geq 0$ and $B \geq 0$, but $AB \ngeq 0$.
3. Let $A, B \geq 0$, both $n \times n$. Is $x^*(AB)x \geq 0$ for all $x \in \mathbb{C}^n$?
4. If $A, B \geq 0$, where $A \in \mathbb{M}_m$, $B \in \mathbb{M}_n$, show that $\begin{pmatrix} A & 0 \\ 0 & B \end{pmatrix} \geq 0$.
5. Let $A, B \geq 0$, both $n \times n$. If $AB = BA$, show that $A^r B^r = B^r A^r$ for all $r > 0$. Conversely, if $A^r B^r = B^r A^r$ for some $r > 0$, show that $AB = BA$. In particular, $AB = BA \Leftrightarrow A^{1/2} B^{1/2} = B^{1/2} A^{1/2}$.
6. (a) Show by an example that $A \geq B \geq 0 \Rightarrow A^2 \geq B^2$ need not be true.

   (b) Show that if $AB = BA$ and $A \geq B \geq 0$, then $A^k \geq B^k$, $k = 1, 2, \ldots$.
7. If $A \in \mathbb{M}_n$ is positive semidefinite and doubly stochastic, show that $A \geq J_n$, where J_n is the $n \times n$ matrix whose entries are all equal to $\frac{1}{n}$.
8. Referring to Theorem 7.2.2, show by example that the matrix P need not be unique. For Hermitian A and B, show that a unitary P exists such that P^*AP and P^*BP are diagonal if and only if $AB = BA$.
9. Show Theorem 7.2.5 for the singular case by a continuity argument, that is, by using $\lim_{\epsilon \to 0^+} (A + \epsilon I)^{1/2} = A^{1/2}$ (entrywise) for $A \geq 0$.
10. Complete the proof of Theorem 7.2.6.
11. Let $A, B \in \mathbb{M}_n$. Show that $A > B$ (i.e., $A - B > 0$) if and only if $X^*AX > X^*BX$ for all $n \times m$ complex matrices X with $\operatorname{rank}(X) = m$.
12. Let A and B be n-square positive semidefinite matrices.
    (a) Show that $A > B \geq 0 \Leftrightarrow \lambda_{\max}(A^{-1}B) < 1$.
    (b) Show that $A \geq B \Rightarrow \operatorname{tr} A \geq \operatorname{tr} B$ with equality only if $A = B$.
    (c) Show that $A > B \geq 0 \Rightarrow \det A > \det B$.
    (d) Find an example: $A \geq B \geq 0$, $\det A = \det B$, but $A \neq B$.

13. Prove each of the following statements.
    (a) Let $A = \left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$, $B = \left(\begin{smallmatrix} 1 & 0 \\ 0 & -1 \end{smallmatrix}\right)$. Then the eigenvalues of AB are $\pm i$.
    (b) Let $A \geq 0$ and $B = B^*$. Then the eigenvalues of AB are real.
    (c) Let $A, B > 0$. Then the eigenvalues of AB are positive.
    (d) Let $A, B, C \geq 0$. Then $\det(ABC) \geq 0$. Is $\operatorname{tr}(ABC) \geq 0$?

14. Let $A \geq 0$ and $B \geq 0$ have the same size. As is known, the eigenvalues of positive semidefinite matrices are the same as the singular values, and the eigenvalues of AB are nonnegative. Are the eigenvalues of AB, where $A \geq 0$ and $B \geq 0$, equal to the singular values of AB?

15. Let $A > 0$ and $B = B^*$, both $n \times n$. Show that there exists a nonsingular matrix P such that $P^*AP = I$, P^*BP is diagonal, and the diagonal entries of P^*BP are the eigenvalues of $A^{-1}B$. Show by example that the assertion is not true in general if $A > 0$ is replaced with $A \geq 0$.

16. Let A, B, and C be three n-square positive semidefinite matrices. Give an example showing that there does not necessarily exist an invertible matrix P such that P^*AP, P^*BP, and P^*CP are all diagonal.

17. Let A and B be n-square real symmetric invertible matrices. If there exists a real n-square invertible matrix P such that P^TAP and P^TBP are both diagonal, show that all the roots of $p(x) = \det(xA - B)$ and $q(x) = \det(xB - A)$ are real. Is the converse true in general?

18. Let $A = \left(\begin{smallmatrix} 1 & 0 \\ 0 & -1 \end{smallmatrix}\right)$ and $B = \left(\begin{smallmatrix} 0 & 1 \\ 1 & 1 \end{smallmatrix}\right)$. Does there exist a real invertible $P \in \mathbb{M}_2$ such that P^TAP and P^TBP are diagonal? Does there exist a complex invertible $P \in \mathbb{M}_2$ such that P^*AP and P^*BP are diagonal?

19. Let $A, B \in \mathbb{M}_{m \times n}$. Show that for any positive definite $X \in \mathbb{M}_m$,
$$|\operatorname{tr}(A^*B)|^2 \leq \operatorname{tr}(A^*XA)\operatorname{tr}(B^*X^{-1}B).$$

20. Let $A = (a_{ij})$ be an $n \times n$ positive semidefinite matrix. Show that
    (a) $\sum_{i=1}^n a_{ii}^2 \leq \operatorname{tr}(A^2) \leq \left(\sum_{i=1}^n a_{ii}\right)^2$,
    (b) $\left(\sum_{i=1}^n a_{ii}\right)^{1/2} \leq \operatorname{tr}(A^{1/2}) \leq \sum_{i=1}^n a_{ii}^{1/2}$, and
    (c) $n^2\left(\sum_{i=1}^n a_{ii}\right)^{-1} \leq \sum_{i=1}^n a_{ii}^{-1} \leq \operatorname{tr}(A^{-1})$ if $A > 0$.

21. Let A and B be $n \times n$ positive semidefinite matrices. Show that
$$\operatorname{tr}(A^{1/2}B^{1/2}) \leq (\operatorname{tr} A)^{1/2}(\operatorname{tr} B)^{1/2}$$
and
$$\left(\operatorname{tr}(A+B)\right)^{1/2} \leq (\operatorname{tr} A)^{1/2} + (\operatorname{tr} B)^{1/2}.$$

22. Let A and B be n-square positive definite matrices. Show that the eigenvalues of $(A^{-1} - B^{-1})(A - B)$ are negative or zero, consequently,

$$\operatorname{tr}\big((A^{-1} - B^{-1})(A - B)\big) \leq 0.$$

23. Let $A > 0$ and $B \geq C \geq 0$, where $A, B, C \in \mathbb{M}_n$. Show that

$$\operatorname{tr}\big((A + B)^{-1}B\big) \geq \operatorname{tr}\big((A + C)^{-1}C\big).$$

24. Let $A = \left(\begin{smallmatrix} 2 & 1 \\ 1 & \frac{9}{8} \end{smallmatrix}\right)$ and $B = \left(\begin{smallmatrix} 1 & 0 \\ 0 & \frac{1}{8} \end{smallmatrix}\right)$. Show that A and B are both positive definite but $AB + BA$ is not, that is, $AB + BA \ngeq 0$. Explain why $A^{1/2}BA^{1/2} \leq \frac{1}{2}(AB + BA)$ need not be true in general for $A, B \geq 0$.

25. Show that for Hermitian matrices A and B of the same order,

$$A^2 + B^2 \geq AB + BA.$$

26. Let $A = \left(\begin{smallmatrix} 1 & 0 \\ 0 & \alpha \end{smallmatrix}\right)$ and $B = \left(\begin{smallmatrix} 1 & \beta \\ \beta & 1 \end{smallmatrix}\right)$. Find a condition on α and β so that

$$A^2 + AB + BA \ngeq 0.$$

27. Let A and B be n-square Hermitian matrices. Show that

$$A > 0,\ AB + BA \geq 0 \quad \Rightarrow \quad B \geq 0$$

and

$$A > 0,\ AB + BA > 0 \quad \Rightarrow \quad B > 0.$$

Show that the condition $A > 0$ cannot be changed to $A \geq 0$.

28. Let $A > 0$ and $B \geq 0$ be of the same order. Show that

$$(A + B)^{-1} \leq A^{-1} \quad (\text{or } B^{-1} \text{ similarly}).$$

29. Let $A > 0$ and $B > 0$ be of the same order. Show that

$$A^{-1} - (A + B)^{-1} - (A + B)^{-1}B(A + B)^{-1} \geq 0.$$

30. Let $A \geq 0$ and $B \geq 0$ be of the same order. Show that

$$2(I + A + B)^{-1} \leq (I + 2A)^{-1} + (I + 2B)^{-1}.$$

31. Let $A \geq 0$ and $B \geq 0$ be of the same order. Prove or disprove that

$$(BAB)^{\alpha} = B^{\alpha}A^{\alpha}B^{\alpha},$$

where $\alpha = 2$, $\frac{1}{2}$, or -1 if A and B are nonsingular.

32. Let $A \geq 0$ and $B \geq 0$ be of the same order. Show that

$$BA^2B \leq I \quad \Rightarrow \quad B^{1/2}AB^{1/2} \leq I.$$

33. Let $A \geq 0$ and $B \geq 0$ be of the same order. Consider the inequalities:

$$\text{(a) } A \geq B \quad \text{(b) } A^2 \geq B^2 \quad \text{(c) } BA^2B \geq B^4 \quad \text{(d) } (BA^2B)^{1/2} \geq B^2.$$

Show that (a) $\not\Rightarrow$ (b), however, it is true that (b) $\Rightarrow$ (c) $\Rightarrow$ (d).

34. Let $A \geq 0$ and $B \geq 0$ be of order n, where $n > 1$. Show that

$$\big(\det(A+B)\big)^{1/n} \geq (\det A)^{1/n} + (\det B)^{1/n}.$$

Equality occurs if and only if $A = 0$ or $B = 0$ or $A + B$ is singular or $B = aA$ for some $a > 0$. Conclude (by rasing to the nth power) that

$$\det(A+B) \geq \det A + \det B$$

with equality if and only if $A=0$ or $B=0$ or $\det(A+B)=0$, and that

$$\det(A+B) \geq \det A$$

with equality if and only if $B=0$ or $\det(A+B)=0$, i.e., $A+B$ is singular.

35. Let $A, B \geq 0$, both $n \times n$. Prove the kth compound matrix inequality

$$C_k(A+B) \geq C_k(A) + C_k(B), \quad k = 1, 2, \ldots, n.$$

36. Let A, B, and C be $n \times n$ positive semidefinite matrices. Show that

$$\det(A+B) + \det(A+C) \leq \det A + \det(A+B+C).$$

[Hint: Use elementary symmetric functions and compound matrices.]

37. Let $A \geq 0$ and $B \geq 0$ be of order n. Show that for any $\lambda, \mu \in \mathbb{C}$,

$$|\det(\lambda A + \mu B)| \leq \det(|\lambda|A + |\mu|B).$$

38. Show by example that $A \geq B \geq 0 \not\Rightarrow (A-B)^{1/2} \geq A^{1/2} - B^{1/2}$.

39. Let $A, B \in \mathbb{M}_n$. If $A \geq B \geq 0$ and A is nonsingular, show that

$$\frac{\operatorname{tr} B}{\operatorname{tr} A} \geq \frac{\det B}{\det A}.$$

40. Let $A \geq 0$ and $B \geq 0$ be of order n. Show that

$$\operatorname{tr}(ABA) \leq \operatorname{tr}(A)\lambda_{\max}(AB)$$

and

$$\operatorname{tr}(A + ABA) \leq \big(n + \operatorname{tr}(AB)\big)\lambda_{\max}(A).$$

41. Show that $A \geq 0$ and $B \geq C \geq 0$ (all $n \times n$) imply the matrix inequality
$$A^{1/2}BA^{1/2} \geq A^{1/2}CA^{1/2} \text{ (but not } B^{1/2}AB^{1/2} \geq C^{1/2}AC^{1/2}).$$

42. Let A, B, and C be $n \times n$ matrices and $C \geq 0$. Find a counterexample:
$$A^*A \leq B^*B \quad \not\Rightarrow \quad A^*CA \leq B^*CB.$$

43. Let A and B be $n \times n$ matrices. Find a counterexample:
$$A^*A \leq B^*B \quad \not\Rightarrow \quad AA^* \leq BB^*.$$

44. Let A and B be $m \times n$ matrices. Recall $|X| = (X^*X)^{1/2}$. Show that
    (a) $A^*A \leq B^*B \Leftrightarrow A = C|B|$ for some contraction $C \in \mathbb{M}_{m\times n}$.
    (b) $A^*A \leq B^*B \Leftrightarrow A = CB$ for some contraction $C \in \mathbb{M}_m$.
    (c) $A^*A \leq B^*B$ but $A \neq BD$ for any matrix D, where $A = \left(\begin{smallmatrix} 1 & 1 \\ 0 & 0 \end{smallmatrix}\right)$ and $B = \left(\begin{smallmatrix} 1 & 1 \\ 1 & 1 \end{smallmatrix}\right)$. Find a contraction $C \in \mathbb{M}_2$ such that $A = CB$.
    (d) $A^*A \leq B^*B \Rightarrow |A| \leq |B|$ (but not $|A^*| \leq |B^*|$).
    (e) If $A = CB$ for some contraction $C \in \mathbb{M}_m$, then $|A| \leq |B|$.
    (f) The converses of (d) and (e) need not be true by verifying that $\sigma_{\max}(AB^{-1}) > 1$ for $A = \left(\begin{smallmatrix} 1 & \varepsilon \\ \varepsilon & \varepsilon \end{smallmatrix}\right)$ and $B = \left(\begin{smallmatrix} 1+\varepsilon & 0 \\ 0 & 2\varepsilon \end{smallmatrix}\right)$, $0 < \varepsilon < 0.1$.

45. Let $A \geq 0$ and $B \geq 0$ be of the same order. For any $t \in [0,1]$ with $\tilde{t} = 1 - t$, assuming that the involved inverses exist, show that
    (a) $(tA + \tilde{t}B)^{-1} \leq tA^{-1} + \tilde{t}B^{-1}$. So, $(\frac{A+B}{2})^{-1} \leq \frac{A^{-1}+B^{-1}}{2}$.
    (b) $(tA + \tilde{t}B)^{-1/2} \leq tA^{-1/2} + \tilde{t}B^{-1/2}$. So, $(\frac{A+B}{2})^{-1/2} \leq \frac{A^{-1/2}+B^{-1/2}}{2}$.
    (c) $(tA + \tilde{t}B)^{1/2} \geq tA^{1/2} + \tilde{t}B^{1/2}$. So, $(\frac{A+B}{2})^{1/2} \geq \frac{A^{1/2}+B^{1/2}}{2}$.
    (d) $(tA + \tilde{t}B)^2 \leq tA^2 + \tilde{t}B^2$. So, $(\frac{A+B}{2})^2 \leq \frac{A^2+B^2}{2}$.
    However, $(\frac{A+B}{2})^3 \leq \frac{A^3+B^3}{2}$ need not be true.
    (e) $\det(tA + \tilde{t}B) \geq (\det A)^t(\det B)^{\tilde{t}}$. So, $\det(\frac{A+B}{2}) \geq \sqrt{\det A \, \det B}$.

46. Let $A_1, \ldots, A_k$ be $n \times n$ positive definite (PD) matrices. Show that
$$k^2(A_1 + \cdots + A_k)^{-1} \leq A_1^{-1} + \cdots + A_k^{-1}, \quad k = 1, 2, \ldots.$$
    [Note: $f(x) = x^{-1}$ is a matrix convex function on PD matrices.]

47. Let A and B be $n \times n$ positive definite matrices. If the eigenvalues of A and B are all contained in the closed interval $[m, M]$, show that
$$\begin{aligned} \frac{2mM}{(m+M)^2}(A+B) &\leq 2(A^{-1}+B^{-1})^{-1} \\ &\leq A^{1/2}(A^{-1/2}BA^{-1/2})^{1/2}A^{1/2} \\ &\leq \frac{1}{2}(A+B). \end{aligned}$$

48. Let A, B, and C be Hermitian matrices of the same order. If $A \geq B$ and if C commutes with both AB and $A + B$, show that C commutes with A and B. What if the condition $A \geq B$ is dropped?

49. Show that the product AB of a positive definite matrix A and a Hermitian matrix B is diagonalizable. What if A is singular?

50. Let $A, B \in \mathbb{M}_{m\times n}$. Show that the following statements are equivalent.

    (a) $\mathrm{Im}(A) \subseteq \mathrm{Im}(B)$.

    (b) $A = BC$ for some matrix $C \in \mathbb{M}_n$.

    (c) $AA^* \leq rBB^*$ for some $r \geq 0$.

51. Let A be an n-square complex matrix. If $A + A^* > 0$, show that every eigenvalue of A has a positive real part. Use this fact to show that $X > 0$ if X is a Hermitian matrix such that for some $Y > 0$

$$XY + YX > 0.$$

52. Let A and B be $n \times n$ Hermitian matrices. If $A^k + B^k = 2I_n$ for some positive integer k, show that $A + B \leq 2I_n$. [Hint: Consider odd k first. For the even case, show that $A^{2p} + B^{2p} \leq 2I \Rightarrow A^p + B^p \leq 2I$.]

53. Let A be an $n \times n$ positive definite matrix and let $r \in [0, 1]$. Show that

$$rA + (1 - r)I \geq A^r$$

    and

$$\langle Au, u\rangle^r \geq \langle A^r u, u\rangle \text{ for all unit vectors } u \in \mathbb{C}^n.$$

54. Let A and C be $n \times n$ matrices. If A is positive semidefinite and C is contractive (i.e., $\sigma_{\max}(C) \leq 1$), show that

$$|\operatorname{tr}(CA)| \leq \operatorname{tr}(A), \quad \operatorname{tr}(C^*CA) \leq \operatorname{tr}(A), \quad \text{and} \quad \operatorname{tr}(C^*AC) \leq \operatorname{tr}(A).$$

55. Let A and C be $n \times n$ matrices. If A is positive semidefinite and C is contractive (i.e., $\sigma_{\max}(C) \leq 1$), show that for any $r \in [0, 1]$,

$$C^*A^rC \leq (C^*AC)^r.$$

56. For a square matrix X, recall $e^X = \sum_{k=0}^{\infty} \frac{X^k}{k!}$ (from Sect. 2.6).

    (a) Find e^X for $X = \frac{1}{2}\begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}$.

    (b) If X is Hermitian, show that $e^X \geq 0$.

    (c) Show by an example that $X \geq Y \geq 0 \not\Rightarrow e^X \geq e^Y$.

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7.3 Partitioned Matrices and The Schur Complement

Large matrices can be divided into smaller, more manageable matrices. Partitioned matrices (or block matrices) play a great role in matrix theory.

(I) If $M \in \mathbb{M}_p$ is a positive semidefinite matrix, then there exists a matrix $N \in \mathbb{M}_p$ such that $M = N^*N$ (Thoerem 7.1.3 (2)). If M is partitioned into blocks, say $M = \left(\begin{smallmatrix} A & B \\ B^* & C \end{smallmatrix}\right) \geq 0$, where $A \in \mathbb{M}_m$ and $C \in \mathbb{M}_n$ with $p = m + n$, we suitably divide N as $N = (X, Y)$, where X consists of the first m columns of N (and Y consists of the remaining columns). We compute

$$\begin{pmatrix} A & B \\ B^* & C \end{pmatrix} = \begin{pmatrix} X^* \\ Y^* \end{pmatrix} (X, Y) = \begin{pmatrix} X^*X & X^*Y \\ Y^*X & Y^*Y \end{pmatrix}. \tag{7.9}$$

It follows that $A = X^*X$, $B = X^*Y$, and $C = Y^*Y$.

(II) Sometimes it is helpful to investigate the block matrix by considering

$$\begin{pmatrix} U^* & 0 \\ 0 & V^* \end{pmatrix} \begin{pmatrix} A & B \\ B^* & C \end{pmatrix} \begin{pmatrix} U & 0 \\ 0 & V \end{pmatrix} = \begin{pmatrix} U^*AU & U^*BV \\ V^*B^*U & V^*CV \end{pmatrix} \tag{7.10}$$

with choices of U and V such that U^*BV is in a desired form (such as SVD).

(III) It is also often useful to express B in terms of A and C in the way that $B = A^{1/2}ZC^{1/2}$ for some contraction $Z \in \mathbb{M}_{m \times n}$ (Theorem 6.3.2, Chap. 6).

These are basic methods and techniques for many problems of partitioned positive semidefinite matrices. We demonstrate the use of these by presenting two trace inequalities for positive semidefinite block matrices.

Theorem 7.3.1 *If* $\left(\begin{smallmatrix} A & B \\ B^* & C \end{smallmatrix}\right) \geq 0$, *where* A, B, *and* C *are* n*-square, then*

$$|\operatorname{tr}(B)|^2 \leq \operatorname{tr}(A)\operatorname{tr}(C) \quad \textit{and} \quad \operatorname{tr}(|B|^2) \leq \operatorname{tr}(A)\operatorname{tr}(C).$$

Proof. By Thoerem 7.1.3 (2), there exist $2n \times n$ matrices X and Y such that $\left(\begin{smallmatrix} A & B \\ B^* & C \end{smallmatrix}\right) = \left(\begin{smallmatrix} X^* \\ Y^* \end{smallmatrix}\right)(X, Y) = \left(\begin{smallmatrix} X^*X & X^*Y \\ Y^*X & Y^*Y \end{smallmatrix}\right)$ with $A = X^*X$, $B = X^*Y$, $C = Y^*Y$.

Recall $\langle P, Q\rangle = \operatorname{tr}(Q^*P)$ (Sect. 1.4). The Cauchy-Schwarz inequality gives

$$|\operatorname{tr}(B)|^2 = |\operatorname{tr}(X^*Y)|^2 = |\langle Y, X\rangle|^2 \leq \langle X, X\rangle\langle Y, Y\rangle = \operatorname{tr}(A)\operatorname{tr}(C).$$

Using the trace cyclic property and applying Theorem 7.2.1 (1), we derive

$$\begin{aligned} \operatorname{tr}(|B|^2) &= \operatorname{tr}(B^*B) = \operatorname{tr}(Y^*XX^*Y) = \operatorname{tr}(XX^*YY^*) \\ &\leq \operatorname{tr}(XX^*)\operatorname{tr}(YY^*) = \operatorname{tr}(X^*X)\operatorname{tr}(Y^*Y) = \operatorname{tr}(A)\operatorname{tr}(C). \end{aligned}$$

Alternatively, if we write $B = A^{1/2}ZC^{1/2}$, where Z is a contraction, then

$$|\operatorname{tr}(B)|^2 = |\langle ZC^{\frac{1}{2}}, A^{\frac{1}{2}}\rangle|^2 \leq \langle ZC^{\frac{1}{2}}, ZC^{\frac{1}{2}}\rangle\langle A^{\frac{1}{2}}, A^{\frac{1}{2}}\rangle \leq \operatorname{tr}(A)\operatorname{tr}(C)$$

(as $\langle ZC^{\frac{1}{2}}, ZC^{\frac{1}{2}}\rangle \leq \langle C^{\frac{1}{2}}, C^{\frac{1}{2}}\rangle$ for $Z^*Z \leq I$; similarly, $\operatorname{tr}(ZCZ^*) \leq \operatorname{tr}(C)$), and

$$\operatorname{tr}(B^*B) = \operatorname{tr}(AZCZ^*) \leq \operatorname{tr}(A)\operatorname{tr}(ZCZ^*) \leq \operatorname{tr}(A)\operatorname{tr}(C). \quad \blacksquare$$

To further study partitioned matrices, we introduce the *Schur complement*. What is a Schur complement good for? It may be used to determine the positive semidefiniteness of a block matrix and to derive a variety of inequalities concerning partitioned positive semidefinite matrices. Utilizing Schur complements (or applying type III elementary operations for partitioned matrices, Sect. 2.1) is an important technique in matrix theory.

Let A be a square matrix partitioned into 2×2 blocks (basic case) as

$$A = \begin{pmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{pmatrix}, \tag{7.11}$$

where A_{11} and A_{22} are square submatrices (of possibly different sizes) of A. If A_{11} is invertible, by block row and column elementary operations, we get

$$\begin{pmatrix} I & 0 \\ -A_{21}A_{11}^{-1} & I \end{pmatrix} A \begin{pmatrix} I & -A_{11}^{-1}A_{12} \\ 0 & I \end{pmatrix} = \begin{pmatrix} A_{11} & 0 \\ 0 & \widetilde{A_{11}} \end{pmatrix}, \tag{7.12}$$

where (the overall tilde of A_{11}, or A/A_{11} as used in some other literature)

$$\widetilde{A_{11}} = A_{22} - A_{21}A_{11}^{-1}A_{12} \tag{7.13}$$

is called the *Schur complement* of A_{11} in A, or the Schur complement of A with respect to A_{11}. In a similar way, if A_{22} is nonsingular, then

$$\widetilde{A_{22}} = A_{11} - A_{12}A_{22}^{-1}A_{21}. \tag{7.14}$$

Note: If $M = \left(\begin{smallmatrix} A & B \\ D & C \end{smallmatrix}\right)$, in which A is square and nonsingular, then the Schur complement of M with respect to A is $\widetilde{A} = M/A = C - DA^{-1}B$. Sometimes it is convenient to label the blocks with indices (or the positions of the blocks) such as A_{11} and A_{ij} especially when there are many blocks. Moreover, the Schur complement may be defined for any nonsingular submatrix.

Theorem 7.3.2 *Let A be a Hermitian matrix partitioned as in* (7.11) *and let A_{11} be nonsingular. Then $A \geq 0$ if and only if $A_{11} > 0$ and $\widetilde{A_{11}} \geq 0$.*

Proof. This is because $P^*AP = \left(\begin{smallmatrix} A_{11} & 0 \\ 0 & \widetilde{A_{11}} \end{smallmatrix}\right)$, where $P = \left(\begin{smallmatrix} I & -A_{11}^{-1}A_{12} \\ 0 & I \end{smallmatrix}\right)$. ∎

Example 7.3.1 Schur complement.

(a) Let $M = \left(\begin{smallmatrix} I & B \\ A & C \end{smallmatrix}\right)$. Then the Schur complement of I in M is $C - AB$.

(b) Let $N = \left(\begin{smallmatrix} 2 & 1 & x \\ 1 & 2 & 1 \\ \bar{x} & 1 & 2 \end{smallmatrix}\right)$, $|x| \leq 1$. Then $N \geq 0$. The Schur complement of the $(1,1)$-entry in N is $\left(\begin{smallmatrix} 2 & 1 \\ 1 & 2 \end{smallmatrix}\right) - \frac{1}{2}\left(\begin{smallmatrix} 1 \\ \bar{x} \end{smallmatrix}\right)(1, x) = \frac{1}{2}\left(\begin{smallmatrix} 3 & 2-x \\ 2-\bar{x} & 4-|x|^2 \end{smallmatrix}\right) \geq 0$.

(c) Let $\left(\begin{smallmatrix} A & u \\ u^* & a \end{smallmatrix}\right) \geq 0$, where $A > 0$, a is a scalar. Then $a - u^*A^{-1}u \geq 0$.

(d) If $A > 0$, then $\left(\begin{smallmatrix} A & I \\ I & A^{-1} \end{smallmatrix}\right) \geq 0$.

(e) For any matrix X, $\left(\begin{smallmatrix} I & X \\ X^* & X^*X \end{smallmatrix}\right) \geq 0$. ∎

Note that if A is positive semidefinite and if A_{11} is nonsingular, then

$$A_{22} \geq \widetilde{A_{11}} \geq 0. \tag{7.15}$$

Theorem 7.3.3 *Let A be a positive definite matrix given as in* (7.11)*. Then*

$$A^{-1} = \begin{pmatrix} \widetilde{A_{22}}^{-1} & X \\ Y & \widetilde{A_{11}}^{-1} \end{pmatrix}, \tag{7.16}$$

where

$$X = -A_{11}^{-1} A_{12} \widetilde{A_{11}}^{-1} = -\widetilde{A_{22}}^{-1} A_{12} A_{22}^{-1}$$

and

$$Y = -A_{22}^{-1} A_{21} \widetilde{A_{22}}^{-1} = -\widetilde{A_{11}}^{-1} A_{21} A_{11}^{-1}.$$

Proof. This follows from Theorem 2.2.4 (Sect. 2.2, Chap. 2) immediately. ∎

Note that $Y = X^*$ in (7.16) as $A^{-1} > 0$ is Hermitian. Moreover, we have

$$A - \begin{pmatrix} \widetilde{A_{22}} & 0 \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} A_{12} A_{22}^{-1} A_{21} & A_{12} \\ A_{21} & A_{22} \end{pmatrix} \geq 0. \tag{7.17}$$

We now turn our attention to the inequalities involving principal submatrices of positive semidefinite matrices. Let A be an $m \times n$ matrix. We have used $A[\alpha|\beta]$ to denote the submatrix of A indexed by α and β (see Sect. 4.4). If $\alpha = \beta$, then $A[\alpha|\beta]$ is abbreviated to $A[\alpha]$, or simply $[A] = A[\alpha]$ if α is immaterial. That is, $[A]$ is the $k \times k$ principal submatrix of A lying in the rows $i_1, \ldots, i_k$ and columns $i_1, \ldots, i_k$, where $\alpha = \{i_1, \ldots, i_k\}$, $1 \leq i_1 < \cdots < i_k \leq \min\{m, n\}$. For an n-square matrix A, $[A] = E_\alpha^T A E_\alpha$, where E_α is the $n \times k$ matrix whose rows $i_1, \ldots, i_k$ form the identity matrix I_k and whose other entries are zeros (Problem 22).

Let A be a positive semidefinite matrix. We are interested in comparing $f([A])$ and $[f(A)]$, where $f(x)$ is the elementary function x^2, $x^{1/2}$, $x^{-1/2}$, or x^{-1}. Here in comparison of $[A]$ and $[B]$, we assume that $[A]$ and $[B]$ are the principal submatrices in the corresponding positions of matrices A and B.

Theorem 7.3.4 *Let $A \geq 0$ and let $[A]$ be a principal submatrix of A. Then*

$$[A^2] \geq [A]^2, \quad [A^{1/2}] \leq [A]^{1/2}, \quad [A^{-1/2}] \geq [A]^{-1/2}, \quad [A^{-1}] \geq [A]^{-1}.$$

In the last two inequalities, the involved inverses are assumed to exist.

Proof. We assume that $[A] = A_{11}$ and A is partitioned as $A = \left(\begin{smallmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{smallmatrix}\right)$. Otherwise we can carry out a similar argument for $P^T A P$, where P is a permutation matrix such that $[A]$ is in the upper-left corner of $P^T A P$.

Partition A^2, $A^{1/2}$, and A^{-1} into blocks conformally with A as

$$A^2 = \begin{pmatrix} E & F \\ F^* & G \end{pmatrix}, \quad A^{1/2} = \begin{pmatrix} P & Q \\ Q^* & R \end{pmatrix}, \quad A^{-1} = \begin{pmatrix} X & Y \\ Y^* & Z \end{pmatrix}.$$

Computing A^2 and noting that $A_{21} = A_{12}^*$, we get the first inequality:

$$[A^2] = E = A_{11}^2 + A_{12}A_{21} \geq A_{11}^2 = [A]^2.$$

This yields $[A]^{1/2} \geq [A^{1/2}]$ by replacing A with $A^{1/2}$ and then taking the square root (Theorem 7.2.5). The third inequality, i.e., $[A^{-1/2}] \geq [A]^{-1/2}$, follows from an application of the last inequality to the second.

We are left to show $[A^{-1}] \geq [A]^{-1}$. By Theorem 2.2.4, we have

$$[A^{-1}] = X = A_{11}^{-1} + A_{11}^{-1}A_{12}\widetilde{A_{11}}^{-1}A_{21}A_{11}^{-1} \geq A_{11}^{-1} = [A]^{-1}. \blacksquare$$

The inequalities in Theorem 7.3.4 may be unified in the form

$$[f(A)] \geq f([A]), \quad \text{where } f(x) = x^2,\ -x^{1/2},\ x^{-1/2},\ x^{-1}.$$

Theorem 7.3.5 *Let A, B, and C be n-square matrices. Then*

$$\begin{pmatrix} A & B \\ B^* & C \end{pmatrix} \geq 0 \quad \Rightarrow \quad \begin{pmatrix} [A] & [B] \\ [B^*] & [C] \end{pmatrix} \geq 0.$$

Proof. This is because a principal submatrix of a positive semidefinite matrix is positive semidefinite. Alternatively, using the fact that $[X] = E_\alpha^T X E_\alpha$, we obtain $\begin{pmatrix} [A] & [B] \\ [B^*] & [C] \end{pmatrix} = \begin{pmatrix} E_\alpha^T A E_\alpha & E_\alpha^T B E_\alpha \\ E_\alpha^T B^* E_\alpha & E_\alpha^T C E_\alpha \end{pmatrix} = \begin{pmatrix} E_\alpha & 0 \\ 0 & E_\alpha \end{pmatrix}^T \begin{pmatrix} A & B \\ B^* & C \end{pmatrix} \begin{pmatrix} E_\alpha & 0 \\ 0 & E_\alpha \end{pmatrix} \geq 0.$

One may use Hadamard product to show this (Problem 29, Sect. 7.5). $\blacksquare$

Observe that if $A \in \mathbb{M}_n$ is positive definite, then for any $B \in \mathbb{M}_{n\times m}$,

$$\begin{pmatrix} A & B \\ B^* & B^*A^{-1}B \end{pmatrix} \geq 0. \tag{7.18}$$

It turns out that $B^*A^{-1}B$ is the "smallest" matrix to make the block matrix in (7.18) positive semidefinite, because if $\begin{pmatrix} A & B \\ B^* & X \end{pmatrix} \geq 0$ for a Hermitian matrix $X \in \mathbb{M}_m$, then $X \geq B^*A^{-1}B$ due to the fact that

$$\begin{pmatrix} I_n & 0 \\ -B^*A^{-1} & I_m \end{pmatrix} \begin{pmatrix} A & B \\ B^* & X \end{pmatrix} \begin{pmatrix} I_n & -A^{-1}B \\ 0 & I_m \end{pmatrix} = \begin{pmatrix} A & 0 \\ 0 & X - B^*A^{-1}B \end{pmatrix}.$$

Using the partitioned matrix in (7.18), we obtain the following result.

Theorem 7.3.6 *If $A \in \mathbb{M}_n$ is positive definite, then for any $B \in \mathbb{M}_n$,*

$$[B^*][A]^{-1}[B] \leq [B^*A^{-1}B].$$

Setting $B = I_n$ gives $[A]^{-1} \leq [A^{-1}]$. Setting $A = I_n$, what is implied?

Example 7.3.2 Denote by X_{ii} the (i,i) diagonal entry of square matrix X. For positive semidefinite matrix A, setting $[A] = A_{ii}$ in Theorem 7.3.4 gives

$$(A_{ii})^2 \leq (A^2)_{ii},\ \ (A^{1/2})_{ii} \leq (A_{ii})^{1/2},\ \ (A_{ii})^{-1} \leq (A^{-1})_{ii}\ \ (\text{if } A > 0)$$

for each i. An inequality above becomes an equality if and only if the ith row and column consist entirely of zeros except A_{ii} (Problem 46, Sect. 7.1). $\blacksquare$

By applying the Schur complement, we derive a matrix identity from which we shall obtain determinant inequalities (in the next section).

Example 7.3.3 Let $A \in \mathbb{M}_n$. Show that for any $B \in \mathbb{M}_n$, the identity holds:

$$\begin{aligned} I + AA^* &= (A+B)(I+B^*B)^{-1}(A+B)^* \\ &\quad + (I-AB^*)(I+BB^*)^{-1}(I-BA^*). \end{aligned} \tag{7.19}$$

Proof 1. We apply the Schur complement to a block matrix. Let

$$X = \begin{pmatrix} I+B^*B & B^*+A^* \\ A+B & I+AA^* \end{pmatrix}.$$

Then the Schur complement of $I+B^*B$ in X is

$$(I+AA^*) - (A+B)(I+B^*B)^{-1}(A+B)^*. \tag{7.20}$$

On the other hand, we write X as a product of two block matrices:

$$X = \begin{pmatrix} I & B^* \\ A & I \end{pmatrix} \begin{pmatrix} I & A^* \\ B & I \end{pmatrix}.$$

By using (7.16), if $I-AB^*$ is invertible (then so is $I-B^*A$), we compute

$$\begin{aligned} X^{-1} &= \begin{pmatrix} I & A^* \\ B & I \end{pmatrix}^{-1} \begin{pmatrix} I & B^* \\ A & I \end{pmatrix}^{-1} \\ &= \begin{pmatrix} (I-A^*B)^{-1} & -(I-A^*B)^{-1}A^* \\ -(I-BA^*)^{-1}B & (I-BA^*)^{-1} \end{pmatrix} \\ &\quad \times \begin{pmatrix} (I-B^*A)^{-1} & -B^*(I-AB^*)^{-1} \\ -A(I-B^*A)^{-1} & (I-AB^*)^{-1} \end{pmatrix}. \end{aligned}$$

Multiplying out the right-hand side and then taking inverse (as in the lower-right corner in (7.16)), we obtain the block in the lower-right corner of X^{-1} :

$$(I-AB^*)(I+BB^*)^{-1}(I-BA^*). \tag{7.21}$$

Equating (7.20) and (7.21) results in (7.19).

The singular case of $I-AB^*$ follows from a (entrywise) continuity argument by replacing A with ϵA such that $I-\epsilon AB^*$ is invertible (because $I-\epsilon AB^*$ is singular for finitely many values of ϵ) and by letting $\epsilon \to 1$.

Proof 2. We present a direct, algebraic proof of (7.19). Noticing that

$$B(I+B^*B) = (I+BB^*)B,$$

we have

$$(I+BB^*)^{-1}B = B(I+B^*B)^{-1} \tag{7.22}$$

and, by taking the conjugate transpose,

$$B^*(I+BB^*)^{-1} = (I+B^*B)^{-1}B^*. \tag{7.23}$$

Furthermore, the identity $I = (I + B^*B)(I + B^*B)^{-1}$ yields

$$I - B^*B(I + B^*B)^{-1} = (I + B^*B)^{-1} \tag{7.24}$$

and $I-(I+B^*B)^{-1}=B^*B(I+B^*B)^{-1}$, which gives, by switching B and B^*,

$$I - (I + BB^*)^{-1} = BB^*(I + BB^*)^{-1}. \tag{7.25}$$

We derive

$$\begin{aligned}
&(I + AA^*) - (I - AB^*)(I + BB^*)^{-1}(I - BA^*) \\
&= AA^* - AB^*(I + BB^*)^{-1}BA^* + AB^*(I + BB^*)^{-1} \\
&\quad + (I + BB^*)^{-1}BA^* + I - (I + BB^*)^{-1} \quad \text{(by expansion)} \\
&= AA^* - AB^*B(I + B^*B)^{-1}A^* + A(I + B^*B)^{-1}B^* \\
&\quad + B(I + B^*B)^{-1}A^* + BB^*(I + BB^*)^{-1} \quad \text{(by (7.22), (7.23), (7.25))} \\
&= A(I + B^*B)^{-1}A^* + A(I + B^*B)^{-1}B^* \\
&\quad + B(I + B^*B)^{-1}A^* + B(I + B^*B)^{-1}B^* \quad \text{(by (7.23), (7.24))} \\
&= (A + B)(I + B^*B)^{-1}(A + B)^* \quad \text{(by factoring).} \ \blacksquare
\end{aligned}$$

We end the section with a useful block positive semidefinite matrix.

Theorem 7.3.7 *Let A be an $m \times n$ matrix with $|A| = (A^*A)^{1/2}$. Then*

$$\begin{pmatrix} |A| & A^* \\ A & |A^*| \end{pmatrix} \geq 0.$$

Proof. Suppose $\operatorname{rank}(A) = r > 0$ (or it is trivial). Let $A = UDV$ be a singular value decomposition of A, where $U \in \mathbb{M}_{m\times r}$ and $V \in \mathbb{M}_{r\times n}$ are partially unitary matrices (i.e., $U^*U = I_r = VV^*$) and $D \in \mathbb{M}_r$ is the diagonal matrix with the positive singular values of A on the diagonal (Problem 39 of Sect. 3.2). Then $|A| = V^*DV$ and $|A^*| = UDU^*$. We have

$$\begin{pmatrix} |A| & A^* \\ A & |A^*| \end{pmatrix} = \begin{pmatrix} V^*DV & V^*DU^* \\ UDV & UDU^* \end{pmatrix} = \begin{pmatrix} V^* & 0 \\ 0 & U \end{pmatrix} \begin{pmatrix} D & D \\ D & D \end{pmatrix} \begin{pmatrix} V & 0 \\ 0 & U^* \end{pmatrix} \geq 0. \ \blacksquare$$

Problems

1. Let $A = \left(\begin{smallmatrix} 2 & 1 & 0 \\ 0 & 5 & 3 \\ 4 & 0 & 6 \end{smallmatrix}\right)$. Find the Schur complements of A with respect to (a) the (1,1)-entry (2) (as a submatrix) and (b) the submatrix $\left(\begin{smallmatrix} 2 & 1 \\ 0 & 5 \end{smallmatrix}\right)$.

2. Let $A \in \mathbb{M}_n$. If $A > 0$, show that $\left(\begin{smallmatrix} A & u \\ u^* & u^*A^{-1}u \end{smallmatrix}\right) \geq 0$ for any $u \in \mathbb{C}^n$.

3. Let $A, B \in \mathbb{M}_n$. If $A \geq B \geq 0$, show that $\left(\begin{smallmatrix} A & B \\ B & A \end{smallmatrix}\right) \geq 0$ and $\left(\begin{smallmatrix} A & B \\ B & B \end{smallmatrix}\right) \geq 0$.

4. Let $X \in \mathbb{M}_n$. If $\begin{pmatrix} I+X & I \\ I & I-X \end{pmatrix} \geq 0$, show that $X = 0$.

5. Let $X \in \mathbb{M}_{m\times n}$. Show that $\begin{pmatrix} X^*X & X^* \\ X & I_m \end{pmatrix} \geq 0$.

6. Show that $\begin{pmatrix} A & A^{1/2}B^{1/2} \\ B^{1/2}A^{1/2} & B \end{pmatrix} \geq 0$ and $\begin{pmatrix} A & A^{1/2} \\ A^{1/2} & A \end{pmatrix} \geq 0$ if $A, B \geq 0$.

7. Let $A \in \mathbb{M}_n$ and let $B \in \mathbb{M}_{n\times m}$. If A is nonsingular, verify that

$$\begin{pmatrix} A & B \\ B^* & B^*A^{-1}B \end{pmatrix} = \begin{pmatrix} I & 0 \\ 0 & B^* \end{pmatrix} \begin{pmatrix} A & I \\ I & A^{-1} \end{pmatrix} \begin{pmatrix} I & 0 \\ 0 & B \end{pmatrix}.$$

8. Let A, B, $X \in \mathbb{M}_n$. If A and B are positive definite, show that

$$\begin{pmatrix} A+B & A \\ A & A+X \end{pmatrix} \geq 0 \;\Leftrightarrow\; X + (A^{-1} + B^{-1})^{-1} \geq 0.$$

9. Let $A \in \mathbb{M}_m$, $B \in \mathbb{M}_n$, $X \in \mathbb{M}_{n\times m}$. If $A > 0$ and $B > 0$, show that

$$A - X^*B^{-1}X > 0 \quad \Leftrightarrow \quad B - XA^{-1}X^* > 0, \;\text{ where } X \in \mathbb{M}_{n\times m}.$$

10. Let H and K be $n \times n$ Hermitian matrices. Show that

$$\begin{pmatrix} H & K \\ K & H \end{pmatrix} \geq 0 \;\;\Leftrightarrow\;\; H \geq \pm K.$$

Is $H \geq \pm K$ equivalent to $H \geq |K|$ (where $|K| = (K^*K)^{1/2}$)?

11. Let H and K be $n \times n$ Hermitian matrices such that $H \geq \pm K$. Show that $H \geq 0$, $\mathrm{tr}H \geq \mathrm{tr}\,|K| \geq |\,\mathrm{tr}K|$, and $\det H \geq \det|K| = |\det K|$.

12. Let $A \in \mathbb{M}_{m\times n}$. Show that $A = |A^*|^{\frac{1}{2}}C|A|^{\frac{1}{2}}$ for some contraction C.

13. Let A be a nonsingular square matrix and let A_{11} be a nonsingular principal submatrix of A. Show that the Schur complement $\widetilde{A_{11}}$ of A relative to A_{11} is nonsingular and show that, by writing $A = (A^{-1})^{-1}$,

$$\widetilde{\widetilde{A_{11}}^{-1}}^{-1} = A_{11} \quad \text{or} \quad \widetilde{\widetilde{A_{11}}^{-1}} = A_{11}^{-1}.$$

14. Let A and B be $n \times n$ positive definite matrices conformally partitioned as in (7.11). Show that

$$4(A_{11} + B_{11})^{-1} \leq \widetilde{A_{22}}^{-1} + \widetilde{B_{22}}^{-1}.$$

15. Let A and B be $n \times n$ positive definite matrices conformally partitioned as in (7.11). If $A \leq B$, show that $\widetilde{A_{11}} \leq \widetilde{B_{11}}$.

16. Let A be an $n \times m$ matrix and let B be an $n \times n$ positive definite matrix. Show that $A = 0$ if $A^*A = 0$ or if $A^*BA = 0$.

17. Let $A \in \mathbb{M}_n$. Denote by A_{ii} the diagonal (i,i)-entry of A, $i = 1, \ldots, n$.
    (a) Show by example that $|A_{ii}| \leq |A|_{ii}$ need not be true.
    (b) Show that $(|A|_{ii})^2 \leq (A^*A)_{ii}$ for every i.
    (c) Show that $|A_{ii}|^2 \leq |A|_{ii}|A^*|_{ii}$ for every i.
    (d) Show that $|A_{ii}|^2 \leq (A^*A)_{ii}$ for every i.
    (e) Is $(A^*A)_{ii} \leq |A|_{ii}|A^*|_{ii}$ for every i?

18. Let $\left(\begin{smallmatrix} A & B \\ B^* & C \end{smallmatrix}\right) \geq 0$, where $A, B, C \in \mathbb{M}_n$. Show that
    (a) $\left(\begin{smallmatrix} \det A & \det B \\ \det B^* & \det C \end{smallmatrix}\right) \geq 0$, i.e., $|\det B|^2 \leq \det A \det C$.
    (b) $\left(\begin{smallmatrix} \operatorname{tr} A & \operatorname{tr} B \\ \operatorname{tr} B^* & \operatorname{tr} C \end{smallmatrix}\right) \geq 0$, i.e., $|\operatorname{tr} B|^2 \leq \operatorname{tr} A \operatorname{tr} C$.
    (c) $\left(\begin{smallmatrix} \operatorname{tr} A & \operatorname{tr}|B| \\ \operatorname{tr}|B^*| & \operatorname{tr} C \end{smallmatrix}\right) \geq 0$, i.e., $(\operatorname{tr}|B|)^2 \leq \operatorname{tr} A \operatorname{tr} C$.
    (d) $\left(\begin{smallmatrix} \operatorname{tr} A & \sqrt{\operatorname{tr}(|B|^2)} \\ \sqrt{\operatorname{tr}(|B|^2)} & \operatorname{tr} C \end{smallmatrix}\right) \geq 0$, i.e., $\operatorname{tr}(B^*B) \leq \operatorname{tr} A \operatorname{tr} C$.
    (e) $\left(\begin{smallmatrix} \sigma_1(A) & \sigma_1(B) \\ \sigma_1(B) & \sigma_1(C) \end{smallmatrix}\right) \geq 0$, i.e., $\sigma_1^2(B) \leq \sigma_1(A)\sigma_1(C)$.
    (f) If A or C is singular, then B is singular.
    (g) $\left(\begin{smallmatrix} B^*AB & |B|^2 \\ |B|^2 & C \end{smallmatrix}\right)$ is positive semidefinite.
    (h) $\left(\begin{smallmatrix} B^*AB & |B|^2 \\ |B^*|^2 & C \end{smallmatrix}\right)$ need not be positive semidefinite.
    (i) Neither $\left(\begin{smallmatrix} A & |B| \\ |B| & C \end{smallmatrix}\right)$ nor $\left(\begin{smallmatrix} A & |B| \\ |B^*| & C \end{smallmatrix}\right)$ need be positive semidefinite.

19. Show that the following 2×2 block matrices are positive semidefinite.
    (a) $\left(\begin{smallmatrix} \sigma_1(A) I_n & A^* \\ A & \sigma_1(A) I_m \end{smallmatrix}\right)$ for any $m \times n$ matrix A.
    (b) $\left(\begin{smallmatrix} A & I \\ I & A^{-1} \end{smallmatrix}\right)$ for any positive definite matrix A.
    (c) $\left(\begin{smallmatrix} A^*A & A^*B \\ B^*A & B^*B \end{smallmatrix}\right)$ for any matrices A and B of the same size.
    (d) $\left(\begin{smallmatrix} I+A^*A & A^*+B^* \\ A+B & I+BB^* \end{smallmatrix}\right)$ for any matrices A and B of the same size.
    (e) $\left(\begin{smallmatrix} \lambda A & A \\ A & \frac{1}{\lambda} A \end{smallmatrix}\right)$ for any $\lambda > 0$ and positive semidefinite matrix A.
    (f) $\left(\begin{smallmatrix} |A^*| & A \\ A^* & |A| \end{smallmatrix}\right)$ for any $m \times n$ matrix A (where $|X| = (X^*X)^{1/2}$).

20. Let $\left(\begin{smallmatrix} A & B \\ B^* & C \end{smallmatrix}\right) \geq 0$, where $A, B, C \in \mathbb{M}_n$ and A is nonsingular. Show that
$$\operatorname{tr}(C) \geq \operatorname{tr}(BB^*A^{-1}) \quad \text{and} \quad \operatorname{tr}(AC) \geq \operatorname{tr}(BAB^*A^{-1}).$$

21. Let $A \in \mathbb{M}_n$, $C \in \mathbb{M}_m$, and $B \in \mathbb{M}_{n \times m}$ be such that $\left(\begin{smallmatrix} A & B \\ B^* & C \end{smallmatrix}\right) \geq 0$. Is it true that $\det(B^*B) \leq \det A \det C$ for $m = n$? For $m < n$? For $m > n$?

22. Let $A \in \mathbb{M}_n$ and let $[A] = A[\alpha]$ be the $k \times k$ principal submatrix of A lying in the intersections of rows $i_1, \ldots, i_k$ and columns $i_1, \ldots, i_k$ of A, where $\alpha = \{i_1, \ldots, i_k\}$, $1 \leq i_1 < \cdots < i_k \leq n$. Show that $[A] = E_\alpha^T A E_\alpha$, where E_α is the $n \times k$ matrix whose rows $i_1, \ldots, i_k$ form the identity matrix I_k and whose other entries are zeros.

23. Let $A \in \mathbb{M}_m$, $B \in \mathbb{M}_{m\times n}$, and $C \in \mathbb{M}_n$ be such that $\left(\begin{smallmatrix} A & B \\ B^* & C \end{smallmatrix}\right) \geq 0$. Show that for any $X \in \mathbb{M}_{m\times p}$ and $Y \in \mathbb{M}_{n\times q}$, $\left(\begin{smallmatrix} X^*AX & X^*BY \\ Y^*B^*X & Y^*CY \end{smallmatrix}\right) \geq 0$ and $\left(\begin{smallmatrix} [X^*AX] & [X^*BY] \\ [Y^*B^*X] & [Y^*CY] \end{smallmatrix}\right) \geq 0$ (where $[X]$ is defined as in Problem 22).

24. Let A and B be $n \times n$ positive definite matrices. Show that

    (a) $[(A+B)^{-1}] \leq [A^{-1}] + [B^{-1}]$,
    (b) $[A+B]^{-1} \leq [A]^{-1} + [B^{-1}]$,
    (c) $[A+B]^{-1} \leq [(A+B)^{-1}]$, and
    (d) $[A]^{-1} + [B]^{-1} \leq [A^{-1}] + [B^{-1}]$.

25. Let A and B be square matrices of the same order. Show that

    (a) $[AB][B^*A^*] \leq [ABB^*A^*]$. In particular, $[A^*][A] \leq [A^*A]$.
    (b) $[A][B][B^*][A^*] \leq [A][BB^*][A^*]$.
    (c) $[A][B][B^*][A^*] \not\leq [ABB^*A^*]$ (by examples for (c)–(f)).
    (d) $[AB][B^*A^*] \not\leq [A][BB^*][A^*]$.
    (e) $[A][BB^*][A^*] \not\leq [ABB^*A^*]$.
    (f) $[B^*][A][B] \not\leq [B^*AB]$, where $A \geq 0$.

26. Let $X \in \mathbb{M}_n$ and $|X| = (X^*X)^{1/2}$. Show that $|[X]|$ and $[|X|]$ are incomparable, i.e., neither $|[X]| \leq [|X|]$ nor $|[X]| \geq [|X|]$ need be true.

27. Let $A \in \mathbb{M}_n$. Show that the following inequalities hold:

    (a) $[|A|^2] \geq [|A|]^2$.
    (b) $[|A|^2] \geq |[A]|^2$.
    (c) $[|A|^{1/2}] \leq [|A|]^{1/2}$.
    (d) $[|A|^{-1/2}] \geq [|A|]^{-1/2}$ (if the inverses exist).
    (e) $[|A|^{-1}] \leq [|A|^{-2}]^{1/2}$ (if the inverses exist).

28. Show that $[A^4] \geq [A]^4$ need not true for Hermitian matrices by taking

$$A = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 0 & 1 \\ 1 & 1 & 1 \end{pmatrix}, \quad [A] = A[\alpha], \quad \alpha = \{1, 2\}.$$

    What is wrong with the proof, using $[A^2] \geq [A]^2$ in Theorem 7.3.4,

$$[A^4] = [\left(A^2\right)^2] \geq [A^2]^2 \geq \left([A]^2\right)^2 = [A]^4?$$

29. Let $[X] = X[\alpha]$ be defined as in Problem 22. For any $A \geq 0$, show that

$$\operatorname{rank}[A^k] = \operatorname{rank}[A]^k = \operatorname{rank}[A], \quad k = 1, 2, \ldots.$$

[Hint: $\operatorname{rank}[AB] \leq \operatorname{rank}[A]$ and $\operatorname{rank}[A^2] = \operatorname{rank}[A]$ for $A, B \geq 0$.]

30. Let $A > 0$ and $B \geq 0$ be of the same order. Show that

(a) $\operatorname{tr}(A^{-1}B) \geq \frac{\operatorname{tr} B}{\lambda_{\max}(A)} \geq \frac{\operatorname{tr} B}{\operatorname{tr} A}$.

(b) $\operatorname{tr}\left((I + AB)^{-1}A\right) \geq \frac{\operatorname{tr} A}{1+\lambda_{\max}(AB)}$.

31. Find $\begin{pmatrix} |A| & A^* \\ A & |A^*| \end{pmatrix}$ for $A = (1, i)$ (where $|A| = (A^*A)^{1/2}$ and $i = \sqrt{-1}$).

32. Find $|A|$, $|A^*|$, $\begin{pmatrix} |A| & A^* \\ A & |A^*| \end{pmatrix}$, $\begin{pmatrix} |A| & A \\ A^* & |A^*| \end{pmatrix}$, and $\begin{pmatrix} |A^*| & A \\ A^* & |A| \end{pmatrix}$ for $A = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$. Are all three block matrices positive semidefinite?

33. Show the inequalities in Theorem 7.3.4 and Example 7.3.2 by Theorem 7.3.5 by extracting respectively a principal submatrix and a diagonal entry from each block of the PSD matrices $\begin{pmatrix} I & A \\ A & A^2 \end{pmatrix}$ and $\begin{pmatrix} A & I \\ I & A^{-1} \end{pmatrix}$.

34. Show that $\begin{pmatrix} 1 & 1 & 0 \\ 1 & 2 & 1 \\ 0 & 1 & 1 \end{pmatrix}$ is a positive semidefinite matrix and $\lambda = 1$ is an eigenvalue. [Note: It is possible that a diagonal entry of a PSD matrix is an eigenvalue and the row (column) of the diagonal entry contains other nonzero entries. This diagonal entry cannot be the largest eigenvalue.]

35. Find examples: A, B, and C are n-square matrices such that

(a) $\begin{pmatrix} A & B \\ B^* & C \end{pmatrix} \geq 0$, $\begin{pmatrix} A & B^* \\ B & C \end{pmatrix} \not\geq 0$.

(b) $\begin{pmatrix} A & B \\ B^* & C \end{pmatrix} \geq 0$, $\begin{pmatrix} A^2 & B^2 \\ (B^*)^2 & C^2 \end{pmatrix} \not\geq 0$.

36. Let A, B, and C be n-square matrices. If $AB = BA$, show that

$$\begin{pmatrix} A & B \\ B^* & C \end{pmatrix} \geq 0 \quad \Rightarrow \quad A^{1/2}CA^{1/2} \geq B^*B.$$

37. Let A, B, and C be n-square matrices. Show that

$$\begin{pmatrix} A & B \\ B^* & C \end{pmatrix} \geq 0 \quad \Rightarrow \quad A \pm (B + B^*) + C \geq 0.$$

In particular, if $\begin{pmatrix} a & b \\ \bar{b} & c \end{pmatrix} \geq 0$, where $a, b, c \in \mathbb{C}$, then $a + c \geq 2|\Re(b)|$.

38. Let $A \in \mathbb{M}_n$, $C \in \mathbb{M}_m$, and $B \in \mathbb{M}_{n\times m}$. Show that

$$\begin{pmatrix} A & B \\ B^* & C \end{pmatrix} \geq 0 \quad \Rightarrow \quad \begin{pmatrix} \Sigma(A) & \Sigma(B) \\ \Sigma(B^*) & \Sigma(C) \end{pmatrix} \geq 0,$$

where $\Sigma(X)$ denotes the sum of all entries of matrix X.

39. Let $A \in \mathbb{M}_n$ be a positive definite matrix. Show that for any $B \in \mathbb{M}_n$, $A + B + B^* + B^*A^{-1}B \geq 0$. In particular, $I + B + B^* + B^*B \geq 0$.

40. Let A be an $n \times n$ positive semidefinite matrix partitioned as

$$A = \begin{pmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{pmatrix}, \text{ where } A_{11} \text{ and } A_{22} \text{ are square.}$$

Show that

$$\mathrm{Col}(A_{12}) \subseteq \mathrm{Col}(A_{11}), \quad \mathrm{Col}(A_{21}) \subseteq \mathrm{Col}(A_{22}),$$

and (recall that $\mathrm{Col}(\cdot)$ is for column space and $\mathrm{Row}(\cdot)$ is for row space)

$$\mathrm{Row}(A_{12}) \subseteq \mathrm{Row}(A_{22}), \quad \mathrm{Row}(A_{21}) \subseteq \mathrm{Row}(A_{11}).$$

Furthermore, show that

$$\mathrm{rank}(A_{11}, A_{12}) = \mathrm{rank}(A_{11}), \quad \mathrm{rank}(A_{21}, A_{22}) = \mathrm{rank}(A_{22}).$$

Thus, $A_{12} = A_{11}P$, $A_{21} = QA_{11}$ for some matrices P and Q, and

$$\mathrm{rank}(A_{12}) = \mathrm{rank}(A_{21}) \leq \min\{\mathrm{rank}(A_{11}), \mathrm{rank}(A_{22})\}.$$

41. If $M = \left(\begin{smallmatrix} A & X \\ X^* & B \end{smallmatrix}\right) \geq 0$, where $A, B, X \in \mathbb{M}_n$, show that there exist unitary matrices $U, V \in \mathbb{M}_{2n}$ such that $\left(\begin{smallmatrix} A & X \\ X^* & B \end{smallmatrix}\right) = U^* \left(\begin{smallmatrix} A & 0 \\ 0 & 0 \end{smallmatrix}\right) U + V^* \left(\begin{smallmatrix} 0 & 0 \\ 0 & B \end{smallmatrix}\right) V$, revealing $\lambda_1(M) \leq \lambda_1(A) + \lambda_1(B)$. [Hint: Use Example 3.2.6, Sect. 3.2.]

42. Let $u_1, u_2, \ldots, u_n \in \mathbb{C}^m$ be n column vectors of m components. Construct four matrices as follows by the vectors. Show that the first three matrices are positive semidefinite, while the last one is not in general.

$$\begin{pmatrix} u_1^*u_1 & u_1^*u_2 & \ldots & u_1^*u_n \\ u_2^*u_1 & u_2^*u_2 & \ldots & u_2^*u_n \\ \ldots & \ldots & \ldots & \ldots \\ u_n^*u_1 & u_n^*u_2 & \ldots & u_n^*u_n \end{pmatrix}, \quad \begin{pmatrix} u_1^*u_1 & u_2^*u_1 & \ldots & u_n^*u_1 \\ u_1^*u_2 & u_2^*u_2 & \ldots & u_n^*u_2 \\ \ldots & \ldots & \ldots & \ldots \\ u_1^*u_n & u_2^*u_n & \ldots & u_n^*u_n \end{pmatrix},$$

$$\begin{pmatrix} u_1u_1^* & u_1u_2^* & \ldots & u_1u_n^* \\ u_2u_1^* & u_2u_2^* & \ldots & u_2u_n^* \\ \ldots & \ldots & \ldots & \ldots \\ u_nu_1^* & u_nu_2^* & \ldots & u_nu_n^* \end{pmatrix}, \quad \begin{pmatrix} u_1u_1^* & u_2u_1^* & \ldots & u_nu_1^* \\ u_1u_2^* & u_2u_2^* & \ldots & u_nu_2^* \\ \ldots & \ldots & \ldots & \ldots \\ u_1u_n^* & u_2u_n^* & \ldots & u_nu_n^* \end{pmatrix}.$$

43. Let x and y be unit (column) vectors in $\mathbb{C}^n$. Let $\gamma = x^*y$ (and $\gamma^* = y^*x$). Show that M and N are $2n \times 2n$ positive semidefinite matrices, where

$$M = \begin{pmatrix} I + xx^* & \gamma I - yx^* \\ \gamma^* I - xy^* & I + yy^* \end{pmatrix}, \quad N = \begin{pmatrix} I - xx^* & \gamma I - yx^* \\ \gamma^* I - xy^* & I - yy^* \end{pmatrix}.$$

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7.4 The Schur Complement and Determinant Inequalities

In this section we derive some determinant inequalities via Schur complement; they are: the Fischer determinant inequality, the Hadamard determinant inequality, and the Hua determinant inequality.

Let A be a matrix partitioned as $A = \left(\begin{smallmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{smallmatrix}\right)$, where A_{11} is a square submatrix of A. If A_{11} is nonsingular, upon computation, we obtain

$$\begin{pmatrix} I & 0 \\ -A_{21}A_{11}^{-1} & I \end{pmatrix} \begin{pmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{pmatrix} \begin{pmatrix} I & -A_{11}^{-1}A_{12} \\ 0 & I \end{pmatrix} = \begin{pmatrix} A_{11} & 0 \\ 0 & \widetilde{A_{11}} \end{pmatrix}, \quad (7.26)$$

where $\widetilde{A_{11}} = A_{22} - A_{21}A_{11}^{-1}A_{12}$ is called the *Schur complement* of A_{11} in A.

Taking the determinants of both sides of (7.26) reveals the identity

$$\det A = \det A_{11} \det \widetilde{A_{11}}. \quad (7.27)$$

If A is Hermitian and A_{11} is nonsingular, then $A_{21} = A_{12}^*$ and

$$P^*AP = \begin{pmatrix} A_{11} & 0 \\ 0 & \widetilde{A_{11}} \end{pmatrix}, \text{ where } P = \begin{pmatrix} I & -A_{11}^{-1}A_{12} \\ 0 & I \end{pmatrix}. \quad (7.28)$$

If A is positive semidefinite and A_{11} is nonsingular, then $P^*AP \geq 0$ and

$$A_{22} \geq \widetilde{A_{11}} \geq 0, \quad (7.29)$$

which implies $\det A_{22} \geq \det \widetilde{A_{11}}$. The Fischer determinant inequality follows.

Theorem 7.4.1 (Fischer Determinant Inequality) *Let $A = \left(\begin{smallmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{smallmatrix}\right)$ be a positive semidefinite matrix in which A_{11} is a square submatrix. Then*

$$\det A \leq \det A_{11} \det A_{22}. \quad (7.30)$$

Equality occurs if and only if both sides are equal to zero or $A_{12} = 0$. Moreover, if A_{11}, A_{12}, A_{21}, and A_{22} are square and have the same order, then

$$|\det A_{12}|^2 \leq \det A_{11} \det A_{22}. \quad (7.31)$$

Proof. If A is nonsingular, then A_{11} is nonsingular. By (7.28), $\det A = \det A_{11} \det \widetilde{A_{11}} \geq \det A_{11} \det A_{22}$. If A is singular, we replace A with $A + \epsilon I$, $\epsilon > 0$, and obtain the desired inequality by a continuity argument.

If equality holds and both A_{11} and A_{22} are nonsingular, then

$$\det \widetilde{A_{11}} = \det A_{22} = \det(A_{22} - A_{21}A_{11}^{-1}A_{12}).$$

Thus, by Theorem 7.2.4 (1), $A_{21}A_{11}^{-1}A_{12} = 0$, implying $A_{12} = 0$.

For the second inequality, notice that $A_{22} \geq A_{21}A_{11}^{-1}A_{12} \geq 0$. The assertion follows immediately by taking the determinants. ∎

An induction on the order of the matrices gives the next result.

Theorem 7.4.2 (Hadamard Determinant Inequality) *Let A be an $n \times n$ positive semidefinite matrix with diagonal entries $a_{11}, a_{22}, \ldots, a_{nn}$. Then*

$$\det A \le a_{11}a_{22}\cdots a_{nn}.$$

Equality holds if and only if A is a diagonal matrix or some $a_{ii} = 0$.

A direct proof goes as follows. If some $a_{ii} = 0$, then the ith row and the ith column of A consist entirely of zeros, and both sides of the inequality vanish. Assume that all $a_{ii} > 0$ and let $D = \operatorname{diag}(a_{11}^{-1/2}, \ldots, a_{nn}^{-1/2})$. Put $B = DAD$. Then B is a positive semidefinite matrix with diagonal entries all equal to 1. By the arithmetic mean–geometric mean inequality, we have

$$n = \operatorname{tr} B = \sum_{i=1}^{n} \lambda_i(B) \ge n\Big(\prod_{i=1}^{n} \lambda_i(B)\Big)^{\frac{1}{n}} = n(\det B)^{\frac{1}{n}}.$$

This implies $\det B \le 1$. Thus,

$$\det A = \det(D^{-1}BD^{-1}) = \Big(\prod_{i=1}^{n} a_{ii}\Big) \det B \le \prod_{i=1}^{n} a_{ii}.$$

Equality occurs if and only if the eigenvalues of B are identical and $\det B = 1$, equivalently, $B = I_n$ and $A = \operatorname{diag}(a_{11}, \ldots, a_{nn})$ is diagonal.

[Note: The inequality has appeared as Problem 46 of Sect. 3.2.]

Example 7.4.1 Applications of the Hadamard determinant inequality.

(a) If A is doubly stochastic and positive semidefinite, then $\det A \le 1$.

(b) For any $m \times n$ matrix $A = (a_{pq})$, A^*A is positive semidefinite and

$$\det(A^*A) \le \prod_{q=1}^{n} \sum_{p=1}^{m} |a_{pq}|^2. \tag{7.32}$$

(c) Let $u, v, w \in \mathbb{C}^n$ be unit vectors. Unless u, v, w are mutually orthogonal,

$$\begin{vmatrix} 1 & \langle v,u\rangle & \langle w,u\rangle \\ \langle u,v\rangle & 1 & \langle w,v\rangle \\ \langle u,w\rangle & \langle v,w\rangle & 1 \end{vmatrix} < 1.$$

(d) Let $A = B + iC \ge 0$, where B and C are real matrices. Then

$$\det A \le \det B.$$

Proof. Since $A^* = A$, B is real symmetric and C is skew-symmetric. Let Q be such a real orthogonal matrix that Q^TBQ is a diagonal matrix with μ_k's on the diagonal. Because Q^TCQ is skew-symmetric, the diagonal entries of Q^TCQ are all equal to zero. Thus, the diagonal entries of Q^TAQ are μ_k's. By the Hadamard determinant inequality, $\det A = \det(Q^TAQ) \le \prod_k \mu_k = \det B$. (See Problem 15.) ∎

We now turn our attention to *Hua's matrix identity* and *Hua's inequality* for contractive matrices. Recall that a matrix X is contractive if $I - X^*X \geq 0$, and strictly contractive if $I - X^*X > 0$. It is readily seen that the product of contractive matrices (say, A and B) is a contractive matrix because

$$I \geq A^*A \Rightarrow B^*B \geq B^*(A^*A)B \Rightarrow I \geq B^*B \geq (AB)^*(AB).$$

(Or alternatively, by the fact that $\sigma_1(AB) \leq \sigma_1(A)\sigma_1(B) \leq 1$. See Sect. 4.1.)

Theorem 7.4.3 (Hua's Matrix Identity) *Let A and B be $m \times n$ matrices and let $I - A^*A$ be nonsingular. Then the Hua matrix identity holds:*

$$(I-B^*A)(I-A^*A)^{-1}(I-A^*B) = (B-A)^*(I-AA^*)^{-1}(B-A)+(I-B^*B).$$

Proof. To prove the Hua matrix identity, we need a few basic identities about converting $(I - AA^*)^{-1}$ to $(I - A^*A)^{-1}$. Note that $I - AA^*$ is invertible if and only if $I - A^*A$ is invertible. [Note: The I in $I - AA^*$ is $m \times m$, i.e., I_m, while the I in $I - A^*A$ is $n \times n$, i.e., I_n. We simply use I for both identity matrices as no confusion will be caused.] The following identities can be verified directly or derived from Problem 25 of Sect. 2.2:

$$(I - A^*A)^{-1} = I + A^*(I - AA^*)^{-1}A, \tag{7.33}$$

$$A(I - A^*A)^{-1} = (I - AA^*)^{-1}A, \tag{7.34}$$

$$(I - A^*A)^{-1}A^* = A^*(I - AA^*)^{-1}. \tag{7.35}$$

Using the above identities, we expand and compute

$$\begin{aligned}
&(I - B^*A)(I - A^*A)^{-1}(I - A^*B) - (I - B^*B) \\
&\quad = (I - A^*A)^{-1} - B^*A(I - A^*A)^{-1} - (I - A^*A)^{-1}A^*B \\
&\qquad +B^*A(I - A^*A)^{-1}A^*B - (I - B^*B) \\
&\quad = I + A^*(I - AA^*)^{-1}A - B^*(I - AA^*)^{-1}A \\
&\qquad -A^*(I - AA^*)^{-1}B + B^*\big((I - AA^*)^{-1} - I\big)B - (I - B^*B) \\
&\quad = A^*(I - AA^*)^{-1}A - B^*(I - AA^*)^{-1}A \\
&\qquad -A^*(I - AA^*)^{-1}B + B^*(I - AA^*)^{-1}B \\
&\quad = (B - A)^*(I - AA^*)^{-1}(B - A). \ \blacksquare
\end{aligned}$$

It follows that both $I - B^*B$ and $(B - A)^*(I - AA^*)^{-1}(B - A)$ are dominated (in the Loewner order sense) by $(I-B^*A)(I-A^*A)^{-1}(I-A^*B)$.

Theorem 7.4.4 (Hua Determinant Inequality) *Let A and B be $n\times n$ strictly contractive matrices, $n \geq 2$. Then*

$$|\det(I - A^*B)|^2 \geq \det(I - A^*A)\det(I - B^*B) + |\det(A - B)|^2. \tag{7.36}$$

Equality holds if and only if $A = B$.

Proof. The determinant inequality (7.36) follows immediately from the fact that $\det(P+Q) \geq \det P + \det Q$ for $P, Q \geq 0$ with equality if and only if $P+Q$ is singular or $P=0$ or $Q=0$ (Theorem 7.2.4 (1)).

If A and B are strictly contractive, then $(I-B^*A)(I-A^*A)^{-1}(I-A^*B)$ in the Hua matrix identity is nonsingular, so is $I-B^*B$, and moreover, $(B-A)^*(I-AA^*)^{-1}(B-A)=0$ if and only if $B-A=0$, i.e., $A=B$. ∎

Notes: If $n=1$, then (7.36) is an identity for any scalars A and B. If A and B are contractive but not strict, the inequality (7.36) is still valid (by a continuity argument, Problem 26), but the equality may hold for $A \neq B$. For instance, it holds for any B when $A=I$. In addition, if A or B is not contractive, the inequality in (7.36) need not be true (Problem 26).

Let A and B be strictly contractive. Hua's matrix identity reveals

$$(I-B^*A)(I-A^*A)^{-1}(I-A^*B) \geq I-B^*B, \tag{7.37}$$

which is equivalent to (by taking the Schur complement of the block matrix)

$$\begin{pmatrix} (I-A^*A)^{-1} & (I-B^*A)^{-1} \\ (I-A^*B)^{-1} & (I-B^*B)^{-1} \end{pmatrix} \geq 0.$$

Question: For strictly contractive matrices A and B of the same order, is

$$\begin{pmatrix} (I-A^*A)^{-1} & (I-A^*B)^{-1} \\ (I-B^*A)^{-1} & (I-B^*B)^{-1} \end{pmatrix} \geq 0?$$

Note: In general $\left(\begin{smallmatrix} X & Z^* \\ Z & Y \end{smallmatrix}\right) \geq 0 \not\Rightarrow \left(\begin{smallmatrix} X & Z \\ Z^* & Y \end{smallmatrix}\right) \geq 0$ (Problem 35, Sect. 7.3).

The next result is viewed as a reversal of Hua's determinant inequality.

Theorem 7.4.5 *For any n-square matrices A and B with $n \geq 2$,*

$$\det(I+AA^*)\det(I+B^*B) \geq |\det(A+B)|^2 + |\det(I-AB^*)|^2. \tag{7.38}$$

Equality occurs if and only if $A+B=0$ or $AB^*=I$.

Proof. The identity (7.19) in Example 7.3.3 of Sect. 7.3 states

$$\begin{aligned} I+AA^* &= (A+B)(I+B^*B)^{-1}(A+B)^* \\ &\quad + (I-AB^*)(I+BB^*)^{-1}(I-BA^*). \end{aligned} \tag{7.39}$$

Note that $I+AA^*$, $I+B^*B$, and $I+BB^*$ are positive definite. Now apply the fact that $\det(P+Q) \geq \det P + \det Q$ for $P, Q \geq 0$ with equality if and only if $P+Q$ is singular or $P=0$ or $Q=0$ (Theorem 7.2.4 (1)). ∎

As consequences of (7.39), we have the matrix inequalities

$$I+AA^* \geq (A+B)(I+B^*B)^{-1}(A+B)^* \geq 0, \tag{7.40}$$

$$I+AA^* \geq (I-AB^*)(I+BB^*)^{-1}(I-AB^*)^* \geq 0. \tag{7.41}$$

In fact, the inequalities (7.40) and (7.41) are obtainable via Schur complements. For example, we get (7.41) by taking the Schur complement with respect to the $(1,1)$-block of $\begin{pmatrix} I+AA^* & I-AB^* \\ I-BA^* & I+BB^* \end{pmatrix} = \begin{pmatrix} I & A \\ I & -B \end{pmatrix} \begin{pmatrix} I & I \\ A^* & -B^* \end{pmatrix} \geq 0$.

We end the section by presenting a result on a partitioned positive semidefinite matrix: If all blocks of the partitioned matrix are square and have the same size, then the resulting matrix by taking determinant of each block is also positive semidefinite. The analog for trace is given as Problem 29.

Theorem 7.4.6 *Let A be a $kn \times kn$ positive semidefinite matrix partitioned as $A = \begin{pmatrix} A_{11} & \dots & A_{1k} \\ \vdots & \ddots & \vdots \\ A_{k1} & \dots & A_{kk} \end{pmatrix}$, where each A_{ij} is an $n \times n$ matrix, $1 \leq i, j \leq k$. Then the $k \times k$ matrix $D = \begin{pmatrix} \det A_{11} & \dots & \det A_{1k} \\ \vdots & \ddots & \vdots \\ \det A_{k1} & \dots & \det A_{kk} \end{pmatrix}$ is positive semidefinite.*

Proof. By Theorem 7.1.3 (2), we can write $A = R^*R$, where R is a $kn \times kn$ matrix. Partition $R = (R_1, \dots, R_k)$, where R_i is a $kn \times n$ matrix, $i = 1, \dots, k$. Then $A_{ij} = R_i^* R_j$ (of size $n \times n$). Applying the Binet-Cauchy formula (Theorem 4.4.1) to $\det A_{ij} = \det(R_i^* R_j)$ with $\alpha = \{1, \dots, n\}$, we have

$$\det A_{ij} = \det(R_i^* R_j) = \sum_{\beta} \det R_i^*[\alpha|\beta] \det R_j[\beta|\alpha],$$

where $\beta = \{j_1, \dots, j_n\}$, $1 \leq j_1 < \cdots < j_n \leq kn$. It follows that

$$\begin{aligned} D = \big(\det A_{ij}\big) &= \Big(\sum_{\beta} \det R_i^*[\alpha|\beta]\ \det R_j[\beta|\alpha]\Big) \\ &= \sum_{\beta} \Big(\det R_i^*[\alpha|\beta]\ \det R_j[\beta|\alpha]\Big) \\ &= \sum_{\beta} \Big(\det(R_i[\beta|\alpha])^*\ \det R_j[\beta|\alpha]\Big) \\ &= \sum_{\beta} T_\beta^* T_\beta \geq 0, \end{aligned}$$

where T_β is the row vector $\big(\det R_1[\beta|\alpha], \dots, \det R_k[\beta|\alpha]\big)$. ∎

Example 7.4.2 If $\begin{pmatrix} A & B \\ B^* & C \end{pmatrix} \geq 0$, $A, B, C \in \mathbb{M}_n$, then $\begin{pmatrix} \det A & \det B \\ \det B^* & \det C \end{pmatrix} \geq 0$. ∎

It would be interesting to ask: What matrix functions f preserve the positivity of a block matrix? That is, characterize the matrix functions f such that $\begin{pmatrix} f(A) & f(B) \\ f(B^*) & f(C) \end{pmatrix} \geq 0$ whenever $\begin{pmatrix} A & B \\ B^* & C \end{pmatrix} \geq 0$, where $A, B, C \in \mathbb{M}_n$. Determinant, spectral norm, and trace are examples of such functions.

Problems

1. Referring to Theorem 7.4.1, show the reversal Fischer inequality
$$\det \widetilde{A_{11}} \det \widetilde{A_{22}} \leq \det A.$$
2. Show the Hadamard determinant inequality by Theorem 7.1.3 (4).
3. Find the conditions on matrix A such that equality in (7.32) holds.
4. Let A be an $m \times n$ matrix.
   (a) Show that $I_n + A^*A$ and $I_m + AA^*$ are positive definite.
   (b) If $I_n - A^*A$ is invertible, show that $I_m - AA^*$ is invertible.
   (c) If $A^*A \geq B$ for some matrix B, is it true that $AA^* \geq B$?
   (d) Is $\operatorname{tr}(A^*A) = \operatorname{tr}(AA^*)$?
   (e) Is $\det(A^*A) = \det(AA^*)$?
   (f) Is $\det(I_n + A^*A) = \det(I_m + AA^*)$?
5. Show that a square matrix A is unitary if and only if each row (column) vector of A has length 1 and $|\det A| = 1$. [Hint: Use Theorem 7.4.2.]
6. For any nonzero vectors $x_1, x_2, \ldots, x_n$ in an inner product space, let
$$G(x_1, x_2, \ldots, x_n) = \big(\langle x_j, x_i \rangle\big).$$
   (This is the *Gram matrix* of $x_1, x_2, \ldots, x_n$. See Sect. 1.4.) Show that
$$G(x_1, x_2, \ldots, x_n) \geq 0$$
   and that
$$\det G(x_1, \ldots, x_n) \leq \prod_{i=1}^{n} \langle x_i, x_i \rangle.$$
   Equality holds if and only if $x_1, x_2, \ldots, x_n$ are orthogonal. Moreover,
$$\det G(x_1, \ldots, x_n) \leq \det G(x_1, \ldots, x_k) \det G(x_{k+1}, \ldots, x_n),\ 1 \leq k \leq n.$$
7. Let $A = (A_{ij})$ be a $kn \times kn$ partitioned positive semidefinite matrix in which each block A_{ij} is an $n \times n$ matrix. Let $X_1, \ldots, X_k$ be $n \times m$ matrices. Show that the $km \times km$ block matrix $B = (X_i^* A_{ij} X_j)$ is also positive semidefinite. Deduce the special case for $m = 1$.
8. Construct a block matrix and use the Schur complement to show that
$$I + AA^* \geq (A + B)(I + B^*B)^{-1}(A + B)^* \geq 0, \text{ where } A, B \in \mathbb{M}_n.$$
   Show by example that $I + AA^*$ cannot be changed to $I + A^*A$.

9. Let A be an $m \times n$ matrix. Show that
   (a) $A(I_n - A^*A) = (I_m - AA^*)A$
   (b) $A(I_n - A^*A)^{-1} = (I_m - AA^*)^{-1}A$ if $I_n - A^*A$ is invertible.
   (c) $(I_n - A^*A)^{-1} = I_n + A^*(I - AA^*)^{-1}A$ if $I_n - A^*A$ is invertible.

10. Use the block matrix $\begin{pmatrix} I & A \\ A^* & I \end{pmatrix}$ to show the matrix identities
$$(I - A^*A)^{-1} = I + A^*(I - AA^*)^{-1}A,$$
$$I + AA^*(I - AA^*)^{-1} = (I - AA^*)^{-1}.$$

11. Let P be type III row and column elementary operations (Sect. 2.1) on
$$A = \begin{pmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{pmatrix}, \text{ with } P = \begin{pmatrix} I & X \\ 0 & I \end{pmatrix}.$$
Show that the Schur complements of A_{11} in A and in P^TAP are equal.

12. Let A, B, C, and D be $n \times n$ nonsingular matrices. Show that
$$\begin{vmatrix} A^{-1} & B^{-1} \\ C^{-1} & D^{-1} \end{vmatrix} = \frac{(-1)^n}{\det(ACBD)} \begin{vmatrix} A & C \\ B & D \end{vmatrix}.$$

13. Use the Hadamard inequality to show the Fischer inequality. [Hint: If B is a principal submatrix of A, where A is Hermitian, then there exists a unitary matrix U such that U^*BU is diagonal.]

14. Show Theorem 7.4.6 by a compound matrix. [Hint: The $k \times k$ matrix $(\det A_{ij})$ is a principal submatrix of the nth compound matrix of A.]

15. Let $A = B + iC \geq 0$, where B and C are real matrices. Show that
   (a) B is positive semidefinite.
   (b) C is skew-symmetric.
   (c) $\det B \geq \det A$. When does equality occur?
   (d) $\operatorname{rank}(B) \geq \max\{\operatorname{rank}(A), \operatorname{rank}(C)\}$.

16. Find a positive semidefinite matrix A partitioned as $A = \begin{pmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{pmatrix}$, where A_{11} and A_{22} are square (of possibly different orders), such that
$$\det A = \det A_{11} \det A_{22} \quad \text{but} \quad A_{12} = A_{21}^* \neq 0.$$
[Note: If A is positive definite and $A_{12} \neq 0$, then $\det A < \det A_{11} \det A_{22}$.]

17. Let A and B be $m \times n$ matrices. Show that
   (a) $\begin{pmatrix} \det(A^*A) & \det(A^*B) \\ \det(B^*A) & \det(B^*B) \end{pmatrix} \geq 0.$
   (b) $\begin{pmatrix} \operatorname{tr}(A^*A) & \operatorname{tr}(A^*B) \\ \operatorname{tr}(B^*A) & \operatorname{tr}(B^*B) \end{pmatrix} \geq 0.$
   (c) $\det \begin{pmatrix} A^*A & A^*B \\ B^*A & B^*B \end{pmatrix} = 0.$

18. Show that for matrices A, B, C, and D of appropriate sizes,

$$|\det(AC+BD)|^2 \le \det(AA^*+BB^*)\det(C^*C+D^*D).$$

In particular, for any two square matrices X and Y of the same size,

$$|\det(X+Y)|^2 \le \det(I+XX^*)\det(I+Y^*Y);$$

$$|\det(I+XY)|^2 \le \det(I+XX^*)\det(I+Y^*Y).$$

19. Find examples of n-square matrices A and B such that

$$\det(A^*A+B^*B) \ne \det(A^*A+BB^*);$$

$$\det(A^*A+B^*B) \ne \det(AA^*+BB^*).$$

20. Let A and B be $m \times n$ matrices. Show that for any $n \times n$ matrix X,

$$\begin{aligned} AA^*+BB^* &= (B+AX)(I+X^*X)^{-1}(B+AX)^* \\ &\quad + (A-BX^*)(I+XX^*)^{-1}(A-BX^*)^*. \end{aligned}$$

21. Denote $\mathcal{H}(X) = \frac{1}{2}(X^*+X)$ for a square matrix X. For any n-square matrices A and B, explain why $(A-B)^*(A-B) \ge 0$ and show that

$$\mathcal{H}(I-A^*B) \ge \tfrac{1}{2}\big((I-A^*A)+(I-B^*B)\big).$$

22. Let $A, B \in \mathbb{M}_n$ and let $H = \begin{pmatrix} I-A^*A & I-A^*B \\ I-B^*A & I-B^*B \end{pmatrix}$. Show that

$$\det H = (-1)^n|\det(A-B)|^2.$$

23. Let A, B, C, and D be square matrices of the same size. Show that

$$\begin{aligned} &I+D^*C-(I+D^*B)(I+A^*B)^{-1}(I+A^*C) \\ &\quad = (D-A)^*(I+BA^*)^{-1}(C-B) \end{aligned}$$

provided that $I+A^*B$ is nonsingular, by considering the block matrix

$$\begin{pmatrix} I+A^*B & I+A^*C \\ I+D^*B & I+D^*C \end{pmatrix}.$$

24. Let $A_1, \ldots, A_m$ be m positive semidefinite matrices of order n and let $\lambda_1, \ldots, \lambda_m$ be m complex numbers. Show that

$$|\det(\lambda_1 A_1+\cdots+\lambda_m A_m)| \le \det(|\lambda_1|A_1+\cdots+|\lambda_m|A_m).$$

25. Let $A \in \mathbb{M}_n$. Denote by α for any index sequence $\alpha_1 < \alpha_2 < \cdots < \alpha_k$, where $k \le n$ and $\alpha_k \le n$. If A is positive semidefinite, show that

$$\prod_{\alpha} \det A[\alpha|\alpha] \ge (\det A)^{\binom{n-1}{k-1}}.$$

26. Referring to Theorem 7.4.4, the Hua determinant inequality states that for $n \times n$ strictly contractive matrices A and B:

$$|\det(I - A^*B)|^2 \geq \det(I - A^*A)\det(I - B^*B) + |\det(A - B)|^2.$$

(a) Show that the inequality holds for singular contractive A and B.
(b) If $n = 1$, or $A = B$, or A or B is unitary, show that equality holds.
(c) If $A = \begin{pmatrix} \frac{1}{2} & 0 \\ 0 & 1 \end{pmatrix}$ and $B = \begin{pmatrix} 1 & 0 \\ 0 & \frac{1}{2} \end{pmatrix}$, show that equality holds.
(d) Does the inequality hold for non-contractive A or B?
(e) Show by example that the inequality need not be true if A^*B on the left-hand side is changed to AB. Can it be changed to AB^*?

27. Let A and B be $m \times n$ contractive matrices. Show that

$$|\det(I - A^*B)|^2 \geq \det(I - A^*A)\det(I - B^*B).$$

The inequality becomes an equality if and only if $A = B$ (including the case $m = n = 1$) or A^*B has an eigenvalue 1 (in which case both sides are equal to 0). In addition, show that the inequality holds true if A^*B on the left-hand side is replaced with AB.

28. Let $A, B \in \mathbb{M}_n$. For brevity, let $a = \|A\|_{\text{sp}}$ and $b = \|B\|_{\text{sp}}$. Show that

$$|\det(abI - A^*B)|^2 \geq \det(aI - A^*A)\det(bI - B^*B) + |\det(bA - aB)|^2.$$

29. Let A be a $kn \times kn$ positive semidefinite matrix partitioned as $A = (A_{ij})$, in which each block A_{ij} is an $n \times n$ submatrix of A, $1 \leq i, j \leq k$. Show that the $k \times k$ matrix $T = (\text{tr}A_{ij})$ is positive semidefinite.

30. Let A be a $kn \times kn$ positive semidefinite matrix partitioned as $A = (A_{ij})$, in which each block A_{ij} is an $n \times n$ submatrix of A, $1 \leq i, j \leq k$. Show that the matrices $C = \big(C_m(A_{ij})\big)$ and $E = \big(E_m(A_{ij})\big)$ are positive semidefinite, where $C_m(X)$ and $E_m(X)$, $1 \leq m \leq n$, denote the mth compound matrix of $X \in \mathbb{M}_n$ and the mth elementary symmetric function of the eigenvalues of $X \in \mathbb{M}_n$ (Sect. 4.4), respectively. Conclude that if $A = (A_{ij})$ is positive semidefinite, then so is $D = (\det A_{ij})$ (Theorem 7.4.6). [Hint: Use compound matrix properties.]

31. Let $A = (A_{ij})$ be a partitioned positive semidefinite matrix in which each block $A_{ij} \in \mathbb{M}_n$, $i, j = 1, 2, \ldots, m$. (A has m^2 blocks.) Show that

$$\max_{1 \leq i \leq m} \|A_{ii}\| \leq \|A\| \leq \sum_{i=1}^{m} \|A_{ii}\|,$$

where $\|\cdot\|$ is either the Frobenius norm or the spectral norm. Derive

$$\lambda_1(A) \leq \lambda_1(A_{11}) + \cdots + \lambda_1(A_{mm}).$$

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7.5 The Kronecker and Hadamard Products and Inequalities

The Kronecker product $\otimes$ and the Hadamard product $\circ$ of matrices were introduced in Sect. 4.3 and some basic properties were presented there. In this section we are interested in the matrix inequalities concerning the Kronecker product and the Hadamard product of positive semidefinite matrices.

Theorem 7.5.1 (Kronecker) *Let $A \geq 0$ and $B \geq 0$. Then $A \otimes B \geq 0$.*

Proof. By the spectral decomposition (Sect. 3.2), let $A = U^*CU$ and $B = V^*DV$, where C and D are diagonal matrices with nonnegative entries on the diagonals, and U and V are unitary matrices. By Theorem 4.3.2,

$$A \otimes B = (U^*CU) \otimes (V^*DV) = (U^* \otimes V^*)(C \otimes D)(U \otimes V) \geq 0. \blacksquare$$

Note that A and B in the above theorem may have different orders. Moreover, if A and B are positive definite, then $A \otimes B$ is positive definite.

Our next theorem due to Schur on Hadamard product (also called the *Schur product*, i.e., entrywise product) is often used to derive matrix inequalities involving the Hadamard product of positive semidefinite matrices.

Theorem 7.5.2 (Schur) *Let A and B be n-square matrices. Then*

$$A \geq 0,\ B \geq 0 \quad \Rightarrow \quad A \circ B \geq 0$$

and

$$A > 0,\ B > 0 \quad \Rightarrow \quad A \circ B > 0.$$

Proof 1. Since the Hadamard product $A \circ B$ is a principal submatrix of the Kronecker product $A \otimes B$ (Theorem 4.3.3), which is positive semidefinite by the preceding theorem, the positive semidefiniteness of $A \circ B$ follows.

For the positive definite case, it is sufficient to notice that a principal submatrix of a positive definite matrix is also positive definite.

Proof 2. By Theorem 7.1.3, we write $A = U^*U$ and $B = V^*V$. Let u_i and v_i be the ith columns of matrices U and V, respectively, $i = 1, \ldots, n$. Then

$$a_{ij} = u_i^* u_j = \langle u_j, u_i \rangle, \quad b_{ij} = v_i^* v_j = \langle v_j, v_i \rangle$$

for each pair of i and j. It follows that (by Problem 11, Sect. 4.3)

$$A \circ B = (a_{ij} b_{ij}) = \big(\langle u_j, u_i \rangle \langle v_j, v_i \rangle\big) = \big(\langle u_j \otimes v_j, u_i \otimes v_i \rangle\big) \geq 0.$$

Proof 3. Let $A = \sum_{i=1}^n \lambda_i u_i u_i^*$, where λ_i's are the eigenvalues of A (thus nonnegative), and u_i's are orthonormal column vectors. Denote by U_i the

diagonal matrix with the components of u_i on the main diagonal of U_i for $i = 1, \ldots, n$. Note that $(u_i u_i^*) \circ B = U_i B U_i^*$ for $i = 1, \ldots, n$. We compute

$$A \circ B = \Big(\sum_{i=1}^{n} \lambda_i u_i u_i^* \Big) \circ B = \sum_{i=1}^{n} \lambda_i (u_i u_i^*) \circ B = \sum_{i=1}^{n} \lambda_i U_i B U_i^* \geq 0.$$

Proof 4. For $x = (x_1, \ldots, x_n)^T \in \mathbb{C}^n$, let $D_x = \operatorname{diag}(x_1, \ldots, x_n)$. We derive

$$\begin{aligned} x^*(A \circ B)x &= \operatorname{tr}(D_x^* A D_x B^T) \\ &= \operatorname{tr}\left((B^{1/2})^T D_x^* A^{1/2} A^{1/2} D_x (B^{1/2})^T\right) \\ &= \operatorname{tr}\left(A^{1/2} D_x (B^{1/2})^T\right)^* \left(A^{1/2} D_x (B^{1/2})^T\right) \geq 0. \ \blacksquare \end{aligned}$$

Example 7.5.1 Let $A = \begin{pmatrix} 1 & -1 \\ -1 & 2 \end{pmatrix} > 0$, $B = \begin{pmatrix} a & c \\ \bar{c} & b \end{pmatrix} \geq 0$, $C = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 2 & i \\ 1 & -i & 3 \end{pmatrix} \geq 0$. Then $A \circ A = \begin{pmatrix} 1 & 1 \\ 1 & 4 \end{pmatrix} > 0$, $A \otimes A = \begin{pmatrix} 1 & -1 & -1 & 1 \\ -1 & 2 & 1 & -2 \\ -1 & 1 & 2 & -2 \\ 1 & -2 & -2 & 4 \end{pmatrix} > 0$, $A \circ B = \begin{pmatrix} a & -c \\ -\bar{c} & 2b \end{pmatrix} \geq 0$,

$$A \otimes B = \begin{pmatrix} a & c & -a & -c \\ \bar{c} & b & -\bar{c} & -b \\ -a & -c & 2a & 2c \\ -\bar{c} & -b & 2\bar{c} & 2b \end{pmatrix} \geq 0, \ A \otimes C = \begin{pmatrix} C & -C \\ -C & 2C \end{pmatrix} \geq 0. \ \blacksquare$$

Example 7.5.2 Let A, B, $C \in \mathbb{M}_n$ be such that $\begin{pmatrix} A & B^* \\ B & C \end{pmatrix} \geq 0$. Let $\operatorname{diag}(X) = X \circ I$ be the diagonal matrix of the diagonal entries of matrix $X \in \mathbb{M}_n$. Since $\begin{pmatrix} I & I \\ I & I \end{pmatrix} \geq 0$, we have $\begin{pmatrix} A & B^* \\ B & C \end{pmatrix} \circ \begin{pmatrix} I & I \\ I & I \end{pmatrix} = \begin{pmatrix} \operatorname{diag}(A) & \operatorname{diag}(B^*) \\ \operatorname{diag}(B) & \operatorname{diag}(C) \end{pmatrix} \geq 0$. $\blacksquare$

Example 7.5.3 3×3 positive semidefinite matrices and vector inequalities.

Let $a, b, c \in \mathbb{C}$ and $f(x) = |x|$, $\Re(x)$, or x^k $(k = 1, 2, \ldots)$. We show

$$\begin{pmatrix} 1 & a & b \\ \bar{a} & 1 & c \\ \bar{b} & \bar{c} & 1 \end{pmatrix} \geq 0 \quad \Rightarrow \quad \begin{pmatrix} 1 & f(a) & f(b) \\ f(\bar{a}) & 1 & f(c) \\ f(\bar{b}) & f(\bar{c}) & 1 \end{pmatrix} \geq 0.$$

The case $f(x) = |x|$: Without loss of generality, we may assume that a and c are nonnegative scalars (otherwise multiply the first row by $e^{i\theta}$ and the first column by $e^{-i\theta}$, where $e^{i\theta} a \geq 0$; and similarly for c, $e^{i\rho} c \geq 0$).

All three 2×2 principal submatrices are positive semidefinite, for example, $\begin{pmatrix} 1 & |b| \\ |b| & 1 \end{pmatrix} \geq 0$ because $\begin{pmatrix} 1 & b \\ \bar{b} & 1 \end{pmatrix} \geq 0$. Thus, it is sufficient (by Theorem 7.1.2) to show $\det \begin{pmatrix} 1 & |a| & |b| \\ |a| & 1 & |c| \\ |b| & |c| & 1 \end{pmatrix} \geq 0$. This is true because $\begin{pmatrix} 1 & a & b \\ a & 1 & c \\ \bar{b} & c & 1 \end{pmatrix} \geq 0 \Rightarrow 1 + ac(b + \bar{b} \geq a^2 + |b|^2 + c^2 \Rightarrow 1 + 2ac|b| \geq a^2 + |b|^2 + c^2$ as $b + \bar{b} \leq 2|b|$.

The case $f(x) = \Re(x)$: This is immediate from the fact that if $A = B + iC \geq 0$, where B and C are real, then $B \geq 0$ (Problem 15, Sect. 7.4).

The case $f(x) = x^k$: Take the Hadamard product k times.

As an application, let u, v, and w be unit vectors in an inner product space over $\mathbb{C}$. Then the following three matrices are all positive semidefinite:

$$G = \begin{pmatrix} 1 & \langle v,u\rangle & \langle w,u\rangle \\ \langle u,v\rangle & 1 & \langle w,v\rangle \\ \langle u,w\rangle & \langle v,w\rangle & 1 \end{pmatrix} \geq 0,$$

$$A = \begin{pmatrix} 1 & |\langle v,u\rangle| & |\langle w,u\rangle| \\ |\langle u,v\rangle| & 1 & |\langle w,v\rangle| \\ |\langle u,w\rangle| & |\langle v,w\rangle| & 1 \end{pmatrix} \geq 0,$$

$$R = \begin{pmatrix} 1 & \Re\langle v,u\rangle & \Re\langle w,u\rangle \\ \Re\langle u,v\rangle & 1 & \Re\langle w,v\rangle \\ \Re\langle u,w\rangle & \Re\langle v,w\rangle & 1 \end{pmatrix} \geq 0.$$

Since $G \geq 0$, we see that $|\langle u,v\rangle| \leq 1$, which is essentially the same as the Cauchy-Schwarz inequality. Taking the determinant of G, we derive

$$\begin{aligned} |\langle u,v\rangle|^2 + |\langle v,w\rangle|^2 + |\langle w,u\rangle|^2 &\leq 1 + 2\Re\big(\langle u,v\rangle\langle v,w\rangle\langle w,u\rangle\big) \\ &\leq 1 + 2|\langle u,v\rangle\langle v,w\rangle\langle w,u\rangle|. \end{aligned}$$

Similarly, because $R \geq 0$, we obtain

$$\Re^2\langle u,v\rangle + \Re^2\langle v,w\rangle + \Re^2\langle w,u\rangle \leq 1 + 2\Re\langle u,v\rangle\Re\langle v,w\rangle\Re\langle w,u\rangle. \tag{7.42}$$

It is interesting to note that the angle inequality $\Theta_{u,v} \leq \Theta_{u,w} + \Theta_{w,v}$, where $\Theta_{u,v} = \cos^{-1}\big(\Re\langle u,v\rangle\big)$ (Sect. 1.4), can be derived from (7.42). In fact, since $\cos^{-1}$ is strictly decreasing on $[0,\pi]$, it is sufficient to show that

$$\cos\Theta_{u,v} \geq \cos(\Theta_{u,w} + \Theta_{w,v}). \tag{7.43}$$

For brevity, denote by $a = \Re\langle u,v\rangle$, $b = \Re\langle v,w\rangle$, and $c = \Re\langle w,u\rangle$. By using some trigonometric identities, (7.43) is expanded and rewritten as

$$a \geq bc - \sqrt{1-b^2}\,\sqrt{1-c^2}, \tag{7.44}$$

i.e.,

$$\sqrt{1-b^2}\,\sqrt{1-c^2} \geq bc - a. \tag{7.45}$$

If $bc - a < 0$, then (7.45) holds trivially. If $bc - a \geq 0$, by squaring both sides, (7.45) is equivalent to (7.42) which is $1 + 2abc \geq a^2 + b^2 + c^2$. ∎

Next we compare $(A \circ B)^2$ to $A^2 \circ B^2$, and $(A \circ B)^{-1}$ to $A^{-1} \circ B^{-1}$.

Theorem 7.5.3 *Let A and B be $n \times n$ positive semidefinite matrices. Then*

$$A^2 \circ B^2 \geq (A \circ B)^2.$$

Moreover, if A and B are positive definite, then

$$A^{-1} \circ B^{-1} \geq (A \circ B)^{-1}.$$

In particular,

$$A \circ A^{-1} \geq I.$$

Proof. Let a_i and b_i denote respectively the ith columns of matrices A and B (of n columns, $i = 1, \ldots, n$). It is verified by direct computations that

$$(AA^*) \circ (BB^*) = (A \circ B)(A^* \circ B^*) + \sum_{i \neq j} (a_i \circ b_j)(a_i^* \circ b_j^*).$$

It follows that for any matrices A and B of the same size

$$(AA^*) \circ (BB^*) \geq (A \circ B)(A^* \circ B^*). \tag{7.46}$$

In particular, if A and B are positive semidefinite, then

$$A^2 \circ B^2 \geq (A \circ B)^2.$$

If A and B are nonsingular matrices, then, by Theorem 4.3.2,

$$(A \otimes B)^{-1} = A^{-1} \otimes B^{-1}.$$

Noticing that $A \circ B$ and $A^{-1} \circ B^{-1}$ are respectively the principal submatrices of $A \otimes B$ and $A^{-1} \otimes B^{-1}$ in the same position, we have, by Theorem 7.3.4, with $[X]$ representing a principal submatrix of X (indexed, say, by α), that

$$A^{-1} \circ B^{-1} = [(A \otimes B)^{-1}] \geq [A \otimes B]^{-1} = (A \circ B)^{-1}.$$

To show $A \circ A^{-1} \geq I$, we replace B with A^{-1} in the above inequality to get $A^{-1} \circ A \geq (A \circ A^{-1})^{-1}$, which implies $(A^{-1} \circ A)^2 \geq I$. Taking the square roots of both sides reveals $A^{-1} \circ A = A \circ A^{-1} \geq I$. ∎

The inequality $A \circ A^{-1} \geq I$ for $A > 0$ is also obtained by considering

$$\begin{pmatrix} A & I \\ I & A^{-1} \end{pmatrix} \circ \begin{pmatrix} A^{-1} & I \\ I & A \end{pmatrix} = \begin{pmatrix} A \circ A^{-1} & I \\ I & A \circ A^{-1} \end{pmatrix} \geq 0$$

or shown by induction on the order of the matrices as follows. Partition

$$A = \begin{pmatrix} a & \alpha \\ \alpha^* & A_1 \end{pmatrix} \quad \text{and} \quad A^{-1} = \begin{pmatrix} b & \beta \\ \beta^* & B_1 \end{pmatrix}, \quad a, b > 0.$$

By Theorem 7.3.3, $b = (a - \alpha A_1^{-1} \alpha^*)^{-1}$, $A_1 = (B_1 - \frac{1}{b}\beta^* \beta)^{-1}$. By (7.17)

$$A - \begin{pmatrix} \frac{1}{b} & 0 \\ 0 & 0 \end{pmatrix} \geq 0, \quad A^{-1} - \begin{pmatrix} 0 & 0 \\ 0 & A_1^{-1} \end{pmatrix} \geq 0,$$

and by Theorem 7.5.1,

$$\left(A - \begin{pmatrix} \frac{1}{b} & 0 \\ 0 & 0 \end{pmatrix} \right) \circ \left(A^{-1} - \begin{pmatrix} 0 & 0 \\ 0 & A_1^{-1} \end{pmatrix} \right) \geq 0,$$

which yields

$$A \circ A^{-1} \geq \begin{pmatrix} 1 & 0 \\ 0 & A_1 \circ A_1^{-1} \end{pmatrix}.$$

An induction hypothesis on $A_1 \circ A_1^{-1}$ (of smaller order) reveals $A \circ A^{-1} \geq I$.

Theorem 7.5.4 *Let* $\left(\begin{smallmatrix} A & B \\ B^* & C \end{smallmatrix}\right) \geq 0$, *where* A, B, $C \in \mathbb{M}_n$. *Then*

$$|\operatorname{tr}(AC) - \operatorname{tr}(B^*B)| \leq (\operatorname{tr} A)(\operatorname{tr} C) - (\operatorname{tr} B^*)(\operatorname{tr} B). \tag{7.47}$$

Proof. We show $\operatorname{tr}(AC) - \operatorname{tr}(B^*B) \leq (\operatorname{tr} A)(\operatorname{tr} C) - (\operatorname{tr} B^*)(\operatorname{tr} B)$ and leave $\operatorname{tr}(B^*B) - \operatorname{tr}(AC) \leq (\operatorname{tr} A)(\operatorname{tr} C) - (\operatorname{tr} B^*)(\operatorname{tr} B)$ to the reader (Problem 27).

The inequality is unitarily invariant, i.e., (7.47) holds if and only if it holds when A, B, and C are replaced with U^*AU, U^*BU, and U^*CU for any unitary $U \in \mathbb{M}_n$. So, we can assume that A is a diagonal matrix. As usual, x_{ij} denotes the (i,j)-entry of matrix X. Inequality (7.47) is equivalent to

$$\sum_i a_{ii}c_{ii} - \sum_{i,j} |b_{ij}|^2 \leq \sum_i a_{ii} \sum_j c_{jj} - \sum_i \overline{b_{ii}} \sum_i b_{ii},$$

which is the same as

$$\sum_{i \neq j} \overline{b_{ii}}\, b_{jj} - \sum_{i \neq j} |b_{ij}|^2 \leq \sum_{i \neq j} a_{ii}c_{jj}.$$

In fact the following stronger inequality holds:

$$\sum_{i \neq j} \overline{b_{ii}}\, b_{jj} \leq \sum_{i \neq j} a_{ii}c_{jj}. \tag{7.48}$$

(7.48) is immediate from the observation that $\left(\begin{smallmatrix} a_{ii}c_{jj} & b_{ii}\overline{b_{jj}} \\ \overline{b_{ii}}\, b_{jj} & c_{ii}a_{jj} \end{smallmatrix}\right) \geq 0$ because it is a principal submatrix of $\left(\begin{smallmatrix} A & B \\ B^* & C \end{smallmatrix}\right) \otimes \left(\begin{smallmatrix} C & B^* \\ B & A \end{smallmatrix}\right)$. Note that for scalars x, y, and z, if $\left(\begin{smallmatrix} x & y \\ \bar{y} & z \end{smallmatrix}\right) \geq 0$, then $y + \bar{y} \leq x + z$ since $(1,-1)\left(\begin{smallmatrix} x & y \\ \bar{y} & z \end{smallmatrix}\right)\left(\begin{smallmatrix} 1 \\ -1 \end{smallmatrix}\right) \geq 0$. ∎

Note that $\left(\begin{smallmatrix} A\circ C & B\circ B^* \\ B^*\circ B & C\circ A \end{smallmatrix}\right)$ is a principal submatrix of $\left(\begin{smallmatrix} A\otimes C & B\otimes B^* \\ B^*\otimes B & C\otimes A \end{smallmatrix}\right)$.

For $A \geq 0$ and $B \geq 0$, we have $(A^{1/2} \otimes B^{1/2})^2 = A \otimes B$, that is,

$$(A \otimes B)^{\frac{1}{2}} = A^{\frac{1}{2}} \otimes B^{\frac{1}{2}}. \tag{7.49}$$

Example 7.5.4 Let A and B be $m \times n$ and $p \times q$ matrices, respectively. Then

$$|A \otimes B| = |A| \otimes |B|, \tag{7.50}$$

where $|X| = (X^*X)^{1/2}$. Reason: By Theorem 4.3.2 (a), $(A \otimes B)^*(A \otimes B) = (A^*A) \otimes (B^*B)$. Now take the square roots of both sides and use (7.49). ∎

Next, we present some inequalities for matrix sums and the Hadamard products using Schur complements. From Theorem 7.2.4 and Theorem 7.5.3, for any positive definite matrices A and B of the same order, we have

$$(A+B)^{-1} \leq \tfrac{1}{4}(A^{-1} + B^{-1}), \quad (A \circ B)^{-1} \leq A^{-1} \circ B^{-1}. \tag{7.51}$$

These are special cases of the next theorem whose proof uses the fact that

$$\begin{pmatrix} A & X \\ X^* & X^*A^{-1}X \end{pmatrix} \geq 0 \quad \text{provided that } A > 0.$$

Theorem 7.5.5 *Let A, $B \in \mathbb{M}_n$ be positive definite. Then for any C, $D \in \mathbb{M}_{m\times n}$, the following inequalities for sum and Hadamard product hold:*

$$(C+D)(A+B)^{-1}(C+D)^* \le CA^{-1}C^* + DB^{-1}D^*, \tag{7.52}$$

$$(C\circ D)(A\circ B)^{-1}(C\circ D)^* \le (CA^{-1}C^*)\circ(DB^{-1}D^*). \tag{7.53}$$

Proof. Note that $X \ge 0$, $Y \ge 0$, $X+Y \ge 0$, and $X \circ Y \ge 0$, where

$$X = \begin{pmatrix} A & C^* \\ C & CA^{-1}C^* \end{pmatrix}, \quad Y = \begin{pmatrix} B & D^* \\ D & DB^{-1}D^* \end{pmatrix}.$$

(7.52) and (7.53) are immediate by taking the Schur complements of $X+Y$ with respect to $A+B$ and $X\circ Y$ with respect to $A\circ B$, respectively. ∎

An alternative approach to proving (7.53) is to use Theorem 7.3.6 with the observation that $X\circ Y$ is a principal submatrix of $X\otimes Y$.

Example 7.5.5 Let A and B be n-square positive definite matrices.

(a) Setting $C = D = I_n$ in the theorem yields the inequalities in (7.51).

(b) Let a, $b > 0$, $a+b=1$, $C = aI_n$, $D = bI_n$. Then $C+D = I_n$, and

$$(A+B)^{-1} \le a^2A^{-1} + b^2B^{-1} \le aA^{-1} + bB^{-1} \le A^{-1} + B^{-1}.$$

Replacing A by aA and B by bB in the first inequality reveals

$$(aA+bB)^{-1} \le aA^{-1} + bB^{-1}. \quad \text{(See also Theorem 7.2.4 (2).)}$$

(c) Putting $A = B = I_n$ (7.53), we have (the inequality in (7.46))

$$(C\circ D)(C\circ D)^* \le (CC^*)\circ(DD^*).$$

(d) Taking $C = (A,0)^*$ and $D = (0,B)^*$ in (7.52), we obtain

$$\begin{pmatrix} A \\ B \end{pmatrix}(A+B)^{-1}(A,B) \le \begin{pmatrix} A & 0 \\ 0 & B \end{pmatrix}. \ ∎$$

Theorem 7.5.6 *Let A and B be positive definite matrices of the same order partitioned conformally (with submatrices A_{11} and B_{11} being square) as*

$$A = \begin{pmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{pmatrix}, \quad B = \begin{pmatrix} B_{11} & B_{12} \\ B_{21} & B_{22} \end{pmatrix}.$$

Then

$$\widetilde{A_{11}+B_{11}} \ge \widetilde{A_{11}} + \widetilde{B_{11}} \tag{7.54}$$

and

$$\widetilde{A_{11}\circ B_{11}} \ge \widetilde{A_{11}} \circ \widetilde{B_{11}}. \tag{7.55}$$

(Recall that the tilde $\widetilde{X_{11}}$ is for the Schur complement of X_{11} in X.)

Proof. The inequality (7.54) is obtained by taking the Schur complement of $A_{11} + B_{11}$ in $A + B$ and by using (7.52). For (7.55), notice that

$$A_{22} \geq A_{21}A_{11}^{-1}A_{12}, \quad B_{22} \geq B_{21}B_{11}^{-1}B_{12}.$$

We have

$$\begin{aligned} &A_{22} \circ (B_{21}B_{11}^{-1}B_{12}) + B_{22} \circ (A_{21}A_{11}^{-1}A_{12}) \\ &\quad \geq 2\big((A_{21}A_{11}^{-1}A_{12}) \circ (B_{21}B_{11}^{-1}B_{12})\big). \end{aligned}$$

It follows that

$$\begin{aligned} \widetilde{A_{11}} \circ \widetilde{B_{11}} &= (A_{22} - A_{21}A_{11}^{-1}A_{12}) \circ (B_{22} - B_{21}B_{11}^{-1}B_{12}) \\ &\leq A_{22} \circ B_{22} - (A_{21}A_{11}^{-1}A_{12}) \circ (B_{21}B_{11}^{-1}B_{12}). \end{aligned}$$

Applying (7.53) to the right-hand side of the above inequality yields

$$\begin{aligned} &A_{22} \circ B_{22} - (A_{21}A_{11}^{-1}A_{12}) \circ (B_{21}B_{11}^{-1}B_{12}) \\ &\quad \leq A_{22} \circ B_{22} - (A_{21} \circ B_{21})(A_{11} \circ B_{11})^{-1}(A_{12} \circ B_{12}) \\ &\quad = \widetilde{A_{11} \circ B_{11}}. \end{aligned}$$

Thus,

$$\widetilde{A_{11}} \circ \widetilde{B_{11}} \leq \widetilde{A_{11} \circ B_{11}}. \ \blacksquare$$

We end this section by showing Oppenheim's determinant inequality about positive semidefinite matrices. The proof uses Schur complement as a tool. Recall from Sect. 7.4 that for a nonsingular principal submatrix A_{11} of A,

$$\det A = \det A_{11} \det \widetilde{A_{11}}.$$

Theorem 7.5.7 (Oppenheim) *Let A and B be $n \times n$ positive semidefinite matrices with diagonal entries a_{ii} and b_{ii}, $i = 1, \ldots, n$, respectively. Then*

$$\prod_{i=1}^{n} a_{ii}b_{ii} \geq \det(A \circ B) \geq a_{11} \cdots a_{nn} \det B \geq \det A \det B. \tag{7.56}$$

Proof. The first and last inequalities follow immediately from the Hadamard determinant inequality. We show the second inequality by induction on n.

If $n = 1$, (7.56) is trivial. Suppose it holds for matrices of orders $< n$.

Let $A = \begin{pmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{pmatrix}$, $B = \begin{pmatrix} B_{11} & B_{12} \\ B_{21} & B_{22} \end{pmatrix}$. Then $\widetilde{B_{11}} = B_{22} - B_{21}B_{11}^{-1}B_{12}$.

Let $\hat{B} = \begin{pmatrix} B_{11} & B_{12} \\ B_{21} & B_{21}B_{11}^{-1}B_{12} \end{pmatrix}$. Since $A \geq 0, \hat{B} \geq 0$, we have $A \circ \hat{B} \geq 0$. Taking the Schur complement of $A \circ \hat{B}$ with respect to $A_{11} \circ B_{11}$, we obtain

$$A_{22} \circ (B_{22} - \widetilde{B_{11}}) - (A_{21} \circ B_{21})(A_{11} \circ B_{11})^{-1}(A_{12} \circ B_{12}) \geq 0.$$

Thus,

$$A_{22} \circ B_{22} - (A_{21} \circ B_{21})(A_{11} \circ B_{11})^{-1}(A_{12} \circ B_{12}) \geq A_{22} \circ \widetilde{B_{11}} \geq 0.$$

Observe that the left-hand side of the above inequality is the Schur complement of $A \circ B$ relative to $A_{11} \circ B_{11}$, i.e., $\widetilde{A_{11} \circ B_{11}}$. Taking determinants gives

$$\det(\widetilde{A_{11} \circ B_{11}}) \geq \det(A_{22} \circ \widetilde{B_{11}}).$$

Multiply both sides by $\det(A_{11} \circ B_{11})$ to obtain

$$\det(A \circ B) \geq \det(A_{11} \circ B_{11}) \det(A_{22} \circ \widetilde{B_{11}}).$$

Note that $\det B = \det B_{11} \det \widetilde{B_{11}}$. Since A_{11} and A_{22} have smaller orders less than n, the desired inequality $\det(A \circ B) \geq a_{11} \cdots a_{nn}$ follows from induction hypotheses on $\det(A_{11} \circ B_{11})$ and $\det(A_{22} \circ \widetilde{B_{11}})$. ∎

Oppenheim's inequality reduces to the Hadamard inequality $\det A \leq \prod_i a_{ii}$ by taking B to be the identity matrix I_n in the last inequality of (7.56).

Let $\lambda_i(X)$ be the eigenvalues of $n \times n$ matrix X, $i = 1, 2, \ldots, n$. The inequalities in Theorem 7.5.7 are rewritten in terms of eigenvalues as

$$\prod_{i=1}^{n} a_{ii} b_{ii} \geq \prod_{i=1}^{n} \lambda_i(A \circ B) \geq \prod_{i=1}^{n} a_{ii} \lambda_i(B) \geq \prod_{i=1}^{n} \lambda_i(A) \lambda_i(B).$$

We will present many more inequalities of similar type in Chap. 10.

Problems

1. Let A be an $n \times n$ positive definite matrix partitioned as $A = \begin{pmatrix} a & \alpha \\ \alpha^* & A_1 \end{pmatrix}$, where $a \in \mathbb{C}$, $\alpha \in \mathbb{C}^{n-1}$ (in rows), and $A \in \mathbb{M}_{n-1}$.
   (a) Show that $A_1 - \frac{1}{a}\alpha^* a$ is positive definite.
   (b) Find A^{-1} explicitly (in terms of a, α, and A_1).
   (c) Show that $A^{-1} - \begin{pmatrix} 0 & 0 \\ 0 & A_1^{-1} \end{pmatrix}$ is positive semidefinite.
   (d) If the (1,1)-entry of $A \circ A^{-1}$ is 1, show that $\alpha = 0$.
   (e) If $A \circ A^{-1} = I$, show that A is a positive diagonal matrix.

2. Let A and B be $n \times n$ positive semidefinite matrices. Show that

$$A \geq B \;\Leftrightarrow\; A \otimes I \geq B \otimes I \;\Leftrightarrow\; A \otimes A \geq B \otimes B.$$

3. Let A and B be $n \times n$ positive semidefinite matrices. Show that

$$\operatorname{rank}(A \circ B) \leq \operatorname{rank}(A) \operatorname{rank}(B).$$

   Show further that if A is positive definite, then $\operatorname{rank}(A \circ B)$ is equal to the number of nonzero diagonal entries of matrix B.

4. Let A and B be $n \times n$ positive semidefinite matrices. Show that
   (a) $\operatorname{tr}(AB) \le \operatorname{tr}(A \otimes B) = \operatorname{tr}A \operatorname{tr}B \le \frac{1}{4}(\operatorname{tr}A + \operatorname{tr}B)^2$.
   (b) $\operatorname{tr}(A \circ B) \le \frac{1}{2}\operatorname{tr}(A \circ A + B \circ B)$.
   (c) $\operatorname{tr}(A \otimes B) \le \frac{1}{2}\operatorname{tr}(A \otimes A + B \otimes B)$.
   (d) $\det(A \otimes B) \le \frac{1}{2}\big(\det(A \otimes A) + \det(B \otimes B)\big)$.

5. Let A, B, and C be $n \times n$ matrices such that $\left(\begin{smallmatrix} A & B \\ B^* & C \end{smallmatrix}\right) \ge 0$. Show that
$$A + C \ge \pm(B + B^*) \quad \text{and} \quad A \circ C \ge \pm B \circ B^*.$$

6. Let $A > 0$ and let λ be any eigenvalue of $A^{-1} \circ A$. Show that $\lambda \ge 1$.

7. Let A be a positive definite matrix. Show that
$$A^{-1} + A \ge 2I \quad \text{and} \quad A^{-1} \circ A^{-1} \ge (A \circ A)^{-1}.$$

8. Let A, B, C, and D be $n \times n$ positive semidefinite matrices. Show that
$$A \ge B \;\Rightarrow\; A \circ C \ge B \circ C,\;\; A \otimes C \ge B \otimes C,$$
$$A \ge B,\; C \ge D \;\Rightarrow\; A \circ C \ge B \circ D,\;\; A \otimes C \ge B \otimes D.$$

9. Let A and B be $n \times n$ positive definite matrices. Show that
$$(A^{-1} \circ B^{-1})^{-1} \le A \circ B \le (A^2 \circ B^2)^{1/2}.$$

10. Prove or disprove that for $A \ge 0$ and $B \ge 0$ of the same order
$$A^3 \circ B^3 \ge (A \circ B)^3 \quad \text{or} \quad A^{1/2} \circ B^{1/2} \le (A \circ B)^{1/2}.$$

11. Show that for any matrices A and B of the same size
$$(A \circ B)(A^* \circ B^*) \le (\sigma_1^2 I) \circ (AA^*),$$
where $\sigma_1 = \sigma_{\max}(B)$ is the largest singular value of B.

12. Let A be a Hermitian matrix. Show that $A \circ A \le A^2 \circ I$.

13. Let A and B be any matrices of the same size. Show that
$$\begin{pmatrix} (AA^*) \circ I & A \circ B \\ A^* \circ B^* & (B^*B) \circ I \end{pmatrix} \ge 0 \;\text{ and }\; \begin{pmatrix} (AA^*) \circ (BB^*) & A \circ B \\ A^* \circ B^* & I \end{pmatrix} \ge 0.$$

14. Let $A \ge 0$ and $B \ge 0$ have the same order. Let λ be the largest eigenvalue of A and let μ be the largest diagonal entry of B. Show that
$$A \circ B \le \lambda I \circ B \le \lambda\mu I \;\text{ and }\; \begin{pmatrix} \lambda\mu I & A \circ B \\ A \circ B & \lambda\mu I \end{pmatrix} \ge 0.$$

15. Let A, B, C, D, X, Y, U, and V be n-square matrices and let

$$M = \begin{pmatrix} A & B \\ C & D \end{pmatrix},\ N = \begin{pmatrix} X & Y \\ U & V \end{pmatrix},\ M \odot N = \begin{pmatrix} A \otimes X & B \otimes Y \\ C \otimes U & D \otimes V \end{pmatrix}.$$

Show that $M \circ N$ is a principal submatrix of $M \odot N$ and that $M \odot N$ is a principal submatrix of $M \otimes N$. Moreover, $M \odot N \geq 0$ if $M, N \geq 0$.

16. Let $|X| = (X^*X)^{1/2}$ (the modulus of matrix X, Sect. 3.2). Verify that it is not true that $|A \circ B| \leq |A| \circ |B|$ for $A = \left(\begin{smallmatrix} 1 & 1 \\ 0 & 0 \end{smallmatrix}\right)$ and $B = \left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$.

17. Show that $\left(\begin{smallmatrix} 1 & 1 & 0 \\ 1 & 2 & 1 \\ 0 & 1 & 1 \end{smallmatrix}\right) \geq 0$ and $\left(\begin{smallmatrix} 1 & 1 & 0 \\ 1 & 2^r & 1 \\ 0 & 1 & 1 \end{smallmatrix}\right) \ngeq 0$ for $r < 1$.

18. A *correlation matrix* is a positive semidefinite matrix whose diagonal entries are all equal to 1. Let A be an $n \times n$ positive semidefinite matrix. Show that $\min_X \det(A \circ X) = \det A$, where the minimal value is taken over all $n \times n$ correlation matrices X.

19. Let $M = \left(\begin{smallmatrix} A & B \\ B^* & C \end{smallmatrix}\right)$ be a partitioned positive semidefinite matrix, where A and C are square matrices (of possibly different orders). Show that

$$\lambda_{\max}(M) \leq \lambda_{\max}(A) + \lambda_{\max}(C).$$

20. Let $A = (a_{ij})$ be an $n \times n$ positive semidefinite matrix.

    (a) Show that $A \circ A = (a_{ij}^2) \geq 0$.

    (b) Show that $A \circ A^T = A \circ \overline{A} = (|a_{ij}|^2) \geq 0$.

    (c) Show that $(a_{ij}^3) \geq 0$ and $(|a_{ij}|^4) \geq 0$.

    (d) Find a 3×3 matrix $A = (a_{ij}) \geq 0$ such that $(\sqrt{|a_{ij}|}\,) \ngeq 0$.

    (e) Show that the following 4×4 (the smallest order of such examples) matrix A is positive semidefinite, but $(|a_{ij}|)$ is not, where

$$A = \begin{pmatrix} 1 & \alpha & 0 & -\alpha \\ \alpha & 1 & \alpha & 0 \\ 0 & \alpha & 1 & \alpha \\ -\alpha & 0 & \alpha & 1 \end{pmatrix}, \quad \alpha = \tfrac{1}{\sqrt{2}}.$$

    (f) Find a matrix such that $(|a_{ij}|^3) \ngeq 0$.

21. Show that $A = \left(\begin{smallmatrix} I_2 & I_2 \\ I_2 & I_2 \end{smallmatrix}\right) = \left(\begin{smallmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \end{smallmatrix}\right)$ is a positive semidefinite matrix. If A is partitioned as $A = \left(\begin{smallmatrix} A_{11} & A_{12} & A_{13} \\ A_{21} & A_{22} & A_{23} \\ A_{31} & A_{32} & A_{33} \end{smallmatrix}\right)$ with $A_{11} = A_{22} = (1)$ and $A_{33} = I_2$, show that the matrix $M = \left(\begin{smallmatrix} \|A_{11}\| & \|A_{12}\| & \|A_{13}\| \\ \|A_{21}\| & \|A_{22}\| & \|A_{23}\| \\ \|A_{31}\| & \|A_{32}\| & \|A_{33}\| \end{smallmatrix}\right)$ with blocks in norms is not positive semidefinite, where $\|\cdot\|$ is the spectral norm.

22. Let $\lambda_1, \lambda_2, \ldots, \lambda_n$ be positive numbers. Use Cauchy matrices (Sect. 5.2) to show that the following matrices are positive semidefinite.

$$\left(\frac{1}{\lambda_i+\lambda_j}\right),\ \left(\frac{1}{\lambda_i\lambda_j}\right),\ \left(\frac{\lambda_i\lambda_j}{\lambda_i+\lambda_j}\right),\ \left(\frac{1}{\lambda_i^2+\lambda_j^2}\right),\ \left(\sqrt{\lambda_i\lambda_j}\right),$$

$$\left(\frac{2}{\lambda_i^{-1}+\lambda_j^{-1}}\right),\ \left(\frac{1}{\lambda_i(\lambda_i+\lambda_j)\lambda_j}\right),\ \left(\frac{\sqrt{\lambda_i\lambda_j}}{\lambda_i+\lambda_j}\right),\ \left(\frac{\lambda_i\lambda_j}{\sqrt{\lambda_i}+\sqrt{\lambda_j}}\right).$$

23. Let A be a positive definite matrix partitioned as $A = \left(\begin{smallmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{smallmatrix}\right)$ (in which A_{11} is square). Use the identity $\det \widetilde{A_{11}} = \frac{\det A}{\det A_{11}}$ to show that

$$\frac{\det(A+B)}{\det(A_{11}+B_{11})} \geq \frac{\det A}{\det A_{11}} + \frac{\det B}{\det B_{11}}.$$

24. Let $A, B \in \mathbb{M}_n$. If A, B, and $A+B$ are invertible, show that $\left(\begin{smallmatrix} A & A \\ A & A+B \end{smallmatrix}\right)$ is invertible. Find the inverse and use Schur complement to show that

$$A - A(A+B)^{-1}A = (A^{-1}+B^{-1})^{-1}.$$

25. Let $A > 0$ and $B > 0$, both $n \times n$. Show that for all $x, y \in \mathbb{C}^n$,

$$(x+y)^*(A+B)^{-1}(x+y) \leq x^*A^{-1}x + y^*B^{-1}y$$

and

$$(x \circ y)^*(A \circ B)^{-1}(x \circ y) \leq (x^*A^{-1}x) \circ (y^*B^{-1}y).$$

26. Let $A, B, C \in \mathbb{M}_n$ be such that $\left(\begin{smallmatrix} A & B \\ B^* & C \end{smallmatrix}\right)$ is a positive semidefinite matrix, and let $\star$ denote the sum $+$ or the Hadamard product $\circ$. Show that

$$\big(\operatorname{tr}(B^* \star B)^2\big)^{1/2} \leq \operatorname{tr}(A \star C) \quad \text{and} \quad \det(B^* \star B) \leq \det(A \star C).$$

27. Let $\left(\begin{smallmatrix} A & B \\ B^* & C \end{smallmatrix}\right) \geq 0$, where $A, B, C \in \mathbb{M}_n$. Show that

$$\operatorname{tr}(B^*B) - \operatorname{tr}(AC) \leq (\operatorname{tr} A)(\operatorname{tr} C) - (\operatorname{tr} B^*)(\operatorname{tr} B).$$

28. Find examples: $A, B, C \in \mathbb{M}_2$ are such that $\left(\begin{smallmatrix} A & B \\ B^* & C \end{smallmatrix}\right) \geq 0$.
    (a) $\operatorname{tr}(AC) < \operatorname{tr}(B^*)\operatorname{tr}(B)$.
    (b) $\operatorname{tr}(AC) < \operatorname{tr}(B^*B)$.

29. Show Theorem 7.3.5 by using the Hadamard product. [Hint: Denote $F = F_\alpha$ the n-square matrix whose principal submatrix $[F] = F[\alpha]$ indexed by α is the all-ones matrix (or order $|\alpha|$) and whose other entries are zeros. Then $\left(\begin{smallmatrix} F & F \\ F & F \end{smallmatrix}\right)$ and $\left(\begin{smallmatrix} [F] & [F] \\ [F] & [F] \end{smallmatrix}\right)$ are both positive semidefinite.]

30. Show that the block matrices $\begin{pmatrix} A^*A & A^* \\ A & I_m \end{pmatrix}$, $\begin{pmatrix} B^*B & B^* \\ B & I_m \end{pmatrix}$, and $\begin{pmatrix} I_n & B^* \\ B & BB^* \end{pmatrix}$ are all positive semidefinite for any $A, B \in \mathbb{M}_{m\times n}$. Derive the inequalities:

$$I_m \geq A(A^*A)^{-1}A^* \text{ if } \operatorname{rank}(A) = n,$$
$$(A^*A) \circ (B^*B) \geq (A^* \circ B^*)(A \circ B),$$
$$A^*A + B^*B \geq \frac{(A^* + B^*)(A + B)}{2},$$
$$\big((A^*A) \circ I\big)\big((BB^*) \circ I\big) \geq A^* \circ A \circ B^* \circ B \text{ if } m = n.$$

31. Let A be a positive definite matrix partitioned as $\begin{pmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{pmatrix}$, where A_{11} and A_{22} are square matrices (of possibly different orders). Show that

$$A \circ A^{-1} \geq \begin{pmatrix} A_{11} \circ A_{11}^{-1} & 0 \\ 0 & \widetilde{A_{11}} \circ \widetilde{A_{11}}^{-1} \end{pmatrix} \geq 0.$$

Conclude that $A \circ A^{-1} \geq I$. Similarly, show that $A^T \circ A^{-1} \geq I$.

32. Let $A \in \mathbb{M}_n$ be a positive definite matrix. Show that for any $B \in \mathbb{M}_n$,

$$\begin{pmatrix} A \circ (B^*A^{-1}B) & B^* \circ B \\ B^* \circ B & A \circ (B^*A^{-1}B) \end{pmatrix} \geq 0.$$

In particular,

$$\begin{pmatrix} A \circ A^{-1} & I \\ I & A \circ A^{-1} \end{pmatrix} \geq 0.$$

Derive the inequalities:

$$(A \circ A^{-1})^{-1} \leq A \circ A^{-1},$$
$$\det(B^* \circ B) \leq \det\big(A \circ (B^*A^{-1}B)\big),$$
$$\big(\operatorname{tr}(B^* \circ B)^2\big)^{1/2} \leq \operatorname{tr}\big(A \circ (B^*A^{-1}B)\big),$$
$$I \circ B^*B \geq (B^* \circ B)(I \circ B^*B)^{-1}(B^* \circ B)$$

if B has no zero row or column. Discuss the analogues for sum $(+)$.

33. Let A_1 and A_2 be $n \times n$ strictly contractive real matrices. Show that

(a) $\det(I - A_i) \geq 0$ for $i = 1, 2$.

(b) $H = (h_{ij}) \geq 0$, where $h_{ij} = \frac{1}{\det(I - A_i^*A_j)}$, $1 \leq i, j \leq 2$.

(c) $L = (l_{ij}) \geq 0$, where $l_{ij} = \frac{1}{\det(I - A_i^*A_j)^k}$, $1 \leq i, j \leq 2$, $k = 1, 2, \ldots$.

[Hint: Use the Hua determinant inequality in Sect. 7.4.]

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7.6 Inequalities of Choi, Kadison, Kantorovich, and Wielandt

The Cauchy-Schwarz inequality (Sect. 1.4) is one of the most useful and fundamental results in mathematics. It states that for any vectors x and y in an inner product space equipped with an inner product $\langle \cdot\,, \cdot \rangle$,

$$|\langle x, y\rangle|^2 \le \langle x, x\rangle\langle y, y\rangle,$$

and equality holds if and only if x and y are linearly dependent. Thus, for column vectors $x, y \in \mathbb{C}^n$ with the standard inner product $\langle x, y\rangle = y^*x$,

$$|y^*x|^2 \le (x^*x)(y^*y),$$

which also follows from the positive semidefiniteness of the 2×2 Gram matrix

$$(x,y)^*(x,y) = \begin{pmatrix} x^* \\ y^* \end{pmatrix}(x,y) = \begin{pmatrix} x^*x & x^*y \\ y^*x & y^*y \end{pmatrix} = \begin{pmatrix} \langle x,x\rangle & \langle y,x\rangle \\ \langle x,y\rangle & \langle y,y\rangle \end{pmatrix} \ge 0.$$

In this section we first prove a refined version of the Cauchy-Schwarz inequality, show a Cauchy-Schwarz type inequality involving matrices, give some related inequalities (i.e., the Wielandt and Kantorovich inequalities), and present the inequalities of Choi and Kadison on positive linear maps.

Lemma 7.6.1 *Let $A = \left(\begin{smallmatrix} a & b \\ \bar{b} & c \end{smallmatrix}\right)$ be a nonzero 2×2 positive semidefinite matrix with eigenvalues α and β, $\alpha \ge \beta$. (Thus, $\alpha > 0$ and $\beta \ge 0$). Then*

$$|b|^2 \le \left(\frac{\alpha-\beta}{\alpha+\beta}\right)^2 ac. \tag{7.57}$$

Equality holds if and only if $a = c$ or $\beta = 0$ (i.e., A is singular or $|b|^2 = ac$).

Proof. Solving the equation $\det(\lambda I - A) = 0$ gives the eigenvalues of A

$$\alpha,\ \beta = \frac{(a+c) \pm \sqrt{(a-c)^2 + 4|b|^2}}{2}.$$

Computing $\frac{\alpha-\beta}{\alpha+\beta}$ and squaring it, we get an equivalent form of (7.57),

$$(a-c)^2(ac - |b|^2) \ge 0.$$

Equality occurs if and only if $a = c$ or $ac = |b|^2$. The conclusions follow. ∎

Theorem 7.6.1 *Let x and y be vectors in an inner product space and let $A = \left(\begin{smallmatrix} \langle x,x\rangle & \langle x,y\rangle \\ \langle y,x\rangle & \langle y,y\rangle \end{smallmatrix}\right)$. If the eigenvalues of A are α and β, $\alpha \ge \beta$, $\alpha > 0$, then*

$$|\langle x, y\rangle|^2 \le \left(\frac{\alpha-\beta}{\alpha+\beta}\right)^2 \langle x, x\rangle\langle y, y\rangle. \tag{7.58}$$

Equality holds if and only if x and y have the same length (i.e., $\|x\| = \|y\|$) or x and y are linearly dependent (equivalently, A is singular, i.e., $\beta = 0$).

Proof. The inequality follows from the lemma at once. If equality occurs in (7.58), then $\langle x, x\rangle = \langle y, y\rangle$ or A is singular. For the latter case, we may assume that $\|y\| = 1$. Then the singularity of A yields $\langle x, x\rangle = |\langle x, y\rangle|^2$, implying $\langle x - ty, x - ty\rangle = 0$, where $t = \langle x, y\rangle$. Thus, $x = ty$. ∎

We proceed to derive more related inequalities concerning matrices. For these inequalities, we use the fact that if $0 < r \le p \le q \le s$, then $\frac{q-p}{q+p} \le \frac{s-r}{s+r}$. This is because $\frac{t-1}{t+1}$ is an increasing function on $(0, +\infty)$.

Theorem 7.6.2 *Let $M = \left(\begin{smallmatrix} A & B^* \\ B & C \end{smallmatrix}\right) \ge 0$, where A, B, $C \in \mathbb{M}_n$. Let α and β be the largest and smallest eigenvalues of M, respectively. If $M \ne 0$, then*

$$|\langle Bx, y\rangle|^2 \le \left(\frac{\alpha - \beta}{\alpha + \beta}\right)^2 \langle Ax, x\rangle\langle Cy, y\rangle \le \langle Ax, x\rangle\langle Cy, y\rangle, \quad x, y \in \mathbb{C}^n. \quad (7.59)$$

Proof. For any $x, y \in \mathbb{C}^n$ (in columns), let

$$N = \begin{pmatrix} x^* & 0 \\ 0 & y^* \end{pmatrix} M \begin{pmatrix} x & 0 \\ 0 & y \end{pmatrix} = \begin{pmatrix} x^*Ax & x^*B^*y \\ y^*Bx & y^*Cy \end{pmatrix}.$$

N is a 2×2 positive semidefinite matrix. If $N = 0$, (7.59) holds. Let $N \ne 0$. Let λ and μ be the eigenvalues of N, $\lambda \ge \mu$, $\lambda > 0$. By Lemma 7.6.1, we have

$$|\langle Bx, y\rangle|^2 \le \left(\frac{\lambda - \mu}{\lambda + \mu}\right)^2 \langle Ax, x\rangle\langle Cy, y\rangle.$$

By scaling, we may assume that x and y in (7.59) are unit vectors. Because $\beta I_{2n} \le M \le \alpha I_{2n}$ and $\left(\begin{smallmatrix} x & 0 \\ 0 & y \end{smallmatrix}\right)^* \left(\begin{smallmatrix} x & 0 \\ 0 & y \end{smallmatrix}\right) = I_2$, pre- and post-multiplying by $\left(\begin{smallmatrix} x & 0 \\ 0 & y \end{smallmatrix}\right)^*$ and $\left(\begin{smallmatrix} x & 0 \\ 0 & y \end{smallmatrix}\right)$, respectively, we obtain that $\beta I_2 \le N \le \alpha I_2$. It follows that $\beta \le \mu \le \lambda \le \alpha$. Consequently, $0 \le \frac{\lambda-\mu}{\lambda+\mu} \le \frac{\alpha-\beta}{\alpha+\beta}$, implying (7.59). ∎

Note: Theorem 7.6.2 with $A = B = C = I_n$ implies Theorem 7.6.1.

Example 7.6.1 Let $M = \left(\begin{smallmatrix} A & B^* \\ B & C \end{smallmatrix}\right) \ge 0$, where A, B, $C \in \mathbb{M}_n$. Let α and β be the largest and smallest eigenvalues of M, respectively. If $M \ne 0$, then

$$\sigma_1^2(B) \le \left(\frac{\alpha - \beta}{\alpha + \beta}\right)^2 \lambda_1(A)\lambda_1(C).$$

This is slightly stronger than $\sigma_1^2(B) \le \lambda_1(A)\lambda_1(C)$ (Problem 18, Sect. 7.3). ∎

Theorem 7.6.3 (Wielandt Inequality) *Let A be an $n \times n$ nonzero positive semidefinite matrix and let λ_1 and λ_n be the largest and smallest eigenvalues of A, respectively. Then for all orthogonal column vectors x and y in $\mathbb{C}^n$,*

$$|x^*Ay|^2 \le \left(\frac{\lambda_1 - \lambda_n}{\lambda_1 + \lambda_n}\right)^2 (x^*Ax)(y^*Ay).$$

Proof 1. Let $N = (x,y)^*A(x,y) = \begin{pmatrix} x^*Ax & x^*Ay \\ y^*Ax & y^*Ay \end{pmatrix}$. Then N is a 2×2 positive semidefinite matrix. We may assume that x and y are orthonormal vectors (by scaling). Note that $(x,y)^*(x,y) = I_2$. Because $\lambda_n I \le A \le \lambda_1 I$, multiplying by $(x,y)^*$ from the left and by (x,y) from the right, we have $\lambda_n I \le N \le \lambda_1 I$. If α and β are the eigenvalues of N with $\alpha \ge \beta$, $\alpha > 0$ (the case of $\alpha = 0$ is trivial), then $\lambda_n \le \beta \le \alpha \le \lambda_1$. Lemma 7.6.1 reveals

$$|y^*Ax|^2 \le \left(\frac{\alpha-\beta}{\alpha+\beta}\right)^2 (x^*Ax)(y^*Ay).$$

Using the fact that $\frac{\alpha-\beta}{\alpha+\beta} \le \frac{\lambda_1-\lambda_n}{\lambda_1+\lambda_n}$, we obtain the inequality.

Proof 2. Let x and y be orthonormal vectors and let θ be a real number such that $e^{i\theta}\langle Ay, x\rangle = |\langle Ay, x\rangle| = |x^*Ay| = |y^*Ax|$. Because $\lambda_n I \le A \le \lambda_1 I$, we have, for any complex number c,

$$\lambda_n \|x + cy\|^2 \le \langle A(x+cy), x+cy\rangle \le \lambda_1 \|x+cy\|^2.$$

By expanding and setting $c = te^{i\theta},\ t \in \mathbb{R}$, we have

$$t^2(y^*Ay - \lambda_n) + 2t|x^*Ay| + x^*Ax - \lambda_n \ge 0 \tag{7.60}$$

and

$$t^2(\lambda_1 - y^*Ay) + 2t|x^*Ay| + \lambda_1 - x^*Ax \ge 0. \tag{7.61}$$

Multiply (7.60) by λ_1 and (7.61) by λ_n, then add the inequalities to get

$$t^2(\lambda_1 - \lambda_n)y^*Ay + 2t(\lambda_1 + \lambda_n)|x^*Ay| + (\lambda_1 - \lambda_n)x^*Ax \ge 0.$$

Since this is true for all real t, by taking the discriminant, we have

$$(\lambda_1 + \lambda_n)^2|x^*Ay|^2 \le (\lambda_1 - \lambda_n)^2(x^*Ax)(y^*Ay).$$

Dividing both sides by $(\lambda_1 + \lambda_n)^2$ reveals the Wielandt inequality. ∎

Example 7.6.2 Let A be an $n \times n$ positive definite matrix with the largest and smallest eigenvalues λ_1 and λ_n, respectively. If x and y are orthogonal (vectors in $\mathbb{C}^n$), then $A^{1/2}x$ and $A^{-1/2}y$ are orthogonal. Thus, for any unit vector $y \in \mathbb{C}^n$ that is orthogonal to x, Theorem 7.6.3 implies

$$|x^*Ay| = |(A^{\frac{1}{2}}x)^*A(A^{-\frac{1}{2}})y| \le \left(\frac{\lambda_1-\lambda_n}{\lambda_1+\lambda_n}\right)\sqrt{x^*A^2x} = \left(\frac{\lambda_1-\lambda_n}{\lambda_1+\lambda_n}\right)\|Ax\|. \ ∎$$

Note that the Cauchy-Schwarz inequality ensures that $|x^*Ay| \le \|Ax\|$ for any $A \in \mathbb{M}_n$, $x \in \mathbb{C}^n$, and $y \in \mathbb{C}^n$ with $\|y\| = 1$. Moreover, for a positive definite A, in the classical Cauchy-Schwarz inequality $|y^*x| \le (x^*x)(y^*y)$, substituting x and y with $A^{1/2}x$ and $A^{-1/2}y$, respectively, we arrive at

$$|y^*x|^2 \le (x^*Ax)(y^*A^{-1}y).$$

In particular, setting $y = x$ and for any unit column vector $x \in \mathbb{C}^n$, we obtain

$$1 \le (x^*Ax)(x^*A^{-1}x).$$

A reversal of the preceding inequality is the well-known *Kantorovich inequality*, which is a special case of our next theorem on positive maps.

Let $f\colon \mathbb{M}_n \to \mathbb{M}_k$ be a linear transformation (map). We say that f is *positive* if $f(A) \geq 0$ whenever $A \geq 0$; f is *strictly positive* if $f(A) > 0$ whenever $A > 0$; and f is *unital* if $f(I_n) = I_k$. We are interested in the positive and (or) unital linear maps. Here are some examples:

(1) $f\colon A \mapsto \mathrm{tr}A$ is strictly positive from $\mathbb{M}_n$ to $\mathbb{M}_1 = \mathbb{C}$.

(2) $g\colon A \mapsto X^*AX$ is positive, where X is a fixed $n \times k$ matrix.

(3) $h\colon A \mapsto A_k$ is positive and unital, where A_k is the $k \times k$ leading principal submatrix of A.

(4) $p\colon A \mapsto A \otimes X$ is positive, where X is a positive semidefinite matrix (of any order). p is strictly positive if $X > 0$, and unital if $X = I$.

(5) $q\colon A \mapsto A \circ Y$ is positive, where Y is a positive semidefinite matrix of the same order as A. q is strictly positive if $Y > 0$, unital if $Y = I$.

Theorem 7.6.4 *Let $f : \mathbb{M}_n \to \mathbb{M}_k$ be a strictly positive and unital linear map. Let $A \in \mathbb{M}_n$ be a positive definite matrix whose eigenvalues all lie in the interval $[a, b]$, $0 < a < b$. Then*

$$f(A^{-1}) \leq \frac{(a+b)^2}{4ab}\big(f(A)\big)^{-1}.$$

Proof. Since the eigenvalues of A are all contained in $[a, b]$, the matrices $A - aI$ and $bI - A$ are both positive semidefinite (here $I = I_n$ is $n \times n$). Because they commute, we have $(A - aI)(bI - A) \geq 0$, implying

$$abI \leq (a+b)A - A^2 \quad \text{or} \quad abA^{-1} \leq (a+b)I - A.$$

Applying f to both sides (as f preserves the inequality), we have

$$abf(A^{-1}) \leq (a+b)I - f(A) \quad (\text{here } I \text{ is } k \times k).$$

For real c and x, $(c - 2x)^2 = c^2 - 4xc + 4x^2 \geq 0$. Thus, $4xc - 4x^2 \leq c^2$. For $x > 0$, dividing both sides by $4x$ implies $c - x \leq \frac{1}{4}c^2x^{-1}$. It follows that

$$abf(A^{-1}) \leq (a+b)I - f(A) \leq \frac{(a+b)^2}{4}\big(f(A)\big)^{-1}. \blacksquare$$

We present the *Kantorovich inequality* as a consequence.

Theorem 7.6.5 (Kantorovich Inequality) *Let $A \in \mathbb{M}_n$ be a positive definite matrix with the largest eigenvalue λ_1 and the smallest eigenvalue λ_n. Then*

$$(x^*Ax)(x^*A^{-1}x) \leq \frac{(\lambda_1 + \lambda_n)^2}{4\lambda_1\lambda_n}(x^*x)^2, \quad x \in \mathbb{C}^n. \tag{7.62}$$

Proof. In Theorem 7.6.4, let $f\colon A \mapsto y^*Ay$, where $y \in \mathbb{C}^n$ is a unit vector. Then f is positive and unital. Now set $a = \lambda_n$, $b = \lambda_1$, $y = \frac{x}{\|x\|}$ for $x \neq 0$. $\blacksquare$

The Kantorovich inequality (7.62) is invariant under unitary similarity, that is, A can be replaced by U^*AU for any unitary matrix $U \in \mathbb{M}_n$. By spectral decomposition of A, without loss of generality, we may assume that A is a diagonal matrix. Taking x to be a unit vector in (7.62), we see that the Kantorovich inequality is essentially the same as the statement:

Let $\lambda_1 \geq \cdots \geq \lambda_n > 0$ and let $t_1, \ldots, t_n \geq 0$, $t_1 + \cdots + t_n = 1$. Then

$$\Big(\sum_{i=1}^n t_i\lambda_i\Big)\Big(\sum_{i=1}^n \frac{t_i}{\lambda_i}\Big) \leq \frac{(\lambda_1+\lambda_n)^2}{4\lambda_1\lambda_n}. \tag{7.63}$$

A direct, simple proof goes as follows. Observe that if we replace every λ_i in (7.63) by $c\lambda_i$, $c > 0$, the inequality remains unchanged. Thus, we can assume $\lambda_1\lambda_n = 1$. So, $\lambda_n = \frac{1}{\lambda_1}$. Note that $t + \frac{1}{t}$ is a convex function for $t > 0$. We have $t + \frac{1}{t} \leq \lambda_1 + \frac{1}{\lambda_1}$ for all t between λ_1 and $\frac{1}{\lambda_1}$ (because the function attains its maximum value on a closed interval at an end point. Problem 2). In particular, $\lambda_i + \frac{1}{\lambda_i} \leq \lambda_1 + \frac{1}{\lambda_1}$ for every λ_i. It follows that $\sum_i t_i\lambda_i + \sum_i \frac{t_i}{\lambda_i} \leq \lambda_1 + \frac{1}{\lambda_1} = \lambda_1 + \lambda_n \leq \frac{(\lambda_1+\lambda_n)^2}{4}$. This completes the proof.

Example 7.6.3 Let A be an $n \times n$ positive definite matrix with the largest eigenvalue λ_1 and the smallest eigenvalue λ_n. (Thus, $\lambda_n > 0$ as $A > 0$). Then by replacing x in the Kantorovich inequality (7.62) with $A^{1/2}x$, we have

$$(x^*A^2x)(x^*x) \leq \frac{(\lambda_1+\lambda_n)^2}{4\lambda_1\lambda_n}(x^*Ax)^2, \quad x \in \mathbb{C}^n,$$

from which we obtain

$$\frac{2\sqrt{\lambda_1\lambda_n}}{\lambda_1+\lambda_n} \leq \frac{x^*Ax}{\|Ax\|}, \quad x \in \mathbb{C}^n,\ \|x\| = 1. \ \blacksquare$$

The Kantorovich inequality has made appearances in a variety of forms. A more general matrix version is as follows. Let $A \in \mathbb{M}_n$ be a positive definite matrix. Then for any $n \times k$ matrix X satisfying $X^*X = I_k$,

$$(X^*AX)^{-1} \leq X^*A^{-1}X \leq \frac{(\lambda_1+\lambda_n)^2}{4\lambda_1\lambda_n}(X^*AX)^{-1}. \tag{7.64}$$

The first inequality is proven by observing that $I - Y(Y^*Y)^{-1}Y^* \geq 0$ for any matrix Y with linearly independent columns. In addition, we have the inequalities for the Hadamard product of positive definite matrices:

$$(A \circ B)^{-1} \leq A^{-1} \circ B^{-1} \leq \frac{(\lambda+\mu)^2}{4\lambda\mu}(A \circ B)^{-1}, \tag{7.65}$$

where λ and μ are the largest and smallest eigenvalues of $A \otimes B$, respectively.

We leave the proofs of (7.64) and (7.65) to the reader (Problems 21 and 22). We end the chapter with two elegant results on positive unital maps.

Theorem 7.6.6 *Let φ be a positive unital linear map from $\mathbb{M}_n$ to $\mathbb{M}_k$. Then for any normal matrix $A \in \mathbb{M}_n$,*

$$\varphi(A^*A) \geq (\varphi(A))^*(\varphi(A)). \qquad \text{(Choi inequality)}$$

In particular, for any positive semidefinite matrix $A \in \mathbb{M}_n$,

$$\varphi(A^2) \geq (\varphi(A))^2. \qquad \text{(Kadison inequality)}$$

Proof. Let $A = \sum_{i=1}^n \lambda_i u_i u_i^*$ be a spectral decomposition of A, where $\lambda_1, \ldots, \lambda_n \in \mathbb{C}$, and $u_1, \ldots, u_n$ are orthonormal (see Problem 31 of Sect. 3.2). We have $A^*A = \sum_{i=1}^n |\lambda_i|^2 u_i u_i^*$ and $\varphi(A^*) = \sum_{i=1}^n \overline{\lambda_i}\varphi(u_i u_i^*) = (\varphi(A))^*$. Moreover, because φ is unital, $\varphi(I_n) = \varphi(\sum_{i=1}^n u_i u_i^*) = \sum_{i=1}^n \varphi(u_i u_i^*) = I_k$.

Since $\begin{pmatrix} |\lambda_i|^2 & \lambda_i \\ \overline{\lambda_i} & 1 \end{pmatrix} \geq 0$ and $\begin{pmatrix} \varphi(u_i u_i^*) & \varphi(u_i u_i^*) \\ \varphi(u_i u_i^*) & \varphi(u_i u_i^*) \end{pmatrix} \geq 0$ for each i, we obtain

$$\sum_{i=1}^n \begin{pmatrix} |\lambda_i|^2\varphi(u_i u_i^*) & \lambda_i\varphi(u_i u_i^*) \\ \overline{\lambda_i}\varphi(u_i u_i^*) & \varphi(u_i u_i^*) \end{pmatrix} = \begin{pmatrix} \sum_i |\lambda_i|^2\varphi(u_i u_i^*) & \sum_i \lambda_i\varphi(u_i u_i^*) \\ \sum_i \overline{\lambda_i}\varphi(u_i u_i^*) & \sum_i \varphi(u_i u_i^*) \end{pmatrix} \geq 0,$$

i.e., $\begin{pmatrix} \varphi(A^*A) & \varphi(A) \\ (\varphi(A))^* & I_k \end{pmatrix} \geq 0$, which reveals the Choi and Kadison inequalities. ∎

Example 7.6.4 Let A be a positive semidefinite matrix. If A_{11} is a principal submatrix of A and $(A^2)_{11}$ is the principal submatrix of A^2 in the corresponding position, then $(A^2)_{11} \geq (A_{11})^2$ because $\varphi(A) = A_{11}$ is positive unital. (See also Theorem 7.3.4 for this result with a different proof.) ∎

Problems

1. Show that $f(t) = \frac{t-1}{t+1}$ is an increasing function of t on $(-1, \infty)$.
2. Let $r > 0$. Show that $t + \frac{1}{t} \leq r + \frac{1}{r}$ for any t between $\frac{1}{r}$ and r.
3. Let $0 < a < b$. Show that $\frac{1}{t} \leq \frac{a+b}{ab} - \frac{t}{ab}$ for $t \in [a, b]$.
4. Let x, $y \in \mathbb{C}^n$ be unit vectors. Show that the eigenvalues of the matrix $\begin{pmatrix} 1 & \langle x,y\rangle \\ \langle y,x\rangle & 1 \end{pmatrix}$ are $1 - \delta$ and $1 + \delta$, where $\delta = |\langle x, y\rangle|$.
5. Let $A = \begin{pmatrix} a & b \\ \bar{b} & c \end{pmatrix}$ be a nonzero 2×2 Hermitian matrix with (necessarily real) eigenvalues α and β, $\alpha \geq \beta$. Show that $2|b| \leq \alpha - \beta$.
6. Let A be an $n \times n$ positive definite matrix, and x, $y \in \mathbb{C}^n$. Show that
$$|y^*x|^2 \leq (x^*Ax)(y^*A^{-1}y).$$
The inequality becomes an equality if and only if $y = 0$ or $Ax = cy$ for some scalar c. Conclude that $|x^*x|^2 \leq (x^*Ax)(x^*A^{-1}x)$ for all $x \in \mathbb{C}^n$. Setting $x = e_i = (0, \ldots, 0, 1, 0 \ldots, 0)^T \in \mathbb{R}^n$, what does one obtain?
7. Let $y \in \mathbb{C}^n$ be a fixed unit vector. Show that $f(A) = y^*Ay$ is a linear map from $\mathbb{M}_n$ to $\mathbb{C}$. Show further that f is positive and unital.

8. Let $\lambda_1, \ldots, \lambda_n$ be positive numbers. Show that for any $x, y \in \mathbb{C}^n$,

$$\left| \sum_{i=1}^n x_i \overline{y_i} \right|^2 \leq \left(\sum_{i=1}^n \lambda_i |x_i|^2 \right) \left(\sum_{i=1}^n \lambda_i^{-1} |y_i|^2 \right).$$

9. Let $a_1, a_2, \ldots, a_n$ be positive numbers. Show that for any $t \in [0,1]$,

$$\left(\sum_{i=1}^n a_i^{1/2} \right)^2 \leq \left(\sum_{i=1}^n a_i^t \right) \left(\sum_{i=1}^n a_i^{1-t} \right).$$

   Equality occurs if and only if $t = \frac{1}{2}$ or all a_i are equal to each other.

10. Let A be an $n \times n$ Hermitian matrix. Show that for any unit $x \in \mathbb{C}^n$,

$$\langle Ax, x \rangle^2 \leq \langle A^2 x, x \rangle.$$

    Setting $x = e_i = (0, \ldots, 0, 1, 0 \ldots, 0)^T \in \mathbb{R}^n$, what inequality does one obtain? Is it true that $|\langle Ax, x \rangle|^2 \leq |\langle A^2 x, x \rangle|$ for normal matrices A?

11. Let A, B, and C be $n \times n$ matrices. Assume that A and C are positive definite. Show that the following statements are equivalent.

    (a) $\left(\begin{smallmatrix} A & B^* \\ B & C \end{smallmatrix} \right) \geq 0$.

    (b) $\lambda_{\max}(BA^{-1}B^*C^{-1}) \leq 1$.

    (c) $|\langle Bx, y \rangle|^2 \leq \langle Ax, x \rangle \langle Cy, y \rangle$ for all $x, y \in \mathbb{C}^n$.

    (d) $|\langle Bx, y \rangle| \leq \frac{1}{2} \big(\langle Ax, x \rangle + \langle Cy, y \rangle \big)$ for all $x, y \in \mathbb{C}^n$.

12. Show that for any $m \times n$ matrices A, B, and any $x \in \mathbb{C}^n$, $y \in \mathbb{C}^m$,

$$\big| \big\langle (A+B)x, y \big\rangle \big|^2 \leq \big\langle (I + A^*A)x, x \big\rangle \big\langle (I + BB^*)y, y \big\rangle.$$

13. Let $A \in \mathbb{M}_n$ and $B \in \mathbb{M}_{m \times n}$. If A is positive definite, show that

$$|\langle Bx, y \rangle|^2 \leq \langle Ax, x \rangle \langle BA^{-1}B^* y, y \rangle \text{ for any } x, y \in \mathbb{C}^n.$$

14. Let A and B be $n \times n$ positive definite matrices with eigenvalues contained in $[m, M]$, where $0 < m < M$. Show that for any $t \in [0,1]$,

$$tA^2 + (1-t)B^2 - \big(tA + (1-t)B \big)^2 \leq \tfrac{1}{4}(M-m)^2 I.$$

15. Prove the Kantorovich inequality by the Wielandt inequality with

$$y = \|x\|^2 (A^{-1}x) - (x^* A^{-1} x)x.$$

16. Let A and B be $n \times n$ Hermitian matrices. If $A \leq B$, show that

$$A \circ B \leq \tfrac{1}{2}(A \circ A + B \circ B).$$

17. Show the Kantorovich inequality following the steps below:

    (a) If $0 < m \le t \le M$, then $0 \le (m + M - t)t - mM$.

    (b) If $0 < m \le \lambda_i \le M$, $i = 1, \ldots, n$, and $\sum_{i=1}^n |x_i|^2 = 1$, then

    $$S\big(\tfrac{1}{\lambda}\big) \le \frac{m + M - S(\lambda)}{mM},$$

    where $S(\frac{1}{\lambda}) = \sum_{i=1}^n \frac{1}{\lambda_i}|x_i|^2$ and $S(\lambda) = \sum_{i=1}^n \lambda_i |x_i|^2$.

    (c) The Kantorovich inequality follows from the inequality

    $$S(\lambda)S\big(\tfrac{1}{\lambda}\big) \le \frac{(m + M)^2}{4mM}.$$

18. Let $\lambda_1 \ge \lambda_2 \ge \cdots \ge \lambda_n$ be positive numbers. Show that

    $$\max_{i,\,j} \frac{(\lambda_i + \lambda_j)^2}{4\lambda_i\lambda_j} = \frac{(\lambda_1 + \lambda_n)^2}{4\lambda_1\lambda_n}.$$

19. Let $0 < a < b$. Let $\lambda_1, \ldots, \lambda_n \in [a, b]$ and let $t_1, \ldots, t_n \ge 0$. Show that

    $$\Big(\sum_{i=1}^n t_i\lambda_i\Big)\Big(\sum_{i=1}^n \frac{t_i}{\lambda_i}\Big) \le \frac{(a + b)^2}{4ab}\Big(\sum_{i=1}^n t_i\Big)^2.$$

20. Let A be an $n \times n$ positive definite matrix having the largest eigenvalue $\lambda_{\max}(A) = \lambda_1$ and the smallest eigenvalue $\lambda_{\min}(A) = \lambda_n$. Show that

    $$1 \le \frac{(x^*Ax)(x^*A^{-1}x)}{(x^*x)^2} \le \frac{\lambda_{\max}(A)}{\lambda_{\min}(A)} \quad \text{for all nonzero } x \in \mathbb{C}^n.$$

21. Let A be an $n \times n$ positive definite matrix with $\lambda_1 = \lambda_{\max}(A)$ and $\lambda_n = \lambda_{\min}(A)$. Show that for all $n \times m$ matrices X with $X^*X = I_m$,

    $$(X^*AX)^{-1} \le X^*A^{-1}X \le \frac{(\lambda_1 + \lambda_n)^2}{4\lambda_1\lambda_n}(X^*AX)^{-1}.$$

    Use the first inequality to derive the inequality for principal submatrices

    $$[A]^{-1} \le [A^{-1}]$$

    and then show the following matrix inequalities:

    (a) $X^*AX - (X^*A^{-1}X)^{-1} \le (\sqrt{\lambda_1} - \sqrt{\lambda_n}\,)^2 I$.

    (b) $(X^*AX)^2 \le X^*A^2X \le \frac{(\lambda_1+\lambda_n)^2}{4\lambda_1\lambda_n}(X^*AX)^2$.

    (c) $X^*A^2X - (X^*AX)^2 \le \frac{(\lambda_1-\lambda_n)^2}{4} I$.

    (d) $X^*AX \le (X^*A^2X)^{1/2} \le \frac{\lambda_1+\lambda_n}{2\sqrt{\lambda_1\lambda_n}}(X^*AX)$.

    (e) $(X^*A^2X)^{1/2} - X^*AX \le \frac{(\lambda_1-\lambda_n)^2}{4(\lambda_1+\lambda_n)} I$.

22. Let A and B be $n \times n$ positive definite matrices. Show that

$$\lambda_{\max}(A \otimes B) = \lambda_{\max}(A)\lambda_{\max}(B) \quad \text{(denoted by } \lambda\text{)}$$

and

$$\lambda_{\min}(A \otimes B) = \lambda_{\min}(A)\lambda_{\min}(B) \quad \text{(denoted by } \mu\text{)}.$$

Derive the following inequalities from the previous problem.

(a) $(A \circ B)^{-1} \le A^{-1} \circ B^{-1} \le \frac{(\lambda+\mu)^2}{4\lambda\mu}(A \circ B)^{-1}$.

(b) $A \circ B - (A^{-1} \circ B^{-1})^{-1} \le (\sqrt{\lambda} - \sqrt{\mu})^2 I$.

(c) $(A \circ B)^2 \le A^2 \circ B^2 \le \frac{(\lambda+\mu)^2}{4\lambda\mu}(A \circ B)^2$.

(d) $A^2 \circ B^2 - (A \circ B)^2 \le \frac{(\lambda-\mu)^2}{4} I$.

(e) $A \circ B \le (A^2 \circ B^2)^{1/2} \le \frac{\lambda+\mu}{2\sqrt{\lambda\mu}} A \circ B$.

(f) $(A^2 \circ B^2)^{1/2} - A \circ B \le \frac{(\lambda-\mu)^2}{4(\lambda+\mu)} I$.

23. Let A be an $n \times n$ positive definite matrix having the largest eigenvalue $\lambda_{\max}(A) = \lambda_1$ and the smallest eigenvalue $\lambda_{\min}(A) = \lambda_n$. Show that

$$I \le A \circ A^{-1} \le \frac{\lambda_1^2 + \lambda_n^2}{2\lambda_1\lambda_n} I.$$

24. Let φ be a positive linear map from $\mathbb{M}_n$ to $\mathbb{M}_k$. If $A \in \mathbb{M}_n$ is positive semidefinite, show that $\begin{pmatrix} \varphi(A) & \varphi(A) \\ \varphi(A) & \varphi(A) \end{pmatrix} \in \mathbb{M}_{2k}$ is positive semidefinite.

25. Determine if each of the following maps f is linear, positive, or unital from $\mathbb{M}_{2n}$ to $\mathbb{M}_{2k}$ (in which the matrices are divided into 2×2 blocks).

(a) $f : \begin{pmatrix} A & B \\ C & D \end{pmatrix} \mapsto \begin{pmatrix} \frac{1}{2}(A+D) & \frac{1}{2}(B+C) \\ \frac{1}{2}(B+C) & \frac{1}{2}(A+D) \end{pmatrix}$ $(k = n)$.

(b) $f : \begin{pmatrix} A & B \\ C & D \end{pmatrix} \mapsto \begin{pmatrix} \operatorname{diag}(A) & \operatorname{diag}(B) \\ \operatorname{diag}(C) & \operatorname{diag}(D) \end{pmatrix}$ $(k = n)$.

(c) $f : \begin{pmatrix} A & B \\ C & D \end{pmatrix} \mapsto \begin{pmatrix} [A] & [B] \\ [C] & [D] \end{pmatrix}$ (k is the order of $[X]$).

(d) $f : \begin{pmatrix} A & B \\ C & D \end{pmatrix} \mapsto \begin{pmatrix} X^*AX & X^*B \\ CX & D \end{pmatrix}$ ($k = n$, fixed $X \in \mathbb{M}_n$).

(e) $f : \begin{pmatrix} A & B \\ C & D \end{pmatrix} \mapsto \begin{pmatrix} U^*AU & U^*BV \\ V^*CU & V^*DV \end{pmatrix}$ ($k = n$, U and V are unitary).

(f) $f : \begin{pmatrix} A & B \\ C & D \end{pmatrix} \mapsto \begin{pmatrix} A \otimes D & B \otimes C \\ C \otimes B & D \otimes A \end{pmatrix}$ $(k = n^2)$.

(g) $f : \begin{pmatrix} A & B \\ C & D \end{pmatrix} \mapsto \begin{pmatrix} \operatorname{tr}(A) & \operatorname{tr}(B) \\ \operatorname{tr}(C) & \operatorname{tr}(D) \end{pmatrix}$ $(k = 1)$.

(h) $f : \begin{pmatrix} A & B \\ C & D \end{pmatrix} \mapsto \begin{pmatrix} \frac{1}{n}\operatorname{tr}(A) & \frac{1}{n}\operatorname{tr}(B) \\ \frac{1}{n}\operatorname{tr}(C) & \frac{1}{n}\operatorname{tr}(D) \end{pmatrix}$ $(k = 1)$.

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CHAPTER 8

Hermitian Matrices

Introduction: This chapter contains the basic results of Hermitian matrices and the techniques used to derive the results. Sect. 8.1 presents equivalent conditions of matrices being Hermitian, Sect. 8.2 gives some trace inequalities and discusses a necessary and sufficient condition for a square matrix to be a product of two Hermitian matrices, and Sect. 8.3 develops the min-max theorem (principle) and the interlacing theorem for eigenvalues. Sect. 8.4 gives some simple eigenvalue and singular value inequalities of Hermitian matrices, Sect. 8.5 shows a few advanced eigenvalue inequalities of the sum of Hermitian matrices by min-max expressions, and Sect. 8.6 presents a matrix version of the triangle inequality.

8.1 Hermitian Matrices and Their Inertias

A square matrix $A = (a_{ij})$ is said to be *Hermitian* if A is equal to its conjugate transpose, that is, $A^* = A$, or precisely, $a_{ji} = \overline{a_{ij}}$ for all i, j.

Example 8.1.1 Hermitian matrices.

(a) $\left(\begin{smallmatrix} 0 & i \\ -i & 0 \end{smallmatrix}\right)$ is Hermitian. $\left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$ is real symmetric and Hermitian.

(b) $\left(\begin{smallmatrix} 0 & i \\ i & 0 \end{smallmatrix}\right)$ and $\left(\begin{smallmatrix} 0 & 1 \\ 1 & i \end{smallmatrix}\right)$ are complex symmetric but not Hermitian.

(c) $A^* + A$ is Hermitian for any square matrix A.

(d) A^*A and $\left(\begin{smallmatrix} 0 & A \\ A^* & 0 \end{smallmatrix}\right)$ are Hermitian for any matrix A.

(e) The diagonal entries of a Hermitian matrix are necessarily real.

(f) The eigenvalues of a Hermitian matrix are necessarily real. ∎

Theorem 8.1.1 (Spectral Decomposition for Hermitian Matrices) *Let A be an n-square matrix. Then A is Hermitian if and only if there exists an n-square unitary matrix U such that*

$$A = U^* \operatorname{diag}(\lambda_1, \ldots, \lambda_n) U, \tag{8.1}$$

where the λ_i's are all real numbers (and they are the eigenvalues of A).

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In other words, matrix A is Hermitian if and only if A is unitarily similar to a real diagonal matrix. This is the Hermitian case of the spectral decomposition theorem (Theorem 3.2.2). The decomposition (8.1) is often used when a trace or norm inequality is under investigation.

Theorem 8.1.2 *Let $A \in \mathbb{M}_n$. The following statements are equivalent.*

(1) *A is Hermitian.*
(2) *$x^*Ax \in \mathbb{R}$ for all $x \in \mathbb{C}^n$.*
(3) *$A^2 = A^*A$ $(= |A|^2)$.*
(4) *$\mathrm{tr}A^2 = \mathrm{tr}(A^*A)$.*

We show that (1) $\Leftrightarrow$ (2) and (1) $\Leftrightarrow$ (3). (1) $\Leftrightarrow$ (4) is similarly shown. It is not difficult to see that (1) and (2) are equivalent, because a complex number a is real if and only if $a^* = a$ and because (Problem 24)

$$A^* = A \quad \Leftrightarrow \quad x^*(A^* - A)x = 0 \text{ for all } x \in \mathbb{C}^n.$$

We present four proofs for (3) $\Rightarrow$ (1), that is, $A^2 = A^*A \Rightarrow A = A^*$, each of which shows a useful technique of linear algebra and matrix theory. Other implications (such as (1) $\Rightarrow$ (3)) are either easy or similar.

Proof 1. Use Schur triangular decomposition. Write $A = U^*TU$, where U is unitary and $T = (t_{ij})$ is an upper-triangular matrix with the eigenvalues $\lambda_1, \ldots, \lambda_n$ of A on the diagonal. Then $A^2 = A^*A$ implies $T^2 = T^*T$.

Comparing the diagonal entries on both sides of $T^2 = T^*T$ reveals

$$\lambda_j^2 = |\lambda_j|^2 + \sum_{i<j} |t_{ij}|^2, \quad j = 1, \ldots, n.$$

It follows that every λ_j is real and that $t_{ij} = 0$ whenever $i < j$. Thus, T is real diagonal, and $A^* = (U^*TU)^* = U^*T^*U = U^*TU = A$.

Proof 2. Use the fact that $\mathrm{tr}(XX^*) = 0 \Leftrightarrow X = 0$ for $X \in \mathbb{M}_{n\times m}$. We show that $\mathrm{tr}(A - A^*)(A - A^*)^* = 0$ to conclude $A - A^* = 0$, i.e., $A = A^*$.

Upon computation and using (3), we have

$$(A - A^*)(A - A^*)^* = AA^* - A^2 + A^*A - (A^*)^2 = AA^* - (A^*)^2.$$

Taking trace and by $\mathrm{tr}(AA^*) = \mathrm{tr}(A^*A)$ and $A^*A = A^2 = (A^*)^2$, we obtain

$$\mathrm{tr}(A - A^*)(A - A^*)^* = 0.$$

Proof 3. Use eigenvalues. Let $B = i(A - A^*)$. Then $B = B^*$ is Hermitian. We show that B has only zero eigenvalues, consequently, $B = 0$.

Suppose that λ is a nonzero eigenvalue of B with eigenvector x:

$$Bx = \lambda x, \quad \lambda \neq 0, \ x \neq 0.$$

Note that the condition $A^2 = A^*A$ implies $BA = 0$. We have

$$\lambda A^*x = A^*(Bx) = (BA)^*x = 0.$$

Thus, $A^*x = 0$. However, $Bx = \lambda x$ yields $A^*x = Ax + i\lambda x$. Therefore,

$$0 = x^*A^*x = x^*Ax + i\lambda x^*x = (x^*A^*x)^* + i\lambda x^*x = i\lambda x^*x.$$

It follows that $\lambda = 0$. This is a contradiction to the assumption $\lambda \neq 0$.

Proof 4. Use inner product. Note that (Problem 27, Sect. 1.4, Chap. 1)

$$\mathbb{C}^n = \operatorname{Ker} A^* \oplus \operatorname{Im} A.$$

Therefore, to show $A^* = A$, it is sufficient to show that $A^*x = Ax$ for all $x \in \operatorname{Ker} A^*$ and all $x \in \operatorname{Im} A$. If $x \in \operatorname{Ker} A^*$, then, by $A^2 = (A^*)^2 = A^*A$,

$$\langle Ax, Ax\rangle = \langle A^*Ax, x\rangle = \langle (A^*)^2x, x\rangle = 0.$$

This forces $Ax = 0$ for all $x \in \operatorname{Ker} A^*$, namely, $Ax = A^*x$ for all $x \in \operatorname{Ker} A^*$. If $x \in \operatorname{Im} A$, write $x = Ay$ for some $y \in \mathbb{C}^n$. We then have

$$A^*x = (A^*A)y = A^2y = A(Ay) = Ax. \blacksquare$$

The eigenvalues of a Hermitian matrix are all necessarily real. Let A be Hermitian. Then $Ax = \lambda x, x \neq 0 \Rightarrow x^*Ax = \lambda x^*x \in \mathbb{R} \Rightarrow \lambda \in \mathbb{R}$. Let $i_+(A)$, $i_-(A)$, and $i_0(A)$ be respectively the numbers of positive, negative, and zero eigenvalues of A (counted according to multiplicities). The ordered triple $(i_+(A), i_-(A), i_0(A))$ is called the *inertia* of A, and denoted by $\operatorname{In}(A)$, i.e., $\operatorname{In}(A) = (i_+(A), i_-(A), i_0(A))$. Obviously, $\operatorname{rank}(A) = i_+(A) + i_-(A)$.

Example 8.1.2 Inertias of Hermitian matrices.

(a) $A = \left(\begin{smallmatrix} 1 & 1 \\ 1 & 0 \end{smallmatrix}\right)$ has one positive eigenvalue $(1.618\cdots)$, one negative eigenvalue $(-0.618\cdots)$, and no zero eigenvalue. Thus, $\operatorname{In}(A) = (1, 1, 0)$.

(b) $B = \left(\begin{smallmatrix} 1 & i \\ -i & 1 \end{smallmatrix}\right)$ has one positive eigenvalue (2), no negative eigenvalue, and one zero eigenvalue. Thus, $\operatorname{In}(B) = (1, 0, 1)$.

(c) $C = \left(\begin{smallmatrix} 0 & i & 0 \\ -i & 1 & 1 \\ 0 & 1 & 0 \end{smallmatrix}\right)$ has one positive eigenvalue (2), one negative eigenvalue (-1), and one zero eigenvalue. Thus, $\operatorname{In}(C) = (1, 1, 1)$.

(d) If H is an $n \times n$ Hermitian matrix, then $i_+(H) + i_-(H) + i_0(H) = n$.

(e) $\operatorname{In}(I_n) = (n, 0, 0)$, $\operatorname{In}(-I_n) = (0, n, 0)$, and $\operatorname{In}(E_n) = (1, 0, n-1)$, where E_n is the all-ones matrix of order n. $\blacksquare$

Observation: Let A be an $n \times n$ Hermitian matrix. Then $A \geq 0$ if and only if $i_-(A) = 0$, and $A > 0$ if and only if $i_+(A) = n$. For $n \times n$ Hermitian matrices A and B, if $A \leq B$, then $i_+(A) \leq i_+(B)$ (Problem 33).

The inertias of a nonsingular Hermitian matrix and its inverse are the same since their (necessarily nonzero) eigenvalues are reciprocals of each other. The inertias of similar Hermitian matrices are the same because their eigenvalues are the same. The inertias of *-congruent matrices are also the same (* means the conjugate transpose operation); this is the so-called *Sylvester's law of inertia*. We say that two matrices $A, B \in \mathbb{M}_n$ are **-congruent* if there is a nonsingular matrix $S \in \mathbb{M}_n$ such that $B = S^*AS$.

Theorem 8.1.3 (Sylvester's Law of Inertia) *Let A and B be Hermitian matrices of the same order. Then A and B are *-congruent if and only if they have the same inertia, that is,* $\text{In}(A) = \text{In}(B)$.

Proof. The spectral theorem ensures that there exist positive diagonal matrices E and F with orders $i_+(A)$ and $i_-(A)$, respectively, such that A is unitarily similar (*-congruent) to $E \oplus (-F) \oplus 0_{i_0(A)}$. Setting $G = E^{-1/2} \oplus F^{-1/2} \oplus I_{i_0(A)}$ and upon computation, we have

$$G^* \left(E \oplus (-F) \oplus 0_{i_0(A)}\right) G = I_{i_+(A)} \oplus (-I_{i_-(A)}) \oplus 0_{i_0(A)}.$$

A similar argument shows that B is *-congruent to $I_{i_+(B)} \oplus (-I_{i_-(B)}) \oplus 0_{i_0(B)}$. If $\text{In}(A) = \text{In}(B)$, transitivity of *-congruence implies that A and B are *-congruent, i.e., $B = S^*AS$ for some nonsingular S.

Conversely, suppose that $B = S^*AS$ for some nonsingular matrix S. Let $A = U^*MU$ and $B = VNV^*$, where U and V are nonsingular matrices, $M = I_{i_+(A)} \oplus (-I_{i_-(A)}) \oplus 0_{i_0(A)}$ and $N = I_{i_+(B)} \oplus (-I_{i_-(B)}) \oplus 0_{i_0(B)}$. Then $B = S^*AS$ implies that $N = W^*MW$, where $W = USV$ is a nonsingular matrix. Denote the first $i_+(A)$ rows of W by W_1 and the rest rows by W_2. Then $N = W^*MW$ reveals $N = {W_1}^*W_1 - W_2^*(I_{i_-(A)} \oplus 0)W_2$. So $N \leq {W_1}^*W_1$. Since $N = I_{i_+(B)} \oplus (-I_{i_-(B)}) \oplus 0_{i_0(B)}$, the leading principal submatrix of ${W_1}^*W_1$ corresponding to $I_{i_+(B)}$ is positive definite. Thus, $i_+(B) = i_+(N) \leq i_+({W_1}^*W_1) = \text{rank}({W_1}^*W_1) = \text{rank}(W_1) = i_+(A)$. So, $i_-(B) \geq i_-(A)$ because A and B have the same rank. On the other hand, applying the above argument to $-A$ and $-B$ and noting that $i_+(-H) = i_-(H)$ for any Hermitian matrix H, we conclude that $\text{In}(A) = \text{In}(B)$. ∎

Below is the inertia additive property with Schur complement.

Theorem 8.1.4 *Let A be a Hermitian matrix. If A_{11} is a nonsingular principal submatrix of A and $\widetilde{A_{11}}$ is the Schur complement of A_{11} in A, then*

$$\text{In}(A) = \text{In}(A_{11}) + \text{In}(\widetilde{A_{11}}).$$

Proof. By simultaneous row and column permutations (i.e., via permutation similarity, if necessary), we can place A_{11} in the upper-left corner, that is, we may assume that $A = \begin{pmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{pmatrix}$. Let $G = \begin{pmatrix} I & -A_{11}^{-1}A_{12} \\ 0 & I \end{pmatrix}$. Then

$$G^*AG = \begin{pmatrix} A_{11} & 0 \\ 0 & \widetilde{A_{11}} \end{pmatrix},$$

implying $i_+(G^*AG) = i_+(A_{11}) + i_+(\widetilde{A_{11}})$, $i_-(G^*AG) = i_-(A_{11}) + i_-(\widetilde{A_{11}})$. Thus, $\text{In}(G^*AG) = \text{In}(A_{11}) + \text{In}(\widetilde{A_{11}})$ (the triples are added as vectors). By Sylvester's law of inertia, $\text{In}(A) = \text{In}(G^*AG) = \text{In}(A_{11}) + \text{In}(\widetilde{A_{11}})$. ∎

Example 8.1.3 Let X be an $m \times n$ matrix with rank r and let

$$D = \begin{pmatrix} I_m & X \\ X^* & X^*X \end{pmatrix}, \; F = \begin{pmatrix} I_m & X \\ X^* & 0 \end{pmatrix}, \; G = \begin{pmatrix} 0 & X \\ X^* & 0 \end{pmatrix}, \; H = \begin{pmatrix} I_m & X \\ X^* & -X^*X \end{pmatrix}.$$

The above block matrices are all $(m+n) \times (m+n)$ and Hermitian. We have

$$\text{In}(D) = (m, 0, n), \; \text{In}(F) = \text{In}(H) = (m, r, n-r), \; \text{In}(G) = (r, r, m+n-2r). \; ∎$$

Problems

1. What are the differences in the spectral decompositions of normal, Hermitian, positive semidefinite, and unitary matrices?
2. Show that the diagonal entries of a Hermitian matrix are all real.
3. Show that $A^* + A$ and $i(A - A^*)$ are Hermitian for square matrices A.
4. If A and B are Hermitian matrices of the same order, show that $A+B$ and $A-B$ are Hermitian matrices. What about AB and ABA?
5. Let A be an $n \times n$ Hermitian matrix. Show that C^*AC is also a Hermitian matrix for any $n \times m$ matrix C.
6. If A and B are Hermitian matrices of the same order, show that $AB = 0$ if and only if $BA = 0$. Is this true for general matrices?
7. Is a matrix similar to a Hermitian matrix necessarily Hermitian? Is a matrix unitarily similar to a Hermitian matrix necessarily Hermitian?
8. Show that a square matrix A is *skew-Hermitian*, that is, $A^* = -A$, if and only if $A = iB$ for some Hermitian matrix B.
9. Let A be a square matrix. If A^2 is Hermitian, show that the eigenvalues of A are real ($a \in \mathbb{R}$) or pure imaginary (bi, $b \in \mathbb{R}$).
10. Show that the eigenvectors associated with distinct eigenvalues of a Hermitian matrix are orthogonal. (Recall from Sect. 1.3, Chap. 1 that the eigenvectors associated with distinct eigenvalues of a square matrix are linearly independent. Orthogonality implies linear independence.)
11. Let A be an $n \times n$ positive definite (real or complex) matrix and let B be an $n \times n$ real skew-symmetric matrix. Show that rank(B^TAB) is an even number. Can the conditions be changed to A being positive semidefinite or B being (complex) skew-Hermitian?

12. If $A \in \mathbb{M}_n$ is unitary and Hermitian, show that $A^2 = I_n$.

13. Let $A \in \mathbb{M}_n$ be a projection (idempotent) matrix, that is, $A^2 = A$. Show that $\|Ax\| \leq \|x\|$ for all $x \in \mathbb{C}^n$ if and only if A is Hermitian.

14. If $\lambda_1, \ldots, \lambda_n$ are the eigenvalues of an $n \times n$ Hermitian matrix A, what are the singular eigenvalues of A? If $A = U^* \operatorname{diag}(\lambda_1, \ldots, \lambda_n)U$ is a spectral decomposition of A, where U is unitary, find $|A| = (A^*A)^{1/2}$.

15. Let $A \in \mathbb{M}_n$. If $A^k = A^{k+1}$ for some positive integer k, show that

$$\operatorname{tr} A = \operatorname{tr} A^2 = \cdots = \operatorname{tr} A^n = \cdots .$$

16. Let A be a nonsingular skew-Hermitian matrix. Show that $A^2 + A^{-1}$ is nonsingular and that $B = (A^2 - A^{-1})(A^2 + A^{-1})^{-1}$ is unitary.

17. Show that a square complex matrix A can be uniquely written as

$$A = B + iC = S - iT,$$

where B and C are Hermitian, and S and T are skew-Hermitian.

18. Let $A = B + iC$, where B and C are $n \times n$ Hermitian matrices such that $BC = CB = 0$. Show that $A^k = B^k + i^k C^k$ for all positive integers k.

19. Show the implication (4) $\Rightarrow$ (1) in Theorem 8.1.2.

20. If A is Hermitian, show that A^2 is positive semidefinite.

21. Find a 2×2 unitary matrix U such that U^*HU is diagonal, where

$$H = \begin{pmatrix} 1 & -i \\ i & 1 \end{pmatrix}.$$

22. Show that *-congruence is an equivalence relation $\sim$ of Hermitian matrices, that is, for $n \times n$ Hermitian matrices A, B, and C, (i) $A \sim A$, (ii) if $A \sim B$, then $B \sim A$, (iii) if $A \sim B$ and $B \sim C$, then $A \sim C$.

23. Let A and B be Hermitian matrices of the same order. If $AB - BA$ and $A - B$ commute, show that A and B commute.

24. Let A be an n-square matrix. Show that A is Hermitian if and only if

$$\langle Ax, y\rangle = \langle x, Ay\rangle \ \text{ (i.e., } y^*Ax = y^*A^*x) \ \text{ for all } x, y \in \mathbb{C}^n,$$

and if and only if

$$\langle Ax, x\rangle = \langle x, Ax\rangle \ \text{ (i.e., } x^*Ax = x^*A^*x) \ \text{ for all } x \in \mathbb{C}^n,$$

Is it true that A is Hermitian if

$$\langle Ax, x\rangle = \langle x, Ax\rangle \ \text{ (i.e., } x^*Ax = x^*A^*x) \ \text{ for all } x \in \mathbb{R}^n?$$

25. Let A be an n-square matrix. Show that

    (a) $\mathrm{tr}(AX) = 0$ for all Hermitian $X \in \mathbb{M}_n(\mathbb{C})$ if and only if $A = 0$.

    (b) $\mathrm{tr}(AX) \in \mathbb{R}$ for all Hermitian $X \in \mathbb{M}_n(\mathbb{C})$ if and only if $A = A^*$.

    (c) If A is a Hermitian matrix such that $\mathrm{tr}A \geq \Re\big(\mathrm{tr}(AU)\big)$ for all unitary $U \in \mathbb{M}_n(\mathbb{C})$, then A is positive semidefinite.

26. Show that the rank of a Hermitian matrix is equal to the number of nonzero eigenvalues of the matrix and that the rank of a general matrix A is equal to the number of nonzero singular values of A which may be strictly greater than the number of nonzero eigenvalues.

27. Show that if A is a Hermitian matrix with rank r, then A has a nonsingular principal submatrix of order r. How about a general matrix?

28. Let A be a Hermitian matrix with rank r. Show that all nonzero $r \times r$ principal minors of A have the same (positive or negative) sign.

29. Let $A = A^*$ and $\mathrm{tr}A = 0$. If the sum of all 2×2 principal minors of A is zero, show that $A = 0$. [Hint: Use Problem 16 of Sect. 5.4.]

30. Show that the numbers of positive, negative, and zero eigenvalues in Theorem 8.1.1 do not depend on the choice of unitary matrix U.

31. Find the rank and inertia of each of the following Hermitian matrices:

$$\begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & 0 \\ 1 & 0 & 0 \end{pmatrix}, \quad \begin{pmatrix} 0 & 0 & i \\ 0 & 2 & 0 \\ -i & 0 & 0 \end{pmatrix}, \quad \begin{pmatrix} 1 & 0 & 1 \\ 0 & 2 & 0 \\ 1 & 0 & 1 \end{pmatrix}.$$

32. Let A be a Hermitian matrix and let B be a principal submatrix of A. Show that $i_\pm(B) \leq i_\pm(A)$. Is it true that $i_0(B) \leq i_0(A)$?

33. Let A and B be Hermitian matrices of the same order. Show that (a) if $A \leq B$, then $i_+(A) \leq i_+(B)$, and (b) if $i_\pm(A) \leq i_\pm(B)$ and if $\mathrm{rank}(A) = \mathrm{rank}(B)$, then A and B are *-congruent.

34. Let $H \in \mathbb{M}_n$ be a Hermitian matrix. Show that for any $X \in \mathbb{M}_{n\times m}$,

    (a) $i_+(X^*HX) \leq i_+(H)$.

    (b) $i_-(X^*HX) \leq i_-(H)$.

    (c) $i_0(X^*HX) \geq i_0(H)$ if $m \geq n$.

    (d) $i_\pm(X^*HX) \leq \mathrm{rank}(X)$.

35. Let H be an $n \times n$ Hermitian matrix with rank r. The *signature* of H is defined to be $s = i_+(H) - i_-(H)$. Show that the inertia $\mathrm{In}(H)$ is uniquely determined by the rank r and the signature s of H.

36. Let $A = \left(\begin{smallmatrix} 0 & 1 \\ 4 & 0 \end{smallmatrix}\right)$ and $S = \left(\begin{smallmatrix} 1 & 1 \\ 0 & i \end{smallmatrix}\right)$. Find the eigenvalues of A, S^*AS, and $S^{-1}AS$. Show that $\det(T^*AT) < 0$ for any nonsingular $T \in \mathbb{M}_2$.

37. Let $A = \left(\begin{smallmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{smallmatrix}\right)$ and $B = \left(\begin{smallmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ -1 & 0 & 0 \end{smallmatrix}\right)$. Find the eigenvalues of B^*AB.

38. Let A be an $n \times n$ (not necessarily Hermitian) matrix. If the eigenvalues of A are all positive, does it follow that the eigenvalues of S^*AS are also all positive (where S is any n-square nonsingular matrix)?

39. Let A and B be $n \times n$ nonsingular Hermitian matrices. If the smallest eigenvalue $\lambda_{\min}(B^{-1}A) > 0$, show that $A \geq B \Leftrightarrow B^{-1} \geq A^{-1}$.

40. Let A be a Hermitian matrix with all eigenvalues in $[-1, 1]$. Show that

$$A^p\big(I - A^q\big)^{1/k} = \big(I - A^q\big)^{1/k} A^p, \quad p,\, q,\, k = 1, 2, \ldots.$$

41. Compute the inertias of the following partitioned matrices in which I is the $n \times n$ identity matrix and A and B are any $n \times n$ matrices:

$$\begin{pmatrix} I & 0 \\ 0 & -I \end{pmatrix}, \quad \begin{pmatrix} I & I \\ I & 0 \end{pmatrix}, \quad \begin{pmatrix} I + A^*A & I + A^*B \\ I + B^*A & I + B^*B \end{pmatrix}.$$

42. Show that for any $m \times n$ matrix A,

$$\operatorname{In}\begin{pmatrix} I_m & A \\ A^* & I_n \end{pmatrix} = (m, 0, 0) + \operatorname{In}(I_n - A^*A) = (n, 0, 0) + \operatorname{In}(I_m - AA^*).$$

43. Let $A, B \in \mathbb{M}_n$ and let $H = \left(\begin{smallmatrix} I - A^*A & I - A^*B \\ I - B^*A & I - B^*B \end{smallmatrix}\right)$. Show that the $2n$ eigenvalues of H are all real. If A and B are strictly contractive, show that exactly n eigenvalues of H are positive (including multiplicities).

44. Let $A \in \mathbb{M}_n$. If $A^2 = AA^*$, show that A is Hermitian.

45. Let $A \in \mathbb{M}_n$.

    (a) If $A^3 = AA^*A$, show that A is Hermitian.

    (b) If $\operatorname{tr}(A^3) = \operatorname{tr}(AA^*A)$, must A be Hermitian?

    (c) If $A^3 = A^*AA^*$, show that A is skew-Hermitian.

    (d) If $A^3 = AAA^*$, must A be Hermitian (or skew-Hermitian)?

    (e) Show that $A^3 = |A|^3$ (where $|A| = (A^*A)^{\frac{1}{2}}$) if and only if A is normal and the eigenvalues of A are $r\omega$, where $r \geq 0$ and $\omega^3 = 1$.

[Note: See more related results via majorization in Sect.10.6, Chap. 10.]

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8.2 The Product of Hermitian Matrices

In this section we study the product of Hermitian matrices. Note that the product of two Hermitian matrices is not necessarily Hermitian in general.

Example 8.2.1 Product of Hermitian matrices.

(a) Let $A = \left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$, $B = \left(\begin{smallmatrix} 1 & 0 \\ 0 & -1 \end{smallmatrix}\right)$. A and B are Hermitian. $AB = \left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right)$. The eigenvalues of AB are $\pm i$ (where $i = \sqrt{-1}$). AB is not Hermitian.

(b) Let $A = \left(\begin{smallmatrix} 1 & 0 \\ 0 & -2 \end{smallmatrix}\right)$, $B = \left(\begin{smallmatrix} 1 & 1 \\ 1 & -1 \end{smallmatrix}\right)$. A and B are Hermitian. $AB = \left(\begin{smallmatrix} 1 & 1 \\ -2 & 2 \end{smallmatrix}\right)$. The eigenvalues of AB are $\frac{1}{2}(3 \pm \sqrt{7}\, i)$. $\operatorname{tr}(AB) = 3$ is real.

(c) Let $A = \left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$, $B = \left(\begin{smallmatrix} 1 & 0 \\ 0 & -1 \end{smallmatrix}\right)$, $C = \left(\begin{smallmatrix} 1 & i \\ -i & 1 \end{smallmatrix}\right)$. A, B, and C are Hermitian. $ABC = \left(\begin{smallmatrix} i & -1 \\ 1 & i \end{smallmatrix}\right)$. The eigenvalues of ABC are $2i$ and 0. $\operatorname{tr}(ABC) = 2i$.

(d) Let $A, B, C \in \mathbb{M}_n$. If A, B, C are Hermitian, then $\det(ABC)$ is real.

(e) Let $A, B \in \mathbb{M}_n$. $(AB)^* = B^*A^*$. When is AB Hermitian? ∎

The above examples (a) and (b) suggest that the trace (sum of all eigenvalues) of the product of two Hermitian matrices is a real number although the product may not be Hermitian. This is in fact true and is seen as follows. If A and B are Hermitian (of the same order), then

$$\operatorname{tr}(AB) = \operatorname{tr}(A^*B^*) = \operatorname{tr}(BA)^* = \overline{\operatorname{tr}(BA)} = \overline{\operatorname{tr}(AB)}.$$

That is, $\operatorname{tr}(AB)$ is real. This need not be true for three Hermitian matrices.

We first show a trace inequality for the product of two Hermitian matrices, and we then discuss when a matrix product is Hermitian.

Theorem 8.2.1 *Let A and B be n-square Hermitian matrices. Then*

$$\operatorname{tr}(AB)^2 \le \operatorname{tr}(A^2B^2). \tag{8.2}$$

Equality occurs if and only if A and B commute, namely, $AB = BA$.

Proof 1. Let $C = AB - BA$. Using the fact that $\operatorname{tr}(XY) = \operatorname{tr}(YX)$ for any square matrices X and Y of the same order, we compute

$$\begin{aligned} \operatorname{tr}(C^*C) &= \operatorname{tr}(AB - BA)^*(AB - BA) = \operatorname{tr}(BA - AB)(AB - BA) \\ &= \operatorname{tr}(BA^2B) + \operatorname{tr}(AB^2A) - \operatorname{tr}(BABA) - \operatorname{tr}(ABAB) \\ &= 2\operatorname{tr}(A^2B^2) - 2\operatorname{tr}(AB)^2. \end{aligned}$$

Note that $\operatorname{tr}(A^2B^2)$ is nonnegative (as A^2 and B^2 are PSD) and $\operatorname{tr}(AB)^2 = \operatorname{tr}(A(BAB))$ is real. The inequality (8.2) follows immediately from the fact that $\operatorname{tr}(C^*C) \ge 0$ with equality if and only if $C = 0$, that is, $AB = BA$.

Proof 2. Since for any unitary matrix $U \in \mathbb{M}_n$, U^*AU is also Hermitian, inequality (8.2) holds for all Hermitian A and B if and only if

$$\operatorname{tr}\big((U^*AU)B\big)^2 \le \operatorname{tr}\big((U^*AU)^2B^2\big).$$

Thus, we assume $A = \operatorname{diag}(a_1, \ldots, a_n)$, where a_i's are all real. Then

$$\begin{aligned} \operatorname{tr}(A^2B^2) - \operatorname{tr}(AB)^2 &= \sum_{i,j} a_i^2|b_{ij}|^2 - \sum_{i,j} a_ia_j|b_{ij}|^2 \\ &= \sum_{i<j} (a_i - a_j)^2|b_{ij}|^2 \ge 0. \end{aligned}$$

Equality occurs if and only if $a_ib_{ij} = a_jb_{ij}$ for all i, j. This implies $AB = BA$.

Alternatively, the equality case can be shown by using the fact that a matrix X is Hermitian if and only if $\operatorname{tr} X^2 = \operatorname{tr}(XX^*)$ (Theorem 8.1.2).

Notice that the Hermiticity of A and B reveals

$$\operatorname{tr}(A^2B^2) = \operatorname{tr}(AABB) = \operatorname{tr}(ABBA) = \operatorname{tr}\big((AB)(AB)^*\big).$$

Thus,

$$\operatorname{tr}(AB)^2 = \operatorname{tr}(A^2B^2) \;\Rightarrow\; \operatorname{tr}(AB)^2 = \operatorname{tr}\big((AB)(AB)^*\big).$$

It follows that AB is Hermitian. Hence, $AB = (AB)^* = B^*A^* = BA$. ∎

Example 8.2.2 Product of Hermitian matrices.

(a) $\left(\begin{smallmatrix} i & 0 \\ 0 & 1 \end{smallmatrix}\right)$ cannot be equal to a product of two Hermitian matrices because the trace of a product of two Hermitian matrices need be real.

(b) $\left(\begin{smallmatrix} 0 & i \\ 1 & 0 \end{smallmatrix}\right)$ cannot be equal to a product of two Hermitian matrices because the determinant of a product of two Hermitian matrices need be real.

(c) $\left(\begin{smallmatrix} i & 0 \\ 0 & -i \end{smallmatrix}\right) = \left(\begin{smallmatrix} 0 & i \\ -i & 0 \end{smallmatrix}\right)\left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$ is a product of two Hermitian matrices.

(d) $\left(\begin{smallmatrix} 1 & 0 \\ -2i & 1 \end{smallmatrix}\right) = \left(\begin{smallmatrix} 0 & i \\ -i & 1 \end{smallmatrix}\right)\left(\begin{smallmatrix} 1 & i \\ -i & 0 \end{smallmatrix}\right)$ is a product of two Hermitian matrices. ∎

The product of two Hermitian matrices is Hermitian if and only if the two matrices commute (Problem 7). We are interested in the question: When can a given matrix be a product of two Hermitian matrices?

Let A be a given square matrix. If $A = BC$ is the product of Hermitian matrices B and C, then $A^* = (BC)^* = CB$. If B is nonsingular, then

$$A = BC = B(CB)B^{-1} = BA^*B^{-1},$$

so, A is similar to A^*. (What if B is singular?) This is in fact also a sufficient condition for a matrix to be a product of two Hermitian matrices. By this, because $A = \left(\begin{smallmatrix} 1 & 0 \\ -2i & 1 \end{smallmatrix}\right)$ and $A^* = \left(\begin{smallmatrix} 1 & 2i \\ 0 & 1 \end{smallmatrix}\right)$ have the same Jordan form $\left(\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}\right)$, A is similar to A^*. Thus, A is a product of two Hermitian matrices.

Theorem 8.2.2 *A square matrix A is a product of two Hermitian matrices if and only if A is similar to A^* (equivalently, A is similar to a real matrix).*

Proof. First observe that if $A \in \mathbb{M}_n$ is similar to a product of two Hermitian matrices, then A is equal to a product of two Hermitian matrices because

$$A = P^{-1}H_1H_2P \quad \Rightarrow \quad A = P^{-1}H_1(P^{-1})^*P^*H_2P. \tag{8.3}$$

Necessity: Let $A = BC$, where B and C are Hermitian. Then

$$AB = BCB = B(BC)^* = BA^*$$

and $A^2B = A(AB) = A(BA^*) = (AB)A^* = B(A^*)^2$. Inductively, we have

$$A^kB = B(A^*)^k, \quad k = 1, 2, \ldots. \tag{8.4}$$

Since $A = BC$, by the above observation and considering a Jordan canonical form of A, without loss of generality (Problem 10), we assume

$$A = \begin{pmatrix} J & 0 \\ 0 & K \end{pmatrix},$$

where J contains the Jordan blocks of eigenvalue 0 (if A is singular, or J is absent), and K contains the Jordan blocks of nonzero eigenvalues (if any, or K is absent). Note that J is nilpotent and K is invertible.

Partition B and C conformally with A as

$$B = \begin{pmatrix} L & M \\ M^* & N \end{pmatrix}, \quad C = \begin{pmatrix} P & Q \\ Q^* & R \end{pmatrix}.$$

Then (8.4) implies that for each positive integer k

$$K^kM^* = M^*(J^*)^k.$$

Notice that $(J^*)^k = 0$ when $k \geq n$ (because J is nilpotent). It follows that $M = 0$ (because K is nonsingular). Thus, $A = BC$ is the same as

$$\begin{pmatrix} J & 0 \\ 0 & K \end{pmatrix} = \begin{pmatrix} L & 0 \\ 0 & N \end{pmatrix} \begin{pmatrix} P & Q \\ Q^* & R \end{pmatrix}.$$

This yields $K = NR$, and hence N and R are nonsingular.

Taking $k = 1$ in (8.4), we have

$$\begin{pmatrix} J & 0 \\ 0 & K \end{pmatrix} \begin{pmatrix} L & 0 \\ 0 & N \end{pmatrix} = \begin{pmatrix} L & 0 \\ 0 & N \end{pmatrix} \begin{pmatrix} J^* & 0 \\ 0 & K^* \end{pmatrix},$$

which gives $KN = NK^*$. Since N is invertible, we obtain

$$N^{-1}KN = K^*.$$

So, K is similar to K^*. Because any square matrix is similar to its transpose (Theorem 3.5.1 (1)), J is similar to $J^T = J^*$. Thus, A is similar to A^*.

Sufficiency: We show that if A is similar to A^*, then A can be expressed as a product of two Hermitian matrices. For this, it is sufficient to show that A is similar to a product of two Hermitian matrices (see (8.3)).

Recall from Theorem 3.5.1 that A is similar to A^* if and only if the Jordan blocks of the non-real eigenvalues λ (if any) of A occur in conjugate pairs. We show that the 2×2 block case of the paired Jordan blocks

$$\begin{pmatrix} J(\lambda) & 0 \\ 0 & J(\overline{\lambda}) \end{pmatrix},$$

where $J(\lambda)$ is a Jordan block with λ's on the diagonal, is similar to a product of two Hermitian matrices. This is seen as follows: The block matrices

$$\begin{pmatrix} J(\lambda) & 0 \\ 0 & J(\overline{\lambda}) \end{pmatrix} \quad \text{and} \quad \begin{pmatrix} J(\lambda) & 0 \\ 0 & (J(\overline{\lambda}))^T \end{pmatrix}$$

are similar (because any square matrix is similar to its transpose), and

$$\begin{pmatrix} J(\lambda) & 0 \\ 0 & (J(\overline{\lambda}))^T \end{pmatrix} = \begin{pmatrix} J(\lambda) & 0 \\ 0 & (J(\lambda))^* \end{pmatrix},$$

which is equal to a product of two Hermitian matrices:

$$\begin{pmatrix} 0 & J(\lambda) \\ (J(\lambda))^* & 0 \end{pmatrix} \begin{pmatrix} 0 & I \\ I & 0 \end{pmatrix}.$$

For the Jordan blocks $J(\lambda)$ with real λ, one verifies for the 3×3 case

$$J(\lambda) = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{pmatrix} \begin{pmatrix} 0 & 0 & \lambda \\ 0 & \lambda & 1 \\ \lambda & 1 & 0 \end{pmatrix}$$

and generalizes this for the Jordan blocks of any order (Problem 20). Thus, if A is similar to A^*, then A is similar to a product of Hermitian matrices.

Equivalently, by Theorem 3.5.3 (Sect. 3.5), A is similar to A^* if and only if A is similar to a real matrix. Therefore, A can be written as a product of two Hermitian matrices if and only if A is similar to a real matrix. ∎

Example 8.2.3 Product of two Hermitian matrices.

(a) $\left(\begin{smallmatrix} 1 & 2i \\ 0 & 1 \end{smallmatrix}\right)$ is a product of two Hermitian matrices. (Use Jordan form.)

(b) $\left(\begin{smallmatrix} 1 & 2i \\ 0 & i \end{smallmatrix}\right)$ is not a product of two Hermitian matrices. (Check eigenvalues.)

(c) For any square matrix A, by Theorem 3.5.3, $\overline{A}A$ is similar to a real matrix. Thus, $\overline{A}A$ is equal to a product of two Hermitian matrices.

(d) If the eigenvalues of $A \in \mathbb{M}_n$ are all real, then A is a product of two Hermitian matrices because the Jordan canonical form of A is real. ∎

Problems

1. Determine if each matrix is a product of two Hermitian matrices:
$$\begin{pmatrix} 0 & i \\ i & 0 \end{pmatrix}, \quad \begin{pmatrix} i & 0 \\ 0 & i \end{pmatrix}, \quad \begin{pmatrix} 1 & i \\ 0 & 1 \end{pmatrix}, \quad \begin{pmatrix} 0 & i \\ 1 & 1 \end{pmatrix}, \quad \begin{pmatrix} 1 & 1 \\ 0 & 2 \end{pmatrix}.$$

2. (a) Show that if $A, B \in \mathbb{M}_n$ are Hermitian and $AB = 0$, then $BA = 0$.
(b) Find example: $A, B \in \mathbb{M}_2$ satisfy $AB = 0$ but BA is not Hermitian.

3. Let A and B be $n \times n$ Hermitian matrices. Show that $AB + BA$ and $ABA - BAB$ are Hermitian, and that $AB - BA$ is skew-Hermitian.

4. Let A, B, and C be $n \times n$ Hermitian matrices. Prove or disprove
$$\operatorname{tr}(ABC) = \operatorname{tr}(BCA) \quad \text{or} \quad \operatorname{tr}(ABC) = \operatorname{tr}(CBA).$$
Is $\operatorname{tr}(ABC)$ necessarily real? How about $\det(ABC)$?

5. Let A and B be $n \times n$ Hermitian matrices. Show that $\operatorname{tr}(A^p B^q)$ and $\operatorname{tr}(AB)^r$ are real for all positive integers p, q, and r.

6. Show that the determinant of a product of (finitely many) Hermitian matrices of the same order is real. How about trace and eigenvalues?

7. Let A and B be $n \times n$ Hermitian matrices. Show that AB is Hermitian if and only if $AB = BA$. Does this hold true for normal matrices?

8. Show that AB and BA are similar, where $A, B \in \mathbb{M}_n$ are Hermitian.

9. Show that A and B below are normal and AB and BA are not similar.
$$A = \begin{pmatrix} 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}, \quad B = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}.$$

10. Give in detail the reason why the matrix A can be assumed to be a Jordan canonical form in the proof of Theorem 8.2.2.

11. Show that for any square matrices A and B of the same order,
$$\operatorname{tr}(AB - BA) = 0 \quad \text{and} \quad \operatorname{tr}(AB - BA)(AB + BA) = 0.$$

12. Let A and B be Hermitian matrices of the same order. Show that
$$|\operatorname{tr}(AB)| \le (\operatorname{tr}A^2)^{1/2} (\operatorname{tr}B^2)^{1/2} \le \operatorname{tr}\left(\frac{A^2 + B^2}{2}\right)$$
and
$$(\operatorname{tr}(A + B)^2)^{1/2} \le (\operatorname{tr}A^2)^{1/2} + (\operatorname{tr}B^2)^{1/2}.$$

13. Let A, B, and C be $n \times n$ Hermitian matrices. Show by example that

$$|\operatorname{tr}(ABC)| \leq |\operatorname{tr}(A^2B^2C^2)|^{1/2}$$

is not true in general. [Hint: Assume that A is a diagonal matrix.]

14. Let A be a square matrix with all eigenvalues being real (note that A need not be Hermitian), k of which are nonzero, $k \geq 1$. Show that

$$\frac{(\operatorname{tr}A)^2}{\operatorname{tr}A^2} \leq k \leq \operatorname{rank}(A).$$

15. Let $A, B \in \mathbb{M}_n$ be Hermitian matrices with positive traces. Show that

$$\frac{\operatorname{tr}(A+B)^2}{\operatorname{tr}(A+B)} \leq \frac{\operatorname{tr}A^2}{\operatorname{tr}A} + \frac{\operatorname{tr}B^2}{\operatorname{tr}B}.$$

16. Let A and B be Hermitian matrices of the same order. Show that there exists a unitary matrix U such that U^*AU and U^*BU are both diagonal if and only if $AB = BA$.

17. Let A and B be Hermitian matrices of the same order. If $AB = BA$, show that for any $a, b \in \mathbb{C}$ the eigenvalues of $aA + bB$ are in the form $a\lambda + b\mu$, where λ and μ are some eigenvalues of A and B, respectively.

18. Let $A \in \mathbb{M}_n$ and suppose that $S \in \mathbb{M}_n$ is an invertible matrix such that $S^{-1}AS = A^*$. Let $H_c = cS + \bar{c}S^*$, where $c \in \mathbb{C}$. Show that H_c and AH_c are Hermitian matrices. Also show that $A = (AH_c)H_c^{-1}$ is a product of two Hermitian matrices for some c such that H_c is invertible. Why does such an invertible matrix H_c exist?

19. Show that a matrix is diagonalizable (not necessarily unitarily diagonalizable) with real eigenvalues if and only if it can be written as a product of a positive definite matrix and a Hermitian matrix. For the singular case, is the product of a singular positive semidefinite matrix and a Hermitian matrix always diagonalizable?

20. Show that any square matrix is a product of two symmetric matrices.

21. Let $A \in \mathbb{M}_n$ be positive definite and let $B \in \mathbb{M}_n$ be Hermitian such that AB is a Hermitian matrix. Show that AB is positive definite if and only if the eigenvalues of B are all positive.

22. Show that a square matrix A is a product of two positive semidefinite matrices if and only if A is similar to a positive semidefinite matrix.

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8.3 The Min-max Principles and Interlacing Theorems

In this section we show some basic techniques by using vector spaces to derive eigenvalue (interlacing) inequalities for Hermitian matrices. The idea is to choose vectors in certain subspaces spanned by eigenvectors in order to obtain the min-max representations that yield the desired inequalities.

Let H be an $n \times n$ Hermitian matrix with (necessarily real) eigenvalues $\lambda_i(H)$, or simply λ_i (if only H is in the discussion), $i = 1, 2, \ldots, n$. By the spectral theorem (Theorem 8.1.1), there exists a unitary matrix U such that

$$U^*HU = \operatorname{diag}(\lambda_1, \lambda_2, \ldots, \lambda_n)$$

or

$$HU = U \operatorname{diag}(\lambda_1, \lambda_2, \ldots, \lambda_n).$$

The column vectors $u_1, u_2, \ldots, u_n$ of U are orthonormal eigenvectors of H corresponding to the eigenvalues $\lambda_1, \lambda_2, \ldots, \lambda_n$, respectively, that is,

$$Hu_i = \lambda_i u_i, \quad u_i^* u_j = \delta_{ij}, \quad i, j = 1, 2, \ldots, n, \tag{8.5}$$

where $\delta_{ij} = 1$ if $i = j$ and $\delta_{ij} = 0$ if $i \neq j$ (the *Kronecker delta*).

We arrange the eigenvalues of Hermitian matrices in decreasing order. If $H \in \mathbb{M}_n$ is a Hermitian matrix with eigenvalues $\lambda_i(H) = \lambda_i$, we have

$$\lambda_{\max} = \lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_n = \lambda_{\min}.$$

The following theorem is of fundamental importance to the rest of this chapter. The result, as well as the idea of the proof, is employed frequently.

Theorem 8.3.1 *Let H be an n-square Hermitian matrix. Let $u_1, u_2, \ldots, u_n$ be orthonormal eigenvectors of H corresponding to the (not necessarily distinct) eigenvalues $\lambda_1(H) \geq \lambda_2(H) \geq \cdots \geq \lambda_n(H)$ of H, respectively. Let $W = \operatorname{Span}\{u_p, \ldots, u_q\}$, $1 \leq p \leq q \leq n$. Then for any unit vector $x \in W$,*

$$\lambda_q(H) \leq x^*Hx \leq \lambda_p(H). \tag{8.6}$$

Proof. Let $x = x_p u_p + \cdots + x_q u_q$, where $x_p, \ldots, x_q \in \mathbb{C}$. By (8.5), we have

$$\begin{aligned} x^*Hx &= x^*(x_p Hu_p + \cdots + x_q Hu_q) \\ &= x^*(\lambda_p x_p u_p + \cdots + \lambda_q x_q u_q) \\ &= \lambda_p x_p x^* u_p + \cdots + \lambda_q x_q x^* u_q \\ &= \lambda_p |x_p|^2 + \cdots + \lambda_q |x_q|^2. \end{aligned}$$

The desired inequality follows at once because $|x_p|^2 + \cdots + |x_q|^2 = 1$. ∎

Example 8.3.1 Let $A = U\,\mathrm{diag}(-1,2,3,4)U^*$, where U is a unitary matrix. Let the columns of U be u_1, u_2, u_3, u_4. Then for any nonzero $v \in \mathrm{Span}\{u_2, u_3\}$ with length (norm) $\|v\|$, $\frac{1}{\|v\|}v$ is a unit vector. It follows that

$$2\|v\|^2 \le v^*Av \le 3\|v\|^2. \ \blacksquare$$

Theorem 8.3.2 (Rayleigh-Ritz Theorem) *If $H \in \mathbb{M}_n$ is Hermitian, then*

$$\lambda_{\min}(H) = \lambda_n(H) = \min_{x^*x=1} x^*Hx \ \text{ and } \ \lambda_{\max}(H) = \lambda_1(H) = \max_{x^*x=1} x^*Hx.$$

Proof. The eigenvectors of H in (8.5) form an orthonormal basis for $\mathbb{C}^n$. By (8.6), it is sufficient to observe that

$$\lambda_{\min}(H) = u_n^*Hu_n \quad \text{and} \quad \lambda_{\max}(H) = u_1^*Hu_1. \ \blacksquare$$

Example 8.3.2 The Rayleigh–Ritz theorem implies the following statements: For Hermitian matrices H, K (both $n \times n$), and M (with $n \times n$ blocks),

(a) $\lambda_{\min}(H) = \min_{x\neq 0} \frac{x^*Hx}{x^*x}$ and $\lambda_{\max}(H) = \max_{x\neq 0} \frac{x^*Hx}{x^*x}$.

(b) $\lambda_n(H) + \lambda_n(K) \le \lambda_n(H+K) \le \lambda_1(H+K) \le \lambda_1(H) + \lambda_1(K)$.

(c) If $M = \left(\begin{smallmatrix} A & B \\ B^* & C \end{smallmatrix}\right)$, then $\lambda_{\min}(M) \le \lambda_{\min}(A) \le \lambda_{\max}(A) \le \lambda_{\max}(M)$. $\blacksquare$

Recall the dimension identity (Theorem 1.1.1 of Sect. 1.1): If S_1 and S_2 are subspaces of a vector space of finite dimension, say n, then

$$\dim(S_1 \cap S_2) = \dim S_1 + \dim S_2 - \dim(S_1 + S_2).$$

It follows that S_1 and S_2 contain a common nonzero vector if

$$\dim S_1 + \dim S_2 > n. \tag{8.7}$$

Note that (Problem 29 of Sect. 1.1) for three subspaces S_1, S_2, and S_3,

$$\dim(S_1 \cap S_2 \cap S_3) \ge \dim S_1 + \dim S_2 + \dim S_3 - 2n. \tag{8.8}$$

We use these inequalities to obtain the Courant-Fischer min-max theorem and derive eigenvalue inequalities for Hermitian matrices.

Theorem 8.3.3 (Courant-Fischer Minimax Principle) *Let H be an n-square Hermitian matrix and let S represent an arbitrary subspace of $\mathbb{C}^n$. Then*

$$\begin{aligned}
\lambda_k(H) &= \max_{(\dim S=k)} \ \min_{(x\in S,\, x^*x=1)} x^*Hx \\
&= \max_{(\dim S=n-k)} \ \min_{(x\in S^\perp,\, x^*x=1)} x^*Hx \\
&= \min_{(\dim S=n-k+1)} \ \max_{(x\in S,\, x^*x=1)} x^*Hx \\
&= \min_{(\dim S=k-1)} \ \max_{(x\in S^\perp,\, x^*x=1)} x^*Hx.
\end{aligned}$$

Proof. We show the first max-min representation of $\lambda_k(H)$. The second one follows from the first one immediately as $\dim S^{\perp} = k$ if $\dim S = n - k$. The representations in the min-max forms are proven similarly.

Let u_i be orthonormal eigenvectors belonging to λ_i, $i = 1, 2, \ldots, n$. Let

$$S_1 = \text{Span}\{u_k, \ldots, u_n\}, \quad \dim S_1 = n - k + 1,$$

and let $S_2 = S$ be any subspace (of $\mathbb{C}^n$) of dimension k. By (8.7), there exists a vector x such that $x \in S_1 \cap S_2$, $x^*x = 1$, and for this x, by (8.6), $\lambda_k \geq x^*Hx$. Thus, for any subspace S of dimension k, we have

$$\lambda_k \geq \min_{v \in S,\ v^*v=1} v^*Hv.$$

It follows that

$$\lambda_k \geq \max_{(\dim S = k)} \ \min_{(v \in S,\ v^*v=1)} v^*Hv.$$

We now show the reverse inequality. Let $S = \text{Span}\{u_1, u_2, \ldots, u_k\}$, which has dimension k. For any unit vector $v \in S$, by (8.6), we have $v^*Hv \geq \lambda_k$ and $u_k^*Hu_k = \lambda_k$. Thus, for $S = \text{Span}\{u_1, \ldots, u_k\}$, we arrive at

$$\min_{v \in S,\ v^*v=1} v^*Hv \geq \lambda_k,$$

implying

$$\max_{(\dim S = k)} \ \min_{(v \in S,\ v^*v=1)} v^*Hv \geq \lambda_k. \ \blacksquare$$

Example 8.3.3 Let $H = \begin{pmatrix} 3 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & -1 \end{pmatrix}$. Then $\lambda_1 = 4$, $\lambda_2 = 3$, $\lambda_3 = -1$, and

$$\begin{aligned}
\lambda_1 &= \max_{(\dim S = 1)} \ \min_{(x \in S,\ x^*x=1)} x^*Hx = \max_{x \in \mathbb{C}^3,\ x^*x=1} x^*Hx. \\
\lambda_2 &= \max_{(\dim S = 2)} \ \min_{(x \in S,\ x^*x=1)} x^*Hx. \\
\lambda_3 &= \max_{(\dim S = 3)} \ \min_{(x \in S,\ x^*x=1)} x^*Hx = \min_{x \in \mathbb{C}^3,\ x^*x=1} x^*Hx. \ \blacksquare
\end{aligned}$$

We next present an important property about the eigenvalues of a Hermitian matrix and its principal submatrices. The theorem is usually referred to as the *Cauchy* (or *Poincaré* or *Sturm*) *eigenvalue interlacing theorem.* It states, simply put, that the eigenvalues of a principal submatrix of a Hermitian matrix interlace those of the Hermitian matrix. This result is used to obtain many eigenvalue and singular value inequalities.

Theorem 8.3.4 (Eigenvalue Interlacing Theorem) *Let $H \in \mathbb{M}_n$ be a Hermitian matrix with eigenvalues $\lambda_n \leq \cdots \leq \lambda_2 \leq \lambda_1$ and let $A \in \mathbb{M}_{n-1}$ be a principal submatrix of H with eigenvalues $\mu_{n-1} \leq \cdots \leq \mu_2 \leq \mu_1$. Then*

$$\lambda_n \leq \mu_{n-1} \leq \lambda_{n-1} \leq \cdots \leq \lambda_2 \leq \mu_1 \leq \lambda_1.$$

More generally, if A is an $m \times m$ principal submatrix of H, $1 \le m \le n$, and if the eigenvalues of A are $\mu_m \le \cdots \le \mu_2 \le \mu_1$, then

$$\lambda_{k+n-m} \le \mu_k \le \lambda_k, \quad k = 1, 2, \ldots, m.$$

Proof. We show the general case where A is an $m \times m$ principal submatrix of H. By permutation similarity (which leaves eigenvalues unchanged), we assume that A is located in the upper-left corner, that is, $H = \left(\begin{smallmatrix} A & B \\ B^* & C \end{smallmatrix}\right)$.

Let $u_1, u_2, \ldots, u_n$ be orthonormal eigenvectors of H associated with the eigenvalues $\lambda_1, \lambda_2, \ldots, \lambda_n$, respectively. Likewise, let $v_1, v_2, \ldots, v_m$ be orthonormal eigenvectors of A corresponding to $\mu_1, \mu_2, \ldots, \mu_m$. Thus,

$$Hu_i = \lambda_i u_i, \quad u_i^* u_j = \delta_{ij}, \quad i, j = 1, 2, \ldots, n, \ u_i \in \mathbb{C}^n,$$

$$Av_i = \mu_i v_i, \quad v_i^* v_j = \delta_{ij}, \quad i, j = 1, 2, \ldots, m, \ v_i \in \mathbb{C}^m,$$

where $\delta_{ij} = 1$ if $i = j$ and 0 otherwise (the Kronecker delta). Let

$$w_i = \begin{pmatrix} v_i \\ 0 \end{pmatrix} \in \mathbb{C}^n, \quad i = 1, 2, \ldots, m.$$

For each k, $1 \le k \le m$, we set

$$S_1 = \operatorname{Span}\{u_k, \ldots, u_n\} \subseteq \mathbb{C}^n$$

and

$$S_2 = \operatorname{Span}\{w_1, \ldots, w_k\} \subseteq \mathbb{C}^n.$$

Then

$$\dim S_1 = n - k + 1 \quad \text{and} \quad \dim S_2 = k.$$

There exists a unit vector $x \in S_1 \cap S_2$. Since $x \in S_2$, we can write $x = \left(\begin{smallmatrix} y \\ 0 \end{smallmatrix}\right)$, where $y \in \operatorname{Span}\{v_1, \ldots, v_k\}$, and y is a unit vector in $\mathbb{C}^m$. By (8.6), we get

$$\lambda_k \ge x^* H x = y^* A y \ge \mu_k. \tag{8.9}$$

An application of inequalities (8.9) to $-H$ reveals $\mu_k \ge \lambda_{k+n-m}$. ∎

Alternatively, one may show the interlacing inequalities by considering the adjugate $\operatorname{adj}(tI - H) = \det(tI - H)(tI - H)^{-1}$ of $tI - H$, where t is a real parameter that is not equal to $\lambda_1, \lambda_2, \ldots, \lambda_n$.

Let $H = U^* \operatorname{diag}(\lambda_1, \lambda_2, \ldots, \lambda_n) U$ be a spectral decomposition of H, where $U = (u_{ij})$ is unitary. Assume that A has order $n - 1$ and is in the upper-left corner of H. Computing the (n, n)-entry of $(tI - H)^{-1}$ yields

$$\frac{\det(tI_{n-1} - A)}{\det(tI_n - H)} = \frac{|u_{1n}|^2}{t - \lambda_1} + \frac{|u_{2n}|^2}{t - \lambda_2} + \cdots + \frac{|u_{nn}|^2}{t - \lambda_n}. \tag{8.10}$$

Analyze the continuous function $\det(tI_{n-1} - A)$ in (8.10). The simplest case is that all eigenvalues λ_i's are simple, i.e., $\lambda_1 > \cdots > \lambda_n$, and the zeros of $\det(tI_{n-1} - A)$ (i.e., the eigenvalues of A) lie in the intervals $(\lambda_{i+1}, \lambda_i)$.

Example 8.3.4 Let $H = \begin{pmatrix} a & x & 1 \\ \bar{x} & 0 & y \\ 1 & \bar{y} & a \end{pmatrix}$ be a Hermitian matrix. Then a is real. $\begin{pmatrix} a & 1 \\ 1 & a \end{pmatrix}$ is a principal submatrix of H with eigenvalues $a+1$ and $a-1$. (0) is also a principal submatrix of H. It follows that (for any values x and y)

$$\lambda_{\max}(H) \geq \max\{a+1, 0\} \quad \text{and} \quad \lambda_{\min}(H) \leq \min\{a-1, 0\}. \blacksquare$$

As another application of the eigenvalue interlacing theorem, we present a result due to Poincaré: If $H \in \mathbb{M}_n$ is Hermitian, then for any (partially unitary) $n \times m$ matrix V ($n \geq m$) satisfying $V^*V = I_m$ and for $k = 1, 2, \ldots, m$,

$$\lambda_{k+n-m}(H) \leq \lambda_k(V^*HV) \leq \lambda_k(H). \tag{8.11}$$

To show this, let $U = (V, W)$ be unitary, where $W \in \mathbb{M}_{n\times(n-m)}$. Then

$$U^*HU = (V,W)^*H(V,W) = \begin{pmatrix} V^*HV & V^*HW \\ W^*HV & W^*HW \end{pmatrix}.$$

Thus, by applying the Cauchy eigenvalue interlacing theorem, we have

$$\begin{aligned} \lambda_{k+n-m}(H) &= \lambda_{k+n-m}(U^*HU) \\ &\leq \lambda_k(V^*HV) \\ &\leq \lambda_k(U^*HU) \\ &= \lambda_k(H). \end{aligned}$$

Problems

1. Let V be an $m \times n$ matrix. If $V^*V = I_n$, show that $m \geq n$.
2. Let $A = \operatorname{diag}(2, -1, 0)$. If the eigenvalues are ordered decreasingly, what are $\lambda_1(A), \lambda_2(A), \lambda_3(A)$? And what are $\lambda_1(|A|), \lambda_2(|A|), \lambda_3(|A|)$?
3. Let A be a principal submatrix of a Hermitian matrix H. Show that
$$\lambda_{\min}(H) \leq \lambda_{\min}(A) \leq \lambda_{\max}(A) \leq \lambda_{\max}(H).$$
4. Let A be an $n \times n$ Hermitian matrix. Show that for $k = 1, 2, \ldots, n$,
$$\lambda_k(-A) = -\lambda_{n-k+1}(A) \quad \text{and} \quad \lambda_k(I - A) = 1 - \lambda_{n-k+1}(A).$$
5. Let A be an $n\times n$ positive definite matrix. Show that for $k = 1, 2, \ldots, n$,
$$\lambda_k(A^{-1}) = \frac{1}{\lambda_{n-k+1}(A)}.$$
6. Let α and β ($\alpha \geq \beta$) be the eigenvalues of $\begin{pmatrix} a & b \\ \bar{b} & c \end{pmatrix}$, $a, c \in \mathbb{R}$. Show that
$$\beta \leq \min\{a, c\} \leq \max\{a, c\} \leq \alpha. \text{ (So, } \alpha - \beta \geq |a - c|.)$$

7. Let $A \in \mathbb{M}_n$ be Hermitian. Show that for any nonzero $x \in \mathbb{C}^n$,

$$\lambda_{\min}(A) \leq \frac{x^*Ax}{x^*x} \leq \lambda_{\max}(A)$$

and for every diagonal entry a_{ii} of A, $i = 1, 2, \ldots, n$,

$$\lambda_{\min}(A) \leq a_{ii} \leq \lambda_{\max}(A).$$

8. For any Hermitian matrices A and B of the same order, show that

$$\lambda_{\max}(A - B) + \lambda_{\min}(B) \leq \lambda_{\max}(A).$$

9. Let $A \in \mathbb{M}_n$ be Hermitian and let $B \in \mathbb{M}_n$ be positive definite. Show that the eigenvalues of AB^{-1} are all real numbers. Moreover, show that

$$\lambda_{\max}(AB^{-1}) = \max_{x \neq 0,\, x \in \mathbb{C}^n} \frac{x^*Ax}{x^*Bx}$$

and

$$\lambda_{\min}(AB^{-1}) = \min_{x \neq 0,\, x \in \mathbb{C}^n} \frac{x^*Ax}{x^*Bx}.$$

10. Let A be an $n \times n$ Hermitian matrix with eigenvalues $\lambda_1(A) \geq \cdots \geq \lambda_n(A)$ and let X be an $n \times p$ matrix such that $X^*X = I_p$. Show that

$$\sum_{i=n-p+1}^{n} \lambda_i(A) \leq \operatorname{tr}(X^*AX) \leq \sum_{i=1}^{p} \lambda_i(A)$$

and if A is positive definite, then

$$\sum_{i=1}^{p} \lambda_i^{-1}(A) \leq \operatorname{tr}(X^*AX)^{-1} \leq \sum_{i=n-p+1}^{n} \lambda_i^{-1}(A).$$

11. Let $A \in \mathbb{M}_n$ be a positive semidefinite matrix with eigenvalues $\lambda_1(A) \geq \cdots \geq \lambda_n(A)$. If V is an $n \times m$ matrix such that $V^*V = I_m$, show that

$$\prod_{i=1}^{m} \lambda_{n-m+i}(A) \leq \det(V^*AV) \leq \prod_{i=1}^{m} \lambda_i(A).$$

12. Let A be a positive semidefinite matrix partitioned as

$$A = \begin{pmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{pmatrix},$$

where A_{11} is square, and let $\widetilde{A_{11}} = A_{22} - A_{21}A_{11}^{-1}A_{12}$ be the Schur complement of A_{11} in A when A_{11} is nonsingular. Show that

$$\lambda_{\min}(A) \leq \lambda_{\min}(\widetilde{A_{11}}) \leq \lambda_{\min}(A_{22}).$$

13. Let $A = \begin{pmatrix} 0 & a & 0 \\ \bar{a} & 1 & b \\ 0 & \bar{b} & 0 \end{pmatrix}$, where $a, b \in \mathbb{C}$. If $a \neq 0$, show that A has one positive eigenvalue, one negative eigenvalue, and one zero eigenvalue.

14. Let $A = U^*DU$, where $U \in \mathbb{M}_n$ is unitary and $D = \operatorname{diag}(\lambda_1, \ldots, \lambda_n)$ is diagonal. Let the columns of U be $u_1, \ldots, u_n$, i.e., $U = (u_1, \ldots, u_n)$. For any λ that is not equal to any of $\lambda_1, \ldots, \lambda_n$, show that

$$(\lambda I_n - A)^{-1} = \sum_{i=1}^{n} \frac{u_i u_i^*}{\lambda - \lambda_i}.$$

15. Let $x_1 \geq x_2 \geq \cdots \geq x_n$ and let A be the $n \times n$ tridiagonal matrix:

$$A = \begin{pmatrix} x_1 & 1 & & & & \\ 1 & x_2 & 1 & & & \\ & 1 & x_3 & 1 & & \\ & & \ddots & \ddots & \ddots & \\ & & & 1 & x_{n-1} & 1 \\ & & & & 1 & x_n \end{pmatrix}.$$

Show that

$$\lambda_{\max}(A) \geq \tfrac{1}{2}(x_1 + x_2) + 1 \quad \text{and} \quad \lambda_{\min}(A) \leq \tfrac{1}{2}(x_{n-1} + x_n) - 1.$$

16. Let $A = (a_{ij})$ be an $n \times n$ Hermitian matrix. Let $\lambda_{\max}$ and $\lambda_{\min}$ be the largest and smallest eigenvalues of A, respectively. Show that

$$\lambda_{\max} - \lambda_{\min} = 2 \max\{|x^*Ay| : \text{orthonormal } x, y \in \mathbb{C}^n\}.$$

Derive that

$$\lambda_{\max} - \lambda_{\min} \geq 2 \max_{i \neq j} |a_{ij}|.$$

Is it true that

$$\lambda_{\max} - \lambda_{\min} \geq 2 \max_{i} |a_{ii}|?$$

17. Let $H = \begin{pmatrix} X & Z \\ Z^* & Y \end{pmatrix}$ be Hermitian, where X is the $k \times k$ leading principal submatrix of H. If $\lambda_i(H) = \lambda_i(X)$, $i = 1, 2, \ldots, k$, i.e., the eigenvalues of X are the first k largest eigenvalues of H, show that $Z = 0$.

18. Let A be an $n \times n$ matrix and let U be an $n \times k$ matrix, $k \leq n$, such that $U^*U = I_k$. Show that $\sigma_i(U^*AU) \leq \sigma_i(A)$, $i = 1, 2, \ldots, k$, where the singular values are ordered decreasingly: $\sigma_1(\cdot) \geq \sigma_2(\cdot) \geq \cdots$.

19. Show the min-max representations in Theorem 8.3.3.

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8.4 Eigenvalue and Singular Value Inequalities

Recall from Sect. 4.1 that any eigenvalue $\lambda(A)$ of a square matrix A satisfies

$$\sigma_{\min}(A) \le |\lambda(A)| \le \sigma_{\max}(A).$$

This section shows some basic eigenvalue and singular value inequalities by using the min-max representation theorem and the eigenvalue interlacing theorem. We assume that the eigenvalues $\lambda_i(A)$, if all real, and the singular values $\sigma_i(A)$ of an $n \times n$ matrix A are arranged in decreasing order:

$$\lambda_1(A) \ge \cdots \ge \lambda_n(A), \quad \sigma_1(A) \ge \cdots \ge \lambda_n(A).$$

Let $A = (a_{ij}) \in \mathbb{M}_n$ be Hermitian. Let $d_1(A), \ldots, d_n(A)$ be the diagonal entries of A arranged decreasingly, that is, $d_1(A), \ldots, d_n(A)$ is a rearrangement (or a re-list) of $a_{11}, \ldots, a_{nn}$ such that $d_1(A) \ge \cdots \ge d_n(A)$.

Example 8.4.1 For $A = \left(\begin{smallmatrix} 1 & 0 & 2 \\ 0 & 3 & 0 \\ 2 & 0 & 1 \end{smallmatrix}\right)$, $a_{11} = 1$, $a_{22} = 3$, $a_{33} = 1$, $d_1(A) = 3$, $d_2(A) = 1$, $d_3(A) = 1$. The characteristic polynomial of A is $(\lambda - 3)^2(\lambda + 1)$. Thus, the eigenvalues of A are $\lambda_1(A) = 3$, $\lambda_2(A) = 3$, $\lambda_3(A) = -1$, and the singular values of A are $\sigma_1(A) = 3$, $\sigma_2(A) = 3$, $\sigma_3(A) = 1$. ∎

Theorem 8.4.1 *Let $A, B \in \mathbb{M}_n$ be Hermitian matrices and $A \ge B$. Then*

(1) $a_{ii} \ge b_{ii}$, $i = 1, 2, \ldots, n$. (Diagonal dominance)

(2) $d_i(A) \ge d_i(B)$, $i = 1, 2, \ldots, n$.

(3) $\lambda_i(A) \ge \lambda_i(B)$, $i = 1, 2, \ldots, n$. (Eigenvalue dominance)

(4) $\lambda_n(A)I \le A \le \lambda_1(A)I$.

Proof. (1) $a_{ii} - b_{ii}$ are the diagonal entries of $A - B \ge 0$. So, $a_{ii} \ge b_{ii}$.

(2) By (1), $a_{ii} \ge b_{ii}$ for all i. Thus, (2) is equivalent to the statement (the case where A and B are diagonal): Let $a_1, a_2, \ldots, a_n$ and $b_1, b_2, \ldots, b_n$ be real numbers. If $a_i \ge b_i$, $i = 1, 2, \ldots, n$, then $a_{[i]} \ge b_{[i]}$, $i = 1, 2, \ldots, n$, where $x_{[1]}, x_{[2]}, \ldots, x_{[n]}$ is a re-list of real numbers $x_1, x_2, \ldots, x_n$ such that $x_{[1]} \ge x_{[2]} \ge \cdots \ge x_{[n]}$. This fact is interesting in its own right.

Without of loss of generality, we assume that $b_1 \ge b_2 \ge \cdots \ge b_n$. Then $b_{[1]} = b_1 \le a_1 \le a_{[1]}$, In general, for each i, $i = 1, 2, \ldots, n$, we have

$$b_{[i]} = \min\{b_1, \ldots, b_i\} \le \min\{a_1, \ldots, a_i\} \le \min\{a_{[1]}, \ldots, a_{[i]}\} = a_{[i]}.$$

(3) follows from the Courant-Fischer theorem immediately because

$$A \ge B \quad \Rightarrow \quad x^*Ax \ge x^*Bx, \ \ x \in \mathbb{C}^n.$$

For (4), by a spectral decomposition of A, $\lambda_n(A)I \le A \le \lambda_1(A)I$ is equivalent to $\lambda_n(A)I \le \mathrm{diag}(\lambda_1(A), \ldots, \lambda_n(A)) \le \lambda_1(A)I$. ∎

Example 8.4.2 Eigenvalue dominance.

(a) Let $A = \left(\begin{smallmatrix} 3 & i \\ -i & 0 \end{smallmatrix}\right)$ and $B = \left(\begin{smallmatrix} 1 & 1 \\ 1 & -1 \end{smallmatrix}\right)$. Then $A - B = \left(\begin{smallmatrix} 2 & i-1 \\ -i-1 & 1 \end{smallmatrix}\right) \geq 0$. Thus, by the above theorem, $\lambda_1(A) \geq \lambda_1(B)$ and $\lambda_2(A) \geq \lambda_2(B)$.

(b) Let $A = \left(\begin{smallmatrix} 2 & 0 \\ 0 & 1 \end{smallmatrix}\right)$ and $B = \left(\begin{smallmatrix} 1 & 1 \\ 1 & 1 \end{smallmatrix}\right)$. $\lambda_i(A) \geq \lambda_i(B)$, $i = 1, 2$, but $A \not\geq B$.

(c) Let A be an $n \times n$ positive semidefinite matrix. Then $A + \varepsilon I_n$ is positive definite for any $\varepsilon > 0$ with $\lambda_{\min}(A + \varepsilon I_n) = \lambda_{\min}(A) + \varepsilon$.

(d) Let H be an $n \times n$ Hermitian matrix. Add ε to every entry of H to get a matrix $H(\varepsilon)$. Then for each i, $\lambda_i(H(\varepsilon)) \geq \lambda_i(H)$ for any $\varepsilon > 0$ because $H(\varepsilon) = H + \varepsilon E$, where E is the $n \times n$ all-ones matrix.

(e) Let $M = \left(\begin{smallmatrix} A & B \\ B^* & C \end{smallmatrix}\right)$ be a partitioned positive semidefinite matrix. If A is nonsingular, then $\lambda_i(C) \geq \lambda_i(B^* A^{-1} B)$ for each i. ∎

Our next result compares the eigenvalues of sum and (usual and entry-wise) products of Hermitian matrices to those of the individual matrices.

Theorem 8.4.2 *Let $A, B \in \mathbb{M}_n$ be Hermitian matrices. Then for each i,*

(1) $\lambda_i(A) + \lambda_n(B) \leq \lambda_i(A + B) \leq \lambda_i(A) + \lambda_1(B)$.
(2) $\lambda_i(A)\lambda_n(B) \leq \lambda_i(AB) \leq \lambda_i(A)\lambda_1(B)$ *if* $A \geq 0$.
(3) $d_i(A)\lambda_n(B) \leq \lambda_i(A \circ B) \leq d_i(A)\lambda_1(B)$ *if* $A \geq 0$ *and* $B \geq 0$.

Proof. Let x be any unit vector in $\mathbb{C}^n$. Then we have

$$x^* Ax + \min_x x^* Bx \leq x^*(A + B)x \leq x^* Ax + \max_x x^* Bx.$$

Thus, by the Rayleigh-Ritz theorem, we have

$$x^* Ax + \lambda_n(B) \leq x^*(A + B)x \leq x^* Ax + \lambda_1(B).$$

Applying the Courant-Fischer minimax principle, we obtain (1).

For (2), since $A \geq 0$, A has a positive semidefinite square root $A^{1/2}$, and $\lambda_i(AB) = \lambda_i(A^{1/2} B A^{1/2})$. Because B is Hermitian, $\lambda_1(B)I - B \geq 0$. Thus,

$$A^{1/2} B A^{1/2} \leq A^{1/2} B A^{1/2} + A^{1/2}\big(\lambda_1(B)I - B\big)A^{1/2} = \lambda_1(B)A.$$

An application of Theorem 8.4.1 results in the second inequality in (2). Replacing B with $-B$ gives the first inequality as $\lambda_1(-B) = -\lambda_n(B)$.

For (3), recall that the Hadamard product of positive semidefinite matrices is positive semidefinite (Schur theorem, Sect. 7.5). Since $\lambda_1(B)I - B \geq 0$ and $B - \lambda_n(B)I \geq 0$, taking the Hadamard product with A, we have

$$A \circ \big(\lambda_1(B)I - B\big) \geq 0 \quad \text{and} \quad A \circ \big(B - \lambda_n(B)I\big) \geq 0$$

implying

$$\lambda_n(B)(I \circ A) \leq A \circ B \leq \lambda_1(B)(I \circ A).$$

Note that $I \circ A$ is the diagonal matrix of the diagonal entries of A: $d_1(A) \geq \cdots \geq d_n(A) \geq 0$ (maybe not in this order on the diagonal). Since $B \geq 0$, the ith largest eigenvalues of $\lambda_n(B)(I \circ A)$ and $\lambda_1(B)(I \circ A)$ are $d_i(A)\lambda_n(B)$ and $d_i(A)\lambda_1(B)$, respectively. Theorem 8.4.1 yields for each i,

$$d_i(A)\lambda_n(B) \leq \lambda_i(A \circ B) \leq d_i(A)\lambda_1(B). \blacksquare$$

In the proof of (3), we see that the condition $A \geq 0$ can be changed to all $d_i(A) \geq 0$. Moreover, it is natural to ask: If $A \geq 0$, $B \geq 0$, is it true that

$$\lambda_i(A)\lambda_n(B) \leq \lambda_i(A \circ B) \leq \lambda_i(A)\lambda_1(B)?$$

The answer is negative (Problem 9). However, we have the following result.

Example 8.4.3 Let A and B be $n \times n$ positive semidefinite matrices. Then

$$\lambda_n(A)\lambda_n(B) \leq \lambda_n(A \circ B) \leq \lambda_1(A \circ B) \leq \lambda_1(A)\lambda_1(B).$$

This is because $\lambda_n(A) \leq d_n(A)$ and $d_1(A) \leq \lambda_1(A)$ due to the facts that $A - \lambda_n(A)I \geq 0$ and $\lambda_1(A)I - A \geq 0$ (so $\lambda_n(A) \leq a_{ii} \leq \lambda_1(A)$ for all i). $\blacksquare$

Can the condition on A and B in the preceding example be relaxed, say A is positive semidefinite and B is Hermitian? Try $B = I_n$ and $B = -I_n$.

We now turn our attention to singular values. Obviously, $A \geq B$ does not imply $\sigma_i(A) \geq \sigma_i(B)$ (unless both $A, B \geq 0$) by taking $A = 0$ and $B = -I$.

We start with the singular values of submatrices of a general matrix.

Theorem 8.4.3 (Singular Value Interlacing Theorem) *Let $A \in \mathbb{M}_{m\times n}$ and let B be a submatrix obtained from A by deleting s rows and t columns ($s = 0$ or $t = 0$ if no row or column is deleted). Denote $r = s + t$. Then*

$$\sigma_{r+i}(A) \leq \sigma_i(B) \leq \sigma_i(A)$$

for each $i = 1, 2, \ldots, , n - t$ with $\sigma_{r+i}(A) = 0$ if $r + i > n$ conventionally.

Proof. Note that B is an $(m - s) \times (n - t)$ matrix, having $n - t$ singular values. B has at most $\min\{m-s, n-t\}$ nonzero singular values. In addition, because singular values (of A and B) remain unchanged via permutations, we can assume that B is located in the upper-left corner of A.

Consider the case where $r = 1$ and B is obtained from A by deleting the last column, say b, that is, $A = (B, b)$. Then $A^*A = \begin{pmatrix} B^*B & B^*b \\ b^*B & b^*b \end{pmatrix}$.

Since $B^*B \in \mathbb{M}_{n-1}$ is a principal submatrix of $A^*A \in \mathbb{M}_n$, by the eigenvalue interlacing theorem (Theorem 8.3.4), we have for each i,

$$\lambda_{i+1}(A^*A) \leq \lambda_i(B^*B) \leq \lambda_i(A^*A).$$

Taking the square roots gives $\sigma_{i+1}(A) \leq \sigma_i(B) \leq \sigma_i(A)$, $i = 1, 2, \ldots, n-1$.

We proceed in this way to delete another column of A, that is, if $A = (B, c, b)$, let $A_1 = (B, c)$. Then $A = (A_1, b)$. The above argument implies

$$\sigma_{i+2}(A) \le \sigma_{i+1}(A_1) \le \sigma_i(B) \le \sigma_i(A_1) \le \sigma_i(A).$$

Inductively, remove the t columns of A that contain no entries of B to get

$$\sigma_{i+t}(A) \le \sigma_i(B) \le \sigma_i(A), \quad i = 1, 2, \ldots, n-t.$$

Repeating the process for the s rows reveals the desired inequalities. ∎

It follows from the theorem that if B is a submatrix of matrix A, then

$$\sigma_{\max}(B) \le \sigma_{\max}(A). \tag{8.12}$$

That is, the spectral norm of B is dominated by the spectral norm of A (here the norms are induced from different spaces as A and B have different sizes):

$$\|B\|_{\text{sp}} \le \|A\|_{\text{sp}}. \tag{8.13}$$

Example 8.4.4 Singular values and the spectral norm of a submatrix.

(a) Let $M = \begin{pmatrix} P & S \\ R & Q \end{pmatrix}$, where P, Q, R, and S are some matrices. Then the spectral norm of any of P, Q, R, and S is no more than that of M.

(b) Let U be an $n \times n$ unitary matrix and let V be a $k \times k$ submatrix of U. If $k > \frac{n}{2}$, then $\sigma_1(V) = 1$, because $r + 1 \le n$ for $r = 2n - 2k$, we have $1 \ge \sigma_1(V) \ge \sigma_{r+1}(U) = 1$. (See also Problem 25, Sect. 6.1.)

(c) Let H be a Hermitian matrix and let K be a submatrix of H that contains no diagonal entries of H. Then

$$2\|K\|_{\text{sp}} = 2\sigma_{\max}(K) \le \lambda_{\max}(H) - \lambda_{\min}(H). \tag{8.14}$$

Proof. Because $\lambda_{\max}(H) - \lambda_{\min}(H) = \lambda_{\max}(H_t) - \lambda_{\min}(H_t)$ for any real scalar t, where $H_t = H + tI$, and because the submatrix is not affected by the shifting with tI, without loss of generality, we assume that $\lambda_{\max}(H) = h$ and $\lambda_{\min}(H) = -h$ (so $h \ge 0$). Since H is Hermitian, $\|H\|_{\text{sp}} = \sigma_{\max}(H) = h$. The norm inequality (8.13) reveals $\|K\|_{\text{sp}} \le \|H\|_{\text{sp}}$. Thus, $2\|K\|_{\text{sp}} \le 2h = \lambda_{\max}(H) - \lambda_{\min}(H)$.

(d) Let $M = \begin{pmatrix} H & A \\ A^* & K \end{pmatrix}$, where $H \in \mathbb{M}_m$ and $K \in \mathbb{M}_n$ are Hermitian and $A \in \mathbb{M}_{m \times n}$. Then $2\sigma_{\max}(A) \le \lambda_{\max}(M) - \lambda_{\min}(M)$. For example, $M = -\begin{pmatrix} I & I \\ I & I \end{pmatrix}$, $\sigma_{\max}(A) = 1$, $\lambda_{\max}(M) = 0$, and $\lambda_{\min}(M) = -2$.

(e) Let $M = \begin{pmatrix} A & B \\ B^* & C \end{pmatrix} \ge 0$, where A, B, and C are $n \times n$ matrices. Then

$$2\sigma_j(B) \le \lambda_j(M), \quad j = 1, 2, \ldots, n. \tag{8.15}$$

Proof. $\begin{pmatrix} A & B \\ B^* & C \end{pmatrix} \ge 0 \Rightarrow \begin{pmatrix} A & -B \\ -B^* & C \end{pmatrix} \ge 0$. Thus, $\begin{pmatrix} A & B \\ B^* & C \end{pmatrix} \ge \begin{pmatrix} 0 & 2B \\ 2B^* & 0 \end{pmatrix}$. An application of Theorem 8.4.1 (3) reveals $\lambda_j \begin{pmatrix} A & B \\ B^* & C \end{pmatrix} \ge \lambda_j \begin{pmatrix} 0 & 2B \\ 2B^* & 0 \end{pmatrix}$.

By Theorem 2.4.2, the eigenvalues of $\left(\begin{smallmatrix} 0 & 2B \\ 2B^* & 0 \end{smallmatrix}\right)$ are $2\sigma_1(B), \ldots, 2\sigma_n(B)$. $-2\sigma_n(B), \ldots, -2\sigma_1(B)$. The inequalities (8.15) follow. ∎

Note that both (8.14) ($H \geq 0$) and (8.15) ($j = 1$) imply the spectral norm inequality $2\|B\|_{\rm sp} \leq \|M\|_{\rm sp}$, where $M = \left(\begin{smallmatrix} A & B \\ B^* & C \end{smallmatrix}\right) \geq 0$.

Theorem 8.4.4 (Singular Value Inequalities for Sum and Product)
(1) *If A and B are $m \times n$ matrices, then for each i,*

$$\sigma_i(A) - \sigma_1(B) \leq \sigma_i(A+B) \leq \sigma_i(A) + \sigma_1(B). \tag{8.16}$$

(2) *If A and B are $m \times n$ and $n \times p$ matrices, respectively, then for each i,*

$$\sigma_i(A)\sigma_n(B^*) \leq \sigma_i(AB) \leq \sigma_i(A)\sigma_1(B). \tag{8.17}$$

Proof. For (1), recall Theorem 2.4.2 that if X is an $m \times n$ matrix of rank r, then $H = \left(\begin{smallmatrix} 0 & X \\ X^* & 0 \end{smallmatrix}\right)$ is a Hermitian matrix of order $m+n$ having eigenvalues

$$\sigma_1(X), \ \ldots, \ \sigma_r(X), \ \overbrace{0, \ldots, 0}^{m+n-2r}, \ -\sigma_r(X), \ \ldots, \ -\sigma_1(X).$$

We use this fact and apply Theorem 8.4.2 to the Hermitian matrices

$$H_1 = \begin{pmatrix} 0 & A \\ A^* & 0 \end{pmatrix}, \ H_2 = \begin{pmatrix} 0 & B \\ B^* & 0 \end{pmatrix}, \ H_3 = \begin{pmatrix} 0 & A+B \\ A^*+B^* & 0 \end{pmatrix}.$$

If $i > \min\{m, n\}$, $\sigma_i(A+B) = 0$. For $i = 1, 2, \ldots, \min\{m, n\}$, we obtain

$$\sigma_i(A+B) = \lambda_i(H_3) \leq \lambda_i(H_1) + \lambda_1(H_2) = \sigma_i(A) + \sigma_1(B)$$

and

$$\sigma_i(A+B) = \lambda_i(H_3) \geq \lambda_i(H_1) + \lambda_{m+n}(H_2) = \sigma_i(A) - \sigma_1(B).$$

For (2), AB is $m \times p$ and has p singular values. For $i = 1, \ldots, p$

$$\sigma_i^2(AB) = \lambda_i(B^*A^*AB) = \lambda_i(A^*ABB^*).$$

Using Theorem 8.4.2 (2), we obtain (8.17) by taking the square roots of

$$\lambda_i(A^*A)\lambda_n(BB^*) \leq \sigma_i^2(AB) \leq \lambda_i(A^*A)\lambda_1(BB^*). \ ∎$$

Note that $\sigma_n(B^*) = \sqrt{\lambda_n(BB^*)}$ in (8.17) cannot be replaced by $\sigma_p(B)$ in general. For instance, take $A = (1, 0)$, $B = (0, 1)^T$, $n = 2$, $m = p = 1$. For $i = 1$, $\sigma_i(AB) = 0 < \sigma_i(A)\sigma_p(B) = \sigma_1(A)\sigma_1(B) = 1$. However, if $n = p$, then B is $n \times n$, and $\sigma_n(B^*) = \sigma_n(B)$. Moreover, $\sigma_n(A)\sigma_i(B^*) \leq \sigma_i(AB) \leq \sigma_1(A)\sigma_i(B^*)$ holds even though $\sigma_i(AB)$ is not equal to $\sigma_i(BA)$ in general.

For $A = -B = I_n$, we see $\sigma_i(A+B) \not\geq \sigma_i(A) + \sigma_n(B)$ for any i.

Example 8.4.5 Let $M = \left(\begin{smallmatrix} A & B \\ D & C \end{smallmatrix}\right)$ be a partitioned matrix. Write M as $M = \left(\begin{smallmatrix} A & 0 \\ 0 & C \end{smallmatrix}\right) + \left(\begin{smallmatrix} 0 & B \\ D & 0 \end{smallmatrix}\right)$. Observing that the singular values of $\left(\begin{smallmatrix} A & 0 \\ 0 & C \end{smallmatrix}\right)$ are those of A and C and the singular values of $\left(\begin{smallmatrix} 0 & B \\ D & 0 \end{smallmatrix}\right)$ are those of B and D, we obtain

$$\sigma_1(M) \leq \max\{\sigma_1(A), \sigma_1(C)\} + \max\{\sigma_1(B), \sigma_1(D)\}. \ ∎$$

By min-max expressions, some results stronger than Theorem 8.4.2 can be obtained for indices r and s, $r+s \le n-1$. We show two such inequalities, one for the sum of Hermitian matrices and one for the product of positive semidefinite matrices, and leave others to the reader (Problem 23).

Theorem 8.4.5 (Weyl) *Let A and B be n-square Hermitian matrices. If r and s are nonnegative integers such that $r+s \le n-1$, then*

$$\lambda_{r+s+1}(A+B) \le \lambda_{r+1}(A) + \lambda_{s+1}(B).$$

Proof. Let $u_1, \ldots, u_n$ be orthonormal eigenvectors of A associated with the eigenvalues $\lambda_1(A), \ldots, \lambda_n(A)$, respectively. Likewise, v_i, $i = 1, \ldots, n$, are orthonormal eigenvectors of B associated with the eigenvalues $\lambda_i(B)$, $i = 1, \ldots, n$, and w_i, $i = 1, \ldots, n$, are orthonormal eigenvectors of $A+B$ corresponding to the eigenvalues $\lambda_i(A+B)$, $i = 1, \ldots, n$. Let

$$\begin{aligned} S_1 &= \text{Span}\{u_{r+1}, \ldots\ldots, u_n\}, & \dim S_1 &= n-r, \\ S_2 &= \text{Span}\{v_{s+1}, \ldots\ldots, v_n\}, & \dim S_2 &= n-s, \\ S_3 &= \text{Span}\{w_1, \ldots\ldots, w_{r+s+1}\}, & \dim S_3 &= r+s+1. \end{aligned}$$

Then, by Problem 29 of Sect. 1.1,

$$\dim(S_1 \cap S_2 \cap S_3) \ge \dim S_1 + \dim S_2 + \dim S_3 - 2n = 1.$$

Thus, there exists a nonzero unit vector $x \in S_1 \cap S_2 \cap S_3$. By (8.6),

$$\lambda_{r+s+1}(A+B) \le x^*(A+B)x = x^*Ax + x^*Bx \le \lambda_{r+1}(A) + \lambda_{s+1}(B). \blacksquare$$

Example 8.4.6 Let $M = \left(\begin{smallmatrix} A & B \\ D & C \end{smallmatrix}\right)$ be a $k \times k$ partitioned Hermitian matrix, where A and C are $m \times m$ and $n \times n$ matrices, respectively, $m+n=k$. Then

$$\lambda_{r+s+1}(M) + \lambda_{\min}(M) \le \lambda_{r+1}(A) + \lambda_{s+1}(C), \quad r+s+1 \le k,\ r+1 \le m,\ s+1 \le n.$$

Proof. This follows from Theorem 8.4.5 by first considering the case where M is positive semidefinite and by writing $M = (S,T)^*(S,T) = \left(\begin{smallmatrix} S^*S & S^*T \\ T^*S & T^*T \end{smallmatrix}\right)$ with $A = S^*S$ and $C = T^*T$. Note that X^*X and XX^* have the same nonzero eigenvalues. For Hermitian M, note that $M - \lambda_{\min}(M) I_k \ge 0$. $\blacksquare$

Theorem 8.4.6 *Let G and H be $n \times n$ positive semidefinite matrices. If r and s are nonnegative integers such that $r+s \le n-1$, then*

$$\lambda_{r+s+1}(GH) \le \lambda_{r+1}(G)\lambda_{s+1}(H).$$

Proof. First consider the case where G or H (or both) is nonsingular, say H. Let u_i and v_i be respectively the orthonormal eigenvectors (in $\mathbb{C}^n$) of G and H corresponding to the eigenvalues $\lambda_i(G)$ and $\lambda_i(H)$, $i = 1, \ldots, n$. Let

$$\begin{aligned} W_1 &= \text{Span}\{u_1, \ldots\ldots, u_r\}, & \dim W_1 &= r, \\ W_2 &= \text{Span}\{v_1, \ldots\ldots, v_s\}, & \dim W_2 &= s, \\ W_3 &= \text{Span}\{H^{-1/2}u_1, \ldots, H^{-1/2}u_r\}, & \dim W_3 &= r. \end{aligned}$$

Then $\dim(W_2 + W_3) \le r + s$. Let S be a subspace of dimension $r + s$ containing $W_2 + W_3$. Then $S^\perp \subseteq (W_2 + W_3)^\perp = W_2^\perp \cap W_3^\perp$. It follows that

$$\begin{aligned}
\lambda_{r+s+1}(GH) &= \lambda_{r+s+1}(H^{\frac{1}{2}}GH^{\frac{1}{2}}) \quad \text{(now use Theorem 8.3.3)} \\
&= \min_{(\dim W = r+s)} \max_{(x \in W^\perp,\, x^*x=1)} x^*(H^{\frac{1}{2}}GH^{\frac{1}{2}})x \\
&\le \max_{x \in S^\perp,\, x^*x=1} x^*(H^{\frac{1}{2}}GH^{\frac{1}{2}})x \\
&\le \max_{x \in W_2^\perp \cap W_3^\perp,\, x^*x=1} \frac{x^*(H^{\frac{1}{2}}GH^{\frac{1}{2}})x}{x^*Hx} \cdot x^*Hx \\
&\le \max_{x \in W_3^\perp,\, x^*x=1} \frac{x^*(H^{\frac{1}{2}}GH^{\frac{1}{2}})x}{x^*Hx} \cdot \max_{x \in W_2^\perp,\, x^*x=1} x^*Hx \\
&\le \max_{y \in W_1^\perp} \frac{y^*Gy}{y^*y} \cdot \max_{x \in W_2^\perp,\, x^*x=1} x^*Hx \\
&= \lambda_{r+1}(G)\lambda_{s+1}(H).
\end{aligned}$$

If both G and H are singular, we can use a continuity argument with H replaced by $H + \varepsilon I$, $\varepsilon > 0$. Note that there exist parameterized eigenvalues of $H + \varepsilon I$ and $G(H + \varepsilon I)$ as continuous functions in $\varepsilon > 0$ (see Sect. 2.6).

Alternatively, let H be nonzero and singular. Since the eigenvalues remain unchanged under unitary similarity, we assume $H = \left(\begin{smallmatrix} D_h & 0 \\ 0 & 0 \end{smallmatrix}\right)$, where h is the rank of H, and D_h is the diagonal matrix with the positive eigenvalues of H on the diagonal, $0 < h < n$. Partition $G = \left(\begin{smallmatrix} G_h & P \\ Q & R \end{smallmatrix}\right)$ conformally as H. Then the ordered eigenvalues of G_h are no more than those of G (Theorem 8.3.4). Since $GH = \left(\begin{smallmatrix} G_hD_h & 0 \\ QD_h & 0 \end{smallmatrix}\right)$, we see that GH and G_hD_h have the same nonzero eigenvalues. G_h and D_h are positive semidefinite, and D_h is nonsingular. Applying the above argument to G_h and D_h gives the desired inequality. ∎

Problems

1. Let $A, B \in \mathbb{M}_n$. If $A \ge B \ge 0$, use Theorem 8.4.1 to show that

$$\operatorname{rank}(A) \ge \operatorname{rank}(B), \quad \operatorname{tr}A \ge \operatorname{tr}B, \quad \det A \ge \det B.$$

   If $A \ge B$ without assuming "≥ 0", which of the above remains true?

2. Find an example: $A, B \ge 0$ and $\lambda_i(A) \ge \lambda_i(B)$ for all i, but $A \not\ge B$.

3. Let A and B be $n \times n$ Hermitian matrices. Show that $A > B$ implies $\lambda_i(A) > \lambda_i(B)$ for each $i = 1, 2, \ldots, n$. Is it true that $\sigma_i(A) \ge \sigma_i(B)$?

4. Let $A, B, X \in \mathbb{M}_n$. If $A \ge B \ge 0$ and $X \ge 0$, show that for every i,

$$\lambda_i(AX) \ge \lambda_i(BX).$$

5. Let $a_i, b_i \in \mathbb{R}$, $i = 1, 2, \ldots, n$. If $a_i^2 \geq b_i^2$ for each i, show that there exist a reordering of a_i's (in magnitude): $|a_{[1]}| \geq \cdots \geq |a_{[n]}|$ and a reordering of b_i's: $|b_{[1]}| \geq \cdots \geq |b_{[n]}|$, such that $|a_{[i]}| \geq |b_{[i]}|$, $i = 1, 2, \ldots, n$.

6. Refer to Theorem 8.4.2 (1) and Theorem 8.4.4 (1). Can the left-hand side of the first inequality $\sigma_i(A) - \sigma_1(B) \leq \sigma_i(A+B) \leq \sigma_i(A) + \sigma_1(B)$ be changed to $\sigma_i(A) + \sigma_1(B)$? Or $\sigma_i(A) + \sigma_n(B)$? Or $\sigma_n(A) + \sigma_n(B)$?

7. Let $A \in \mathbb{M}_n$ be Hermitian, $\lambda_1 = \lambda_{\max}(A)$, and $\lambda_n = \lambda_{\min}(A)$. Show that
   (a) $\lambda_1 I - A \geq 0 \geq \lambda_n I - A$.
   (b) $(\lambda_1 + \lambda_n)A \geq \lambda_1 \lambda_n I + A^2$.
   (c) $(\lambda_1 + \lambda_n)I \geq \lambda_1 \lambda_n A^{-1} + A$ if A is nonsingular.

8. Let A and B be $n \times n$ positive semidefinite matrices. Show that
$$\lambda_n(A) + \lambda_n(B) \leq \lambda_n(A+B) \leq \lambda_1(A+B) \leq \lambda_1(A) + \lambda_1(B),$$
$$\lambda_n(A)\lambda_n(B) \leq \lambda_n(AB) \leq \lambda_1(AB) \leq \lambda_1(A)\lambda_1(B),$$
$$\lambda_n(A)\lambda_n(B) \leq \lambda_n(A \circ B) \leq \lambda_1(A \circ B) \leq \lambda_1(A)\lambda_1(B).$$
   (Recall that $A \circ B$ is the Hadamard (entrywise) product of A and B.)

9. Let A and B be $n \times n$ positive semidefinite matrices, and let $d_i(A)$ be the ith largest diagonal entry of $A = (a_{st})$, $i = 1, 2, \ldots, n$. Show that
$$d_i(A)\lambda_n(B) \leq \lambda_i(A \circ B) \leq d_i(A)\lambda_1(B).$$
   Show by examples that the following inequalities need not be true:
$$\lambda_i(A)\lambda_n(B) \leq \lambda_i(A \circ B) \leq \lambda_i(A)\lambda_1(B),$$
$$a_{ii}\lambda_n(B) \leq \lambda_i(A \circ B) \leq a_{ii}\lambda_1(B).$$

10. Show by (separate) examples that the following inequalities may both occur for some $n \times n$ positive semidefinite matrices $A = (a_{ij})$ and B
$$\lambda_n(AB) > (\min_i a_{ii})\lambda_n(B), \quad \lambda_n(AB) < (\min_i a_{ii})\lambda_n(B).$$

11. Let A and B be $n \times n$ Hermitian matrices. Is it true that for every i,
$$\lambda_i(A+B) \leq \lambda_i(A) + \lambda_i(B)?$$

12. Let A and B be $n \times n$ Hermitian matrices. Show that for any $\alpha \in [0, 1]$,
$$\lambda_{\min}\big(\alpha A + (1-\alpha)B\big) \geq \alpha\lambda_{\min}(A) + (1-\alpha)\lambda_{\min}(B)$$
   and
$$\lambda_{\max}\big(\alpha A + (1-\alpha)B\big) \leq \alpha\lambda_{\max}(A) + (1-\alpha)\lambda_{\max}(B).$$

13. Let $M = \left(\begin{smallmatrix} A & B \\ B^* & C \end{smallmatrix}\right)$ be a partitioned Hermitian matrix. Show that
$$\lambda_{\max}(M) \leq \max\{\lambda_{\max}(A), \lambda_{\max}(C)\} + \sigma_{\max}(B).$$

14. Let A be Hermitian and let $B \geq 0$, both $n \times n$. Show that for every i,
$$\lambda_{\min}(B)\lambda_i(A^2) \leq \lambda_i(ABA) \leq \lambda_{\max}(B)\lambda_i(A^2).$$

15. Let $A \in \mathbb{M}_n$ be Hermitian. Show that for any row vector $u \in \mathbb{C}^n$,
$$\lambda_{i+2}(A \pm u^*u) \leq \lambda_{i+1}(A) \leq \lambda_i(A \pm u^*u), \quad i = 1, 2, \ldots, n$$
and
$$\lambda_{i+2}(A) \leq \lambda_{i+1}(A \pm u^*u) \leq \lambda_i(A), \quad i = 1, 2, \ldots, n.$$

16. Let A be an n-square matrix.
    (a) Show that the diagonal entries of A are exactly the singular values of A if and only if A is nonnegative diagonal.
    (b) Show that the eigenvalues of A are exactly the singular values of A if and only if A is positive semidefinite.
    (c) If A is Hermitian, show that the diagonal entries of A are exactly the eigenvalues of A if and only if A is real diagonal.
    (d) Find an example of non-symmetric real matrix such that the diagonal entries are exactly the eigenvalues.

17. Let A be an $n \times n$ positive semidefinite matrix. If V is an $n \times m$ matrix such that $V^*V = \operatorname{diag}(\delta_1, \ldots, \delta_m)$, where all $\delta_i > 0$, show that
$$\lambda_n(A) \min_i \delta_i \leq \lambda_i(V^*AV) \leq \lambda_1(A) \max_i \delta_i \text{ for each } i.$$

18. Let A and B be n-square matrices. Show that every eigenvalue $\lambda_i(AB)$ of AB is bounded above by the spectral norms of A and B in the sense
$$|\lambda_i(AB)| \leq \min\{\sigma_1(AB), \sigma_1(BA)\} \leq \sigma_1(A)\sigma_1(B).$$
Show the analogous inequalities for the lower bound and deduce that
$$\sigma_n(A)\sigma_n(B) \leq |\lambda_i(AB)| \leq \sigma_1(A)\sigma_1(B).$$

19. Let A, X, B be $m \times p$, $p \times q$, $q \times n$ matrices, respectively. Show that
$$\sigma_i(AXB) \leq \sigma_1(A)\sigma_i(X)\sigma_1(B) \text{ for every } i \leq \min\{m, p, q, n\}.$$

20. Let X and Y be $n \times n$ matrices, $t \in [0, 1]$, and $\tilde{t} = 1 - t$. Show that
$$\sigma_i(tX + \tilde{t}Y) \leq \sigma_i(X \oplus Y).$$
[Hint: Consider $(aI, bI)(X \oplus Y)(aI, bI)^T$ and use Problem 19.]

21. (**Bhatia-Kittaneh**) Let A and B be n-square matrices. Show that
$$2\sigma_i(A^*B) \leq \sigma_i(AA^* + BB^*), \quad i = 1, 2, \ldots, n.$$
Show by example that AA^* cannot be replaced with A^*A, that is, it does not hold that $2\sigma_i(A^*B) \leq \sigma_i(A^*A + BB^*)$ for some A, B, and i.

22. (a) Let $A = (a_{ij})$ be an $m \times n$ matrix. Show that $|a_{ij}| \le \sigma_{\max}(A)$ for all i, j, i.e., all entries are dominated by the largest singular value.

    (b) Let $A = (a_{ij})$ be an $m \times n$ matrix. If some entry's absolute value equals the largest singular value of A, say $|a_{st}| = \sigma_{\max}(A)$, show that the entries except a_{st} in row s and column t are all zeros. In particular, for $A = \left(\begin{smallmatrix} a & \alpha \\ \beta & B \end{smallmatrix}\right)$, where $a \in \mathbb{C}$ and $B \in \mathbb{M}_{(m-1)\times(n-1)}$, if $|a|$ equals the largest singular value of A, then $\alpha = 0$ and $\beta = 0$.

    (c) Let $P = \gamma I_3 + \left(\begin{smallmatrix} 0 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{smallmatrix}\right)$, where $\gamma > \sqrt{3} - 1$. Show that P is positive definite, γ is a diagonal entry of P, and γ is an eigenvalue of P.

    (d) Does the example in (c) contradict the statement in (b)?

23. Let A and B be $n \times n$ matrices and $0 \le r + s \le n - 1$. Show that

    (a) If A and B are Hermitian, then

$$\lambda_{n-r-s}(A + B) \ge \lambda_{n-r}(A) + \lambda_{n-s}(B).$$

    (b) If A and B are positive semidefinite, then

$$\lambda_{n-r-s}(AB) \ge \lambda_{n-r}(A)\lambda_{n-s}(B).$$

    (c) For singular values, the following inequalities hold:

$$\sigma_{r+s+1}(A + B) \le \sigma_{r+1}(A) + \sigma_{s+1}(B),$$
$$\sigma_{r+s+1}(AB) \le \sigma_{r+1}(A)\sigma_{s+1}(B),$$
$$\sigma_{n-r-s}(AB) \ge \sigma_{n-r}(A)\sigma_{n-s}(B),$$

    but it need not be true that

$$\sigma_{n-r-s}(A + B) \ge \sigma_{n-r}(A) + \sigma_{n-s}(B).$$

24. Let $M = \begin{pmatrix} A_1 & & \star \\ & \ddots & \\ \star & & A_m \end{pmatrix}$ be a partitioned Hermitian matrix, where the diagonal blocks $A_1, \ldots, A_m$ are square (Hermitian) matrices. Show that

$$\lambda_1(M) + (m-1)\lambda_{\min}(M) \le \lambda_1(A_1) + \cdots + \lambda_1(A_m) \le m\lambda_1(M).$$

    In particular, if M is positive semidefinite and $m = 2$, show that

$$\lambda_1(M) \le \lambda_1(M) + \lambda_{\min}(M) \le \lambda_1(A_1) + \lambda_1(A_2) \le 2\lambda_1(M).$$

25. Let $M = \left(\begin{smallmatrix} A & B \\ D & C \end{smallmatrix}\right)$ be a partitioned complex matrix. Show that

$$\sigma_{r+s+1}(M) \le \sigma_{i+1}(A) + \sigma_{j+1}(B) + \sigma_{s-j+1}(C) + \sigma_{r-i+1}(D)$$

    for all possible indices, where r, s, i, and j are nonnegative integers.

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8.5 Sum of Eigenvalues of Hermitian Matrices A, B, and $A + B$

This section is devoted to the relations between the eigenvalues of Hermitian matrices A and B and the eigenvalues of the sum $A + B$. Analogous results for product AB which usually requires A and B to be positive semidefinite will be studied in Chapter 10 as a special topic. It has been evident that min-max representations play an important role in the study. We start with a simple and neat result of min-max type due to Fan.

Theorem 8.5.1 (Fan) *Let H be an $n \times n$ Hermitian matrix. Then*

$$\sum_{i=1}^{k} \lambda_i(H) = \max_{\substack{x_1,\ldots,x_k \in \mathbb{C}^n \\ x_i^* x_j = \delta_{ij}}} \sum_{i=1}^{k} x_i^* H x_i = \max_{\substack{U \in \mathbb{M}_{n\times k} \\ U^*U = I_k}} \operatorname{tr} U^* H U, \ 1 \le k \le n. \quad (8.18)$$

Proof. For orthonormal vectors $x_1, x_2, \ldots, x_k \in \mathbb{C}^n$ (i.e., $x_i^* x_j = \delta_{ij}$), let $V = (x_1, x_2, \ldots, x_k)$. Then V is an $n \times k$ matrix and $V^*V = I_k$. By (8.11)

$$\sum_{i=1}^{k} x_i^* H x_i = \operatorname{tr}(V^* H V) = \sum_{i=1}^{k} \lambda_i(V^* H V) \le \sum_{i=1}^{k} \lambda_i(H).$$

The maximum (identity) is attained by taking orthonormal eigenvectors v_i of H associated with the eigenvalues $\lambda_i(H)$, $i = 1, 2, \ldots, k$:

$$\sum_{i=1}^{k} v_i^* H v_i = \sum_{i=1}^{k} \lambda_i(H).$$

For the max in trace, set $U = (x_1, \ldots, x_k) \in \mathbb{M}_{n\times k}$. Then $U^*U = I_k$. ∎

A dual form of (8.18) is the minimum for the sum of smallest eigenvalues

$$\sum_{i=1}^{k} \lambda_{n-i+1}(H) = \min_{\substack{x_1,\ldots,x_k \in \mathbb{C}^n \\ x_i^* x_j = \delta_{ij}}} \sum_{i=1}^{k} x_i^* H x_i = \min_{\substack{U \in \mathbb{M}_{n\times k} \\ U^*U = I_k}} \operatorname{tr} U^* H U. \quad (8.19)$$

Rayleigh–Ritz's formula (Theorem 8.3.2) is Fan's theorem of $k = 1$.

Our main result of this section is the following theorem, which gives rise to a number of matrix inequalities (of majorization type, Chapter 10).

Theorem 8.5.2 (Thompson) *Let A and B be $n \times n$ Hermitian matrices and let $C = A + B$. Let $\alpha_1 \ge \cdots \ge \alpha_n$, $\beta_1 \ge \cdots \ge \beta_n$, and $\gamma_1 \ge \cdots \ge \gamma_n$ be the eigenvalues of A, B, and C, respectively. Then, for any positive integer $k \le n$ and for any index sequence $1 \le i_1 < \cdots < i_k \le n$,*

$$\sum_{t=1}^{k} \alpha_{i_t} + \sum_{t=1}^{k} \beta_{n-k+t} \le \sum_{t=1}^{k} \gamma_{i_t} \le \sum_{t=1}^{k} \alpha_{i_t} + \sum_{t=1}^{k} \beta_t. \quad (8.20)$$

Example 8.5.1 Special cases of Thompson's theorem.

(a) Take $k = 1$. Then $\lambda_i(A) + \lambda_n(B) \le \lambda_i(A + B) \le \lambda_i(A) + \lambda_1(B)$.

(b) Take $i_t = t$. Then $\sum_{t=1}^{k} \lambda_t(A + B) \le \sum_{t=1}^{k} \lambda_t(A) + \sum_{t=1}^{k} \lambda_t(B)$.

(c) Setting $n = 3$, $k = 2$, $i_1 = 1$, $i_2 = 3$, reveals the upper bound inequality $\lambda_1(A + B) + \lambda_3(A + B) \le \lambda_1(A) + \lambda_3(A) + \lambda_1(B) + \lambda_2(B)$. ∎

Lower bound inequalities for (b) and (c) can be similarly deduced.

Thompson's theorem is proved by using the Wielandt max-min expression for the sum of (arbitrarily selected or indexed) eigenvalues. Wielandt's theorem is shown based on several lemmas. In the following, Lemma 8.5.1 is used in the proof of Lemma 8.5.2 which is needed for Lemma 8.5.3.

Theorem 8.5.3 (Wielandt Minimax Principle) *Let H be an $n \times n$ Hermitian matrix. Then, for any index sequence $1 \le i_1 < \cdots < i_k \le n$,*

$$\sum_{t=1}^{k} \lambda_{i_t}(H) = \max_{\substack{W_1 \subset \cdots \subset W_k \\ \dim W_t = i_t}} \min_{\substack{x_t \in W_t \\ x_i^* x_j = \delta_{ij}}} \sum_{t=1}^{k} x_t^* H x_t. \tag{8.21}$$

The max is taken over all subspaces $W_1, \ldots, W_k$ satisfying $W_1 \subset \cdots \subset W_k$ with $\dim W_t = i_t$ for $t = 1, \ldots, k$, and (for the $W_1 \subseteq \cdots \subseteq W_k$) the min is taken over all orthonormal vectors $x_1, \ldots, x_k$, where $x_t \in W_t$ for $t = 1, \ldots, k$.

Lemma 8.5.1 *Let $U_1, U_2, \ldots, U_k$ be subspaces of a finite-dimensional inner product space such that $\dim U_t \ge t$, $t = 1, 2, \ldots, k$. If $u_2, \ldots, u_k$ are linearly independent vectors in $U_2, \ldots, U_k$, respectively, then there exist linearly independent vectors $y_1, y_2, \ldots, y_k$ in $U_1, U_2, \ldots, U_k$, respectively, such that $\mathrm{Span}\{u_2, \ldots, u_k\} \subseteq \mathrm{Span}\{y_1, y_2, \ldots, y_k\}$. Moreover, if $U_1 \subset U_2 \subset \cdots \subset U_k$, then $y_1, y_2, \ldots, y_k$ can be chosen to be orthonormal.*

Proof. We use induction on k. Let $k = 2$ and let $u_2 \in U_2$ be nonzero. If $u_2 \in U_1$, then we set $y_1 = u_2$ and take y_2 from U_2 so that y_2 is not a multiple of u_2 (i.e., y_1 and y_2 are linearly independent). This is possible because $\dim U_2 \ge 2$. If $u_2 \notin U_1$, there must exist a nonzero vector $y_1 \in U_1$ that is not a multiple of u_2. Now set $y_2 = u_2$. y_1 and y_2 are linearly independent. In either case, $\mathrm{Span}\{u_2\} \subseteq \mathrm{Span}\{y_1, y_2\}$.

Suppose that the statement is true for the case of $k - 1$, that is, for $u_2, \ldots, u_{k-1}$, there exist linearly independent $y_1, y_2, \ldots, y_{k-1}$ such that the span of the u's is contained in the span of the y's. We show the case of k. If $u_k \notin \mathrm{Span}\{y_1, y_2, \ldots, y_{k-1}\}$, then we take $y_k = u_k$. Otherwise, we take a nonzero $y_k \in U_k \cap (\mathrm{Span}\{y_1, y_2, \ldots, y_{k-1}\})^{\perp}$ (the intersection has dimension at least 1 by the dimension identity by Theorem 1.1.1). In any case, we have $\mathrm{Span}\{u_2, \ldots, u_k\} \subseteq \mathrm{Span}\{y_1, y_2, \ldots, y_k\}$, where $y_t \in U_t$ are linearly independent, $t = 1, 2, \ldots, k$. By the Gram–Schmidt process (Sect. 1.4), $y_1, y_2, \ldots, y_k$ can be chosen to be orthonormal (and each y_t lies in U_t). ∎

Setting $X_t = U_{k-t+1}$ for each t in the lemma, we obtain an equivalent statement: If $X_1, \ldots, X_k$ are subspaces of an inner product space, $\dim X_t \geq k-t+1$ for each t, and if $v_1, \ldots, v_{k-1}$ are linearly independent vectors, where $v_t \in X_t$, $t = 1, 2, \ldots, k-1$, then there exist linearly independent vectors $y_1, y_2, \ldots, y_k$, where $y_t \in X_t$, $t = 1, 2, \ldots, k$, such that $\text{Span}\{v_1, \ldots, v_{k-1}\} \subseteq \text{Span}\{y_1, y_2, \ldots, y_k\}$. In addition, the vectors $y_1, \ldots, y_k$ can be chosen to be orthonormal if $X_1 \supset X_2 \supset \cdots \supset X_k$.

Lemma 8.5.2 *Let $1 \leq i_1 < \cdots < i_k \leq n$. Let $W_1, \ldots, W_k$ and $V_1, \ldots, V_k$ be subspaces of an inner product space of dimension n such that $\dim W_t \geq i_t$, $\dim V_t \geq n - i_t + 1$, $t = 1, \ldots, k$, and $V_1 \supseteq \cdots \supseteq V_k$. Then there exist orthonormal set $\{x_1, \ldots, x_k\}$ and orthonormal set $\{y_1, \ldots, y_k\}$, where $x_t \in W_t$ and $y_t \in V_t$ for each t, such that $\text{Span}\{x_1, \ldots, x_k\} = \text{Span}\{y_1, \ldots, y_k\}$.*

Proof. Since $\dim W_1 + \dim V_1 = n + 1$, by the dimension identity (Theorem 1.1.1, Chap. 1), $W_1 \cap V_1$ has dimension at least 1. Thus, a unit vector $x_1 = y_1 \in W_1 \cap V_1$ exists. So the assertion is true for $k = 1$. Suppose that the statement is true for $k-1$ subspaces, that is, there exist orthonormal sets of x_t's and v_t's, where $x_t \in W_t$ and $v_t \in V_t$, $t = 1, \ldots, k-1$, such that $\text{Span}\{x_1, \ldots, x_{k-1}\} = \text{Span}\{v_1, \ldots, v_{k-1}\}$. Now let

$$W = \text{Span}\{x_1, \ldots, x_{k-1}\} + W_k \cap (\text{Span}\{x_1, \ldots, x_{k-1}\})^{\perp}. \tag{8.22}$$

Then $\dim W \geq \dim W_k \geq i_k$ (Problem 7). Because $i_k - i_t \geq k - t$, we have $\dim W + \dim V_t \geq i_k + n - i_t + 1 \geq n + k - t + 1$ for every t.

Set $X_t = W \cap V_t$, $t = 1, \ldots, k$. The dimension identity theorem ensures $\dim X_t \geq k - t + 1$. Moreover, $X_1 \supseteq \cdots \supseteq X_k$, and $v_t \in X_t$, $t = 1, \ldots, k-1$. By the discussion following Lemma 8.5.1, there exist orthonormal $y_t \in X_t \subseteq V_t$, $t = 1, \ldots, k$, such that $\text{Span}\{x_1, \ldots, x_{k-1}\} = \text{Span}\{v_1, \ldots, v_{k-1}\} \subseteq \text{Span}\{y_1, \ldots, y_k\} \subseteq W$. Choose a unit vector x_k from $(\text{Span}\{x_1, \ldots, x_{k-1}\})^{\perp} \cap \text{Span}\{y_1, \ldots, y_k\} \neq \{0\}$. By (8.22), $x_k \in W_k$. So, $x_1, \ldots, x_k$ are orthonormal and $\text{Span}\{x_1, \ldots, x_k\} = \text{Span}\{y_1, \ldots, y_k\}$. ∎

Lemma 8.5.3 *Let H be an $n \times n$ Hermitian matrix and let $1 \leq i_1 < \cdots < i_k \leq n$. Then for any subspaces $W_1, \ldots, W_k$ of $\mathbb{C}^n$ such that $\dim W_t = i_t$, $t = 1, \ldots, k$, there exist orthonormal $x_t \in W_t$, $t = 1, \ldots, k$, such that*

$$\sum_{t=1}^{k} x_t^* H x_t \leq \sum_{t=1}^{k} \lambda_{i_t}(H).$$

Proof. Let $V_t = \text{Span}\{u_{i_t}, u_{i_t+1}, \ldots, u_n\}$, where u_j are the orthonormal eigenvectors of H belonging to the eigenvalues λ_j, respectively. Then $\dim V_t = n - i_t + 1$, $t = 1, \ldots, k$, and $V_1 \supseteq \cdots \supseteq V_k$. By Lemma 8.5.2, there exist orthonormal vectors $x_1, \ldots, x_k$ and orthonormal vectors $y_1, \ldots, y_k$, where $x_t \in W_t$ and $y_t \in V_t$, $t = 1, \ldots, k$, such that $\text{Span}\{x_1, \ldots, x_k\} = \text{Span}\{y_1, \ldots, y_k\}$.

By Theorem 8.3.1, $y_t^* H y_t \le \lambda_{i_t}(H)$. By Problem 8, we obtain

$$\sum_{t=1}^{k} x_t^* H x_t = \sum_{t=1}^{k} y_t^* H y_t \le \sum_{t=1}^{k} \lambda_{i_t}(H). \blacksquare$$

We are now ready to prove Theorem 8.5.3 and Theorem 8.5.2.

Proof of (Wielandt) Theorem 8.5.3. Let $W_t = \mathrm{Span}\{u_1, u_2, \ldots, u_{i_t}\}$, where u_j are orthonormal eigenvectors of H belonging to the eigenvalues $\lambda_j(H)$, respectively. Then $\dim W_t = i_t$, $t = 1, \ldots, k$, and for unit vectors $x_t \in W_t$,

$$\sum_{t=1}^{k} x_t^* H x_t \ge \sum_{t=1}^{k} \lambda_{i_t}(H).$$

Combining this with Lemma 8.5.3, we complete the proof. $\blacksquare$

Proof of (Thompson) Theorem 8.5.2. By Theorem 8.5.3, we assume that $\sum_{t=1}^{k} \gamma_{i_t}$ is attained by subspaces S_t, $\dim S_t = i_t$, $S_1 \subset \cdots \subset S_k$,

$$\sum_{t=1}^{k} \gamma_{i_t} = \sum_{t=1}^{k} \lambda_{i_t}(A+B) = \min_{\substack{x_t \in S_t \\ x_i^* x_j = \delta_{ij}}} \sum_{t=1}^{k} x_t^* (A+B) x_t.$$

Lemma 8.5.3 ensures that for some orthonormal $y_t \in S_t$, $t = 1, \ldots, k$,

$$\sum_{t=1}^{k} y_t^* A y_t \le \sum_{t=1}^{k} \lambda_{i_t}(A) = \sum_{t=1}^{k} \alpha_{i_t}.$$

It follows that

$$\begin{aligned} \sum_{t=1}^{k} \gamma_{i_t} &\le \min_{\substack{x_t \in S_t \\ x_i^* x_j = \delta_{ij}}} \sum_{t=1}^{k} x_t^* A x_t + \max_{\substack{x_t \in S_t \\ x_i^* x_j = \delta_{ij}}} \sum_{t=1}^{k} x_t^* B x_t \\ &\le \sum_{t=1}^{k} y_t^* A y_t + \sum_{t=1}^{k} \lambda_t(B) \quad \text{(by Theorem 8.5.1)} \\ &\le \sum_{t=1}^{k} \alpha_{i_t} + \sum_{t=1}^{k} \beta_t. \end{aligned}$$

This is the inequality on the right-hand side of (8.20). For the first inequality, let $D = -B$ and denote the eigenvalues of D by $\delta_1 \ge \cdots \ge \delta_n$. Then $A = C + D$. The above inequality which we just proved reveals

$$\sum_{t=1}^{k} \alpha_{i_t} \le \sum_{t=1}^{k} \gamma_{i_t} + \sum_{t=1}^{k} \delta_t.$$

Note that $\delta_t = -\beta_{n-t+1}$, $t = 1, \ldots, n$. The desired inequality follows. $\blacksquare$

Apparently, the Wielandt minimax theorem (of the extremal case) is more general than Fan's theorem, i.e., (8.18) and (8.19). In addition, one can show the following dual form of Wielandt's max-min expression in min-max

$$\sum_{t=1}^{k} \lambda_{i_t}(H) = \min_{\substack{W_1 \supset \cdots \supset W_k \\ \dim W_t = n - i_t + 1}} \max_{\substack{x_t \in W_t \\ x_i^* x_j = \delta_{ij}}} \sum_{t=1}^{k} x_t^* H x_t. \tag{8.23}$$

Problems

1. Let A be an $n \times n$ Hermitian matrix with eigenvalues $\mu_i(A)$'s ordered (decreasingly in moduli) as $|\mu_1(A)| \geq \cdots \geq |\mu_n(A)|$. Show that

$$|\mu_1(A)| = \max_{x^* x = 1} |x^* A x|,$$

   but $|\mu_n(A)| = \min_{x^* x = 1} |x^* A x|$ need not be true.

2. Let $A = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix}$ and $B = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{pmatrix}$. Show that

$$1 \leq \lambda_{\max}(A + B) + \lambda_{\min}(A + B) \leq 3.$$

3. Let H and X_1 be $n \times n$ Hermitian matrices. Denote $X_2 = H + X_1$. By using Thompson's inequality (Theorem 8.5.2), deduce the inequalities:

$$\big(\lambda_1(X_2) - \lambda_1(X_1)\big) + \big(\lambda_n(X_2) - \lambda_n(X_1)\big) \leq \lambda_1(H) + \lambda_2(H),$$

$$\big(\lambda_1(X_2) - \lambda_1(X_1)\big) + \big(\lambda_n(X_2) - \lambda_n(X_1)\big) \geq \lambda_{n-1}(H) + \lambda_n(H).$$

4. Let A and B be $n \times n$ Hermitian matrices. If $n = 2k$ is even, show that

$$\lambda_k(A + B) + \lambda_{k+1}(A + B) \leq \lambda_k(A) + \lambda_{k+1}(A) + \lambda_1(B) + \lambda_2(B),$$

$$\lambda_k(A + B) + \lambda_{k+1}(A + B) \geq \lambda_k(A) + \lambda_{k+1}(A) + \lambda_{n-1}(B) + \lambda_n(B).$$

5. Let A and B be $n \times n$ Hermitian matrices. Show that

$$\sum_{t=1}^{k} \lambda_t(A + B) \leq \sum_{t=1}^{k} \lambda_t(A) + \sum_{t=1}^{k} \lambda_t(B), \quad k = 1, 2, \ldots, n.$$

   Equality holds when $k = n$.

6. Let $\sigma_i(A)$ be the ith largest singular value of $A \in \mathbb{M}_{m \times n}$. Show that

$$\begin{aligned} \sigma_i(A) &= \max_{(\dim W = i)} \min_{(x \in W,\, x^* x = 1)} x^* (A^* A)^{1/2} x \\ &= \max_{(\dim W = i)} \min_{(x \in W,\, x^* x = 1)} (x^* A^* A x)^{1/2}. \end{aligned}$$

   Deduce the max- and min- representations for $\sigma_1(A)$ and $\sigma_n(A)$.

7. Let V and S be subspaces of an inner product space of finite dimension. Let $W = S + (S^{\perp} \cap V)$. Show that $\dim W \geq \dim V$.

8. Let $\{x_1, \ldots, x_m\}$ and $\{y_1, \ldots, y_m\}$ be orthonormal sets in $\mathbb{C}^n$ such that

$$\operatorname{Span}\{x_1, \ldots, x_m\} = \operatorname{Span}\{y_1, \ldots, y_m\}$$

and let $A \in \mathbb{M}_n$. Show that

$$\sum_{t=1}^{m} x_t^* A x_t = \sum_{t=1}^{m} y_t^* A y_t.$$

9. Let H be an $n \times n$ Hermitian matrix. For $k = 1, 2, \ldots, n$, show that

$$\sum_{i=1}^{k} \lambda_{n-i+1}(H) = \min_{U^*U = I_k} \operatorname{tr} U^* H U = \min_{\substack{x_1, \ldots, x_k \in \mathbb{C}^n \\ x_i^* x_j = \delta_{ij}}} \sum_{i=1}^{k} x_i^* H x_i.$$

[Note: The size of U is not explicitly given in the display. However, it is easily seen from the context that U is an $n \times k$ matrix because it needs to have n rows in HU and k columns in $U^*U = I_k$.]

10. Let $H \in \mathbb{M}_n$ be positive semidefinite. For $k = 1, 2, \ldots, n$, show that

$$\prod_{i=1}^{k} \lambda_i(H) = \max_{U^*U = I_k} \det U^* H U = \max_{\substack{x_1, \ldots, x_k \in \mathbb{C}^n \\ x_i^* x_j = \delta_{ij}}} \det(x_i^* H x_j),$$

$$\prod_{i=1}^{k} \lambda_{n-i+1}(H) = \min_{U^*U = I_k} \det U^* H U = \min_{\substack{x_1, \ldots, x_k \in \mathbb{C}^n \\ x_i^* x_j = \delta_{ij}}} \det(x_i^* H x_j).$$

11. Let $A, B \in \mathbb{M}_n$ and let $\mu_1 \geq \cdots \geq \mu_n$ be the zeros of $\det(A - \mu B) = 0$. If A is Hermitian and B is positive definite, show that

$$\mu_1 = \max_{x \neq 0} \frac{x^* A x}{x^* B x}, \quad \mu_n = \min_{x \neq 0} \frac{x^* A x}{x^* B x},$$

and more generally, for $k = 1, 2, \ldots, n$,

$$\sum_{i=1}^{k} \mu_i = \max_{\substack{x_1, \ldots, x_k \in \mathbb{C}^n \\ x_i^* B x_j = 0,\ i \neq j}} \sum_{i=1}^{k} \frac{x_i^* A x_i}{x_i^* B x_i},$$

$$\prod_{i=n-k+1}^{n} \mu_i = \min_{\substack{x_1, \ldots, x_k \in \mathbb{C}^n \\ x_i^* B x_j = 0,\ i \neq j}} \prod_{i=1}^{k} \frac{x_i^* A x_i}{x_i^* B x_i}.$$

12. Let $A, B \in \mathbb{M}_n$. Apply Theorem 8.5.2 to the Hermitian block matrices $\left(\begin{smallmatrix} 0 & A \\ A^* & 0 \end{smallmatrix}\right)$ and $\left(\begin{smallmatrix} 0 & B \\ B^* & 0 \end{smallmatrix}\right)$ (see Theorem 2.4.2 for their eigenvalues) to prove

$$\sum_{t=1}^{k} \sigma_t(A+B) \le \sum_{t=1}^{k} \sigma_t(A) + \sum_{t=1}^{k} \sigma_t(B), \quad k \le n.$$

13. Let A be an $m \times n$ matrix. For $k = 1, 2, \ldots, n$, show that

$$\prod_{i=1}^{k} \sigma_i(A) = \max_{U^*U = I_k} \left(\det(U^*A^*AU)\right)^{\frac{1}{2}},$$

$$\prod_{i=1}^{k} \sigma_{n-i+1}(A) = \min_{U^*U = I_k} \left(\det(U^*A^*AU)\right)^{\frac{1}{2}}.$$

14. Let A be an $m \times n$ matrix. For $k = 1, 2, \ldots, n$, show that

$$\sum_{i=1}^{k} \sigma_i(A) = \max_{U^*U = V^*V = I_k} |\operatorname{tr}(U^*AV)| = \max_{U^*U = V^*V = I_k} \Re\left(\operatorname{tr}(U^*AV)\right)$$

(U is $m \times k$ and V is $n \times k$). However, neither of the following holds:

$$\sum_{i=1}^{k} \sigma_{n-i+1}(A) = \min_{U^*U = V^*V = I_k} |\operatorname{tr}(U^*AV)|,$$

$$\sum_{i=1}^{k} \sigma_{n-i+1}(A) = \min_{U^*U = V^*V = I_k} \Re(\operatorname{tr}(U^*AV)).$$

15. Let A and B be $n \times n$ Hermitian matrices such that $\operatorname{tr}(A+B)^k = \operatorname{tr}(A^k) + \operatorname{tr}(B^k)$ for all positive integers k. Show that

    (a) $\operatorname{rank}(A+B) = \operatorname{rank}(A) + \operatorname{rank}(B)$.

    (b) $\operatorname{Im}(A+B) = \operatorname{Im} A + \operatorname{Im} B$.

    (c) $AB = 0$.

    [Hint: For (a), use Problem 7 of Sect. 5.4. For (c), view A and B as linear transformations on $\operatorname{Im}(A+B)$. Find the matrix of $A+B$ on $\operatorname{Im}(A+B)$ and use the equality case of the Hadamard inequality.]

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8.6 A Triangle Inequality for the Matrix $|A| = (A^*A)^{1/2}$

Let A be an $m \times n$ matrix. This section studies the positive semidefinite matrix $(A^*A)^{1/2}$, denoted by $|A|$ and called the *modulus* of matrix A (Sect. 3.2). The main result is the matrix triangle inequality, i.e., for any square matrices A and B of the same order, there exist unitary matrices U and V such that

$$|A + B| \leq U^*|A|U + V^*|B|V.$$

It is known (Theorem 8.4.1) that for Hermitian matrices A and B,

$$A \geq B \quad \Rightarrow \quad \lambda_i(A) \geq \lambda_i(B)$$

for each i, where the eigenvalues are ordered decreasingly. The converse is false, that is, the inequalities $\lambda_i(A) \geq \lambda_i(B)$ for all i do not imply $A \geq B$.

Example 8.6.1 Let A and B be the following positive definite matrices:

$$A = \begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix}, \quad B = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}.$$

Then $\lambda_1(A) = 2 \geq \lambda_1(B) = 2$ and $\lambda_2(A) = 1 \geq \lambda_2(B) = 0$. But $A \not\geq B$ because $A - B = \left(\begin{smallmatrix} 1 & -1 \\ -1 & 0 \end{smallmatrix}\right)$ is obviously not positive semidefinite. ∎

We have, however, the following result.

Theorem 8.6.1 *Let $A, B \in \mathbb{M}_n$ be Hermitian matrices. If*

$$\lambda_i(A) \geq \lambda_i(B)$$

for $i = 1, \ldots, n$, then there exists a unitary matrix $U \in \mathbb{M}_n$ such that

$$U^*AU \geq B.$$

Proof. By spectral decomposition, let P and Q be unitary matrices such that

$$A = P^* \operatorname{diag}(\lambda_1(A), \ldots, \lambda_n(A))P,$$

$$B = Q^* \operatorname{diag}(\lambda_1(B), \ldots, \lambda_n(B))Q.$$

The condition $\lambda_i(A) \geq \lambda_i(B)$, $i = 1, \ldots, n$, implies that

$$\operatorname{diag}(\lambda_1(A), \ldots, \lambda_n(A)) - \operatorname{diag}(\lambda_1(B), \ldots, \lambda_n(B)) \geq 0.$$

Multiply both sides by Q^* from the left and by Q from the right to get

$$Q^* \operatorname{diag}(\lambda_1(A), \ldots, \lambda_n(A))Q - B \geq 0.$$

Take $U = P^*Q$. Then we have $U^*AU - B \geq 0$, as desired. ∎

We now turn our attention to the positive semidefinite matrix $|A|$.

Note that $|A|$ is the unique positive semidefinite matrix satisfying

$$|A|^2 = A^*A.$$

Moreover,

$$A = W|A| \tag{8.24}$$

is a polar decomposition of A, where W is some unitary matrix (Sect. 3.2). Because of such relations between $|A|$ and A, matrix $|A|$ has drawn much attention, and many interesting results have been obtained.

The eigenvalues of $|A|$ are the singular values of A, and they are equal to the (positive) square roots of the eigenvalues of A^*A. So, we have for each i

$$\lambda_i(|A|) = \sigma_i(A) = \lambda_i\big((A^*A)^{\frac{1}{2}}\big) = \big(\lambda_i(A^*A)\big)^{\frac{1}{2}}.$$

If A is an $n \times n$ matrix and if $A = UDV$ is a singular value decomposition of A, where U and V are unitary, and $D = \operatorname{diag}(\sigma_1(A), \ldots, \sigma_n(A))$, then

$$|A| = V^*DV \quad \text{and} \quad |A^*| = UDU^*.$$

It is easy to check that $|PA| = |A|$ and $|AQ| = Q^*|A|Q$ for unitary P and Q. If A is a square matrix, then $|A^*| = R^*|A|R$ for some unitary R.

The next result compares the eigenvalues of $\frac{1}{2}(A^* + A)$ and $(A^*A)^{1/2}$.

Theorem 8.6.2 (Fan-Hoffman) *Let A be an n-square matrix. Then*

$$\lambda_i\left(\frac{A^* + A}{2}\right) \le \lambda_i(|A|), \quad i = 1, \ldots, n.$$

Proof. Let $u_1, \ldots, u_n$ be orthonormal eigenvectors of $\frac{A^*+A}{2}$ associated with $\lambda_1\big(\frac{A^*+A}{2}\big), \ldots, \lambda_n\big(\frac{A^*+A}{2}\big)$ and let $v_1, \ldots, v_n$ be orthonormal eigenvectors of A^*A associated with $\lambda_1(A^*A), \ldots, \lambda_n(A^*A)$, respectively. Thus, for each i

$$\big(\tfrac{A^*+A}{2}\big)u_i = \lambda_i\big(\tfrac{A^*+A}{2}\big)u_i, \quad (A^*A)v_i = \big(\lambda_i(|A|)\big)^2 v_i.$$

For each fixed positive integer i, $1 \le i \le n$, let

$$S_1 = \operatorname{Span}\{u_1, \ldots, u_i\}, \quad S_2 = \operatorname{Span}\{v_i, \ldots, v_n\}.$$

Since $\dim S_1 + \dim S_2 = n + 1$, a unit $w \in S_1 \cap S_2$ exists. By Theorem 8.3.1,

$$w^*\big(\tfrac{A^*+A}{2}\big)w \ge \lambda_i\big(\tfrac{A^*+A}{2}\big) \quad \text{and} \quad w^*(A^*A)w \le \big(\lambda_i(|A|)\big)^2.$$

We derive, for each $i = 1, \ldots, n$,

$$\begin{aligned} \lambda_i\big(\tfrac{A^*+A}{2}\big) &\le w^*\big(\tfrac{A^*+A}{2}\big)w = \Re(w^*Aw) \le |w^*Aw| \\ &\le \|w\|\,\|Aw\| = \sqrt{w^*A^*Aw} \le \lambda_i(|A|). \end{aligned}$$

∎

Combining Theorems 8.6.1 and 8.6.2, we see that for any $n \times n$ matrix A, there exists an $n \times n$ unitary matrix U such that

$$\frac{A^* + A}{2} \leq U^*|A|U. \tag{8.25}$$

It follows immediately that $\operatorname{rank}(\frac{A^*+A}{2}) \leq \operatorname{rank}(A)$ (Problem 30).

We now present a matrix version of the triangle inequality.

Theorem 8.6.3 (Thompson Matrix Triangle Inequality) *Let A and B be $n \times n$ matrices. Then there are $n \times n$ unitary matrices U and V such that*

$$|A + B| \leq U^*|A|U + V^*|B|V.$$

Proof. By the polar decomposition (8.24), we can write

$$A + B = W|A + B|,$$

where W is an $n \times n$ unitary matrix. By (8.25), for some unitary U and V,

$$\begin{aligned} |A+B| &= W^*(A+B) \\ &= \frac{1}{2}\big(W^*(A+B) + (A+B)^*W\big) \\ &= \frac{1}{2}(A^*W + W^*A) + \frac{1}{2}(B^*W + W^*B) \\ &\leq U^*|W^*A|U + V^*|W^*B|V \\ &= U^*|A|U + V^*|B|V. \ \blacksquare \end{aligned}$$

Example 8.6.2 Notes about the Fan-Hoffman and Thompson inequalities.

(a) In the Fan-Hoffman eigenvalue inequality, $|A|$ cannot be replaced by A, nor can $\frac{A^*+A}{2}$ be replaced by $|\frac{A^*+A}{2}|$. Denote $\mathcal{H}(A) = \frac{A^*+A}{2}$.

It is not true that $\lambda_1(\mathcal{H}(A)) \leq \lambda_1(A)$, nor $\lambda_2(|\mathcal{H}(A)|) \leq \lambda_2(|A|)$ for $A = \left(\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right)$ with $\mathcal{H}(A) = \frac{1}{2}\left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$, $|\mathcal{H}(A)| = \frac{1}{2}I_2$, and $|A| = \left(\begin{smallmatrix} 0 & 0 \\ 0 & 1 \end{smallmatrix}\right)$.

Note that $\sigma_1(\mathcal{H}(A)) = \frac{1}{2}\sigma_1(A^* + A) \leq \frac{1}{2}\sigma_1(A^*) + \frac{1}{2}\sigma_1(A) = \sigma_1(A)$. However, $\sigma_i(\mathcal{H}(A)) \not\leq \sigma_i(A)$ for $i > 1$ in general (Problem 31).

(b) In Thompson's triangle inequality, the unitary matrices U and V are needed. For example, take $A = \left(\begin{smallmatrix} 1 & 0 \\ 0 & 0 \end{smallmatrix}\right)$ and $B = \left(\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right)$. Then $|A + B| = |\left(\begin{smallmatrix} 1 & 1 \\ 0 & 0 \end{smallmatrix}\right)| = \left(\begin{smallmatrix} 1 & 1 \\ 1 & 1 \end{smallmatrix}\right)^{\frac{1}{2}} = \frac{1}{\sqrt{2}}\left(\begin{smallmatrix} 1 & 1 \\ 1 & 1 \end{smallmatrix}\right)$, $|A| = \left(\begin{smallmatrix} 1 & 0 \\ 0 & 0 \end{smallmatrix}\right)$, $|B| = \left(\begin{smallmatrix} 0 & 0 \\ 0 & 1 \end{smallmatrix}\right)$, and $|A+B| \not\leq |A|+|B|$. So, U and V cannot be absent. However, $|A+B| \leq U^*|A|U + V^*|B|V$, where $U = \frac{1}{\sqrt{2}}\left(\begin{smallmatrix} 1 & 1 \\ -1 & 1 \end{smallmatrix}\right)$ and $V = \frac{1}{\sqrt{2}}\left(\begin{smallmatrix} 1 & -1 \\ 1 & 1 \end{smallmatrix}\right)$.

(c) $|A + A^*| \not\leq |A| + |A^*|$ for $A = \left(\begin{smallmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{smallmatrix}\right)$ because $|A| = \left(\begin{smallmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{smallmatrix}\right)$, $|A^*| = \left(\begin{smallmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{smallmatrix}\right)$, $|A + A^*| = \frac{1}{\sqrt{2}}\left(\begin{smallmatrix} 1 & 0 & 1 \\ 0 & 2 & 0 \\ 1 & 0 & 1 \end{smallmatrix}\right)$, and $\det(|A|+|A^*|-|A+A^*|) < 0$. $\blacksquare$

Problems

1. Let $A \in \mathbb{M}_n$ and $\mathcal{H}(A) = \frac{A^*+A}{2}$. Show that
   (a) $\mathcal{H}(A)$ is Hermitian.
   (b) $\mathcal{H}(A) \geq 0$ if and only if $\Re(x^*Ax) \geq 0$ for all $x \in \mathbb{C}^n$.
   (c) $\mathcal{H}(A) \geq 0$ if and only if the numerical range $W(A) = \{x^*Ax : x^*x = 1\}$ of A lies in in the closed right-half plane $\{z : \Re(z) \geq 0\}$.
2. Let $A \in \mathbb{M}_{m\times n}$. Show that $A^*A \geq 0$. What is the rank of $(A^*A)^{1/2}$?
3. Let A be a square matrix and $|A| = (A^*A)^{1/2}$. Show that
   (a) A is positive semidefinite if and only if $|A| = A$.
   (b) A is normal if and only if $|A^*| = |A|$.
   (c) $|A|$ and $|A^*|$ are similar.
   (d) $|A| = U^*PU$ and $|A^*| = P$ if $A = PU$ is a polar decomposition of A, where $P \geq 0$ and U is unitary.
4. Find $|A|$ and $|A^*|$ for each of the following matrices A.
$$\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \quad \begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix}, \quad \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}, \quad \begin{pmatrix} 1 & 1 \\ -1 & -1 \end{pmatrix}.$$
5. Find $|A|$ and $|A^*|$ for $A = \left(\begin{smallmatrix} 1 & a \\ 0 & 0 \end{smallmatrix}\right)$, where $a > 0$.
6. Let $M = \left(\begin{smallmatrix} 0 & A^* \\ A & 0 \end{smallmatrix}\right)$, where $A \in \mathbb{M}_{m\times n}$. Show that $|M| = \left(\begin{smallmatrix} |A| & 0 \\ 0 & |A^*| \end{smallmatrix}\right)$.
7. Let $A, B \in \mathbb{M}_n$ be such that A^2 and B^2 are Hermitian and $A^2 \geq B^2 \geq 0$.
   (a) Show that the eigenvalues of A and B are all real.
   (b) Show that $|\nu_i(A)| \geq |\nu_i(B)|$ for each i, where $\nu_i(X)$'s are the eigenvalues of $X \in \mathbb{M}_n$ ordered as $|\nu_1(X)| \geq |\nu_2(X)| \geq \cdots \geq |\nu_n(X)|$.
   (c) Show that $|\det A| \geq |\det B|$.
   (d) Show that $|A| \geq |B|$ if A and B are Hermitian and $A^2 \geq B^2$.
   (e) Are (a), (b), (c), and (d) all true for $A, B \in \mathbb{M}_n$ with $A^2 \geq B^2$?
8. Let A be an $n \times n$ normal matrix with the spectral decomposition
$$A = U \operatorname{diag}(\lambda_1, \ldots, \lambda_n)U^* = \sum_{i=1}^{n} \lambda_i u_i u_i^*,$$
   where $u_1, \ldots, u_n \in \mathbb{C}^n$ and $U = (u_1, \ldots, u_n)$ is unitary. Find $|A|$.
9. Let A be an $m \times n$ matrix. Show that $|UAV| = V^*|A|V$ for any $m \times m$ unitary matrix $U \in \mathbb{M}_m$ and any $n \times n$ unitary matrix V.
10. Let $A \in \mathbb{M}_n$. Show that A and UAV have the same singular values for unitary $U, V \in \mathbb{M}_n$. Do they have the same eigenvalues? (As is known, A and W^*AW have the same eigenvalues for any unitary $W \in \mathbb{M}_n$.)

11. Let H and K be $n \times n$ Hermitian matrices such that $H \geq \pm K$.
    (a) Show that $\mathrm{tr} H \geq \mathrm{tr}\,|K|$.
    (b) If H and K commute, show that $H \geq |K|$.
    (c) Is (b) necessarily true if H and K do not commute?

12. Let A be a square matrix and let $|A| = (A^*A)^{1/2}$.
    (a) Find an example: $A|A| \neq |A|A$.
    (b) Show that $A|A| = |A|A \Rightarrow |A|^2 = |A^2|$. ($A$ is normal; Sect. 9.1).

13. Show that for any matrix A there exist matrices X and Y such that
$$A = (AA^*)X \quad \text{and} \quad A = (AA^*)^{1/2}Y.$$

14. Let A be an $m \times n$ matrix, where $m \geq n$. Show that there exists an $m \times m$ unitary matrix U such that $AA^* = U^* \left(\begin{smallmatrix} A^*A & 0 \\ 0 & 0 \end{smallmatrix}\right) U$.

15. Let $f(X) = |X|$, X^2, $X^{1/2}$, or X^{-1}. Show that for matrices A and B
$$f(A \otimes B) = f(A) \otimes f(B)$$
(assuming that $A, B \geq 0$ for $X^{1/2}$ and the inverses exist for X^{-1}).

16. If A is an invertible square matrix, show that $|A^{-1}| = |A^*|^{-1}$.

17. Is it true that $|A^2| = |A|^2$ for all square matrices A?

18. Show that $|A + A^*| \leq |A| + |A^*|$ for all normal matrices A.

19. Show that $|A + A^*| \not\leq |A|^2 + I$ for $A = \left(\begin{smallmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{smallmatrix}\right)$.

20. Show that $|A \circ A^*| \not\leq |A|^2 \circ I$ for $A = \left(\begin{smallmatrix} 0 & 1 & 0 \\ -1 & 1 & 0 \\ 0 & 0 & 1 \end{smallmatrix}\right)$.

21. Show that $|A \circ A^*| \not\leq |A| \circ |A^*|$ for $A = \left(\begin{smallmatrix} 0 & 0 & 1 \\ 0 & 0 & 1 \\ 1 & 1 & 0 \end{smallmatrix}\right)$.

22. Show that $4|A + B|^{-1} \not\leq |A|^{-1} + |B|^{-1}$ for $A = \left(\begin{smallmatrix} 1 & 0 \\ -1 & 1 \end{smallmatrix}\right)$ and $B = A^*$.

23. Find an example showing that it is not true that for $A, B \in \mathbb{M}_n$
$$|A + B| \leq |A| + |B|.$$
Show that the following trace inequality, however, always holds:
$$\mathrm{tr}\,|A + B| \leq \mathrm{tr}(|A| + |B|).$$

24. Show that $\frac{A^2+B^2}{2} \geq \left(\frac{A+B}{2}\right)^2$ for all n-square Hermitian matrices A and B and show by example that $\frac{A^3+B^3}{2} \geq \left(\frac{A+B}{2}\right)^3$ need not be true.

25. Let $A, B \in \mathbb{M}_n$, $p, q > 1$, and $1/p + 1/q = 1$. Show that
$$|A - B|^2 + \left|\sqrt{p/q}\, A + \sqrt{q/p}\, B\right|^2 = p|A|^2 + q|B|^2.$$

26. Let $A, B \in \mathbb{M}_n$, $a, b > 0$, $c \in \mathbb{R}$, and $ab \geq c^2$. Show that
$$a|A|^2 + b|B|^2 + c(A^*B + B^*A) \geq 0.$$

27. Let $A_1, \ldots, A_k \in \mathbb{M}_n$, $a_1, \ldots, a_k \geq 0$, $a_1 + \cdots + a_k = 1$. Show that
$$|a_1 A_1 + \cdots + a_k A_k|^2 \leq a_1 |A_1|^2 + \cdots + a_k |A_k|^2.$$

28. Let A, B, C, D be n-square matrices. If $CC^* + DD^* \leq I_n$, show that
$$|CA + DB| \leq (|A|^2 + |B|^2)^{1/2}.$$

29. Let A and B be $n \times n$ Hermitian matrices. Show that
$$\begin{pmatrix} A^2 & B^2 \\ B^2 & I \end{pmatrix} \geq 0 \quad \Rightarrow \quad \begin{pmatrix} |A| & |B| \\ |B| & I \end{pmatrix} \geq 0.$$

30. Let A be a square matrix and let $H = \mathcal{H}(A) = \frac{A^*+A}{2}$. Show that
$$\max\{i_+(H), i_-(H)\} \leq \operatorname{rank}(|A|) = \operatorname{rank}(A).$$
[Recall that $i_+(H)$ and $i_-(H)$ are respectively the numbers of positive eigenvalues and negative eigenvalues of Hermitian matrix H. Sect. 8.1.]

31. Let A be a square matrix and let $\mathcal{H}(A) = \frac{1}{2}(A^* + A)$. Show that the following inequalities hold for $i = 1$ and they need not be true for $i > 1$:
$$\sigma_i(\mathcal{H}(A)) \leq \sigma_i(A), \quad |\lambda_i(\mathcal{H}(A))| \leq \sigma_i(A).$$

32. Let A be a square matrix (with real or complex entries).
    (a) Show that $|\operatorname{tr}(\frac{A^*+A}{2})| \leq \operatorname{tr}\left((A^*A)^{\frac{1}{2}}\right)$, where the inequality becomes an equality if and only if A or $-A$ is positive semidefinite.
    (b) If A is normal, show that $|\det(\frac{A+A^*}{2})| \leq |\det A|$, where equality holds if and only if A is (normal) singular or Hermitian.
    (c) If $A + A^* \geq 0$, show that $|\det(\frac{A+A^*}{2})| \leq |\det A|$, where equality holds if and only if A is Hermitian (i.e., PSD as $A + A^* \geq 0$).
    (d) Do the inequalities in (b) and (c) hold for general matrices A?
    (e) Show that $\frac{A^*+A}{2} = (A^*A)^{\frac{1}{2}}$ if and only if A is positive semidefinite.

33. Let $A \in \mathbb{M}_n$. Show that the following statements are equivalent.
    (a) $|A| = A$, i.e., $A \geq 0$ (where $|A| = (A^*A)^{1/2}$).
    (b) $\operatorname{tr}|A| = |\operatorname{tr} A|$.
    (c) $\operatorname{tr}|A| = \operatorname{tr} A$.

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CHAPTER 9

Normal Matrices

Introduction: A great deal of elegant work has been witnessed for normal matrices. The goal of this chapter is to present basic results and methods about normal matrices. Sect. 9.1 gives conditions equivalent to the normality of matrices, Sect. 9.2 focuses on a special type of normal matrix with entries consisting of zeros and ones, Sect. 9.3 studies the positive semidefinite matrix $(A^*A)^{1/2}$ associated with a matrix A (see also Sect. 8.6), and finally Sect. 9.4 compares two normal matrices in terms of norms.

9.1 Equivalent Conditions

Recall (from Sect. 1.2 and Sect. 3.2) that a square matrix A is *normal* if it commutes with its conjugate transpose A^*, that is, A is normal if and only if

$$A^*A = AA^*.$$

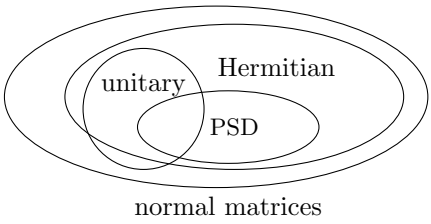


Fig. 9.1. Normal matrix family

Matrix normality is one of the most interesting topics in linear algebra and matrix theory because normal matrices have not only simple structures under unitary similarity but also many applications.

This section presents equivalent conditions about matrix normality.

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Theorem 9.1.1 *Let $A = (a_{ij})$ be an n-square matrix and let $\lambda_1, \ldots, \lambda_n$ be the eigenvalues of A. The following statements are equivalent.*

(1) *A is normal, that is, $A^*A = AA^*$.*

(2) *A is unitarily diagonalizable, that is, for some unitary $U \in \mathbb{M}_n$,*

$$U^*AU = \operatorname{diag}(\lambda_1, \ldots, \lambda_n). \tag{9.1}$$

(3) *There exists a polynomial $p(x)$ such that $A^* = p(A)$.*

(4) *There are eigenvectors of A that form an orthonormal basis for $\mathbb{C}^n$.*

(5) *A has n linearly independent eigenvectors, and the eigenvectors corresponding to distinct eigenvalues are mutually orthogonal.*

(6) *Every eigenvector of A is an eigenvector of A^*.*

(7) *Every eigenvector of A is an eigenvector of $A + A^*$ (or $A - A^*$).*

(8) *$A = B + iC$ for some Hermitian B and C satisfying $BC = CB$.*

(9) *If U is a unitary matrix such that $U^*AU = \left(\begin{smallmatrix} B & C \\ 0 & D \end{smallmatrix}\right)$, where B and D are square, then B and D are normal and $C = 0$.*

(10) *If $W \subseteq \mathbb{C}^n$ is an invariant subspace of A, then so is $W^\perp$.*

(11) *If x is an eigenvector of A, then $x^\perp$ is an invariant subspace of A.*

(12) *A can be written as $A = \sum_{i=1}^n \lambda_i E_i$, where $\lambda_i \in \mathbb{C}$ and $E_i \in \mathbb{M}_n$ satisfy $E_i^2 = E_i = E_i^*$, $E_iE_j = 0$ if $i \neq j$, and $\sum_{i=1}^n E_i = I_n$.*

(13) *$\sum_{i=1}^n \sigma_i^2 = \sum_{i=1}^n |\lambda_i|^2$, where σ_i's are the singular values of A.*

(14) *$\sum_{i=1}^n \sigma_i = \sum_{i=1}^n |\lambda_i|$, where σ_i's are the singular values of A.*

(15) *The singular values of A are $|\lambda_1|, \ldots, |\lambda_n|$.*

(16) $\sum_{i=1}^n (\Re(\lambda_i))^2 = \frac{1}{4}\operatorname{tr}(A + A^*)^2$.

(17) $\sum_{i=1}^n (\Im(\lambda_i))^2 = -\frac{1}{4}\operatorname{tr}(A - A^*)^2$.

(18) *The eigenvalues of $A + A^*$ are $\lambda_1 + \overline{\lambda_1}, \ldots, \lambda_n + \overline{\lambda_n}$.*

(19) *The eigenvalues of AA^* (or A^*A) are $\delta_1\lambda_1, \ldots, \delta_n\lambda_n$ for some scalars $\delta_1, \ldots, \delta_n$, where $|\delta_i| \leq |\lambda_i|$, $i = 1, \ldots, n$.*

(20) $\operatorname{tr}(A^*A)^2 = \operatorname{tr}\left((A^*)^2A^2\right)$.

(21) $(A^*A)^2 = (A^*)^2A^2$.

(22) *$\|Ax\| = \|A^*x\|$ for all $x \in \mathbb{C}^n$.*

(23) *$\langle Ax, Ay\rangle = \langle A^*x, A^*y\rangle$ for all $x, y \in \mathbb{C}^n$.*

(24) *$|A| = |A^*|$, where $|A| = (A^*A)^{1/2}$.*

(25) *$A^* = AU$ for some unitary $U \in \mathbb{M}_n$.*

(26) *$A^* = VA$ for some unitary $V \in \mathbb{M}_n$.*

(27) *$UP = PU$ if $A = UP$ is a polar decomposition of A.*

(28) *$AU = UA$ if $A = UP$ is a polar decomposition of A.*

(29) *$AP = PA$ if $A = UP$ is a polar decomposition of A.*
[Note: U is unitary and P is positive semidefinite in (27)–(29).]

(30) *If $A = UDV$ is a singular decomposition of A, where U and V are unitary, and D is nonnegative diagonal, then $VUD = DVU$.*

(31) *A commutes with some normal matrix of no duplicate eigenvalues.*

(32) *A commutes with $A + A^*$ (or $A - A^*$).*

(33) *$A + A^*$ and $A - A^*$ commute.*

(34) *A commutes with A^*A.*

(35) *A commutes with $|A|$, where $|A| = (A^*A)^{1/2}$.*

(36) *There exists a polynomial $p(x)$ such that $|A| = p(A)$.*

(37) *A commutes with $AA^* - A^*A$.*

(38) *$A^*B = BA^*$ whenever $AB = BA$.*

(39) *$A^*A - AA^*$ is a positive semidefinite matrix.*

(40) *$|\langle Ax, x\rangle| \le \langle |A|x, x\rangle$ for all $x \in \mathbb{C}^n$, where $|A| = (A^*A)^{1/2}$.*

Proof. (2) $\Leftrightarrow$ (1): This is the spectral decomposition (Theorem 3.2.2).

(3) $\Leftrightarrow$ (2): To show that (2) implies (3), we choose a polynomial $p(x)$ of degree at most $n-1$ (by interpolation; see Problem 35 of Sect. 3.1) such that

$$p(\lambda_i) = \overline{\lambda_i}, \quad i = 1, \ldots, n.$$

Thus, if $A = U^* \operatorname{diag}(\lambda_1, \ldots, \lambda_n)U$ for some unitary matrix U, then

$$\begin{aligned} A^* &= U^* \operatorname{diag}(\overline{\lambda_1}, \ldots, \overline{\lambda_n})U = U^* \operatorname{diag}(p(\lambda_1), \ldots, p(\lambda_n))U \\ &= U^* p(\operatorname{diag}(\lambda_1, \ldots, \lambda_n))U = p(U^* \operatorname{diag}(\lambda_1, \ldots, \lambda_n)U) = p(A). \end{aligned}$$

For the other direction, if $A^* = p(A)$ for some polynomial p, then

$$A^*A = p(A)A = Ap(A) = AA^*.$$

(4) $\Leftrightarrow$ (2): If (9.1) holds, then multiplying by U from the left gives

$$AU = U \operatorname{diag}(\lambda_1, \ldots, \lambda_n).$$

Partition $U = (u_1, \ldots, u_n)$ according to its columns. Then

$$Au_i = \lambda_i u_i, \quad i = 1, \ldots, n,$$

Thus, the columns of U are eigenvectors of A and they form an orthonormal basis of $\mathbb{C}^n$ because U is unitary. Conversely, if A has a set of eigenvectors that form an orthonormal basis for $\mathbb{C}^n$, then the matrix U consisting of these vectors as columns is unitary and satisfies (9.1).

$(5) \Leftrightarrow (4)$: Apparently, $(4) \Rightarrow (5)$. We show $(5) \Rightarrow (4)$. For the n linearly independent eigenvectors, we regroup them so that the eigenvectors associated with the same eigenvalue are in the same group. Thus, the eigenvectors from different groups are mutually orthogonal. By the Gram–Schmidt orthonormalization (Sect. 1.4), we convert the eigenvector(s) in each group to orthonormal vectors. Then, all these n orthonormal vectors are eigenvectors and they comprise an orthonormal basis for $\mathbb{C}^n$.

$(6) \Leftrightarrow (1)$: For $(1) \Rightarrow (6)$, let A be normal and let u be a unit eigenvector of A corresponding to eigenvalue λ. Extend u to a unitary matrix with u as the first column (Theorem 1.4.5, Sect. 1.4). Then

$$U^*AU = \begin{pmatrix} \lambda & \alpha \\ 0 & A_1 \end{pmatrix}, \tag{9.2}$$

where α is some row vector. The normality of A forces $\alpha = 0$.

Taking the conjugate transpose and by a simple computation, we see that u is an eigenvector of A^* corresponding to the eigenvalue $\overline{\lambda}$ of A^*.

To see the other way around, we use induction on n. Note that

$$Ax = \lambda x \quad \Leftrightarrow \quad (U^*AU)(U^*x) = \lambda(U^*x)$$

for any n-square unitary matrix U. Thus, when considering $Ax = \lambda x$, we may assume that A is upper-triangular by Schur decomposition.

Take $e_1 = (1, 0, \ldots, 0)^T$. Then e_1 is an eigenvector of A. Hence, by the assumption, e_1 is an eigenvector of A^*. Let $A^*e_1 = \mu e_1$ for some scalar μ. A direct computation yields that the first column of A^* (thus the first row of A) consists of zeros except the first component. Thus, if we write

$$A = \begin{pmatrix} \lambda_1 & 0 \\ 0 & B \end{pmatrix}, \quad \text{then} \quad A^* = \begin{pmatrix} \overline{\lambda_1} & 0 \\ 0 & B^* \end{pmatrix},$$

where B is a matrix of order $n-1$. As every eigenvector of A is an eigenvector of A^*, this property is inherited by B and B^*. An induction hypothesis on B shows that A is diagonal. It follows that A is normal.

$(7) \Rightarrow (6)$: Let $(A + A^*)u = \lambda u$ and $Au = \mu u$, $u \neq 0$. Then $A^*u = \lambda u - Au = (\lambda - \mu)u$, that is, u is an eigenvector of A^*. Conversely, for $(6) \Rightarrow (7)$, let $Au = \lambda u$ and $A^*u = \mu u$. Then $(A + A^*)u = (\lambda + \mu)u$.

$(8) \Leftrightarrow (1)$: Let $B = \frac{A+A^*}{2}$ and $C = \frac{A-A^*}{2i}$. Then B and C are Hermitian, and $A = B + iC$ (*Cartesian decomposition*). Upon computation, we have

$$A^*A = AA^* \quad \Leftrightarrow \quad BC = CB.$$

$(9) \Leftrightarrow (1)$: We show that (1) implies (9). The other direction is easy. By computation, we have that $A^*A = AA^*$ implies

$$\begin{pmatrix} B^*B & B^*C \\ C^*B & C^*C + D^*D \end{pmatrix} = \begin{pmatrix} BB^* + CC^* & CD^* \\ DC^* & DD^* \end{pmatrix}.$$

Thus,

$$B^*B = BB^* + CC^* \quad \text{and} \quad C^*C + D^*D = DD^*.$$

By taking the trace for both sides of the first identity and noticing that $\mathrm{tr}(BB^*) = \mathrm{tr}(B^*B)$, we obtain $\mathrm{tr}(CC^*) = 0$. This reveals $C = 0$. Thus, B is normal. For the same reason from the second identity, D is normal.

We have shown that the first ten conditions are all equivalent.

(9) $\Rightarrow$ (10): Note that $\mathbb{C}^n = W \oplus W^\perp$ and that the vectors in an orthonormal basis of W and the vectors in an orthonormal basis of $W^\perp$ form an orthonormal basis of $\mathbb{C}^n$. Consider A as a linear map on $\mathbb{C}^n$. Because W is invariant under A, i.e., $A(w) \in W$ for all $w \in W$, the matrix representation of A with respect to the orthonormal basis of $\mathbb{C}^n$ has the form $\left(\begin{smallmatrix} B & C \\ 0 & D \end{smallmatrix}\right)$. To be explicit, $AU = U \left(\begin{smallmatrix} B & C \\ 0 & D \end{smallmatrix}\right)$, where U is the unitary matrix composed of the orthonormal basis vectors of W and $W^\perp$. By the assumption, $C = 0$. Thus, $W^\perp$ is invariant under A.

(10) $\Rightarrow$ (11) $\Rightarrow$ (4): (11) is a special case of (10) with $W = \{\mu x : \mu \in \mathbb{C}\}$. For (11) $\Rightarrow$ (4), let (λ_1, x_1) be an eigenpair (which exists over $\mathbb{C}$), where x_1 is a unit vector. By (11), $x_1^\perp$ is invariant under A. Thus, A has an eigenpair (λ_2, x_2), where $x_2 \in x_1^\perp$ is a unit vector. Note that x_1, x_2 are orthonormal. Since $x_1^\perp$ and $x_2^\perp$ are invariant under A, we see that $\{x_1, x_2\}^\perp = x_1^\perp \cap x_2^\perp$ is invariant under A. A has an eigenpair (λ_3, x_3), where x_3 is a unit vector in $\{x_1, x_2\}^\perp$. Repeating this process, we obtain a set of eigenvectors $x_1, x_2, x_3, \ldots, x_n$ of A that form an orthonormal basis of $\mathbb{C}^n$.

(12) $\Rightarrow$ (1): It is by direct computations:

$$A^*A = \sum_i \overline{\lambda_i} E_i^* \cdot \sum_j \lambda_j E_j = \sum_{i,\,j} \overline{\lambda_i} \lambda_j E_i E_j = \sum_i |\lambda_i|^2 E_i = AA^*.$$

(2) $\Rightarrow$ (12): Let $U = (u_1, \ldots, u_n)$, where u_i is the ith column of U. Then

$$A = U\,\mathrm{diag}(\lambda_1, \ldots, \lambda_n)U^* = \lambda_1 u_1 u_1^* + \cdots + \lambda_n u_n u_n^*.$$

Take $E_i = u_i u_i^*$, $i = 1, \ldots, n$. It is easy to check the conditions in (12).

(13) $\Leftrightarrow$ (2): This is Example 3.2.3 of Chap. 3.

(14) $\Leftrightarrow$ (1): It is easy to see that (1) $\Rightarrow$ (2) $\Rightarrow$ (14). We show (14) $\Rightarrow$ (1).

If $A = 0$, it is trivial. Let $\mathrm{rank}(A) = r > 0$ and let $\sigma_1, \ldots, \sigma_r$ be the nonzero singular values of A. Let $A = UTU^*$ be a Schur decomposition of A, where $U \in \mathbb{M}_n$ is unitary and $T = \begin{pmatrix} \lambda_1 & & \star \\ & \ddots & \\ 0 & & \lambda_n \end{pmatrix} \in \mathbb{M}_n$ is upper-triangular ($\star$ for unimportant entries). Let $|\lambda_k| = \lambda_k \rho_k$, $|\rho_k| = 1$, $k = 1, \ldots, n$, and let $P = \mathrm{diag}(\rho_1, \ldots, \rho_n)$. Then P is (diagonal) unitary, and $TP = \begin{pmatrix} |\lambda_1| & & \star \\ & \ddots & \\ 0 & & |\lambda_n| \end{pmatrix}$.

Since U and P are unitary, A, T, and TP have the same nonzero singular values $\sigma_1, \ldots, \sigma_r$. Let $TP = V^*DW$ be a singular value decomposition of TP, where $V, W \in \mathbb{M}_n$ are unitary, and $D = \left(\begin{smallmatrix} D_r & 0 \\ 0 & 0 \end{smallmatrix}\right)$ with $D_r = \mathrm{diag}(\sigma_1, \ldots, \sigma_r)$.

Let $Q = WV^*$. Then Q is unitary. We compute the trace of TP and get

$$|\lambda_1| + |\lambda_2| + \cdots + |\lambda_n| = \operatorname{tr}(V^*DW) = \operatorname{tr}(DWV^*) = \operatorname{tr}(DQ).$$

Partition Q as $Q = \left(\begin{smallmatrix} S & \star \\ \star & \star \end{smallmatrix}\right)$, where $S = (s_{ij})$ is an $r \times r$ matrix, $|s_{ij}| \le 1$ for all i, j. Then $\operatorname{tr}(DQ) = \operatorname{tr}(D_r S) = s_{11}\sigma_1 + \cdots + s_{rr}\sigma_r$. It follows that

$$|\lambda_1| + |\lambda_2| + \cdots + |\lambda_n| = s_{11}\sigma_1 + \cdots + s_{rr}\sigma_r \le \sigma_1 + \cdots + \sigma_r. \tag{9.3}$$

If equality holds in (9.3), then $s_{11} = \cdots = s_{rr} = 1$. Since Q is unitary, it must be in the form $Q = \left(\begin{smallmatrix} I_r & 0 \\ 0 & \star \end{smallmatrix}\right)$. We have $DQ = D$. It follows that $DWV^* = D$, implying $DW = DV$, thus $TP = V^*DW = V^*DV$ which is Hermitian (positive semidefinite). So, TP is diagonal (as TP is upper-triangular) and T is diagonal (as P is diagonal). Thus, $A = UTU^*$ is normal.

[Note: See a similar proof for (14) $\Rightarrow$ (1) in Theorem 9.3.2 of Sect. 9.3.]

(15) $\Rightarrow$ (14) $\Leftrightarrow$ (2) $\Rightarrow$ (15): This is nearly obvious.

(16) $\Rightarrow$ (13): Notice that

$$\operatorname{tr}(A + A^*)^2 = \operatorname{tr} A^2 + 2\operatorname{tr}(A^*A) + \operatorname{tr}(A^*)^2.$$

If (16) holds, then

$$4\sum_{i=1}^{n}(\Re(\lambda_i))^2 = \sum_{i=1}^{n}\lambda_i^2 + 2\operatorname{tr}(A^*A) + \sum_{i=1}^{n}\overline{\lambda_i}^2.$$

Since $4(\Re(\lambda_i))^2 = (\lambda_i + \overline{\lambda_i})^2 = \lambda_i^2 + 2|\lambda_i|^2 + \overline{\lambda_i}^2$, the above displayed identity yields $\operatorname{tr}(A^*A) = \sum_i |\lambda_i|^2$, which is (13). Note that (2) $\Rightarrow$ (16).

Similarly, (17) $\Rightarrow$ (13) and (2) $\Rightarrow$ (17).

(18) $\Rightarrow$ (16): If the eigenvalues of $A + A^*$ are $\lambda_1 + \overline{\lambda_1}, \ldots, \lambda_n + \overline{\lambda_n}$, then their squares $(\lambda_i + \overline{\lambda_i})^2$, $i = 1, \ldots, n$, are the eigenvalues of $(A + A^*)^2$. Thus,

$$\operatorname{tr}(A + A^*)^2 = \sum_{i=1}^{n}(\lambda_i + \overline{\lambda_i})^2 = 4\sum_{i=1}^{n}(\Re(\lambda_i))^2,$$

which is (16). Note that (18) follows from (2) at once.

(19) $\Leftrightarrow$ (2): Obviously (2) $\Rightarrow$ (15) $\Rightarrow$ (19) by taking $\delta_i = \overline{\lambda_i}$ for each i. For the converse, we may assume that $|\lambda_1| \ge \cdots \ge |\lambda_n|$ and that $A = (a_{ij})$ is upper-triangular, $a_{ii} = \lambda_i$ for $i = 1, \ldots, n$, and $a_{ij} = 0$ if $i > j$. Compute the $(1,1)$-entry of AA^* to get $|\lambda_1|^2 + \sum_{j>1}|a_{1j}|^2$, which cannot exceed the largest eigenvalue of AA^*: $\max_k\{\delta_k\lambda_k\} \le |\lambda_1|^2$. Thus, all $a_{1j} = 0$ for $j > 1$, and $A = \left(\begin{smallmatrix} \lambda_1 & 0 \\ 0 & B \end{smallmatrix}\right)$, where $B \in \mathbb{M}_{n-1}$. $|\lambda_1|^2$ is an (largest) eigenvalue of AA^*. We can list the eigenvalues of AA^* as $|\lambda_1|^2, \delta_2\lambda_2, \ldots, \delta_n\lambda_n$. The eigenvalues of B are $\lambda_2, \ldots, \lambda_n$, and the eigenvalues of BB^* are $\delta_2\lambda_2, \ldots, \delta_n\lambda_n$. By induction, B is diagonal. Thus, A is diagonal. So, (19) implies (2).

(20) $\Leftrightarrow$ (1): (20) is immediate from (1). To see that (20) implies (1), we use the facts that for any square matrices X and Y of the same size,

$$\operatorname{tr}(XY) = \operatorname{tr}(YX) \quad \text{and} \quad \operatorname{tr}(X^*X) = 0 \;\Rightarrow\; X = 0.$$

Upon computation and noticing that $\operatorname{tr}(AA^*)^2 = \operatorname{tr}(A^*A)^2$, we have

$$\begin{aligned} &\operatorname{tr}\left((A^*A - AA^*)^*(A^*A - AA^*)\right) \\ &\quad = \operatorname{tr}(A^*A - AA^*)^2 \\ &\quad = \operatorname{tr}(A^*A)^2 - \operatorname{tr}\left((A^*)^2A^2\right) - \operatorname{tr}\left(A^2(A^*)^2\right) + \operatorname{tr}(AA^*)^2, \end{aligned}$$

which is equal to 0 by assumption. Thus, $A^*A - AA^* = 0$.

(21) $\Leftrightarrow$ (1) because (21) $\Rightarrow$ (20) $\Rightarrow$ (1) $\Rightarrow$ (21).

(22) $\Leftrightarrow$ (1): Rewrite the norm identity as inner product

$$\langle Ax, Ax\rangle = \langle A^*x, A^*x\rangle,$$

equivalently, $\langle x, A^*Ax\rangle = \langle x, AA^*x\rangle$, or $\langle x, (A^*A - AA^*)x\rangle = 0$, which holds for all $x \in \mathbb{C}^n$ if and only if $A^*A - AA^* = 0$ (Problem 24 of Sect. 8.1).

(23) $\Rightarrow$ (22) by setting $x = y$; (22) $\Rightarrow$ (1) $\Rightarrow$ (23).

(24) $\Leftrightarrow$ (1): Use the uniqueness of the positive semidefinite square root.

(25) $\Rightarrow$ (1) $\Leftrightarrow$ (2) $\Rightarrow$ (25): If $A^* = AU$ for some unitary matrix U, then

$$A^*A = A^*(A^*)^* = (AU)(AU)^* = AA^*,$$

which is (1). We now show (2) $\Rightarrow$ (25). Let $A = V^* \operatorname{diag}(\lambda_1, \ldots, \lambda_n)V$, where V is unitary. Take $U = V^* \operatorname{diag}(l_1, \ldots, l_n)V$, where $l_i = \frac{\overline{\lambda_i}}{\lambda_i}$ if $\lambda_i \neq 0$, and $l_i = 1$ otherwise, for $i = 1, \ldots, n$. Then U is unitary and

$$\begin{aligned} A^* &= V^* \operatorname{diag}(\,\overline{\lambda_1}, \ldots, \overline{\lambda_n}\,)V \\ &= V^* \operatorname{diag}(\lambda_1, \ldots, \lambda_n)V \cdot V^* \operatorname{diag}(l_1, \ldots, l_n)V = AU. \end{aligned}$$

Similarly, (26) is equivalent to (1).

(27) $\Leftrightarrow$ (1): Let $A = UP$ be a polar decomposition of A, where U is unitary and P is positive semidefinite. For (1) $\Rightarrow$ (27), $A^*A = AA^*$ implies

$$P^*P = UPP^*U^*, \quad \text{i.e.,} \quad P^2 = UP^2U^*.$$

By taking square roots, we have

$$P = UPU^*, \quad \text{i.e.,} \quad PU = UP.$$

The other direction (27) $\Rightarrow$ (1) is easy to check: $A^*A = P^2 = AA^*$.

(28) $\Leftrightarrow$ (27): Note that U is invertible and $U^{-1}A = P$.

(29) $\Leftrightarrow$ (27): We show that (29) implies (27). The other direction is immediate by multiplying P from the right-hand side.

Suppose that $AP = PA$, that is, $UP^2 = PUP$. If P is nonsingular, then obviously $UP = PU$. Let $r = \text{rank}(A) = \text{rank}(P)$, $0 < r < n$, and let

$$P = V^* \begin{pmatrix} D & 0 \\ 0 & 0 \end{pmatrix} V,$$

where V is unitary and D is an $r \times r$ positive definite diagonal matrix. $UP^2 = PUP$ yields $(VUV^*)(VP^2V^*) = (VPV^*)(VUV^*)(VPV^*)$, that is,

$$W \begin{pmatrix} D^2 & 0 \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} D & 0 \\ 0 & 0 \end{pmatrix} W \begin{pmatrix} D & 0 \\ 0 & 0 \end{pmatrix}, \quad \text{where } W = VUV^*.$$

Partition W as

$$\begin{pmatrix} W_1 & W_2 \\ W_3 & W_4 \end{pmatrix}, \quad \text{where } W_1 \text{ is } r \times r.$$

Then

$$W_1 D^2 = DW_1 D \quad \text{and} \quad W_3 D^2 = 0,$$

implying

$$W_1 D = DW_1 \quad \text{and} \quad W_3 = 0.$$

It follows that $W_2 = 0$ because W is unitary and that

$$W \begin{pmatrix} D & 0 \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} D & 0 \\ 0 & 0 \end{pmatrix} W.$$

This results in $UP = PU$, that is, (29) $\Rightarrow$ (27).

(30) $\Leftrightarrow$ (24): Since $A = UDV$, we have $|A| = V^*DV$ and $|A^*| = UDU^*$. Thus, $|A| = |A^*|$ if and only if $V^*DV = UDU^*$, i.e., $DVU = VUD$.

(31) $\Rightarrow$ (2): Let A commute with B, where B is normal and all the eigenvalues of B are distinct. Write $B = V^*CV$, where V is unitary and $C = \text{diag}(c_1, \ldots, c_n)$ is diagonal and all c_i are distinct. Then $AB = BA$ implies $WC = CW$, where $W = (w_{ij}) = VAV^*$. It follows that $w_{ij}c_i = w_{ij}c_j$, i.e., $w_{ij}(c_i - c_j) = 0$ for all i, j. Since $c_i \neq c_j$ if $i \neq j$, we have $w_{ij} = 0$ if $i \neq j$. Thus, VAV^* is diagonal, that is, A is unitarily diagonalizable, and it is normal. (2) $\Rightarrow$ (31): For $A = U \,\text{diag}(\lambda_1, \lambda_2, \ldots, \lambda_n)U^*$, take $B = U \,\text{diag}(1, 2, \ldots, n)U^*$. Then B is normal and $AB = BA$. Note that A does not commute with all normal matrices of distinct eigenvalues in general.

(32) and (33) are equivalent to (1) by direct verifications.

(34) $\Rightarrow$ (21) $\Leftrightarrow$ (1) $\Rightarrow$ (34): If A commutes with A^*A, then $AA^*A = A^*A^2$. Multiply both sides by A^* from the left to get $(A^*A)^2 = (A^*)^2A^2$.

(37) $\Rightarrow$ (20) $\Leftrightarrow$ (1) $\Rightarrow$ (37) is similarly proven.

(35) $\Rightarrow$ (34): $A|A| = |A|A \Rightarrow A(A^*A) = A|A|^2 = |A|^2A = (A^*A)A$. For the converse, (34) $\Rightarrow$ (2) $\Rightarrow$ (35). (See also Problem 38 of Sect. 7.1.)

(36) $\Rightarrow$ (35) is obvious because A commutes with any polynomial in A. For the converse, first, (35) $\Rightarrow$ (2). As for the proof of (2) $\Rightarrow$ (3), we choose a polynomial $p(x)$ of degree at most $n-1$ (by interpolation; see Problem 35 of Sect. 3.1) such that $p(\lambda_i) = |\lambda_i|, \quad i = 1, \ldots, n$. Then $|A| = p(A)$.

(38) $\Rightarrow$ (1) $\Leftrightarrow$ (2): Take $B = A$. For (2) $\Rightarrow$ (38), let $A = UDU^*$, where U is unitary and $D = \operatorname{diag}(\lambda_1, \ldots, \lambda_n)$. If$AB = BA$, where $B \in \mathbb{M}_n$, then

$$(UDU^*)B = B(UDU^*) \quad \Rightarrow \quad D(U^*BU) = (U^*BU)D.$$

Denote $T = U^*BU = (t_{ij})$. Then $DT = TD$, or explicitly,

$$\operatorname{diag}(\lambda_1, \ldots, \lambda_n)T = T \operatorname{diag}(\lambda_1, \ldots, \lambda_n),$$

implying $(\lambda_i - \lambda_j)t_{ij} = 0$, so, $(\overline{\lambda_i} - \overline{\lambda_j})t_{ij} = 0$ for all i, j, which yield $\operatorname{diag}(\overline{\lambda_1}, \ldots, \overline{\lambda_n})(U^*BU) = (U^*BU) \operatorname{diag}(\overline{\lambda_1}, \ldots, \overline{\lambda_n})$, revealing $A^*B = BA^*$.

(39) $\Leftrightarrow$ (1): This is a combination of two facts: $\operatorname{tr}(XY - YX) = 0$ for all square matrices X and Y of the same order (thus $\operatorname{tr}(A^*A - AA^*) = 0$), and if matrix X is positive semidefinite, then $\operatorname{tr} X = 0$ if and only if $X = 0$.

For (40), notice that $|\langle Ax, x\rangle| \leq \langle |A|x, x\rangle$ is unitarily invariant, that is, the inequality holds if and only if $|\langle U^*AUx, x\rangle| \leq \langle |U^*AU|x, x\rangle$, where U is any unitary matrix. (Bear in mind that $|U^*AU| = U^*|A|U$.) By the Schur decomposition (Theorem 3.2.1), we may assume that A is upper-triangular.

(1) $\Rightarrow$ (40): If A is normal, then A is unitarily diagonalizable. We may assume that $A = \operatorname{diag}(\lambda_1, \ldots, \lambda_n)$. Then the inequality is the same as saying that $|\sum_i \lambda_i |x_i|^2| \leq \sum_i |\lambda_i||x_i|^2$, which is obvious.

(40) $\Rightarrow$ (1): Suppose that $|\langle Ax, x\rangle| \leq \langle |A|x, x\rangle$ for all $x \in \mathbb{C}^n$, where A is upper-triangular. We show that A is in fact a diagonal matrix. We demonstrate the proof for the case of $n = 2$; the general case is similarly proven by induction. Let $A = \left(\begin{smallmatrix} \lambda_1 & \alpha \\ 0 & \lambda_2 \end{smallmatrix}\right)$, where $\alpha \in \mathbb{C}$. We show $\alpha = 0$.

If $\lambda_1 = \lambda_2 = 0$ and $\alpha \neq 0$, then we take positive numbers s, t, $s > t$, and set $x = \binom{s}{t} \in \mathbb{R}^2$. Then $|x^*Ax| = st|\alpha|$ and $x^*|A|x = t^2|\alpha|$. The inequality $|x^*Ax| \leq x^*|A|x$ implies $s \leq t$, a contradiction to $s > t$.

If λ_1 (or λ_2) is not 0. Let $|A| = \left(\begin{smallmatrix} a & b \\ \bar{b} & c \end{smallmatrix}\right)$. Putting $x = \binom{1}{0}$ in $|x^*Ax| \leq x^*|A|x$ gives $|\lambda_1| \leq a$. Computing $|A|^2 = A^*A$ and comparing their $(1,1)$-entries, we have $a^2 + |b|^2 = |\lambda_1|^2$. So $b = 0$ and $|A| = \left(\begin{smallmatrix} a & 0 \\ 0 & c \end{smallmatrix}\right)$. Inspecting the $(1,2)$-entries of $|A|^2$ and A^*A, we obtain $\bar{\lambda}_1\alpha = (a + c)b = 0$. Thus, $\alpha = 0$. ∎

About 100 equivalent conditions of normal matrices have been observed in the literature. More are shown as we proceed. For example (Problem 26, Sect. 10.6), $A \in \mathbb{M}_n$ is normal if (and only if) for a positive integer $k \geq 2$,

$$\operatorname{tr}(AA^*)^k = \operatorname{tr}(A^k(A^*)^k).$$

Example 9.1.1 Let $A = (a_{ij})$ be an n-square matrix. Let $\lambda_1, \ldots, \lambda_n$ and $\sigma_1, \ldots, \sigma_n$ be the eigenvalues and the singular values of A, respectively.

(a) For what matrices are the eigenvalues exactly the singular values? Positive semidefinite matrices, because A is normal as $\sum_i \sigma_i = \sum_i \lambda_i$ by Theorem 9.1.1 (15) and the eigenvalues of A are nonnegative.

(b) For what matrices are the diagonal entries exactly the singular values? Nonnegative diagonal matrices, since $\mathrm{tr}(A^*A) = \sum_i |a_{ii}|^2 + \sum_{i \neq j} |a_{ij}|^2$ and $\mathrm{tr}(A^*A) = \sum_i \sigma_i^2$. If a_{ii}'s are σ_i's, then $a_{ij} = 0$ for all $i \neq j$.

(c) If A is a normal (or Hermitian in particular) matrix whose eigenvalues are exactly the diagonal entries, then A is a diagonal matrix.

Proof. By spectral decomposition, let $U^*AU = \mathrm{diag}(\lambda_1, \ldots, \lambda_n)$, where U is unitary. Then $U^*A^*AU = \mathrm{diag}(|\lambda_1|^2, \ldots, |\lambda_n|^2)$, and $\mathrm{tr}(A^*A) = \sum_i |\lambda_i|^2$. On the other hand, $\mathrm{tr}(A^*A) = \sum_i |a_{ii}|^2 + \sum_{i \neq j} |a_{ij}|^2$. If a_{ii}'s are λ_i's, then $a_{ij} = 0$ for all $i \neq j$. (If $A = A^*$, then $a_{ii} \in \mathbb{R}$.) ∎

Normal matrices have simple structures (via unitary similarity). Characterizing normal matrices in terms of their eigenvalues (eigenvectors), numerical range, norm, and related properties is always interesting and important in matrix theory. Below, we derive another necessary and sufficient condition.

Theorem 9.1.2 *$A \in \mathbb{M}_n$ is normal $\Leftrightarrow \|Ax\|^2 \leq \|A^2x\|$ for all unit $x \in \mathbb{C}^n$.*

Proof. If A is normal, then $(A^*A)^2 = (A^*)^2A^2$. By Problem 35 of Sect. 4.1,

$$\|Ax\|^4 = (x^*A^*Ax)^2 \leq x^*(A^*A)^2x = x^*(A^*)^2A^2x = \|A^2x\|^2.$$

Taking the square roots of both sides, we have $\|Ax\|^2 \leq \|A^2x\|$, as desired.

Conversely, suppose that $\|Ax\|^2 \leq \|A^2x\|$ for all $x \in \mathbb{C}^n$ with $\|x\| = 1$. We show by induction on the order of the matrix that A is normal.

If $n = 1$ or $A = 0$, then A is trivially normal. Let $n > 1$. Let $\sigma_1 > 0$ be the largest singular value of A (i.e., the spectral norm $\|A\|_{\mathrm{sp}}$) and let V_1 be the eigenspace of A^*A with respect to σ_1^2, i.e.,

$$V_1 = \{x \in \mathbb{C}^n : (A^*A)x = \sigma_1^2 x\}.$$

We show that V_1 is invariant under A, that is, $Ax \in V_1$ for all $x \in V_1$.

Let $x \in V_1$ (and $x \neq 0$). We need to show that $(A^*A)(Ax) = \sigma_1^2(Ax)$. We first prove that $(A^*A)(Ax)$ and Ax are linearly dependent by considering the inner product $\langle (A^*A)(Ax), Ax \rangle$ and by using the Cauchy-Schwarz inequality.

By scaling, we assume that $\|x\| = 1$. Note that $\|Ax\| = \sqrt{x^*A^*Ax} = \sigma_1$. On the one hand, for (any) unit $x \in \mathbb{C}^n$,

$$0 \leq \langle (A^*A)(Ax), Ax \rangle \leq \|(A^*A)(Ax)\| \|Ax\| \leq \sigma_1^4.$$

On the other hand, by assumption $\|A^2x\| \geq \|Ax\|^2$, we have

$$|\langle (A^*A)(Ax), Ax \rangle| = \langle A^2x, A^2x \rangle = \|A^2x\|^2 \geq \|Ax\|^4 = \sigma_1^4.$$

Thus, $\langle (A^*A)(Ax), Ax\rangle = \|(A^*A)(Ax)\|\|Ax\| = \sigma_1^4$. By the equality case of the Cauchy-Schwarz inequality, $(A^*A)(Ax)$ and Ax are linearly dependent.

For nonzero $x \in V_1$, Ax is nonzero because $(A^*A)x = \sigma_1^2 x \neq 0$. Let $(A^*A)(Ax) = tAx$. Then $\sigma_1^4 = x^*(A^*)^2A^2x = tx^*A^*Ax = t\sigma_1^2$ (for $\|x\| = 1$). Hence, $t = \sigma_1^2$, and $(A^*A)(Ax) = \sigma_1^2(Ax)$, that is, V_1 is invariant under A.

As a linear map on V_1 (over $\mathbb{C}$), A has eigenvalue(s) and eigenvectors. Thus, there exists a unit vector $u \in V_1$ such that $Au = \lambda u$, where $\lambda \in \mathbb{C}$. This gives $(Au)^*(Au) = |\lambda|^2$. Because $u \in V_1$, we have $(A^*A)u = \sigma_1^2 u$. So, $u^*(A^*A)u = \sigma_1^2$. We obtain $|\lambda| = \sigma_1$. It follows that $|u^*Au| = \sigma_1$.

Let U be a unitary matrix with u as its first column. Then $AU = U \left(\begin{smallmatrix} \lambda & a \\ 0 & B \end{smallmatrix}\right)$ for some $a \in \mathbb{C}$ and some $B \in \mathbb{M}_{n-1}$. a must be equal to 0 because A and $U^*AU = \left(\begin{smallmatrix} \lambda & a \\ 0 & B \end{smallmatrix}\right)$ have the same singular values, while the largest singular value of U^*AU is at least $\sqrt{|\lambda|^2 + |a|^2}$. So, $\sigma_1 \geq \sqrt{\sigma_1^2 + |a|^2}$, yielding $a = 0$.

Therefore, A is unitarily similar to $\left(\begin{smallmatrix} \lambda & 0 \\ 0 & B \end{smallmatrix}\right)$. By induction on B, we conclude that A is unitarily similar to a diagonal matrix, i.e., A is normal. ∎

Problems

1. Is matrix $\left(\begin{smallmatrix} 1 & i \\ i & 1 \end{smallmatrix}\right)$ normal, Hermitian, or symmetric? How about $\left(\begin{smallmatrix} 1 & i \\ -i & 1 \end{smallmatrix}\right)$?
2. Let A be an $n \times n$ normal matrix and P be an $n \times n$ permutation matrix. Show that PAP^T is normal but PAP (or AP) need not be normal.
3. If A is a normal matrix, show that A^k is also a normal matrix for any positive integer k. Find a pair of normal matrices A and B such that neither AB nor $A + B$ is normal.
4. If $\left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)$ is normal, where $a, b, c, d \in \mathbb{R}$, show that either $c = b$, or $c = -b$ and $d = a$, i.e., a 2×2 real normal matrix is in the form $\left(\begin{smallmatrix} a & b \\ b & d \end{smallmatrix}\right)$ or $\left(\begin{smallmatrix} a & b \\ -b & a \end{smallmatrix}\right)$.
5. Let $A = \left(\begin{smallmatrix} a & b \\ 0 & c \end{smallmatrix}\right)$, where $a, b, c \in \mathbb{C}$. Show that
   (a) A is normal if and only if $b = 0$.
   (b) A is Hermitian if and only if $b = 0$ and $a, c \in \mathbb{R}$.
   (c) A is PSD if and only if $b = 0$ and $a, c \geq 0$.
   (d) A is unitary if and only if $b = 0$ and $|a| = |c| = 1$.
   (e) A is nilpotent if and only if $a = c = 0$, i.e., $A = \left(\begin{smallmatrix} 0 & b \\ 0 & 0 \end{smallmatrix}\right)$.
   (f) A is idempotent if and only if $A = 0$, I_2, $\left(\begin{smallmatrix} 1 & b \\ 0 & 0 \end{smallmatrix}\right)$, or $\left(\begin{smallmatrix} 0 & b \\ 0 & 1 \end{smallmatrix}\right)$.
6. Give an example showing that a unitary matrix U in a spectral decomposition $A = U^* \operatorname{diag}(\lambda_1, \ldots, \lambda_n)U$ of a normal matrix $A \in \mathbb{M}_n$ need not be unique. Show that A^p defined to be $A^p = U^* \operatorname{diag}(\lambda_1^p, \ldots, \lambda_n^p)U$ for real $p \geq 0$ is independent of the choices of unitary matrices U and the orders of λ_i's (see Problem 50 of Sect. 3.2). Why is this definition consistent with $A^p = A \cdots A$ (p A's) when p is a nonnegative integer?

7. Find examples $A, B \in \mathbb{M}_2$: The columns of A are orthogonal, but the rows are not; both B^*B and BB^* are diagonal, but B is not normal.

8. Let $A \in \mathbb{M}_n$. Show that A is normal if and only if

   (a) $I_n + A$ is normal.
   (b) A commutes with $A - A^*$.
   (c) Every eigenvector of A is an eigenvector of $A - A^*$.

9. If A is a normal matrix such that $A + A^* + I = 0$, show that the eigenvalues of A all lie on the line $x = -\frac{1}{2}$ in the xy-plane.

10. Let A be an $n \times n$ matrix and let $x \in \mathbb{C}^n$ be a unit vector. Show that

$$|\langle Ax, x\rangle|^2 \le \langle |A|^2 x, x\rangle.$$

    When does equality hold for all unit $x \in \mathbb{C}^n$? For some unit $x \in \mathbb{C}^n$?

11. Let $A \in \mathbb{M}_n$ and let $\lambda_1, \ldots, \lambda_n$ be the eigenvalues of A. Show that $\lambda_1^2, \ldots, \lambda_n^2$ are the eigenvalues of A^*A if and only if A is Hermitian.

12. Let $\lambda_1, \lambda_2, \ldots, \lambda_n$ be the eigenvalues of matrix $A \in \mathbb{M}_n$. Show that

$$|\lambda_i \lambda_j| \le \lambda_{\max}(A^*A)$$

    for all i, j, where $\lambda_{\max}(A^*A)$ is the largest eigenvalue of A^*A.

13. Let $A \in \mathbb{M}_n$ have eigenvalues λ_i's and singular values σ_i's ordered as $|\lambda_1| \ge |\lambda_2| \ge \cdots \ge |\lambda_n|$ and $\sigma_1 \ge \sigma_2 \ge \cdots \ge \sigma_n$. Show that

$$\sigma_1\sigma_2\cdots\sigma_k = |\lambda_1\lambda_2\cdots\lambda_k| \text{ for } k = 1, 2, \ldots, n \quad \Leftrightarrow \quad A^*A = AA^*.$$

    [Note: $\sigma_1\sigma_2\cdots\sigma_n = |\lambda_1\lambda_2\cdots\lambda_n|$ always holds; Problem 31 of Sect. 2.4.]

14. Let A be a normal matrix. Show that $Ax = 0$ if and only if $A^*x = 0$.

15. Let A be a normal matrix. If x is an eigenvector of A, show that A^*x is also an eigenvector of A associated with the same eigenvalue.

16. If a square matrix A commutes with some normal matrix with distinct eigenvalues, show that A is normal. Is the converse true?

17. Show that unitary matrices, PSD matrices, Hermitian matrices, skew-Hermitian matrices, real orthogonal matrices, and permutation matrices are all normal. Is a complex orthogonal matrix normal?

18. When is a normal matrix Hermitian? Positive semidefinite? Skew-Hermitian? Unitary? Nilpotent? Or idempotent?

19. When is a triangular matrix normal?

20. Let A be a square matrix. Show that if A is a normal matrix, then $f(A)$ is normal for any polynomial f. If $f(A)$ is normal for some nonzero polynomial f, does it follow that A is normal?

21. Let $A = \left(\begin{smallmatrix} 1 & 0 \\ 0 & i \end{smallmatrix}\right)$. Find a polynomial $p(x)$ such that $A^* = p(A)$.

22. Show that (29) $\Leftrightarrow$ (34) in Theorem 9.1.1 by Problem 38 of Sect. 7.1.

23. Show that two normal matrices are similar if and only if they have the same eigenvalues and if and only if they are unitarily similar.

24. Let A and B be $n \times n$ normal matrices such that $AB = BA$. Show that

    (a) There exists an $n \times n$ unitary matrix U such that U^*AU and U^*BU are diagonal, that is, U diagonalizes both A and B.

    (b) $A+B$, AB, and A^pB^q are normal, where p, q are positive integers.

25. Let A be a nonsingular square matrix and let $M = A^{-1}A^*$. Show that A is normal if and only if M is unitary.

26. Let $A \in \mathbb{M}_n$ be a normal matrix. Show that for any unitary $U \in \mathbb{M}_n$,

$$\min_i\{|\lambda_i(A)|\} \leq |\lambda_i(AU)| \leq \max_i\{|\lambda_i(A)|\}.$$

    [Note: $\lambda_i(X)$ denotes any eigenvalue of matrix X.]

27. Complete the proof of (40) in Theorem 9.1.1 for the case $n > 2$.

28. Let A be a normal matrix and let k be a positive integer.

    (a) If $A^k = A$, show that the eigenvalues λ of A satisfy $\lambda^k = \lambda$.

    (b) If A^k is unitary, show that A is unitary.

    (c) If $A^k = 0$, show that $A = 0$.

29. Let $C = A + B$, where $A \in \mathbb{M}_n$ is Hermitian and $B \in \mathbb{M}_n$ is skew-Hermitian. Show that the following statements are equivalent.

    (a) C is normal. (b) $AB = BA$. (c) AB is skew-Hermitian.

30. Let B be an n-square matrix and let A be the block matrix

$$A = \begin{pmatrix} B & B^* \\ B^* & B \end{pmatrix}.$$

    Show that A is normal. Moreover, if B is a normal matrix having eigenvalues $\lambda_t(B) = x_t + y_t i$, where $x_t, y_t \in \mathbb{R}$, $t = 1, \ldots, n$, show that

$$\det A = 4i \prod_{t=1}^{n} x_t y_t.$$

31. Show that $\mathbb{C}^n = \operatorname{Ker} A \oplus \operatorname{Im} A$ for any n-square normal matrix A.

32. Let A and B be $n \times n$ normal matrices. If $\operatorname{Im} A \perp \operatorname{Im} B$, that is, $\langle x, y\rangle = 0$ for all $x \in \operatorname{Im} A$ and $y \in \operatorname{Im} B$ (over $\mathbb{R}$ or $\mathbb{C}$), show that $A+B$ is normal.

33. Let A be a normal matrix. Show that $A\overline{A} = 0 \Leftrightarrow AA^T = A^TA = 0$.

34. Let A and B be $n \times n$ normal matrices. Show that $\operatorname{rank}(AB) = \operatorname{rank}(BA)$. In particular, if $AB = 0$, show that $BA = 0$ and $AB^* = 0$.

35. Show that for any n-square matrix A

$$\operatorname{tr}(A^*A)^2 \geq \operatorname{tr}\left((A^*)^2A^2\right)$$

and that equality holds if and only if A is normal. Is it true that

$$(A^*A)^2 \geq (A^*)^2A^2?$$

36. Show that $\|Ax\| = \||A|x\|$, where $A \in \mathbb{M}_n$, $x \in \mathbb{C}^n$, and $|A| = (A^*A)^{1/2}$.

37. Let $A = \left(\begin{smallmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{smallmatrix}\right)$.

    (a) Show that $\max_{\|x\|=1} \|Ax\|^2 = \max_{\|x\|=1} \|A^2x\|$.

    (b) Find a unit $y \in \mathbb{R}^3$ such that $\|Ay\|^2 < \|A^2y\|$.

    (c) Find a unit $z \in \mathbb{R}^3$ such that $\|Az\|^2 > \|A^2z\|$.

    (d) Does (b) or (c) contradict (a)?

38. Let $A \in \mathbb{M}_n$ and $x \in \mathbb{C}^n$. If $\|Ax\| = \sigma_1\|x\|$, show that $(A^*A)x = \sigma_1^2x$, where σ_1 is the largest singular value (i.e., the spectral norm) of A.

39. Let A be an n-square matrix. Show that A is normal if and only if $\|A^kx\|^2 \leq \|A^{2k}x\|$ for all unit vectors $x \in \mathbb{C}^n$ and for all $k = 1, 2, \ldots$.

40. Let A be an n-square matrix. Show that A is normal if and only if $\|A^{p+q}x\|^2 \leq \|A^{2p}x\|\|A^{2q}x\|$ for all $x \in \mathbb{C}^n$ and for all $p, q = 0, 1, 2, \ldots$.

41. Let A and B be $n \times n$ matrices. The matrix $AB - BA$ is called the *commutator* of A and B, and it is denoted by $[A, B]$. Show that

    (a) $\operatorname{tr}[A, B] = 0$.

    (b) $[A, B]^* = [B^*, A^*]$.

    (c) $[A, B + C] = [A, B] + [A, C]$.

    (d) $[A, [B, C]] + [B, [C, A]] + [C, [A, B]] = 0$.

    (e) $[A, B]$ is never similar to the identity matrix I_n.

    (f) If A and B are both Hermitian (or skew-Hermitian), then $[A, B]$ is skew-Hermitian.

    (g) If A and B are Hermitian, then the real part of every eigenvalue of $[A, B]$ is zero.

    (h) A is normal if and only if $[A, A^*] = 0$.

    (i) A is normal if and only if $[A, [A, A^*]] = 0$.

42. Let $E \in \mathbb{M}_n$. If $EE^* = \left(\begin{smallmatrix} I_r & 0 \\ 0 & 0 \end{smallmatrix}\right)$ and $E^2 = E$, show that $E = \left(\begin{smallmatrix} I_r & 0 \\ 0 & 0 \end{smallmatrix}\right)$.

43. Let $A = \left(\begin{smallmatrix} 2\pi i & 1 \\ 0 & 0 \end{smallmatrix}\right)$ with $i = \sqrt{-1}$. Show that e^A is normal but A is not. [Note: $e^A e^B = e^B e^A \not\Rightarrow AB = BA$. See Problem 26 of Sect. 2.6.]

44. Verify with the following matrix A that $A^*A - AA^*$ is an entrywise nonnegative matrix, but A is not normal (i.e., $A^*A - AA^* \neq 0$):

$$A = \begin{pmatrix} 0 & 1 & 3 \\ -\sqrt{2} & 0 & -\sqrt{3} \\ 2\sqrt{2} & 2 & 0 \end{pmatrix}.$$

45. Let $N = \left(\begin{smallmatrix} A & B \\ C & D \end{smallmatrix}\right)$ be a block matrix. Prove the following statements.

    (a) If $C = 0$, then N is normal $\Leftrightarrow$ A and D are normal and $B = 0$.

    (b) If $D = 0$, then the normality of N gives $|B| = |C^*|$. Thus, B and C have the same nonzero singular values (and the same rank).

    (c) If N and A are normal, then $|B^*| = |C|$. Consequently, B and C have the same nonzero singular values (and the same rank).

46. Let $\lambda_1, \ldots, \lambda_n$ be the eigenvalues of $A \in \mathbb{M}_n$. Show that A is normal, i.e., $A^*A = AA^*$, if and only if there are scalars $\delta_1, \ldots, \delta_n$ with moduli no more than 1 such that $\delta_1\lambda_1, \ldots, \delta_n\lambda_n$ are the singular values of A.

47. Let A and B be $n \times n$ normal matrices and let $M \in \mathbb{M}_n$. Show that

    (a) If $AM = MA$, then $A^*M = MA^*$.

    (b) If $AM = MB$, then $A^*M = MB^*$.

48. Let A and B be n-square matrices.

    (a) If A and B are Hermitian, show that AB is Hermitian if and only if A and B commute, i.e., $AB = BA$. (Problem 7 of Sect. 8.2.)

    (b) If A and B are normal and $AB = BA$, show that AB is normal.

    (c) Find an example: A, B, and AB are all normal, but $AB \neq BA$.

49. Let A and B be normal matrices such that AB is normal. Show that

    (a) BA is normal.

    (b) $A^*AB = BAA^*$.

    (c) $|A|B = B|A|$, where $|A| = (A^*A)^{1/2}$.

. ——————— ⊙ ——————— .

9.2 Normal Matrices with Zero and One Entries

Matrices with zero and one entries are referred to as $(0,1)$-*matrices*. $(0,1)$-matrices have applications in graph theory and combinatorics, etc. The zero matrices and identity matrices are $(0,1)$-matrices; permutation matrices are $(0,1)$-matrices. There are 2^{mn} $(0,1)$-matrices of size $m \times n$.

One may ask many questions about $(0,1)$-matrices such as: what are the ranks of $(0,1)$-matrices? what are the determinants of square $(0,1)$-matrices? what are the eigenvalues of $(0,1)$-matrices? when are $(0,1)$-matrices invertible? and so on. This section presents three theorems about $(0,1)$-matrices: the first one shows how to construct a symmetric (normal) $(0,1)$-matrix from a given $(0,1)$-matrix, the second one gives a sufficient condition on the normality of $(0,1)$-matrices, and the last one is about commutativity.

Given an $m \times n$ $(0,1)$-matrix, say A, we add the 1s in each row to get row sums $r_1, r_2, \ldots, r_m$. We call $R = (r_1, r_2, \ldots, r_m)$ the *row sum vector* of A and denote it by $R(A)$. Similarly, we can define the *column sum vector* by adding the entries in columns of A and denote it by $S(A)$. For example,

$$A = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 1 \\ 1 & 1 & 0 & 0 \end{pmatrix}, \quad \begin{aligned} R(A) &= (r_1, r_2, r_3, r_4) = (1, 2, 3, 2), \\ S(A) &= (s_1, s_2, s_3, s_4) = (2, 3, 1, 2). \end{aligned}$$

The sum of the components of $R(A)$ equals the total number of 1s in A. The same is true of $S(A)$. Given vectors R and S of nonnegative integers, there may not exist a $(0,1)$-matrix that has R as its row sum vector and S as its column sum vector. For instance, no 3×3 $(0,1)$-matrix A satisfies $R(A) = (3,1,1)$ and $S(A) = (3,2,0)$. For what R and S does there exist a $(0,1)$-matrix with R and S as its row and column sum vectors? This is an intriguing problem and is well studied in combinatorial matrix theory.

Example 9.2.1 $(0,1)$-matrices with the same row and column sum vectors.

(a) $\begin{pmatrix} 1 & 1 & 1 \\ 1 & 0 & 0 \\ 1 & 0 & 0 \end{pmatrix}$ is symmetric, $R = S = (3,1,1)$.

(b) $\begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 0 \\ 1 & 0 & 0 \end{pmatrix}$ is symmetric, $R = S = (3,2,1)$.

(c) $\begin{pmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \end{pmatrix}$ is symmetric, $R = S = (2,2,2)$.

(d) $\begin{pmatrix} 1 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix}$ is not symmetric, $R = S = (2,1,1)$.

(e) There is no 3×3 symmetric $(0,1)$-matrix with $R = (3,3,1)$; nor is there 3×3 symmetric $(0,1)$-matrix with $R = (3,1,0)$. ∎

Apparently, for a symmetric $(0,1)$-matrix, the row sum vector and column sum vector are the same. The following result says the converse is also true in some sense (via reconstruction). Thus, there exists a $(0,1)$-matrix that has the given vector R as its row and column sum vectors if and only if there exists such a symmetric $(0,1)$-matrix.

Theorem 9.2.1 *If there exists a (0, 1)-matrix A with $R(A) = S(A) = R$, then there exists a symmetric (0, 1)-matrix B that has the same row and column vector as A, that is, $R(B) = S(B) = R$.*

Proof. We use induction on n. If $n = 1$ or 2, it is obvious. Let $n > 2$. We can assume that A contains no zero row (or column). Suppose that the assertion is true for the matrices of order less than n. If A is symmetric, there is nothing to prove. We assume that $A = (a_{ij})$ is not symmetric.

If the first column is the transpose of the first row, then by induction hypothesis on the matrices of order $(n-1)$, we can replace the $(n-1)$-square submatrix in the lower-right corner by a symmetric $(0,1)$-matrix of order $n-1$, then we are done. Suppose the first column is not the transpose of the first row. Because A is not symmetric and $R(A) = S(A)$, we have $a_{1p} = 1$, $a_{1q} = 0$, $a_{p1} = 0$, and $a_{q1} = 1$ for some p and q.

Now consider the rows p and q. With $a_{p1} = 0$, $a_{q1} = 1$, if $a_{pt} \leq a_{qt}$, $t = 2, \ldots, n$, then $r_p < r_q$. For the columns p and q, with $a_{1p} = 1$, $a_{1q} = 0$, if $a_{tp} \geq a_{tq}$, $t = 2, \ldots, n$, then $s_p > s_q$. Both cases cannot happen at the same time, or $r_p < r_q = s_q < s_p = r_p$, a contradiction. Thus, there exists a t such that either $a_{pt} = 1$ and $a_{qt} = 0$, or $a_{tp} = 0$ and $a_{tq} = 1$.

Now interchange the 0s and 1s in the intersections of rows p, q and columns $1, t$, or columns p, q and rows $1, t$. Notice that such an interchange reduces the number of different entries in the first row and first column of A without affecting the row and column sum vectors of A. Repeating this process results in a matrix in which the first column is the transpose of the first row. By induction, a symmetric matrix is obtained from A. ∎

Take the following matrix A as an example. In rows 2 and 4, replacing the 0s by 1s and 1s by 0s (all $\times$s remain unchanged) results in the matrix A_1 that has the same row and column sum vector as A, whereas the first column of A_1 is "closer" to the transpose of the first row of A_1 (or A).

$$A = \begin{pmatrix} \times & 1 & \times & 0 \\ 0 & \times & 1 & \times \\ \times & \times & \times & \times \\ 1 & \times & 0 & \times \end{pmatrix}, \quad A_1 = \begin{pmatrix} \times & 1 & \times & 0 \\ 1 & \times & 0 & \times \\ \times & \times & \times & \times \\ 0 & \times & 1 & \times \end{pmatrix}.$$

The next theorem often appears in combinatorics when a configuration of subsets is under investigation.

Theorem 9.2.2 *Let A be an n-square (0, 1)-matrix and let J_n be the n-square all-ones matrix. If*

$$AA^T = tI_n + J_n \tag{9.4}$$

for some positive integer t, then A is a normal matrix, i.e., $AA^T = A^TA$.

Proof. Denote $J_n = J$. By considering the diagonal entries of AA^T, we see that (9.4) implies that every row sum of A is equal to $t + 1$, that is,

$$AJ = (t + 1)J. \tag{9.5}$$

Matrix A is nonsingular, because the determinant

$$(\det A)^2 = \det(AA^T) = \det(tI + J) = (t + n)t^{n-1}$$

is nonzero. Thus, from (9.5), we have

$$A^{-1}J = (t + 1)^{-1}J.$$

Multiplying both sides of (9.4) by J from the right reveals

$$AA^TJ = tJ + J^2 = (t + n)J.$$

It follows by multiplying A^{-1} from the left that

$$A^TJ = (t + 1)^{-1}(t + n)J.$$

By taking the transpose, we have

$$JA = (t + 1)^{-1}(t + n)J. \tag{9.6}$$

Multiply both sides by J from the right to get

$$JAJ = n(t + 1)^{-1}(t + n)J.$$

Multiply both sides of (9.5) by J from the left to get

$$JAJ = n(t + 1)J.$$

Comparing the right-hand sides of the two identities above yields

$$(t + 1)^2 = t + n \quad \text{or} \quad n = t^2 + t + 1.$$

Substituting this into (9.6), one gets

$$JA = (t + 1)J.$$

From (9.5),

$$AJ = JA \quad \text{or} \quad A^{-1}JA = J.$$

We then have

$$\begin{aligned} A^T A &= A^{-1}(AA^T)A = A^{-1}(tI + J)A \\ &= tI + A^{-1}JA = tI + J = AA^T. \ \blacksquare \end{aligned}$$

(9.4) states that if A is a $(0,1)$-matrix such that $AA^T = tI + J$, then $A^T A = tI + J$. Is this true for a general nonnegative square matrix?

The $(0,1)$-matrices that satisfy (9.4) have applications in (block designs) combinatorics. For a general version of the theorem, see Problem 25.

Our next result asserts that if the product of two $(0,1)$-matrices is a matrix whose diagonal entries are all 0s and whose off-diagonal entries are all 1s, then the two matrices commute. The proof of this assertion uses the following determinant identity ((2.5), Sect. 2.2), for $A \in \mathbb{M}_{m\times n}$, $B \in \mathbb{M}_{n\times m}$,

$$\det(I_m + AB) = \det(I_n + BA).$$

Theorem 9.2.3 *Let A and B be n-square (0, 1)-matrices such that*

$$AB = J_n - I_n. \tag{9.7}$$

where J_n is the n-square all-ones matrix. Then A and B commute, namely,

$$AB = BA.$$

Proof. Let a_i and b_i be the columns of A and B^T, respectively, $i = 1, \ldots, n$,

$$A = (a_1, \ldots, a_n), \quad B^T = (b_1, \ldots, b_n).$$

By computation,

$$0 = \operatorname{tr}(AB) = \operatorname{tr}(BA) = \sum_{i=1}^{n} b_i^T a_i.$$

Thus, $b_i^T a_i = 0$ for each i, because A and B have nonnegative entries.

Rewrite equation (9.7) as

$$I_n = J_n - AB \quad \text{or} \quad I_n = J_n - \sum_k a_k b_k^T,$$

which implies, by moving the terms $a_i b_i^T$ and $a_j b_j^T$ to the left-hand side,

$$I_n + a_i b_i^T + a_j b_j^T = J_n - \sum_{k\neq i,\, j} a_k b_k^T.$$

Note that the right-hand side contains $n-1$ matrices (including J_n), each of which has rank one. Thus, by the rank formula for sum (Problem 5), the

matrix on the right-hand side has rank at most $n-1$. Hence, the matrix on the left-hand side is singular. It follows that, for $i \neq j$,

$$\begin{aligned} 0 &= \det(I_n + a_i b_i^T + a_j b_j^T) \\ &= \det\left(I_n + (a_i, a_j)(b_i, b_j)^T\right) \\ &= \det\left(I_2 + (b_i, b_j)^T (a_i, a_j)\right) \\ &= \det\left(I_2 + \begin{pmatrix} 0 & b_i^T a_j \\ b_j^T a_i & 0 \end{pmatrix}\right) \\ &= 1 - (b_i^T a_j)(b_j^T a_i). \end{aligned}$$

This reveals $b_i^T a_j = 1$ for each pair of i and j, $i \neq j$, because A and B are $(0,1)$-matrices. Since $b_i^T a_i = 0$ for every i as shown previously, we have

$$BA = (b_i^T a_j) = J_n - I_n = AB. \blacksquare$$

Example 9.2.2 Matrices in Theorem 9.2.2 and Theorem 9.2.3.

(a) Let $A = \begin{pmatrix} 0&1&1&0&1&0&0 \\ 1&0&0&1&1&0&0 \\ 1&0&1&0&0&1&0 \\ 0&1&0&1&0&1&0 \\ 1&1&0&0&0&0&1 \\ 0&0&1&1&0&0&1 \\ 0&0&0&0&1&1&1 \end{pmatrix}$. Then $AA^T = 2I_7 + J_7$.

(b) Let $A = \begin{pmatrix} 1&1&0&0&0 \\ 0&1&1&0&0 \\ 0&0&1&1&0 \\ 0&0&0&1&1 \\ 1&0&0&0&1 \end{pmatrix}$ and $B = \begin{pmatrix} 0&1&0&1&0 \\ 0&0&1&0&1 \\ 1&0&0&1&0 \\ 0&1&0&0&1 \\ 1&0&1&0&0 \end{pmatrix}$. Then $AB = J_5 - I_5$. $\blacksquare$

Problems

1. Why does every square $(0,1)$-matrix have at least one nonnegative real eigenvalue? Show by example that a square $(0,1)$-matrix may have a negative real eigenvalue and (or) a non-real complex eigenvalue, and the non-real complex eigenvalues occur in conjugate pair(s).

2. Let $A = \begin{pmatrix} 1&0&0&1 \\ 1&1&0&0 \\ 0&0&1&1 \\ 1&1&1&1 \end{pmatrix}$. Find row and column sum vectors $R(A)$ and $S(A)$.

3. Show that there does not exist a 3×3 $(0,1)$-matrix A having respectively the row and column sum vectors $R(A) = (3,1,1)$ and $S(A) = (3,2,0)$.

4. Evaluate the determinants of the following matrices.

$$\begin{pmatrix} 0&1&1&1 \\ 1&0&1&1 \\ 1&1&0&1 \\ 1&1&1&0 \end{pmatrix}, \quad \begin{pmatrix} 0&1&1&1 \\ -1&0&1&1 \\ -1&-1&0&1 \\ -1&-1&-1&0 \end{pmatrix}.$$

5. Show that $\operatorname{rank}(A_1 + \cdots + A_n) \leq n$, where each $A_i \in \mathbb{M}_m$ has rank one.

6. Let $A = (a_{ij})$ be an $n \times n$ $(0,1)$-matrix with diagonal entries 0 and off-diagonal entries 1, i.e., $a_{ii} = 0$ and $a_{ij} = 1$ if $i \neq j$. Compute $\det A$.

7. Let $A = (a_{ij})$ be an $n \times n$ skew-symmetric matrix whose off-diagonal entries are ± 1, i.e., $a_{ii} = 0$ and $a_{ij} = -a_{ji}$ $(=1$ or $-1)$ if $i \neq j$. Show that $\det A$ is an odd integer if n is even, and $\det A = 0$ if n is odd.

8. Let A be an n-square $(0,1)$-matrix. Denote the number of 1s in row i by r_i and in column j by c_j. If A is normal, show that $r_i = c_i$ for each i and that $J - A$ is normal, where J is the $n \times n$ all-ones matrix.

9. Construct a nonsymmetric 4×4 normal matrix of zeros and ones such that each row sum as well as each column sum is equal to 2.

10. Consider the 3×3 $(0,1)$-matrices. (a) How many 3×3 $(0,1)$-matrices are there? (b) What is the largest possible determinant of a 3×3 $(0,1)$-matrix? (c) How many of the matrices are nonsingular? (d) How many 1s as entries can such a nonsingular matrix have at most?

11. Let A be an $m \times m$ symmetric $(0,1)$-matrix with row sum vector $R = (r_1, \ldots, r_m)$, where $r_1 \geq \cdots \geq r_m$. Show that, for $1 \leq k \leq t \leq m$,

$$\sum_{i=1}^{k} r_i \leq kt + \sum_{j=t+1}^{m} r_j.$$

12. Does there exist a normal matrix with real entries of the sign pattern

$$\begin{pmatrix} + & + & 0 \\ + & 0 & + \\ + & + & + \end{pmatrix}?$$

13. If A is an $n \times n$ matrix with integer entries, show that $2Ax = x$ has no non-trivial solutions, that is, the only solution is $x = 0$.

14. If A is a $(0,1)$ (or real) matrix such that $AA^T = J - I$, what is A?

15. Let A be a matrix and let $AA^T = B$. Show that $CC^T = B$, where C is obtained from A by permuting the columns of A (in any ways).

16. Let $A = \left(\begin{smallmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \end{smallmatrix}\right)$. Show that $AA^T = I_3 + J_3$.

17. Let $A = \left(\begin{smallmatrix} 1 & 1 & 0 & 1 \\ 0 & 1 & 1 & 1 \\ 1 & 1 & 1 & 0 \\ 1 & 0 & 1 & 1 \end{smallmatrix}\right)$. Show that $AA^T = I_4 + 2J_4$.

18. Let $A = \left(\begin{smallmatrix} 1 & 0 \\ 1 & 0 \end{smallmatrix}\right)$. Show that $AA^T = J_2 = 0I_2 + J_2$ and that A is not a normal matrix. Does this contradict Theorem 9.2.2?

19. Let $A = \left(\begin{smallmatrix} \sqrt{2} & 0 \\ \frac{1}{\sqrt{2}} & \frac{\sqrt{3}}{\sqrt{2}} \end{smallmatrix}\right)$. Show that $AA^T = I_2 + J_2$. Is A a normal matrix?

20. Sow that $AA^T = 2I_7 + J_7$, where A is the 7×7 $(0,1)$-matrix

$$A = \begin{pmatrix} 1 & 1 & 1 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 1 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 & 1 & 1 & 0 \end{pmatrix}.$$

21. Let A be an $n \times n$ $(0,1)$-matrix. Show that the trace, eigenvalues λ_i (in moduli), and singular values σ_i of A are no more than n, that is,

$$\mathrm{tr}A \leq n, \quad |\lambda_i| \leq n, \quad \sigma_i \leq n.$$

22. Let A be an $n \times n$ $(0,1)$-matrix and let r_i be the number of 1s in row i, $i = 1, \ldots, n$. Show that $|\det A| \leq (r_1 \cdots r_n)^{1/2}$ and that equality occurs if and only if A has a zero row or A is a permutation matrix.

23. Let A be an n-square $(0,1)$-matrix such that $AA^T = tI + J$ for some positive integer t, where J is the $n \times n$ all-ones matrix, and let $C = J - A$. Show that A commutes with C and C^T. Compute CC^T.

24. Let A be an n-square real matrix such that $AA^T = tI + J$ for some real nonnegative number t, where J is the $n \times n$ all-ones matrix.

    (a) Find the singular values of A in terms of n and t.

    (b) Show that all matrices X satisfying $XX^T = tI + J$ have the same singular values as A. Do they have the same eigenvalues?

    (c) Does A^T (in place of X) satisfy the equation $XX^T = tI + J$?

    (d) When is A nonsingular?

25. Let A be a $(0,1)$-matrix and let J be the all-ones matrix, both $n \times n$. If

$$AA^T = (k-h)I + hJ,$$

    where k and h are some positive integers such that $0 < h < k < n$,

    (a) show that A is normal,

    (b) show that $h = \frac{1}{n-1}k(k-1)$,

    (c) show that $A^{-1} = \frac{1}{k(k-h)}(kA^T - hJ)$, and

    (d) find the eigenvalues of AA^T.

. ——————— ⊙ ——————— .

9.3 Normality and Cauchy-Schwarz Type Inequalities

We present in this section some basic inequalities involving eigenvalues and singular values of matrices (there will be more related results in Chap. 10); we also study the Cauchy-Schwarz type inequalities. The equality cases of these inequalities will result in the normality of the matrices.

Theorem 9.3.1 (Schur Eigenvalue Inequality) *Let $A = (a_{ij})$ be an n-square matrix and let $\lambda_1, \ldots, \lambda_n$ be the eigenvalues of A. Then*

$$\sum_{i=1}^{n} |\lambda_i|^2 \le \sum_{i,\, j=1}^{n} |a_{ij}|^2. \tag{9.8}$$

Equality occurs if and only if A is a normal matrix, i.e., $A^*A = AA^*$.

Proof. Let $A = U^*TU$ be a Schur decomposition of A, where U is unitary, $T = (t_{ij})$ is upper-triangular, and $\lambda_1, \ldots, \lambda_n$ are on the diagonal of T. Then $A^*A = U^*T^*TU$. Thus, $\mathrm{tr}(A^*A) = \mathrm{tr}(T^*T)$. By computation, we have

$$\mathrm{tr}(A^*A) = \sum_{i,\, j=1}^{n} |a_{ij}|^2 \quad \text{and} \quad \mathrm{tr}(T^*T) = \sum_{i=1}^{n} |\lambda_i|^2 + \sum_{i<j} |t_{ij}|^2.$$

The inequality is immediate. Equality holds if and only if $t_{ij} = 0$, $i < j$, that is, T is diagonal, namely, A is unitarily diagonalizable, i.e., normal. ∎

Let $\sigma_1, \ldots, \sigma_n$ be the singular values of A. We can rewrite (9.8) as

$$\sum_{i=1}^{n} |\lambda_i|^2 \le \sum_{i=1}^{n} \sigma_i^2. \tag{9.9}$$

Theorem 9.3.2 *Let $A = (a_{ij})$ be an n-square matrix. Let $\lambda_1, \ldots, \lambda_n$ and $\sigma_1, \ldots, \sigma_n$ be the eigenvalues and singular values of A, respectively. Then*

$$|\lambda_1| + \cdots + |\lambda_n| \le \sigma_1 + \cdots + \sigma_n, \tag{9.10}$$

where equality holds if and only if A is normal, and

$$|\lambda_1 + \cdots + \lambda_n| \le \sigma_1 + \cdots + \sigma_n, \tag{9.11}$$

where equality holds if and only if $A = uP$ with $|u| = 1$ $(u \in \mathbb{C})$ and $P \ge 0$.

Proof. Inequality (9.11) (i.e., $|\,\mathrm{tr}A| \le \mathrm{tr}|A|$) is immediate from (9.10). (9.11) is Example 3.2.7 of Sect. 3.2 and (9.10) is shown in (14) ⇒ (1) of Theorem 9.1.1. In the following, we give another direct proof for (9.11), then use (9.11) to show (9.10). Note that $\lambda_1 + \cdots + \lambda_n = \mathrm{tr}A$ and $\sigma_1 + \cdots + \sigma_n = \mathrm{tr}|A|$.

Without loss of generality, we assume that $A \neq 0$. Let $r = \operatorname{rank}(A)$ and let $\sigma_1, \ldots, \sigma_r$ be the nonzero singular values of A. Let $A = UDV$ be a singular value decomposition of A, where $D = \left(\begin{smallmatrix} D_r & 0 \\ 0 & 0 \end{smallmatrix}\right)$ with $D_r = \operatorname{diag}(\sigma_1, \ldots, \sigma_r)$, and U and V are $n \times n$ unitary matrices. Let U_r consist of the first r columns $u_1, \ldots, u_r$ of U and let V_r consist of the first r rows $v_1, \ldots, v_r$ of V. Then $A = UDV = U_r D_r V_r = \sigma_1 u_1 v_1 + \cdots + \sigma_r u_r v_r$. By computation, we have

$$\begin{aligned} |\operatorname{tr} A| &= |\operatorname{tr}(\sigma_1 u_1 v_1 + \cdots + \sigma_r u_r v_r)| \\ &= |\sigma_1 \operatorname{tr}(u_1 v_1) + \cdots + \sigma_r \operatorname{tr}(u_r v_r)| \\ &= |\sigma_1 \langle v_1, u_1^* \rangle + \cdots + \sigma_r \langle v_r, u_r^* \rangle| \\ &\leq \sigma_1 |\langle v_1, u_1^* \rangle| + \cdots + \sigma_r |\langle v_r, u_r^* \rangle| \qquad (9.12) \\ &\leq \sigma_1 \|v_1\| \|u_1^*\| + \cdots + \sigma_r \|v_r\| \|u_r^*\| \qquad (9.13) \\ &= \sigma_1 + \cdots + \sigma_r = \operatorname{tr}|A|. \end{aligned}$$

If $|\operatorname{tr} A| = \operatorname{tr}|A|$, then both equalities in (9.12) and (9.13) hold. For the equality in (9.13) to hold, v_j and u_j^* are linearly dependent for each j. Since they are unit vectors, we can write $v_j = c_j u_j^*$ with $|c_j| = 1$ for $j = 1, \ldots, r$. For the equality in (9.12) to hold, all $\langle v_j, u_j^* \rangle = c_j \langle u_j^*, u_j^* \rangle = c_j$ have the same argument. Thus, $c_1 = \cdots = c_r$. Let $u = c_1$ and $P = UDU^*$. We obtain

$$\begin{aligned} A &= \sigma_1 u_1 v_1 + \cdots + \sigma_r u_r v_r \\ &= c_1 \sigma_1 u_1 u_1^* + \cdots + c_1 \sigma_r u_r u_r^* \\ &= c_1 U_r D_r U_r^* = c_1 UDU^* = uP. \end{aligned}$$

Conversely, if $A = uP$ with $|u| = 1$ and $P \geq 0$, then the singular values of A are just the eigenvalues of P. So, $|\operatorname{tr} A| = |\operatorname{tr}(uP)| = |u| \operatorname{tr} P = \sigma_1 + \cdots + \sigma_n$.

Now we show inequality (9.10). Let $A = R^* S R$ be a Schur decomposition of A, where R is unitary, S is upper-triangular, and the diagonal entries of S are $\lambda_1, \ldots, \lambda_n$. Let $W = \operatorname{diag}(e^{i\theta_1}, \ldots, e^{i\theta_n})$, where $\theta_k \in \mathbb{R}$ and $e^{i\theta_k} \lambda_k \geq 0$ for each $k = 1, \ldots, n$. Then $WS = WRAR^*$ is an upper-triangular matrix with $|\lambda_1|, \ldots, |\lambda_n|$ on the diagonal. Note that A, S, and WS have the same singular values σ_i's. Applying inequality (9.11) to WS, we arrive at

$$|\operatorname{tr}(WS)| = |\lambda_1| + \cdots + |\lambda_n| \leq \sum\nolimits_i \sigma_i(WS) = \sum\nolimits_i \sigma_i(A).$$

Equality holds in (9.10) if and only if A is normal (Theorem 9.1.1 (14)). ∎

We will present many more similar inequalities in Chap. 10.

There is a variety of Cauchy-Schwarz type inequalities in mathematics. The Cauchy-Schwarz inequalities involving matrices have been obtained by different methods and techniques. Below we give some Cauchy-Schwarz type matrix inequalities concerning the modulus $|A|$ of a matrix A (see Sect. 3.2):

$$|A| = (A^* A)^{1/2}.$$

It is easy to see that if $A = UDV$ is a singular value decomposition of A, where U and V are unitary and D is nonnegative diagonal, then

$$|A| = V^*DV \quad \text{and} \quad |A^*| = UDU^*.$$

Theorem 9.3.3 *Let A be an n-square matrix. Then for any $u, v \in \mathbb{C}^n$,*

$$|\langle Au, v\rangle|^2 \le \langle |A|u, u\rangle\langle |A^*|v, v\rangle, \tag{9.14}$$

$$\big|\langle (A \circ A^*)u, u\rangle\big| \le \langle (|A \circ A^*|)u, u\rangle, \tag{9.15}$$

$$\big|\langle (A \circ A^*)u, u\rangle\big| \le \langle (|A| \circ |A^*|)u, u\rangle. \tag{9.16}$$

Proof. For (9.14), by Theorem 7.6.2, it is sufficient to observe that

$$\begin{pmatrix} |A| & A^* \\ A & |A^*| \end{pmatrix} = \begin{pmatrix} V^* & 0 \\ 0 & U \end{pmatrix} \begin{pmatrix} D & D \\ D & D \end{pmatrix} \begin{pmatrix} V & 0 \\ 0 & U^* \end{pmatrix} \ge 0.$$

Inequality (9.15) follows from (9.14) with $A \circ A^*$ for A and $u = v$. Note that $A \circ A^*$ is Hermitian. Inequality (9.16) holds because (by Theorem 7.5.2)

$$\begin{pmatrix} |A| \circ |A^*| & A^* \circ A \\ A \circ A^* & |A^*| \circ |A| \end{pmatrix} = \begin{pmatrix} |A| & A^* \\ A & |A^*| \end{pmatrix} \circ \begin{pmatrix} |A^*| & A \\ A^* & |A| \end{pmatrix} \ge 0. \ \blacksquare$$

In view of the right-hand sides of (9.15) and (9.16), it is tempting to have

$$\langle (|A \circ A^*|)u, u\rangle \le \langle (|A| \circ |A^*|)u, u\rangle.$$

However, this is untrue because $|A \circ A^*| \not\le |A| \circ |A^*|$ (Problem 21 of Sect. 8.6).

For more results, let A be an $n \times n$ Hermitian matrix decomposed as

$$A = U^* \operatorname{diag}(\lambda_1, \ldots, \lambda_n)U,$$

where $U \in \mathbb{M}_n$ is unitary. For $\alpha \in \mathbb{R}$, if all λ_i^α are well-defined, we denote

$$A^\alpha = U^* \operatorname{diag}(\lambda_1^\alpha, \ldots, \lambda_n^\alpha)U.$$

Theorem 9.3.4 *Let $A \in \mathbb{M}_n$. Then for any $\alpha \in (0, 1)$ and $u, v \in \mathbb{C}^n$,*

$$|\langle Au, v\rangle| \le \||A|^\alpha u\| \, \||A^*|^{1-\alpha} v\|. \tag{9.17}$$

Proof. Let $A = W|A|$ be a polar decomposition of A (W is unitary). Then

$$A = W|A|^{1-\alpha}W^*W|A|^\alpha = |A^*|^{1-\alpha}W|A|^\alpha.$$

By the Cauchy-Schwarz inequality, we have

$$|\langle Au, v\rangle| = |\langle W|A|^\alpha u, |A^*|^{1-\alpha} v\rangle| \le \||A|^\alpha u\| \, \||A^*|^{1-\alpha} v\|. \ \blacksquare$$

Note that (9.14) is a special case of (9.17) by taking $\alpha = \frac{1}{2}$.

Theorem 9.3.5 *Let $A \in \mathbb{M}_n$ and let α be a positive real number. Then*

$$|\langle Au, u\rangle| \leq \langle |A|u, u\rangle^{\alpha} \langle |A^*|u, u\rangle^{1-\alpha} \quad \text{for all } u \in \mathbb{C}^n \tag{9.18}$$

if and only if A is a normal matrix or $\alpha = \frac{1}{2}$.

Proof. If A is normal, then $|A| = |A^*|$, and the inequality reduces to $|\langle Au, u\rangle| \leq \langle |A|u, u\rangle$, which is true for normal matrices by Theorem 9.1.1 (40). If $\alpha = \frac{1}{2}$, then it is the case of (9.14) with $u = v$.

We show that if $\alpha \neq \frac{1}{2}$, then A is normal by induction on n. If $n = 1$, we have nothing to prove. Suppose that the assertion is true for $(n-1)$-square matrices. For n-square matrices, first consider the case where A is nonsingular. Let $A = UDV$ be a singular value decomposition of A, where D is diagonal and invertible, and U and V are unitary. Observe that $|A| = V^*DV$ and $|A^*| = UDU^*$. The inequality in (9.18) becomes

$$|\langle UDVu, u\rangle| \leq \langle V^*DVu, u\rangle^{\alpha} \langle UDU^*u, u\rangle^{1-\alpha}.$$

Thus,

$$|\langle D^{1/2}Vu, D^{1/2}U^*u\rangle| \leq \langle D^{1/2}Vu, D^{1/2}Vu\rangle^{\alpha} \langle D^{1/2}U^*u, D^{1/2}U^*u\rangle^{1-\alpha}.$$

Notice that $D^{1/2}U^*$ is nonsingular. For any nonzero $u \in \mathbb{C}^n$, we set

$$y = \frac{1}{||D^{1/2}U^*u||} D^{1/2}U^*u.$$

Then $||y|| = 1$, and y ranges over all unit vectors as u runs over all nonzero vectors. By putting $\hat{A} = D^{1/2}VUD^{-1/2}$, we have, for all unit $y \in \mathbb{C}^n$,

$$|\langle \hat{A}y, y\rangle| \leq \langle \hat{A}y, \hat{A}y\rangle^{\alpha}.$$

Note that all the eigenvalues of $\hat{A}$ are equal to 1 in absolute value. Applying Problem 32 of Sect. 6.1 to $\hat{A}$ (as $\alpha \neq \frac{1}{2}$), we see that $\hat{A}$ is unitary. Thus,

$$D^{-1/2}U^*V^*DVUD^{-1/2} = I.$$

It follows that

$$V^*DV = UDU^*,$$

i.e., $|A| = |A^*|$. Squaring both sides reveals $A^*A = AA^*$, and A is normal.

We now deal with the case where A is singular (so A has zero eigenvalues). Noticing that (9.18) still holds when A is replaced by W^*AW for any n-square unitary matrix W, we assume, without loss of generality, that

$$A = \begin{pmatrix} A_1 & b \\ 0 & 0 \end{pmatrix}, \quad A_1 \in \mathbb{M}_{n-1}, \ \ b \in \mathbb{C}^{n-1}.$$

We show that $b = 0$, thus A is normal by induction hypothesis on A_1.

Suppose that $b \neq 0$. Then $\langle b, b\rangle > 0$. Let $u = \begin{pmatrix} b \\ t \end{pmatrix}$, where $t > 0$. Then

$$|\langle Au, u\rangle| = |\langle A_1 b, b\rangle + \langle b, b\rangle t|$$

and

$$\langle |A^*|u, u\rangle = \langle (A_1 A_1^* + bb^*)^{1/2} b, b\rangle$$

which is independent of t. To compute $\langle |A|u, u\rangle$, we write

$$|A| = \begin{pmatrix} C & d \\ d^* & \beta \end{pmatrix}.$$

Then

$$b^* b = d^* d + \beta^2.$$

Hence, $\beta \neq 0$; otherwise $d = 0$ and thus $b = 0$. It follows that $\beta > 0$ and

$$\langle |A|u, u\rangle = \langle Cb, b\rangle + t(\langle d, b\rangle + \langle b, d\rangle) + \beta t^2.$$

Thus, (9.18) becomes

$$|\langle A_1 b, b\rangle + \langle b, b\rangle t| \leq \big(\langle Cb, b\rangle + 2t\,\Re(\langle b, d\rangle) + \beta t^2\big)^{\alpha} \langle (A_1 A_1^* + bb^*)^{1/2} b, b\rangle^{1-\alpha}.$$

Considering the terms of variable t on both sides and letting $t \to \infty$, we arrive at $1 \leq 2\alpha$. It follows that $\alpha > \frac{1}{2}$ (as $\alpha \neq \frac{1}{2}$ is assumed).

Note that (9.18) is the same as

$$|\langle A^* u, u\rangle| \leq \langle |A^*|u, u\rangle^{\gamma}\, \langle |A|u, u\rangle^{1-\gamma}, \quad \gamma = 1 - \alpha.$$

If A is singular and non-normal, then so is A^*. We can write (under unitary similarity) $A^* = \left(\begin{smallmatrix} A_2 & c \\ 0 & 0 \end{smallmatrix}\right)$, $c \neq 0$, and repeat the above argument with A^* in place of A to get $\gamma > \frac{1}{2}$, so $\alpha < \frac{1}{2}$, contradicting $\alpha > \frac{1}{2}$.

To conclude, if $\alpha \neq \frac{1}{2}$, then A is normal. ∎

Problems

1. For what value(s) of $x \in \mathbb{C}$ is the matrix $A = \left(\begin{smallmatrix} 0 & 1 \\ x & 0 \end{smallmatrix}\right)$ normal? Hermitian? Or diagonalizable? Show that A^n is always normal if n is even.

2. Show that the eigenvalues of $\left(\begin{smallmatrix} 6 & 2 & 2 \\ 1 & 4 & 1 \\ -2 & 2 & 2 \end{smallmatrix}\right)$ are 6, 4, 2 (diagonal entries).

3. Let $P \in \mathbb{M}_n$ be a positive definite matrix and let $U \in \mathbb{M}_n$ be a unitary matrix. If $|\operatorname{tr}(PU)| = \operatorname{tr}P$, show that $U = dI_n$ for some $d \in \mathbb{C}$, $|d| = 1$.

4. Let A be a square matrix and let $|A| = (A^* A)^{1/2}$. Show that

$$\det|A| = \det|A^*| = |\det A| \quad \text{and} \quad \operatorname{tr}|A| = \operatorname{tr}|A^*| \geq |\operatorname{tr}A|.$$

When is $\operatorname{tr}|A| = |\operatorname{tr}A|$? $\operatorname{tr}|A| = \operatorname{tr}A$? Or $|\operatorname{tr}A| = \operatorname{tr}A$?

5. If A is a normal matrix such that the (unordered) eigenvalues of A are exactly the diagonal entries of A, show that A is diagonal.

6. Let $A \in \mathbb{M}_n$. Let $\lambda_1, \ldots, \lambda_n$ and $\sigma_1, \ldots, \sigma_n$ be the eigenvalues and singular values of A, respectively. If $\lambda_1 + \cdots + \lambda_n = \sigma_1 + \cdots + \sigma_n$, show that A normal. That is, if the sum of the eigenvalues equals the sum of the singular values, then the matrix is normal. Is the converse true?

7. Let $A \in \mathbb{M}_n$. Let $\lambda_1, \ldots, \lambda_n$ and $\sigma_1, \ldots, \sigma_n$ be the eigenvalues and singular values of A, respectively. If $\lambda_1^2 + \cdots + \lambda_n^2 = \sigma_1^2 + \cdots + \sigma_n^2$, show that A normal. That is, if the sum of the squares of the eigenvalues is equal to the sum of the squares of the singular values, then the matrix is normal. Is the converse true?

8. Let $A \in \mathbb{M}_n$ have eigenvalues λ_i's and singular values σ_i's ordered as $|\lambda_1| \geq \cdots \geq |\lambda_n|$ and $\sigma_1 \geq \cdots \geq \sigma_n$, respectively. Use Theorem 9.3.2 and the singular interlacing theorem (Theorem 8.4.3) to show that

$$|\lambda_1| + \cdots + |\lambda_k| \leq \sigma_1 + \cdots + \sigma_k, \quad k = 1, \ldots, n.$$

   When does equality holds for $k = 1$? For $k = n$?

9. Let $A = (a_{ij})$ be an n-square matrix. Let $\lambda_1, \ldots, \lambda_n$ and $\sigma_1, \ldots, \sigma_n$ be the eigenvalues and singular values of A, respectively. Show that

$$|\operatorname{tr}(A^2)| = \left| \sum_i \lambda_i^2 \right| \leq \sum_i |\lambda_i|^2 \leq \sum_i \sigma_i^2 = \sum_{i,j} |a_{ij}|^2 = \operatorname{tr}(A^* A).$$

   Discuss when the inequalities become equalities.

10. Show that an idempotent normal matrix is positive semidefinite.

11. If $A \in \mathbb{M}_n$ is normal and A^2 is Hermitian, what can be said about the eigenvalues of A? Show that such an A can be written as $A = B + C$, where B is Hermitian, C is skew-Hermitian, and $BC = CB = 0$.

12. If A is a normal matrix such that $A^3 = A^2$, show that $A^2 = A = |A|$.

13. Let $A = (a_{ij})$ be a normal matrix. Show that there is an eigenvalue λ of A such that $|\lambda| \geq \max_{i,j} |a_{ij}|$. Is this true for non-normal matrices?

14. For any n-square matrices A and B, show that

$$\begin{pmatrix} |A| + |B| & A^* + B^* \\ A + B & |A^*| + |B^*| \end{pmatrix} \geq 0.$$

   Derive the determinant inequality

$$(\det |A + B|)^2 \leq \det(|A| + |B|) \det(|A^*| + |B^*|).$$

   In particular,

$$\det |A + A^*| \leq \det(|A| + |A^*|).$$

   Discuss the analogue for the Hadamard product.

15. Show that $\begin{pmatrix} |A| & A^* \\ A & |A^*| \end{pmatrix} \geq 0$ for any matrix A but $\begin{pmatrix} |A| & A \\ A^* & |A^*| \end{pmatrix} \geq 0$ need not be true (by a counterexample). Conclude that $\begin{pmatrix} A & B \\ B^* & C \end{pmatrix} \geq 0 \not\Rightarrow \begin{pmatrix} A & B^* \\ B & C \end{pmatrix} \geq 0$. Explain why it is true that $\begin{pmatrix} A & B \\ B^* & A \end{pmatrix} \geq 0 \Rightarrow \begin{pmatrix} A & B^* \\ B & A \end{pmatrix} \geq 0$.

16. Let A be an $n \times n$ matrix. Show that for any $n \times n$ matrices X and Y,

$$|\operatorname{tr}(X^*A^*Y)|^2 \leq \operatorname{tr}(X^*|A|X)\operatorname{tr}(Y^*|A^*|Y)$$

and

$$\operatorname{tr}\left(|X^*A^*Y|^2\right) \leq \operatorname{tr}(X^*|A|X)\operatorname{tr}(Y^*|A^*|Y).$$

17. Let A be an $n \times n$ matrix. Show that for all unit $u \in \mathbb{C}^n$ (in columns),

$$|u^*Au|^2 \leq u^*|A|^2u.$$

Is $|u^*A^2u|$ comparable to $|u^*Au|^2$? (I.e., is $|u^*A^2u| \leq |u^*Au|^2$ or $|u^*A^2u| \geq |u^*Au|^2$ for all unit u?) Is $|u^*A^2u|$ comparable to $u^*|A|^2u$?

18. Let A be an $n \times n$ matrix and $\alpha \in [0, 1]$. Show that

$$\begin{pmatrix} |A|^{2\alpha} & A^* \\ A & |A^*|^{2(1-\alpha)} \end{pmatrix} \geq 0.$$

19. Let A and B be $n \times n$ matrices. Show that

$$A^*A = B^*B \quad \Leftrightarrow \quad |A| = |B|.$$

Is it true that

$$A^*A \geq B^*B \quad \Leftrightarrow \quad |A| \geq |B|?$$

Is it true that

$$|A| \geq |B| \quad \Leftrightarrow \quad |A^*| \geq |B^*|?$$

20. Let $A \in \mathbb{M}_m$, $C \in \mathbb{M}_n$, and $B \in \mathbb{M}_{m\times n}$. If $B = UDV$ is a singular value decomposition of B, where $U \in \mathbb{M}_m$ and $V \in \mathbb{M}_n$ are unitary, and $D \in \mathbb{M}_{m\times n}$ has the singular values of B on the diagonal, show that

$$\begin{pmatrix} A & B \\ B^* & C \end{pmatrix} \geq 0 \;\Rightarrow\; \begin{pmatrix} U^*AU & D \\ D^T & VCV^* \end{pmatrix} \geq 0.$$

21. Let A and B be $n \times n$ normal matrices. Suppose that A (or B) is such a matrix that its different eigenvalues have different moduli. Show that AB is normal if and only if A and B commute, that is, $AB = BA$. Consequently, if A is positive semidefinite and B is normal, then AB is normal if and only if A and B commute.

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9.4 Spectral Perturbation of Normal Matrices

Given two square matrices of the same order, how "close" are the matrices in terms of their eigenvalues (spectrum)? More interestingly, if a matrix is "perturbed" a little bit, how would the eigenvalues of the matrix change? In this section we present three results about normal matrix perturbations. The first one is on the comparison of $|A| - |B|$ and $A - B$ in terms of the Frobenius norm (or 2-norm), the second one is on the difference (closeness) of suitably ordered eigenvalues of the normal matrices, and the third result is the same kind except that only one matrix is required to be normal.

Theorem 9.4.1 (Kittaneh) *If A and B are $n \times n$ normal matrices, then*

$$\| \, |A| - |B| \, \|_F \le \|A - B\|_F. \tag{9.19}$$

Proof. By spectral decomposition, we let $A = U^*\Lambda U$ and $B = V^*\Gamma V$, where $\Lambda = \mathrm{diag}(\lambda_1, \ldots, \lambda_n)$, $\Gamma = \mathrm{diag}(\gamma_1, \ldots, \gamma_n)$, and U and V are unitary matrices. Then $|A| = U^*|\Lambda|U$ and $|B| = V^*|\Gamma|V$, where $|\Lambda| = \mathrm{diag}(|\lambda_1|, \ldots, |\lambda_n|)$, $|\Gamma| = \mathrm{diag}(|\gamma_1|, \ldots, |\gamma_n|)$, and $W = (w_{ij}) = UV^*$. Recall the Frobenius norm $\|X\|_F = (\sum_{i,j} |x_{ij}|^2)^{1/2}$, where $X = (x_{ij})$, and note that this norm is unitarily invariant, i.e., $\|PXQ\|_F = \|X\|_F$, where P and Q are unitary matrices (of appropriate orders). By computation, we have

$$\begin{aligned}
\| \, |A| - |B| \, \|_F &= \|U^*|\Lambda|U - V^*|\Gamma|V\|_F \\
&= \|U^*(|\Lambda|UV^* - UV^*|\Gamma|)V\|_F \\
&= \| \, |\Lambda|W - W|\Gamma| \, \|_F \\
&= \Big(\sum_{i,j=1}^{n} \big(|\lambda_i| - |\gamma_j| \big)^2 \cdot |w_{ij}|^2 \Big)^{1/2} \\
&\le \Big(\sum_{i,j=1}^{n} |\lambda_i - \gamma_j|^2 \cdot |w_{ij}|^2 \Big)^{1/2} \\
&= \Big(\sum_{i,j=1}^{n} |(\lambda_i - \gamma_j) w_{ij}|^2 \Big)^{1/2} \\
&= \|\Lambda W - W\Gamma\|_F = |\Lambda UV^* - UV^*\Gamma\|_F \\
&= \|U^*\Lambda U - V^*\Gamma V\|_F = \|A - B\|_F. \ \blacksquare
\end{aligned}$$

Corollary 9.4.1 *Let A and B be any $n \times n$ matrices. Then*

$$\| \, |A| - |B| \, \|_F \le \sqrt{2} \, \|A - B\|_F. \tag{9.20}$$

Proof. Let $\hat{A} = \begin{pmatrix} 0 & A \\ A^* & 0 \end{pmatrix}$ and $\hat{B} = \begin{pmatrix} 0 & B \\ B^* & 0 \end{pmatrix}$. Then $\hat{A}$ and $\hat{B}$ are Hermitian (thus normal) matrices. Theorem 9.4.1 ensures that $\| \, |\hat{A}| - |\hat{B}| \, \|_F \le \|\hat{A} - \hat{B}\|_F$.

Upon computation, we obtain $|\hat{A}| = \begin{pmatrix} |A^*| & 0 \\ 0 & |A| \end{pmatrix}$ and $|\hat{B}| = \begin{pmatrix} |B^*| & 0 \\ 0 & |B| \end{pmatrix}$. It is immediate that $|\hat{A}| - |\hat{B}| = \begin{pmatrix} |A^*|-|B^*| & 0 \\ 0 & |A|-|B| \end{pmatrix}$ and $\hat{A} - \hat{B} = \begin{pmatrix} 0 & A-B \\ A^*-B^* & 0 \end{pmatrix}$.

Note that $\|X\|_F = (\sum_{i,j} |x_{ij}|^2)^{1/2} = \|X^*\|_F$ and that if $M = \begin{pmatrix} P & Q \\ R & S \end{pmatrix}$ is a partitioned matrix, then $\|M\|_F^2 = \|P\|_F^2 + \|Q\|_F^2 + \|R\|_F^2 + \|S\|_F^2$. We have

$$\| \, |\hat{A}| - |\hat{B}| \, \|_F^2 = \| \, |A^*| - |B^*| \, \|_F^2 + \| \, |A| - |B| \, \|_F^2$$

and

$$\|\hat{A} - \hat{B}\|_F^2 = \|A^* - B^*\|_F^2 + \|A - B\|_F^2 = 2\|A - B\|_F^2.$$

It follows that

$$\| \, |A| - |B| \, \|_F^2 \leq \| \, |\hat{A}| - |\hat{B}| \, \|_F^2 \leq 2\|A - B\|_F^2.$$

The desired inequality follows immediately by taking the square roots. ∎

Example 9.4.1 Norms of $|A| - |B|$ and $A - B$.

(a) Let $A = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$ and $B = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$. Then A and B are normal matrices, and $|A| = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$, $|B| = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$. We have $\| \, |A|-|B| \, \|_F = 1 < \|A-B\|_F = \sqrt{3}$.

(b) Let $A = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$ and $B = \begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix}$. Then $|A| = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$ and $|B| = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$. We have $\|A - B\|_F^2 = 1$ and $\| \, |A| - |B| \, \|_F^2 = (1 - \frac{1}{\sqrt{2}})^2 + \frac{3}{2} = 3 - \sqrt{2}$. Thus, $\|A - B\|_F = 1 < \| \, |A| - |B| \, \|_F = \sqrt{3 - \sqrt{2}} < \sqrt{2}\|A - B\|_F = \sqrt{2}$. ∎

This example shows that $\| \, |A| - |B| \, \|_F \not\leq \|A - B\|_F$. However, we have $| \, \|A\| - \|B\| \, | \leq \|A - B\|$ for any norm $\|\cdot\|$ by norm continuity (Theorem 4.2.1).

Note: The constant $\sqrt{2}$ in (9.20) is best possible (Problem 19).

Theorem 9.4.2 (Hoffman-Wielandt) *Let A and B be $n \times n$ normal matrices and let $\lambda_1, \ldots, \lambda_n$ and $\mu_1, \ldots, \mu_n$ be the eigenvalues of A and B, respectively. Then there exists a permutation p on $\{1, 2, \ldots, n\}$ such that*

$$\left(\sum_{i=1}^{n} |\lambda_i - \mu_{p(i)}|^2 \right)^{1/2} \leq \|A - B\|_F. \tag{9.21}$$

Proof. Let $A = U^* \operatorname{diag}(\lambda_1, \ldots, \lambda_n) U$ and $B = V^* \operatorname{diag}(\mu_1, \ldots, \mu_n) V$ be spectral decompositions of A and B, respectively, where U and V are n-square unitary matrices. For simplicity, denote $E = \operatorname{diag}(\lambda_1, \ldots, \lambda_n)$, $F = \operatorname{diag}(\mu_1, \ldots, \mu_n)$, and $W = (w_{ij}) = UV^*$. We compute

$$\begin{aligned} \|A - B\|_F^2 &= \|U^*(EUV^* - UV^*F)V\|_F^2 \\ &= \|EW - WF\|_F^2 \\ &= \sum_{i,\,j=1}^{n} |\lambda_i - \mu_j|^2 |w_{ij}|^2. \end{aligned} \tag{9.22}$$

Set $G = (|\lambda_i - \mu_j|^2)$ and $S = (|w_{ij}|^2) \in \mathbb{M}_n$. Then (9.22) is rewritten as

$$\|A - B\|_F^2 = \sum_{i,\,j=1}^{n} |\lambda_i - \mu_j|^2 |w_{ij}|^2 = e^T (G \circ S) e,$$

where $G \circ S$ is the Hadamard (entrywise) product of G and S, and $e \in \mathbb{R}^n$ is the all-ones column vector. Note that G is entrywise nonnegative and S is a doubly stochastic matrix. By the Birkhoff theorem (Theorem 5.6.7), S is a convex combination of permutation matrices: $S = \sum_{i=1}^{m} t_i P_i$, where all t_i are positive numbers adding up to 1, and P_i are permutation matrices. Among all $e^T(G \circ P_i)e$, $i = 1, \ldots, m$, suppose that $e^T(G \circ P_k)e$ is the smallest for some k. Consider this P_k as a permutation p on the set $\{1, 2, \ldots, n\}$, that is, the $(i, p(i))$-entry of P_k is 1 and all other entries are 0, $i = 1, \ldots, n$. Then $G \circ P_k$ has $(i, p(i))$-entry $|\lambda_i - \mu_{p(i)}|$ and all other entries 0. It follows that

$$\begin{aligned} \|A - B\|_F^2 &= e^T (G \circ S) e = \sum_{i=1}^{m} t_i e^T (G \circ P_i) e \\ &\geq \sum_{i=1}^{m} t_i e^T (G \circ P_k) e \\ &= e^T (G \circ P_k) e = \sum_{i=1}^{n} |\lambda_i - \mu_{p(i)}|^2. \end{aligned}$$ ∎

The Hoffman-Wielandt inequality is about rearrangements (optimal matching or coupling) of the eigenvalues compared with the Frobenius norm of the difference of the two matrices. Note that an optimal matching is necessary in general (check with Example 9.4.1 (a)). Besides, the theorem requires both matrices to be normal; it need not be true for non-normal matrices.

Example 9.4.2 Let $A = \left(\begin{smallmatrix} 5 & 0 \\ 0 & 0 \end{smallmatrix}\right)$ and $B = \left(\begin{smallmatrix} 1 & -1 \\ 1 & -1 \end{smallmatrix}\right)$. Then A is normal, but B is not. The eigenvalues of A are $\lambda_1 = 5$, $\lambda_2 = 0$; the eigenvalues of B are $\mu_1 = \mu_2 = 0$. $\|A - B\|_F^2 = 19$, while $|\lambda_1 - \mu_{p(1)}|^2 + |\lambda_2 - \mu_{p(2)}|^2 = 25$. ∎

Note that the Frobenius norm $\|\cdot\|_F$ in the Hoffman-Wielandt inequality (9.21) cannot be replaced by the spectral norm $\|\cdot\|_{\mathrm{sp}}$. For example, take $A = 2I_n$ and $B = I_n$. Then the left-hand side of (9.21) is equal to n, while the spectral norm of $A - B$ is 1, i.e., $\|A - B\|_{\mathrm{sp}} = \sigma_{\max}(A - B) = 1$. In addition, the left-hand sides of (9.19) and (9.21) are incomparable.

In what follows we present a result in which one matrix is normal and the other is arbitrary, and the 2-norm need to be enlarged. For this, we need a lemma. If A is a square matrix, for convenience, we write $A = U_A + D_A + L_A$, where U_A, D_A, and L_A are the upper part, diagonal, and lower part of A, respectively. For instance, $A = \left(\begin{smallmatrix} 1 & 2 \\ 3 & 4 \end{smallmatrix}\right) = \left(\begin{smallmatrix} 0 & 2 \\ 0 & 0 \end{smallmatrix}\right) + \left(\begin{smallmatrix} 1 & 0 \\ 0 & 4 \end{smallmatrix}\right) + \left(\begin{smallmatrix} 0 & 0 \\ 3 & 0 \end{smallmatrix}\right)$.

Lemma 9.4.1 *Let A be an n-square normal matrix. Then*

$$\|U_A\|_F \le \sqrt{n-1}\,\|L_A\|_F \quad \textit{and} \quad \|L_A\|_F \le \sqrt{n-1}\,\|U_A\|_F.$$

Proof. We show the first inequality. The second inequality follows from the first one by replacing A with A^T. Upon computation, we have

$$\begin{aligned}
\|U_A\|_F^2 &= \sum_{i=1}^{n-1}\sum_{j=i+1}^{n} |a_{ij}|^2 \le \sum_{i=1}^{n-1}\sum_{j=i+1}^{n} (j-i)|a_{ij}|^2 \\
&= \sum_{j=1}^{n-1}\sum_{i=j+1}^{n} (i-j)|a_{ij}|^2 \quad \text{(Problem 17)} \\
&\le (n-1)\sum_{j=1}^{n-1}\sum_{i=j+1}^{n} |a_{ij}|^2 = (n-1)\|L_A\|_F^2. \ \blacksquare
\end{aligned}$$

Theorem 9.4.3 (Sun) *Let A be an $n \times n$ normal matrix having eigenvalues $\lambda_1, \ldots, \lambda_n$ and let B be any $n \times n$ matrix having eigenvalues $\mu_1, \ldots, \mu_n$. Then there exists a permutation p on $\{1, 2, \ldots, n\}$ such that*

$$\Big(\sum_{i=1}^{n} |\lambda_i - \mu_{p(i)}|^2\Big)^{1/2} \le \sqrt{n}\,\|A - B\|_F. \tag{9.23}$$

Proof. By the Schur triangularization theorem (Theorem 3.2.1), there exists a unitary n-square matrix U such that U^*BU is upper-triangular. Without loss of generality, we assume that B is upper-triangular. Thus, $D_B = \operatorname{diag}(\mu_1, \ldots, \mu_n)$ and L_B contains only 0's. Let $C = A - B$. Then

$$A - D_B = C + U_B, \quad U_B = U_A - U_C, \quad L_A = L_C.$$

Because A and D_B are both normal, by the Hoffman–Wielandt theorem, there exists a permutation p on the index set $\{1, 2, \ldots, n\}$ such that

$$\Big(\sum_{i=1}^{n} |\lambda_i - \mu_{p(i)}|^2\Big)^{1/2} \le \|A - D_B\|_F = \|C + U_B\|_F.$$

Now we apply Lemma 9.4.1 to compute

$$\begin{aligned}
\|C + U_B\|_F^2 &= \|C + U_A - U_C\|_F^2 \\
&= \|L_C + D_C + U_A\|_F^2 \\
&= \|L_C\|_F^2 + \|D_C\|_F^2 + \|U_A\|_F^2 \\
&\le \|L_C\|_F^2 + \|D_C\|_F^2 + (n-1)\|L_A\|_F^2 \\
&= \|L_C\|_F^2 + \|D_C\|_F^2 + (n-1)\|L_C\|_F^2 \\
&\le n\|C\|_F^2 = n\|A - B\|_F^2.
\end{aligned}$$

Taking square roots of both sides yields the desired inequality. $\blacksquare$

Problems

1. Let $A = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & -1 \\ 0 & 0 & -1 \end{pmatrix}$. Find the eigenvalues λ_i's and singular values σ_i's of A, then rearrange λ_i's, $|\lambda_i|$'s, and σ_i's in decreasing orders.

2. Find $|X| = (X^*X)^{1/2}$, where X is each of the following matrices:
$$\begin{pmatrix} 0 & x \\ 0 & 0 \end{pmatrix}, \quad \begin{pmatrix} 1 & x \\ 0 & 0 \end{pmatrix}, \quad \begin{pmatrix} 1 & 0 \\ x & 0 \end{pmatrix}, \quad \begin{pmatrix} 1 & x \\ x & x^2 \end{pmatrix}, \quad \text{where } x \geq 0.$$

3. Let $A = \left(\begin{smallmatrix} 1 & \varepsilon \\ 0 & 0 \end{smallmatrix}\right)$, $\varepsilon \geq 0$. Find $\| \, |A| - |A^*| \, \|_F$ and $\|A - A^*\|_F$.

4. Let $B = \left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right)$, $\varepsilon \geq 0$. Find $\| \, |B| - |B^*| \, \|_F$ and $\|B - B^*\|_F$.

5. Let $A = \left(\begin{smallmatrix} 1 & 0 \\ 0 & 0 \end{smallmatrix}\right)$ and $B = \left(\begin{smallmatrix} 1 & 1 \\ 0 & 0 \end{smallmatrix}\right)$. Show that $\sigma_1(A - B) < \sigma_1(|A| - |B|)$.

6. Let A be a square matrix. Show that A is normal if and only if $\| \, |A| - |A^*| \, \|_F = 0$, that A is Hermitian if and only if $\|A - A^*\|_F = 0$, and that if A is normal, but not Hermitian, then $\| \, |A| - |A^*| \, \|_F < \|A - A^*\|_F$.

7. Let A be an n-square matrix. Show that there exists an n-square unitary matrix W such that $\| \, |A| - |A^*| \, \|_F = \|A - W^*AW\|_F$.

8. Let $A = \left(\begin{smallmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{smallmatrix}\right)$, $B = \left(\begin{smallmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{smallmatrix}\right)$. Let p be a permutation on $\{1, 2, \ldots, n\}$.
   (a) Apply the Kittaneh Theorem to A and B.
   (b) Apply the Hoffman-Wielandt Theorem to A and B.
   (c) Show that $\| \, |A| - |B| \, \|_F < \left(\sum_{i=1}^n |\lambda_i - \mu_{p(i)}|^2 \right)^{1/2}$ for all p.
   (d) Does it hold that $\left(\sum_{i=1}^n |\lambda_i - \mu_{p(i)}|^2 \right)^{1/2} \leq \|A - B\|_F$ for all p?

9. Let $A = (a_{ij})$ be a normal matrix. For a $k \times k$ principal submatrix K of A, we denote by $\Sigma(K)$ the sum of all entries of K. Show that there exists at least one eigenvalue λ of A such that $k|\lambda| \geq \max_K |\Sigma(K)|$.

10. Let $A \in \mathbb{M}_n$, $H = \frac{1}{2}(A + A^*)$, and $S = \frac{1}{2}(A - A^*)$. Let $\lambda_t = a_t + b_t i$ be the eigenvalues of A, where a_t and b_t are real, $t = 1, \ldots, n$. Show that
$$\sum_{t=1}^n |a_t|^2 \leq \|H\|_F^2 \quad \text{and} \quad \sum_{t=1}^n |b_t|^2 \leq \|S\|_F^2.$$

11. Let $A \in \mathbb{M}_n$ be a normal matrix with eigenvalues $\lambda_1, \ldots, \lambda_n$. Show that
$$\|A\|_{\text{sp}} = \max_i \{|\lambda_i|\}, \quad \|A\|_F = \sqrt{\sum_i |\lambda_i|^2}.$$

12. Let $A = (a_{pq}) \in \mathbb{M}_n$ and let $m = \max_{p,\,q} |a_{pq}|$. Show that
$$|\det A| \leq (m\sqrt{n}\,)^n.$$

13. Let $A = (a_{pq})$ be an n-square matrix and let

$$m_1 = \max_{p,\,q} |a_{pq}|,\ m_2 = \max_{p,\,q} |a_{pq} + \overline{a_{qp}}|/2,\ m_3 = \max_{p,\,q} |a_{pq} - \overline{a_{qp}}|/2.$$

For any eigenvalue $\lambda = a + bi$ of A, where a and b are real, show that

$$|\lambda| \le nm_1, \quad |a| \le nm_2, \quad |b| \le nm_3.$$

14. Let $A = (a_{pq})$ be an $n \times n$ real matrix and $d = \max_{p,q}\{|a_{pq} - a_{qp}|/2\}$. Let $\lambda = a + bi$ be any eigenvalue of A, where a, b are real. Show that

$$|b| \le \sqrt{\frac{n(n-1)}{2}}\, d.$$

15. Let $A = (a_{pq})$ be a normal matrix of order n having eigenvalues $\lambda_t = a_t + b_t i$, where a_t and b_t are real, $t = 1, \ldots, n$. Show that

$$\max_t |a_t| \ge \max_{p,\,q} |a_{pq} + \overline{a_{qp}}|/2, \quad \max_t |b_t| \ge \max_{p,\,q} |a_{pq} - \overline{a_{qp}}|/2.$$

16. Let A be a normal matrix partitioned into blocks as $A = \begin{pmatrix} E & F \\ G & H \end{pmatrix}$, where E and H are square matrices (of possibly different sizes).
    (a) Show that $\|F\|_{_F} = \|G\|_{_F}$.
    (b) Does $\|E\|_{_F} = \|H\|_{_F}$?
    (c) Does $\|F\|_{\mathrm{sp}} = \|G\|_{\mathrm{sp}}$?
    (d) Discuss (b) and (c) for a partitioned unitary matrix A.
    (e) Show that $\begin{pmatrix} H & G \\ F & E \end{pmatrix}$ is also a normal matrix. Is $\begin{pmatrix} F & E \\ H & G \end{pmatrix}$ normal?

17. Use Problem 16 (a) to show that for any $n \times n$ normal matrix $X = (x_{ij})$,

$$\sum_{i=1}^{n-1} \sum_{j=i+1}^{n} (j-i)|x_{ij}|^2 = \sum_{j=1}^{n-1} \sum_{i=j+1}^{n} (i-j)|x_{ij}|^2.$$

18. Find examples.
    (a) Theorem 9.4.1 need not be true if A or B is non-normal.
    (b) Theorem 9.4.3 need not be true if A and B are non-normal.

19. Show that the constant $\sqrt{2}$ in Corollary 9.4.1 is best possible by considering $\frac{\||A|-|B|\|_{_F}}{\|A-B\|_{_F}}$ with $A = \left(\begin{smallmatrix} 1 & 0 \\ 0 & 0 \end{smallmatrix}\right)$ and $B = \left(\begin{smallmatrix} 1 & x \\ 0 & 0 \end{smallmatrix}\right)$, $x > 0$, as $x \to 0$.

20. Let $A = \left(\begin{smallmatrix} 0 & 1 \\ x & 0 \end{smallmatrix}\right)$ (where $x \ge 0$) and $B = \left(\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right)$. Let $\lambda_{1,2} = \pm\sqrt{x}$ be the eigenvalues of A and let $\mu_1 = \mu_2 = 0$ be the eigenvalues of B. For $i = 1, 2$, and $0 < x < \frac{1}{2}$, show that $|\lambda_i - \mu_i| > \sqrt{2}\,\|A - B\|_{_F}$.

21. Show that the constant $\sqrt{n}$ in Theorem 9.4.3 is best possible by considering the $n \times n$ matrices

$$A = \begin{pmatrix} 0 & 1 & 0 & \dots & 0 \\ 0 & 0 & 1 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \dots & 1 \\ 1 & 0 & 0 & \dots & 0 \end{pmatrix}, \quad B = \begin{pmatrix} 0 & 1 & 0 & \dots & 0 \\ 0 & 0 & 1 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \dots & 1 \\ 0 & 0 & 0 & \dots & 0 \end{pmatrix}.$$

22. Let A and B be $n \times n$ normal matrices. Let $\lambda_1, \dots, \lambda_n$ and $\mu_1, \dots, \mu_n$ be the eigenvalues of A and B, respectively. Show that

$$\min_p \max_i |\lambda_i - \mu_{p(i)}| \le n \|A - B\|_{\text{sp}},$$

where p represents permutation(s) on $\{1, 2, \dots, n\}$ and $\|A-B\|_{\text{sp}}$ is the spectral norm of $A - B$, i.e., the largest singular value of $A - B$.

23. Let A and B be $n \times n$ normal matrices and let $\lambda_1, \dots, \lambda_n$ and $\mu_1, \dots, \mu_n$ be the eigenvalues of A and B, respectively. Show that there exists a permutation q on $\{1, 2, \dots, n\}$ such that

$$\|A - B\|_F \le \left(\sum_{i=1}^n |\lambda_i - \mu_{q(i)}|^2 \right)^{1/2}.$$

Conclude that there do not exist normal matrices A and B such that $\| \, |A| - |B| \, \|_F > \left(\sum_{i=1}^n |\lambda_i - \mu_{s(i)}|^2 \right)^{1/2}$ for all permutations s.

Note: Combining with the Hoffman-Wielandt inequality, we can write

$$\min_{i,j} \sum |\lambda_i - \mu_j|^2 \le \|A - B\|_F^2 \le \max_{i,j} \sum |\lambda_i - \mu_j|^2.$$

24. Let A and B be $m \times n$ matrices with singular values $\sigma_j(\cdot)$ arranged decreasingly: $\sigma_1(\cdot) \ge \cdots \ge \sigma_n(\cdot)$. Let $r = \min\{m, n\}$. Show that

$$\left(\sum_{j=1}^r \big(\sigma_j(A) - \sigma_j(B) \big)^2 \right)^{1/2} \le \|A - B\|_F.$$

25. Let $A = (a_{ij}) \in \mathbb{M}_n$ have eigenvalues $\lambda_1, \dots, \lambda_n$. The *spread* of A, denoted by $s(A)$, is defined to be $s(A) = \max_{i,j} |\lambda_i - \lambda_j|$. Show that

(a) $s(A) \le \sqrt{2} \, \|A\|_F$ for any $A \in \mathbb{M}_n$.
(b) $s(A) \ge \sqrt{3} \max_{i \ne j} |a_{ij}|$ if A is normal.
(c) $s(A) \ge 2 \max_{i \ne j} |a_{ij}|$ if A is Hermitian.

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CHAPTER 10

Majorization and Matrix Inequalities

Introduction: As a special topic in matrix analysis, majorization is an important tool in deriving inequalities of eigenvalues, singular values, and norms of matrices. It has a wide range of application in other fields such as optimization, statistics, etc. We introduce the concept of majorization, present its basic properties, and show some matrix inequalities in majorization.

10.1 Basic Properties of Majorization

Two vectors (in $\mathbb{R}^n$) can be compared in different ways. For instance, one vector may be longer than the other vector when measured in terms of length (norm); one vector may dominate the other vector componentwise. We introduce the notion of *majorization*, with which we compare two real vectors (sequences) and see which vector's components are "less spread-out" or if one vector "contains" or "controls" the other. For example, the components of $(0, -1, 1)$ and $(3, -2, -1)$ have the same average (mean) 0. The components of the first vector are closer to the mean, while the second vector's components are more spread-out. We shall see much more about vectors.

Let $x = (x_1, x_2, \ldots, x_n) \in \mathbb{R}^n$. We rearrange the components $x_1, x_2, \ldots, x_n$ in a decreasing (going-down) order and list them as $x_1^{\downarrow} \geq x_2^{\downarrow} \geq \cdots \geq x_n^{\downarrow}$. Let

$$x^{\downarrow} = (x_1^{\downarrow}, x_2^{\downarrow}, \ldots, x_n^{\downarrow}).$$

That is, $x_i^{\downarrow} = (x^{\downarrow})_i$ is the ith largest component of x for $i = 1, 2, \ldots, n$. Similarly, let $x_1^{\uparrow} \leq x_2^{\uparrow} \leq \cdots \leq x_n^{\uparrow}$ be the components of x in an increasing (going-up) order and $x^{\uparrow} = (x_1^{\uparrow}, x_2^{\uparrow}, \ldots, x_n^{\uparrow})$. Obviously, $x_1^{\uparrow} = x_n^{\downarrow}$ and $x_n^{\uparrow} = x_1^{\downarrow}$.

Example 10.1.1 Let $x = (x_1, x_2, \ldots, x_n) \in \mathbb{R}^n$.

(a) For $x = (-1, -2, 3, 0)$, $x^{\downarrow} = (3, 0, -1, -2)$ and $x^{\uparrow} = (-2, -1, 0, 3)$.

(b) $(-x)^{\downarrow} = -(x^{\uparrow})$ and $x_i^{\downarrow} = x_{n-i+1}^{\uparrow}$ for each $i = 1, 2, \ldots, n$.

(c) $x^{\downarrow} = x$ if the components of x are already in decreasing order. ∎

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For real vectors $x = (x_1, x_2, \ldots, x_n)$ and $y = (y_1, y_2, \ldots, y_n)$, if

$$\sum_{i=1}^{k} x_i^{\downarrow} \le \sum_{i=1}^{k} y_i^{\downarrow}, \quad k = 1, 2, \ldots, n, \tag{10.1}$$

we say that y *weakly majorizes* x or x is *weakly majorized* by y, denoted by

$$x \prec_w y \quad \text{or} \quad y \succ_w x.$$

If $x \prec_w y$ and the equality in (10.1) holds for $k = n$, i.e., $x_1 + \cdots + x_n = y_1 + \cdots + y_n$, we say that y *majorizes* x or x is *majorized* by y, written as

$$x \prec y \quad \text{or} \quad y \succ x.$$

Obviously, $x \prec_w x$, $x \prec x$, $x \prec x^{\downarrow}$, and $x^{\uparrow} \prec x^{\downarrow}$. In addition, if $x \le y$ (componentwise, i.e., $x_i \le y_i$ for all i), then $x \prec_w y$; if $x \prec y$, then $x \prec_w y$.

Example 10.1.2 Majorization and weak majorization.

(a) Let $x = (0, 0)$ and $y = (1, -1)$. Then $x \prec y$, $-x \prec -y$, $-y \not\prec -x$. Thus, it is incorrect to state that $x \prec y \Leftrightarrow -y \prec -x$.

(b) Let $x = (0, 0)$ and $y = (1, 0)$. Then $x \prec_w y$ and $-y \prec_w -x$.

(c) Let $x = (-1, 0, 1)$, $y = (3, -2, -1)$, $z = (3, 0, 0)$, and $v = (2, 2, 0)$. Then $x \prec y$, $y \prec_w z$, and $x \prec_w z$; neither $z \prec_w v$ nor $v \prec_w z$ holds.

(d) $(\frac{a+b+c}{3}, \frac{a+b+c}{3}, \frac{a+b+c}{3}) \prec (a, b, c)$, for any $a, b, c \in \mathbb{R}$.

(e) $(\frac{1}{n}, \ldots, \frac{1}{n}) \prec (1, 0, \ldots, 0)$ (both vectors have n components). ∎

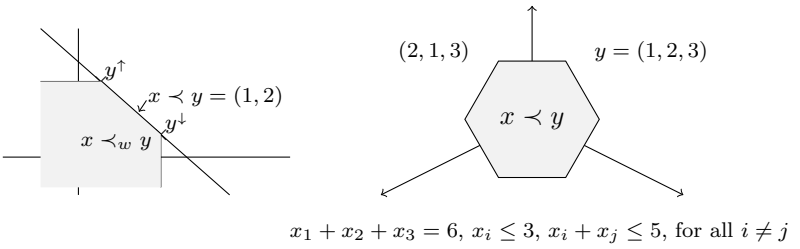


Fig. 10.1. Example 10.1.3

Example 10.1.3 Majorization and weak majorization.

(a) Let $y = (1, 2)$. The set $\{x \in \mathbb{R}^2 : x \prec y\}$ is the line segment joining the points $(2, 1)$ and $(1, 2)$, while the set of x weakly majorized by y is the unbounded region $\{(s, t) \in \mathbb{R}^2 : s \le 2, t \le 2, s + t \le 3\}$. One checks: $(1.9, 1.1) \prec (2, 1)$, $(0, 0) \prec_w (2, 1)$, $(0, 2) \prec_w (2, 1)$, $(-1, -1) \prec_w (2, 1)$.

(b) Let $y = (1, 2, 3)$. The set $\{x \in \mathbb{R}^3 : x \prec y\}$ is the hexagon with vertices $(1, 2, 3)$, $(1, 3, 2)$, $(2, 3, 1)$, $(3, 2, 1)$, $(3, 1, 2)$, $(2, 1, 3)$ in $\mathbb{R}^3$. ∎

Note that the positions of the components in the vectors are unimportant for majorization. More precisely, if a vector x is majorized by y, then any vector obtained by reordering the components of x is also majorized by y.

Let $z = (z_1, z_2, \ldots, z_n) \in \mathbb{R}^n$. Then for each $k = 1, 2, \ldots, n$, we have

$$\sum_{i=1}^{k} z_i^{\downarrow} = \max_{1 \le i_1 < \cdots < i_k \le n} \sum_{t=1}^{k} z_{i_t}. \tag{10.2}$$

For $x = (x_1, x_2, \ldots, x_n)$, $y = (y_1, y_2, \ldots, y_n) \in \mathbb{R}^n$, the weak majorization $x \prec_w y$, i.e., the inequalities (10.1), is restated in the equivalent form:

$$\max_{1 \le i_1 < \cdots < i_k \le n} \sum_{t=1}^{k} x_{i_t} \le \max_{1 \le i_1 < \cdots < i_k \le n} \sum_{t=1}^{k} y_{i_t}, \quad k = 1, 2, \ldots, n. \tag{10.3}$$

It is easily checked from the definitions that majorization $\prec$ and weak majorization $\prec_w$ are transitive binary relations of the elements in $\mathbb{R}^n$, that is, $x \prec y$ and $y \prec z$ imply $x \prec z$; $x \prec_w y$ and $y \prec_w z$ imply $x \prec_w z$, namely,

$$x \prec y,\ y \prec z \ \Rightarrow\ x \prec z; \quad x \prec_w y,\ y \prec_w z \ \Rightarrow\ x \prec_w z.$$

(For simplicity, "$x \prec y,\ y \prec z \Rightarrow x \prec z$" is for "$x \prec y$ and $y \prec z \Rightarrow x \prec z$".)

Theorem 10.1.1 *Let $x = (x_1, x_2, \ldots, x_n)$, $y = (y_1, y_2, \ldots, y_n) \in \mathbb{R}^n$. Then*

(1) $x \prec_w y \Rightarrow x_1^{\downarrow} \le y_1^{\downarrow}$.
(2) $x \prec y \Rightarrow y_n^{\downarrow} \le x_i \le y_1^{\downarrow}$ *for all* x_i.
(3) $x \prec y \Leftrightarrow x \prec_w y$ *and* $-x \prec_w -y$.
(4) $x \prec y,\ y \prec x \Leftrightarrow x = yP$ *for some* $n \times n$ *permutation matrix* P.
(5) $x \prec_w y,\ y \prec_w x \Leftrightarrow x = yQ$ *for some* $n \times n$ *permutation matrix* Q.

Proof. (1) is obvious from the definition of majorization.

For (2), it is sufficient to show $y_n^{\downarrow} \le x_n^{\downarrow}$. Notice that $x \prec y$ reveals

$$\sum_{i=1}^{n} x_i^{\downarrow} = \sum_{i=1}^{n} y_i^{\downarrow} \quad \text{and} \quad \sum_{i=1}^{n-1} x_i^{\downarrow} \le \sum_{i=1}^{n-1} y_i^{\downarrow}.$$

Subtracting the inequality from the equality, we obtain $x_n^{\downarrow} \ge y_n^{\downarrow}$.

We show (3). If $x \prec y$, then $\sum_{i=1}^{n} x_i^{\downarrow} = \sum_{i=1}^{n} y_i^{\downarrow}$ and $\sum_{i=1}^{k} x_i^{\downarrow} \le \sum_{i=1}^{k} y_i^{\downarrow}$ for $k = 1, 2, \ldots, n-1$. So $x \prec_w y$. To show $-x \prec_w -y$, subtracting the inequality from the equality, we obtain $\sum_{i=1}^{n-k} x_{n-i+1}^{\downarrow} \ge \sum_{i=1}^{n-k} y_{n-i+1}^{\downarrow}$. It follows that $\sum_{i=1}^{n-k} -(x_{n-i+1}^{\downarrow}) \le \sum_{i=1}^{n-k} -(y_{n-i+1}^{\downarrow})$. Because $-(x_{n-i+1}^{\downarrow}) = (-x)_i^{\downarrow}$ and $-(y_{n-i+1}^{\downarrow}) = (-y)_i^{\downarrow}$, setting $m = n - k$, we have $\sum_{i=1}^{m}(-x)_i^{\downarrow} \le \sum_{i=1}^{m}(-y)_i^{\downarrow}$ for $m = 1, 2, \ldots, n-1$ as $k = n-1, \ldots, 2, 1$. With $\sum_{i=1}^{n}(-x)_i^{\downarrow} = \sum_{i=1}^{n}(-y)_i^{\downarrow}$, we conclude that $-x \prec_w -y$. For the converse, it suffices to show

that the x_i's and y_i's have the same sum. This is immediate because $x \prec_w y$ implies $\sum_{i=1}^n x_i \le \sum_{i=1}^n y_i$, and $-x \prec_w -y$ yields $-\sum_{i=1}^n x_i \le -\sum_{i=1}^n y_i$. Thus, $\sum_{i=1}^n x_i = \sum_{i=1}^n y_i$, which, together with $x \prec_w y$, results in $x \prec y$.

(4) and (5) are similar. We show (4). If $x \prec y$ and $y \prec x$, then $x_1^\downarrow \le y_1^\downarrow$ and $y_1^\downarrow \le x_1^\downarrow$. Thus $x_1^\downarrow = y_1^\downarrow$. Let $\tilde{x}$ and $\tilde{y}$ be the vectors obtained from x and y by omitting $x_1^\downarrow$ and $y_1^\downarrow$, respectively. Then $\tilde{x} \prec \tilde{y}$ and $\tilde{y} \prec \tilde{x}$. The above argument reveals $\tilde{x}_1^\downarrow = \tilde{y}_1^\downarrow$, i.e., $x_2^\downarrow = y_2^\downarrow$. Inductively, $x_i^\downarrow = y_i^\downarrow$ for all i. Thus, the components of x are some rearrangements of the components of y, i.e., $x = yP$ for some permutation matrix P. The converse is trivial. ∎

By (4) and (5), $x \prec y$ and $y \prec x \Leftrightarrow x^\downarrow = y^\downarrow \Leftrightarrow x \prec_w y$ and $y \prec_w x$.

Our next theorem best characterizes the relationship between the weak majorization $\prec_w$ and the majorization $\prec$ via the componentwise dominance $\le$. This theorem is used repeatedly in later sections.

Theorem 10.1.2 *Let $x, y \in \mathbb{R}^n$. The following statements are equivalent.*

(1) $x \prec_w y$.

(2) $x \le z$ *and* $z \prec y$ *for some* $z \in \mathbb{R}^n$.

(3) $x \prec u$ *and* $u \le y$ *for some* $u \in \mathbb{R}^n$.

Proof. (1) $\Leftrightarrow$ (2): (2) $\Rightarrow$ (1) is easy. We show (1) $\Rightarrow$ (2) by induction on n. Suppose that the statement is true for vectors with less than n components.

If $n = 1$, it is trivial. Let $n > 1$. Without loss of generality, we assume that the components of x and y are already in decreasing order. Let $\epsilon = \min_k\{\sum_{i=1}^k (y_i - x_i)\} \ge 0$ and $\tilde{x} = x + (\epsilon, 0, \ldots, 0)$. Then $\tilde{x} = \tilde{x}^\downarrow$ and $x \le \tilde{x}$. We have $\tilde{x} \prec_w y$ because for each $p = 1, 2, \ldots, n$,

$$\sum_{i=1}^p \tilde{x}_i = \sum_{i=1}^p x_i + \epsilon \le \sum_{i=1}^p x_i + \sum_{i=1}^p (y_i - x_i) = \sum_{i=1}^p y_i.$$

If ϵ is attained when $k = m$, then $\sum_{i=1}^m \tilde{x}_i = \sum_{i=1}^m y_i$. This yields

$$(\tilde{x}_1, \ldots, \tilde{x}_m) \prec (y_1, \ldots, y_m), \quad (\tilde{x}_{m+1}, \ldots, \tilde{x}_n) \prec_w (y_{m+1}, \ldots, y_n).$$

If $m = n$, the weak majorization in the above display is absent. Let $m < n$. By induction on $k = n - m$, there is a real vector $(z_{m+1}, \ldots, z_n)$ such that

$$(\tilde{x}_{m+1}, \ldots, \tilde{x}_n) \le (z_{m+1}, \ldots, z_n) \prec (y_{m+1}, \ldots, y_n).$$

Set $z = (\tilde{x}_1, \ldots, \tilde{x}_m, z_{m+1}, \ldots, z_n)$. Then $x \le z$ and $z \prec y$, as desired.

(1) $\Leftrightarrow$ (3): One obtains (3) $\Rightarrow$ (1) as $u \le y \Rightarrow u \prec_w y$. We show (1) $\Rightarrow$ (3).

Let y_k be the smallest component of y. Let $\delta = \sum_{i=1}^n y_i - \sum_{i=1}^n x_i$ and $u = y - \delta e_k$, where $e_k \in \mathbb{R}^n$ has 1 in the kth position and 0 elsewhere. Then,

the smallest component of u is $u_k = y_k - \delta$, $u \le y$, and $\sum_{i=1}^{n} x_i = \sum_{i=1}^{n} u_i$. Moreover, for $p < n$, $\sum_{i=1}^{p} x_i^{\downarrow} \le \sum_{i=1}^{p} y_i^{\downarrow} = \sum_{i=1}^{p} u_i^{\downarrow}$. Thus, $x \prec u$. ∎

We now proceed to discuss majorization involving the sum of two vectors. First, note that for $x, y \in \mathbb{R}^n$, $(x+y)^{\downarrow}$ and $x^{\downarrow} + y^{\downarrow}$ need not be equal as one checks with $x = (0,1)$, $y = (1,0)$: $(x+y)^{\downarrow} = (1,1)$, while $x^{\downarrow} + y^{\downarrow} = (2,0)$.

Theorem 10.1.3 *Let $x, y \in \mathbb{R}^n$. Then*

$$x + y \prec x^{\downarrow} + y^{\downarrow}.$$

Proof. Let $z = x + y$. Note that the sum of the (all) components of $z = x + y$ is equal to the sum of the components of $x^{\downarrow} + y^{\downarrow}$. We show the inequalities.

For each fixed k $(1 \le k \le n)$, let $z_1^{\downarrow} = (x+y)_{j_1}, \ldots, z_k^{\downarrow} = (x+y)_{j_k}$. Then

$$\sum_{i=1}^{k} z_i^{\downarrow} = \sum_{i=1}^{k} (x+y)_{j_i} = \sum_{i=1}^{k} x_{j_i} + \sum_{i=1}^{k} y_{j_i} \le \sum_{i=1}^{k} x_i^{\downarrow} + \sum_{i=1}^{k} y_i^{\downarrow}.$$

Observe that the components of $x^{\downarrow} + y^{\downarrow}$ are already in decreasing order, i.e., $(x^{\downarrow} + y^{\downarrow})^{\downarrow} = x^{\downarrow} + y^{\downarrow}$, because $x_i^{\downarrow} + y_i{}^{\downarrow} \ge x_{i+1}^{\downarrow} + y_{i+1}^{\downarrow}$ for every i. ∎

Note that $x + y \not\prec x^{\downarrow} + y$ by taking $x = (0,1)$ and $y = (-1,0)$ as a counterexample. Moreover, $u \prec v$ does not imply $u + y \prec v + y$ in general.

Example 10.1.4 Let $x, y \in \mathbb{R}^n$. Then $(x+y)^{\downarrow} \prec x + y \prec x^{\downarrow} + y^{\downarrow}$. Thus,

$$\sum_{i=1}^{k} (x+y)_i^{\downarrow} \le \sum_{i=1}^{k} (x_i^{\downarrow} + y_i^{\downarrow}), \quad k = 1, 2, \ldots, n.$$

However, $(x+y)_i^{\downarrow} \not\le x_i^{\downarrow} + y_i^{\downarrow}$ (if $i \ge 2$) as one easily finds a counterexample. ∎

Theorem 10.1.4 *Let $x, y, z \in \mathbb{R}^n$. Then*

(1) $x \prec z$, $y \prec z \Rightarrow px + qy \prec z$, *where* $p, q \ge 0$, $p + q = 1$.

(2) $x \prec_w z$, $y \prec_w z \Rightarrow px + qy \prec_w z$, *where* $p, q \ge 0$, $p + q = 1$.

Proof. The proofs of (1) and (2) are similar. We only show (2). Note that $(px)^{\downarrow} = p(x^{\downarrow})$ for $p \ge 0$. Let $u = px + qy$. Then $u \prec_w z$ is derived from the following inequalities, for $k = 1, 2, \ldots, n$,

$$\begin{aligned} \sum_{i=1}^{k} u_i^{\downarrow} &= \sum_{i=1}^{k} (px+qy)_i^{\downarrow} \le \sum_{i=1}^{k} p\, x_i^{\downarrow} + \sum_{i=1}^{k} q\, y_i^{\downarrow} \\ &= p \sum_{i=1}^{k} x_i^{\downarrow} + q \sum_{i=1}^{k} y_i^{\downarrow} \le (p+q) \sum_{i=1}^{k} z_i^{\downarrow} = \sum_{i=1}^{k} z_i^{\downarrow}. \end{aligned}$$

∎

Theorem 10.1.5 *Let $x, y \in \mathbb{R}^m$ and $u, v \in \mathbb{R}^n$. Then*

(1) $x \prec y,\ u \prec v \Rightarrow (x,u) \prec (y,v)$.
(2) $x \prec_w y,\ u \prec_w v \Rightarrow (x,u) \prec_w (y,v)$.
(3) $x \prec y,\ u \prec v \Rightarrow x + u \prec y^{\downarrow} + v^{\downarrow}$ *(when $m = n$)*.
(4) $x \prec_w y,\ u \prec_w v \Rightarrow x + u \prec_w y^{\downarrow} + v^{\downarrow}$ *(when $m = n$)*.

Proof. The proofs of (1) and (2) are similar; so are the proofs of (3) and (4). We show (2) and (4). Let $\tilde{x} = (x,u)$ and $\tilde{y} = (y,v)$. For each positive integer $k \le n$, suppose that the first k largest components of $\tilde{x}$ consist of $x_1^{\downarrow}, \ldots, x_r^{\downarrow}$ and $u_1^{\downarrow}, \ldots, u_s^{\downarrow}$, $r + s = k$. Since $x \prec_w y$ and $u \prec_w v$, we have

$$\sum_{i=1}^{k} \tilde{x}_i^{\downarrow} = \sum_{i=1}^{r} x_i^{\downarrow} + \sum_{i=1}^{s} u_i^{\downarrow} \le \sum_{i=1}^{r} y_i^{\downarrow} + \sum_{i=1}^{s} v_i^{\downarrow} \le \sum_{i=1}^{k} \tilde{y}_i^{\downarrow}.$$

This says that $\tilde{x}$ is weakly majorized by $\tilde{y}$, i.e., $(x,u) \prec_w (y,v)$.

(4) is proven by verifying that for each $k = 1, 2, \ldots, n$,

$$\begin{aligned} \sum_{i=1}^{k} (x+u)_i^{\downarrow} &\le \sum_{i=1}^{k} x_i^{\downarrow} + \sum_{i=1}^{k} u_i^{\downarrow} \le \sum_{i=1}^{k} y_i^{\downarrow} + \sum_{i=1}^{k} v_i^{\downarrow} \\ &= \sum_{i=1}^{k} (y_i^{\downarrow} + v_i^{\downarrow}) = \sum_{i=1}^{k} (y^{\downarrow} + v^{\downarrow})_i \\ &= \sum_{i=1}^{k} (y^{\downarrow} + v^{\downarrow})_i^{\downarrow}. \ \blacksquare \end{aligned}$$

Example 10.1.5 One should be careful about operations on majorization. "Addition" and "subtraction" are not legitimate operations. For example, $x \prec y + z \not\Rightarrow x - y \prec z$ for $x = (0,0)$, $y = (1,-1)$, and $z = (0,0)$. $\blacksquare$

Example 10.1.6 For $x = (x_1, x_2, \ldots, x_n) \in \mathbb{R}^n$, let $|x| = (|x_1|, |x_2|, \ldots, |x_n|)$. Since $x_i \le |x_i|$ for each i, we have $x \le |x|$, consequently $x \prec_w |x|$.

Note that $|x^{\downarrow}|_i \not\le |x|_i^{\downarrow}$ in general. Take $x = (3,-2,0,1) \in \mathbb{R}^4$. Then

$$|x| = (3,2,0,1),\ x^{\downarrow} = (3,1,0,-2),\ |x^{\downarrow}| = (3,1,0,2),\ |x|^{\downarrow} = (3,2,1,0).$$

It is obvious that $|x^{\downarrow}|_4 = 2 > |x|_4^{\downarrow} = 0$. However, $|x^{\downarrow}| \prec |x| \prec |x^{\downarrow}|$ is true. $\blacksquare$

Theorem 10.1.6 *Let $x = (x_1, x_2, \ldots, x_n)$, $y = (y_1, y_2, \ldots, y_n) \in \mathbb{R}^n$. Then*

(1) $|x + y| \prec_w |x|^{\downarrow} + |y|^{\downarrow}$.
(2) $x^{\downarrow} + y^{\uparrow} \prec x + y \prec x^{\downarrow} + y^{\downarrow}$.
(3) $\sum_{i=1}^{n} x_i^{\downarrow} y_i^{\uparrow} \le \sum_{i=1}^{n} x_i y_i \le \sum_{i=1}^{n} x_i^{\downarrow} y_i^{\downarrow}$.

Proof. (1). $|x + y| \le |x| + |y| \prec_w |x|^{\downarrow} + |y|^{\downarrow}$ (by Theorem 10.1.3).

Note: it is not true in general that $|x + y|_i^{\downarrow} \le |x|_i^{\downarrow} + |y|_i^{\downarrow}$ (when $i > 1$).

(2). The second majorization inequality is Theorem 10.1.3. To show the first inequality, we may assume that $x = x^{\downarrow}$. Consider the case $n = 2$ first. If $y_1 \le y_2$, then $y = y^{\uparrow}$ and $x^{\downarrow} + y^{\uparrow} = x + y \prec x + y$. If $y_1 > y_2$, then $y = y^{\downarrow}$ and $x + y = x^{\downarrow} + y^{\downarrow}$. Since $y^{\uparrow} \prec y^{\downarrow}$, by Theorem 10.1.5 (3), we obtain $x^{\downarrow} + y^{\uparrow} \prec x^{\downarrow} + y^{\downarrow} = x + y$. We next deal with the case $n > 2$.

Our goal is to obtain $y^{\uparrow}$ by repeatedly exchanging the components y so that they are in increasing order. If $y = y^{\uparrow}$, there is nothing to prove. Suppose $y_i > y_j$ for some i and j, $i < j$. Switch the components y_i and y_j in y and denote the resulting vector by $\tilde{y}$. Note that a pair of components y_i, y_j in y now are in the increasing order y_j, y_i in $\tilde{y}$ (say, $y = (3, 1, 0, 2)$, $\tilde{y} = (3, 0, 1, 2)$). Observe that $x+\tilde{y}$ and $x+y$ differ by two components. By Theorem 10.1.5 (1) and by the above argument for the case of $n = 2$, we see $x + \tilde{y} \prec x + y$. If $\tilde{y} = y^{\uparrow}$, we are done. Otherwise, by the same argument, we have $\hat{y}$ so that $x + \hat{y} \prec x + \tilde{y}$ and $\hat{y}$ has two more components than $\tilde{y}$ that are in increasing order. Repeating the process gives $x^{\downarrow} + y^{\uparrow} \prec \cdots \prec x + \hat{y} \prec x + \tilde{y} \prec x + y$.

For (3), we again assume that $x = x^{\downarrow}$. We show the first inequality; the proof for the second one is similar. If $y = y^{\uparrow}$, then we have nothing to prove. Otherwise, let $\tilde{y}$ be the vector as in the above discussion. Compute

$$x_i y_j + x_j y_i - x_i y_i - x_j y_j = (x_i - x_j)(y_j - y_i) \le 0$$

or

$$x_i \tilde{y}_i + x_j \tilde{y}_j = x_i y_j + x_j y_i \le x_i y_i + x_j y_j.$$

This yields $\sum_{t=1}^{n} x_t \tilde{y}_t \le \sum_{t=1}^{n} x_t y_t$. Similarly, we get $\sum_{t=1}^{n} x_t \hat{y}_t \le \sum_{t=1}^{n} x_t \tilde{y}_t$. Repeat this process until $y^{\uparrow}$ is obtained. Thus, (3) follows. ∎

Example 10.1.7 Referring to Theorem 10.1.6 (3), the sum $\sum_{i=1}^{n}(\cdot)$ of all n terms cannot be replaced by partial sums of k terms $\sum_{i=1}^{k}(\cdot)$, $k < n$, in other words, the inequalities $\sum_{i=1}^{k} x_i^{\downarrow} y_i^{\uparrow} \le \sum_{i=1}^{k} x_i y_i \le \sum_{i=1}^{k} x_i^{\downarrow} y_i^{\downarrow}$ need not be true. For example, if $x = (-1, -2, -3)$, $y = (-2, -1, -3)$, $k = 1$, then $x_1^{\downarrow} y_1^{\uparrow} = 3 > x_1 y_1 = 2 > x_1^{\downarrow} y_1^{\downarrow} = 1$. However, the partial sum inequalities do hold true when the components of x and y are all nonnegative (Theorem 10.1.8 (3)). ∎

Theorem 10.1.7 *Let $x = (x_1, \ldots, x_n)$, $y = (y_1, \ldots, y_n) \in \mathbb{R}^n$. Then*

$$x \prec_w y \;\Leftrightarrow\; \sum_{i=1}^{n} x_i^{\downarrow} u_i \le \sum_{i=1}^{n} y_i^{\downarrow} u_i \;\text{ for all } u_1 \ge \cdots \ge u_n \ge 0 \tag{10.4}$$

and

$$x \prec y \;\Leftrightarrow\; \sum_{i=1}^{n} x_i^{\downarrow} u_i \le \sum_{i=1}^{n} y_i^{\downarrow} u_i \;\text{ for all real } u_1 \ge \cdots \ge u_n. \tag{10.5}$$

Proof. For the weak majorization, if the right-hand side of (10.4) holds, then for each $k \le n$, we set $u_i = 1$ for $i \le k$ and 0 otherwise, concluding

that $x \prec_w y$. Conversely, let $t_i = y_i^{\downarrow} - x_i^{\downarrow}$ for each i. Then $x \prec_w y$ implies $t_1 + \cdots + t_k = (y_1 + \cdots + y_k) - (x_1 + \cdots + x_k) \geq 0$ for every $k \leq n$. Since $u_1 \geq \cdots \geq u_n$ and $u_n \geq 0$, upon computation, we have

$$\begin{aligned}\sum_{i=1}^{n} y_i^{\downarrow} u_i - \sum_{i=1}^{n} x_i^{\downarrow} u_i &= \sum_{i=1}^{n} t_i u_i \\ &= t_1(u_1 - u_2) + (t_1 + t_2)(u_2 - u_3) + \cdots \\ &+ (t_1 + \cdots + t_{n-1})(u_{n-1} - u_n) + (t_1 + \cdots + t_n)u_n \geq 0.\end{aligned}$$

For the majorization, if the right-hand side of (10.5) holds, then $x \prec_w y$ by (10.5). Putting all $u_i = -1$ reveals the identity $x_1 + \cdots + x_n = y_1 + \cdots + y_n$. Thus, $x \prec y$. For the converse, with $u_1 \geq \cdots \geq u_n$, it is sufficient to notice in the above display that $t_1 + \cdots + t_n = 0$ because $x \prec y$. ∎

Denote by $\mathbb{R}^n_+$ the set of all vectors in $\mathbb{R}^n$ with nonnegative components.

Theorem 10.1.8 *Let $x, y, u, v \in \mathbb{R}^n_+$. Then (for componentwise product $\circ$)*

(1) $x \prec_w y \;\Rightarrow\; x \circ u \prec_w y^{\downarrow} \circ u^{\downarrow}$.

(2) $x \prec_w u,\ y \prec_w v \;\Rightarrow\; x \circ y \prec_w u^{\downarrow} \circ v^{\downarrow}$.

(3) $x \circ y \prec_w x^{\downarrow} \circ y^{\downarrow}$. *Thus,* $\sum_{t=1}^{k} x_t y_t \leq \sum_{t=1}^{k} x_t^{\downarrow} y_t^{\downarrow}$, $k = 1, \ldots, n$.

Proof. (1) implies (2) by applying (1) twice (as $x \circ y \prec_w u^{\downarrow} \circ y^{\downarrow} \prec_w u^{\downarrow} \circ v^{\downarrow}$), and (3) follows from (2) because $x \prec x^{\downarrow}$ and $y \prec y^{\downarrow}$; or see the argument below for a direct proof of (3). Thus, we only show (1).

For $u = (u_1, \ldots, u_n)$, taking $u_1^{\downarrow} \geq \cdots \geq u_k^{\downarrow} \geq 0 = \cdots = 0$ in (10.4) gives

$$\sum_{i=1}^{k} x_i^{\downarrow} u_i^{\downarrow} \leq \sum_{i=1}^{k} y_i^{\downarrow} u_i^{\downarrow}, \quad k = 1, \ldots, n. \tag{10.6}$$

Note that the components of $x^{\downarrow} \circ u^{\downarrow}$ and $y^{\downarrow} \circ u^{\downarrow}$ are already in decreasing order. (10.6) says $x^{\downarrow} \circ u^{\downarrow} \prec_w y^{\downarrow} \circ u^{\downarrow}$. We show below that $x \circ u \prec_w x^{\downarrow} \circ u^{\downarrow}$.

For any index sequence $1 \leq i_1 < \cdots < i_k \leq n$, let $\hat{x} = (x_{i_1}, \ldots, x_{i_k})$ and $\hat{u} = (u_{i_1}, \ldots, u_{i_k})$. Note that $0 \leq \hat{x}_t^{\downarrow} \leq x_t^{\downarrow}$ and $0 \leq \hat{u}_t^{\downarrow} \leq u_t^{\downarrow}$ for $t = 1, \ldots, k$. Applying Theorem 10.1.6 (3) to $\hat{x}$ and $\hat{u}$ and by inequalities (10.6), we have

$$\sum_{t=1}^{k} x_{i_t} u_{i_t} = \sum_{t=1}^{k} \hat{x}_t \hat{u}_t \leq \sum_{t=1}^{k} \hat{x}_t^{\downarrow} \hat{u}_t^{\downarrow} \leq \sum_{t=1}^{k} x_t^{\downarrow} u_t^{\downarrow}. \tag{10.7}$$

Thus, as the components of $x^{\downarrow} \circ u^{\downarrow}$ are already in decreasing order, we have

$$\max_{1 \leq i_1 < \cdots < i_k \leq n} \sum_{t=1}^{k} x_{i_t} u_{i_t} \leq \sum_{t=1}^{k} x_t^{\downarrow} u_t^{\downarrow} = \sum_{t=1}^{k} (x^{\downarrow} \circ u^{\downarrow})_t. \tag{10.8}$$

Using (10.3), as $x^{\downarrow} \circ u^{\downarrow} \prec_w y^{\downarrow} \circ u^{\downarrow}$, we obtain $x \circ u \prec_w x^{\downarrow} \circ u^{\downarrow} \prec_w y^{\downarrow} \circ u^{\downarrow}$. ∎

Questions: Can $y^{\downarrow} \circ u^{\downarrow}$ on the right-hand side of Theorem 10.1.8 (1) be replaced by $y \circ u$? Does the theorem hold true for all real $x, y, u, v \in \mathbb{R}^n$?

Problems

1. Find two vectors $x, y \in \mathbb{R}^3$ such that neither $x \prec y$ nor $x \succ y$ holds.
2. Let $x, y \in \mathbb{R}^n$. If $x \prec_w y$ and $y \prec_w x$, show that $x^{\downarrow} = y^{\downarrow}$.
3. Let $x, y \in \mathbb{R}^n$. Is it true that $x \prec_w y \Leftrightarrow -y \prec_w -x$?
4. Let $x, y \in \mathbb{R}^n$. If $x \leq y$ (componentwise), show that $x \prec_w y$.
5. Let $y = (1, 0) \in \mathbb{R}^2$. Sketch the following sets in the xy-plane $\mathbb{R}^2$:
$$\{x \in \mathbb{R}^2 : x \prec y\} \quad \text{and} \quad \{x \in \mathbb{R}^2 : x \prec_w y\}.$$
6. Let $x_1, \ldots, x_n$ be real numbers in the interval $[0, 1]$. Show that
$$(x_1^2, \ldots, x_n^2) \prec_w (x_1, \ldots, x_n) \prec_w (\sqrt{x_1}, \ldots, \sqrt{x_n}).$$
7. Let $x, y \in \mathbb{R}^n$. Show that $x \prec y$ if and only if $ax \prec ay$, where $a > 0$, and if and only if $x + be \prec y + be$, where $e = (1, \ldots, 1) \in \mathbb{R}^n$ and $b \in \mathbb{R}$.
8. Let $x, y \in \mathbb{R}^n$. Show that $(x^{\downarrow} + y^{\downarrow})^{\downarrow} = x^{\downarrow} + y^{\downarrow}$ and $(x^{\downarrow} - y^{\uparrow})^{\downarrow} = x^{\downarrow} - y^{\uparrow}$. If x and y are nonnegative vectors, show that $(x^{\downarrow} \circ y^{\downarrow})^{\downarrow} = x^{\downarrow} \circ y^{\downarrow}$.
9. Let $x \in \mathbb{R}^n$. Show that $x^{\downarrow} + (-x)^{\uparrow} = 0$ and $x^{\uparrow} + (-x)^{\downarrow} = 0$.
10. Let $x, y \in \mathbb{R}^n$. Show that $x^{\downarrow} - y^{\downarrow} \prec x - y \prec x^{\downarrow} - y^{\uparrow}$.
11. Let $x = (x_1, x_2, \ldots, x_n)$, $y = (y_1, y_2, \ldots, y_n) \in \mathbb{R}^n$. Show that $x \prec_w y$ if and only if $(x_1^{\downarrow}, x_2^{\downarrow}, \ldots, x_k^{\downarrow}) \prec_w (y_1^{\downarrow}, y_2^{\downarrow}, \ldots, y_k^{\downarrow})$, $k = 1, 2, \ldots, n$.
12. Let $x = (x_1, x_2, \ldots, x_n) \in \mathbb{R}^n$. If $x_1 \geq x_2 \geq \cdots \geq x_n$, show that $\frac{1}{m} \sum_{i=1}^{m} x_i \geq \frac{1}{n} \sum_{i=1}^{n} x_i$ for every positive integer $m \leq n$.
13. Let $e = (1, 1, \ldots, 1) \in \mathbb{R}^n$. Find all $x \in \mathbb{R}^n$ such that $x \prec e$.
14. Let $x = (x_1, x_2, \ldots, x_n) \in \mathbb{R}^n$ and $\bar{x} = \frac{1}{n}(x_1 + x_2 + \cdots + x_n)$. Show that $\bar{x}e \prec x$, where $e = (1, 1, \ldots, 1) \in \mathbb{R}^n$. State the case of $n = 2$.
15. Let $x, y \in \mathbb{R}^n$. If $x \prec y$, show that $x + z \prec x^{\downarrow} + z^{\downarrow} \prec y^{\downarrow} + z^{\downarrow}$ for any $z \in \mathbb{R}^n$. Conversely, if $x^{\downarrow} + z^{\downarrow} \prec y^{\downarrow} + z^{\downarrow}$ for some $z \in \mathbb{R}^n$, show that $x \prec y$. If $x + z^{\downarrow} \prec y^{\downarrow} + z^{\downarrow}$ for some $z \in \mathbb{R}^n$, does it follow that $x \prec y$?
16. Find examples to show that each of the following need not be true.
    (a) $(x + y)_i^{\downarrow} \leq x_i^{\downarrow} + y_i^{\downarrow}$, where $x, y \in \mathbb{R}_+^n$.
    (b) $(x \circ y)_i^{\downarrow} \leq x_i^{\downarrow} y_i^{\downarrow}$, where $x, y \in \mathbb{R}_+^n$.
    (c) $x \circ y \prec_w x \circ y^{\downarrow}$, where $x, y \in \mathbb{R}_+^n$.
    (d) $x \circ y \prec_w x^{\downarrow} \circ y^{\downarrow}$, where $x \in \mathbb{R}_+^n$ and $y \in \mathbb{R}^n$.
    (e) $x + z \prec y + z \Rightarrow x \prec y$, where $x, y, z \in \mathbb{R}_+^n$.

17. Let $x, y \in \mathbb{R}^n$. Show that $x \prec y$ if and only if $-x \prec -y$. If $x \prec_w y$, does it necessarily follow that $-x \prec_w -y$ (or $-y \prec_w -x$)? Or $x - y \prec_w 0$?

18. Let $x, y, z \in \mathbb{R}^n$. If $x + y \prec_w z$, show that $x \prec_w z^{\downarrow} - y^{\uparrow}$.

19. Let $x = (x_1, x_2, \ldots, x_n) \in \mathbb{R}^n$ and let $1 \leq i_1 < \cdots < i_k \leq n$ be an index sequence. Show that $\min\{x_{i_1}, \ldots, x_{i_k}\} \leq x_k^{\downarrow}$ and $x_{i_t}^{\downarrow} \leq x_t^{\downarrow}$ for each t.

20. Let $x, y \in \mathbb{R}^n$ and $x \leq y$ (componentwise). Show that $xP \leq yP$ for any $n \times n$ permutation matrix P and that $x^{\downarrow} \leq y^{\downarrow}$ (see Theorem 8.4.1 (2)).

21. Let $x, y \in \mathbb{R}^n_+$. Show that $(x, y) \prec_w (x + y, 0)$. Is this true if $x, y \in \mathbb{R}^n$?

22. Let $x, y \in \mathbb{R}^n$. If $x \prec y$ and if $0 \leq \alpha \leq \beta \leq 1$, show that

$$\beta x + (1-\beta)y \prec \beta x^{\downarrow} + (1-\beta)y^{\downarrow} \prec \alpha x^{\downarrow} + (1-\alpha)y^{\downarrow}.$$

23. Let $x, y \in \mathbb{R}^n$. Show that $x \prec y \Leftrightarrow (x, z) \prec (y, z)$ for all $z \in \mathbb{R}^m$.

24. Let $x, y \in \mathbb{R}^n$. If $(x, z) \prec_w (y, z)$ for some $z \in \mathbb{R}^m$, show that $x \prec_w y$. Consider the more general case: If $(x, u) \prec (y, v)$ for some $u, v \in \mathbb{R}^m$ satisfying $u \prec v$, does it necessarily follow that $x \prec y$ or $x \prec_w y$?

25. Let x, $y \in \mathbb{R}^n$ such that $x \prec_w y$. Show that (i) there exists $\tilde{y} \in \mathbb{R}^n$ which differs from y by at most one component such that $x \prec \tilde{y}$ and (ii) there exist $a, b \in \mathbb{R}$ such that $(x, a) \prec (y, b)$.

26. Let $x, y \in \mathbb{R}^n$. If $2x \prec_w y + z$, show that $(x, x) \prec_w (y, z)$.

27. If $(x_1, \ldots, x_n) \prec (y_1, \ldots, y_n)$, show that $y_m^{\downarrow} \geq x_m^{\downarrow} \geq y_{m+1}^{\downarrow}$ for some m.

28. Let $t \in \mathbb{R}$ and denote $t^+ = \max\{t, 0\}$. If $(x_1, \ldots, x_n) \prec (y_1, \ldots, y_n)$, show that $(x_1^+, \ldots, x_n^+) \prec_w (y_1^+, \ldots, y_n^+)$. Is the converse true?

29. Let $x = (x_1, \ldots, x_n) \in \mathbb{R}^n$ and $\beta = x_1 + \cdots + x_n$. Let α be a real number such that $x_n^{\downarrow} \leq \alpha \leq x_1^{\downarrow}$. Show that $(\alpha, \frac{\beta-\alpha}{n-1}, \ldots, \frac{\beta-\alpha}{n-1}) \prec x$.

30. For the given n and $y \in \mathbb{R}^n$ in (a)–(e), determine and sketch the sets:

    (1) $S_1 = \{x \in \mathbb{R}^n : x \prec y\}$.
    (2) $S_2 = \{x \in \mathbb{R}^n : x \prec_w y\}$.
    (3) $S_3 = \{x \in \mathbb{R}^n : y \prec x\}$.
    (4) $S_4 = \{x \in \mathbb{R}^n : y \prec_w x\}$.

    (a) $n = 1$, $y = 0 \in \mathbb{R}$.
    (b) $n = 2$, $y = (1, 1) \in \mathbb{R}^2$.
    (c) $n = 2$, $y = (2, 0) \in \mathbb{R}^2$.
    (d) $n = 3$, $y = (0, 0, 0) \in \mathbb{R}^3$.
    (e) $n = 3$, $y = (1, 0, 0) \in \mathbb{R}^3$.

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10.2 Majorization and Stochastic Matrices

Recall (from Sect. 5.6, Chap. 5) that a *doubly stochastic matrix* is a square nonnegative matrix whose row sums and column sums are each equal to 1, i.e., $A \in \mathbb{M}_n$ is doubly stochastic if A is nonnegative and for $e = (1, \ldots, 1) \in \mathbb{R}^n$,

$$eA = e \quad \text{and} \quad Ae^T = e^T. \tag{10.9}$$

This section shows a close relation between majorization and doubly stochastic matrices. Two fundamental results are: (1) on majorization, for $x, y \in \mathbb{R}^n$, $x \prec y$ if and only if $x = yP$ for some doubly stochastic matrix P, and (2) on weak majorization, for $x, y \in \mathbb{R}^n_+$, $x \prec_w y$ if and only if $x = yQ$ for some doubly substochastic matrix Q. A *doubly substochastic matrix* is a square nonnegative matrix, say A, whose row and column sums are each less than or equal to 1, i.e., $eA \le e$ and $Ae^T \le e^T$, where $e = (1, \ldots, 1) \in \mathbb{R}^n$.

Theorem 10.2.1 *Let A be an n-square nonnegative matrix. Then A is doubly stochastic if and only if $xA \prec x$ for all (row vectors) $x \in \mathbb{R}^n$.*

Proof. For necessity, by Theorem 5.6.7 (Sect. 5.6, Chap. 5), we write A as a convex combination of some n-square permutation matrices $P_1, P_2, \ldots, P_m$:

$$A = \sum_{i=1}^{m} \alpha_i P_i, \quad \sum_{i=1}^{m} \alpha_i = 1, \ \ \alpha_i > 0.$$

Because $xP_i \prec x$ for each i, by Theorem 10.1.5 (3), we have

$$xA = \sum_{i=1}^{m} \alpha_i x P_i \prec \sum_{i=1}^{m} \alpha_i x^{\downarrow} = x^{\downarrow} \prec x.$$

For sufficiency, take $x = e = (1, \ldots, 1)$. Then $eA \prec e$ says that each column sum of A is at most 1, and the sum of all entries (in columns) of A equals $n = 1 + \cdots + 1$. Thus, every column sum of A is equal to 1.

Now set $x = e_i = (0, \ldots, 1, \ldots, 0) \in \mathbb{R}^n$, where 1 is the ith component of e_i and all other components are equal to 0. Then $e_i A \prec e_i$ reveals that the ith row sum of A is equal to 1 for $i = 1, 2, \ldots, n$. ∎

Example 10.2.1 (a) For any $\theta \in \mathbb{R}$, $A = \begin{pmatrix} \sin^2\theta & \cos^2\theta \\ \cos^2\theta & \sin^2\theta \end{pmatrix}$ is doubly stochastic. We have $(a, b)A = (a\sin^2\theta + b\cos^2\theta,\ a\cos^2\theta + b\sin^2\theta) \prec (a, b)$ for all $a, b \in \mathbb{R}$.

(b) $B = \begin{pmatrix} \frac{1}{3} & \frac{1}{3} & \frac{1}{3} \\ \frac{1}{3} & \frac{1}{3} & \frac{1}{3} \\ \frac{1}{3} & \frac{1}{3} & \frac{1}{3} \end{pmatrix}$ is doubly stochastic. For any $a, b, c \in \mathbb{R}$, we have

$$(a, b, c)B = \left(\tfrac{a+b+c}{3}, \tfrac{a+b+c}{3}, \tfrac{a+b+c}{3}\right) \prec (a, b, c). \ ∎$$

We next take a closer look at the relation between two vectors when one is majorized by the other. Specifically, we want to see how to obtain one vector from the other if they are related via majorization. This can be accomplished by successively applying (a finite number of) the so-called *T-transforms*.

A *basic T-transformation*, or *transform* for short, is a 2×2 matrix of the form $\begin{pmatrix} t & 1-t \\ 1-t & t \end{pmatrix}$, where $0 \le t \le 1$. For higher orders, we call a matrix a *T-transform* if it is obtained from the identity matrix I by replacing a 2×2 principal submatrix of I with a basic T-transform. It is readily seen that a T-transform is a doubly stochastic matrix and it can be written as $tI + (1-t)P$, where $0 \le t \le 1$ and P is a permutation matrix that exchanges two rows (or columns) of I. Note that when $t = 0$, the T-transform is a permutation matrix. Since a permutation on the set $\{1, 2, \ldots, n\}$ can be obtained by a sequence of exchanges (Problem 37, Sect. 5.6), every permutation matrix can be factorized as a product of T-transforms that are obtained from the identity matrix by one exchange of two rows (or columns).

Example 10.2.2 $\begin{pmatrix} \frac{1}{3} & \frac{2}{3} \\ \frac{2}{3} & \frac{1}{3} \end{pmatrix}$, $\begin{pmatrix} \frac{1}{3} & \frac{2}{3} & 0 \\ \frac{2}{3} & \frac{1}{3} & 0 \\ 0 & 0 & 1 \end{pmatrix}$, and $\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & \frac{1}{3} & 0 & \frac{2}{3} \\ 0 & 0 & 1 & 0 \\ 0 & \frac{2}{3} & 0 & \frac{1}{3} \end{pmatrix}$ are T-transforms. ∎

Theorem 10.2.2 *Let $x, y \in \mathbb{R}^n$. Then $x \prec y$ if and only if there exist T-transforms $T_1, \ldots, T_m$ of order n such that $x = yT_1 \cdots T_m$. Consequently,*

$$x \prec y \;\Leftrightarrow\; x = yD \text{ for some doubly stochastic matrix } D.$$

Proof. Sufficiency: This follows from the previous theorem. To prove the necessity, we use induction on n. If $n = 1$, it is trivial. Let $n > 1$. We first show the case of $n = 2$ which will lead to the more general case.

Let $(x_1, x_2) \prec (y_1, y_2)$. By exchanges ($T$-transforms), we can assume that $x_1 \ge x_2$ and $y_1 \ge y_2$. Because $(x_1, x_2) \prec (y_1, y_2)$, we have $x_1 + x_2 = y_1 + y_2$ and $y_2 \le x_1 \le y_1$. There exists an $s \in [0, 1]$ such that $x_1 = sy_1 + (1-s)y_2$, with which we obtain $x_2 = y_1 + y_2 - x_1 = (1-s)y_1 + sy_2$. It follows that

$$(x_1, x_2) = (y_1, y_2)\begin{pmatrix} s & 1-s \\ 1-s & s \end{pmatrix}.$$

Suppose that $n > 2$ and the assertion is true for $n - 1$ or less.

If $x_1 = y_1$, then $(x_2, \ldots, x_n) \prec (y_2, \ldots, y_n)$. By induction hypothesis, there exist T-transforms $S_1, S_2, \ldots, S_m$, all of order $n-1$, such that $(x_2, \ldots, x_n) = (y_2, \ldots, y_n)S_1S_2 \cdots S_m$. Let $T_i = \begin{pmatrix} 1 & 0 \\ 0 & S_i \end{pmatrix}$. Apparently, T_i is also a T-transform, $i = 1, 2, \ldots, m$. We arrive at $x = yT_1T_2 \cdots T_m$, as desired.

If $x_i = y_j$ for some i and j, we may apply permutations (or a series of exchanges) on x and y so that x_i and y_j are the first components of the resulting vectors. Since every permutation matrix is a product of T-transforms, the argument reduces to the case $x_1 = y_1$ that we have settled.

Let $x_i \neq y_j$ for all i, j. We further assume that x and y are in decreasing order (via permutations). By Problem 27 of Sect. 10.1, $x \prec y$ implies

$$y_k > x_k > y_{k+1}, \quad \text{for some } k,\ 1 \leq k < n.$$

So

$$x_k = ty_k + (1-t)y_{k+1}, \quad \text{for some } t \in (0,1).$$

Let T_0 be the T-transform with $\begin{pmatrix} t & 1-t \\ 1-t & t \end{pmatrix}$ as the submatrix lying in rows k and $k+1$ and columns k and $k+1$. Set $z = yT_0$. Then z and y have the same components except the kth and $(k+1)$th components:

$$z_k = ty_k + (1-t)y_{k+1} = x_k$$

and

$$z_{k+1} = (1-t)y_k + ty_{k+1} = y_k + y_{k+1} - x_k.$$

Now we show $x \prec z$. For any positive integer $r \leq n$, if $r < k$, we have

$$\sum_{i=1}^{r} x_i \leq \sum_{i=1}^{r} y_i = \sum_{i=1}^{r} z_i \leq \sum_{i=1}^{r} z_i^{\downarrow}.$$

If $r = k$, then

$$\sum_{i=1}^{k} x_i \leq \sum_{i=1}^{k-1} y_i + x_k = \sum_{i=1}^{k} z_i \leq \sum_{i=1}^{k} z_i^{\downarrow}.$$

If $r > k$, because $z_k + z_{k+1} = y_k + y_{k+1}$, we also have

$$\sum_{i=1}^{r} x_i \leq \sum_{i=1}^{r} y_i = \sum_{i=1}^{r} z_i \leq \sum_{i=1}^{r} z_i^{\downarrow}.$$

Thus, $x \prec_w z$. Since $\sum_{i=1}^{n} x_i = \sum_{i=1}^{n} y_i = \sum_{i=1}^{n} z_i$, we arrive at $x \prec_w z$.

Note that x and z have the same kth component $x_k = z_k$. By the above argument (or the induction hypothesis on $n-1$), we have that $x = zT_1T_2 \cdots T_m = yT_0T_1T_2 \cdots T_m$, where $T_0, T_1, \ldots, T_m$ are T-transforms. ∎

Every doubly stochastic matrix is a convex combination of permutation matrices (Theorem 5.6.7, Sect. 5.6). Thus, we may restate the second part of the theorem as: $x \prec y$ if and only if there exist permutation matrices $P_1, \ldots, P_m$ such that $x = t_1yP_1 + \cdots + t_myP_m$, where $t_1 + \cdots + t_m = 1$, and all $t_i > 0$. Since each yP_i is a vector of rearrangements of the components of y, given $y \in \mathbb{R}^n$, we see that $\{x : x \prec y\}$ is the convex hull of the (up to $n!$) points in $\mathbb{R}^n$ obtained from y by permuting its components.

Question: In view of Theorem 10.2.2, can we say that every doubly stochastic matrix is a product of some T-transforms? (See Problem 7.)

Example 10.2.3 Let $y = (1, 2, 3) \in \mathbb{R}^3$. The vectors majorized by y form the convex set $\{x \in \mathbb{R}^3 : x \prec y\} = \{yD : D \text{ is a } 3 \times 3 \text{ doubly stochastic matrix}\}$; it is the hexagon of vertices $(1, 2, 3), (1, 3, 2), (2, 3, 1), (2, 1, 3), (3, 1, 2), (3, 2, 1)$ (Example 10.1.3) since D is a convex combination of permutation matrices, i.e., if $P_1, \ldots, P_6$ are the six permutation matrices of order 3, then $D = t_1P_1 + \cdots + t_6P_6$ for some nonnegative t_i's of sum 1, each yP_i is a vertex. So,

$$yD = y(t_1P_1 + \cdots + t_6P_6) = t_1yP_1 + \cdots + t_6yP_6. \blacksquare$$

Example 10.2.4 Majorization and doubly stochastic matrices.

(a) Let $a_1 \geq \cdots \geq a_n \geq 0$ and suppose that $a_1 + \cdots + a_n = 1$. Then

$$(\tfrac{1}{n}, \ldots, \tfrac{1}{n}) \prec (a_1, \ldots, a_n) \prec (1, 0, \ldots, 0),$$

that is, $\frac{k}{n} \leq a_1 + \cdots + a_k \leq 1$ for $k = 1, \ldots, n$. The first majorization is because $(\frac{1}{n}, \ldots, \frac{1}{n}) = (a_1, \ldots, a_n)N$, where N is the $n \times n$ matrix whose entries are all equal to $\frac{1}{n}$. Obviously, N is doubly stochastic.

(b) Let $c_1, c_2, c_3 \geq 0$ be such that $c_1 + c_2 + c_3 = 1$. Then $C = \begin{pmatrix} c_1 & c_3 & c_2 \\ c_2 & c_1 & c_3 \\ c_3 & c_2 & c_1 \end{pmatrix}$ is (circulant) doubly stochastic. For any $x = (x_1, x_2, x_3) \in \mathbb{R}^3$, $xC \prec x$:

$$(c_1x_1 + c_2x_2 + c_3x_3, c_3x_1 + c_1x_2 + c_2x_3, c_2x_1 + c_3x_2 + c_1x_3) \prec x.$$

(c) Let $x, y \in \mathbb{R}^n$. Then $x \prec y \Leftrightarrow -x \prec -y$. This is because $x = yD$ if and only if $(-x) = (-y)D$, where D is a doubly stochastic matrix.

(d) Let $x, y \in \mathbb{R}^n$. Then $x \prec y \Leftrightarrow (x, z) \prec (y, z)$ for any $z \in \mathbb{R}^m$. This is because if $x = yD$ for some doubly stochastic matrix D, then $(x, z) = (y, z)S$, where $S = \begin{pmatrix} D & 0 \\ 0 & I_m \end{pmatrix}$ is a doubly stochastic matrix. $\blacksquare$

We now study the analogue of Theorem 10.2.2 for weak majorization. Note that $x \leq y$ (componentwise) easily implies $x \prec_w y$, and $x \prec y$ implies $x \prec_w y$. We show that a weak majorization $\prec_w$ becomes the componentwise inequality $\leq$ via T-transforms. When the vectors are nonnegative, the weak majorization can be characterized by doubly substochastic matrices.

Theorem 10.2.3 *Let $x, y \in \mathbb{R}^n$. Then $x \prec_w y$ if and only if there exist some T-transforms $T_1, T_2, \ldots, T_m$ of order n such that $x \leq yT_1T_2 \cdots T_m$.*

Proof. If $x \prec_w y$, by Theorem 10.1.2, there exists $z \in \mathbb{R}^n$ such that $x \leq z$ and $z \prec y$. By Theorem 10.2.2, $z = yT_1T_2 \cdots T_m$, where $T_1, \ldots, T_m$ are some T-transforms. It follows that $x \leq yT_1T_2 \cdots T_m$.

Conversely, let $u = yT_1T_2 \cdots T_m$. Then $x \leq u$ implies $x \prec_w u$. On the other hand, $u \prec y$ because $T_1T_2 \cdots T_m$ is doubly stochastic. Thus, $x \prec_w y$. $\blacksquare$

Next we present a result analogous to Theorem 10.2.2 for weak majorization. Note that an $n \times n$ nonnegative matrix S is doubly substochastic if and only if $S \leq D$ (entrywise) for some doubly stochastic matrix D (Problem 3). It follows that $\begin{pmatrix} S & D-S \\ D-S & S \end{pmatrix}$ is a doubly stochastic matrix of order $2n$.

Theorem 10.2.4 *Let $x, y \in \mathbb{R}^n_+$. Then $x \prec_w y$ if and only if $x = yS$ for some doubly substochastic matrix S.*

Proof. Let $x = yS$ for some doubly substochastic matrix S. Then there exists a doubly stochastic matrix P such that $P - S$ is a nonnegative matrix. Thus, $yS \le yP$ for $y \in \mathbb{R}^n_+$. Then $yP \prec y \Rightarrow x = yS \le yP \prec y \Rightarrow x \prec_w y$.

For the converse, because $x \prec_w y$, by Theorem 10.2.3, there exist some T-transforms $T_1, T_2, \ldots, T_m$ such that $x \le yT$, where $T = T_1 T_2 \cdots T_m$. Let $z = yT$. Then $x \le z$. Note that x is nonnegative. By scaling the components of z to get x, i.e., choosing r_i so that $x_i = z_i r_i$, $0 \le r_i \le 1$, $i = 1, 2, \ldots, n$, we have a diagonal matrix $R = \mathrm{diag}(r_1, r_2, \ldots, r_n)$ such that $x = zR$. Set $S = TR$. It is easy to see that $S(\le T)$ is doubly substochastic and $x = yS$. ∎

Question: Does Theorem 10.2.4 hold true for all real $x, y \in \mathbb{R}^n$?

As is known (Birkhoff's theorem, Theorem 5.6.7, Chap. 5), the set of $n \times n$ doubly stochastic matrices is the convex hull of the $n!$ permutation matrices of order n. The following is an analog for the doubly substochastic matrices. We call a square $(0,1)$-matrix a *sub-permutation* matrix if each row and each column have at most one 1. That is, a sub-permutation matrix is obtained from a permutation matrix by changing (some or all) 1's to 0's.

Theorem 10.2.5 *Let Ω_n be the set of $n \times n$ doubly substochastic matrices. Then Ω_n is the convex hull of $\sum_{k=0}^{n} \binom{n}{k}^2 k!$ sub-permutation matrices.*

Proof. There is only one zero sub-permutation matrix. For each positive integer k, there are $k!$ permutation matrices (which are obtained from I_k by permuting rows and columns). For $k \le n$, choose k rows and k columns (there are $\binom{n}{k}\binom{n}{k}$ ways), and place a $k \times k$ permutation matrix in the selected rows and columns to get an $n \times n$ sub-permutation matrix with k 1's (rank k). Thus, there are $\binom{n}{k}^2 k!$ sub-permutation matrices of rank k. It follows that there are $\sum_{k=0}^{n} \binom{n}{k}^2 k!$ sub-permutation matrices in total. ∎

Every doubly substochastic matrix is a submatrix of some doubly stochastic matrix of higher order (Problem 11) and every doubly stochastic matrix is a convex combination of permutation matrices. Thus, every doubly substochastic matrix is a convex combination of partial (sub-) permutation matrices. For given $y \in \mathbb{R}^n_+$, $\{x \in \mathbb{R}^n_+ : x \prec_w y\}$ is the convex hull of all points $(t_1 y_{p(1)}, t_2 y_{p(2)}, \ldots, t_n y_{p(n)})$, where p runs over all permutations and each $t_i \in \{0, 1\}$. Note that both x and y are required to be nonnegative vectors in Theorem 10.2.4. The statement need not be true otherwise (Problem 19).

Example 10.2.5 Let $y = (1, 2, 3) \in \mathbb{R}^3$. Then $\{x \in \mathbb{R}^3_+ : x \prec_w y\}$ is the convex hull of the points (x_1, x_2, x_3), where $x_i = 0, 1, 2, 3$, and $1, 2, 3$ each appear no more than once. There are 1+9+18+6=34 such points, 16 of which comprise the vertices of the convex hull (Problem 26). ∎

Problems

1. Find a doubly stochastic matrix P such that $(1,2,3) = (6,0,0)P$.
2. Let $x = (x_1, \ldots, x_n)$, $y = (y_1, \ldots, y_n) \in \mathbb{R}^n$ and $a, b \in \mathbb{R}$. Show that
$$x \prec y \;\Rightarrow\; (ax_1 + b, \ldots, ax_n + b) \prec (ay_1 + b, \ldots, ay_n + b).$$
3. Let $Q = (q_{ij})$ be an n-square nonnegative matrix. Show that Q is doubly substochastic if and only if there exists an n-square doubly stochastic matrix $D = (d_{ij})$ such that $q_{ij} \le d_{ij}$ for all i, j.
4. If $U = (u_{ij}) \in \mathbb{M}_n$ is unitary, show that $V = (|u_{ij}|^2)$ is doubly stochastic and $S = (|u_{ij}|^3)$ is doubly substochastic. How about $W = (|u_{ij}|)$?
5. Let $A = (a_{ij})$ and $B = (b_{ij})$ be $n \times n$ matrices. Show that
   (a) If A and B are (nonnegative) doubly substochastic, then so are $C = (a_{ij}b_{ij})$, $D = (a_{ij}^2)$, $E = (\sqrt{a_{ij}b_{ij}})$, and $F = \left(\frac{a_{ij}+b_{ij}}{2}\right)$.
   (b) If A and B are unitary, then $G = (|a_{ij}b_{ij}|)$ is doubly substochastic, but $H = (\sqrt{|a_{ij}b_{ij}|})$ need not be doubly substochastic.
6. Let $A = \left(\begin{smallmatrix} 0 & 1 & 1 \\ 1 & 0 & 0 \\ 1 & 0 & 0 \end{smallmatrix}\right)$ and $e = (1,1,1)$. Show that $eA \ge e$, $Ae^T \ge e^T$, and that $A - Q$ is never nonnegative for any doubly stochastic matrix Q.
7. Let $A = \left(\begin{smallmatrix} 0 & \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & 0 & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} & 0 \end{smallmatrix}\right)$. Show that A is doubly stochastic and that A cannot be expressed as a product of T-transforms.
8. Let P be a square matrix. If both P and its inverse P^{-1} are doubly stochastic, show that P is a permutation matrix.
9. Prove each of the following statements.
   (a) The (ordinary) product of two doubly stochastic matrices is a doubly stochastic matrix.
   (b) The (ordinary) product of two doubly substochastic matrices is a doubly substochastic matrix.
   (c) The Hadamard product of two doubly substochastic matrices is a doubly substochastic matrix.
   (d) The Kronecker product of two doubly substochastic matrices is a doubly substochastic matrix.
   (e) The convex combination of doubly stochastic matrices is a doubly stochastic matrix.
   (f) The convex combination of doubly substochastic matrices is a doubly substochastic matrix.
10. Show that exchanging two columns (or two rows) of the identity matrix (of any given order) results in a symmetric matrix.

11. Show that any square submatrix of a doubly stochastic matrix is doubly substochastic and that every doubly substochastic matrix can be regarded as a square submatrix of a doubly stochastic matrix.

12. How many 3×3 partial (i.e., sub-) permutation matrices are there?

13. Show that a (nonnegative) matrix is doubly substochastic if and only if it is a convex combination of sub-permutation matrices.

14. Let C be a nonnegative circulant matrix with a row sum $c > 0$. Show that $\frac{1}{c}C$ and $\frac{1}{c}C^T$ are doubly stochastic.

15. Let $A = (a_{ij})$ be an $n \times n$ doubly stochastic matrix. Show that n entries of A can be chosen from different rows and columns such that their product is positive, that is, there exists a permutation p such that $\prod_{i=1}^{n} a_{ip(i)} > 0$. [Hint: Use the (Frobenius-König) Theorem 5.6.6.]

16. A matrix A of order $n \geq 2$ is said to be *reducible* if $P^T AP = \begin{pmatrix} B & C \\ 0 & D \end{pmatrix}$ for some permutation matrix P, where B and D are some square matrices; A is *irreducible* if it is not reducible. A matrix A is said to be *decomposable* if $PAQ = \begin{pmatrix} B & C \\ 0 & D \end{pmatrix}$ for some permutation matrices P and Q, where B and D are some square matrices; A is *indecomposable* if it is not decomposable. Prove each of the following statements.

    (a) If A is a nonnegative indecomposable matrix of order n, then the entries of A^{n-1} are all positive.

    (b) The product of two nonnegative indecomposable matrices is indecomposable.

    (c) The product of two nonnegative irreducible matrices need not be irreducible.

17. Let $x, y \in \mathbb{R}^n_+$. Show that $x \prec_w y$ if and only if there exist subpermutation matrices $Q_1, Q_2, \ldots, Q_m$ such that x is a convex combination of the vectors $yQ_1, yQ_2, \ldots, yQ_m$.

18. Let A be an $n \times n$ nonnegative matrix. Show that A is doubly substochastic if and only if $Ax \prec_w x$ for all column vectors $x \in \mathbb{R}^n_+$.

19. Give an example of some positive $x \in \mathbb{R}^n$, real $y \in \mathbb{R}^n$, and an $n \times n$ substochastic matrix S such that $x = yS$, but $x \prec_w y$ does not hold.

20. Let $x = (1, -1), y = (2, 0)$. Show that $x \prec_w y$ but there does not exist a substochastic matrix S such that $x = yS$. (See Theorem 10.2.4.)

21. Let $a_1, a_2, \ldots, a_n$ be nonnegative numbers such that $\sum_{i=1}^{n} a_i = 1$. Show that $(\frac{1}{n}, \frac{1}{n}, \ldots, \frac{1}{n}) \prec (a_1, a_2, \ldots, a_n) \prec (1, 0, \ldots, 0)$ and that

$$v_n \prec (v_{n-1}, 0) \prec \cdots \prec (v_2, 0, \ldots, 0) \prec (1, 0, \ldots, 0),$$

    where $v_k = (\frac{1}{k}, \frac{1}{k}, \ldots, \frac{1}{k})$ with k copies of $\frac{1}{k}$ for $k = 1, 2, \ldots, n$.

22. Let $(x_1, \ldots, x_n), (y_1, \ldots, y_n), (u_1, \ldots, u_n) \in \mathbb{R}^n$. If $x \prec y$, show that

$$\sum_{i=1}^n u_i^{\downarrow} y_i^{\uparrow} \le \sum_{i=1}^n u_i^{\downarrow} x_i^{\uparrow} \le \sum_{i=1}^n u_i x_i \le \sum_{i=1}^n u_i^{\downarrow} x_i^{\downarrow} \le \sum_{i=1}^n u_i^{\downarrow} y_i^{\downarrow}.$$

23. Let $T = \begin{pmatrix} 1-a & b \\ a & 1-b \end{pmatrix}$, $S = \begin{pmatrix} -a & b \\ a & -b \end{pmatrix}$, $0 < a < 1$, $0 < b < 1$. Show that

$$T^n = I_2 + \frac{1 - r^n}{1 - r} S, \quad \text{where } r = 1 - (a + b),\ n = 1, 2, \ldots.$$

Find the (entrywise) limit of T^n as $n \to \infty$.

24. Let $y \in \mathbb{R}^n$ and denote the vectors obtained by permuting the components of y by $\tilde{y}_1, \ldots, \tilde{y}_{n!}$. Show that $x \prec y$ if and only if $x = t_1 \tilde{y}_1 + \cdots + t_{n!} \tilde{y}_{n!}$, where all $t_i \ge 0$ and $t_1 + \cdots + t_{n!} = 1$. That is, $\{x \in \mathbb{R}^n : x \prec y\}$ is the convex hull of (at most $n!$ points) $\tilde{y}_1, \ldots, \tilde{y}_{n!}$.

25. Let $y = (y_1, \ldots, y_n) \in \mathbb{R}^n_+$. Show that $\{x \in \mathbb{R}^n_+ : x \prec_w y\}$ is the convex hull of the points $(\epsilon_1 y_{p(1)}, \ldots, \epsilon_n y_{p(n)})$, where p varies over all permutations and each ϵ_i is 0 or 1. Is this true if $y \in \mathbb{R}^n$?

26. Let $y = (1, 2, 3) \in \mathbb{R}^3$. Find all extreme points of $\{x \in \mathbb{R}^3_+ : x \prec_w y\}$.

27. Let $y \in \mathbb{R}^n_+$ (in row) and let P be any permutation matrix of order n. Show that y and yP are extreme points of $\{x \in \mathbb{R}^n_+ : x \prec_w y\}$.

28. Let $y = (y_1, y_2, \ldots, y_n) \in \mathbb{R}^n$, and let

(1) $S_1 = \{x \in \mathbb{R}^n : x \prec y\}$, $S_1^+ = \{x \in \mathbb{R}^n_+ : x \prec y\}$ if $y \in \mathbb{R}^n_+$,
(2) $S_2 = \{x \in \mathbb{R}^n : x \prec_w y\}$, $S_2^+ = \{x \in \mathbb{R}^n_+ : x \prec_w y\}$ if $y \in \mathbb{R}^n_+$,
(3) $S_3 = \{x \in \mathbb{R}^n : y \prec x\}$, $S_3^+ = \{x \in \mathbb{R}^n_+ : y \prec x\}$ if $y \in \mathbb{R}^n_+$, and
(4) $S_4 = \{x \in \mathbb{R}^n : y \prec_w x\}$, $S_4^+ = \{x \in \mathbb{R}^n_+ : y \prec_w x\}$ if $y \in \mathbb{R}^n_+$.

(a) If $y \in \mathbb{R}^n_+$, show that $S_1 = S_1^+$, $S_2^+ \subset S_2$, $S_3^+ \subset S_3$, and $S_4^+ \subset S_4$.
(b) Show that S_1, S_2, and S_2^+ are convex sets.
(c) Are S_3, S_4, S_3^+, and S_4^+ convex sets?
(d) Show that S_1, S_2^+, and S_3^+ are bounded, while S_2, S_3, and S_4^+ are unbounded (as a consequence, S_4 is unbounded).
(e) Find each of the eight sets for $y = (2, 1)$.
(f) Find S_1, S_2, S_3, and S_4 for $y = (2, -1)$.
(g) Find the convex hull of the points in the forms $(\epsilon_1 y_1, \epsilon_2 y_2)$ and $(\epsilon_1 y_2, \epsilon_2 y_1)$, where each ϵ_i is 0 or 1, and $y = (y_1, y_2) = (2, -1)$.

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10.3 Majorization and Convex Functions

We study majorization and convex functions in this section. A real-valued function $f(t)$ defined on a real interval $\mathbb{I} \subseteq \mathbb{R}$ is said to be *increasing* if $f(a) \leq f(b)$ for all $a \leq b$, $a, b \in \mathbb{I}$; $f(t)$ is *strictly increasing* if $a < b \Rightarrow f(a) < f(b)$.

$f(t)$ is said to be *convex* on $\mathbb{I}$ if for all $a, b \in \mathbb{I}$ and $\alpha, \beta \geq 0$ with $\alpha + \beta = 1$,

$$f(\alpha a + \beta b) \leq \alpha f(a) + \beta f(b). \tag{10.10}$$

$f(t)$ is *strictly convex* if the strict inequality holds for $a \neq b$, $\alpha, \beta > 0$.

$f(t)$ is *decreasing* if $-f(t)$ is increasing; $f(t)$ is *concave* if $-f(t)$ is convex.

A convex function on an open interval is continuous (Problem 34).

If $f(t)$ is a differentiable function, from calculus we know that $f(t)$ is increasing if the first derivative is nonnegative, i.e., $f'(t) \geq 0$ on the interval, and $f(t)$ is convex if the second derivative is nonnegative, i.e., $f''(t) \geq 0$. If $f'(t) > 0$, then f is strictly increasing; if $f''(t) > 0$, then f is strictly convex. Geometrically, the graph of a convex function is concave upwards, that is, a secant line of the function lies above the graph of the function.

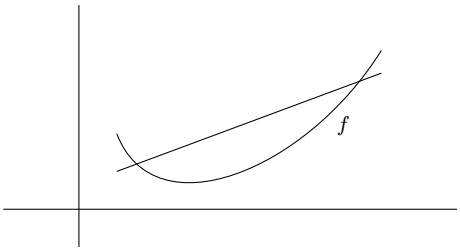


Fig. 10.2. Convex function

Example 10.3.1 Convex functions of one variable.

(a) $|t|$ is convex on $(-\infty, \infty)$.
(b) $\sin t$ is strictly concave on $[0, \pi]$.
(c) t^2 is strictly convex on $(-\infty, \infty)$; t^3 is strictly convex on $(0, \infty)$.
(d) $\sqrt{t}$ is strictly increasing and strictly concave on $[0, \infty)$.
(e) $\frac{1}{t}$ is strictly decreasing and strictly convex on $(0, \infty)$.
(f) e^t is strictly increasing and strictly convex on $(-\infty, \infty)$.
(g) $\ln t$ is strictly increasing and strictly concave on $(0, \infty)$.
(h) $t \ln t$ is strictly convex on $(0, \infty)$ and strictly increasing on $(\frac{1}{e}, \infty)$.
(i) $\frac{1}{t} \ln t$ is increasing on $(0, e)$, decreasing on (e, ∞), convex on $(e^{3/2}, \infty)$, and concave on $(0, e^{3/2})$, all strictly. [Note: $(e, \frac{1}{e})$ is the critical point.]
(j) t^t is strictly convex on $(0, \infty)$ and strictly increasing on $(\frac{1}{e}, \infty)$. ∎

A general and equivalent form of (10.10) is known as *Jensen's inequality.*

Theorem 10.3.1 (Jensen's Inequality) *Let f be a convex function defined on an interval $\mathbb{I} \subseteq \mathbb{R}$. Then, for any convex combination of finitely many points in $\mathbb{I}$, say $\sum_{i=1}^n t_i x_i$, where all $x_i \in \mathbb{I}$ and all $t_i \geq 0$ with $\sum_{i=1}^n t_i = 1$,*

$$f\Big(\sum_{i=1}^n t_i x_i\Big) \leq \sum_{i=1}^n t_i f(x_i). \tag{10.11}$$

If f is strictly convex, then (10.11) is an equality if and only if $t_p t_q = 0$ for each pair $p, q \in \{1, 2, \ldots, n\}$ such that $x_p \neq x_q$.

Proof. For $n = 2$, the inequality holds for convex functions by (10.10). Suppose the claim is true for n (or less). We show that it is true for $n+1$.

Let $t_1, \ldots, t_n, t_{n+1}$ be positive numbers (or it reduces to the case n or less, which is true by induction hypothesis) with sum equal to 1. Note that

$$\sum_{i=1}^{n+1} t_i x_i = (1 - t_{n+1}) \sum_{i=1}^{n} \frac{t_i}{1 - t_{n+1}} x_i + t_{n+1} x_{n+1}.$$

Let $t_i' = \frac{t_i}{1-t_{n+1}}$, $i = 1, \ldots, n$. All $t_i' > 0$, $\sum_{i=1}^n t_i' = 1$, and $\sum_{i=1}^n t_i' x_i \in \mathbb{I}$. Applying f to both sides reveals (by the case of 2 and the hypothesis on n)

$$\begin{aligned}
f\Big(\sum_{i=1}^{n+1} t_i x_i\Big) &= f\Big((1 - t_{n+1}) \sum_{i=1}^{n} t_i' x_i + t_{n+1} x_{n+1}\Big) \\
&\leq (1 - t_{n+1}) f\Big(\sum_{i=1}^{n} t_i' x_i\Big) + t_{n+1} f(x_{n+1}) \\
&\leq (1 - t_{n+1}) \sum_{i=1}^{n} t_i' f(x_i) + t_{n+1} f(x_{n+1}) = \sum_{i=1}^{n+1} t_i f(x_i).
\end{aligned}$$

Now consider the strict inequality case. Suppose that f is strictly convex and there exist $x_p \neq x_q$ with t_p, $t_q > 0$. Let $\alpha = \frac{t_p}{t_p + t_q}$ and $\beta = \frac{t_q}{t_p + t_q}$. Then

$$\begin{aligned}
f\big(\textstyle\sum_i t_i x_i\big) &= f\big(t_p x_p + t_q x_q + \sum_{i \neq p,\, q} t_i x_i\big) \\
&= f\big((t_p + t_q)(\alpha x_p + \beta x_q) + \textstyle\sum_{i \neq p,\, q} t_i x_i\big) \\
&\leq (t_p + t_q) f(\alpha x_p + \beta x_q) + \textstyle\sum_{i \neq p,\, q} t_i f(x_i) \\
&< (t_p + t_q)\big(\alpha f(x_p) + \beta f(x_q)\big) + \textstyle\sum_{i \neq p,\, q} t_i f(x_i) \\
&= t_p f(x_p) + t_q f(x_q) + \textstyle\sum_{i \neq p,\, q} t_i f(x_i) = \sum_i t_i f(x_i). \quad \blacksquare
\end{aligned}$$

As a result of the theorem, if f is strictly convex and if all $t_i > 0$, then the inequality is strict except the extreme case where all x_i's are equal.

Example 10.3.2 Let $x_1, \ldots, x_n \in (0,1)$ be such that $x_1 + \cdots + x_n = 1$. Applying the Jensen inequality with the convex function $f(t) = -\ln t$, $t_1 = x_1, \ldots, t_n = x_n$, and $\frac{1}{x_1}, \ldots, \frac{1}{x_n}$ for $x_1, \ldots, x_n$ in (10.11), we obtain

$$-\ln n \le x_1\big(-\ln \tfrac{1}{x_1}\big) + \cdots + x_n\big(-\ln \tfrac{1}{x_n}\big),$$

which implies

$$x_1^{x_1} \cdots x_n^{x_n} \ge \tfrac{1}{n}.$$

The inequality becomes an equality if and only if $x_1 = \cdots = x_n = \frac{1}{n}$. ∎

Theorem 10.3.2 *Let $A = (a_{ij})$ be an $n \times n$ Hermitian matrix and let f be a convex function on a real interval that contains all eigenvalues of A. Then*

$$f(a_{ii}) \le \big(f(A)\big)_{ii} \tag{10.12}$$

for $i = 1, 2, \ldots, n$, where $\big(f(A)\big)_{ii}$ denotes the diagonal (i,i)-entry of $f(A)$. If f is strictly convex, then the inequality becomes an equality (for some i) if and only if a_{ii} is the only (possible) nonzero entry in row (column) i of A.

Proof. Let $A = UDU^*$ be a spectral decomposition, where $U = (u_{ij})$ is unitary and $D = \operatorname{diag}(\lambda_1, \ldots, \lambda_n)$ is the diagonal matrix of the eigenvalues λ_j's of A. We have $f(A) = Uf(D)U^*$ (Problem 50 of Sect. 3.2, Chap. 3) and

$$a_{ii} = (UDU^*)_{ii} = \sum\nolimits_j \lambda_j |u_{ij}|^2, \;\; (f(A))_{ii} = (Uf(D)U^*)_{ii} = \sum\nolimits_j f(\lambda_j)|u_{ij}|^2.$$

Note that $\sum_j |u_{ij}|^2 = 1$. By Jensen's inequality, we obtain for $i = 1, 2, \ldots, n$,

$$f(a_{ii}) = f\Big(\sum\nolimits_j \lambda_j |u_{ij}|^2\Big) \le \sum\nolimits_j f(\lambda_j)|u_{ij}|^2 = \big(f(A)\big)_{ii}. \tag{10.13}$$

Now we consider the equality case with strictly convex function f. The sufficiency is easy to verify. We show the necessity. If $n = 1$, there is nothing to show. We assume that $n \ge 2$ and that the equality holds in (10.13).

Let u_i be the ith row of U. Then $a_{ij} = u_i D u_j^*$. We show $a_{ij} = 0$ if $j \ne i$.

By the equality case of Jensen's inequality (Theorem 10.3.1), the equality occurs in (10.13) only if $\lambda_1 = \cdots = \lambda_n$ or $u_{ip}\, u_{iq} = 0$ whenever $\lambda_p \ne \lambda_q$.

If $\lambda_j u_{ij} = 0$ for all j, then $(u_{i1}, u_{i2}, \ldots, u_{in})D = (\lambda_1 u_{i1}, \lambda_2 u_{i2}, \ldots, \lambda_n u_{in})$ is a zero row vector, i.e., $u_i D = 0$. It follows that $u_i D u_j^* = 0$ for all j, implying that the ith row of A consists entirely of zeros.

If some $\lambda_k u_{ik} \ne 0$, then all other $u_{ij} = 0$ unless $\lambda_j = \lambda_k$. This implies that $(u_{i1}, u_{i2}, \ldots, u_{in})D = (\lambda_1 u_{i1}, \lambda_2 u_{i2}, \ldots, \lambda_n u_{in}) = \lambda_k(u_{i1}, u_{i2}, \ldots, u_{in}) = \lambda_k u_i$. Thus, $a_{ij} = u_i D u_j^* = \lambda_k u_i u_j^* = 0$ if $j \ne i$. So, the off-diagonal entries in row i of A are all zeros. (The same is true for column i as $A^* = A$.) ∎

Note: Taking $f(t) = t^2$, $-t^{1/2}$, and t^{-1} reveals the inequalities in Example 7.3.2. Moreover, under the conditions in Theorem 10.3.2, one can show (as the proof of the theorem) that $f(x^*Ax) \le x^* f(A) x$ for unit vectors $x \in \mathbb{C}^n$. Setting $x = e_i = (0, \ldots, 0, 1, 0, \ldots, 0)^T \in \mathbb{C}^n$ gives the inequalities in (10.12).

Below is a reversal inequality of the Jensen-type inequality.

Theorem 10.3.3 *Let $f : [0, \infty) \to \mathbb{R}$ be a strictly convex function such that $f(0) \le 0$. Then for any finite sequence of nonnegative numbers, say $x_1, \ldots, x_n \in [0, \infty)$, in which at least two of x_i's are positive, the strict inequality holds:*

$$\sum_{i=1}^{n} f(x_i) < f\Big(\sum_{i=1}^{n} x_i\Big).$$

Proof. We prove the case for $n = 2$. The general case is left as Problem 23. Let x_1 and x_2 be both positive. Write $x_1 = \frac{x_1}{x_1+x_2}(x_1 + x_2) + \frac{x_2}{x_1+x_2}\, 0$. Then

$$f(x_1) < \frac{x_1}{x_1 + x_2} f(x_1 + x_2) + \frac{x_2}{x_1 + x_2} f(0).$$

In a similar way, we have

$$f(x_2) < \frac{x_1}{x_1 + x_2} f(0) + \frac{x_2}{x_1 + x_2} f(x_1 + x_2).$$

Adding both sides of the above inequalities, we arrive at

$$f(x_1) + f(x_2) < f(x_1 + x_2) + f(0).$$

The desired inequality follows at once because $f(0) \le 0$. ∎

Can the condition $f(0) \le 0$ in the theorem be dropped?

Next we present a result about majorization preserving, with which we shall show more majorization inequalities via convex functions as we proceed.

For simplicity, we write $x \in \mathbb{R}^n$ to mean $x = (x_1, \ldots, x_n)$ with all $x_i \in \mathbb{R}$. Let $\mathbb{I}$ be an interval of $\mathbb{R}$. In most cases, $\mathbb{I} = (-\infty, \infty)$, $(0, \infty)$, or $(0, 1)$.

Theorem 10.3.4 (Convexity Preserves Majorization) *Let $x, y \in \mathbb{R}^n$ and let f be a function defined on an interval $\mathbb{I} \subseteq \mathbb{R}$ that contains all x_i and y_i.*

(1) *If f is convex, then $x \prec y \;\Rightarrow\; f(x) \prec_w f(y)$, or explicitly*

$$x \prec y \quad \Rightarrow \quad \big(f(x_1), \ldots, f(x_n)\big) \prec_w \big(f(y_1), \ldots, f(y_n)\big). \tag{10.14}$$

(2) *If f is increasing and convex, then $x \prec_w y \;\Rightarrow\; f(x) \prec_w f(y)$, or explicitly*

$$x \prec_w y \quad \Rightarrow \quad \big(f(x_1), \ldots, f(x_n)\big) \prec_w \big(f(y_1), \ldots, f(y_n)\big). \tag{10.15}$$

Proof. Since $x \prec y$, by Theorem 10.2.2, there exists a doubly stochastic matrix $A = (a_{ij})$ such that $x = yA$. This reveals

$$x_i = \sum_{j=1}^{n} a_{ji} y_j, \quad i = 1, \ldots, n.$$

Because f is convex, applying f to both sides, we obtain

$$f(x_i) \le \sum_{j=1}^{n} a_{ji} f(y_j), \quad i = 1, \ldots, n.$$

Therefore,

$$(f(x_1), \ldots, f(x_n)) \le (f(y_1), \ldots, f(y_n))A.$$

By the fact that "$\le$" implies "$\prec_w$" and by using Theorem 10.2.2, we have

$$(f(x_1), \ldots, f(x_n)) \prec_w (f(y_1), \ldots, f(y_n))A \prec (f(y_1), \ldots, f(y_n)).$$

For the second part, Theorem 10.1.2 ensures the existence of some $z \in \mathbb{R}^n$ such that $x \le z \prec y$. So, $f(x) \le (\prec_w) f(z) \prec_w f(y)$ by part (1). ∎

Note that for (10.15), the condition that f be increasing on $\mathbb{I}$ is needed. One may check with $x = (-1, -1) \prec_w (0, 0) = y$, and $f(t) = t^2$ on $\mathbb{R}$.

Example 10.3.3 $-t$, $|t|$, and t^2 are convex on $\mathbb{R}$, $\frac{1}{t}$ is convex on $(0, \infty)$, and e^t is increasing and convex on $\mathbb{R}$. Let $x, y \in \mathbb{R}^n$. Theorem 10.3.4 reveals

(a) $x \prec y \Rightarrow -x \prec -y$. [Note: $x \prec_w y \not\Rightarrow -x \prec_w -y$.]
(b) $x \prec y \Rightarrow |x| \prec_w |y|$, i.e., $(|x_1|, \ldots, |x_n|) \prec_w (|y_1|, \ldots, |y_n|)$.
(c) $x \prec y \Rightarrow x^2 \prec_w y^2$, i.e., $(x_1^2, \ldots, x_n^2) \prec_w (y_1^2, \ldots, y_n^2)$.
(d) $x \prec y \Rightarrow (\frac{1}{x_1}, \ldots, \frac{1}{x_n}) \prec_w (\frac{1}{y_1}, \ldots, \frac{1}{y_n})$, where all $x_i, y_i > 0$.
(e) $x \prec_w y \Rightarrow e^x \prec_w e^y$, where $e^z = (e^{z_1}, \ldots, e^{z_n})$, $z \in \mathbb{R}^n$.
(f) $\ln x \prec_w \ln y \Rightarrow x \prec_w y$, where $\ln z = (\ln z_1, \ldots, \ln z_n)$, all $z_i > 0$. ∎

Example 10.3.4 Let α, β, and γ be the angles of a triangle. Note that $(\frac{\pi}{3}, \frac{\pi}{3}, \frac{\pi}{3}) \prec (\alpha, \beta, \gamma) \prec (\pi, 0, 0)$. As $f(t) = -\sin t$ is convex on $[0, \pi]$, we have

$$\big(-\sin\tfrac{\pi}{3}, -\sin\tfrac{\pi}{3}, -\sin\tfrac{\pi}{3}\big) \prec_w (-\sin\alpha, -\sin\beta, -\sin\gamma) \prec_w (-\sin\pi, 0, 0).$$

In particular, we obtain that for any nonnegative α, β, γ with $\alpha + \beta + \gamma = \pi$,

$$\sin\alpha + \sin\beta + \sin\gamma \le \frac{3\sqrt{3}}{2}. \quad ∎$$

Majorization can be characterized in terms of convex functions.

Theorem 10.3.5 *Let $x, y \in \mathbb{R}^n$. Let $\mathbb{I} \subseteq \mathbb{R}$ be an interval that contains all components of x and y and let f be a function defined on $\mathbb{I}$. Then*

(1) $x \prec y \Leftrightarrow \sum_{i=1}^{n} f(x_i) \le \sum_{i=1}^{n} f(y_i)$ *for all convex functions f.*

(2) $x \prec_w y \Leftrightarrow \sum_{i=1}^{n} f(x_i) \le \sum_{i=1}^{n} f(y_i)$ *for all increasing and convex functions f.*

Proof. Necessities are immediate from Theorem 10.3.4.

For sufficiencies, we show that $\sum_{i=1}^{k} x_i^{\downarrow} \le \sum_{i=1}^{k} y_i^{\downarrow}$, $k = 1, \ldots, n$. For a fixed k, let $f(r) = (r - y_k^{\downarrow})^+$, where $t^+ = \max\{t, 0\}$, i.e., $f(r) = r - y_k^{\downarrow}$ if $r \ge y_k^{\downarrow}$, $f(r) = 0$ otherwise. Then $f(r)$ is a convex function in r. Note that $\sum_{i=1}^{n} f(y_i) = \sum_{i=1}^{n} f(y_i^{\downarrow})$. The condition $\sum_{i=1}^{n} f(y_i) \ge \sum_{i=1}^{n} f(x_i)$ reveals

$$\begin{aligned}
\sum_{i=1}^{k} (y_i^{\downarrow} - y_k^{\downarrow}) &= \sum_{i=1}^{k} (y_i^{\downarrow} - y_k^{\downarrow})^+ = \sum_{i=1}^{n} (y_i^{\downarrow} - y_k^{\downarrow})^+ \\
&\ge \sum_{i=1}^{n} (x_i^{\downarrow} - y_k^{\downarrow})^+ \ge \sum_{i=1}^{k} (x_i^{\downarrow} - y_k^{\downarrow})^+ \\
&\ge \sum_{i=1}^{k} (x_i^{\downarrow} - y_k^{\downarrow}),
\end{aligned}$$

for each $k \le n$. This implies (2): $\sum_{i=1}^{k} x_i^{\downarrow} \le \sum_{i=1}^{k} y_i^{\downarrow}$, i.e., $x \prec_w y$.

For (1), an application of the convex function $f(t) = -t$ yields $-\sum_{i=1}^{n} x_i \le -\sum_{i=1}^{n} y_i$. Since $x \prec_w y$, we have $\sum_{i=1}^{n} x_i = \sum_{i=1}^{n} y_i$. Thus, $x \prec y$. ∎

Theorem 10.3.6 *Let $f(t)$ be a convex function on an interval $\mathbb{I} \subseteq \mathbb{R}$. Let*

$$\phi(x) = f(x_1) + \cdots + f(x_n), \quad \textit{where } x = (x_1, \ldots, x_n), \textit{ all } x_i \in \mathbb{I}.$$

If $x \prec y$ (where y's components are all contained in $\mathbb{I}$), then $\phi(x) \le \phi(y)$.

Proof. This is immediate from Theorem 10.3.5. ∎

The preceding theorem leads to the so-called *Schur-convex functions*. A function ϕ defined on a convex subset $\mathbb{D}$ of $\mathbb{R}^n$ is said to be *Schur-convex* if

$$x \prec y \quad \Rightarrow \quad \phi(x) \le \phi(y).$$

Since majorizations are unaffected by changing the orders of the components of vectors, in addition to convexity, we require $\mathbb{D}$ to be *symmetric* in the sense that if x is in $\mathbb{D}$, then the vectors obtained from x by permuting the components are also in $\mathbb{D}$. That is, the Schur-convex functions are usually studied on convex and symmetric subsets of $\mathbb{R}^n$ with $n > 1$. These sets $\mathbb{D}$ are mostly $\mathbb{R}^n$, $\mathbb{R}_+^n$, $\{(x_1, \ldots, x_n) : x_i \in (0, 1)\}$, or similar type.

A function $\phi : \mathbb{D} \to \mathbb{R}$ is *Schur-concave* if $-\phi$ is Schur-convex. Monotonicity and convexity of ϕ on $\mathbb{D}$ can be similarly defined as in (10.10).

Note: The definitions of convexity and monotonicity of functions of a single variable are naturally extended to functions of multivariables on $\mathbb{D}$.

Example 10.3.5 Schur-convex functions of multivariables.

(a) $\varphi(z) = \max\{z_1, \ldots, z_n\}$ is Schur-convex because $x \prec y$ implies $x_1^{\downarrow} \le y_1^{\downarrow}$, and $\chi(z) = \min\{z_1, \ldots, z_n\}$ is Schur-concave because $x_n^{\downarrow} \ge y_n^{\downarrow}$.

(b) $\psi(z) = |z_1| + \cdots + |z_n|$ is Schur-convex because $f(t) = |t|$ is convex. Below is an alternative proof. If $x \prec y$, by Theorem 10.2.2, we write $x = yA$, where $A = (a_{ij})$ is an $n \times n$ doubly stochastic matrix. Then

$$\begin{aligned} \psi(x) &= \sum_{i=1}^{n} |x_i| = \sum_{i=1}^{n} \Big| \sum_{j=1}^{n} a_{ji} y_j \Big| \\ &\le \sum_{i=1}^{n} \sum_{j=1}^{n} a_{ji} |y_j| = \sum_{j=1}^{n} \Big(\sum_{i=1}^{n} a_{ji} \Big) |y_j| \\ &= \sum_{j=1}^{n} |y_j| = \psi(y). \ \blacksquare \end{aligned}$$

The Schur-convex functions have applications in various areas such as optimization and statistics. Studying majorizations and Schur-convex functions is a rich research area in mathematics that one may further explore.

The next theorem shows a strict inequality when a strictly convex and (strictly) increasing function is applied to a majorization. Note that an increasing and strictly convex function is strictly increasing (Problem 35).

Theorem 10.3.7 *Let $x, y \in \mathbb{R}^n$ and suppose that y is not a permutation of x (that is, $x^{\downarrow}$ and $y^{\downarrow}$ are different). Then for any strictly convex and increasing function f defined on an interval that contains all components of x and y,*

$$x \prec_w y \quad \Rightarrow \quad \sum_{i=1}^{n} f(x_i) < \sum_{i=1}^{n} f(y_i).$$

Proof. Since $x \prec_w y$, by Theorem 10.2.1, there exists a vector $z \in \mathbb{R}^n$ such that $x \prec z \le y$. Let $z = (z_1, \ldots, z_n)$. By Theorem 10.3.5, we have

$$\sum_{i=1}^{n} f(x_i) \le \sum_{i=1}^{n} f(z_i) \le \sum_{i=1}^{n} f(y_i).$$

We show that of the two inequalities at least one strict inequality holds.

If $z \ne y$, then $z_k < y_k$ for some k. Since f is strictly increasing, $f(z_k) < f(y_k)$. The second inequality is strict as $f(z_i) \le f(y_i)$ for $i \ne k$.

If $z = y$, then $x \prec y = z$. By Theorem 10.2.2, there exists a doubly stochastic matrix $A = (a_{ij})$ such that $x = yA$. It follows that

$$x_i = \sum_{j=1}^{n} a_{ji} y_j, \quad i = 1, \ldots, n.$$

Applying f to both sides, and because of the strict convexity of f, we have

$$f(x_i) \le \sum_{j=1}^{n} a_{ji} f(y_j), \quad i = 1, \ldots, n. \tag{10.16}$$

Since x is not a permutation of y, the components of y cannot be all the same (otherwise $x = y$), and A is not a permutation matrix. By Jensen's (strict) inequality, at least one strict inequality in (10.16) holds. Therefore,

$$\sum_{i=1}^{n} f(x_i) < \sum_{i=1}^{n}\sum_{j=1}^{n} a_{ji} f(y_j) = \sum_{j=1}^{n}\sum_{i=1}^{n} a_{ji} f(y_j) = \sum_{j=1}^{n} f(y_j). \blacksquare$$

Considering $f(t) = t$, it is convex (but not strictly convex) and strictly increasing on $\mathbb{R}$. The strict inequality in the theorem does not hold for $x \prec y$. So, the assumption of strict convexity of f is needed. In addition, the theorem can be restated as: If $x \prec_w y$ and $\sum_{i=1}^{n} f(x_i) = \sum_{i=1}^{n} f(y_i)$ for some strictly convex and increasing function f (such as e^t), then $x^{\downarrow} = y^{\downarrow}$.

Our next result about majorization in product form is useful in deriving matrix inequalities and is used repeatedly in the later sections. For this purpose, we introduce *log-majorization*. Let $x = (x_1, \ldots, x_n)$, $y = (y_1, \ldots, y_n) \in \mathbb{R}^n_+$. We say that x is *weakly log-majorized* by y and write it as $x \prec_{\text{wlog}} y$ if

$$\prod_{i=1}^{k} x_i^{\downarrow} \le \prod_{i=1}^{k} y_i^{\downarrow}, \quad k = 1, \ldots, n.$$

If the equality holds when $k = n$, we say that x is *log-majorized* by y and write it as $x \prec_{\log} y$. If all components of x and y are positive, because $\ln t$ and e^t are strictly increasing functions, with $\ln z = (\ln z_1, \ldots, \ln z_n)$, we have

$$x \prec_{\text{wlog}} y \quad \Leftrightarrow \quad \ln x \prec_w \ln y \tag{10.17}$$

and

$$x \prec_{\log} y \quad \Leftrightarrow \quad \ln x \prec \ln y. \tag{10.18}$$

Observe that the weak log-majorization $x \prec_{\text{wlog}} y$ is equivalent to

$$\max_{i_1 < \cdots < i_k} \prod_{j=1}^{k} x_{i_j} \le \max_{i_1 < \cdots < i_k} \prod_{j=1}^{k} y_{i_j}, \quad k = 1, \ldots, n. \tag{10.19}$$

Example 10.3.6 Log-majorization.

$$(2,0,3) \prec_{\text{wlog}} (6,1,1), \quad (2,1,3) \prec_{\log} (6,1,1),$$
$$(2,0,3) \prec_{\log} (7,1,0), \quad (1,1,1) \prec_{\log} (3,1,\frac{1}{3}). \blacksquare$$

Theorem 10.3.8 *Let $x, y \in \mathbb{R}^n_+$ (i.e., all components $x_i, y_i \ge 0$). Then*

$$x \prec_{\text{wlog}} y \quad \Rightarrow \quad x \prec_w y,$$

that is (log-majorization implies weak-majorization),

$$\prod_{i=1}^{k} x_i^{\downarrow} \le \prod_{i=1}^{k} y_i^{\downarrow},\ k = 1, \ldots, n \ \Rightarrow \ \sum_{i=1}^{k} x_i^{\downarrow} \le \sum_{i=1}^{k} y_i^{\downarrow},\ k = 1, \ldots, n.$$

Proof. If all components of x are positive, then all components of y are positive. In this case, by (10.17) and by Example 10.3.3 (f), we have

$$x \prec_{\log} y \quad \Leftrightarrow \quad \ln x \prec_w \ln y \quad \Rightarrow \quad x \prec_w y.$$

If x contains zero components, say, $x_k^{\downarrow} = 0$ and $x_i^{\downarrow} > 0$ for all $i < k$, then, by the above argument, $(x_1^{\downarrow}, \ldots, x_{k-1}^{\downarrow}) \prec_w (y_1^{\downarrow}, \ldots, y_{k-1}^{\downarrow})$. It follows that $x = (x_1^{\downarrow}, \ldots, x_{k-1}^{\downarrow}, 0, \ldots, 0) \prec_w (y_1^{\downarrow}, \ldots, y_{k-1}^{\downarrow}, y_k^{\downarrow} \ldots, y_n^{\downarrow}) = y$. ∎

Note that the converse of Theorem 10.3.8 is not true. For example, take $x = (3, 2, 1)$, $y = (4, 1, 1)$. Then $x \prec y$, but $x \not\prec_{\log} y$ because $3 \cdot 2 > 4 \cdot 1$. So, log-majorization is stronger than majorization in general. If $x \prec y$, then $x_1^{\downarrow} \le y_1^{\downarrow}$ and $x_n^{\downarrow} \ge y_n^{\downarrow}$. Moreover, we have (Problem 20) for $x, y \in \mathbb{R}_+^n$:

$$x \prec y \quad \Rightarrow \quad \prod_{i=1}^{n} x_i \ge \prod_{i=1}^{n} y_i. \tag{10.20}$$

Example 10.3.7 Let $x, y \in \mathbb{R}_+^n$. If $x \prec_{\log} y$ and $x \prec y$, then, by (10.20), $x \prec_{\log} y$. It can be shown that y is a permutation of x (Problem 37). ∎

Theorem 10.1.6 (2) states that $x^{\downarrow} + y^{\uparrow} \prec x + y \prec x^{\downarrow} + y^{\downarrow}$ for $x, y \in \mathbb{R}^n$. It is immediate from (10.20) that, for any $x, y \in \mathbb{R}^n$ of positive components,

$$\prod_{i=1}^{n}(x_i^{\downarrow} + y_i^{\uparrow}) \ge \prod_{i=1}^{n}(x_i + y_i) \ge \prod_{i=1}^{n}(x_i^{\downarrow} + y_i^{\downarrow}). \tag{10.21}$$

The inequalities in (10.21) do not hold in general if n is replaced by a positive integer $k < n$. However, it can be slightly generalized as follows.

Applying the convex function $f(t) = -\ln t$ to $x^{\downarrow} + y^{\uparrow} \prec x + y \prec x^{\downarrow} + y^{\downarrow}$ (where x, y have positive components) and using Theorem 10.3.4, we have

$$-\ln(x^{\downarrow} + y^{\uparrow}) \prec_w -\ln(x + y) \prec_w -\ln(x^{\downarrow} + y^{\downarrow}). \tag{10.22}$$

Because $\big(\ln(x^{\downarrow} + y^{\downarrow})\big)^{\downarrow} = \ln(x^{\downarrow} + y^{\downarrow})$, the second $\prec_w$ implies that for $k \le n$,

$$\ln(x_k + y_k) + \cdots + \ln(x_n + y_n) \ge \ln(x_k^{\downarrow} + y_k^{\downarrow}) + \cdots + \ln(x_n^{\downarrow} + y_n^{\downarrow}).$$

Thus,

$$\prod_{i=k}^{n}(x_i + y_i) \ge \prod_{i=k}^{n}(x_i^{\downarrow} + y_i^{\downarrow}). \tag{10.23}$$

Does (10.22) imply $\prod_{i=k}^{n}(x_i^{\downarrow} + y_i^{\uparrow}) \ge \prod_{i=k}^{n}(x_i + y_i)$? (See Problem 22.)

In view of $x^{\downarrow} + y^{\uparrow} \prec x + y \prec x^{\downarrow} + y^{\downarrow}$, would the stronger majorizations $x^{\downarrow} + y^{\uparrow} \prec_{\log} x + y \prec_{\log} x^{\downarrow} + y^{\downarrow}$ hold for $x, y \in \mathbb{R}^n$ of positive components? The answer is negative as one checks with $x = (1, 2, 1), y = (2, 1, 1)$.

The log-majorizations hold for the componentwise product $x \circ y$.

Theorem 10.3.9 *Let $x, y \in \mathbb{R}^n_+$ (i.e., all components $x_i, y_i \geq 0$). Then*

$$x^{\downarrow} \circ y^{\uparrow} \prec_{\log} x \circ y \prec_{\log} x^{\downarrow} \circ y^{\downarrow}. \tag{10.24}$$

Consequently,

$$x^{\downarrow} \circ y^{\uparrow} \prec_{w} x \circ y \prec_{w} x^{\downarrow} \circ y^{\downarrow}. \tag{10.25}$$

Proof. We may only consider the case of positive x and y; otherwise we replace the zero components of x and y with small positive numbers and use a continuity argument. Note that $(\ln x)^{\downarrow} = \ln x^{\downarrow}$ and $(\ln x)^{\uparrow} = \ln x^{\uparrow}$. By Theorem 10.1.6 (2), $\ln x^{\downarrow} + \ln y^{\uparrow} \prec \ln x + \ln y \prec \ln x^{\downarrow} + \ln y^{\downarrow}$ or $\ln(x^{\downarrow} \circ y^{\uparrow}) \prec \ln(x \circ y) \prec \ln(x^{\downarrow} \circ y^{\downarrow})$, i.e., (10.24). Theorem 10.3.8 reveals (10.25). ∎

We point out that if x or y contains negative components, then the inequalities in (10.25) need not be true. For example, take $x = (-1, -2)$ and $y = (1, 2)$. Then it does not hold that $x \circ y \prec_w x^{\downarrow} \circ y^{\downarrow}$.

Problems

1. Let $x, y \in \mathbb{R}^n$. If $x_i^2 \geq y_i^2$ for $i = 1, \ldots, n$, show that $|x|^{\downarrow} \geq |y|^{\downarrow}$.

2. Let $x_1 \geq \cdots \geq x_n > 0$. For any $k \leq n$ and index set $1 \leq i_1 < \cdots < i_k \leq n$, show that $x_{i_j} \leq x_j$ for every j and $x_{i_1} \cdots x_{i_k} \leq x_1 \cdots x_k$.

3. If $f(t)$ is a convex function, show that $e^{f(t)}$ is a convex function. Conclude that if $\ln g(t)$ is convex (where $g(t) > 0$), then $g(t)$ is convex. If $h(t)$ is convex and $h(t) > 0$, must $\ln h(t)$ be convex? (Assume that f, g, and h are twice differentiable functions on their domains.)

4. Prove each of the following statements for functions of t:

   (a) $|t|^{\alpha}$, $\alpha > 1$, is strictly convex on $\mathbb{R}$ and increasing on $\mathbb{R}_+ = [0, \infty)$.

   (b) t^{α}, $0 < \alpha < 1$, is strictly concave and increasing on $\mathbb{R}_+$.

   (c) $t^{-\alpha}$, $\alpha > 0$, is strictly convex and decreasing on $\mathbb{R}_+$.

   (d) $e^{\alpha t}$, $\alpha \neq 0$, is strictly convex on $\mathbb{R}$, strictly increasing if $\alpha > 0$, and strictly decreasing if $\alpha < 0$.

   (e) e^{t^2} is convex on $\mathbb{R}$; e^{-t^2} is convex on $(1/e, \infty)$; e^{1/t^2} is convex and decreasing on $\mathbb{R}_+$; and $e^{1/(1-t^2)}$ is convex on $(-1, 1)$.

   (f) a^t, $a > 0$, $a \neq 1$, is strictly convex on $\mathbb{R}$, strictly increasing if $a > 1$, and strictly decreasing if $0 < a < 1$.

   (g) $\sqrt{1 + t^2}$ is strictly convex on $\mathbb{R}$ and increasing on $\mathbb{R}_+$.

   (h) $\sqrt{1 - t^2}$ is strictly concave on $[-1, 1]$ and increasing on $[-1, 0]$.

   (i) $\cos t$ is strictly concave on $[-\frac{\pi}{2}, \frac{\pi}{2}]$ and increasing on $[0, \frac{\pi}{2}]$.

   (j) $e^t - t - 1$, is strictly convex on $\mathbb{R}$ and increasing on $\mathbb{R}_+$.

(k) $t^2 + C$, $C \in \mathbb{R}$, is strictly convex on $\mathbb{R}$ and increasing on $\mathbb{R}_+$.
(l) $\frac{1+t}{\sqrt{t}}$ is strictly convex on $\mathbb{R}_+$ and increasing on $(1, \infty)$.
(m) $\frac{1}{t} - t$ is strictly convex and deceasing on $\mathbb{R}_+$.
(n) $-\ln(1+t)$ is strictly convex and decreasing on $(-1, \infty)$.
(o) $t\ln(1+t)$ is strictly convex on $(-1, \infty)$ and increasing on $\mathbb{R}_+$.
(p) $\ln(1+e^t)$ is strictly convex and increasing on $\mathbb{R}$.
(q) $\ln(1+e^{-t})$ is strictly convex and decreasing on $\mathbb{R}$.
(r) $\ln(1-e^t)$ is strictly concave and decreasing on $(-\infty, 0)$.
(s) $\ln\frac{1+t}{1-t}$ is strictly convex and increasing on $[0, 1)$.

5. Let $a, b > 0$, $\lambda, \mu \geq 0$, $\lambda + \mu = 1$, and $p \geq 1$. Show that

$$a^\lambda b^\mu \leq \lambda a + \mu b \quad \text{(Young's inequality)}$$

with equality if and only if $a = b$; or $\lambda = 0$; or $\mu = 0$, and that

$$(\lambda a + \mu b)^p \leq \lambda a^p + \mu b^p$$

with equality if and only if $a = b$; or $\lambda = 0$; or $\mu = 0$; or $p = 1$.

6. Show that $f(t) = \ln(\frac{1}{t} - 1)$ is convex on $(0, \frac{1}{2})$ but not on $(\frac{1}{2}, 1)$.

7. Show that $f(t) = \begin{cases} -\sqrt{t}, \text{ if } t > 0 \\ 1 - t, \text{ if } t \leq 0 \end{cases}$ is convex on $\mathbb{R}_+$ but not on $\mathbb{R}$.

8. If $g(t)$ is convex and increasing on $\mathbb{R}$, show that $g(|t|)$ is convex on $\mathbb{R}$.

9. Use Jensen's inequality to show the *arithmetic mean–geometric mean* (AM-GM) *inequality*: If all $a_i \geq 0$, $p_i > 0$, and $\sum_{i=1}^n p_i = 1$, then

$$\prod_{i=1}^n a_i^{p_i} \leq \sum_{i=1}^n p_i a_i, \quad \text{(Hint: use } f(t) = -\ln t.\text{)}$$

which is an equality if and only if all a_i are equal to each other.

10. Use Young's inequality to show the *Hölder inequality*: If $p, q > 1$ such that $\frac{1}{p} + \frac{1}{q} = 1$, then for any complex numbers $a_1, \ldots, a_n$, $b_1, \ldots, b_n$,

$$\left| \sum_{j=1}^n a_j b_j \right| \leq \left(\sum_{j=1}^n |a_j|^p \right)^{1/p} \left(\sum_{j=1}^n |b_j|^q \right)^{1/q},$$

which is an equality if and only if all $a_j = 0$; or all $b_j = 0$; or there exist $\lambda > 0$, $\theta \in \mathbb{R}$ such that $|b_j|^q = \lambda |a_j|^p$ and $a_j b_j = e^{i\theta} |a_j b_j|$ for all j. In the case where all a_j, b_j are nonnegative, the equality holds if and only if $(a_1^p, \ldots, a_n^p)$ and $(b_1^q, \ldots, b_n^q)$ are linearly dependent (proportional).

11. Let $a_1, \ldots, a_n, b_1, \ldots, b_n$ be complex numbers and let $p > 1$. Show that (the *Minkowski inequality*)

$$\left(\sum_{i=1}^n |a_i + b_i|^p\right)^{1/p} \le \left(\sum_{i=1}^n |a_i|^p\right)^{1/p} + \left(\sum_{i=1}^n |b_i|^p\right)^{1/p}.$$

The inequality becomes an equality if and only if all $a_i = 0$; or all $b_i = 0$; or $(b_1, \ldots, b_n) = \lambda(a_1, \ldots, a_n)$ for some $\lambda > 0$ (positively proportional).

12. Find an example of a convex, nonincreasing function $f(t)$ such that $x \prec_w y$ but $(f(x_1), \ldots, f(x_n)) \not\prec_w (f(y_1), \ldots, f(y_n))$ for some $x, y \in \mathbb{R}^n$.

13. If $f(t)$ is increasing and convex on $\mathbb{R}$ and $g(t)$ is convex on $\mathbb{R}$, show that $h(x) = f(g(x_1)) + \cdots + f(g(x_n))$ is Schur-convex on $\mathbb{R}^n$.

14. Let $f(t)$ be a positive continuous function defined on an interval $\mathbb{I} \subseteq \mathbb{R}$. If $\ln f$ is (strictly) convex on $\mathbb{I}$, show that $F(x) = \prod_{i=1}^n f(x_i)$ is (strictly) Schur-convex on $\mathbb{I}^n = \{(x_1, \ldots, x_n) : x_1, \ldots, x_n \in \mathbb{I}\} \subseteq \mathbb{R}^n$.

15. Let a, b, c be the lengths of the sides of a triangle. Let $m = \frac{1}{3}(a+b+c)$. Show that $(m, m, m) \prec (a, b, c) \prec (\frac{3m}{2}, \frac{3m}{2}, 0)$. Use this to show that

$$\frac{1}{3} \le \frac{a^2 + b^2 + c^2}{(a+b+c)^2} < \frac{1}{2}.$$

16. Let $x, y \in \mathbb{R}^n_+$. If $x \prec_{\text{wlog}} y$, show that $(e + x) \prec_{\text{wlog}} (e + y)$, where $e = (1, \ldots, 1) \in \mathbb{R}^n$. Is this true if only $\prod_{i=1}^n x_i \le \prod_{i=1}^n y_i$ is assumed?

17. Let $x_i > 0$, $i = 1, \ldots, n$. If $\prod_{i=1}^n x_i \ge 1$, show that $\prod_{i=1}^n (1 + x_i) \ge 2^n$.

18. Let $x, y \in \mathbb{R}^n_+$. If $x \prec_{\text{wlog}} y$, show that $\sqrt{x} \prec_w \sqrt{y}$. If $\prec_{\text{wlog}}$ is changed to $\prec_w$ or $\prec$, does $x \prec y$ necessarily imply $\sqrt{x} \prec_w \sqrt{y}$ (as for $x^2 \prec_w y^2$)?

19. Let $x, y \in \mathbb{R}^n_+$. Show that $x \prec y \Rightarrow \sum_{i=k}^n (x_i^{\downarrow})^{\frac{1}{2}} \ge \sum_{i=k}^n (y_i^{\downarrow})^{\frac{1}{2}}$, $k \le n$.

20. Let $x, y \in \mathbb{R}^n_+$. If $x \prec y$, show that $\prod_{i=1}^n x_i \ge \prod_{i=1}^n y_i$ and that the strict inequality holds if $x, y > 0$ and if y is not a permutation of x.

21. Let $x, y \in \mathbb{R}^n_+$. Show that the sum inequalities $\sum_{i=1}^k x_i^{\uparrow} \le \sum_{i=1}^k y_i^{\uparrow}$ (for all $k \le n$) imply the product inequalities $\prod_{i=1}^k x_i^{\uparrow} \le \prod_{i=1}^k y_i^{\uparrow}$.

22. Let $x, y \in \mathbb{R}^n_+$. Show that $\prod_{i=k}^n (x_i^{\downarrow} + y_i^{\uparrow}) \ge \prod_{i=k}^n (x_i^{\downarrow} + y_i^{\downarrow})$ for $k \le n$. But it is not true in general that $\prod_{i=k}^n (x_i^{\downarrow} + y_i^{\uparrow}) \ge \prod_{i=k}^n (x_i + y_i)$.

23. Show the general case (for $n > 2$) of Theorem 10.3.3 by induction on n. Can the condition $f(0) \le 0$ in the theorem be dropped?

24. Refer to Jensen's inequality. Take $t_1 = t_2 = 0$, $t_3 + t_4 = 1$ and $x_1 = x_2 \ne 0$, $x_3 = x_4 = 0$. Show that $f(t_1x_1 + t_2x_2 + t_3x_3 + t_4x_4) = t_1f(x_1) + t_2f(x_2) + t_3f(x_3) + t_4f(x_4)$. [Note: Not all x_i's are the same.]

25. Let $A \in \mathbb{M}_n$ and define $f(x) = x^*Ax$ for $x \in \mathbb{C}^n$. Show that f is a convex function from $\mathbb{C}^n$ to $\mathbb{C}$ if and only if A is positive semidefinite.

26. Let $z = (z_1, \ldots, z_n) \in \mathbb{R}^n$ and denote $\varphi(z) = (\varphi(z_1), \ldots, \varphi(z_n))$. Prove the following majorization preserving inequalities on $\mathbb{R}^n$ or $\mathbb{R}^n_+$:

    (a) $x \prec y \Rightarrow |x| \prec_w |y|$ on $\mathbb{R}^n$. Does $x \prec_w y \Rightarrow |x| \prec_w |y|$?
    (b) $|x| \prec_w |y| \Rightarrow x^2 \prec_w y^2$ on $\mathbb{R}^n$. Does $x \prec_w y \Rightarrow x^2 \prec_w y^2$?
    (c) $x \prec y \Rightarrow -\sqrt{|x|} \prec_w -\sqrt{|y|}$ on $\mathbb{R}_+$ and on $-\mathbb{R}_+ = (-\infty, 0]$.
    (d) $x \prec y \Rightarrow |x|^p \prec_w |y|^p$ on $\mathbb{R}^n$, $p \geq 1$. Does $x \prec y \Rightarrow x^3 \prec_w y^3$?
    (e) $x \prec y \Rightarrow -\ln x \prec_w -\ln y$ on $\mathbb{R}^n_+$. Does $x \prec y \Rightarrow \ln y \prec_w \ln x$?
    (f) $x \prec y \Rightarrow e^x \prec_w e^y$ on $\mathbb{R}^n$ and $x \prec y \Rightarrow e^{\frac{1}{x}} \prec_w e^{\frac{1}{y}}$ on $\mathbb{R}^n_+$.

27. Show that the following functions are Schur-convex on their domains.

    (a) $f(x) = \max_i |x_i|$ on $\mathbb{R}^n$.
    (b) $g(x) = \sum_{i=1}^n |x_i|^p$, $p \geq 1$, on $\mathbb{R}^n$.
    (c) $h(x) = \left(\sum_{i=1}^n |x_i|^p \right)^{1/p}$, $p \geq 1$, on $\mathbb{R}^n$.
    (d) $p(x) = \sum_{i=1}^n \frac{1}{x_i}$ for $x > 0$ (i.e., $\mathbb{R}^n_+$ with all $x_i > 0$).
    (e) $q(x) = \sum_{i=1}^n x_i \log_b x_i$ for $x > 0$, $b > 1$.
    (f) $r(x) = -\prod_{i=1}^n x_i$ for $x > 0$.

28. Let $x_1, \ldots, x_n$ be positive numbers such that $x_1 + \cdots + x_n = 1$. Prove

    (a) $\sum_{i=1}^n \frac{1-x_i}{x_i} \geq n(n-1)$ and $\sum_{i=1}^n \frac{x_i}{1-x_i} \geq \frac{n}{(n-1)}$.
    (b) $\sum_{i=1}^n \frac{1+x_i}{x_i} \geq n(n+1)$ and $\frac{n}{(n+1)} \geq \sum_{i=1}^n \frac{x_i}{1+x_i} \geq \frac{1}{2}$.
    (c) $\sum_{i=1}^n \frac{1+x_i}{1-x_i} \geq \frac{n(n+1)}{n-1}$ and $n - 1 \geq \sum_{i=1}^n \frac{1-x_i}{1+x_i} \geq \frac{n(n-1)}{n+1}$.
    (d) $(1 + \frac{1}{n})^n \geq \prod_{i=1}^n (1 + x_i) > 2 > \prod_{i=1}^n (1 + x_i)^{x_i} \geq 1 + \frac{1}{n}$.

29. Let $x_1, \ldots, x_n$ be positive numbers such that $x_1 + \cdots + x_n = 1$ and let f be a convex function on $[0, 1]$. Show that (by Jensen's inequality)

$$f(x_1^2 + \cdots + x_n^2) \leq x_1 f(x_1) + \cdots + x_n f(x_n),$$

$$f(x_1 x_n + \cdots + x_n x_1) \leq x_1 f(x_n) + \cdots + x_n f(x_1),$$

    and

$$nf(\tfrac{1}{n}) \leq f(x_1) + \cdots + f(x_n) \leq f(1) + (n-1)f(0).$$

    Deduce the following inequalities (which are strict unless all $x_i = \frac{1}{n}$):

    (a) $\frac{1}{n} \leq x_1^2 + \cdots + x_n^2 \leq \sqrt{x_1^3 + \cdots + x_n^3}$.
    (b) $\frac{1}{\sqrt{n}} \leq \sqrt{x_1^3} + \cdots + \sqrt{x_n^3} \leq \sqrt{x_1^2 + \cdots + x_n^2}$.
    (c) $n^2 \leq \frac{1}{x_1} + \cdots + \frac{1}{x_n}$.
    (d) $n^{3/2} \leq \frac{1}{\sqrt{x_1}} + \cdots + \frac{1}{\sqrt{x_n}}$.

30. Let $a = (a_1, \ldots, a_n)$, $b = (b_1, \ldots, b_n) \in \mathbb{R}^n$.

    (a) Show that $a^\downarrow - b^\downarrow \prec a - b \prec a^\downarrow - b^\uparrow$, that is,

    $$\big(a_1^\downarrow - b_1^\downarrow, \ldots, a_n^\downarrow - b_n^\downarrow\big) \prec \big(a_1 - b_1, \ldots, a_n - b_n\big) \prec \big(a_1^\downarrow - b_n^\downarrow, \ldots, a_n^\downarrow - b_1^\downarrow\big).$$

    (b) Show that $|a^\downarrow - b^\downarrow| \prec_w |a - b| \prec_w |a^\downarrow - b^\uparrow|$. [Note: but not for "$\prec$"]

    (c) Derive that for $p \geq 1$

    $$\Big(\sum_{i=1}^n |a_i^\downarrow - b_i^\downarrow|^p\Big)^{1/p} \leq \Big(\sum_{i=1}^n |a_i - b_i|^p\Big)^{1/p} \leq \Big(\sum_{i=1}^n |a_i^\downarrow - b_{n-i+1}^\downarrow|^p\Big)^{1/p}.$$

31. Let $a_1, \ldots, a_n$ and $b_1, \ldots, b_n$ be positive numbers. Show that

    $$\Big(\frac{a_1^\downarrow}{b_1^\downarrow}, \ldots, \frac{a_n^\downarrow}{b_n^\downarrow}\Big) \prec_w \Big(\frac{a_1}{b_1}, \ldots, \frac{a_n}{b_n}\Big) \prec_w \Big(\frac{a_1^\downarrow}{b_n^\downarrow}, \ldots, \frac{a_n^\downarrow}{b_1^\downarrow}\Big).$$

32. Let $a_1, \ldots, a_n$ and $b_1, \ldots, b_n$ be positive numbers. Show that

    $$\sum_{i=1}^k a_i^\downarrow b_i^\uparrow \leq \sum_{i=1}^k a_i b_i \leq \sum_{i=1}^k a_i^\downarrow b_i^\downarrow \leq \frac{r+s}{2\sqrt{rs}} \sum_{i=1}^n a_i b_i, \ k = 1, \ldots, n,$$

    where r and s are the numbers such that $r \geq \frac{a_i}{b_i} \geq s > 0$, $i = 1, \ldots, n$. Can the upper limit n in $\sum_{i=1}^n a_i b_i$ on the right be replaced by k? [Hint: Use the Kantorovich inequality for $A = \operatorname{diag}(\frac{a_1}{b_1}, \ldots, \frac{a_n}{b_n})$.]

33. For a real number r, denote $r^+ = \max\{r, 0\}$. Let $x, y \in \mathbb{R}^n$. Show that

    (a) $x \prec_w y \Leftrightarrow \sum_{i=1}^n (x_i - t)^+ \leq \sum_{i=1}^n (y_i - t)^+$ for all $t \in \mathbb{R}$.

    (b) $x \prec y \Leftrightarrow \sum_{i=1}^n |x_i - t| \leq \sum_{i=1}^n |y_i - t|$ for all $t \in \mathbb{R}$.

34. Show that a convex function on an open interval is continuous.

35. Let $f(t)$ be a function defined on an interval $\mathbb{I} \subseteq \mathbb{R}$. If $f(t)$ is increasing and strictly convex on $\mathbb{I}$, show that $f(t)$ is strictly increasing on $\mathbb{I}$.

36. Let g be a differentiable function on an interval $\mathbb{I} \subseteq \mathbb{R}$. Show that

    (a) g is convex if and only if $g(\frac{a+b}{2}) \leq \frac{1}{2}(g(a) + g(b))$ for all $a, b \in \mathbb{I}$.

    (b) g is linear if and only if $g(\frac{a+b}{2}) = \frac{1}{2}(g(a) + g(b))$ for all $a, b \in \mathbb{I}$.

    (c) g preserves the majorization, that is, $(g(x_1), \ldots, g(x_n)) \prec (g(y_1), \ldots, g(y_n))$ whenever $x \prec y$, $x, y \in \mathbb{R}^n$, if and only if g is linear.

37. Let $x, y \in \mathbb{R}^n_+$. If $x \prec_{\log} y$ and $x \prec y$, show that $x^\downarrow = y^\downarrow$ (i.e., x is a permutation of y). If $x \prec_{\log} y$ and $x \prec_w y$, is it true that $x^\downarrow = y^\downarrow$?

38. Let $x, y \in \mathbb{R}^n_+$ and let $x \prec_w y$. Show that $x^m \prec_w y^m$ for all real numbers $m \geq 1$, where $z^m = (z_1^m, \ldots, z_n^m)$ for $z = (z_1, \ldots, z_n) \in \mathbb{R}^n$. If the majorization $x^m \prec y^m$ holds for some $m > 1$, show that $x^\downarrow = y^\downarrow$.

39. If $x, y, u, v \in \mathbb{R}^n_+$, show that $x \prec_{\text{wlog}} u$, $y \prec_{\text{wlog}} v \Rightarrow x \circ y \prec_{\text{wlog}} u^{\downarrow} \circ v^{\downarrow}$.

40. Let A be an $n \times n$ positive semidefinite matrix, $n > 1$. Show that $\operatorname{tr} e^A \leq e^{\operatorname{tr} A} + (n-1)$ with equality if and only if $A = 0$ or $\operatorname{rank}(A) = 1$.

41. Let $r, s, t \in [0, 1]$. If $r + s \leq 1 + t$, show that

$$\sqrt{1-t^2} \leq \sqrt{1-r^2} + \sqrt{1-s^2}.$$

The strict inequality holds if $r, s, t \in (0, 1)$.

42. Reprove the inequality in Example 6.5.1 (Sect. 6.5) via $(p, q) \prec_w (n, x)$ with the increasing and convex function $f(t) = -\sqrt{n^2 - t^2}$.

43. Let $t_1, \ldots, t_n$ and $\sigma_1, \ldots, \sigma_n$ be positive numbers and suppose that the sum of $t_1, \ldots, t_n$ is equal to 1. Prove the following statements:

(a) $f(t) = t \ln t$ is convex for $t > 0$. Is it increasing?
(b) $g(t) = \frac{1}{t} \ln\big(\sum_{i=1}^n t_i \sigma_i^t\big)$ is increasing for $t > 0$. Is it convex?
(c) $p(t) = \sum_{i=1}^n \sigma_i^t$ is convex for $t > 0$. Is it increasing?
(d) $q(t) = (\sum_{i=1}^n \sigma_i^t)^{1/t}$ is decreasing for $t > 0$.
[Note: $q(t)$ is also convex, a result of Beckenbach.]

44. Let A be an $n \times n$ matrix and denote $|A| = (A^*A)^{1/2}$. Let $k \geq 1$.

(a) Show that $\max_{\|x\|=1,\, x \in \mathbb{C}^n} |x^* A^k x| \leq \max_{\|x\|=1,\, x \in \mathbb{C}^n} x^* |A|^k x$.
(b) Does $|x^* A^2 x| \leq x^* |A|^2 x$ for every unit vector $x \in \mathbb{C}^n$?
(c) By the Cauchy-Schwarz inequality, $(x^*|A|x)^2 \leq x^*|A|^2 x$ for every unit vector $x \in \mathbb{C}^n$. Show that for every unit vector $x \in \mathbb{C}^n$,

$$(x^*|A|x)^k \leq x^*|A|^k x.$$

(d) By the Cauchy-Schwarz inequality, $|x^*Ax|^2 \leq x^*|A|^2 x$ for every unit vector $x \in \mathbb{C}^n$. Show that $|x^*Ax|^k \leq x^*|A|^k x$ every unit vector $x \in \mathbb{C}^n$, if $k \geq 2$, while $|x^*Ax| \leq x^*|A|x$ need not be true.

45. Let A an $n \times n$ positive semidefinite matrix and let $\alpha \geq 1$. Show that

$$(x^*Ax)^\alpha \leq x^* A^\alpha x \text{ for all unit vectors } x \in \mathbb{C}^n.$$

46. Let A be an $n \times n$ positive semidefinite matrix and let X_{ii} denote the (i, i)-entry of a square matrix X on the diagonal. For $m > 1$, show that

$$(A_{ii})^m \leq (A^m)_{ii}, \quad i = 1, 2, \ldots, n,$$

where equality occurs (for some i) if and only if A_{ii} is the only (possible) nonzero entry in row (column) i. Does this hold for Hermitian matrices?

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10.4 Majorization of Diagonal Entries, Eigenvalues, and Singular Values

Majorization is a very useful tool in deriving matrix inequalities. This section presents some elegant matrix inequalities involving diagonal entries, eigenvalues, and singular values in terms of majorization. Especially, we show that the relation between the diagonal entries and eigenvalues of a Hermitian matrix is precisely characterized by majorization.

We denote the vectors of the diagonal entries, eigenvalues, and singular values of an n-square (real or complex) matrix A, respectively, by

$$\begin{aligned} d(A) &= (d_1(A), d_2(A), \ldots, d_n(A)), \\ \lambda(A) &= (\lambda_1(A), \lambda_2(A), \ldots, \lambda_n(A)), \\ \sigma(A) &= (\sigma_1(A), \sigma_2(A), \ldots, \sigma_n(A)). \end{aligned}$$

The singular values are always arranged in decreasing order. In the case where A is Hermitian, all $d_i(A)$ and $\lambda_i(A)$ are real, we usually (unless otherwise stated) order the components of $d(A)$ and $\lambda(A)$ decreasingly.

We point out that sometimes when the absolute values of eigenvalues are in the study, we may (re-)order them as $|\lambda_1(A)| \geq |\lambda_2(A)| \geq \cdots \geq |\lambda_n(A)|$. The singular values of Hermitian A are simply $\sigma_i(A) = |\lambda_i(A)|$, $1 \leq i \leq n$. When it comes to majorization, say $|\lambda(A)| \prec_w \sigma(A)$ (see (10.28)), the order of components is unimportant because majorization takes the components as a whole, regardless of the order (arrangement) of the components.

Example 10.4.1 Let $A = \begin{pmatrix} -1 & 1 \\ -1 & 1 \end{pmatrix}$, $B = \begin{pmatrix} -1 & 1 \\ 1 & 1 \end{pmatrix}$, $C = \begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix}$, $D = \begin{pmatrix} 0 & 0 \\ 0 & -1 \end{pmatrix}$.

$$\begin{array}{lll} d(A) = (1,-1), & \lambda(A) = (0,0), & \sigma(A) = (2,0), \\ d(B) = (1,-1), & \lambda(B) = (\sqrt{2},-\sqrt{2}), & \sigma(B) = (\sqrt{2},\sqrt{2}), \\ d(C) = (1,1), & \lambda(C) = (1-i,1+i), & \sigma(C) = (\sqrt{2},\sqrt{2}), \\ d(D) = (0,-1), & \lambda(D) = (0,-1), & \sigma(D) = (1,0), \\ |d(D)| = (0,1), & |\lambda(D)|^{\downarrow} = (1,0), & \sigma(D) = \lambda(|D|). \ \blacksquare \end{array}$$

Theorem 10.4.1 (Schur) *Let H be an n-square Hermitian matrix. Then*

$$d(H) \prec \lambda(H).$$

Proof 1. Let H_k, $1 \leq k \leq n$, be the $k \times k$ principal submatrix of H of diagonal entries $d_1(H), \ldots, d_k(H)$. By eigenvalue interlacing Theorem 8.3.4,

$$\sum_{i=1}^{k} d_i(H) = \operatorname{tr} H_k = \sum_{i=1}^{k} \lambda_i(H_k) \leq \sum_{i=1}^{k} \lambda_i(H).$$

When $k = n$, both sides of the inequality equal $\operatorname{tr} H$, so the equality holds.

Proof 2. Use spectral decomposition. Let U be a unitary matrix such that

$$H = U^* \operatorname{diag}(\lambda_1(H), \ldots, \lambda_n(H))U.$$

Let $H = (h_{ij})$ and $U = (u_{ij})$. Then for each $j = 1, \ldots, n$,

$$h_{jj} = \sum_i \overline{u_{ij}} \lambda_i(H) u_{ij} = \sum_i \lambda_i(H) |u_{ij}|^2.$$

Let $P = (|u_{ij}|^2)$. P is doubly stochastic (Problem 24, Sect. 5.6), and

$$(h_{11}, \ldots, h_{nn}) = (\lambda_1(H), \ldots, \lambda_n(H))P.$$

It follows that $(h_{11}, \ldots, h_{nn}) \prec \lambda(H)$, that is, $d(H) \prec \lambda(H)$. ∎

Example 10.4.2 Let $H = \begin{pmatrix} -1 & i & 3 \\ -i & 1 & 2 \\ 3 & 2 & 0 \end{pmatrix}$. H is Hermitian. Let $\lambda_1 \geq \lambda_2 \geq \lambda_3$ be the (necessarily real) eigenvalues of H. Then $(-1, 1, 0) \prec (\lambda_1, \lambda_2, \lambda_3)$, implying $\lambda_3 \leq -1$, $\lambda_1 \geq 1$, and $\lambda_1 + \lambda_2 + \lambda_3 = 0$. ∎

Example 10.4.3 If $H \in \mathbb{M}_n$ is Hermitian and $U \in \mathbb{M}_n$ is unitary, then

$$d(U^*HU) \prec \lambda(H). \text{ ∎} \tag{10.26}$$

Example 10.4.4 (Fan) Let $A \in \mathbb{M}_n$ have eigenvalues $\lambda_1(A), \ldots, \lambda_n(A)$. Let $\Re\big(\lambda(A)\big) = \big(\Re(\lambda_1(A)), \ldots, \Re(\lambda_n(A))\big)$ and let $\mathcal{H}(A) = \frac{A+A^*}{2}$. Then

$$\Re\big(\lambda(A)\big) \prec \lambda\big(\mathcal{H}(A)\big).$$

Proof. By Schur triangularization (Theorem 3.2.1), we write $A = V^*TV$, where V is unitary and T is upper-triangular. Then $\mathcal{H}(A) = V^*\mathcal{H}(T)V$, and $d(\mathcal{H}(T)) = \Re(\lambda(A))$. Since $\mathcal{H}(T)$ is Hermitian, we obtain

$$\Re\big(\lambda(A)\big) = d\big(\mathcal{H}(T)\big) \prec \lambda\big(\mathcal{H}(T)\big) = \lambda\big(\mathcal{H}(A)\big). \text{ ∎}$$

The next result shows the validity of the converse of Theorem 10.4.1. Hence, the relation between the diagonal entries and eigenvalues of a Hermitian matrix is precisely characterized through majorization.

Theorem 10.4.2 (Horn-Mirsky) *Let $(d_1, d_2, \ldots, d_n), (\lambda_1, \lambda_2, \ldots, \lambda_n) \in \mathbb{R}^n$. If $(d_1, d_2, \ldots, d_n) \prec (\lambda_1, \lambda_2, \ldots, \lambda_n)$, then there is an $n \times n$ real symmetric matrix that has diagonal entries $d_1, d_2, \ldots, d_n$ and eigenvalues $\lambda_1, \lambda_2, \ldots, \lambda_n$.*

Proof. Denote $d = (d_1, d_2, \ldots, d_n)$ and $\lambda = (\lambda_1, \lambda_2, \ldots, \lambda_n)$. We show by induction that there exists an $n \times n$ real symmetric matrix H that has diagonal entries $d_1, d_2, \ldots, d_n$ and eigenvalues $\lambda_1, \lambda_2, \ldots, \lambda_n$. We first deal with the case of $n = 2$. For $n > 2$, we construct a new vector (sequence) λ' that differs from λ by at most two components one of which is some d_i, i.e., λ' is a step "closer" to d. An application of induction results in the desired conclusion. We may assume that $d_1, d_2, \ldots, d_n$ and $\lambda_1, \lambda_2, \ldots, \lambda_n$ are arranged in decreasing order. Because $d \prec \lambda$, we have $\lambda_n \leq d_n$ and $d_1 \leq \lambda_1$.

Denote $\operatorname{diag} d = \operatorname{diag}(d_1, d_2, \ldots, d_n)$ and $\operatorname{diag} \lambda = \operatorname{diag}(\lambda_1, \lambda_2, \ldots, \lambda_n)$.

If $n = 2$ and $\lambda_1 = \lambda_2$, then $d_1 = d_2$. The statement is obviously true. If $n = 2$ and $\lambda_1 > \lambda_2$, because $(d_1, d_2) \prec (\lambda_1, \lambda_2)$, we have $\lambda_2 \le d_2 \le d_1 \le \lambda_1$ and $d_1 + d_2 = \lambda_1 + \lambda_2$. Construct the 2×2 matrix

$$U = (\lambda_1 - \lambda_2)^{-1/2} \begin{pmatrix} (d_1 - \lambda_2)^{1/2} & -(\lambda_1 - d_1)^{1/2} \\ (\lambda_1 - d_1)^{1/2} & (d_1 - \lambda_2)^{1/2} \end{pmatrix}.$$

Then U is real orthogonal and $H = U^T(\operatorname{diag} \lambda)\, U = \begin{pmatrix} d_1 & \star \\ \star & d_2 \end{pmatrix}$ is a 2×2 real symmetric matrix having diagonal entries d_1, d_2 and eigenvalues λ_1, λ_2.

Now suppose $n > 2$. If $d_1 = \lambda_1$, then $(d_2, \ldots, d_n) \prec (\lambda_2, \ldots, \lambda_n)$. By induction hypothesis on $n - 1$, there exists an $(n-1)$-square real symmetric matrix G having diagonal entries $d_2, \ldots, d_n$ and eigenvalues $\lambda_2, \ldots, \lambda_n$. Then $H = \begin{pmatrix} \lambda_1 & 0 \\ 0 & G \end{pmatrix}$ is an n-square real symmetric matrix as desired.

If $d_1 < \lambda_1$, then there is a d_k such that $d_1 < \lambda_1, \ldots, d_k < \lambda_k$, and $d_{k+1} \ge \lambda_{k+1}$ for some $k \ge 1$. Then $\lambda_{k+1} \le d_{k+1} \le d_k < \lambda_k$. Put $\lambda'_{k+1} = \lambda_{k+1} + \lambda_k - d_k$, $\lambda'_k = d_k$. It follows that $(\lambda'_k, \lambda'_{k+1}) \prec (\lambda_k, \lambda_{k+1})$. By the above argument for $n = 2$, there exists a 2×2 real orthogonal matrix U_2 such that the diagonal entries of $U_2^T \operatorname{diag}(\lambda_k, \lambda_{k+1}) U_2$ are $\lambda'_k, \lambda'_{k+1}$.

Now replace the 2×2 principal submatrix in rows k and $k+1$ of the identity matrix I_n by U_2 to get an $n \times n$ matrix V. This V is real orthogonal. Set $F = V^T(\operatorname{diag} \lambda)\, V$. Then F is real symmetric, having eigenvalues $\lambda_1, \ldots, \lambda_k, \lambda_{k+1}, \ldots, \lambda_n$ and diagonal entries $\lambda_1, \ldots \lambda'_k, \lambda'_{k+1}, \ldots, \lambda_n$. Let $\lambda' = (\lambda_1, \ldots \lambda'_k, \lambda'_{k+1}, \ldots, \lambda_n)$. It is immediate that $\lambda' \prec \lambda$. It is not difficult to show that $d \prec \lambda'$. (Note that λ'_k need not be greater or equal to λ'_{k+1}.) Thus, $d \prec \lambda' \prec \lambda$; λ' and λ differ by only two elements, and d and λ' both contain $\lambda'_k = d_k$. Let $\tilde{d}$ be the vector by deleting d_k from d and let $\tilde{\lambda}$ be the vector by deleting d_k from λ'. Then $\tilde{d} \prec \tilde{\lambda}$. By induction hypothesis, there exists an $(n-1)$-square real orthogonal matrix U_{n-1} such that $U_{n-1}^T(\operatorname{diag} \tilde{\lambda})\, U_{n-1}$ has diagonal entries $\tilde{d}$. Now construct an n-square matrix $W = (w_{ij})$ by inserting a row and a column in U_{n-1} so that $w_{kk} = 1$ and all other w's are 0 in the row and column. Then W is real orthogonal and $H = W^T F W$ has diagonal entries $d_1, d_2, \ldots, d_n$, as desired. ∎

Example 10.4.5 (a) Let $d = (3, 2, 1, -1)$ and $\lambda = (5, 4, 0, -4)$. Then $d \prec \lambda$. There exists a 4×4 real symmetric matrix S such that the diagonal entries of S are 3, 2, 1, -1, and the eigenvalues of S are 5, 4, 0, -4.

(b) Let $a_1, \ldots, a_n \in \mathbb{R}$ and $m = \frac{1}{n}(a_1 + \cdots + a_n)$. Because $(m, \ldots, m) \prec (a_1, \ldots, a_n)$, there exists a real symmetric matrix S such that the diagonal entries of S are the m's and the eigenvalues of S are $a_1, \ldots, a_n$. ∎

Theorem 10.4.3 *Let $A \in \mathbb{M}_n$ and $d(A)$, $\lambda(A)$, and $\sigma(A)$ be the vectors of the diagonal entries, eigenvalues, and singular values of A, respectively. Then*

$$|d(A)| \prec_w \sigma(A) \tag{10.27}$$

and

$$|\lambda(A)| \prec_w \sigma(A). \tag{10.28}$$

Proof. We assume that the absolute values of the diagonal entries of A are in decreasing order: $|a_{11}| \geq \cdots \geq |a_{nn}|$. For each a_{ii}, let t_i be such that

$$t_i a_{ii} = |a_{ii}|, \quad |t_i| = 1, \quad i = 1, \ldots, n.$$

Let $B = A \operatorname{diag}(t_1, \ldots, t_n)$ and let C be the leading $k \times k$ principal submatrix of B. Then B has the same singular values as A, and

$$d(C) = (|a_{11}|, \ldots, |a_{kk}|).$$

For $i \leq k$, by Theorem 8.4.3, $\sigma_i(C) \leq \sigma_i(B)$, and by Theorem 9.3.2

$$\begin{aligned} |\operatorname{tr} C| &= |a_{11}| + \cdots + |a_{kk}| \\ &\leq \sigma_1(C) + \cdots + \sigma_k(C) \\ &\leq \sigma_1(B) + \cdots + \sigma_k(B) \\ &= \sigma_1(A) + \cdots + \sigma_k(A). \end{aligned}$$

This gives (10.27). For (10.28), let $A = U^*TU$ be a Schur decomposition of A (Theorem 3.2.1), where U is unitary and T is upper-triangular. Then

$$|\lambda(A)| = |d(T)| \quad \text{and} \quad \sigma(A) = \sigma(T).$$

Applying (10.27) reveals (10.28). (See also Problem 8 of Sect. 9.3.) ∎

Note that $|d(A)|$ and $|\lambda(A)|$ are not comparable in general (Problem 7). In addition, the "stronger inequality" $|d(A)| \prec_{\text{wlog}} \sigma(A)$ is false. However, $|\lambda(A)| \prec_{\log} \sigma(A)$ holds (see Problem 19 of Sect. 4.4 or Weyl's Theorem 10.4.5).

Example 10.4.6 Let $A = (a_{ij})$ be an $n \times n$ matrix with diagonal entries a_{ii}'s, eigenvalues λ_i's, and singular values σ_i's. The following statements hold.

(a) $\sum_i |a_{ii}|^2 = \sum_i \sigma_i^2$ if and only if A is a diagonal matrix.

If A is diagonal, then obviously the singular values of A are the absolute values of the diagonal entries. Conversely, if $\sum_i |a_{ii}|^2 = \sum_i \sigma_i^2$, because $\sum_i \sigma_i^2 = \operatorname{tr}(A^*A) = \sum_{i,j} |a_{ij}|^2$, we see that $a_{ij} = 0$ if $i \neq j$.

As a result, all $|a_{ii}|$'s are exactly the singular values $\Leftrightarrow$ A is diagonal; and all a_{ii}'s are exactly the singular values $\Leftrightarrow$ A is nonnegative diagonal.

(b) $\sum_i |a_{ii}| = \sum_i \sigma_i$ if and only if $A = PU$ for some positive semidefinite matrix $P \in \mathbb{M}_n$ and some unitary diagonal matrix $U \in \mathbb{M}_n$.

$\Leftarrow$: Let $A = PU$, $P = (p_{ij}) \geq 0$ and $U = \operatorname{diag}(\delta_1, \ldots, \delta_n)$ with all $|\delta_i| = 1$. Apparently, $a_{ii} = p_{ii}\delta_i$. Since $P \geq 0$, we have $|a_{ii}| = p_{ii}$ for each i. Note that $A^*A = U^*P^2U$ and the eigenvalues of P are the singular values of A. We arrive at $\sum_i \sigma_i = \sum_i \lambda_i(P) = \operatorname{tr} P = \sum_i p_{ii} = \sum_i |a_{ii}|$.

$\Rightarrow$: This is left to the reader as Problem 11.

This says that the weak majorization $\prec_w$ in $|d(A)| \prec_w \sigma(A)$ in (10.27) becomes majorization $\prec$ if and only if A is the product of a positive semidefinite matrix and a unitary diagonal matrix. Take, for example, $A = \left(\begin{smallmatrix} 1 & 1 \\ -1 & -1 \end{smallmatrix}\right) = \left(\begin{smallmatrix} 1 & -1 \\ -1 & 1 \end{smallmatrix}\right)\left(\begin{smallmatrix} 1 & 0 \\ 0 & -1 \end{smallmatrix}\right)$. We have $|d(A)| = (1, 1) \prec (2, 0) = \sigma(A)$. Note that this A is not Hermitian (nor normal).

By this, we obtain $|\operatorname{tr} A| = \operatorname{tr}|A| \Leftrightarrow A = uP$, where $|u| = 1$ and $P \geq 0$ (Theorem 9.3.2); and $\operatorname{tr} A = \operatorname{tr}|A| \Leftrightarrow A \geq 0$ (Problem 33 of Sect. 8.6).

(c) $\sum_i |\lambda_i| = \sum_i \sigma_i$ if and only if A is normal. This is Theorem 9.3.2.

In other words, the weak majorization $\prec_w$ in $|\lambda(A)| \prec_w \sigma(A)$ in (10.28) becomes a majorization $\prec$ if and only if A is normal.

As a result, all $|\lambda_i|$'s are exactly the singular values $\Leftrightarrow$ A is normal; all λ_i's are exactly the singular values $\Leftrightarrow$ A is positive semidefinite.

(d) If A is a triangular matrix, then the diagonal entries of A are obviously the eigenvalues of A. The converse need not be true. For example, let $A = \left(\begin{smallmatrix} 2 & 2 & -2 \\ 1 & 4 & 1 \\ 2 & 2 & 6 \end{smallmatrix}\right)$. The eigenvalues of A are $2, 4, 6$; they are exactly the same as the diagonal entries of A. A is neither triangular nor normal.

If A is normal, then the diagonal entries of A are exactly the eigenvalues of A if and only if A is a diagonal matrix (Problem 15). ∎

We now present a result of majorization about the eigenvalues and singular values of the blocks of a partitioned positive semidefinite matrix.

Theorem 10.4.4 *Let $A, B, C \in \mathbb{M}_n$ be such that $\left(\begin{smallmatrix} A & B \\ B^* & C \end{smallmatrix}\right) \geq 0$. Then*

$$\sigma(B) \prec_{\text{wlog}} \lambda^{1/2}(A) \circ \lambda^{1/2}(C) \tag{10.29}$$

and

$$|\lambda(B)| \prec_{\text{wlog}} \lambda^{1/2}(A) \circ \lambda^{1/2}(C). \tag{10.30}$$

Proof. First, for any $n \times p$ matrix U and any $n \times q$ matrix V, we have

$$\begin{pmatrix} U^* & 0 \\ 0 & V^* \end{pmatrix}\begin{pmatrix} A & B \\ B^* & C \end{pmatrix}\begin{pmatrix} U & 0 \\ 0 & V \end{pmatrix} = \begin{pmatrix} U^*AU & U^*BV \\ V^*B^*U & V^*CV \end{pmatrix} \geq 0.$$

If $B = 0$, it is trivial. Let $\operatorname{rank}(B) = r > 0$. By Problem 39 of Sect. 3.2, $B = UDV^*$ is a (full rank) singular value decomposition, where D is the $r \times r$

diagonal matrix $\operatorname{diag}(\sigma_1(B), \ldots, \sigma_r(B))$, and U and V are $n \times r$ partially unitary matrices, that is, $U^*U = V^*V = I_r$. Note that $U^*BV = D$.

Denote by $[X]_k$ the $k \times k$ leading principal submatrix of matrix X. Extracting such a submatrix from each block of $\left(\begin{smallmatrix} U^*AU & U^*BV \\ V^*B^*U & V^*CV \end{smallmatrix}\right)$ for $k \le r$:

$$\begin{pmatrix} [U^*AU]_k & [D]_k \\ [D]_k & [V^*CV]_k \end{pmatrix} \ge 0.$$

Taking the determinant of each block results in a 2×2 positive semidefinite matrix which has a nonnegative determinant. It follows that

$$\big(\det[D]_k\big)^2 \le \det\big([U^*AU]_k\big) \det\big([V^*CV]_k\big),$$

or equivalently, for each $k = 1, 2, \ldots, r$,

$$\prod_{i=1}^{k} \sigma_i^2(B) \le \prod_{i=1}^{k} \lambda_i\big([U^*AU]_k\big)\lambda_i\big([V^*CV]_k\big).$$

By the eigenvalue interlacing theorem (Sect. 8.3), we arrive at

$$\prod_{i=1}^{k} \sigma_i^2(B) \le \prod_{i=1}^{k} \lambda_i(A)\lambda_i(C).$$

(10.29) follows by taking the square roots of both sides.

(10.30) is similarly obtained by letting $B = WTW^*$, where W is unitary and T is an upper-triangular matrix with eigenvalues $\lambda_1(B), \lambda_2(B), \ldots, \lambda_n(B)$ on the diagonal (Schur triangularization, Theorem 3.2.1). ∎

Alternatively, one can show (10.29) by using compound matrices and the fact that if $\left(\begin{smallmatrix} A & B \\ B^* & C \end{smallmatrix}\right) \ge 0$, then $\sigma_1^2(B) \le \lambda_1(A)\lambda_1(C)$. (This is the case of $k = 1$ in the preceding proof; see also Problem 18 of Sect. 7.3 or Example 7.6.1 of Sect. 7.6.) Note that $\left(\begin{smallmatrix} A & B \\ B^* & C \end{smallmatrix}\right) \ge 0$ implies $\left(\begin{smallmatrix} C_k(A) & C_k(B) \\ C_k(B^*) & C_k(C) \end{smallmatrix}\right) \ge 0$ for $k \le n$, where $C_k(X)$ denotes the kth compound matrix of X. Therefore, $\sigma_1^2\big(C_k(B)\big) \le \lambda_1\big(C_k(A)\big)\lambda_1\big(C_k(C)\big) = \prod_{i=1}^{k} \lambda_i(A)\lambda_i(C)$, which yields (10.29) because $\sigma_1\big(C_k(B)\big) = \prod_{i=1}^{k} \sigma_i(B)$ (see Sect. 4.4, Chap. 4).

One may also prove (10.29) by writing $B = A^{1/2}XC^{1/2}$ for some contractive matrix X (see Theorem 6.3.2) and by using later Theorem 10.4.6.

Example 10.4.7 Let $\left(\begin{smallmatrix} A & B \\ B^* & C \end{smallmatrix}\right) \ge 0$, where A and C are square matrices. The following are immediate from Theorem 10.4.4 (see also Problem 18 of Sect. 7.3).

(a) $k = 1$, $\sigma_1(B) = \|B\|_{\mathrm{sp}}$ (spectral norm), $\left(\begin{smallmatrix} \|A\|_{\mathrm{sp}} & \|B\|_{\mathrm{sp}} \\ \|B\|_{\mathrm{sp}} & \|C\|_{\mathrm{sp}} \end{smallmatrix}\right) \ge 0$.

(b) $k = n$, $\sigma_1(B) \cdots \sigma_n(B) = |\det B|$, $\left(\begin{smallmatrix} \det A & \det B \\ \det B^* & \det C \end{smallmatrix}\right) \ge 0$.

(c) If $\left(\begin{smallmatrix} A & b \\ b^* & c \end{smallmatrix}\right) \ge 0$, where b is a column vector, then $\|b\| \le \sqrt{c\,\|A\|_{\mathrm{sp}}}$. ∎

Theorem 10.4.5 (Weyl) *Let A be a square matrix. Then*

$$|\lambda(A)| \prec_{\log} \sigma(A). \tag{10.31}$$

Proof 1. Let $A \in \mathbb{M}_n$. Note that $|\lambda_1(A)\cdots\lambda_n(A)| = \sigma_1(A)\cdots\sigma_n(A) = |\det A|$. To show the log-majorization, let $A = UDV$ be a singular value decomposition of A, where U and V are unitary, and D is diagonal. Then

$$\begin{pmatrix} |A^*| & A \\ A^* & |A| \end{pmatrix} = \begin{pmatrix} U & 0 \\ 0 & V^* \end{pmatrix} \begin{pmatrix} D & D \\ D & D \end{pmatrix} \begin{pmatrix} U^* & 0 \\ 0 & V \end{pmatrix} \geq 0.$$

Applying (10.30) to the block matrix on the left gives the inequality (10.31).

Proof 2. Let $|\lambda_{\max}(X)|$ represent the largest modulus of the eigenvalues of X (i.e., spectral radius). Then $|\lambda_{\max}(X)| \leq \sigma_{\max}(X)$. For any positive integer $k \leq n$, consider the compound matrix $C_k(A)$. For any $j_1 < \cdots < j_k$,

$$\prod_{i=1}^{k} |\lambda_{j_i}(A)| \leq |\lambda_{\max}\big(C_k(A)\big)| \leq \sigma_{\max}\big(C_k(A)\big) = \prod_{i=1}^{k} \sigma_i(A). \ \blacksquare$$

Note: The log-majorization (10.31) has appeared as Problem 19 of Sect. 4.4.

As log-majorization implies (weak) majorization (Theorem 10.3.8), we have

$$|\lambda(A)| \prec_w \sigma(A), \quad \text{i.e.,} \quad |\lambda(A)| \prec_w \lambda(|A|),$$

which is (10.28). For $\left(\begin{smallmatrix} A & B \\ B^* & C \end{smallmatrix}\right) \geq 0$, we can combine (10.29) and (10.30) as

$$|\lambda(B)| \prec_{\log} \sigma(B) \prec_{\text{wlog}} \lambda^{1/2}(A) \circ \lambda^{1/2}(C). \tag{10.32}$$

Theorem 10.4.6 (Horn) *Let A and B be n-square matrices. Then*

$$\sigma(AB) \prec_{\log} \sigma(A) \circ \sigma(B). \tag{10.33}$$

Consequently, $\sigma(A^m) \prec_{\log} \sigma^m(A)$ for every positive integer k, and

$$\sigma(AB) \prec_w \sigma(A) \circ \sigma(B). \tag{10.34}$$

Proof. This is immediate from (10.29) by observing that

$$(A^*, B)^*(A^*, B) = \begin{pmatrix} AA^* & AB \\ B^*A^* & B^*B \end{pmatrix} \geq 0. \tag{10.35}$$

Note that for any n-square matrix X, $\prod_{i=1}^{n} \sigma_i(X) = |\det X|$. $\blacksquare$

It is important to note that the components of $\sigma(A)$ and $\sigma(B)$ in (10.33) and (10.34), as well as in later sections, are ordered decreasingly.

The block matrix in (10.35) is useful and worth more attention. For instance, it is used to show the singular value inequalities (see Sect. 8.4)

$$2\sigma_i(AB) \leq \sigma_i(A^*A + BB^*) \ \text{ for each } i. \tag{10.36}$$

Using Horn's inequality (10.33), one can easily show (Problem 23) that if $\left(\begin{smallmatrix} A & B \\ B^* & C \end{smallmatrix}\right) \geq 0$, where $A, B, C \in \mathbb{M}_n$, then exists a contraction $X \in \mathbb{M}_n$ such that

$$\sigma(B) \prec_{\text{wlog}} \lambda^{1/2}(A) \circ \sigma(X) \circ \lambda^{1/2}(C).$$

Following is a list of some notations we have used and will use often. Let A be an n-square matrix with diagonal entries a_{ii}, eigenvalues $\lambda_i(A)$, and singular values $\sigma_i(A)$, $i = 1, \ldots, n$. Let $|A| = (A^*A)^{1/2}$ (the modulus of A).

- $d(A) = (a_{11}, \ldots, a_{nn})$,
- $|d(A)| = (|a_{11}|, \ldots, |a_{nn}|)$,
- $\lambda(A) = (\lambda_1(A), \ldots, \lambda_n(A))$,
- $|\lambda(A)| = (|\lambda_1(A)|, \ldots, |\lambda_n(A)|)$,
- $\lambda^m(A) = (\lambda_1^m(A), \ldots, \lambda_n^m(A))$, where $\lambda_i^m(A) = (\lambda_i(A))^m$ for each i,
- $\sigma(A) = (\sigma_1(A), \ldots, \sigma_n(A)) = \lambda(|A|)$, where $\sigma_1(A) \geq \cdots \geq \sigma_n(A)$, and
- $\sigma^m(A) = (\sigma_1^m(A), \ldots, \sigma_n^m(A))$, where $\sigma_i^m(A) = (\sigma_i(A))^m$ for each i.

Problems

1. Let $A \in \mathbb{M}_n$. Determine if each of the following identities holds.
   (a) $\sigma(A) = \sigma(A^*)$.
   (b) $\sigma(|A|) = \sigma(|A^*|)$.
   (c) $\lambda(A) = \lambda(A^*)$.
   (d) $\lambda(|A|) = \lambda(|A^*|)$.
   (e) $d(A) = d(A^*)$.
   (f) $d(|A|) = d(|A^*|)$.
   (g) $\sigma(A) = \lambda(|A|)$.
   (h) $\sigma(A) = \sigma(|A|)$.
2. Let $A \in \mathbb{M}_n$. Show that $\lambda(A^m) = \lambda^m(A)$. Does $\sigma(A^m) = \sigma^m(A)$?
3. Let $A = \left(\begin{smallmatrix} 1 & 0 \\ 0 & 0 \end{smallmatrix}\right)$. Show that $\sigma_1(P^{-1}AP)$ can be arbitrarily large for suitably selected P while the eigenvalues $\lambda(P^{-1}AP)$ remain unchanged.
4. Let $A \in \mathbb{M}_{m \times n}$. Show that $\sigma(PAQ) = \sigma(A)$ for any permutation matrices $P \in \mathbb{M}_m$ and $Q \in \mathbb{M}_n$. Does $\lambda(PAQ) = \lambda(A)$ (when $m = n$)?
5. Let $A \in \mathbb{M}_n$. Show that $|d(PAQ)| \prec_w \sigma(A)$ and $|\lambda(PAQ)| \prec_w \sigma(A)$ for any unitary (especially for permutation) matrices P, $Q \in \mathbb{M}_n$.
6. Let X be a square matrix and let k be a positive integer. Show that $|\lambda(X^k)| \prec_{\log} \lambda(|X^k|) \prec_{\log} \lambda(|X|^k)$. Consequently, $|\operatorname{tr}(X^k)| \leq \operatorname{tr}(|X|^k)$.
7. Show that $|d(A)| \prec_w |\lambda(A)|$ for all normal matrices A and that it is not true for non-normal matrix $B = \left(\begin{smallmatrix} 2 & 1 \\ 1 & 1 \end{smallmatrix}\right)^{-1} \left(\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right) \left(\begin{smallmatrix} 2 & 1 \\ 1 & 1 \end{smallmatrix}\right) = \left(\begin{smallmatrix} 1 & 1 \\ -1 & -1 \end{smallmatrix}\right)$.

8. Let $A \geq 0$ and $m \geq 1$. Show that $(d(A))^m \leq d(A^m) \prec \lambda^m(A)$.

9. Show by an example that $|d(A)| \prec_{\log} \sigma(A)$ need not be true (even though $|d(A)| \prec_w \sigma(A)$ holds by Theorem 10.4.3). Is $|d(A)| \prec_w d(|A|)$?

10. Let $A \in \mathbb{M}_n$ have eigenvalues $\lambda_1, \ldots, \lambda_n$ and singular values $\sigma_1, \ldots, \sigma_n$. If $|\lambda_1 + \cdots + \lambda_n| = \sigma_1 + \cdots + \sigma_n$ (i.e., $|\operatorname{tr} A|$ equals $\operatorname{tr}|A|$, the trace norm of A), show that $A = uP$, where $u \in \mathbb{C}$ with $|u| = 1$ and $P \geq 0$.

11. Let $A = (a_{ij}) \in \mathbb{M}_n$ and let p be a permutation on $\{1, 2, \ldots, n\}$. Denote $d_p(A) = (a_{1p(1)}, a_{2p(2)}, \ldots, a_{np(n)})$ (called a *general diagonal*).
    (a) Show that $|d_p(A)| \prec_w \sigma(A)$.
    (b) Show that $\prec_w$ becomes $\prec$, i.e., $|d_p(A)| \prec \sigma(A)$, equivalently, $\sum_i |a_{ip(i)}| = \sum_i \sigma_i(A)$, if and only if $A = SUP$, where S is a positive semidefinite matrix, U is a unitary diagonal matrix, and P is the permutation matrix associated to p (Problem 36, Sect. 5.6).
    (c) Show that $\sum_i |a_{ip(i)}|^2 = \sum_i \sigma_i^2(A)$ if and only if all a_{st} other than $a_{ip(i)}$, $i = 1, 2, \ldots, n$, are equal to zero.

12. Verify that $\begin{pmatrix} 1 & 0 \\ c & 1 \end{pmatrix}^{-1} \begin{pmatrix} \lambda_1 & 1 \\ 0 & \lambda_2 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ c & 1 \end{pmatrix} = \begin{pmatrix} \lambda_1 + c & \star \\ \star & \lambda_2 - c \end{pmatrix}$ for $\lambda_1, \lambda_2, c \in \mathbb{C}$.

13. Find a 2×2 nondiagonal matrix $A = \begin{pmatrix} d_1 & \star \\ \star & d_2 \end{pmatrix}$ such that the sum of the diagonal entries in moduli are equal to the sum of the singular values, that is, $|d_1| + |d_2| = \sigma_1(A) + \sigma_2(A)$. Are $|d_1| = \sigma_1(A)$ and $|d_2| = \sigma_2(A)$?

14. Show that every Hermitian matrix is unitarily similar to a real symmetric matrix in which the diagonal entries are all equal to each other.

15. Let A be a normal matrix. Show that the diagonal entries of A are exactly the eigenvalues of A if and only if A is a diagonal matrix.

16. Let A be a square matrix having the properties: $|\lambda(A)| = 1$ for every eigenvalue $\lambda(A)$ of A and $\sigma_1(A) = 1$. Show that A is a unitary matrix.

17. Let $d_1, \ldots, d_n$ and $\sigma_1, \ldots, \sigma_n$ be respectively the the diagonal entries and singular values of $A \in \mathbb{M}_n$. Show that $|d_1|^p + \cdots + |d_n|^p = \sigma_1^p + \cdots + \sigma_n^p$ for some $p > 1$ if and only if A is a diagonal matrix.

18. Let $A \in \mathbb{M}_n$ be a positive definite matrix and let $d_1 \geq d_2 \geq \cdots \geq d_n$ and $\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_n$ be the diagonal entries and eigenvalues of A, respectively. Show that for each $k = 1, 2, \ldots, n$,

$$\sum_{i=1}^{k} d_i \leq \sum_{i=1}^{k} \lambda_i, \quad \prod_{i=k}^{n} d_i \geq \prod_{i=k}^{n} \lambda_i, \quad \sum_{i=k}^{n} \frac{1}{d_i} \leq \sum_{i=k}^{n} \frac{1}{\lambda_i}.$$

19. Prove Theorem 10.4.4 for square matrices A and C of different orders. (Add 0's so that $\sigma(A)$ and $\sigma(C)$ have the same number of components.)

20. Let A, B, and C be $n \times n$ matrices such that $\left(\begin{smallmatrix} A & B \\ B^* & C \end{smallmatrix}\right) \geq 0$. Show that

$$|d(B)|^2 \prec_{\text{wlog}} d(A) \circ d(C),$$

in which the components of $d(A)$ and $d(C)$ are in decreasing order. Is $|d(B)|^2 \prec_{\text{wlog}} \lambda(A) \circ \lambda(C)$? Is $|d(B)|^2 \prec_w \lambda(A) \circ \lambda(C)$?

21. Let $A = \left(\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}\right)$. Show that $|d(A)|^2 \leq d(|A|) \circ d(|A^*|)$ does not hold if the components of $d(|A|)$ and $d(|A^*|)$ are arranged in decreasing order. (Recall from Problem 17 of Sect. 7.3 that there exist arrangements of the components of $d(|A|)$ and $d(|A^*|)$ such that $|d(A)|^2 \leq d(|A|) \circ d(|A^*|)$.)

22. Let A be an $n \times n$ matrix. Show that

(a) $|\lambda(A)|^2 = |\lambda(A^2)| \prec_{\log} \sigma(A^2) \prec_{\log} \sigma^2(A) = \lambda(|A|^2) = \lambda^2(|A|)$.
(b) $\lambda(|A^2|) = \sigma(|A^2|) \prec_{\log} \sigma(|A|^2) = \lambda(|A|^2) = \lambda^2(|A|) = \sigma^2(A)$.
(c) $|d(A)|^2 \prec_{\text{wlog}} d^{\downarrow}(|A|) \circ d^{\downarrow}(|A^*|) \prec_w \sigma^2(A)$.
(d) $|\lambda(A + A^*)| \prec_{\text{wlog}} \lambda(|A| + |A^*|)$.
(e) $|\lambda(A \circ A^*)| \prec_{\text{wlog}} \lambda(|A| \circ |A^*|)$.

23. Show that Theorem 10.4.6 implies Theorem 10.4.4 via Theorem 6.3.2. That is, if A, B, and C are $n \times n$ matrices such that $\left(\begin{smallmatrix} A & B \\ B^* & C \end{smallmatrix}\right) \geq 0$, then there exists an $n \times n$ contractive matrix X (i.e., $\sigma_1(X) \leq 1$) such that

$$\sigma(B) \prec_{\text{wlog}} \lambda^{1/2}(A) \circ \sigma(X) \circ \lambda^{1/2}(C) \leq \lambda^{1/2}(A) \circ \lambda^{1/2}(C).$$

24. Let A, B, and C be $n \times n$ matrices such that $\left(\begin{smallmatrix} A & B \\ B^* & C \end{smallmatrix}\right) \geq 0$. Show that

$$\begin{pmatrix} \|A\|_{\text{sp}} & \|B\|_{\text{sp}} \\ \|B^*\|_{\text{sp}} & \|C\|_{\text{sp}} \end{pmatrix} \geq 0 \quad \text{and} \quad \begin{pmatrix} \rho(A) & \rho(B) \\ \rho(B^*) & \rho(C) \end{pmatrix} \geq 0,$$

where $\|X\|_{\text{sp}}$ and $\rho(X)$ are the spectral norm and spectral radius of square matrix X, respectively. Are these true for 3×3 block matrices?

25. Let $A \in \mathbb{M}_n$ and let $x = (x_1, \ldots, x_n) \in \mathbb{R}^n$. If $x \prec_w \sigma(A)$, show that

$$\sum_{i=1}^{n} \sqrt{1 + |x_i|^2} \leq \operatorname{tr}\left((I + A^*A)^{1/2}\right).$$

Derive the inequalities of diagonal entries $d_i(A)$ and eigenvalues $\lambda_i(A)$:

$$\sum_{i=1}^{n} \sqrt{1 + |d_i(A)|^2} \leq \operatorname{tr}\left((I + A^*A)^{1/2}\right)$$

and

$$\sum_{i=1}^{n} \sqrt{1 + |\lambda_i(A)|^2} \leq \operatorname{tr}\left((I + A^*A)^{1/2}\right).$$

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10.5 Majorization for Matrix Sum

Now we turn our attention to the majorization inequalities of the eigenvalues and singular values of the sum of Hermitian matrices. We arrange the components of the vectors of (real) eigenvalues in decreasing order, that is,

$$\lambda(X) = \lambda^{\downarrow}(X) = \big(\lambda_1(X), \ldots, \lambda_n(X)\big),$$

where $\lambda_1(X) \geq \cdots \geq \lambda_n(X)$ are the eigenvalues of Hermitian $X \in \mathbb{M}_n$.

Theorem 10.5.1 (Fan) *Let A and B be $n \times n$ Hermitian matrices. Then*

$$\lambda(A+B) \prec \lambda(A) + \lambda(B). \tag{10.37}$$

Proof. For each positive integer $k \leq n$, by Theorem 8.5.1, we have

$$\begin{aligned} \sum_{i=1}^{k} \lambda_i(A+B) &= \max_{U^*U=I_k} \operatorname{tr}\big(U^*(A+B)U\big) \\ &\leq \max_{U^*U=I_k} \operatorname{tr}(U^*AU) + \max_{U^*U=I_k} \operatorname{tr}(U^*BU) \\ &= \sum_{i=1}^{k} \lambda_i(A) + \sum_{i=1}^{k} \lambda_i(B). \end{aligned}$$

Since $\operatorname{tr}(A+B) = \operatorname{tr} A + \operatorname{tr} B$, the desired majorization (10.37) follows. ∎

Example 10.5.1 Majorization for Hermitian matrices.

(a) Let $A = \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}$, $B = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}$. Then $A+B = \begin{pmatrix} 0 & -1 \\ -1 & 2 \end{pmatrix}$, $\lambda(A) = (2,0)$, $\lambda(B) = (1,-1)$, and $\lambda(A+B) = (1+\sqrt{2}, 1-\sqrt{2})$. (10.37) is

$$\lambda(A+B) = (1+\sqrt{2}, 1-\sqrt{2}) \prec (2,0) + (1,-1) = \lambda(A) + \lambda(B).$$

(b) Let $H = \begin{pmatrix} H_1 & K \\ K^* & H_2 \end{pmatrix}$ be a block Hermitian matrix with $H_1, H_2, K \in \mathbb{M}_n$. Since $H = \begin{pmatrix} H_1 & 0 \\ 0 & H_2 \end{pmatrix} + \begin{pmatrix} 0 & K \\ K^* & 0 \end{pmatrix}$ and $2\begin{pmatrix} H_1 & 0 \\ 0 & H_2 \end{pmatrix} = \begin{pmatrix} H_1 & K \\ K^* & H_2 \end{pmatrix} + \begin{pmatrix} H_1 & -K \\ -K^* & H_2 \end{pmatrix}$, and because $\begin{pmatrix} H_1 & -K \\ -K^* & H_2 \end{pmatrix}$ is similar to $\begin{pmatrix} H_1 & K \\ K^* & H_2 \end{pmatrix}$ via $\begin{pmatrix} I_n & 0 \\ 0 & -I_n \end{pmatrix}$, we get

$$\big(\lambda(H_1), \lambda(H_2)\big) \prec \lambda(H) \prec \big(\lambda(H_1), \lambda(H_2)\big)^{\downarrow} + \big(\sigma(K), -\sigma(K)\big).$$

[Note: By Sect. 2.4, the eigenvalues of $\begin{pmatrix} 0 & K \\ K^* & 0 \end{pmatrix}$ are $\pm\sigma_i(K)$, $i = 1, \ldots, n$.]

(c) Let A and B be $n \times n$ positive semidefinite matrices. Let $X = \begin{pmatrix} A^{1/2}, & B^{1/2} \end{pmatrix}$. Then $X^*X = \begin{pmatrix} A & A^{1/2}B^{1/2} \\ B^{1/2}A^{1/2} & B \end{pmatrix}$ and $XX^* = A+B$. Since X^*X and XX^* have the same nonzero eigenvalues, by (b), we obtain

$$\big(\lambda(A), \lambda(B)\big) \prec \lambda(X^*X) = \big(\lambda(XX^*), 0\big) = \big(\lambda(A+B), 0\big). \quad ∎$$

Theorem 10.5.2 *Let A and B be n-square Hermitian matrices. Then*

$$\lambda(A) - \lambda(B) \prec \lambda(A - B). \tag{10.38}$$

Proof. Let k be a positive integer, $1 \le k \le n$, and let $1 \le i_1 < \cdots < i_k \le n$. Writing $A = B + (A - B)$ and applying Theorem 8.5.2, we have

$$\sum_{t=1}^{k} \lambda_{i_t}(A) \le \sum_{t=1}^{k} \lambda_{i_t}(B) + \sum_{j=1}^{k} \lambda_j(A - B),$$

which yields

$$\max_{1 \le i_1 < \cdots < i_k \le n} \sum_{t=1}^{k} \big(\lambda_{i_t}(A) - \lambda_{i_t}(B)\big) \le \sum_{j=1}^{k} \lambda_j(A - B)$$

for $k = 1, 2, \ldots, n$, that is,

$$\lambda(A) - \lambda(B) \prec_w \lambda(A - B).$$

Since equality holds when $k = n$, the majorization (10.38) follows. ∎

Explain what is wrong with the following proof of (10.38):

Because $\lambda(A) = \lambda(B + (A - B)) \prec \lambda(B) + \lambda(A - B)$, subtracting $\lambda(B)$ from both sides gives $\lambda(A) - \lambda(B) \prec \lambda(A - B)$. [Hint: See Example 10.1.5.]

Example 10.5.2 Let A and B be n-square Hermitian matrices having eigenvalues $\lambda_1(A) \ge \cdots \ge \lambda_n(A)$ and $\lambda_1(B) \ge \cdots \ge \lambda_n(B)$, respectively. Because $\lambda(A) - \lambda(B) \prec \lambda(A - B)$ and since $f(t) = |t|$ is a convex function, by Theorem 10.3.4, we have $|\lambda(A) - \lambda(B)| \prec_w |\lambda(A - B)|$. It is immediate that

$$\sum_{i=1}^{n} |\lambda_i(A) - \lambda_i(B)| \le \sum_{i=1}^{n} |\lambda_i(A - B)|. \ ∎$$

Note that $\lambda^{\uparrow}(H) = (\lambda_n(H), \ldots, \lambda_1(H)) = -\lambda(-H)$ for any $n \times n$ Hermitian matrix H. By the above two theorems, we have the majorizations

$$\lambda^{\downarrow}(A) + \lambda^{\uparrow}(B) \prec \lambda(A + B) \prec \lambda^{\downarrow}(A) + \lambda^{\downarrow}(B); \tag{10.39}$$

$$\lambda^{\downarrow}(A) - \lambda^{\downarrow}(B) \prec \lambda(A - B) \prec \lambda^{\downarrow}(A) - \lambda^{\uparrow}(B). \tag{10.40}$$

Note that the components of $\lambda^{\downarrow}(A)+\lambda^{\downarrow}(B)$ and $\lambda^{\downarrow}(A)-\lambda^{\uparrow}(B)$ are already in decreasing order, while the components of $\lambda^{\downarrow}(A)+\lambda^{\uparrow}(B)$ and $\lambda^{\downarrow}(A)-\lambda^{\downarrow}(B)$ are not in decreasing order in general.

Example 10.5.3 Let $A = \operatorname{diag}(1, 0, -1)$ and $B = \operatorname{diag}(0, 2, 0)$. Then

$$\lambda^{\downarrow}(A) + \lambda^{\uparrow}(B) = (1, 0, 1), \qquad \lambda^{\downarrow}(A) + \lambda^{\downarrow}(B) = (3, 0, -1),$$
$$\lambda^{\downarrow}(A) - \lambda^{\downarrow}(B) = (-1, 0, -1), \quad \lambda^{\downarrow}(A) - \lambda^{\uparrow}(B) = (1, 0, -3). \ ∎$$

Theorem 10.5.3 *Let A and B be $n \times n$ matrices having only real eigenvalues ordered decreasingly as $\lambda_1 \geq \cdots \geq \lambda_n$ and $\mu_1 \geq \cdots \geq \mu_n$, respectively. Then*

(1) *For any $k \geq 1$ and any permutation p on $\{1, 2, \ldots, n\}$,*

$$\Big(\sum_{i=1}^{n} |\lambda_i - \mu_i|^k \Big)^{\frac{1}{k}} \leq \Big(\sum_{i=1}^{n} |\lambda_i - \mu_{p(i)}|^k \Big)^{\frac{1}{k}} \leq \Big(\sum_{i=1}^{n} |\lambda_i - \mu_{n-i+1}|^k \Big)^{\frac{1}{k}}.$$

(2) *If A is Hermitian, then $|\lambda_i - \mu_i| \leq \sqrt{n}\, \|A - B\|_F$ for $i = 1, \ldots, n$.*

Proof. (1) By Problem 10 (or by Theorem 10.1.6) of Sect. 10.1, we have

$$(\lambda_1 - \mu_1, \ldots, \lambda_n - \mu_n) \prec (\lambda_1 - \mu_{p(1)}, \ldots, \lambda_n - \mu_{p(n)}) \prec (\lambda_1 - \mu_n, \ldots, \lambda_n - \mu_1).$$

Applying the Schur-convex function $\phi(x) = \big(\sum_i |x_i|^k | \big)^{1/k}$ (see Problem 27, Sect. 10.3) to the above majorizations reveals the desired inequalities.

(2) By Sun's theorem (Theorem 9.4.3), there is a permutation p such that

$$\Big(\sum_{i} |\lambda_i - \mu_{p(i)}|^2 \Big)^{1/2} \leq \sqrt{n}\, \|A - B\|_F.$$

By part (1) with $k = 2$, $|\lambda_i - \mu_i| \leq \big(\sum_i |\lambda_i - \mu_{p(i)}|^2 \big)^{1/2} \leq \sqrt{n}\, \|A - B\|_F$. ∎

Example 10.5.4 A (or B) need be Hermitian in Theorem 10.5.3 (2). Let $A = \left(\begin{smallmatrix} 0 & 1 \\ x & 0 \end{smallmatrix}\right)$, $x > 0$, $B = \left(\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right)$. The eigenvalues of A are $\lambda_{1,2} = \pm\sqrt{x}$, and the eigenvalues of B are $\mu_1 = \mu_2 = 0$. For $k \geq 1$ and $0 < x < \frac{1}{2}$, we have $(\sum_j |\lambda_j - \mu_j|^k)^{1/k} \geq |\lambda_i - \mu_i| = \sqrt{x} > \sqrt{2}x = \sqrt{2}\|A - B\|_F$, $i = 1, 2$.

In fact, for any constant $K > 0$, $|\lambda_i - \mu_i| > K\|A - B\|_F$ if x is small enough, i.e., there does not exist constant K such that $|\lambda_i - \mu_i| \leq K\|A - B\|_F$. ∎

Example 10.5.5 Let A and B be $n \times n$ Hermitian matrices. Then

$$|\lambda_i(A) - \lambda_i(B)| \leq \sqrt{n}\, \|A - B\|_F$$

for $i = 1, 2, \ldots, n$, where the eigenvalues are ordered monotonically.

Corollary 10.5.1 *Let $A(t)$ be an $n \times n$ Hermitian matrix for each parameter t (in its domain). Let $\lambda_1(t) \geq \cdots \geq \lambda_n(t)$ be the (necessarily real) eigenvalues of $A(t)$. If all the entries of $A(t)$ are continuous functions of t, then every parameterized eigenvalue $\lambda_i(t)$ is a continuous function of t, $i = 1, \ldots, n$.*

Proof. For small ε, the eigenvalues of $A(t+\varepsilon)$ are $\lambda_1(t+\varepsilon) \geq \cdots \geq \lambda_n(t+\varepsilon)$. Let $B = A(t + \varepsilon)$. By (3) of Theorem 10.5.3, $|\lambda_i(t) - \lambda_i(t + \varepsilon)| \leq \sqrt{n}\|A(t) - B\|_F = \sqrt{n}\, \|A(t) - A(t+\varepsilon)\|_F$. Because matrix norm is a continuous function, we have $\|A(t) - A(t + \varepsilon)\|_F \to 0$ as $\varepsilon \to 0$. So, $\lambda_i(t + \varepsilon) \to \lambda_i(t)$ as $\varepsilon \to 0$. ∎

Now we discuss the majorization inequalities of the singular values of matrix sum. Let A be an $m\times n$ matrix. The singular values of A are the eigenvalues of $|A| = (A^*A)^{1/2}$. If the rank of A is r (> 0), then A has r nonzero singular values $\sigma_1(A) \geq \cdots \geq \sigma_r(A)$. Let $\sigma(A) = (\sigma_1(A), \ldots, \sigma_r(A), 0, \ldots, 0) \in \mathbb{R}^n$ be the vector of the singular values of A. By Theorem 2.4.2 of Sect. 2.4, the block matrix $\left(\begin{smallmatrix} 0 & A \\ A^* & 0 \end{smallmatrix}\right)$ is Hermitian and has eigenvalues $\sigma_1(A), \ldots, \sigma_r(A)$, 0, $\ldots$, 0, $-\sigma_r(A), \ldots, -\sigma_1(A)$. This is an important and useful fact.

Theorem 10.5.4 *Let A and B be $m \times n$ matrices. Then*

$$\sigma(A+B) \prec_w \sigma(A) + \sigma(B). \tag{10.41}$$

Proof. Apply Theorem 10.5.1 to $\hat{A} = \left(\begin{smallmatrix} 0 & A \\ A^* & 0 \end{smallmatrix}\right)$ and $\hat{B} = \left(\begin{smallmatrix} 0 & B \\ B^* & 0 \end{smallmatrix}\right)$. ∎

(10.41) is analogous to (10.37) for singular values. Does the stronger log-majorization $\sigma(A+B) \prec_{\log} \sigma(A) + \sigma(B)$ hold? The answer is negative by considering $A = \left(\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right)$ and $B = \left(\begin{smallmatrix} 0 & 0 \\ 1 & 0 \end{smallmatrix}\right)$. Below is a similar result to (10.38).

Theorem 10.5.5 *Let A and B be $m \times n$ matrices. Then*

$$|\sigma(A) - \sigma(B)| \prec_w \sigma(A-B). \tag{10.42}$$

Proof. Let $\hat{A} = \left(\begin{smallmatrix} 0 & A \\ A^* & 0 \end{smallmatrix}\right)$ and $\hat{B} = \left(\begin{smallmatrix} 0 & B \\ B^* & 0 \end{smallmatrix}\right)$. Then $\hat{A} - \hat{B} = \left(\begin{smallmatrix} 0 & A-B \\ A^*-B^* & 0 \end{smallmatrix}\right)$. Applying Theorem 10.5.2 to $\hat{A}$ and $\hat{B}$ implies $\lambda(\hat{A}) - \lambda(\hat{B}) \prec \lambda(\hat{A} - \hat{B})$, i.e.,

$$(\sigma_1(A) - \sigma_1(B), \ldots, 0, \ldots, 0, \ldots, \sigma_1(B) - \sigma_1(A)) \prec \lambda(\hat{A} - \hat{B}).$$

(Note that $\sigma_1(A) - \sigma_1(B)$ need not be the largest component on the left.) Thus, for $k = 1, 2, \ldots, n$, and for any index sequence $1 \leq i_1 < \cdots < i_k \leq n$,

$$\max_{1\leq i_1<\cdots<i_k\leq n} \sum_{t=1}^{k} \left|\sigma_{i_t}(A) - \sigma_{i_t}(B)\right| \leq \sum_{j=1}^{k} \sigma_j(A-B),$$

which is essentially the same as (10.42). (See Problem 4.) ∎

The following results are on the log-majorization (due to Zhan) and weak majorization (due to Bhatia and Kittaneh) of the singular values of a linear combination of two positive semidefinite matrices.

Theorem 10.5.6 *Let A and B be n-square positive semidefinite matrices and let z be any complex number. Then*

$$\sigma(A - |z|B) \prec_{\log} \sigma(A + zB) \prec_{\log} \sigma(A + |z|B). \tag{10.43}$$

Proof. The second log-majorization follows from Theorem 10.4.4 because

$$\begin{pmatrix} A + |z|B & A + zB \\ A + z^*B & A + |z|B \end{pmatrix} \geq 0.$$

Now we show the first log-majorization in (10.43). If A and B are nonnegative diagonal matrices, invoking the elementary inequality $|a-|z|b| \le |a+zb|$, where $z \in \mathbb{C}$, $a, b \ge 0$, we arrive at

$$|\det(A - |z|B)| \le |\det(A + zB)|. \tag{10.44}$$

Inequality (10.44) actually holds for all positive semidefinite A and B due to the fact that there exists an invertible matrix P such that P^*AP and P^*BP are both nonnegative diagonal (Theorem 7.2.2, Sect. 7.2).

Rewrite (10.44) in terms of singular values (Problem 31 of Sect. 2.4) as

$$\prod_{i=1}^{n} \sigma_i(A - |z|B) \le \prod_{i=1}^{n} \sigma_i(A + zB). \tag{10.45}$$

We claim that the upper limit n for the products in (10.45) can be replaced by any positive integer $k \le n$. Since A and B are positive semidefinite, $A - |z|B$ is Hermitian. Thus, there exists an n-square unitary matrix U such that $U^*(A - |z|B)U = \operatorname{diag}(\lambda_1, \ldots, \lambda_n)$, where $|\lambda_i| = \sigma_i(A - |z|B)$, $i = 1, 2, \ldots, n$. Partition U as $U = (U_1, U_2)$, where U_1 consists of the first k columns of U. Then $U_1^*AU_1$ and $U_1^*BU_1$ are $k \times k$ matrices. Note that $U_1^*(A - |z|B)U_1 = \operatorname{diag}(\lambda_1, \ldots, \lambda_k)$. It follows that

$$\begin{aligned}
\prod_{i=1}^{k} \sigma_i(A - |z|B) &= \prod_{i=1}^{k} |\lambda_i| = \prod_{i=1}^{k} \sigma_i\big(U_1^*(A - |z|B)U_1\big) \\
&= \prod_{i=1}^{k} \sigma_i(U_1^*AU_1 - |z|U_1^*BU_1) \\
&\le \prod_{i=1}^{k} \sigma_i(U_1^*AU_1 + zU_1^*BU_1) \quad \text{(by (10.45))} \\
&= \prod_{i=1}^{k} \sigma_i(U_1^*(A + zB)U_1) \\
&\le \prod_{i=1}^{k} \sigma_i(A + zB).
\end{aligned}$$

The last inequality is because U_1 is partially unitary (see Sect. 8.3). ∎

The following result is immediate since "$\prec_{\text{wlog}}$" implies "$\prec_w$".

Corollary 10.5.2 *Let A and B be n-square positive semidefinite matrices and let z be any complex number with $|z| = 1$. Then*

$$\sigma(A - B) \prec_w \sigma(A + zB) \prec_w \sigma(A + B). \tag{10.46}$$

Obviously, the positive semidefiniteness of both A and B in (10.46) is needed; or taking $B = -A$ results in zero on the right-hand side for any A.

Example 10.5.6 Let A and B be n-square positive semidefinite matrices. By writing $2(A^2+B^2)=(A+B)^2+(A-B)^2$ and using (10.46), we obtain

$$2\sigma(A^2+B^2) \prec_w \sigma((A+B)^2)+\sigma((A-B)^2)) \prec_w 2\sigma((A+B)^2)),$$

which gives immediately

$$\lambda(A^2+B^2) \prec_w \lambda((A+B)^2)). \tag{10.47}$$

Note that $A^2+B^2 \le (A+B)^2$ need not be true (Problem 24, Sect. 7.2). ∎

Let A be an $n\times n$ matrix. The *Cartesian decomposition* of A is

$$A=\mathcal{H}(A)+\mathfrak{I}(A)i, \text{ where } \mathcal{H}A)=\tfrac{A+A^*}{2} \text{ and } \mathfrak{I}(A)=\tfrac{A-A^*}{2i}.$$

Note that $\mathcal{H}(A)$ and $\mathfrak{I}(A)$ are Hermitian and the decomposition is unique.

Theorem 10.5.7 *Let A be an $n\times n$ matrix. $\mathcal{H}(A)=\frac{A+A^*}{2}$ is the closest to A in the sense of majorization, i.e., for any $n\times n$ Hermitian matrix G,*

$$\sigma(A-\mathcal{H}(A)) \prec_w \sigma(A-G). \tag{10.48}$$

Proof. $A-\mathcal{H}(A)=\frac{A-A^*}{2}=\frac{A-G}{2}-\frac{(A-G)^*}{2}$. By Theorem 10.5.4, we derive

$$\begin{aligned}\sigma(A-\mathcal{H}(A)) &= \sigma\Big(\frac{A-G}{2}-\frac{(A-G)^*}{2}\Big)\\ &\prec_w \sigma\Big(\frac{A-G}{2}\Big)+\sigma\Big(\frac{(A-G)^*}{2}\Big)\\ &= \sigma(A-G).\end{aligned}$$ ∎

Theorem 10.5.8 *Let A be an $n\times n$ positive semidefinite matrix. To A, every $n\times n$ unitary matrix U is between $-I_n$ and I_n in the sense that*

$$\sigma(A-I_n) \prec_w \sigma(A-U) \prec_w \sigma(A+I_n). \tag{10.49}$$

Proof. Because (10.49) holds if and only if it holds when A is replaced with W^*AW, where W is unitary, we assume that A is a diagonal matrix with nonnegative diagonal entries. An application of Theorem 10.5.5 results in

$$\begin{aligned}\sigma(A-I_n) &= |\sigma(A)-\sigma(U)| \prec_w \sigma(A-U)\\ &\prec_w \sigma(A)+\sigma(U)=\sigma(A+I_n).\end{aligned}$$ ∎

Corollary 10.5.3 *Let $A\in\mathbb{M}_n$ have polar decomposition $A=U|A|$, where $U\in\mathbb{M}_n$ is unitary and $|A|=(A^*A)^{1/2}$. Then for any unitary $V\in\mathbb{M}_n$*

$$\sigma(A-U) \prec_w \sigma(A-V) \prec_w \sigma(A+U). \tag{10.50}$$

Proof. We show the first inequality; the second one is similar. Notice that

$$\sigma(A-U)=\sigma(U|A|-U)=\sigma(|A|-I_n).$$

Invoking Theorem 10.5.8 reveals the desired weak majorization. ∎

Problems

1. Let z be a complex number. Show that $|1-|z|| \le |1-z| \le 1+|z|$. Show that the first equality occurs if and only if $z \in [0,\infty)$ and that the second equality occurs if and only if $z \in (-\infty, 0]$.

2. Let $A = \left(\begin{smallmatrix} 1 & 1 \\ 1 & 1 \end{smallmatrix}\right)$ and $B = \left(\begin{smallmatrix} -2 & 0 \\ 0 & 0 \end{smallmatrix}\right)$. Then $A+B = \left(\begin{smallmatrix} -1 & 1 \\ 1 & 1 \end{smallmatrix}\right)$. The eigenvalues of A, B, and $A+B$ are, respectively, $\lambda(A) : 2, 0$, $\lambda(B) : -2, 0$, and $\lambda(A+B) : \sqrt{2}, -\sqrt{2}$. By Theorem 10.5.1, $\lambda(A+B) = (\sqrt{2}, -\sqrt{2}) \prec \lambda(A) + \lambda(B) = (0,0)$ which is obviously false. What is wrong with it?

3. Let $x \in \mathbb{R}^n$ and $y, z \in \mathbb{R}^n_+$. If $|x-y| \prec_w z$, show that $|x| \prec_w z^{\downarrow} + y^{\downarrow}$.

4. Let $r_1, \dots, r_n \in (-\infty, \infty)$, $p_1, \dots, p_n \in [0,\infty)$, $q_1, \dots, q_n \in (-\infty, 0]$. If $(r_1, \dots, r_n, -r_n, \dots, -r_1) \prec_w (p_1, \dots, p_n, q_1, \dots, q_n)$, show that

$$(|r_1|, \dots, |r_n|) \prec_w (p_1, \dots, p_n).$$

5. Show by the example with $z = i = \sqrt{-1}$, $A = \left(\begin{smallmatrix} 4 & 2 \\ 2 & 1 \end{smallmatrix}\right)$, and $B = \left(\begin{smallmatrix} 1 & 2 \\ 2 & 4 \end{smallmatrix}\right)$ that the following matrix inequalities do not hold in general:

$$|A - |z|B| \le |A - zB| \le A + |z|B.$$

6. Find two 2×2 Hermitian matrices A and B with real eigenvalues λ_i's and μ_i's ordered as $|\lambda_1| \ge |\lambda_2|$ and $|\mu_1| \ge |\mu_2|$ such that $|\lambda_1 - \mu_1| > \sqrt{2}\|A-B\|_F$. (Check the ordering of eigenvalues in Theorem 10.5.3.)

7. Find two $n \times n$ (non-Hermitian) matrices A and B with real eigenvalues λ_i's and μ_i's ordered decreasingly such that $|\lambda_1 - \mu_1| > \sqrt{n}\,\|A-B\|_F$.

8. Let $A, B \in \mathbb{M}_n$. Let $\lambda_1, \dots, \lambda_n$ and $\mu_1, \dots, \mu_n$ be the eigenvalues of A and B, respectively. If A is normal, show that there is a rearrangement $\mu_{i_1}, \dots, \mu_{i_n}$ of $\mu_1, \dots, \mu_n$ such that $|\lambda_j - \mu_{i_j}| \le \sqrt{n}\,\|A-B\|_F$ for each j.

9. Do the majorizations in Theorem 10.5.6 hold for Hermitian A and B?

10. Let $M = \left(\begin{smallmatrix} A & B \\ C & D \end{smallmatrix}\right)$. Show that $\sigma(M) \prec_w \big(\sigma(A), \sigma(D)\big)^{\downarrow} + \big(\sigma(B), \sigma(C)\big)^{\downarrow}$. In particular, $\sigma_1(M) \le \max\{\sigma_1(A), \sigma_1(D)\} + \max\{\sigma_1(B), \sigma_1(C)\}$.

11. Find examples of $A, B \in \mathbb{M}_n$ such that the following do not hold:

$$\sigma_i(A+B) \ge \sigma_i(A) + \sigma_n(B), \quad i = 1, 2, \dots, n;$$

$$\sum_{i=1}^{k} \sigma_i(A+B) \ge \sum_{i=1}^{k} \sigma_i(A) + \sum_{i=1}^{k} \sigma_{n-i+1}(B), \quad k = 1, 2, \dots, n.$$

12. Let A and B be $n \times n$ positive semidefinite matrices. Show that

$$\big(\operatorname{tr}(A+B)^p\big)^{1/p} \le (\operatorname{tr} A^p)^{1/p} + (\operatorname{tr} B^p)^{1/p} \quad \text{for } p > 1.$$

13. Let $A \in \mathbb{M}_n$ be Hermitian and let $x \in \mathbb{C}^n$ be a row vector. Show that

$$\lambda(A + x^*x) \prec \big(\lambda_1(A) + xx^*,\, \lambda_2(A), \ldots,\, \lambda_n(A)\big).$$

14. Let A be an $n \times n$ matrix. Denote $\mathcal{S}(A) = \frac{A - A^*}{2}$. Show that $\mathcal{S}(A)$ is skew-Hermitian and for any $n \times n$ skew-Hermitian matrix G,

$$\sigma(A - \mathcal{S}(A)) \prec_w \sigma(A - G).$$

15. Let $A \in \mathbb{M}_n$ have a singular value decomposition $A = UDV$, where U and V are unitary. Show that for any $n \times n$ unitary matrix W,

$$\sigma(A - UV) \prec_w \sigma(A - W) \prec_w \sigma(A + UV).$$

16. If $A = (a_{ij}) \in \mathbb{M}_n$ is normal and $d(A) = (a_{11}, \ldots, a_{nn})$, show that

$$\Re\big(d(A)\big) \prec \Re\big(\lambda(A)\big).$$

17. Let A be a square matrix and let $\mathcal{H}(A) = \frac{A + A^*}{2}$. Show that

$$\lambda\big(\mathcal{H}(A)\big) \le \big|\lambda\big(\mathcal{H}(A)\big)\big| \prec_w \sigma(A).$$

18. **(Fan-Hoffman)** Let $A \in \mathbb{M}_n$. Order and denote the eigenvalues of the Hermitian matrix $\mathcal{H}(A) = \frac{A + A^*}{2}$ by $h_1 \ge h_2 \ge \cdots \ge h_n$. Show that

$$h_i \le \sigma_i(A), \quad i = 1, 2, \ldots, n.$$

However, if we rearrange the absolute values of these eigenvalues of $\mathcal{H}(A)$ in the decreasing order $|h_{i_1}| \ge |h_{i_2}| \ge \cdots \ge |h_{i_n}|$, show by an example that $|h_{i_t}| \le \sigma_t(A)$ does not hold in general.

19. Let A and B be n-square matrices. Show that

$$\big|\sigma\big((A + B)^2\big) - \sigma(A^2 + B^2)\big| \prec_w \sigma(AB + BA).$$

20. Let A and B be n-square matrices. Discuss and determine which ones (if not all) of the following majorization inequalities hold(s) in

$$\sigma(A + B) \prec_w \sigma(|A + B|) \prec_w \sigma(|A| + |B|) \prec_w \sigma(|A|) + \sigma(|B|).$$

21. Let A and B be $n \times n$ positive semidefinite matrices. Show that

$$|\lambda(A - B)| \prec_{\text{wlog}} \lambda(A + B).$$

22. Let A and B be $n \times n$ positive semidefinite matrices. Show that

$$\sigma_i(A - B) \le \sigma_i(A \oplus B), \quad i = 1, 2, \ldots, n.$$

23. Show by an example that $\sigma(|A| - |B|) \prec_w \sigma(A - B)$ is false.

24. Let A and B be $n \times n$ positive semidefinite matrices. Show that

$$\big(\lambda^2(A), \lambda^2(B)\big) \prec \big(\lambda(A^2 + B^2), 0\big) \prec_w \big(\lambda^2(A + B), 0\big).$$

25. Let $A, B \in \mathbb{M}_{m\times n}$. Show that for all nonnegative r, s with $r + s + 1 \le n$

$$2\lambda_{r+s+1}(A^*A + B^*B) \le \sigma^2_{r+1}(A - B) + \sigma^2_{s+1}(A + B)$$

and

$$2\lambda(A^*A + B^*B) \prec \sigma^2(A - B) + \sigma^2(A + B).$$

26. Let A and B be $n \times n$ positive definite matrices. Show that

$$\sum_{i=1}^{n} \frac{1}{\lambda_i(A) + \lambda_{n-i+1}(B)} \le \operatorname{tr}\big((A + B)^{-1}\big) \le \sum_{i=1}^{n} \frac{1}{\lambda_i(A) + \lambda_i(B)}$$

and

$$\prod_{i=1}^{n} \big(\lambda_i(A) + \lambda_{n-i+1}(B)\big) \ge \det(A + B) \ge \prod_{i=1}^{n} \big(\lambda_i(A) + \lambda_i(B)\big).$$

27. Let A and B be $n \times n$ strictly contractive matrices, and let

$$H = \begin{pmatrix} (I - A^*A)^{-1} & (I - A^*B)^{-1} \\ (I - B^*A)^{-1} & (I - B^*B)^{-1} \end{pmatrix}.$$

Show that

(a) The diagonal entries of H are greater than 1, i.e., $d(H) > (1, \dots, 1)$.

(b) $d(H) \prec \big(\lambda((I - A^*A)^{-1}), \lambda((I - B^*B)^{-1})\big) \prec \lambda(H)$.

(c) $H \ge \left(\begin{smallmatrix} I & I \\ I & I \end{smallmatrix}\right)$, concluding that H has at least n eigenvalues that are greater than or equal to 2. [Hint: Use Hua's matrix; Sect. 7.4.]

28. Let $A = (a_{ij})$ and $B = (b_{ij})$ be $n \times n$ matrices having singular values $\sigma_1(A) \ge \cdots \ge \sigma_n(A)$ and $\sigma_1(B) \ge \cdots \ge \sigma_n(B)$, respectively. Prove

$$\sum_{i=1}^{n} \big(\sigma_i(A) - \sigma_i(B)\big)^2 \le \sum_{i,j=1}^{n} |a_{ij}|^2 + \sum_{i,j=1}^{n} |b_{ij}|^2 - 2\Re\Big(\sum_{i,j=1}^{n} \overline{a_{ij}} b_{ij}\Big).$$

29. Let $L(X) = -\ln\big(\det(X)\big)$, where X is a positive definite matrix. Show that $L(\alpha X_1 + \beta X_2) \le \alpha L(X_1) + \beta L(X_2)$ for all positive definite matrices X_1 and X_2 (of the same order), and $\alpha, \beta > 0$ with $\alpha + \beta = 1$. That is, L is a convex function on the domain of positive definite matrices.

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10.6 Majorization for Matrix Product

We showed in the previous section that for $m \times n$ matrices A and B

$$\sigma(A+B) \prec_w \sigma(A) + \sigma(B).$$

The analogue of this for product is (Corollary 10.6.2): For $A, B \in \mathbb{M}_n$,

$$\sigma(AB) \prec_w \sigma(A) \circ \sigma(B). \tag{10.51}$$

In particular, for $n \times n$ positive semidefinite matrices A and B, we have

$$\lambda(AB) \prec_w \sigma(AB) \prec_w \lambda(A) \circ \lambda(B). \tag{10.52}$$

Note that the eigenvalues and singular values of AB need not be the same.

Example 10.6.1 Let $A = \left(\begin{smallmatrix} 1 & 0 \\ 0 & 0 \end{smallmatrix}\right)$ and $B = \left(\begin{smallmatrix} 1 & 1 \\ 1 & 1 \end{smallmatrix}\right)$. Then A and B are positive semidefinite. A computation shows $\sigma(AB) = (\sqrt{2}, 0) \neq \lambda(AB) = (1, 0)$.

Note that $\lambda(XY) = \lambda(YX)$ for all $X, Y \in \mathbb{M}_n$, while for $X = \left(\begin{smallmatrix} 1 & 0 \\ 0 & 0 \end{smallmatrix}\right)$, $Y = \left(\begin{smallmatrix} 0 & 0 \\ 1 & 0 \end{smallmatrix}\right)$, $\sigma(XY) \neq \sigma(YX)$ (unless X and Y are normal. Problem 11). ∎

Keep in mind that the components of $\lambda(\cdot)$ and $\sigma(\cdot)$, that is, the nonnegative eigenvalues and singular values, are arranged in decreasing order.

This section is devoted to the majorizations of the eigenvalues and singular values of matrix product AB, the power of product $(AB)^m$, and the product of the powers $A^m B^m$. For the case of eigenvalues, A and B are assumed to be positive semidefinite (otherwise AB may have non-real eigenvalues).

Theorem 10.6.1 (Lidskii Product Inequality) *Let $A, B \in \mathbb{M}_n$ be positive semidefinite and let $1 \leq i_1 < \cdots < i_k \leq n$ be any index set. Then*

$$\prod_{t=1}^{k} \lambda_{i_t}(AB) \leq \prod_{t=1}^{k} \lambda_{i_t}(A)\lambda_t(B). \tag{10.53}$$

The inequality becomes an equality when $k = n$.

Proof. Note that $\lambda_i(XY) \leq \lambda_i(X)\lambda_1(Y)$ for any positive semidefinite matrices X and Y of the same order and for any index i (Theorem 8.4.2). Apply this to the compound matrix $C_k(AB) = C_k(A)C_k(B)$. Because $\prod_{t=1}^{k} \lambda_{i_t}(\cdot)$ is an eigenvalue of $C_k(\cdot)$, say, αth or indexed by $\alpha = (i_1, \ldots, i_k)$, we have

$$\begin{aligned} \prod_{t=1}^{k} \lambda_{i_t}(AB) &= \lambda_\alpha\big(C_k(AB)\big) = \lambda_\alpha\big(C_k(A)C_k(B)\big) \\ &\leq \lambda_\alpha\big(C_k(A)\big)\lambda_1\big(C_k(B)\big) = \prod_{t=1}^{k} \lambda_{i_t}(A)\lambda_t(B). \end{aligned}$$

Equality holds when $k = n$ because $\det(AB) = \det A \det B$. ∎

As AB and BA have the same eigenvalues, AB on the left-hand side of (10.53) can be changed to BA, that is, $\prod_{t=1}^{k} \lambda_{i_t}(BA) \le \prod_{t=1}^{k} \lambda_{i_t}(A)\lambda_t(B)$.

Example 10.6.2 Let $A, B \in \mathbb{M}_n$ be positive semidefinite. Then

$$\lambda_i(AB)\lambda_j(AB) \le \lambda_i(A)\lambda_j(A)\lambda_1(B)\lambda_2(B), \quad i \ne j.$$

Note that the above inequality need not be true if $i = j$. ∎

Setting $i_t = t$ in Lidskii's inequality, we obtain for each $k \le n$,

$$\prod_{t=1}^{k} \lambda_t(AB) \le \prod_{t=1}^{k} \lambda_t(A)\lambda_t(B). \tag{10.54}$$

(10.53) and (10.54) result in the following majorizations.

Corollary 10.6.1 *Let $A, B \in \mathbb{M}_n$ be positive semidefinite. Then*

$$\lambda(A) \circ \lambda^{\uparrow}(B) \prec_{\log} \lambda(AB) \prec_{\log} \lambda(A) \circ \lambda(B) \tag{10.55}$$

and

$$\lambda(A) \circ \lambda^{\uparrow}(B) \prec_{w} \lambda(AB) \prec_{w} \lambda(A) \circ \lambda(B). \tag{10.56}$$

Proof. The weak majorizations in (10.56) are immediate from the log-majorizations in (10.55) (by Theorem 10.3.8). $\lambda(AB) \prec_{\log} \lambda(A) \circ \lambda(B)$ is the same as (10.54). So, we only need to show $\lambda(A) \circ \lambda^{\uparrow}(B) \prec_{\log} \lambda(AB)$.

We show that $\lambda(AB) \prec_{\log} \lambda(A) \circ \lambda(B)$ implies $\lambda(A) \circ \lambda^{\uparrow}(B) \prec_{\log} \lambda(AB)$.

First assume that A (or B) is nonsingular. (10.53) is rewritten as

$$\prod_{t=1}^{k} \lambda_{i_t}(AB)/\lambda_{i_t}(A) \le \prod_{t=1}^{k} \lambda_t(B). \tag{10.57}$$

Observe that $1/\lambda_{i_t}(A) = \lambda_{n-i_t+1}(A^{-1})$ for each t. (10.57) is the same as $\prod_{t=1}^{k} \lambda_{i_t}(AB)\lambda_{n-i_t+1}(A^{-1}) \le \prod_{t=1}^{k} \lambda_t(B)$. Replacing A with A^{-1} yields $\prod_{t=1}^{k} \lambda_{i_t}(A^{-1}B)\lambda_{n-i_t+1}(A) \le \prod_{t=1}^{k} \lambda_t(B)$. Replacing B with AB reveals $\prod_{t=1}^{k} \lambda_{i_t}(B)\lambda_{n-i_t+1}(A) \le \prod_{t=1}^{k} \lambda_t(AB)$, that is, for every $k = 1, \ldots, n$,

$$\max_{1 \le i_1 < \cdots < i_k \le n} \prod_{t=1}^{k} \lambda_{i_t}(B)\lambda_{n-i_t+1}(A) \le \prod_{t=1}^{k} \lambda_t(AB).$$

Equality holds if $k = n$. So, $\lambda^{\uparrow}(A) \circ \lambda^{\downarrow}(B) \prec_{\log} \lambda(AB)$ for nonsingular A.

If A is singular, we use $A_\varepsilon = A + \varepsilon I$, $0 < \varepsilon < 1$, to replace A in the above argument to get $\lambda^{\uparrow}(A_\varepsilon) \circ \lambda^{\downarrow}(B) \prec_{\log} \lambda(A_\varepsilon B)$. Because $\lambda(A_\varepsilon) \to \lambda(A)$ and $\lambda(A_\varepsilon B) \to \lambda(AB)$ as $\varepsilon \to 0^+$ (see Sect. 2.6), by taking the limits of both sides, we obtain $\lambda^{\uparrow}(A) \circ \lambda^{\downarrow}(B) \prec_{\log} \lambda(AB)$ for singular A.

The proof is complete as $\lambda(A) \circ \lambda^{\uparrow}(B)$ is a permutation of $\lambda^{\uparrow}(A) \circ \lambda^{\downarrow}(B)$. ∎

Note: If one uses (10.54) instead of (10.53) in the above proof, one would arrive at $\prod_{t=1}^{k} \lambda_t(B)\lambda_{n-t+1}(A) \le \prod_{t=1}^{k} \lambda_t(AB)$. However, this is insufficient to conclude $\lambda^{\uparrow}(A) \circ \lambda^{\downarrow}(B) \prec_{\log} \lambda(AB)$ because $\lambda_t(B)\lambda_{n-t+1}(A)$ (i.e., the components of $\lambda^{\uparrow}(A) \circ \lambda^{\downarrow}(B)$) may not decrease as t increases. For example, take $\lambda(A) = (3, 2, 1)$ and $\lambda(B) = (2, 2, 1)$. Then $\lambda^{\uparrow}(A) \circ \lambda^{\downarrow}(B) = (2, 4, 3)$.

We next study the analogous majorizations for singular values.

For positive semidefinite matrices A and B (of the same order), the eigenvalues of AB are necessarily nonnegative because AB and $A^{1/2}BA^{1/2}$ have the same eigenvalues and the latter is positive semidefinite. However, the eigenvalues and singular values of AB need not be the same (Example 10.6.1). So, the inequalities of eigenvalues and singular values of the product AB are generally different for positive semidefinite matrices A and B. In summary, for $A, B \ge 0$, $\lambda(A) = \sigma(A)$, $\lambda(AB) \ne \sigma(AB)$; for normal A and B, $\sigma(AB) = \sigma(BA)$; for general A and B, $\lambda(AB) = \lambda(BA)$, $\sigma(AB) \ne \sigma(BA)$.

Theorem 10.6.2 (Gelfand-Naimark Inequality) *Let A and B be $n \times n$ matrices and let $1 \le i_1 < \cdots < i_k \le n$ be any index set. Then*

$$\prod_{t=1}^{k} \sigma_{i_t}(AB) \le \prod_{t=1}^{k} \sigma_{i_t}(A)\sigma_t(B). \tag{10.58}$$

The inequality becomes an equality when $k = n$.

Proof. Since $\sigma_j^2(X) = \lambda_j(X^*X)$ for any matrix X and index j, we have

$$\prod_{t=1}^{k} \sigma_{i_t}^2(AB) = \prod_{t=1}^{k} \lambda_{i_t}(B^*A^*AB) = \prod_{t=1}^{k} \lambda_{i_t}(A^*ABB^*).$$

An application of Lidskii's inequality gives the desired inequality. ∎

Setting $i_t = t$ in the Gelfand-Naimark inequality, we obtain

$$\prod_{t=1}^{k} \sigma_t(AB) \le \prod_{t=1}^{k} \sigma_t(A)\sigma_t(B). \tag{10.59}$$

This is the same as $\sigma(AB) \prec_{\log} \sigma(A) \circ \sigma(B)$ (see Horn's Theorem 10.4.6).

Corollary 10.6.2 *Let A and B be $n \times n$ matrices. Then*

$$\sigma(A) \circ \sigma^{\uparrow}(B) \prec_{\log} \sigma(AB) \prec_{\log} \sigma(A) \circ \sigma(B) \tag{10.60}$$

and

$$\sigma(A) \circ \sigma^{\uparrow}(B) \prec_{w} \sigma(AB) \prec_{w} \sigma(A) \circ \sigma(B). \tag{10.61}$$

Proof. (10.60) is immediate from (10.55) of Corollary 10.6.1 by writing

$$\sigma^2(AB) = \lambda(B^*A^*AB) = \lambda(A^*A \cdot BB^*).$$

(10.61) follows from (10.60) as log-majorization implies majorization.

Below is an alternative proof of the first inequality in (10.60) by (10.58).

Assume that A and B are nonsingular matrices by continuity (Sect. 2.6).

Take the logarithm for both sides of (10.58) and move the terms with A to the left-hand side. As $\sum \ln \sigma_t(B)$ is independent of the indices i's, we obtain

$$\max_{1\le i_1<\cdots<i_k\le n} \sum_{i=1}^{k} \big(\ln \sigma_{i_t}(AB) - \ln \sigma_{i_t}(A)\big) \le \sum_{t=1}^{k} \ln \sigma_t(B).$$

It follows that $\ln \sigma(AB) - \ln \sigma(A) \prec \ln \sigma(B)$. Replacing A by A^{-1}, then B by AB, reveals $\ln \sigma(B) - \ln \sigma(A^{-1}) \prec \ln \sigma(AB)$, which is the same as $\ln\big(\sigma(B) \circ \sigma^{\uparrow}(A)\big) \prec \ln \sigma(AB)$, that is, $\sigma(B) \circ \sigma^{\uparrow}(A) \prec_{\log} \sigma(AB)$. Note that $\sigma(B) \circ \sigma^{\uparrow}(A)$ is a permutation of $\sigma(A) \circ \sigma^{\uparrow}(B)$. So, $\sigma(A) \circ \sigma^{\uparrow}(B) \prec_{\log} \sigma(AB)$. ∎

We now study the majorizations concerning the power of product $(AB)^m$ and the product of the powers A^mB^m, where $A, B \in \mathbb{M}_n$ are positive semidefinite. Note that for Hermitian A and B, AB need not be Hermitian. For $A = \left(\begin{smallmatrix} 1 & 0 \\ 0 & -1 \end{smallmatrix}\right)$, $B = \left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$, $AB = \left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right)$ has eigenvalues $\pm i$. However, if A and B are positive semidefinite, then for any positive integer m, the eigenvalues of A^mB^m and $(AB)^m$ are all nonnegative because the eigenvalues of the product of two positive semidefinite matrices are nonnegative. Note that $(AB)^m = (AB \cdots A)B$ is a product of two positive semidefinite matrices. Moreover, $\lambda(A^m) = \lambda^m(A)$ for any $A \in \mathbb{M}_n$ and positive integer m, while $\sigma(A^m) \prec_{\log} \sigma^m(A)$ as $\sigma(AB) \prec_{\log} \sigma(AB)$ for $A, B \in \mathbb{M}_n$ (Horn's singular value majorization, Theorem 10.4.6). Furthermore, by Weyl's inequality (Theorem 10.4.5), $|\lambda(AB)| \prec_{\log} \sigma(AB)$ for all $A, B \in \mathbb{M}_n$.

Example 10.6.3 Let $A = \left(\begin{smallmatrix} 1 & 0 \\ 0 & 0 \end{smallmatrix}\right)$ and $B = \left(\begin{smallmatrix} 1 & 1 \\ 1 & 1 \end{smallmatrix}\right)$. Both A and B are positive semidefinite matrices. Upon computations, we obtain

$$\lambda(AB) = (1, 0), \quad \lambda(A^3B^3) = (4, 0), \quad \lambda(A) \circ \lambda(B) = (2, 0),$$

$$\sigma^2(AB) = (2, 0), \quad \sigma\big((AB)^2\big) = (\sqrt{2}, 0), \quad \sigma(A^2B^2) = (2\sqrt{2}, 0).$$ ∎

Let A and B be $n \times n$ positive semidefinite matrices and let m be a positive integer. By (10.55), we have $\lambda((AB)^m) \prec_{\log} \lambda^m(A) \circ \lambda^m(B)$ and

$$\lambda(A^mB^m) \prec_{\log} \lambda^m(A) \circ \lambda^m(B).$$

(As usual, $\lambda^t(X) = \big(\lambda_1^t(X), \ldots, \lambda_n^t(X)\big) = \big((\lambda_1(X))^t, \ldots, (\lambda_n(X))^t\big)$.) Thus

$$\lambda^{1/m}(A^mB^m) \prec_{\log} \lambda(A) \circ \lambda(B).$$

This says that $\lambda^{1/m}(A^mB^m)$ is bounded above by $\lambda(A)\circ\lambda(B)$ in majorization. In what follows we show that it is bounded below by $\lambda(AB)$, or equivalently, $\lambda\big((AB)^m\big) = \lambda^m(AB) \prec_{\log} \lambda(A^mB^m)$. Putting these together, we have

$$\lambda(AB) \prec_{\log} \lambda^{1/m}(A^mB^m) \prec_{\log} \lambda(A)\circ\lambda(B). \tag{10.62}$$

In fact, more can be said about $\lambda^{1/m}(A^mB^m)$ (see Theorem 10.6.3).

Recall that for $n\times n$ positive semidefinite matrices A and B (Sect. 8.4),

$$\lambda_1(AB) \le \lambda_1(A)\lambda_1(B), \quad \lambda_n(AB) \ge \lambda_n(A)\lambda_n(B),$$

and that for arbitrary $n\times n$ matrices X and Y (Sect. 4.1 and Sect. 4.2),

$$\max_j |\lambda_j(XY)| \le \sigma_1(XY) \le \sigma_1(X)\sigma_1(Y)$$

and

$$\min_j |\lambda_j(XY)| \ge \sigma_n(XY) \ge \sigma_n(X)\sigma_n(Y).$$

Lemma 10.6.1 *Let A and B be $n\times n$ positive semidefinite matrices. Then for any positive integers p and q with $p\le q$,*

$$\lambda_1(AB) \le \lambda_1^{1/p}(A^pB^p) \le \lambda_1^{1/q}(A^qB^q) \le \lambda_1(A)\lambda_1(B) \tag{10.63}$$

and

$$\lambda_n(AB) \ge \lambda_n^{1/p}(A^pB^p) \ge \lambda_n^{1/q}(A^qB^q) \ge \lambda_n(A)\lambda_n(B). \tag{10.64}$$

Proof. We first use induction on positive integers m to show that

$$\lambda_1^{1/m}(A^mB^m) \le \lambda_1^{1/m+1}(A^{m+1}B^{m+1}). \tag{10.65}$$

If $m=1$, then

$$\lambda_1(AB) \le \sigma_1(AB) = \lambda_1^{1/2}(ABBA) = \lambda_1^{1/2}(A^2B^2).$$

Suppose that the inequalities hold for $m-1$. We show it holds for m. Put

$$X = A^{(m+1)/2}B^{(m+1)/2}, \quad Y = B^{(m-1)/2}A^{(m-1)/2}.$$

Then $\lambda_1(XY) = \lambda_1(A^mB^m)$. Because $\lambda_1(XY) \le \sigma_1(XY) \le \sigma_1(X)\sigma_1(Y)$,

$$\lambda_1(A^mB^m) \le \sigma_1(X)\sigma_1(Y) = \lambda_1^{1/2}(A^{m+1}B^{m+1})\lambda_1^{1/2}(A^{m-1}B^{m-1}).$$

The induction hypothesis on $m-1$ for (10.65) ensures

$$\lambda_1(A^{m-1}B^{m-1}) \le \lambda_1^{(m-1)/m}(A^mB^m).$$

It follows that

$$\lambda_1(A^m B^m) \le \lambda_1^{1/2}(A^{m+1}B^{m+1})\lambda_1^{(m-1)/2m}(A^m B^m).$$

Thus,

$$\lambda_1^{(m+1)/2m}(A^m B^m) \le \lambda_1^{1/2}(A^{m+1}B^{m+1}),$$

that is,

$$\lambda_1^{1/m}(A^m B^m) \le \lambda_1^{1/(m+1)}(A^{m+1}B^{m+1}).$$

This proves (10.65), which immediately implies (10.63) by observing that

$$\lambda_1^{1/m}(A^m B^m) \le \lambda_1^{1/m}(A^m)\lambda_1^{1/m}(B^m) = \lambda_1(A)\lambda_1(B).$$

The second part (10.64) for the smallest eigenvalues is similarly proven. ∎

Example 10.6.4 Let A and B be $n \times n$ positive semidefinite matrices. Then

$$\lambda_1(AB) \le \lambda_1^{1/2}(A^2B^2) \le \lambda_1^{1/3}(A^3B^3) \le \cdots \le \lambda_1(A)\lambda_1(B). \blacksquare$$

The following theorem states that $\lambda^{1/t}(A^t B^t)$, bounded below by $\lambda(AB)$ and above by $\lambda(A) \circ \lambda(B)$, increases in log-majorization as t increases.

Theorem 10.6.3 (Wang-Gong) *Let A and B be $n \times n$ positive semidefinite matrices. Then for any positive integers p and q with $p \le q$,*

$$\lambda(AB) \prec_{\log} \lambda^{1/p}(A^p B^p) \prec_{\log} \lambda^{1/q}(A^q B^q) \prec_{\log} \lambda(A) \circ \lambda(B).$$

Proof. It is sufficient to show that for any positive integer m,

$$\lambda^{1/m}(A^m B^m) \prec_{\log} \lambda^{1/m+1}(A^{m+1}B^{m+1}).$$

Considering the compound matrix $C_k(A^m B^m)$, where $1 \le k \le n$, we have

$$C_k(A^m B^m) = C_k(A^m)C_k(B^m) = \big(C_k(A)\big)^m \big(C_k(B)\big)^m.$$

Note that

$$\lambda_1\big(\big(C_k(A)\big)^m \big(C_k(B)\big)^m\big) = \lambda_1\big(C_k(A^m B^m)\big) = \prod_{i=1}^{k} \lambda_i(A^m B^m)$$

and that $C_k(A^m)$ and $C_k(B^m)$ are positive semidefinite. By (10.65),

$$\begin{aligned}
\prod_{i=1}^{k} \lambda_i^{1/m}(A^m B^m) &= \lambda_1^{1/m}\Big(\big(C_k(A)\big)^m \big(C_k(B)\big)^m\Big) \\
&\le \lambda_1^{1/m+1}\Big(\big(C_k(A)\big)^{m+1} \big(C_k(B)\big)^{m+1}\Big) \\
&= \prod_{i=1}^{k} \lambda_i^{1/m+1}(A^{m+1}B^{m+1}).
\end{aligned}$$

When $k = n$, equality holds for any positive integer m because

$$\prod_{i=1}^{n} \lambda_i^{1/m}(A^m B^m) = \big(\det(A^m B^m)\big)^{1/m} = \det A \det B. \ \blacksquare$$

Theorem 10.6.3 refines $\lambda(AB) \prec_{\log} \lambda(A) \circ \lambda(B)$ (see (10.55)). The next result, equivalent to Theorem 10.6.3, in reverse form, shows a refinement of the analogous inequality $\lambda(A) \circ \lambda^{\uparrow}(B) \prec_{\log} \lambda(AB)$.

Corollary 10.6.3 *Let A and B be $n \times n$ positive semidefinite matrices. Then for any positive integers p and q with $p \le q$,*

$$\lambda(A) \circ \lambda^{\uparrow}(B) \prec_{\log} \lambda^q(A^{1/q} B^{1/q}) \prec_{\log} \lambda^p(A^{1/p} B^{1/p}) \prec_{\log} \lambda(AB).$$

Proof. Theorem 10.6.3 ensures that $\lambda^{1/p}(A^p B^p) \prec_{\log} \lambda^{1/q}(A^q B^q)$, implying $\lambda^q(A^p B^p) \prec_{\log} \lambda^p(A^q B^q)$. Replacing A by $A^{1/pq}$ and B by $B^{1/pq}$, we obtain the desired log-majorization in the middle.

Set $p = 1$ to get the last log-majorization; the first log-majorization is due to the fact that $\lambda(A^{1/q}) \circ \lambda^{\uparrow}(B^{1/q}) \prec_{\log} \lambda(A^{1/q} B^{1/q})$ (see (10.55)). $\blacksquare$

The following majorizations are immediate. For instance, the first log-majorization in Theorem 10.6.3 reveals $\lambda^m(AB) \prec_{\log} \lambda(A^m B^m)$.

Corollary 10.6.4 *Let A and B be $n \times n$ positive semidefinite matrices and let m be a positive integer. Then*

$$\lambda^m(A) \circ (\lambda^m(B))^{\uparrow} \prec_{\log} \lambda^m(AB) \prec_{\log} \lambda(A^m B^m) \prec_{\log} \lambda^m(A) \circ \lambda^m(B).$$

Consequently,

$$\lambda^m(A) \circ (\lambda^m(B))^{\uparrow} \prec_{w} \lambda^m(AB) \prec_{w} \lambda(A^m B^m) \prec_{w} \lambda^m(A) \circ \lambda^m(B).$$

The analogous majorizations for singular values of $A, B \in \mathbb{M}_n$ would be

$$\sigma^m(A) \circ (\sigma^m(B))^{\uparrow} \prec_{\log} \sigma^m(AB) \prec_{\log} \sigma(A^m B^m) \prec_{\log} \sigma^m(A) \circ \sigma^m(B).$$

It is not difficult to show that the first and the third log-majorizations are indeed true. However, the one in the middle is not true in general unless A and B are normal. For example, take $A = \left(\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right)$, $B = I_2$, and $m = 2$. Then $\sigma^m(AB) = \sigma^2(A) = (1, 0) \not\prec_{\log} (0, 0) = \sigma(A^2) = \sigma(A^m B^m)$. Moreover, unlike $\lambda^m(AB) = \lambda((AB)^m)$, $\sigma^m(AB)$ and $\sigma((AB)^m)$ need not be the same even for positive semidefinite A and B. For example, take $A = \left(\begin{smallmatrix} 1 & 0 \\ 0 & 0 \end{smallmatrix}\right)$ and $B = \left(\begin{smallmatrix} 1 & 1 \\ 1 & 1 \end{smallmatrix}\right)$. Then $\sigma^2(AB) = (2, 0) \ne \sigma((AB)^2) = (\sqrt{2}, 0)$.

Corollary 10.6.5 (Lieb-Thirring) *Let A and B be $n \times n$ positive semidefinite matrices. Then for any positive integer m,*

$$\operatorname{tr}(AB)^m \le \operatorname{tr}(A^m B^m). \tag{10.66}$$

The inequality becomes an equality if and only if $m = 1$ or $AB = BA$.

Proof. The inequality is immediate from $\lambda^m(AB) \prec_w \lambda(A^m B^m)$ in Corollary 10.6.4. We show that equality holds if and only if $m = 1$ or $AB = BA$.

Sufficiency is obvious. For necessity, we first consider the case $m = 2$, that is, $\operatorname{tr}(AB)^2 = \operatorname{tr}(A^2B^2)$. Without loss of generality, we may assume that A is a diagonal matrix with diagonal entries $a_1, \ldots, a_n$. Then

$$\begin{aligned} \operatorname{tr}(A^2B^2) - \operatorname{tr}(AB)^2 &= \sum\nolimits_{i,j} a_i^2 |b_{ij}|^2 - \sum\nolimits_{i,j} a_i a_j |b_{ij}|^2 \\ &= \sum\nolimits_{i<j} (a_i - a_j)^2 |b_{ij}|^2 = 0. \end{aligned}$$

Thus, $a_i b_{ij} = a_j b_{ij}$ for every pair of i and j, that is, $AB = BA$.

Let $m > 2$ be given. We show that $\operatorname{tr}(AB)^m = \operatorname{tr}(A^m B^m)$ implies $\operatorname{tr}(AB)^2 = \operatorname{tr}(A^2B^2)$, which leads to $AB = BA$, as we have just proven.

If $\operatorname{tr}(AB)^2 \neq \operatorname{tr}(A^2B^2)$, then $\lambda^2(AB)$ is not equal to $\lambda(A^2B^2)$. Applying the strictly convex and increasing function $f(t) = t^{m/2}$, $t \geq 0$, to the weak majorization $\lambda^2(AB) \prec_w \lambda(A^2B^2)$, and using Theorem 10.3.7, we have

$$\operatorname{tr}(AB)^m = \sum\nolimits_i \lambda_i^{m/2}((AB)^2) < \sum\nolimits_i \lambda_i^{m/2}(A^2B^2).$$

On the other hand, since $x \prec_w y \Rightarrow x^m \prec_w y^m$ for $x, y \in \mathbb{R}^n_+$ (Problem 38, Sect. 10.3) and $\lambda^{1/2}(A^2B^2) \prec_w \lambda^{1/m}(A^m B^m)$ (Theorem 10.6.3), which results in $\lambda^{m/2}(A^2B^2) \prec_w \lambda(A^m B^m)$, we have

$$\sum\nolimits_i \lambda_i^{m/2}(A^2B^2) \leq \sum\nolimits_i \lambda_i(A^m B^m) = \operatorname{tr}(A^m B^m).$$

So, if $\operatorname{tr}(AB)^2 \neq \operatorname{tr}(A^2B^2)$, then $\operatorname{tr}(AB)^m < \operatorname{tr}(A^m B^m)$, equivalently, if $\operatorname{tr}(AB)^m = \operatorname{tr}(A^m B^m)$, then $\operatorname{tr}(AB)^2 = \operatorname{tr}(A^2B^2)$, yielding $AB = BA$. ∎

Inequality(10.66) reveals immediately that for $A, B \geq 0$ and $m \geq 1$,

$$\operatorname{tr}(AB)^m \leq \operatorname{tr}(A^m B^m) \leq \lambda_1(A^m) \operatorname{tr}(B^m) = \|A\|_{\mathrm{sp}}^m \operatorname{tr}(B^m).$$

Problems

1. Let $A, B \in \mathbb{M}_n$. Recall from Sect. 2.4 that AB and BA have the same (set of) eigenvalues, or $\lambda(AB) = \lambda(BA)$. Does $\sigma(AB) = \sigma(BA)$?

2. Find example: $A, B \in \mathbb{M}_n$, the eigenvalues of A, B, and AB are all real, but $\lambda_1(AB) \not\leq \lambda_1(A)\lambda_1(B)$, where $\lambda_1(X)$ is the largest eigenvalue of X.

3. Assume that the matrices are $n \times n$. $\lambda_i(\cdot)$ denotes any eigenvalue.
   (a) If $A, B \geq 0$, show that $\lambda_i(AB) \geq 0$.
   (b) If $A \geq 0$ and B is Hermitian, show that $\lambda_i(AB) \in \mathbb{R}$.
   (c) Find an example: A and B are Hermitian, $\lambda_i(AB)$ is non-real.

4. Show that $\max_i |\lambda_i(AB)| \leq (\max_i |\lambda_i(A)|)(\max_i |\lambda_i(B)|)$ for normal (in particular, Hermitian) matrices $A, B \in \mathbb{M}_n$, where $\lambda_i(X)$ are the eigenvalues of $X \in \mathbb{M}_n$. Is this true if A or B is non-normal?

5. Show that the inequalities in Lemma 10.6.1 hold for singular value σ_1 (and σ_n) in place of λ_1 (resp., λ_n) when A and B are normal matrices. Find an example: A or B is not normal and $\sigma_1(AB) \not\leq \sqrt{\sigma_1(A^2B^2)}$.

6. Let A be an $n \times n$ nonsingular matrix. Show that for $j = 1, \ldots, n$,

$$\sigma_j(A^{-1}) = \frac{1}{\sigma_{n-j+1}(A)} = \frac{1}{\sigma_j^{\uparrow}(A)}, \quad \text{i.e.,} \quad \sigma_j^{\downarrow}(A^{-1})\sigma_j^{\uparrow}(A) = 1.$$

7. Let A and B be positive semidefinite matrices of the same order, and let p and q be positive numbers such that $\frac{1}{p} + \frac{1}{q} = 1$. Show that

$$\mathrm{tr}(A^{1/p}B^{1/q}) \leq (\mathrm{tr}A)^{1/p}(\mathrm{tr}B)^{1/q}.$$

8. Let A be an $n \times n$ positive semidefinite matrix, and let p and q be positive numbers such that $\frac{1}{p} + \frac{1}{q} = 1$. For all $n \times n$ positive semidefinite matrices B with $\mathrm{tr}B^q = 1$, show that the trace inequality holds:

$$\mathrm{tr}(AB) \leq (\mathrm{tr}A^p)^{1/p}.$$

   If $A \neq 0$, show that equality occurs if and only if $B^q = \frac{1}{\mathrm{tr}A^p}A^p$.

9. Let A and B be positive semidefinite matrices of the same order. Show that $\lambda_1^m(A^{1/m}B^{1/m})$ decreases as m increases, that is,

$$\lambda_1^{m+1}(A^{1/m+1}B^{1/m+1}) \leq \lambda_1^m(A^{1/m}B^{1/m}).$$

10. Let $|X| = (X^*X)^{1/2}$ for $X \in \mathbb{M}_n$. For $A, B \in \mathbb{M}_n$, show that

$$|\lambda(AB)| \prec_{\mathrm{wlog}} \lambda(|A|) \circ \lambda(|B|).$$

11. Show that $\sigma(AB) = \sigma(BA) = \sigma(|A|\,|B|)$ for normal matrices A and B of the same order. Is this true in general if A or B is not normal?

12. Find examples (of some matrices and some m) for the statements:

    (a) $|AB| \neq |A|\,|B|$ (where $|X| = (A^*A)^{1/2}$).
    (b) $\sigma(PQ) \neq \lambda(PQ)$, where $P, Q \in \mathbb{M}_n$ are positive semidefinite.
    (c) $\sigma(XY) \neq \sigma(YX)$ (unlike $\lambda(XY) = \lambda(YX)$), where $X, Y \in \mathbb{M}_n$.
    (d) $\sigma(Z^m) \neq \sigma^m(Z)$ (unlike $\lambda(Z^m) = \lambda^m(Z)$), where $Z \in \mathbb{M}_n$.
    (e) $\sigma^m(XY) \neq \sigma\big((XY)^m\big) \neq \sigma(X^mY^m)$, where $X, Y \in \mathbb{M}_n$.

13. Prove Theorem 10.6.3 for the singular values of normal A and B, namely,

$$\sigma(AB) \prec_{\log} \sigma^{1/p}(A^pB^p) \prec_{\log} \sigma^{1/q}(A^qB^q) \prec_{\log} \sigma(A) \circ \sigma(B), \quad p \leq q.$$

14. If $A, B, C \in \mathbb{M}_n$ are such that $\left(\begin{smallmatrix} A & B \\ B^* & C \end{smallmatrix}\right)$ is positive semidefinite, show that

$$|\lambda(B^m)| \prec_{\log} \sigma(B^m) \prec_{\log} \sigma^m(B) \prec_{\mathrm{wlog}} \lambda^{m/2}(A) \circ \lambda^{m/2}(C), \ m = 1, 2, \ldots.$$

15. Let $A, B \in \mathbb{M}_n$ and let $m \geq 2$ be a positive integer. Show that

    (a) $\sigma(A^m B^m) \prec_{\log} \sigma^m(A) \circ \sigma^m(B)$ for all A, B.

    (b) $\sigma((AB)^m) \prec_{\log} \sigma^m(AB)$ for all A, B.

    (c) $\sigma^m(AB) \prec_{\log} \sigma(A^m B^m)$ for all normal A, B.

    (d) $\sigma^m(AB) \not\prec_{\log} \sigma(A^m B^m)$ for some non-normal A, B.

    (e) $\sigma^m(A) \circ (\sigma^m(B))^{\uparrow} \not\prec_{\log} \sigma((AB)^m)$ for some non-normal A, B.

16. Let A and B be $n \times n$ positive semidefinite matrices. If $(AB)^m = A^m B^m$ for some integer $m > 1$, show that $AB = BA$. Show by an example that the statement need not be true for Hermitian matrices A and B.

17. Let $A \circ B$ be the Hadamard (entrywise) product of $A, B \in \mathbb{M}_n$. Show by the example $A = \left(\begin{smallmatrix} 1 & 0 \\ 0 & 1 \end{smallmatrix}\right)$ and $B = \left(\begin{smallmatrix} 1 & 1 \\ 1 & 1 \end{smallmatrix}\right)$ that $\lambda(A \circ B) \not\prec_{\text{wlog}} \lambda(A) \circ \lambda(B)$ and $\sum_{i=1}^n \lambda_i^m(A) \lambda_{n-i+1}^m(B) \not\leq \operatorname{tr}((A \circ B)^m)$, where $m > 1$.

    [Note: $\lambda(A \circ B) \prec_w \lambda(A) \circ \lambda(B)$ is indeed true. One may pursue more on the Hadamard product of positive semidefinite matrices.]

18. Let A be a square matrix. Show that the following are equivalent:

    (a) A is normal.

    (b) $\operatorname{tr}(A^* A)^2 = \operatorname{tr}((A^*)^2 A^2)$.

    (c) $\sigma^2(A) = \sigma(A^2)$.

    (d) $\sigma^2(A) \prec \sigma(A^2)$.

    (e) $\sigma(A^2) \prec \sigma^2(A)$.

19. Referring to (10.60), show that for all n-square matrices A and B, $\sigma(A) \circ \sigma^{\uparrow}(B) \prec_{\log} \sigma(BA)$. [Note: $\sigma(AB) \neq \sigma(BA)$ in general.]

20. Let A and B be $n \times n$ positive semidefinite matrices. Show that

$$\lambda_1(A^r B^r) \leq \lambda_1^r(AB)$$

    for $0 \leq r \leq 1$, and the inequality is reversed for $r \geq 1$.

21. Show that the Gelfand-Naimark inequality (10.58) holds for rectangular matrices $A \in \mathbb{M}_{p \times n}$ and $B \in \mathbb{M}_{n \times q}$ (with all $i_t \leq \min\{p, q, n\}$), and use (10.59) to show the rank inequality $\operatorname{rank}(AB) \leq \min\{\operatorname{rank}(A), \operatorname{rank}(B)\}$.

22. Let A and B be $n \times n$ positive semidefinite matrices and let k be a positive integer, $k \leq n$. If $1 \leq i_1 < i_2 < \cdots < i_k \leq n$, show that

    (a) $\prod_{t=1}^k \lambda_t(AB) \geq \prod_{t=1}^k \lambda_{i_t}(A) \lambda_{n-i_t+1}(B)$.

    (b) $\sum_{t=1}^k \lambda_t(AB) \geq \sum_{t=1}^k \lambda_{i_t}(A) \lambda_{n-i_t+1}(B)$.

    (c) $\operatorname{tr}(AB) \geq \sum_{t=1}^n \lambda_t(A) \lambda_{n-t+1}(B)$.

23. Let A and B be $n \times n$ matrices and let k be a positive integer, $k \le n$. If $1 \le i_1 < i_2 < \cdots < i_k \le n$, show that

$$\prod_{t=1}^{k} \sigma_t(AB) \ge \prod_{t=1}^{k} \sigma_{i_t}(A)\sigma_{n-i_t+1}(B)$$

and

$$\prod_{t=1}^{k} \sigma_{i_t}(AB) \ge \prod_{t=1}^{k} \sigma_{i_t}(A)\sigma_{n-t+1}(B).$$

24. Let A be an $n \times n$ matrix and let $\sigma_i(A)$ and $\sigma_j(A)$ be two singular values of A, $i \ne j$. Show that for any nonsingular $n \times n$ matrix B,

$$\sigma_i(A)\sigma_j(A) \le \sigma_1(AB)\sigma_2(AB)\sigma_i(B^{-1})\sigma_j(B^{-1})$$

and

$$\sigma_i(A)\sigma_j(A) \le \sigma_i(AB)\sigma_j(AB)\sigma_1(B^{-1})\sigma_2(B^{-1}).$$

25. Let A, B, and C be $n \times n$ positive definite matrices. Show that

$$\ln \lambda(A^{-1}C) \prec \ln \lambda(A^{-1}B) + \ln \lambda(B^{-1}C).$$

26. Let A be an $n \times n$ matrix. Show that for any positive integer k,

$$\sigma(A^k) \prec_{\log} \sigma^k(A) \quad \text{and} \quad \lambda\big((A^*)^k A^k\big) \prec_{\log} \lambda\big((A^*A)^k\big).$$

Deduce that

$$\operatorname{tr}\big((A^*)^k A^k\big) \le \operatorname{tr}(A^*A)^k.$$

Show that equality holds if and only if $k = 1$ or A is a normal matrix.

27. Let $A \in \mathbb{M}_n$. Show that $A^k = |A|^k$ (where $|A| = (A^*A)^{1/2}$) for some positive integer k if and only if A is a normal matrix with eigenvalues in the form $r\xi^k$, where $r \ge 0$ and $\xi^k = 1$.

28. Let A and B be $n \times n$ positive semidefinite matrices. Let k and m be positive integers such that $k \le m$. Show that

$$\operatorname{tr}(A^k B^k)^m \le \operatorname{tr}(A^m B^m)^k$$

and the inequality is an equality if and only if $m = k$ or $AB = BA$.

29. Let $A, B \in \mathbb{M}_n$ be Hermitian. For any positive integer m, show that

$$|\operatorname{tr}(AB)^{2m}| \le \operatorname{tr}(A^{2m}B^{2m})$$

and

$$|\operatorname{tr}(A^m B^m)^2| \le \operatorname{tr}(A^{2m}B^{2m}).$$

Show by the example with $A = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$ and $B = \begin{pmatrix} -1 & 1 \\ 1 & 0 \end{pmatrix}$ that in general

$$|\operatorname{tr}(AB)^3| \not\le \operatorname{tr}(A^3B^3).$$

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10.7 Majorization and Unitarily Invariant Norms

This section shows a close relation between the (weak) majorization of singular values of matrices and unitarily invariant norms of matrices. To be precise, for $A, B \in \mathbb{M}_{m\times n}$, we show that $\sigma(A) \prec_w \sigma(B)$ if and only if $\|A\| \le \|B\|$ for all unitarily invariant norms $\|\cdot\|$ on $\mathbb{M}_{m\times n}$. This is done through the so-called symmetric gauge functions (ϕ) which connect the singular values (vectors) and the norms in the way that $\|A\| = \phi(\sigma(A))$ for all $A \in \mathbb{M}_{m\times n}$.

A *symmetric gauge function* is a function defined on $\mathbb{R}^n$ that satisfies the vector norm conditions (positivity, homogeneity, and triangle inequality; see Sect. 1.4 and Sect. 4.2) and it is invariant under permutations of the components of the vectors and changes of signs of the components. To be exact, $\phi : \mathbb{R}^n \mapsto \mathbb{R}$ is a *symmetric gauge function* if it satisfies the following.

(i) $\phi(x) \ge 0$ (for all x). And $\phi(x) = 0$ if and only if $x = 0$. (Positivity)

(ii) $\phi(cx) = |c|\phi(x)$, $c \in \mathbb{R}$. (Homogeneity)

(iii) $\phi(x + y) \le \phi(x) + \phi(y)$. (Triangle inequality)

(iv) $\phi(x_1, x_2, \ldots, x_n) = \phi(x_{p(1)}, x_{p(2)}, \ldots, x_{p(n)})$, p is any permutation.

(v) $\phi(x_1, x_2, \ldots, x_n) = \phi(\epsilon_1 x_1, \epsilon_2 x_2, \ldots, \epsilon_n x_n)$, every $\epsilon_i = 1$ or -1.

Note: A symmetric gauge function can be defined over $\mathbb{C}^n$ or over a symmetric convex set of $\mathbb{R}^n$ (such as $\mathbb{R}^n_+$. By symmetry of a set we mean the set is closed under the permutations of the components of its vectors).

Example 10.7.1 Symmetric gauge functions. Let $x = (x_1, \ldots, x_n) \in \mathbb{R}^n$.

(a) $\varphi(x) = \max_i |x_i|$.

(b) $\psi(x) = \sum_i |x_i|$.

(c) $\|x\|_{(k)} = \sum_{t=1}^k (|x|^{\downarrow})_t = \max_{1\le i_1<\cdots<i_k\le n} \sum_{t=1}^k |x_{i_t}|$. (Fan k-norm)

(d) $\|x\|_p = \Big(\sum_{i=1}^n |x_i|^p\Big)^{1/p}$, $p \ge 1$. (ℓ_p-*norm*) (see Problem 6.) ∎

Three basic properties of symmetric gauge functions ϕ on $\mathbb{R}^n$ follow.

(1) $\phi(x) = \phi(|x|)$ (for all x), by (v).

(2) $\phi(x)$ is convex, by (ii) and (iii).

(3) $0 \le x \le y \Rightarrow \phi(x) \le \phi(y)$.

We demonstrate the proof of (3) for the case $n = 2$. Let $x, y \in \mathbb{R}^2_+$ and

$x = (x_1, x_2) \le y = (y_1, y_2)$. Let $t_1 \in [0,1]$ be such that $t_1 y_1 = x_1$. Then

$$\begin{aligned}
\phi(x) &= \phi(x_1, x_2) = \phi(t_1 y_1, x_2) \\
&= \phi\left(\frac{1+t_1}{2}(y_1, x_2) + \frac{1-t_1}{2}(-y_1, x_2)\right) \\
&\le \frac{1+t_1}{2}\phi(y_1, x_2) + \frac{1-t_1}{2}\phi(-y_1, x_2) \\
&= \phi(y_1, x_2) \quad \text{because } \phi(-y_1, x_2) = \phi(y_1, x_2).
\end{aligned}$$

Likewise, $\phi(y_1, x_2) \le \phi(y_1, y_2) = \phi(y)$. It follows that $\phi(x) \le \phi(y)$.

Theorem 10.7.1 *Symmetric gauge functions are Schur-convex on $\mathbb{R}^n$, i.e.,*

$$x \prec y \quad \Rightarrow \quad \phi(x) \le \phi(y).$$

Proof. Let $x, y \in \mathbb{R}^n$ and $x \prec y$. We show that $\phi(x) \le \phi(y)$ for any symmetric gauge function ϕ on $\mathbb{R}^n$. By Theorem 10.2.2, there exists a doubly stochastic matrix A such that $x = yA$. By Theorem 5.6.7, every doubly stochastic matrix is a convex combination of permutation matrices.

Let $A = \sum_{i=1}^m t_i P_i$, where all $t_i > 0$ with sum 1 and P_i are permutation matrices. Then $x = yA = \sum_{i=1}^m t_i y P_i$. Since ϕ is convex, by (iv), we get

$$\phi(x) = \phi\Big(\sum_{i=1}^m t_i y P_i\Big) \le \sum_{i=1}^m t_i \phi(y P_i) = \sum_{i=1}^m t_i \phi(y) = \phi(y). \ \blacksquare$$

Theorem 10.7.2 *Let $x, y \in \mathbb{R}^n$. Then $|x| \prec_w |y|$ if and only if $\phi(x) \le \phi(y)$ for all symmetric gauge functions ϕ on $\mathbb{R}^n$.*

Proof. For sufficiency, take ϕ to be the Fan k-norm, $k = 1, 2, \ldots, n$ (Example 10.7.1 (c)). Then $\phi(x) \le \phi(y)$ yields $|x| \prec_w |y|$. For necessity, since $|x| \prec_w |y|$, there exists a nonnegative vector u such that $|x| \prec u \le |y|$. Theorem 10.7.1 implies that $\phi(x) = \phi(|x|) \le \phi(u) \le \phi(|y|) = \phi(y)$. $\blacksquare$

Now we turn our attention to the connection between the majorizations of singular values (in vectors) and the unitarily invariant norms of matrices. Recall that a *norm* or a *matrix-vector norm* (see Sect. 4.2) on the vector space $\mathbb{M}_{m\times n}$ is a function $\|\cdot\|$ from $\mathbb{M}_{m\times n}$ to $\mathbb{R}_+$ that satisfies

(i) $\|A\| \ge 0$ (for all A). And $\|A\| = 0$ if and only if $A = 0$.
(ii) $\|cA\| = |c|\|A\|$, $c \in \mathbb{C}$.
(iii) $\|A + B\| \le \|A\| + \|B\|$.

A norm on $\mathbb{M}_{m\times n}$ is further said to be a *unitarily invariant norm* if

$$\|UAV\| = \|A\| \text{ for all } A \in \mathbb{M}_{m\times n} \text{ and unitary } U \in \mathbb{M}_m \text{ and } V \in \mathbb{M}_n.$$

Example 10.7.2 A few well-known unitarily invariant norms.

(a) The spectral norm $\|A\|_{\rm sp} = \sigma_{\max}(A) = \sigma_1(A)$.

(b) The Frobenius norm $\|A\|_{_F} = \big(\operatorname{tr}(A^*A)\big)^{1/2} = \big(\sum_{i,j} |a_{ij}|^2\big)^{1/2}$.

(c) The trace (nuclear) norm $\|A\|_{\rm tr} = \operatorname{tr}\big((A^*A)^{1/2}\big) = \sum_i \sigma_i(A)$. ∎

Note that the submultiplicative condition $\|AB\| \le \|A\|\|B\|$ (on $\mathbb{M}_n$) is not required (in this section) for a matrix-vector norm on $\mathbb{M}_{m\times n}$. Example 4.2.1 shows that ℓ_1-norm is submultiplicative, while ℓ_∞-norm is not. Neither ℓ_1-norm nor ℓ_∞-norm is a unitarily invariant norm.

If ϕ is a symmetric gauge function on $\mathbb{R}^n$ and A is an $m \times n$ matrix, then $\sigma(A) \in \mathbb{R}^n_+$ and $\phi(\sigma(A))$ is well-defined. Moreover, $\phi(\sigma(UAV)) = \phi(\sigma(A))$ because $\sigma(UAV) = \sigma(A)$ for all $m \times m$ unitary U and $n \times n$ unitary V. The next two theorems best characterize the relation between unitarily invariant norms and singular values through symmetric gauge functions.

Theorem 10.7.3 *If $\phi : \mathbb{R}^n \to \mathbb{R}$ is a symmetric gauge function, then $\|A\|_\phi = \phi(\sigma(A))$, where $A \in \mathbb{M}_{m\times n}$, is a unitarily invariant norm on $\mathbb{M}_{m\times n}$.*

Proof. Conditions (i) and (ii) are obviously satisfied. To show (iii), let A and B be $m \times n$ matrices. Because $\sigma(A + B) \prec_w \sigma(A) + \sigma(B)$, we have

$$\begin{aligned} \|A + B\|_\phi &= \phi\big(\sigma(A + B)\big) \le \phi\big(\sigma(A) + \sigma(B)\big) \\ &\le \phi\big(\sigma(A)\big) + \phi\big(\sigma(B)\big) = \|A\|_\phi + \|B\|_\phi. \end{aligned}$$

Thus, $\|\cdot\|_\phi$ is a matrix-vector norm. It is unitarily invariant because $\sigma(UAV) = \sigma(A)$ for all $m \times m$ unitary U and $n \times n$ unitary V. ∎

Roughly speaking, the above theorem states that a symmetric gauge function is a unitarily invariant norm. The next theorem says that a unitarily invariant norm is (represented as) a symmetric gauge function.

Theorem 10.7.4 *If $\|\cdot\|$ is a unitarily invariant norm (which may not be submultiplicative) on $\mathbb{M}_{m\times n}$, then there exists a symmetric gauge function ϕ on $\mathbb{R}^n$ such that $\|A\| = \phi(\sigma(A))$ for all $m \times n$ matrices A.*

Proof. If $m \ge n$, for $x = (x_1, \ldots, x_n) \in \mathbb{R}^n$, we define an $m \times n$ matrix M_x to have (i, i)-entry x_i, $i = 1, 2, \ldots, n$, and 0 elsewhere. Let $\phi(x) = \|M_x\|$. Then ϕ is a function from $\mathbb{R}^n$ to $\mathbb{R}_+$. It is readily seen that ϕ satisfies (i) and (ii). For (iv) and (v), it suffices to notice that $\|PM_xQ\| = \|M_x\|$ for all matrices $P \in \mathbb{M}_m$ and $Q \in \mathbb{M}_n$, where P and Q are permutation matrices or diagonal matrices with 1 or -1 on the main diagonal. For (iii), we have

$$\begin{aligned} \phi(x + y) &= \|M_{x+y}\| = \|M_x + M_y\| \\ &\le \|M_x\| + \|M_y\| = \phi(x) + \phi(y). \end{aligned}$$

Thus, $\phi : \mathbb{R}^n \to \mathbb{R}_+$ is a symmetric gauge function.

Now for $A \in \mathbb{M}_{m\times n}$, let $A = UA_\sigma V$ be a singular value decomposition of A, where $A_\sigma = M_{\sigma(A)}$, $U \in \mathbb{M}_m$ and $V \in \mathbb{M}_n$ are unitary. Then

$$\|A\| = \|A_\sigma\| = \|M_{\sigma(A)}\| = \phi(\sigma(A)).$$

If $m < n$, we define $\|X\|\tilde{} = \|X^T\|$, where X is $n \times m$. Then $\|\cdot\|\tilde{}$ is a unitarily invariant norm on $\mathbb{M}_{n\times m}$. The preceding argument about $\mathbb{M}_{n\times m}$ with $n \geq m$ ensures a symmetric gauge function $\varphi : \mathbb{R}^m \to \mathbb{R}_+$ such that $\varphi(\sigma(X)) = \|X\|\tilde{} = \|X^T\|$. Define $\phi : \mathbb{R}^n \to \mathbb{R}_+$ by $\phi(x) = \varphi(\tilde{x})$, where $\tilde{x}$ is the vector consisting of the first m largest components of x in absolute value. Then ϕ is a symmetric gauge function on $\mathbb{R}^n$ (Problem 8).

Now for $A \in \mathbb{M}_{m\times n}$, $A^T \in \mathbb{M}_{n\times m}$, $\widetilde{\sigma(A)} \in \mathbb{R}^m$ and $\sigma(A) \in \mathbb{R}^n$ differ by $n - m$ zeros. Notice that A^*A, AA^*, and $\overline{AA^*} = (A^T)^*A^T$ have the same nonzero (positive) eigenvalues. Thus, $\widetilde{\sigma(A)} = \sigma(A^T)$. It follows that

$$\phi(\sigma(A)) = \varphi\big(\widetilde{\sigma(A)}\big) = \varphi(\sigma(A^T)) = \|A^T\|\tilde{} = \|A\|. \blacksquare$$

Theorem 10.7.5 (von Neumann) *Let $A, B \in \mathbb{M}_{m\times n}$. Then*

$$\sigma(A) \prec_w \sigma(B) \quad \Leftrightarrow \quad \|A\| \leq \|B\|$$

for all unitarily invariant norms $\|\cdot\|$ on $\mathbb{M}_{m\times n}$.

Proof. Theorem 10.7.4 says that unitarily invariant norms are essentially the same as symmetric gauge functions (via the vectors of singular values). On the other hand, Theorem 10.7.2 ensures that $\sigma(A) \prec_w \sigma(B)$ if and only if $\phi(\sigma(A)) \leq \phi(\sigma(B))$ for all symmetric gauge functions ϕ. $\blacksquare$

Example 10.7.3 A unitarily invariant (matrix-vector) norm need not be submultiplicative. For example, $\|A\| = \frac{1}{2}\|A\|_{\text{sp}} = \frac{1}{2}\sigma_1(A)$, where $A \in \mathbb{M}_n$. However, if $\|\cdot\|$ is unitarily invariant and $\|\operatorname{diag}(1, 0, \ldots, 0)\| = 1$ (or ≥ 1, see Problem 25), then $\|\cdot\|$ is indeed submultiplicative. Here is why:

Because $(\sigma_1(A), 0 \ldots, 0) \prec_w \sigma(A)$, by the von Neumann theorem, we have $\sigma_1(A) = \|\operatorname{diag}(\sigma_1(A), 0, \ldots, 0)\| \leq \|A\|$. On the other hand, $\sigma(AB) \prec_w \sigma(A) \circ \sigma(B) \leq \sigma_1(A)\sigma(B)$, we obtain $\|AB\| \leq \sigma_1(A)\|B\| \leq \|A\|\|B\|$. $\blacksquare$

Some matrix inequalities on unitarily invariant norms follow.

Corollary 10.7.1 *Let $\|\cdot\|$ be a unitarily invariant norm on $\mathbb{M}_n$. Then for all positive semidefinite matrices $A, B \in \mathbb{M}_n$ and complex numbers z*

$$\|A - |z|B\| \leq \|A + zB\| \leq \|A + |z|B\|. \tag{10.67}$$

Proof. By Theorem 10.5.6, $\sigma(A - |z|B) \prec_w \sigma(A + zB) \prec_w \sigma(A + |z|B)$. $\blacksquare$

Note: It is easy to see that (10.67) does not hold for Hermitian matrices in general. As an example, take $A = I_n$, $B = -I_n$, and $z = \pm 1$.

Corollary 10.7.2 *Let $\|\cdot\|$ be a unitarily invariant norm on $\mathbb{M}_n$. Then for all normal matrices $A, B \in \mathbb{M}_n$ and all positive integers m,*

$$\|(AB)^m\| \le \|A^m B^m\|. \tag{10.68}$$

Proof. By Theorem 10.7.5, it suffices to show that $\sigma((AB)^m) \prec_w \sigma(A^m B^m)$. Note that $\sigma((AB)^m) \prec_{\log} \sigma^m(AB)$ as $\sigma(X^m) \prec_{\log} \sigma^m(X)$ for any $X \in \mathbb{M}_n$. We show the stronger log-majorization inequality: $\sigma^m(AB) \prec_{\log} \sigma(A^m B^m)$.

Since A is normal, $(A^*A)^m = (A^*)^m A^m$; the same is true of B.

$$\begin{aligned}
\sigma^m(AB) &= \lambda^{m/2}(B^*A^*AB) = \big(\lambda^m(BB^*A^*A)\big)^{1/2} \text{ (now use (10.62))} \\
&\prec_{\log} \big(\lambda((BB^*)^m(A^*A)^m)\big)^{1/2} = \big(\lambda(B^m(B^*)^m(A^*)^m A^m)\big)^{1/2} \\
&= \big(\lambda((B^*)^m(A^*)^m A^m B^m)\big)^{1/2} = \sigma(A^m B^m). \ \blacksquare
\end{aligned}$$

In fact, Problem 15 (b) and (c) of Sect. 10.6 show that for normal A, B,

$$\sigma((AB)^m) \prec_{\log} \sigma^m(AB) \prec_{\log} \sigma(A^m B^m). \tag{10.69}$$

Note: The second inequality in (10.69) requires that A and B be normal.

Example 10.7.4 Let $A = \left(\begin{smallmatrix} 1 & 1 \\ 1 & -1 \end{smallmatrix}\right)$ and $B = \left(\begin{smallmatrix} -1 & 1 \\ 1 & 0 \end{smallmatrix}\right)$. A and B are real symmetric (thus, normal) matrices. By Corollary 10.7.2, $\|(AB)^m\| \le \|A^m B^m\|$ for any unitarily invariant norm $\|\cdot\|$ on $\mathbb{M}_2$ and any positive integer m.

For $m = 3$, by computation, we have $A^3B^3 = \left(\begin{smallmatrix} -2 & 2 \\ -10 & 6 \end{smallmatrix}\right)$ with $\sigma(A^3B^3)$ $= (11.98..., 0.66...)$ and $(AB)^3 = \left(\begin{smallmatrix} -2 & -1 \\ 2 & -3 \end{smallmatrix}\right)$ with $\sigma\big((AB)^3\big) = (3.62..., 2.20...)$. It is obvious that $\sigma\big((AB)^3\big) \prec_w \sigma(A^3B^3)$. [Note: $|\operatorname{tr}(AB)^3| \not\le \operatorname{tr}(A^3B^3)$.] $\blacksquare$

Example 10.7.5 Let $A = \left(\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right)$ (non-normal), $B = \left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$, and $m = 2$. Then $(AB)^2 = \left(\begin{smallmatrix} 1 & 0 \\ 0 & 0 \end{smallmatrix}\right)$ and $A^2B^2 = 0$. Thus, $\|(AB)^2\| > \|A^2B^2\|$ for any norm. This says that (10.68) does not hold for non-normal matrices in general. $\blacksquare$

Example 10.7.6 Let $A = \left(\begin{smallmatrix} 1 & 0 \\ 0 & 2 \end{smallmatrix}\right)$, $B = \left(\begin{smallmatrix} 10 & 1 \\ 1 & 0 \end{smallmatrix}\right)$, and $m = 2$. Both A and B are normal. We compute to get $(AB)^2 = \left(\begin{smallmatrix} 102 & 10 \\ 20 & 2 \end{smallmatrix}\right)$ and $A^2B^2 = \left(\begin{smallmatrix} 101 & 10 \\ 40 & 4 \end{smallmatrix}\right)$. Thus, $\|(AB)^2\|_{\ell_\infty} = 102 > \|A^2B^2\|_{\ell_\infty} = 101$ for the ℓ_∞-norm (Sect. 4.2). This says that (10.68) need not be true for non-unitarily invariant norms in general. $\blacksquare$

Theorem 10.7.6 (Fan Dominance Theorem) *Let $A, B \in \mathbb{M}_{m\times n}$. Then $\|A\| \le \|B\|$ for all unitarily invariant norms $\|\cdot\|$ on $\mathbb{M}_{m\times n}$ if and only if $\|A\|_{(k)} \le \|B\|_{(k)}$ for every k, where $\|A\|_{(k)}$ denotes the Fan k-norms:*

$$\|A\|_{(k)} = \sum_{i=1}^{k} \sigma_i(A), \quad k = 1, \dots, q = \min\{m, n\}.$$

Proof. The Fan k-norms (depending solely on singular values) are unitarily invariant, so necessity is obvious. Conversely, that $\|A\|_{(k)} \le \|B\|_{(k)}$ for every k is the same as $\sigma(A) \prec_w \sigma(B)$. Theorem 10.7.5 yields the sufficiency. $\blacksquare$

Problems

1. Sow that a symmetric vector norm is Schur-convex, that is, if $\|\cdot\|$ is a vector norm on $\mathbb{R}^n$ such that $\|xP\| = \|x\|$ for all $x \in \mathbb{R}^n$ and $n \times n$ permutation matrices P, show that $x \prec y$ implies $\|x\| \le \|y\|$.

2. Show that a symmetric gauge function is convex, that is, if φ is a symmetric gauge function on $\mathbb{R}^n$, show that $\varphi(\alpha x + \beta y) \le \alpha\varphi(x) + +\beta\varphi(y)$, where $x, y \in \mathbb{R}^n$, $\alpha + \beta = 1$, $\alpha, \beta \ge 0$.

3. Let $\|\cdot\|$ be a vector norm on $\mathbb{R}^n$. Show that $\|\,|x|\,\| = \|x\|$ for all $x \in \mathbb{R}^n$ if and only if $|x| \le |y| \Rightarrow \|x\| \le \|y\|$ for all $x, y \in \mathbb{R}^n$.

4. Let $\|\cdot\|$ be a unitarily invariant norm on $\mathbb{M}_n$. Show that

$$\|I_n\| \le \|\operatorname{diag}(\rho_1, \ldots, \rho_n)\|,$$

where $\rho_i = \sigma_i(A)/\sigma_{n-i+1}(A)$, $i = 1, \ldots, n$, for any invertible $A \in \mathbb{M}_n$.

5. For $m \times n$ matrix A, let $\|A\|_{(k)}$ be given as in Theorem 10.7.6; for $x \in \mathbb{R}^n$, let $\|x\|_{(k)}$ be defined as in Example 10.7.1 (c). Show that

$$\|A\|_{(k)} = \|\sigma(A)\|_{(k)}.$$

6. Show that the l_p-norm $\|\cdot\|_p$ (or the *Schatten p-norm*) and the Fan k-norm $\|\cdot\|_{(k)}$ defined on $\mathbb{R}^n$ are symmetric gauge functions, where

$$\|x\|_p = \Big(\sum_{i=1}^{n} |x_i|^p\Big)^{1/p}, \quad p \ge 1; \qquad \|x\|_{(k)} = \max_{1 \le i_1 < \cdots < i_k \le n} \sum_{t=1}^{k} |x_{i_t}|.$$

7. We have used $\|\cdot\|_{(k)}$ to denote the Fan k-norms for matrices (in Theorem 10.7.6) and for vectors (in Problem 6). Let $A = \left(\begin{smallmatrix} 1 & 0 \\ 1 & 0 \end{smallmatrix}\right)$. Compute $\|A\|_{(1)}$ for matrix A. If A is considered as a vector in $\mathbb{R}^4$ (by stacking the columns) $v(A) = (1, 1, 0, 0)^T$, find $\|v(A)\|_{(1)}$. Is $\|A\|_{(1)} = \|v(A)\|_{(1)}$?

8. Let φ be a symmetric gauge function on $\mathbb{R}^m$. Let $n > m$. For $x \in \mathbb{R}^n$, let $\tilde{x}$ be the vector of the first m largest components of x in absolute value. Show that $\phi(x) = \varphi(\tilde{x})$ is a symmetric gauge function on $\mathbb{R}^n$.

9. Extend the definition of symmetric gauge functions for $\mathbb{C}^n$ as follows. Replace the conditions (ii) by (ii') $\phi(cx) = |c|\phi(x)$, $c \in \mathbb{C}$ and (v) by (v') $\phi(x) = \phi(|x|)$, where $|x| = (|x_1|, \ldots, |x_n|)$. (Other conditions remain the same.) For $x, y \in \mathbb{C}^n$, show that $|x| \prec_w |y|$ if and only if $\phi(x) \le \phi(y)$ for all symmetric gauge functions ϕ on $\mathbb{C}^n$.

10. Show that $\|\cdot\|$ defined below is a unitarily invariant norm on $\mathbb{M}_n$:

$$\|A\| = \max\{|\operatorname{tr}(AX)| : X \in \mathbb{M}_n,\ \operatorname{tr}(X^*X) = 1\}.$$

11. Let $\|\cdot\|$ be a unitarily invariant norm on $\mathbb{M}_n$. Show that $\|A\| = \||A|\|$ for any $A \in \mathbb{M}_n$, where $|A| = (A^*A)^{1/2}$. Extend this to $\mathbb{M}_{m\times n}$.

12. Let $\alpha_1 \geq \alpha_2 \geq \cdots \geq \alpha_n \geq 0$ be given, not all α_i's are zero. Define

$$\|A\|_\alpha = \sum_{i=1}^{n} \alpha_i \sigma_i(A), \quad A \in \mathbb{M}_n.$$

Show that $\|\cdot\|_\alpha$ is a unitarily invariant norm on $\mathbb{M}_n$.

13. Let $A \in \mathbb{M}_n$ and let $\lambda_1, \lambda_2, \ldots, \lambda_n$ be the eigenvalues of A. Denote by Λ the diagonal matrix $\Lambda = \operatorname{diag}(\lambda_1, \lambda_2, \ldots, \lambda_n)$. If $\|\cdot\|$ is a unitarily invariant norm on $\mathbb{M}_n$, show that $\|\Lambda\| \leq \|A\|$.

14. Let A and B be $n\times n$ matrices with singular values $\sigma_1(A) \geq \cdots \geq \sigma_n(A)$ and $\sigma_1(B) \geq \cdots \geq \sigma_n(B)$, respectively. Show that

$$\|\operatorname{diag}(\sigma_1(A) - \sigma_1(B), \ldots, \sigma_n(A) - \sigma_n(B))\| \leq \|A - B\|$$

for unitarily invariant norms $\|\cdot\|$ on $\mathbb{M}_n$. Generalize this to $\mathbb{M}_{m\times n}$.

15. Let $A, B \in \mathbb{M}_n$. Show that for unitarily invariant norms $\|\cdot\|$ on $\mathbb{M}_n$,

$$2\|AB\| \leq \|A^*A\| + \|B^*B\|.$$

16. Let $A, B \in \mathbb{M}_n$. As is known, $|A+B| \leq |A| + |B|$ is not true in general (Sect. 8.6), where $|X| = (X^*X)^{1/2}$. However, show that the following inequality holds for all unitarily invariant norms $\|\cdot\|$ on $\mathbb{M}_n$,

$$\| \, |A+B| \, \| \leq \| \, |A| \, \| + \| \, |B| \, \|.$$

17. Show by example that $\| \, |A| - |B| \, \| \leq \|A - B\|$ need not be true for matrices, where $\|\cdot\|$ is either the spectral norm or the Frobenius norm.

18. Let $A, B \in \mathbb{M}_n$. Show that for unitarily invariant norms $\|\cdot\|$ on $\mathbb{M}_{2n}$,

$$\left\| \frac{A+B}{2} \oplus \frac{A+B}{2} \right\| \leq \|A \oplus B\| = \| \, |A| \oplus |B| \, \| \leq \|(|A| + |B|) \oplus 0\|.$$

19. Let A and B be $n \times n$ normal matrices. Let $A \circ B$ be the Hadamard (entrywise) product of A and B and $|A| = (A^*A)^{1/2}$. Show that

$$\|A \circ B\| \leq \| \, |A| \circ |B| \, \|$$

for unitarily invariant norms on $\mathbb{M}_n$. [Hint: Note that $\begin{pmatrix} |A| & A^* \\ A & |A^*| \end{pmatrix} \geq 0$.]

20. Let A and B be $n \times n$ matrices. If the product AB is normal, show that $\|AB\| \leq \|BA\|$ for all unitarily invariant norms $\|\cdot\|$ on $\mathbb{M}_n$.

21. Let p and q be positive real numbers such that $\frac{1}{p} + \frac{1}{q} = 1$. Show that

$$\|AB\|^{pq} = \|AB\|^{p+q} \leq \| \, |A|^p \| \, \| \, |B|^q \|$$

for all $A, B \in \mathbb{M}_n$ and for all unitarily invariant norms $\|\cdot\|$ on $\mathbb{M}_n$.

22. Let $A = U \begin{pmatrix} D & 0 \\ 0 & 0 \end{pmatrix} V$ be a singular value decomposition of $A \in \mathbb{M}_{m\times n}$, where U and V are $m \times m$ and $n \times n$ unitary matrices, respectively, D is an $r \times r$ positive diagonal matrix, and $r = \operatorname{rank}(A)$. (Assume $r > 0$ to avoid triviality.) Let $A^\dagger$ be the *Moore-Penrose inverse* of A, i.e., $A^\dagger = V^* \begin{pmatrix} D^{-1} & 0 \\ 0 & 0 \end{pmatrix} U^*$ by definition. Let $A, B \in \mathbb{M}_{m\times n}$. Show that for any $n \times n$ matrix X and unitarily invariant norm $\|\cdot\|$ on $\mathbb{M}_{m\times n}$,

$$\|A(A^\dagger B) - B\| \le \|AX - B\|.$$

23. Let A be an $m \times n$ matrix. Show that the Moore-Penrose inverse $A^\dagger$ of A is the only matrix in $\mathbb{M}_{m\times n}$ that satisfies all the equations

$$AXA = A, \;\; XAX = X, \;\; (AX)^* = AX, \;\; (XA)^* = XA.$$

24. Let $A \in \mathbb{M}_n$ and $\|\cdot\|$ denote a unitarily invariant norm on $\mathbb{M}_n$. Prove

    (a) For any Hermitian $X \in \mathbb{M}_n$, $\|A - \frac{1}{2}(A + A^*)\| \le \|A - X\|$.

    (b) For any skew-Hermitian $X \in \mathbb{M}_n$, $\|A - \frac{1}{2}(A - A^*)\| \le \|A - X\|$.

    (c) If $A = UP$ is a polar decomposition of A, where $U \in \mathbb{M}_n$ is unitary and $P \in \mathbb{M}_n$ is positive semidefinite, then for any unitary $X \in \mathbb{M}_n$, $\|A - U\| \le \|A - X\| \le \|A + U\|$.

    (d) If $A = UDV$ is a singular value decomposition of A, where $U, V \in \mathbb{M}_n$ are unitary, and $D \in \mathbb{M}_n$ is nonnegative diagonal, then for any unitary $X \in \mathbb{M}_n$, $\|A - UV\| \le \|A - X\| \le \|A + UV\|$.

    (e) If A is normal, then $\|A\| \le \|X^{-1}AX\|$ for all invertible $X \in \mathbb{M}_n$.

    (f) If A is positive semidefinite, then $\|A - I\| \le \|A - X\| \le \|A + I\|$ for any unitary $X \in \mathbb{M}_n$.

25. **(Horn-Johnson)** Let $\|\cdot\|$ be a unitarily invariant (matrix-vector) norm on $\mathbb{M}_n$. Show that the following statements are equivalent.

    (a) $\|\operatorname{diag}(1, 0, \ldots, 0)\| \ge 1$.

    (b) $\|A\| \ge \sigma_{\max}(A)$ for all matrices $A \in \mathbb{M}_n$.

    (c) $\|A\| \ge \rho(A)$ (the spectral radius of A) for all matrices $A \in \mathbb{M}_n$.

    (d) $\|AB\| \le \|A\|\,\|B\|$ for all matrices $A, B \in \mathbb{M}_n$.

26. Let $A, B \in \mathbb{M}_n$ be positive semidefinite and let $\|\cdot\|$ be a unitarily invariant norm on $\mathbb{M}_n$. Show that for every positive integer m,

$$\|A^m + B^m\| \le \|(A + B)^m\|.$$

[Hint: Use Problem 27 of Sect. 1.2 and $\|(A + zB)^m\| \le \|(A + |z|B)^m\|$.]

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Notation

$\lvert z\rvert$, 6	absolute value (or modulus) of complex number z
$\Re(z)$, 6	real part of complex number z
$\Im(z)$, 6	imaginary part of complex number z
$\mathbb{F}$, 1	field of numbers, i.e., $\mathbb{C}$ or $\mathbb{R}$ in this book
$\mathbb{C}$, 2	complex numbers
$\mathbb{R}$, 2	real numbers
$\mathbb{R}_+$, 428	nonnegative real numbers, i.e., $\mathbb{R}_+ = [0, \infty)$
$\mathbb{Q}$, 6	rational numbers
$\mathbb{F}^n$, 3	n-tuples, i.e., vectors with n components in $\mathbb{F}$
$\mathbb{C}^n$, 3	vectors (mostly in columns) with n complex components
$\mathbb{R}^2$, 2	ordered real pairs, i.e., $\mathbb{R}^2 = \{(x, y) : x, y \in \mathbb{R}\}$
$\mathbb{R}^3$, 2	ordered real triples, i.e., $\mathbb{R}^3 = \{(x, y, z) : x, y, z \in \mathbb{R}\}$
$\mathbb{R}^n$, 3	vectors (mostly in columns) with n real components
$\mathbb{R}_+^n$, 428	vectors with n nonnegative components
$\mathbb{R}_n[x]$, 3	real polynomials with degree at most n
$\mathbb{R}[x]$, 3	real polynomials with any finite degree and zero
$\mathbb{M}_n$, 12	$n \times n$ (i.e., n-square) complex matrices
$\mathbb{M}_{m\times n}$, 12	$m \times n$ complex matrices
$\mathbb{M}_n(\mathbb{F})$, 12	$n \times n$ matrices over field $\mathbb{F}$
$\mathbb{M}_{m\times n}(\mathbb{F})$, 12	$m \times n$ matrices over field $\mathbb{F}$
$\mathcal{C}(\mathbb{R})$, 8	continuous functions on $\mathbb{R}$
$\mathcal{C}[a, b]$, 10	continuous functions on interval $[a, b]$
$\mathcal{C}'(\mathbb{R})$, 19	functions with continuous derivatives on $\mathbb{R}$
$\mathcal{C}^\infty[a, b]$, 31	infinitely differentiable functions on interval $[a, b]$
e_i, 6	vector in $\mathbb{C}^n$ with ith component 1 and 0 elsewhere
δ_{ij}, 49	Kronecker delta, i.e., $\delta_{ij} = 1$ if $i = j$, and 0 otherwise
ω, 28	nth primitive root of unity
$\operatorname{Span} S$, 4	vector space spanned by the vectors in S
$\dim V$, 5	dimension of the vector space V
$V \cap W$, 6	intersection of sets V and W
$V \cup W$, 6	union of sets V and W
$V + W$, 6	sum of subspaces V and W
$V \oplus W$, 6	direct sum of subspaces V and W
$V \times W$, 11	$V \times W = \{(v, w) : v \in V, w \in W\}$, the Cartesian product of V and W
$\mathcal{D}_x$, 31	differential operator
S_n, 21	nth symmetric group, i.e., all permutations on $\{1, 2, \ldots, n\}$
$\langle u, v\rangle$, 46	inner product of vectors u and v (mostly v^*u in this book)
$\operatorname{Proj}_v u$, 52	orthogonal projection of u onto v
$\Theta_{u,v}$, 54	angle between vectors u and v, i.e., $\Theta_{u,\,v} = \cos^{-1}(\langle u, v\rangle / \lVert u\rVert\, \lVert v\rVert)$
$\theta_{u,v}$, 58	angle between lines of u and v, i.e., $\theta_{u,\,v} = \cos^{-1}(\lvert\langle u, v\rangle\rvert / \lVert u\rVert\, \lVert v\rVert)$
$\lvert x\rvert$, 426	absolute value vector $\lvert x\rvert = (\lvert x_1\rvert, \ldots, \lvert x_n\rvert)$ for $x = (x_1, \ldots, x_n)$
$\lVert x\rVert$, 50	length or norm of vector x, mostly Euclidean norm (ℓ_2-norm)
$\lVert x\rVert_p$, 484	l_p-norm of vector x, i.e., $\lVert x\rVert_p = \left(\sum_{i=1}^n \lvert x_i\rvert^p\right)^{1/p}$
$\lVert x\rVert_{(k)}$, 484	Fan k-norm of vector x, i.e., $\lVert x\rVert_{(k)} = \max_{i_1<\cdots<i_k} \sum_{t=1}^k \lvert x_{i_t}\rvert$

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x^T, 3	transpose of x; it is a row vector if x is a column vector in $\mathbb{C}^n$
$x^\perp$, 51	vectors orthogonal to vector x
$S^\perp$, 51	vector space orthogonal to set S, i.e., the orthogonal complement of S
$S_1 \perp S_2$, 51	$\langle x, y\rangle = 0$ for all $x \in S_1$ and $y \in S_2$
V_λ, 35	eigenspace of the eigenvalue λ
I_n, I, 13	identity matrix of order n
$A = (a_{ij})$, 12	matrix A with (i,j)-entry a_{ij}
A^T, 3,14	transpose of matrix (or vector) A
$\overline{A}$, 14	conjugate of matrix A
A^*, 14	conjugate transpose of matrix A
A^{-1}, 13, 22	inverse of matrix A
$A^\dagger$, 491	Moore-Penrose inverse of matrix A
A^0, A^k, 13	$A^0 = I$, $A^k = A \cdots A$ (k copies of A)
e^A, 110	$e^A = \sum_{k=0}^{\infty} \frac{1}{k!} A^k = I + A + \frac{A^2}{2!} + \frac{A^3}{3!} + \cdots$
$p(A)$, 29, 132	matrix polynomial in matrix A, where p is a polynomial
$f(A)$, A^r 131	function of normal matrix A, where f is a function, $r \in \mathbb{R}$
A_{11}, 61	principal submatrix of matrix A in the upper-left corner
$A(i\|j)$, 20	matrix by deleting the ith row and jth column of matrix A
E_{ij}, 26	matrix with (i,j)-entry 1, and 0 elsewhere
adj(A), 22	adjugate matrix of matrix A
det A, $\|A\|$, 20	determinant of matrix A ($\|A\|$ is used if A has a large size)
rank(A), 18	rank of matrix A
tr A, 38	trace of matrix A
diag S, 14	diagonal matrix with the elements of S on the diagonal
$\left\vert \begin{smallmatrix} A & B \\ C & D \end{smallmatrix} \right\vert$, 11	determinant of the 2×2 block matrix $\left(\begin{smallmatrix} A & B \\ C & D \end{smallmatrix}\right)$
Im(A), 16, 31	image of matrix or linear transformation A, i.e., Im $A = \{Ax\}$
Ker(A), 16, 31	kernel or null space of A, i.e., Ker $A = \{x : Ax = 0\}$
Nul(A), 15	null space of matrix A
Col(A), 16	column space of matrix A
Row(A), 80	row space of matrix A
$\mathcal{A}(W)$, 40	$\{\mathcal{A}(w) : w \in W\}$, the image of W under $\mathcal{A}$
$\mathcal{H}(A)$, 27, 471	Hermitian part of matrix A, i.e., $\mathcal{H}(A) = \frac{1}{2}(A + A^*)$
$\mathfrak{I}(A)$, 27, 471	imaginary part of A, i.e., $\mathfrak{I}(A) = \frac{A - A^*}{2i}$
$V_n(a_i)$, 217	n-square Vandermonde matrix of $a_1, \ldots, a_n$
$G(x_i)$, 49	Gram matrix of $x_1, \ldots, x_n$
$s_k(a_i)$, 220	kth elementary symmetric function of $a_1, \ldots, a_n$
$W(A)$, 161	numerical range of matrix A
$w(A)$, 166	numerical radius of matrix A
$\rho(A)$, 166	spectral radius of matrix A
$i_+(A)$, 343	number of positive eigenvalues of Hermitian matrix A
$i_-(A)$, 343	number of negative eigenvalues of Hermitian matrix A
$i_0(A)$, 343	number of zero eigenvalues of Hermitian matrix A
In(A), 343	inertia of Hermitian matrix A, i.e., In$(A) = (i_+(A), i_-(A), i_0(A))$
$\lambda_{\max}(A)$, 170	largest eigenvalue of matrix A
$\sigma_{\max}(A)$, 90	largest singular value of matrix A, i.e., the spectral norm of A
$\sigma_1(A)$, 90	largest singular value of matrix A; the same as $\sigma_{\max}(A)$
$\lambda_{\min}(A)$, 170	smallest eigenvalue of matrix A
$\sigma_{\min}(A)$, 90	smallest singular value of matrix A
$\lambda_i(A)$, 38	eigenvalue of matrix A; $\lambda_i^k(A) = (\lambda_i(A))^k$
$\sigma_i(A)$, 90	singular value of matrix A; $\sigma_i^k(A) = (\sigma_i(A))^k$
$\lambda(A)$, 454	eigenvalue vector of $A \in \mathbb{M}_n$, i.e., $\lambda(A) = (\lambda_1(A), \ldots, \lambda_n(A))$
$\sigma(A)$, 454	singular value vector of $A \in \mathbb{M}_{m\times n}$, i.e., $\sigma(A) = (\sigma_1(A), \ldots, \sigma_n(A))$
$\lambda^m(A)$, 461	$\lambda^m(A) = (\lambda_1^m(A), \ldots, \lambda_n^m(A)) = ((\lambda_1(A))^m, \ldots, (\lambda_n(A))^m)$
$\sigma^m(A)$, 461	$\sigma^m(A) = (\sigma_1^m(A), \ldots, \sigma_n^m(A)) = ((\sigma_1(A))^m, \ldots, (\sigma_n(A))^m)$
$d_i(\lambda)$, 141	invariant factors of λ-matrix $\lambda I - A$

$m_A(\lambda)$, 133	minimal polynomial of matrix A
$p_A(\lambda)$, 38	characteristic polynomial of matrix A, i.e., $p_A(\lambda) = \det(\lambda I - A)$
$A \geq 0$, 120, 241	A is positive semidefinite (or a nonnegative matrix in Sect. 5.7, 241)
$A > 0$, 120	A is positive definite (or a positive matrix in Sect. 5.7, 241)
$A \geq B$, 120	$A - B$ is positive semidefinite (or $a_{ij} \geq b_{ij}$ in Sect. 5.7, 241)
$A^{1/2}$, 122, 283	square root of positive semidefinite matrix A
$A[\alpha\|\beta]$, 189	submatrix of A indexed by α and β
$[A] = A[\alpha]$, 301	principal submatrix of A indexed by α
$\|\underline{A}\|$, 124,241	$\|A\| = (A^*A)^{1/2}$ (or $(\|a_{ij}\|)$ in Sect. 5.7, p. 241)
$\widetilde{A_{11}}$, 299	Schur complement of A_{11} in matrix A
$C_k(A)$, 189	kth compound matrix of matrix A
$\|A\|$, 173	norm of matrix A
$\|A\|_{\rm sp}$, 174	spectral norm of matrix A, i.e., $\|A\|_{\rm sp} = \sup_{\|x\|=1} \|Ax\| = \sigma_1(A)$
$\|A\|_F$, 47, 174	Frobenious norm of matrix A, i.e., $\|A\|_F = \left(\sum_{i=1}^n \sigma_i^2(A)\right)^{1/2}$
$\|A\|_{(k)}$, 177	Fan k-norm of matrix A, i.e., $\|A\|_{(k)} = \sum_{i=1}^k \sigma_i(A)$
$\|A\|_p$, 177	Schatten p-norm of matrix A, i.e., $\|A\|_p = \left(\sum_{i=1}^n \sigma_i^p(A)\right)^{1/p}$
$\|A\|_{\rm tr}$, 177	trace norm, $\|A\|_{\rm tr} = {\rm tr}(\|A\|) = \sigma_1(A) + \cdots + \sigma_r(A)$
$\|A\|_2$, 177	$\|A\|_2 = \|A\|_F = \left(\sum_{i,\,j} \|a_{ij}\|^2\right)^{1/2} = \left(\sum_{i=1}^n \sigma_i^2(A)\right)^{1/2}$
$[A, B]$, 398	commutator of A and B, i.e., $[A, B] = AB - BA$
$A \oplus B$, 68	direct sum of matrices A and B, i.e., $A \oplus B = \left(\begin{smallmatrix} A & 0 \\ 0 & B \end{smallmatrix}\right)$
$A \otimes B$, 182	Kronecker product of matrices A and B, i.e., $A \otimes B = (a_{ij}B)$
$A \circ B$, 182	Hadamard product of matrices A and B, i.e., $A \circ B = (a_{ij}b_{ij})$
$x \circ y$, 182	$x \circ y = (x_1y_1, \ldots, x_ny_n)$
x^m, 443	$x^m = (x_1^m, \ldots, x_n^m)$ if $x = (x_1, \ldots, x_n)$
$x^\downarrow$, 421	$x^\downarrow = (x_1^\downarrow, x_2^\downarrow, \ldots, x_n^\downarrow)$, where $x_1^\downarrow \geq x_2^\downarrow \geq \cdots \geq x_n^\downarrow$
$x^\uparrow$, 421	$x^\uparrow = (x_1^\uparrow, x_2^\uparrow, \ldots, x_n^\uparrow)$, where $x_1^\uparrow \leq x_2^\uparrow \leq \cdots \leq x_n^\uparrow$
$x \prec_w y$, 422	x is weakly majorized by y, i.e., $\sum_{i=1}^k x_i^\downarrow \leq \sum_{i=1}^k y_i^\downarrow$, $k \leq n$
$x \prec y$, 422	x is majorized by y, i.e., $x \prec_w y$ and $\sum_{i=1}^n x_i = \sum_{i=1}^n y_i$
$x \prec_{\rm wlog} y$, 446	x is weakly log-majorized by y, i.e., $\prod_{i=1}^k x_i^\downarrow \leq \prod_{i=1}^k y_i^\downarrow$, $k \leq n$
$x \prec_{\rm log} y$, 446	x is log-majorized by y, i.e., $x \prec_{\rm wlog} y$ and $\prod_{i=1}^n x_i = \prod_{i=1}^n y_i$

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